

NURTURE COURSE

FUNCTIONS
&
INVERSE TRIGONOMETRIC FUNCTIONS

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FUNCTIONS

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INVERSE TRIGONOMETRIC FUNCTIONS

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JEE (Advanced) Syllabus :

Real valued functions of a real variable, into, onto and one-to-one functions, sum, difference, product and quotient of two functions, composite functions, absolute value, polynomial, rational, trigonometric, exponential and logarithmic functions. Inverse trigonometric functions (principal value only).

JEE (Main) Syllabus :

Real valued functions, algebra of functions, polynomials, rational, trigonometric, logarithmic and exponential functions, inverse functions. Graphs of simple functions. Oneone, into and onto functions, composition of functions.

Inverse trigonometrical functions and their properties.

FUNCTION

1. CARTESIAN PRODUCT OF TWO SETS :

Given two non-empty sets A and B. The cartesian product $A \times B$ is the set of all ordered pairs of the form (a, b) where the first entry comes from set A & second comes from set B.

$$A \times B = \{(a, b) \mid a \in A, b \in B\}$$

e.g. $A = \{1, 2, 3\}$ $B = \{p, q\}$

$$A \times B = \{(1, p), (1, q), (2, p), (2, q), (3, p), (3, q)\}$$

Note :

- (i) If either A or B is the null set, then $A \times B$ will also be empty set, i.e. $A \times B = \phi$
- (ii) If $n(A) = p, n(B) = q$, then $n(A \times B) = pq$, where $n(X)$ denotes the number of elements in set X.
- (iii) A **Relation** R from set A to B is any subset of $A \times B$. If $A R B$ & $(a, b) \in R$ then b is image of a under R and a is preimage of b under R.

Note : If $n(A) = m, n(B) = n$, then number of relations defined from set A to B are $2^{mn} - 1$.

2. FUNCTION :

A relation R from set A to set B is called a function if each element of A is uniquely associated with some element of B. It is denoted by the symbol :

$$f : A \rightarrow B \text{ or } A \xrightarrow{f} B$$

which reads 'f' is a function from A to B 'or' f maps A to B,

If an element $a \in A$ is associated with an element $b \in B$, then b is called 'the f image of a' or 'image of a under f' or 'the value of the function f at a'. Also a is called the 'pre-image of b' or 'argument of b under the function f'. We write it as

$$b = f(a) \text{ or } f : a \rightarrow b \text{ or } f : (a, b)$$

Thus a function 'f' from set A to set B is subset of $A \times B$ in which each a belonging to A appears in one and only one ordered pair belonging to f.

Representation of Funtion :

(a) **Ordered pair :** Every function from $A \rightarrow B$ satisfies the following conditions :

(i) $f \subset A \times B$ (ii) $\forall a \in A$ there exist $b \in B$ and (iii) $(a, b) \in f \text{ \& } (a, c) \in f \Rightarrow b = c$

(b) **Formula based (uniformly/nonuniformly) :**

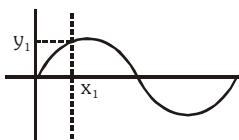
(i) $f : \mathbb{R} \rightarrow \mathbb{R}, y = f(x) = 4x, f(x) = x^2$ (uniformly defined)

(ii) $f(x) = \begin{cases} x+1 & -1 \leq x < 4 \\ -x & 4 \leq x < 7 \end{cases}$ (non-uniformly defined)

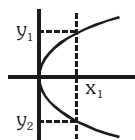
(iii) $f(x) = \begin{cases} x^2 & x \geq 0 \\ -x-1 & x < 0 \end{cases}$ (non-uniformly defined)

(c) **Graphical representation :**

If a vertical line cuts a given graph at more than one point then it can not be the graph of a function.



Graph (1)



Graph (2)

Graph(1) represent a function but graph(2) does not represent a function.

Every function is a relation but every relation is not necessarily a function.

3. DOMAIN, CO-DOMAIN & RANGE OF A FUNCTION :

Let $f: A \rightarrow B$, then the set A is known as the domain of f & the set B is known as co-domain of f . The set of f images of all the elements of A is known as the range of f .

Domain of $f = \{a \mid a \in A, (a, f(a)) \in f\}$

Range of $f = \{f(a) \mid a \in A, f(a) \in B\}$

- (a) If only the rule of function is given then the domain of the function is the set of those real numbers, where function is defined.
- (b) For a continuous function, the interval from minimum to maximum value of a function gives the range
- (c) It should be noted that range is a subset of co-domain.

Note :

- (i) The complete set of all positive real numbers is denoted by $\mathbb{R}^+$.
- (ii) The complete set of all negative real numbers is denoted by $\mathbb{R}^-$.
- (iii) The complete set of all real numbers other than zero is denoted by $\mathbb{R}_0$.
- (iv) The complete set of all integers is denoted by $\mathbb{Z}$.

Illustration 1 : Find the domain of following functions :

$$(i) y = \sqrt{5-2x} \quad (ii) y = \frac{1}{\sqrt{x-|x|}}$$

Solution : (i) $5 - 2x \geq 0 \Rightarrow x \leq \frac{5}{2} \therefore$ Domain is $(-\infty, 5/2]$

(ii) $x - |x| > 0 \Rightarrow |x| < x \Rightarrow x$ cannot take any real values $\therefore$ Domain is ϕ

Illustration 2 : Find the range of following functions :

$$(i) f(x) = \log_{\sqrt{2}}((x-1)^2 + 4) \quad (ii) f(x) = 3 - \cos x$$

Solution : (i) $f(x) = \log_{\sqrt{2}}((x-1)^2 + 4)$
 $4 \leq (x-1)^2 + 4 < \infty$
 $\Rightarrow \log_{\sqrt{2}} 4 \leq \log_{\sqrt{2}}((x-1)^2 + 4) < \infty$
 $\Rightarrow 4 \leq \log_{\sqrt{2}}((x-1)^2 + 4) < \infty$
 $\therefore$ Range of $f(x) = [4, \infty)$
 (ii) $f(x) = 3 - \cos x$
 $-1 \leq \cos x \leq 1$
 $2 \leq 3 - \cos x \leq 4$
 $\therefore$ Range of $f(x) = [2, 4]$

Do yourself - 1 :

(i) Find the domain of following functions :

$$(a) y = 1 - \log_{10} x \quad (b) y = \frac{1}{\sqrt{x^2 - 4x}}$$

(ii) Find the range of the following function :

$$(a) \log_4 \left| x + \frac{1}{x} \right| \quad (b) f(x) = \sin(3x^2 + 1)$$

$$(c) f(x) = 2 \sin \left(2x + \frac{\pi}{4} \right) \quad (d) f(x) = \cos \left(2x + \frac{\pi}{4} \right)$$

4. IMPORTANT TYPES OF FUNCTIONS :

(a) Polynomial Function :

If a function f is defined by $f(x) = a_0 x^n + a_1 x^{n-1} + a_2 x^{n-2} + \dots + a_{n-1} x + a_n$ where n is a non negative integer and $a_0, a_1, a_2, \dots, a_n$ are real numbers and $a_0 \neq 0$, then f is called a polynomial function of degree n . If n is odd, then polynomial is of odd degree, if n is even, then polynomial is of even degree.

Note :

- (i) Range of odd degree polynomial is always $\mathbb{R}$.
- (ii) Range of even degree polynomial is never $\mathbb{R}$.
- (iii) A polynomial of degree one with no constant term is called an odd linear function.
i.e. $f(x) = ax$, $a \neq 0$
- (iv) $f(x) = ax + b$, $a \neq 0$ is a linear polynomial
- (v) $f(x) = c$ is non linear polynomial (its degree is zero)
- (vi) $f(x) = 0$ is a polynomial but its degree is **not** defined
- (vii) There are two polynomial functions, satisfying the relation ;
 $f(x) \cdot f(1/x) = f(x) + f(1/x)$. They are :
(a) $f(x) = x^n + 1$ & (b) $f(x) = 1 - x^n$, where n is a positive integer.

(b) Algebraic Function :

A function ' f ' is called an algebraic function if it can be constructed using algebraic operations (such as addition, subtraction, multiplication, division, and taking radicals) within polynomials.

Example : $f(x) = \sqrt{x^2 + 1}$; $g(x) = \frac{x^4 - 16x^2}{x + \sqrt{x}} + (x - 2) \sqrt[3]{x + 1}$

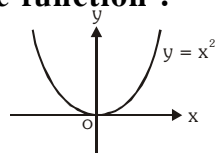
y is an algebraic function of x , if it is a function that satisfies an algebraic equation of the form $P_0(x) y^n + P_1(x) y^{n-1} + \dots + P_{n-1}(x) y + P_n(x) = 0$. Where n is a positive integer and $P_0(x), P_1(x), \dots$ are Polynomials in x . e.g. $x^3 + y^3 - 3xy = 0$.

Note :

- (i) All polynomial functions are Algebraic but not the converse.
- (ii) A function that is not algebraic is called **Transcendental Function**.

Basic algebraic function :

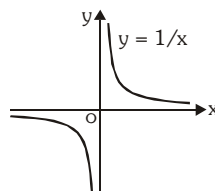
(i) $y = x^2$



Domain : $\mathbb{R}$

Range : $\mathbb{R}^+ \cup \{0\}$ or $[0, \infty)$

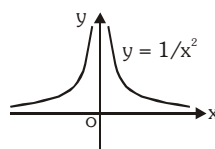
(ii) $y = \frac{1}{x}$



Domain : $\mathbb{R} - \{0\}$ or $\mathbb{R}_0$

Range : $\mathbb{R} - \{0\}$

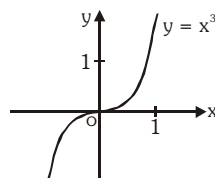
(iii) $y = \frac{1}{x^2}$



Domain : $\mathbb{R}_0$

Range : $\mathbb{R}^+$ or $(0, \infty)$

(iv) $y = x^3$



Domain : $\mathbb{R}$

Range : $\mathbb{R}$

(c) Rational function :

A rational function is a function of the form $y = f(x) = \frac{g(x)}{h(x)}$, where $g(x)$ & $h(x)$ are polynomials

&

$h(x) \neq 0$, **Domain :** $\mathbb{R} - \{x \mid h(x)=0\}$

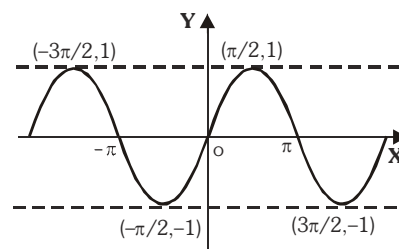
Any rational function is automatically an algebraic function.

(d) Trigonometric functions :**(i) Sine function**

$$f(x) = \sin x$$

Domain : $\mathbb{R}$

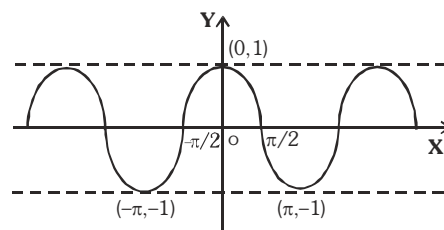
Range : $[-1, 1]$, period 2π

**(ii) Cosine function**

$$f(x) = \cos x$$

Domain : $\mathbb{R}$

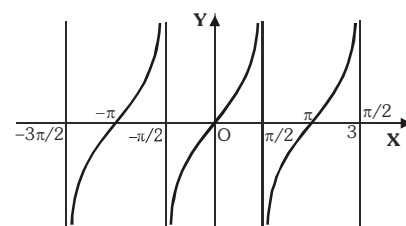
Range : $[-1, 1]$, period 2π

**(iii) Tangent function**

$$f(x) = \tan x$$

$$\text{Domain : } \mathbb{R} - \left\{ x \mid x = \frac{(2n+1)\pi}{2}, n \in \mathbb{I} \right\}$$

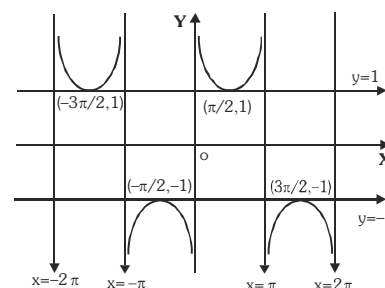
Range : $\mathbb{R}$, period π

**(iv) Cosecant function**

$$f(x) = \operatorname{cosec} x$$

Domain : $\mathbb{R} - \{x \mid x = n\pi, n \in \mathbb{I}\}$

Range : $\mathbb{R} - (-1, 1)$, period 2π

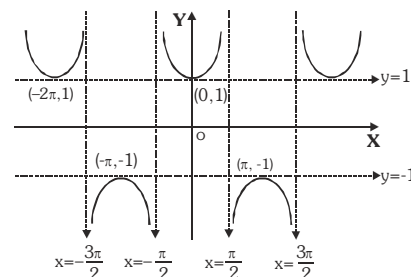


(v) Secant function

$$f(x) = \sec x$$

$$\text{Domain : } \mathbb{R} - \{x | x = (2n + 1) \pi/2 : n \in \mathbb{I}\}$$

$$\text{Range : } \mathbb{R} - (-1, 1), \text{ period } 2\pi$$

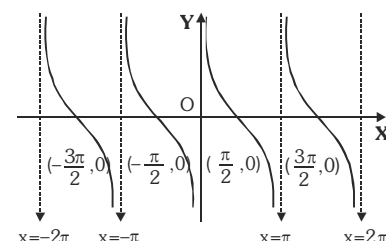


(vi) Cotangent function

$$f(x) = \cot x$$

$$\text{Domain : } \mathbb{R} - \{x | x = n\pi, n \in \mathbb{I}\}$$

$$\text{Range : } \mathbb{R}, \text{ period } \pi$$



(e) Exponential and Logarithmic Function :

A function $f(x) = a^x (a > 0, a \neq 1, x \in \mathbb{R})$ is called an exponential function. The inverse of the exponential function is called the logarithmic function, i.e. $g(x) = \log_a x$.

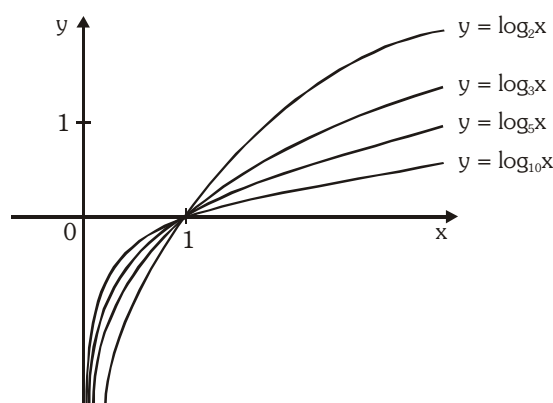
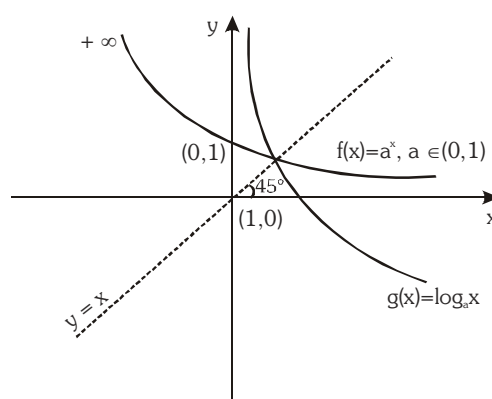
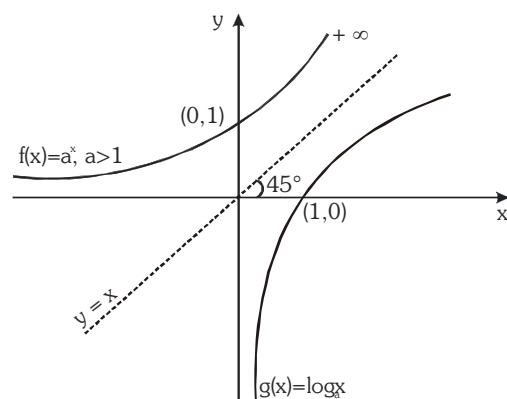
Note that $f(x)$ & $g(x)$ are inverse of each other & their graphs are as shown. (If functions are mirror image of each other about the line $y = x$)

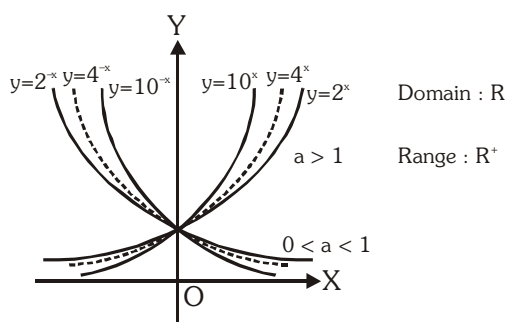
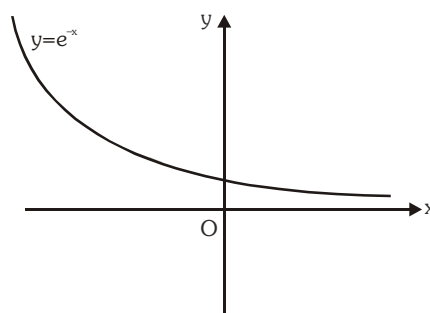
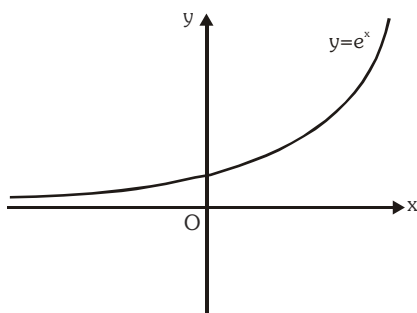
Domain of a^x is $\mathbb{R}$

Range $\mathbb{R}^+$

Domain of $\log_a x$ is $\mathbb{R}^+$

Range $\mathbb{R}$





Note-1 : $f(x) = a^{1/x}$, $a > 0$ **Domain :** $\mathbb{R} - \{0\}$ **Range :** $\mathbb{R}^+ - \{1\}$

Note-2 : $f(x) = \log_x a = \frac{1}{\log_a x}$ **Domain :** $\mathbb{R}^+ - \{1\}$ **Range :** $\mathbb{R} - \{0\}$
 $(a > 0) (a \neq 1)$

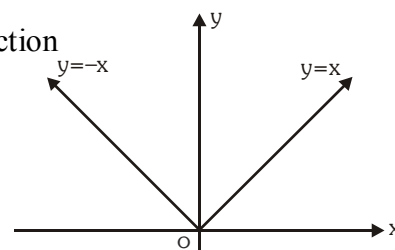
(f) Absolute Value Function :

A function $y = f(x) = |x|$ is called the absolute value function or Modulus function. It is defined as :

$$y = |x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$

For $f(x) = |x|$, domain is $\mathbb{R}$ and range is $[0, \infty)$

For $f(x) = \frac{1}{|x|}$, domain is $\mathbb{R} - \{0\}$ and range is $\mathbb{R}^+$.



(g) Signum Function :

A function $y = f(x) = \text{Sgn}(x)$ is defined as follows :

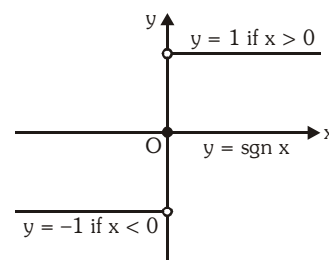
$$y = f(x) = \begin{cases} 1 & \text{for } x > 0 \\ 0 & \text{for } x = 0 \\ -1 & \text{for } x < 0 \end{cases}$$

It is also written as $\text{Sgn } x = |x|/x$; $x \neq 0$;
 $= 0$; $x = 0$

Note : $f(x) = (\text{sgn}(x))x \Rightarrow f(x) = |x|$

Domain : $\mathbb{R}$

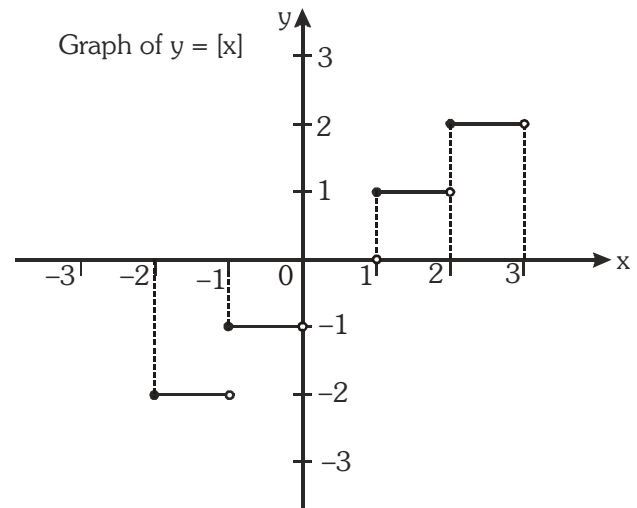
Range : $\{-1, 0, 1\}$



(h) **Greatest integer or step up function :**

The function $y = f(x) = [x]$ is called the greatest integer function where $[x]$ denotes the greatest integer less than or equal to x . Note that for :

x	$[x]$
$[-2, -1)$	-2
$[-1, 0)$	-1
$[0, 1)$	0
$[1, 2)$	1



Domain : $\mathbb{R}$

Range : $\mathbb{I}$

Properties of greatest integer function :

(i) $[x] \leq x < [x] + 1$ and $x - 1 < [x] \leq x$, $0 \leq x - [x] < 1$

(ii) $[x + m] = [x] + m$, if m is an integer.

(iii) $[x] + [-x] = \begin{cases} 0, & x \in \mathbb{I} \\ -1, & x \notin \mathbb{I} \end{cases}$

Illustration 3 : If $y = 2[x] + 3$ & $y = 3[x - 2] + 5$, then find $[x + y]$ where $[.]$ denotes greatest integer function.

Solution : $y = 3[x - 2] + 5 = 3[x] - 1$

so $3[x] - 1 = 2[x] + 3$

$[x] = 4 \Rightarrow 4 \leq x < 5$

then $y = 11$

so $x + y$ will lie in the interval $[15, 16)$

so $[x + y] = 15$

Illustration 4 : Find the value of $\left[\frac{1}{2}\right] + \left[\frac{1}{2} + \frac{1}{1000}\right] + \dots + \left[\frac{1}{2} + \frac{2946}{1000}\right]$ where $[.]$ denotes greatest integer function ?

Solution : $\left[\frac{1}{2}\right] + \left[\frac{1}{2} + \frac{1}{1000}\right] + \dots + \left[\frac{1}{2} + \frac{499}{1000}\right] + \left[\frac{1}{2} + \frac{500}{1000}\right] + \dots + \left[\frac{1}{2} + \frac{1499}{1000}\right] + \left[\frac{1}{2} + \frac{1500}{1000}\right] + \dots$

$+ \left[\frac{1}{2} + \frac{2499}{1000}\right] + \left[\frac{1}{2} + \frac{2500}{1000}\right] + \dots + \left[\frac{1}{2} + \frac{2946}{1000}\right]$

$= 0 + 1 \times 1000 + 2 \times 1000 + 3 \times 447 = 3000 + 1341 = 4341$

Ans.

(i) Fractional part function :

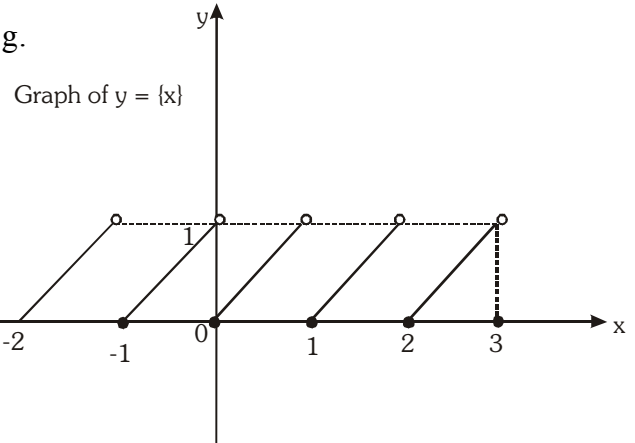
It is defined as : $g(x) = \{x\} = x - [x]$ e.g.

the fractional part of the number 2.1 is $2.1 - 2 = 0.1$ and the fractional part of -3.7 is 0.3 . The period of this function is 1 and graph of this function is as shown.

x	$\{x\}$
$[-2, -1)$	$x+2$
$[-1, 0)$	$x+1$
$[0, 1)$	x
$[1, 2)$	$x-1$

Domain : $\mathbb{R}$

Range : $[0, 1)$

**Properties of fractional part function :**

$$(i) \quad 0 \leq \{x\} < 1 \quad (ii) \quad \{[x]\} = [\{x\}] = 0 \quad (iii) \quad \{\{x\}\} = \{x\}$$

$$(iv) \quad \{x+m\} = \{x\}, \quad m \in \mathbb{I} \quad (v) \quad \{x\} + \{-x\} = \begin{cases} 1, & x \notin \mathbb{I} \\ 0, & x \in \mathbb{I} \end{cases}$$

Illustration 5 : Solve the equation $|2x - 1| = 3[x] + 2\{x\}$ where $[.]$ denotes greatest integer and $\{.\}$ denotes fractional part function.

Solution : We are given that, $|2x - 1| = 3[x] + 2\{x\}$

Let, $2x - 1 \leq 0$ i.e. $x \leq \frac{1}{2}$. The given equation yields.

$$1 - 2x = 3[x] + 2\{x\}$$

$$\Rightarrow 1 - 2[x] - 2\{x\} = 3[x] + 2\{x\} \Rightarrow 1 - 5[x] = 4\{x\} \Rightarrow \{x\} = \frac{1 - 5[x]}{4}$$

$$\Rightarrow 0 \leq \frac{1 - 5[x]}{4} < 1 \Rightarrow 0 \leq 1 - 5[x] < 4 \Rightarrow -\frac{3}{5} < [x] \leq \frac{1}{5}$$

Now, $[x] = 0$ as zero is the only integer lying between $-\frac{3}{5}$ and $\frac{1}{5}$

$$\Rightarrow \{x\} = \frac{1}{4} \Rightarrow x = \frac{1}{4} \text{ which is less than } \frac{1}{2}, \text{ Hence } \frac{1}{4} \text{ is one solution.}$$

Now, let $2x - 1 > 0$ i.e. $x > \frac{1}{2}$

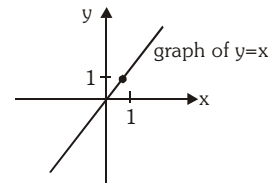
$$\Rightarrow 2x - 1 = 3[x] + 2\{x\} \Rightarrow 2[x] + 2\{x\} - 1 = 3[x] + 2\{x\}$$

$$\Rightarrow [x] = -1 \Rightarrow -1 \leq x < 0 \text{ which is not a solution as } x > \frac{1}{2}$$

$$\Rightarrow x = \frac{1}{4} \text{ is the only solution.}$$

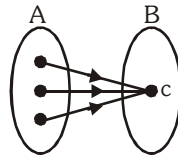
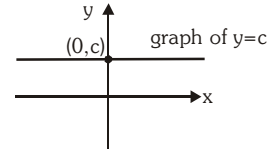
(j) **Identity function :**

The function $f : A \rightarrow A$ defined by $f(x) = x \quad \forall x \in A$ is called the identity of A and is denoted by I_A . It is easy to observe that identity function defined on R is a bijection.



(k) **Constant function :**

A function $f : A \rightarrow B$ is said to be a constant function if every element of A has the same f image in B . Thus $f : A \rightarrow B ; f(x) = c, \quad \forall x \in A, c \in B$ is a constant function. Note that the range of a constant function is a singleton.



Do yourself - 2 :

- (i) Let $\{x\}$ & $[x]$ denotes the fraction and integral part of a real number x respectively, then match the column.

Column-I

- (A) $[x^2] \geq 4$
 (B) $[x]^2 - 5[x] + 6 = 0$
 (C) $x = \{x\}$
 (D) $[x] < -5$

Column-II

- (p) $x \in [2, 4]$
 (q) $x \in (-\infty, -2] \cup [2, \infty)$
 (r) $x \in (-\infty, -5)$
 (s) $x \in \{-2\}$
 (t) $x \in [0, 1]$

5. ALGEBRAIC OPERATIONS ON FUNCTIONS :

If f & g are real valued functions of x with domain set A, B respectively, $f + g, f - g, (f \cdot g)$ & (f/g) as follows :

- (a) $(f \pm g)(x) = f(x) \pm g(x)$ domain in each case is $A \cap B$
 (b) $(f \cdot g)(x) = f(x) \cdot g(x)$ domain is $A \cap B$
 (c) $\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)}$ domain $A \cap B - \{x | g(x) = 0\}$

Illustration 6 : Find the domain of the following function :

(i) $y = \log_{(x-4)}(x^2 - 11x + 24)$ (ii) $f(x) = \log_2 \left(-\log_{1/2} \left(1 + \frac{1}{\sqrt[4]{x}} \right) - 1 \right)$

Solution :

(i) $y = \log_{(x-4)}(x^2 - 11x + 24)$

Here 'y' would assume real value if,

$$x - 4 > 0 \text{ and } \neq 1, x^2 - 11x + 24 > 0 \Rightarrow x > 4 \text{ and } \neq 5, (x - 3)(x - 8) > 0 \\ \Rightarrow x > 4 \text{ and } \neq 5, x < 3 \text{ or } x > 8 \Rightarrow x > 8 \Rightarrow \text{Domain } (y) = (8, \infty)$$

(ii) We have $f(x) = \log_2 \left(-\log_{1/2} \left(1 + \frac{1}{\sqrt[4]{x}} \right) - 1 \right)$

$f(x)$ is defined if $-\log_{1/2} \left(1 + \frac{1}{\sqrt[4]{x}} \right) - 1 > 0$

or if $\log_{1/2} \left(1 + \frac{1}{\sqrt[4]{x}} \right) < -1$ or if $\left(1 + \frac{1}{\sqrt[4]{x}} \right) > (1/2)^{-1}$

or if $1 + \frac{1}{\sqrt[4]{x}} > 2$ or if $\frac{1}{\sqrt[4]{x}} > 1$ or if $x^{1/4} < 1$ or if $0 < x < 1$

$\therefore D(f) = (0, 1)$

Illustration 7 : Find the domain $f(x) = \frac{1}{\sqrt{[|x| - 5] - 11}}$ where $[.]$ denotes greatest integer function.

Solution : $[|x| - 5] > 11$

so $[|x| - 5] > 11$ or $[|x| - 5] < -11$

$[|x|] > 16$ $[|x|] < -6$

$|x| \geq 17$ or $|x| < -6$ (Not Possible)

$\Rightarrow x \leq -17$ or $x \geq 17$

so $x \in (-\infty, -17] \cup [17, \infty)$

Illustration 8 : Find the range of following functions :

(i) $f(x) = \frac{1}{8 - 3 \sin x}$ (ii) $f(x) = \log_{\sqrt{2}} (2 - \log_2 (16 \sin^2 x + 1))$

Solution : (i) $f(x) = \frac{1}{8 - 3 \sin x}$

$-1 \leq \sin x \leq 1$

$\therefore$ Range of $f = \left[\frac{1}{11}, \frac{1}{5} \right]$

(ii) $f(x) = \log_{\sqrt{2}} (2 - \log_2 (16 \sin^2 x + 1))$

$1 \leq 16 \sin^2 x + 1 \leq 17$

$\therefore 0 \leq \log_2 (16 \sin^2 x + 1) \leq \log_2 17$

$\therefore 2 - \log_2 17 \leq 2 - \log_2 (16 \sin^2 x + 1) \leq 2$

Now consider $0 < 2 - \log_2 (16 \sin^2 x + 1) \leq 2$

$\therefore -\infty < \log_{\sqrt{2}} [2 - \log_2 (16 \sin^2 x + 1)] \leq \log_{\sqrt{2}} 2 = 2$

$\therefore$ the range is $(-\infty, 2]$

Illustration 9 : Find the range of $f(x) = \frac{x - [x]}{1 + x - [x]}$, where $[.]$ denotes greatest integer function.

Solution : $y = \frac{x - [x]}{1 + x - [x]} = \frac{\{x\}}{1 + \{x\}}$
 $\therefore \frac{1}{y} = \frac{1}{\{x\}} + 1 \Rightarrow \frac{1}{\{x\}} = \frac{1-y}{y} \Rightarrow \{x\} = \frac{y}{1-y}$
 $0 \leq \{x\} < 1 \Rightarrow 0 \leq \frac{y}{1-y} < 1$
 Range = $[0, 1/2)$

Do yourself - 3 :

(i) Find domain of following functions :

(a) $f(x) = \sin(\sqrt{1-x^2}) + \sqrt{x+2} + \frac{1}{\log_{10}^{(x+1)}}$

(b) $f(x) = \sqrt{\frac{(2x+1)}{x^3 - 3x^2 + 2x}}$

(ii) Find range of following functions :

(a) $f(x) = \log_2(\log_{1/2}(x^2 + 4x + 4))$

(b) $f(x) = \frac{1}{2 - \cos 3x}$

6. EQUAL OR IDENTICAL FUNCTION :

Two function f & g are said to be equal if :

- (a) The domain of f = the domain of g
- (b) The co-domain of f = co-domain of g and
- (c) $f(x) = g(x)$, for every x belonging to their common domain (i.e. should have the same graph)

Illustration 10 : The functions $f(x) = \log(x-1) - \log(x-2)$ and $g(x) = \log\left(\frac{x-1}{x-2}\right)$ are identical when x

lies in the interval

- (A) $[1, 2]$ (B) $[2, \infty)$ (C) $(2, \infty)$ (D) $(-\infty, \infty)$

Solution : Since $f(x) = \log(x-1) - \log(x-2)$.
 Domain of $f(x)$ is $x > 2$ or $x \in (2, \infty)$ (i)
 $g(x) = \log\left(\frac{x-1}{x-2}\right)$ is defined if $\frac{x-1}{x-2} > 0 \Rightarrow x \in (-\infty, 1) \cup (2, \infty)$ (ii)
 From (i) and (ii), $x \in (2, \infty)$. **Ans. (C)**

Do yourself - 4 :

(i) Are the following functions identical ?

$$(a) f(x) = \frac{x}{x^2} \text{ \& } \phi(x) = \frac{x^2}{x} \quad (b) f(x) = x \text{ \& } \phi(x) = \sqrt{x^2} \quad (c) f(x) = \log_{10} x^2 \text{ \& } \phi(x) = 2 \log_{10} |x|$$

7. HOMOGENEOUS FUNCTIONS :

A function is said to be homogeneous with respect to any set of variables when each of its terms is of the same degree with respect to those variables.

For examples $5x^2 + 3y^2 - xy$ is homogenous in x & y . Symbolically if, $f(tx, ty) = t^n f(x, y)$ then $f(x, y)$ is homogeneous function of degree n .

Illustration 11 : Which of the following function is not homogeneous ?

$$(A) x^3 + 8x^2y + 7y^3 \quad (B) y^2 + x^2 + 5xy \quad (C) \frac{xy}{x^2 + y^2} \quad (D) \frac{2x - y + 1}{2y - x + 1}$$

Solution : It is clear that (D) does not have the same degree in each term. **Ans. (D)**

8. BOUNDED FUNCTION :

A function is said to be bounded if there exists a finite M such that $|f(x)| \leq M, \forall x \in D_f$.

9. IMPLICIT & EXPLICIT FUNCTION :

A function defined by an equation not solved for the dependent variable is called an **implicit function**. e.g. the equations $x^3 + y^3 = 1$ & $x^y = y^x$, defines y as an implicit function. If y has been expressed in terms of x alone then it is called an **Explicit function**.

Illustration 12 : Which of the following function is implicit function ?

$$(A) y = \frac{x^2 + e^x + 5}{\sqrt{1 - \cos^{-1} x}} \quad (B) y = x^2 \quad (C) xy - \sin(x + y) = 0 \quad (D) y = \frac{x^2 \log x}{\sin x}$$

Solution : It is clear that in (C) y is not clearly expressed in x . **Ans. (C)**

Do yourself - 5 :

(i) Find the boundness of the function $f(x) = \frac{x^2}{x^4 + 1}$

(ii) Which of the following function is implicit function ?

$$(A) xy - \cos(x + y) = 0$$

$$(B) y = x^3$$

$$(C) y = \log(x^2 + x + 1)$$

$$(D) y = |x|$$

(iii) Convert the implicit form into the explicit function :

$$(a) xy = 1 \quad (b) x^2y = 1.$$

10. APPLICATIONS OF FUNCTIONAL RULE :

Illustration 13 : Determine all functions f satisfying the functional relation.

$$f(x) + f\left(\frac{1}{1-x}\right) = \frac{2(1-2x)}{x(1-x)} \text{ where } x \in \mathbb{R} - \{0, 1\}$$

Solution : Given $f(x) + f\left(\frac{1}{1-x}\right) = \frac{2(1-2x)}{x(1-x)} = \frac{2}{x} - \frac{2}{1-x}$... (i)

Replacing x by $\frac{1}{1-x}$ we obtain

$$f\left(\frac{1}{1-x}\right) + f\left(\frac{1}{1-\frac{1}{1-x}}\right) = 2(1-x) - \frac{2}{1-\frac{1}{1-x}}$$

$$\Rightarrow f\left(\frac{1}{1-x}\right) + f\left(1 - \frac{1}{x}\right) = -2x + \frac{2}{x}$$
 ... (ii)

Again replacing x by $\left(1 - \frac{1}{x}\right)$ in (i) we obtain

$$\Rightarrow f\left(1 - \frac{1}{x}\right) + f\left(\frac{1}{1-\left(1-\frac{1}{x}\right)}\right) = \frac{2}{1-\frac{1}{x}} - \frac{2}{1-\left(1-\frac{1}{x}\right)}$$

$$\Rightarrow f\left(1 - \frac{1}{x}\right) + f(x) = \frac{2x}{x-1} - 2x$$
 (iii)

subtracting (ii) from (i) then

$$f(x) - f\left(1 - \frac{1}{x}\right) = 2x - \frac{2}{1-x}$$

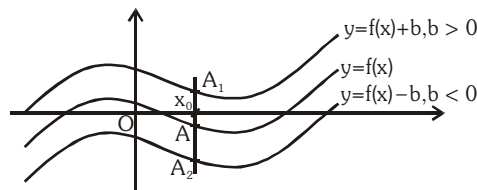
Now adding (iii) and (iv) we get

$$2f(x) = \frac{2x}{x-1} - \frac{2}{1-x}$$

$$\Rightarrow f(x) = \frac{x+1}{x-1}$$

11. BASIC TRANSFORMATIONS ON GRAPHS :

- (i) Drawing the graph of
- $y = f(x) + b$
- ,
- $b \in \mathbb{R}$
- , from the known graph of
- $y = f(x)$



It is obvious that domain of $f(x)$ and $f(x) + b$ are the same. Let us take any point x_0 in the domain of $f(x)$. $y|_{x=x_0} = f(x_0)$.

The corresponding point on $f(x) + b$ would be $f(x_0) + b$.

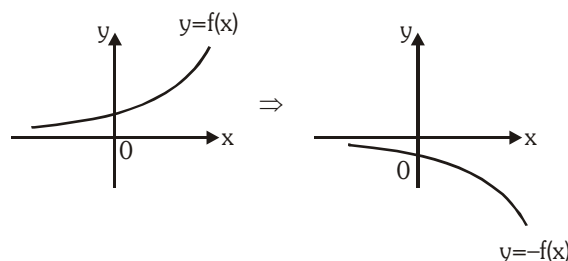
For $b > 0 \Rightarrow f(x_0) + b > f(x_0)$ it means that the corresponding point on $f(x) + b$ would be lying at a distance 'b' units above the point on $f(x)$.

For $b < 0 \Rightarrow f(x_0) + b < f(x_0)$ it means that the corresponding point on $f(x) + b$ would be lying at a distance 'b' units below the point on $f(x)$.

Accordingly the graph of $f(x) + b$ can be obtained by translating the graph of $f(x)$ either in the positive y-axis direction (if $b > 0$) or in the negative y-axis direction (if $b < 0$), through a distance $|b|$ units.

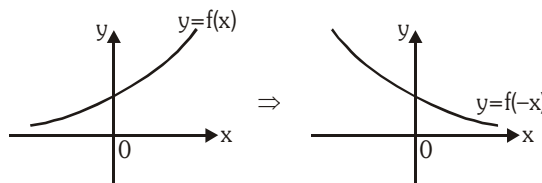
- (ii) Drawing the graph of
- $y = -f(x)$
- from the known graph of
- $y = f(x)$

To draw $y = -f(x)$, take the image of the curve $y = f(x)$ in the x-axis as plane mirror.



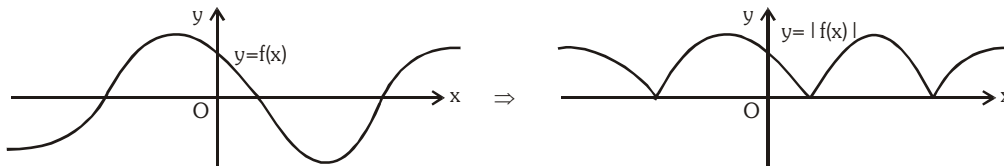
- (iii) Drawing the graph of
- $y = f(-x)$
- from the known graph of
- $y = f(x)$

To draw $y = f(-x)$, take the image of the curve $y = f(x)$ in the y-axis as plane mirror.



- (iv) Drawing the graph of
- $y = |f(x)|$
- from the known graph of
- $y = f(x)$

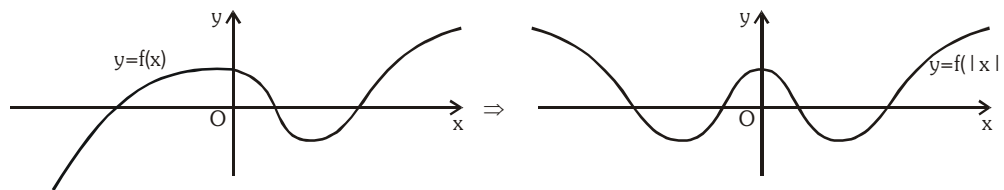
$|f(x)| = f(x)$ if $f(x) \geq 0$ and $|f(x)| = -f(x)$ if $f(x) < 0$. It means that the graph of $f(x)$ and $|f(x)|$ would coincide if $f(x) \geq 0$ and for the portions where $f(x) < 0$ graph of $|f(x)|$ would be image of $y = f(x)$ in x-axis.



(v) Drawing the graph of $y = f(|x|)$ from the known graph of $y = f(x)$

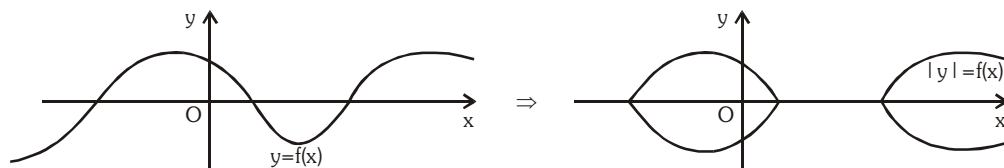
It is clear that, $f(|x|) = \begin{cases} f(x), & x \geq 0 \\ f(-x), & x < 0 \end{cases}$. Thus $f(|x|)$ would be an even function, graph of $f(|x|)$ and $f(x)$

would be identical in the first and the fourth quadrants (as $x \geq 0$) and as such the graph of $f(|x|)$ would be symmetric about the y-axis (as $(|x|)$ is even).

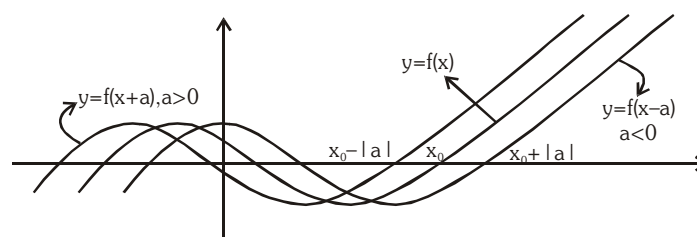


(vi) Drawing the graph of $|y| = f(x)$ from the known graph of $y = f(x)$

Clearly $|y| \geq 0$. If $f(x) < 0$, graph of $|y| = f(x)$ would not exist. And if $f(x) \geq 0$, $|y| = f(x)$ would give $y = \pm f(x)$. Hence graph of $|y| = f(x)$ would exist only in the regions where $f(x)$ is non-negative and will be reflected about the x-axis only in those regions.



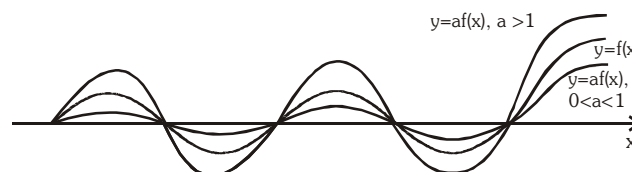
(vii) Drawing the graph of $y = f(x + a)$, $a \in \mathbb{R}$ from the known graph of $y = f(x)$



(i) If $a > 0$, shift the graph of $f(x)$ through 'a' units towards left of $f(x)$.

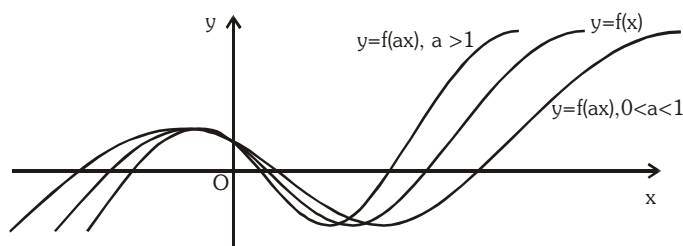
(ii) If $a < 0$, shift the graph of $f(x)$ through 'a' units towards right of $f(x)$.

(viii) Drawing the graph of $y = af(x)$ from the known graph of $y = f(x)$



It is clear that the corresponding points (points with same x co-ordinates) would have their ordinates in the ratio of 1 : a.

(ix) Drawing the graph of $y = f(ax)$ from the known graph of $y = f(x)$.



Let us take any point $x_0 \in \text{domain of } f(x)$. Let $ax = x_0$ or $x = \frac{x_0}{a}$.

Clearly if $0 < a < 1$, then $x > x_0$ and $f(x)$ will stretch by $\frac{1}{a}$ units along the x-axis and if $a > 1$, $x < x_0$, then $f(x)$ will compress by 'a' units along the x-axis.

Note :

- (i) A function $h(x)$ is defined as $h(x) = \max. \{f(x), g(x)\}$ then

$$h(x) = \begin{cases} f(x) & f(x) \geq g(x) \\ g(x) & g(x) > f(x) \end{cases}$$

- (ii) A function $h(x)$ is defined as $h(x) = \min. \{f(x), g(x)\}$ then

$$h(x) = \begin{cases} f(x) & f(x) \leq g(x) \\ g(x) & g(x) < f(x) \end{cases}$$

Illustration 14: Find $f(x) = \max \{1 + x, 1 - x, 2\}$.

Solution : From the graph it is clear that

$$f(x) = \begin{cases} 1 - x; & x < -1 \\ 2 & -1 \leq x \leq 1 \\ 1 + x; & x > 1 \end{cases}$$

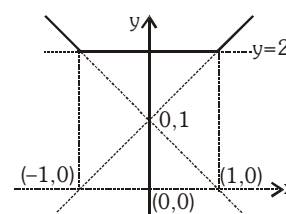
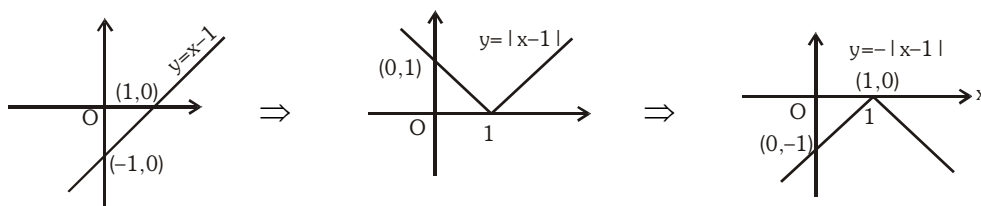


Illustration 15: Draw the graph of $y = |2 - |x - 1||$.

Solution :



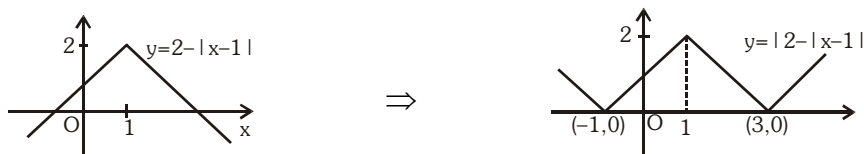


Illustration 16: Draw the graph of $y = 2 - \frac{4}{|x-1|}$

Solution :

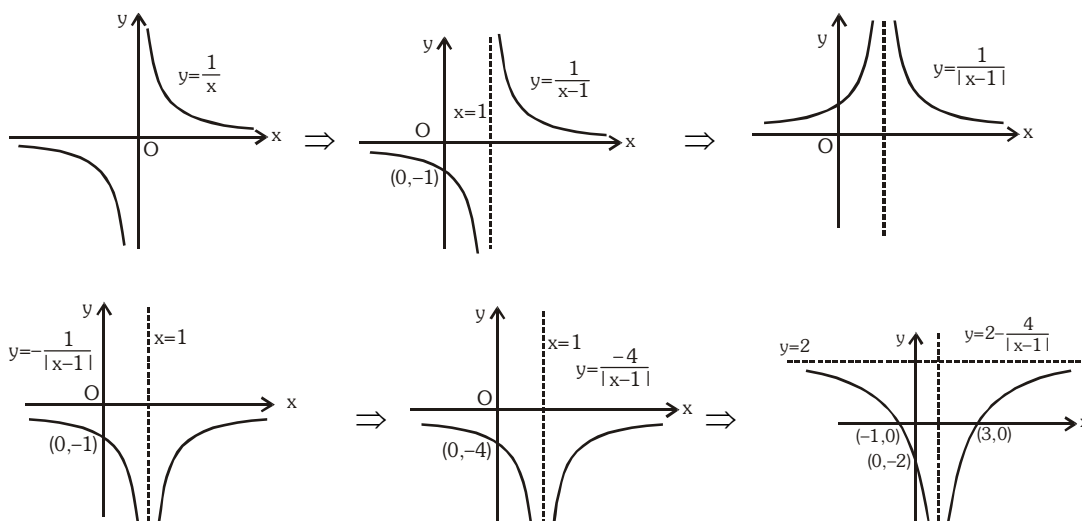


Illustration 17: Draw the graph of $y = |e^{|x|} - 2|$

Solution :

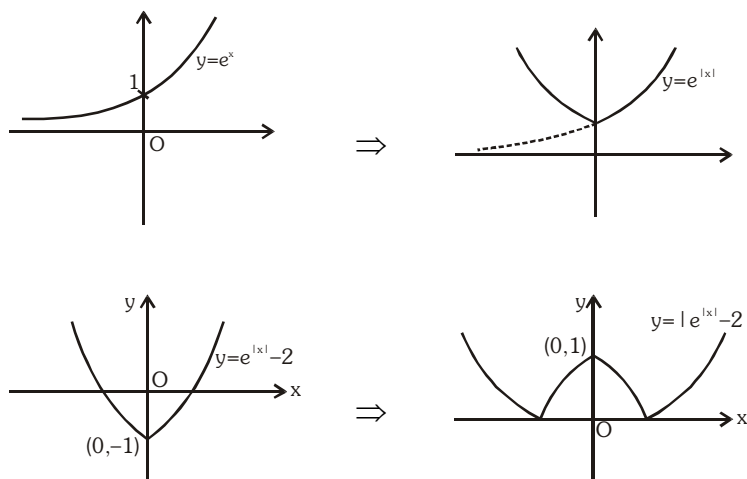
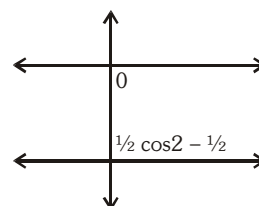


Illustration 18: Draw the graph of $f(x) = \cos x \cos(x+2) - \cos^2(x+1)$.

Solution :

$$\begin{aligned}
 f(x) &= \cos x \cos(x+2) - \cos^2(x+1) \\
 &= \frac{1}{2} [\cos(2x+2) + \cos 2] - \frac{1}{2} [\cos(2x+2) + 1] \\
 &= \frac{1}{2} \cos 2 - \frac{1}{2} < 0.
 \end{aligned}$$



Do yourself - 6 :

(i) Draw graph of following functions :

(a) $y = \lfloor \ln |x| \rfloor + 1$

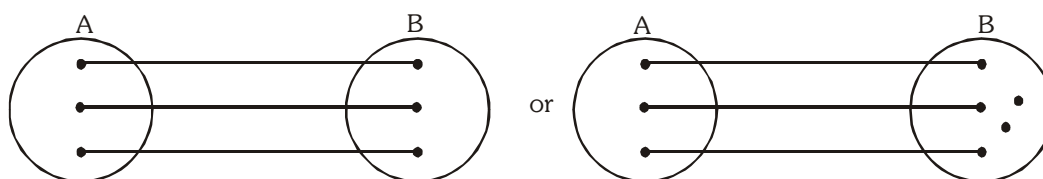
(b) $y = \min. \{x^2 + 1, 3 - x\}$

12. CLASSIFICATION OF FUNCTIONS :**One-One Function (Injective mapping) :**

A function $f : A \rightarrow B$ is said to be a one-one function or injective mapping if different elements of A have different f images in B.

Thus there exist $x_1, x_2 \in A$ & $f(x_1), f(x_2) \in B$, $f(x_1) = f(x_2) \Leftrightarrow x_1 = x_2$ or $x_1 \neq x_2 \Leftrightarrow f(x_1) \neq f(x_2)$.

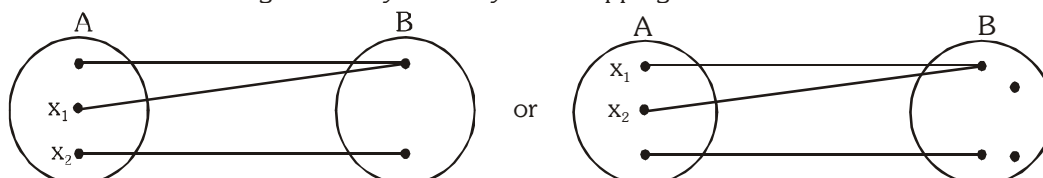
Diagrammatically an injective mapping can be shown as

**Many-one function (not injective) :**

A function $f : A \rightarrow B$ is said to be a many one function if two or more elements of A have the same f image in B.

Thus $f : A \rightarrow B$ is many one there exist $x_1, x_2 \in A$, $f(x_1) = f(x_2)$ but $x_1 \neq x_2$.

Diagrammatically a many one mapping can be shown as

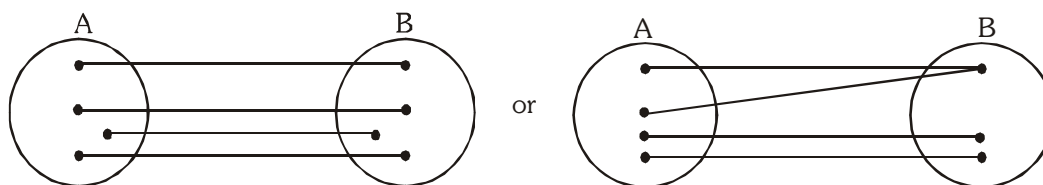
**Note :**

- (i) If a line parallel to x-axis cuts the graph of the function atmost at one point, then the function is one-one.
- (ii) If any line parallel to x-axis cuts the graph of the function atleast at two points, then f is many-one.
- (iii) If continuous function $f(x)$ is always increasing or decreasing in whole domain, then $f(x)$ is one-one.
- (iv) All linear functions are one-one.
- (v) All trigonometric functions in their domain are many one
- (vi) All even degree polynomials are many one
- (vii) Linear by Linear is one-one
- (viii) Quadratic by quadratic with no common factor is many one.

Onto function (Surjective mapping) :

If the function $f : A \rightarrow B$ is such that each element in B (co-domain) is the f image of atleast one element in A, then we say that f is a function from A 'onto' B. Thus $f : A \rightarrow B$ is surjective iff $\forall b \in B, \exists$ some $a \in A$ such that $f(a) = b$.

Diagrammatically surjective mapping can be shown as

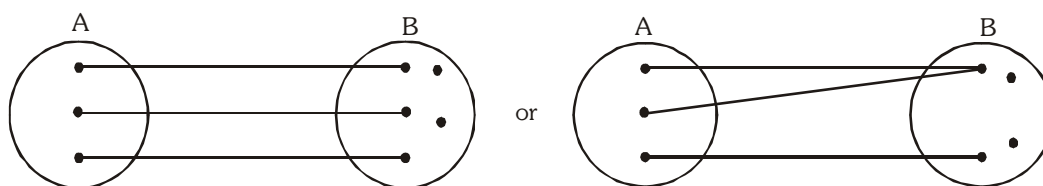


Note that : if range is same as co-domain, then $f(x)$ is onto.

Into function :

If $f : A \rightarrow B$ is such that there exists atleast one element in co-domain which is not the image of any element in domain, then $f(x)$ is into.

Diagrammatically into function can be shown as



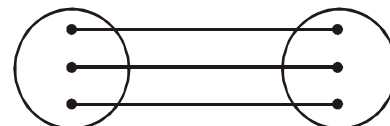
Note :

- A polynomial function of degree even define from $\mathbb{R} \rightarrow \mathbb{R}$ will always be into.
- A polynomial function of degree odd defined from $\mathbb{R} \rightarrow \mathbb{R}$ will always be onto.
- Quadratic by quadratic without any common factor define from $\mathbb{R} \rightarrow \mathbb{R}$ is always an into function.

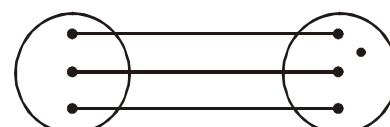
Thus a function can be one of these four types :

- one-one onto (injective & surjective)

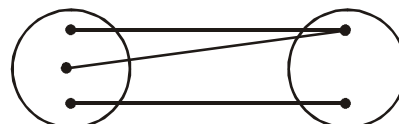
(also known as **Bijjective** mapping)



- one-one into (injective but not surjective)



- many-one onto (surjective but not injective)



- many-one into (neither surjective nor injective)

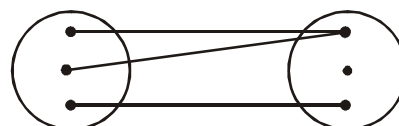


Illustration19 : Let $A = \{x : -1 \leq x \leq 1\} = B$ be a mapping $f : A \rightarrow B$. For each of the following functions from A to B, find whether it is bijective or non-bijective.

(a) $f(x) = x|x|$

(b) $f(x) = x^3$

(c) $f(x) = \sin \frac{\pi x}{2}$

Solution :

$$(a) \quad f(x) = x|x| = \begin{cases} -x^2, & -1 < x < 0 \\ x^2, & 0 \leq x < 1 \end{cases},$$

Graphically,

The graph shows $f(x)$ is one-one, as the straight line parallel to x-axis cuts only at one point.

Here, range

$$f(x) \in [-1, 1]$$

Thus, range = co-domain

Hence, onto.

Therefore, $f(x)$ is one-one onto or (Bijective).

$$(b) \quad f(x) = x^3,$$

Graphically;

Graph shows $f(x)$ is one-one onto

(i.e. Bijective)

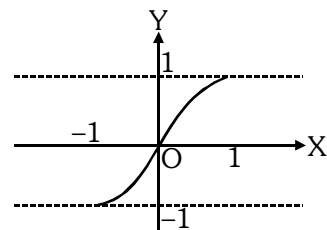
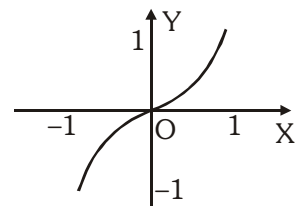
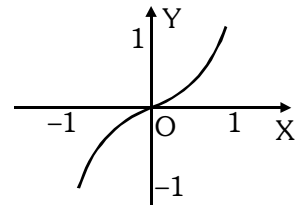
[as explained in above example]

$$(c) \quad f(x) = \sin \frac{\pi x}{2}$$

Graphically;

Which shows $f(x)$ is one-one and onto as range

= co-domain.

Therefore, $f(x)$ is bijective.**Illustration 20 :**Let $f : \mathbb{N} \rightarrow \mathbb{I}$ be a function defined as $f(x) = x - 1000$. Show that $f(x)$ is an into function.**Solution :**

$$\text{Let } f(x) = y = x - 1000 \Rightarrow x = y + 1000 = g(y) \text{ (say)}$$

here $g(y)$ is defined for each $y \in \mathbb{I}$, but $g(y) \notin \mathbb{N}$ for $y \leq -1000$.Hence $f(x)$ is into.**Illustration 21 :**Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function defined by $f(x) = x + \sqrt{x^2}$, then f is

(A) injective (B) surjective (C) bijective (D) None of these

Solution :

$$\text{We have, } f(x) = x + \sqrt{x^2} = x + |x|$$

Clearly, f is not one-one as $f(-1) = f(-2) = 0$ and $-1 \neq -2$ Also, f is not onto as $f(x) \geq 0 \forall x \in \mathbb{R}$ $\therefore$ range of $f = (0, \infty) \subset \mathbb{R}$ **Ans.(D)****Illustration 22 :**Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function defined as $f(x) = 2x^3 + 6x^2 + 12x + 3 \cos x - 4 \sin x$; then f is -

(A) Injective (B) Surjective (C) Bijective (D) Not Surjective

Solution :

$$\text{We have } f(x) = 2x^3 + 6x^2 + 12x + 3 \cos x - 4 \sin x$$

$$\Rightarrow f'(x) = 6x^2 - 12x + 12 - 3 \sin x - 4 \cos x$$

$$f'(x) = \underbrace{6(x-1)^2 + 6}_{g(x)} - \underbrace{(3 \sin x + 4 \cos x)}_{h(x)}$$

$$\text{range of } g(x) = [6, \infty)$$

$$\text{range of } h(x) = [-5, 5]$$

hence $f'(x)$ always lies in the interval $[1, \infty)$

$$\Rightarrow f'(x) > 0$$

Hence $f(x)$ is increasing i.e. one-one function

Now $x \rightarrow \infty \Rightarrow f \rightarrow \infty$ & $x \rightarrow -\infty \Rightarrow f \rightarrow -\infty$ & $f(x)$ is continuous

hence its range is $\mathbb{R} \Rightarrow f$ is onto so f is bijective.

Ans. (C)

Illustration 23 : Let $f(x) = \frac{x^2 + 3x + a}{x^2 + x + 1}$, where $f: \mathbb{R} \rightarrow \mathbb{R}$. Find the value of parameter 'a' so that the given function is one-one.

Solution : $f(x) = \frac{x^2 + 3x + a}{x^2 + x + 1}$

$$f(x) = \frac{(x^2 + x + 1)(2x + 3) - (x^2 + 3x + a)(2x + 1)}{(x^2 + x + 1)^2} = \frac{-2x^2 + 2x(1 - a) + (3 - a)}{(x^2 + x + 1)^2}$$

Let, $g(x) = -2x^2 + 2x(1 - a) + (3 - a)$

$g(x)$ will be negative if $4(1 - a)^2 + 8(3 - a) < 0$

$$\Rightarrow 1 + a^2 - 2a + 6 - 2a < 0 \Rightarrow (a - 2)^2 + 3 < 0$$

which is not possible. Therefore function is not monotonic.

Hence, no value of a is possible.

Do yourself - 7 :

- (i) Is the function $f: \mathbb{N} \rightarrow \mathbb{N}$ (the set of natural numbers) defined by $f(x) = 2x + 3$ surjective ?
- (ii) Let $A = \mathbb{R} - \{3\}$, $B = \mathbb{R} - \{1\}$ and let $f: A \rightarrow B$ defined by $f(x) = \frac{x-2}{x-3}$. Check whether the function $f(x)$ is bijective or not.
- (iii) A mapping $f: A \rightarrow [-1, 1]$ defined by $f(x) = \sin x$, $\forall x \in \mathbb{R}$, where A is a subset of $\mathbb{R}$ (the set of all real numbers) is one-one and onto if A is the interval, then A belongs to

- (A) $[0, 2\pi]$ (B) $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ (C) $[-\pi, \pi]$ (D) $[0, \pi]$

13. COMPOSITE OF UNIFORMLY & NON-UNIFORMLY DEFINED FUNCTION:

Let $f: A \rightarrow B$ & $g: B \rightarrow C$ be two functions. Then the function $g \circ f: A \rightarrow C$ defined by $(g \circ f)(x) = g(f(x))$ $\forall x \in A$ is called the composite of the two functions f & g .

Diagrammatically $\xrightarrow{x} \boxed{f} \xrightarrow{f(x)} \boxed{g} \xrightarrow{g(f(x))}$

Thus the image of every $x \in A$ under the function $g \circ f$ is the g -image of f -image of x .

Note that $g \circ f$ is defined only if $\forall x \in A$, $f(x)$ is an element of the domain of ' g ' so that we can take its g -image.

Properties of composite functions:

- (a) In general composite of functions is not commutative i.e. $g \circ f \neq f \circ g$.

- (b) The composition of functions is associative i.e. if f, g, h are three functions such that $fo(goh)$ & $(fog)oh$ are defined, then $fo(goh) = (fog)oh$.
- (c) The composition of two bijections is a bijection i.e. if f & g are two bijections such that gof is defined, then gof is also a bijection.

Illustration 24: If $f(x) = x^2 + 1$, $g(x) = \frac{1}{x-1}$, then find $(fog)(x)$ and $(gof)(x)$.

Solution : Given, $f(x) = x^2 + 1$ (1) $g(x) = \frac{1}{x-1}$... (2)

$$\text{Now } (fog)(x) = f(g(x)) = f\left(\frac{1}{x-1}\right) = f(z), \text{ where } z = \frac{1}{x-1}$$

$$= z^2 + 1 \quad [\because f(x) = x^2 + 1]$$

$$= \left(\frac{1}{x-1}\right)^2 + 1 = \frac{1}{(x-1)^2} + 1$$

Note : Domain of $fog(x)$ is $x \in \mathbb{R} - \{1\}$

$$(gof)(x) = g(f(x)) = g(x^2 + 1) = g(u), \text{ where } u = x^2 + 1$$

$$= \frac{1}{u-1} = \frac{1}{x^2 + 1 - 1} = \frac{1}{x^2}$$

Note : Domain of $gof(x)$ is $x \in \mathbb{R} - \{0\}$

Illustration 25: If f be the greatest integer function and g be the modulus function, then

$$(gof)\left(-\frac{5}{3}\right) - (fog)\left(-\frac{5}{3}\right) =$$

(A) 1

(B) -1

(C) 2

(D) 4

Solution : Given $(gof)\left(-\frac{5}{3}\right) - (fog)\left(-\frac{5}{3}\right) = g\left\{f\left(-\frac{5}{3}\right)\right\} - f\left\{g\left(-\frac{5}{3}\right)\right\} = g(-2) - f\left(\frac{5}{3}\right) = 2 - 1 = 1$

Ans.(A)

Illustration 26: Let $f(x) = \begin{cases} x+1, & x \leq 1 \\ 2x+1, & 1 < x \leq 2 \end{cases}$ and $g(x) = \begin{cases} x^2, & -1 \leq x < 2 \\ x+2, & 2 \leq x \leq 3 \end{cases}$, find (fog)

Solution : $f(g(x)) = \begin{cases} g(x)+1, & g(x) \leq 1 \\ 2g(x)+1, & 1 < g(x) \leq 2 \end{cases}$

Here, $g(x)$ becomes the variable that means we should draw the graph.

It is clear that $g(x) \leq 1$; $\forall x \in [-1, 1]$

and $1 < g(x) \leq 2$; $\forall x \in (1, \sqrt{2}]$

$$\Rightarrow f(g(x)) = \begin{cases} x^2 + 1, & -1 \leq x \leq 1 \\ 2x^2 + 1, & 1 < x \leq \sqrt{2} \end{cases}$$

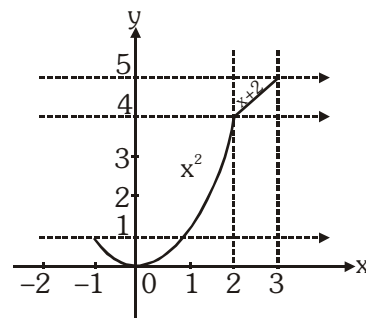


Illustration 27: Find the domain and range of $h(x) = g(f(x))$, where

$$f(x) = \begin{cases} [x], & -2 \leq x \leq -1 \\ |x| + 1, & -1 < x \leq 2 \end{cases} \text{ and } g(x) = \begin{cases} [x], & -\pi \leq x \leq 0 \\ \sin x, & 0 \leq x \leq \pi \end{cases}, [\cdot] \text{ denotes the greatest integer function.}$$

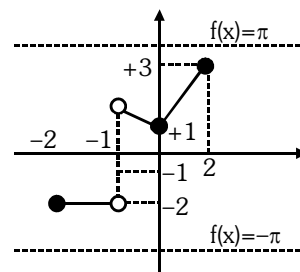
Solution : $h(x) = g(f(x)) = \begin{cases} [f(x)], & -\pi \leq f(x) < 0 \\ \sin(f(x)), & 0 \leq f(x) \leq \pi \end{cases}$

From graph of $f(x)$, we get

$$h(x) = \begin{cases} [[x]], & -2 \leq x \leq -1 \\ \sin(|x| + 1), & -1 < x \leq 2 \end{cases}$$

$\Rightarrow$ Domain of $h(x)$ is $[-2, 2]$

and Range of $h(x)$ is $\{-2, -1\} \cup [\sin 3, 1]$



Do yourself - 8 :

(i) $f(x) = x^3 - x$ & $g(x) = \sin 2x$, find

- | | | |
|---|----------------|---|
| (a) $f(f(1))$ | (b) $f(f(-1))$ | (c) $f\left(g\left(\frac{\pi}{2}\right)\right)$ |
| (d) $f\left(g\left(\frac{\pi}{4}\right)\right)$ | (e) $g(f(1))$ | (f) $g\left(g\left(\frac{\pi}{2}\right)\right)$ |

(ii) If $f(x) = \begin{cases} x+1; & 0 \leq x < 2 \\ |x|; & 2 \leq x < 3 \end{cases}$, then find $f \circ f(x)$.

14. INVERSE OF A FUNCTION :

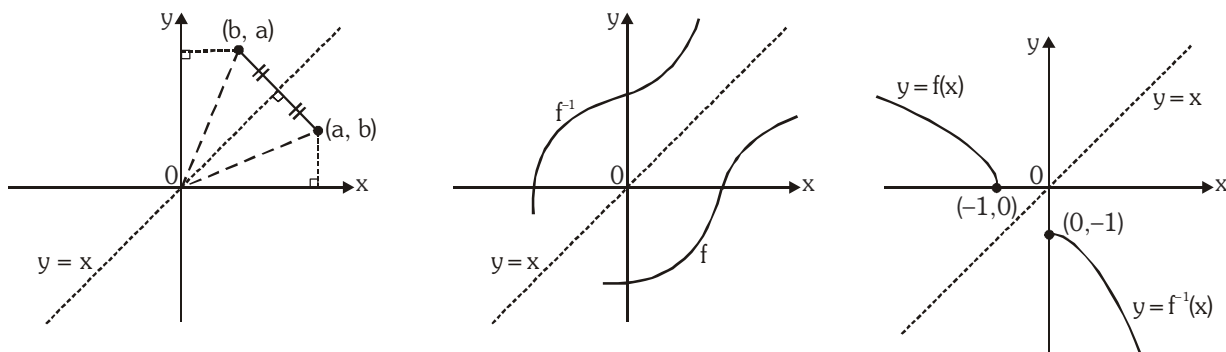
Let $f: A \rightarrow B$ be a one-one & onto function, then there exists a unique function $g: B \rightarrow A$ such that $f(x) = y \Leftrightarrow g(y) = x, \forall x \in A \text{ \& } y \in B$. Then g is said to be inverse of f .

Thus $g = f^{-1}: B \rightarrow A = \{(f(x), x) | (x, f(x)) \in f\}$.

Properties of inverse function :

- The inverse of a bijection is unique.
- If $f: A \rightarrow B$ is a bijection & $g: B \rightarrow A$ is the inverse of f , then $f \circ g = I_B$ and $g \circ f = I_A$, where I_A & I_B are identity functions on the sets A & B respectively. If $f \circ f = I$, then f is inverse of itself.
- The inverse of a bijection is also a bijection.
- If f & g are two bijections $f: A \rightarrow B, g: B \rightarrow C$ then the inverse of $g \circ f$ exists and $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$.

- (e) Since $f(a) = b$ if and only if $f^{-1}(b) = a$, the point (a, b) is on the graph of ' f ' if and only if the point (b, a) is on the graph of f^{-1} . But we get the point (b, a) from (a, b) by reflecting about the line $y = x$.



The graph of f^{-1} is obtained by reflecting the graph of f about the line $y = x$.

Drawing the graph of $y = f^{-1}(x)$ from the known graph of $y = f(x)$

For drawing the graph of $y = f^{-1}(x)$ take the reflection of $y = f(x)$ about the line $y = x$. The reflected part would give us the graph of $y = f^{-1}(x)$.

e.g. let us draw the graph of $y = \sin^{-1}x$. We know that $y = f(x) = \sin x$ is invertible if

$$f: \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \rightarrow [-1, 1] \Rightarrow \text{the inverse mapping would be } f^{-1}: [-1, 1] \rightarrow \left[-\frac{\pi}{2}, \frac{\pi}{2}\right].$$

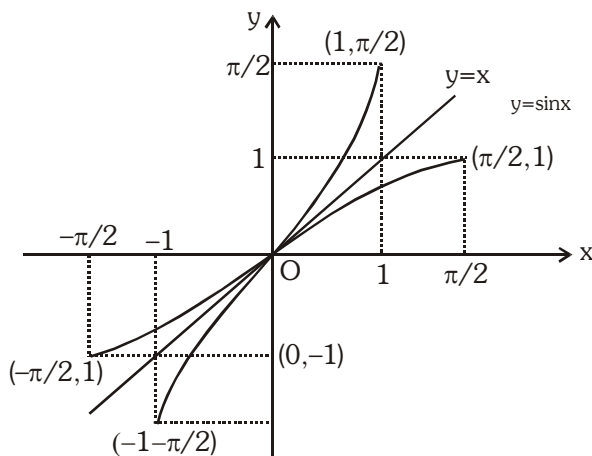


Illustration 28: Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = (e^x - e^{-x})/2$. Is $f(x)$ invertible? If so, find its inverse.

Solution : Let us check for invertibility of $f(x)$:

(a) One-One :

$$f(x) = \frac{1}{2}(e^x - e^{-x}) \Rightarrow f'(x) = \frac{1}{2}(e^x + e^{-x})$$

$\Rightarrow f'(x) > 0$, $f(x)$ is increasing function

$\therefore f(x)$ is one-one function.

(b) Onto :

As x tends to larger and larger values so does $f(x)$ and

when $x \rightarrow \infty$, $f(x) \rightarrow \infty$.

Similarly as $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$ i.e. $-\infty < f(x) < \infty$ so long as $x \in (-\infty, \infty)$

Hence the range of f is same as the set $\mathbb{R}$. Therefore $f(x)$ is onto.

Since $f(x)$ is both one-one and onto, $f(x)$ is invertible.

(c) To find $f^{-1}(x)$: Interchange x & y

$$\frac{e^y - e^{-y}}{2} = x \Rightarrow e^{2y} - 2xe^y - 1 = 0$$

$$\Rightarrow e^y = \frac{2x \pm \sqrt{4x^2 + 4}}{2} \Rightarrow e^y = x \pm \sqrt{1 + x^2}$$

Since $e^y > 0$, hence negative sign is ruled out and

$$\text{Hence } e^y = x + \sqrt{1 + x^2}$$

Taking logarithm, we have $y = \ln(x + \sqrt{1 + x^2})$ or $f^{-1}(x) = \ln(x + \sqrt{1 + x^2})$

Illustration 29 : Find the inverse of the function $f(x) = \log_a \left(x + \sqrt{x^2 + 1} \right)$; $a > 1$ and assuming it to be an onto function.

Solution : Given $f(x) = \log_a \left(x + \sqrt{x^2 + 1} \right)$

$$\therefore f'(x) = \frac{\log_a e}{\sqrt{1 + x^2}} > 0$$

which is a strictly increasing functions.

Thus, $f(x)$ is injective, given that $f(x)$ is onto. Hence the given function $f(x)$ is invertible.

Interchanging x & y

$$\Rightarrow \log_a \left(y + \sqrt{y^2 + 1} \right) = x$$

$$\Rightarrow y + \sqrt{y^2 + 1} = a^x \quad \dots(i)$$

$$\text{and } \sqrt{y^2 + 1} - y = a^{-x} \quad \dots(ii)$$

$$\text{From (i) and (ii), we get } y = \frac{1}{2}(a^x - a^{-x}) \text{ or } f^{-1}(x) = \frac{1}{2}(a^x - a^{-x})$$

Illustration 30 : Find the inverse of the function $f(x) = \ln(x^2 + 3x + 1)$; $x \in [1, 3]$ and assuming it to be an onto function.

Solution : Given $f(x) = \ln(x^2 + 3x + 1)$

$$\therefore f'(x) = \frac{2x + 3}{x^2 + 3x + 1} > 0 \forall x \in [1, 3]$$

Which is a strictly increasing function. Thus $f(x)$ is injective, given that $f(x)$ is onto. Hence the given function $f(x)$ is invertible.

Interchanging x & y

$$\Rightarrow (y)^2 + 3(y) + 1 - e^x = 0$$

$$\therefore y = \frac{-3 \pm \sqrt{9 - 4 \cdot (1 - e^x)}}{2} = \frac{-3 \pm \sqrt{(5 + 4e^x)}}{2}$$

$$\Rightarrow y = \frac{-3 + \sqrt{(5 + 4e^x)}}{2} \quad (\text{as } y \in [1, 3])$$

$$\text{Hence } f^{-1}(x) = \frac{-3 + \sqrt{(5 + 4e^x)}}{2}$$

Do yourself - 9 :

(i) Let $f: [-1, 1] \rightarrow [-1, 1]$ defined by $f(x) = x|x|$, find $f^{-1}(x)$.

(ii) $f(x) = 1 + \ln(x+2)$, find $f^{-1}(x)$.

15. ODD & EVEN FUNCTIONS :

Consider a function $f(x)$ such that both x and $-x$ are in its domain then

If $\begin{cases} f(-x) = f(x) & \text{then } f \text{ is said to be an even function} \\ f(-x) = -f(x) & \text{then } f \text{ is said to be an odd function} \end{cases}$

Note :

- (i) $f(x) - f(-x) = 0 \Rightarrow f(x)$ is even & $f(x) + f(-x) = 0 \Rightarrow f(x)$ is odd.
- (ii) A function may neither be odd nor even.
- (iii) The only function which is defined on the entire number line & is even and odd at the same time is $f(x) = 0$.
- (iv) Every constant function is even function.
- (v) Inverse of an even function is not defined.
- (vi) Every even function is symmetric about the y-axis & every odd function is symmetric about the origin.

Special Note : If a function $f(x)$ is defined as $f(a+x) = f(a-x)$ then this function is symmetric about line $x = a$

(vii) Every function which has ' $-x$ ' in its domain whenever ' x ' is in its domain, can be expressed as the sum of an even & an odd function.

$$\text{i.e. } f(x) = \underbrace{\frac{f(x) + f(-x)}{2}}_{\text{EVEN}} + \underbrace{\frac{f(x) - f(-x)}{2}}_{\text{ODD}}$$

(viii) If $f(x)$ is odd and defined at $x = 0$, then $f(0) = 0$.

$f(x)$	$g(x)$	$f(x) + g(x)$	$f(x) - g(x)$	$f(x) \cdot g(x)$	$f(x)/g(x)$	$(g \circ f)(x)$	$(f \circ g)(x)$
odd	odd	odd	odd	even	even	odd	odd
even	even	even	even	even	even	even	even
odd	even	neither odd nor even	neither odd nor even	odd	odd	even	even
even	odd	neither odd nor even	neither odd nor even	odd	odd	even	even

Illustration 31 : Which of the following functions is (are) even, odd or neither :

(i) $f(x) = x^2 \sin x$ (ii) $f(x) = \sin x - \cos x$ (iii) $f(x) = \frac{e^x + e^{-x}}{2}$

Solution :

(i) $f(-x) = (-x)^2 \sin(-x) = -x^2 \sin x = -f(x)$. Hence $f(x)$ is odd.

(ii) $f(-x) = \sin(-x) - \cos(-x) = -\sin x - \cos x$.

Hence $f(x)$ is neither even nor odd.

(iii) $f(-x) = \frac{e^{-x} + e^{-(-x)}}{2} = \frac{e^{-x} + e^x}{2} = f(x)$. Hence $f(x)$ is even

Illustration 32: Identify the given functions as odd, even or neither :

(i) $f(x) = \frac{x}{e^x - 1} + \frac{x}{2} + 1$ (ii) $f(x + y) = f(x) + f(y)$ for all $x, y \in \mathbb{R}$

Solution :

(i) $f(x) = \frac{x}{e^x - 1} + \frac{x}{2} + 1$

Clearly domain of $f(x)$ is $\mathbb{R} \sim \{0\}$. We have,

$$\begin{aligned} f(-x) &= \frac{-x}{e^{-x} - 1} - \frac{x}{2} + 1 = \frac{-e^x \cdot x}{1 - e^x} - \frac{x}{2} + 1 = \frac{(e^x - 1 + 1)x}{(e^x - 1)} - \frac{x}{2} + 1 \\ &= x + \frac{x}{e^x - 1} - \frac{x}{2} + 1 = \frac{x}{e^x - 1} + \frac{x}{2} + 1 = f(x) \end{aligned}$$

Hence $f(x)$ is an even function.

(ii) $f(x + y) = f(x) + f(y)$ for all $x, y \in \mathbb{R}$

Replacing x, y by zero, we get $f(0) = 2f(0) \Rightarrow f(0) = 0$

Replacing y by $-x$, we get $f(x) + f(-x) = f(0) = 0 \Rightarrow f(x) = -f(-x)$

Hence $f(x)$ is an odd function.

Do yourself - 10 :

(i) Which of the following functions is (are) even, odd or neither :

(a) $f(x) = x^3 \sin 3x$ (b) $f(x) = \frac{e^{x^2} + e^{-x^2}}{2x}$

(c) $f(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$ (d) $f(x) = x^2 + 2^x$

16. PERIODIC FUNCTION :

A function $f(x)$ is called periodic if there exists a positive number T ($T > 0$), where T is the smallest such value called the period of the function such that $f(x+T) = f(x)$, for all values of x , $x+T$ within the domain of f .

Note :

- (i) Odd powers of $\sin x$, $\cos x$, $\sec x$, $\csc x$ are periodic with period 2π .
- (ii) None zero integral powers of $\tan x$, $\cot x$ are periodic with period π .
- (iii) Non zero even powers or modulus of $\sin x$, $\cos x$, $\sec x$, $\csc x$ are periodic with period π .
- (iv) $f(T) = f(0) = f(-T)$, where ' T ' is the period.
- (v) if $f(x)$ has a period T then **$f(ax+b)$ has a period $T/|a|$ ($a \neq 0$).**

Proof : Let $f(x+T) = f(x)$ and $f[a(x+T') + b] = f(ax+b)$
 $f(ax+b+aT') = f(ax+b)$

$$f(y+aT') = f(y) = f(y+T) \Rightarrow T = aT' \Rightarrow T' = \frac{T}{a}$$

(vi) If $f(x)$ & $g(x)$ are periodic with period T_1 & T_2 respectively, then period of $f(x) \pm g(x)$ is L.C.M. of (T_1, T_2) .

(a) LCM of T_1 & T_2 is defined when T_1/T_2 is rational.

$$(b) \text{ LCM of } \left\{ \frac{a}{b}, \frac{p}{q} \right\} = \frac{\text{LCM of } (a, p)}{\text{HCF of } (b, q)}$$

In case if there exists a finite K such that $K < \text{LCM of } T_1 \text{ and } T_2$ and overall function repeats itself after every K , then period of the function will be K .

- (vii) Every constant function is always periodic, whose period is undefined.
- (viii) Inverse of a periodic function does not exist.

Illustration 33 : Find the periods (if periodic) of the following functions, where $[.]$ denotes the greatest integer function

$$(i) f(x) = e^{\ln(\sin x)} + \tan^3 x - \csc(3x-5) \quad (ii) f(x) = x - [x-b], b \in \mathbb{R}$$

$$(iii) f(x) = \frac{|\sin x + \cos x|}{|\sin x| + |\cos x|} \quad (iv) f(x) = \tan \frac{\pi}{2} [x]$$

$$(v) f(x) = \cos(\sin x) + \cos(\cos x) \quad (vi) f(x) = \frac{(1 + \sin x)(1 + \sec x)}{(1 + \cos x)(1 + \csc x)}$$

$$(vii) f(x) = e^{x - [x] + |\cos \pi x| + |\cos 2\pi x| + \dots + |\cos n\pi|}$$

Solution :

$$(i) f(x) = e^{\ln(\sin x)} + \tan^3 x - \csc(3x-5)$$

Period of $e^{\ln \sin x} = 2\pi$, $\tan^3 x = \pi$

$$\csc(3x-5) = \frac{2\pi}{3}$$

$$\therefore \text{Period} = 2\pi$$

$$(ii) f(x) = x - [x-b] = b + \{x-b\}$$

$\therefore \text{Period} = 1$

$$(iii) f(x) = \frac{|\sin x + \cos x|}{|\sin x| + |\cos x|}$$

Since period of $|\sin x + \cos x| = \pi$ and period of $|\sin x| + |\cos x|$ is $\frac{\pi}{2}$. Hence $f(x)$ is periodic with π as its period

$$(iv) f(x) = \tan \frac{\pi}{2} [x]$$

$$\tan \frac{\pi}{2} [x + T] = \tan \frac{\pi}{2} [x] \Rightarrow \frac{\pi}{2} [x + T] = n\pi + \frac{\pi}{2} [x]$$

$$\therefore T = 2$$

$$\therefore \text{Period} = 2$$

(v) Let $f(x)$ is periodic then $f(x + T) = f(x)$

$$\Rightarrow \cos(\sin(x + T)) + \cos(\cos(x + T)) = \cos(\sin x) + \cos(\cos x)$$

If $x = 0$ then $\cos(\sin T) + \cos(\cos T)$

$$= \cos(0) + \cos(1) = \cos\left(\cos \frac{\pi}{2}\right) + \cos\left(\sin \frac{\pi}{2}\right)$$

$$\text{On comparing } T = \frac{\pi}{2}$$

$$(vi) f(x) = \frac{(1 + \sin x)(1 + \sec x)}{(1 + \cos x)(1 + \csc x)} = \frac{(1 + \sin x)(1 + \cos x) \sin x}{\cos x (1 + \sin x)(1 + \cos x)}$$

$$\Rightarrow f(x) = \tan x$$

Hence $f(x)$ has period π .

$$(vii) f(x) = e^{x - [x] + |\cos \pi x| + |\cos 2\pi x| + \dots + |\cos n\pi x|}$$

$$\text{Period of } x - [x] = 1$$

$$\text{Period of } |\cos \pi x| = 1$$

$$\text{Period of } |\cos 2\pi x| = \frac{1}{2}$$

.....

$$\text{Period of } |\cos n\pi x| = \frac{1}{n}$$

So period of $f(x)$ will be L.C.M. of all period = 1

Illustration 34: Find the periods (if periodic) of the following functions, where $[.]$ denotes the greatest integer function

$$(i) f(x) = e^{x-[x]} + \sin x \quad (ii) f(x) = \sin \frac{\pi x}{\sqrt{2}} + \cos \frac{\pi x}{\sqrt{3}} \quad (iii) f(x) = \sin \frac{\pi x}{\sqrt{3}} + \cos \frac{\pi x}{2\sqrt{3}}$$

Solution :

$$(i) \text{ Period of } e^{x-[x]} = 1$$

$$\text{period of } \sin x = 2\pi$$

$\therefore$ L.C.M. of rational and an irrational number does not exist.

$\therefore$ not periodic.

$$(ii) \quad \text{Period of } \sin \frac{\pi x}{\sqrt{2}} = \frac{2\pi}{\pi/\sqrt{2}} = 2\sqrt{2}$$

$$\text{Period of } \cos \frac{\pi x}{\sqrt{3}} = \frac{2\pi}{\pi/\sqrt{3}} = 2\sqrt{3}$$

$\therefore$ L.C.M. of two different kinds of irrational number does not exist.
 $\therefore$ not periodic.

$$(iii) \quad \text{Period of } \sin \frac{\pi x}{\sqrt{3}} = \frac{2\pi}{\pi/\sqrt{3}} = 2\sqrt{3}$$

$$\text{Period of } \cos \frac{\pi x}{2\sqrt{3}} = \frac{2\pi}{\pi/2\sqrt{3}} = 4\sqrt{3}$$

$\therefore$ L.C.M. of two similar irrational number exist.

$\therefore$ Periodic with period = $4\sqrt{3}$

Ans.

Do yourself - 11 :

(i) Find the periods (if periodic) of the following functions.

(a) $f(x) = \ell n(\cos x) + \tan^3 x$.

(b) $f(x) = e^{x-[x]}$, $[.]$ denotes greatest integer function

(c) $f(x) = \left| \sin \frac{x}{2} \right| + \left| \cos \frac{x}{2} \right|$

17. GENERAL :

If x, y are independent variables, then :

(a) $f(xy) = f(x) + f(y) \Rightarrow f(x) = k \ell n x$

(b) $f(xy) = f(x) \cdot f(y) \Rightarrow f(x) = x^n, n \in \mathbb{R}$ or $f(x) = 0$

(c) $f(x+y) = f(x) \cdot f(y) \Rightarrow f(x) = a^{kx}$ or $f(x) = 0$

(d) $f(x+y) = f(x) + f(y) \Rightarrow f(x) = kx$, where k is a constant.

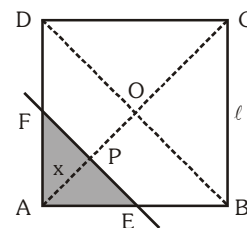
Miscellaneous Illustration :

Illustration 35: ABCD is a square of side ℓ . A line parallel to the diagonal BD at a distance 'x' from the vertex A cuts two adjacent sides. Express the area of the segment of the square with A at a vertex, as a function of x. Find this area at $x = 1/\sqrt{2}$ and at $x = 2$, when $\ell = 2$.

Solution : There are two different situations

Case-I : when $x = AP \leq OA$, i.e., $x \leq \frac{\ell}{\sqrt{2}}$

$$\text{ar}(\triangle AEF) = \frac{1}{2} x \cdot 2x = x^2 \quad (\because PE = PF = AP = x)$$



Case-II : when $x = AP > OA$, i.e., $x > \frac{\ell}{\sqrt{2}}$ but $x \leq \sqrt{2}\ell$

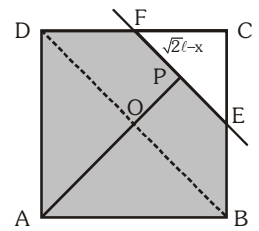
$$\text{ar(ABEFDA)} = \text{ar(ABCD)} - \text{ar}(\Delta CFE)$$

$$= \ell^2 - \frac{1}{2}(\sqrt{2}\ell - x) \cdot 2(\sqrt{2}\ell - x) \quad [\because CP = \sqrt{2}\ell - x]$$

$$= \ell^2 - (2\ell^2 + x^2 - 2\sqrt{2}\ell x) = 2\sqrt{2}\ell x - x^2 - \ell^2$$

$\therefore$ the required function $s(x)$ is as follows :

$$s(x) = \begin{cases} x^2, & 0 \leq x \leq \frac{\ell}{\sqrt{2}} \\ 2\sqrt{2}\ell x - x^2 - \ell^2, & \frac{\ell}{\sqrt{2}} < x \leq \sqrt{2}\ell \end{cases} ; \text{ area of } s(x) = \begin{cases} \frac{1}{2} & \text{at } x = \frac{1}{\sqrt{2}} \\ 8(\sqrt{2} - 1) & \text{at } x = 2 \end{cases}$$



Ans.

Illustration 36 : If the function $f(x)$ satisfies the functional rule, $f(x + y) = f(x) + f(y) \forall x, y \in \mathbb{R}$ & $f(1) = 5$, then find $\sum_{n=1}^m f(n)$.

Solution :

Here, $f(x + y) = f(x) + f(y)$; put $x = t - 1, y = 1$
 $f(t) = f(t - 1) + f(1)$ (1)

$$\begin{aligned} \therefore f(t) &= f(t - 1) + 5 \\ \Rightarrow f(t) &= \{f(t - 2) + 5\} + 5 \\ \Rightarrow f(t) &= f(t - 2) + 2(5) \\ \Rightarrow f(t) &= f(t - 3) + 3(5) \end{aligned}$$

$$\begin{aligned} \Rightarrow f(t) &= f\{t - (t - 1)\} + (t - 1)5 \\ \Rightarrow f(t) &= f(1) + (t - 1)5 \\ \Rightarrow f(t) &= 5 + (t - 1)5 \\ \Rightarrow f(t) &= 5t \end{aligned}$$

$$\therefore \sum_{n=1}^m f(n) = \sum_{n=1}^m (5n) = 5[1 + 2 + 3 + \dots + m] = \frac{5m(m + 1)}{2}$$

$$\text{Hence, } \sum_{n=1}^m f(n) = \frac{5m(m + 1)}{2}.$$

ANSWERS FOR DO YOURSELF

1: (i) (a) $x \in (0, \infty)$ (b) $x \in (-\infty, 0) \cup (4, \infty)$

(ii) (a) $\left[\frac{1}{2}, \infty\right)$ (b) $[-1, 1]$ (c) $[-2, 2]$ (d) $[-1, 1]$

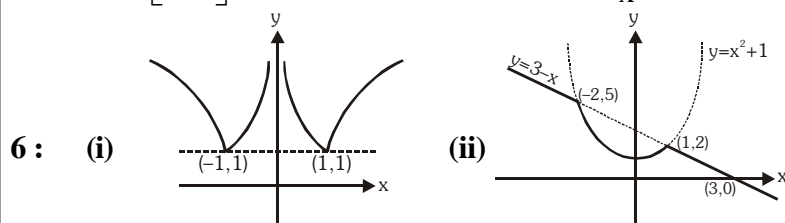
2: (i) (A) $\rightarrow (p, q, r, s)$, (B) $\rightarrow (p)$, (C) $\rightarrow (t)$, (D) $\rightarrow (r)$

3: (i) (a) $(-1, 0) \cup (0, 1]$ (b) $\left(-\infty, -\frac{1}{2}\right] \cup (0, 1) \cup (2, \infty)$

(ii) (a) $(-\infty, \infty)$ (b) $\left[\frac{1}{3}, 1\right]$

4: (i) (a) no (b) no (c) yes

5: (i) $\left[0, \frac{1}{2}\right]$ (ii) A (iii) (a) $y = \frac{1}{x}$ (b) $y = \frac{1}{x^2}$



7: (i) not onto (ii) yes (iii) B

8: (i) (a) 0 (b) 0 (c) 0 (d) 0 (e) 0 (f) 0 (ii) $\begin{cases} x+2, & 0 \leq x < 1 \\ x+1, & 1 \leq x < 2 \\ x, & 2 \leq x < 3 \end{cases}$

9: (i) $f^{-1}(x) = \begin{cases} -\sqrt{-x}, & -1 \leq x \leq 0 \\ \sqrt{x}, & 0 \leq x \leq 1 \end{cases}$ (ii) $y = -2 + e^{x-1}$

10: (i) (a) even (b) odd (c) odd (d) neither even nor odd

11: (i) (a) 2π (b) 1 (c) π

EXERCISE (O-1)

Subjective Type Questions

1. Find the domain of definition of the given functions :

$$(i) y = \sqrt{-px} \ (p > 0) \quad (ii) y = \frac{1}{x^2 + 1} \quad (iii) y = \frac{1}{x^3 - x} \quad (iv) y = \frac{1}{\sqrt{x^2 - 4x}}$$

$$(v) y = \sqrt{x^2 - 4x + 3} \quad (vi) y = \frac{x}{\sqrt{x^2 - 3x + 2}} \quad (vii) y = \sqrt{1 - |x|} \quad (viii) y = \log_x 2.$$

$$(ix) y = \frac{1}{\log_{10}(1-x)} + \sqrt{x+2} \quad (x) y = \sqrt{x} + \sqrt[3]{\frac{1}{x-2}} - \log_{10}(2x-3)$$

$$(xi) y = \frac{3}{4-x^2} + \log_{10}(x^3 - x) \quad (xii) y = \frac{1}{\sqrt{\sin x}} + \sqrt[3]{\sin x}$$

$$(xiii) y = \log_{10}(\sqrt{x-4} + \sqrt{6-x}) \quad (xiv) y = \log_{10}[1 - \log_{10}(x^2 - 5x + 16)]$$

2. Find the range of the following functions :

$$(i) f(x) = \frac{x-1}{x+2} \quad (ii) f(x) = \frac{2}{x} \quad (iii) f(x) = \frac{1}{x^2 - x + 1} \quad (iv) f(x) = \frac{x^2 - x + 1}{x^2 + x + 1}$$

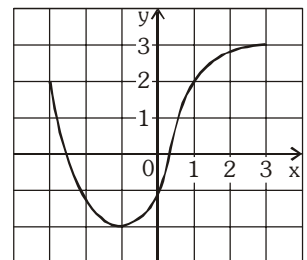
$$(v) f(x) = e^{(x-1)^2} \quad (vi) f(x) = x^3 - x^2 + x + 1$$

$$(vii) f(x) = \log(x^8 + x^4 + x^2 + 1) \quad (viii) f(x) = \sin^2 x - 2\sin x + 4$$

$$(ix) f(x) = \sin(\log_2 x) \quad (x) f(x) = 2^{x^2} + 1 \quad (xi) f(x) = \frac{e^{2x} - e^x + 1}{e^{2x} + e^x + 1} \quad (xii) f(x) = \frac{1}{8 - 3\sin x}$$

3. The graph of a function f is given.

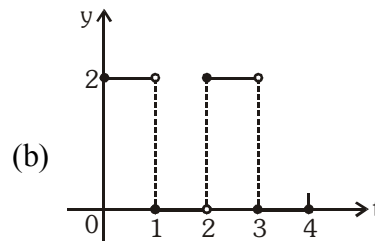
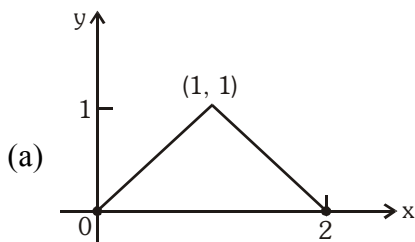
- State the value of $f(-1)$.
- For what values of x is $f(x) = 2$
- State the domain and range of f .
- On what interval is f increasing ?
- Estimated value of $f(2)$ is -



- 2.2
 - 2.8
 - 2.5
 - 3
- (f) Estimated value of x such that $f(x) = 0$, is -
- 2.5
 - 0.8
 - 2.9
 - 0.3

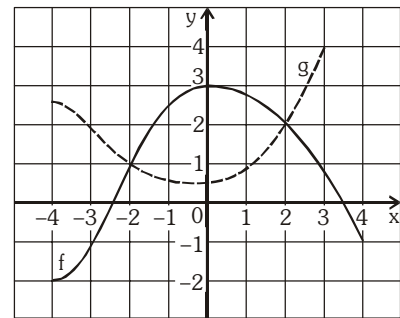
4. Graph the function $F(x) = \begin{cases} 3-x, & x \leq 1 \\ 2x, & x > 1 \end{cases}$

5. Find a formula for each function graphed



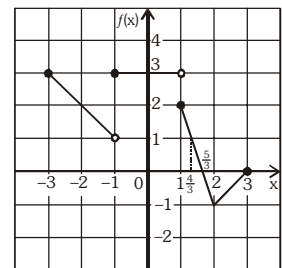
6. The graphs of f and g are given.

- (a) State the value of $f(-4)$ and $g(3)$
 (b) For what value of x is $f(x) = g(x)$?
 (c) Estimate the solution of the equation $f(x) = -1$.
 (d) On what interval is f decreasing?
 (e) State the domain and range of f .
 (f) State the domain and range of g .



7. Solve the following inequalities using graph of $f(x)$:

- (a) $0 \leq f(x) \leq 1$
 (b) $-1 \leq f(x) \leq 2$
 (c) $2 \leq f(x) \leq 3$
 (d) $f(x) > -1$ & $f(x) < 0$



Straight Objective Type

8. If $[a]$ denotes the greatest integer less than or equal to a and $-1 \leq x < 0$, $0 \leq y < 1$, $1 \leq z < 2$, then

$$\begin{vmatrix} [x]+1 & [y] & [z] \\ [x] & [y]+1 & [z] \\ [x] & [y] & [z]+1 \end{vmatrix} \text{ is equal to -}$$

- (A) $[x]$ (B) $[y]$ (C) $[z]$ (D) none of these
9. If $[x]$ and $\{x\}$ denotes the greatest integer function less than or equal to x and fractional part function respectively, then the number of real x , satisfying the equation $(x-2)[x] = \{x\} - 1$, is-
- (A) 0 (B) 1 (C) 2 (D) infinite
10. The range of the function $f(x) = \operatorname{sgn}\left(\frac{\sin^2 x + 2 \sin x + 4}{\sin^2 x + 2 \sin x + 3}\right)$ is (where $\operatorname{sgn}(\cdot)$ denotes signum function)-
- (A) $\{-1, 0, 1\}$ (B) $\{-1, 0\}$ (C) $\{1\}$ (D) $\{0, 1\}$
11. If $2f(x) - 3f\left(\frac{1}{x}\right) = x^2$, x is not equal to zero, then $f(2)$ is equal to-

- (A) $-\frac{7}{4}$ (B) $\frac{5}{2}$ (C) -1 (D) none of these

12. The number of integers lying in the domain of the function $f(x) = \sqrt{\log_{0.5} \left(\frac{5-2x}{x} \right)}$ is -
 (A) 3 (B) 2 (C) 1 (D) 0
13. The range of the function $f: \mathbb{N} \rightarrow \mathbb{I}; f(x) = (-1)^{x-1}$, is -
 (A) $[-1, 1]$ (B) $\{-1, 1\}$ (C) $\{0, 1\}$ (D) $\{0, 1, -1\}$
14. The range of the function $f(x) = e^{-x} + e^x$, is -
 (A) $f(x) \geq 1$ (B) $f(x) \leq 1$ (C) $f(x) \geq 2$ (D) $f(x) \leq 2$
15. If $f(x) = \frac{4^x}{4^x + 2}$, then $f(x) + f(1-x)$ is equal to -
 (A) 0 (B) -1 (C) 1 (D) 4
16. The range of the function $f(x) = \sqrt{4-x^2} + \sqrt{x^2-1}$ is
 (A) $[\sqrt{3}, \sqrt{7}]$ (B) $[\sqrt{3}, \sqrt{5}]$ (C) $[\sqrt{2}, \sqrt{3}]$ (D) $[\sqrt{3}, \sqrt{6}]$
17. A function f has domain $[-1, 2]$ and range $[0, 1]$. The domain and range respectively of the function g defined by $g(x) = 1 - f(x+1)$ is
 (A) $[-1, 1]; [-1, 0]$ (B) $[-2, 1]; [0, 1]$ (C) $[0, 2]; [-1, 0]$ (D) $[1, 3]; [-1, 0]$
18. For the function $f(x) = \frac{e^x + 1}{e^x - 1}$, if $n(d)$ denotes the number of integers which are not in its domain and $n(r)$ denotes the number of integers which are not in its range, then $n(d) + n(r)$ is equal to -
 (A) 2 (B) 3 (C) 4 (D) Infinite
19. If $x^4 f(x) - \sqrt{1 - \sin 2\pi x} = |f(x)| - 2f(x)$, then $f(-2)$ equals
 (A) $\frac{1}{17}$ (B) $\frac{1}{11}$ (C) $\frac{1}{19}$ (D) 0
20. Let $f: \mathbb{R} - \left\{ \frac{-15}{2} \right\} \rightarrow \mathbb{R} - \left\{ \frac{1}{2} \right\}$ be defined by $f(x) = \frac{x+10}{2x+15}$ then $f(x)$ is -
 (A) one-one but not onto (B) many one but not-onto
 (C) one-one and onto (D) many one and onto
21. $f: \mathbb{R} \rightarrow \mathbb{R}$ $f(x) = \frac{2x^2 - 5x + 3}{8x^2 + 9x + 11}$, then f is -
 (A) one-one onto (B) many-one onto (C) one-one into (D) many one into
22. If $f: \mathbb{R} \rightarrow \mathbb{R}$ & $f(x) = \frac{\sin([x]\pi)}{x^2 + 2x + 3} + 2x - 1 + \sqrt{x(x-1) + \frac{1}{4}}$ (where $[x]$ denotes integral part of x), then $f(x)$ is -
 (A) one-one but not onto (B) one-one & onto
 (C) onto but not one-one (D) neither one-one nor onto

23. Which of the following function is surjective but not injective

(A) $f: \mathbb{R} \rightarrow \mathbb{R} \quad f(x) = x^4 + 2x^3 - x^2 + 1$

(B) $f: \mathbb{R} \rightarrow \mathbb{R} \quad f(x) = x^3 + x + 1$

(C) $f: \mathbb{R} \rightarrow \mathbb{R}^+ \quad f(x) = \sqrt{1+x^2}$

(D) $f: \mathbb{R} \rightarrow \mathbb{R} \quad f(x) = x^3 + 2x^2 - x + 1$

24. If $f(x) = x|x|$ then $f^{-1}(x)$ equals-

(A) $\sqrt{|x|}$

(B) $(\operatorname{sgn} x) \cdot \sqrt{|x|}$

(C) $-\sqrt{|x|}$

(D) Does not exist

(where $\operatorname{sgn}(x)$ denotes signum function of x)

25. If $f: (-\infty, 3] \rightarrow [7, \infty)$; $f(x) = x^2 - 6x + 16$, then which of the following is true -

(A) $f^{-1}(x) = 3 + \sqrt{x-7}$

(B) $f^{-1}(x) = 3 - \sqrt{x-7}$

(C) $f^{-1}(x) = \frac{1}{x^2 - 6x + 16}$

(D) f is many-one

26. $f: \mathbb{R} \rightarrow \mathbb{R}$ such that $f(x) = \ln(x + \sqrt{x^2 + 1})$. Another function $g(x)$ is defined such that $g \circ f(x) = x \quad \forall x \in \mathbb{R}$. Then $g(2)$ is -

(A) $\frac{e^2 + e^{-2}}{2}$

(B) e^2

(C) $\frac{e^2 - e^{-2}}{2}$

(D) e^{-2}

27. Let $P(x) = kx^3 + 2k^2x^2 + k^3$. The sum of all real numbers k for which $(x-2)$ is a factor of $P(x)$, is

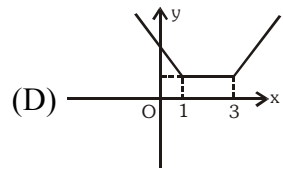
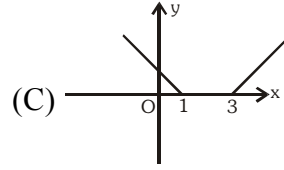
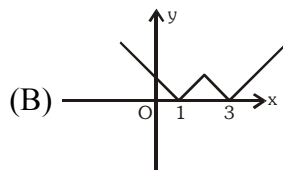
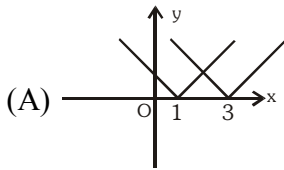
(A) 4

(B) 8

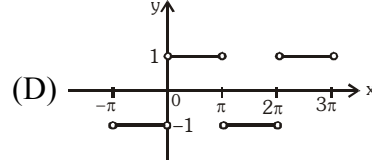
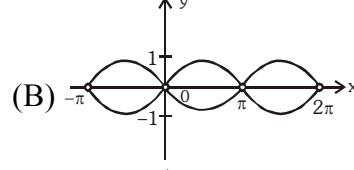
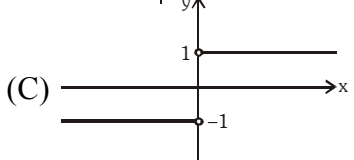
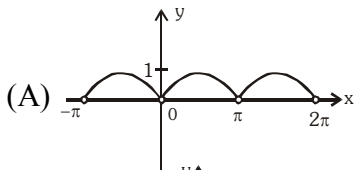
(C) -4

(D) -8

28. Which of the following is the graph of $y = |x-1| + |x-3|$?



29. Which of the following is the graph of $y = \frac{|\sin x|}{\sin x}$?



30. Period of function $f(x) = \min\{\sin x, |x|\} + \frac{x}{\pi} - \left[\frac{x}{\pi}\right]$ (where $[.]$ denotes greatest integer function) is -

(A) $\pi/2$

(B) π

(C) 2π

(D) 4π

Multiple Correct Answer

31. Which of the following function(s) have the same domain and range?

(A) $f(x) = \sqrt{1-x^2}$

(B) $g(x) = \frac{1}{x}$

(C) $h(x) = \sqrt{x}$

(D) $l(x) = \sqrt{4-x}$

32. Let $f(x) = x^2 + 3x + 2$, then number of solutions to -
 (A) $f(|x|) = 2$ is 1 (B) $f(|x|) = 2$ is 3 (C) $|f(x)| = 0.125$ is 4 (D) $|f(|x|)| = 0.125$ is 8
33. Which of the following pair(s) of function have same graphs?
 (A) $f(x) = \frac{\sec x}{\cos x} - \frac{\tan x}{\cot x}$, $g(x) = \frac{\cos x}{\sec x} + \frac{\sin x}{\csc x}$
 (B) $f(x) = \operatorname{sgn}(x^2 - 4x + 5)$, $g(x) = \operatorname{sgn}\left(\cos^2 x + \sin^2\left(x + \frac{\pi}{3}\right)\right)$ where sgn denotes signum function.
 (C) $f(x) = e^{\ln(x^2 + 3x + 3)}$, $g(x) = x^2 + 3x + 3$
 (D) $f(x) = \frac{\sin x}{\sec x} + \frac{\cos x}{\csc x}$, $g(x) = \frac{2\cos^2 x}{\cot x}$
34. If a function is defined by an implicit equation $2^{|x|+|y|} + 2^{|x|-|y|} = 2$, then -
 (A) Domain of function is singleton
 (B) Range of function is singleton
 (C) graph of the function intersects the line $y = x$
 (D) maximum value of function is 2
35. For each real x , let $f(x) = \max\{x, x^2, x^3, x^4\}$, then $f(x)$ is -
 (A) x^4 for $x \leq -1$ (B) x^2 for $-1 < x \leq 0$ (C) $f\left(\frac{1}{2}\right) = \frac{1}{2}$ (D) $f\left(\frac{1}{2}\right) = \frac{1}{4}$
36. Let $f(x) = \sin^6 x + \cos^6 x$, then -
 (A) $f(x) \in [0, 1] \forall x \in \mathbb{R}$ (B) $f(x) = 0$ has no solution
 (C) $f(x) \in \left[\frac{1}{4}, 1\right] \forall x \in \mathbb{R}$ (D) $f(x)$ is an injective function
37. Let $f(x) = \begin{cases} x^2 - 3x + 4 & ; x < 3 \\ x + 7 & ; x \geq 3 \end{cases}$ and $g(x) = \begin{cases} x + 6 & ; x < 4 \\ x^2 + x + 2 & ; x \geq 4 \end{cases}$, then which of the following is/are true -
 (A) $(f + g)(1) = 9$ (B) $(f - g)(3.5) = 1$ (C) $(f \cdot g)(0) = 24$ (D) $\left(\frac{f}{g}\right)(5) = \frac{8}{3}$

Matrix Match Type

38. Match the functions given in column-I correctly with mappings given in column-II.

Column-I

(A) $f: \left[-\frac{1}{2}, \frac{1}{2}\right] \rightarrow \left[\frac{4}{7}, \frac{4}{3}\right]$

$f(x) = \frac{1}{x^2 + x + 1}$

(B) $f: [-2, 2] \rightarrow [-1, 1]$

$f(x) = \sin x$

(C) $f: \mathbb{R} - \mathbb{I} \rightarrow \mathbb{R}$

$f(x) = \ell n\{x\}$, (where $\{.\}$ represents fractional part function)

(D) $f: (-\infty, 0] \rightarrow [1, \infty)$, $f(x) = (1 + \sqrt{-x}) + (\sqrt{-x} - x)$

Column-II

(P) Injective mapping

(Q) Non-injective mapping

(R) Surjective mapping

(S) Non-surjective mapping

(T) Bijective mapping

Linked Comprehension Type**Paragraph for Question 39 & 40**

$$\text{Let } f(x) = \begin{cases} x & ; x < 0 \\ 1-x & ; x \geq 0 \end{cases} \quad \& \quad g(x) = \begin{cases} x^2 & ; x < -1 \\ 2x+3 & ; -1 \leq x \leq 1 \\ x & ; x > 1 \end{cases}$$

On the basis of above information, answer the following questions :

39. Range of $f(x)$ is -
 (A) $(-\infty, 1]$ (B) $(-\infty, \infty)$ (C) $(-\infty, 0]$ (D) $(-\infty, 2]$
40. Range of $g(f(x))$ is -
 (A) $(-\infty, \infty)$ (B) $[1, 3) \cup (3, \infty)$ (C) $[1, \infty)$ (D) $[0, \infty)$

EXERCISE (O-2)**Straight Objective Type**

1. The number of integral values of x satisfying the inequality $[x-5][x-3] + 2 < [x-5] + 2[x-3]$ (where $[.]$ represents greatest integer function) is -
 (A) 0 (B) 1 (C) 2 (D) 3
2. Range of function $f(x) = \log_2 \left(\frac{4}{\sqrt{x+2} + \sqrt{2-x}} \right)$ is given by
 (A) $(0, \infty)$ (B) $\left[\frac{1}{2}, 1 \right]$ (C) $[1, 2]$ (D) $\left[\frac{1}{4}, 1 \right]$
3. A lion moves in the region given by the graph $y - |y| - x + |x| = 0$. Then on which of the following curve a person can move so that he does not encounter lion -
 (A) $y = e^{-|x|}$ (B) $y = \frac{1}{x}$ (C) $y = \text{signum}(x)$ (D) $y = -|4 + |x||$
4. Suppose, $f(x, n) = \sum_{k=1}^n \log_x \left(\frac{k}{x} \right)$, then the value of x satisfying the equation $f(x, 10) = f(x, 11)$, is
 (A) 9 (B) 10 (C) 11 (D) none
5. Range of $f(x) = \frac{\sec x + \tan x - 1}{\tan x - \sec x + 1}$; $x \in \left(0, \frac{\pi}{2} \right)$ is-
 (A) $(0, 1)$ (B) $(1, \infty)$ (C) $(-1, 0)$ (D) $(-\infty, -1)$
6. If $f(x, y) = \max(x, y) + \min(x, y)$ and $g(x, y) = \max(x, y) - \min(x, y)$, then the value of $f\left(g\left(-\frac{2}{3}, -\frac{3}{2}\right), g(-3, -4)\right)$ is greater than -
 (A) 1 (B) 2 (C) 3 (D) 4

7. If functions $f(x)$ and $g(x)$ are defined on $\mathbb{R} \rightarrow \mathbb{R}$ such that $f(x) = \begin{cases} x+3 & , x \in \text{rational} \\ 4x & , x \in \text{irrational} \end{cases}$,
 $g(x) = \begin{cases} x + \sqrt{5} & , x \in \text{irrational} \\ -x & , x \in \text{rational} \end{cases}$ then $(f - g)(x)$ is-
 (A) one-one & onto (B) neither one-one nor onto
 (C) one-one but not onto (D) onto but not one-one
8. Let $f: A \rightarrow B$ be an onto function such that $f(x) = \sqrt{x-2-2\sqrt{x-3}} - \sqrt{x-2+2\sqrt{x-3}}$, then set 'B' is-
 (A) $[-2,0]$ (B) $[0,2]$ (C) $[-3,0]$ (D) $[-1,0]$
9. Let $f(x) = \frac{x}{1-x}$ and let α be a real number. If $x_0 = \alpha, x_1 = f(x_0), x_2 = f(x_1), \dots$ & $x_{2011} = -\frac{1}{2012}$ then the value of α is -
 (A) $\frac{2011}{2012}$ (B) 1 (C) 2011 (D) -1
10. If $f_1(x) = 2^{f_2(x)}$, where $f_2(x) = 2012^{f_3(x)}$, where $f_3(x) = \left(\frac{1}{2013}\right)^{f_4(x)}$, where $f_4(x) = \log_{2013} \log_x 2012$, then the range of $f_1(x)$ is -
 (A) $(2, \infty)$ (B) $(2012, \infty)$ (C) $(0, \infty)$ (D) $(-\infty, \infty)$
11. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = \ln(x + \sqrt{x^2 + 1})$, then number of solutions of $|f^{-1}(x)| = e^{-|x|}$ is :-
 (A) 1 (B) 2 (C) 3 (D) Infinite
12. $f(x) = [x - 1] + \{x\}^{[x]}$, $x \in (1, 3)$, then $f^{-1}(x)$ is -
 (where $[.]$ denotes greatest integer function and $\{.\}$ denotes fractional part function)
 (A) $\begin{cases} x+1 & x \in (1, 2) \\ 2+\sqrt{x-1} & x \in [2, 3) \end{cases}$ (B) $\begin{cases} x-1 & x \in (1, 2) \\ 2-\sqrt{x-1} & x \in [2, 3) \end{cases}$
 (C) $\begin{cases} x-1 & x \in (0, 1) \\ 2-\sqrt{x-1} & x \in [1, 2) \end{cases}$ (D) $\begin{cases} x+1 & x \in (0, 1) \\ 2+\sqrt{x-1} & x \in [1, 2) \end{cases}$
13. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ and $f(x) = x^3 + ax^2 + bx - 8$. If $f(x) = 0$ has three real roots & $f(x)$ is a bijective function, then $(a + b)$ is equal to
 (A) 0 (B) 6 (C) -6 (D) 12
14. Which of the following functions is an odd function ?
 (A) $|x - 2| + (x + 2) \operatorname{sgn}(x + 2)$ (B) $\frac{1}{x(e^x - 1)} + \frac{1}{2x}$
 (C) $\log(\sin x + \sqrt{1 + \sin^2 x})$ (D) $e^{-4x} (e^{2x} - 1)^4$
 (where $\operatorname{sgn}(x)$ denotes signum function of x)

15. Period of $f(x) = \{x\} + \left\{x + \frac{1}{3}\right\} + \left\{x + \frac{2}{3}\right\}$ is equal to (where $\{.\}$ denotes fractional part function)
- (A) 1 (B) $\frac{2}{3}$ (C) $\frac{1}{2}$ (D) $\frac{1}{3}$
16. Let $f(x) = 2x - \left\{\frac{x}{\pi}\right\}$ and $g(x) = \cos x$, where $\{.\}$ denotes fractional part function, then period of $g \circ f(x)$ is -
- (A) $\frac{\pi}{2}$ (B) π (C) $\frac{3\pi}{2}$ (D) $\frac{\pi}{4}$
17. The period of the function $\frac{\sin x + \sin 5x}{\cos x + \cos 5x}$ is -
- (A) $\frac{\pi}{3}$ (B) $\frac{\pi}{2}$ (C) π (D) 2π
18. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a real valued function such that $f(10+x) = f(10-x) \forall x \in \mathbb{R}$ and $f(20+x) = -f(20-x) \forall x \in \mathbb{R}$. Then which of the following statements is true -
- (A) $f(x)$ is odd and periodic (B) $f(x)$ is odd and aperiodic
(C) $f(x)$ is even and periodic (D) $f(x)$ is even and aperiodic

Multiple Correct Answer Type

19. Let $f(x) = \begin{cases} x^2 & ; 0 < x < 2 \\ 2x-3 & ; 2 \leq x < 3 \\ x+2 & ; x \geq 3 \end{cases}$ Then :-
- (A) $f\left\{f\left(f\left(\frac{3}{2}\right)\right)\right\} = f\left(\frac{3}{2}\right)$ (B) $1 + f\left\{f\left(f\left(\frac{5}{2}\right)\right)\right\} = f\left(\frac{5}{2}\right)$
(C) $f\{f(1)\} = f(1) = 1$ (D) none of these
20. The range of the function $f(\theta) = \sqrt{8\sin^2 \theta + 4\cos^2 \theta - 8\sin \theta \cos \theta}$ is -
- (A) $[\sqrt{5}-1, \sqrt{5}+1]$ (B) $[0, \sqrt{5}+1]$
(C) $[\sqrt{6-\sqrt{20}}, \sqrt{6+\sqrt{20}}]$ (D) none of these
21. For the function $f(x) = |x+3| - |x+1| - |x-1| + |x-3|$, identify correct option(s)
- (A) Range of $f(x)$ is $(-\infty, 4]$ (B) maximum value of $f(x)$ is 4
(C) $f(x) = 4$ has infinite solutions (D) $f(x) = 0$ has infinite solutions
22. If $\{x\} = \frac{2}{3}$ & $\left[x + \left\{x + \left[x + \left\{x + \dots \dots 100 \text{ times} \right\}\right]\right\}\right] = 5$, then -
- (A) $x = \frac{14}{3}$ (B) $[x] = 5$ (C) $x = \frac{17}{3}$ (D) $[x] = 4$
(where $[.]$ & $\{.\}$ denotes greatest integer function & fractional part function respectively)

23. The values of x in $[-2\pi, 2\pi]$, for which the graph of the function $y = \sqrt{\frac{1+\sin x}{1-\sin x}} - \sec x$ and $y = -\sqrt{\frac{1-\sin x}{1+\sin x}} + \sec x$, coincide are
- (A) $\left[-2\pi, -\frac{3\pi}{2}\right) \cup \left(\frac{3\pi}{2}, 2\pi\right]$ (B) $\left(-\frac{3\pi}{2}, -\frac{\pi}{2}\right) \cup \left(\frac{\pi}{2}, \frac{3\pi}{2}\right)$
- (C) $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ (D) $[-2\pi, 2\pi] - \left\{\pm\frac{\pi}{2}, \pm\frac{3\pi}{2}\right\}$
24. A parabola of the form $y = ax^2 + bx + c$ with $a > 0$ intersects the graph of $f(x) = \frac{1}{x^2 - 4}$. Number of possible distinct intersection(s) of these graph is
- (A) 0 (B) 2 (C) 3 (D) 4
25. Let $f: \{1, 2, 3, 4, 5\} \rightarrow \{1, 2, 3, 4, 5\}$ is such that $f(x)$ is a one-one function satisfying following condition $f(x) = x + 1$ **if and only if** x is even (i.e. $f(3) \neq 4$, $f(4) = 5$ etc). Then $f^{-1}(2)$ can be-
- (A) 1 (B) 3 (C) 5 (D) 2
26. Which of the following statement(s) is(are) correct ?
- (A) If f is a one-one mapping from set A to A , then f is onto.
- (B) If f is an onto mapping from set A to A , then f is one-one
- (C) Let f and g be two functions defined from $\mathbb{R} \rightarrow \mathbb{R}$ such that $g \circ f$ is injective, then f must be injective.
- (D) If set A contains 3 elements while set B contains 2 elements, then total number of functions from A to B is 8.
27. If $f(x) = ax + b$ and $f(f(f(x))) = 27x + 13$ where a and b are real numbers, then-
- (A) $a + b = 3$ (B) $a + b = 4$ (C) $f'(x) = 3$ (D) $f'(x) = -3$
28. Given $f(1) = 2$ and $f(n+1) = \frac{f(n)-1}{f(n)+1} \forall n \in \mathbb{N}$, then-
- (A) $f(2015) = -\frac{1}{2}$ (B) $(f(2012))^{f(2013)} = 9$
- (C) $f(1001) = 2$ (D) $f(2015) = -3$
29. Which of the following is/are true?
- (A) $f(x) = e^x$ and $g(x) = \ln x$, then $f(g(x)) = x$ (wherever $f(g(x))$ is defined)
- (B) $f(x) = \frac{2+x}{3-x}$ and $g(x) = \frac{(3x-2)}{(x+1)}$, then $f(g(x)) = x$ (wherever $f(g(x))$ is defined)
- (C) $f(x) = 4x + 7$ and $g(x) = \frac{(x-7)}{4}$, then $f(g(x)) = x$ (wherever $f(g(x))$ is defined)
- (D) $f(x) = x^3 + 1$ and $g(x) = (x-1)^{1/3}$, then $f(g(x)) = x$ (wherever $f(g(x))$ is defined)

30. If $g(x) = x^2 - x + 1$ and $f(x) = \sqrt{\frac{1}{x} - x}$, then -

- (A) Domain of $f(g(x))$ is $[0, 1]$ (B) Range of $f(g(x))$ is $\left(0, \frac{7}{2\sqrt{3}}\right]$
 (C) $f(g(x))$ is many-one function (D) $f(g(x))$ is unbounded function

Linked Comprehension Type

Paragraph for Question 31 to 33

$$f(x) = \frac{x^2 + 1}{ax} \quad (a \neq 0); \quad g(x) = 3 \sec x; \quad h(x) = \frac{x + 3}{x - 4}.$$

On the basis of above information, answer the following questions :

31. If range of $f(x)$ and $g(x)$ are equal sets then 'a' is equal to -
 (A) 3 (B) $-2/3$ (C) $3/2$ (D) $-3/2$
32. $f(x)$ is one-one if-
 (A) $x \in (0, \infty)$ (B) $x \in (-\infty, 0)$ (C) $x \in (1, \infty)$ (D) $x \in (-\infty, 1) - \{0\}$
33. Which the following is always false ?
 (A) $h(x)$ is one-one
 (B) $f(x)$ is one-one if $x > 10$
 (C) $g(x)$ is many-one if $x \in \left(0, \frac{\pi}{2}\right) \cup \left(\frac{\pi}{2}, 3\right)$
 (D) The values of k for which $f(x) = k$ has exactly one solution is $k = 2$ or $k = -2$

Matrix Match Type

- | 34. Column-I | Column-II |
|--|-----------------|
| Number of integers in | |
| (A) Domain of $f(x) = \ln\{x\}$ | (P) 0 |
| (B) Domain of $f(x) = \sqrt{\sec(\sin x)} + \sqrt{\left[x + \frac{1}{x}\right]} + \sqrt{10 - [x]^2}$ | (Q) 2 |
| (C) Range of $f(x) = x^2 - 2x + 2, x \in [0, 2]$ | (R) 3 |
| (D) Range of $f(x) = \sqrt{25 - [x]^2}$ | (S) less than 3 |
| | (T) more than 3 |

(where $[.]$ and $\{.\}$ denote greatest integer function and fractional part function respectively)

35. Match the function mentioned in column-I with the respective classification given in column-II.
 (where $[.]$ and $\{.\}$ denote greatest integer function and fractional part function respectively)

- | Column-I | Column-II |
|---|---------------|
| (A) $f: \mathbb{R} \rightarrow \mathbb{R}^+ \quad f(x) = (e^{[x]})(e^{\{x\}})$ | (P) one-one |
| (B) $f: (-\infty, -2) \cup (0, \infty) \rightarrow \mathbb{R} \quad f(x) = \ln(x^2 + 2x)$ | (Q) many-one |
| (C) $f: [-2, 2] \rightarrow [-1, 1] \quad f(x) = \sin x$ | (R) onto |
| (D) $f: \mathbb{R} \rightarrow \mathbb{R} \quad f(x) = x^3 - 3x^2 + 3x - 7$ | (S) periodic |
| | (T) aperiodic |

EXERCISE (S-1)

1. Find the domains of definitions of the following functions :

(Read the symbols $[*]$ and $\{*\}$ as greatest integers and fractional part functions respectively.)

(i) $f(x) = \sqrt{\cos 2x} + \sqrt{16 - x^2}$ (ii) $f(x) = \log_7 \log_5 \log_3 \log_2 (2x^3 + 5x^2 - 14x)$

(iii) $f(x) = \ln(\sqrt{x^2 - 5x - 24} - x - 2)$ (iv) $f(x) = \sqrt{\frac{1 - 5^x}{7^{-x} - 7}}$

(v) $y = \log_{10} \sin(x - 3) + \sqrt{16 - x^2}$ (vi) $f(x) = \log_{100x} \left(\frac{2 \log_{10} x + 1}{-x} \right)$

(vii) $f(x) = \sqrt{x^2 - |x|} + \frac{1}{\sqrt{9 - x^2}}$ (viii) $f(x) = \sqrt{(x^2 - 3x - 10) \ln^2(x - 3)}$

(ix) $f(x) = \sqrt{(5x - 6 - x^2) \left[\{ \ln \{x\} \} \right]} + \sqrt{(7x - 5 - 2x^2)} + \left(\ln \left(\frac{7}{2} - x \right) \right)^{-1}$

(x) $f(x) = \log_{\left[\frac{x+1}{x} \right]} \left| x^2 - x - 6 \right| + {}^{16-x}C_{2x-1} + {}^{20-3x}P_{2x-5}$

2. Find the domain & range of the following functions.

(i) $y = \log_{\sqrt{5}} (\sqrt{2} (\sin x - \cos x) + 3)$ (ii) $y = \frac{2x}{1+x^2}$ (iii) $f(x) = \frac{x^2 - 3x + 2}{x^2 + x - 6}$

(iv) $f(x) = \frac{x}{1+|x|}$ (v) $y = \sqrt{2-x} + \sqrt{1+x}$ (vi) $f(x) = \frac{\sqrt{x+4} - 3}{x-5}$

3. (a) Draw graphs of the following function, where $[]$ denotes the greatest integer function.

(i) $f(x) = x + [x]$ (ii) $y = (x)^{[x]}$ where $x = [x] + (x)$ & $x > 0$ & $x \leq 3$

(iii) $y = \operatorname{sgn}[x]$ (iv) $\operatorname{sgn}(x - |x|)$

(b) Identify the pair(s) of functions which are identical ?

(where $[x]$ denotes greatest integer and $\{x\}$ denotes fractional part function)

(i) $f(x) = \operatorname{sgn}(x^2 - 3x + 4)$ and $g(x) = e^{\{x\}}$ (ii) $f(x) = \sqrt{\frac{1 - \cos 2x}{1 + \cos 2x}}$ and $g(x) = \tan x$

(iii) $f(x) = \ln(1+x) + \ln(1-x)$ and $g(x) = \ln(1-x^2)$ (iv) $f(x) = \frac{\cos x}{1 - \sin x}$ and $g(x) = \frac{1 + \sin x}{\cos x}$

4. Classify the following functions $f(x)$ defined in $\mathbb{R} \rightarrow \mathbb{R}$ as injective, surjective, both or none.

(a) $f(x) = \frac{x^2 + 4x + 30}{x^2 - 8x + 18}$ (b) $f(x) = x^3 - 6x^2 + 11x - 6$ (c) $f(x) = (x^2 + x + 5)(x^2 + x - 3)$

5. Solve the following problems from (a) to (d) on functional equation :

(a) The function $f(x)$ defined on the real numbers has the property that $f(f(x)) \cdot (1 + f(x)) = -f(x)$ for all x in the domain of f . If the number 3 is in the domain and range of f , compute the value of $f(3)$.

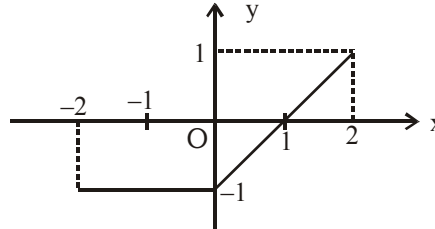
(b) Suppose f is a real function satisfying $f(x + f(x)) = 4f(x)$ and $f(1) = 4$. Find the value of $f(21)$.

(c) Let f be function defined from $\mathbb{R}^+ \rightarrow \mathbb{R}^+$. If $[f(xy)]^2 = x(f(y))^2$ for all positive numbers x and y and $f(2) = 6$, find the value of $f(50)$.

(d) Let f be a function such that $f(3) = 1$ and $f(3x) = x + f(3x - 3)$ for all x . Then find the value of $f(300)$.

6. Suppose $f(x) = \sin x$ and $g(x) = 1 - \sqrt{x}$. Then find the domain and range of the following functions.
 (a) fog (b) gof (c) fof (d) gog
7. A function $f : \mathbb{R} \rightarrow \mathbb{R}$ is such that $f\left(\frac{1-x}{1+x}\right) = x$ for all $x \neq -1$. Prove the following.
 (a) $f(f(x)) = x$ (b) $f(1/x) = -f(x)$, $x \neq 0$ (c) $f(-x-2) = -f(x)-2$.
8. (a) Find the formula for the function fogoh, given $f(x) = \frac{x}{x+1}$; $g(x) = x^{10}$ and $h(x) = x+3$. Find also the domain of this function. Also compute (fogoh)(-1).
 (b) If $f(x) = \max(x, 1/x)$ for $x > 0$ where $\max(a, b)$ denotes the greater of the two real numbers a and b . Define the function $g(x) = f(x) f(1/x)$ and plot its graph.
9. Let f be a one-one function with domain $\{x, y, z\}$ and range $\{1, 2, 3\}$. It is given that exactly one of the following statements is true and the remaining two are false.
 $f(x) = 1$; $f(y) \neq 1$; $f(z) \neq 2$. Determine $f^{-1}(1)$
10. $f(x) = \begin{cases} 1-x & \text{if } x \leq 0 \\ x^2 & \text{if } x > 0 \end{cases}$ and $g(x) = \begin{cases} -x & \text{if } x < 1 \\ 1-x & \text{if } x \geq 1 \end{cases}$ find (fog)(x) and (gof)(x).
11. Find whether the following functions are even or odd or none :
 (a) $f(x) = \log(x + \sqrt{1+x^2})$ (b) $f(x) = \frac{x(a^x+1)}{a^x-1}$ (c) $f(x) = \sin x + \cos x$
 (d) $f(x) = x \sin^2 x - x^3$ (e) $f(x) = \sin x - \cos x$ (f) $f(x) = \frac{(1+2^x)^2}{2^x}$
 (g) $f(x) = \frac{x}{e^x-1} + \frac{x}{2} + 1$ (h) $f(x) = [(x+1)^2]^{1/3} + [(x-1)^2]^{1/3}$
12. (i) Write explicitly, functions of y defined by the following equations and also find the domains of definition of the given implicit functions :
 (a) $10^x + 10^y = 10$ (b) $x + |y| = 2y$
 (ii) The function $f(x)$ is defined on the interval $[0, 1]$. Find the domain of definition of the functions.
 (a) $f(\sin x)$ (b) $f(2x+3)$
 (iii) Given that $y = f(x)$ is a function whose domain is $[4, 7]$ and range is $[-1, 9]$. Find the range and domain of
 (a) $g(x) = \frac{1}{3}f(x)$ (b) $h(x) = f(x-7)$
13. Compute the inverse of the functions :
 (a) $f(x) = \ln(x + \sqrt{x^2+1})$ (b) $f(x) = 2^{\frac{x}{x-1}}$ (c) $y = \frac{10^x - 10^{-x}}{10^x + 10^{-x}}$

14. Find the inverse of $f(x) = 2^{\log_{10} x} + 8$ and hence solve the equation $f(x) = f^{-1}(x)$.
15. (a) Function f & g are defined by $f(x) = \sin x$, $x \in \mathbb{R}$; $g(x) = \tan x$, $x \in \mathbb{R} - \left(K + \frac{1}{2}\right)\pi$ where $K \in \mathbb{I}$. Find (i) periods of $f \circ g$ & $g \circ f$ (ii) range of the function $f \circ g$ & $g \circ f$
- (b) Suppose that f is an even, periodic function with period 2, and that $f(x) = x$ for all x in interval $[0, 1]$. Find the value of $f(3.14)$.
16. The graph of the function $y = f(x)$ is as follows :

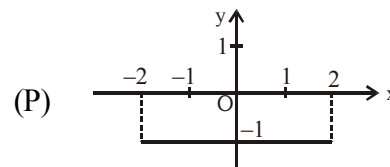


Match the function mentioned in **Column-I** with the respective graph given in **Column-II**.

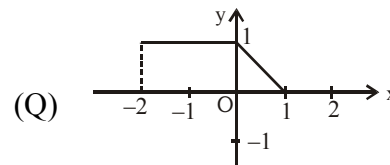
Column-I

Column-II

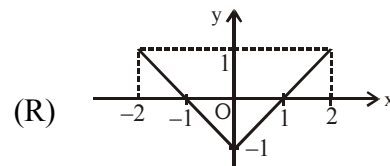
(A) $y = |f(x)|$



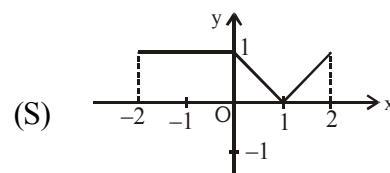
(B) $y = f(|x|)$



(C) $y = f(-|x|)$



(D) $y = \frac{1}{2}(|f(x)| - f(x))$



17. **Column-I**

Column-II

f(x)

Range

(A) $\frac{\cos^2 x + \cos x + 2}{\cos^2 x + \cos x + 1}$

(P) $\left[0, \frac{7}{3}\right]$

(B) $\left| \frac{(\sqrt{\cos x} - \sqrt{\sin x})(\sqrt{\cos x} + \sqrt{\sin x})}{3(\cos x + \sin x)} \right|$

(Q) $\left[\frac{4}{3}, \frac{7}{3}\right]$

(C) $\frac{7}{3(x^6 + 2x^4 + 3x^2 + 1)}$

(R) $\left[0, \frac{1}{3}\right]$

(D) $\log_8(x^2 + 2x + 2)$

(S) $[0, \infty)$

18. The sum of integral values of the elements in the domain of $f(x) = \sqrt{\log_{\frac{1}{2}} |3-x|}$ is
19. Number of integers in range of $f(x) = x(x+2)(x+4)(x+6) + 7$, $x \in [-4, 2]$ is
20. The number of even integral value(s) in the range of the function $f(x) = \frac{\tan^2 x + 8 \tan x + 15}{1 + \tan^2 x}$ is
21. If $f(x) = a \log\left(\frac{1+x}{1-x}\right) + bx^3 + c \sin x + 5$ and $f(\log_3 2) = 4$, then $f\left(\log_3\left(\frac{1}{2}\right)\right)$ is equal to
22. Let $P(x) = x^4 + ax^3 + bx^2 + cx + d$ be a polynomial such that $P(1) = 1$, $P(2) = 8$, $P(3) = 27$, $P(4) = 64$, then find $P(10)$.
23. If $f(x) = \begin{cases} 0 & x < 1 \\ 2x - 2 & x \geq 1 \end{cases}$; then the number of solutions of the equation $f(f(f(x))) = x$ is
24. Let 'f' be an even periodic function with period '4' such that $f(x) = 2^x - 1$, $0 \leq x \leq 2$.
The number of solutions of the equation $f(x) = 1$ in $[-10, 20]$ are
25. Let $f(x) = \frac{2x-1}{x+3}$. If $f^{-1} = \frac{ax+b}{c-x}$, then $a + b + c$ is
26. Let $f(x)$ be a periodic function with period 'p' satisfying $f(x) + f(x+3) + f(x+6) + \dots + f(x+42) = \text{constant} \forall x \in \mathbb{R}$, then sum of digits of 'p' is

EXERCISE (S-2)

1. Find the domains of definitions of the following functions :
(Read the symbols $[*]$ as greatest integers function)
- (i) $f(x) = \sqrt{\log_x (\cos 2\pi x)}$ (ii) $f(x) = \frac{[x]}{2x - [x]}$
2. (a) Let $P(x) = x^6 + ax^5 + bx^4 + cx^3 + dx^2 + ex + f$ be a polynomial such that $P(1) = 1$; $P(2) = 2$; $P(3) = 3$; $P(4) = 4$; $P(5) = 5$ and $P(6) = 6$, then find the value of $P(7)$.
- (b) Let a and b be real numbers and let $f(x) = a \sin x + b \sqrt[3]{x} + 4$, $\forall x \in \mathbb{R}$. If $f(\log_{10}(\log_3 10)) = 5$ then find the value of $f(\log_{10}(\log_3 3))$.
3. Suppose $p(x)$ is a polynomial with integer coefficients. The remainder when $p(x)$ is divided by $x - 1$ is 1 and the remainder when $p(x)$ is divided by $x - 4$ is 10. If $r(x)$ is the remainder when $p(x)$ is divided by $(x-1)(x-4)$, find the value of $r(2006)$.

4. A function f , defined for all $x, y \in \mathbb{R}$ is such that $f(1) = 2$; $f(2) = 8$ & $f(x+y) - kxy = f(x) + 2y^2$, where k is some constant. Find $f(x)$ & show that :

$$f(x+y)f\left(\frac{1}{x+y}\right) = k \text{ for } x+y \neq 0.$$

5. If $f(x) = -1 + |x - 2|$, $0 \leq x \leq 4$
 $g(x) = 2 - |x|$, $-1 \leq x \leq 3$

Then find $\text{fog}(x)$ & $\text{gof}(x)$. Draw rough sketch of the graphs of $\text{fog}(x)$ & $\text{gof}(x)$.

6. Let $f(x) = x^{135} + x^{125} - x^{115} + x^5 + 1$. If $f(x)$ is divided by $x^3 - x$ then the remainder is some function of x say $g(x)$. Find the value of $g(10)$.
7. Let $\{x\}$ & $[x]$ denote the fractional and integral part of a real number x respectively. Solve $4\{x\} = x + [x]$.
8. Let $f(x) = \frac{9^x}{9^x + 3}$ then find the value of the sum $f\left(\frac{1}{2006}\right) + f\left(\frac{2}{2006}\right) + f\left(\frac{3}{2006}\right) + \dots + f\left(\frac{2005}{2006}\right)$
9. Let $f(x) = (x+1)(x+2)(x+3)(x+4) + 5$ where $x \in [-6, 6]$. If the range of the function is $[a, b]$ where $a, b \in \mathbb{N}$ then find the value of $(a+b)$.
10. The set of real values of ' x ' satisfying the equality $\left[\frac{3}{x}\right] + \left[\frac{4}{x}\right] = 5$ (where $[]$ denotes the greatest integer function) belongs to the interval $\left[a, \frac{b}{c}\right]$ where $a, b, c \in \mathbb{N}$ and $\frac{b}{c}$ is in its lowest form. Find the value of $a + b + c + abc$.
11. $f(x)$ and $g(x)$ are linear function such that for all x , $f(g(x))$ and $g(f(x))$ are Identity functions. If $f(0) = 4$ and $g(5) = 17$, compute $f(2006)$.

Comprehension

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ is a function satisfying $f(2-x) = f(2+x)$ and $f(20-x) = f(x)$, $\forall x \in \mathbb{R}$.

On the basis of above information, answer the following questions :

12. If $f(0) = 5$, then minimum possible number of values of x satisfying $f(x) = 5$, for $x \in [0, 170]$ is -
 (A) 21 (B) 12 (C) 11 (D) 22
13. Graph of $y = f(x)$ is -
 (A) symmetrical about $x = 18$ (B) symmetrical about $x = 5$
 (C) symmetrical about $x = 8$ (D) symmetrical about $x = 20$
14. If $f(2) \neq f(6)$, then
 (A) fundamental period of $f(x)$ is 1 (B) fundamental period of $f(x)$ may be 1
 (C) period of $f(x)$ can't be 1 (D) fundamental period of $f(x)$ is 8
15. The function $f(x)$ has the property that for each real number x in its domain, $1/x$ is also in its domain and $f(x) + f(1/x) = x$. Find the largest set of real numbers that can be in the domain of $f(x)$?

EXERCISE (JM)

1. For real x , let $f(x) = x^3 + 5x + 1$, then :- [AIEEE - 2009]

- (1) f is one-one and onto $\mathbb{R}$ (2) f is neither one-one nor onto $\mathbb{R}$
 (3) f is one-one but not onto $\mathbb{R}$ (4) f is onto $\mathbb{R}$ but not one-one

2. The domain of the function $f(x) = \frac{1}{\sqrt{|x| - x}}$ is :- [AIEEE - 2011]

- (1) $(-\infty, 0)$ (2) $(-\infty, \infty) - \{0\}$ (3) $(-\infty, \infty)$ (4) $(0, \infty)$

3. Let f be a function defined by $f(x) = (x - 1)^2 + 1$, ($x \geq 1$) [AIEEE - 2009, 2011]

Statement - 1 : The set $\{x : f(x) = f^{-1}(x)\} = \{1, 2\}$

Statement - 2 : f is bijection and $f^{-1}(x) = 1 + \sqrt{x - 1}$, $x \geq 1$.

- (1) Statement-1 is true, Statement-2 is false.
 (2) Statement-1 is false, Statement-2 is true.
 (3) Statement-1 is true, Statement-2 is true ; Statement-2 is a correct explanation for Statement-1.
 (4) Statement-1 is true, Statement-2 is true ; Statement-2 is not a correct explanation for statement-1.
4. If $a \in \mathbb{R}$ and the equation $-3(x - [x])^2 + 2(x - [x]) + a^2 = 0$ (where $[x]$ denotes the greatest integer $\leq x$) has no integral solution, then all possible values of a lie in the interval :

[JEE(Main)-2014]

- (1) $(-1, 0) \cup (0, 1)$ (2) $(1, 2)$ (3) $(-2, -1)$ (4) $(-\infty, -2) \cup (2, \infty)$

5. If $f(x) + 2f\left(\frac{1}{x}\right) = 3x$, $x \neq 0$, and [JEE(Main)-2016]

$S = \{x \in \mathbb{R} : f(x) = f(-x)\}$; then S :

- (1) contains more than two elements. (2) is an empty set.
 (3) contains exactly one element (4) contains exactly two elements

6. The function $f : \mathbb{R} \rightarrow \left[-\frac{1}{2}, \frac{1}{2}\right]$ defined as $f(x) = \frac{x}{1+x^2}$, is : [JEE(Main)-2017]

- (1) neither injective nor surjective. (2) invertible.
 (3) injective but not surjective. (4) surjective but not injective

7. Let $S = \{x \in \mathbb{R} : x \geq 0 \text{ and } 2|\sqrt{x} - 3| + \sqrt{x}(\sqrt{x} - 6) + 6 = 0\}$. Then S : [JEE(Main)-2018]

- (1) contains exactly one element. (2) contains exactly two elements.
 (3) contains exactly four elements. (4) is an empty set.

EXERCISE (JA)

1. If functions $f(x)$ and $g(x)$ are defined on $\mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(x) = \begin{cases} 0, & x \in \text{rational} \\ x, & x \in \text{irrational} \end{cases}, \quad g(x) = \begin{cases} 0, & x \in \text{irrational} \\ x, & x \in \text{rational} \end{cases}, \text{ then } (f - g)(x) \text{ is -}$$

- (A) one-one and onto (B) neither one-one nor onto
(C) one-one but not onto (D) onto but not one-one [JEE 2005 (Scr.)]

2. Let $f(x) = x^2$ and $g(x) = \sin x$ for all $x \in \mathbb{R}$. Then the set of all x satisfying $(f \circ g \circ g \circ f)(x) = (g \circ g \circ f)(x)$, where $(f \circ g)(x) = f(g(x))$, is- [JEE 2011, 3, (-1)]

- (A) $\pm\sqrt{n\pi}, n \in \{0, 1, 2, \dots\}$ (B) $\pm\sqrt{n\pi}, n \in \{1, 2, \dots\}$
(C) $\frac{\pi}{2} + 2n\pi, n \in \{\dots, -2, -1, 0, 1, 2, \dots\}$ (D) $2n\pi, n \in \{\dots, -2, -1, 0, 1, 2, \dots\}$

3. The function $f : [0, 3] \rightarrow [1, 29]$, defined by $f(x) = 2x^3 - 15x^2 + 36x + 1$, is : [JEE 2012, 3, (-1)]

- (A) one-one and onto (B) onto but not one-one
(C) one-one but not onto (D) neither one-one nor onto

4. Let $f : (-1, 1) \rightarrow \mathbb{R}$ be such that $f(\cos 4\theta) = \frac{2}{2 - \sec^2 \theta}$ for $\theta \in \left(0, \frac{\pi}{4}\right) \cup \left(\frac{\pi}{4}, \frac{\pi}{2}\right)$. Then the value(s) of

$f\left(\frac{1}{3}\right)$ is (are) - [JEE 2012, 4]

- (A) $1 - \sqrt{\frac{3}{2}}$ (B) $1 + \sqrt{\frac{3}{2}}$ (C) $1 - \sqrt{\frac{2}{3}}$ (D) $1 + \sqrt{\frac{2}{3}}$

5. Let $f : \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \rightarrow \mathbb{R}$ be given by $f(x) = (\log(\sec x + \tan x))^3$. Then :-

- (A) $f(x)$ is an odd function (B) $f(x)$ is a one-one function
(C) $f(x)$ is an onto function (D) $f(x)$ is an even function
[JEE(Advanced)-2014, 3]

INVERSE TRIGONOMETRIC FUNCTION

1. INTRODUCTION :

The inverse trigonometric functions, denoted by $\sin^{-1}x$ or $(\arcsin x)$, $\cos^{-1}x$ etc., denote the angles whose sine, cosine etc, is equal to x . The angles are usually the numerically smallest angles, except in the case of $\cot^{-1}x$ and if positive & negative angles have same numerical value, the positive angle has been chosen.

It is worthwhile noting that the functions $\sin x$, $\cos x$ etc are in general not invertible. Their inverse is defined by choosing an appropriate domain & co-domain so that they become invertible. For this reason the chosen value is usually the simplest and easy to remember.

2. DOMAIN & RANGE OF INVERSE TRIGONOMETRIC FUNCTIONS :

S.No	$f(x)$	Domain	Range
(1)	$\sin^{-1}x$	$ x \leq 1$	$\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$
(2)	$\cos^{-1}x$	$ x \leq 1$	$[0, \pi]$
(3)	$\tan^{-1}x$	$x \in \mathbb{R}$	$\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$
(4)	$\sec^{-1}x$	$ x \geq 1$	$[0, \pi] - \left\{\frac{\pi}{2}\right\}$ or $\left[0, \frac{\pi}{2}\right) \cup \left(\frac{\pi}{2}, \pi\right]$
(5)	$\operatorname{cosec}^{-1}x$	$ x \geq 1$	$\left[-\frac{\pi}{2}, \frac{\pi}{2}\right] - \{0\}$
(6)	$\cot^{-1}x$	$x \in \mathbb{R}$	$(0, \pi)$

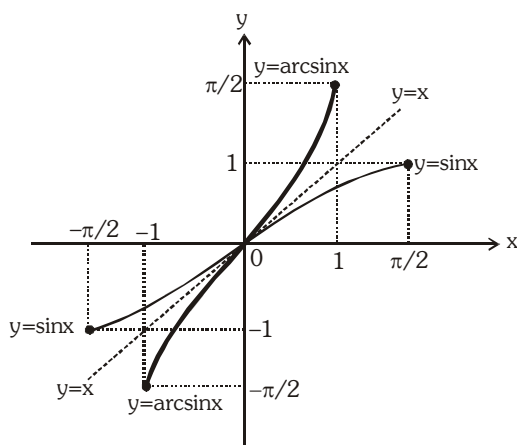
3. GRAPH OF INVERSE TRIGONOMETRIC FUNCTIONS :

(a) $f: \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \rightarrow [-1, 1]$

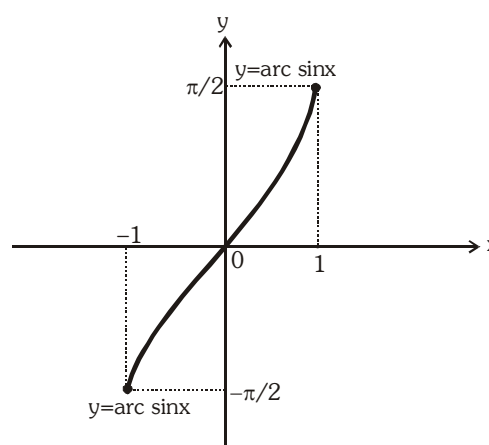
$f(x) = \sin x$

$f^{-1}: [-1, 1] \rightarrow [-\pi/2, \pi/2]$

$f^{-1}(x) = \sin^{-1}(x)$



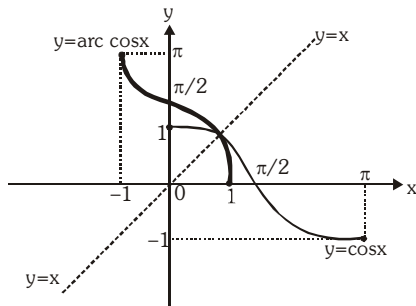
(Taking image of $\sin x$ about $y = x$ to get $\sin^{-1}x$)



$(y = \sin^{-1}x)$

(b) $f: [0, \pi] \rightarrow [-1, 1]$

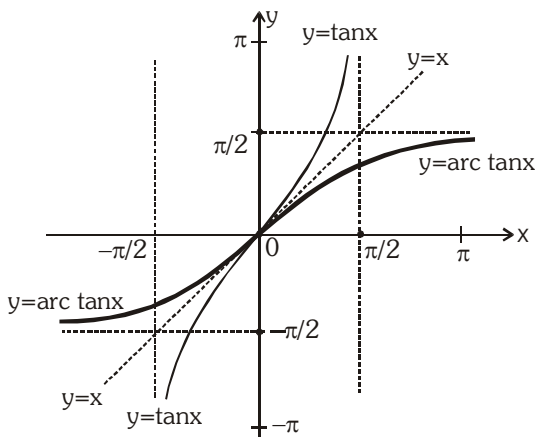
$f(x) = \cos x$



(Taking image of $\cos x$ about $y = x$)

(c) $f: (-\pi/2, \pi/2) \rightarrow \mathbb{R}$

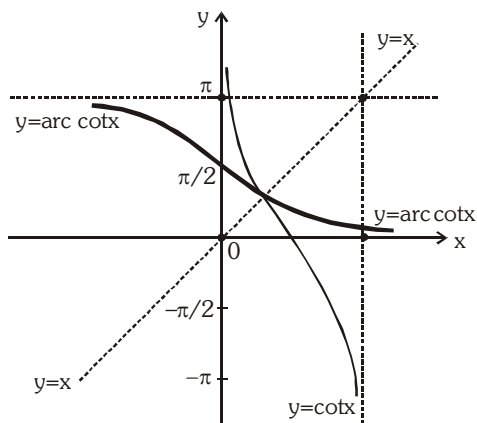
$f(x) = \tan x$



(Taking image of $\tan x$ about $y = x$)

(d) $f: (0, \pi) \rightarrow \mathbb{R}$

$f(x) = \cot x$



(Taking image of $\cot x$ about $y = x$)

(e) $f: [0, \pi/2) \cup (\pi/2, \pi] \rightarrow (-\infty, -1] \cup [1, \infty)$

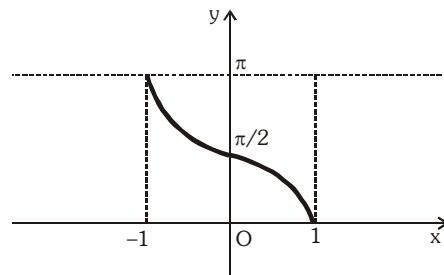
$f(x) = \sec x$

$f^{-1}: (-\infty, -1] \cup [1, \infty) \rightarrow [0, \pi/2) \cup (\pi/2, \pi]$

$f^{-1}(x) = \sec^{-1} x$

$f^{-1}: [-1, 1] \rightarrow [0, \pi]$

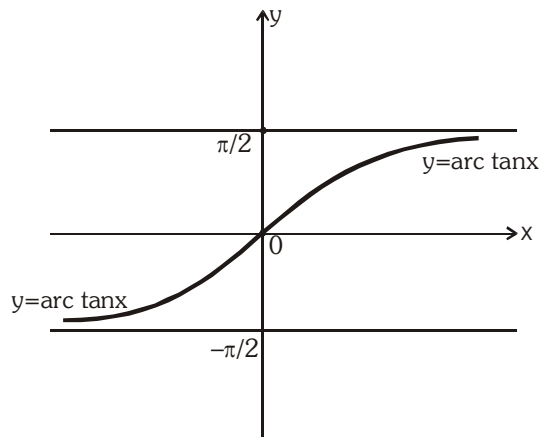
$f^{-1}(x) = \cos^{-1} x$



$(y = \cos^{-1} x)$

$f^{-1}: \mathbb{R} \rightarrow (-\pi/2, \pi/2)$

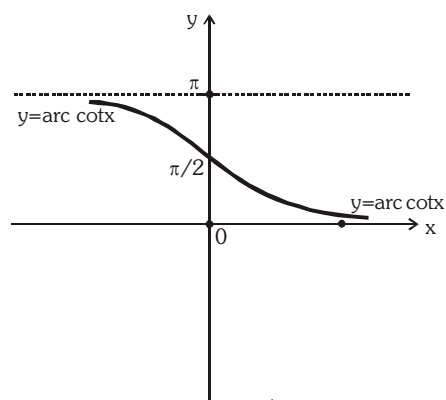
$f^{-1}(x) = \tan^{-1} x$



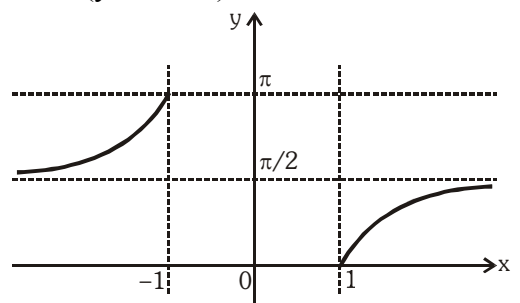
$(y = \tan^{-1} x)$

$f^{-1}: \mathbb{R} \rightarrow (0, \pi)$

$f^{-1}(x) = \cot^{-1} x$



$(y = \cot^{-1} x)$

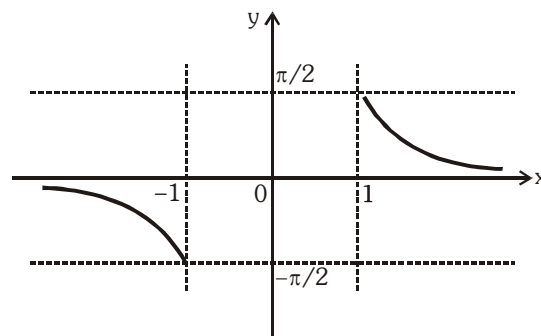


$$(f) \quad f: [-\pi/2, 0) \cup (0, \pi/2] \rightarrow (-\infty, -1] \cup [1, \infty)$$

$$f(x) = \operatorname{cosec} x$$

$$f^{-1}: (-\infty, -1] \cup [1, \infty) \rightarrow [-\pi/2, 0) \cup (0, \pi/2]$$

$$f^{-1}(x) = \operatorname{cosec}^{-1} x$$



From the above discussions following IMPORTANT points can be concluded :

- (i) All the inverse trigonometric functions represent an angle.
- (ii) If $x > 0$, then all six inverse trigonometric functions viz $\sin^{-1} x$, $\cos^{-1} x$, $\tan^{-1} x$, $\sec^{-1} x$, $\operatorname{cosec}^{-1} x$, $\cot^{-1} x$ represent an acute angle.
- (iii) If $x < 0$, then $\sin^{-1} x$, $\tan^{-1} x$ & $\operatorname{cosec}^{-1} x$ represent an angle from $-\pi/2$ to 0 (IVth quadrant)
- (iv) If $x < 0$, then $\cos^{-1} x$, $\cot^{-1} x$ & $\sec^{-1} x$ represent an obtuse angle. (IInd quadrant)
- (v) IIIrd quadrant is never used in range of inverse trigonometric function.

Illustration 1 : The value of $\tan^{-1}(1) + \cos^{-1}\left(-\frac{1}{2}\right) + \sin^{-1}\left(-\frac{1}{2}\right)$ is equal to

- (A) $\frac{\pi}{4}$ (B) $\frac{5\pi}{12}$ (C) $\frac{3\pi}{4}$ (D) $\frac{13\pi}{12}$

Solution : $\tan^{-1}(1) + \cos^{-1}\left(-\frac{1}{2}\right) + \sin^{-1}\left(-\frac{1}{2}\right) = \frac{\pi}{4} + \frac{2\pi}{3} - \frac{\pi}{6} = \frac{\pi}{4} + \frac{\pi}{2} = \frac{3\pi}{4}$ **Ans.(C)**

Illustration 2 : If $\sum_{i=1}^{2n} \cos^{-1} x_i = 0$ then find the value of $\sum_{i=1}^{2n} x_i$

Solution : We know, $0 \leq \cos^{-1} x \leq \pi$

Hence, each value $\cos^{-1} x_1, \cos^{-1} x_2, \cos^{-1} x_3, \dots, \cos^{-1} x_{2n}$ are non-negative their sum is zero only when each value is zero.

i.e., $\cos^{-1} x_i = 0$ for all i

$\Rightarrow x_i = 1$ for all i

$$\therefore \sum_{i=1}^{2n} x_i = x_1 + x_2 + x_3 + \dots + x_{2n} = \underbrace{\{1 + 1 + 1 + \dots + 1\}}_{2n \text{ times}} = 2n \quad \{\text{using (i)}\}$$

$$\Rightarrow \sum_{i=1}^{2n} x_i = 2n$$

Ans.

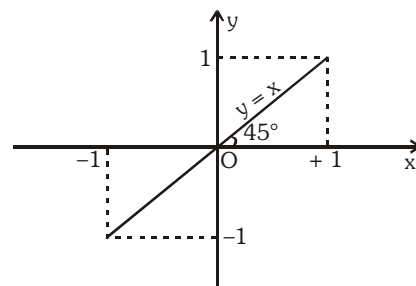
Do yourself - 1 :

- (i) If α, β are roots of the equation $6x^2 + 11x + 3 = 0$, then
 (A) both $\cos^{-1}\alpha$ and $\cos^{-1}\beta$ are real (B) both $\operatorname{cosec}^{-1}\alpha$ and $\operatorname{cosec}^{-1}\beta$ are real
 (C) both $\cot^{-1}\alpha$ and $\cot^{-1}\beta$ are real (D) none of these
- (ii) If $\sin^{-1}x + \sin^{-1}y = \pi$ and $x = ky$, then find the value of $39^{2k} + 5^k$.

3. PROPERTIES OF INVERSE CIRCULAR FUNCTIONS :

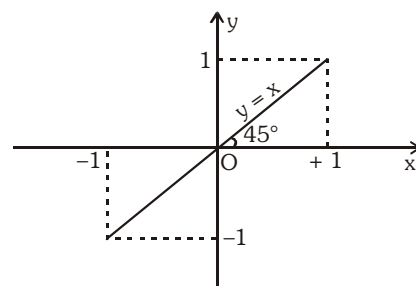
P-1 (i) $y = \sin(\sin^{-1}x) = x$

$x \in [-1, 1], y \in [-1, 1]$



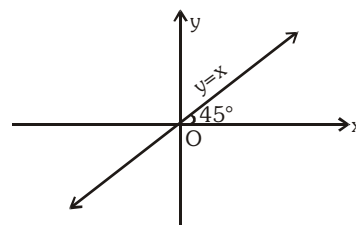
(ii) $y = \cos(\cos^{-1}x) = x$

$x \in [-1, 1], y \in [-1, 1]$



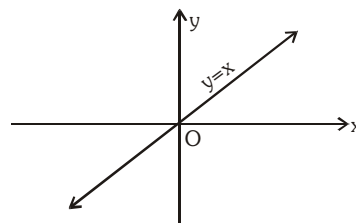
(iii) $y = \tan(\tan^{-1}x) = x$

$x \in \mathbb{R}, y \in \mathbb{R}$



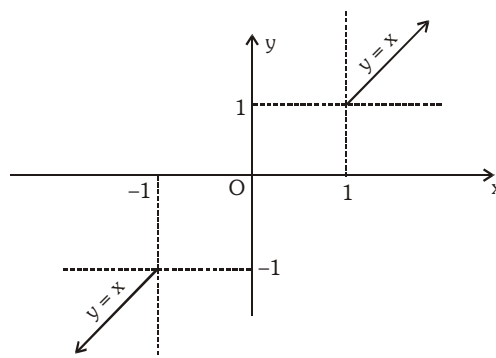
(iv) $y = \cot(\cot^{-1}x) = x$,

$x \in \mathbb{R}; y \in \mathbb{R}$



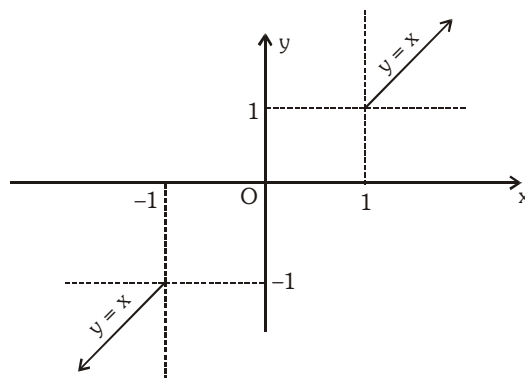
(v) $y = \operatorname{cosec}(\operatorname{cosec}^{-1}x) = x$,

$|x| \geq 1, |y| \geq 1$



$$(vi) \quad y = \sec(\sec^{-1} x) = x$$

$$|x| \geq 1 ; |y| \geq 1$$



Note : All the above functions are aperiodic.

Illustration 3 : Evaluate the following :

$$(i) \quad \sin(\cos^{-1} 3/5) \quad (ii) \quad \cos(\tan^{-1} 3/4) \quad (iii) \quad \sin\left(\frac{\pi}{2} - \sin^{-1}\left(-\frac{1}{2}\right)\right)$$

$$(iv) \quad \tan\left\{2 \tan^{-1} \frac{1}{5} - \frac{\pi}{4}\right\}$$

Solution :

$$(i) \quad \text{Let } \cos^{-1} 3/5 = \theta. \text{ Then,} \\ \cos \theta = 3/5 \Rightarrow \sin \theta = 4/5 \\ \therefore \sin(\cos^{-1} 3/5) = \sin \theta = 4/5$$

$$(ii) \quad \text{Let } \tan^{-1} 3/4 = \theta. \text{ Then,} \\ \tan \theta = 3/4 \\ \Rightarrow \cos \theta = \frac{4}{5} \quad \left\{ \because \text{ as } \cos^2 \theta = \frac{1}{1 + \tan^2 \theta} \right\} \\ \therefore \cos(\tan^{-1} 3/4) = \cos \theta = 4/5$$

$$(iii) \quad \sin\left(\frac{\pi}{2} - \sin^{-1}\left(-\frac{1}{2}\right)\right) = \sin\left(\frac{\pi}{2} - \left(-\frac{\pi}{6}\right)\right) = \sin \frac{2\pi}{3} = \frac{\sqrt{3}}{2}$$

$$(iv) \quad \text{Let } \tan^{-1}\left(\frac{1}{5}\right) = \theta \Rightarrow \tan \theta = \frac{1}{5}$$

$$\tan\left(2\theta - \frac{\pi}{4}\right) = \frac{\tan(2\theta) - 1}{1 + \tan 2\theta} \text{ and } \tan 2\theta = \frac{2 \times \frac{1}{5}}{1 - \frac{1}{25}} = \frac{5}{12} \quad (\because \tan \theta = \frac{1}{5})$$

$$\Rightarrow \tan\left(2\theta - \frac{\pi}{4}\right) = \frac{\frac{5}{12} - 1}{1 + \frac{5}{12}} = -\frac{7}{17}$$

Ans.

Illustration 4 : Prove that $\sec^2(\tan^{-1}2) + \operatorname{cosec}^2(\cot^{-1}3) = 15$

Solution :

We have,

$$\sec^2(\tan^{-1}2) + \operatorname{cosec}^2(\cot^{-1}3)$$

$$= \left\{ \sec\left(\tan^{-1}2\right) \right\}^2 + \left\{ \operatorname{cosec}\left(\cot^{-1}3\right) \right\}^2 = \left\{ \sec\left(\tan^{-1}\frac{2}{1}\right) \right\}^2 + \left\{ \operatorname{cosec}\left(\cot^{-1}\frac{3}{1}\right) \right\}^2$$

$$= \left\{ \sec\left(\sec^{-1}\sqrt{5}\right) \right\}^2 + \left\{ \operatorname{cosec}\left(\operatorname{cosec}^{-1}\sqrt{10}\right) \right\}^2 = (\sqrt{5})^2 + (\sqrt{10})^2 = 15$$

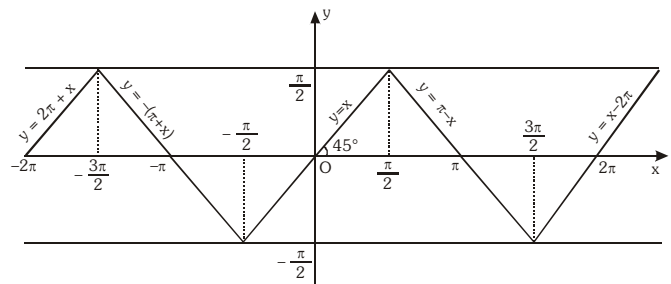
Do yourself - 2 :

Evaluate the following :

(i) $\tan\left(\cos^{-1}\left(\frac{8}{17}\right)\right)$ (ii) $\sin\left(\frac{1}{2}\cos^{-1}\left(\frac{4}{5}\right)\right)$ (iii) $\sin\left(\frac{\pi}{3} - \sin^{-1}\left(\frac{-1}{2}\right)\right)$ (iv) $\sin\left(\cos^{-1}\frac{3}{5}\right)$

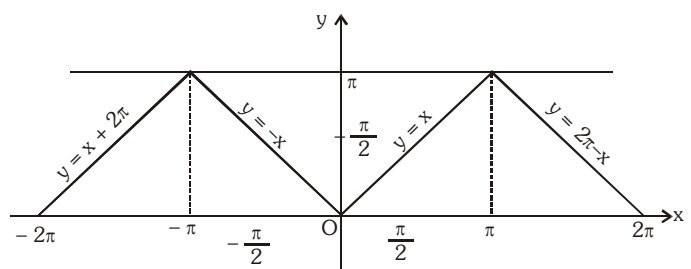
P-2 (i) $y = \sin^{-1}(\sin x)$, $x \in \mathbb{R}$, $y \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ periodic with period 2π and it is an odd function.

$$\sin^{-1}(\sin x) = \begin{cases} -\pi - x & , -\pi \leq x \leq -\frac{\pi}{2} \\ x & , -\frac{\pi}{2} < x < \frac{\pi}{2} \\ \pi - x & , \frac{\pi}{2} \leq x \leq \pi \end{cases}$$



(ii) $y = \cos^{-1}(\cos x)$, $x \in \mathbb{R}$, $y \in [0, \pi]$, periodic with period 2π and it is an even function.

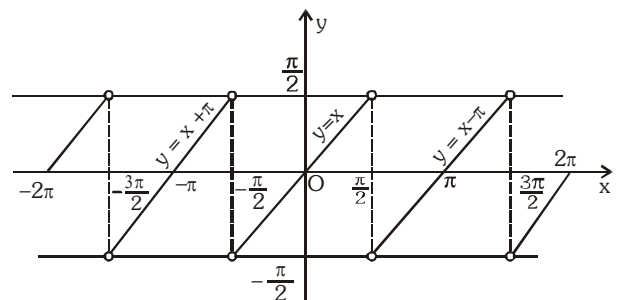
$$\cos^{-1}(\cos x) = \begin{cases} -x & , -\pi \leq x \leq 0 \\ x & , 0 < x \leq \pi \end{cases}$$



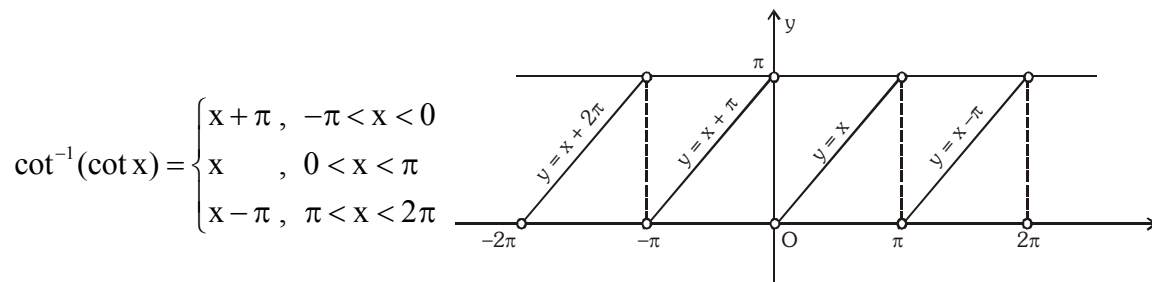
(iii) $y = \tan^{-1}(\tan x)$

$x \in \mathbb{R} - \left\{ (2n-1)\frac{\pi}{2}, n \in \mathbb{I} \right\}$; $y \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, periodic with period π and it is an odd function.

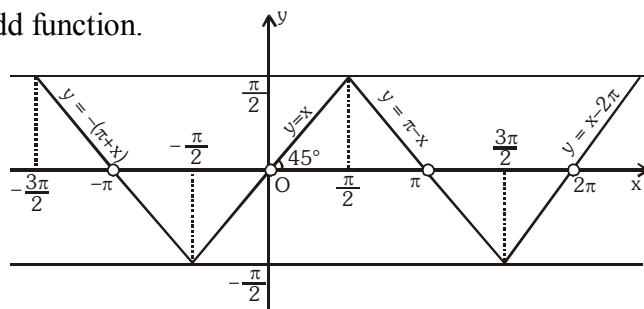
$$\tan^{-1}(\tan x) = \begin{cases} x + \pi & , -\frac{3\pi}{2} < x < -\frac{\pi}{2} \\ x & , -\frac{\pi}{2} < x < \frac{\pi}{2} \\ x - \pi & , \frac{\pi}{2} < x < \frac{3\pi}{2} \end{cases}$$



- (iv) $y = \cot^{-1}(\cot x)$, $x \in \mathbb{R} - \{n\pi, n \in \mathbb{I}\}$, $y \in (0, \pi)$, periodic with period π and neither even nor odd function.



- (v) $y = \operatorname{cosec}^{-1}(\operatorname{cosec} x)$, $x \in \mathbb{R} - \{n\pi, n \in \mathbb{I}\}$, $y \in \left[-\frac{\pi}{2}, 0\right) \cup \left(0, \frac{\pi}{2}\right]$, is periodic with period 2π and it is an odd function.



- (vi) $y = \sec^{-1}(\sec x)$, y is periodic with period 2π and it is an even function.

$$x \in \mathbb{R} - \left\{(2n-1)\frac{\pi}{2}, n \in \mathbb{I}\right\}, y \in \left[0, \frac{\pi}{2}\right] \cup \left[\frac{\pi}{2}, \pi\right]$$

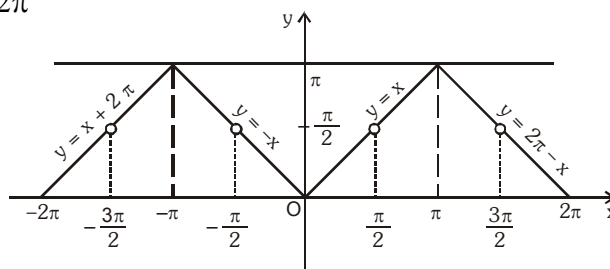


Illustration 5 : The value of $\sin^{-1}(-\sqrt{3}/2) + \cos^{-1}(\cos(7\pi/6))$ is -

- (A) $5\pi/6$ (B) $\pi/2$ (C) $3\pi/2$ (D) none of these

Solution : $\sin^{-1}\left(-\sqrt{3}/2\right) = -\sin^{-1}\left(\sqrt{3}/2\right) = -\pi/3$

and $\cos^{-1}(\cos(7\pi/6)) = \cos^{-1}(\cos(2\pi - 5\pi/6)) = \cos^{-1}(\cos(5\pi/6)) = 5\pi/6$

Hence $\sin^{-1}(-\sqrt{3}/2) + \cos^{-1}(\cos 7\pi/6) = -\frac{\pi}{3} + \frac{5\pi}{6} = \frac{\pi}{2}$

Ans.(B)

Illustration 6 : Evaluate the following :

- (i) $\sin^{-1}(\sin 10)$ (ii) $\tan^{-1}(\tan(-6))$ (iii) $\cot^{-1}(\cot 4)$

Solution :

- (i) We know that $\sin^{-1}(\sin\theta) = \theta$, if $-\pi/2 \leq \theta \leq \pi/2$

Here, $\theta = 10$ radians which does not lie between $-\pi/2$ and $\pi/2$

But, $3\pi - \theta$ i.e., $3\pi - 10$ lie between $-\frac{\pi}{2}$ and $\frac{\pi}{2}$

Also, $\sin(3\pi - 10) = \sin 10$

$$\therefore \sin^{-1}(\sin 10) = \sin^{-1}(\sin(3\pi - 10)) = (3\pi - 10)$$

- (ii) We know that,

$\tan^{-1}(\tan\theta) = \theta$, if $-\pi/2 < \theta < \pi/2$. Here, $\theta = -6$, radians which does not lie between $-\pi/2$ and $\pi/2$. We find that $2\pi - 6$ lies between $-\pi/2$ and $\pi/2$ such that;

$$\tan(2\pi - 6) = -\tan 6 = \tan(-6)$$

$$\therefore \tan^{-1}(\tan(-6)) = \tan^{-1}(\tan(2\pi - 6)) = (2\pi - 6)$$

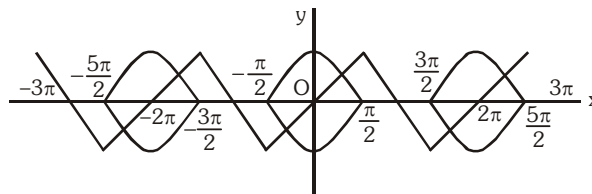
- (iii) $\cot^{-1}(\cot 4) = \cot^{-1}(\cot(\pi + (4 - \pi))) = \cot^{-1}(\cot(4 - \pi)) = (4 - \pi)$

Ans.

Illustration 7 : Find the number of solutions of (x, y) which satisfy $|y| = \cos x$ and $y = \sin^{-1}(\sin x)$, where $|x| \leq 3\pi$.

Solution :

Graphs of $y = \sin^{-1}(\sin x)$ and $|y| = \cos x$ meet exactly six times in $[-3\pi, 3\pi]$.



Do yourself - 3 :

Evaluate the following :

- (i) $\cos^{-1}\left(\cos\frac{13\pi}{6}\right)$ (ii) $\tan^{-1}\left(\tan\left(\frac{7\pi}{6}\right)\right)$ (iii) $\sin^{-1}\left(\sin\left(\frac{5\pi}{6}\right)\right)$

P-3 (i) $\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2} \quad |x| \leq 1$

(ii) $\tan^{-1} x + \cot^{-1} x = \frac{\pi}{2} \quad x \in \mathbb{R}$

(iii) $\operatorname{cosec}^{-1} x + \sec^{-1} x = \frac{\pi}{2} \quad |x| \geq 1$

- P-4** (i) $\sin^{-1}(-x) = -\sin^{-1} x$, $|x| \leq 1$
 (ii) $\operatorname{cosec}^{-1}(-x) = -\operatorname{cosec}^{-1} x$, $|x| \geq 1$
 (iii) $\tan^{-1}(-x) = -\tan^{-1} x$, $x \in \mathbb{R}$
 (iv) $\cot^{-1}(-x) = \pi - \cot^{-1} x$, $x \in \mathbb{R}$
 (v) $\cos^{-1}(-x) = \pi - \cos^{-1} x$, $|x| \leq 1$
 (vi) $\sec^{-1}(-x) = \pi - \sec^{-1} x$, $|x| \geq 1$

- P-5** (i) $\operatorname{cosec}^{-1} x = \sin^{-1} \frac{1}{x}$; $|x| \geq 1$
 (ii) $\sec^{-1} x = \cos^{-1} \frac{1}{x}$; $|x| \geq 1$
 (iii) $\cot^{-1} x = \begin{cases} \tan^{-1} \frac{1}{x} & ; x > 0 \\ \pi + \tan^{-1} \frac{1}{x} & ; x < 0 \end{cases}$

Illustration 8 : Find value of x if $\cos^{-1}(-x) + \tan^{-1}(-x) - 2\sin^{-1}(x) + \sec^{-1}\left(-\frac{1}{x}\right) = \frac{\pi}{4}$ for $|x| \leq 1$.

Solution :

$$\pi - \cos^{-1}(x) - \tan^{-1}(x) - 2\sin^{-1}(x) + \cos^{-1}(-x) = \frac{\pi}{4}$$

$$\pi - \cos^{-1}(x) - \tan^{-1}(x) - 2\sin^{-1}(x) + \pi - \cos^{-1}(x) = \frac{\pi}{4}$$

$$2\pi - 2(\sin^{-1}x + \cos^{-1}x) - \frac{\pi}{4} = \tan^{-1}x$$

$$2\pi - \pi - \frac{\pi}{4} = \tan^{-1}x \Rightarrow \tan^{-1}x = \frac{3\pi}{4} \text{ Hence no solution}$$

Ans.

Do yourself - 4 :

(i) Prove the following :

(a) $\cos^{-1}\left(\frac{5}{13}\right) = \tan^{-1}\left(\frac{12}{5}\right)$ (b) $\sin^{-1}\left(-\frac{4}{5}\right) = \tan^{-1}\left(-\frac{4}{3}\right) = \cos^{-1}\left(-\frac{3}{5}\right) - \pi$

(ii) Find the value of $\sin(\tan^{-1}a + \tan^{-1}\frac{1}{a})$; $a \neq 0$

P-6 (i) (a) $\tan^{-1} x + \tan^{-1} y = \begin{cases} \tan^{-1} \frac{x+y}{1-xy} & \text{where } x > 0, y > 0 \text{ \& } xy < 1 \\ \pi + \tan^{-1} \frac{x+y}{1-xy} & \text{where } x > 0, y > 0 \text{ \& } xy > 1 \\ \frac{\pi}{2}, & \text{where } x > 0, y > 0 \text{ \& } xy = 1 \end{cases}$

- (b) $\tan^{-1} x - \tan^{-1} y = \tan^{-1} \frac{x-y}{1+xy}$ where $x > 0, y > 0$
- (c) $\tan^{-1} x + \tan^{-1} y + \tan^{-1} z = \tan^{-1} \left[\frac{x+y+z-xyz}{1-xy-yz-zx} \right]$ if $x > 0, y > 0, z > 0$ & $xy + yz + zx < 1$
- (ii) (a) $\sin^{-1} x + \sin^{-1} y = \begin{cases} \sin^{-1} [x\sqrt{1-y^2} + y\sqrt{1-x^2}] & \text{where } x > 0, y > 0 \text{ \& } (x^2 + y^2) \leq 1 \\ \pi - \sin^{-1} [x\sqrt{1-y^2} + y\sqrt{1-x^2}] & \text{where } x > 0, y > 0 \text{ \& } x^2 + y^2 > 1 \end{cases}$
- (b) $\sin^{-1} x - \sin^{-1} y = \sin^{-1} [x\sqrt{1-y^2} - y\sqrt{1-x^2}]$ where $x > 0, y > 0$
- (iii) (a) $\cos^{-1} x + \cos^{-1} y = \cos^{-1} [xy - \sqrt{1-x^2}\sqrt{1-y^2}]$ where $x > 0, y > 0$
- (b) $\cos^{-1} x - \cos^{-1} y = \begin{cases} \cos^{-1} (xy + \sqrt{1-x^2}\sqrt{1-y^2}) & ; \quad x < y, \quad x, y > 0 \\ -\cos^{-1} (xy + \sqrt{1-x^2}\sqrt{1-y^2}) & ; \quad x > y, \quad x, y > 0 \end{cases}$

Note : In the above results x & y are taken positive. In case if these are given as negative, we first apply **P-4** and then use above results.

Illustration 9 : Prove that

$$(i) \quad \tan^{-1} \frac{1}{7} + \tan^{-1} \frac{1}{13} = \tan^{-1} \frac{2}{9} \quad (ii) \quad \tan^{-1} \frac{1}{5} + \tan^{-1} \frac{1}{7} + \tan^{-1} \frac{1}{3} + \tan^{-1} \frac{1}{8} = \frac{\pi}{4}$$

Solution :

$$(i) \quad \text{L.H.S.} = \tan^{-1} \frac{1}{7} + \tan^{-1} \frac{1}{13}$$

$$= \tan^{-1} \left\{ \frac{\frac{1}{7} + \frac{1}{13}}{1 - \frac{1}{7} \times \frac{1}{13}} \right\} \quad \left\{ \because \tan^{-1} x + \tan^{-1} y = \tan^{-1} \left(\frac{x+y}{1-xy} \right); \text{ if } xy < 1 \right\}$$

$$= \tan^{-1} \left(\frac{20}{90} \right) = \tan^{-1} \left(\frac{2}{9} \right) = \text{R.H.S.}$$

$$(ii) \quad \left(\tan^{-1} \frac{1}{5} + \tan^{-1} \frac{1}{7} \right) + \left(\tan^{-1} \frac{1}{3} + \tan^{-1} \frac{1}{8} \right)$$

$$= \tan^{-1} \left(\frac{\frac{1}{5} + \frac{1}{7}}{1 - \frac{1}{5} \times \frac{1}{7}} \right) + \tan^{-1} \left(\frac{\frac{1}{3} + \frac{1}{8}}{1 - \frac{1}{3} \times \frac{1}{8}} \right) = \tan^{-1} \left(\frac{6}{17} \right) + \tan^{-1} \left(\frac{11}{23} \right)$$

$$= \tan^{-1} \left(\frac{\frac{6}{17} + \frac{11}{23}}{1 - \frac{6}{17} \times \frac{11}{23}} \right) = \tan^{-1} \left(\frac{325}{325} \right) = \tan^{-1}(1) = \frac{\pi}{4}$$

Ans.

Illustration 10 : Prove that $\sin^{-1} \frac{12}{13} + \cot^{-1} \frac{4}{3} + \tan^{-1} \frac{63}{16} = \pi$

Solution : We have,

$$\begin{aligned}
 & \sin^{-1} \frac{12}{13} + \cos^{-1} \frac{4}{5} + \tan^{-1} \frac{63}{16} \\
 &= \tan^{-1} \frac{12}{5} + \tan^{-1} \frac{3}{4} + \tan^{-1} \frac{63}{16} \quad \left[\because \sin^{-1} \frac{12}{13} = \tan^{-1} \frac{12}{5} \text{ and } \cos^{-1} \frac{4}{5} = \tan^{-1} \frac{3}{4} \right] \\
 &= \pi + \tan^{-1} \left\{ \frac{\frac{12}{5} + \frac{3}{4}}{1 - \frac{12}{5} \times \frac{3}{4}} \right\} + \tan^{-1} \frac{63}{16} \quad \left[\because \tan^{-1} x + \tan^{-1} y = \pi + \tan^{-1} \left(\frac{x+y}{1-xy} \right), \text{ if } xy > 1 \right] \\
 &= \pi + \tan^{-1} \left(\frac{63}{-16} \right) + \tan^{-1} \left(\frac{63}{16} \right) \\
 &= \pi - \tan^{-1} \frac{63}{16} + \tan^{-1} \frac{63}{16} \quad \left[\because \tan^{-1}(-x) = -\tan^{-1} x \right] \\
 &= \pi
 \end{aligned}$$

Illustration 11 : Prove that : $\cos^{-1} \frac{12}{13} + \sin^{-1} \frac{3}{5} = \sin^{-1} \frac{56}{65}$

Solution : We have, L.H.S. = $\cos^{-1} \left(\frac{12}{13} \right) + \sin^{-1} \left(\frac{3}{5} \right) = \tan^{-1} \left(\frac{5}{12} \right) + \tan^{-1} \left(\frac{3}{4} \right)$

$$\left[\because \cos^{-1} \left(\frac{12}{13} \right) = \tan^{-1} \left(\frac{5}{12} \right) \text{ \& } \sin^{-1} \left(\frac{3}{5} \right) = \tan^{-1} \left(\frac{3}{4} \right) \right]$$

$$\text{L.H.S.} = \tan^{-1} \left(\frac{\frac{5}{12} + \frac{3}{4}}{1 - \frac{5}{12} \times \frac{3}{4}} \right) = \tan^{-1} \left(\frac{56}{33} \right)$$

$$\text{R.H.S.} = \sin^{-1} \left(\frac{56}{65} \right) = \tan^{-1} \left(\frac{56}{33} \right)$$

L.H.S. = R.H.S. Hence Proved

Do yourself - 5 :

Prove the following :

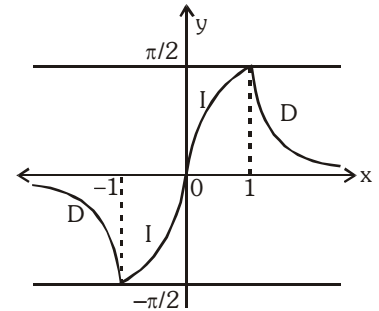
(i) $\sin^{-1}\left(\frac{3}{5}\right) + \sin^{-1}\left(\frac{8}{17}\right) = \cos^{-1}\left(\frac{36}{85}\right)$

(ii) $\tan^{-1}\left(\frac{3}{4}\right) + \tan^{-1}\left(\frac{3}{5}\right) - \tan^{-1}\left(\frac{8}{19}\right) = \frac{\pi}{4}$

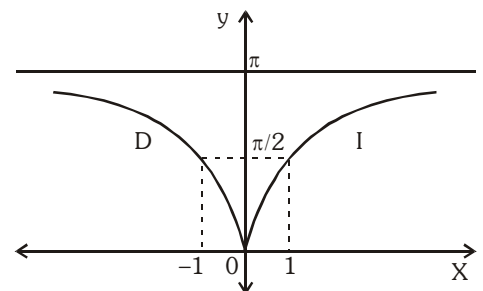
(iii) $\tan^{-1}\left(\frac{2}{11}\right) + \tan^{-1}\left(\frac{7}{24}\right) = \tan^{-1}\left(\frac{1}{2}\right)$

4. SIMPLIFIED INVERSE TRIGONOMETRIC FUNCTIONS :

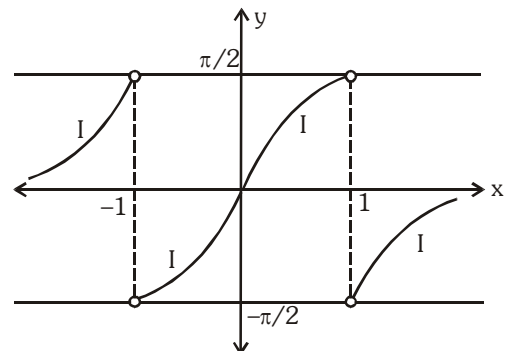
(a) $y = f(x) = \sin^{-1}\left(\frac{2x}{1+x^2}\right) = \begin{cases} 2 \tan^{-1} x & \text{if } |x| \leq 1 \\ \pi - 2 \tan^{-1} x & \text{if } x > 1 \\ -(\pi + 2 \tan^{-1} x) & \text{if } x < -1 \end{cases}$



(b) $y = f(x) = \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right) = \begin{cases} 2 \tan^{-1} x & \text{if } x \geq 0 \\ -2 \tan^{-1} x & \text{if } x < 0 \end{cases}$

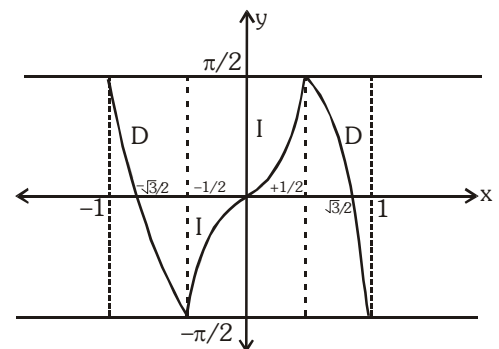


(c) $y = f(x) = \tan^{-1} \frac{2x}{1-x^2} = \begin{cases} 2 \tan^{-1} x & \text{if } |x| < 1 \\ \pi + 2 \tan^{-1} x & \text{if } x < -1 \\ -(\pi - 2 \tan^{-1} x) & \text{if } x > 1 \end{cases}$



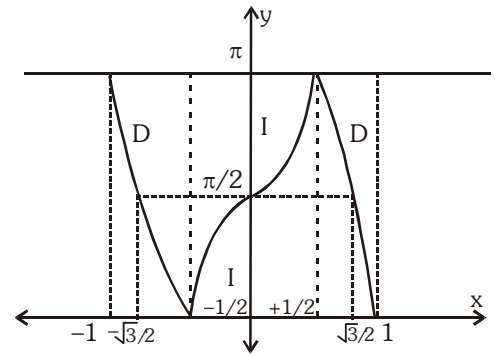
(d) $y = f(x) = \sin^{-1}(3x - 4x^3)$

$$= \begin{cases} -(\pi + 3 \sin^{-1} x) & \text{if } -1 \leq x \leq -\frac{1}{2} \\ 3 \sin^{-1} x & \text{if } -\frac{1}{2} \leq x \leq \frac{1}{2} \\ \pi - 3 \sin^{-1} x & \text{if } \frac{1}{2} \leq x \leq 1 \end{cases}$$

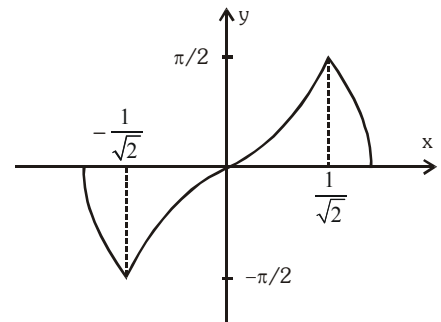


(e) $y = f(x) = \cos^{-1}(4x^3 - 3x)$

$$= \begin{cases} 3\cos^{-1}x - 2\pi & \text{if } -1 \leq x \leq -\frac{1}{2} \\ 2\pi - 3\cos^{-1}x & \text{if } -\frac{1}{2} \leq x \leq \frac{1}{2} \\ 3\cos^{-1}x & \text{if } \frac{1}{2} \leq x \leq 1 \end{cases}$$



(f) $\sin^{-1}(2x\sqrt{1-x^2}) = \begin{cases} -(\pi + 2\sin^{-1}x) & -1 \leq x \leq -\frac{1}{\sqrt{2}} \\ 2\sin^{-1}x & -\frac{1}{\sqrt{2}} \leq x \leq \frac{1}{\sqrt{2}} \\ \pi - 2\sin^{-1}x & \frac{1}{\sqrt{2}} \leq x \leq 1 \end{cases}$



(g) $\cos^{-1}(2x^2 - 1) = \begin{cases} 2\cos^{-1}x & 0 \leq x \leq 1 \\ 2\pi - 2\cos^{-1}x & -1 \leq x \leq 0 \end{cases}$

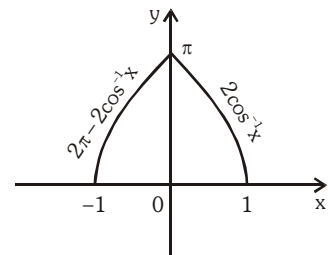


Illustration 12 : Prove that : $2\tan^{-1}\frac{1}{2} + \tan^{-1}\frac{1}{7} = \tan^{-1}\frac{31}{17}$

Solution : We have, $2\tan^{-1}\frac{1}{2} + \tan^{-1}\frac{1}{7}$

$$= \tan^{-1}\left\{\frac{2 \times \frac{1}{2}}{1 - \left(\frac{1}{2}\right)^2}\right\} + \tan^{-1}\frac{1}{7} \quad \left[\because 2\tan^{-1}x = \tan^{-1}\left(\frac{2x}{1-x^2}\right), \text{ if } -1 < x < 1 \right]$$

$$= \tan^{-1}\frac{4}{3} + \tan^{-1}\frac{1}{7} = \tan^{-1}\left\{\frac{\frac{4}{3} + \frac{1}{7}}{1 - \frac{4}{3} \times \frac{1}{7}}\right\} = \tan^{-1}\frac{31}{17}$$

Illustration 13 : Prove that $\tan^{-1} \sqrt{x} = \frac{1}{2} \cos^{-1} \left(\frac{1-x}{1+x} \right), x \in [0, 1]$

Solution : We have, $\frac{1}{2} \cos^{-1} \left(\frac{1-x}{1+x} \right) = \frac{1}{2} \cos^{-1} \left\{ \frac{1-(\sqrt{x})^2}{1+(\sqrt{x})^2} \right\} = \frac{1}{2} \times 2 \tan^{-1} \sqrt{x} = \tan^{-1} \sqrt{x}.$

Alter : Putting $\sqrt{x} = \tan \theta$, we have $\Rightarrow \theta \in \left[0, \frac{\pi}{4} \right]$

$$\begin{aligned} \text{RHS} &= \frac{1}{2} \cos^{-1} \left(\frac{1-x}{1+x} \right) = \frac{1}{2} \cos^{-1} \left(\frac{1-\tan^2 \theta}{1+\tan^2 \theta} \right) = \frac{1}{2} \cos^{-1} (\cos 2\theta) = \theta \quad \because \left(2\theta \in \left[0, \frac{\pi}{2} \right] \right) \\ &= \tan^{-1} \sqrt{x} = \text{LHS} \end{aligned}$$

Illustration 14 : Prove that : (i) $4 \tan^{-1} \frac{1}{5} - \tan^{-1} \frac{1}{70} + \tan^{-1} \frac{1}{99} = \frac{\pi}{4}$

$$(ii) \quad 2 \tan^{-1} \frac{1}{5} + \sec^{-1} \frac{5\sqrt{2}}{7} + 2 \tan^{-1} \frac{1}{8} = \frac{\pi}{4}$$

Solution : (i) $4 \tan^{-1} \frac{1}{5} - \tan^{-1} \frac{1}{70} + \tan^{-1} \frac{1}{99} = 2 \left\{ 2 \tan^{-1} \frac{1}{5} \right\} - \tan^{-1} \frac{1}{70} + \tan^{-1} \frac{1}{99}$

$$= 2 \left\{ \tan^{-1} \frac{2 \times 1/5}{1-(1/5)^2} \right\} - \tan^{-1} \frac{1}{70} + \tan^{-1} \frac{1}{99} \quad \left[\begin{array}{l} \because 2 \tan^{-1} x \\ = \tan^{-1} \frac{2x}{1-x^2}, \text{ if } |x| < 1 \end{array} \right]$$

$$= 2 \tan^{-1} \frac{5}{12} - \left\{ \tan^{-1} \frac{1}{70} - \tan^{-1} \frac{1}{99} \right\} = \tan^{-1} \left\{ \frac{2 \times 5/12}{1-(5/12)^2} \right\} - \tan^{-1} \left\{ \frac{\frac{1}{70} - \frac{1}{99}}{1 + \frac{1}{70} \times \frac{1}{99}} \right\}$$

$$= \tan^{-1} \frac{120}{119} - \tan^{-1} \frac{29}{6931} = \tan^{-1} \frac{120}{119} - \tan^{-1} \frac{1}{239} = \tan^{-1} \left\{ \frac{\frac{120}{119} - \frac{1}{239}}{1 + \frac{120}{119} \times \frac{1}{239}} \right\} = \tan^{-1} 1 = \frac{\pi}{4}$$

$$(ii) \quad 2 \tan^{-1} \frac{1}{5} + \sec^{-1} \frac{5\sqrt{2}}{7} + 2 \tan^{-1} \frac{1}{8} = 2 \left\{ \tan^{-1} \frac{1}{5} + \tan^{-1} \frac{1}{8} \right\} + \sec^{-1} \frac{5\sqrt{2}}{7}$$

$$= 2 \tan^{-1} \left\{ \frac{\frac{1}{5} + \frac{1}{8}}{1 - \frac{1}{5} \times \frac{1}{8}} \right\} + \tan^{-1} \sqrt{\left(\frac{5\sqrt{2}}{7} \right)^2 - 1} \quad \left[\because \sec^{-1} x = \tan^{-1} \sqrt{x^2 - 1} \right]$$

$$= 2 \tan^{-1} \frac{13}{39} + \tan^{-1} \frac{1}{7} = 2 \tan^{-1} \frac{1}{3} + \tan^{-1} \frac{1}{7}$$

$$= \tan^{-1} \left\{ \frac{2 \times 1/3}{1 - (1/3)^2} \right\} + \tan^{-1} \frac{1}{7} \quad \left[\because 2 \tan^{-1} x = \tan^{-1} \frac{2x}{1-x^2}, \text{ if } |x| < 1 \right]$$

$$= \tan^{-1} \frac{3}{4} + \tan^{-1} \frac{1}{7} = \tan^{-1} \left\{ \frac{\frac{3}{4} + \frac{1}{7}}{1 - \frac{3}{4} \times \frac{1}{7}} \right\} = \tan^{-1} 1 = \frac{\pi}{4}$$

Do yourself - 6 :

Prove the following results :

$$(i) \quad 2 \tan^{-1} \left(\frac{1}{5} \right) + \tan^{-1} \left(\frac{1}{8} \right) = \tan^{-1} \left(\frac{4}{7} \right)$$

$$(ii) \quad 2 \sin^{-1} \left(\frac{3}{5} \right) - \tan^{-1} \left(\frac{17}{31} \right) = \frac{\pi}{4}$$

6. EQUATIONS INVOLVING INVERSE TRIGONOMETRIC FUNCTIONS :

Illustration 15 : The equation $2\cos^{-1}x + \sin^{-1}x = \frac{11\pi}{6}$ has

- (A) no solution (B) only one solution (C) two solutions (D) three solutions

Solution : Given equation is $2 \cos^{-1} x + \sin^{-1} x = \frac{11\pi}{6}$

$$\Rightarrow \cos^{-1} x + (\cos^{-1} x + \sin^{-1} x) = \frac{11\pi}{6} \Rightarrow \cos^{-1} x + \frac{\pi}{2} = \frac{11\pi}{6} \Rightarrow \cos^{-1} x = 4\pi/3$$

which is not possible as $\cos^{-1} x \in [0, \pi]$

Ans.(A)

Illustration 16 : If $(\tan^{-1} x)^2 + (\cot^{-1} x)^2 = 5\pi^2/8$, then x is equal to-

- (A) -1 (B) 0 (C) 1 (D) none of these

Solution : The given equation can be written as $(\tan^{-1} x + \cot^{-1} x)^2 - 2 \tan^{-1} x \cot^{-1} x = 5\pi^2/8$

Since $\tan^{-1} x + \cot^{-1} x = \pi/2$ we have

$$(\pi/2)^2 - 2 \tan^{-1} x (\pi/2 - \tan^{-1} x) = 5\pi^2/8$$

$$\Rightarrow 2(\tan^{-1} x)^2 - 2(\pi/2) \tan^{-1} x - 3\pi^2/8 = 0 \Rightarrow \tan^{-1} x = -\pi/4 \Rightarrow x = -1 \quad \text{Ans. (A)}$$

Illustration 17 : Solve the equation : $\tan^{-1} \frac{x-1}{x-2} + \tan^{-1} \frac{x+1}{x+2} = \frac{\pi}{4}$

Solution : $\tan^{-1} \frac{x-1}{x-2} + \tan^{-1} \frac{x+1}{x+2} = \frac{\pi}{4}$

taking tangent on both sides

$$\Rightarrow \tan \left(\tan^{-1} \left(\frac{x-1}{x-2} \right) + \tan^{-1} \left(\frac{x+1}{x+2} \right) \right) = 1 \Rightarrow \frac{\tan \left(\tan^{-1} \left(\frac{x-1}{x-2} \right) \right) + \tan \left(\tan^{-1} \left(\frac{x+1}{x+2} \right) \right)}{1 - \tan \left(\tan^{-1} \left(\frac{x-1}{x-2} \right) \right) \tan \left(\tan^{-1} \left(\frac{x+1}{x+2} \right) \right)} = 1$$

$$\Rightarrow \frac{\frac{x-1}{x-2} + \frac{x+1}{x+2}}{1 - \frac{x-1}{x-2} \cdot \frac{x+1}{x+2}} = 1 \Rightarrow \frac{(x-1)(x+2) + (x-2)(x+1)}{x^2 - 4 - (x^2 - 1)} = 1 \Rightarrow 2x^2 - 4 = -3 \Rightarrow x = \pm \frac{1}{\sqrt{2}}$$

Now verify $x = \frac{1}{\sqrt{2}}$

$$= \tan^{-1} \left(\frac{\frac{1}{\sqrt{2}} - 1}{\frac{1}{\sqrt{2}} - 2} \right) + \tan^{-1} \left(\frac{\frac{1}{\sqrt{2}} + 1}{\frac{1}{\sqrt{2}} + 2} \right) = \tan^{-1} \left(\frac{\sqrt{2} - 1}{2\sqrt{2} - 1} \right) + \tan^{-1} \left(\frac{\sqrt{2} + 1}{2\sqrt{2} + 1} \right)$$

$$= \tan^{-1} \left(\frac{(2\sqrt{2} + 1)(\sqrt{2} - 1) + (2\sqrt{2} - 1)(\sqrt{2} + 1)}{(2\sqrt{2} - 1)(2\sqrt{2} + 1) - (\sqrt{2} - 1)(\sqrt{2} + 1)} \right) = \tan^{-1} \left(\frac{6}{6} \right) = \tan^{-1}(1) = \frac{\pi}{4}$$

$$x = -\frac{1}{\sqrt{2}}$$

$$= \tan^{-1} \left(\frac{-\frac{1}{\sqrt{2}} - 1}{-\frac{1}{\sqrt{2}} - 2} \right) + \tan^{-1} \left(\frac{-\frac{1}{\sqrt{2}} + 1}{-\frac{1}{\sqrt{2}} + 2} \right) = \tan^{-1} \left(\frac{\sqrt{2} + 1}{2\sqrt{2} + 1} \right) + \tan^{-1} \left(\frac{\sqrt{2} - 1}{2\sqrt{2} - 1} \right) \text{ \{same as above\}}$$

$$= \tan^{-1}(1) = \frac{\pi}{4}$$

$$\therefore x = \pm \frac{1}{\sqrt{2}} \text{ are solutions}$$

Ans.

Illustration 18 : Solve the equation : $2 \tan^{-1}(2x + 1) = \cos^{-1}x$.

Solution : Here, $2 \tan^{-1}(2x + 1) = \cos^{-1}x$

$$\text{or } \cos(2 \tan^{-1}(2x + 1)) = x \quad \left\{ \text{We know } \cos 2\theta = \frac{1 - \tan^2 \theta}{1 + \tan^2 \theta} \right\}$$

$$\therefore \frac{1 - (2x + 1)^2}{1 + (2x + 1)^2} = x \Rightarrow (1 - 2x - 1)(1 + 2x + 1) = x(4x^2 + 4x + 2)$$

$$\Rightarrow -2x \cdot 2(x+1) = 2x(2x^2 + 2x + 1) \Rightarrow 2x(2x^2 + 2x + 1 + 2x + 2) = 0$$

$$\Rightarrow 2x(2x^2 + 4x + 3) = 0$$

$$\Rightarrow x = 0 \text{ or } 2x^2 + 4x + 3 = 0 \quad \{\text{No solution}\}$$

$$\text{Verify } x = 0$$

$$2\tan^{-1}(1) = \cos^{-1}(1) \Rightarrow \frac{\pi}{2} = \frac{\pi}{2}$$

$$\therefore x = 0 \text{ is only the solution}$$

Ans.

Do yourself - 7 :Solve the following equation for x :

$$(i) \quad \sin \left[\sin^{-1} \left(\frac{1}{5} \right) + \cos^{-1} x \right] = 1$$

$$(ii) \quad \cos^{-1} x + \sin^{-1} \frac{x}{2} = \frac{\pi}{6}$$

7. INEQUATIONS INVOLVING INVERSE TRIGONOMETRIC FUNCTION :**Illustration 19 :** Find the complete solution set of $\sin^{-1}(\sin 5) > x^2 - 4x$.

$$\text{Solution : } \sin^{-1}(\sin 5) > x^2 - 4x \Rightarrow \sin^{-1}[\sin(5 - 2\pi)] > x^2 - 4x$$

$$\Rightarrow x^2 - 4x < 5 - 2\pi \Rightarrow x^2 - 4x + (2\pi - 5) < 0$$

$$\Rightarrow 2 - \sqrt{9 - 2\pi} < x < 2 + \sqrt{9 - 2\pi} \Rightarrow x \in (2 - \sqrt{9 - 2\pi}, 2 + \sqrt{9 - 2\pi})$$

Ans.

Illustration 20 : Find the complete solution set of $[\cot^{-1}x]^2 - 6[\cot^{-1}x] + 9 \leq 0$, where $[.]$ denotes the greatest integer function.

$$\text{Solution : } [\cot^{-1}x]^2 - 6[\cot^{-1}x] + 9 \leq 0$$

$$\Rightarrow ([\cot^{-1}x] - 3)^2 \leq 0 \Rightarrow [\cot^{-1}x] = 3 \Rightarrow 3 \leq \cot^{-1}x < 4 \Rightarrow x \in (-\infty, \cot 3]$$

Illustration 21 : If $\cot^{-1} \frac{n}{\pi} > \frac{\pi}{6}$, $n \in \mathbb{N}$, then the maximum value of n is -

(A) 1

(B) 5

(C) 9

(D) none of these

$$\text{Solution : } \cot^{-1} \frac{n}{\pi} > \frac{\pi}{6}$$

$$\Rightarrow \cot \left(\cot^{-1} \left(\frac{n}{\pi} \right) \right) < \cot \left(\frac{\pi}{6} \right) \Rightarrow \frac{n}{\pi} < \sqrt{3}$$

$$\Rightarrow n < \pi\sqrt{3} \Rightarrow n < 5.5 \text{ (approx)}$$

$$\Rightarrow n = 5 \quad \because (n \in \mathbb{N})$$

Ans. (B)

Do yourself - 8 :

- (i) Solve the inequality $\tan^{-1}x > \cot^{-1}x$.
 (ii) Complete solution set of inequation $(\cos^{-1}x)^2 - (\sin^{-1}x)^2 > 0$, is

- (A) $\left[0, \frac{1}{\sqrt{2}}\right)$ (B) $\left[-1, \frac{1}{\sqrt{2}}\right)$ (C) $(-1, \sqrt{2})$ (D) none of these

8. SUMMATION OF SERIES :

Illustration 22 : Prove that :

$$\tan^{-1}\left(\frac{c_1x-y}{c_1y+x}\right) + \tan^{-1}\left(\frac{c_2-c_1}{1+c_2c_1}\right) + \tan^{-1}\left(\frac{c_3-c_2}{1+c_3c_2}\right) + \dots + \tan^{-1}\left(\frac{c_n-c_{n-1}}{1+c_nc_{n-1}}\right) + \tan^{-1}\left(\frac{1}{c_n}\right) = \tan^{-1}\left(\frac{x}{y}\right)$$

Solution : L.H.S.

$$\tan^{-1}\left(\frac{c_1x-y}{c_1y+x}\right) + \tan^{-1}\left(\frac{c_2-c_1}{1+c_2c_1}\right) + \tan^{-1}\left(\frac{c_3-c_2}{1+c_3c_2}\right) + \dots + \tan^{-1}\left(\frac{c_n-c_{n-1}}{1+c_nc_{n-1}}\right) + \tan^{-1}\left(\frac{1}{c_n}\right)$$

$$= \tan^{-1}\left(\frac{\frac{x}{y} - \frac{1}{c_1}}{1 + \frac{x}{y} \cdot \frac{1}{c_1}}\right) + (\tan^{-1}c_2 - \tan^{-1}c_1) + (\tan^{-1}c_3 - \tan^{-1}c_2) + \dots + (\tan^{-1}c_n - \tan^{-1}c_{n-1}) + \tan^{-1}\left(\frac{1}{c_n}\right)$$

$$= \tan^{-1}\left(\frac{x}{y}\right) - \tan^{-1}\left(\frac{1}{c_1}\right) - \tan^{-1}c_1 + \tan^{-1}c_n + \tan^{-1}\left(\frac{1}{c_n}\right)$$

$$= \tan^{-1}\left(\frac{x}{y}\right) - (\cot^{-1}c_1 + \tan^{-1}c_1) + (\tan^{-1}c_n + \cot^{-1}c_n)$$

$$= \tan^{-1}\left(\frac{x}{y}\right) - \frac{\pi}{2} + \frac{\pi}{2} = \tan^{-1}\left(\frac{x}{y}\right) = \text{R.H.S.}$$

Do yourself - 9 :

- (i) Evaluate : $\sum_{r=1}^{\infty} \tan^{-1}\left(\frac{2}{1+(2r+1)(2r-1)}\right)$

Miscellaneous Illustrations :

Illustration 23 : If $\tan^{-1} y = 4 \tan^{-1} x, \left(|x| < \tan \frac{\pi}{8} \right)$, find y as an algebraic function of x and hence prove

that $\tan \frac{\pi}{8}$ is a root of the equation $x^4 - 6x^2 + 1 = 0$.

Solution :

We have $\tan^{-1} y = 4 \tan^{-1} x$

$$\Rightarrow \tan^{-1} y = 2 \tan^{-1} \frac{2x}{1-x^2} \quad (\text{as } |x| < 1)$$

$$= \tan^{-1} \frac{\frac{4x}{1-x^2}}{1 - \frac{4x^2}{(1-x^2)^2}} = \tan^{-1} \frac{4x(1-x^2)}{x^4 - 6x^2 + 1} \quad \left(\text{as } \left| \frac{2x}{1-x^2} \right| < 1 \right)$$

$$\Rightarrow y = \frac{4x(1-x^2)}{x^4 - 6x^2 + 1}$$

$$\text{If } x = \tan \frac{\pi}{8}$$

$$\Rightarrow \tan^{-1} y = 4 \tan^{-1} x = \frac{\pi}{2} \Rightarrow y \text{ is not defined} \Rightarrow x^4 - 6x^2 + 1 = 0 \quad \text{Ans.}$$

Illustration 24 : If $A = 2 \tan^{-1}(2\sqrt{2}-1)$ and $B = 3 \sin^{-1}(1/3) + \sin^{-1}(3/5)$, then show $A > B$.

Solution :

We have, $A = 2 \tan^{-1}(2\sqrt{2}-1) = 2 \tan^{-1}(1.828)$

$$\Rightarrow A > 2 \tan^{-1}(\sqrt{3}) \Rightarrow A > \frac{2\pi}{3} \quad \dots\dots (i)$$

$$\text{also we have, } \sin^{-1}\left(\frac{1}{3}\right) < \sin^{-1}\left(\frac{1}{2}\right) \Rightarrow \sin^{-1}\left(\frac{1}{3}\right) < \frac{\pi}{6}$$

$$\Rightarrow 3 \sin^{-1}\left(\frac{1}{3}\right) < \frac{\pi}{2}$$

$$\text{also, } 3 \sin^{-1}\left(\frac{1}{3}\right) = \sin^{-1}\left(3 \cdot \frac{1}{3} - 4 \left(\frac{1}{3}\right)^3\right) = \sin^{-1}\left(\frac{23}{27}\right) = \sin^{-1}(0.852)$$

$$\Rightarrow 3 \sin^{-1}(1/3) < \sin^{-1}(\sqrt{3}/2) \Rightarrow 3 \sin^{-1}(1/3) < \pi/3$$

$$\text{also, } \sin^{-1}(3/5) = \sin^{-1}(0.6) < \sin^{-1}(\sqrt{3}/2) \Rightarrow \sin^{-1}(3/5) < \pi/3$$

$$\text{Hence, } B = 3 \sin^{-1}(1/3) + \sin^{-1}(3/5) < \frac{2\pi}{3} \quad \dots\dots (ii)$$

From (i) and (ii), we have $A > B$.

Illustration 25 : If $\theta = \tan^{-1}(2 \tan^2 \theta) - \frac{1}{2} \sin^{-1} \left(\frac{3 \sin 2\theta}{5 + 4 \cos 2\theta} \right)$ then find the sum of all possible values of $\tan \theta$.

Solution :

$$\theta = \tan^{-1}(2 \tan^2 \theta) - \frac{1}{2} \sin^{-1} \left(\frac{3 \sin 2\theta}{5 + 4 \cos 2\theta} \right) \Rightarrow \theta = \tan^{-1}(2 \tan^2 \theta) - \frac{1}{2} \sin^{-1} \left(\frac{6 \tan \theta}{9 + \tan^2 \theta} \right)$$

$$\Rightarrow \theta = \tan^{-1}(2 \tan^2 \theta) - \frac{1}{2} \sin^{-1} \left[\frac{2 \left(\frac{1}{3} \tan \theta \right)}{1 + \left(\frac{1}{3} \tan \theta \right)^2} \right] \Rightarrow \theta = \tan^{-1}(2 \tan^2 \theta) - \frac{2}{2} \tan^{-1} \left(\frac{1}{3} \tan \theta \right)$$

$$\Rightarrow \theta = \tan^{-1}(2 \tan^2 \theta) - \tan^{-1} \left(\frac{1}{3} \tan \theta \right) \quad \dots\dots\dots (i)$$

taking tangent on both sides

$$\Rightarrow \tan \theta = \frac{6 \tan^2 \theta - \tan \theta}{3 + 2 \tan^3 \theta} \Rightarrow 2 \tan^4 \theta - 6 \tan^2 \theta + 4 \tan \theta = 0$$

$$\Rightarrow 2 \tan \theta (\tan^3 \theta - 3 \tan \theta + 2) = 0 \Rightarrow 2 \tan \theta (\tan \theta - 1)^2 (\tan \theta + 2) = 0$$

$$\Rightarrow \tan \theta = 0, 1, -2 \quad \text{which satisfy equation (i)}$$

$$\therefore \text{sum} = 0 + 1 - 2 = -1$$

Ans.

ANSWERS FOR DO YOURSELF

- | | | | | | | |
|------------|-------------|---|-------------|-----------------------|--------------|-----------------------------|
| 1 : | (i) | C | (ii) | 1526 | | |
| 2 : | (i) | $\frac{15}{8}$ | (ii) | $\frac{1}{\sqrt{10}}$ | (iii) | 1 (iv) $\frac{4}{5}$ |
| 3 : | (i) | $\frac{\pi}{6}$ | (ii) | $\frac{\pi}{6}$ | (iii) | $\frac{\pi}{6}$ |
| 4 : | (ii) | $\begin{cases} 1, & \text{if } a > 0 \\ -1, & \text{if } a < 0 \end{cases}$ | | | | |
| 7 : | (i) | $\frac{1}{5}$ | (ii) | 1 | (iii) | $\sqrt{3}$ |
| 8 : | (i) | $(1, \infty)$ | (ii) | B | | |
| 9 : | (i) | $\pi/4$ | | | | |

EXERCISE (O-1)

Straight Objective Type

- The domain of the function $\sin^{-1}\left(\log_2\left(\frac{x}{3}\right)\right)$ is-
 (A) $\left[\frac{1}{2}, 3\right]$ (B) $\left[\frac{1}{2}, 3\right]$ (C) $\left[\frac{3}{2}, 6\right]$ (D) $\left[\frac{1}{2}, 2\right]$
- Domain of the function $f(x) = \log_e \cos^{-1}\{\sqrt{x}\}$ is, where $\{.\}$ represents fractional part function -
 (A) $x \in \mathbb{R}$ (B) $x \in [0, \infty)$ (C) $x \in (0, \infty)$ (D) $x \in \mathbb{R} - \{x \mid x \in \mathbb{I}\}$
- $\tan^{-1}\left(1 - x^2 - \frac{1}{x^2}\right) + \sin^{-1}\left(x^2 + \frac{1}{x^2} - 1\right)$ (where $x \neq 0$) is equal to
 (A) $\frac{\pi}{2}$ (B) $\frac{\pi}{4}$ (C) $\frac{3\pi}{4}$ (D) π
- The value of $\tan^2(\sec^{-1}3) + \cot^2(\operatorname{cosec}^{-1}4)$ is -
 (A) 9 (B) 16 (C) 25 (D) 23
- If $x > 0$ $\cos^{-1}\left(\frac{12}{x}\right) = \frac{\pi}{2} - \cos^{-1}\left(\frac{16}{x}\right)$ then x is -
 (A) 12 (B) 16 (C) 20 (D) 320
- If $2 \leq a < 3$, then the value of $\cos^{-1} \cos[a] + \operatorname{cosec}^{-1} \operatorname{cosec}[a] + \cot^{-1} \cot[a]$, (where $[.]$ denotes greatest integer less than equal to x) is equal to
 (A) $2 - \pi$ (B) $2 + \pi$ (C) π (D) 6
- If $\cos^{-1}(2x^2 - 1) = 2\pi - 2\cos^{-1}x$, then -
 (A) $x \in [-1, 0]$ (B) $x \in [0, 1]$ (C) $x \in \left[0, \frac{1}{\sqrt{2}}\right]$ (D) $x \in \left[-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right]$
- Number of integral ordered pairs (a, b) for which $\sin^{-1}(1 + b + b^2 + \dots \infty) + \cos^{-1}\left(a - \frac{a^2}{3} + \frac{a^3}{9} - \dots \infty\right) = \frac{\pi}{2}$ is-
 (A) 0 (B) 4 (C) 9 (D) Infinitely many
- $\lim_{n \rightarrow \infty} \sum_{r=1}^n \tan^{-1} \frac{2r+1}{r^4 + 2r^3 + r^2 + 1}$ is equal to -
 (A) $\frac{\pi}{4}$ (B) $\frac{3\pi}{4}$ (C) $\frac{\pi}{2}$ (D) $-\frac{\pi}{8}$

Multiple Correct Answer Type

10. If $\sin\left(2\cos^{-1}\left(\frac{1}{\sqrt{5}}\right)\right) + \cos\left(2\tan^{-1}\left(\frac{1}{3}\right)\right) = \frac{p}{q}$, where p & q are relatively prime then digit at units place of $(p - q)^{2k+1}$, $k \in \mathbb{N}$, can be -
 (A) 1 (B) 3 (C) 7 (D) 9
11. Consider the function $f(x) = e^x$ and $g(x) = \sin^{-1}x$, then which of the following is/are necessarily true.
 (A) Domain of $g \circ f$ = Domain of f (B) Range of $g \circ f \subset$ Range of g
 (C) Domain of $g \circ f$ is $(-\infty, 0]$ (D) Range of $g \circ f$ is $\left[-\frac{\pi}{2}, 0\right]$
12. Let $f(x) = \sin^{-1}(\tan x) + \cos^{-1}(\cot x)$ then
 (A) $f(x) = \frac{\pi}{2}$ wherever defined (B) domain of $f(x)$ is $x = n\pi \pm \frac{\pi}{4}$, $n \in \mathbb{I}$
 (C) period of $f(x)$ is $\frac{\pi}{2}$ (D) $f(x)$ is many one function
13. Let $f(x) = e^{x^3 - x^2 + x}$ be an invertible function such that $f^{-1} = g$, then -
 (A) $g(e) = 0$ (B) Domain of 'g' is $\mathbb{R}^+$
 (C) Range of 'g' is $\mathbb{R}$ (D) $f(g(e)) = e$
14. Value of $3\tan^{-1}\left(\frac{1}{3}\right) + \tan^{-1}\left(\frac{1}{2}\right) + \sin^{-1}\left(\frac{1}{\sqrt{5}}\right) + \cos^{-1}\left(\frac{2}{\sqrt{5}}\right)$ is greater than
 (A) $\frac{\pi}{2}$ (B) $\frac{2\pi}{3}$ (C) $\frac{3\pi}{4}$ (D) $\frac{5\pi}{6}$
15. Which of the following is/are correct ?
 (A) $\cot^{-1}(x) = \tan^{-1}\left(\frac{1}{x}\right) \forall x \in \mathbb{R} - \{0\}$
 (B) If $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $f(x) = \operatorname{sgn}(e^x)$ then $f(x)$ is an into function.
 (C) If $f : \mathbb{R}^+ \rightarrow \mathbb{R}$ such that $f(x) = \sin x + x$ then $f(x)$ is an odd function.
 (D) If $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $f(x) = \frac{e^x}{e^{[x]}}$ then $f(x)$ is a periodic function .
 (where $[.]$ represents greatest integer function)

EXERCISE (O-2)

Straight Objective Type

- The range of the function $f(x) = \sin^{-1}(\log_2(-x^2 + 2x + 3))$ is -
 (A) $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ (B) $\left[-\frac{\pi}{2}, 0\right]$ (C) $\left[0, \frac{\pi}{2}\right]$ (D) $[-1, 1]$
- Range of $f(x) = \cot^{-1}(\log_e(1 - x^2))$ is -
 (A) $(0, \pi)$ (B) $\left(0, \frac{\pi}{2}\right)$ (C) $\left[\frac{\pi}{2}, \pi\right)$ (D) $\left(0, \frac{\pi}{2}\right]$
- The value of $\sin^{-1}\left\{\cot\left(\sin^{-1}\sqrt{\frac{2-\sqrt{3}}{4}} + \cos^{-1}\frac{\sqrt{12}}{4} + \sec^{-1}\sqrt{2}\right)\right\}$ is -
 (A) 5 (B) 6 (C) 0 (D) 10
- Number of solution(s) of the equation $\cos^{-1}\sqrt{x} - \sin^{-1}\sqrt{x-1} + \cos^{-1}\sqrt{1-x} - \sin^{-1}\frac{1}{\sqrt{x}} = \frac{\pi}{2}$ is -
 (A) 0 (B) 1 (C) 2 (D) 4
- $\sum_{r=0}^{\infty} \tan^{-1}\left(\frac{r((r+1)!)}{(r+1) + ((r+1)!)^2}\right)$ is equal to -
 (A) $\frac{\pi}{2}$ (B) $\frac{\pi}{4}$ (C) $\cot^{-1}3$ (D) $\tan^{-1}2$

Multiple Correct Answer Type

- Let $f(x) = \sqrt{\cos^{-1}\sqrt{1-x^2} - \sin^{-1}x}$ then which of the following statement/s is/are correct -
 (A) Domain of $f(x)$ is $[-1, 1]$ (B) Domain of $f(x)$ is $[0, 1]$
 (C) Range of $f(x)$ is $\{0\}$ (D) Range of $f(x)$ is $[0, \sqrt{\pi}]$
- If $\alpha = 2\tan^{-1}(\sqrt{3-2\sqrt{2}}) + \sin^{-1}\left(\frac{1}{\sqrt{6}-\sqrt{2}}\right)$, $\beta = \cot^{-1}(\sqrt{3}-2) + \frac{1}{8}\sec^{-1}(-2)$ & $\gamma = \tan^{-1}\frac{1}{\sqrt{2}} + \cos^{-1}\frac{1}{\sqrt{3}}$, then
 (A) $\alpha = \beta$ (B) $\alpha + \beta = 3\gamma$ (C) $4(\beta - \gamma) = \alpha$ (D) $\beta = \gamma$
- If α is only real root of the equation $x^3 + (\cos 1)x^2 + (\sin 1)x + 1 = 0$, then $\left(\tan^{-1}\alpha + \tan^{-1}\frac{1}{\alpha}\right)$ cannot be equal to -
 (A) 0 (B) $\frac{\pi}{2}$ (C) $-\frac{\pi}{2}$ (D) π
- Let $f(x) = \begin{cases} x^2 - 4 & \text{if } |x| \leq 3 \\ 5\operatorname{sgn}|x-3| & \text{if } |x| > 3 \end{cases}$ and $g(x) = 2\tan^{-1}(e^x) - \frac{\pi}{2}$ for all $x \in \mathbb{R}$, then which of the following is wrong?
 (where $\operatorname{sgn}(x)$ denotes signum function of x)
 (A) $f(x)$ is an even function (B) $g \circ f(x)$ is an even function
 (C) $g(x)$ is an odd function (D) $f \circ g(x)$ is an odd function

Linked Comprehension Type

Paragraph for Question 10 to 12

Consider a continuous function such that each image has atmost three preimage & atleast one image has exactly three preimages. This type of function is to be called as three-one function.

On the basis of above information, answer the following questions :

10. Which of the following function is a three-one function ?
 (A) $|\ln|x||$ (B) $e^{|x|}$ (C) $x^3 + 3x^2 - 7x + 6$ (D) $\cos(\cos^{-1}x)$
11. If $f(x) = \sin^{-1}(\sin x)$ is a three-one function, then possible interval of x is -
 (A) $[-\pi, \pi]$ (B) $\left[-\frac{\pi}{2}, \frac{3\pi}{2}\right]$ (C) $(-2\pi, 0]$ (D) $\left[\frac{-3\pi}{2}, \frac{\pi}{2}\right]$
12. If $f(x)$ is a three-one function such that $f(a) = f(b)$ (where $a \neq b$), then number of maximum possible values of b is -
 (A) 1 (B) 2 (C) 3 (D) 4

Subjective Type Questions

13. Least integral value of x for which inequality $\sin^{-1}\left(\sin\left(\frac{2e^x + 3}{e^x + 1}\right)\right) > \pi - \frac{5}{2}$ holds, is
14. The product of all real values of x satisfying the equation

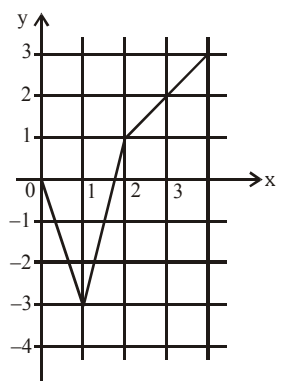
$$\sin^{-1} \cos\left(\frac{2x^2 + 10|x| + 4}{x^2 + 5|x| + 3}\right) = \cot\left(\cot^{-1}\left(\frac{2 - 18|x|}{9|x|}\right)\right) + \frac{\pi}{2}$$
 is
15. Number of integral solutions of the equation $2\sin^{-1}\sqrt{x^2 - x + 1} + \cos^{-1}\sqrt{x^2 - x} = \frac{3\pi}{2}$ is
16. If $\sin^{-1}\sin\left(\frac{10\pi}{3}\right) + \cos^{-1}\cos\frac{22\pi}{3} + \tan^{-1}\tan 10 = a\pi + b$, then $\left|\frac{3ab}{80}\right|$ is equal to

EXERCISE (S-1)

1. Given is a partial graph of an even periodic function f whose period is 8.

If $[*]$ denotes greatest integer function then find the value of the expression.

$$f(-3) + 2|f(-1)| + \left[f\left(\frac{7}{8}\right)\right] + f(0) + \arccos(f(-2)) + f(-7) + f(20)$$



2. (a) Find the following :

$$(i) \quad \tan\left[\cos^{-1}\frac{1}{2} + \tan^{-1}\left(\frac{-1}{\sqrt{3}}\right)\right]$$

$$(ii) \quad \cos^{-1}\left(\cos\frac{7\pi}{6}\right)$$

$$(iii) \quad \cos\left(\tan^{-1}\frac{3}{4}\right)$$

$$(iv) \quad \tan\left(\sin^{-1}\frac{3}{5} + \cot^{-1}\frac{3}{2}\right)$$

(b) Find the following :

$$(i) \quad \sin \left[\frac{\pi}{2} - \sin^{-1} \left(\frac{-\sqrt{3}}{2} \right) \right]$$

$$(ii) \quad \cos \left[\cos^{-1} \left(\frac{-\sqrt{3}}{2} \right) + \frac{\pi}{6} \right]$$

$$(iii) \quad \tan^{-1} \left(\tan \frac{3\pi}{4} \right)$$

$$(iv) \quad \sin \left(\frac{1}{4} \arcsin \frac{\sqrt{63}}{8} \right)$$

3. Find the domain of definition the following functions.

(Read the symbols $[*]$ and $\{*\}$ as greatest integers and fractional part functions respectively)

$$(i) \quad f(x) = \arccos \frac{2x}{1+x}$$

$$(ii) \quad f(x) = \sqrt{\cos(\sin x)} + \sin^{-1} \frac{1+x^2}{2x}$$

$$(iii) \quad f(x) = \sin^{-1} \left(\frac{x-3}{2} \right) - \log_{10}(4-x)$$

$$(iv) \quad f(x) = \sin^{-1}(2x + x^2)$$

$$(v) \quad f(x) = \frac{\sqrt{1-\sin x}}{\log_5(1-4x^2)} + \cos^{-1}(1-\{x\}), \text{ where } \{x\} \text{ is the fractional part of } x.$$

$$(vi) \quad f(x) = \sqrt{3-x} + \cos^{-1} \left(\frac{3-2x}{5} \right) + \log_6(2|x|-3) + \sin^{-1}(\log_2 x)$$

$$(vii) \quad f(x) = e^{\sin^{-1} \left(\frac{x}{2} \right)} + \tan^{-1} \left[\frac{x}{2} - 1 \right] + \ln(\sqrt{x - [x]})$$

4. Identify the pair(s) of functions which are identical. Also plot the graphs in each case.

$$(a) \quad y = \tan(\cos^{-1} x); y = \frac{\sqrt{1-x^2}}{x}$$

$$(b) \quad y = \tan(\cot^{-1} x); y = \frac{1}{x}$$

$$(c) \quad y = \sin(\arcsin x); y = \frac{x}{\sqrt{1+x^2}}$$

$$(d) \quad y = \cos(\arcsin x); y = \sin(\arcsin x)$$

5. Let $y = \sin^{-1}(\sin 8) - \tan^{-1}(\tan 10) + \cos^{-1}(\cos 12) - \sec^{-1}(\sec 9) + \cot^{-1}(\cot 6) - \operatorname{cosec}^{-1}(\operatorname{cosec} 7)$. If y simplifies to $a\pi + b$, then find $(a-b)$.

$$6. \quad \text{Show that : } \sin^{-1} \left(\sin \frac{33\pi}{7} \right) + \cos^{-1} \left(\cos \frac{46\pi}{7} \right) + \tan^{-1} \left(-\tan \frac{13\pi}{8} \right) + \cot^{-1} \left(\cot \left(-\frac{19\pi}{8} \right) \right) = \frac{13\pi}{7}$$

$$7. \quad \text{Prove that : (a) } 2\cos^{-1} \frac{3}{\sqrt{13}} + \cot^{-1} \frac{16}{63} + \frac{1}{2}\cos^{-1} \frac{7}{25} = \pi \quad (b) \quad \arccos \sqrt{\frac{2}{3}} - \arccos \frac{\sqrt{6}+1}{2\sqrt{3}} = \frac{\pi}{6}$$

8. If α and β are the roots of the equation $x^2 + 5x - 49 = 0$, then find the value of $\cot(\cot^{-1}\alpha + \cot^{-1}\beta)$.

$$9. \quad \text{If } a > b > c > 0, \text{ then find the value of : } \cot^{-1} \left(\frac{1+ab}{a-b} \right) + \cot^{-1} \left(\frac{1+bc}{b-c} \right) + \cot^{-1} \left(\frac{1+ca}{c-a} \right).$$

10. Find all values of k for which there is a triangle whose angles have measure $\tan^{-1} \left(\frac{1}{2} \right), \tan^{-1} \left(\frac{1}{2} + k \right)$ and $\tan^{-1} \left(\frac{1}{2} + 2k \right)$.

11. Find the simplest value of

(a) $f(x) = \arccos x + \arccos \left(\frac{x}{2} + \frac{1}{2} \sqrt{3-3x^2} \right), x \in \left(\frac{1}{2}, 1 \right)$

(b) $f(x) = \tan^{-1} \left(\frac{\sqrt{1+x^2}-1}{x} \right), x \in \mathbb{R} - \{0\}$

12. Prove that the identities :

(a) $\sin^{-1} \cos(\sin^{-1} x) + \cos^{-1} \sin(\cos^{-1} x) = \frac{\pi}{2}, |x| \leq 1$

(d) $\tan(\tan^{-1} x + \tan^{-1} y + \tan^{-1} z) = \cot(\cot^{-1} x + \cot^{-1} y + \cot^{-1} z)$

13. (a) Solve the inequality : $(\arccos x)^2 - 6(\arccos x) + 8 > 0$

(b) If $\sin^2 x + \sin^2 y < 1; x, y \in \mathbb{R}$ then prove that $\sin^{-1}(\tan x \cdot \tan y) \in (-\pi/2, \pi/2)$.

14. Solve the following :

(a) $\sin^{-1} x + \sin^{-1} 2x = \frac{\pi}{3}$

(b) $\tan^{-1} \frac{1}{1+2x} + \tan^{-1} \frac{1}{1+4x} = \tan^{-1} \frac{2}{x^2}$

(c) $\tan^{-1}(x-1) + \tan^{-1}(x) + \tan^{-1}(x+1) = \tan^{-1}(3x)$ (d) $3\cos^{-1} x = \sin^{-1}(\sqrt{1-x^2}(4x^2-1))$

(e) $\sin^{-1} x + \sin^{-1} y = \frac{2\pi}{3}$ & $\cos^{-1} x - \cos^{-1} y = \frac{\pi}{3}$

15. Find the sum of the series :

(a) $\cot^{-1} 7 + \cot^{-1} 13 + \cot^{-1} 21 + \cot^{-1} 31 + \dots$ to n terms.

(b) $\tan^{-1} \frac{1}{3} + \tan^{-1} \frac{2}{9} + \dots + \tan^{-1} \frac{2^{n-1}}{1+2^{2n-1}} + \dots \infty$

(c) $\tan^{-1} \frac{1}{x^2+x+1} + \tan^{-1} \frac{1}{x^2+3x+3} + \tan^{-1} \frac{1}{x^2+5x+7} + \tan^{-1} \frac{1}{x^2+7x+13}$ to n terms.

(d) $\sin^{-1} \frac{1}{\sqrt{5}} + \sin^{-1} \frac{1}{\sqrt{65}} + \sin^{-1} \frac{1}{\sqrt{325}} + \dots + \sin^{-1} \frac{1}{\sqrt{4n^4+1}} + \dots \infty$ terms.

EXERCISE (S-2)

1. If $\alpha = 2 \arctan \left(\frac{1+x}{1-x} \right)$ & $\beta = \arcsin \left(\frac{1-x^2}{1+x^2} \right)$ for $0 < x < 1$, then prove that $\alpha + \beta = \pi$, what the value of $\alpha + \beta$ will be if $x > 1$.

2. Solve the following :

(a) $\tan^{-1} \frac{x-1}{x+1} + \tan^{-1} \frac{2x-1}{2x+1} = \tan^{-1} \frac{23}{36}$

(b) $2 \tan^{-1} x = \cos^{-1} \frac{1-a^2}{1+a^2} - \cos^{-1} \frac{1-b^2}{1+b^2}$ ($a > 0, b > 0$)

3. If α and β are the roots of the equation $x^2 - 4x + 1 = 0$ ($\alpha > \beta$) then find the value of

$f(\alpha, \beta) = \frac{\beta^3}{2} \operatorname{cosec}^2 \left(\frac{1}{2} \tan^{-1} \frac{\beta}{\alpha} \right) + \frac{\alpha^3}{2} \sec^2 \left(\frac{1}{2} \tan^{-1} \frac{\alpha}{\beta} \right)$.

4. Find all the positive integral solutions of, $\tan^{-1} x + \cos^{-1} \frac{y}{\sqrt{1+y^2}} = \sin^{-1} \frac{3}{\sqrt{10}}$.

5. Prove that the equation, $(\sin^{-1}x)^3 + (\cos^{-1}x)^3 = \alpha\pi^3$ has no roots for $\alpha < \frac{1}{32}$ and $\alpha > \frac{7}{8}$.
6. Solve the following inequalities :
 (a) $\arccot^2 x - 5 \arccot x + 6 > 0$ (b) $\arcsin x > \arccos x$ (c) $\tan^2(\arcsin x) > 1$
7. Solve the following system of inequalities :
 $4 \arctan^2 x - 8 \arctan x + 3 < 0$ & $4 \arccot x - \arccot^2 x - 3 \geq 0$
8. If the sum $\sum_{n=1}^{10} \sum_{m=1}^{10} \tan^{-1} \left(\frac{m}{n} \right) = k\pi$, find the value of k .
9. Show that the roots r, s and t of the cubic $x(x-2)(3x-7)=2$, are real and positive. Also compute the value of $\tan^{-1}(r) + \tan^{-1}(s) + \tan^{-1}(t)$.
10. Solve for x : $\sin^{-1} \left(\sin \left(\frac{2x^2+4}{1+x^2} \right) \right) < \pi - 3$.
11. Find the set of values of 'a' for which the equation $2\cos^{-1}x = a + a^2(\cos^{-1}x)^{-1}$ posses a solution.

EXERCISE (JM)

1. Let $f : (-1, 1) \rightarrow B$, be a function defined by $f(x) = \tan^{-1} \frac{2x}{1-x^2}$, then f is both one-one and onto when B is the interval- [AIEEE-2005]
 (1) $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ (2) $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ (3) $\left(0, \frac{\pi}{2}\right)$ (4) $\left[0, \frac{\pi}{2}\right]$
2. If $\cos^{-1}x - \cos^{-1} \frac{y}{2} = \alpha$, then $4x^2 - 4xy \cos \alpha + y^2$ is equal to - [AIEEE-2005]
 (1) $2 \sin 2\alpha$ (2) 4 (3) $4 \sin^2 \alpha$ (4) $-4 \sin^2 \alpha$
3. If $\sin^{-1} \left(\frac{x}{5} \right) + \operatorname{cosec}^{-1} \left(\frac{5}{4} \right) = \frac{\pi}{2}$, then a value of x is- [AIEEE-2007]
 (1) 1 (2) 3 (3) 4 (4) 5
4. The value of $\cot \left(\operatorname{cosec}^{-1} \frac{5}{3} + \tan^{-1} \frac{2}{3} \right)$ is equal to- [AIEEE-2008]
 (1) $\frac{6}{17}$ (2) $\frac{3}{17}$ (3) $\frac{4}{17}$ (4) $\frac{5}{17}$
5. If x, y, z are in A.P. and $\tan^{-1}x, \tan^{-1}y$ and $\tan^{-1}z$ are also in A.P., then [JEE (Main)-2013]
 (1) $x = y = z$ (2) $2x = 3y = 6z$ (3) $6x = 3y = 2z$ (4) $6x = 4y = 3z$
6. Let $\tan^{-1} y = \tan^{-1} x + \tan^{-1} \left(\frac{2x}{1-x^2} \right)$, where $|x| < \frac{1}{\sqrt{3}}$. Then a value of y is : [JEE (Main)-2015]
 (1) $\frac{3x-x^3}{1+3x^2}$ (2) $\frac{3x+x^3}{1+3x^2}$ (3) $\frac{3x-x^3}{1-3x^2}$ (4) $\frac{3x+x^3}{1-3x^2}$

EXERCISE (JA)

1. Let (x, y) be such that $\sin^{-1}(ax) + \cos^{-1}(y) + \cos^{-1}(bxy) = \frac{\pi}{2}$ [JEE 2007, 6]

Match the statements in column-I with statements in column-II and indicate your answer by darkening the appropriate bubbles in the 4×4 matrix given in the ORS.

Column-I

Column-II

- (A) If $a = 1$ and $b = 0$, then (x, y) (p) lies on the circle $x^2 + y^2 = 1$
 (B) If $a = 1$ and $b = 1$, then (x, y) (q) lies on $(x^2 - 1)(y^2 - 1) = 0$
 (C) If $a = 1$ and $b = 2$, then (x, y) (r) lies on $y = x$
 (D) If $a = 2$ and $b = 2$, then (x, y) (s) lies on $(4x^2 - 1)(y^2 - 1) = 0$
2. If $0 < x < 1$, then $\sqrt{1+x^2} [\{x \cos(\cot^{-1}x) + \sin(\cot^{-1}x)\}^2 - 1]^{1/2} =$ [JEE 2008, 3]
 (A) $\frac{x}{\sqrt{1+x^2}}$ (B) x (C) $x\sqrt{1+x^2}$ (D) $\sqrt{1+x^2}$

3. The value of $\cot\left(\sum_{n=1}^{23} \cot^{-1}\left(1 + \sum_{k=1}^n 2k\right)\right)$ is [JEE(Advanced) 2013, 2]
 (A) $\frac{23}{25}$ (B) $\frac{25}{23}$ (C) $\frac{23}{24}$ (D) $\frac{24}{23}$

4. Match List-I with List-II and select the correct answer using the code given below the lists.

List-I

List-II

P. $\left(\frac{1}{y^2} \left(\frac{\cos(\tan^{-1}y) + y \sin(\tan^{-1}y)}{\cot(\sin^{-1}y) + \tan(\sin^{-1}y)}\right)^2 + y^4\right)^{1/2}$ takes value 1. $\frac{1}{2}\sqrt{\frac{5}{3}}$

Q. If $\cos x + \cos y + \cos z = 0 = \sin x + \sin y + \sin z$, 2. $\sqrt{2}$

then possible value of $\cos \frac{x-y}{2}$ is

R. If $\cos\left(\frac{\pi}{4} - x\right) \cos 2x + \sin x \sin 2x \sec x$ 3. $\frac{1}{2}$

$= \cos x \sin 2x \sec x + \cos\left(\frac{\pi}{4} + x\right) \cos 2x$,

then possible value of $\sec x$ is

S. If $\cot(\sin^{-1}\sqrt{1-x^2}) = \sin(\tan^{-1}(x\sqrt{6}))$, $x \neq 0$, 4. 1

then possible value of x is

Codes :

	P	Q	R	S
(A)	4	3	1	2
(B)	4	3	2	1
(C)	3	4	2	1
(D)	3	4	1	2

[JEE(Advanced) 2013, 3, (-1)]

5. Let $f : [0, 4\pi] \rightarrow [0, \pi]$ be defined by $f(x) = \cos^{-1}(\cos x)$. The number of points $x \in [0, 4\pi]$ satisfying the equation $f(x) = \frac{10-x}{10}$ is [JEE(Advanced)-2014, 3]

6. If $\alpha = 3 \sin^{-1}\left(\frac{6}{11}\right)$ and $\beta = 3 \cos^{-1}\left(\frac{4}{9}\right)$ where the inverse trigonometric functions take only the principal values, then the correct option(s) is(are) [JEE(Advanced)-2015, 4]

- (A) $\cos \beta > 0$ (B) $\sin \beta < 0$ (C) $\cos(\alpha + \beta) > 0$ (D) $\cos \alpha < 0$

7. The number of real solutions of the equation

$$\sin^{-1}\left(\sum_{i=1}^{\infty} x^{i+1} - x \sum_{i=1}^{\infty} \left(\frac{x}{2}\right)^i\right) = \frac{\pi}{2} - \cos^{-1}\left(\sum_{i=1}^{\infty} \left(-\frac{x}{2}\right)^i - \sum_{i=1}^{\infty} (-x)^i\right) \text{ lying in the interval } \left(-\frac{1}{2}, \frac{1}{2}\right) \text{ is } ______$$

(Here, the inverse trigonometric functions $\sin^{-1}x$ and $\cos^{-1}x$ assume value in $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ and $[0, \pi]$, respectively.) [JEE(Advanced)-2018]

8. Let $E_1 = \left\{x \in \mathbb{R} : x \neq 1 \text{ and } \frac{x}{x-1} > 0\right\}$ and $E_2 = \left\{x \in E_1 : \sin^{-1}\left(\log_e\left(\frac{x}{x-1}\right)\right) \text{ is a real number}\right\}$.

(Here, the inverse trigonometric function $\sin^{-1}x$ assumes values in $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$.)

Let $f : E_1 \rightarrow \mathbb{R}$ be the function defined by $f(x) = \log_e\left(\frac{x}{x-1}\right)$

and $g : E_2 \rightarrow \mathbb{R}$ be the function defined by $g(x) = \sin^{-1}\left(\log_e\left(\frac{x}{x-1}\right)\right)$.

LIST-I

- P. The range of f is
Q. The range of g contains
R. The domain of f contains
S. The domain of g is

LIST-II

1. $\left(-\infty, \frac{1}{1-e}\right] \cup \left[\frac{e}{e-1}, \infty\right)$
2. $(0, 1)$
3. $\left[-\frac{1}{2}, \frac{1}{2}\right]$
4. $(-\infty, 0) \cup (0, \infty)$
5. $\left(-\infty, \frac{e}{e-1}\right]$
6. $(-\infty, 0) \cup \left(\frac{1}{2}, \frac{e}{e-1}\right]$

The correct option is :

- (A) P $\rightarrow$ 4; Q $\rightarrow$ 2; R $\rightarrow$ 1; S $\rightarrow$ 1
(B) P $\rightarrow$ 3; Q $\rightarrow$ 3; R $\rightarrow$ 6; S $\rightarrow$ 5
(C) P $\rightarrow$ 4; Q $\rightarrow$ 2; R $\rightarrow$ 1; S $\rightarrow$ 6
(D) P $\rightarrow$ 4; Q $\rightarrow$ 3; R $\rightarrow$ 6; S $\rightarrow$ 5

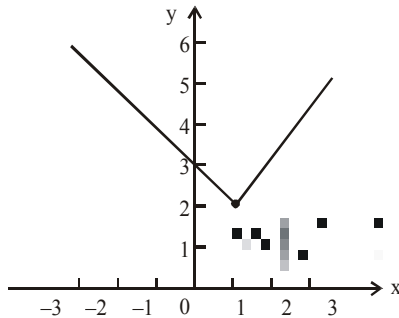
[JEE(Advanced)-2018]

ANSWER KEY

FUNCTIONS

EXERCISE (O-1)

1. (i) $-\infty < x \leq 0$ (ii) $x \in \mathbb{R}$ (iii) $x \in \mathbb{R} - \{-1, 0, 1\}$
 (iv) $-\infty < x < 0$ & $4 < x < \infty$ (v) $-\infty < x \leq 1$ and $3 \leq x < \infty$ (vi) $-\infty < x < 1$ and $2 < x < \infty$
 (vii) $-1 \leq x \leq 1$ (viii) $0 < x < 1$ and $1 < x < \infty$ (ix) $-2 \leq x < 0$ and $0 < x < 1$
 (x) $\frac{3}{2} < x < 2$ and $2 < x < \infty$ (xi) $-1 < x < 0$ and $1 < x < 2$; $2 < x < \infty$
 (xii) $2k\pi < x < (2k+1)\pi$, where k is an integer. (xiii) $4 \leq x \leq 6$ (xiv) $2 < x < 3$
2. (i) $\mathbb{R} - \{1\}$ (ii) $\mathbb{R} - \{0\}$ (iii) $(0, 4/3]$ (iv) $[1/3, 3]$ (v) $[1, \infty)$ (vi) $\mathbb{R}$
 (vii) $[0, \infty)$ (viii) $[3, 7]$ (ix) $[-1, 1]$ (x) $[2, \infty)$ (xi) $[1/3, 1]$ (xii) $\left[\frac{1}{11}, \frac{1}{5}\right]$
3. (a) -2, (b) -3, 1, (c) $[-3, 3]$, $[-2, 3]$, (d) $(-1, 3]$, (e) B, (f) A, D



- 4.
5. (a) $y = \begin{cases} x, & 0 \leq x \leq 1 \\ 2-x, & 1 < x \leq 2 \end{cases}$ (b) $y = \begin{cases} 2, & 0 \leq x < 1 \text{ and } 2 \leq x < 3 \\ 0, & 1 \leq x < 2 \text{ and } 3 \leq x \leq 4 \end{cases}$
6. (a) -2, 4, (b) -2, 2, (c) -3, 4, (d) $(0, 4]$ (e) $[-4, 4]$, $[-2, 3]$ (f) $[-4, 3]$, $[0.5, 4]$
7. (a) $\left[\frac{4}{3}, \frac{5}{3}\right] \cup \{3\}$ (b) $[-2, -1) \cup [1, 3]$ (c) $[-3, -2] \cup [-1, 1]$ (d) $\left(\frac{5}{3}, 3\right) - \{2\}$
8. C 9. D 10. C 11. A 12. C 13. B 14. C
 15. C 16. D 17. B 18. C 19. A 20. C 21. D
 22. B 23. D 24. B 25. B 26. C 27. D 28. D
 29. D 30. C 31. B, C 32. A, C 33. A, B, C, D 34. A, B, C
 35. A, B, C 36. A, B, C 37. A, B, C
 38. (A) $\rightarrow$ (P, R, T); (B) $\rightarrow$ (Q, R); (C) $\rightarrow$ (Q, S); (D) $\rightarrow$ (P, R, T) 39. A 40. C

EXERCISE (O-2)

1. C 2. B 3. D 4. C 5. B 6. A 7. B
 8. A 9. D 10. A 11. B 12. D 13. B 14. C
 15. D 16. B 17. C 18. A 19. A, B, C 20. A, C
 21. B, C, D 22. A, D 23. A, C 24. B, C, D 25. B, C
 26. C, D 27. B, C 28. A, B, C 29. A, B, C, D 30. A, C
 31. B 32. C 33. C 34. (A) $\rightarrow$ (P, S); (B) $\rightarrow$ (R); (C) $\rightarrow$ (Q, S); (D) $\rightarrow$ (T)
 35. (A) $\rightarrow$ (P, R, T); (B) $\rightarrow$ (Q, R, T); (C) $\rightarrow$ (Q, R, T); (D) $\rightarrow$ (P, R, T)

EXERCISE (S-1)

1. (i) $\left[-\frac{5\pi}{4}, -\frac{3\pi}{4}\right] \cup \left[-\frac{\pi}{4}, \frac{\pi}{4}\right] \cup \left[\frac{3\pi}{4}, \frac{5\pi}{4}\right]$ (ii) $\left(-4, -\frac{1}{2}\right) \cup (2, \infty)$ (iii) $(-\infty, -3]$
 (iv) $(-\infty, -1) \cup [0, \infty)$ (v) $(3 - 2\pi < x < 3 - \pi) \cup (3 < x \leq 4)$
 (vi) $\left(0, \frac{1}{100}\right) \cup \left(\frac{1}{100}, \frac{1}{\sqrt{10}}\right)$
 (vii) $(-3, -1] \cup \{0\} \cup [1, 3)$ (viii) $\{4\} \cup [5, \infty)$ (ix) $(1, 2) \cup (2, 5/2)$ (x) $x \in \{4, 5\}$
2. (i) $D : x \in \mathbb{R} \quad R : [0, 2]$ (ii) $D = \mathbb{R}; \text{range } [-1, 1]$
 (iii) $D : \{x \mid x \in \mathbb{R}; x \neq -3; x \neq 2\} \quad R : \{f(x) \mid f(x) \in \mathbb{R}, f(x) \neq 1/5; f(x) \neq 1\}$
 (iv) $D : \mathbb{R}, R : (-1, 1)$ (v) $D : -1 \leq x \leq 2 \quad R : [\sqrt{3}, \sqrt{6}]$
 (vi) $D : [-4, \infty) - \{5\}; R : \left(0, \frac{1}{6}\right) \cup \left(\frac{1}{6}, \frac{1}{3}\right]$
3. (b) (i), (iii) are identical
4. (a) neither surjective nor injective (b) surjective but not injective
 (c) neither injective nor surjective
5. (a) $-3/4$; (b) 64; (c) 30, (d) 5050
6. (a) domain is $x \geq 0$; range $[-1, 1]$; (b) domain $2k\pi \leq x \leq 2k\pi + \pi$; range $[0, 1]$
 (c) domain $x \in \mathbb{R}$; range $[-\sin 1, \sin 1]$ (d) domain is $0 \leq x \leq 1$; range is $[0, 1]$
8. (a) $\frac{(x+3)^{10}}{(x+3)^{10}+1}$, domain is $\mathbb{R}$, $\frac{1024}{1025}$; (b) $g(x) = \begin{cases} \frac{1}{x^2} & \text{if } 0 < x \leq 1 \\ x^2 & \text{if } x > 1 \end{cases}$
9. $f^{-1}(1) = y$ 10. $(\text{gof})(x) = \begin{cases} x & \text{if } x \leq 0 \\ -x^2 & \text{if } 0 < x < 1 \\ 1-x^2 & \text{if } x \geq 1 \end{cases}$ $(\text{fog})(x) = \begin{cases} x^2 & \text{if } x < 0 \\ 1+x & \text{if } 0 \leq x < 1 \\ x & \text{if } x \geq 1 \end{cases}$
11. (a) odd, (b) even, (c) neither odd nor even, (d) odd, (e) neither odd nor even,
 (f) even, (g) even, (h) even
12. (i) (a) $y = \log(10 - 10^x)$, $-\infty < x < 1$
 (b) $y = x/3$ when $-\infty < x < 0$ & $y = x$ when $0 \leq x < +\infty$
 (ii) (a) $2K\pi \leq x \leq 2K\pi + \pi$ where $K \in \mathbb{I}$ (b) $[-3/2, -1]$
 (iii) (a) Range : $[-1/3, 3]$, Domain = $[4, 7]$; (b) Range $[-1, 9]$ and domain $[11, 14]$
13. (a) $\frac{e^x - e^{-x}}{2}$; (b) $\frac{\log_2 x}{\log_2 x - 1}$; (c) $\frac{1}{2} \log \frac{1+x}{1-x}$
14. $x = 10$; $f^{-1}(x) = 10^{\log_2(x-8)}$
15. (a) (i) Period of fog is π , period of gof is 2π (ii) range of fog is $[-1, 1]$, range of gof is $[-\tan 1, \tan 1]$
 (b) 0.86
16. (A) S; (B) R; (C) P; (D) Q 17. (A) $\rightarrow$ (Q); (B) $\rightarrow$ (R); (C) $\rightarrow$ (P); (D) $\rightarrow$ (S)
18. 6 19. 401 20. 9 21. 6 22. 4024 23. 2 24. 15
25. 6 26. 9

EXERCISE (S-2)

1. (i) $(0, 1/4) \cup (3/4, 1) \cup \{x : x \in \mathbb{N}, x \geq 2\}$ (ii) $\mathbb{R} - \left\{-\frac{1}{2}, 0\right\}$
2. (a) 727, (b) 3 3. 6016 4. $f(x) = 2x^2$
5. $f \circ g(x) = \begin{cases} -(1+x), & -1 \leq x \leq 0 \\ x-1, & 0 < x \leq 2 \end{cases}; g \circ f(x) = \begin{cases} x+1, & 0 \leq x < 1 \\ 3-x, & 1 \leq x \leq 2 \\ x-1, & 2 < x \leq 3 \\ 5-x, & 3 < x \leq 4 \end{cases}$
- $f \circ f(x) = \begin{cases} x, & 0 \leq x \leq 1 \\ 4-x, & 1 < x \leq 2 \end{cases}; g \circ g(x) = \begin{cases} -x, & -1 \leq x \leq 0 \\ x, & 0 < x \leq 2 \\ 4-x, & 2 < x \leq 3 \end{cases}$
6. 21
7. $x = 0$ या $5/3$ 8. 1002.5 9. 5049 10. 20
11. 122 12. D 13. A 14. C 15. $\{-1, 1\}$

EXERCISE (JM)

- 1.** 1 **2.** 1 **3.** 3 **4.** 1 **5.** 4 **6.** 4 **7.** 2

EXERCISE (JA)

- 1. A 2. A 3. B 4. (zero marks to all) 5. A,B,C**

INVERSE TRIGONOMETRIC FUNCTIONS

EXERCISE (O-1)

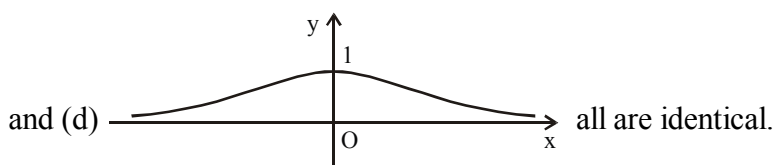
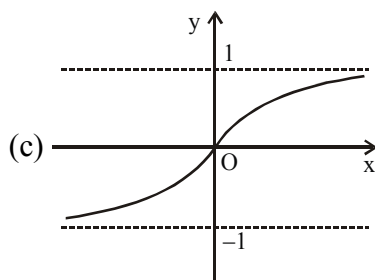
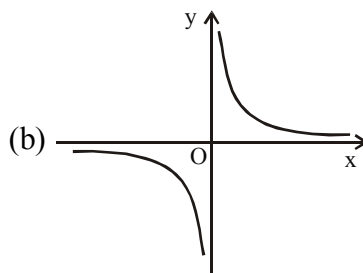
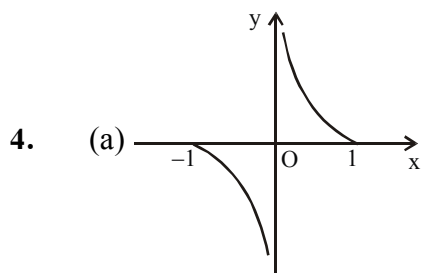
- | | | | | | | | | | | | | | |
|------------|-----|------------|-----|------------|-----|------------|-----|------------|---------|------------|-------|-----------|---|
| 1. | C | 2. | B | 3. | B | 4. | D | 5. | C | 6. | B | 7. | A |
| 8. | A | 9. | A | 10. | B,C | 11. | B,C | 12. | A,B,C,D | 13. | B,C,D | | |
| 14. | A,B | 15. | B,D | | | | | | | | | | |

EXERCISE (0-2)

- | | | | | | | | | | | | | | |
|-----|-----|-----|-------|-----|---|-----|---|-----|---|-----|-----|-----|---|
| 1. | A | 2. | C | 3. | C | 4. | A | 5. | B | 6. | A,D | | |
| 7. | A,C | 8. | A,B,D | | | 9. | D | 10. | C | 11. | A | 12. | B |
| 13. | 1 | 14. | 9 | 15. | 2 | 16. | 1 | | | | | | |

EXERCISE (S-1)

1. 5
2. (a) (i) $\frac{1}{\sqrt{3}}$, (ii) $\frac{5\pi}{6}$, (iii) $\frac{4}{5}$, (iv) $\frac{17}{6}$; (b) (i) $\frac{1}{2}$, (ii) -1 , (iii) $-\frac{\pi}{4}$, (iv) $\frac{\sqrt{2}}{4}$
3. (i) $-1/3 \leq x \leq 1$ (ii) $\{1, -1\}$ (iii) $1 \leq x < 4$ (iv) $[-(1 + \sqrt{2}), (\sqrt{2} - 1)]$
(v) $x \in (-1/2, 1/2), x \neq 0$ (vi) $(3/2, 2]$ (vii) $(-2, 2) - \{-1, 0, 1\}$



5. 53 8. 10 9. π 10. $k = \frac{11}{4}$ 11. (a) $\frac{\pi}{3}$; (b) $\frac{1}{2}\tan^{-1}x$

13. (a) $(-\infty, \sec 2) \cup [1, \infty)$

14. (a) $x = \frac{1}{2}\sqrt{\frac{3}{7}}$; (b) $x = 3$; (c) $x = 0, \frac{1}{2}, -\frac{1}{2}$; (d) $\left[\frac{\sqrt{3}}{2}, 1\right]$; (e) $x = \frac{1}{2}, y = 1$

15. (a) $\operatorname{arccot}\left[\frac{2n+5}{n}\right]$, (b) $\frac{\pi}{4}$, (c) $\arctan(x+n) - \arctan x$, (d) $\frac{\pi}{4}$

EXERCISE (S-2)

1. $-\pi$ 2. (a) $x = \frac{4}{3}$; (b) $x = \frac{a-b}{1+ab}$; 3. 56 4. $x = 1; y = 2$ & $x = 2; y = 7$

6. (a) $(\cot 2, \infty) \cup (-\infty, \cot 3)$ (b) $\left(\frac{\sqrt{2}}{2}, 1\right]$ (c) $\left(\frac{\sqrt{2}}{2}, 1\right) \cup \left(-1, -\frac{\sqrt{2}}{2}\right)$

7. $\left(\tan \frac{1}{2}, \cot 1\right]$ 8. $k = 25$ 9. $\frac{3\pi}{4}$ 10. $x \in (-1, 1)$

11. $a \in [-2\pi, \pi] - \{0\}$

EXERCISE (JM)

1. 2 2. 3 3. 2 4. 1 5. 1 6. 3

EXERCISE (JA)

1. (A) P; (B) Q; (C) P; (D) S 2. C 3. B 4. B 5. 3
6. B, C, D 7. 2 8. A

Important Notes

Important Notes

NURTURE COURSE

COMPOUND ANGLES

CONTENTS

COMPOUND ANGLES

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JEE (Main) Syllabus :

Trigonometric Identities

JEE (Advanced) Syllabus :

Trigonometric functions, their periodicity and graphs, addition and subtraction formulae, formulae involving multiple and sub-multiple angles.

TRIGONOMETRIC RATIOS & IDENTITIES

1. INTRODUCTION TO TRIGONOMETRY :

The word 'trigonometry' is derived from the Greek words 'trigon' and 'metron' and it means 'measuring the sides of a triangle'. The subject was originally developed to solve geometric problems involving triangles. It was studied by sea captains for navigation, surveyor to map out the new lands, by engineers and others. Currently, trigonometry is used in many areas such as the science of seismology, designing electric circuits, describing the state of an atom, predicting the heights of tides in the ocean, analysing a musical tone and in many other areas.

(a) **Measurement of angles :** Commonly two systems of measurement of angles are used.

(i) **Sexagesimal or English System :** Here 1 right angle = 90° (degrees)

$$1^\circ = 60' \text{ (minutes)}$$

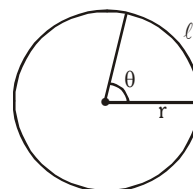
$$1' = 60'' \text{ (seconds)}$$

(ii) **Circular system :** Here an angle is measured in radians. One radian corresponds to the angle subtended by an arc of length 'r' at the centre of the circle of radius r. It is a constant quantity and does not depend upon the radius of the circle.

(b) Relation between the these systems : $\frac{D}{90} = \frac{R}{\pi/2}$

(c) If θ is the angle subtended at the centre of a circle of radius 'r',

by an arc of length ' ℓ ' then $\frac{\ell}{r} = \theta$.



Note that here ℓ , r are in the same units and θ is always in radians.

Illustration 1 : If the arcs of same length in two circles subtend angles of 60° and 75° at their centres. Find the ratio of their radii.

Solution : Let r_1 and r_2 be the radii of the given circles and let their arcs of same length 's' subtend angles of 60° and 75° at their centres.

$$\text{Now, } 60^\circ = \left(60 \times \frac{\pi}{180}\right)^\circ = \left(\frac{\pi}{3}\right)^\circ \text{ and } 75^\circ = \left(75 \times \frac{\pi}{180}\right)^\circ = \left(\frac{5\pi}{12}\right)^\circ$$

$$\therefore \frac{\pi}{3} = \frac{s}{r_1} \text{ and } \frac{5\pi}{12} = \frac{s}{r_2}$$

$$\Rightarrow \frac{\pi}{3} r_1 = s \text{ and } \frac{5\pi}{12} r_2 = s \Rightarrow \frac{\pi}{3} r_1 = \frac{5\pi}{12} r_2 \Rightarrow 4r_1 = 5r_2 \Rightarrow r_1 : r_2 = 5 : 4 \quad \text{Ans.}$$

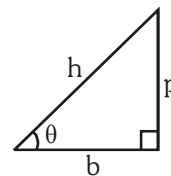
Do yourself - 1 :

(i) The radius of a circle is 30 cm. Find the length of an arc of this circle if the length of the chord of the arc is 30 cm.

2. T-RATIOS (or Trigonometric functions) :

In a right angle triangle

$$\sin \theta = \frac{p}{h}; \cos \theta = \frac{b}{h}; \tan \theta = \frac{p}{b}; \operatorname{cosec} \theta = \frac{h}{p}; \sec \theta = \frac{h}{b} \text{ and } \cot \theta = \frac{b}{p}$$



'p' is perpendicular ; 'b' is base and 'h' is hypotenuse.

Note : The quantity by which the cosine falls short of unity i.e. $1 - \cos \theta$, is called the versed sine θ of θ and also by which the sine falls short of unity i.e. $1 - \sin \theta$ is called the covered sine of θ .

3. BASIC TRIGONOMETRIC IDENTITIES :

(1) $\sin \theta \cdot \operatorname{cosec} \theta = 1$

(2) $\cos \theta \cdot \sec \theta = 1$

(3) $\tan \theta \cdot \cot \theta = 1$

(4) $\tan \theta = \frac{\sin \theta}{\cos \theta} \text{ \& } \cot \theta = \frac{\cos \theta}{\sin \theta}$

(5) $\sin^2 \theta + \cos^2 \theta = 1$ or $\sin^2 \theta = 1 - \cos^2 \theta$ or $\cos^2 \theta = 1 - \sin^2 \theta$

(6) $\sec^2 \theta - \tan^2 \theta = 1$ or $\sec^2 \theta = 1 + \tan^2 \theta$ or $\tan^2 \theta = \sec^2 \theta - 1$

(7) $\sec \theta + \tan \theta = \frac{1}{\sec \theta - \tan \theta}$

(8) $\operatorname{cosec}^2 \theta - \cot^2 \theta = 1$ or $\operatorname{cosec}^2 \theta = 1 + \cot^2 \theta$ or $\cot^2 \theta = \operatorname{cosec}^2 \theta - 1$

(9) $\operatorname{cosec} \theta + \cot \theta = \frac{1}{\operatorname{cosec} \theta - \cot \theta}$

(10) Expressing trigonometrical ratio in terms of each other :

	$\sin \theta$	$\cos \theta$	$\tan \theta$	$\cot \theta$	$\sec \theta$	$\operatorname{cosec} \theta$
$\sin \theta$	$\sin \theta$	$\sqrt{1 - \cos^2 \theta}$	$\frac{\tan \theta}{\sqrt{1 + \tan^2 \theta}}$	$\frac{1}{\sqrt{1 + \cot^2 \theta}}$	$\frac{\sqrt{\sec^2 \theta - 1}}{\sec \theta}$	$\frac{1}{\operatorname{cosec} \theta}$
$\cos \theta$	$\sqrt{1 - \sin^2 \theta}$	$\cos \theta$	$\frac{1}{\sqrt{1 + \tan^2 \theta}}$	$\frac{\cot \theta}{\sqrt{1 + \cot^2 \theta}}$	$\frac{1}{\sec \theta}$	$\frac{\sqrt{\operatorname{cosec}^2 \theta - 1}}{\operatorname{cosec} \theta}$
$\tan \theta$	$\frac{\sin \theta}{\sqrt{1 - \sin^2 \theta}}$	$\frac{\sqrt{1 - \cos^2 \theta}}{\cos \theta}$	$\tan \theta$	$\frac{1}{\cot \theta}$	$\sqrt{\sec^2 \theta - 1}$	$\frac{1}{\sqrt{\operatorname{cosec}^2 \theta - 1}}$
$\cot \theta$	$\frac{\sqrt{1 - \sin^2 \theta}}{\sin \theta}$	$\frac{\cos \theta}{\sqrt{1 - \cos^2 \theta}}$	$\frac{1}{\tan \theta}$	$\cot \theta$	$\frac{1}{\sqrt{\sec^2 \theta - 1}}$	$\sqrt{\operatorname{cosec}^2 \theta - 1}$
$\sec \theta$	$\frac{1}{\sqrt{1 - \sin^2 \theta}}$	$\frac{1}{\cos \theta}$	$\sqrt{1 + \tan^2 \theta}$	$\frac{\sqrt{1 + \cot^2 \theta}}{\cot \theta}$	$\sec \theta$	$\frac{\operatorname{cosec} \theta}{\sqrt{\operatorname{cosec}^2 \theta - 1}}$
$\operatorname{cosec} \theta$	$\frac{1}{\sin \theta}$	$\frac{1}{\sqrt{1 - \cos^2 \theta}}$	$\frac{\sqrt{1 + \tan^2 \theta}}{\tan \theta}$	$\sqrt{1 + \cot^2 \theta}$	$\frac{\sec \theta}{\sqrt{\sec^2 \theta - 1}}$	$\operatorname{cosec} \theta$

Illustration 2 : If $\sin \theta + \sin^2 \theta = 1$, then prove that $\cos^{12} \theta + 3 \cos^{10} \theta + 3 \cos^8 \theta + \cos^6 \theta - 1 = 0$

Solution : Given that $\sin \theta = 1 - \sin^2 \theta = \cos^2 \theta$

$$\text{L.H.S.} = \cos^6 \theta (\cos^2 \theta + 1)^3 - 1 = \sin^3 \theta (1 + \sin \theta)^3 - 1 = (\sin \theta + \sin^2 \theta)^3 - 1 = 1 - 1 = 0$$

Illustration 3 : $4(\sin^6 \theta + \cos^6 \theta) - 6(\sin^4 \theta + \cos^4 \theta)$ is equal to

- (A) 0 (B) 1 (C) -2 (D) none of these

Solution : $4[(\sin^2 \theta + \cos^2 \theta)^3 - 3 \sin^2 \theta \cos^2 \theta (\sin^2 \theta + \cos^2 \theta)] - 6[(\sin^2 \theta + \cos^2 \theta)^2 - 2 \sin^2 \theta \cos^2 \theta]$
 $= 4[1 - 3 \sin^2 \theta \cos^2 \theta] - 6[1 - 2 \sin^2 \theta \cos^2 \theta]$
 $= 4 - 12 \sin^2 \theta \cos^2 \theta - 6 + 12 \sin^2 \theta \cos^2 \theta = -2$

Ans.(C)

Do yourself - 2 :

- (i) If $\cot \theta = \frac{4}{3}$, then find the value of $\sin \theta$, $\cos \theta$ and $\operatorname{cosec} \theta$ in first quadrant.
 (ii) If $\sin \theta + \operatorname{cosec} \theta = 2$, then find the value of $\sin^8 \theta + \operatorname{cosec}^8 \theta$

4. NEW DEFINITION OF T-RATIOS :

By using rectangular coordinates the definitions of trigonometric functions can be extended to angles of any size in the following way (see diagram). A point P is taken with coordinates (x, y). The radius vector OP has length r and the angle θ is taken as the directed angle measured anticlockwise from the x-axis. The three main trigonometric functions are then defined in terms of r and the coordinates x and y.

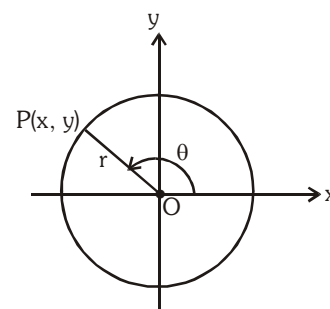
$$\sin \theta = y/r,$$

$$\cos \theta = x/r$$

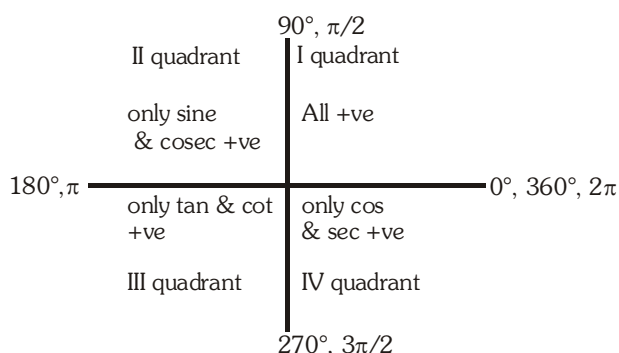
$$\tan \theta = y/x,$$

(The other function are reciprocals of these)

This can give negative values of the trigonometric functions.



5. SIGNS OF TRIGONOMETRIC FUNCTIONS IN DIFFERENT QUADRANTS :



6. TRIGONOMETRIC FUNCTIONS OF ALLIED ANGLES :

(a) $\sin(2n\pi + \theta) = \sin \theta$, $\cos(2n\pi + \theta) = \cos \theta$, where $n \in \mathbb{I}$

(b)	$\sin(-\theta) = -\sin \theta$	$\cos(-\theta) = \cos \theta$
	$\sin(90^\circ - \theta) = \cos \theta$	$\cos(90^\circ - \theta) = \sin \theta$
	$\sin(90^\circ + \theta) = \cos \theta$	$\cos(90^\circ + \theta) = -\sin \theta$
	$\sin(180^\circ - \theta) = \sin \theta$	$\cos(180^\circ - \theta) = -\cos \theta$
	$\sin(180^\circ + \theta) = -\sin \theta$	$\cos(180^\circ + \theta) = -\cos \theta$
	$\sin(270^\circ - \theta) = -\cos \theta$	$\cos(270^\circ - \theta) = -\sin \theta$
	$\sin(270^\circ + \theta) = -\cos \theta$	$\cos(270^\circ + \theta) = \sin \theta$
	$\sin(360^\circ - \theta) = -\sin \theta$	$\cos(360^\circ - \theta) = \cos \theta$
	$\sin(360^\circ + \theta) = \sin \theta$	$\cos(360^\circ + \theta) = \cos \theta$

7. VALUES OF T-RATIOS OF SOME STANDARD ANGLES :

Angles	0°	30°	45°	60°	90°	180°	270°
T-ratio	0	$\pi/6$	$\pi/4$	$\pi/3$	$\pi/2$	π	$3\pi/2$
$\sin \theta$	0	1/2	$1/\sqrt{2}$	$\sqrt{3}/2$	1	0	-1
$\cos \theta$	1	$\sqrt{3}/2$	$1/\sqrt{2}$	1/2	0	-1	0
$\tan \theta$	0	$1/\sqrt{3}$	1	$\sqrt{3}$	N.D.	0	N.D.
$\cot \theta$	N.D.	$\sqrt{3}$	1	$1/\sqrt{3}$	0	N.D.	0
$\sec \theta$	1	$2/\sqrt{3}$	$\sqrt{2}$	2	N.D.	-1	N.D.
$\operatorname{cosec} \theta$	N.D.	2	$\sqrt{2}$	$2/\sqrt{3}$	1	N.D.	-1

N.D. $\rightarrow$ Not Defined

(a) $\sin n\pi = 0$; $\cos n\pi = (-1)^n$; $\tan n\pi = 0$ where $n \in \mathbb{I}$

(b) $\sin(2n+1)\frac{\pi}{2} = (-1)^n$; $\cos(2n+1)\frac{\pi}{2} = 0$ where $n \in \mathbb{I}$

Illustration 4 : If $\sin \theta = -\frac{1}{2}$ and $\tan \theta = \frac{1}{\sqrt{3}}$ then θ is equal to -

- (A) 30° (B) 150° (C) 210° (D) none of these

Solution : Let us first find out θ lying between 0 and 360°.

Since $\sin \theta = -\frac{1}{2} \Rightarrow \theta = 210^\circ$ or 330° and $\tan \theta = \frac{1}{\sqrt{3}} \Rightarrow \theta = 30^\circ$ or 210°

Hence, $\theta = 210^\circ$ or $\frac{7\pi}{6}$ is the value satisfying both.

Ans. (C)

Do yourself - 3 :

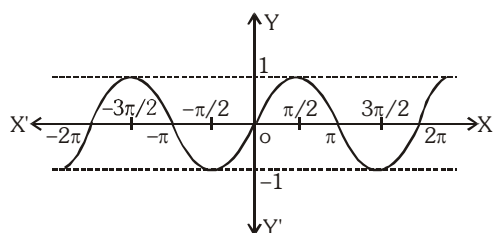
(i) If $\cos\theta = -\frac{1}{2}$ and $\pi < \theta < \frac{3\pi}{2}$, then find the value of $4\tan^2\theta - 3\operatorname{cosec}^2\theta$.

(ii) Prove that : (a) $\cos 570^\circ \sin 510^\circ + \sin(-330^\circ) \cos(-390^\circ) = 0$

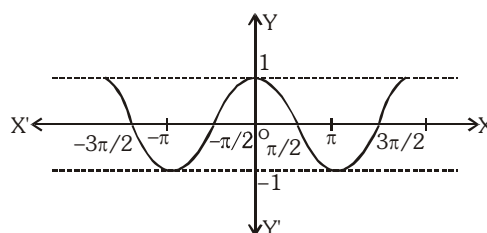
$$(b) \quad \tan \frac{11\pi}{3} - 2 \sin \frac{9\pi}{3} - \frac{3}{4} \operatorname{cosec}^2 \frac{\pi}{4} + 4 \cos^2 \frac{17\pi}{6} = \frac{3-2\sqrt{3}}{2}$$

8. GRAPH OF TRIGONOMETRIC FUNCTIONS :

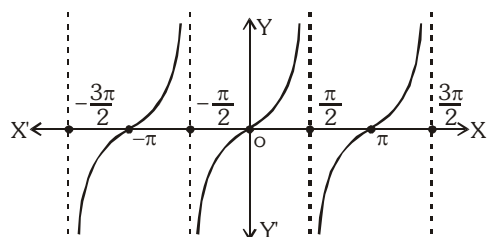
(i) $y = \sin x$



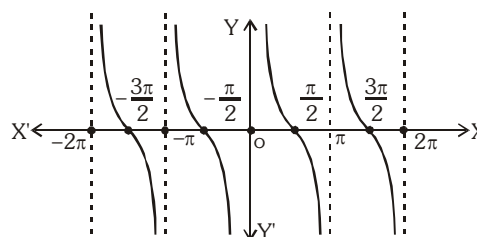
(ii) $y = \cos x$



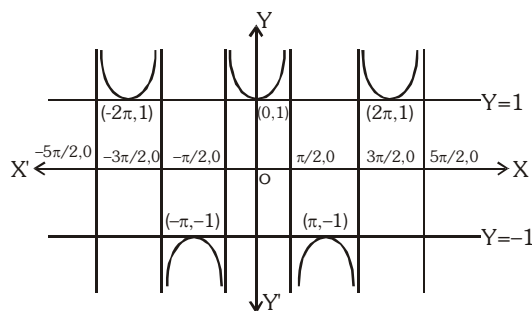
(iii) $y = \tan x$



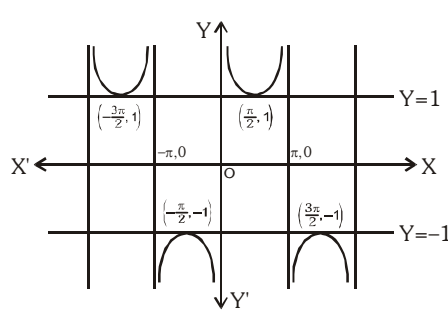
(iv) $y = \cot x$



(v) $y = \sec x$



(vi) $y = \operatorname{cosec} x$



9. DOMAINS, RANGES AND PERIODICITY OF TRIGONOMETRIC FUNCTIONS :

T-Ratio	Domain	Range	Period
$\sin x$	$\mathbb{R}$	$[-1, 1]$	2π
$\cos x$	$\mathbb{R}$	$[-1, 1]$	2π
$\tan x$	$\mathbb{R} - \{(2n+1)\pi/2 ; n \in \mathbb{I}\}$	$\mathbb{R}$	π
$\cot x$	$\mathbb{R} - \{n\pi : n \in \mathbb{I}\}$	$\mathbb{R}$	π
$\sec x$	$\mathbb{R} - \{(2n+1)\pi/2 : n \in \mathbb{I}\}$	$(-\infty, -1] \cup [1, \infty)$	2π
$\operatorname{cosec} x$	$\mathbb{R} - \{n\pi : n \in \mathbb{I}\}$	$(-\infty, -1] \cup [1, \infty)$	2π

10. TRIGONOMETRIC RATIOS OF THE SUM & DIFFERENCE OF TWO ANGLES :

(i) $\sin (A+B)=\sin A \cos B+\cos A \sin B$. **(ii)** $\sin (A-B)=\sin A \cos B-\cos A \sin B$.

(iii) $\cos (A+B)=\cos A \cos B-\sin A \sin B$ (iv) $\cos (A-B)=\cos A \cos B+\sin A \sin B$

$$\text{(v)} \quad \tan(A+B) = \frac{\tan A + \tan B}{1 - \tan A \tan B} \qquad \text{(vi)} \quad \tan(A-B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$$

$$\text{(vii)} \quad \cot (A+B)=\frac{\cot B \cot A-1}{\cot B+\cot A} \qquad \text{(viii)} \quad \cot (A-B)=\frac{\cot B \cot A+1}{\cot B-\cot A}$$

Some more results :

(i) $\sin^2 A - \sin^2 B = \sin(A+B) \cdot \sin(A-B) = \cos^2 B - \cos^2 A$.

(ii) $\cos^2 A - \sin^2 B = \cos (A+B). \cos (A-B).$

Illustration 5 : Prove that $\sqrt{3} \operatorname{cosec} 20^\circ - \sec 20^\circ = 4$.

Solution :

$$\begin{aligned} \text{L.H.S.} &= \frac{\sqrt{3}}{\sin 20^\circ} - \frac{1}{\cos 20^\circ} = \frac{\sqrt{3} \cos 20^\circ - \sin 20^\circ}{\sin 20^\circ \cdot \cos 20^\circ} \\ &= \frac{4 \left(\frac{\sqrt{3}}{2} \cos 20^\circ - \frac{1}{2} \sin 20^\circ \right)}{2 \sin 20^\circ \cos 20^\circ} = \frac{4(\sin 60^\circ \cdot \cos 20^\circ - \cos 60^\circ \cdot \sin 20^\circ)}{\sin 40^\circ} \\ &= 4 \cdot \frac{\sin(60^\circ - 20^\circ)}{\sin 40^\circ} = 4 \cdot \frac{\sin 40^\circ}{\sin 40^\circ} = 4 = \text{R.H.S.} \end{aligned}$$

Illustration 6: Prove that $\tan 70^\circ = \cot 70^\circ + 2\cot 40^\circ$.

Solution : L.H.S. = $\tan 70^\circ = \tan(20^\circ + 50^\circ) = \frac{\tan 20^\circ + \tan 50^\circ}{1 - \tan 20^\circ \tan 50^\circ}$

$$\text{or } \tan 70^\circ - \tan 20^\circ \tan 50^\circ \tan 70^\circ = \tan 20^\circ + \tan 50^\circ$$

$$\text{or } \tan 70^\circ = \tan 70^\circ \tan 50^\circ \tan 20^\circ + \tan 20^\circ + \tan 50^\circ = 2 \tan 50^\circ + \tan 20^\circ$$

$$= \cot 70^\circ + 2 \cot 40^\circ = \text{R.H.S.}$$

Do yourself - 4 :

(i) If $\sin A = \frac{3}{5}$ and $\cos B = \frac{9}{41}$, $0 < A \& B < \frac{\pi}{2}$, then find the value of the following :

(a) $\sin(A + B)$ (b) $\sin(A - B)$ (c) $\cos(A + B)$ (d) $\cos(A - B)$

(ii) If $x + y = 45^\circ$, then prove that :

(a) $(1 + \tan x)(1 + \tan y) = 2$ (b) $(\cot x - 1)(\cot y - 1) = 2$

(Remember these results)

11. FORMULAE TO TRANSFORM THE PRODUCT INTO SUM OR DIFFERENCE :

(i) $2 \sin A \cos B = \sin (A+B) + \sin (A-B)$. **(ii)** $2 \cos A \sin B = \sin (A+B) - \sin (A-B)$.

(iii) $2 \cos A \cos B = \cos (A + B) + \cos (A - B)$ (iv) $2 \sin A \sin B = \cos (A - B) - \cos (A + B)$

Illustration 7: If $\sin 2A = \lambda \sin 2B$, then prove that $\frac{\tan(A+B)}{\tan(A-B)} = \frac{\lambda+1}{\lambda-1}$.

Solution : Given $\sin 2A = \lambda \sin 2B$

$$\Rightarrow \frac{\sin 2A}{\sin 2B} = \frac{\lambda}{1}$$

Applying componendo & dividendo,

$$\frac{\sin 2A + \sin 2B}{\sin 2B - \sin 2A} = \frac{\lambda + 1}{1 - \lambda}$$

$$\Rightarrow \frac{2 \sin \left(\frac{2A+2B}{2} \right) \cos \left(\frac{2A-2B}{2} \right)}{2 \cos \left(\frac{2B+2A}{2} \right) \sin \left(\frac{2B-2A}{2} \right)} = \frac{\lambda + 1}{1 - \lambda}$$

$$\Rightarrow \frac{\sin(A+B) \cos(A-B)}{\cos(A+B) \sin\{-(A-B)\}} = \frac{\lambda + 1}{1 - \lambda} \Rightarrow \frac{\sin(A+B) \cos(A-B)}{\cos(A+B) \times -\sin(A-B)} = \frac{\lambda + 1}{-(\lambda - 1)}$$

$$\Rightarrow \frac{\sin(A+B) \cos(A-B)}{\cos(A+B) \sin(A-B)} = \frac{\lambda + 1}{\lambda - 1} \Rightarrow \tan(A+B) \cot(A-B) = \frac{\lambda + 1}{\lambda - 1}$$

$$\Rightarrow \frac{\tan(A+B)}{\tan(A-B)} = \frac{\lambda + 1}{\lambda - 1}$$

12. FORMULAE TO TRANSFORM SUM OR DIFFERENCE INTO PRODUCT :

$$(i) \quad \sin C + \sin D = 2 \sin \left(\frac{C+D}{2} \right) \cos \left(\frac{C-D}{2} \right)$$

$$(ii) \quad \sin C - \sin D = 2 \cos \left(\frac{C+D}{2} \right) \sin \left(\frac{C-D}{2} \right)$$

$$(iii) \quad \cos C + \cos D = 2 \cos \left(\frac{C+D}{2} \right) \cos \left(\frac{C-D}{2} \right)$$

$$(iv) \quad \cos C - \cos D = 2 \sin \left(\frac{C+D}{2} \right) \sin \left(\frac{D-C}{2} \right)$$

Illustration 8 : $\frac{\sin 5\theta + \sin 2\theta - \sin \theta}{\cos 5\theta + 2 \cos 3\theta + 2 \cos^2 \theta + \cos \theta}$ is equal to -

(A) $\tan \theta$

(B) $\cos \theta$

(C) $\cot \theta$

(D) none of these

Solution :

$$\begin{aligned} \text{L.H.S.} &= \frac{2 \sin 2\theta \cos 3\theta + \sin 2\theta}{2 \cos 3\theta \cdot \cos 2\theta + 2 \cos 3\theta + 2 \cos^2 \theta} = \frac{\sin 2\theta [2 \cos 3\theta + 1]}{2 [\cos 3\theta (\cos 2\theta + 1) + (\cos^2 \theta)]} \\ &= \frac{\sin 2\theta [2 \cos 3\theta + 1]}{2 [\cos 3\theta (2 \cos^2 \theta) + \cos^2 \theta]} = \frac{\sin 2\theta (2 \cos 3\theta + 1)}{2 \cos^2 \theta (2 \cos 3\theta + 1)} = \tan \theta \end{aligned}$$

Ans. (A)

Illustration 9: Show that $\sin 12^\circ \cdot \sin 48^\circ \cdot \sin 54^\circ = 1/8$

Solution :

$$\begin{aligned} \text{L.H.S.} &= \frac{1}{2} [\cos 36^\circ - \cos 60^\circ] \sin 54^\circ = \frac{1}{2} \left[\cos 36^\circ \sin 54^\circ - \frac{1}{2} \sin 54^\circ \right] \\ &= \frac{1}{4} [2 \cos 36^\circ \sin 54^\circ - \sin 54^\circ] = \frac{1}{4} [\sin 90^\circ + \sin 18^\circ - \sin 54^\circ] \\ &= \frac{1}{4} [1 - (\sin 54^\circ - \sin 18^\circ)] = \frac{1}{4} [1 - 2 \sin 18^\circ \cos 36^\circ] \\ &= \frac{1}{4} \left[1 - \frac{2 \sin 18^\circ}{\cos 18^\circ} \cos 18^\circ \cos 36^\circ \right] = \frac{1}{4} \left[1 - \frac{\sin 36^\circ \cos 36^\circ}{\cos 18^\circ} \right] \\ &= \frac{1}{4} \left[1 - \frac{2 \sin 36^\circ \cos 36^\circ}{2 \cos 18^\circ} \right] = \frac{1}{4} \left[1 - \frac{\sin 72^\circ}{2 \sin 72^\circ} \right] = \frac{1}{4} \left[1 - \frac{1}{2} \right] = \frac{1}{8} = \text{R.H.S.} \end{aligned}$$

Do yourself - 5 :

(i) Simplify $\frac{\sin 75^\circ - \sin 15^\circ}{\cos 75^\circ + \cos 15^\circ}$

(ii) Prove that

(a) $(\sin 3A + \sin A) \sin A + (\cos 3A - \cos A) \cos A = 0$

(b) $\cos 20^\circ \cos 40^\circ \cos 60^\circ \cos 80^\circ = \frac{1}{16}$

(c) $\frac{\sin 8\theta \cos \theta - \sin 6\theta \cos 3\theta}{\cos 2\theta \cos \theta - \sin 3\theta \sin 4\theta} = \tan 2\theta$

13. TRIGONOMETRIC RATIOS OF SUM OF MORE THAN TWO ANGLES :

(i) $\sin (A+B+C) = \sin A \cos B \cos C + \sin B \cos A \cos C + \sin C \cos A \cos B - \sin A \sin B \sin C$
 $= \Sigma \sin A \cos B \cos C - \Pi \sin A$

$$= \cos A \cos B \cos C [\tan A + \tan B + \tan C - \tan A \tan B \tan C]$$

(ii) $\cos (A+B+C) = \cos A \cos B \cos C - \sin A \sin B \cos C - \sin A \cos B \sin C - \cos A \sin B \sin C$
 $= \Pi \cos A - \Sigma \sin A \sin B \cos C$

$$= \cos A \cos B \cos C [1 - \tan A \tan B - \tan B \tan C - \tan C \tan A]$$

(iii) $\tan (A+B+C) = \frac{\tan A + \tan B + \tan C - \tan A \tan B \tan C}{1 - \tan A \tan B - \tan B \tan C - \tan C \tan A} = \frac{S_1 - S_3}{1 - S_2}$

14. TRIGONOMETRIC RATIOS OF MULTIPLE ANGLES :

(a) Trigonometrical ratios of an angle 2θ in terms of the angle θ :

$$(i) \quad \sin 2\theta = 2 \sin \theta \cos \theta = \frac{2 \tan \theta}{1 + \tan^2 \theta}$$

$$(ii) \quad \cos 2\theta = \cos^2 \theta - \sin^2 \theta = 2 \cos^2 \theta - 1 = 1 - 2 \sin^2 \theta = \frac{1 - \tan^2 \theta}{1 + \tan^2 \theta}$$

$$(iii) \quad 1 + \cos 2\theta = 2 \cos^2 \theta$$

$$(iv) \quad 1 - \cos 2\theta = 2 \sin^2 \theta$$

$$(v) \quad \tan \theta = \frac{1 - \cos 2\theta}{\sin 2\theta} = \frac{\sin 2\theta}{1 + \cos 2\theta}$$

$$(vi) \quad \tan 2\theta = \frac{2 \tan \theta}{1 - \tan^2 \theta}$$

Illustration 10: Prove that : $\frac{2 \cos 2A + 1}{2 \cos 2A - 1} = \tan(60^\circ + A) \tan(60^\circ - A)$.

Solution : R.H.S. = $\tan(60^\circ + A) \tan(60^\circ - A)$

$$\begin{aligned} &= \left(\frac{\tan 60^\circ + \tan A}{1 - \tan 60^\circ \tan A} \right) \left(\frac{\tan 60^\circ - \tan A}{1 + \tan 60^\circ \tan A} \right) = \left(\frac{\sqrt{3} + \tan A}{1 - \sqrt{3} \tan A} \right) \left(\frac{\sqrt{3} - \tan A}{1 + \sqrt{3} \tan A} \right) \\ &= \frac{3 - \tan^2 A}{1 - 3 \tan^2 A} = \frac{3 - \frac{\sin^2 A}{\cos^2 A}}{1 - 3 \frac{\sin^2 A}{\cos^2 A}} = \frac{3 \cos^2 A - \sin^2 A}{\cos^2 A - 3 \sin^2 A} = \frac{2 \cos^2 A + \cos^2 A - 2 \sin^2 A + \sin^2 A}{2 \cos^2 A - 2 \sin^2 A - \sin^2 A - \cos^2 A} \\ &= \frac{2(\cos^2 A - \sin^2 A) + \cos^2 A + \sin^2 A}{2(\cos^2 A - \sin^2 A) - (\sin^2 A + \cos^2 A)} = \frac{2 \cos 2A + 1}{2 \cos 2A - 1} = \text{L.H.S.} \end{aligned}$$

Do yourself - 6 :

(i) Prove that :

$$(a) \quad \frac{\sin 2\theta}{1 + \cos 2\theta} = \tan \theta$$

$$(b) \quad \frac{1 + \sin 2\theta + \cos 2\theta}{1 + \sin 2\theta - \cos 2\theta} = \cot \theta$$

(b) Trigonometrical ratios of an angle 3θ in terms of the angle θ :

$$(i) \quad \sin 3\theta = 3 \sin \theta - 4 \sin^3 \theta.$$

$$(ii) \quad \cos 3\theta = 4 \cos^3 \theta - 3 \cos \theta.$$

$$(iii) \quad \tan 3\theta = \frac{3 \tan \theta - \tan^3 \theta}{1 - 3 \tan^2 \theta}$$

Illustration 11: Prove that : $\tan A + \tan(60^\circ + A) + \tan(120^\circ + A) = 3 \tan 3A$

Solution :

$$\begin{aligned}
 \text{L.H.S.} &= \tan A + \tan(60^\circ + A) + \tan(120^\circ + A) \\
 &= \tan A + \tan(60^\circ + A) + \tan\{180^\circ - (60^\circ - A)\} \\
 &= \tan A + \tan(60^\circ + A) - \tan(60^\circ - A) \quad [\because \tan(180^\circ - \theta) = -\tan \theta] \\
 &= \tan A + \frac{\tan 60^\circ + \tan A}{1 - \tan 60^\circ \tan A} - \frac{\tan 60^\circ - \tan A}{1 + \tan 60^\circ \tan A} = \tan A + \frac{\sqrt{3} + \tan A}{1 - \sqrt{3} \tan A} - \frac{\sqrt{3} - \tan A}{1 + \sqrt{3} \tan A} \\
 &= \tan A + \frac{\sqrt{3} + \tan A + 3 \tan A + \sqrt{3} \tan^2 A - \sqrt{3} + \tan A + 3 \tan A - \sqrt{3} \tan^2 A}{(1 - \sqrt{3} \tan A)(1 + \sqrt{3} \tan A)} \\
 &= \tan A + \frac{8 \tan A}{1 - 3 \tan^2 A} = \frac{\tan A - 3 \tan^3 A + 8 \tan A}{1 - 3 \tan^2 A} \\
 &= \frac{9 \tan A - 3 \tan^3 A}{1 - 3 \tan^2 A} = 3 \left(\frac{3 \tan A - \tan^3 A}{1 - 3 \tan^2 A} \right) = 3 \tan 3A = \text{R.H.S.}
 \end{aligned}$$

Do yourself - 7 :

(i) Prove that :

$$\begin{aligned}
 \text{(a)} \quad & \cot \theta \cot(60^\circ - \theta) \cot(60^\circ + \theta) = \cot 3\theta \quad \text{(b)} \quad \cos 5\theta = 16 \cos^5 \theta - 20 \cos^3 \theta + 5 \cos \theta \\
 \text{(c)} \quad & \sin 4\theta = 4 \sin \theta \cos^3 \theta - 4 \cos \theta \sin^3 \theta
 \end{aligned}$$

15. TRIGONOMETRIC RATIOS OF SUB MULTIPLE ANGLES :

Since the trigonometric relations are true for all values of angle θ , they will be true if instead of θ be substitute $\frac{\theta}{2}$

$$\text{(i)} \quad \sin \theta = 2 \sin \frac{\theta}{2} \cos \frac{\theta}{2} = \frac{2 \tan \frac{\theta}{2}}{1 + \tan^2 \frac{\theta}{2}}$$

$$\text{(ii)} \quad \cos \theta = \cos^2 \frac{\theta}{2} - \sin^2 \frac{\theta}{2} = 2 \cos^2 \frac{\theta}{2} - 1 = 1 - 2 \sin^2 \frac{\theta}{2} = \frac{1 - \tan^2 \frac{\theta}{2}}{1 + \tan^2 \frac{\theta}{2}}$$

$$\text{(iii)} \quad 1 + \cos \theta = 2 \cos^2 \frac{\theta}{2}$$

$$\text{(iv)} \quad 1 - \cos \theta = 2 \sin^2 \frac{\theta}{2}$$

$$\text{(v)} \quad \tan \frac{\theta}{2} = \frac{1 - \cos \theta}{\sin \theta} = \frac{\sin \theta}{1 + \cos \theta}$$

$$\text{(vi)} \quad \tan \theta = \frac{2 \tan \frac{\theta}{2}}{1 - \tan^2 \frac{\theta}{2}}$$

$$(vii) \quad \sin \frac{\theta}{2} = \pm \sqrt{\frac{1 - \cos \theta}{2}}$$

$$(viii) \quad \cos \frac{\theta}{2} = \pm \sqrt{\frac{1 + \cos \theta}{2}}$$

$$(ix) \quad \tan \frac{\theta}{2} = \pm \sqrt{\frac{1 - \cos \theta}{1 + \cos \theta}}$$

$$(x) \quad 2 \sin \frac{\theta}{2} = \pm \sqrt{1 + \sin \theta} \pm \sqrt{1 - \sin \theta}$$

$$(xi) \quad 2 \cos \frac{\theta}{2} = \pm \sqrt{1 + \sin \theta} \mp \sqrt{1 - \sin \theta}$$

$$(xii) \quad \tan \frac{\theta}{2} = \frac{\pm \sqrt{1 + \tan^2 \theta} - 1}{\tan \theta}$$

for (vii) to (xii), we decide the sign of ratio according to value of θ .

Illustration 12: $\sin 67\frac{1}{2}^\circ + \cos 67\frac{1}{2}^\circ$ is equal to

(A) $\frac{1}{2}\sqrt{4+2\sqrt{2}}$ (B) $\frac{1}{2}\sqrt{4-2\sqrt{2}}$ (C) $\frac{1}{4}(\sqrt{4+2\sqrt{2}})$ (D) $\frac{1}{4}(\sqrt{4-2\sqrt{2}})$

Solution : $\sin 67\frac{1}{2}^\circ + \cos 67\frac{1}{2}^\circ = \sqrt{1 + \sin 135^\circ} = \sqrt{1 + \frac{1}{\sqrt{2}}} \quad (\text{using } \cos A + \sin A = \sqrt{1 + \sin 2A})$
 $= \frac{1}{2}\sqrt{4+2\sqrt{2}}$ **Ans.(A)**

Do yourself - 8 :

(i) Find the value of

(a) $\sin \frac{\pi}{8}$

(b) $\cos \frac{\pi}{8}$

(c) $\tan \frac{\pi}{8}$

16. TRIGONOMETRIC RATIOS OF SOME STANDARD ANGLES :

(i) $\sin 18^\circ = \sin \frac{\pi}{10} = \frac{\sqrt{5}-1}{4} = \cos 72^\circ = \cos \frac{2\pi}{5}$

(ii) $\cos 36^\circ = \cos \frac{\pi}{5} = \frac{\sqrt{5}+1}{4} = \sin 54^\circ = \sin \frac{3\pi}{10}$

(iii) $\sin 72^\circ = \sin \frac{2\pi}{5} = \frac{\sqrt{10+2\sqrt{5}}}{4} = \cos 18^\circ = \cos \frac{\pi}{10}$

(iv) $\sin 36^\circ = \sin \frac{\pi}{5} = \frac{\sqrt{10-2\sqrt{5}}}{4} = \cos 54^\circ = \cos \frac{3\pi}{10}$

(v) $\sin 15^\circ = \sin \frac{\pi}{12} = \frac{\sqrt{3}-1}{2\sqrt{2}} = \cos 75^\circ = \cos \frac{5\pi}{12}$

(vi) $\cos 15^\circ = \cos \frac{\pi}{12} = \frac{\sqrt{3}+1}{2\sqrt{2}} = \sin 75^\circ = \sin \frac{5\pi}{12}$

$$(vii) \quad \tan 15^\circ = \tan \frac{\pi}{12} = 2 - \sqrt{3} = \frac{\sqrt{3}-1}{\sqrt{3}+1} = \cot 75^\circ = \cot \frac{5\pi}{12}$$

$$(viii) \quad \tan 75^\circ = \tan \frac{5\pi}{12} = 2 + \sqrt{3} = \frac{\sqrt{3}+1}{\sqrt{3}-1} = \cot 15^\circ = \cot \frac{\pi}{12}$$

$$(ix) \quad \tan(22.5^\circ) = \tan \frac{\pi}{8} = \sqrt{2} - 1 = \cot(67.5^\circ) = \cot \frac{3\pi}{8}$$

$$(x) \quad \tan(67.5^\circ) = \tan \frac{3\pi}{8} = \sqrt{2} + 1 = \cot(22.5^\circ) = \cot \frac{\pi}{8}$$

Illustration 13: Evaluate $\sin 78^\circ - \sin 66^\circ - \sin 42^\circ + \sin 6^\circ$.

Solution : The expression $= (\sin 78^\circ - \sin 42^\circ) - (\sin 66^\circ - \sin 6^\circ) = 2\cos(60^\circ) \sin(18^\circ) - 2\cos 36^\circ \cdot \sin 30^\circ$

$$= \sin 18^\circ - \cos 36^\circ = \left(\frac{\sqrt{5}-1}{4} \right) - \left(\frac{\sqrt{5}+1}{4} \right) = -\frac{1}{2}$$

Do yourself - 9 :

(i) Find the value of

$$(a) \quad \sin \frac{\pi}{10} + \sin \frac{13\pi}{10}$$

$$(b) \quad \cos^2 48^\circ - \sin^2 12^\circ$$

17. CONDITIONAL TRIGONOMETRIC IDENTITIES :

If $A + B + C = 180^\circ$, then

$$(i) \quad \tan A + \tan B + \tan C = \tan A \tan B \tan C$$

$$(ii) \quad \cot A \cot B + \cot B \cot C + \cot C \cot A = 1$$

$$(iii) \quad \tan \frac{A}{2} \tan \frac{B}{2} + \tan \frac{B}{2} \tan \frac{C}{2} + \tan \frac{C}{2} \tan \frac{A}{2} = 1$$

$$(iv) \quad \cot \frac{A}{2} + \cot \frac{B}{2} + \cot \frac{C}{2} = \cot \frac{A}{2} \cot \frac{B}{2} \cot \frac{C}{2}$$

$$(v) \quad \sin 2A + \sin 2B + \sin 2C = 4 \sin A \sin B \sin C$$

$$(vi) \quad \cos 2A + \cos 2B + \cos 2C = -1 - 4 \cos A \cos B \cos C$$

$$(vii) \quad \sin A + \sin B + \sin C = 4 \cos \frac{A}{2} \cos \frac{B}{2} \cos \frac{C}{2}$$

$$(viii) \quad \cos A + \cos B + \cos C = 1 + 4 \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2}$$

Illustration 14: In any triangle ABC, $\sin A - \cos B = \cos C$, then angle B is

$$(A) \pi/2$$

$$(B) \pi/3$$

$$(C) \pi/4$$

$$(D) \pi/6$$

Solution : We have, $\sin A - \cos B = \cos C$
 $\sin A = \cos B + \cos C$

$$\begin{aligned} \Rightarrow 2 \sin \frac{A}{2} \cos \frac{A}{2} &= 2 \cos \left(\frac{B+C}{2} \right) \cos \left(\frac{B-C}{2} \right) \\ \Rightarrow 2 \sin \frac{A}{2} \cos \frac{A}{2} &= 2 \cos \left(\frac{\pi-A}{2} \right) \cos \left(\frac{B-C}{2} \right) \quad \because A+B+C=\pi \\ \Rightarrow 2 \sin \frac{A}{2} \cos \frac{A}{2} &= 2 \sin \frac{A}{2} \cos \left(\frac{B-C}{2} \right) \\ \Rightarrow \cos \frac{A}{2} &= \cos \frac{B-C}{2} \quad \text{or } A=B-C \quad ; \quad \text{But } A+B+C=\pi \end{aligned}$$

Therefore $2B = \pi \Rightarrow B = \pi/2$

Ans.(A)

Illustration 15: If $A+B+C = \frac{3\pi}{2}$, then $\cos 2A + \cos 2B + \cos 2C$ is equal to-

- (A) $1 - 4\cos A \cos B \cos C$ (B) $4 \sin A \sin B \sin C$
(C) $1 + 2\cos A \cos B \cos C$ (D) $1 - 4 \sin A \sin B \sin C$

Solution :

$$\begin{aligned} \cos 2A + \cos 2B + \cos 2C &= 2 \cos (A+B) \cos (A-B) + \cos 2C \\ &= 2 \cos \left(\frac{3\pi}{2} - C \right) \cos (A-B) + \cos 2C \quad \because A+B+C = \frac{3\pi}{2} \\ &= -2 \sin C \cos (A-B) + 1 - 2 \sin^2 C = 1 - 2 \sin C [\cos (A-B) + \sin C] \\ &= 1 - 2 \sin C [\cos (A-B) + \sin \left(\frac{3\pi}{2} - (A+B) \right)] \\ &= 1 - 2 \sin C [\cos (A-B) - \cos (A+B)] = 1 - 4 \sin A \sin B \sin C \end{aligned}$$

Ans.(D)

Do yourself - 10 :

- (i) If ABCD is a cyclic quadrilateral, then find the value of $\sin A + \sin B - \sin C - \sin D$
(ii) If $A+B+C = \frac{\pi}{2}$, then find the value of $\tan A \tan B + \tan B \tan C + \tan C \tan A$

18. MAXIMUM & MINIMUM VALUES OF TRIGONOMETRIC EXPRESSIONS :

- (i) $a \cos \theta + b \sin \theta$ will always lie in the interval $[-\sqrt{a^2+b^2}, \sqrt{a^2+b^2}]$ i.e. the maximum and minimum values are $\sqrt{a^2+b^2}, -\sqrt{a^2+b^2}$ respectively.
(ii) Minimum value of $a^2 \tan^2 \theta + b^2 \cot^2 \theta = 2ab$ where $a, b > 0$
(iii) $-\sqrt{a^2+b^2+2ab \cos(\alpha-\beta)} \leq a \cos(\alpha+\theta) + b \cos(\beta+\theta) \leq \sqrt{a^2+b^2+2ab \cos(\alpha-\beta)}$ where α and β are known angles.
(iv) In case a quadratic in $\sin \theta$ & $\cos \theta$ is given then the maximum or minimum values can be obtained by making perfect square.

Illustration 16: Prove that : $-4 \leq 5 \cos \theta + 3 \cos \left(\theta + \frac{\pi}{3} \right) + 3 \leq 10$, for all values of θ .

Solution : We have, $5 \cos \theta + 3 \cos \left(\theta + \frac{\pi}{3} \right) = 5 \cos \theta + 3 \cos \theta \cos \frac{\pi}{3} - 3 \sin \theta \sin \frac{\pi}{3} = \frac{13}{2} \cos \theta - \frac{3\sqrt{3}}{2} \sin \theta$

$$\text{Since, } -\sqrt{\left(\frac{13}{2}\right)^2 + \left(-\frac{3\sqrt{3}}{2}\right)^2} \leq \frac{13}{2} \cos \theta - \frac{3\sqrt{3}}{2} \sin \theta \leq \sqrt{\left(\frac{13}{2}\right)^2 + \left(-\frac{3\sqrt{3}}{2}\right)^2}$$

$$\Rightarrow -7 \leq \frac{13}{2} \cos \theta - \frac{3\sqrt{3}}{2} \sin \theta \leq 7$$

$$\Rightarrow -7 \leq 5 \cos \theta + 3 \cos \left(\theta + \frac{\pi}{3} \right) \leq 7 \quad \text{for all } \theta.$$

$$\Rightarrow -7 + 3 \leq 5 \cos \theta + 3 \cos \left(\theta + \frac{\pi}{3} \right) + 3 \leq 7 + 3 \quad \text{for all } \theta.$$

$$\Rightarrow -4 \leq 5 \cos \theta + 3 \cos \left(\theta + \frac{\pi}{3} \right) + 3 \leq 10 \quad \text{for all } \theta.$$

Illustration 17: Find the maximum value of $1 + \sin \left(\frac{\pi}{4} + \theta \right) + 2 \cos \left(\frac{\pi}{4} - \theta \right)$ -
 (A) 1 (B) 2 (C) 3 (D) 4

Solution : We have $1 + \sin \left(\frac{\pi}{4} + \theta \right) + 2 \cos \left(\frac{\pi}{4} - \theta \right)$

$$= 1 + \frac{1}{\sqrt{2}} (\cos \theta + \sin \theta) + \sqrt{2} (\cos \theta + \sin \theta) = 1 + \left(\frac{1}{\sqrt{2}} + \sqrt{2} \right) (\cos \theta + \sin \theta)$$

$$= 1 + \left(\frac{1}{\sqrt{2}} + \sqrt{2} \right) \cdot \sqrt{2} \cos \left(\theta - \frac{\pi}{4} \right)$$

$$\therefore \text{maximum value} = 1 + \left(\frac{1}{\sqrt{2}} + \sqrt{2} \right) \cdot \sqrt{2} = 4 \quad \text{Ans. (D)}$$

Do yourself - 11 :

- (i) Find maximum and minimum value of $5 \cos \theta + 3 \sin \left(\theta + \frac{\pi}{6} \right)$ for all real values of θ .
- (ii) Find the minimum value of $\cos \theta + \cos 2\theta$ for all real values of θ .
- (iii) Find maximum and minimum value of $\cos^2 \theta - 6 \sin \theta \cos \theta + 3 \sin^2 \theta + 2$.

19. IMPORTANT RESULTS :

- (i) $\sin \theta \sin (60^\circ - \theta) \sin (60^\circ + \theta) = \frac{1}{4} \sin 3\theta$
- (ii) $\cos \theta \cdot \cos (60^\circ - \theta) \cos (60^\circ + \theta) = \frac{1}{4} \cos 3\theta$
- (iii) $\tan \theta \tan (60^\circ - \theta) \tan (60^\circ + \theta) = \tan 3\theta$
- (iv) $\cot \theta \cot (60^\circ - \theta) \cot (60^\circ + \theta) = \cot 3\theta$

- (v) (a) $\sin^2 \theta + \sin^2 (60^\circ + \theta) + \sin^2 (60^\circ - \theta) = \frac{3}{2}$
 (b) $\cos^2 \theta + \cos^2 (60^\circ + \theta) + \cos^2 (60^\circ - \theta) = \frac{3}{2}$
 (c) $\tan \theta + \tan(60^\circ + \theta) + \tan(120^\circ + \theta) = 3 \tan 3\theta$
- (vi) (a) If $\tan A + \tan B + \tan C = \tan A \tan B \tan C$, then $A + B + C = n\pi$, $n \in \mathbb{I}$
 (b) If $\tan A \tan B + \tan B \tan C + \tan C \tan A = 1$, then $A + B + C = (2n + 1) \frac{\pi}{2}$, $n \in \mathbb{I}$
- (vii) $\cos \theta \cos 2\theta \cos 4\theta \dots \cos (2^{n-1} \theta) = \frac{\sin(2^n \theta)}{2^n \sin \theta}$
- (viii) (a) $\cot A - \tan A = 2 \cot 2A$ (b) $\cot A + \tan A = 2 \operatorname{cosec} 2A$
- (ix) $\sin \alpha + \sin (\alpha + \beta) + \sin (\alpha + 2\beta) + \dots + \sin (\alpha + (n-1)\beta) = \frac{\sin \left\{ \alpha + \left(\frac{n-1}{2} \right) \beta \right\} \sin \left(\frac{n\beta}{2} \right)}{\sin \left(\frac{\beta}{2} \right)}$
- (x) $\cos \alpha + \cos (\alpha + \beta) + \cos (\alpha + 2\beta) + \dots + \cos (\alpha + (n-1)\beta) = \frac{\cos \left\{ \alpha + \left(\frac{n-1}{2} \right) \beta \right\} \sin \left(\frac{n\beta}{2} \right)}{\sin \left(\frac{\beta}{2} \right)}$

Do yourself - 12 :

- (i) Evaluate $\sin \frac{\pi}{n} + \sin \frac{3\pi}{n} + \sin \frac{5\pi}{n} + \dots$ to n terms

Miscellaneous Illustration :

Illustration 18: Prove that

$$\tan \alpha + 2 \tan 2\alpha + 2^2 \tan^2 \alpha + \dots + 2^{n-1} \tan 2^{n-1} \alpha + 2^n \cot 2^n \alpha = \cot \alpha$$

Solution : We know $\tan \theta = \cot \theta - 2 \cot 2\theta$ (i)

Putting $\theta = \alpha, 2\alpha, 2^2\alpha, \dots$ in (i), we get

$$\tan \alpha = (\cot \alpha - 2 \cot 2\alpha)$$

$$2 (\tan 2\alpha) = 2(\cot 2\alpha - 2 \cot 2^2\alpha)$$

$$2^2 (\tan 2^2 \alpha) = 2^2 (\cot 2^2 \alpha - 2 \cot 2^3 \alpha)$$

.....

$$2^{n-1} (\tan 2^{n-1} \alpha) = 2^{n-1} (\cot 2^{n-1} \alpha - 2 \cot 2^n \alpha)$$

Adding,

$$\tan \alpha + 2 \tan 2\alpha + 2^2 \tan^2 \alpha + \dots + 2^{n-1} \tan 2^{n-1} \alpha = \cot \alpha - 2^n \cot 2^n \alpha$$

$$\therefore \tan \alpha + 2 \tan 2\alpha + 2^2 \tan^2 \alpha + \dots + 2^{n-1} \tan 2^{n-1} \alpha + 2^n \cot 2^n \alpha = \cot \alpha$$

Illustration 19: If A, B, C and D are angles of a quadrilateral and $\sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2} \sin \frac{D}{2} = \frac{1}{4}$, prove that $A = B = C = D = \pi/2$.

Solution :

$$\left(2 \sin \frac{A}{2} \sin \frac{B}{2}\right) \left(2 \sin \frac{C}{2} \sin \frac{D}{2}\right) = 1$$

$$\Rightarrow \left\{ \cos \left(\frac{A-B}{2} \right) - \cos \left(\frac{A+B}{2} \right) \right\} \left\{ \cos \left(\frac{C-D}{2} \right) - \cos \left(\frac{C+D}{2} \right) \right\} = 1$$

Since, $A + B = 2\pi - (C + D)$, the above equation becomes,

$$\Rightarrow \left\{ \cos \left(\frac{A-B}{2} \right) - \cos \left(\frac{A+B}{2} \right) \right\} \left\{ \cos \left(\frac{C-D}{2} \right) + \cos \left(\frac{A+B}{2} \right) \right\} = 1$$

$$\Rightarrow \cos^2 \left(\frac{A+B}{2} \right) - \cos \left(\frac{A+B}{2} \right) \left\{ \cos \left(\frac{A-B}{2} \right) - \cos \left(\frac{C-D}{2} \right) \right\} + 1 - \cos \left(\frac{A-B}{2} \right) \cos \left(\frac{C-D}{2} \right) = 0$$

This is a quadratic equation in $\cos \left(\frac{A+B}{2} \right)$ which has real roots.

$$\Rightarrow \left\{ \cos \left(\frac{A-B}{2} \right) - \cos \left(\frac{C-D}{2} \right) \right\}^2 - 4 \left\{ 1 - \cos \left(\frac{A-B}{2} \right) \cos \left(\frac{C-D}{2} \right) \right\} \geq 0$$

$$\left(\cos \frac{A-B}{2} + \cos \frac{C-D}{2} \right)^2 \geq 4$$

$$\Rightarrow \cos \frac{A-B}{2} + \cos \frac{C-D}{2} \geq 2, \text{ Now both } \cos \frac{A-B}{2} \text{ and } \cos \frac{C-D}{2} \leq 1$$

$$\Rightarrow \cos \frac{A-B}{2} = 1 \text{ \& } \cos \frac{C-D}{2} = 1$$

$$\Rightarrow \frac{A-B}{2} = 0 = \frac{C-D}{2}$$

$$\Rightarrow A = B, C = D.$$

Similarly $A = C, B = D \Rightarrow A = B = C = D = \pi/2$

ANSWERS FOR DO YOURSELF

- | | | |
|--|---|--------|
| 1: (i) 10π cm | 2: (i) $\frac{3}{5}, \frac{4}{5}, \frac{5}{3}$ | (ii) 2 |
| 3: (i) 8 | 4: (i) (a) $\frac{187}{205}$ (b) $\frac{-133}{205}$ (c) $\frac{-84}{205}$ (d) $\frac{156}{205}$ | |
| 5: (i) $\frac{1}{\sqrt{3}}$ | 8: (i) (a) $\sqrt{\frac{\sqrt{2}-1}{2\sqrt{2}}}$ (b) $\sqrt{\frac{\sqrt{2}+1}{2\sqrt{2}}}$ (c) $\sqrt{2}-1$ | |
| 9: (i) (a) $-\frac{1}{2}$ (b) $\frac{\sqrt{5}+1}{8}$ | 10: (i) 0 (ii) 1 | |
| 11: (i) 7 & -7 (ii) $-\frac{9}{8}$ | (iii) $4+\sqrt{10}$ & $4-\sqrt{10}$ 12: (i) 0 | |

EXERCISE (O-1)

1. If $\sin x + \sin^2 x = 1$, then the value of $\cos^2 x + \cos^4 x$ is -
 (A) 0 (B) 2 (C) 1 (D) 3
2. $2(\sin^6 \theta + \cos^6 \theta) - 3(\sin^4 \theta + \cos^4 \theta) + 1$ is equal to -
 (A) 2 (B) 0 (C) 4 (D) 6
3. If $\tan A = -\frac{1}{2}$ and $\tan B = -\frac{1}{3}$, (where $A, B > 0$), then $A + B$ can be
 (A) $\frac{\pi}{4}$ (B) $\frac{3\pi}{4}$ (C) $\frac{5\pi}{4}$ (D) $\frac{7\pi}{4}$
4. $\cos^2 48^\circ - \sin^2 12^\circ$ is equal to -
 (A) $\frac{\sqrt{5}-1}{4}$ (B) $\frac{\sqrt{5}+1}{8}$ (C) $\frac{\sqrt{3}-1}{4}$ (D) $\frac{\sqrt{3}+1}{2\sqrt{2}}$
5. The expression $\frac{\sin 8\theta \cos \theta - \sin 6\theta \cos 3\theta}{\cos 2\theta \cos \theta - \sin 3\theta \sin 4\theta}$ is equals -
 (A) $\tan \theta$ (B) $\tan 2\theta$ (C) $\sin 2\theta$ (D) $\cos 2\theta$
6. If $3 \sin \alpha = 5 \sin \beta$, then $\frac{\tan \frac{\alpha+\beta}{2}}{\tan \frac{\alpha-\beta}{2}} =$
 (A) 1 (B) 2 (C) 3 (D) 4
7. $\frac{\sin(A-C) + 2\sin A + \sin(A+C)}{\sin(B-C) + 2\sin B + \sin(B+C)}$ is equal to -
 (A) $\tan A$ (B) $\frac{\sin A}{\sin B}$ (C) $\frac{\cos A}{\cos B}$ (D) $\frac{\sin C}{\cos B}$
8. $\frac{1 + \sin 2\theta + \cos 2\theta}{1 + \sin 2\theta - \cos 2\theta} =$
 (A) $\frac{1}{2} \tan \theta$ (B) $\frac{1}{2} \cot \theta$ (C) $\tan \theta$ (D) $\cot \theta$
9. If $A = \tan 6^\circ \tan 42^\circ$ and $B = \cot 66^\circ \cot 78^\circ$, then -
 (A) $A = 2B$ (B) $A = 1/3B$ (C) $A = B$ (D) $3A = 2B$
10. If $x = y \cos \frac{2\pi}{3} = z \cos \frac{4\pi}{3}$, then $xy + yz + zx =$
 (A) -1 (B) 0 (C) 1 (D) 2
11. If $\tan \alpha = (1+2^{-x})^{-1}$, $\tan \beta = (1+2^{x+1})^{-1}$, then $\alpha + \beta =$
 (A) $\pi/6$ (B) $\pi/4$ (C) $\pi/3$ (D) $\pi/2$
12. If $\tan A + \tan B + \tan C = \tan A \cdot \tan B \cdot \tan C$, then -
 (A) A,B,C must be angles of a triangle
 (B) the sum of any two of A,B,C is equal to the third
 (C) $A+B+C$ must be n integral multiple of π
 (D) None of these

13. The value of $\sin 10^\circ + \sin 20^\circ + \sin 30^\circ + \dots + \sin 360^\circ$ is equal to -
 (A) 0 (B) 1 (C) $\sqrt{3}$ (D) 2
14. The number of real solutions of the equation $\sin(e^x) = 2^x + 2^{-x}$ is -
 (A) 1 (B) 0 (C) 2 (D) Infinite
15. If $f(x) = \frac{\sin 3x}{\sin x}$, $x \neq n\pi$, then the range of values of $f(x)$ for real values of x is -
 (A) $[-1, 3]$ (B) $(-\infty, -1]$ (C) $(3, +\infty)$ (D) $[-1, 3]$
16. If $\cos x + \cos y + \cos \alpha = 0$ and $\sin x + \sin y + \sin \alpha = 0$, then $\cot\left(\frac{x+y}{2}\right) =$
 (A) $\sin \alpha$ (B) $\cos \alpha$ (C) $\cot \alpha$ (D) $2 \sin \alpha$
17. The value of $\sin \frac{\pi}{14} \sin \frac{3\pi}{14} \sin \frac{5\pi}{14}$ is :-
 (A) $\frac{1}{16}$ (B) $\frac{1}{8}$ (C) $\frac{1}{2}$ (D) 1
18. Maximum and minimum value of $2\sin^2\theta - 3\sin\theta + 2$ is -
 (A) $\frac{1}{4}, -\frac{7}{4}$ (B) $\frac{1}{4}, \frac{21}{4}$ (C) $\frac{21}{4}, -\frac{3}{4}$ (D) $7, \frac{7}{8}$
19. For $\theta \in (0, \pi/2)$, the maximum value of $\sin\left(\theta + \frac{\pi}{6}\right) + \cos\left(\theta + \frac{\pi}{6}\right)$ is attained at $\theta =$
 (A) $\frac{\pi}{12}$ (B) $\frac{\pi}{6}$ (C) $\frac{\pi}{3}$ (D) $\frac{\pi}{4}$
20. Minimum value of the expression $\cos^2\theta - (6 \sin \theta \cos \theta) + 3 \sin^2 \theta + 2$, is -
 (A) $4 + \sqrt{10}$ (B) $4 - \sqrt{10}$ (C) 0 (D) 4

EXERCISE (O-2)

1. If $x + y = 3 - \cos 4\theta$ and $x - y = 4 \sin 2\theta$ then
 (A) $x^4 + y^4 = 9$ (B) $\sqrt{x} + \sqrt{y} = 16$ (C) $x^3 + y^3 = 2(x^2 + y^2)$ (D) $\sqrt{x} + \sqrt{y} = 2$
2. If $\tan B = \frac{n \sin A \cos A}{1 - n \cos^2 A}$ then $\tan(A + B)$ equals
 (A) $\frac{\sin A}{(1-n)\cos A}$ (B) $\frac{(n-1)\cos A}{\sin A}$ (C) $\frac{\sin A}{(n-1)\cos A}$ (D) $\frac{\sin A}{(n+1)\cos A}$
3. If $A = \sin \frac{2\pi}{7} + \sin \frac{4\pi}{7} + \sin \frac{8\pi}{7}$ and $B = \cos \frac{2\pi}{7} + \cos \frac{4\pi}{7} + \cos \frac{8\pi}{7}$ then $\sqrt{A^2 + B^2}$ is equal to
 (A) 1 (B) $\sqrt{2}$ (C) 2 (D) $\sqrt{3}$
4. If $\cos \alpha = \frac{2 \cos \beta - 1}{2 - \cos \beta}$ then $\tan \frac{\alpha}{2} \cdot \cot \frac{\beta}{2}$ has the value equal to $\{ \text{where } \alpha, \beta \in (0, \pi) \}$
 (A) 2 (B) $\sqrt{2}$ (C) 3 (D) $\sqrt{3}$

5. If $\frac{\cos 3x}{\cos x} = \frac{1}{3}$ for some angle x , $0 \leq x \leq \frac{\pi}{2}$, then the value of $\frac{\sin 3x}{\sin x}$ for some x , is

(A) $\frac{7}{3}$ (B) $\frac{5}{3}$ (C) 1 (D) $\frac{2}{3}$

6. If $\frac{5\pi}{2} < x < 3\pi$, then the value of the expression $\frac{\sqrt{1-\sin x} + \sqrt{1+\sin x}}{\sqrt{1-\sin x} - \sqrt{1+\sin x}}$ is

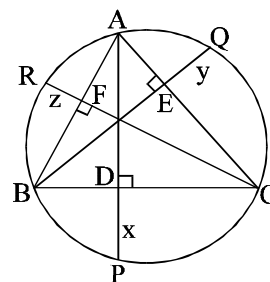
(A) $-\cot \frac{x}{2}$ (B) $\cot \frac{x}{2}$ (C) $\tan \frac{x}{2}$ (D) $-\tan \frac{x}{2}$

7. As shown in the figure AD is the altitude on BC and AD produced meets the circumcircle of $\triangle ABC$ at P where $DP = x$. Similarly $EQ = y$ and $FR = z$. If a ,

b , c respectively denotes the sides BC, CA and AB then $\frac{a}{2x} + \frac{b}{2y} + \frac{c}{2z}$

has the value equal to

(A) $\tan A + \tan B + \tan C$ (B) $\cot A + \cot B + \cot C$
(C) $\cos A + \cos B + \cos C$ (D) $\operatorname{cosec} A + \operatorname{cosec} B + \operatorname{cosec} C$



8. The exact value of $\frac{96 \sin 80^\circ \sin 65^\circ \sin 35^\circ}{\sin 20^\circ + \sin 50^\circ + \sin 110^\circ}$ is equal to

(A) 12 (B) 24 (C) -12 (D) 48

9. The value of $\cot x + \cot (60^\circ + x) + \cot (120^\circ + x)$ is equal to :

(A) $\cot 3x$ (B) $\tan 3x$ (C) $3 \tan 3x$ (D) $\frac{3 - 9 \tan^2 x}{3 \tan x - \tan^3 x}$

10. The value of $\operatorname{cosec} \frac{\pi}{18} - \sqrt{3} \sec \frac{\pi}{18}$ is a

(A) surd (B) rational which is not integral
(C) negative integer (D) natural number

11. If $\tan x + \tan y = 25$ and $\cot x + \cot y = 30$, then the value of $\tan(x + y)$ is

(A) 150 (B) 200 (C) 250 (D) 100

12. If $\cos(\alpha + \beta) + \sin(\alpha - \beta) = 0$ and $2010 \tan \beta + 1 = 0$, then $\tan \alpha$ is equal to

(A) 1 (B) -1 (C) 2010 (D) $\frac{1}{2010}$

13. In a right angled triangle the hypotenuse is $2\sqrt{2}$ times the perpendicular drawn from the opposite vertex. Then the other acute angles of the triangle are

(A) $\frac{\pi}{3}$ and $\frac{\pi}{6}$ (B) $\frac{\pi}{8}$ and $\frac{3\pi}{8}$ (C) $\frac{\pi}{4}$ and $\frac{\pi}{4}$ (D) $\frac{\pi}{5}$ and $\frac{3\pi}{10}$

14. The value of $\cot 7\frac{1}{2}^\circ + \tan 67\frac{1}{2}^\circ - \cot 67\frac{1}{2}^\circ - \tan 7\frac{1}{2}^\circ$ is :

(A) a rational number (B) irrational number
(C) $2(3 + 2\sqrt{3})$ (D) $2(3 - \sqrt{3})$

15. If m and n are positive integers satisfying

$$1 + \cos 2\theta + \cos 4\theta + \cos 6\theta + \cos 8\theta + \cos 10\theta = \frac{\cos m\theta \cdot \sin n\theta}{\sin \theta} \text{ then } (m+n) \text{ is equal to}$$

- (A) 9 (B) 10 (C) 11 (D) 12

Paragraph for Question Nos. 16 to 18

Consider the polynomial $P(x) = (x - \cos 36^\circ)(x - \cos 84^\circ)(x - \cos 156^\circ)$

16. The coefficient of x^2 is

- (A) 0 (B) 1 (C) $-\frac{1}{2}$ (D) $\frac{\sqrt{5}-1}{2}$

17. The coefficient of x is

- (A) $\frac{3}{2}$ (B) $-\frac{3}{2}$ (C) $-\frac{3}{4}$ (D) zero

18. The absolute term in $P(x)$ has the value equal to

- (A) $\frac{\sqrt{5}-1}{4}$ (B) $\frac{\sqrt{5}-1}{16}$ (C) $\frac{\sqrt{5}+1}{16}$ (D) $\frac{1}{16}$

Multiple Objective Type :

19. Let $y = \frac{\cos x + \cos 2x + \cos 3x + \cos 4x + \cos 5x + \cos 6x + \cos 7x}{\sin x + \sin 2x + \sin 3x + \sin 4x + \sin 5x + \sin 6x + \sin 7x}$, then which of the following hold good?

- (A) The value of y when $x = \pi/8$ is not defined. (B) The value of y when $x = \pi/16$ is 1.
(C) The value of y when $x = \pi/32$ is $\sqrt{2}-1$. (D) The value of y when $x = \pi/48$ is $2+\sqrt{3}$.

20. Two parallel chords are drawn on the same side of the centre of a circle of radius R . It is found that they subtend an angle of θ and 2θ at the centre of the circle. The perpendicular distance between the chords is

- (A) $2R \sin \frac{3\theta}{2} \sin \frac{\theta}{2}$ (B) $\left(1 - \cos \frac{\theta}{2}\right) \left(1 + 2 \cos \frac{\theta}{2}\right) R$
(C) $\left(1 + \cos \frac{\theta}{2}\right) \left(1 - 2 \cos \frac{\theta}{2}\right) R$ (D) $2R \sin \frac{3\theta}{4} \sin \frac{\theta}{4}$

EXERCISE (S-1)

- Prove that : $\cos^2 \alpha + \cos^2 (\alpha + \beta) - 2 \cos \alpha \cos \beta \cos (\alpha + \beta) = \sin^2 \beta$
- Prove that : $\cos 2\alpha = 2 \sin^2 \beta + 4 \cos (\alpha + \beta) \sin \alpha \sin \beta + \cos 2(\alpha + \beta)$
- Prove that : $\tan \alpha + 2 \tan 2\alpha + 4 \tan 4\alpha + 8 \cot 8\alpha = \cot \alpha$.
- Prove that : (a) $\tan 20^\circ \cdot \tan 40^\circ \cdot \tan 60^\circ \cdot \tan 80^\circ = 3$
(b) $\tan 9^\circ - \tan 27^\circ - \tan 63^\circ + \tan 81^\circ = 4$. (c) $\sin^4 \frac{\pi}{16} + \sin^4 \frac{3\pi}{16} + \sin^4 \frac{5\pi}{16} + \sin^4 \frac{7\pi}{16} = \frac{3}{2}$
- If $X = \sin\left(\theta + \frac{7\pi}{12}\right) + \sin\left(\theta - \frac{\pi}{12}\right) + \sin\left(\theta + \frac{3\pi}{12}\right)$, $Y = \cos\left(\theta + \frac{7\pi}{12}\right) + \cos\left(\theta - \frac{\pi}{12}\right) + \cos\left(\theta + \frac{3\pi}{12}\right)$, then prove that $\frac{X}{Y} - \frac{Y}{X} = 2 \tan 2\theta$.

6. If $m \tan(\theta - 30^\circ) = n \tan(\theta + 120^\circ)$, show that $\cos 2\theta = \frac{m+n}{2(m-n)}$.
7. If $\cos(\alpha + \beta) = \frac{4}{5}$; $\sin(\alpha - \beta) = \frac{5}{13}$ & α, β lie between 0 & $\frac{\pi}{4}$, then find the value of $\tan 2\alpha$.
8. If the value of the expression $\sin 25^\circ \cdot \sin 35^\circ \cdot \sin 85^\circ$ can be expressed as $\frac{\sqrt{a} + \sqrt{b}}{c}$, where $a, b, c \in \mathbb{N}$ and are in their lowest form, find the value of $(a + b + c)$.
9. Prove that $(4 \cos^2 29^\circ - 3)(4 \cos^2 27^\circ - 3) = \tan 9^\circ$.
10. If $A + B + C = \pi$, prove that $\sum \left(\frac{\tan A}{\tan B \cdot \tan C} \right) = \sum (\tan A) - 2 \sum (\cot A)$.
11. If $\alpha + \beta = \gamma$, prove that $\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1 + 2 \cos \alpha \cos \beta \cos \gamma$.
12. Let $P(k) = \left(1 + \cos \frac{\pi}{4k}\right) \left(1 + \cos \frac{(2k-1)\pi}{4k}\right) \left(1 + \cos \frac{(2k+1)\pi}{4k}\right) \left(1 + \cos \frac{(4k-1)\pi}{4k}\right)$, then find the value of (a) $P(5)$ and (b) $P(6)$.
13. Calculate without using trigonometric tables :
- (a) $4 \cos 20^\circ - \sqrt{3} \cot 20^\circ$ (b) $\frac{2 \cos 40^\circ - \cos 20^\circ}{\sin 20^\circ}$
- (c) $\cos^6 \frac{\pi}{16} + \cos^6 \frac{3\pi}{16} + \cos^6 \frac{5\pi}{16} + \cos^6 \frac{7\pi}{16}$ (d) $\tan 10^\circ - \tan 50^\circ + \tan 70^\circ$
14. Given that $(1 + \tan 1^\circ)(1 + \tan 2^\circ) \dots (1 + \tan 45^\circ) = 2^n$, find n .
15. In a right angled triangle, acute angles A and B satisfy
 $\tan A + \tan B + \tan^2 A + \tan^2 B + \tan^3 A + \tan^3 B = 70$
 find the angle A and B in radians.

EXERCISE (S-2)

1. (a) If $y = 10 \cos^2 x - 6 \sin x \cos x + 2 \sin^2 x$, then find the greatest & least value of y .
 (b) If $y = 1 + 2 \sin x + 3 \cos^2 x$, find the maximum & minimum values of $y \forall x \in \mathbb{R}$.
 (c) If $y = 9 \sec^2 x + 16 \operatorname{cosec}^2 x$, find the minimum value of y for all permissible value of x .
 (d) If $a \leq 3 \cos \left(\theta + \frac{\pi}{3} \right) + 5 \cos \theta + 3 \leq b$, find a and b , where a is the minimum value & b is the maximum value.
2. Let $k = 1^\circ$, then prove that $\sum_{n=0}^{88} \frac{1}{\cos nk \cdot \cos(n+1)k} = \frac{\cos k}{\sin^2 k}$
3. If $A + B + C = \pi$; prove that $\tan^2 \frac{A}{2} + \tan^2 \frac{B}{2} + \tan^2 \frac{C}{2} \geq 1$.
4. (a) If $4 \sin x \cdot \cos y + 2 \sin x + 2 \cos y + 1 = 0$ where $x, y \in [0, 2\pi]$ find the largest possible value of the sum $(x + y)$.
 (b) If M and m denotes maximum and minimum value of $\sqrt{49 \cos^2 \theta + \sin^2 \theta} + \sqrt{49 \sin^2 \theta + \cos^2 \theta}$ then find the value of $(M + m)$.

5. Determine the smallest positive value of x (in degrees) for which $\tan(x + 100^\circ) = \tan(x + 50^\circ) \tan x \tan(x - 50^\circ)$.
6. Find the positive integers p, q, r, s satisfying $\tan \frac{\pi}{24} = (\sqrt{p} - \sqrt{q})(\sqrt{r} - s)$.
7. If the product $(\sin 1^\circ)(\sin 3^\circ)(\sin 5^\circ)(\sin 7^\circ)\dots\dots(\sin 89^\circ) = \frac{1}{2^n}$, then find the value of n .
8. If $f(\theta) = \sum_{n=1}^6 \operatorname{cosec}\left(\theta + \frac{(n-1)\pi}{4}\right) \operatorname{cosec}\left(\theta + \frac{n\pi}{4}\right)$, where $0 < \theta < \frac{\pi}{2}$, then find the minimum value of $f(\theta)$.
9. Let $x_1 = \prod_{r=1}^5 \cos \frac{r\pi}{11}$ and $x_2 = \sum_{r=1}^5 \cos \frac{r\pi}{11}$, then show that $x_1 \cdot x_2 = \frac{1}{64} \left(\operatorname{cosec} \frac{\pi}{22} - 1 \right)$, where Π denotes the continued product.
10. If x and y are real number such that $x^2 + 2xy - y^2 = 6$, find the minimum value of $(x^2 + y^2)^2$.
11. Find the exact value of $\tan^2 \frac{\pi}{16} + \tan^2 \frac{3\pi}{16} + \tan^2 \frac{5\pi}{16} + \tan^2 \frac{7\pi}{16}$.
12. If ' θ ' is eliminated from the equations $\cos \theta - \sin \theta = b$ and $\cos 3\theta + \sin 3\theta = a$, find the eliminant.
13. Given that $3 \sin x + 4 \cos x = 5$ where $x \in (0, \pi/2)$. Find the value of $2 \sin x + \cos x + 4 \tan x$.

EXERCISE (JM)

1. If $0 < x < \pi$, and $\cos x + \sin x = \frac{1}{2}$, then $\tan x$ is- [AIEEE-2006]
 (1) $(4 - \sqrt{7})/3$ (2) $-(4 + \sqrt{7})/3$ (3) $(1 + \sqrt{7})/4$ (4) $(1 - \sqrt{7})/4$
2. Let $\cos(\alpha + \beta) = \frac{4}{5}$ and let $\sin(\alpha - \beta) = \frac{5}{13}$, where $0 \leq \alpha, \beta \leq \frac{\pi}{4}$. Then $\tan 2\alpha =$ [AIEEE-2010]
 (1) $\frac{25}{16}$ (2) $\frac{56}{33}$ (3) $\frac{19}{12}$ (4) $\frac{20}{7}$
3. If $A = \sin^2 x + \cos^4 x$, then for all real x :- [AIEEE-2011]
 (1) $1 \leq A \leq 2$ (2) $\frac{3}{4} \leq A \leq \frac{13}{16}$ (3) $\frac{3}{4} \leq A \leq 1$ (4) $\frac{13}{16} \leq A \leq 1$
4. In a ΔPQR , if $3 \sin P + 4 \cos Q = 6$ and $4 \sin Q + 3 \cos P = 1$, then the angle R is equal to : [AIEEE-2012]
 (1) $\frac{3\pi}{4}$ (2) $\frac{5\pi}{6}$ (3) $\frac{\pi}{6}$ (4) $\frac{\pi}{4}$
5. The expression $\frac{\tan A}{1 - \cot A} + \frac{\cot A}{1 - \tan A}$ can be written as [JEE-MAIN 2013]
 (1) $\sin A \cos A + 1$ (2) $\sec A \operatorname{cosec} A + 1$ (3) $\tan A + \cot A$ (4) $\sec A + \operatorname{cosec} A$

6. ABCD is a trapezium such that AB and CD are parallel and $BC \perp CD$. If $\angle ADB = \theta$, $BC = p$ and $CD = q$, then AB is equal to [JEE-MAIN 2013]

(1) $\frac{(p^2 + q^2)\sin \theta}{p \cos \theta + q \sin \theta}$ (2) $\frac{p^2 + q^2 \cos \theta}{p \cos \theta + q \sin \theta}$ (3) $\frac{p^2 + q^2}{p^2 \cos \theta + q^2 \sin \theta}$ (4) $\frac{(p^2 + q^2)\sin \theta}{(p \cos \theta + q \sin \theta)^2}$

7. Let $f_k(x) = \frac{1}{k}(\sin^k x + \cos^k x)$ where $x \in \mathbb{R}$ and $k \geq 1$. Then $f_4(x) - f_6(x)$ equals : [JEE-MAIN 2014]

(1) $\frac{1}{6}$ (2) $\frac{1}{3}$ (3) $\frac{1}{4}$ (4) $\frac{1}{12}$

8. If $5(\tan^2 x - \cos^2 x) = 2\cos 2x + 9$, then the value of $\cos 4x$ is :- [JEE-MAIN 2017]

(1) $-\frac{7}{9}$ (2) $-\frac{3}{5}$ (3) $\frac{1}{3}$ (4) $\frac{2}{9}$

EXERCISE (JA)

1. Let $\theta \in \left(0, \frac{\pi}{4}\right)$ and $t_1 = (\tan \theta)^{\tan \theta}$, $t_2 = (\tan \theta)^{\cot \theta}$, $t_3 = (\cot \theta)^{\tan \theta}$, $t_4 = (\cot \theta)^{\cot \theta}$, then -

[JEE 06,3M,-1M]

(A) $t_1 > t_2 > t_3 > t_4$ (B) $t_4 > t_3 > t_1 > t_2$ (C) $t_3 > t_1 > t_2 > t_4$ (D) $t_2 > t_3 > t_1 > t_4$

One or more than one is/are correct : [Q.5(a) & (b)]

2. (a) If $\frac{\sin^4 x}{2} + \frac{\cos^4 x}{3} = \frac{1}{5}$, then

[JEE 2009, 4 + 4]

(A) $\tan^2 x = \frac{2}{3}$

(B) $\frac{\sin^8 x}{8} + \frac{\cos^8 x}{27} = \frac{1}{125}$

(C) $\tan^2 x = \frac{1}{3}$

(D) $\frac{\sin^8 x}{8} + \frac{\cos^8 x}{27} = \frac{2}{125}$

- (b) For $0 < \theta < \frac{\pi}{2}$, the solution(s) of $\sum_{m=1}^6 \operatorname{cosec}\left(\theta + \frac{(m-1)\pi}{4}\right) \operatorname{cosec}\left(\theta + \frac{m\pi}{4}\right) = 4\sqrt{2}$ is (are) -

(A) $\frac{\pi}{4}$

(B) $\frac{\pi}{6}$

(C) $\frac{\pi}{12}$

(D) $\frac{5\pi}{12}$

3. (a) The maximum value of the expression $\frac{1}{\sin^2 \theta + 3 \sin \theta \cos \theta + 5 \cos^2 \theta}$ is

- (b) Two parallel chords of a circle of radius 2 are at a distance $\sqrt{3} + 1$ apart. If the chords subtend at the center, angles of $\frac{\pi}{k}$ and $\frac{2\pi}{k}$ where $k > 0$, then the value of $[k]$ is -

[Note : $[k]$ denotes the largest integer less than or equal to k]

[JEE 2010, 3+3]

4. Let $P = \{\theta : \sin \theta - \cos \theta = \sqrt{2} \cos \theta\}$ and $Q = \{\theta : \sin \theta + \cos \theta = \sqrt{2} \sin \theta\}$ be two sets. Then
 (A) $P \subset Q$ and $Q - P \neq \emptyset$ (B) $Q \not\subset P$
 (C) $P \not\subset Q$ (D) $P = Q$ [JEE 2011,3]
5. The value of $\sum_{k=1}^{13} \frac{1}{\sin\left(\frac{\pi}{4} + \frac{(k-1)\pi}{6}\right) \sin\left(\frac{\pi}{4} + \frac{k\pi}{6}\right)}$ is equal to [JEE(Advanced)-2016, 3(-1)]
 (A) $3 - \sqrt{3}$ (B) $2(3 - \sqrt{3})$ (C) $2(\sqrt{3} - 1)$ (D) $2(2 + \sqrt{3})$

ANSWER KEY

EXERCISE (O-1)

1. C 2. B 3. D 4. B 5. B 6. D 7. B 8. D
 9. C 10. B 11. B 12. C 13. A 14. B 15. D 16. C
 17. B 18. D 19. A 20. B

EXERCISE (O-2)

1. D 2. A 3. B 4. D 5. A 6. D 7. A 8. B
 9. D 10. D 11. A 12. B 13. B 14. B 15. C 16. A
 17. C 18. B 19. B,D 20. B,D

EXERCISE (S-1)

7. $\frac{56}{33}$ 8. 24 12. (a) $\frac{3-\sqrt{5}}{32}$; (b) $\frac{2-\sqrt{3}}{16}$ 13. (a) -1, (b) $\sqrt{3}$, (c) $\frac{5}{4}$, (d) $\sqrt{3}$
 14. $n = 23$ 15. $\frac{\pi}{12}$ and $\frac{5\pi}{12}$

EXERCISE (S-2)

1. (a) $y_{\max} = 11, y_{\min} = 1$; (b) $y_{\max} = \frac{13}{3}, y_{\min} = -1$; (c) 49; (d) $a = -4$ & $b = 10$ 4. (a) $\frac{23\pi}{6}$ (b) 18
 5. $x = 30^\circ$ 6. $p = 3, q = 2; r = 2; s = 1$ 7. $\frac{89}{2}$ 8. $2\sqrt{2}$ 10. 18
 11. 28 12. $a = 3b - 2b^3$ 13. 5

EXERCISE (JM)

1. 2 2. 2 3. 3 4. 3 5. 2 6. 1 7. 4 8. 1

EXERCISE (JA)

1. B 2. (a) A, B; (b) C,D 3. (a) 2; (b) $k = 3$ 4. D 5. C

Important Notes

NURTURE COURSE
TRIGONOMETRIC EQUATION

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IIT-JEE Syllabus :

General solution of trigonometric equations.

TRIGONOMETRIC EQUATION

1. TRIGONOMETRIC EQUATION :

An equation involving one or more trigonometrical ratios of unknown angles is called a trigonometrical equation.

2. SOLUTION OF TRIGONOMETRIC EQUATION :

A value of the unknown angle which satisfies the given equation is called a solution of the trigonometric equation.

- (a) **Principal solution :-** The solution of the trigonometric equation lying in the interval $[0, 2\pi)$.
- (b) **General solution :-** Since all the trigonometric functions are many one & periodic, hence there are infinite values of θ for which trigonometric functions have the same value. All such possible values of θ for which the given trigonometric function is satisfied is given by a general formula. Such a general formula is called general solution of trigonometric equation.
- (c) **Particular solution :-** The solution of the trigonometric equation lying in the given interval.

3. GENERAL SOLUTIONS OF SOME TRIGONOMETRIC EQUATIONS (TO BE REMEMBERED) :

- (a) If $\sin \theta = 0$, then $\theta = n\pi$, $n \in I$ (set of integers)
- (b) If $\cos \theta = 0$, then $\theta = (2n+1)\frac{\pi}{2}$, $n \in I$
- (c) If $\tan \theta = 0$, then $\theta = n\pi$, $n \in I$
- (d) If $\sin \theta = \sin \alpha$, then $\theta = n\pi + (-1)^n \alpha$ where $\alpha \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$, $n \in I$
- (e) If $\cos \theta = \cos \alpha$, then $\theta = 2n\pi \pm \alpha$, $n \in I$, $\alpha \in [0, \pi]$
- (f) If $\tan \theta = \tan \alpha$, then $\theta = n\pi + \alpha$, $n \in I$, $\alpha \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$
- (g) If $\sin \theta = 1$, then $\theta = 2n\pi + \frac{\pi}{2} = (4n+1)\frac{\pi}{2}$, $n \in I$
- (h) If $\cos \theta = 1$ then $\theta = 2n\pi$, $n \in I$
- (i) If $\sin^2 \theta = \sin^2 \alpha$ or $\cos^2 \theta = \cos^2 \alpha$ or $\tan^2 \theta = \tan^2 \alpha$, then $\theta = n\pi \pm \alpha$, $n \in I$
- (j) For $n \in I$, $\sin n\pi = 0$ and $\cos n\pi = (-1)^n$, $n \in I$
 $\sin(n\pi + \theta) = (-1)^n \sin \theta$ and $\cos(n\pi + \theta) = (-1)^n \cos \theta$
- (k) $\cos n\pi = (-1)^n$, $n \in I$

If n is an odd integer, then $\sin \frac{n\pi}{2} = (-1)^{\frac{n-1}{2}}$, $\cos \frac{n\pi}{2} = 0$,

$$\sin\left(\frac{n\pi}{2} + \theta\right) = (-1)^{\frac{n-1}{2}} \cos \theta$$

$$\cos\left(\frac{n\pi}{2} + \theta\right) = (-1)^{\frac{n+1}{2}} \sin \theta$$

Illustration 1 : Find the set of values of x for which $\frac{\tan 3x - \tan 2x}{1 + \tan 3x \cdot \tan 2x} = 1$.

Solution : We have, $\frac{\tan 3x - \tan 2x}{1 + \tan 3x \cdot \tan 2x} = 1 \Rightarrow \tan(3x - 2x) = 1 \Rightarrow \tan x = 1$
 $\Rightarrow \tan x = \tan \frac{\pi}{4} \Rightarrow x = n\pi + \frac{\pi}{4}, n \in \mathbb{I}$ {using $\tan \theta = \tan \alpha \Leftrightarrow \theta = n\pi + \alpha$ }
 But for this value of x , $\tan 2x$ is not defined.
 Hence the solution set for x is ϕ . **Ans.**

Do yourself-1 :

(i) Find general solutions of the following equations :

(a) $\sin \theta = \frac{1}{2}$	(b) $\cos\left(\frac{3\theta}{2}\right) = 0$	(c) $\tan\left(\frac{3\theta}{4}\right) = 0$
(d) $\cos^2 2\theta = 1$	(e) $\sqrt{3} \sec 2\theta = 2$	(f) $\operatorname{cosec}\left(\frac{\theta}{2}\right) = -1$

4. IMPORTANT POINTS TO BE REMEMBERED WHILE SOLVING TRIGONOMETRIC EQUATIONS :

- (a) For equations of the type $\sin \theta = k$ or $\cos \theta = k$, one must check that $|k| \leq 1$.
- (b) Avoid squaring the equations, if possible, because it may lead to extraneous solutions. Reject extra solutions if they do not satisfy the given equation.
- (c) Do not cancel the common variable factor from the two sides of the equations which are in a product because we may lose some solutions.
- (d) The answer should not contain such values of θ , which make any of the terms undefined or infinite.
 - (i) Check that denominator is not zero at any stage while solving equations.
 - (ii) If $\tan \theta$ or $\sec \theta$ is involved in the equations, θ should not be odd multiple of $\frac{\pi}{2}$.
 - (iii) If $\cot \theta$ or $\operatorname{cosec} \theta$ is involved in the equation, θ should not be multiple of π or 0 .

5. DIFFERENT STRATEGIES FOR SOLVING TRIGONOMETRIC EQUATIONS :

(a) **Solving trigonometric equations by factorisation.**

e.g. $(2 \sin x - \cos x)(1 + \cos x) = \sin^2 x$

$\therefore (2 \sin x - \cos x)(1 + \cos x) - (1 - \cos^2 x) = 0$

$\therefore (1 + \cos x)(2 \sin x - \cos x - 1 + \cos x) = 0$

$\therefore (1 + \cos x)(2 \sin x - 1) = 0$

$\Rightarrow \cos x = -1$ or $\sin x = \frac{1}{2}$

$\Rightarrow \cos x = -1 = \cos \pi \Rightarrow x = 2n\pi + \pi = (2n + 1)\pi, n \in \mathbb{I}$

or $\sin x = \frac{1}{2} = \sin \frac{\pi}{6} \Rightarrow x = k\pi + (-1)^k \frac{\pi}{6}, k \in \mathbb{I}$

Illustration 2 : If $\frac{1}{6} \sin \theta$, $\cos \theta$ and $\tan \theta$ are in G.P. then the general solution for θ is -

- (A) $2n\pi \pm \frac{\pi}{3}$ (B) $2n\pi \pm \frac{\pi}{6}$ (C) $n\pi \pm \frac{\pi}{3}$ (D) none of these

Solution : Since, $\frac{1}{6} \sin \theta$, $\cos \theta$, $\tan \theta$ are in G.P.

$$\Rightarrow \cos^2 \theta = \frac{1}{6} \sin \theta \cdot \tan \theta \Rightarrow 6\cos^3 \theta + \cos^2 \theta - 1 = 0$$

$$\therefore (2\cos \theta - 1)(3\cos^2 \theta + 2\cos \theta + 1) = 0$$

$$\Rightarrow \cos \theta = \frac{1}{2} \quad (\text{other values of } \cos \theta \text{ are imaginary})$$

$$\Rightarrow \cos \theta = \cos \frac{\pi}{3} \Rightarrow \theta = 2n\pi \pm \frac{\pi}{3}, n \in \mathbb{I}. \quad \text{Ans. (A)}$$

(b) Solving of trigonometric equation by reducing it to a quadratic equation.

e.g. $6 - 10\cos x = 3\sin^2 x$

$$\therefore 6 - 10\cos x = 3 - 3\cos^2 x \Rightarrow 3\cos^2 x - 10\cos x + 3 = 0$$

$$\Rightarrow (3\cos x - 1)(\cos x - 3) = 0 \Rightarrow \cos x = \frac{1}{3} \text{ or } \cos x = 3$$

Since $\cos x = 3$ is not possible as $-1 \leq \cos x \leq 1$

$$\therefore \cos x = \frac{1}{3} = \cos\left(\cos^{-1} \frac{1}{3}\right) \Rightarrow x = 2n\pi \pm \cos^{-1}\left(\frac{1}{3}\right), n \in \mathbb{I}$$

Illustration 3 : Solve $\sin^2 \theta - \cos \theta = \frac{1}{4}$ for θ and write the values of θ in the interval $0 \leq \theta \leq 2\pi$.

Solution : The given equation can be written as

$$1 - \cos^2 \theta - \cos \theta = \frac{1}{4} \Rightarrow \cos^2 \theta + \cos \theta - 3/4 = 0$$

$$\Rightarrow 4\cos^2 \theta + 4\cos \theta - 3 = 0 \Rightarrow (2\cos \theta - 1)(2\cos \theta + 3) = 0$$

$$\Rightarrow \cos \theta = \frac{1}{2}, -\frac{3}{2}$$

Since, $\cos \theta = -3/2$ is not possible as $-1 \leq \cos \theta \leq 1$

$$\therefore \cos \theta = \frac{1}{2} \Rightarrow \cos \theta = \cos \frac{\pi}{3} \Rightarrow \theta = 2n\pi \pm \frac{\pi}{3}, n \in \mathbb{I}$$

For the given interval, $n = 0$ and $n = 1$.

$$\Rightarrow \theta = \frac{\pi}{3}, \frac{5\pi}{3}$$

Ans.

Illustration 4 : Find the number of solutions of $\tan x + \sec x = 2\cos x$ in $[0, 2\pi]$.

Solution : Here, $\tan x + \sec x = 2\cos x \Rightarrow \sin x + 1 = 2\cos^2 x$

$$\Rightarrow 2\sin^2 x + \sin x - 1 = 0 \Rightarrow \sin x = \frac{1}{2}, -1$$

But $\sin x = -1 \Rightarrow x = \frac{3\pi}{2}$ for which $\tan x + \sec x = 2\cos x$ is not defined.

$$\text{Thus } \sin x = \frac{1}{2} \Rightarrow x = \frac{\pi}{6}, \frac{5\pi}{6}$$

$\Rightarrow$ number of solutions of $\tan x + \sec x = 2\cos x$ is 2.

Ans.

Illustration 5 : Solve the equation $5\sin^2 x - 7\sin x \cos x + 16\cos^2 x = 4$

Solution : To solve this equation we use the fundamental formula of trigonometric identities,

$$\sin^2 x + \cos^2 x = 1$$

writing the equation in the form,

$$5\sin^2 x - 7\sin x \cos x + 16\cos^2 x = 4(\sin^2 x + \cos^2 x)$$

$$\Rightarrow \sin^2 x - 7\sin x \cos x + 12\cos^2 x = 0$$

dividing by $\cos^2 x$ on both side we get,

$$\tan^2 x - 7\tan x + 12 = 0$$

Now it can be factorized as :

$$(\tan x - 3)(\tan x - 4) = 0$$

$$\Rightarrow \tan x = 3, 4$$

$$\text{i.e., } \tan x = \tan(\tan^{-1} 3) \text{ or } \tan x = \tan(\tan^{-1} 4)$$

$$\Rightarrow x = n\pi + \tan^{-1} 3 \text{ or } x = n\pi + \tan^{-1} 4, n \in \mathbb{I}.$$

Ans.

Illustration 6 : If $x \neq \frac{n\pi}{2}$, $n \in \mathbb{I}$ and $(\cos x)^{\sin^2 x - 3\sin x + 2} = 1$, then find the general solutions of x .

Solution : As $x \neq \frac{n\pi}{2} \Rightarrow \cos x \neq 0, 1, -1$

$$\text{So, } (\cos x)^{\sin^2 x - 3\sin x + 2} = 1 \Rightarrow \sin^2 x - 3\sin x + 2 = 0$$

$$\therefore (\sin x - 2)(\sin x - 1) = 0 \Rightarrow \sin x = 1, 2$$

where $\sin x = 2$ is not possible and $\sin x = 1$ which is also not possible as $x \neq \frac{n\pi}{2}$

$\therefore$ no general solution is possible.

Ans.

Illustration 7 : Solve the equation $\sin^4 x + \cos^4 x = \frac{7}{2} \sin x \cos x$.

Solution : $\sin^4 x + \cos^4 x = \frac{7}{2} \sin x \cos x \Rightarrow (\sin^2 x + \cos^2 x)^2 - 2\sin^2 x \cos^2 x = \frac{7}{2} \sin x \cos x$

$$\Rightarrow 1 - \frac{1}{2}(\sin 2x)^2 = \frac{7}{4}(\sin 2x) \Rightarrow 2\sin^2 2x + 7\sin 2x - 4 = 0$$

$$\Rightarrow (2\sin 2x - 1)(\sin 2x + 4) = 0 \Rightarrow \sin 2x = \frac{1}{2} \text{ or } \sin 2x = -4 \text{ (which is not possible)}$$

$$\Rightarrow 2x = n\pi + (-1)^n \frac{\pi}{6}, n \in I$$

$$\text{i.e., } x = \frac{n\pi}{2} + (-1)^n \frac{\pi}{12}, n \in I$$

Ans.

Do yourself-2 :

(i) Solve the following equations :

(a) $3\sin x + 2\cos^2 x = 0$

(b) $\sec^2 2\alpha = 1 - \tan 2\alpha$

(c) $7\cos^2 \theta + 3\sin^2 \theta = 4$

(d) $4\cos \theta - 3\sec \theta = \tan \theta$

(ii) Solve the equation : $2\sin^2 \theta + \sin^2 2\theta = 2$ for $\theta \in (-\pi, \pi)$.

(c) **Solving trigonometric equations by introducing an auxiliary argument.**

Consider, $a \sin \theta + b \cos \theta = c$ (i)

$$\therefore \frac{a}{\sqrt{a^2 + b^2}} \sin \theta + \frac{b}{\sqrt{a^2 + b^2}} \cos \theta = \frac{c}{\sqrt{a^2 + b^2}}$$

equation (i) has a solution only if $|c| \leq \sqrt{a^2 + b^2}$

$$\text{let } \frac{a}{\sqrt{a^2 + b^2}} = \cos \phi, \quad \frac{b}{\sqrt{a^2 + b^2}} = \sin \phi \quad \& \quad \phi = \tan^{-1} \frac{b}{a}$$

by introducing this auxiliary argument ϕ , equation (i) reduces to

$$\sin(\theta + \phi) = \frac{c}{\sqrt{a^2 + b^2}} \quad \text{Now this equation can be solved easily.}$$

Illustration 8 : Find the number of distinct solutions of $\sec x + \tan x = \sqrt{3}$, where $0 \leq x \leq 3\pi$.

Solution : Here, $\sec x + \tan x = \sqrt{3} \Rightarrow 1 + \sin x = \sqrt{3} \cos x$

$$\text{or } \sqrt{3} \cos x - \sin x = 1$$

dividing both sides by $\sqrt{a^2 + b^2}$ i.e. $\sqrt{4} = 2$, we get

$$\Rightarrow \frac{\sqrt{3}}{2} \cos x - \frac{1}{2} \sin x = \frac{1}{2}$$

$$\Rightarrow \cos \frac{\pi}{6} \cos x - \sin \frac{\pi}{6} \sin x = \frac{1}{2} \Rightarrow \cos \left(x + \frac{\pi}{6} \right) = \frac{1}{2}$$

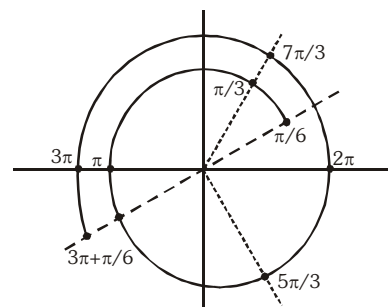
As $0 \leq x \leq 3\pi$

$$\frac{\pi}{6} \leq x + \frac{\pi}{6} \leq 3\pi + \frac{\pi}{6}$$

$$\Rightarrow x + \frac{\pi}{6} = \frac{\pi}{3}, \frac{5\pi}{3}, \frac{7\pi}{3} \Rightarrow x = \frac{\pi}{6}, \frac{3\pi}{2}, \frac{13\pi}{6}$$

But at $x = \frac{3\pi}{2}$, $\tan x$ and $\sec x$ is not defined.

$\therefore$ Total number of solutions are 2.



Ans.

Illustration 9 : Prove that the equation $k \cos x - 3 \sin x = k + 1$ possess a solution iff $k \in (-\infty, 4]$.

Solution : Here, $k \cos x - 3 \sin x = k + 1$, could be re-written as :

$$\frac{k}{\sqrt{k^2 + 9}} \cos x - \frac{3}{\sqrt{k^2 + 9}} \sin x = \frac{k+1}{\sqrt{k^2 + 9}}$$

$$\text{or } \cos(x + \phi) = \frac{k+1}{\sqrt{k^2 + 9}}, \text{ where } \tan \phi = \frac{3}{k}$$

$$\text{which possess a solution only if } -1 \leq \frac{k+1}{\sqrt{k^2 + 9}} \leq 1$$

$$\text{i.e., } \left| \frac{k+1}{\sqrt{k^2 + 9}} \right| \leq 1$$

$$\text{i.e., } (k+1)^2 \leq k^2 + 9$$

$$\text{i.e., } k^2 + 2k + 1 \leq k^2 + 9$$

$$\text{or } k \leq 4$$

$\Rightarrow$ The interval of k for which the equation $(k \cos x - 3 \sin x = k + 1)$ has a solution is $(-\infty, 4]$. **Ans.**

Do yourself-3 :

(i) Solve the following equations :

(a) $\sin x + \sqrt{2} = \cos x.$

(b) $\operatorname{cosec} \theta = 1 + \cot \theta.$

(d) **Solving trigonometric equations by transforming sum of trigonometric functions into product.**

e.g. $\cos 3x + \sin 2x - \sin 4x = 0$

$$\cos 3x - 2 \sin x \cos 3x = 0$$

$$\Rightarrow (\cos 3x)(1 - 2 \sin x) = 0$$

$$\Rightarrow \cos 3x = 0 \quad \text{or} \quad \sin x = \frac{1}{2}$$

$$\Rightarrow \cos 3x = 0 = \cos \frac{\pi}{2} \quad \text{or} \quad \sin x = \frac{1}{2} = \sin \frac{\pi}{6}$$

$$\Rightarrow 3x = 2n\pi \pm \frac{\pi}{2} \quad \text{or} \quad x = m\pi + (-1)^m \frac{\pi}{6}$$

$$\Rightarrow x = \frac{2n\pi}{3} \pm \frac{\pi}{6} \quad \text{or} \quad x = m\pi + (-1)^m \frac{\pi}{6}; (n, m \in \mathbb{I})$$

Illustration 10 : Solve : $\cos\theta + \cos3\theta + \cos5\theta + \cos7\theta = 0$

Solution :

We have $\cos\theta + \cos7\theta + \cos3\theta + \cos5\theta = 0$

$$\Rightarrow 2\cos4\theta\cos3\theta + 2\cos4\theta\cos\theta = 0 \Rightarrow \cos4\theta(\cos3\theta + \cos\theta) = 0$$

$$\Rightarrow \cos4\theta(2\cos2\theta\cos\theta) = 0$$

$$\Rightarrow \text{Either } \cos\theta = 0 \Rightarrow \theta = (2n_1 + 1)\pi/2, n_1 \in I$$

$$\text{or } \cos2\theta = 0 \Rightarrow \theta = (2n_2 + 1)\frac{\pi}{4}, n_2 \in I$$

$$\text{or } \cos4\theta = 0 \Rightarrow \theta = (2n_3 + 1)\frac{\pi}{8}, n_3 \in I$$

Ans.

(e) **Solving trigonometric equations by transforming a product into sum.**

e.g. $\sin5x \cdot \cos3x = \sin6x \cdot \cos2x$

$$\sin8x + \sin2x = \sin8x + \sin4x$$

$$\therefore 2\sin2x \cdot \cos2x - \sin2x = 0$$

$$\Rightarrow \sin2x(2\cos2x - 1) = 0$$

$$\Rightarrow \sin2x = 0 \quad \text{or} \quad \cos2x = \frac{1}{2}$$

$$\Rightarrow \sin2x = 0 = \sin0 \quad \text{or} \quad \cos2x = \frac{1}{2} = \cos\frac{\pi}{3}$$

$$\Rightarrow 2x = n\pi + (-1)^n \times 0, n \in I \quad \text{or} \quad 2x = 2m\pi \pm \frac{\pi}{3}, m \in I$$

$$\Rightarrow x = \frac{n\pi}{2}, n \in I \quad \text{or} \quad x = m\pi \pm \frac{\pi}{6}, m \in I$$

Illustration 11 : Solve : $\cos\theta \cos2\theta \cos3\theta = \frac{1}{4}$; where $0 \leq \theta \leq \pi$.

Solution :

$$\frac{1}{2}(2\cos\theta \cos3\theta) \cos2\theta = \frac{1}{4} \Rightarrow (\cos2\theta + \cos4\theta) \cos2\theta = \frac{1}{2}$$

$$\Rightarrow \frac{1}{2}[2\cos^22\theta + 2\cos4\theta \cos2\theta] = \frac{1}{2} \Rightarrow 1 + \cos4\theta + 2\cos4\theta \cos2\theta = 1$$

$$\therefore \cos4\theta(1 + 2\cos2\theta) = 0$$

$$\cos4\theta = 0 \quad \text{or} \quad (1 + 2\cos2\theta) = 0$$

Now from the first equation : $2\cos4\theta = 0 = \cos(\pi/2)$

$$\therefore 4\theta = \left(n + \frac{1}{2}\right)\pi \Rightarrow \theta = (2n + 1)\frac{\pi}{8}, n \in I$$

$$\text{for } n = 0, \theta = \frac{\pi}{8}; n = 1, \theta = \frac{3\pi}{8}; n = 2, \theta = \frac{5\pi}{8}; n = 3, \theta = \frac{7\pi}{8} \quad (\because 0 \leq \theta \leq \pi)$$

and from the second equation :

$$\cos 2\theta = -\frac{1}{2} = -\cos(\pi/3) = \cos(\pi - \pi/3) = \cos(2\pi/3)$$

$$\therefore 2\theta = 2k\pi \pm 2\pi/3 \quad \therefore \theta = k\pi \pm \pi/3, k \in \mathbb{I}$$

$$\text{again for } k=0, \theta = \frac{\pi}{3}; k=1, \theta = \frac{2\pi}{3} \quad (\because 0 \leq \theta \leq \pi)$$

$$\therefore \theta = \frac{\pi}{8}, \frac{\pi}{3}, \frac{3\pi}{8}, \frac{5\pi}{8}, \frac{2\pi}{3}, \frac{7\pi}{8}$$

Ans.

Do yourself-4 :

(i) Solve $4\sin\theta \sin 2\theta \sin 4\theta = \sin 3\theta$.

(ii) Solve for x : $\sin x + \sin 3x + \sin 5x = 0$.

(f) Solving equations by a change of variable :

(i) Equations of the form $P(\sin x \pm \cos x, \sin x \cdot \cos x) = 0$, where $P(y, z)$ is a polynomial, can be solved by the substitution :

$$\cos x \pm \sin x = t \quad \Rightarrow \quad 1 \pm 2 \sin x \cdot \cos x = t^2.$$

Illustration 12 : Solve : $\sin x + \cos x = 1 + \sin x \cdot \cos x$.

Solution :

$$\text{put } \sin x + \cos x = t$$

$$\Rightarrow \sin^2 x + \cos^2 x + 2\sin x \cdot \cos x = t^2$$

$$\Rightarrow 2\sin x \cos x = t^2 - 1 \quad (\because \sin^2 x + \cos^2 x = 1)$$

$$\Rightarrow \sin x \cdot \cos x = \left(\frac{t^2 - 1}{2} \right)$$

Substituting above result in given equation, we get :

$$t = 1 + \frac{t^2 - 1}{2}$$

$$\Rightarrow 2t = t^2 + 1 \Rightarrow t^2 - 2t + 1 = 0$$

$$\Rightarrow (t - 1)^2 = 0 \Rightarrow t = 1$$

$$\Rightarrow \sin x + \cos x = 1$$

Dividing both sides by $\sqrt{1^2 + 1^2}$ i.e. $\sqrt{2}$, we get

$$\Rightarrow \frac{1}{\sqrt{2}} \sin x + \frac{1}{\sqrt{2}} \cos x = \frac{1}{\sqrt{2}} \Rightarrow \cos x \cos \frac{\pi}{4} + \sin x \sin \frac{\pi}{4} = \frac{1}{\sqrt{2}}$$

$$\Rightarrow \cos \left(x - \frac{\pi}{4} \right) = \cos \frac{\pi}{4} \Rightarrow x - \frac{\pi}{4} = 2n\pi \pm \frac{\pi}{4}$$

$$\Rightarrow x = 2n\pi \text{ or } x = 2n\pi + \frac{\pi}{2} = (4n + 1) \frac{\pi}{2}, n \in \mathbb{I}$$

(ii) Equations of the form of $a\sin x + b\cos x + d = 0$, where a, b & d are real numbers can be solved by changing $\sin x$ & $\cos x$ into their corresponding tangent of half the angle.

Illustration 13 : Solve : $3 \cos x + 4 \sin x = 5$

Solution :

$$\Rightarrow 3 \left(\frac{1 - \tan^2 x/2}{1 + \tan^2 x/2} \right) + 4 \left(\frac{2 \tan x/2}{1 + \tan^2 x/2} \right) = 5$$

$$\Rightarrow \frac{3 - 3 \tan^2 \frac{x}{2}}{1 + \tan^2 \frac{x}{2}} + \frac{8 \tan \frac{x}{2}}{1 + \tan^2 \frac{x}{2}} = 5$$

$$\Rightarrow 3 - 3 \tan^2 \frac{x}{2} + 8 \tan \frac{x}{2} = 5 + 5 \tan^2 \frac{x}{2} \Rightarrow 8 \tan^2 \frac{x}{2} - 8 \tan \frac{x}{2} + 2 = 0$$

$$\Rightarrow 4 \tan^2 \frac{x}{2} - 4 \tan \frac{x}{2} + 1 = 0 \Rightarrow \left(2 \tan \frac{x}{2} - 1 \right)^2 = 0$$

$$\Rightarrow 2 \tan \frac{x}{2} - 1 = 0 \Rightarrow \tan \frac{x}{2} = \frac{1}{2} = \tan \left(\tan^{-1} \frac{1}{2} \right)$$

$$\Rightarrow \frac{x}{2} = n\pi + \tan^{-1} \left(\frac{1}{2} \right), n \in I \Rightarrow x = 2n\pi + 2 \tan^{-1} \frac{1}{2}, n \in I$$

(g) Solving trigonometric equations with the use of the boundness of the functions involved.

Illustration 14 : Solve the equation $(\sin x + \cos x)^{1+\sin 2x} = 2$, when $0 \leq x \leq \pi$.

Solution : We know, $-\sqrt{a^2 + b^2} \leq a \sin \theta + b \cos \theta \leq \sqrt{a^2 + b^2}$ and $-1 \leq \sin \theta \leq 1$.

$\therefore (\sin x + \cos x)$ admits the maximum value as $\sqrt{2}$

and $(1 + \sin 2x)$ admits the maximum value as 2.

Also $(\sqrt{2})^2 = 2$.

$\therefore$ the equation could hold only when, $\sin x + \cos x = \sqrt{2}$ and $1 + \sin 2x = 2$

$$\text{Now, } \sin x + \cos x = \sqrt{2} \Rightarrow \cos \left(x - \frac{\pi}{4} \right) = 1$$

$$\Rightarrow x = 2n\pi + \pi/4, n \in I \quad \dots (i)$$

$$\text{and } 1 + \sin 2x = 2 \Rightarrow \sin 2x = 1 = \sin \frac{\pi}{2}$$

$$\Rightarrow 2x = m\pi + (-1)^m \frac{\pi}{2}, m \in I \Rightarrow x = \frac{m\pi}{2} + (-1)^m \frac{\pi}{4} \quad \dots (ii)$$

The value of x in $[0, \pi]$ satisfying equations (i) and (ii) is $x = \frac{\pi}{4}$ (when $n = 0$ & $m = 0$)

Ans.

Note : $\sin x + \cos x = -\sqrt{2}$ and $1 + \sin 2x = 2$ also satisfies but as $x \geq 0$, this solution is not in domain.

Illustration 15 : Solve for x and y : $2^{\frac{1}{\cos^2 x}} \sqrt{y^2 - y + 1/2} \leq 1$

Solution : $2^{\frac{1}{\cos^2 x}} \sqrt{y^2 - y + 1/2} \leq 1$ (i)

$$2^{\frac{1}{\cos^2 x}} \sqrt{\left(y - \frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^2} \leq 1$$

Minimum value of $2^{\frac{1}{\cos^2 x}} = 2$

Minimum value of $\sqrt{\left(y - \frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^2} = \frac{1}{2}$

$\Rightarrow$ Minimum value of $2^{\frac{1}{\cos^2 x}} \sqrt{y^2 - y + \frac{1}{2}}$ is 1

$\Rightarrow$ (i) is possible when $2^{\frac{1}{\cos^2 x}} \sqrt{\left(y - \frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^2} = 1$

$\Rightarrow \cos^2 x = 1$ and $y = 1/2 \Rightarrow \cos x = \pm 1 \Rightarrow x = n\pi$, where $n \in \mathbb{I}$.

Hence $x = n\pi$, $n \in \mathbb{I}$ and $y = 1/2$.

Ans.

Illustration 16 : The number of solution(s) of $2\cos^2\left(\frac{x}{2}\right) \sin^2 x = x^2 + \frac{1}{x^2}$, $0 \leq x \leq \pi/2$, is/are -

(A) 0

(B) 1

(C) infinite

(D) none of these

Solution : Let $y = 2\cos^2\left(\frac{x}{2}\right) \sin^2 x = x^2 + \frac{1}{x^2} \Rightarrow y = (1 + \cos x) \sin^2 x$ and $y = x^2 + \frac{1}{x^2}$

when $y = (1 + \cos x) \sin^2 x = (\text{a number} < 2)(\text{a number} \leq 1) \Rightarrow y < 2$ (i)

and when $y = x^2 + \frac{1}{x^2} = \left(x - \frac{1}{x}\right)^2 + 2 \geq 2 \Rightarrow y \geq 2$ (ii)

No value of y can be obtained satisfying (i) and (ii), simultaneously

$\Rightarrow$ No real solution of the equation exists.

Ans. (A)

Note: If L.H.S. of the given trigonometric equation is always less than or equal to k and RHS is always greater than k, then no solution exists. If both the sides are equal to k for same value of θ , then solution exists and if they are equal for different values of θ , then solution does not exist.

Do yourself-5 :

(i) If $x^2 - 4x + 5 - \sin y = 0$, $y \in [0, 2\pi]$, then -

(A) $x = 1$, $y = 0$

(B) $x = 1$, $y = \pi/2$

(C) $x = 2$, $y = 0$

(D) $x = 2$, $y = \pi/2$

(ii) If $\sin x + \cos x = \sqrt{y + \frac{1}{y}}$, $y > 0$, $x \in [0, \pi]$, then find the least positive value of x satisfying the given condition.

6. TRIGONOMETRIC INEQUALITIES :

There is no general rule to solve trigonometric inequations and the same rules of algebra are valid provided the domain and range of trigonometric functions should be kept in mind.

Illustration 17 : Find the solution set of inequality $\sin x > 1/2$.

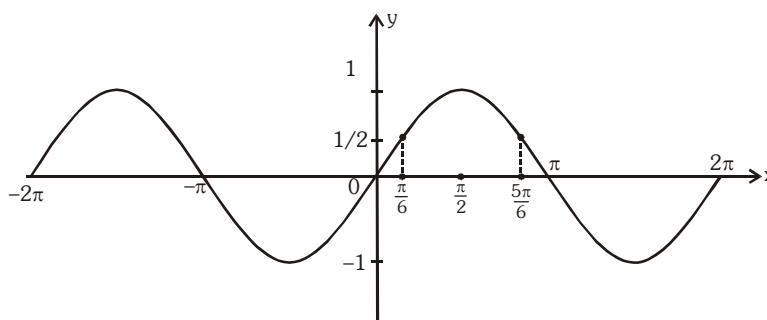
Solution : When $\sin x = \frac{1}{2}$, the two values of x between 0 and 2π are $\pi/6$ and $5\pi/6$.

From the graph of $y = \sin x$, it is obvious that between 0 and 2π ,

$$\sin x > \frac{1}{2} \text{ for } \pi/6 < x < 5\pi/6$$

Hence, $\sin x > 1/2$

$$\Rightarrow 2n\pi + \pi/6 < x < 2n\pi + 5\pi/6, n \in \mathbb{I}$$



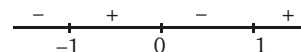
Thus, the required solution set is $\bigcup_{n \in \mathbb{I}} \left(2n\pi + \frac{\pi}{6}, 2n\pi + \frac{5\pi}{6} \right)$

Ans.

Illustration 18 : Find the values of α lying between 0 and π for which the inequality : $\tan \alpha > \tan^3 \alpha$ is valid.

Solution : We have : $\tan \alpha - \tan^3 \alpha > 0 \Rightarrow \tan \alpha (1 - \tan^2 \alpha) > 0$

$$\Rightarrow (\tan \alpha)(\tan \alpha + 1)(\tan \alpha - 1) < 0$$



So $\tan \alpha < -1, 0 < \tan \alpha < 1$

$$\therefore \text{ Given inequality holds for } \alpha \in \left(0, \frac{\pi}{4} \right) \cup \left(\frac{\pi}{2}, \frac{3\pi}{4} \right)$$

Ans.

Do yourself - 6 :

- (i) Find the solution set of the inequality : $\cos x \geq -1/2$.
- (ii) Find the values of x in the interval $[0, 2\pi]$ for which $4\sin^2 x - 8\sin x + 3 \leq 0$.

Miscellaneous Illustration :

Illustration 19 : Solve the following equation : $\tan^2\theta + \sec^2\theta + 3 = 2(\sqrt{2}\sec\theta + \tan\theta)$

Solution : We have $\tan^2\theta + \sec^2\theta + 3 = 2\sqrt{2}\sec\theta + 2\tan\theta$

$$\Rightarrow \tan^2\theta - 2\tan\theta + \sec^2\theta - 2\sqrt{2}\sec\theta + 3 = 0$$

$$\Rightarrow \tan^2\theta + 1 - 2\tan\theta + \sec^2\theta - 2\sqrt{2}\sec\theta + 2 = 0$$

$$\Rightarrow (\tan\theta - 1)^2 + (\sec\theta - \sqrt{2})^2 = 0 \quad \Rightarrow \quad \tan\theta = 1 \text{ and } \sec\theta = \sqrt{2}$$

As the periodicity of $\tan\theta$ and $\sec\theta$ are not same, we get

$$\theta = 2n\pi + \frac{\pi}{4}, n \in \mathbb{I}$$

Ans.

Illustration 20 : Find the solution set of equation $5^{(1 + \log_5 \cos x)} = 5/2$.

Solution : Taking log to base 5 on both sides in given equation :

$$(1 + \log_5 \cos x) \cdot \log_5 5 = \log_5 (5/2) \Rightarrow \log_5 5 + \log_5 \cos x = \log_5 5 - \log_5 2$$

$$\Rightarrow \log_5 \cos x = -\log_5 2 \Rightarrow \cos x = 1/2 \Rightarrow x = 2n\pi \pm \pi/3, n \in \mathbb{I}$$

Ans.

Illustration 21 : If the set of all values of x in $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ satisfying $|4\sin x + \sqrt{2}| < \sqrt{6}$ is $\left(\frac{a\pi}{24}, \frac{b\pi}{24}\right)$ then

find the value of $\left|\frac{a-b}{3}\right|$.

Solution : $|4\sin x + \sqrt{2}| < \sqrt{6}$

$$\Rightarrow -\sqrt{6} < 4\sin x + \sqrt{2} < \sqrt{6} \quad \Rightarrow \quad -\sqrt{6} - \sqrt{2} < 4\sin x < \sqrt{6} - \sqrt{2}$$

$$\Rightarrow \frac{-(\sqrt{6} + \sqrt{2})}{4} < \sin x < \frac{\sqrt{6} - \sqrt{2}}{4} \quad \Rightarrow \quad -\frac{5\pi}{12} < x < \frac{\pi}{12} \text{ for } x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$$

Comparing with $\frac{a\pi}{24} < x < \frac{b\pi}{24}$, we get, $a = -10, b = 2$

$$\therefore \left|\frac{a-b}{3}\right| = \left|\frac{-10-2}{3}\right| = 4$$

Ans.

Illustration 22 : The number of values of x in the interval $[0, 5\pi]$ satisfying the equation $3 \sin^2 x - 7 \sin x + 2 = 0$ is - [JEE 98]

- (A) 0 (B) 5 (C) 6 (D) 10

Solution :

$$3 \sin^2 x - 7 \sin x + 2 = 0$$

$$\Rightarrow (3 \sin x - 1)(\sin x - 2) = 0$$

$$\therefore \sin x \neq 2$$

$$\Rightarrow \sin x = \frac{1}{3} = \sin \alpha \text{ (say)}$$

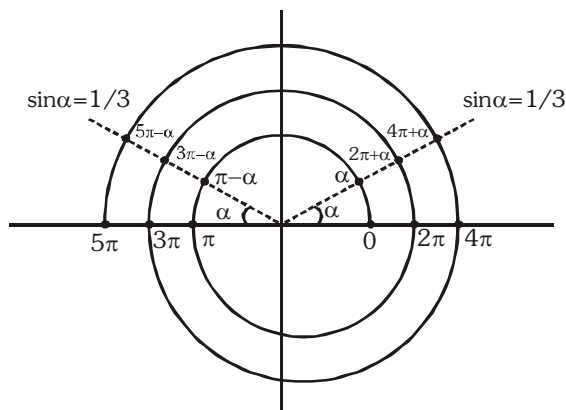
where α is the least positive value of x

$$\text{such that } \sin \alpha = \frac{1}{3}.$$

Clearly $0 < \alpha < \frac{\pi}{2}$. We get the solution,

$$x = \alpha, \pi - \alpha, 2\pi + \alpha, 3\pi - \alpha, 4\pi + \alpha \text{ and } 5\pi - \alpha.$$

Hence total six values in $[0, 5\pi]$



Ans. (C)

ANSWERS FOR DO YOURSELF

1: (i) (a) $\theta = n\pi + (-1)^n \frac{\pi}{6}, n \in I$ (b) $\theta = (2n+1)\frac{\pi}{3}, n \in I$ (c) $\theta = \frac{4n\pi}{3}, n \in I$

(d) $\theta = \frac{n\pi}{2}, n \in I$ (e) $\theta = n\pi \pm \frac{\pi}{12}, n \in I$

(f) $\theta = 2n\pi + (-1)^{n+1} \pi, n \in I$

2: (i) (a) $x = n\pi + (-1)^{n+1} \frac{\pi}{6}, n \in I$ (b) $\alpha = \frac{n\pi}{2}$ or $\alpha = \frac{k\pi}{2} + \frac{3\pi}{8}, n, k \in I$

(c) $\theta = n\pi \pm \frac{\pi}{3}, n \in I$

(d) $\theta = n\pi + (-1)^n \alpha$, where $\alpha = \sin^{-1}\left(\frac{\sqrt{17}-1}{8}\right)$ or $\sin^{-1}\left(\frac{-1-\sqrt{17}}{8}\right), n \in I$

(ii) $\theta = \left\{-\frac{\pi}{4}, -\frac{3\pi}{4}, -\frac{\pi}{2}, \frac{\pi}{4}, \frac{3\pi}{4}, \frac{\pi}{2}\right\}$

3: (i) (a) $x = 2n\pi - \frac{\pi}{4}, n \in I$ (b) $2m\pi + \frac{\pi}{2}, m \in I$

4: (i) $\theta = n\pi$ or $\theta = \frac{m\pi}{3} \pm \frac{\pi}{9}; n, m \in I$ (ii) $x = \frac{n\pi}{3}, n \in I$ and $k\pi \pm \frac{\pi}{3}, k \in I$

5: (i) D (ii) $x = \frac{\pi}{4}$

6: (i) $\bigcup_{n \in I} \left[2n\pi - \frac{2\pi}{3}, 2n\pi + \frac{2\pi}{3}\right]$ (ii) $\left[\frac{\pi}{6}, \frac{5\pi}{6}\right]$

EXERCISE # (O-1)

- The number of solutions of the equation $\sin 2x - 2\cos x + 4 \sin x = 4$ in the interval $[0, 5\pi]$ is -
 (A) 6 (B) 4 (C) 3 (D) 5
- Let $A = \{\theta : \sin(\theta) = \tan(\theta)\}$ and $B = \{\theta : \cos(\theta) = 1\}$ be two sets. Then -
 (A) $A = B$ (B) $A \subset B$ and $B - A \neq \phi$
 (C) $A \not\subset B$ (D) $B \not\subset A$
- The complete solution set of the inequality $\tan^2 x - 2\sqrt{2} \tan x + 1 \leq 0$ is -
 (A) $n\pi + \frac{\pi}{8} \leq x \leq \frac{3\pi}{8} + n\pi, n \in I$ (B) $n\pi + \frac{\pi}{4} \leq x \leq \frac{3\pi}{4} + n\pi, n \in I$
 (C) $n\pi + \frac{\pi}{16} \leq x \leq \frac{3\pi}{8} + n\pi, n \in I$ (D) $n\pi + \frac{\pi}{3} \leq x \leq \frac{2\pi}{3} + n\pi, n \in I$
- The general solution of the equation $\tan^2 \alpha + 2\sqrt{3} \tan \alpha = 1$ is given by -
 (A) $\alpha = \frac{n\pi}{2} (n \in I)$ (B) $\alpha = (2n+1) \frac{\pi}{2} (n \in I)$
 (C) $\alpha = (6n+1) \frac{\pi}{12} (n \in I)$ (D) $\alpha = \frac{n\pi}{12} (n \in I)$
- If $2 \tan^2 \theta = \sec^2 \theta$, then the general solution of θ -
 (A) $n\pi + \frac{\pi}{4} (n \in I)$ (B) $n\pi - \frac{\pi}{4} (n \in I)$ (C) $n\pi \pm \frac{\pi}{4} (n \in I)$ (D) $2n\pi \pm \frac{\pi}{4} (n \in I)$
- Number of principal solution(s) of the equation $4 \cdot 16^{\sin^2 x} = 2^{6 \sin x}$ is
 (A) 1 (B) 2 (C) 3 (D) 4
- The general solution of equation $4 \cos^2 x + 6 \sin^2 x = 5$ is -
 (A) $x = n\pi \pm \frac{\pi}{2} (n \in I)$ (B) $x = n\pi \pm \frac{\pi}{4} (n \in I)$ (C) $x = n\pi \pm \frac{3\pi}{2} (n \in I)$ (D) None of these
- If $\tan \theta + \tan 4\theta + \tan 7\theta = \tan \theta \tan 4\theta \tan 7\theta$, then $\theta =$
 (A) $\frac{n\pi}{4}$ (B) $\frac{n\pi}{7}$ (C) $\frac{n\pi}{12}$ (D) $n\pi$
 where $n \in I$
- If $\frac{1 - \cos 2\theta}{1 + \cos 2\theta} = 3$, then the general solution of θ is -
 (A) $2n\pi \pm \pi/6$ (B) $n\pi \pm \pi/6$ (C) $2n\pi \pm \pi/3$ (D) $n\pi \pm \pi/3$
 where $n \in I$
- The number of solutions of the equation $2\cos\left(\frac{x}{2}\right) = 3^x + 3^{-x}$ is -
 (A) 1 (B) 2 (C) 3 (D) None

EXERCISE # (O-2)

- Number of values of x satisfying the equation $\log_2(\sin x) + \log_{1/2}(-\cos x) = 0$ in the interval $(-\pi, \pi]$ is equal to-
 (A) 0 (B) 1 (C) 2 (D) 3
- Given $a^2 + 2a + \operatorname{cosec}^2\left(\frac{\pi}{2}(a+x)\right) = 0$ then, which of the following holds good?
 (A) $a = 1; \frac{x}{2} \in I$ (B) $a = -1; \frac{x}{2} \in I$
 (C) $a \in \mathbb{R}; x \in \phi$ (D) a, x are finite but not possible to find
- If the equation $\cot^4 x - 2 \operatorname{cosec}^2 x + a^2 = 0$ has atleast one solution then, sum of all possible integral values of 'a' is equal to
 (A) 4 (B) 3 (C) 2 (D) 0
- The set of angles between 0 and 2π satisfying the equation $4 \cos^2 \theta - 2\sqrt{2} \cos \theta - 1 = 0$ is
 (A) $\left\{\frac{\pi}{12}, \frac{5\pi}{12}, \frac{19\pi}{12}, \frac{23\pi}{12}\right\}$ (B) $\left\{\frac{\pi}{12}, \frac{7\pi}{12}, \frac{17\pi}{12}, \frac{23\pi}{12}\right\}$
 (C) $\left\{\frac{5\pi}{12}, \frac{13\pi}{12}, \frac{19\pi}{12}\right\}$ (D) $\left\{\frac{\pi}{12}, \frac{7\pi}{12}, \frac{19\pi}{12}, \frac{23\pi}{12}\right\}$
- In which one of the following intervals the inequality, $\sin x < \cos x < \tan x < \cot x$ can hold good?
 (A) $\left(0, \frac{\pi}{4}\right)$ (B) $\left(\frac{3\pi}{4}, \pi\right)$ (C) $\left(\frac{5\pi}{4}, \frac{3\pi}{2}\right)$ (D) $\left(\frac{7\pi}{4}, 2\pi\right)$
- If the equation $\sin^4 x - (k+2)\sin^2 x - (k+3) = 0$ has a solution then k must lie in the interval :
 (A) $(-4, -2)$ (B) $[-3, 2)$ (C) $(-4, -3)$ (D) $[-3, -2]$
- The smallest positive angle satisfying the equation $1 + \cos 3x - 2\cos 2x = 0$, is equal to
 (A) 15° (B) 22.5° (C) 30° (D) 45°
- Statement-1:** If $\sin \frac{3x}{2} \cos \frac{5y}{3} = k^8 - 4k^4 + 5$, where $x, y \in \mathbb{R}$ then exactly four distinct real values of k are possible.
because
Statement-2: $\sin \frac{3x}{2}$ and $\cos \frac{5y}{3}$ both are less than or equal to one and greater than or equal to -1 .
 (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
 (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
 (C) Statement-1 is true, statement-2 is false.
 (D) Statement-1 is false, statement-2 is true.
- The equation $2\cos^2\left(\frac{x}{2}\right)\sin^2 x = x^2 + x^{-2}$, $0 < x \leq \frac{\pi}{2}$ has
 (A) one real solutions (B) more than one real solutions
 (C) no real solution (D) none of the above

10. The number of solutions of the equation $\sin x = x^2 + x + 1$ is-
 (A) 0 (B) 1 (C) 2 (D) None
11. Number of integral solution(s) of the inequality $2\sin^2 x - 5\sin x + 2 > 0$ in $x \in [0, 2\pi]$, is-
 (A) 3 (B) 4 (C) 5 (D) 6
12. If $\tan \theta - \sqrt{2} \sec \theta = \sqrt{3}$, then the general solution of θ is -
 (A) $n\pi + (-1)^n \frac{\pi}{4} - \frac{\pi}{3}$ (B) $n\pi + (-1)^n \frac{\pi}{3} - \frac{\pi}{4}$ (C) $n\pi + (-1)^n \frac{\pi}{3} + \frac{\pi}{4}$ (D) $n\pi + (-1)^n \frac{\pi}{4} + \frac{\pi}{3}$
 where $n \in \mathbb{I}$

EXERCISE # (S-1)

- Solve the equation for x , $5^{\frac{1}{2}} + 5^{\frac{1}{2} + \log_5(\sin x)} = 15^{\frac{1}{2} + \log_{15} \cos x}$
- Find all the values of θ satisfying the equation; $\sin \theta + \sin 5\theta = \sin 3\theta$ such that $0 \leq \theta \leq \pi$.
- Solve the equality: $2 \sin 11x + \cos 3x + \sqrt{3} \sin 3x = 0$
- Find all value of θ , between 0 & π , which satisfy the equation; $\cos \theta \cdot \cos 2\theta \cdot \cos 3\theta = 1/4$.
- Solve for x , the equation $\sqrt{13 - 18 \tan x} = 6 \tan x - 3$, where $-2\pi < x < 2\pi$.
- Determine the smallest positive value of x which satisfy the equation, $\sqrt{1 + \sin 2x} - \sqrt{2} \cos 3x = 0$.
- Find the number of principal solution of the equation, $\sin x - \sin 3x + \sin 5x = \cos x - \cos 3x + \cos 5x$.
- Find the general solution of the trigonometric equation $3^{\left(\frac{1}{2} + \log_3(\cos x + \sin x)\right)} - 2^{\log_2(\cos x - \sin x)} = \sqrt{2}$.
- Find all values of θ between 0° & 180° satisfying the equation; $\cos 6\theta + \cos 4\theta + \cos 2\theta + 1 = 0$.
- Find the general solution of the equation, $\sin \pi x + \cos \pi x = 0$. Also find the sum of all solutions in $[0, 100]$.
- Find the range of y such that the equation, $y + \cos x = \sin x$ has a real solution. For $y = 1$, find x such that $0 < x < 2\pi$.
- Find the general values of θ for which the quadratic function $(\sin \theta) x^2 + (2 \cos \theta) x + \frac{\cos \theta + \sin \theta}{2}$ is the square of a linear function.
- Prove that the equations
 (a) $\sin x \cdot \sin 2x \cdot \sin 3x = 1$ (b) $\sin x \cdot \cos 4x \cdot \sin 5x = -1/2$
 have no solution.
- Let $f(x) = \sin^6 x + \cos^6 x + k(\sin^4 x + \cos^4 x)$ for some real number k . Determine
 (a) all real numbers k for which $f(x)$ is constant for all values of x .
 (b) all real numbers k for which there exists a real number 'c' such that $f(c) = 0$.
 (c) If $k = -0.7$, determine all solutions to the equation $f(x) = 0$.
- If the set of values of x satisfying the inequality $\tan x \cdot \tan 3x < -1$ in the interval $\left(0, \frac{\pi}{2}\right)$ is (a, b) , then
 the value of $\left(\frac{36(b-a)}{\pi}\right)$ is

EXERCISE (S-2)

1. Solve the equation : $\sin 5x = 16 \sin^5 x$.
2. Find all the solutions of $4 \cos^2 x \sin x - 2 \sin^2 x = 3 \sin x$.
3. Solve for x , $(-\pi \leq x \leq \pi)$ the equation; $2(\cos x + \cos 2x) + \sin 2x(1 + 2 \cos x) = 2 \sin x$.
4. Find the general solution of the following equation :
 $2(\sin x - \cos 2x) - \sin 2x(1 + 2 \sin x) + 2 \cos x = 0$.
5. Find the values of x , between 0 & 2π , satisfying the equation $\cos 3x + \cos 2x = \sin \frac{3x}{2} + \sin \frac{x}{2}$.
6. Solve: $\tan^2 2x + \cot^2 2x + 2 \tan 2x + 2 \cot 2x = 6$.

7. Solve the equation: $1 + 2 \operatorname{cosec} x = -\frac{\sec^2 \frac{x}{2}}{2}$.

8. Solve: $\tan^2 x \cdot \tan^2 3x \cdot \tan 4x = \tan^2 x - \tan^2 3x + \tan 4x$.

9. Find the set of values of x satisfying the equality

$$\sin\left(x - \frac{\pi}{4}\right) - \cos\left(x + \frac{3\pi}{4}\right) = 1 \text{ and the inequality } \frac{2 \cos 7x}{\cos 3 + \sin 3} > 2^{\cos 2x}.$$

10. Find the solution set of the equation, $\log_{\frac{-x^2-6x}{10}}(\sin 3x + \sin x) = \log_{\frac{-x^2-6x}{10}}(\sin 2x)$.

EXERCISE (JM)

1. Let A and B denote the statements

$$\mathbf{A} : \cos \alpha + \cos \beta + \cos \gamma = 0$$

$$\mathbf{B} : \sin \alpha + \sin \beta + \sin \gamma = 0$$

If $\cos(\beta - \gamma) + \cos(\gamma - \alpha) + \cos(\alpha - \beta) = -\frac{3}{2}$, then -

[AIEEE 2009]

- (1) Both **A** and **B** are true (2) Both **A** and **B** are false
 (3) **A** is true and **B** is false (4) **A** is false and **B** is true
2. The possible values of $\theta \in (0, \pi)$ such that $\sin(\theta) + \sin(4\theta) + \sin(7\theta) = 0$ are: [AIEEE 2011]

(1) $\frac{2\pi}{9}, \frac{\pi}{4}, \frac{4\pi}{9}, \frac{\pi}{2}, \frac{3\pi}{4}, \frac{8\pi}{9}$

(2) $\frac{\pi}{4}, \frac{5\pi}{12}, \frac{\pi}{2}, \frac{2\pi}{3}, \frac{3\pi}{4}, \frac{8\pi}{9}$

(3) $\frac{2\pi}{9}, \frac{\pi}{4}, \frac{\pi}{2}, \frac{2\pi}{3}, \frac{3\pi}{4}, \frac{35\pi}{36}$

(4) $\frac{2\pi}{9}, \frac{\pi}{4}, \frac{\pi}{2}, \frac{2\pi}{3}, \frac{3\pi}{4}, \frac{8\pi}{9}$

3. If $0 \leq x < 2\pi$, then the number of real values of x , which satisfy the equation

$$\cos x + \cos 2x + \cos 3x + \cos 4x = 0, \text{ is :-}$$

[JEE(Main) 2016]

(1) 9

(2) 3

(3) 5

(4) 7

4. If sum of all the solutions of the equation $8 \cos x \cdot \left(\cos\left(\frac{\pi}{6} + x\right) \cdot \cos\left(\frac{\pi}{6} - x\right) - \frac{1}{2} \right) = 1$ in $[0, \pi]$ is $k\pi$, then k is equal to :

[JEE(Main) 2018]

(1) $\frac{13}{9}$

(2) $\frac{8}{9}$

(3) $\frac{20}{9}$

(4) $\frac{2}{3}$

EXERCISE (JA)

1. The number of values of θ in the interval $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ such that $\theta \neq \frac{n\pi}{5}$ for $n = 0, \pm 1, \pm 2$ and $\tan \theta = \cot 5\theta$ as well as $\sin 2\theta = \cos 4\theta$, is [JEE 2010, 3]

2. The positive integer value of $n > 3$ satisfying the equation

$$\frac{1}{\sin\left(\frac{\pi}{n}\right)} = \frac{1}{\sin\left(\frac{2\pi}{n}\right)} + \frac{1}{\sin\left(\frac{3\pi}{n}\right)} \text{ is [JEE 2011, 4]}$$

3. Let $\theta, \varphi \in [0, 2\pi]$ be such that

$$2 \cos \theta (1 - \sin \varphi) = \sin^2 \theta \left(\tan \frac{\theta}{2} + \cot \frac{\theta}{2} \right) \cos \varphi - 1, \quad \tan(2\pi - \theta) > 0 \text{ and } -1 < \sin \theta < -\frac{\sqrt{3}}{2}.$$

Then φ **cannot** satisfy-

[JEE 2012, 4M]

(A) $0 < \varphi < \frac{\pi}{2}$ (B) $\frac{\pi}{2} < \varphi < \frac{4\pi}{3}$ (C) $\frac{4\pi}{3} < \varphi < \frac{3\pi}{2}$ (D) $\frac{3\pi}{2} < \varphi < 2\pi$

4. For $x \in (0, \pi)$, the equation $\sin x + 2\sin 2x - \sin 3x = 3$ has

(A) infinitely many solutions

(B) three solutions

(C) one solution

(D) no solution

[JEE(Advanced)-2014, 3(-1)]

5. The number of distinct solutions of equation $\frac{5}{4} \cos^2 2x + \cos^4 x + \sin^4 x + \cos^6 x + \sin^6 x = 2$ in the interval $[0, 2\pi]$ is [JEE 2015, 4M, -0M]

6. Let $S = \left\{ x \in (-\pi, \pi) : x \neq 0, \pm \frac{\pi}{2} \right\}$. The sum of all distinct solution of the equation

$$\sqrt{3} \sec x + \operatorname{cosec} x + 2(\tan x - \cot x) = 0 \text{ in the set } S \text{ is equal to -}$$

[JEE(Advanced)-2016, 3(-1)]

(A) $-\frac{7\pi}{9}$ (B) $-\frac{2\pi}{9}$ (C) 0 (D) $\frac{5\pi}{9}$

7. Let a, b, c be three non-zero real numbers such that the equation

$$\sqrt{3}a \cos x + 2b \sin x = c, \quad x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$$

has two distinct real roots α and β with $\alpha + \beta = \frac{\pi}{3}$. Then the value of $\frac{b}{a}$ is _____

[JEE(Advanced)-2018, 3(0)]

ANSWERS**EXERCISE (O-1)**

1. C 2. C 3. A 4. C 5. C 6. C 7. B 8. C
 9. D 10. A 11. A 12. B 13. C 14. A 15. C 16. D
 17. A 18. A 19. B 20. C

EXERCISE (O-2)

1. B 2. B 3. D 4. B 5. A 6. D 7. C 8. D
 9. C 10. A 11. C 12. D

EXERCISE (S-1)

1. $x = 2n\pi + \frac{\pi}{6}, n \in I$ 2. $0, \frac{\pi}{6}, \frac{\pi}{3}, \frac{2\pi}{3}, \frac{5\pi}{6} \text{ \& } \pi$ 3. $x = \frac{n\pi}{7} - \frac{\pi}{84}$ or $x = \frac{n\pi}{4} + \frac{7\pi}{48}, n \in I$
 4. $\frac{\pi}{8}, \frac{\pi}{3}, \frac{3\pi}{8}, \frac{5\pi}{8}, \frac{2\pi}{3}, \frac{7\pi}{8}$ 5. $\alpha - 2\pi; \alpha - \pi, \alpha, \alpha + \pi$, where $\tan \alpha = \frac{2}{3}$ 6. $x = \pi/16$
 7. 10 solutions 8. $x = 2n\pi + \frac{\pi}{12}$ 9. $30^\circ, 45^\circ, 90^\circ, 135^\circ, 150^\circ$
 10. $x = n - \frac{1}{4}, n \in I$; sum = 5025 11. $-\sqrt{2} \leq y \leq \sqrt{2}; \frac{\pi}{2}, \pi$ 12. $2n\pi + \frac{\pi}{4}$ or $(2n+1)\pi - \tan^{-1}2, n \in I$
 14. (a) $-\frac{3}{2}$; (b) $k \in \left[-1, -\frac{1}{2}\right]$; (c) $x = \frac{n\pi}{2} \pm \frac{\pi}{6}$ 15. 3

EXERCISE (S-2)

1. $x = n\pi$ or $x = n\pi \pm \frac{\pi}{6}$ 2. $n\pi; n\pi + (-1)^n \frac{\pi}{10}$ or $n\pi + (-1)^n \left(-\frac{3\pi}{10}\right)$ 3. $\frac{\pm\pi}{3}, \frac{-\pi}{2}, \pm\pi$
 4. $x = 2n\pi$ or $x = n\pi + (-1)^n \left(-\frac{\pi}{2}\right)$ or $x = n\pi + (-1)^n \frac{\pi}{6}$ 5. $\frac{\pi}{7}, \frac{5\pi}{7}, \pi, \frac{9\pi}{7}, \frac{13\pi}{7}$
 6. $x = \frac{n\pi}{4} + (-1)^n \frac{\pi}{8}$ or $\frac{n\pi}{4} + (-1)^{n+1} \frac{\pi}{24}$ 7. $x = 2n\pi - \frac{\pi}{2}$
 8. $\frac{(2n+1)\pi}{4}, k\pi$, where $n, k \in I$ 9. $x = 2n\pi + \frac{3\pi}{4}, n \in I$ 10. $x = -\frac{5\pi}{3}$

EXERCISE (JM)

1. 1 2. 1 3. 4 4. 1

EXERCISE (JA)

1. 3 2. 7 3. A, C, D 4. D 5. 8 6. C 7. 0.5

Important Notes

CONTENTS

MATRICES

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JEE (Main) Syllabus :

Matrices, algebra of matrices, types of matrices, matrices of order two and three. Adjoint and evaluation of inverse of a square matrix using determinants and elementary transformations. Test of consistency and solution of simultaneous linear equations in two or three variables using matrices.

JEE (Advanced) Syllabus :

Matrices as a rectangular array of real numbers, equality of matrices, addition, multiplication by a scalar and product of matrices, transpose of a matrix, determinant of a square matrix of order up to three, inverse of a square matrix of order up to three, properties of these matrix operations, diagonal, symmetric and skew-symmetric matrices and their properties, solutions of simultaneous linear equations in two or three variables.

MATRIX

1. INTRODUCTION :

A rectangular array of mn numbers (which may be **real or complex**) in the form of ' m ' horizontal lines (called **rows**) and ' n ' vertical lines (called **columns**), is called a matrix of order m by n , written as $m \times n$ matrix.

Such an array is enclosed by $[]$ or $()$ or $\|$. An $m \times n$ matrix is usually written as

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \dots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \dots & a_{2n} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \dots & a_{mn} \end{bmatrix}$$

In compact form, the above matrix is represented by $A = [a_{ij}]_{m \times n}$. The number $a_{11}, a_{12}, \dots$ etc are known as the elements of the matrix A , a_{ij} belongs to the i^{th} row and j^{th} column and is called the **$(i, j)^{\text{th}}$ element** of the matrix $A = [a_{ij}]$.

e.g., $A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & -1 & 9 \end{bmatrix}$ is a matrix having 2 rows and 3 columns. Its order is 2×3 and it has 6 elements :

$$a_{11} = 1, a_{12} = 2, a_{13} = 3, a_{21} = 0, a_{22} = -1, a_{23} = 9.$$

2. SPECIAL TYPE OF MATRICES :

(a) **Row Matrix (Row vector)** : $A = [a_{11}, a_{12}, \dots, a_{1n}]$ i.e. row matrix has exactly one row.

(b) **Column Matrix (Column vector)** : $A = \begin{bmatrix} a_{11} \\ a_{12} \\ \vdots \\ a_{m1} \end{bmatrix}$ i.e. column matrix has exactly one column.

(c) **Zero or Null Matrix** : ($A = O_{m \times n}$) An $m \times n$ matrix whose all entries are zero.

$$A = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \text{ is a } 3 \times 2 \text{ null matrix \& } B = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \text{ is } 3 \times 3 \text{ null matrix}$$

(d) **Horizontal Matrix** : A matrix of order $m \times n$ is a horizontal matrix if $n > m$ e.g. $\begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 5 & 1 & 1 \end{bmatrix}$

(e) **Vertical Matrix** : A matrix of order $m \times n$ is a vertical matrix if $m > n$ e.g. $\begin{bmatrix} 2 & 5 \\ 1 & 1 \\ 3 & 6 \\ 2 & 4 \end{bmatrix}$

(f) **Square Matrix** : If number of rows = number of columns $\Rightarrow$ matrix is a square matrix. If number of rows = number of columns = n then, matrix is of the **order 'n'**.

Note : The pair of elements a_{ij} & a_{ji} are called **Conjugate Elements**.

3. TRACE OF MATRIX :

The sum of the elements of a **square matrix** A lying along the principal diagonal is called the trace of A i.e. $\text{tr}(A)$. Thus, if $A = [a_{ij}]_{n \times n}$, then $\text{tr}(A) = \sum_{i=1}^n a_{ii} = a_{11} + a_{22} + \dots + a_{nn}$

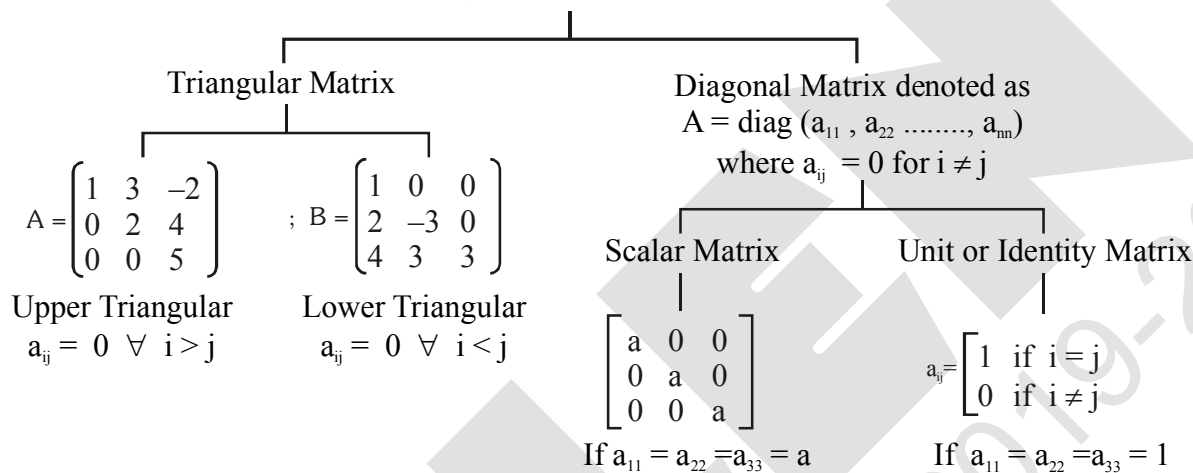
Properties of trace of a matrix :

Let $A = [a_{ij}]_{n \times n}$ and $B = [b_{ij}]_{n \times n}$ and λ be a scalar then

$$(i) \quad \text{tr}(\lambda A) = \lambda \text{tr}(A) \quad (ii) \quad \text{tr}(A + B) = \text{tr}(A) + \text{tr}(B) \quad (iii) \quad \text{tr}(AB) = \text{tr}(BA)$$

4.

SQUARE MATRICES



Note :

- (i) Minimum number of zeros in triangular matrix of order $n = n(n-1)/2$.
- (ii) Minimum number of zero in a diagonal matrix of order $n = n(n-1)$.

5. EQUALITY OF MATRICES :

Let $A = [a_{ij}]$ & $B = [b_{ij}]$ are equal if

- (a) both have the same order.
- (b) $a_{ij} = b_{ij}$ for each pair of i & j .

Illustration 1 : Find the value of x, y, z and w which satisfy the matrix equation

$$\begin{bmatrix} x+3 & 2y+x \\ z-1 & 4w-8 \end{bmatrix} = \begin{bmatrix} -x-1 & 0 \\ 3 & 2w \end{bmatrix}$$

Solution : As the given matrices are equal so their corresponding elements are equal.

$$\begin{aligned} x+3 &= -x-1 & \Rightarrow & 2x = -4 \\ \therefore x &= -2 & & \dots\dots\dots(i) \\ 2y+x &= 0 & \Rightarrow & 2y-2=0 \quad [\text{from (i)}] \\ \Rightarrow y &= 1 & & \dots\dots\dots(ii) \\ z-1 &= 3 & \Rightarrow & z=4 & \dots\dots\dots(iii) \\ 4w-8 &= 2w & \Rightarrow & 2w=8 \\ \therefore w &= 4 & & \dots\dots\dots(iv) \end{aligned}$$

Ans.

Do yourself -1 :

- (i) Find 2×3 matrix $[a_{ij}]_{2 \times 3}$, where $a_{ij} = i + 2j$
- (ii) Find the minimum number of zeroes in a triangular matrix of order 4.
- (iii) Find minimum number of zeros in a diagonal matrix of order 6.
- (iv) If $\begin{bmatrix} 2x+y & 2 & x-2y \\ a-b & 2a+b & -3 \end{bmatrix} = \begin{bmatrix} 3 & 2 & 4 \\ 4 & -1 & -3 \end{bmatrix}$, then find the values of x,y,a and b.

6. ALGEBRA OF MATRICES :

Addition : $A + B = [a_{ij} + b_{ij}]$ where A & B are of the same order.

- (a) **Addition of matrices is commutative :** i.e. $A + B = B + A$
- (b) **Matrix addition is associative :** $(A + B) + C = A + (B + C)$
- (c) **Additive inverse :** If $A + B = \mathbf{O} = B + A$, then B is called **additive inverse** of A.
- (d) **Existence of additive identity :** Let $A = [a_{ij}]$ be an $m \times n$ matrix and $\mathbf{O}$ be an $m \times n$ zero matrix, then $A + \mathbf{O} = \mathbf{O} + A = A$. In other words, $\mathbf{O}$ is the **additive identity** for matrix addition.
- (e) **Cancellation laws** hold good in case of addition of matrices. If A,B,C are matrices of the same order, then $A + B = A + C \Rightarrow B = C$ (**left cancellation law**) and $B + A = C + A \Rightarrow B = C$ (**right-cancellation law**)

Note : The zero matrix plays the same role in matrix addition as the number zero does in addition of numbers.

Illustration 2 : If $A = \begin{bmatrix} 1 & 3 \\ 3 & 2 \\ 2 & 5 \end{bmatrix}$ & $B = \begin{bmatrix} -1 & -2 \\ 0 & 5 \\ 3 & 1 \end{bmatrix}$ and $A + B - D = \mathbf{O}$ (zero matrix), then D matrix will be-

(A) $\begin{bmatrix} 0 & 2 \\ 3 & 7 \\ 6 & 5 \end{bmatrix}$ (B) $\begin{bmatrix} 0 & 2 \\ 3 & 7 \\ 5 & 6 \end{bmatrix}$ (C) $\begin{bmatrix} 0 & 1 \\ 3 & 7 \\ 5 & 6 \end{bmatrix}$ (D) $\begin{bmatrix} 0 & -2 \\ -3 & -7 \\ -5 & -6 \end{bmatrix}$

Solution :

Let $D = \begin{bmatrix} a & b \\ c & d \\ e & f \end{bmatrix}$

$$\therefore A + B - D = \begin{bmatrix} 1 & 3 \\ 3 & 2 \\ 2 & 5 \end{bmatrix} + \begin{bmatrix} -1 & -2 \\ 0 & 5 \\ 3 & 1 \end{bmatrix} - \begin{bmatrix} a & b \\ c & d \\ e & f \end{bmatrix} \Rightarrow \begin{bmatrix} 1-1-a & 3-2-b \\ 3+0-c & 2+5-d \\ 2+3-e & 5+1-f \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\begin{aligned} \Rightarrow -a &= 0 \Rightarrow a = 0, & 1-b &= 0 \Rightarrow b = 1, \\ 3-c &= 0 \Rightarrow c = 3, & 7-d &= 0 \Rightarrow d = 7, \\ 5-e &= 0 \Rightarrow e = 5, & 6-f &= 0 \Rightarrow f = 6 \end{aligned}$$

$$\therefore D = \begin{bmatrix} 0 & 1 \\ 3 & 7 \\ 5 & 6 \end{bmatrix}$$

Ans. (C)

Do yourself-2 :

(i) If $A = \begin{bmatrix} 2 & 3 & 9 \\ 8 & -2 & 5 \end{bmatrix}$ and $B = \begin{bmatrix} -5 & -7 & 2 \\ 6 & 4 & 8 \end{bmatrix}$, then find a matrix C such that $A - B + C = O$ and also find the order of the matrix C.

(ii) If $A = \begin{bmatrix} 8 & 9 \\ 7/2 & 8 \\ 1 & -1 \end{bmatrix}$, then find the additive inverse of A and show that additive inverse of additive inverse will be the matrix itself.

7. MULTIPLICATION OF A MATRIX BY A SCALAR :

$$\text{If } A = \begin{bmatrix} a & b & c \\ b & c & a \\ c & a & b \end{bmatrix}; kA = \begin{bmatrix} ka & kb & kc \\ kb & kc & ka \\ kc & ka & kb \end{bmatrix}$$

Properties of scalar multiplication :

- (a) If A and B are two matrices of the same order and 'k' be a scalar then $k(A + B) = kA + kB$.
 (b) If k_1 and k_2 are two scalars and 'A' is a matrix, then $(k_1 + k_2)A = k_1A + k_2A$.
 (c) If k_1 and k_2 are two scalars and 'A' is a matrix, then $(k_1 k_2)A = k_1(k_2A) = k_2(k_1A)$

8. MULTIPLICATION OF MATRICES (Row by Column) :

Let A be a matrix of order $m \times n$ and B be a matrix of order $p \times q$, then the matrix multiplication AB is possible if and only if $n = p$ and matrices are said to be **conformable** for multiplication.

In the product AB, A is called pre-factor and B is called post factor.

$\Rightarrow$ AB is possible if and only if number of columns in pre-factor = number of rows in post-factor.

Let $A_{m \times n} = [a_{ij}]$ and $B_{n \times p} = [b_{ij}]$, then order of AB is $m \times p$ & $(AB)_{ij} = \sum_{r=1}^n a_{ir} b_{rj}$

e.g. $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix}_{2 \times 3}$ and $B = \begin{bmatrix} b_{11} & b_{12} & b_{13} & b_{14} \\ b_{21} & b_{22} & b_{23} & b_{24} \\ b_{31} & b_{32} & b_{33} & b_{34} \end{bmatrix}_{3 \times 4}$

Then order of AB is 2×4 .

$$(AB)_{11} = a_{11}b_{11} + a_{12}b_{21} + a_{13}b_{31} = \sum_{r=1}^3 a_{1r}b_{r1}$$

$$(AB)_{23} = a_{21}b_{13} + a_{22}b_{23} + a_{23}b_{33} = \sum_{r=1}^3 a_{2r}b_{r3}$$

$$\text{In general } (ab)_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + a_{i3}b_{3j} = \sum_{r=1}^3 a_{ir}b_{rj}$$

Illustration 3 : If $[1 \times 2] \begin{bmatrix} 2 & 3 & 1 \\ 0 & 4 & 2 \\ 0 & 3 & 2 \end{bmatrix} \begin{bmatrix} x \\ 1 \\ -1 \end{bmatrix} = \mathbf{O}$, then the value of x is :-

- (A) -1 (B) 0 (C) 1 (D) 2

Solution : The LHS of the equation

$$= [2 \quad 4x + 9 \quad 2x + 5] \begin{bmatrix} x \\ 1 \\ -1 \end{bmatrix} = [2x + 4x + 9 - 2x - 5] = 4x + 4$$

$$\text{Thus } 4x + 4 = 0 \Rightarrow x = -1$$

Ans. (A)

Illustration 4 : If A, B are two matrices such that $A + B = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$, $A - B = \begin{bmatrix} 3 & 2 \\ -2 & 0 \end{bmatrix}$, then find AB.

Solution : Given $A + B = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$ (i) & $A - B = \begin{bmatrix} 3 & 2 \\ -2 & 0 \end{bmatrix}$ (ii)

Adding (i) & (ii)

$$2A = \begin{bmatrix} 4 & 4 \\ 0 & 4 \end{bmatrix} \Rightarrow A = \begin{bmatrix} 2 & 2 \\ 0 & 2 \end{bmatrix}$$

Subtracting (ii) from (i)

$$2B = \begin{bmatrix} -2 & 0 \\ 4 & 4 \end{bmatrix} \Rightarrow B = \begin{bmatrix} -1 & 0 \\ 2 & 2 \end{bmatrix}$$

$$\text{Now } AB = \begin{bmatrix} 2 & 2 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 2 & 2 \end{bmatrix} = \begin{bmatrix} 2 & 4 \\ 4 & 4 \end{bmatrix}$$

Ans.

9. PROPERTIES OF MATRIX MULTIPLICATION :

(a) **Matrix multiplication is not commutative :** i.e. $AB \neq BA$

Here both AB & BA exist and also they are of the same type but $AB \neq BA$.

Example :

$$\text{Let } A = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \text{ \& } B = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}; \text{ then } AB = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}; \quad BA = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$$

$$\Rightarrow AB \neq BA \text{ (in general)}$$

(b) $AB = \mathbf{O} \nRightarrow A = \mathbf{O} \text{ or } B = \mathbf{O}$ (in general)

$$\text{Let } A = \begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix} \text{ \& } B = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}, \text{ then } AB = \begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix} \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Note :

If A and B are two non - zero matrices such that $AB = \mathbf{O}$ then A and B are called the divisors of zero. If A and B are two matrices such that

(i) $AB = BA$ then A and B are said to commute

(ii) $AB = -BA$ then A and B are said to anticommute

(c) Matrix Multiplication Is Associative :

If A, B & C are conformable for the product AB & BC, then $(AB)C = A(BC)$

(d) Distributivity :

$$\left. \begin{aligned} A(B+C) &= AB+AC \\ (A+B)C &= AC+BC \end{aligned} \right\} \text{ Provided A, B \& C are conformable for respective products}$$

Illustration 5 : Let $A = \begin{bmatrix} 1 & -3 & 2 \\ 2 & 1 & -3 \\ 4 & -3 & -1 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 4 & 1 & 0 \\ 2 & 1 & 1 & 1 \\ 1 & -2 & 1 & 2 \end{bmatrix}$ & $C = \begin{bmatrix} 1 & 1 & -1 & -2 \\ 3 & -2 & -1 & -1 \\ 2 & -5 & -1 & 0 \end{bmatrix}$ be the matrices then, prove that in matrix multiplication cancellation law does not hold.

Solution : We have to show that $AB = AC$; though B is not equal to C.

$$\text{We have } AB = \begin{bmatrix} 1 & -3 & 2 \\ 2 & 1 & -3 \\ 4 & -3 & -1 \end{bmatrix} \begin{bmatrix} 1 & 4 & 1 & 0 \\ 2 & 1 & 1 & 1 \\ 1 & -2 & 1 & 2 \end{bmatrix} = \begin{bmatrix} -3 & -3 & 0 & 1 \\ 1 & 15 & 0 & -5 \\ -3 & 15 & 0 & -5 \end{bmatrix}_{3 \times 4}$$

$$\text{Now, } AC = \begin{bmatrix} 1 & -3 & 2 \\ 2 & 1 & -3 \\ 4 & -3 & -1 \end{bmatrix} \begin{bmatrix} 2 & 1 & -1 & -2 \\ 3 & -2 & -1 & -1 \\ 2 & -5 & -1 & 0 \end{bmatrix} = \begin{bmatrix} -3 & -3 & 0 & 1 \\ 1 & 15 & 0 & -5 \\ -3 & 15 & 0 & -5 \end{bmatrix}_{3 \times 4}$$

Here, $AB = AC$ though B is not equal to C. Thus cancellation law does not hold in general.

Do yourself - 3 :

(i) If $A = \begin{bmatrix} 2 & 9 \\ -4 & 5 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 5 \\ 6 & 8 \end{bmatrix}$ and $C = \begin{bmatrix} 9 & -7 \\ -2 & 4 \end{bmatrix}$, then show that $A(B+C) = AB+AC$.

(ii) If $A = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 4 \\ -1 & 1 \end{bmatrix}$, then prove that $(A-B)^2 \neq A^2 - 2AB + B^2$.

(iii) Find the value of x : $2 \begin{bmatrix} 3 & 1 & -2 \\ -1 & -3 & 4 \end{bmatrix} + x \begin{bmatrix} 1 & 0 & -1 \\ 3 & 4 & 5 \end{bmatrix} = \begin{bmatrix} 4 & 2 & -2 \\ -8 & -14 & -2 \end{bmatrix}$

10. POSITIVE INTEGRAL POWERS OF A SQUARE MATRIX :

For a square matrix A, $A^n = \underbrace{A \cdot A \cdot A \cdots A}_{\text{upto n times}}$, where $n \in \mathbb{N}$

Note :

(i) $A^m \cdot A^n = A^{m+n}$

(ii) $(A^m)^n = A^{mn}$, where $m, n \in \mathbb{N}$

(iii) If A and B are square matrices of same order and $AB = BA$ then

$$(A+B)^n = {}^nC_0 A^n + {}^nC_1 A^{n-1} B + {}^nC_2 A^{n-2} B^2 + \cdots + {}^nC_n B^n$$

Note that for a unit matrix I of any order, $I^m = I$ for all $m \in \mathbb{N}$.

Do yourself -4 :

(i) If $A = \begin{bmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{bmatrix}$, then prove that $A^n = \begin{bmatrix} \cos n\alpha & \sin n\alpha \\ -\sin n\alpha & \cos n\alpha \end{bmatrix}$ where n is positive integer.

(ii) If $A = \begin{bmatrix} i & 0 \\ 0 & i \end{bmatrix}$, where $i = \sqrt{-1}$ and $x \in \mathbb{N}$, then A^{4x} equals -

(A) $\begin{bmatrix} 0 & i \\ i & 0 \end{bmatrix}$

(B) $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

(C) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

(D) $\begin{bmatrix} i & 0 \\ 0 & i \end{bmatrix}$

11. SPECIAL SQUARE MATRICES :

(a) **Idempotent Matrix :** A square matrix is idempotent provided $A^2 = A$.

For idempotent matrix note the following :

(i) $A^n = A \quad \forall n \geq 2, n \in \mathbb{N}$.

(ii) determinant value of idempotent matrix is either 0 or 1

(b) **Periodic Matrix :** A square matrix which satisfies the relation $A^{k+1} = A$, for some positive integer K , is a periodic matrix. The period of the matrix is the least value of K for which this holds true.

Note that period of an idempotent matrix is 1.

(c) **Nilpotent Matrix :** A square matrix of the order ' n ' is said to be nilpotent matrix of order m , $m \in \mathbb{N}$, if $A^m = \mathbf{O}$ & $A^{m-1} \neq \mathbf{O}$.

(d) **Involutory Matrix :** If $A^2 = I$, the matrix is said to be an involutory matrix. i.e. square roots of identity matrix is involutory matrix.

Note : The determinant value of involutory matrix is 1 or -1.

Illustration 6 : Show that the matrix $A = \begin{bmatrix} 2 & -2 & -4 \\ -1 & 3 & 4 \\ 1 & -2 & -3 \end{bmatrix}$ is idempotent.

Solution : $A^2 = A.A = \begin{bmatrix} 2 & -2 & -4 \\ -1 & 3 & 4 \\ 1 & -2 & -3 \end{bmatrix} \times \begin{bmatrix} 2 & -2 & -4 \\ -1 & 3 & 4 \\ 1 & -2 & -3 \end{bmatrix}$

$$= \begin{bmatrix} 2.2 + (-2).(-1) + (-4).1 & 2(-2) + (-2).3 + (-4).(-2) & 2.(-4) + (-2).4 + (-4).(-3) \\ (-1).2 + 3.(-1) + 4.1 & (-1).(-2) + 3.3 + 4.(-2) & (-1).(-4) + 3.4 + 4.(-3) \\ 1.2 + (-2).(-1) + (-3).1 & 1.(-2) + (-2).3 + (-3).(-2) & 1.(-4) + (-2).4 + (-3).(-3) \end{bmatrix}$$

$$= \begin{bmatrix} 2 & -2 & -4 \\ -1 & 3 & 4 \\ 1 & -2 & -3 \end{bmatrix} = A$$

Hence the matrix A is idempotent.

Illustration 7 : Show that $\begin{bmatrix} 1 & 1 & 3 \\ 5 & 2 & 6 \\ -2 & -1 & -3 \end{bmatrix}$ is nilpotent matrix of order 3.

Solution : Let $A = \begin{bmatrix} 1 & 1 & 3 \\ 5 & 2 & 6 \\ -2 & -1 & -3 \end{bmatrix}$

$$A^2 = A.A = \begin{bmatrix} 1 & 1 & 3 \\ 5 & 2 & 6 \\ -2 & -1 & -3 \end{bmatrix} \times \begin{bmatrix} 1 & 1 & 3 \\ 5 & 2 & 6 \\ -2 & -1 & -3 \end{bmatrix} = \begin{bmatrix} 1+5-6 & 1+2-3 & 3+6-9 \\ 5+10-12 & 5+4-6 & 15+12-18 \\ -2-5+6 & -2-2+3 & -6-6+9 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 & 0 \\ 3 & 3 & 9 \\ -1 & -1 & -3 \end{bmatrix}$$

$$A^3 = A^2.A = \begin{bmatrix} 0 & 0 & 0 \\ 3 & 3 & 9 \\ -1 & -1 & -3 \end{bmatrix} \times \begin{bmatrix} 1 & 1 & 3 \\ 5 & 2 & 6 \\ -2 & -1 & -3 \end{bmatrix} = \begin{bmatrix} 0+0+0 & 0+0+0 & 0+0+0 \\ 3+15-18 & 3+6-9 & 9+18-37 \\ -1-5+6 & -1-2+3 & -3-6+9 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \mathbf{O}$$

$$\therefore A^3 = \mathbf{O} \quad \text{i.e.,} \quad A^k = \mathbf{O}$$

Here $k = 3$

Hence A is nilpotent of order 3.

Illustration 8 : Show that the matrix $A = \begin{bmatrix} -5 & -8 & 0 \\ 3 & 5 & 0 \\ 1 & 2 & -1 \end{bmatrix}$ is involutory.

Solution :

$$A^2 = A.A = \begin{bmatrix} -5 & -8 & 0 \\ 3 & 5 & 0 \\ 1 & 2 & -1 \end{bmatrix} \times \begin{bmatrix} -5 & -8 & 0 \\ 3 & 5 & 0 \\ 1 & 2 & -1 \end{bmatrix} = \begin{bmatrix} 25-24+0 & 40-40+0 & 0+0+0 \\ -15+15+0 & -24+25+0 & 0+0+0 \\ -5+6-1 & -8+10-2 & 0+0+1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I$$

Hence the given matrix A is involutory.

Illustration 9: Show that a square matrix A is involutory, iff $(I - A)(I + A) = O$

Solution : Let A be involutory

$$\text{Then } A^2 = I$$

$$(I - A)(I + A) = I^2 + IA - AI - A^2 = I + A - A - A^2 = I - A^2 = O$$

Conversely, let $(I - A)(I + A) = O$

$$\Rightarrow I^2 + IA - AI - A^2 = O \quad \Rightarrow I + A - A - A^2 = O$$

$$\Rightarrow I - A^2 = O \quad \Rightarrow A \text{ is involutory}$$

Do yourself - 5 :

(i) The matrix $A = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \\ -1 & -2 & -3 \end{bmatrix}$ is

(A) idempotent matrix

(B) involutory matrix

(C) nilpotent matrix

(D) periodic matrix

(ii) If $A = \begin{bmatrix} x & 1 \\ 1 & 0 \end{bmatrix}$ and A^2 is the identity matrix, then find the value of x

12. THE TRANSPOSE OF A MATRIX : (Changing rows & columns)

Let A be any matrix of order $m \times n$. Then A^T or $A' = [a_{ji}]$ for $1 \leq i \leq m$ & $1 \leq j \leq n$ of order $n \times m$

Properties of transpose :

If A^T & B^T denote the transpose of A and B ,

(a) $(A+B)^T = A^T + B^T$; note that A & B have the same order.

(b) $(A B)^T = B^T A^T$ (Reversal law) A & B are conformable for matrix product AB

Note : In general : $(A_1, A_2, \dots, A_n)^T = A_n^T \cdot \dots \cdot A_2^T \cdot A_1^T$ (reversal law for transpose)

(c) $(A^T)^T = A$

(d) $(kA)^T = kA^T$, k is a scalar.

Illustration 10 : If A and B are matrices of order $m \times n$ and $n \times m$ respectively, then order of matrix $B^T(A^T)^T$ is -

(A) $m \times n$

(B) $m \times m$

(C) $n \times n$

(D) Not defined

Solution : Order of B is $n \times m$ so order of B^T will be $m \times n$

Now $(A^T)^T = A$ & its order is $m \times n$. For the multiplication $B^T(A^T)^T$

Number of columns in prefactor $\neq$ Number of rows in post factor.

Hence this multiplication is not defined.

Ans. (D)

13. ORTHOGONAL MATRIX

A square matrix is said to be orthogonal matrix if $A A^T = I$

Note :

(i) The determinant value of orthogonal matrix is either 1 or -1.

$$(ii) \text{ Let } A = \begin{bmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{bmatrix} \Rightarrow A^T = \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix}$$

$$AA^T = \begin{bmatrix} a_1^2 + a_2^2 + a_3^2 & a_1b_1 + a_2b_2 + a_3b_3 & a_1c_1 + a_2c_2 + a_3c_3 \\ b_1a_1 + b_2a_2 + b_3a_3 & b_1^2 + b_2^2 + b_3^2 & b_1c_1 + b_2c_2 + b_3c_3 \\ c_1a_1 + c_2a_2 + c_3a_3 & c_1b_1 + c_2b_2 + c_3b_3 & c_1^2 + c_2^2 + c_3^2 \end{bmatrix}$$

If $AA^T = I$, then

$$\sum_{i=1}^3 a_i^2 = \sum_{i=1}^3 b_i^2 = \sum_{i=1}^3 c_i^2 = 1 \quad \text{and} \quad \sum_{i=1}^3 a_i b_i = \sum_{i=1}^3 b_i c_i = \sum_{i=1}^3 c_i a_i = 0$$

Illustration 11 : Determine the values of α , β , γ , when $\begin{bmatrix} 0 & 2\beta & \gamma \\ \alpha & \beta & -\gamma \\ \alpha & -\beta & \gamma \end{bmatrix}$ is orthogonal.

Solution :

$$\text{Let } A = \begin{bmatrix} 0 & 2\beta & \gamma \\ \alpha & \beta & -\gamma \\ \alpha & -\beta & \gamma \end{bmatrix}$$

$$\therefore A' = \begin{bmatrix} 0 & \alpha & \alpha \\ 2\beta & \beta & -\beta \\ \gamma & -\gamma & \gamma \end{bmatrix}$$

But given A is orthogonal.

$$\therefore AA^T = I$$

$$\Rightarrow \begin{bmatrix} 0 & 2\beta & \gamma \\ \alpha & \beta & -\gamma \\ \alpha & -\beta & \gamma \end{bmatrix} \begin{bmatrix} 0 & \alpha & \alpha \\ 2\beta & \beta & -\beta \\ \gamma & -\gamma & \gamma \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 4\beta^2 + \gamma^2 & 2\beta^2 - \gamma^2 & -2\beta^2 + \gamma^2 \\ 2\beta^2 - \gamma^2 & \alpha^2 + \beta^2 + \gamma^2 & \alpha^2 - \beta^2 - \gamma^2 \\ -2\beta^2 + \gamma^2 & \alpha^2 - \beta^2 - \gamma^2 & \alpha^2 + \beta^2 + \gamma^2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Equating the corresponding elements, we have

$$4\beta^2 + \gamma^2 = 1 \quad \dots\dots(i)$$

$$2\beta^2 - \gamma^2 = 0 \quad \dots\dots(ii)$$

$$\alpha^2 + \beta^2 + \gamma^2 = 1 \quad \dots\dots(iii)$$

From (i) and (ii), $6\beta^2 = 1 \therefore \beta^2 = \frac{1}{6}$ and $\gamma^2 = \frac{1}{3}$

From (iii) $\alpha^2 = 1 - \beta^2 - \gamma^2 = 1 - \frac{1}{6} - \frac{1}{3} = \frac{1}{2}$

Hence, $\alpha = \pm \frac{1}{\sqrt{2}}, \beta = \pm \frac{1}{\sqrt{6}}$ and $\gamma = \pm \frac{1}{\sqrt{3}}$

Ans.

Do yourself - 6 :

(i) If $A = \begin{bmatrix} 4 & 2 & -5 \\ 1 & 0 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 6 & -7 & 0 \\ -1 & 2 & 5 \\ 1 & 0 & 3 \end{bmatrix}$, then show that $(AB)^T = B^T \cdot A^T$.

(ii) If $A = \begin{bmatrix} 2 & 3 & -4 \\ -1 & 2 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} -1 & 2 \\ 3 & 4 \\ -5 & -6 \end{bmatrix}$, then find $A + B^T$.

(iii) If $A = \begin{bmatrix} 9 & -3 & 6 \\ 8 & \frac{1}{2} & 7 \\ -1 & 0 & 0 \end{bmatrix}$, then, show that $(A^T)^T = A$.

(iv) Show that the matrix $A = \begin{bmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{bmatrix}$ is an orthogonal matrix.

14. SYMMETRIC & SKEW SYMMETRIC MATRIX :

(a) Symmetric matrix :

A square matrix $A = [a_{ij}]$ is said to be, symmetric if, $a_{ij} = a_{ji} \forall i \& j$ (conjugate elements are equal).
Hence for symmetric matrix $A = A^T$.

Note : Max. number of distinct entries in any symmetric matrix of order n is $\frac{n(n+1)}{2}$.

(b) Skew symmetric matrix :

Square matrix $A = [a_{ij}]$ is said to be skew symmetric if $a_{ij} = -a_{ji} \forall i \& j$ (the pair of conjugate elements are additive inverse of each other). For a skew symmetric matrix $A = -A^T$.

Note :

- (i) If A is skew symmetric, then $a_{ii} = -a_{ii} \Rightarrow a_{ii} = 0 \forall i$. Thus the diagonal elements of a skew square matrix are all zero, but not the converse.
- (ii) The determinant value of odd order skew symmetric matrix is zero.

(c) Properties of symmetric & skew symmetric matrix :

- (i) A is symmetric if $A^T = A$ & A is skew symmetric if $A^T = -A$
- (ii) Let A be any square matrix then, $A + A^T$ is a symmetric matrix & $A - A^T$ is a skew symmetric matrix.
- (iii) The sum of two symmetric matrix is a symmetric matrix and the sum of two skew symmetric matrix is a skew symmetric matrix.

- (iv) If A & B are symmetric matrices then,
 (1) $AB + BA$ is a symmetric matrix
 (2) $AB - BA$ is a skew symmetric matrix.
 (v) Every square matrix can be uniquely expressed as a sum or difference of a symmetric and a skew symmetric matrix.

$$A = \underbrace{\frac{1}{2}(A + A^T)}_{\text{symmetric}} + \underbrace{\frac{1}{2}(A - A^T)}_{\text{skew symmetric}} \quad \text{and} \quad A = \frac{1}{2}(A^T + A) - \frac{1}{2}(A^T - A)$$

Illustration 12: If A is symmetric as well as skew symmetric matrix, then A is -
 (A) diagonal matrix (B) null matrix (C) triangular matrix (D) none of these

Solution : Let $A = [a_{ij}]$ Since A is skew symmetric $a_{ij} = -a_{ji}$
 for $i = j$, $a_{ii} = -a_{ii} \Rightarrow a_{ii} = 0$
 for $i \neq j$, $a_{ij} = -a_{ji}$ [$\because$ A is skew symmetric] & $a_{ij} = a_{ji}$ [$\because$ A is symmetric]
 $\therefore a_{ij} = 0$ for all $i \neq j$
 so, $a_{ij} = 0$ for all 'i' and 'j' i.e. A is null matrix.

Ans. (B)

Do yourself - 7 :

(i) If $A = \begin{bmatrix} -2 & -1 & 1 \\ -1 & 7 & 4 \\ 1 & -x & -3 \end{bmatrix}$ be symmetric matrix then find the value of x.

(ii) Express matrix $A = \begin{bmatrix} 2 & 5 & 7 \\ 9 & -7 & 2 \\ 1 & -1 & 0 \end{bmatrix}$ as a sum of a symmetric and a skew symmetric matrix.

15. ADJOINT OF A SQUARE MATRIX :

Let $A = [a_{ij}]$ be a square matrix of order n and let C_{ij} be cofactor of a_{ij} in A then the adjoint of A, denoted by $\text{adj}A$, is defined as the transpose of the cofactor matrix.

$$\text{Then, } \text{adj}A = [C_{ij}]^T \Rightarrow \text{adj}A = \begin{bmatrix} C_{11} & C_{12} & C_{13} \\ C_{23} & C_{22} & C_{23} \\ C_{31} & C_{32} & C_{33} \end{bmatrix}^T$$

Theorem : $A (\text{adj. } A) = (\text{adj. } A) \cdot A = |A| I_n$.

$$\text{Proof : } A(\text{adj. } A) = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} \begin{pmatrix} C_{11} & C_{21} & C_{31} \\ C_{12} & C_{22} & C_{32} \\ C_{13} & C_{23} & C_{33} \end{pmatrix}$$

$$\begin{pmatrix} |A| & 0 & 0 \\ 0 & |A| & 0 \\ 0 & 0 & |A| \end{pmatrix} = |A| \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \Rightarrow A(\text{Adj. } A) = |A| I$$

(whatever may be the value only |A| will come out as a common element)

If $|A| \neq 0$, then $\frac{A(\text{adj. } A)}{|A|} = I = \text{unit matrix of the same order as that of A}$

Properties of adjoint matrix :

If A be a square matrix of order n , then

- (i) $|\text{adj } A| = |A|^{n-1}$
- (ii) $\text{adj}(\text{adj } A) = |A|^{n-2} A$, where $|A| \neq 0$
- (iii) $|\text{adj}(\text{adj } A)| = |A|^{(n-1)^2}$, where $|A| \neq 0$
- (iv) $\text{adj}(AB) = (\text{adj } B)(\text{adj } A)$
- (v) $\text{adj}(KA) = K^{n-1}(\text{adj } A)$, K is a scalar
- (vi) $\text{adj } A^T = (\text{adj } A)^T$

Method to find adjoint of a 2×2 square matrix, directly :

Let A be a 2×2 square matrix. In order to find the adjoint simply interchange the diagonal elements and reverse the sign of off diagonal elements (rest of the elements).

e.g. If $A = \begin{bmatrix} p & q \\ r & s \end{bmatrix} \Rightarrow \text{adj } A = \begin{bmatrix} s & -q \\ -r & p \end{bmatrix}$

Illustration 13: If $A = \begin{bmatrix} 1 & 3 & 5 \\ 3 & 5 & 1 \\ 5 & 1 & 3 \end{bmatrix}$, then $\text{adj } A$ is equal to -

(A) $\begin{bmatrix} 14 & -4 & -22 \\ -4 & -22 & 14 \\ -22 & 14 & -4 \end{bmatrix}$

(B) $\begin{bmatrix} -14 & 4 & 22 \\ 4 & 22 & -14 \\ 22 & -14 & 4 \end{bmatrix}$

(C) $\begin{bmatrix} 14 & 4 & -22 \\ 4 & -22 & -14 \\ -22 & -14 & -4 \end{bmatrix}$

(D) none of these

Solution : $\text{adj. } A = \begin{bmatrix} 14 & -4 & -22 \\ -4 & -22 & 14 \\ -22 & 14 & -4 \end{bmatrix}^T = \begin{bmatrix} 14 & -4 & -22 \\ -4 & -22 & 14 \\ -22 & 14 & -4 \end{bmatrix}$

Ans. (A)

Illustration 14: If $A = \begin{bmatrix} 2 & 0 & 0 \\ 2 & 2 & 0 \\ 2 & 2 & 2 \end{bmatrix}$, then $\text{adj}(\text{adj } A)$ is equal to -

(A) $8 \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$

(B) $16 \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$

(C) $64 \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$

(D) none of these

Solution : $|A| = \begin{vmatrix} 2 & 0 & 0 \\ 2 & 2 & 0 \\ 2 & 2 & 2 \end{vmatrix} = 8$

Now $\text{adj}(\text{adj } A) = |A|^{3-2} A$

$= 8 \begin{vmatrix} 2 & 0 & 0 \\ 2 & 2 & 0 \\ 2 & 2 & 2 \end{vmatrix} = 16 \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$

Ans. (B)

Do yourself - 8 :

- (i) For any 2×2 matrix, if $A(\text{Adj}A) = \begin{bmatrix} 15 & 0 \\ 0 & 15 \end{bmatrix}$, then $|A|$ is equal -
 (A) 20 (B) 625 (C) 15 (D) 0
- (ii) Which of the following is/are incorrect ?
 (A) Adjoint of a symmetric matrix is skew symmetric matrix.
 (B) Adjoint of a diagonal matrix is a diagonal matrix.
 (C) $A(\text{Adj}A) = (\text{Adj}A)A = |A|I$
 (D) Adjoint of a unit matrix is a diagonal matrix
- (iii) If A be a square matrix of the order 5 and $B = \text{Adj}(A)$ then find $\text{Adj}(5A)$.
- (iv) If A be a square matrix of order 4 and $|A| = 3$ then find $\text{adj}(\text{adj}A)$.

16. INVERSE OF A MATRIX (Reciprocal Matrix) :

A square matrix A said to be invertible if and only if it is non-singular (i.e. $|A| \neq 0$) and there exists a matrix B such that, $AB = I = BA$.

B is called the **inverse** (reciprocal) of A and is denoted by A^{-1} . Thus

$$A^{-1} = B \Leftrightarrow AB = I = BA$$

We have, $A(\text{adj } A) = |A| I_n$

$$A^{-1} \cdot A(\text{adj } A) = A^{-1} |A| I_n$$

$$I_n(\text{adj } A) = A^{-1} |A| I_n$$

$$\therefore A^{-1} = \frac{(\text{adj } A)}{|A|}$$

Note : The necessary and sufficient condition for a square matrix A to be invertible is that $|A| \neq 0$

Properties of inverse :

- (i) If A & B are invertible matrices of the same order, then $(AB)^{-1} = B^{-1}A^{-1}$.

Note: If $A_1, A_2, \dots, A_n$ are all invertible square matrices of order n

$$\text{then } (A_1 A_2 \dots A_n)^{-1} = A_n^{-1} A_{n-1}^{-1} \dots A_2^{-1} A_1^{-1}$$

- (ii) If A be an invertible matrix, then A^T is also invertible & $(A^T)^{-1} = (A^{-1})^T$.

- (iii) If A is invertible, (a) $(A^{-1})^{-1} = A$ (b) $(A^k)^{-1} = (A^{-1})^k = A^{-k}$; $k \in \mathbb{N}$

- (iv) If A is non-singular matrix, then $|A^{-1}| = |A|^{-1}$

- (v) If idempotent matrix is invertible then its inverse will be identity matrix.

- (vi) A nilpotent matrix will not be invertible because its determinant value is zero.

- (vii) Orthogonal matrix A is always invertible and $A^{-1} = A^T$.

- (viii) $A = A^{-1}$ for an involutory matrix.

Cancellation law : Let A, B, C be square matrices of the same order 'n'.

If A is a non-singular matrix, then

- (a) $AB = AC \Rightarrow B = C$ (Left cancellation law)
 (b) $BA = CA \Rightarrow B = C$ (Right cancellation law)

Note that these cancellation laws hold only if the matrix ' A ' is **non-singular** (i.e. $|A| \neq 0$).

Illustration 15 : Prove that if A is non-singular matrix such that A is symmetric then A^{-1} is also symmetric.

Solution : $A^T = A$ [$\because$ A is a symmetric matrix]

$$(A^T)^{-1} = A^{-1} \text{ [since A is non-singular matrix]}$$

$$\Rightarrow (A^{-1})^T = A^{-1} \text{ Hence proved}$$

Illustration 16 : $\begin{bmatrix} 1 & -\tan \theta/2 \\ \tan \theta/2 & 1 \end{bmatrix} \begin{bmatrix} 1 & \tan \theta/2 \\ -\tan \theta/2 & 1 \end{bmatrix}^{-1}$ is equal to -

(A) $\begin{bmatrix} \sin \theta & -\cos \theta \\ \cos \theta & \sin \theta \end{bmatrix}$ (B) $\begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}$ (C) $\begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$ (D) none of these

Solution : $\begin{bmatrix} 1 & \tan \theta/2 \\ \tan \theta/2 & 1 \end{bmatrix}^{-1} = \frac{1}{\sec^2 \theta/2} \begin{bmatrix} 1 & -\tan \theta/2 \\ \tan \theta/2 & 1 \end{bmatrix}$

$$\therefore \text{Product} = \frac{1}{\sec^2 \theta/2} \begin{bmatrix} 1 & -\tan \theta/2 \\ \tan \theta/2 & 1 \end{bmatrix} \begin{bmatrix} 1 & -\tan \theta/2 \\ \tan \theta/2 & 1 \end{bmatrix}$$

$$= \frac{1}{\sec^2 \theta/2} \begin{bmatrix} 1 - \tan^2 \theta/2 & -2 \tan \theta/2 \\ 2 \tan \theta/2 & 1 - \tan^2 \theta/2 \end{bmatrix}$$

$$= \begin{bmatrix} \cos^2 \theta/2 & \sin^2 \theta/2 & -2 \sin \theta/2 \cos \theta/2 \\ 2 \sin \theta/2 \cos \theta/2 & \cos^2 \theta/2 & \cos^2 \theta/2 - \sin^2 \theta/2 \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$

Ans. (C)

Illustration 17 : If $A = \begin{bmatrix} 0 & -1 & 2 \\ 2 & -2 & 0 \end{bmatrix}$, $B = \begin{bmatrix} 0 & 1 \\ 1 & 0 \\ 1 & 1 \end{bmatrix}$ and $M = AB$, then M^{-1} is equal to-

(A) $\begin{bmatrix} 2 & -2 \\ 2 & 1 \end{bmatrix}$ (B) $\begin{bmatrix} 1/3 & 1/3 \\ -1/3 & 1/6 \end{bmatrix}$ (C) $\begin{bmatrix} 1/3 & -1/3 \\ 1/3 & 1/6 \end{bmatrix}$ (D) $\begin{bmatrix} 1/3 & -1/3 \\ -1/3 & 1/6 \end{bmatrix}$

Solution : $M = \begin{bmatrix} 0 & -1 & 2 \\ 2 & -2 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ -2 & 2 \end{bmatrix}$

$$|M| = 6, \text{adj } M = \begin{bmatrix} 2 & -2 \\ 2 & 1 \end{bmatrix}$$

$$\therefore M^{-1} = \frac{1}{6} \begin{bmatrix} 2 & -2 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} 1/3 & -1/3 \\ 1/3 & 1/6 \end{bmatrix}$$

Ans. (C)

Do yourself -9 :

- (i) If 'A' is a square matrix such that $A^2 = I$ then A^{-1} is equal to -
 (A) $A + I$ (B) A (C) 0 (D) $2A$
- (ii) If 'A' is an orthogonal matrix, then A^{-1} equals -
 (A) A (B) A^T (C) A^2 (D) none of these
- (iii) If $A = \begin{bmatrix} 3 & 2 \\ 0 & 1 \end{bmatrix}$, then $(A^{-1})^3$ is equal to -
 (A) $\frac{1}{27} \begin{bmatrix} 1 & -26 \\ 0 & 27 \end{bmatrix}$ (B) $\frac{1}{27} \begin{bmatrix} 1 & 26 \\ 0 & 27 \end{bmatrix}$ (C) $\frac{1}{27} \begin{bmatrix} 1 & -26 \\ 0 & -27 \end{bmatrix}$ (D) $\frac{1}{27} \begin{bmatrix} -1 & -26 \\ 0 & -27 \end{bmatrix}$

17. MATRIX POLYNOMIAL :

If $f(x) = a_0x^n + a_1x^{n-1} + a_2x^{n-2} + \dots + a_nx^0$, then we define a matrix polynomial

$$f(A) = a_0A^n + a_1A^{n-1} + a_2A^{n-2} + \dots + a_nI^n.$$

where A is the given square matrix. If $f(A)$ is the null matrix, then A is called the zero or root of the polynomial $f(x)$.

18. CHARACTERISTIC EQUATION :

Let A be a square matrix. Then the polynomial $|A - xI|$ is called as characteristic polynomial of A & the equation $|A - xI| = 0$ is called as characteristic equation of A. After solving the characteristic polynomial the values of 'x' are said to be characteristic roots of the polynomial.

Note : (i) Sum of the roots of the characteristic equation is equal to trace of the matrix.

(ii) Product of the roots of the characteristic equation is equal to the determinant value.

(iii) The degree of characteristic equation is same as the order of the matrix.

Illustration 18: If $f(x) = x^2 - 3x + 3$ and $A = \begin{bmatrix} 2 & 1 \\ -1 & 1 \end{bmatrix}$ be a square matrix then prove that $f(A) = O$.

Hence find A^4 .

Solution :

$$A^2 = A.A = \begin{bmatrix} 2 & 1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 3 & 3 \\ -3 & 0 \end{bmatrix}$$

$$\text{Hence } A^2 - 3A + 3I = \begin{bmatrix} 3 & 3 \\ -3 & 0 \end{bmatrix} - 3 \begin{bmatrix} 2 & 1 \\ -1 & 1 \end{bmatrix} + 3 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = O$$

$$\text{Aliter : } \because |A - XI| = 0 \Rightarrow \begin{vmatrix} 2-x & 1 \\ -1 & 1-x \end{vmatrix} = 0$$

$$\Rightarrow (2-x)(1-x) + 1 = 0 \Rightarrow x^2 - 3x + 3 = 0 \quad (\text{characteristic polynomial})$$

by Cayley-Hamilton Theorem $A^2 - 3A + 3I = O$. Hence proved.

$$\text{Now } A^2 = 3A - 3I$$

squaring on both the sides

$$\begin{aligned}
 A^4 &= 9(A^2 - 2A + I) \\
 &= 9\left(\begin{bmatrix} 3 & 3 \\ -3 & 0 \end{bmatrix} - \begin{bmatrix} 4 & 2 \\ -2 & 2 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\right) = 9\begin{bmatrix} 3-4+1 & 3-2 \\ -3+2 & -2+1 \end{bmatrix} \\
 &= 9\begin{bmatrix} 0 & 1 \\ -1 & -1 \end{bmatrix} = \begin{bmatrix} 0 & 9 \\ -9 & -9 \end{bmatrix}
 \end{aligned}$$

19. CAYLEY - HAMILTON THEOREM :

Every square matrix A satisfy its characteristic equation

i.e. $a_0x^n + a_1x^{n-1} + \dots + a_{n-1}x + a_n = 0$ is the characteristic equation of A, then

$$a_0A^n + a_1A^{n-1} + \dots + a_{n-1}A + a_nI = \mathbf{O}$$

Note : This theorem is helpful to find the inverse of any non-singular square matrix.

$$\text{i.e. } a_0A^n + a_1A^{n-1} + \dots + a_{n-1}A + a_nI = \mathbf{O}$$

On multiplying by A^{-1} on both the sides of above equation, we get

$$A^{-1} = -\frac{1}{a_n}(a_0A^{n-1} + a_1A^{n-2} + \dots + a_{n-1}I)$$

Illustration 19 : If $A = \begin{bmatrix} 1 & 2 & 0 \\ 2 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$, show that $5A^{-1} = A^2 + A - 5I$

Solution :

We have the characteristic equation of A.

$$|A - xI| = 0$$

$$\text{i.e. } \begin{vmatrix} 1-x & 2 & 0 \\ 2 & -1-x & 0 \\ 0 & 0 & -1-x \end{vmatrix} = 0$$

$$\text{i.e. } x^3 + x^2 - 5x - 5 = 0$$

Using Cayley – Hamilton theorem

$$A^3 + A^2 - 5A - 5I = \mathbf{O} \Rightarrow 5I = A^3 + A^2 - 5A$$

Multiplying by A^{-1} , we get $5A^{-1} = A^2 + A - 5I$

Do yourself -10 :

(i) Determine the characteristic roots of the matrix A. Hence find the trace and determinant value of A.

Where $A = \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$ and also prove that $A^3 - 18A^2 + 45A = \mathbf{O}$.

20. SYSTEM OF EQUATION & CRITERIA FOR CONSISTENCY

Gauss - Jordan method :

$$a_1x + b_1y + c_1z = d_1$$

$$a_2x + b_2y + c_2z = d_2$$

$$a_3x + b_3y + c_3z = d_3$$

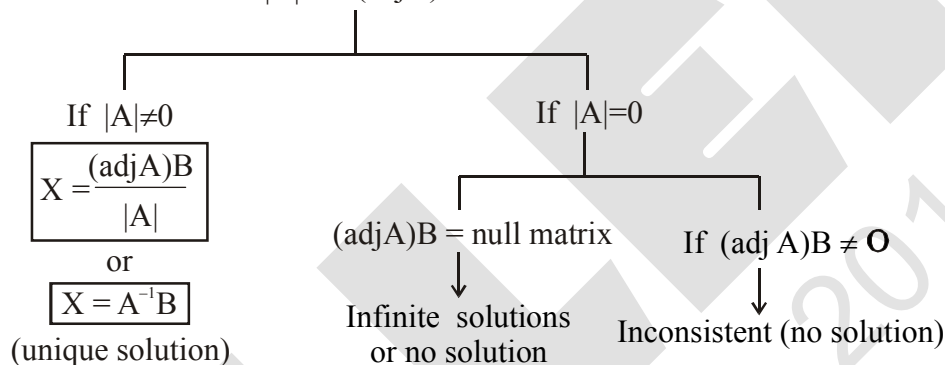
$$\Rightarrow \begin{bmatrix} a_1x + b_1y + c_1z \\ a_2x + b_2y + c_2z \\ a_3x + b_3y + c_3z \end{bmatrix} = \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix} \Rightarrow \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix}$$

$$\Rightarrow A X = B \quad \dots(i)$$

Multiplying $\text{adj}A$ on both the sides of (i)

$$\Rightarrow (\text{adj}A) AX = (\text{adj}A)B \Rightarrow |A|X = (\text{adj}A)B$$

$$|A|X = (\text{adj}A)B$$



$$x + y + z = 16$$

Illustration 20: Solve the system $x - y + z = 2$ using matrix method.

$$2x + y - z = 1$$

Solution :

$$\text{Let } A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 2 & 1 & -1 \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \text{ \& } B = \begin{bmatrix} 6 \\ 2 \\ 1 \end{bmatrix}$$

Then the system is $AX = B$. $|A| = 6$, hence A is non singular,

$$\text{Cofactor } A = \begin{bmatrix} 0 & 3 & 3 \\ 2 & -3 & 1 \\ 2 & 0 & -2 \end{bmatrix}$$

$$\text{adj } A = \begin{bmatrix} 0 & 2 & 2 \\ 3 & -3 & 0 \\ 3 & 1 & -2 \end{bmatrix}$$

$$A^{-1} = \frac{1}{|A|} \text{adj}A = \frac{1}{6} \begin{bmatrix} 0 & 2 & 2 \\ 3 & -3 & 0 \\ 3 & 1 & -2 \end{bmatrix} = \begin{bmatrix} 0 & 1/3 & 1/3 \\ 1/2 & -1/2 & 0 \\ 1/2 & 1/6 & -1/3 \end{bmatrix}$$

$$X = A^{-1} B = \begin{bmatrix} 0 & 1/3 & 1/3 \\ 1/2 & -1/2 & 0 \\ 1/2 & 1/6 & -1/3 \end{bmatrix} \begin{bmatrix} 6 \\ 2 \\ 1 \end{bmatrix} \quad \text{i.e.} \quad \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

$$\Rightarrow x = 1, y = 2, z = 3$$

Ans.

Do yourself -11 :

(i) The system of equations $x + 2y - 3z = 1$, $x - y + 4z = 0$, $2x + y + z = 1$ has -

- (A) only two solutions (B) only one solution
(C) no solution (D) infinitely many solutions

(ii) The system of equations $x + y + z = 8$, $x - y + 2z = 6$, $3x + 5y - 7z = 14$ has-

- (A) Unique solution (B) infinite number of solutions
(C) no solution (D) none of these

ANSWERS FOR DO YOURSELF

1: (i) $\begin{bmatrix} 3 & 5 & 7 \\ 4 & 6 & 8 \end{bmatrix}$ (ii) 6 (iii) 30 (iv) $x = 2, y = -1, a = 1, b = -3$

2: (i) $\begin{bmatrix} -7 & -10 & -7 \\ -2 & 6 & 3 \end{bmatrix}$ & 2×3 (ii) $\begin{bmatrix} -8 & -9 \\ -7/2 & -8 \\ -1 & 1 \end{bmatrix}$

3: (iii) $x = -2$

4: (ii) C

5: (i) C (ii) $x = 0$

6: (ii) $\begin{bmatrix} 1 & 6 & -9 \\ 1 & 6 & -3 \end{bmatrix}$

7: (i) -4 (ii) $\begin{bmatrix} 2 & 7 & 4 \\ 7 & -7 & \frac{1}{2} \\ 4 & \frac{1}{2} & 0 \end{bmatrix} + \begin{bmatrix} 0 & -2 & 3 \\ 2 & 0 & \frac{3}{2} \\ -3 & -\frac{3}{2} & 0 \end{bmatrix}$

8: (i) C (ii) A (iii) 625 B (iv) 9A

9: (i) B (ii) B (iii) A

10: (i) $\lambda = 0, 3$ and $15 \text{tr}(A) = 18, |A| = 0$

11: (i) D (ii) A

EXERCISE (O-1)

1. Let $A + 2B = \begin{bmatrix} 1 & 2 & 0 \\ 6 & -3 & 3 \\ -5 & 3 & 1 \end{bmatrix}$ and $2A - B = \begin{bmatrix} 2 & -1 & 5 \\ 2 & -1 & 6 \\ 0 & 1 & 2 \end{bmatrix}$, then $\text{Tr}(A) - \text{Tr}(B)$ has the value equal to
- (A) 0 (B) 1 (C) 2 (D) none
2. If $\begin{bmatrix} x & 3x-y \\ zx+z & 3y-w \end{bmatrix} = \begin{bmatrix} 3 & 2 \\ 4 & 7 \end{bmatrix}$, then
- (A) $x = 3, y = 7, z = 1, w = 14$ (B) $x = 3, y = -5, z = -1, w = -4$
 (C) $x = 3, y = 6, z = 2, w = 7$ (D) None of these
3. The matrix $A^2 + 4A - 5I$, where I is identity matrix and $A = \begin{bmatrix} 1 & 2 \\ 4 & -3 \end{bmatrix}$ equals : [JEE-MAIN Online 2013]
- (A) $32 \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$ (B) $4 \begin{bmatrix} 2 & 1 \\ 2 & 0 \end{bmatrix}$ (C) $4 \begin{bmatrix} 0 & -1 \\ 2 & 2 \end{bmatrix}$ (D) $32 \begin{bmatrix} 2 & 1 \\ 2 & 0 \end{bmatrix}$
4. If $M = \begin{bmatrix} 0 & 2 \\ 5 & 0 \end{bmatrix}$ and $N = \begin{bmatrix} 0 & 5 \\ 2 & 0 \end{bmatrix}$, then M^{2011} is -
- (A) $10^{1005}M$ (B) $10^{1005}N$ (C) $10^{2010}M$ (D) $10^{2011}M$
5. If $A = \begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix}$ and $A^2 - kA - I_2 = 0$, then value of k is-
- (A) 4 (B) 2 (C) 1 (D) -4
6. Let three matrices are $A = \begin{bmatrix} 2 & 1 \\ 4 & 1 \end{bmatrix}$; $B = \begin{bmatrix} 3 & 4 \\ 2 & 3 \end{bmatrix}$ and $C = \begin{bmatrix} 3 & -4 \\ -2 & 3 \end{bmatrix}$, then
- $t_r(A) + t_r\left(\frac{ABC}{2}\right) + t_r\left(\frac{A(BC)^2}{4}\right) + t_r\left(\frac{A(BC)^3}{8}\right) + \dots + \infty$ is equal to-
- (A) 6 (B) 9 (C) 12 (D) none
7. For a matrix $A = \begin{bmatrix} 1 & 2r-1 \\ 0 & 1 \end{bmatrix}$, the value of $\prod_{r=1}^{50} \begin{bmatrix} 1 & 2r-1 \\ 0 & 1 \end{bmatrix}$ is equal to -
- (A) $\begin{bmatrix} 1 & 100 \\ 0 & 1 \end{bmatrix}$ (B) $\begin{bmatrix} 1 & 4950 \\ 0 & 1 \end{bmatrix}$ (C) $\begin{bmatrix} 1 & 5050 \\ 0 & 1 \end{bmatrix}$ (D) $\begin{bmatrix} 1 & 2500 \\ 0 & 1 \end{bmatrix}$

8. A and B are two given matrices such that the order of A is 3×4 , if $A'B$ and BA' are both defined then
- (A) order of B' is 3×4 (B) order of $B'A$ is 4×4
 (C) order of $B'A$ is 3×3 (D) $B'A$ is undefined
9. If the product of n matrices $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix} \dots \begin{bmatrix} 1 & n \\ 0 & 1 \end{bmatrix}$ is equal to the matrix $\begin{bmatrix} 1 & 378 \\ 0 & 1 \end{bmatrix}$ then the value of n is equal to -
- (A) 26 (B) 27 (C) 377 (D) 378
10. Consider a matrix $A(\theta) = \begin{bmatrix} \sin \theta & \cos \theta \\ -\cos \theta & \sin \theta \end{bmatrix}$ then
- (A) $A(\theta)$ is symmetric (B) $A(\theta)$ is skew symmetric
 (C) $A^{-1}(\theta) = A(\pi - \theta)$ (D) $A^2(\theta) = A\left(\frac{\pi}{2} - 2\theta\right)$
11. If p, q, r are 3 real number satisfying the matrix equation, $\begin{bmatrix} p & q & r \end{bmatrix} \begin{bmatrix} 3 & 4 & 1 \\ 3 & 2 & 3 \\ 2 & 0 & 2 \end{bmatrix} = \begin{bmatrix} 3 & 0 & 1 \end{bmatrix}$, then $2p + q - r$ equals :-
- (A) -1 (B) 4 (C) -3 (D) 2 [JEE-MAIN Online 2013]
12. If A, B and C are $n \times n$ matrices and $\det(A) = 2$, $\det(B) = 3$ and $\det(C) = 5$, then the value of the $\det(A^2BC^{-1})$ is equal to
- (A) $\frac{6}{5}$ (B) $\frac{12}{5}$ (C) $\frac{18}{5}$ (D) $\frac{24}{5}$
13. Which of the following is an orthogonal matrix -
- (A) $\begin{bmatrix} 6/7 & 2/7 & -3/7 \\ 2/7 & 3/7 & 6/7 \\ 3/7 & -6/7 & 2/7 \end{bmatrix}$ (B) $\begin{bmatrix} 6/7 & 2/7 & 3/7 \\ 2/7 & -3/7 & 6/7 \\ 3/7 & 6/7 & -2/7 \end{bmatrix}$
 (C) $\begin{bmatrix} -6/7 & -2/7 & -3/7 \\ 2/7 & 3/7 & 6/7 \\ -3/7 & 6/7 & 2/7 \end{bmatrix}$ (D) $\begin{bmatrix} 6/7 & -2/7 & 3/7 \\ 2/7 & 2/7 & -3/7 \\ -6/7 & 2/7 & 3/7 \end{bmatrix}$
14. Matrix $A = \begin{bmatrix} x & 3 & 2 \\ 1 & y & 4 \\ 2 & 2 & z \end{bmatrix}$, if $xyz = 60$ and $8x + 4y + 3z = 20$, then $A(\text{adj } A)$ is equal to -
- (A) $\begin{bmatrix} 64 & 0 & 0 \\ 0 & 64 & 0 \\ 0 & 0 & 64 \end{bmatrix}$ (B) $\begin{bmatrix} 88 & 0 & 0 \\ 0 & 88 & 0 \\ 0 & 0 & 88 \end{bmatrix}$ (C) $\begin{bmatrix} 68 & 0 & 0 \\ 0 & 68 & 0 \\ 0 & 0 & 68 \end{bmatrix}$ (D) $\begin{bmatrix} 34 & 0 & 0 \\ 0 & 34 & 0 \\ 0 & 0 & 34 \end{bmatrix}$

15. The matrix $\begin{bmatrix} 2 & -2 & -4 \\ -1 & 3 & 4 \\ 1 & -2 & -3 \end{bmatrix}$ is a
 (A) non-singular (B) Idempotent (C) Nilpotent (D) Orthogonal
16. If $A = \begin{bmatrix} ab & b^2 \\ -a^2 & -ab \end{bmatrix}$, then A is
 (A) Involutory matrix (B) Idempotent matrix (C) Nilpotent matrix (D) none of these
17. If A and B are symmetric matrices, then ABA is -
 (A) symmetric matrix (B) skew symmetric matrix
 (C) diagonal matrix (D) scalar matrix
18. Let $A = \begin{pmatrix} 0 & \sin \alpha & \sin \alpha \sin \beta \\ -\sin \alpha & 0 & \cos \alpha \cos \beta \\ -\sin \alpha \sin \beta & -\cos \alpha \cos \beta & 0 \end{pmatrix}$, then -
 (A) $|A|$ is independent of α and β (B) A^{-1} depends only on α
 (C) A^{-1} depends only on β (D) none of these
19. Number of real values of λ for which the matrix $A = \begin{bmatrix} \lambda-1 & \lambda & \lambda+1 \\ 2 & -1 & 3 \\ \lambda+3 & \lambda-2 & \lambda+7 \end{bmatrix}$ has no inverse
 (A) 0 (B) 1 (C) 2 (D) infinite
20. If $A = [a_{ij}]_{2 \times 2}$ where $a_{ij} = \begin{cases} i+j & i \neq j \\ i^2-2j & i = j \end{cases}$, then A^{-1} is equal to -
 (A) $\frac{1}{9} \begin{bmatrix} 0 & 3 \\ 3 & 1 \end{bmatrix}$ (B) $\frac{1}{9} \begin{bmatrix} 0 & -3 \\ 3 & -1 \end{bmatrix}$ (C) $\frac{1}{9} \begin{bmatrix} 0 & -3 \\ -3 & -1 \end{bmatrix}$ (D) $\frac{1}{3} \begin{bmatrix} 0 & 3 \\ 3 & 1 \end{bmatrix}$

EXERCISE (O-2)

1. Let A , other than I or $-I$, be a 2×2 real matrix such that $A^2 = I$, I being the unit matrix. Let $\text{Tr}(A)$ be the sum of diagonal elements of A . [JEE-MAIN Online 2013]

Statement-1 : $\text{Tr}(A) = 0$

Statement-2 : $\det(A) = -1$

- (A) Statement-1 and Statement-2 are true and Statement-2 is not the correct explanation for Statement-1
(B) Statement-1 and Statement-2 are true and Statement-2 is a correct explanation for Statement-1.
(C) Statement-1 is true and Statement-2 is false.
(D) Statement-1 is false and Statement-2 is true.

2. Let $S = \left\{ \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} : a_{ij} \in \{0, 1, 2\}, a_{11} = a_{22} \right\}$. Then the number of non-singular matrices in the set S is : [JEE-MAIN Online 2013]
(A) 24 (B) 10 (C) 20 (D) 27

3. If $A = \begin{bmatrix} \cos^2 \alpha & \sin \alpha \cos \alpha \\ \sin \alpha \cos \alpha & \sin^2 \alpha \end{bmatrix}$; $B = \begin{bmatrix} \cos^2 \beta & \sin \beta \cos \beta \\ \sin \beta \cos \beta & \sin^2 \beta \end{bmatrix}$ are such that, AB is a null matrix, then which of the following should necessarily be an odd integral multiple of $\frac{\pi}{2}$.
(A) α (B) β (C) $\alpha - \beta$ (D) $\alpha + \beta$

4. If A and B are invertible matrices, which one of the following statement is/are incorrect -
(A) $\text{Adj}(A) = |A|A^{-1}$ (B) $\det(A^{-1}) = |\det(A)|^{-1}$
(C) $(A + B)^{-1} = B^{-1} + A^{-1}$ (D) $(AB)^{-1} = B^{-1}A^{-1}$
5. Identify the incorrect statement in respect of two square matrices A and B conformable for sum and product -
(A) $t_r(A + B) = t_r(A) + t_r(B)$ (B) $t_r(\alpha A) = \alpha t_r(A)$, $\alpha \in \mathbb{R}$
(C) $t_r(A^T) = t_r(A)$ (D) $t_r(AB) \neq t_r(BA)$

6. Let $A = \begin{bmatrix} 2 & 2 & 1 \\ 2 & 5 & 2 \\ 1 & 2 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} -x & -y & z \\ 0 & y & 2z \\ x & -y & z \end{bmatrix}$ where $x, y, z \in \mathbb{R}$. If $B^T A B = \begin{bmatrix} 8 & 0 & 0 \\ 0 & 27 & 0 \\ 0 & 0 & 42 \end{bmatrix}$ then the number of ordered triplet (x, y, z) is -
(A) 2 (B) 6 (C) 8 (D) 9

7. Let $A = \begin{bmatrix} 1 & -1 & 1 \\ 2 & 1 & -3 \\ 1 & 1 & 1 \end{bmatrix}$ and $10B = \begin{bmatrix} 4 & 2 & 2 \\ -5 & 0 & \alpha \\ 1 & -2 & 3 \end{bmatrix}$. If B is the inverse of matrix A , then α is -
(A) -2 (B) -1 (C) 2 (D) 5

8. $A = \begin{bmatrix} 1 & \tan x \\ -\tan x & 1 \end{bmatrix}$ then let us define a function $f(x) = \det. (A^T A^{-1})$ then which of the following can not be the value of $\underbrace{f(f(f(\dots f(x))))}_{n \text{ times}}$ is ($n \geq 2$)
(A) $f^n(x)$ (B) 1 (C) $f^{n-1}(x)$ (D) $n f(x)$

[ONE OR MORE THAN ONE ARE CORRECT]

9. Let $\det(\text{adj}(\text{adj}A)) = 14^4$ where $A = \begin{bmatrix} x & 2 & -1 \\ -1 & 1 & 2 \\ 2 & -1 & 1 \end{bmatrix}$, $x \neq -\frac{11}{3}$, then

- (A) $x = 1$ (B) $\det(2A) = 112$ (C) $x = 2$ (D) $\det(2A) = 256$

10. Let $A^{-1} = \begin{bmatrix} 1 & -1 & 2 \\ -1 & 2 & 1 \\ 2 & 1 & -1 \end{bmatrix}$, then -

(A) $7|A| = \frac{1}{2}$

(B) $|\text{adj} A| = \frac{1}{196}$

(C) $\text{trace}(\text{adj}A) = -\frac{1}{7}$

(D) Matrix A is a symmetric matrix

11. If $A = \begin{bmatrix} 3 & 0 & 0 \\ 2 & 2 & 0 \\ 4 & 5 & 3 \end{bmatrix}$, then which of the following is(are) true ?

(trace of A denotes sum of principal diagonal elements of A)

(A) A is invertible

(B) $\text{trace}(\text{adj}(\text{adj}(A))) = 144$

(C) $\text{trace}(\text{adj}(\text{adj}(A))) = 8$

(D) $|\text{adj} A|$ is less than 400

12. Let $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 2 & -1 \\ 3 & 0 & k \end{bmatrix}$ and $f(x) = x^3 - 2x^2 - \alpha x + \beta = 0$. If A satisfies $f(x) = 0$, then-

(A) $k = 1, \alpha = 14$

(B) $\alpha = 14, \beta = 22$

(C) $k = -1, \beta = 22$

(D) $\alpha = -14, \beta = -22$

13. If A and B are 3×3 matrices and $|A| \neq 0$, then which of the following are true?

(A) $|AB| = 0 \Rightarrow |B| = 0$

(B) $|AB| = 0 \Rightarrow B = 0$

(C) $|A^{-1}| = |A|^{-1}$

(D) $|A + A| = 2|A|$

14. If D_1 and D_2 are two 3×3 diagonal matrices where none of the diagonal element is zero, then -

(A) $D_1 D_2$ is a diagonal matrix

(B) $D_1 D_2 = D_2 D_1$

(C) $D_1^2 + D_2^2$ is a diagonal matrix

(D) none of these

15. If A and B are two 3×3 matrices such that their product AB is a null matrix then

(A) $\det. A \neq 0 \Rightarrow B$ must be a null matrix.

(B) $\det. B \neq 0 \Rightarrow A$ must be a null matrix.

(C) If none of A and B are null matrices then atleast one of the two matrices must be singular.

(D) If neither $\det. A$ nor $\det. B$ is zero then the given statement is not possible.

16. Let $A = a_{ij}$ be a matrix of order 3 where $a_{ij} = \begin{cases} x & \text{if } i = j, x \in \mathbb{R} \\ 1 & \text{if } |i - j| = 1 \\ 0 & \text{otherwise} \end{cases}$, then which of the following hold(s)

good ?

- (A) for $x = 2$, A is a diagonal matrix. (B) A is a symmetric matrix
(C) for $x = 2$, $\det A$ has the value equal to 6
(D) Let $f(x) = \det A$, then the function $f(x)$ has both the maxima and minima.
17. If A & B are square matrices of order 2 such that $A + \text{adj}(B^T) = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ & $A^T - \text{adj}(B) = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, then-
- (A) B is symmetric matrix (B) $A^n = A \forall n \in \mathbb{N}$
(C) $|A + A^2 + A^3 + A^4 + A^5| = 0$ (D) $|B + B^2 + B^3 + B^4 + B^5| = 0$
18. If $A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$ & $A^n = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, (where $n \geq 2$ & $n \in \mathbb{N}$), then -
- (A) $a = d$ (B) $b = c$
(C) $b = a + 1$ if n is odd (D) $b = a - 1$ if n is even
19. If A and B are two orthogonal matrices of order 3, then -
- (A) A and B both will be invertible matrices (B) matrix ABA will also be orthogonal
(C) matrix A^2B^2 will also be orthogonal (D) maximum value of $\det\left(\frac{A}{2}\text{adj}(2B)\right)$ is 8.
20. If A & B are two non singular matrices of order 3×3 such that $A^T + B = I$ & $BA^T = -B$, then which is/are always true (where X^T denotes transpose of X and I denotes unit matrix)-
- (A) $|B| = 2$ (B) $|B| = 8$ (C) $|A| = -1$ (D) $|A| = 1$
(where $|X|$ denotes determinant value of X)

Paragraph for Question 21 to 22

Consider the system $AX = B$, where $A = \begin{bmatrix} 1 & 1 & -1 \\ 2 & 1 & 1 \\ 3 & 2 & 2 \end{bmatrix}$, $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$, $B = \begin{bmatrix} -1 \\ 2 \\ 3 \end{bmatrix}$.

21. Sum of elements of $(\text{adj}A) B$ is-
(A) -1 (B) 2 (C) -2 (D) -4
22. Value of $\text{tr}(XB^T)$ is (where $\text{tr}(A)$ denotes trace of matrix A)-
(A) 0 (B) 1 (C) 2 (D) 3

Paragraph for question nos. 23 to 25

If A is a symmetric and B skew symmetric matrix and $A + B$ is non singular and $C = (A + B)^{-1}(A - B)$ then

23. $C^T(A + B)C =$
(A) $A + B$ (B) $A - B$ (C) A (D) B
24. $C^T(A - B)C =$
(A) $A + B$ (B) $A - B$ (C) A (D) B
25. $C^TAC =$
(A) $A + B$ (B) $A - B$ (C) A (D) B

EXERCISE (S-1)

- Let $M = \begin{bmatrix} a & -360 \\ b & c \end{bmatrix}$, where a, b and c are integers. Find the smallest positive value of b such that $M^2 = \mathbf{O}$, where $\mathbf{O}$ denotes 2×2 null matrix.
- Find the number of 2×2 matrix satisfying following conditions :
(i) a_{ij} is 1 or -1 ; (ii) $a_{11}a_{21} + a_{12}a_{22} = 0$
- Find the value of x and y that satisfy the equations

$$\begin{bmatrix} 3 & -2 \\ 3 & 0 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} y & y \\ x & x \end{bmatrix} = \begin{bmatrix} 3 & 3 \\ 3y & 3y \\ 10 & 10 \end{bmatrix}$$

- Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ and $B = \begin{bmatrix} p \\ q \end{bmatrix} \neq \begin{bmatrix} 0 \\ 0 \end{bmatrix}$. Such that $AB = B$ and $a + d = 5050$. Find the value of $(ad - bc)$.
- Define $A = \begin{bmatrix} 0 & 1 \\ 3 & 0 \end{bmatrix}$. Find a vertical vector V such that $(A^8 + A^6 + A^4 + A^2 + I)V = \begin{bmatrix} 0 \\ 11 \end{bmatrix}$ (where I is the 2×2 identity matrix).

- If $A = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 2 & 1 \\ 2 & 0 & 3 \end{bmatrix}$, then show that the matrix A is a root of the polynomial $f(x) = x^3 - 6x^2 + 7x + 2$.

- If the matrices $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$

(a, b, c, d not all simultaneously zero) commute, find the value of $\frac{d-b}{a+c-b}$. Also show that the

matrix which commutes with A is of the form $\begin{bmatrix} \alpha - \beta & 2\beta/3 \\ \beta & \alpha \end{bmatrix}$

- If $\begin{bmatrix} a & b \\ c & 1-a \end{bmatrix}$ is an idempotent matrix. Find the value of $f(a)$, where $f(x) = x - x^2$, when $bc = 1/4$. Hence otherwise evaluate a .

- If the matrix A is involutory, show that $\frac{1}{2}(I + A)$ and $\frac{1}{2}(I - A)$ are idempotent and $\frac{1}{2}(I + A) \cdot \frac{1}{2}(I - A) = \mathbf{O}$.

- Show that the matrix $A = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$ can be decomposed as a sum of a unit and a nilpotent matrix.

Hence evaluate the matrix $\begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}^{2007}$.

11. $A = \begin{pmatrix} 3 & a & -1 \\ 2 & 5 & c \\ b & 8 & 2 \end{pmatrix}$ is Symmetric and $B = \begin{pmatrix} d & 3 & a \\ b-a & e & -2b-c \\ -2 & 6 & -f \end{pmatrix}$ is Skew Symmetric, then find AB.

Is AB a symmetric, Skew Symmetric or neither of them. Justify your answer.

12. Express the matrix $\begin{bmatrix} 1 & 2 & 5 \\ 2 & 3 & -6 \\ -1 & 0 & 4 \end{bmatrix}$ as a sum of a lower triangular matrix & an upper triangular matrix

with zero in its leading diagonal. Also express the matrix as a sum of a symmetric and a skew symmetric matrix.

13. (a) A is a square matrix of order n.
 ℓ = maximum number of distinct entries if A is a triangular matrix
 m = maximum number of distinct entries if A is a diagonal matrix
 p = minimum number of zeroes if A is a triangular matrix.
 If $\ell + 5 = p + 2m$, find the order of the matrix.
- (b) Let A be the set of all 3×3 skew symmetric matrices whose entries are either -1 , 0 or 1 . If there are exactly three 0's, three 1's and three (-1) 's, then find the number of such matrices.
14. If A is an idempotent non-zero matrix and I is an identity matrix of the same order, find the value of n, $n \in \mathbb{N}$, such that $(A + I)^n = I + 127A$.

15. Let $A = \begin{bmatrix} 2 & 0 & 7 \\ 0 & 1 & 0 \\ 1 & -2 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} -x & 14x & 7x \\ 0 & 1 & 0 \\ x & -4x & -2x \end{bmatrix}$ are two matrices such that $AB = (AB)^{-1}$ and $AB \neq I$ (where I is an identity matrix of order 3×3).

Find the value of $\text{Tr.} (AB + (AB)^2 + (AB)^3 + \dots + (AB)^{100})$,

where $\text{Tr.} (A)$ denotes the trace of matrix A.

16. If $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ then prove that value of f and g satisfying the matrix equation $A^2 + fA + gI = O$ are equal to $-t_r(A)$ and determinant of A respectively. Given a, b, c, d are non zero reals and $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$; $O = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$.

17. Let $A = \begin{bmatrix} 3x^2 \\ 1 \\ 6x \end{bmatrix}$, $B = [a \ b \ c]$ and $C = \begin{bmatrix} (x+2)^2 & 5x^2 & 2x \\ 5x^2 & 2x & (x+2)^2 \\ 2x & (x+2)^2 & 5x^2 \end{bmatrix}$ be three given matrices, where

a, b, c and $x \in \mathbb{R}$. Given that $\text{tr}(AB) = \text{tr}(C) \forall x \in \mathbb{R}$, where $\text{tr}(A)$ denotes trace of A. Find the value of $(a + b + c)$

EXERCISE (S-2)

1. Let A be the 2×2 matrices given by $A = [a_{ij}]$, where $a_{ij} \in \{0, 1, 2, 3, 4\}$ such that $a_{11} + a_{12} + a_{21} + a_{22} = 4$
- Find the number of matrices A such that the trace of A is equal to 4.
 - Find the number of matrices A such that A is invertible.
 - Find the absolute value of the difference between maximum value and minimum value of $\det(A)$.
 - Find the number of matrices A such that A is either symmetric or skew-symmetric or both and $\det(A)$ is divisible by 2.

2. For the matrix $A = \begin{bmatrix} 4 & -4 & 5 \\ -2 & 3 & -3 \\ 3 & -3 & 4 \end{bmatrix}$ find A^{-2} .

3. (a) Given $A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 4 & 1 \\ 2 & 3 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 2 & 3 \\ 3 & 4 \end{bmatrix}$. Find P such that $BPA = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$.

(b) Find the matrix A satisfying the matrix equation, $\begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix} \cdot A \cdot \begin{bmatrix} 3 & 2 \\ 5 & -3 \end{bmatrix} = \begin{bmatrix} 2 & 4 \\ 3 & -1 \end{bmatrix}$.

4. If $F(x) = \begin{bmatrix} \cos x & -\sin x & 0 \\ \sin x & \cos x & 0 \\ 0 & 0 & 1 \end{bmatrix}$ then show that $F(x) \cdot F(y) = F(x+y)$. Hence prove that $[F(x)]^{-1} = F(-x)$.

5. Let A be a 3×3 matrix such that $a_{11} = a_{33} = 2$ and all the other $a_{ij} = 1$. Let $A^{-1} = xA^2 + yA + zI$, then find the value of $(x + y + z)$ where I is a unit matrix of order 3.

6. Given that $A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 2 & 3 \\ 1 & -1 & 3 \end{bmatrix}$, $C = \begin{bmatrix} 2 & 1 & 1 \\ 2 & 2 & 1 \\ 1 & 1 & 1 \end{bmatrix}$, $D = \begin{bmatrix} 10 \\ 13 \\ 9 \end{bmatrix}$ and that $Cb = D$.

Solve the matrix equation $Ax = b$.

7. Let $A = \begin{bmatrix} 1 & \frac{3}{2} \\ 1 & 2 \end{bmatrix}$, $B = \begin{bmatrix} 4 & -3 \\ -2 & 2 \end{bmatrix}$ and $C_r = \begin{bmatrix} r \cdot 3^r & 2^r \\ 0 & (r-1)3^r \end{bmatrix}$ be 3 given matrices.

Compute the value of $\sum_{r=1}^{50} \text{tr}((AB)^r C_r)$. (where $\text{tr}(A)$ denotes trace of matrix A)

8. Let X be the solution set of the equation $A^x = I$, where $A = \begin{bmatrix} 0 & 1 & -1 \\ 4 & -3 & 4 \\ 3 & -3 & 4 \end{bmatrix}$ and I is the corresponding

unit matrix and $x \subseteq \mathbb{N}$ then find the minimum value of $\sum (\cos^x \theta + \sin^x \theta)$, $\theta \in \mathbb{R}$.

9. Consider the two matrices A and B where $A = \begin{bmatrix} 1 & 2 \\ 4 & 3 \end{bmatrix}$; $B = \begin{bmatrix} 5 \\ -3 \end{bmatrix}$. Let $n(A)$ denotes the number of elements in A and $n(XY) = \mathbf{O}$, when the two matrices X and Y are not conformable for multiplication.

If $C = (AB)(B'A)$; $D = (B'A)(AB)$ then, find the value of $\left(\frac{n(C)((|D|^2 + n(D)))}{n(A) - n(B)} \right)$.

10. Given $A = \begin{bmatrix} 2 & 1 \\ 2 & 1 \end{bmatrix}$; $B = \begin{bmatrix} 9 & 3 \\ 3 & 1 \end{bmatrix}$. I is a unit matrix of order 2. Find all possible matrix X in the following cases.

(a) $AX = A$ (b) $XA = I$ (c) $XB = \mathbf{O}$ but $BX \neq \mathbf{O}$.

11. Find the product of two matrices A & B, where $A = \begin{bmatrix} -5 & 1 & 3 \\ 7 & 1 & -5 \\ 1 & -1 & 1 \end{bmatrix}$ & $B = \begin{bmatrix} 1 & 1 & 2 \\ 3 & 2 & 1 \\ 2 & 1 & 3 \end{bmatrix}$ and use it to

solve the following system of linear equations,

$$x + y + 2z = 1; 3x + 2y + z = 7; 2x + y + 3z = 2.$$

12. Determine the values of a and b for which the system $\begin{bmatrix} 3 & -2 & 1 \\ 5 & -8 & 9 \\ 2 & 1 & a \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} b \\ 3 \\ -1 \end{bmatrix}$

(a) has a unique solution ; (b) has no solution and (c) has infinitely many solutions

13. If $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$; $B = \begin{bmatrix} 3 & 1 \\ 1 & 0 \end{bmatrix}$; $C = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$ and $X = \begin{bmatrix} x_1 & x_2 \\ x_3 & x_4 \end{bmatrix}$, then solve the following matrix equation.

(a) $AX = B - I$ (b) $(B - I)X = IC$ (c) $CX = A$

14. $A_{3 \times 3}$ is a matrix such that $|A| = a$, $B = (\text{adj } A)$ such that $|B| = b$. Find the value of $(ab^2 + a^2b + 1)S$ where $\frac{1}{2}S = \frac{a}{b} + \frac{a^2}{b^3} + \frac{a^3}{b^5} + \dots$ up to ∞ , and $a = 3$.

15. If A and B are square matrices of order 3, where $|A| = -2$ and $|B| = 1$, then find $\left| (A^{-1}) \text{adj}(B^{-1}) \text{adj}(2A^{-1}) \right|$.

EXERCISE (JM)

1. Let A be a 2×2 matrix [AIEEE- 2009]
Statement-1 : $\text{adj}(\text{adj } A) = A$
Statement-2 : $|\text{adj } A| = |A|$
 (1) Statement-1 is true, Statement-2 is false.
 (2) Statement-1 is false, Statement-2 is true.
 (3) Statement-1 is true, Statement-2 is true; Statement-2 is a correct explanation for Statement-1.
 (4) Statement-1 is true, Statement-2 is true; Statement-2 is not a correct explanation for statement-1.
2. The number of 3×3 non-singular matrices, with four entries as 1 and all other entries as 0, is :- [AIEEE-2010]
 (1) Less than 4 (2) 5 (3) 6 (4) At least 7
3. Let A be a 2×2 matrix with non-zero entries and let $A^2 = I$, where I is 2×2 identity matrix. Define $\text{Tr}(A)$ = sum of diagonal elements of A and $|A|$ = determinant of matrix A . [AIEEE-2010]
Statement-1 : $\text{Tr}(A) = 0$.
Statement-2 : $|A| = 1$.
 (1) Statement-1 is true, Statement-2 is true; Statement-2 is a correct explanation for Statement-1.
 (2) Statement-1 is true, Statement-2 is true; Statement-2 is not a correct explanation for statement-1.
 (3) Statement-1 is true, Statement-2 is false.
 (4) Statement-1 is false, Statement-2 is true.
4. Let A and B be two symmetric matrices of order 3.
Statement-1 : $A(BA)$ and $(AB)A$ are symmetric matrices.
Statement-2 : AB is symmetric matrix if matrix multiplication of A with B is commutative. [AIEEE-2011]
 (1) Statement-1 is true, Statement-2 is false.
 (2) Statement-1 is false, Statement-2 is true
 (3) Statement-1 is true, Statement-2 is true; Statement-2 is a correct explanation for Statement-1
 (4) Statement-1 is true, Statement-2 is true; Statement-2 is **not** a correct explanation for Statement-1.
5. **Statement-1** : Determinant of a skew-symmetric matrix of order 3 is zero.
Statement-1 : For any matrix A , $\det(A^T) = \det(A)$ and $\det(-A) = -\det(A)$.
 Where $\det(B)$ denotes the determinant of matrix B . Then : [AIEEE-2011]
 (1) Statement-1 is true and statement-2 is false
 (2) Both statements are true
 (3) Both statements are false
 (4) Statement-1 is false and statement-2 is true.
6. Let $A = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & 2 & 1 \end{pmatrix}$. If u_1 and u_2 are column matrices such that $Au_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$ and $Au_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$, then [AIEEE-2012]
 $u_1 + u_2$ is equal to :
 (1) $\begin{pmatrix} 1 \\ -1 \\ -1 \end{pmatrix}$ (2) $\begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}$ (3) $\begin{pmatrix} -1 \\ 1 \\ -1 \end{pmatrix}$ (4) $\begin{pmatrix} -1 \\ -1 \\ 0 \end{pmatrix}$

7. If $P = \begin{bmatrix} 1 & \alpha & 3 \\ 1 & 3 & 3 \\ 2 & 4 & 4 \end{bmatrix}$ is the adjoint of a 3×3 matrix A and $|A| = 4$, then α is equal to
- [JEE(Main) - 2013]
- (1) 4 (2) 11 (3) 5 (4) 0
8. If A is an 3×3 non-singular matrix such that $AA' = A'A$ and $B = A^{-1} A'$, the BB' equals :
- [JEE(Main) - 2014]
- (1) $I + B$ (2) I (3) B^{-1} (4) $(B^{-1})'$
9. If $A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ a & 2 & b \end{bmatrix}$ is a matrix satisfying the equation $AA^T = 9I$, where I is 3×3 identity matrix, then the ordered pair (a, b) is equal to :
- [JEE(Main)-2015]
- (1) (2, 1) (2) $(-2, -1)$ (3) (2, -1) (4) $(-2, 1)$
10. If $A = \begin{bmatrix} 5a & -b \\ 3 & 2 \end{bmatrix}$ and $A \text{ adj } A = AA^T$, then $5a + b$ is equal to :
- [JEE(Main)-2016]
- (1) 13 (2) -1 (3) 5 (4) 4
11. If $A = \begin{bmatrix} 2 & -3 \\ -4 & 1 \end{bmatrix}$, then $\text{adj } (3A^2 + 12A)$ is equal to :-
- [JEE(Main)-2017]
- (1) $\begin{bmatrix} 72 & -63 \\ -84 & 51 \end{bmatrix}$ (2) $\begin{bmatrix} 72 & -84 \\ -63 & 51 \end{bmatrix}$ (3) $\begin{bmatrix} 51 & 63 \\ 84 & 72 \end{bmatrix}$ (4) $\begin{bmatrix} 51 & 84 \\ 63 & 72 \end{bmatrix}$
12. Let A and B be two invertible matrices of order 3×3 . If $\det(ABA^T) = 8$ and $\det(AB^{-1}) = 8$, then $\det(BA^{-1} B^T)$ is equal to :-
- [JEE(Main) Jan-2019]
- (1) 16 (2) $\frac{1}{16}$ (3) $\frac{1}{4}$ (4) 1
13. Let $P = \begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 9 & 3 & 1 \end{bmatrix}$ and $Q = [q_{ij}]$ be two 3×3 matrices such that $Q - P^5 = I_3$. Then $\frac{q_{21} + q_{31}}{q_{32}}$ is equal to :
- [JEE(Main) Jan-2019]
- (1) 15 (2) 9 (3) 135 (4) 10
14. Let $A = \begin{pmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{pmatrix}$, $(\alpha \in \mathbb{R})$ such that $A^{32} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$. Then a value of α is
- [JEE(Main) Apr-2019]
- (1) $\frac{\pi}{16}$ (2) 0 (3) $\frac{\pi}{32}$ (4) $\frac{\pi}{64}$
15. If $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix} \cdots \begin{bmatrix} 1 & n-1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 78 \\ 0 & 1 \end{bmatrix}$, then the inverse of $\begin{bmatrix} 1 & n \\ 0 & 1 \end{bmatrix}$ is
- [JEE(Main) Apr-2019]
- (1) $\begin{bmatrix} 1 & -13 \\ 0 & 1 \end{bmatrix}$ (2) $\begin{bmatrix} 1 & 0 \\ 12 & 1 \end{bmatrix}$ (3) $\begin{bmatrix} 1 & -12 \\ 0 & 1 \end{bmatrix}$ (4) $\begin{bmatrix} 1 & 0 \\ 13 & 1 \end{bmatrix}$

16. If A is a symmetric matrix and B is a skew-symmetric matrix such that $A + B = \begin{bmatrix} 2 & 3 \\ 5 & -1 \end{bmatrix}$, then AB is equal to :

[JEE(Main) Apr-2019]

- (1) $\begin{bmatrix} -4 & 2 \\ 1 & 4 \end{bmatrix}$ (2) $\begin{bmatrix} -4 & -2 \\ -1 & 4 \end{bmatrix}$ (3) $\begin{bmatrix} 4 & -2 \\ -1 & -4 \end{bmatrix}$ (4) $\begin{bmatrix} 4 & -2 \\ 1 & -4 \end{bmatrix}$

EXERCISE (JA)

Comprehension (3 questions)

1. Let $\mathcal{A}$ be the set of all 3×3 symmetric matrices all of whose entries are either 0 or 1. Five of these entries are 1 and four of them are 0.

- (a) The number of matrices in $\mathcal{A}$ is -

- (A) 12 (B) 6 (C) 9 (D) 3

- (b) The number of matrices A in $\mathcal{A}$ for which the system of linear equations

$$A \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

has a unique solution, is -

- (A) less than 4 (B) at least 4 but less than 7
(C) at least 7 but less than 10 (D) at least 10

- (c) The number of matrices A in $\mathcal{A}$ for which the system of linear equations

$$A \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

is inconsistent, is -

- (A) 0 (B) more than 2 (C) 2 (D) 1

[JEE 2009, 4+4+4]

2. (a) The number of 3×3 matrices A whose entries are either 0 or 1 and for which the system

$$A \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \text{ has exactly two distinct solutions, is}$$

- (A) 0 (B) $2^9 - 1$ (C) 168 (D) 2

- (b) Let k be a positive real number and let

$$A = \begin{bmatrix} 2k-1 & 2\sqrt{k} & 2\sqrt{k} \\ 2\sqrt{k} & 1 & -2k \\ -2\sqrt{k} & 2k & -1 \end{bmatrix} \text{ and } B = \begin{bmatrix} 0 & 2k-1 & \sqrt{k} \\ 1-2k & 0 & 2\sqrt{k} \\ -\sqrt{k} & -2\sqrt{k} & 0 \end{bmatrix}.$$

If $\det(\text{adj } A) + \det(\text{adj } B) = 10^6$, then $[k]$ is equal to

[Note : $\text{adj } M$ denotes the adjoint of a square matrix M and $[k]$ denotes the largest integer less than or equal to k].

(c) Let p be an odd prime number and T_p be the following set of 2×2 matrices :

$$T_p = \left\{ A = \begin{bmatrix} a & b \\ c & a \end{bmatrix} : a, b, c \in \{0, 1, 2, \dots, p-1\} \right\}$$

- (i) The number of A in T_p such that A is either symmetric or skew-symmetric or both, and $\det(A)$ divisible by p is -
 (A) $(p-1)^2$ (B) $2(p-1)$ (C) $(p-1)^2 + 1$ (D) $2p-1$
- (ii) The number of A in T_p such that the trace of A is not divisible by p but $\det(A)$ is divisible by p is -

[Note : The trace of a matrix is the sum of its diagonal entries.]

- (A) $(p-1)(p^2-p+1)$ (B) $p^3-(p-1)^2$
 (C) $(p-1)^2$ (D) $(p-1)(p^2-2)$
- (iii) The number of A in T_p such that $\det(A)$ is not divisible by p is -
 (A) $2p^2$ (B) p^3-5p (C) p^3-3p (D) p^3-p^2

[JEE 2010, 3+3+3+3+3]

3. Let M and N be two 3×3 non-singular skew-symmetric matrices such that $MN = NM$. If P^T denotes the transpose of P , then $M^2N^2(M^TN)^{-1}(MN^{-1})^T$ is equal to - [JEE 2011, 4]
 (A) M^2 (B) $-N^2$ (C) $-M^2$ (D) MN

4. Let $\omega \neq 1$ be a cube root of unity and S be the set of all non-singular matrices of the form $\begin{bmatrix} 1 & a & b \\ \omega & 1 & c \\ \omega^2 & \omega & 1 \end{bmatrix}$,

where each of a, b and c is either ω or ω^2 . Then the number of distinct matrices in the set S is -
 (A) 2 (B) 6 (C) 4 (D) 8

[JEE 2011, 3, (-1)]

5. Let M be 3×3 matrix satisfying $M \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \\ 3 \end{bmatrix}$, $M \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$ and $M \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 12 \end{bmatrix}$

Then the sum of the diagonal entries of M is

[JEE 2011, 4]

6. Let $P = [a_{ij}]$ be a 3×3 matrix and let $Q = [b_{ij}]$, where $b_{ij} = 2^{i+j}a_{ij}$ for $1 \leq i, j \leq 3$. If the determinant of P is 2, then the determinant of the matrix Q is - [JEE 2012, 3M, -1M]
 (A) 2^{10} (B) 2^{11} (C) 2^{12} (D) 2^{13}
7. If P is a 3×3 matrix such that $P^T = 2P + I$, where P^T is the transpose of P and I is the 3×3 identity

matrix, then there exists a column matrix $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \neq \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$ such that [JEE 2012, 3M, -1M]

- (A) $PX = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$ (B) $PX = X$ (C) $PX = 2X$ (D) $PX = -X$

14. Let $P = \begin{bmatrix} 1 & 0 & 0 \\ 4 & 1 & 0 \\ 16 & 4 & 1 \end{bmatrix}$ and I be the identity matrix of order 3. If $Q = [q_{ij}]$ is a matrix such that

$$P^{50} - Q = I, \text{ then } \frac{q_{31} + q_{32}}{q_{21}} \text{ equals}$$

[JEE(Advanced)-2016, 3(-1)]

- (A) 52 (B) 103 (C) 201 (D) 205

15. Which of the following is(are) NOT the square of a 3×3 matrix with real entries ?

[JEE(Advanced)-2017, 4(-2)]

(A) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$ (B) $\begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$ (C) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ (D) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$

16. How many 3×3 matrices M with entries from $\{0,1,2\}$ are there, for which the sum of the diagonal entries of $M^T M$ is 5 ?

[JEE(Advanced)-2017, 3(-1)]

- (A) 198 (B) 126 (C) 135 (D) 162

17. For a real number α , if the system

$$\begin{bmatrix} 1 & \alpha & \alpha^2 \\ \alpha & 1 & \alpha \\ \alpha^2 & \alpha & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$$

of linear equations, has infinitely many solutions, then $1 + \alpha + \alpha^2 =$ [JEE(Advanced)-2017, 3]

18. Let S be the set of all column matrices $\begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$ such that $b_1, b_2, b_3 \in \mathbb{R}$ and the system of equations (in real variables)

$$\begin{aligned} -x + 2y + 5z &= b_1 \\ 2x - 4y + 3z &= b_2 \\ x - 2y + 2z &= b_3 \end{aligned}$$

has at least one solution. Then, which of the following system(s) (in real variables) has (have) at

least one solution of each $\begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} \in S$?

[JEE(Advanced)-2018, 4(-2)]

- (A) $x + 2y + 3z = b_1$, $4y + 5z = b_2$ and $x + 2y + 6z = b_3$
 (B) $x + y + 3z = b_1$, $5x + 2y + 6z = b_2$ and $-2x - y - 3z = b_3$
 (C) $-x + 2y - 5z = b_1$, $2x - 4y + 10z = b_2$ and $x - 2y + 5z = b_3$
 (D) $x + 2y + 5z = b_1$, $2x + 3z = b_2$ and $x + 4y - 5z = b_3$

19. Let P be a matrix of order 3×3 such that all the entries in P are from the set $\{-1, 0, 1\}$. Then, the maximum possible value of the determinant of P is _____ [JEE(Advanced)-2018, 3(0)]

ANSWER KEY

EXERCISE (O-1)

1. C 2. A 3. B 4. A 5. A 6. A 7. D
 8. B 9. B 10. C 11. C 12. B 13. A 14. C
 15. B 16. C 17. A 18. A 19. D 20. A

EXERCISE (O-2)

1. A 2. C 3. C 4. C 5. D 6. C 7. D
 8. D 9. A,B 10. B,C,D 11. A,B,D 12. B,C 13. A,C 14. A,B,C
 15. A,B,C,D 16. B,D 17. A,C 18. A,B,C,D 19. A,B,C,D
 20. B,C 21. C 22. A 23. A 24. B 25. C

EXERCISE (S-1)

1. 10 2. 8 3. $x = \frac{3}{2}, y = 2$ 4. 5049 5. $V = \begin{bmatrix} 0 \\ 1 \\ 11 \end{bmatrix}$ 7. 1
 8. $f(a) = 1/4, a = 1/2$ 10. $\begin{bmatrix} 1 & 0 \\ 4014 & 1 \end{bmatrix}$ 11. AB is neither symmetric nor skew symmetric
 12. $\begin{bmatrix} 1 & 0 & 0 \\ 2 & 3 & 0 \\ -1 & 0 & 4 \end{bmatrix} + \begin{bmatrix} 0 & 2 & 5 \\ 0 & 0 & -6 \\ 0 & 0 & 0 \end{bmatrix}; \begin{bmatrix} 1 & 2 & 2 \\ 2 & 3 & -3 \\ 2 & -3 & 4 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 3 \\ 0 & 0 & -3 \\ -3 & 3 & 0 \end{bmatrix}$ 13. (a) 4, (b) 8
 14. $n = 7$ 15. 100 16. $f = -(a + d); g = ad - bc$ 17. 7

EXERCISE (S-2)

1. (i) 5, (ii) 18, (iii) 8, (iv) 5 2. $\begin{bmatrix} 17 & 4 & -19 \\ -10 & 0 & 13 \\ -21 & -3 & 25 \end{bmatrix}$ 3. (a) $\begin{bmatrix} -4 & 7 & -7 \\ 3 & -5 & 5 \end{bmatrix};$ (b) $\frac{1}{19} \begin{bmatrix} 48 & -25 \\ -70 & 42 \end{bmatrix}$
 5. 1 6. $x_1 = 1, x_2 = -1, x_3 = 1$ 7. $3(49.3^{50} + 1)$ 8. 2 9. 650

10. (i) $X = \begin{bmatrix} a & b \\ 2-2a & 1-2b \end{bmatrix}$ for $a, b \in \mathbb{R}$; (ii) X does not exist;

(iii) $X = \begin{bmatrix} a & -3a \\ c & -3c \end{bmatrix}$ $a, c \in \mathbb{R}$ and $3a + c \neq 0$; $3b + d \neq 0$

11. $x = 2, y = 1, z = -1$ 12. (i) $a \neq -3, b \in \mathbb{R}$; (ii) $a = -3$ and $b \neq 1/3$; (iii) $a = -3, b = 1/3$

13. (a) $X = \begin{bmatrix} -3 & -3 \\ 5 & 2 \\ 2 & 2 \end{bmatrix}$, (b) $X = \begin{bmatrix} 1 & 2 \\ -1 & -2 \end{bmatrix}$, (c) no solution 14. 225 15. -8

EXERCISE (JM)

1. 4 2. 4 3. 3 4. 4 5. 1 6. 1 7. 2 8. 2 9. 2 10. 3
11. 3 12. 2 13. 4 14. 4 15. 1 16. 3

EXERCISE (JA)

1. (a) A, (b) B, (c) B 2. (a) A, (b) 4; (c) (i) D, (ii) C, (iii) D 3. Bonus 4. A
5. 9 6. D 7. D 8. A, D 9. C, D 10. C, D 11. A, B
12. C, D 13. B, C 14. B 15. A, B 16. A 17. 1 18. A, D 19. 4

NURTURE COURSE

DETERMINANT

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DETERMINANT

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JEE (Main) Syllabus :

Matrices, algebra of matrices, types of matrices, determinants and matrices of order two and three. Properties of determinants, evaluation of determinants, area of triangles using determinants. Adjoint and evaluation of inverse of a square matrix using determinants and elementary transformations. Test of consistency and solution of simultaneous linear equations in two or three variables using determinants and matrices.

DETERMINANT

1. INTRODUCTION :

If the equations $a_1x + b_1 = 0$, $a_2x + b_2 = 0$ are satisfied by the same value of x , then $a_1b_2 - a_2b_1 = 0$. The expression $a_1b_2 - a_2b_1$ is called a determinant of the second order, and is denoted by :

$$\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}$$

A determinant of second order consists of two rows and two columns.

Next consider the system of equations $a_1x + b_1y + c_1 = 0$, $a_2x + b_2y + c_2 = 0$, $a_3x + b_3y + c_3 = 0$

If these equations are satisfied by the same values of x and y , then on eliminating x and y we get.

$$a_1(b_2c_3 - b_3c_2) + b_1(c_2a_3 - c_3a_2) + c_1(a_2b_3 - a_3b_2) = 0$$

The expression on the left is called a determinant of the third order, and is denoted by

$$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$$

A determinant of third order consists of three rows and three columns.

2. VALUE OF A DETERMINANT :

$$D = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = a_1 \begin{vmatrix} b_2 & c_2 \\ b_3 & c_3 \end{vmatrix} - b_1 \begin{vmatrix} a_2 & c_2 \\ a_3 & c_3 \end{vmatrix} + c_1 \begin{vmatrix} a_2 & b_2 \\ a_3 & b_3 \end{vmatrix} = a_1(b_2c_3 - b_3c_2) - b_1(a_2c_3 - a_3c_2) + c_1(a_2b_3 - a_3b_2)$$

Note : Sarrus diagram to get the value of determinant of order three :

$$D = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} \begin{matrix} \nearrow \nearrow \nearrow \\ \searrow \searrow \searrow \\ \nearrow \nearrow \nearrow \\ \searrow \searrow \searrow \end{matrix} = (a_1b_2c_3 + a_2b_3c_1 + a_3b_1c_2) - (a_3b_2c_1 + a_2b_1c_3 + a_1b_3c_2)$$

-ve -ve -ve
+ve +ve +ve

Note that the product of the terms in first bracket (i.e. $a_1a_2a_3b_1b_2b_3c_1c_2c_3$) is same as the product of the terms in second bracket.

Illustration 1 : The value of $\begin{vmatrix} 1 & 2 & 3 \\ -4 & 3 & 6 \\ 2 & -7 & 9 \end{vmatrix}$ is -

(A) 213

(B) -231

(C) 231

(D) 39

Solution :

$$\begin{vmatrix} 1 & 2 & 3 \\ -4 & 3 & 6 \\ 2 & -7 & 9 \end{vmatrix} = 1 \begin{vmatrix} 3 & 6 \\ -7 & 9 \end{vmatrix} - 2 \begin{vmatrix} -4 & 6 \\ 2 & 9 \end{vmatrix} + 3 \begin{vmatrix} -4 & 3 \\ 2 & -7 \end{vmatrix}$$

$$= (27 + 42) - 2(-36 - 12) + 3(28 - 6) = 231$$

Alternative : By sarrus diagram

$$\begin{vmatrix} 1 & 2 & 3 \\ -4 & 3 & 6 \\ 2 & -7 & 9 \end{vmatrix} = \begin{vmatrix} 1 & 2 & 3 & 1 & 2 & 3 \\ -4 & 3 & 6 & -4 & 3 & 6 \\ 2 & -7 & 9 & 2 & -7 & 9 \end{vmatrix}$$

$$= (27 + 24 + 84) - (18 - 42 - 72) = 135 - (18 - 114) = 231$$

Ans. (C)

3. MINORS & COFACTORS :

The minor of a given element of determinant is the determinant obtained by deleting the row & the column in which the given element stands.

For example, the minor of a_1 in $\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$ is $\begin{vmatrix} b_2 & c_2 \\ b_3 & c_3 \end{vmatrix}$ & the minor of b_2 is $\begin{vmatrix} a_1 & c_1 \\ a_3 & c_3 \end{vmatrix}$.

Hence a determinant of order three will have "9 minors".

If M_{ij} represents the minor of the element belonging to i^{th} row and j^{th} column then the cofactor of that element is given by : $C_{ij} = (-1)^{i+j} \cdot M_{ij}$

Illustration 2 : Find the minors and cofactors of elements '-3', '5', '-1' & '7' in the determinant

$$\begin{vmatrix} 2 & -3 & 1 \\ 4 & 0 & 5 \\ -1 & 6 & 7 \end{vmatrix}$$

Solution :

$$\text{Minor of } -3 = \begin{vmatrix} 4 & 5 \\ -1 & 7 \end{vmatrix} = 33 ; \text{ Cofactor of } -3 = -33$$

$$\text{Minor of } 5 = \begin{vmatrix} 2 & -3 \\ -1 & 6 \end{vmatrix} = 9 ; \text{ Cofactor of } 5 = -9$$

$$\text{Minor of } -1 = \begin{vmatrix} -3 & 1 \\ 0 & 5 \end{vmatrix} = -15 ; \text{ Cofactor of } -1 = -15$$

$$\text{Minor of } 7 = \begin{vmatrix} 2 & -3 \\ 4 & 0 \end{vmatrix} = 12 ; \text{ Cofactor of } 7 = 12$$

4. EXPANSION OF A DETERMINANT IN TERMS OF THE ELEMENTS OF ANY ROW OR COLUMN:

$$\text{Let } D = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$$

- (i) The sum of the product of elements of any row (column) with their corresponding cofactors is always equal to the value of the determinant.

D can be expressed in any of the six forms :

$$a_1A_1 + b_1B_1 + c_1C_1, \quad a_1A_1 + a_2A_2 + a_3A_3,$$

$$a_2A_2 + b_2B_2 + c_2C_2, \quad b_1B_1 + b_2B_2 + b_3B_3,$$

$$a_3A_3 + b_3B_3 + c_3C_3, \quad c_1C_1 + c_2C_2 + c_3C_3,$$

where A_i, B_i & C_i ($i = 1, 2, 3$) denote cofactors of a_i, b_i & c_i respectively.

- (ii) The sum of the product of elements of any row (column) with the cofactors of other row (column) is always equal to zero.

Hence,

$$a_2A_1 + b_2B_1 + c_2C_1 = 0,$$

$$b_1A_1 + b_2A_2 + b_3A_3 = 0 \text{ and so on.}$$

where A_i, B_i & C_i ($i = 1, 2, 3$) denote cofactors of a_i, b_i & c_i respectively.

Do yourself -1 :

- (i) Find minors & cofactors of elements '6', '5', '0' & '4' of the determinant $\begin{vmatrix} 2 & 1 & 3 \\ 6 & 5 & 7 \\ 3 & 0 & 4 \end{vmatrix}$.

- (ii) Calculate the value of the determinant $\begin{vmatrix} 5 & -3 & 7 \\ -2 & 4 & -8 \\ 9 & 3 & -10 \end{vmatrix}$

- (iii) The value of the determinant $\begin{vmatrix} a & b & 0 \\ 0 & a & b \\ b & 0 & a \end{vmatrix}$ is equal to -

(A) $a^3 - b^3$

(B) $a^3 + b^3$

(C) 0

(D) none of these

- (iv) Find the value of 'k', if $\begin{vmatrix} 1 & 2 & 0 \\ 2 & 3 & 1 \\ 3 & k & 2 \end{vmatrix} = 4$

5. PROPERTIES OF DETERMINANTS :

- (a) The value of a determinant remains unaltered, if the rows & columns are inter-changed,

e.g. if $D = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$

- (b) If any two rows (or columns) of a determinant be interchanged, the value of determinant is changed in sign only. e.g.

Let $D = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$ & $D_1 = \begin{vmatrix} a_2 & b_2 & c_2 \\ a_1 & b_1 & c_1 \\ a_3 & b_3 & c_3 \end{vmatrix}$. Then $D_1 = -D$.

- (c) If all the elements of a row (or column) are zero, then the value of the determinant is zero.
(d) If all the elements of any row (or column) are multiplied by the same number, then the determinant is multiplied by that number.

e.g. If $D = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$ and $D_1 = \begin{vmatrix} Ka_1 & Kb_1 & Kc_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$. Then $D_1 = KD$

- (e) If all the elements of a row (or column) are proportional (or identical) to the element of any other row, then the determinant vanishes, i.e. its value is zero.

e.g. If $D = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_1 & b_1 & c_1 \\ a_3 & b_3 & c_3 \end{vmatrix} \Rightarrow D = 0$; If $D_1 = \begin{vmatrix} a_1 & b_1 & c_1 \\ ka_1 & kb_1 & kc_1 \\ a_3 & b_3 & c_3 \end{vmatrix} \Rightarrow D_1 = 0$

Illustration 3 : Prove that $\begin{vmatrix} a & b & c \\ x & y & z \\ p & q & r \end{vmatrix} = \begin{vmatrix} y & b & q \\ x & a & p \\ z & c & r \end{vmatrix}$

Solution : $D = \begin{vmatrix} a & b & c \\ x & y & z \\ p & q & r \end{vmatrix} = \begin{vmatrix} a & x & p \\ b & y & q \\ c & z & r \end{vmatrix}$ (By interchanging rows & columns)

$$= - \begin{vmatrix} x & a & p \\ y & b & q \\ z & c & r \end{vmatrix} \quad (C_1 \leftrightarrow C_2)$$

$$= \begin{vmatrix} y & b & q \\ x & a & p \\ z & c & r \end{vmatrix} \quad (R_1 \leftrightarrow R_2)$$

Illustration 4 : Find the value of the determinant $\begin{vmatrix} a^2 & ab & ac \\ ab & b^2 & bc \\ ac & bc & c^2 \end{vmatrix}$

Solution : $D = \begin{vmatrix} a^2 & ab & ac \\ ab & b^2 & bc \\ ac & bc & c^2 \end{vmatrix} = a \begin{vmatrix} a & b & c \\ ab & b^2 & bc \\ ac & bc & c^2 \end{vmatrix} = abc \begin{vmatrix} a & b & c \\ a & b & c \\ a & b & c \end{vmatrix} = 0$

Since all rows are same, hence value of the determinant is zero.

Do yourself -2 :

(i) Without expanding the determinant prove that $\begin{vmatrix} a & p & \ell \\ b & q & m \\ c & r & n \end{vmatrix} + \begin{vmatrix} r & n & c \\ q & m & b \\ p & \ell & a \end{vmatrix} = 0$

(ii) If $D = \begin{vmatrix} \alpha & \beta \\ \gamma & \delta \end{vmatrix}$, then $\begin{vmatrix} 2\alpha & 2\beta \\ 2\gamma & 2\delta \end{vmatrix}$ is equal to -

(A) D

(B) 2D

(C) 4D

(D) 16D

- (f) If each element of any row (or column) is expressed as a sum of two (or more) terms, then the determinant can be expressed as the sum of two (or more) determinants.

$$\text{e.g. } \begin{vmatrix} a_1 + x & b_1 + y & c_1 + z \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} + \begin{vmatrix} x & y & z \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$$

Note that : If $D_r = \begin{vmatrix} f(r) & g(r) & h(r) \\ a & b & c \\ a_1 & b_1 & c_1 \end{vmatrix}$

where $r \in \mathbb{N}$ and a, b, c, a_1, b_1, c_1 are constants, then

$$\sum_{r=1}^n D_r = \begin{vmatrix} \sum_{r=1}^n f(r) & \sum_{r=1}^n g(r) & \sum_{r=1}^n h(r) \\ a & b & c \\ a_1 & b_1 & c_1 \end{vmatrix}$$

- (g) **Row - column operation :** The value of a determinant remains unaltered under a column (C_i) operation of the form $C_i \rightarrow C_i + \alpha C_j + \beta C_k$ ($j, k \neq i$) or row (R_i) operation of the form $R_i \rightarrow R_i + \alpha R_j + \beta R_k$ ($j, k \neq i$). In other words, the value of a determinant is not altered by adding the elements of any row (or column) to the same multiples of the corresponding elements of any other row (or column)

$$\text{e.g. Let } D = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$$

$$D = \begin{vmatrix} a_1 + \alpha a_2 & b_1 + \alpha b_2 & c_1 + \alpha c_2 \\ a_2 & b_2 & c_2 \\ a_3 + \beta a_1 & b_3 + \beta b_1 & c_3 + \beta c_1 \end{vmatrix} \quad (R_1 \rightarrow R_1 + \alpha R_2; R_3 \rightarrow R_3 + \beta R_2)$$

Note :

- (i) By using the operation $R_i \rightarrow xR_i + yR_j + zR_k$ ($j, k \neq i$), the value of the determinant becomes x times the original one.
- (ii) While applying this property **ATLEAST ONE ROW (OR COLUMN)** must remain unchanged.

Illustration 5 : If $D_r = \begin{vmatrix} r & r^3 & 2 \\ n & n^3 & 2n \\ \frac{n(n+1)}{2} & \left(\frac{n(n+1)}{2}\right)^2 & 2(n+1) \end{vmatrix}$, find $\sum_{r=0}^n D_r$.

Solution :

$$\sum_{r=0}^n D_r = \begin{vmatrix} \sum_{r=0}^n r & \sum_{r=0}^n r^3 & \sum_{r=0}^n 2 \\ n & n^3 & 2n \\ \frac{n(n+1)}{2} & \left(\frac{n(n+1)}{2}\right)^2 & 2(n+1) \end{vmatrix} = \begin{vmatrix} \frac{n(n+1)}{2} & \left(\frac{n(n+1)}{2}\right)^2 & 2(n+1) \\ n & n^3 & 2n \\ \frac{n(n+1)}{2} & \left(\frac{n(n+1)}{2}\right)^2 & 2(n+1) \end{vmatrix} = 0 \quad \text{Ans.}$$

Illustration 6 : If $\begin{vmatrix} 3^2+k & 4^2 & 3^2+3+k \\ 4^2+k & 5^2 & 4^2+4+k \\ 5^2+k & 6^2 & 5^2+5+k \end{vmatrix} = 0$, then the value of k is-

- (A) 2 (B) 1 (C) -1 (D) 0

Solution : Applying $(C_3 \rightarrow C_3 - C_1)$

$$D = \begin{vmatrix} 3^2+k & 4^2 & 3 \\ 4^2+k & 5^2 & 4 \\ 5^2+k & 6^2 & 5 \end{vmatrix} = 0$$

$$\Rightarrow \begin{vmatrix} 9+k & 16 & 3 \\ 7 & 9 & 1 \\ 9 & 11 & 1 \end{vmatrix} = 0 \quad (R_3 \rightarrow R_3 - R_2; R_2 \rightarrow R_2 - R_1)$$

$$\Rightarrow k - 1 = 0 \Rightarrow k = 1$$

Ans. (B)

Do yourself - 3 :

(i) Find the value of $\begin{vmatrix} 53 & 106 & 159 \\ 52 & 65 & 91 \\ 102 & 153 & 221 \end{vmatrix}$.

(ii) Solve for x : $\begin{vmatrix} x & 2 & 0 \\ 2+x & 5 & -1 \\ 5-x & 1 & 2 \end{vmatrix} = 0$

(iii) If $D_r = \begin{vmatrix} 2r & 1 & n \\ 1 & -2 & 3 \\ 3 & 2 & 1 \end{vmatrix}$, then find the value of $\sum_{r=1}^n D_r$.

(h) **Factor theorem :** If the elements of a determinant D are rational integral functions of x and two rows (or columns) become identical when $x = a$ then $(x - a)$ is a factor of D .

Note that if r rows become identical when a is substituted for x , then $(x - a)^{r-1}$ is a factor of D .

Illustration 7 : Prove that $\begin{vmatrix} a & a & x \\ m & m & m \\ b & x & b \end{vmatrix} = m(x - a)(x - b)$

Solution : Using factor theorem,
Put $x = a$

$$D = \begin{vmatrix} a & a & a \\ m & m & m \\ b & a & b \end{vmatrix} = 0$$

Since R_1 and R_2 are proportional which makes $D = 0$, therefore $(x - a)$ is a factor of D .

Similarly, by putting $x = b$, D becomes zero, therefore $(x - b)$ is a factor of D .

$$D = \begin{vmatrix} a & a & x \\ m & m & m \\ b & x & b \end{vmatrix} = \lambda(x - a)(x - b) \quad \dots\dots\dots(i)$$

To get the value of λ , put $x = 0$ in equation (i)

$$\begin{vmatrix} a & a & 0 \\ m & m & m \\ b & 0 & b \end{vmatrix} = \lambda ab$$

$$amb = \lambda ab \Rightarrow \lambda = m$$

$$\therefore D = m(x-a)(x-b)$$

Do yourself - 4 :

(i) Without expanding the determinant prove that $\begin{vmatrix} 1 & a & bc \\ 1 & b & ca \\ 1 & c & ab \end{vmatrix} = (a-b)(b-c)(c-a)$

(ii) Using factor theorem, find the solution set of the equation $\begin{vmatrix} 1 & 4 & 20 \\ 1 & -2 & 5 \\ 1 & 2x & 5x^2 \end{vmatrix} = 0$

6. MULTIPLICATION OF TWO DETERMINANTS :

$$\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix} \times \begin{vmatrix} l_1 & m_1 \\ l_2 & m_2 \end{vmatrix} = \begin{vmatrix} a_1 l_1 + b_1 l_2 & a_1 m_1 + b_1 m_2 \\ a_2 l_1 + b_2 l_2 & a_2 m_1 + b_2 m_2 \end{vmatrix}$$

Similarly two determinants of order three are multiplied.

- (a) Here we have multiplied row by column. We can also multiply row by row, column by row and column by column.
- (b) If D_1 is the determinant formed by replacing the elements of determinant D of order n by their corresponding cofactors then $D_1 = D^{n-1}$

Illustration 8 : Let α & β be the roots of equation $ax^2 + bx + c = 0$ and $S_n = \alpha^n + \beta^n$ for $n \geq 1$. Evaluate

the value of the determinant $\begin{vmatrix} 3 & 1+S_1 & 1+S_2 \\ 1+S_1 & 1+S_2 & 1+S_3 \\ 1+S_2 & 1+S_3 & 1+S_4 \end{vmatrix}$.

Solution :

$$D = \begin{vmatrix} 3 & 1+S_1 & 1+S_2 \\ 1+S_1 & 1+S_2 & 1+S_3 \\ 1+S_2 & 1+S_3 & 1+S_4 \end{vmatrix} = \begin{vmatrix} 1+1+1 & 1+\alpha+\beta & 1+\alpha^2+\beta^2 \\ 1+\alpha+\beta & 1+\alpha^2+\beta^2 & 1+\alpha^3+\beta^3 \\ 1+\alpha^2+\beta^2 & 1+\alpha^3+\beta^3 & 1+\alpha^4+\beta^4 \end{vmatrix}$$

$$= \begin{vmatrix} 1 & 1 & 1 \\ 1 & \alpha & \alpha^2 \\ 1 & \beta & \beta^2 \end{vmatrix} \times \begin{vmatrix} 1 & 1 & 1 \\ 1 & \alpha & \beta \\ 1 & \alpha^2 & \beta^2 \end{vmatrix} = \begin{vmatrix} 1 & 1 & 1 \\ 1 & \alpha & \alpha^2 \\ 1 & \beta & \beta^2 \end{vmatrix}^2 = [(1-\alpha)(1-\beta)(\alpha-\beta)]^2$$

$$D = (\alpha - \beta)^2 (\alpha + \beta - \alpha\beta - 1)^2$$

$\therefore \alpha$ & β are roots of the equation $ax^2 + bx + c = 0$

$$\Rightarrow \alpha + \beta = \frac{-b}{a} \quad \& \quad \alpha\beta = \frac{c}{a} \Rightarrow |\alpha - \beta| = \frac{\sqrt{b^2 - 4ac}}{|a|}$$

$$D = \frac{(b^2 - 4ac)}{a^2} \left(\frac{a+b+c}{a} \right)^2 = \frac{(b^2 - 4ac)(a+b+c)^2}{a^4}$$

Ans.

Do yourself - 5 :

(i) If the determinant $D = \begin{vmatrix} 1 & 1 & 1 \\ \alpha + \beta & \alpha^2 + \beta^2 & 2\alpha\beta \\ \alpha + \beta & 2\alpha\beta & \alpha^2 + \beta^2 \end{vmatrix}$ and $D_1 = \begin{vmatrix} 1 & 0 & 0 \\ 0 & \alpha & \beta \\ 0 & \beta & \alpha \end{vmatrix}$, then find the determinant

D_2 such that $D_2 = \frac{D}{D_1}$.

(ii) If $D_1 = \begin{vmatrix} ab^2 - ac^2 & bc^2 - a^2b & a^2c - b^2c \\ ac - ab & ab - bc & bc - ac \\ c - b & a - c & b - a \end{vmatrix}$ & $D_2 = \begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ bc & ac & ab \end{vmatrix}$, then $D_1 D_2$ is equal to -

(A) 0

(B) D_1^2

(C) D_2^2

(D) D_2^3

7. SPECIAL DETERMINANTS :**(a) Cyclic Determinant :**

The elements of the rows (or columns) are in cyclic arrangement.

$$\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} = -(a^3 + b^3 + c^3 - 3abc) = -(a + b + c)(a^2 + b^2 + c^2 - ab - bc - ac)$$

$$= -\frac{1}{2}(a + b + c) \times \{(a - b)^2 + (b - c)^2 + (c - a)^2\}$$

$$= -(a + b + c)(a + b\omega + c\omega^2)(a + b\omega^2 + c\omega), \text{ where } \omega, \omega^2 \text{ are cube roots of unity}$$

(b) Other Important Determinants :

(i) $\begin{vmatrix} 0 & b & -c \\ -b & 0 & a \\ c & -a & 0 \end{vmatrix} = 0$

(ii) $\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ bc & ac & ab \end{vmatrix} = \begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^2 & b^2 & c^2 \end{vmatrix} = (a - b)(b - c)(c - a)$

(iii) $\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^3 & b^3 & c^3 \end{vmatrix} = (a - b)(b - c)(c - a)(a + b + c)$

(iv) $\begin{vmatrix} 1 & 1 & 1 \\ a^2 & b^2 & c^2 \\ a^3 & b^3 & c^3 \end{vmatrix} = (a - b)(b - c)(c - a)(ab + bc + ca)$

(v) $\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^4 & b^4 & c^4 \end{vmatrix} = (a - b)(b - c)(c - a)(a^2 + b^2 + c^2 - ab - bc - ca)$

Illustration 9 : Prove that $\begin{vmatrix} 1 & \alpha & \alpha^2 \\ \alpha & \alpha^2 & 1 \\ \alpha^2 & 1 & \alpha \end{vmatrix} = -(1 - \alpha^3)^2$.

Solution : This is a cyclic determinant.

$$\begin{aligned} \Rightarrow \begin{vmatrix} 1 & \alpha & \alpha^2 \\ \alpha & \alpha^2 & 1 \\ \alpha^2 & 1 & \alpha \end{vmatrix} &= -(1 + \alpha + \alpha^2)(1 + \alpha^2 + \alpha^4 - \alpha - \alpha^2 - \alpha^3) \\ &= -(1 + \alpha + \alpha^2)(-\alpha + 1 - \alpha^3 + \alpha^4) = -(1 + \alpha + \alpha^2)(1 - \alpha)^2(1 + \alpha + \alpha^2) \\ &= -(1 - \alpha)^2(1 + \alpha + \alpha^2)^2 = -(1 - \alpha^3)^2 \end{aligned}$$

Do yourself - 6 :

(i) The value of the determinant $\begin{vmatrix} ka & k^2 + a^2 & 1 \\ kb & k^2 + b^2 & 1 \\ kc & k^2 + c^2 & 1 \end{vmatrix}$ is

(A) $k(a + b)(b + c)(c + a)$

(B) $kabc(a^2 + b^2 + c^2)$

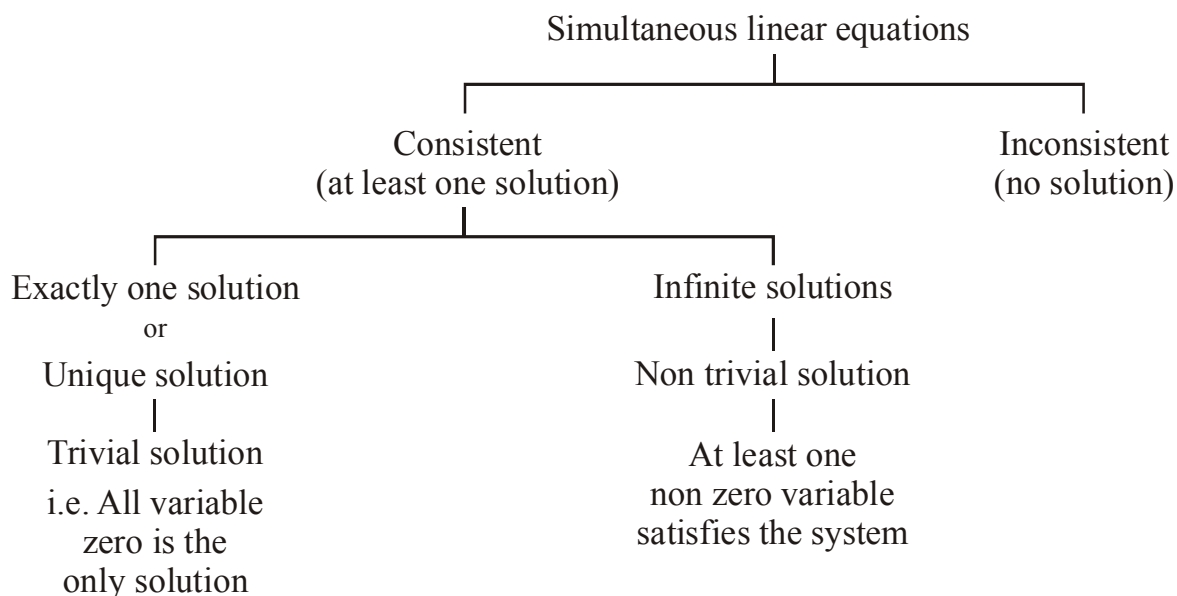
(C) $k(a - b)(b - c)(c - a)$

(D) $k(a + b - c)(b + c - a)(c + a - b)$

(ii) Find the value of the determinant $\begin{vmatrix} a^2 + b^2 & a^2 - c^2 & a^2 - c^2 \\ -a^2 & 0 & c^2 - a^2 \\ b^2 & -c^2 & b^2 \end{vmatrix}$.

(iii) Prove that $\begin{vmatrix} a & b & c \\ bc & ca & ab \\ b + c & c + a & a + b \end{vmatrix} = (a + b + c)(a - b)(b - c)(c - a)$.

8. CRAMER'S RULE (SYSTEM OF LINEAR EQUATIONS) :



(a) Equations involving two variables :

- (i) Consistent Equations : Definite & unique solution (Intersecting lines)
 (ii) Inconsistent Equations : No solution (Parallel lines)
 (iii) Dependent Equations : Infinite solutions (Identical lines)

Let, $a_1x + b_1y + c_1 = 0$
 $a_2x + b_2y + c_2 = 0$ then :

- (1) $\frac{a_1}{a_2} \neq \frac{b_1}{b_2} \Rightarrow$ Given equations are consistent with unique solution
 (2) $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2} \Rightarrow$ Given equations are inconsistent
 (3) $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2} \Rightarrow$ Given equations are consistent with infinite solutions

(b) Equations Involving Three variables :

Let $a_1x + b_1y + c_1z = d_1$ (i)

$a_2x + b_2y + c_2z = d_2$ (ii)

$a_3x + b_3y + c_3z = d_3$ (iii)

Then, $x = \frac{D_1}{D}$, $y = \frac{D_2}{D}$, $z = \frac{D_3}{D}$.

$$\text{Where } D = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}; D_1 = \begin{vmatrix} d_1 & b_1 & c_1 \\ d_2 & b_2 & c_2 \\ d_3 & b_3 & c_3 \end{vmatrix}; D_2 = \begin{vmatrix} a_1 & d_1 & c_1 \\ a_2 & d_2 & c_2 \\ a_3 & d_3 & c_3 \end{vmatrix} \text{ \& } D_3 = \begin{vmatrix} a_1 & b_1 & d_1 \\ a_2 & b_2 & d_2 \\ a_3 & b_3 & d_3 \end{vmatrix}$$

Note :

- (i) If $D \neq 0$ and atleast one of $D_1, D_2, D_3 \neq 0$, then the given system of equations is consistent and has unique non trivial solution.
 (ii) If $D \neq 0$ & $D_1 = D_2 = D_3 = 0$, then the given system of equations is consistent and has trivial solution only.
 (iii) If $D = 0$ but atleast one of D_1, D_2, D_3 is not zero then the equations are inconsistent and have no solution.
 (iv) If $D = D_1 = D_2 = D_3 = 0$, then the given system of equations may have infinite or no solution.

Note that In case $\left. \begin{matrix} a_1x + b_1y + c_1z = d_1 \\ a_1x + b_1y + c_1z = d_2 \\ a_1x + b_1y + c_1z = d_3 \end{matrix} \right\}$ (Atleast two of d_1, d_2 & d_3 are not equal)

$D = D_1 = D_2 = D_3 = 0$. But these three equations represent three parallel planes. Hence the system is inconsistent.

(c) **Homogeneous system of linear equations :**

If x, y, z are not all zero, the condition for

$$a_1x + b_1y + c_1z = 0$$

$$a_2x + b_2y + c_2z = 0$$

$$a_3x + b_3y + c_3z = 0$$

to be consistent in x, y, z is that
$$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = 0.$$

Remember that if a given system of linear equations have **Only Zero** Solution for all its variables then the given equations are said to have **TRIVIAL SOLUTION**.

9. APPLICATION OF DETERMINANTS IN GEOMETRY :

(a) The lines : $a_1x + b_1y + c_1 = 0$ (i)

$$a_2x + b_2y + c_2 = 0$$
 (ii)

$$a_3x + b_3y + c_3 = 0$$
 (iii)

are concurrent or all three parallel if
$$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = 0.$$

This is the necessary condition for consistency of three simultaneous linear equations in 2 variables but may not be sufficient.

(b) Equation $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ represents a pair of straight lines if :

$$abc + 2fgh - af^2 - bg^2 - ch^2 = 0 = \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix}$$

(c) Area of a triangle whose vertices are (x_r, y_r) ; $r = 1, 2, 3$ is $D = \frac{1}{2} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$

If $D = 0$, then the three points are collinear.

(d) Equation of a straight line passing through points (x_1, y_1) & (x_2, y_2) is
$$\begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix} = 0$$

Illustration 10 : Find the nature of solution for the given system of equations :

$$x + 2y + 3z = 1; 2x + 3y + 4z = 3; 3x + 4y + 5z = 0$$

Solution :

$$D = \begin{vmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 3 & 4 & 5 \end{vmatrix} = 0$$

$$\text{Now, } D_1 = \begin{vmatrix} 1 & 2 & 3 \\ 3 & 3 & 4 \\ 0 & 4 & 5 \end{vmatrix} = 5$$

$\therefore D = 0$ but $D_1 \neq 0$
Hence no solution.

Ans.

Illustration 11 : Find the value of λ , if the following equations are consistent :

$$x + y - 3 = 0; (1 + \lambda)x + (2 + \lambda)y - 8 = 0; x - (1 + \lambda)y + (2 + \lambda) = 0$$

Solution : The given equations in two unknowns are consistent, then $\Delta = 0$

$$\text{i.e. } \begin{vmatrix} 1 & 1 & -3 \\ 1+\lambda & 2+\lambda & -8 \\ 1 & -(1+\lambda) & 2+\lambda \end{vmatrix} = 0$$

Applying $C_2 \rightarrow C_2 - C_1$ and $C_3 \rightarrow C_3 + 3C_1$

$$\therefore \begin{vmatrix} 1 & 0 & 0 \\ 1+\lambda & 1 & 3\lambda-5 \\ 1 & -2-\lambda & 5+\lambda \end{vmatrix} = 0$$

$$\Rightarrow (5+\lambda) - (3\lambda-5)(-2-\lambda) = 0 \Rightarrow 3\lambda^2 + 2\lambda - 5 = 0$$

$$\therefore \lambda = 1, -5/3$$

Illustration 12 : If the system of equations $x + \lambda y + 1 = 0$, $\lambda x + y + 1 = 0$ & $x + y + \lambda = 0$ is consistent, then find the value of λ .

Solution : For consistency of the given system of equations

$$D = \begin{vmatrix} 1 & \lambda & 1 \\ \lambda & 1 & 1 \\ 1 & 1 & \lambda \end{vmatrix} = 0$$

$$\Rightarrow 3\lambda = 1 + 1 + \lambda^3 \text{ or } \lambda^3 - 3\lambda + 2 = 0$$

$$\Rightarrow (\lambda-1)^2 (\lambda+2) = 0 \Rightarrow \lambda = 1 \text{ or } \lambda = -2$$

Ans.

Do yourself -7 :

(i) Find nature of solution for given system of equations

$$2x + y + z = 3; \quad x + 2y + z = 4; \quad 3x + z = 2$$

(ii) If the system of equations $x + y + z = 2$, $2x + y - z = 3$ & $3x + 2y + kz = 4$ has a unique solution, then

$$(A) k \neq 0 \quad (B) -1 < k < 1 \quad (C) -2 < k < 1 \quad (D) k = 0$$

(iii) The system of equations $\lambda x + y + z = 0$, $-x + \lambda y + z = 0$ & $-x - y + \lambda z = 0$ has a non-trivial solution, then possible values of λ are -

$$(A) 0 \quad (B) 1 \quad (C) -3 \quad (D) \sqrt{3}$$

ANSWERS FOR DO YOURSELF

1. (i) minors : 4, -1, -4, 4 ; cofactors : -4, -1, 4, 4 (ii) -98 (iii) B (iv) 0

2. (ii) C

3. (i) 0 (ii) 2 (iii) 0

4. (ii) $x = -1, 2$

5. (i) $\begin{vmatrix} 1 & 1 & 1 \\ 1 & \alpha & \beta \\ 1 & \beta & \alpha \end{vmatrix}$ (ii) D

6. (i) C (ii) 0

7. (i) infinite solutions (ii) A (iii) A

EXERCISE (O-1)

1. $\begin{vmatrix} y+z & x & x \\ y & z+x & y \\ z & z & x+y \end{vmatrix}$ equals-

- (A) $x^2y^2z^2$ (B) $4x^2y^2z^2$ (C) xyz (D) $4xyz$

2. If $\begin{vmatrix} 1 & 3 & 4 \\ 1 & x-1 & 2x+2 \\ 2 & 5 & 9 \end{vmatrix} = 0$, then x is equal to-

- (A) 2 (B) 1 (C) 4 (D) 0

3. If a, b, c are in AP, then $\begin{vmatrix} x+1 & x+2 & x+a \\ x+2 & x+3 & x+b \\ x+3 & x+4 & x+c \end{vmatrix}$ equals -

- (A) $a + b + c$ (B) $x + a + b + c$ (C) 0 (D) none of these

4. If $px^4 + qx^3 + rx^2 + sx + t = \begin{vmatrix} x^2+3x & x-1 & x+3 \\ x+1 & 2-x & x-3 \\ x-3 & x+4 & 3x \end{vmatrix}$ then t is equal to -

- (A) 33 (B) 0 (C) 21 (D) none

5. For positive numbers x, y and z , the numerical value of the determinant $\begin{vmatrix} 1 & \log_x y & \log_x z \\ \log_y x & 1 & \log_y z \\ \log_z x & \log_z y & 1 \end{vmatrix}$ is-

- (A) 0 (B) $\log xyz$ (C) $\log(x + y + z)$ (D) $\log x \log y \log z$

6. Let a determinant is given by $A = \begin{vmatrix} a & b & c \\ p & q & r \\ x & y & z \end{vmatrix}$ and suppose $\det. A = 6$. If $B = \begin{vmatrix} p+x & q+y & r+z \\ a+x & b+y & c+z \\ a+p & b+q & c+r \end{vmatrix}$

then

- (A) $\det. B = 6$ (B) $\det. B = -6$ (C) $\det. B = 12$ (D) $\det. B = -12$

7. If $\Delta = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$ and A_2, B_2, C_2 are respectively cofactors of a_2, b_2, c_2 then $a_1A_2 + b_1B_2 + c_1C_2$ is

equal to-

- (A) $-\Delta$ (B) 0 (C) Δ (D) none of these

8. The value of an odd order determinant in which $a_{ij} + a_{ji} = 0 \forall i, j; i \neq j$ is -

(A) perfect square (B) negative (C) ± 1 (D) 0

9. If in the determinant $\Delta = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$, A_1, B_1, C_1 etc. be the co-factors of a_1, b_1, c_1 etc., then which

of the following relations is incorrect-

(A) $a_1A_1 + b_1B_1 + c_1C_1 = \Delta$ (B) $a_2A_2 + b_2B_2 + c_2C_2 = \Delta$
(C) $a_3A_3 + b_3B_3 + c_3C_3 = \Delta$ (D) $a_1A_2 + b_1B_2 + c_1C_2 = \Delta$

10. If $S_r = \begin{vmatrix} 2r & x & n(n+1) \\ 6r^2 - 1 & y & n^2(2n+3) \\ 4r^3 - 2nr & z & n^3(n+1) \end{vmatrix}$ then $\sum_{r=1}^n S_r$ does not depend on-

(A) x (B) y (C) n (D) all of these

11. If a, b, c are sides of a scalene triangle, then the value of $\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}$ is : [JEE-MAIN Online 2013]

(A) non-negative (B) negative (C) positive (D) non-positive

12. The value of k for which the set of equations $3x + ky - 2z = 0$, $x + ky + 3z = 0$ and $2x + 3y - 4z = 0$ has a non-trivial solution is-

(A) 15 (B) 16 (C) $31/2$ (D) $33/2$

13. If the system of linear equations [JEE-MAIN Online 2013]

$$\begin{aligned} x_1 + 2x_2 + 3x_3 &= 6 \\ x_1 + 3x_2 + 5x_3 &= 9 \\ 2x_1 + 5x_2 + ax_3 &= b \end{aligned}$$

is consistent and has infinite number of solutions, then :-

(A) $a \in \mathbb{R} - \{8\}$ and $b \in \mathbb{R} - \{15\}$ (B) $a = 8$, b can be any real number
(C) $a = 8$, $b = 15$ (D) $b = 15$, a can be any real number

14. Consider the system of equations : $x + ay = 0$, $y + az = 0$ and $z + ax = 0$. Then the set of all real values of 'a' for which the system has a unique solution is : [JEE-MAIN Online 2013]

(A) $\{1, -1\}$ (B) $\mathbb{R} - \{-1\}$
(C) $\{1, 0, -1\}$ (D) $\mathbb{R} - \{1\}$

15. If a, b, c > 0 and x, y, z $\in \mathbb{R}$, then the determinant $\begin{vmatrix} (a^x + a^{-x})^2 & (a^x - a^{-x})^2 & 1 \\ (b^y + b^{-y})^2 & (b^y - b^{-y})^2 & 1 \\ (c^z + c^{-z})^2 & (c^z - c^{-z})^2 & 1 \end{vmatrix}$ is equal to -

(A) $a^x b^y c^z$ (B) $a^{-x} b^{-y} c^{-z}$ (C) $a^{2x} b^{2y} c^{2z}$ (D) zero

16. Let a, b, c be any real numbers. Suppose that there are real numbers x, y, z not all zero such that $x = cy + bz, y = az + cx$ and $z = bx + ay$, then $a^2 + b^2 + c^2 + 2abc$ is equal to [AIEEE - 2008]
 (1) 2 (2) -1 (3) 0 (4) 1

17. There are two numbers x making the value of the determinant $\begin{vmatrix} 1 & -2 & 5 \\ 2 & x & -1 \\ 0 & 4 & 2x \end{vmatrix}$ equal to 86. The sum of these two numbers, is-
 (A) -4 (B) 5 (C) -3 (D) 9

18. If $\Delta = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$ and A_1, B_1, C_1 denote the co-factors of a_1, b_1, c_1 respectively, then the value of

the determinant $\begin{vmatrix} A_1 & B_1 & C_1 \\ A_2 & B_2 & C_2 \\ A_3 & B_3 & C_3 \end{vmatrix}$ is -

- (A) Δ (B) Δ^2 (C) Δ^3 (D) 0

EXERCISE (O-2)

1. Let $f(x) = \begin{vmatrix} 1 + \sin^2 x & \cos^2 x & 4 \sin 2x \\ \sin^2 x & 1 + \cos^2 x & 4 \sin 2x \\ \sin^2 x & \cos^2 x & 1 + 4 \sin 2x \end{vmatrix}$, then the maximum value of $f(x)$, is-
 (A) 2 (B) 4 (C) 6 (D) 8

2. The determinant $\begin{vmatrix} \cos(\theta + \phi) & -\sin(\theta + \phi) & \cos 2\phi \\ \sin \theta & \cos \theta & \sin \phi \\ -\cos \theta & \sin \theta & \cos \phi \end{vmatrix}$ is -
 (A) 0 (B) independent of θ
 (C) independent of ϕ (D) independent of θ & ϕ both

3. If the determinant $\begin{vmatrix} a+p & 1+x & u+f \\ b+q & m+y & v+g \\ c+r & n+z & w+h \end{vmatrix}$ splits into exactly K determinants of order 3, each

element of which contains only one term, then the value of K , is-

- (A) 6 (B) 8 (C) 9 (D) 12

4. Let $D_1 = \begin{vmatrix} a & b & a+b \\ c & d & c+d \\ a & b & a-b \end{vmatrix}$ and $D_2 = \begin{vmatrix} a & c & a+c \\ b & d & b+d \\ a & c & a+b+c \end{vmatrix}$ then the value of $\frac{D_1}{D_2}$ where $b \neq 0$ and

$ad \neq bc$, is

- (A) -2 (B) 0 (C) $-2b$ (D) $2b$

5. If $a^2 + b^2 + c^2 = -2$ and $f(x) = \begin{vmatrix} 1+a^2x & (1+b^2)x & (1+c^2)x \\ (1+a^2)x & 1+b^2x & (1+c^2)x \\ (1+a^2)x & (1+b^2)x & 1+c^2x \end{vmatrix}$ then $f(x)$ is a polynomial of degree-

- (A) 0 (B) 1 (C) 2 (D) 3

6. If the system of equation, $a^2x - ay = 1 - a$ & $bx + (3 - 2b)y = 3 + a$ possess a unique solution $x = 1, y = 1$ than :

- (A) $a = 1; b = -1$ (B) $a = -1, b = 1$ (C) $a = 0, b = 0$ (D) none

7. The number of real values of x satisfying $\begin{vmatrix} x & 3x+2 & 2x-1 \\ 2x-1 & 4x & 3x+1 \\ 7x-2 & 17x+6 & 12x-1 \end{vmatrix} = 0$ is -

- (A) 3 (B) 0 (C) 1 (D) infinite

[ONE OR MORE THAN ONE ARE CORRECT]

8. The determinant $\begin{vmatrix} a^2 & a^2 - (b-c)^2 & bc \\ b^2 & b^2 - (c-a)^2 & ca \\ c^2 & c^2 - (a-b)^2 & ab \end{vmatrix}$ is divisible by -

- (A) $a + b + c$ (B) $(a + b)(b + c)(c + a)$
(C) $a^2 + b^2 + c^2$ (D) $(a - b)(b - c)(c - a)$

9. The value of θ lying between $-\frac{\pi}{4}$ & $\frac{\pi}{2}$ and $0 \leq A \leq \frac{\pi}{2}$ and satisfying the equation

$$\begin{vmatrix} 1 + \sin^2 A & \cos^2 A & 2 \sin 4\theta \\ \sin^2 A & 1 + \cos^2 A & 2 \sin 4\theta \\ \sin^2 A & \cos^2 A & 1 + 2 \sin 4\theta \end{vmatrix} = 0 \text{ are -}$$

- (A) $A = \frac{\pi}{4}, \theta = -\frac{\pi}{8}$ (B) $A = \frac{3\pi}{8} = \theta$
(C) $A = \frac{\pi}{5}, \theta = -\frac{\pi}{8}$ (D) $A = \frac{\pi}{6}, \theta = \frac{3\pi}{8}$

10. Which of the following determinant(s) vanish(es) ?

$$(A) \begin{vmatrix} 1 & bc & bc(b+c) \\ 1 & ca & ca(c+a) \\ 1 & ab & ab(a+b) \end{vmatrix}$$

$$(B) \begin{vmatrix} 1 & ab & \frac{1}{a} + \frac{1}{b} \\ 1 & bc & \frac{1}{b} + \frac{1}{c} \\ 1 & ca & \frac{1}{c} + \frac{1}{a} \end{vmatrix}$$

$$(C) \begin{vmatrix} 0 & a-b & a-c \\ b-a & 0 & b-c \\ c-a & c-b & 0 \end{vmatrix}$$

$$(D) \begin{vmatrix} \log_x xyz & \log_x y & \log_x z \\ \log_y xyz & 1 & \log_y z \\ \log_z xyz & \log_z y & 1 \end{vmatrix}$$

11. The determinant $\begin{vmatrix} a & b & a\alpha + b \\ b & c & b\alpha + c \\ a\alpha + b & b\alpha + c & 0 \end{vmatrix}$ is equal to zero, if -

(A) a, b, c are in AP

(B) a, b, c are in GP

(C) α is a root of the equation $ax^2 + bx + c = 0$

(D) $(x - \alpha)$ is a factor of $ax^2 + 2bx + c$

12. System of linear equations in x, y, z

$$2x + y + z = 1$$

$$x - 2y + z = 2$$

$$3x - y + 2z = 3$$

(A) can be written as $(-3\lambda - 1, \lambda, 5\lambda + 3) \forall \lambda \in \mathbb{R}$

(B) can be written as $(3\lambda - 1, -\lambda, -5\lambda + 3) \forall \lambda \in \mathbb{R}$

(C) are such that every solution satisfy $x - 3y + 1 = 0$

(D) are such that none of them satisfy $5x + 3z = 1$

13. System of equation $x + y + az = b$, $2x + 3y = 2a$ & $3x + 4y + a^2z = ab + 2$ has

(A) unique solution when $a \neq 0, b \in \mathbb{R}$

(B) no solution when $a = 0, b = 1$

(C) infinite solution when $a = 0, b = 2$

(D) infinite solution when $a = 1, b \in \mathbb{R}$

[MATRIX MATCH TYPE]

14. Consider a system of linear equations $a_i x + b_i y + c_i z = d_i$ (where $a_i, b_i, c_i \neq 0$ and $i = 1, 2, 3$) & (α, β, γ) is its unique solution, then match the following conditions.

Column-I

(A) If $a_i = k, d_i = k^2, (k \neq 0)$ and $\alpha + \beta + \gamma = 2$, then k is

(B) If $a_i = d_i = k \neq 0$, then $\alpha + \beta + \gamma$ is

(C) If $a_i = k > 0, d_i = k + 1$, then $\alpha + \beta + \gamma$ can be

(D) If $a_i = k < 0, d_i = k + 1$, then $\alpha + \beta + \gamma$ can be

Column-II

(P) 1

(Q) 2

(R) 0

(S) 3

(T) -1

EXERCISE (S-1)

1. (a) On which one of the parameter out of a, p, d or x the value of the determinant

$$\begin{vmatrix} 1 & a & a^2 \\ \cos(p-d)x & \cos px & \cos(p+d)x \\ \sin(p-d)x & \sin px & \sin(p+d)x \end{vmatrix} \text{ does not depend.}$$

(b) If $\begin{vmatrix} x^3+1 & x^2 & x \\ y^3+1 & y^2 & y \\ z^3+1 & z^2 & z \end{vmatrix} = 0$ and x, y, z are all different then, prove that $xyz = -1$.

2. Prove that :

(a) $\begin{vmatrix} a^2+2a & 2a+1 & 1 \\ 2a+1 & a+2 & 1 \\ 3 & 3 & 1 \end{vmatrix} = (a-1)^3$ (b) $\begin{vmatrix} 1 & 1 & 1 \\ x & y & z \\ x^3 & y^3 & z^3 \end{vmatrix} = [(x-y)(y-z)(z-x)(x+y+z)]$

3. (a) Let $f(x) = \begin{vmatrix} x & 1 & \frac{-3}{2} \\ 2 & 2 & 1 \\ \frac{1}{x-1} & 0 & \frac{1}{2} \end{vmatrix}$. Find the minimum value of $f(x)$ (given $x > 1$).

- (b) If $a^2 + b^2 + c^2 + ab + bc + ca \leq 0 \forall a, b, c \in \mathbb{R}$, then find the value of the determinant

$$\begin{vmatrix} (a+b+2)^2 & a^2+b^2 & 1 \\ 1 & (b+c+2)^2 & b^2+c^2 \\ c^2+a^2 & 1 & (c+a+2)^2 \end{vmatrix}.$$

4. If $D = \begin{vmatrix} a & b & c \\ c & a & b \\ b & c & a \end{vmatrix}$ and $D' = \begin{vmatrix} b+c & c+a & a+b \\ a+b & b+c & c+a \\ c+a & a+b & b+c \end{vmatrix}$, then prove that $D' = 2D$.

5. Prove that $\begin{vmatrix} (\beta + \gamma - \alpha - \delta)^4 & (\beta + \gamma - \alpha - \delta)^2 & 1 \\ (\gamma + \alpha - \beta - \delta)^4 & (\gamma + \alpha - \beta - \delta)^2 & 1 \\ (\alpha + \beta - \gamma - \delta)^4 & (\alpha + \beta - \gamma - \delta)^2 & 1 \end{vmatrix} = -64(\alpha - \beta)(\alpha - \gamma)(\alpha - \delta)(\beta - \gamma)(\beta - \delta)(\gamma - \delta)$

6. (a) Solve for x, $\begin{vmatrix} x+2 & 2x+3 & 3x+4 \\ 2x+3 & 3x+4 & 4x+5 \\ 3x+5 & 5x+8 & 10x+17 \end{vmatrix} = 0$. (b) $\begin{vmatrix} x-2 & 2x-3 & 3x-4 \\ x-4 & 2x-9 & 3x-16 \\ x-8 & 2x-27 & 3x-64 \end{vmatrix} = 0$

7. If $a+b+c=0$, solve for x: $\begin{vmatrix} a-x & c & b \\ c & b-x & a \\ b & a & c-x \end{vmatrix} = 0$.

8. Let a, b, c are the solutions of the cubic $x^3 - 5x^2 + 3x - 1 = 0$, then find the value of the determinant

$$\begin{vmatrix} a & b & c \\ a-b & b-c & c-a \\ b+c & c+a & a+b \end{vmatrix}.$$

9. Show that, $\begin{vmatrix} a^2 + \lambda & ab & ac \\ ab & b^2 + \lambda & bc \\ ac & bc & c^2 + \lambda \end{vmatrix}$ is divisible by λ^2 and find the other factor.

10. Prove that : $\begin{vmatrix} a^2 & b^2 & c^2 \\ (a+1)^2 & (b+1)^2 & (c+1)^2 \\ (a-1)^2 & (b-1)^2 & (c-1)^2 \end{vmatrix} = 4 \begin{vmatrix} a^2 & b^2 & c^2 \\ a & b & c \\ 1 & 1 & 1 \end{vmatrix}.$

11. If $\Delta(x) = \begin{vmatrix} 0 & 2x-2 & 2x+8 \\ x-1 & 4 & x^2+7 \\ 0 & 0 & x+4 \end{vmatrix}$ and $f(x) = \sum_{j=1}^3 \sum_{i=1}^3 a_{ij}c_{ij}$, where a_{ij} is the element of i^{th} and j^{th} column

in $\Delta(x)$ and c_{ij} is the cofactor $a_{ij} \forall i$ and j , then find the greatest value of $f(x)$, where $x \in [-3, 18]$

12. If $S_r = \alpha^r + \beta^r + \gamma^r$ then show that $\begin{vmatrix} S_0 & S_1 & S_2 \\ S_1 & S_2 & S_3 \\ S_2 & S_3 & S_4 \end{vmatrix} = (\alpha - \beta)^2 (\beta - \gamma)^2 (\gamma - \alpha)^2.$

13. Solve the following sets of equations using Cramer's rule and remark about their consistency.

$$\begin{array}{lll} x + y + z - 6 = 0 & x + 2y + z = 1 & 7x - 7y + 5z = 3 \\ \text{(a)} \quad 2x + y - z - 1 = 0 & \text{(b)} \quad 3x + y + z = 6 & \text{(c)} \quad 3x + y + 5z = 7 \\ x + y - 2z + 3 = 0 & x + 2y = 0 & 2x + 3y + 5z = 5 \end{array}$$

14. For what value of K do the following system of equations $x + Ky + 3z = 0$, $3x + Ky - 2z = 0$, $2x + 3y - 4z = 0$ possess a non trivial (i.e. not all zero) solution over the set of rationals Q .

For that value of K , find all the solutions of the system.

15. If the equations $a(y + z) = x$, $b(z + x) = y$, $c(x + y) = z$ (where $a, b, c \neq -1$) have nontrivial solutions, then find the value of $\frac{1}{1+a} + \frac{1}{1+b} + \frac{1}{1+c}.$

16. Show that the system of equations $3x - y + 4z = 3$, $x + 2y - 3z = -2$ and $6x + 5y + \lambda z = -3$ has atleast one solution for any real number λ . Find the set of solutions of $\lambda = -5$.

EXERCISE (S-2)

- Prove that
$$\begin{vmatrix} 1+a^2-b^2 & 2ab & -2b \\ 2ab & 1-a^2+b^2 & 2a \\ 2b & -2a & 1-a^2-b^2 \end{vmatrix} = (1+a^2+b^2)^3.$$
- In a ΔABC , determine condition under which
$$\begin{vmatrix} \cot \frac{A}{2} & \cot \frac{B}{2} & \cot \frac{C}{2} \\ \tan \frac{B}{2} + \tan \frac{C}{2} & \tan \frac{C}{2} + \tan \frac{A}{2} & \tan \frac{A}{2} + \tan \frac{B}{2} \\ 1 & 1 & 1 \end{vmatrix} = 0$$
- Prove that :
$$\begin{vmatrix} (a-p)^2 & (a-q)^2 & (a-r)^2 \\ (b-p)^2 & (b-q)^2 & (b-r)^2 \\ (c-p)^2 & (c-q)^2 & (c-r)^2 \end{vmatrix} = \begin{vmatrix} (1+ap)^2 & (1+aq)^2 & (1+ar)^2 \\ (1+bp)^2 & (1+bq)^2 & (1+br)^2 \\ (1+cp)^2 & (1+cq)^2 & (1+cr)^2 \end{vmatrix}$$
- Given $x = cy + bz$; $y = az + cx$; $z = bx + ay$, where x, y, z are not all zero, then prove that $a^2 + b^2 + c^2 + 2abc = 1$.
- Investigate for what values of λ, μ the simultaneous equations $x + y + z = 6$; $x + 2y + 3z = 10$ & $x + 2y + \lambda z = \mu$ have :
(a) A unique solution. (b) An infinite number of solutions.
(c) No solution.
- For what values of p , the equations : $x + y + z = 1$; $x + 2y + 4z = p$ & $x + 4y + 10z = p^2$ have a solution ? Solve them completely in each case.
- Solve the equations : $Kx + 2y - 2z = 1$, $4x + 2Ky - z = 2$, $6x + 6y + Kz = 3$ considering specially the case when $K = 2$.
- Find the sum of all positive integral values of a for which every solution to the system of equation $x + ay = 3$ and $ax + 4y = 6$ satisfy the inequalities $x > 1, y > 0$.
- Given $a = \frac{x}{y-z}$; $b = \frac{y}{z-x}$; $c = \frac{z}{x-y}$, where x, y, z are not all zero, prove that : $1 + ab + bc + ca = 0$.
- Solve the system of equations :
$$\begin{cases} z + ay + a^2x + a^3 = 0 \\ z + by + b^2x + b^3 = 0 \\ z + cy + c^2x + c^3 = 0 \end{cases} \text{ where } a \neq b \neq c.$$

EXERCISE (JM)

- Let a, b, c be such that $b(a+c) \neq 0$. If
$$\begin{vmatrix} a & a+1 & a-1 \\ -b & b+1 & b-1 \\ c & c-1 & c+1 \end{vmatrix} + \begin{vmatrix} a+1 & b+1 & c-1 \\ a-1 & b-1 & c+1 \\ (-1)^{n+2}a & (-1)^{n+1}b & (-1)^n c \end{vmatrix} = 0,$$
 then the value of n is :- [AIEEE - 2009]
(1) Any odd integer (2) Any integer (3) Zero (4) Any even integer
- Consider the system of linear equations : $x_1 + 2x_2 + x_3 = 3$, $2x_1 + 3x_2 + x_3 = 3$, $3x_1 + 5x_2 + 2x_3 = 1$
The system has [AIEEE - 2010]
(1) Infinite number of solutions (2) Exactly 3 solutions
(3) A unique solution (4) No solution

3. The number of values of k for which the linear equations
 $4x + ky + 2z = 0$, $kx + 4y + z = 0$, $2x + 2y + z = 0$ possess a non-zero solution is : [AIEEE - 2011]
 (1) 1 (2) zero (3) 3 (4) 2
4. If the trivial solution is the only solution of the system of equations
 $x - ky + z = 0$, $kx + 3y - kz = 0$, $3x + y - z = 0$ Then the set of all values of k is: [AIEEE - 2011]
 (1) $\{2, -3\}$ (2) $\mathbb{R} - \{2, -3\}$ (3) $\mathbb{R} - \{2\}$ (4) $\mathbb{R} - \{-3\}$
5. The number of values of k , for which the system of equations : [JEE(Main)-2013]
 $(k + 1)x + 8y = 4k$, $kx + (k + 3)y = 3k - 1$ has no solution, is -
 (1) infinite (2) 1 (3) 2 (4) 3
6. If $\alpha, \beta \neq 0$, and $f(n) = \alpha^n + \beta^n$ and $\begin{vmatrix} 3 & 1+f(1) & 1+f(2) \\ 1+f(1) & 1+f(2) & 1+f(3) \\ 1+f(2) & 1+f(3) & 1+f(4) \end{vmatrix} = K(1 - \alpha)^2 (1 - \beta)^2 (\alpha - \beta)^2$, then
 K is equal to : [JEE(Main)-2014]
 (1) $\alpha\beta$ (2) $\frac{1}{\alpha\beta}$ (3) 1 (4) -1
7. The set of all values of λ for which the system of linear equations :
 $2x_1 - 2x_2 + x_3 = \lambda x_1$, $2x_1 - 3x_2 + 2x_3 = \lambda x_2$, $-x_1 + 2x_2 = \lambda x_3$
 has a non-trivial solution [JEE(Main)-2015]
 (1) contains two elements (2) contains more than two elements
 (3) is an empty set (4) is a singleton
8. The system of linear equations $x + \lambda y - z = 0$, $\lambda x - y - z = 0$, $x + y - \lambda z = 0$ has a non-trivial solution
 for : [JEE(Main)-2016]
 (1) exactly three values of λ . (2) infinitely many values of λ .
 (3) exactly one value of λ . (4) exactly two values of λ .
9. If S is the set of distinct values of 'b' for which the following system of linear equations
 $x + y + z = 1$
 $x + ay + z = 1$
 $ax + by + z = 0$
 has no solution, then S is : [JEE(Main)-2017]
 (1) a singleton (2) an empty set
 (3) an infinite set (4) a finite set containing two or more elements
10. If $\begin{vmatrix} x-4 & 2x & 2x \\ 2x & x-4 & 2x \\ 2x & 2x & x-4 \end{vmatrix} = (A + Bx)(x - A)^2$, then the ordered pair (A, B) is equal to : [JEE(Main)-2018]
 (1) $(-4, 3)$ (2) $(-4, 5)$ (3) $(4, 5)$ (4) $(-4, -5)$
11. If the system of linear equations
 $x + ky + 3z = 0$
 $3x + ky - 2z = 0$
 $2x + 4y - 3z = 0$
 has a non-zero solution (x, y, z) , then $\frac{xz}{y^2}$ is equal to : [JEE(Main)-2018]
 (1) 10 (2) -30 (3) 30 (4) -10

EXERCISE (JA)

1. The number of all possible values of θ , where $0 < \theta < \pi$, for which the system of equations

$$(y+z)\cos 3\theta = (xyz)\sin 3\theta$$

$$x \sin 3\theta = \frac{2\cos 3\theta}{y} + \frac{2\sin 3\theta}{z}$$

$$(xyz)\sin 3\theta = (y+2z)\cos 3\theta + y\sin 3\theta$$

have a solution (x_0, y_0, z_0) with $y_0 z_0 \neq 0$, is

[JEE 2010, 3]

2. Which of the following values of α satisfy the equation $\begin{vmatrix} (1+\alpha)^2 & (1+2\alpha)^2 & (1+3\alpha)^2 \\ (2+\alpha)^2 & (2+2\alpha)^2 & (2+3\alpha)^2 \\ (3+\alpha)^2 & (3+2\alpha)^2 & (3+3\alpha)^2 \end{vmatrix} = -648\alpha$?

[JEE(Advanced)-2015, 4M, -2M]

(A) -4

(B) 9

(C) -9

(D) 4

3. The total number of distinct $x \in \mathbb{R}$ for which $\begin{vmatrix} x & x^2 & 1+x^3 \\ 2x & 4x^2 & 1+8x^3 \\ 3x & 9x^2 & 1+27x^3 \end{vmatrix} = 10$ is

[JEE(Advanced)-2016, 3(0)]

4. Let $a, \lambda, \mu \in \mathbb{R}$. Consider the system of linear equations

$$ax + 2y = \lambda$$

$$3x - 2y = \mu$$

Which of the following statement(s) is(are) correct ?

- (A) If $a = -3$, then the system has infinitely many solutions for all values of λ and μ
 (B) If $a \neq -3$, then the system has a unique solution for all values of λ and μ
 (C) If $\lambda + \mu = 0$, then the system has infinitely many solutions for $a = -3$
 (D) If $\lambda + \mu \neq 0$, then the system has no solution for $a = -3$

[JEE(Advanced)-2016, 4(-2)]

ANSWER KEY

EXERCISE (O-1)

1. D 2. A 3. C 4. C 5. A 6. C 7. B
8. D 9. D 10. D 11. B 12. D 13. C 14. B
15. D 16. 4 17. A 18. B

EXERCISE (O-2)

1. C 2. B 3. B 4. A 5. C 6. A 7. D
8. A,C,D 9. A,B,C,D 10. A,B,C,D 11. B,D 12. A,B,D 13. B,C,D
14. (A)→(Q); (B)→(P); (C)→(Q,S); (D)→(R,T)

EXERCISE (S-1)

1. (a) p 3. (a) 4, (b) 65 6. (a) $x = -1$ or $x = -2$; (b) $x = 4$
7. $x = 0$ or $x = \pm \sqrt{\frac{3}{2}(a^2 + b^2 + c^2)}$ 8. 80 9. $\lambda^2(a^2 + b^2 + c^2 + \lambda)$ 11. 0
13. (a) $x = 1, y = 2, z = 3$; consistent (b) $x = 2, y = -1, z = 1$; consistent (c) inconsistent
14. $K = \frac{33}{2}, x : y : z = -\frac{15}{2} : 1 : -3$ 15. 2
16. If $\lambda \neq -5$ then $x = \frac{4}{7}, y = -\frac{9}{7}$ & $z = 0$; If $\lambda = -5$ then $x = \frac{4-5K}{7}, y = \frac{13K-9}{7}$ and $z = K$, where $K \in \mathbb{R}$

EXERCISE (S-2)

2. Triangle ABC is isosceles.
5. (a) $\lambda \neq 3$ (b) $\lambda = 3, \mu = 10$ (c) $\lambda = 3, \mu \neq 10$
6. $x = 1 + 2k, y = -3K, z = K$, when $p = 1$; $x = 2K, y = 1 - 3K, z = K$ when $p = 2$; where $K \in \mathbb{R}$
7. If $K \neq 2, \frac{x}{2(K+6)} = \frac{y}{2K+3} = \frac{z}{6(K-2)} = \frac{1}{2(K^2 + 2K + 15)}$

If $K = 2$, then $x = \lambda, y = \frac{1-2\lambda}{2}$ and $z = 0$ where $\lambda \in \mathbb{R}$

8. 4 10. $x = -(a + b + c), y = ab + bc + ca, z = -abc$

EXERCISE (JM)

1. 1 2. 4 3. 4 4. 2 5. 2 6. 3 7. 1
8. 1 9. 1 10. 2 11. 1

EXERCISE (JA)

1. 3 2. B,C 3. 2 4. B,C,D

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Important Notes

NURTURE COURSE

STRAIGHT LINE

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STRAIGHT LINE

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JEE (Main) Syllabus :

Various forms of equations of a line, intersection of lines, angles between two lines, conditions for concurrence of three lines, distance of a point from a line, equations of internal and external bisectors of angles between two lines, coordinates of centroid, orthocentre and circumcentre of a triangle, equation of family of lines passing through the point of intersection of two lines.

JEE (Advanced) Syllabus :

Cartesian coordinates, distance between two points, section formulae, shift of origin. Equation of a straight line in various forms, angle between two lines, distance of a point from a line; Lines through the point of intersection of two given lines, equation of the bisector of the angle between two lines, concurrency of lines; Centroid, orthocentre, incentre and circumcentre of a triangle.

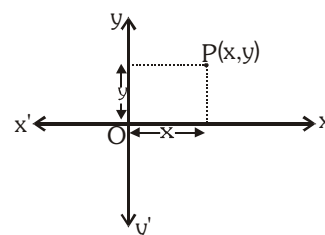
POINT & STRAIGHT LINE

1. INTRODUCTION OF COORDINATE GEOMETRY :

Coordinate geometry is the combination of algebra and geometry. A systematic study of geometry by the use of algebra was first carried out by celebrated French philosopher and mathematician René Descartes. The resulting combination of analysis and geometry is referred as **analytical geometry**.

2. CARTESIAN CO-ORDINATES SYSTEM :

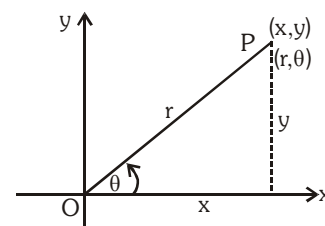
In two dimensional coordinate system, two lines are used; the lines are at right angles, forming a rectangular coordinate system. The horizontal axis is the x-axis and the vertical axis is y-axis. The point of intersection O is the origin of the coordinate system. Distances along the x-axis to the right of the origin are taken as positive, distances to the left as negative.



Distances along the y-axis above the origin are positive; distances below are negative. The position of a point anywhere in the plane can be specified by two numbers, the coordinates of the point, written as (x, y). The x-coordinate (or abscissa) is the distance of the point from the y-axis in a direction parallel to the x-axis (i.e. horizontally). The y-coordinate (or ordinate) is the distance from the x-axis in a direction parallel to the y-axis (vertically). The origin O is the point (0, 0).

3. POLAR CO-ORDINATES SYSTEM :

A coordinate system in which the position of a point is determined by the length of a line segment from a fixed origin together with the angle that the line segment makes with a fixed line. The origin is called the pole and the line segment is the radius vector (r).



The angle θ between the polar axis and the radius vector is called the vectorial angle. By convention, positive values of θ are measured in an anticlockwise sense, negative values in clockwise sense. The coordinates of the point are then specified as (r, θ).

If (x, y) are cartesian co-ordinates of a point P, then : $x = r \cos \theta$, $y = r \sin \theta$

$$\text{and } r = \sqrt{x^2 + y^2} \quad , \quad \theta = \tan^{-1} \left(\frac{y}{x} \right)$$

4. DISTANCE FORMULA AND ITS APPLICATIONS :

If $A(x_1, y_1)$ and $B(x_2, y_2)$ are two points, then $AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

Note :

- (i) Three given points A, B and C are collinear, when sum of any two distances out of AB, BC, CA is equal to the remaining third otherwise the points will be the vertices of a triangle.
- (ii) Let A, B, C & D be the four given points in a plane. Then the quadrilateral will be :

- (a) Square if $AB = BC = CD = DA$ & $AC = BD$; $AC \perp BD$
 (b) Rhombus if $AB = BC = CD = DA$ and $AC \neq BD$; $AC \perp BD$
 (c) Parallelogram if $AB = DC$, $BC = AD$; $AC \neq BD$; $AC \not\perp BD$
 (d) Rectangle if $AB = CD$, $BC = DA$, $AC = BD$; $AC \not\perp BD$

Illustration 1 : The number of points on x-axis which are at a distance c ($c < 3$) from the point $(2, 3)$ is
 (A) 2 (B) 1 (C) infinite (D) no point

Solution : Let a point on x-axis is $(x_1, 0)$ then its distance from the point $(2, 3)$

$$= \sqrt{(x_1 - 2)^2 + 9} = c \quad \text{or} \quad (x_1 - 2)^2 = c^2 - 9$$

$$\therefore x_1 - 2 = \pm \sqrt{c^2 - 9} \quad \text{since } c < 3 \Rightarrow c^2 - 9 < 0$$

$\therefore x_1$ will be imaginary.

Ans. (D)

Illustration 2 : The distance between the point $P(a \cos \alpha, a \sin \alpha)$ and $Q(a \cos \beta, a \sin \beta)$, where $a > 0$ & $\alpha > \beta$, is -

- (A) $4a \sin \frac{\alpha - \beta}{2}$ (B) $2a \sin \frac{\alpha + \beta}{2}$ (C) $2a \sin \frac{\alpha - \beta}{2}$ (D) $2a \cos \frac{\alpha - \beta}{2}$

Solution : $d^2 = (a \cos \alpha - a \cos \beta)^2 + (a \sin \alpha - a \sin \beta)^2 = a^2 (\cos \alpha - \cos \beta)^2 + a^2 (\sin \alpha - \sin \beta)^2$

$$= a^2 \left\{ 2 \sin \frac{\alpha + \beta}{2} \sin \frac{\beta - \alpha}{2} \right\}^2 + a^2 \left\{ 2 \cos \frac{\alpha + \beta}{2} \sin \frac{\alpha - \beta}{2} \right\}^2$$

$$= 4a^2 \sin^2 \frac{\alpha - \beta}{2} \left\{ \sin^2 \frac{\alpha + \beta}{2} + \cos^2 \frac{\alpha + \beta}{2} \right\} = 4a^2 \sin^2 \frac{\alpha - \beta}{2} \Rightarrow d = 2a \sin \frac{\alpha - \beta}{2}$$

Ans. (C)

Do yourself - 1 :

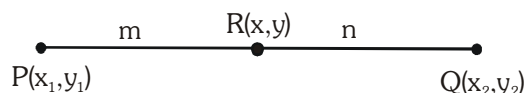
- (i) Find the distance between the points $P(-3, 2)$ and $Q(2, -1)$.
 (ii) If the distance between the points $P(-3, 5)$ and $Q(-x, -2)$ is $\sqrt{58}$, then find the value(s) of x .
 (iii) A line segment is of the length 15 units and one end is at the point $(3, 2)$, if the abscissa of the other end is 15, then find possible ordinates.

5. SECTION FORMULA :

The co-ordinates of a point dividing a line joining the points $P(x_1, y_1)$ and $Q(x_2, y_2)$ in the ratio $m:n$ is given by :

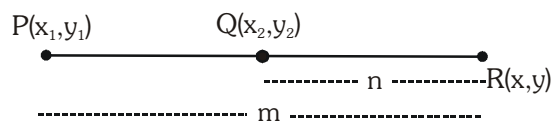
(a) **For internal division :** $P - R - Q \Rightarrow R$ divides line segment PQ , internally.

$$(x, y) \equiv \left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n} \right)$$



- (b) **For external division :** $R - P - Q$ or $P - Q - R \Rightarrow R$ divides line segment PQ , externally.

$$(x, y) \equiv \left(\frac{mx_2 - nx_1}{m - n}, \frac{my_2 - ny_1}{m - n} \right)$$



$$\frac{(PR)}{(QR)} < 1 \Rightarrow R \text{ lies on the left of } P \quad \& \quad \frac{(PR)}{(QR)} > 1 \Rightarrow R \text{ lies on the right of } Q$$

- (c) **Harmonic conjugate :** If P divides AB internally in the ratio $m : n$ & Q divides AB externally in the ratio $m : n$ then P & Q are said to be harmonic conjugate of each other w.r.t. AB .

Mathematically ; $\frac{2}{AB} = \frac{1}{AP} + \frac{1}{AQ}$ i.e. AP, AB & AQ are in **H.P.**

Illustration 3 : Determine the ratio in which $y - x + 2 = 0$ divides the line joining $(3, -1)$ and $(8, 9)$.

Solution : Suppose the line $y - x + 2 = 0$ divides the line segment joining $A(3, -1)$ and $B(8, 9)$

in the ratio $\lambda : 1$ at a point P , then the co-ordinates of the point P are $\left(\frac{8\lambda + 3}{\lambda + 1}, \frac{9\lambda - 1}{\lambda + 1} \right)$

But P lies on $y - x + 2 = 0$ therefore $\left(\frac{9\lambda - 1}{\lambda + 1} \right) - \left(\frac{8\lambda + 3}{\lambda + 1} \right) + 2 = 0$

$$\Rightarrow 9\lambda - 1 - 8\lambda - 3 + 2\lambda + 2 = 0$$

$$\Rightarrow 3\lambda - 2 = 0 \text{ or } \lambda = \frac{2}{3}$$

So, the required ratio is $\frac{2}{3} : 1$, i.e., $2 : 3$ (internally) since here λ is positive.

Do yourself - 2 :

- (i) Find the co-ordinates of the point dividing the join of $A(1, -2)$ and $B(4, 7)$:
- (a) Internally in the ratio $1 : 2$ (b) Externally in the ratio of $2 : 1$
- (ii) In what ratio is the line joining $A(8, 9)$ and $B(-7, 4)$ divided by
- (a) the point $(2, 7)$ (b) the x -axis (c) the y -axis.

6. CO-ORDINATES OF SOME PARTICULAR POINTS :

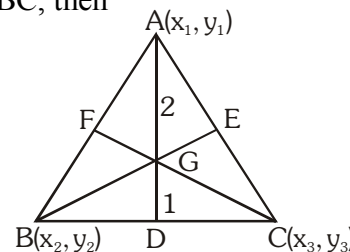
Let $A(x_1, y_1)$, $B(x_2, y_2)$ and $C(x_3, y_3)$ are vertices of any triangle ABC , then

(a) **Centroid :**

The centroid is the point of intersection of the medians (line joining the mid point of sides and opposite vertices).

Centroid divides each median in the ratio of $2 : 1$.

Co-ordinates of centroid $G \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$



(b) Incenter :

The incenter is the point of intersection of internal bisectors of the angles of a triangle. Also it is a centre of the circle touching all the sides of a triangle.

$$\text{Co-ordinates of incenter } I \left(\frac{ax_1 + bx_2 + cx_3}{a + b + c}, \frac{ay_1 + by_2 + cy_3}{a + b + c} \right)$$

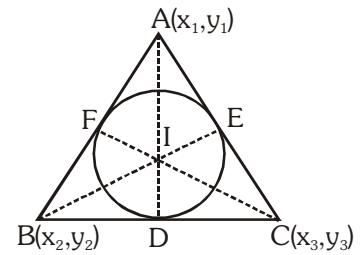
where a, b, c are the sides of triangle ABC .

Note :

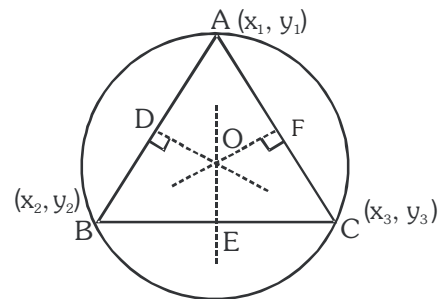
(i) Angle bisector divides the opposite sides in the ratio of remaining sides. e.g.

$$\frac{BD}{DC} = \frac{AB}{AC} = \frac{c}{b}$$

(ii) Incenter divides the angle bisectors in the ratio $(b + c) : a, (c + a) : b, (a + b) : c$.

**(c) Circumcenter :**

It is the point of intersection of perpendicular bisectors of the sides of a triangle. If O is the circumcenter of any triangle ABC , then $OA^2 = OB^2 = OC^2$. Also it is a centre of a circle touching all the vertices of a triangle.

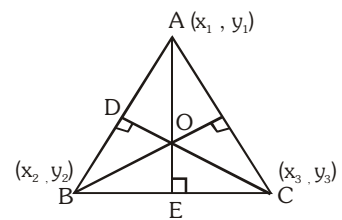
**Note :**

(i) If the triangle is right angled, then its circumcenter is the mid point of hypotenuse.

$$(ii) \text{ Co-ordinates of circumcenter } \left(\frac{x_1 \sin 2A + x_2 \sin 2B + x_3 \sin 2C}{\sin 2A + \sin 2B + \sin 2C}, \frac{y_1 \sin 2A + y_2 \sin 2B + y_3 \sin 2C}{\sin 2A + \sin 2B + \sin 2C} \right)$$

(d) Orthocenter :

It is the point of intersection of perpendiculars drawn from vertices on the opposite sides of a triangle and it can be obtained by solving the equation of any two altitudes.

**Note :**

(i) If a triangle is right angled, then orthocenter is the point where right angle is formed.

$$(ii) \text{ Co-ordinates of circumcenter } \left(\frac{x_1 \tan A + x_2 \tan B + x_3 \tan C}{\tan A + \tan B + \tan C}, \frac{y_1 \tan A + y_2 \tan B + y_3 \tan C}{\tan A + \tan B + \tan C} \right)$$

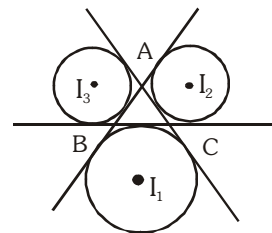
Remarks :

- (i) If the triangle is equilateral, then centroid, incentre, orthocenter, circumcenter, coincide.
- (ii) Orthocentre, centroid and circumcentre are always collinear and centroid divides the line joining orthocentre and circumcentre in the ratio 2 : 1
- (iii) In an isosceles triangle centroid, orthocentre, incentre & circumcentre lie on the same line.

(e) **Ex-centers :**

The centre of a circle which touches side BC and the extended portions of sides AB and AC is called the ex-centre of $\triangle ABC$ with respect to the vertex A. It is denoted by I_1 and its coordinates

$$\text{are } I_1 \left(\frac{-ax_1 + bx_2 + cx_3}{-a + b + c}, \frac{-ay_1 + by_2 + cy_3}{-a + b + c} \right)$$



Similarly ex-centers of $\triangle ABC$ with respect to vertices B and C are denoted by I_2 and I_3 respectively, and

$$I_2 \left(\frac{ax_1 - bx_2 + cx_3}{a - b + c}, \frac{ay_1 - by_2 + cy_3}{a - b + c} \right), I_3 \left(\frac{ax_1 + bx_2 - cx_3}{a + b - c}, \frac{ay_1 + by_2 - cy_3}{a + b - c} \right)$$

Illustration 4 : If $\left(\frac{5}{3}, 3\right)$ is the centroid of a triangle and its two vertices are (0, 1) and (2, 3), then find its third vertex, circumcentre, circumradius & orthocentre.

Solution : Let the third vertex of triangle be (x, y), then

$$\frac{5}{3} = \frac{x + 0 + 2}{3} \Rightarrow x = 3 \text{ and } 3 = \frac{y + 1 + 3}{3} \Rightarrow y = 5. \text{ So third vertex is } (3, 5).$$

Now three vertices are A(0, 1), B(2, 3) and C(3, 5)

Let circumcentre be P(h, k),

$$\text{then } AP = BP = CP = R \text{ (circumradius)} \Rightarrow AP^2 = BP^2 = CP^2 = R^2$$

$$h^2 + (k - 1)^2 = (h - 2)^2 + (k - 3)^2 = (h - 3)^2 + (k - 5)^2 = R^2 \quad \dots\dots\dots (i)$$

from the first two equations, we have

$$h + k = 3 \quad \dots\dots\dots (ii)$$

from the first and third equation, we obtain

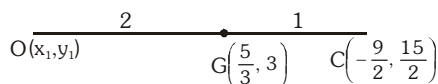
$$6h + 8k = 33 \quad \dots\dots\dots (iii)$$

On solving, (ii) & (iii), we get

$$h = -\frac{9}{2}, \quad k = \frac{15}{2}$$

Substituting these values in (i), we have

$$R = \frac{5}{2}\sqrt{10}$$



$$\text{Let } O(x_1, y_1) \text{ be the orthocentre, then } \frac{x_1 + 2\left(-\frac{9}{2}\right)}{3} = \frac{5}{3} \Rightarrow x_1 = 14, \frac{y_1 + 2\left(\frac{15}{2}\right)}{3} = 3$$

$$\Rightarrow y_1 = -6. \text{ Hence orthocentre of the triangle is } (14, -6).$$

Illustration 5 : The vertices of a triangle are A(0, -6), B(-6, 0) and C(1, 1) respectively, then coordinates of the ex-centre opposite to vertex A is :

- (A) $\left(\frac{-3}{2}, \frac{-3}{2}\right)$ (B) $\left(-4, \frac{3}{2}\right)$ (C) $\left(\frac{-3}{2}, \frac{3}{2}\right)$ (D) (-4, 6)

Solution : $a = BC = \sqrt{(-6-1)^2 + (0-1)^2} = \sqrt{50} = 5\sqrt{2}$

$b = CA = \sqrt{(1-0)^2 + (1+6)^2} = \sqrt{50} = 5\sqrt{2}$

$c = AB = \sqrt{(0+6)^2 + (-6-0)^2} = \sqrt{72} = 6\sqrt{2}$

coordinates of ex-centre opposite to vertex A will be :

$$x = \frac{-ax_1 + bx_2 + cx_3}{-a + b + c} = \frac{-5\sqrt{2} \cdot 0 + 5\sqrt{2}(-6) + 6\sqrt{2}(1)}{-5\sqrt{2} + 5\sqrt{2} + 6\sqrt{2}} = \frac{-24\sqrt{2}}{6\sqrt{2}} = -4$$

$$y = \frac{-ay_1 + by_2 + cy_3}{-a + b + c} = \frac{-5\sqrt{2}(-6) + 5\sqrt{2} \cdot 0 + 6\sqrt{2}(1)}{-5\sqrt{2} + 5\sqrt{2} + 6\sqrt{2}} = \frac{36\sqrt{2}}{6\sqrt{2}} = 6$$

Hence coordinates of ex-centre is (-4, 6)

Ans. (D)

Do yourself - 3 :

- (i) The coordinates of the vertices of a triangle are (0, 1), (2, 3) and (3, 5) :
- Find centroid of the triangle.
 - Find circumcentre & the circumradius.
 - Find orthocentre of the triangle.

7. AREA OF TRIANGLE :

Let A(x_1, y_1), B(x_2, y_2) and C(x_3, y_3) are vertices of a triangle, then

$$\text{Area of } \triangle ABC = \left| \frac{1}{2} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} \right| = \frac{1}{2} |[x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)]|$$

To remember the above formula, take the help of the following method :

$$= \frac{1}{2} \begin{vmatrix} x_1 & x_2 & x_3 & x_1 \\ y_1 & y_2 & y_3 & y_1 \end{vmatrix} = \frac{1}{2} [(x_1 y_2 - x_2 y_1) + (x_2 y_3 - x_3 y_2) + (x_3 y_1 - x_1 y_3)]$$

Remarks :

- (i) If the area of triangle joining three points is zero, then the points are collinear.

- (ii) **Area of Equilateral triangle :** If altitude of any equilateral triangle is P, then its area = $\frac{P^2}{\sqrt{3}}$. If

'a' be the side of equilateral triangle, then its area = $\left(\frac{a^2 \sqrt{3}}{4}\right)$.

(iii) Area of quadrilateral with given vertices $A(x_1, y_1)$, $B(x_2, y_2)$, $C(x_3, y_3)$, $D(x_4, y_4)$

$$\text{Area of quad. ABCD} = \frac{1}{2} \left| \begin{array}{cccc} x_1 & x_2 & x_3 & x_4 \\ y_1 & y_2 & y_3 & y_4 \end{array} \right|$$

Note : Area of a polygon can be obtained by dividing the polygon into disjointed triangles and then adding their areas.

Illustration 6 : If the vertices of a triangle are $(1, 2)$, $(4, -6)$ and $(3, 5)$ then its area is

- (A) $\frac{25}{2}$ sq. units (B) 12 sq. units (C) 5 sq. units (D) 25 sq. units

Solution : $\Delta = \frac{1}{2} [1(-6-5) + 4(5-2) + 3(2+6)] = \frac{1}{2} [-11 + 12 + 24] = \frac{25}{2}$ square units **Ans. (A)**

Illustration 7 : The point A divides the join of the points $(-5, 1)$ and $(3, 5)$ in the ratio $k:1$ and coordinates of points B and C are $(1, 5)$ and $(7, -2)$ respectively. If the area of ΔABC be 2 units, then k equals -

- (A) 7, 9 (B) 6, 7 (C) $7, \frac{31}{9}$ (D) $9, \frac{31}{9}$

Solution : $A \equiv \left(\frac{3k-5}{k+1}, \frac{5k+1}{k+1} \right)$

$$\text{Area of } \Delta ABC = 2 \text{ units} \Rightarrow \frac{1}{2} \left[\frac{3k-5}{k+1} (5+2) + 1 \left(-2 - \frac{5k+1}{k+1} \right) + 7 \left(\frac{5k+1}{k+1} - 5 \right) \right] = \pm 2$$

$$\Rightarrow 14k - 66 = \pm 4(k+1) \Rightarrow k = 7 \text{ or } \frac{31}{9} \quad \text{Ans. (C)}$$

Illustration 8 : Prove that the co-ordinates of the vertices of an equilateral triangle can not all be rational.

Solution : Let $A(x_1, y_1)$, $B(x_2, y_2)$ and $C(x_3, y_3)$ be the vertices of a triangle ABC. If possible let $x_1, y_1, x_2, y_2, x_3, y_3$ be all rational.

$$\text{Now area of } \Delta ABC = \frac{1}{2} |x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)| = \text{Rational} \quad \dots\dots (i)$$

Since ΔABC is equilateral

$$\therefore \text{Area of } \Delta ABC = \frac{\sqrt{3}}{4} (\text{side})^2 = \frac{\sqrt{3}}{4} (AB)^2 = \frac{\sqrt{3}}{4} \{(x_1 - x_2)^2 + (y_1 - y_2)^2\} = \text{Irrational} \quad \dots\dots (ii)$$

From (i) and (ii),

Rational = Irrational

which is contradiction.

Hence $x_1, y_1, x_2, y_2, x_3, y_3$ cannot all be rational.

8. CONDITIONS FOR COLLINEARITY OF THREE GIVEN POINTS :

Three given points A (x_1, y_1), B (x_2, y_2), C (x_3, y_3) are collinear if any one of the following conditions are satisfied.

(a) Area of triangle ABC is zero i.e. $\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = 0$

(b) Slope of AB = slope of BC = slope of AC. i.e. $\frac{y_2 - y_1}{x_2 - x_1} = \frac{y_3 - y_2}{x_3 - x_2} = \frac{y_3 - y_1}{x_3 - x_1}$

(c) Find the equation of line passing through 2 given points, if the third point satisfies the given equation of the line, then three points are collinear.

Do yourself - 4 :

- (i) Find the area of the triangle whose vertices are A(1,1), B(7, -3) and C(12, 2)
- (ii) Find the area of the quadrilateral whose vertices are A(1,1) B(7, -3), C(12,2) and D(7, 21)
- (iii) Prove that the points A(a, b + c), B(b, c + a) and C(c, a + b) are collinear (By determinant method)

9. LOCUS :

The locus of a moving point is the path traced out by that point under one or more geometrical conditions.

(a) Equation of Locus :

The equation to a locus is the relation which exists between the coordinates of any point on the path, and which holds for no other point except those lying on the path.

(b) Procedure for finding the equation of the locus of a point :

- (i) If we are finding the equation of the locus of a point P, assign coordinates (h, k) to P.
- (ii) Express the given condition as equations in terms of the known quantities to facilitate calculations. We sometimes include some unknown quantities known as parameters.
- (iii) Eliminate the parameters, so that the eliminant contains only h, k and known quantities.
- (iv) Replace h by x, and k by y, in the eliminant. The resulting equation would be the equation of the locus of P.

Illustration 9 : The ends of the rod of length ℓ moves on two mutually perpendicular lines, find the locus of the point on the rod which divides it in the ratio $m_1 : m_2$

(A) $m_1^2 x^2 + m_2^2 y^2 = \frac{\ell^2}{(m_1 + m_2)^2}$

(B) $(m_2 x)^2 + (m_1 y)^2 = \left(\frac{m_1 m_2 \ell}{m_1 + m_2} \right)^2$

(C) $(m_1 x)^2 + (m_2 y)^2 = \left(\frac{m_1 m_2 \ell}{m_1 + m_2} \right)^2$

(D) none of these

Solution : Let (h, k) be the point that divide the rod $AB = \ell$, in the ratio $m_1 : m_2$, and $OA = a$, $OB = b$ say

$$\therefore a^2 + b^2 = \ell^2 \quad \dots (i)$$

$$\text{Now } h = \left(\frac{m_2 a}{m_1 + m_2} \right) \Rightarrow a = \left(\frac{m_1 + m_2}{m_2} \right) h$$

$$k = \left(\frac{m_1 b}{m_1 + m_2} \right) \Rightarrow b = \left(\frac{m_1 + m_2}{m_1} \right) k$$

$$\text{putting these values in (i)} \quad \frac{(m_1 + m_2)^2}{m_2^2} h^2 + \frac{(m_1 + m_2)^2}{m_1^2} k^2 = \ell^2$$

$$\therefore \text{Locus of } (h, k) \text{ is } m_1^2 x^2 + m_2^2 y^2 = \left(\frac{m_1 m_2 \ell}{m_1 + m_2} \right)^2 \quad \text{Ans. (C)}$$

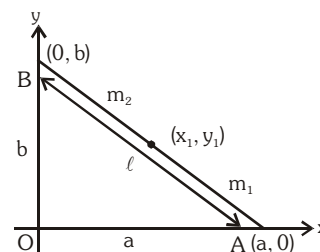


Illustration 10 : $A(a, 0)$ and $B(-a, 0)$ are two fixed points of $\triangle ABC$. If its vertex C moves in such a way that $\cot A + \cot B = \lambda$, where λ is a constant, then the locus of the point C is -

- (A) $y\lambda = 2a$ (B) $y = \lambda a$ (C) $ya = 2\lambda$ (D) none of these

Solution : Given that coordinates of two fixed points A and B are $(a, 0)$ and $(-a, 0)$ respectively. Let variable point C is (h, k) . From the adjoining figure

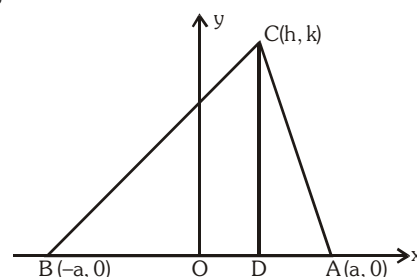
$$\cot A = \frac{DA}{CD} = \frac{a - h}{k}$$

$$\cot B = \frac{BD}{CD} = \frac{a + h}{k}$$

But $\cot A + \cot B = \lambda$, so we have

$$\frac{a - h}{k} + \frac{a + h}{k} = \lambda \Rightarrow \frac{2a}{k} = \lambda$$

Hence locus of C is $y\lambda = 2a$



Ans. (A)

Do yourself - 5 :

- (i) Find the locus of a variable point which is at a distance of 2 units from the y-axis.
- (ii) Find the locus of a point which is equidistant from both the axes.

10. STRAIGHT LINE :

Introduction : A relation between x and y which is satisfied by co-ordinates of every point lying on a line is called equation of the straight line. Here, remember that every one degree equation in variable x and y always represents a straight line i.e. $ax + by + c = 0$; a & $b \neq 0$ simultaneously.

- (a) Equation of a line parallel to x-axis at a distance 'a' is $y = a$ or $y = -a$.
- (b) Equation of x-axis is $y = 0$.
- (c) Equation of a line parallel to y-axis at a distance 'b' is $x = b$ or $x = -b$.
- (d) Equation of y-axis is $x = 0$.

Illustration 11 : Prove that every first degree equation in x, y represents a straight line.

Solution : Let $ax + by + c = 0$ be a first degree equation in x, y where a, b, c are constants.

Let $P(x_1, y_1)$ & $Q(x_2, y_2)$ be any two points on the curve represented by $ax + by + c = 0$. Then $ax_1 + by_1 + c = 0$ and $ax_2 + by_2 + c = 0$

Let R be any point on the line segment joining P & Q

Suppose R divides PQ in the ratio $\lambda : 1$. Then, the coordinates of R are

$$\left(\frac{\lambda x_2 + x_1}{\lambda + 1}, \frac{\lambda y_2 + y_1}{\lambda + 1} \right)$$

$$\text{We have } a \left(\frac{\lambda x_2 + x_1}{\lambda + 1} \right) + b \left(\frac{\lambda y_2 + y_1}{\lambda + 1} \right) + c = \lambda \cdot 0 + 0 = 0$$

$$\therefore R \left(\frac{\lambda x_2 + x_1}{\lambda + 1}, \frac{\lambda y_2 + y_1}{\lambda + 1} \right) \text{ lies on the curve represented by } ax + by + c = 0. \text{ Thus every}$$

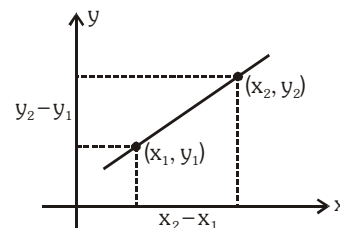
point on the line segment joining P & Q lies on $ax + by + c = 0$.

Hence $ax + by + c = 0$ represents a straight line.

11. SLOPE OF LINE :

If a given line makes an angle θ ($0^\circ \leq \theta < 180^\circ$, $\theta \neq 90^\circ$) with the positive direction of x -axis, then slope of this line will be $\tan \theta$ and is usually denoted by the letter m i.e. $m = \tan \theta$. If $A(x_1, y_1)$ and $B(x_2, y_2)$

$$\& x_1 \neq x_2 \text{ then slope of line } AB = \frac{y_2 - y_1}{x_2 - x_1}$$



Remark :

- (i) If $\theta = 90^\circ$, m does not exist and line is parallel to **y-axis**.
- (ii) If $\theta = 0^\circ$, $m = 0$ and the line is parallel to **x-axis**.
- (iii) Let m_1 and m_2 be slopes of two given lines (none of them is parallel to y -axis)
 - (a) If lines are parallel, $m_1 = m_2$ and vice-versa.
 - (b) If lines are perpendicular, $m_1 m_2 = -1$ and vice-versa

12. STANDARD FORMS OF EQUATIONS OF A STRAIGHT LINE :

- (a) **Slope Intercept form :** Let m be the slope of a line and c its intercept on y -axis. Then the equation of this straight line is written as : $y = mx + c$

If the line passes through origin, its equation is written as $y = mx$

- (b) **Point Slope form :** If m be the slope of a line and it passes through a point (x_1, y_1) , then its equation is written as : $y - y_1 = m(x - x_1)$

(c) **Two point form :** Equation of a line passing through two points (x_1, y_1) and (x_2, y_2) is written as :

$$y - y_1 = \frac{y_2 - y_1}{x_2 - x_1} (x - x_1) \quad \text{or} \quad \begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix} = 0$$

(d) **Intercept form :** If a and b are the intercepts made by a line on the axes of x and y , its equation

is written as : $\frac{x}{a} + \frac{y}{b} = 1$

(i) Length of intercept of line between the coordinate axes $= \sqrt{a^2 + b^2}$

(ii) Area of triangle AOB $= \frac{1}{2} OA \cdot OB = \left| \frac{1}{2} ab \right|$

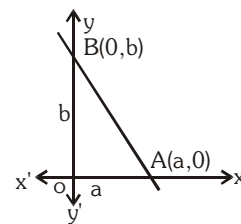


Illustration 12 : The equation of the lines which passes through the point $(3, 4)$ and the sum of its intercepts on the axes is 14 is -

(A) $4x - 3y = 24$, $x - y = 7$

(B) $4x + 3y = 24$, $x + y = 7$

(C) $4x + 3y + 24 = 0$, $x + y + 7 = 0$

(D) $4x - 3y + 24 = 0$, $x - y + 7 = 0$

Solution :

Let the equation of the line be $\frac{x}{a} + \frac{y}{b} = 1$ (i)

This passes through $(3, 4)$, therefore $\frac{3}{a} + \frac{4}{b} = 1$ (ii)

It is given that $a + b = 14 \Rightarrow b = 14 - a$. Putting $b = 14 - a$ in (ii), we get

$$\frac{3}{a} + \frac{4}{14 - a} = 1 \Rightarrow a^2 - 13a + 42 = 0 \Rightarrow (a - 7)(a - 6) = 0 \Rightarrow a = 7, 6$$

For $a = 7$, $b = 14 - 7 = 7$ and for $a = 6$, $b = 14 - 6 = 8$

Putting the values of a and b in (i), we get the equations of the lines

$$\frac{x}{7} + \frac{y}{7} = 1 \quad \text{and} \quad \frac{x}{6} + \frac{y}{8} = 1 \quad \text{or} \quad x + y = 7 \quad \text{and} \quad 4x + 3y = 24$$

Ans. (B)

Illustration 13 : Two points A and B move on the positive direction of x -axis and y -axis respectively, such that $OA + OB = K$. Show that the locus of the foot of the perpendicular from the origin O on the line AB is $(x + y)(x^2 + y^2) = Kxy$.

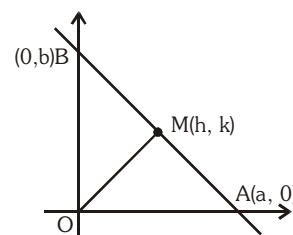
Solution :

Let the equation of AB be $\frac{x}{a} + \frac{y}{b} = 1$ (i)

given, $a + b = K$ (ii)

now, $m_{AB} \times m_{OM} = -1 \Rightarrow ah = bk$ (iii)

from (ii) and (iii),



$$a = \frac{kK}{h+k} \text{ and } b = \frac{hK}{h+k}$$

$$\therefore \text{ from (i) } \frac{x(h+k)}{k.K} + \frac{y(h+k)}{h.K} = 1$$

as it passes through (h, k)

$$\frac{h(h+k)}{k.K} + \frac{k(h+k)}{h.K} = 1 \Rightarrow (h+k)(h^2 + k^2) = Khk$$

$$\therefore \text{ locus of } (h, k) \text{ is } (x+y)(x^2 + y^2) = Kxy.$$

- (e) **Normal form :** If p is the length of perpendicular on a line from the origin, and α the angle which this perpendicular makes with positive x -axis, then the equation of this line is written as : **$x \cos \alpha + y \sin \alpha = p$** (p is always positive) where $0 \leq \alpha < 2\pi$.

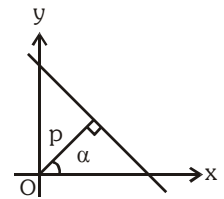


Illustration 14 : Find the equation of the straight line on which the perpendicular from origin makes an angle 30° with positive x -axis and which forms a triangle of area $\left(\frac{50}{\sqrt{3}}\right)$ sq. units with the co-ordinates axes.

Solution :

$$\angle NOA = 30^\circ$$

$$\text{Let } ON = p > 0, OA = a, OB = b$$

$$\text{In } \triangle ONA, \cos 30^\circ = \frac{ON}{OA} = \frac{p}{a} \Rightarrow \frac{\sqrt{3}}{2} = \frac{p}{a}$$

$$\text{or } a = \frac{2p}{\sqrt{3}}$$

$$\text{and in } \triangle ONB, \cos 60^\circ = \frac{ON}{OB} = \frac{p}{b} \Rightarrow \frac{1}{2} = \frac{p}{b}$$

$$\text{or } b = 2p$$

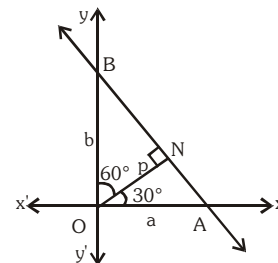
$$\therefore \text{ Area of } \triangle OAB = \frac{1}{2} ab = \frac{1}{2} \left(\frac{2p}{\sqrt{3}} \right) (2p) = \frac{2p^2}{\sqrt{3}}$$

$$\therefore \frac{2p^2}{\sqrt{3}} = \frac{50}{\sqrt{3}} \Rightarrow p^2 = 25$$

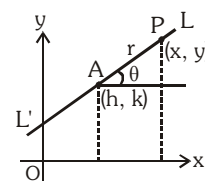
$$\text{or } p = 5$$

$$\therefore \text{ Using } x \cos \alpha + y \sin \alpha = p, \text{ the equation of the line AB is } x \cos 30^\circ + y \sin 30^\circ = 5$$

$$\text{or } x\sqrt{3} + y = 10$$



- (f) **Parametric form :** To find the equation of a straight line which passes through a given point $A(h, k)$ and makes a given angle θ with the positive direction of the x -axis. $P(x, y)$ is any point on the line LAL' .



Let $AP = r$, then $x - h = r \cos \theta$, $y - k = r \sin \theta$ & $\frac{x-h}{\cos \theta} = \frac{y-k}{\sin \theta} = r$ is the equation of the straight line LAL' .

Any point P on the line will be of the form $(h + r \cos \theta, k + r \sin \theta)$, where $|r|$ gives the distance of the point P from the fixed point (h, k) .

Illustration 15 : Equation of a line which passes through point $A(2, 3)$ and makes an angle of 45° with x axis. If this line meet the line $x + y + 1 = 0$ at point P then distance AP is -

- (A) $2\sqrt{3}$ (B) $3\sqrt{2}$ (C) $5\sqrt{2}$ (D) $2\sqrt{5}$

Solution : Here $x_1 = 2$, $y_1 = 3$ and $\theta = 45^\circ$ hence $\frac{x-2}{\cos 45^\circ} = \frac{y-3}{\sin 45^\circ} = r$

from first two parts $\Rightarrow x - 2 = y - 3 \Rightarrow x - y + 1 = 0$

Co-ordinate of point P on this line is $\left(2 + \frac{r}{\sqrt{2}}, 3 + \frac{r}{\sqrt{2}}\right)$.

If this point is on line $x + y + 1 = 0$ then

$$\left(2 + \frac{r}{\sqrt{2}}\right) + \left(3 + \frac{r}{\sqrt{2}}\right) + 1 = 0 \Rightarrow r = -3\sqrt{2} ; |r| = 3\sqrt{2} \quad \text{Ans. (B)}$$

Illustration 16 : A variable line is drawn through O , to cut two fixed straight lines L_1 and L_2 in A_1 and A_2 ,

respectively. A point A is taken on the variable line such that $\frac{m+n}{OA} = \frac{m}{OA_1} + \frac{n}{OA_2}$.

Show that the locus of A is a straight line passing through the point of intersection of L_1 and L_2 where O is being the origin.

Solution : Let the variable line passing through the origin is $\frac{x}{\cos \theta} = \frac{y}{\sin \theta} = r_1$ (i)

Let the equation of the line L_1 is $p_1x + q_1y = 1$ (ii)

Equation of the line L_2 is $p_2x + q_2y = 1$ (iii)

the variable line intersects the line (ii) at A_1 and (iii) at A_2 .

Let $OA_1 = r_1$.

Then $A_1 = (r_1 \cos \theta, r_1 \sin \theta) \Rightarrow A_1$ lies on L_1

$$\Rightarrow r_1 = OA_1 = \frac{1}{p_1 \cos \theta + q_1 \sin \theta}$$

$$\text{Similarly, } r_2 = OA_2 = \frac{1}{p_2 \cos \theta + q_2 \sin \theta}$$

Let $OA = r$

Let co-ordinate of A are $(h, k) \Rightarrow (h, k) = (r \cos \theta, r \sin \theta)$

$$\text{Now } \frac{m+n}{r} = \frac{m}{OA_1} + \frac{n}{OA_2} \Rightarrow \frac{m+n}{r} = \frac{m}{r_1} + \frac{n}{r_2}$$

$$\Rightarrow m + n = m(p_1 r \cos \theta + q_1 r \sin \theta) + n(p_2 r \cos \theta + q_2 r \sin \theta)$$

$$\Rightarrow (p_1 h + q_1 k - 1) + \frac{n}{m}(p_2 h + q_2 k - 1) = 0$$

$$\text{Therefore, locus of A is } (p_1 x + q_1 y - 1) + \frac{n}{m}(p_2 x + q_2 y - 1) = 0$$

$$\Rightarrow L_1 + \lambda L_2 = 0 \text{ where } \lambda = \frac{n}{m}.$$

This is the equation of the line passing through the intersection of L_1 and L_2 .

Illustration 17 : A straight line through $P(-2, -3)$ cuts the pair of straight lines $x^2 + 3y^2 + 4xy - 8x - 6y - 9 = 0$ in Q and R. Find the equation of the line if $PQ \cdot PR = 20$.

Solution :

$$\text{Let line be } \frac{x+2}{\cos \theta} = \frac{y+3}{\sin \theta} = r$$

$$\Rightarrow x = r \cos \theta - 2, y = r \sin \theta - 3 \quad \dots\dots (i)$$

$$\text{Now, } x^2 + 3y^2 + 4xy - 8x - 6y - 9 = 0 \quad \dots\dots (ii)$$

Taking intersection of (i) with (ii) and considering terms of r^2 and constant (as we need $PQ \cdot PR = r_1 \cdot r_2 = \text{product of the roots}$)

$$r^2(\cos^2 \theta + 3 \sin^2 \theta + 4 \sin \theta \cos \theta) + (\text{some terms})r + 80 = 0$$

$$\therefore r_1 \cdot r_2 = PQ \cdot PR = \frac{80}{\cos^2 \theta + 4 \sin \theta \cos \theta + 3 \sin^2 \theta}$$

$$\therefore \cos^2 \theta + 4 \sin \theta \cos \theta + 3 \sin^2 \theta = 4 \quad (\because PQ \cdot PR = 20)$$

$$\therefore \sin^2 \theta - 4 \sin \theta \cos \theta + 3 \cos^2 \theta = 0$$

$$\Rightarrow (\sin \theta - \cos \theta)(\sin \theta - 3 \cos \theta) = 0$$

$$\therefore \tan \theta = 1, \tan \theta = 3$$

$$\text{hence equation of the line is } y + 3 = 1(x + 2) \Rightarrow x - y = 1$$

$$\text{and } y + 3 = 3(x + 2) \Rightarrow 3x - y + 3 = 0.$$

Illustration 18 : If the line $y - \sqrt{3}x + 3 = 0$ cuts the parabola $y^2 = x + 2$ at A and B, then find the value of $PA \cdot PB$ {where $P \equiv (\sqrt{3}, 0)$ }

Solution :

$$\text{Slope of line } y - \sqrt{3}x + 3 = 0 \text{ is } \sqrt{3}.$$

$$\text{If line makes an angle } \theta \text{ with x-axis, then } \tan \theta = \sqrt{3}$$

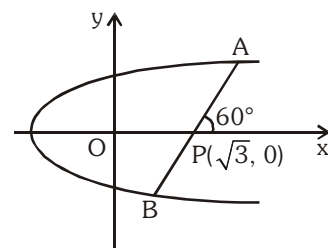
$$\therefore \theta = 60^\circ$$

$$\frac{x - \sqrt{3}}{\cos 60^\circ} = \frac{y - 0}{\sin 60^\circ} = r \Rightarrow \left(\sqrt{3} + \frac{r}{2}, \frac{r\sqrt{3}}{2} \right) \text{ be a point on}$$

$$\text{the parabola } y^2 = x + 2$$

$$\text{then } \frac{3}{4}r^2 = \sqrt{3} + \frac{r}{2} + 2 \Rightarrow 3r^2 - 2r - 4(2 + \sqrt{3}) = 0$$

$$\therefore PA \cdot PB = r_1 r_2 = \left| \frac{-4(2 + \sqrt{3})}{3} \right| = \frac{4(2 + \sqrt{3})}{3}$$



Do yourself - 6 :

- (i) Reduce the line $2x - 3y + 5 = 0$,
- In slope- intercept form and hence find slope & Y-intercept
 - In intercept form and hence find intercepts on the axes.
 - In normal form and hence find perpendicular distance from the origin and angle made by the perpendicular with the positive x-axis.
- (ii) Find distance of point A (2, 3) measured parallel to the line $x - y = 5$ from the line $2x + y + 6 = 0$.

(g) **General form :** We know that a first degree equation in x and y, $ax + by + c = 0$ always represents a straight line. This form is known as general form of straight line.

- (i) Slope of this line $= \frac{-a}{b} = -\frac{\text{coeff. of } x}{\text{coeff. of } y}$
- (ii) Intercept by this line on x-axis $= -\frac{c}{a}$ and intercept by this line on y-axis $= -\frac{c}{b}$
- (iii) To change the general form of a line to normal form, first take c to right hand side and make it positive, then divide the whole equation by $\sqrt{a^2 + b^2}$.

13. EQUATION OF LINES PARALLEL AND PERPENDICULAR TO A GIVEN LINE :

- (a) Equation of line parallel to line $ax + by + c = 0$

$$ax + by + \lambda = 0$$

- (b) Equation of line perpendicular to line $ax + by + c = 0$

$$bx - ay + k = 0$$

Here λ, k , are parameters and their values are obtained with the help of additional information given in the problem.

14. ANGLE BETWEEN TWO LINES :

- (a) If θ be the angle between two lines : $y = m_1x + c_1$ and $y = m_2x + c_2$, then $\tan \theta = \pm \left(\frac{m_1 - m_2}{1 + m_1 m_2} \right)$

Note :

- (i) There are two angles formed between two lines but usually the acute angle is taken as the angle between the lines. So we shall find θ from the above formula only by taking positive value of $\tan \theta$.
- (ii) Let m_1, m_2, m_3 are the slopes of three lines $L_1 = 0$; $L_2 = 0$; $L_3 = 0$ where $m_1 > m_2 > m_3$ then the interior angles of the ΔABC found by these formulas are given by,

$$\tan A = \frac{m_1 - m_2}{1 + m_1 m_2} ; \tan B = \frac{m_2 - m_3}{1 + m_2 m_3} \quad \& \quad \tan C = \frac{m_3 - m_1}{1 + m_3 m_1}$$

(b) If equation of lines are $a_1x+b_1y+c_1=0$ and $a_2x+b_2y+c_2=0$, then these lines are -

- (i) Parallel $\Leftrightarrow \frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$
 (ii) Perpendicular $\Leftrightarrow a_1a_2+b_1b_2=0$
 (iii) Coincident $\Leftrightarrow \frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$

Illustration 19 : If $x + 4y - 5 = 0$ and $4x + ky + 7 = 0$ are two perpendicular lines then k is -

- (A) 3 (B) 4 (C) -1 (D) -4

Solution :

$$m_1 = -\frac{1}{4} \quad m_2 = -\frac{4}{k}$$

Two lines are perpendicular if $m_1 m_2 = -1$

$$\Rightarrow \left(-\frac{1}{4}\right) \times \left(-\frac{4}{k}\right) = -1 \Rightarrow k = -1$$

Ans. (C)

Illustration 20 : A line L passes through the points $(1, 1)$ and $(0, 2)$ and another line M which is perpendicular to L passes through the point $(0, -1/2)$. The area of the triangle formed by these lines with y -axis is -

- (A) 25/8 (B) 25/16 (C) 25/4 (D) 25/32

Solution :

$$\text{Equation of the line } L \text{ is } y - 1 = \frac{-1}{1}(x - 1) \Rightarrow y = -x + 2$$

$$\text{Equation of the line } M \text{ is } y = x - 1/2.$$

If these lines meet y -axis at P and Q , then $PQ = 5/2$.

Also x -coordinate of their point of intersection $R = 5/4$

$$\therefore \text{ area of the } \Delta PQR = \frac{1}{2} \left(\frac{5}{2} \times \frac{5}{4} \right) = 25/16.$$

Ans. (B)

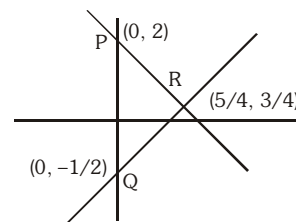


Illustration 21 : If the straight line $3x + 4y + 5 - k(x + y + 3) = 0$ is parallel to y -axis, then the value of k is -

- (A) 1 (B) 2 (C) 3 (D) 4

Solution :

A straight line is parallel to y -axis, if its y -coefficient is zero, i.e. $4 - k = 0$ i.e. $k = 4$ **Ans. (D)**

15. STRAIGHT LINE MAKING A GIVEN ANGLE WITH A LINE :

Equation of line passing through a point (x_1, y_1) and making an angle α , with the line $y = mx + c$ is written as :

$$y - y_1 = \frac{m \pm \tan \alpha}{1 \mp m \tan \alpha} (x - x_1)$$

Illustration 22 : Find the equation to the sides of an isosceles right-angled triangle, the equation of whose hypotenuse is $3x + 4y = 4$ and the opposite vertex is the point $(2, 2)$.

Solution : The problem can be restated as :

Find the equations of the straight lines passing through the given point $(2, 2)$ and making equal angles of 45° with the given straight line $3x + 4y - 4 = 0$. Slope of the line $3x + 4y - 4 = 0$ is $m_1 = -3/4$.

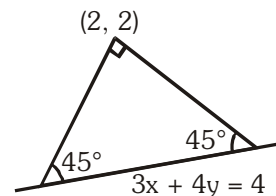
$$\Rightarrow \tan 45^\circ = \pm \frac{m - m_1}{1 + m_1 m}, \text{ i.e., } 1 = \pm \frac{m + 3/4}{1 - \frac{3}{4}m}$$

$$m_A = \frac{1}{7}, \text{ and } m_B = -7$$

Hence the required equations of the two lines are

$$y - 2 = m_A(x - 2) \text{ and } y - 2 = m_B(x - 2)$$

$$\Rightarrow 7y - x - 12 = 0 \text{ and } 7x + y = 16$$



Ans.

Do yourself - 7 :

- (i) Find the angle between the lines $3x + y - 7 = 0$ and $x + 2y - 9 = 0$.
- (ii) Find the line passing through the point $(2, 3)$ and perpendicular to the straight line $4x - 3y = 10$.
- (iii) Find the equation of the line which has positive y-intercept 4 units and is parallel to the line $2x - 3y - 7 = 0$. Also find the point where it cuts the x-axis.
- (iv) Classify the following pairs of lines as coincident, parallel or intersecting :
 - (a) $x + 2y - 3 = 0$ & $-3x - 6y + 9 = 0$
 - (b) $x + 2y + 1 = 0$ & $2x + 4y + 3 = 0$
 - (c) $3x - 2y + 5 = 0$ & $2x + y - 5 = 0$

16. LENGTH OF PERPENDICULAR FROM A POINT ON A LINE :

Length of perpendicular from a point (x_1, y_1) on the line $ax + by + c = 0$ is $\left| \frac{ax_1 + by_1 + c}{\sqrt{a^2 + b^2}} \right|$

In particular, the length of the perpendicular from the origin on the line $ax + by + c = 0$ is $P = \frac{|c|}{\sqrt{a^2 + b^2}}$

Illustration 23 : If the algebraic sum of perpendiculars from n given points on a variable straight line is zero then prove that the variable straight line passes through a fixed point.

Solution : Let n given points be (x_i, y_i) where $i = 1, 2, \dots, n$ and the variable straight line is $ax + by + c = 0$.

$$\text{Given that } \sum_{i=1}^n \left(\frac{ax_i + by_i + c}{\sqrt{a^2 + b^2}} \right) = 0 \Rightarrow a \sum x_i + b \sum y_i + cn = 0 \Rightarrow a \frac{\sum x_i}{n} + b \frac{\sum y_i}{n} + c = 0.$$

Hence the variable straight line always passes through the fixed point $\left(\frac{\sum x_i}{n}, \frac{\sum y_i}{n} \right)$.

Ans.

Illustration 24 : Prove that no line can be drawn through the point $(4, -5)$ so that its distance from $(-2, 3)$ will be equal to 12.

Solution : Suppose, if possible.

Equation of line through $(4, -5)$ with slope m is $y + 5 = m(x - 4)$

$$\Rightarrow mx - y - 4m - 5 = 0$$

$$\text{Then } \frac{|m(-2) - 3 - 4m - 5|}{\sqrt{m^2 + 1}} = 12$$

$$\Rightarrow |-6m - 8| = 12\sqrt{m^2 + 1}$$

$$\text{On squaring, } (6m + 8)^2 = 144(m^2 + 1)$$

$$\Rightarrow 4(3m + 4)^2 = 144(m^2 + 1) \Rightarrow (3m + 4)^2 = 36(m^2 + 1)$$

$$\Rightarrow 27m^2 - 24m + 20 = 0 \quad \dots\dots\dots (i)$$

Since the discriminant of (i) is $(-24)^2 - 4 \cdot 27 \cdot 20 = -1584$ which is negative, there is no real value of m . Hence no such line is possible.

17. DISTANCE BETWEEN TWO PARALLEL LINES :

(a) The distance between two parallel lines $ax + by + c_1 = 0$ and $ax + by + c_2 = 0$ is $\frac{|c_1 - c_2|}{\sqrt{a^2 + b^2}}$

(Note : The coefficients of x & y in both equations should be same)

(b) The area of the parallelogram $= \frac{p_1 p_2}{\sin \theta}$, where p_1 & p_2 are distances between two pairs of opposite sides & θ is the angle between any two adjacent sides. Note that area of the parallelogram bounded by the lines $y = m_1x + c_1$, $y = m_1x + c_2$ and $y = m_2x + d_1$,

$$y = m_2x + d_2 \text{ is given by } \left| \frac{(c_1 - c_2)(d_1 - d_2)}{m_1 - m_2} \right|.$$

Illustration 25 : Three lines $x + 2y + 3 = 0$, $x + 2y - 7 = 0$ and $2x - y - 4 = 0$ form 3 sides of two squares. Find the equation of remaining sides of these squares.

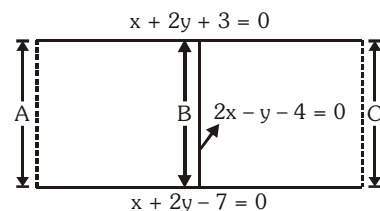
Solution : Distance between the two parallel lines is $\frac{|7 + 3|}{\sqrt{5}} = 2\sqrt{5}$.

The equations of sides A and C are of the form $2x - y + k = 0$.

Since distance between sides A and B = distance between sides B and C

$$\Rightarrow \frac{|k - (-4)|}{\sqrt{5}} = 2\sqrt{5} \Rightarrow \frac{k + 4}{\sqrt{5}} = \pm 2\sqrt{5} \Rightarrow k = 6, -14.$$

Hence the fourth sides of the two squares are (i) $2x - y + 6 = 0$ (ii) $2x - y - 14 = 0$.



Ans.

Do yourself - 8 :

(i) Find the distances between the following pair of parallel lines :

(a) $3x + 4y = 13$, $3x + 4y = 3$

(b) $3x - 4y + 9 = 0$, $6x - 8y - 15 = 0$

(ii) Find the points on the x-axis such that their perpendicular distance from the line $\frac{x}{a} + \frac{y}{b} = 1$ is 'a', a, $b > 0$.

(iii) Show that the area of the parallelogram formed by the lines

$2x - 3y + a = 0$, $3x - 2y - a = 0$, $2x - 3y + 3a = 0$ and $3x - 2y - 2a = 0$ is $\frac{2a^2}{5}$ square units.

18. POSITION OF TWO POINTS WITH RESPECT TO A GIVEN LINE :

Let the given line be $ax + by + c = 0$ and $P(x_1, y_1)$, $Q(x_2, y_2)$ be two points. If the expressions $ax_1 + by_1 + c$ and $ax_2 + by_2 + c$ have the same signs, then both the points P and Q lie on the same side of the line $ax + by + c = 0$. If the quantities $ax_1 + by_1 + c$ and $ax_2 + by_2 + c$ have opposite signs, then they lie on the opposite sides of the line.

19. CONCURRENCY OF LINES :

(a) Three lines $a_1x + b_1y + c_1 = 0$; $a_2x + b_2y + c_2 = 0$ and $a_3x + b_3y + c_3 = 0$ are concurrent,

if $\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = 0$

(b) To test the concurrency of three lines, first find out the point of intersection of any two of the three lines. If this point lies on the remaining line (i.e. coordinates of the point satisfy the equation of the line) then the three lines are concurrent otherwise not concurrent.

Illustration 27 : If the lines $ax + by + p = 0$, $x \cos \alpha + y \sin \alpha - p = 0$ ($p \neq 0$) and $x \sin \alpha - y \cos \alpha = 0$ are

concurrent and the first two lines include an angle $\frac{\pi}{4}$, then $a^2 + b^2$ is equal to -

(A) 1

(B) 2

(C) 4

(D) p^2

Solution :

Since the given lines are concurrent,

$\begin{vmatrix} a & b & p \\ \cos \alpha & \sin \alpha & -p \\ \sin \alpha & -\cos \alpha & 0 \end{vmatrix} = 0$

$\Rightarrow a \cos \alpha + b \sin \alpha + 1 = 0$ (i)

As $ax + by + p = 0$ and $x \cos \alpha + y \sin \alpha - p = 0$ include an angle $\frac{\pi}{4}$.

$\pm \tan \frac{\pi}{4} = \frac{-\frac{a}{b} + \frac{\cos \alpha}{\sin \alpha}}{1 + \frac{a \cos \alpha}{b \sin \alpha}}$

$$\Rightarrow -a \sin \alpha + b \cos \alpha = \pm (b \sin \alpha + a \cos \alpha)$$

$$\Rightarrow -a \sin \alpha + b \cos \alpha = \pm 1 \text{ [from (i)]} \quad \dots\dots\dots (ii)$$

Squaring and adding (i) & (ii), we get

$$a^2 + b^2 = 2.$$

Ans. (B)

Do yourself - 9 :

- (i) Examine the positions of the points (3, 4) and (2, -6) w.r.t. $3x - 4y = 8$
- (ii) If (2, 9), (-2, 1) and (1, -3) are the vertices of a triangle, then prove that the origin lies inside the triangle.
- (iii) Find the equation of the line joining the point (2, -9) and the point of intersection of lines $2x + 5y - 8 = 0$ and $3x - 4y - 35 = 0$.
- (iv) Find the value of λ , if the lines $3x - 4y - 13 = 0$, $8x - 11y - 33 = 0$ and $2x - 3y + \lambda = 0$ are concurrent.

20. REFLECTION OF A POINT :

Let $P(x, y)$ be any point, then its image with respect to

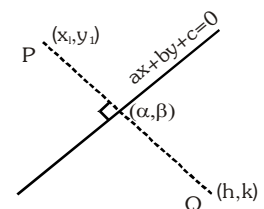
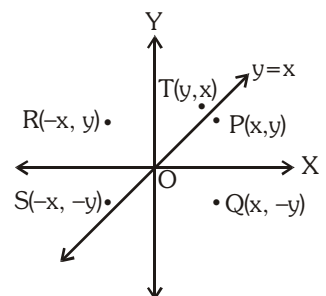
- (a) x-axis is $Q(x, -y)$
- (b) y-axis is $R(-x, y)$
- (c) origin is $S(-x, -y)$
- (d) line $y = x$ is $T(y, x)$
- (e) Reflection of a point about any arbitrary line : The image (h,k) of a

point $P(x_1, y_1)$ about the line $ax + by + c = 0$ is given by following formula.

$$\frac{h - x_1}{a} = \frac{k - y_1}{b} = -2 \frac{(ax_1 + by_1 + c)}{a^2 + b^2}$$

and the foot of perpendicular (α, β) from a point (x_1, y_1) on the line $ax + by + c = 0$ is given by following formula.

$$\frac{\alpha - x_1}{a} = \frac{\beta - y_1}{b} = -\frac{ax_1 + by_1 + c}{a^2 + b^2}$$



21. TRANSFORMATION OF AXES

(a) Shifting of origin without rotation of axes :

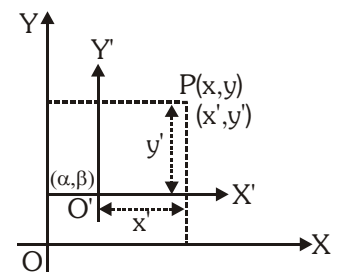
Let $P(x, y)$ with respect to axes OX and OY .

Let $O'(\alpha, \beta)$ is new origin with respect to axes OX and OY and let $P(x', y')$ with respect to axes $O'X'$ and $O'Y'$, where OX and $O'X'$ are parallel and OY and $O'Y'$ are parallel.

$$\text{Then } x = x' + \alpha, \quad y = y' + \beta$$

$$\text{or } x' = x - \alpha, \quad y' = y - \beta$$

Thus if origin is shifted to point (α, β) without rotation of axes, then new equation of curve can be obtained by putting $x + \alpha$ in place of x and $y + \beta$ in place of y .



(b) **Rotation of axes without shifting the origin :**

Let O be the origin. Let P (x, y) with respect to axes OX and OY and let P (x', y') with respect to axes OX' and OY' where $\angle X'OX = \angle YOY' = \theta$, where θ is measured in anticlockwise direction.

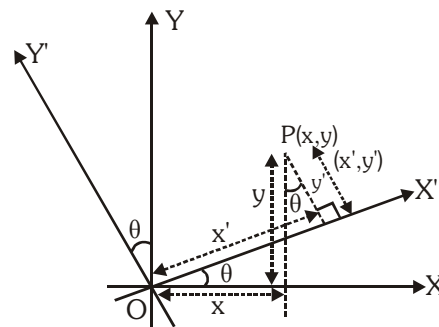
then $x = x' \cos \theta - y' \sin \theta$

$y = x' \sin \theta + y' \cos \theta$

and $x' = x \cos \theta + y \sin \theta$

$y' = -x \sin \theta + y \cos \theta$

The above relation between (x, y) and (x', y') can be easily obtained with the help of following table



New \ Old	x ↓	y ↓
x' →	cos θ	sin θ
y' →	-sin θ	cos θ

Illustration 28 : Through what angle should the axes be rotated so that the equation

$9x^2 - 2\sqrt{3}xy + 7y^2 = 10$ may be changed to $3x^2 + 5y^2 = 5$?

Solution :

Let angle be θ then replacing (x, y) by $(x \cos \theta - y \sin \theta, x \sin \theta + y \cos \theta)$

then $9x^2 - 2\sqrt{3}xy + 7y^2 = 10$ becomes

$9(x \cos \theta - y \sin \theta)^2 - 2\sqrt{3}(x \cos \theta - y \sin \theta)(x \sin \theta + y \cos \theta) + 7(x \sin \theta + y \cos \theta)^2 = 10$

$\Rightarrow x^2(9\cos^2\theta - 2\sqrt{3}\sin\theta\cos\theta + 7\sin^2\theta) + 2xy(-9\sin\theta\cos\theta - \sqrt{3}\cos 2\theta + 7\sin\theta\cos\theta) + y^2(9\cos^2\theta + 2\sqrt{3}\sin\theta\cos\theta + 7\sin^2\theta) = 10$

On comparing with $3x^2 + 5y^2 = 5$ (coefficient of $xy = 0$)

We get $-9\sin\theta\cos\theta - \sqrt{3}\cos 2\theta + 7\sin\theta\cos\theta = 0$

or $\sin 2\theta = -\sqrt{3}\cos 2\theta$ or $\tan 2\theta = -\sqrt{3} = \tan(180^\circ - 60^\circ)$

or $2\theta = 120^\circ$ $\therefore \theta = 60^\circ$

Do yourself - 10 :

(i) The point (4, 1) undergoes the following transformations, then the match the correct alternatives:

Column-I

(A) Reflection about x-axis is

(B) Reflection about y-axis is

(C) Reflection about origin is

(D) Reflection about the line $y = x$ is

(E) Reflection about the line $4x + 3y - 5 = 0$ is

Column-II

(p) (4, -1)

(q) (-4, -1)

(r) $\left(-\frac{12}{25}, -\frac{59}{25}\right)$

(s) (-4, 1)

(t) (1, 4)

(ii) To what point must the origin be shifted, so that the coordinates of the point (4, 5) become (-3, 9).

(iii) If the axes be turned through an angle $\tan^{-1}2$ (in anticlockwise direction), what does the equation $4xy - 3x^2 = a^2$ become ?

22. EQUATION OF BISECTORS OF ANGLES BETWEEN TWO LINES :

If equation of two intersecting lines are $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$, then equation of bisectors of the angles between these lines are written as :

$$\frac{a_1x + b_1y + c_1}{\sqrt{a_1^2 + b_1^2}} = \pm \frac{a_2x + b_2y + c_2}{\sqrt{a_2^2 + b_2^2}} \quad \dots\dots\dots(i)$$

(a) Equation of bisector of angle containing origin :

If the equation of the lines are written with constant terms c_1 and c_2 positive, then the equation of the bisectors of the angle containing the origin is obtained by taking positive sign in (i)

(b) Equation of bisector of acute/obtuse angles :

To find the equation of the bisector of the acute or obtuse angle :

- let ϕ be the angle between one of the two bisectors and one of two given lines. Then if $\tan\phi < 1$ i.e. $\phi < 45^\circ$ i.e. $2\phi < 90^\circ$, the angle bisector will be bisector of acute angle.
- See whether the constant terms c_1 and c_2 in the two equation are +ve or not. If not then multiply both sides of given equation by -1 to make the constant terms positive.

Determine the sign of $a_1a_2 + b_1b_2$

If sign of $a_1a_2 + b_1b_2$	For obtuse angle bisector	For acute angle bisector
+	use + sign in eq. (1)	use - sign in eq. (1)
-	use - sign in eq. (1)	use + sign in eq. (1)

i.e. if $a_1a_2 + b_1b_2 > 0$, then the bisector corresponding to + sign gives obtuse angle bisector

$$\frac{a_1x + b_1y + c_1}{\sqrt{a_1^2 + b_1^2}} = \frac{a_2x + b_2y + c_2}{\sqrt{a_2^2 + b_2^2}}$$

- Another way of identifying an acute and obtuse angle bisector is as follows :

Let $L_1 = 0$ & $L_2 = 0$ are the given lines & $u_1 = 0$ and $u_2 = 0$ are the bisectors between $L_1 = 0$ & $L_2 = 0$. Take a point P on any one of the lines $L_1 = 0$ or $L_2 = 0$ and drop perpendicular on $u_1 = 0$ & $u_2 = 0$ as shown . If,

$$|p| < |q| \Rightarrow u_1 \text{ is the acute angle bisector .}$$

$$|p| > |q| \Rightarrow u_1 \text{ is the obtuse angle bisector .}$$

$$|p| = |q| \Rightarrow \text{the lines } L_1 \text{ \& } L_2 \text{ are perpendicular .}$$

Note : Equation of straight lines passing through $P(x_1, y_1)$ & equally inclined with the lines $a_1x + b_1y + c_1 = 0$ & $a_2x + b_2y + c_2 = 0$ are those which are parallel to the bisectors between these two lines & passing through the point P.

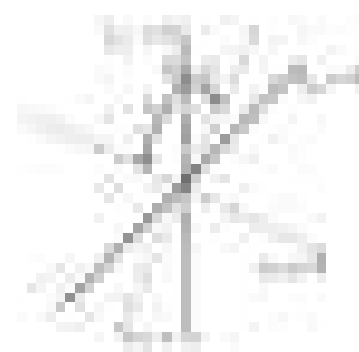


Illustration 29 : For the straight lines $4x + 3y - 6 = 0$ and $5x + 12y + 9 = 0$, find the equation of the

- (i) bisector of the obtuse angle between them.
- (ii) bisector of the acute angle between them.
- (iii) bisector of the angle which contains origin.
- (iv) bisector of the angle which contains $(1, 2)$.

Solution : Equations of bisectors of the angles between the given lines are

$$\frac{4x + 3y - 6}{\sqrt{4^2 + 3^2}} = \pm \frac{5x + 12y + 9}{\sqrt{5^2 + 12^2}} \Rightarrow 9x - 7y - 41 = 0 \text{ and } 7x + 9y - 3 = 0$$

If θ is the acute angle between the line $4x + 3y - 6 = 0$ and the bisector

$$9x - 7y - 41 = 0, \text{ then } \tan \theta = \left| \frac{-\frac{4}{3} - \frac{9}{7}}{1 + \left(-\frac{4}{3}\right)\frac{9}{7}} \right| = \frac{11}{3} > 1$$

Hence

- (i) bisector of the obtuse angle is $9x - 7y - 41 = 0$
- (ii) bisector of the acute angle is $7x + 9y - 3 = 0$
- (iii) bisector of the angle which contains origin

$$\frac{-4x - 3y + 6}{\sqrt{(-4)^2 + (-3)^2}} = \frac{5x + 12y + 9}{\sqrt{5^2 + 12^2}} \Rightarrow 7x + 9y - 3 = 0$$

- (iv) $L_1(1, 2) = 4 \times 1 + 3 \times 2 - 6 = 4 > 0$
 $L_2(1, 2) = 5 \times 1 + 12 \times 2 + 9 = 38 > 0$

$$\begin{aligned} &\text{+ve sign will give the required bisector, } \frac{4x + 3y - 6}{5} = + \frac{5x + 12y + 9}{13} \\ &\Rightarrow 9x - 7y - 41 = 0. \end{aligned}$$

Alternative :

Making c_1 and c_2 positive in the given equation, we get $-4x - 3y + 6 = 0$ and $5x + 12y + 9 = 0$

Since $a_1a_2 + b_1b_2 = -20 - 36 = -56 < 0$, so the origin will lie in the acute angle.

Hence bisector of the acute angle is given by

$$\frac{-4x - 3y + 6}{\sqrt{4^2 + 3^2}} = \frac{5x + 12y + 9}{\sqrt{5^2 + 12^2}} \Rightarrow 7x + 9y - 3 = 0$$

Similarly bisector of obtuse angle is $9x - 7y - 41 = 0$.

Illustration 30 : A ray of light is sent along the line $x - 2y - 3 = 0$. Upon reaching the line mirror $3x - 2y - 5 = 0$, the ray is reflected from it. Find the equation of the line containing the reflected ray.

Solution :

Let Q be the point of intersection of the incident ray and the line mirror, then

$$x_1 - 2y_1 - 3 = 0 \quad \& \quad 3x_1 - 2y_1 - 5 = 0$$

on solving these equations, we get

$$x_1 = 1 \quad \& \quad y_1 = -1$$

Since P(-1, -2) be a point lies on the incident ray, so we can find the image of the point P on the reflected ray about the line mirror (by property of reflection).

Let P'(h, k) be the image of point P about line mirror, then

$$\frac{h+1}{3} = \frac{k+2}{-2} = \frac{-2(-3+4-5)}{13} \Rightarrow h = \frac{11}{13} \text{ and } k = \frac{-42}{13}.$$

$$\text{So } P'\left(\frac{11}{13}, \frac{-42}{13}\right)$$

Then equation of reflected ray will be

$$(y+1) = \frac{\left(\frac{-42}{13}+1\right)(x-1)}{\left(\frac{11}{13}-1\right)}$$

$$\Rightarrow 2y - 29x + 31 = 0.$$

23. FAMILY OF LINES :

If equation of two lines be $P \equiv a_1x + b_1y + c_1 = 0$ and $Q \equiv a_2x + b_2y + c_2 = 0$, then the equation of the lines passing through the point of intersection of these lines is : $P + \lambda Q = 0$ or $a_1x + b_1y + c_1 + \lambda(a_2x + b_2y + c_2) = 0$. The value of λ is obtained with the help of the additional informations given in the problem.

Illustration 31 : Prove that each member of the family of straight lines

$(3\sin\theta + 4\cos\theta)x + (2\sin\theta - 7\cos\theta)y + (\sin\theta + 2\cos\theta) = 0$ (θ is a parameter) passes through a fixed point.

Solution :

The given family of straight lines can be rewritten as

$$(3x + 2y + 1)\sin\theta + (4x - 7y + 2)\cos\theta = 0$$

$$\text{or, } (4x - 7y + 2) + \tan\theta(3x + 2y + 1) = 0 \text{ which is of the form } L_1 + \lambda L_2 = 0$$

Hence each member of it will pass through a fixed point which is the intersection of

$$4x - 7y + 2 = 0 \text{ and } 3x + 2y + 1 = 0 \text{ i.e. } \left(\frac{-11}{29}, \frac{2}{29}\right).$$

Do yourself - 11 :

- (i) Find the equations of bisectors of the angle between the lines $4x + 3y = 7$ and $24x + 7y - 31 = 0$. Also find which of them is (a) the bisector of the angle containing origin (b) the bisector of the acute angle.
- (ii) Find the equations of the line which pass through the point of intersection of the lines $4x - 3y = 1$ and $2x - 5y + 3 = 0$ and is equally inclined to the coordinate axes.
- (iii) Find the equation of the line through the point of intersection of the lines $3x - 4y + 1 = 0$ & $5x + y - 1 = 0$ and perpendicular to the line $2x - 3y = 10$.

24. PAIR OF STRAIGHT LINES :

(a) Homogeneous equation of second degree :

(i) Let us consider the homogeneous equation of 2nd degree as

$$ax^2 + 2hxy + by^2 = 0 \quad \dots\dots(i)$$

which represents pair of straight lines passing through the origin.

Now, we divide by x^2 , we get

$$a + 2h\left(\frac{y}{x}\right) + b\left(\frac{y}{x}\right)^2 = 0$$

$$\frac{y}{x} = m \quad (\text{say})$$

$$\text{then } a + 2hm + bm^2 = 0 \quad \dots\dots(ii)$$

if m_1 & m_2 are the roots of equation (ii), then $m_1 + m_2 = -\frac{2h}{b}$, $m_1 m_2 = \frac{a}{b}$

$$\text{and also, } \tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right| = \left| \frac{\sqrt{(m_1 + m_2)^2 - 4m_1 m_2}}{1 + m_1 m_2} \right| = \left| \frac{\sqrt{\frac{4h^2}{b^2} - \frac{4a}{b}}}{1 + \frac{a}{b}} \right| = \pm \frac{2\sqrt{h^2 - ab}}{a + b}$$

These line will be :

(1) **Real and different**, if $h^2 - ab > 0$ (2) **Real and coincident**, if $h^2 - ab = 0$

(3) **Imaginary**, if $h^2 - ab < 0$

(ii) The condition that these lines are :

(1) At **right angles** to each other is **$a + b = 0$** . i.e. coefficient of x^2 + coefficient of $y^2 = 0$.

(2) **Coincident** is **$h^2 = ab$** .

(3) **Equally inclined** to the axes of x is **$h = 0$** . i.e. coefficient of $xy = 0$.

(iii) Homogeneous equation of 2nd degree **$ax^2 + 2hxy + by^2 = 0$** always represent a pair of straight lines whose equations are

$$y = \left(\frac{-h \pm \sqrt{h^2 - ab}}{b} \right) x \equiv y = m_1 x \text{ \& } y = m_2 x \text{ and } m_1 + m_2 = -\frac{2h}{b}; m_1 m_2 = \frac{a}{b}$$

These straight lines passes through the origin.

(iv) Pair of straight lines perpendicular to the lines $ax^2 + 2hxy + by^2 = 0$ and through origin are given by $bx^2 - 2hxy + ay^2 = 0$.

(v) The product of the perpendiculars drawn from the point (x_1, y_1) on the lines $ax^2 + 2hxy + by^2 = 0$ is

$$by^2 = 0 \text{ is } \left| \frac{ax_1^2 + 2hx_1y_1 + by_1^2}{\sqrt{(a-b)^2 + 4h^2}} \right|$$

Note : A homogeneous equation of degree n represents n straight lines passing through **origin**.

(b) The combined equation of angle bisectors :

The combined equation of angle bisectors between the lines represented by homogeneous equation

of 2nd degree is given by $\frac{x^2 - y^2}{a - b} = \frac{xy}{h}$, $a \neq b$, $h \neq 0$.

Note :

- (i) If $a = b$, the bisectors are $x^2 - y^2 = 0$ i.e. $x - y = 0$, $x + y = 0$
- (ii) If $h = 0$, the bisectors are $xy = 0$ i.e. $x = 0$, $y = 0$.
- (iii) The two bisectors are always at **right angles**, since we have coefficient of x^2 + coefficient of $y^2 = 0$

(c) General Equation and Homogeneous Equation of Second Degree :

(i) The general equation of second degree $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ represents

a pair of straight lines, if $\Delta = abc + 2fgh - af^2 - bg^2 - ch^2 = 0$ i.e. $\begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} = 0$

(ii) If θ be the angle between the lines, then $\tan \theta = \pm \frac{2\sqrt{h^2 - ab}}{a + b}$

Obviously these lines are

- (1) Parallel, if $\Delta = 0$, $h^2 = ab$ or if $h^2 = ab$ and $bg^2 = af^2$
- (2) Perpendicular, if $a + b = 0$ i.e. coeff. of x^2 + coeff. of $y^2 = 0$.

(iii) The product of the perpendiculars drawn from the origin to the lines

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0 \text{ is } \left| \frac{c}{\sqrt{(a-b)^2 + 4h^2}} \right|$$

Illustration 32 : If $\lambda x^2 - 10xy + 12y^2 + 5x - 16y - 3 = 0$ represents a pair of straight lines, then λ is equal to -

- (A) 4 (B) 3 (C) 2 (D) 1

Solution :

Here $a = \lambda$, $b = 12$, $c = -3$, $f = -8$, $g = 5/2$, $h = -5$

Using condition $abc + 2fgh - af^2 - bg^2 - ch^2 = 0$, we have

$$\lambda(12)(-3) + 2(-8)(5/2)(-5) - \lambda(64) - 12(25/4) + 3(25) = 0$$

$$\Rightarrow -36\lambda + 200 - 64\lambda - 75 + 75 = 0 \Rightarrow 100\lambda = 200$$

$$\therefore \lambda = 2$$

Ans. (C)

Do yourself - 12 :

- (i) Prove that the equation $x^2 - 5xy + 4y^2 = 0$ represents two lines passing through the origin. Also find their equations.
- (ii) If the equation $3x^2 + kxy - 10y^2 + 7x + 19y = 6$ represents a pair of lines, find the value of k .
- (iii) If the equation $6x^2 - 11xy - 10y^2 - 19y - 6 = 0$ represents a pair of lines, find their equations. Also find the angle between the two lines.
- (iv) Find the point of intersection and the angle between the lines given by the equation : $2x^2 - 3xy - 2y^2 + 10x + 5y = 0$.

25. EQUATIONS OF LINES JOINING THE POINTS OF INTERSECTION OF A LINE AND A CURVE TO THE ORIGIN :

- (a) Let the equation of curve be :

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0 \quad \dots(i)$$

and straight line be

$$lx + my + n = 0 \quad \dots(ii)$$

Now joint equation of line OP and OQ joining the origin and points of intersection P and Q can be obtained by making the equation (i) homogenous with the help of equation of the line.

Thus required equation is given by

$$ax^2 + 2hxy + by^2 + 2(gx + fy) \left(\frac{lx + my}{-n} \right) + c \left(\frac{lx + my}{-n} \right)^2 = 0$$

$$\Rightarrow (an^2 + 2gln + cl^2)x^2 + 2(hn^2 + gmn + fln + clm)xy + (bn^2 + 2fmn + cm^2)y^2 = 0 \quad \dots(iii)$$

All points which satisfy (i) and (ii) simultaneously, will satisfy (iii)

- (b) Any second degree curve through the four points of intersection of $f(x, y) = 0$ & $xy = 0$ is given by $f(x, y) + \lambda xy = 0$ where $f(x, y) = 0$ is also a second degree curve.

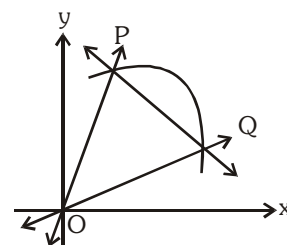


Illustration 35 : The chord $\sqrt{6}y = \sqrt{8}px + \sqrt{2}$ of the curve $py^2 + 1 = 4x$ subtends a right angle at origin then find the value of p .

Solution : $\sqrt{3}y - 2px = 1$ is the given chord. Homogenizing the equation of the curve, we get,

$$py^2 - 4x(\sqrt{3}y - 2px) + (\sqrt{3}y - 2px)^2 = 0$$

$$\Rightarrow (4p^2 + 8p)x^2 + (p + 3)y^2 - 4\sqrt{3}xy - 4\sqrt{3}pxy = 0$$

Now, angle at origin is 90°

$$\therefore \text{coefficient of } x^2 + \text{coefficient of } y^2 = 0$$

$$\therefore 4p^2 + 8p + p + 3 = 0 \Rightarrow 4p^2 + 9p + 3 = 0$$

$$\therefore p = \frac{-9 \pm \sqrt{81 - 48}}{8} = \frac{-9 \pm \sqrt{33}}{8}.$$

Do yourself - 13 :

- (i) Find the angle subtended at the origin by the intercept made on the curve

$$x^2 - y^2 - xy + 3x - 6y + 18 = 0 \text{ by the line } 2x - y = 3.$$

- (ii) Find the equation of the lines joining the origin to the points of intersection of the curve

$$2x^2 + 3xy - 4x + 1 = 0 \text{ and the line } 3x + y = 1.$$

ANSWERS FOR DO YOURSELF

1 : (i) $PQ = \sqrt{34}$ units ; (ii) $x = 6$ or $x = 0$ (iii) $11, -7$

2 : (i) (a) $(2, 1)$; (b) $(7, 16)$; (ii) (a) $2 : 3$ (internally) ; (b) $9 : 4$ (externally) ; (c) $8 : 7$ (internally)

3 : (i) (a) $\left(\frac{5}{3}, 3\right)$; (b) $\left(-\frac{9}{2}, \frac{15}{2}\right), \frac{5\sqrt{10}}{2}$, (c) $(14, -6)$

4 : (i) 25 square units ; (ii) 132 square units ;

5 : (i) $x = \pm 2$; (ii) $y = \pm x$;

6 : (i) (a) $y = \frac{2x}{3} + \frac{5}{3}, \frac{2}{3}, \frac{5}{3}$; (b) $\frac{x}{(-5/2)} + \frac{y}{(5/3)} = 1, -\frac{5}{2}, \frac{5}{3}$;

(c) $-\frac{2x}{\sqrt{13}} + \frac{3y}{\sqrt{13}} = \frac{5}{\sqrt{13}}, \frac{5}{\sqrt{13}}, \alpha = \pi - \tan^{-1}\left(\frac{3}{2}\right)$; (ii) $13\sqrt{2/3}$ units

7 : (i) $\theta = 135^\circ$ or 45° ; (ii) $3x + 4y = 18$; (iii) $2x - 3y + 12 = 0, (-6, 0)$

(iv) (a) Coincident, (b) Parallel, (c) Intersecting

8 : (i) (a) 2 (b) $33/10$; (ii) $\left(\frac{a}{b}(b \pm \sqrt{a^2 + b^2}), 0\right)$

9 : (i) opposite sides of the line ; (iii) $-y + x = 11$; (iv) $\lambda = -7$

10 : (i) (A) $\rightarrow$ (p), (B) $\rightarrow$ (s), (C) $\rightarrow$ (q), (D) $\rightarrow$ (t), (E) $\rightarrow$ (r), ; (ii) $(7, -4)$; (iv) $x^2 - 4y^2 = a^2$

11 : (i) $x - 2y + 1 = 0$ & $2x + y - 3 = 0$; (a) $x - 2y + 1 = 0$; (b) $2x + y - 3 = 0$

(ii) $x + y = 2, x = y$; (iii) $69x + 46y - 25 = 0$

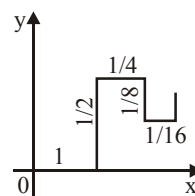
12 : (i) $x - y = 0$ & $x - 4y = 0$; (ii) $k = -1$, or $-\frac{127}{6}$;

(iii) $2x - 5y - 2 = 0$ & $3x + 2y + 3 = 0$; $\pm \tan^{-1}\left(\frac{19}{4}\right)$ (iv) $(-1, 2), 90^\circ$

13 : (i) $\theta = \pm \tan^{-1} \frac{4}{7}$; (ii) $x^2 - y^2 - 5xy = 0$

EXERCISE (O-1)

- Coordinates of the vertices of a triangle ABC are (12,8), (-2,6) and (6,0) then the **correct** statement is-
 (A) triangle is right but not isosceles
 (B) triangle is isosceles but not right
 (C) triangle is obtuse
 (D) the product of the abscissa of the centroid, orthocenter and circumcenter is 160.
- Find the value of x_1 if the distance between the points $(x_1, 2)$ and $(3, 4)$ be 8.
 (A) $3 \pm 2\sqrt{15}$ (B) $3 \pm \sqrt{15}$ (C) $2 \pm 3\sqrt{15}$ (D) $2 \pm \sqrt{15}$
- If P(1,2), Q(4,6), R(5,7) & S(a,b) are the vertices of a parallelogram PQRS, then :
 (A) $a = 2, b = 4$ (B) $a = 3, b = 4$ (C) $a = 2, b = 3$ (D) $a = 3, b = 5$
- The four points whose co-ordinates are (2,1),(1,4),(4,5),(5,2) form :
 (A) a rectangle which is not a square (B) a trapezium which is not a parallelogram
 (C) a square (D) a rhombus which is not a square
- The length of a line segment AB is 10 units. If the coordinates of one extremity are (2,-3) and the abscissa of the other extremity is 10 then the sum of all possible values of the ordinate of the other extremity is -
 (A) 3 (B) -4 (C) 12 (D) -6
- If A and B are the points (-3,4) and (2,1), then the co-ordinates of the point C on AB produced such that $AC = 2BC$ are :
 (A) (2,4) (B) (3,7) (C) (7,-2) (D) $\left(-\frac{1}{2}, \frac{5}{2}\right)$
- The orthocenter of the triangle ABC is 'B' and the circumcenter is 'S' (a,b). If A is the origin then the co-ordinates of C are :
 (A) (2a,2b) (B) $\left(\frac{a}{2}, \frac{b}{2}\right)$ (C) $\left(\sqrt{a^2 + b^2}, 0\right)$ (D) none
- A particle begins at the origin and moves successively in the following manner as shown,
 1 unit to the right, $\frac{1}{2}$ unit up, $\frac{1}{4}$ unit to the right,
 $\frac{1}{8}$ unit down, $\frac{1}{16}$ unit to the right etc.
 The length of each move is half the length of the previous move and movement continues in the 'zigzag' manner infinitely. The co-ordinates of the point to which the 'zigzag' converges is -
 (A) $(4/3, 2/3)$ (B) $(4/3, 2/5)$ (C) $(3/2, 2/3)$ (D) $(2, 2/5)$



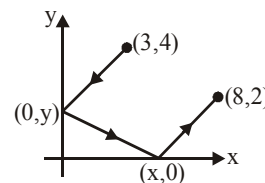
9. The medians of a triangle meet at $(0, -3)$ and its two vertices are at $(-1, 4)$ and $(5, 2)$. Then the third vertex is at -
(A) $(4, 15)$ (B) $(-4, -15)$ (C) $(-4, 15)$ (D) $(4, -15)$
10. If the two vertices of a triangle are $(7, 2)$ and $(1, 6)$ and its centroid is $(4, 6)$ then the coordinate of the third vertex are (a, b) . The value of $(a + b)$, is-
(A) 13 (B) 14 (C) 15 (D) 16
11. If in triangle ABC, $A \equiv (1, 10)$, circumcenter $\equiv \left(-\frac{1}{3}, \frac{2}{3}\right)$ and orthocenter $\equiv \left(\frac{11}{3}, \frac{4}{3}\right)$ then the co-ordinates of mid-point of side opposite to A is-
(A) $(1, -11/3)$ (B) $(1, 5)$ (C) $(1, -3)$ (D) $(1, 6)$
12. Suppose ABC is a triangle with 3 acute angles A, B and C. The point whose coordinates are $(\cos B - \sin A, \sin B - \cos A)$ can be in the -
(A) first and 2nd quadrant (B) second the 3rd quadrant
(C) third and 4th quadrant (D) second quadrant only
13. Consider the points $P(2, -4)$; $Q(4, -2)$ and $R(7, 1)$. The points P, Q, R -
(A) form an equilateral triangle
(B) form a right angled triangle
(C) form an isosceles triangle which is not equilateral
(D) are collinear.
14. A triangle has two of its vertices at $(0, 1)$ and $(2, 2)$ in the cartesian plane. Its third vertex lies on the x-axis. If the area of the triangle is 2 square units then the sum of the possible abscissae of the third vertex, is-
(A) -4 (B) 0 (C) 5 (D) 6
15. A point $P(x, y)$ moves so that the sum of the distance from P to the coordinate axes is equal to the distance from P to the point $A(1, 1)$. The equation of the locus of P in the first quadrant is -
(A) $(x + 1)(y + 1) = 1$ (B) $(x + 1)(y + 1) = 2$
(C) $(x - 1)(y - 1) = 1$ (D) $(x - 1)(y - 1) = 2$
16. Let $A(2, -3)$ and $B(-2, 1)$ be vertices of a $\triangle ABC$. If the centroid of $\triangle ABC$ moves on the line $2x + 3y = 1$, then the locus of the vertex C is-
(A) $2x + 3y = 9$ (B) $2x - 3y = 7$ (C) $3x + 2y = 5$ (D) $3x - 2y = 3$
17. A stick of length 10 units rests against the floor and a wall of a room. If the stick begins to slide on the floor then the locus of its middle point is :
(A) $x^2 + y^2 = 2.5$ (B) $x^2 + y^2 = 25$ (C) $x^2 + y^2 = 100$ (D) none
18. Given the points $A(0, 4)$ and $B(0, -4)$, the equation of the locus of the point P such that $|AP - BP| = 6$ is-
(A) $9x^2 - 7y^2 + 63 = 0$ (B) $9x^2 - 7y^2 - 63 = 0$
(C) $7x^2 - 9y^2 + 63 = 0$ (D) $7x^2 - 9y^2 - 63 = 0$

19. A line passes through (2,2) and cuts a triangle of area 9 square units from the first quadrant. The sum of all possible values for the slope of such a line, is-
 (A) -2.5 (B) -2 (C) -1.5 (D) -1
20. The diagonals of a parallelogram PQRS are along the lines $x + 3y = 4$ and $6x - 2y = 7$. Then PQRS must be a :
 (A) rectangle (B) square
 (C) cyclic quadrilateral (D) rhombus
21. A and B are any two points on the positive x and y axis respectively satisfying $2(OA) + 3(OB) = 10$. If P is the middle point of AB then the locus of P is-
 (A) $2x + 3y = 5$ (B) $2x + 3y = 10$ (C) $3x + 2y = 5$ (D) $3x + 2y = 10$
22. The co-ordinates of the orthocentre of the triangle bounded by the lines, $4x - 7y + 10 = 0$; $x + y = 5$ and $7x + 4y = 15$ is-
 (A) (2,1) (B) (-1,2) (C) (1,2) (D) (1,-2)
23. If the x intercept of the line $y = mx + 2$ is greater than $1/2$ then the gradient of the line lies in the interval-
 (A) (-1,0) (B) $(-1/4, 0)$ (C) $(-\infty, -4)$ (D) $(-4, 0)$
24. The greatest slope along the graph represented by the equation $4x^2 - y^2 + 2y - 1 = 0$, is-
 (A) -3 (B) -2 (C) 2 (D) 3
25. The extremities of the base of an isosceles triangle ABC are the points A(2,0) and B(0,1). If the equation of the side AC is $x = 2$ then the slope of the side BC is -
 (A) $\frac{3}{4}$ (B) $\frac{4}{3}$ (C) $\frac{3}{2}$ (D) $\sqrt{3}$
26. Number of lines that can be drawn through the point(4,-5) so that its distance from (-2,3) will be equal to 12 is equal to-
 (A) 0 (B) 1 (C) 2 (D) 3
27. Two mutually perpendicular straight lines through the origin from an isosceles triangle with the line $2x + y = 5$. Then the area of the triangle is :
 (A) 5 (B) 3 (C) $5/2$ (D) 1
28. Point 'P' lies on the line $l \{(x,y) | 3x + 5y = 15\}$. If 'P' is also equidistant from the coordinate axes, then P can be located in which of the four quadrants -
 (A) I only (B) II only (C) I or II only (D) IV only
29. The area of the parallelogram formed by the lines $3x + 4y = 7a$; $3x + 4y = 7b$; $4x + 3y = 7c$ and $4x + 3y = 7d$ is-
 (A) $\frac{|(a-b)(c-d)|}{7}$ (B) $|(a-b)(c-d)|$ (C) $\frac{|(a-b)(c-d)|}{49}$ (D) $7|(a-b)(c-d)|$
30. If x_1, y_1 are the roots of $x^2 + 8x - 20 = 0$, x_2, y_2 are the roots of $4x^2 + 32x - 57 = 0$ and x_3, y_3 are the roots of $9x^2 + 72x - 112 = 0$, then the points, (x_1, y_1) , (x_2, y_2) and (x_3, y_3) -
 (A) are collinear (B) form an equilateral triangle
 (C) form a right angled isosceles triangle (D) are concyclic

31. If the lines $\lambda x + (\sin \alpha)y + \cos \alpha = 0$, $x + (\cos \alpha)y + \sin \alpha = 0$, $x - (\sin \alpha)y + \cos \alpha = 0$ pass through the same point where $\alpha \in \mathbb{R}$ then λ lies in the interval -
- (A) $[-1, 1]$ (B) $[-\sqrt{2}, \sqrt{2}]$ (C) $[-2, 2]$ (D) $(-\infty, \infty)$
32. Family of lines represented by the equation $(\cos \theta + \sin \theta)x + (\cos \theta - \sin \theta)y - 3(3 \cos \theta + \sin \theta) = 0$ passes through a fixed point M for all real values of θ . The reflection of M in the line $x - y = 0$, is-
- (A) (6,3) (B) (3,6) (C) (-6,3) (D) (3,-6)
33. m,n are integer with $0 < n < m$. A is the point (m,n) on the cartesian plane. B is the reflection of A in the line $y = x$. C is the reflection of B in the y-axis, D is the reflection of C in the x-axis and E is the reflection of D in the y-axis. The area of the pentagon ABCDE is -
- (A) $2m(m + n)$ (B) $m(m + 3n)$ (C) $m(2m + 3n)$ (D) $2m(m + 3n)$
34. Area of the quadrilateral formed by the lines $|x| + |y| = 2$ is :
- (A) 8 (B) 6 (C) 4 (D) none
35. Consider a parallelogram whose sides are represented by the lines $2x + 3y = 0$; $2x + 3y - 5 = 0$; $3x - 4y = 0$ and $3x - 4y = 3$. The equation of the diagonal not passing through the origin, is-
- (A) $21x - 11y + 15 = 0$ (B) $9x - 11y + 15 = 0$
(C) $21x - 29y - 15 = 0$ (D) $21x - 11y - 15 = 0$
36. If the straight lines, $ax + amy + 1 = 0$, $bx + (m + 1)by + 1 = 0$ and $cx + (m + 2)cy + 1 = 0$, $m \neq 0$ are concurrent then a,b,c are in :
- (A) A.P. only for $m = 1$ (B) A.P. for all m
(C) G.P. for all m (D) H.P. for all m
37. A ray of light passing through the point A(1,2) is reflected at a point B on the x-axis and then passes through (5,3). Then the equation of AB is :
- (A) $5x + 4y = 13$ (B) $5x - 4y = -3$
(C) $4x + 5y = 14$ (D) $4x - 5y = -6$
38. If the line $y = mx$ bisects the angle between the lines $ax^2 + 2hxy + by^2 = 0$ then m is a root of the quadratic equation :
- (A) $hx^2 + (a - b)x - h = 0$ (B) $x^2 + h(a - b)x - 1 = 0$
(C) $(a - b)x^2 + hx - (a - b) = 0$ (D) $(a - b)x^2 - hx - (a - b) = 0$
39. If the equation $ax^2 - 6xy + y^2 + 2gx + 2fy + c = 0$ represents a pair of lines whose slopes are m and m^2 , then sum of all possible values of a is-
- (A) 17 (B) -19 (C) 19 (D) -17

40. Suppose that a ray of light leaves the point (3,4), reflects off the y-axis towards the x-axis, reflects off the x-axis, and finally arrives at the point (8,2). The value of x, is-

- (A) $x = 4\frac{1}{2}$ (B) $x = 4\frac{1}{3}$
(C) $x = 4\frac{2}{3}$ (D) $x = 5\frac{1}{3}$



41. Let $S = \{(x,y) | x^2 + 2xy + y^2 - 3x - 3y + 2 = 0\}$, then S -

- (A) consists of two coincident lines.
(B) consists of two parallel lines which are not coincident.
(C) consists of two intersecting lines.
(D) is a parabola.

[MATRIX LIST TYPE]

42. Find the equation to the straight line :

Column-I

- (P) passing through the point (2, 3) and perpendicular to the straight line $4x - 3y = 10$.
(Q) passing through the point (-6, 10) and perpendicular to the straight line $7x + 8y = 5$.
(R) passing through the point (2, -3) and perpendicular to the straight line joining the points (5, 7) and (-6, 3).
(S) passing through the point (-4, -3) and perpendicular to the straight line joining (1, 3) and (2, 7).

Column-II

- (1) $4y + 11x = 10$
(2) $4y + 3x = 18$
(3) $x + 4y + 16 = 0$
(4) $9y - 8x = 118$

Codes :

- | | P | Q | R | S |
|-----|---|---|---|---|
| (A) | 1 | 2 | 3 | 4 |
| (B) | 2 | 4 | 1 | 3 |
| (C) | 4 | 3 | 2 | 1 |
| (D) | 1 | 3 | 4 | 2 |

[MATRIX MATCH]

43. Column-I

- (A) The ratio in which the point $\left(\frac{10}{7}, \frac{33}{7}\right)$ divides the line segment joining the points (1, 3) and (2, 7) is $p : q$, then $(p + q)$ is (p & q are coprime)
(B) The ratio in which the point (-2, -9) divides The line segment joining the points (1, 3) and (2, 7) is $p : q$, then $(p + q)$ is (p & q are coprime)
(C) If (p, q) divides internally the line joining (-1, 2) and (4, -5) in ratio 2 : 3 then $(p + q)$ is
(D) If (p, q) divides externally the line joining (-1, 2) (4, -5) in ratio 2 : 3 then $(p + q)$ is

Column-II

- (P) $\frac{1}{5}$
(Q) -1
(R) 5
(S) 7

44. Column-I

- (A) The points $(2, -2)$, $(8, 4)$, $(5, 7)$ and $(-1, 1)$ taken in order constitute the vertices of a
- (B) The points $(0, -1)$, $(2, 1)$, $(0, 3)$ and $(-2, 1)$ taken in order are the vertices of a
- (C) The points $(3, -5)$, $(-5, -4)$, $(7, 10)$, $(15, 9)$ taken in order are the vertices of a
- (D) The points $(-3, 4)$, $(-1, 0)$, $(1, 0)$ and $(3, 4)$ taken in order are the vertices of a

Column-II

- (P) square
- (Q) rectangle
- (R) trapezium
- (S) parallelogram
- (T) cyclic quadrilateral

45. Column-I

- (A) The points $(2, -2)$, $(-2, 1)$ and $(5, 2)$
- (B) The points $(1, -2)$, $(-3, 0)$ and $(5, 6)$
- (C) The points $(3, 7)$, $(6, 5)$ and $(15, -1)$
- (D) The points $(2, 2)$, $(-2, -2)$ and $(-2\sqrt{3}, 2\sqrt{3})$

Column-II

- (P) are the vertices of a right angled triangle
- (Q) are the vertices of a right angle isosceles triangle
- (R) are the vertices of an equilateral triangle
- (S) do not form a triangle

46. Equation of Straight Line

Column-I

- (A) Which cuts-off an intercept 4 on the x-axis and passes through the point $(2, -3)$.
- (B) Which cuts-off equal intercepts on the co-ordinate axes and passes through $(2, 5)$
- (C) Which makes an angle of 135° with the axis of x and which cuts the axis of y at a distance -8 from the origin and
- (D) Through the point $(4, 1)$ and making with the axes in the first quadrant a triangle whose area is 8.

Column-II

- (P) $2x + y + 1 = 0$
- (Q) $x + y = 7$
- (R) $3x - 2y = 12$
- (S) $x + 4y = 8$
- (T) $x + y + 8 = 0$

47. Column-I

- (A) The four lines $3x - 4y + 11 = 0$; $3x - 4y - 9 = 0$; $4x + 3y + 3 = 0$ and $4x + 3y - 17 = 0$ enclose a figure which is
- (B) The lines $2x + y = 1$, $x + 2y = 1$, $2x + y = 3$ and $x + 2y = 3$ form a figure which is
- (C) If 'O' is the origin, P is the intersection of the lines $2x^2 - 7xy + 3y^2 + 5x + 10y - 25 = 0$, A and B are the points in which these lines are cut by the line $x + 2y - 5 = 0$, then the points O, A, P, B (in some order) are the vertices of

Column-II

- (P) a quadrilateral which is neither a parallelogram nor a trapezium nor a kite.
- (Q) a parallelogram which is neither a rectangle nor a rhombus
- (R) a rhombus which is not a square
- (S) a square

EXERCISE (O-2)

- The line $x = c$ cuts the triangle with corners $(0,0)$; $(1,1)$ and $(9,1)$ into two regions. For the area of the two regions to be the same c must be equal to-
(A) $5/2$ (B) 3 (C) $7/2$ (D) 3 or 15
- If m and b are real numbers and $mb > 0$, then the line whose equation is $y = mx + b$ cannot contain the point-
(A) $(0,2009)$ (B) $(2009,0)$ (C) $(0,-2009)$ (D) $(20,-100)$
- In a triangle ABC, side AB has the equation $2x + 3y = 29$ and the side AC has the equation, $x + 2y = 16$. If the mid-point of BC is $(5,6)$ then the equation of BC is :
(A) $x - y = -1$ (B) $5x - 2y = 13$ (C) $x + y = 11$ (D) $3x - 4y = -9$
- The vertex of the right angle of a right angled triangle lies on the straight line $2x - y - 10 = 0$ and the two other vertices, at points $(2,-3)$ and $(4,1)$ then the area of triangle in sq. units is-
(A) $\sqrt{10}$ (B) 3 (C) $\frac{33}{5}$ (D) 11
- The line $(k + 1)^2x + ky - 2k^2 - 2 = 0$ passes through a point regardless of the value k . Which of the following is the line with slope 2 passing through the point ?
(A) $y = 2x - 8$ (B) $y = 2x - 5$ (C) $y = 2x - 4$ (D) $y = 2x + 8$
- A triangle ABC is formed by the lines $2x - 3y - 6 = 0$; $3x - y + 3 = 0$ and $3x + 4y - 12 = 0$. If the points $P(\alpha,0)$ and $Q(0,\beta)$ always lie on or inside the ΔABC , then :
(A) $\alpha \in [-1,2]$ and $\beta \in [-2,3]$ (B) $\alpha \in [-1,3]$ and $\beta \in [-2,4]$
(C) $\alpha \in [-2,4]$ and $\beta \in [-3,4]$ (D) $\alpha \in [-1,3]$ and $\beta \in [-2,3]$
- The co-ordinates of a point P on the line $2x - y + 5 = 0$ such that $|PA - PB|$ is maximum where A is $(4,-2)$ and B is $(2,-4)$ will be :
(A) $(11,27)$ (B) $(-11,-17)$ (C) $(-11,17)$ (D) $(0,5)$
- If the straight lines joining the origin and the points of intersection of the curve $5x^2 + 12xy - 6y^2 + 4x - 2y + 3 = 0$ and $x + ky - 1 = 0$ are equally inclined to the x-axis then the value of k :
(A) is equal to 1 (B) is equal to -1
(C) is equal to 2 (D) does not exist in the set of real numbers
- Through a point A on the x-axis a straight line is drawn parallel to y-axis so as to meet the pair of straight lines $ax^2 + 2hxy + by^2 = 0$ in B and C. If $AB = BC$ then-
(A) $h^2 = 4ab$ (B) $8h^2 = 9ab$ (C) $9h^2 = 8ab$ (D) $4h^2 = ab$

[MULTIPLE CHOICE]

- The area of triangle ABC is 20 square units. The co-ordinates of vertex A are $(-5,0)$ and B are $(3,0)$. The vertex C lies on the line, $x - y = 2$. The co-ordinates of C are -
(A) $(5,3)$ (B) $(-3,-5)$ (C) $(-5,-7)$ (D) $(7,5)$

11. Three vertices of a triangle are $A(4,3)$; $B(1,-1)$ and $C(7,k)$. Value(s) of k for which centroid, orthocentre, incentre and circumcentre of the $\triangle ABC$ lie on the same straight line is/are-
 (A) 7 (B) -1 (C) $-19/8$ (D) none
12. Line $\frac{x}{a} + \frac{y}{b} = 1$ cuts the co-ordinate axes at $A(a,0)$ and $B(0,b)$ and the line $\frac{x}{a'} + \frac{y}{b'} = -1$ at $A'(-a',0)$ and $B'(0,-b')$. If the points A, B, A', B' are concyclic then the orthocentre of the triangle ABA' is -
 (A) $(0,0)$ (B) $(0,b')$ (C) $\left(0, \frac{aa'}{b}\right)$ (D) $\left(0, \frac{bb'}{a}\right)$
13. A line passes through the origin and makes an angle of $\pi/4$ with the line $x + y + 1 = 0$. Then :
 (A) equation of the line is $x = 0$
 (B) the equation of the line is $y = 0$
 (C) the point of intersection of the line with the given line is $(-1,0)$
 (D) the point of intersection of the line with the given line is $(0,1)$
14. Equation of a straight line passing through the point $(2,3)$ and inclined at an angle of $\arctan \frac{1}{2}$ with the line $y + 2x = 5$ is-
 (A) $y = 3$ (B) $x = 2$ (C) $3x + 4y - 18 = 0$ (D) $4x + 3y - 17 = 0$
15. Consider the equation $y - y_1 = m(x - x_1)$. If m and x_1 are fixed and different lines are drawn for different values of y_1 , then :
 (A) the lines will pass through a fixed point (B) there will be a set of parallel lines
 (C) all the lines intersect the line $x = x_1$ (D) all the lines will be parallel to the line $y = x_1$.
16. If one vertex of an equilateral triangle of side 'a' lies at the origin and the other lies on the line $x - \sqrt{3}y = 0$ then the co-ordinates of the third vertex are :
 (A) $(0,a)$ (B) $\left(\frac{\sqrt{3}a}{2}, -\frac{a}{2}\right)$ (C) $(0,-a)$ (D) $\left(-\frac{\sqrt{3}a}{2}, \frac{a}{2}\right)$
17. The sides of a triangle are the straight lines $x + y = 1$; $7y = x$ and $\sqrt{3}y + x = 0$. Then which of the following is an interior point of the triangle ?
 (A) circumcentre (B) centroid
 (C) incentre (D) orthocentre
18. If $a^2 + 9b^2 - 4c^2 = 6ab$ then the family of lines $ax + by + c = 0$ are concurrent at :
 (A) $(1/2, 3/2)$ (B) $(-1/2, -3/2)$ (C) $(-1/2, 3/2)$ (D) $(1/2, -3/2)$
19. The x-coordinates of the vertices of a square of unit area are the roots of the equation $x^2 - 3|x| + 2 = 0$ and the y-coordinates of the vertices are the roots of the equation $y^2 - 3y + 2 = 0$ then the possible vertices of the square is/are-
 (A) $(1,1), (2,1), (2,2), (1,2)$ (B) $(-1,1), (-2,1), (-2,2), (-1,2)$
 (C) $(2,1), (1,-1), (1,2), (2,2)$ (D) $(-2,1), (-1,-1), (-1,2), (-2,2)$
20. If the vertices P, Q, R of a triangle PQR are rational points, which of the following points of the triangle PQR is/are always rational point(s) ?
 (A) centroid (B) incentre (C) circumcentre (D) orthocentre

21. The lines L_1 and L_2 denoted by $3x^2 + 10xy + 8y^2 + 14x + 22y + 15 = 0$ intersect at the point P and have gradients m_1 and m_2 respectively. The acute angles between them is θ . Which of the following relations hold good ?

- (A) $m_1 + m_2 = 5/4$
(B) $m_1 m_2 = 3/8$

(C) acute angle between L_1 and L_2 is $\sin^{-1}\left(\frac{2}{5\sqrt{5}}\right)$.

(D) sum of the abscissa and ordinate of the point P is -1 .

22. Let $B(1, -3)$ and $D(0, 4)$ represent two vertices of rhombus ABCD in (x,y) plane, then coordinates of vertex A if $\angle BAD = 60^\circ$ can be equal to-

(A) $\left(\frac{1-7\sqrt{3}}{2}, \frac{1-\sqrt{3}}{2}\right)$

(B) $\left(\frac{1+7\sqrt{3}}{2}, \frac{1+\sqrt{3}}{2}\right)$

(C) $\left(\frac{1-14\sqrt{3}}{2}, \frac{1-2\sqrt{3}}{2}\right)$

(D) $\left(\frac{1+14\sqrt{3}}{2}, \frac{1+2\sqrt{3}}{2}\right)$

Paragraph for Question Nos. 23 to 25

Let ABCD is a square with sides of unit length. Points E and F are taken on sides AB and AD respectively so that $AE = AF$. Let P be a point inside the square ABCD.

23. The maximum possible area of quadrilateral CDFE is -

- (A) $\frac{1}{8}$ (B) $\frac{1}{4}$ (C) $\frac{5}{8}$ (D) $\frac{3}{8}$

24. The value of $(PA)^2 - (PB)^2 + (PC)^2 - (PD)^2$ is equal to-

- (A) 3 (B) 2 (C) 1 (D) 0

25. Let a line passing through point A divides the square ABCD into two parts so that area of one portion is double the other, then the length of portion of line inside the square is -

- (A) $\frac{\sqrt{10}}{3}$ (B) $\frac{\sqrt{13}}{3}$ (C) $\frac{\sqrt{11}}{3}$ (D) $\frac{2}{\sqrt{3}}$

Paragraph for Question Nos. 26 to 28

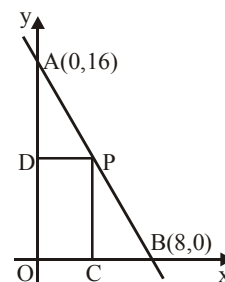
In the diagram, a line is drawn through the points $A(0, 16)$ and $B(8, 0)$. Point P is chosen in the first quadrant on the line through A and B. Points C and D are chosen on the x and y axis respectively, so that PDOC is a rectangle.

26. Perpendicular distance of the line AB from the point $(2, 2)$ is -

- (A) $\sqrt{4}$ (B) $\sqrt{10}$
(C) $\sqrt{20}$ (D) $\sqrt{50}$

27. Sum of the coordinates of the point P if PDOC is a square is -

- (A) $\frac{32}{3}$ (B) $\frac{16}{3}$ (C) 16 (D) 11



28. Number of possible ordered pair(s) of all positions of the point P on AB so that the area of the rectangle PDOC is 30 sq. units, is-
- (A) three (B) two (C) one (D) zero

Paragraph for question nos. 29 and 30

An equilateral triangle ABC has its centroid at the origin and the base BC lies along the line $x + y = 1$.

29. Area of the equilateral $\triangle ABC$ is -
- (A) $\frac{3\sqrt{3}}{2}$ (B) $\frac{3\sqrt{3}}{4}$ (C) $\frac{3\sqrt{2}}{2}$ (D) $\frac{2\sqrt{3}}{4}$
30. Gradient of the other two lines are -
- (A) $\sqrt{3}, \sqrt{2}$ (B) $\sqrt{3}, \frac{1}{\sqrt{3}}$ (C) $\sqrt{2} + 1, \sqrt{2} - 1$ (D) $2 + \sqrt{3}, 2 - \sqrt{3}$

Paragraph for Question Nos. 31 to 33

Consider two points $A \equiv (1, 2)$ and $B \equiv (3, -1)$. Let M be a point on the straight line $L \equiv x + y = 0$.

31. If M be a point on the line $L = 0$ such that $AM + BM$ is minimum, then the reflection of M in the line $x = y$ is -
- (A) $(1, -1)$ (B) $(-1, 1)$ (C) $(2, -2)$ (D) $(-2, 2)$
32. If M be a point on the line $L = 0$ such that $|AM - BM|$ is maximum, then the distance of M from $N \equiv (1, 1)$ is-
- (A) $5\sqrt{2}$ (B) 7 (C) $3\sqrt{5}$ (D) 10
33. If M be a point on the line $L = 0$ such that $|AM - BM|$ is minimum, then the area of $\triangle AMB$ equals-
- (A) $\frac{13}{4}$ (B) $\frac{13}{2}$ (C) $\frac{13}{6}$ (D) $\frac{13}{8}$

EXERCISE (S-1)

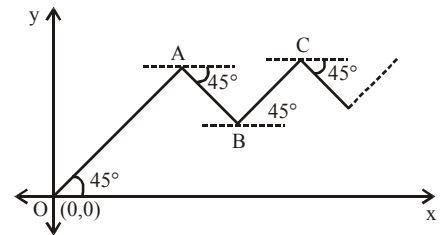
1. Line $\frac{x}{6} + \frac{y}{8} = 1$ intersects the x and y axes at M and N respectively. If the coordinates of the point P lying inside the triangle OMN (where 'O' is origin) are (a, b) such that the areas of the triangle POM, PON and PMN are equal. Find
- (a) the coordinates of the point P and
- (b) the radius of the circle escribed opposite to the angle N.
2. Two vertices of a triangle are $(4, -3)$ & $(-2, 5)$. If the orthocentre of the triangle is at $(1, 2)$, find the coordinates of the third vertex.
3. The point A divides the join of $P(-5, 1)$ & $Q(3, 5)$ in the ratio $K : 1$. Find the two values of K for which the area of triangle ABC, where B is $(1, 5)$ & C is $(7, -2)$, is equal to 2 units in magnitude.
4. Determine the ratio in which the point $P(3, 5)$ divides the join of $A(1, 3)$ & $B(7, 9)$. Find the harmonic conjugate of P w.r.t. A & B.
5. A line is such that its segment between the straight lines $5x - y - 4 = 0$ and $3x + 4y - 4 = 0$ is bisected at the point $(1, 5)$. Obtain the equation.

6. The area of a triangle is 5. Two of its vertices are $(2, 1)$ & $(3, -2)$. The third vertex lies on $y = x + 3$. Find the third vertex.
7. A variable line, drawn through the point of intersection of the straight lines $\frac{x}{a} + \frac{y}{b} = 1$ & $\frac{x}{b} + \frac{y}{a} = 1$, meets the coordinate axes in A & B. Find the locus of the mid point of AB.
8. Consider the family of lines $(x - y - 6) + \lambda(2x + y + 3) = 0$ and $(x + 2y - 4) + \lambda(3x - 2y - 4) = 0$. If the lines of these 2 families are at right angle to each other then find the locus of their point of intersection.
9. Two consecutive sides of a parallelogram are $4x + 5y = 0$ & $7x + 2y = 0$. If the equation to one diagonal is $11x + 7y = 9$, find the equation to the other diagonal.
10. The line $3x + 2y = 24$ meets the y-axis at A & the x-axis at B. The perpendicular bisector of AB meets the line through $(0, -1)$ parallel to x-axis at C. Find the area of the triangle ABC.
11. If the straight line drawn through the point $P(\sqrt{3}, 2)$ & inclined at an angle $\frac{\pi}{6}$ with the x-axis, meets the line $\sqrt{3}x - 4y + 8 = 0$ at Q. Find the length PQ.
12. A triangle has side lengths 18, 24 and 30. Find the area of the triangle whose vertices are the incentre, circumcentre and centroid of the triangle.
13. The points $(1, 3)$ & $(5, 1)$ are two opposite vertices of a rectangle. The other two vertices lie on the line $y = 2x + c$. Find c & the remaining vertices.
14. A straight line L is perpendicular to the line $5x - y = 1$. The area of the triangle formed by the line L & the coordinate axes is 5. Find the equation of the line.
15. Two equal sides of an isosceles triangle are given by the equations $7x - y + 3 = 0$ and $x + y - 3 = 0$ & its third side passes through the point $(1, -10)$. Determine the equation of the third side.
16. The equations of the perpendicular bisectors of the sides AB & AC of a triangle ABC are $x - y + 5 = 0$ & $x + 2y = 0$, respectively. If the point A is $(1, -2)$ find the equation of the line BC.
17. Given vertices A $(1, 1)$, B $(4, -2)$ & C $(5, 5)$ of a triangle, find the equation of the perpendicular dropped from C to the interior bisector of the angle A.
18. Consider a triangle ABC with sides AB and AC having the equations $L_1 = 0$ and $L_2 = 0$. Let the centroid, orthocentre and circumcentre of the $\triangle ABC$ are G, H and S respectively. $L = 0$ denotes the equation of sides BC.
 - (a) If $L_1 : 2x - y = 0$ and $L_2 : x + y = 3$ and $G(2, 3)$ then find the slope of the line $L = 0$.
 - (b) If $L_1 : 2x + y = 0$ and $L_2 : x - y + 2 = 0$ and $H(2, 3)$ then find the y-intercept of $L = 0$.
 - (c) If $L_1 : x + y - 1 = 0$ and $L_2 : 2x - y + 4 = 0$ and $S(2, 1)$ then find the x-intercept of the line $L = 0$.
19. Let P be the point $(3, 2)$. Let Q be the reflection of P about the x-axis. Let R be the reflection of Q about the line $y = -x$ and let S be the reflection of R through the origin. PQRS is a convex quadrilateral. Find the area of PQRS.
20. Let $O(0, 0)$, $A(6, 0)$ and $B(3, \sqrt{3})$ be the vertices of $\triangle OAB$. Let R be the region consisting of all those points P inside $\triangle OAB$ which satisfy $d(P, OA) \leq \text{minimum}(d(P, OB), d(P, AB))$ where $d(P, OA)$, $d(P, OB)$ and $d(P, AB)$ represent the distance of P from the sides OA, OB and AB respectively. If the area of region R is $9(a - \sqrt{b})$ where a and b are coprime, then find the value of $(a + b)$

21. Consider a line pair $2x^2 + 3xy - 2y^2 - 10x + 15y - 28 = 0$ and another line L passing through origin with gradient 3. The line pair and line L form a triangle whose vertices are A , B and C .
- Find the sum of the cotangents of the interior angles of the triangle ABC .
 - Find the area of triangle ABC
 - Find the radius of the circle touching all the 3 sides of the triangle.
22. A straight line is drawn from the point $(1, 0)$ to the curve $x^2 + y^2 + 6x - 10y + 1 = 0$, such that the intercept made on it by the curve subtends a right angle at the origin. Find the equations of the line.

EXERCISE (S-2)

1. Point O , A , B , C are shown in figure where $OA = 2AB = 4BC = \dots$ so on. Let A is the centroid of a triangle whose orthocentre and circumcenter are $(2, 4)$ and $\left(\frac{7}{2}, \frac{5}{2}\right)$ respectively. If an insect starts moving from the point $O(0,0)$ along the straight line in zig-zag fashion and terminates ultimately at point $P(\alpha, \beta)$ then find the value of $(\alpha + \beta)$



2. Let ABC be a triangle such that the coordinates of A are $(-3, 1)$. Equation of the median through B is $2x + y - 3 = 0$ and equation of the angular bisector of C is $7x - 4y - 1 = 0$. Then match the entries of column-I with their corresponding correct entries of column-II.

Column-I

- Equation of the line AB is
- Equation of the line BC is
- Equation of CA is

Column-II

- $2x + y - 3 = 0$
- $2x - 3y + 9 = 0$
- $4x + 7y + 5 = 0$
- $18x - y - 49 = 0$

3. The equations of perpendiculars of the sides AB & AC of triangle ABC are $x - y - 4 = 0$ and $2x - y - 5 = 0$ respectively. If the vertex A is $(-2, 3)$ and point of intersection of perpendiculars bisectors is $\left(\frac{3}{2}, \frac{5}{2}\right)$, find the equation of medians to the sides AB & AC respectively.
4. The interior angle bisector of angle A for the triangle ABC whose coordinates of the vertices are $A(-8, 5)$; $B(-15, -19)$ and $C(1, -7)$ has the equation $ax + 2y + c = 0$. Find 'a' and 'c'.
5. Find the equation of the straight lines passing through $(-2, -7)$ & having an intercept of length 3 between the straight lines $4x + 3y = 12$, $4x + 3y = 3$.
6. Two sides of a rhombus $ABCD$ are parallel to the lines $y = x + 2$ & $y = 7x + 3$. If the diagonals of the rhombus intersect at the point $(1, 2)$ & the vertex A is on the y -axis, find the possible coordinates of A .
7. Find the equations of the sides of a triangle having $(4, -1)$ as a vertex, if the lines $x - 1 = 0$ and $x - y - 1 = 0$ are the equations of two internal bisectors of its angles.
8. P is the point $(-1, 2)$, a variable line through P cuts the x & y axes at A & B respectively Q is the point on AB such that PA, PQ, PB are H.P. Find the locus of Q .

9. Consider a ΔABC whose sides AB, BC and CA are represented by the straight lines $2x + y = 0$, $x + py = q$ and $x - y = 3$ respectively. The point P is (2, 3).
- If P is the centroid, then find the value of $(p + q)$.
 - If P is the orthocentre, then find the value of $(p + q)$.
 - If P is the circumcentre, then find the value of $(p + q)$.
10. The sides of a triangle have the combined equation $x^2 - 3y^2 - 2xy + 8y - 4 = 0$. The third side, which is variable always passes through the point $(-5, -1)$. If the range of values of the slope of the third line so that the origin is an interior point of the triangle, lies in the interval (a, b) , then find $\left(a + \frac{1}{b^2}\right)$.
11. Show that all the chords of the curve $3x^2 - y^2 - 2x + 4y = 0$ which subtend a right angle at the origin are concurrent. Does this result also hold for the curve, $3x^2 + 3y^2 - 2x + 4y = 0$? If yes, what is the point of concurrency & if not, give reasons.
12. The two line pairs $y^2 - 4y + 3 = 0$ and $x^2 + 4xy + 4y^2 - 5x - 10y + 4 = 0$ enclose a 4 sided convex polygon find (i) area of the polygon; (ii) length of its diagonals.

EXERCISE (JM)

1. The lines $p(p^2 + 1)x - y + q = 0$ and $(p^2 + 1)^2x + (p^2 + 1)y + 2q = 0$ are [AIEEE 2009]
Perpendicular to a common line for :
- Exactly two values of p
 - More than two values of p
 - No value of p
 - Exactly one value of p

2. The line L given by $\frac{x}{5} + \frac{y}{b} = 1$ passes through the point (13, 32). The line K is parallel to L and has

the equation $\frac{x}{c} + \frac{y}{3} = 1$. Then the distance between L and K is : [AIEEE-2010]

- $\frac{23}{\sqrt{15}}$
- $\sqrt{17}$
- $\frac{17}{\sqrt{15}}$
- $\frac{23}{\sqrt{17}}$

3. The lines $L_1 : y - x = 0$ and $L_2 : 2x + y = 0$ intersect the line $L_3 : y + 2 = 0$ at P and Q respectively. The bisector of the acute angle between L_1 and L_2 intersects L_3 at R. [AIEEE 2011]

Statement - 1 : The ratio PR : RQ equals $2\sqrt{2} : \sqrt{5}$

Statement - 2 : In any triangle, bisector of an angle divides the triangle into two similar triangles.

- Statement-1 is true, Statement-2 is false.
- Statement-1 is false, Statement-2 is true
- Statement-1 is true, Statement-2 is true; Statement-2 is a correct explanation for Statement-1
- Statement-1 is true, Statement-2 is true; Statement-2 is **not** a correct explanation for Statement-1.

4. The lines $x + y = |a|$ and $ax - y = 1$ intersect each other in the first quadrant. Then the set of all possible values of a is the interval : [AIEEE 2011]

- $(-1, 1]$
- $(0, \infty)$
- $[1, \infty)$
- $(-1, \infty)$

5. A line is drawn through the point (1, 2) to meet the coordinate axes at P and Q such that it forms a triangle OPQ, where O is the origin. If the area of the triangle OPQ is least, then the slope of the line PQ is : **[AIEEE 2012]**
- (1) $-\frac{1}{2}$ (2) $-\frac{1}{4}$ (3) -4 (4) -2
6. If the line $2x + y = k$ passes through the point which divides the line segment joining the points (1, 1) and (2, 4) in the ratio 3 : 2, then k equals : **[AIEEE 2012]**
- (1) $\frac{11}{5}$ (2) $\frac{29}{5}$ (3) 5 (4) 6
7. A ray of light along $x + \sqrt{3}y = \sqrt{3}$ gets reflected upon reaching x-axis, the equation of the reflected ray is : **[JEE-MAIN 2013]**
- (1) $y = x + \sqrt{3}$ (2) $\sqrt{3}y = x - \sqrt{3}$ (3) $y = \sqrt{3}x - \sqrt{3}$ (4) $\sqrt{3}y = x - 1$
8. The x-coordinate of the incentre of the triangle that has the coordinates of mid points of its sides as (0, 1), (1, 1) and (1, 0) is : **[JEE-MAIN 2013]**
- (1) $2 + \sqrt{2}$ (2) $2 - \sqrt{2}$ (3) $1 + \sqrt{2}$ (4) $1 - \sqrt{2}$
9. A light ray emerging from the point source placed at P(1, 3) is reflected at a point Q in the axis of x. If the reflected ray passes through the point R(6, 7), then the abscissa of Q is : **[JEE-MAIN Online 2013]**
- (1) 3 (2) $\frac{7}{2}$ (3) 1 (4) $\frac{5}{2}$
10. If the three lines $x - 3y = p$, $ax + 2y = q$ and $ax + y = r$ form a right – angled triangle then: **[JEE-MAIN Online 2013]**
- (1) $a^2 - 6a - 12 = 0$ (2) $a^2 - 9a + 12 = 0$
 (3) $a^2 - 9a + 18 = 0$ (4) $a^2 - 6a - 18 = 0$
11. If the x-intercept of some line L is double as that of the line, $3x + 4y = 12$ and the y-intercept of L is half as that of the same line, then the slope of L is :- **[JEE-MAIN Online 2013]**
- (1) -3 (2) $-3/2$ (3) $-3/8$ (4) $-3/16$
12. If the extremities of the base of an isosceles triangle are the points (2a, 0) and (0, a) and the equation of one of the sides is $x = 2a$, then the area of the triangle, in square units, is : **[JEE-MAIN Online 2013]**
- (1) $\frac{5}{2}a^2$ (2) $\frac{5}{4}a^2$ (3) $\frac{25a^2}{4}$ (4) $5a^2$

13. Let θ_1 be the angle between two lines $2x + 3y + c_1 = 0$ and $-x + 5y + c_2 = 0$, and θ_2 be the angle between two lines $2x + 3y + c_1 = 0$ and $-x + 5y + c_3 = 0$, where c_1, c_2, c_3 are any real numbers :

[JEE-MAIN Online 2013]

Statement-1 : If c_2 and c_3 are proportional, then $\theta_1 = \theta_2$.

Statement-2 : $\theta_1 = \theta_2$ for all c_2 and c_3 .

- (1) Statement-1 is true and Statement - 2 is true, Statement-2 is not a correct explanation for Statement-1.
 (2) Statement-1 is false and Statement-2 is true.
 (3) Statement-1 is true and Statement-2 is false.
 (4) Statement-1 is true and Statement - 2 is true, Statement-2 is a correct explanation for Statement-1.
14. Let A (-3, 2) and B (-2, 1) be the vertices of a triangle ABC. If the centroid of this triangle lies on the line $3x + 4y + 2 = 0$, then the vertex C lies on the line : [JEE-MAIN Online 2013]
- (1) $4x + 3y + 5 = 0$ (2) $3x + 4y + 5 = 0$ (3) $3x + 4y + 3 = 0$ (4) $4x + 3y + 3 = 0$
15. If the image of point P(2, 3) in a line L is Q (4, 5) then, the image of point R (0, 0) in the same line is : [JEE-MAIN Online 2013]

- (1) (4, 5) (2) (2, 2) (3) (3, 4) (4) (7, 7)

16. Let a, b, c and d be non-zero numbers. If the point of intersection of the lines $4ax + 2ay + c = 0$ and $5bx + 2by + d = 0$ lies in the fourth quadrant and is equidistant from the two axes then : [JEE(Main)-2014]

- (1) $2bc - 3ad = 0$ (2) $2bc + 3ad = 0$ (3) $3bc - 2ad = 0$ (4) $3bc + 2ad = 0$

17. Let PS be the median of the triangle with vertices P (2, 2), Q (6, -1) and R (7, 3). The equation of the line passing through (1, -1) and parallel to PS is : [JEE(Main)-2014]

- (1) $4x - 7y - 11 = 0$ (2) $2x + 9y + 7 = 0$ (3) $4x + 7y + 3 = 0$ (4) $2x - 9y - 11 = 0$

18. Locus of the image of the point (2, 3) in the line $(2x - 3y + 4) + k(x - 2y + 3) = 0$, $k \in \mathbb{R}$, is a

- (1) circle of radius $\sqrt{2}$ (2) circle of radius $\sqrt{3}$
 (3) straight line parallel to x-axis (4) straight line parallel to y-axis

[JEE(Main)-2015]

19. Two sides of a rhombus are along the lines, $x - y + 1 = 0$ and $7x - y - 5 = 0$. If its diagonals intersect at (-1, -2), then which one of the following is a vertex of this rhombus ? [JEE(Main)-2016]

- (1) $\left(-\frac{10}{3}, -\frac{7}{3}\right)$ (2) (-3, -9) (3) (-3, -8) (4) $\left(\frac{1}{3}, -\frac{8}{3}\right)$

20. Let k be an integer such that triangle with vertices (k, -3k), (5, k) and (-k, 2) has area 28 sq. units. Then the orthocentre of this triangle is at the point : [JEE(Main)-2017]

- (1) $\left(2, \frac{1}{2}\right)$ (2) $\left(2, -\frac{1}{2}\right)$ (3) $\left(1, \frac{3}{4}\right)$ (4) $\left(1, -\frac{3}{4}\right)$

21. Let the orthocentre and centroid of a triangle be A(-3, 5) and B(3, 3) respectively. If C is the circumcentre of this triangle, then the radius of the circle having line segment AC as diameter, is: [JEE(Main)-2018]

- (1) $2\sqrt{10}$ (2) $3\sqrt{\frac{5}{2}}$ (3) $\frac{3\sqrt{5}}{2}$ (4) $\sqrt{10}$

22. A straight line through a fixed point (2, 3) intersects the coordinate axes at distinct points P and Q. If O is the origin and the rectangle OPRQ is completed, then the locus of R is : [JEE(Main)-2018]
- (1) $2x + 3y = xy$ (2) $3x + 2y = xy$ (3) $3x + 2y = 6xy$ (4) $3x + 2y = 6$

EXERCISE (JA)

1. Consider the lines given by

$$L_1 = x + 3y - 5 = 0$$

$$L_2 = 3x - ky - 1 = 0$$

$$L_3 = 5x + 2y - 12 = 0$$

Match the statements / Expression in **Column-I** with the statements / Expressions in **Column-II** and indicate your answer by darkening the appropriate bubbles in the 4×4 matrix given in OMR.

Column-I

(A) L_1, L_2, L_3 are concurrent, if

(B) One of L_1, L_2, L_3 is parallel to at least one of the other two, if

(C) L_1, L_2, L_3 form a triangle, if

(D) L_1, L_2, L_3 do not form a triangle, if

Column-II

(P) $k = -9$

(Q) $k = -\frac{6}{5}$

(R) $k = \frac{5}{6}$

(S) $k = 5$

[JEE 2008, 6]

2. Let P, Q, R and S be the points on the plane with position vectors $-2\hat{i} - \hat{j}$, $4\hat{i}, 3\hat{i} + 3\hat{j}$ and $-3\hat{i} + 2\hat{j}$ respectively. The quadrilateral PQRS must be a
- (A) parallelogram, which is neither a rhombus nor a rectangle
(B) square
(C) rectangle, but not a square
(D) rhombus, but not a square

[JEE 2010, 3]

3. A straight line L through the point (3, -2) is inclined at an angle 60° to the line $\sqrt{3}x + y = 1$. If L also intersect the x-axis, then the equation of L is [JEE 2011, 3 (-1)]

(A) $y + \sqrt{3}x + 2 - 3\sqrt{3} = 0$

(B) $y - \sqrt{3}x + 2 + 3\sqrt{3} = 0$

(C) $\sqrt{3}y - x + 3 + 2\sqrt{3} = 0$

(D) $\sqrt{3}y + x - 3 + 2\sqrt{3} = 0$

4. For $a > b > c > 0$, the distance between (1, 1) and the point of intersection of the lines $ax + by + c = 0$ and $bx + ay + c = 0$ is less than $2\sqrt{2}$. Then [JEE-Advanced 2013, 2]

(A) $a + b - c > 0$

(B) $a - b + c < 0$

(C) $a - b + c > 0$

(D) $a + b - c < 0$

5. For a point P in the plane, let $d_1(P)$ and $d_2(P)$ be the distances of the point P from the lines $x - y = 0$ and $x + y = 0$ respectively. The area of the region R consisting of all points P lying in the first quadrant of the plane and satisfying $2 \leq d_1(P) + d_2(P) \leq 4$, is

[JEE(Advanced)-2014, 3]

ANSWER KEY

EXERCISE (O-1)

1. D 2. A 3. C 4. C 5. D 6. C 7. A 8. B
9. B 10. B 11. A 12. D 13. D 14. A 15. B 16. A
17. B 18. A 19. A 20. D 21. A 22. C 23. D 24. C
25. A 26. A 27. A 28. C 29. D 30. A 31. B 32. B
33. B 34. A 35. D 36. D 37. A 38. A 39. B 40. B
41. B 42. B 43. (A) S; (B) S; (C) P; (D) R 44. (A) Q,S,T; (B) P,Q,S,T; (C) S; (D) R,T
45. (A) P,Q; (B) P; (C) S; (D) R 46. (A) R, (B) Q (C) T (D) S 47. (A) S; (B) R ; (C) Q

EXERCISE (O-2)

1. B 2. B 3. C 4. B 5. A 6. D 7. B 8. B
9. B 10. B,D 11. B,C 12. B,C 13. A,B,C,D 14. B,C 15. B,C
16. A,B,C,D 17. B,C 18. C,D 19. A,B 20. A,C,D 21. B,C,D 22. A,B
23. C 24. D 25. B 26. C 27. A 28. B 29. A 30. D
31. B 32. D 33. A

EXERCISE (S-1)

1. (a) $\left(2, \frac{8}{3}\right)$; (b) 4 2. (33, 26) 3. $K = 7$ or $\frac{31}{9}$ 4. $1 : 2$; $Q(-5, -3)$
5. $83x - 35y + 92 = 0$ 6. $\left(\frac{7}{2}, \frac{13}{2}\right)$ or $\left(-\frac{3}{2}, \frac{3}{2}\right)$ 7. $2xy(a + b) = ab(x + y)$
8. $x^2 + y^2 - 3x + 4y - 3 = 0$ 9. $x - y = 0$ 10. 91 sq.units 11. 6 units
12. 3 units 13. $c = -4$; $B(2, 0)$; $D(4, 4)$ 14. $x + 5y + 5\sqrt{2} = 0$ or $x + 5y - 5\sqrt{2} = 0$
15. $x - 3y - 31 = 0$ or $3x + y + 7 = 0$ 16. $14x + 23y = 40$ 17. $x - 5 = 0$
18. (a) 5; (b) 2; (c) $\frac{3}{2}$ 19. 15 20. 5 21. (a) $\frac{50}{7}$; (b) $\frac{63}{10}$; (c) $\frac{3}{10}(8\sqrt{5} - 5\sqrt{10})$
22. $x + y = 1$; $x + 9y = 1$

EXERCISE (S-2)

1. 8 2. (A) R; (B) S; (C) Q 3. $x + 4y = 4$; $5x + 2y = 8$ 4. $a = 11$, $c = 78$
5. $7x + 24y + 182 = 0$ or $x = -2$ 6. $(0, 0)$ or $\left(0, \frac{5}{2}\right)$
7. $2x - y + 3 = 0$, $2x + y - 7 = 0$, $x - 2y - 6 = 0$ 8. $y = 2x$ 9. (a) 74; (b) 50; (c) 47
10. 24 11. $(1, -2)$, yes $\left(\frac{1}{3}, -\frac{2}{3}\right)$ 12. (i) area = 6 sq. units, (ii) diagonals are $\sqrt{5}$ & $\sqrt{53}$

EXERCISE (JM)

1. 4 2. 4 3. 1 4. 3 5. 4 6. 4 7. 2 8. 2 9. 4 10. 3 11. 4 12. 1
13. 4 14. 3 15. 4 16. 3 17. 2 18. 1 19. 4 20. 1 21. 2 22. 2

EXERCISE (JA)

1. (A) S; (B) P,Q; (C) R ; (D) P,Q,S 2. A 3. B 4. A or C or A,C 5. 6

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JEE (Main) Syllabus :

Limits, continuity and differentiability.

JEE (Advanced) Syllabus :

Limit and continuity of a function, limit and continuity of the sum, difference, product and quotient of two functions. Continuity of composite functions, intermediate value property of continuous functions.

LIMIT

1. INTRODUCTION :

The concept of limit of a function is one of the fundamental ideas that distinguishes calculus from algebra and trigonometry. We use limits to describe the way a function f varies. Some functions vary continuously; small changes in x produce only small changes in $f(x)$. Other functions can have values that jump or vary erratically. We also use limits to define tangent lines to graphs of functions. This geometric application leads at once to the important concept of derivative of a function.

2. DEFINITION :

Let $f(x)$ be defined on an open interval about 'a' except possibly at 'a' itself. If $f(x)$ gets arbitrarily close to L (a finite number) for all x sufficiently close to 'a' we say that $f(x)$ approaches the limit L as x approaches 'a' and we write $\lim_{x \rightarrow a} f(x) = L$ and say "the limit of $f(x)$, as x approaches a , equals L ".

This implies if we can make the value of $f(x)$ arbitrarily close to L (as close to L as we like) by taking x to be sufficiently close to a (on either side of a) but not equal to a .

3. LEFT HAND LIMIT AND RIGHT HAND LIMIT OF A FUNCTION :

The value to which $f(x)$ approaches, as x tends to 'a' from the left hand side ($x \rightarrow a^-$) is called left hand limit of $f(x)$ at $x = a$. Symbolically, $LHL = \lim_{x \rightarrow a^-} f(x) = \lim_{h \rightarrow 0} f(a - h)$.

The value to which $f(x)$ approaches, as x tends to 'a' from the right hand side ($x \rightarrow a^+$) is called right hand limit of $f(x)$ at $x = a$. Symbolically, $RHL = \lim_{x \rightarrow a^+} f(x) = \lim_{h \rightarrow 0} f(a + h)$.

Limit of a function $f(x)$ is said to exist as, $x \rightarrow a$ when $\lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^+} f(x) = \text{Finite quantity}$.

Example :

Graph of $y = f(x)$

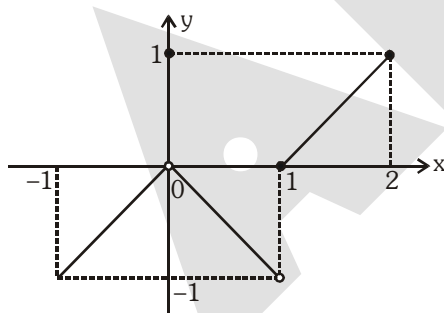


Fig. 1

$$\lim_{x \rightarrow -1^+} f(x) = \lim_{h \rightarrow 0} f(-1 + h) = f(-1^+) = -1$$

$$\lim_{x \rightarrow 0^-} f(x) = \lim_{h \rightarrow 0} f(0 - h) = f(0^-) = 0$$

$$\lim_{x \rightarrow 0^+} f(x) = \lim_{h \rightarrow 0} f(0 + h) = f(0^+) = 0$$

$$\lim_{x \rightarrow 1^-} f(x) = \lim_{h \rightarrow 0} f(1 - h) = f(1^-) = -1$$

$$\lim_{x \rightarrow 1^+} f(x) = \lim_{h \rightarrow 0} f(1 + h) = f(1^+) = 0$$

$$\lim_{x \rightarrow 2^-} f(x) = \lim_{h \rightarrow 0} f(2 - h) = f(2^-) = 1$$

$$\lim_{x \rightarrow 0} f(x) = 0 \text{ and } \lim_{x \rightarrow 1} f(x) \text{ does not exist.}$$

Important note :

In $\lim_{x \rightarrow a} f(x)$, $x \rightarrow a$ necessarily implies $x \neq a$. That is while evaluating limit at $x = a$, we are not concerned with the value of the function at $x = a$. In fact the function may or may not be defined at $x = a$.

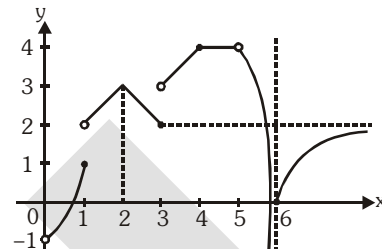
Also it is necessary to note that if $f(x)$ is defined only on one side of ' $x = a$ ', one sided limits are good enough to establish the existence of limits, & if $f(x)$ is defined on either side of ' a ' both sided limits are to be considered.

As in $\lim_{x \rightarrow 1} \cos^{-1} x = 0$, though $f(x)$ is not defined for $x > 1$, even in it's immediate vicinity.

Illustration 1 : Consider the adjacent graph of $y = f(x)$

Find the following :

- (a) $\lim_{x \rightarrow 0^-} f(x)$ (b) $\lim_{x \rightarrow 0^+} f(x)$ (c) $\lim_{x \rightarrow 1^-} f(x)$
 (d) $\lim_{x \rightarrow 1^+} f(x)$ (e) $\lim_{x \rightarrow 2^-} f(x)$ (f) $\lim_{x \rightarrow 2^+} f(x)$
 (g) $\lim_{x \rightarrow 3^-} f(x)$ (h) $\lim_{x \rightarrow 3^+} f(x)$ (i) $\lim_{x \rightarrow 4^-} f(x)$
 (j) $\lim_{x \rightarrow 4^+} f(x)$ (k) $\lim_{x \rightarrow \infty} f(x) = 2$ (l) $\lim_{x \rightarrow 6^-} f(x) = -\infty$



Solution : (a) As $x \rightarrow 0^-$: limit does not exist (the function is not defined to the left of $x = 0$)

(b) As $x \rightarrow 0^+ : f(x) \rightarrow -1 \Rightarrow \lim_{x \rightarrow 0^+} f(x) = -1$. (c) As $x \rightarrow 1^- : f(x) \rightarrow 1 \Rightarrow \lim_{x \rightarrow 1^-} f(x) = 1$.

(d) As $x \rightarrow 1^+ : f(x) \rightarrow 2 \Rightarrow \lim_{x \rightarrow 1^+} f(x) = 2$. (e) As $x \rightarrow 2^- : f(x) \rightarrow 3 \Rightarrow \lim_{x \rightarrow 2^-} f(x) = 3$.

(f) As $x \rightarrow 2^+ : f(x) \rightarrow 2 \Rightarrow \lim_{x \rightarrow 2^+} f(x) = 2$. (g) As $x \rightarrow 3^- : f(x) \rightarrow 2 \Rightarrow \lim_{x \rightarrow 3^-} f(x) = 2$.

(h) As $x \rightarrow 3^+ : f(x) \rightarrow 3 \Rightarrow \lim_{x \rightarrow 3^+} f(x) = 3$. (i) As $x \rightarrow 4^- : f(x) \rightarrow 4 \Rightarrow \lim_{x \rightarrow 4^-} f(x) = 4$.

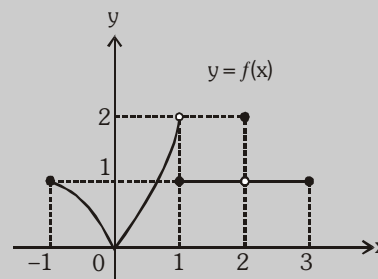
(j) As $x \rightarrow 4^+ : f(x) \rightarrow 4 \Rightarrow \lim_{x \rightarrow 4^+} f(x) = 4$. (k) As $x \rightarrow \infty : f(x) \rightarrow 2 \Rightarrow \lim_{x \rightarrow \infty} f(x) = 2$.

(l) As $x \rightarrow 6^- : f(x) \rightarrow -\infty \Rightarrow \lim_{x \rightarrow 6^-} f(x) = -\infty$ limit does not exist because it is not finite.

Do yourself - 1 :

(i) Which of the following statements about the function $y = f(x)$ graphed here are true, and which are false ?

- (a) $\lim_{x \rightarrow -1^+} f(x) = 1$ (b) $\lim_{x \rightarrow 2} f(x)$ does not exist
 (c) $\lim_{x \rightarrow 2} f(x) = 2$ (d) $\lim_{x \rightarrow 1^-} f(x) = 2$
 (e) $\lim_{x \rightarrow 1} f(x)$ does not exist (f) $\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^-} f(x)$
 (g) $\lim_{x \rightarrow c} f(x)$ exists at every $c \in (-1, 1)$
 (h) $\lim_{x \rightarrow c} f(x)$ exists at every $c \in (1, 3)$
 (i) $\lim_{x \rightarrow 1^-} f(x) = 0$ (j) $\lim_{x \rightarrow 3^+} f(x)$ does not exist.



4. FUNDAMENTAL THEOREMS ON LIMITS :

Let $\lim_{x \rightarrow a} f(x) = l$ & $\lim_{x \rightarrow a} g(x) = m$. If l & m exist finitely then :

- (a) Sum rule : $\lim_{x \rightarrow a} \{f(x) + g(x)\} = l + m$ (b) Difference rule : $\lim_{x \rightarrow a} \{f(x) - g(x)\} = l - m$
- (c) Product rule : $\lim_{x \rightarrow a} f(x) \cdot g(x) = l \cdot m$ (d) Quotient rule : $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{l}{m}$, provided $m \neq 0$
- (e) Constant multiple rule : $\lim_{x \rightarrow a} k f(x) = k \lim_{x \rightarrow a} f(x)$; where k is constant.
- (f) Power rule : If m and n are integers then $\lim_{x \rightarrow a} [f(x)]^{m/n} = l^{m/n}$ provided $l^{m/n}$ is a real number.
- (g) $\lim_{x \rightarrow a} f[g(x)] = f\left(\lim_{x \rightarrow a} g(x)\right) = f(m)$; provided $f(x)$ is continuous at $x = m$.

For example : $\lim_{x \rightarrow a} \ell n(g(x)) = \ell n[\lim_{x \rightarrow a} g(x)]$

$= \ell n(m)$; provided $\ell n x$ is continuous at $x = m$, $m = \lim_{x \rightarrow a} g(x)$.

5. INDETERMINATE FORMS :

$$\frac{0}{0}, \frac{\infty}{\infty}, \infty - \infty, 0 \times \infty, 1^\infty, 0^0, \infty^0$$

Initially we will deal with first five forms only and the other two forms will come up after we have gone through differentiation.

Note : (i) Here $0, 1$ are not exact, infact both are approaching to their corresponding values.

(ii) We cannot plot ∞ on the paper. Infinity (∞) is a symbol & not a number It does not obey the laws of elementary algebra,

$$(a) \infty + \infty \rightarrow \infty \quad (b) \infty \times \infty \rightarrow \infty \quad (c) \infty^\infty \rightarrow \infty \quad (d) 0^\infty \rightarrow 0$$

6. GENERAL METHODS TO BE USED TO EVALUATE LIMITS :

(a) Factorization :

Important factors :

- (i) $x^n - a^n = (x - a)(x^{n-1} + ax^{n-2} + \dots + a^{n-1})$, $n \in \mathbb{N}$
- (ii) $x^n + a^n = (x + a)(x^{n-1} - ax^{n-2} + \dots + a^{n-1})$, n is an odd natural number.

Note : $\lim_{x \rightarrow a} \frac{x^n - a^n}{x - a} = na^{n-1}$

Illustration 2 : Evaluate : $\lim_{x \rightarrow 2} \left[\frac{1}{x-2} - \frac{2(2x-3)}{x^3 - 3x^2 + 2x} \right]$

Solution : We have

$$\begin{aligned} \lim_{x \rightarrow 2} \left[\frac{1}{x-2} - \frac{2(2x-3)}{x^3 - 3x^2 + 2x} \right] &= \lim_{x \rightarrow 2} \left[\frac{1}{x-2} - \frac{2(2x-3)}{x(x-1)(x-2)} \right] = \lim_{x \rightarrow 2} \left[\frac{x(x-1) - 2(2x-3)}{x(x-1)(x-2)} \right] \\ &= \lim_{x \rightarrow 2} \left[\frac{x^2 - 5x + 6}{x(x-1)(x-2)} \right] = \lim_{x \rightarrow 2} \left[\frac{(x-2)(x-3)}{x(x-1)(x-2)} \right] = \lim_{x \rightarrow 2} \left[\frac{x-3}{x(x-1)} \right] = -\frac{1}{2} \end{aligned}$$

Do yourself - 2 :

(i) Evaluate : $\lim_{x \rightarrow 1} \frac{x-1}{2x^2-7x+5}$

(b) Rationalization or double rationalization :

Illustration 3 : Evaluate : $\lim_{x \rightarrow 1} \frac{4-\sqrt{15x+1}}{2-\sqrt{3x+1}}$

Solution :

$$\begin{aligned} \lim_{x \rightarrow 1} \frac{4-\sqrt{15x+1}}{2-\sqrt{3x+1}} &= \lim_{x \rightarrow 1} \frac{(4-\sqrt{15x+1})(2+\sqrt{3x+1})(4+\sqrt{15x+1})}{(2-\sqrt{3x+1})(4+\sqrt{15x+1})(2+\sqrt{3x+1})} \\ &= \lim_{x \rightarrow 1} \frac{(15-15x)}{(3-3x)} \times \frac{2+\sqrt{3x+1}}{4+\sqrt{15x+1}} = \frac{5}{2} \end{aligned}$$

Illustration 4 : Evaluate : $\lim_{x \rightarrow 1} \left(\frac{\sqrt{x^2+8}-\sqrt{10-x^2}}{\sqrt{x^2+3}-\sqrt{5-x^2}} \right)$

Solution : This is of the form $\frac{3-3}{2-2} = \frac{0}{0}$ if we put $x = 1$

To eliminate the $\frac{0}{0}$ factor, multiply by the conjugate of numerator and the conjugate of the denominator

$$\begin{aligned} \therefore \text{Limit} &= \lim_{x \rightarrow 1} \left(\sqrt{x^2+8}-\sqrt{10-x^2} \right) \frac{(\sqrt{x^2+8}+\sqrt{10-x^2})}{(\sqrt{x^2+8}+\sqrt{10-x^2})} \times \frac{(\sqrt{x^2+3}+\sqrt{5-x^2})}{(\sqrt{x^2+3}+\sqrt{5-x^2})(\sqrt{x^2+3}-\sqrt{5-x^2})} \\ &= \lim_{x \rightarrow 1} \frac{\sqrt{x^2+3}+\sqrt{5-x^2}}{\sqrt{x^2+8}+\sqrt{10-x^2}} \times \frac{(x^2+8)-(10-x^2)}{(x^2+3)-(5-x^2)} = \lim_{x \rightarrow 1} \left(\frac{\sqrt{x^2+3}+\sqrt{5-x^2}}{\sqrt{x^2+8}+\sqrt{10-x^2}} \right) \times 1 = \frac{2+2}{3+3} = \frac{2}{3} \end{aligned}$$

Do yourself - 3 :

(i) Evaluate : $\lim_{x \rightarrow 0} \frac{\sqrt{p+x}-\sqrt{p-x}}{\sqrt{q+x}-\sqrt{q-x}}$

(ii) Evaluate : $\lim_{x \rightarrow a} \frac{\sqrt{a+2x}-\sqrt{3x}}{\sqrt{3a+x}-2\sqrt{x}}, a \neq 0$

(iii) If $G(x) = -\sqrt{25-x^2}$, then find the $\lim_{x \rightarrow 1} \left(\frac{G(x)-G(1)}{x-1} \right)$

(c) Limit when $x \rightarrow \infty$:

- Divide by greatest power of x in numerator and denominator.
- Put $x = 1/y$ and apply $y \rightarrow 0$

Illustration 5 : Evaluate : $\lim_{x \rightarrow \infty} \frac{x^2 + x + 1}{3x^2 + 2x - 5}$

Solution : $\lim_{x \rightarrow \infty} \frac{x^2 + x + 1}{3x^2 + 2x - 5}, \left(\frac{\infty}{\infty} \text{ form} \right)$

Put $x = \frac{1}{y}$

$$\text{Limit} = \lim_{y \rightarrow 0} \frac{1 + y + y^2}{3 + 2y - 5y^2} = \frac{1}{3}$$

Illustration 6 : If $\lim_{x \rightarrow \infty} \left(\frac{x^3 + 1}{x^2 + 1} - (ax + b) \right) = 2$, then

- (A) $a = 1, b = 1$ (B) $a = 1, b = 2$ (C) $a = 1, b = -2$ (D) none of these

Solution : $\lim_{x \rightarrow \infty} \left(\frac{x^3 + 1}{x^2 + 1} - (ax + b) \right) = 2 \Rightarrow \lim_{x \rightarrow \infty} \frac{x^3(1-a) - bx^2 - ax + (1-b)}{x^2 + 1} = 2$

$$\Rightarrow \lim_{x \rightarrow \infty} \frac{x(1-a) - b - \frac{a}{x} + \frac{(1-b)}{x^2}}{1 + \frac{1}{x^2}} = 2 \Rightarrow 1-a=0, -b=2 \Rightarrow a=1, b=-2 \quad \text{Ans. (C)}$$

Do yourself - 4 :

(i) Evaluate : $\lim_{n \rightarrow \infty} \frac{\frac{n+2}{n} + \frac{n+1}{n}}{\frac{n+2}{n} - \frac{n+1}{n}}$

(ii) Evaluate : $\lim_{n \rightarrow \infty} (n - \sqrt{n^2 + n})$

(d) **Squeeze play theorem (Sandwich theorem) :**

Statement : If $f(x) \leq g(x) \leq h(x); \forall x$ in the neighbourhood at $x = a$ and

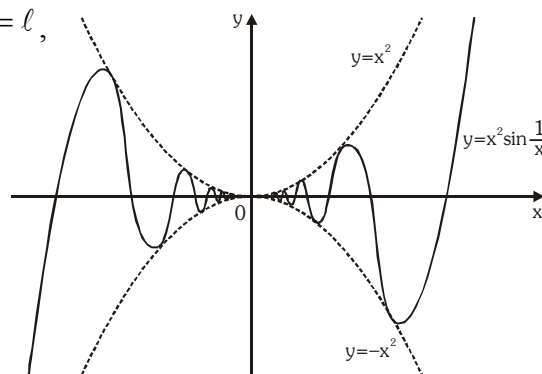
$$\lim_{x \rightarrow a} f(x) = \ell = \lim_{x \rightarrow a} h(x) \quad \text{then} \quad \lim_{x \rightarrow a} g(x) = \ell,$$

Ex.1 $\lim_{x \rightarrow 0} x^2 \sin \frac{1}{x} = 0,$

$\therefore \sin \left(\frac{1}{x} \right)$ lies between -1 & 1

$$\Rightarrow -x^2 \leq x^2 \sin \frac{1}{x} \leq x^2$$

$$\Rightarrow \lim_{x \rightarrow 0} x^2 \sin \frac{1}{x} = 0 \text{ as } \lim_{x \rightarrow 0} (-x^2) = \lim_{x \rightarrow 0} x^2 = 0$$



Ex.2 $\lim_{x \rightarrow 0} x \sin \frac{1}{x} = 0$

$\therefore \sin\left(\frac{1}{x}\right)$ lies between -1 & 1

$\Rightarrow -x \leq x \sin \frac{1}{x} \leq x$

$\Rightarrow \lim_{x \rightarrow 0} x \sin \frac{1}{x} = 0$ as $\lim_{x \rightarrow 0} (-x) = \lim_{x \rightarrow 0} x = 0$

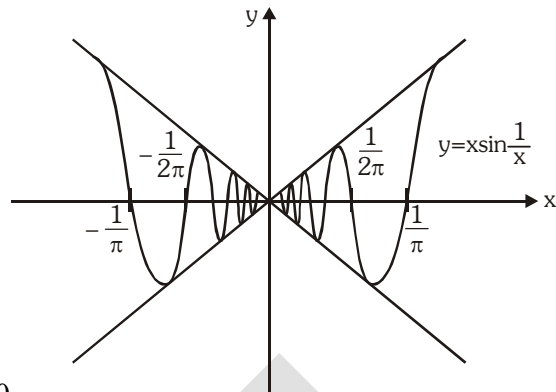


Illustration 7 : Evaluate : $\lim_{n \rightarrow \infty} \frac{[x] + [2x] + [3x] + \dots + [nx]}{n^2}$ (Where $[.]$ denotes the greatest integer function.)

Solution : We know that $x - 1 < [x] \leq x$

$\Rightarrow x + 2x + \dots + nx - n < \sum_{r=1}^n [rx] \leq x + 2x + \dots + nx$

$\Rightarrow \frac{xn}{2}(n+1) - n < \sum_{r=1}^n [rx] \leq \frac{x \cdot n(n+1)}{2} \Rightarrow \frac{x}{2} \left(1 + \frac{1}{n}\right) - \frac{1}{n} < \frac{1}{n^2} \sum_{r=1}^n [rx] \leq \frac{x}{2} \left(1 + \frac{1}{n}\right)$

Now, $\lim_{n \rightarrow \infty} \frac{x}{2} \left(1 + \frac{1}{n}\right) = \frac{x}{2}$ and $\lim_{n \rightarrow \infty} \frac{x}{2} \left(1 + \frac{1}{n}\right) - \frac{1}{n} = \frac{x}{2}$

Thus, $\lim_{n \rightarrow \infty} \frac{[x] + [2x] + \dots + [nx]}{n^2} = \frac{x}{2}$

7. LIMIT OF TRIGONOMETRIC FUNCTIONS :

$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1 = \lim_{x \rightarrow 0} \frac{\tan x}{x} = \lim_{x \rightarrow 0} \frac{\tan^{-1} x}{x} = \lim_{x \rightarrow 0} \frac{\sin^{-1} x}{x}$ [where x is measured in radians]

(a) If $\lim_{x \rightarrow a} f(x) = 0$, then $\lim_{x \rightarrow a} \frac{\sin f(x)}{f(x)} = 1$, e.g. $\lim_{x \rightarrow 1} \frac{\sin(\ln x)}{(\ln x)} = 1$

Illustration 8 : Evaluate : $\lim_{x \rightarrow 0} \frac{x^3 \cot x}{1 - \cos x}$

Solution : $\lim_{x \rightarrow 0} \frac{x^3 \cos x}{\sin x (1 - \cos x)} = \lim_{x \rightarrow 0} \frac{x^3 \cos x (1 + \cos x)}{\sin x \cdot \sin^2 x} = \lim_{x \rightarrow 0} \frac{x^3}{\sin^3 x} \cdot \cos x (1 + \cos x) = 2$

Illustration 9 : Evaluate : $\lim_{x \rightarrow 0} \frac{(2+x)\sin(2+x) - 2\sin 2}{x}$

Solution :

$$\lim_{x \rightarrow 0} \frac{2(\sin(2+x) - \sin 2) + x \sin(2+x)}{x} = \lim_{x \rightarrow 0} \left(\frac{2 \cdot 2 \cdot \cos\left(2 + \frac{x}{2}\right) \sin \frac{x}{2}}{x} + \sin(2+x) \right)$$

$$= \lim_{x \rightarrow 0} \frac{2 \cos\left(2 + \frac{x}{2}\right) \sin \frac{x}{2}}{\frac{x}{2}} + \lim_{x \rightarrow 0} \sin(2+x) = 2\cos 2 + \sin 2$$

Illustration 10 : Evaluate : $\lim_{n \rightarrow \infty} \frac{\sin \frac{a}{n}}{\tan \frac{b}{n+1}}$

Solution : As $n \rightarrow \infty$, $\frac{1}{n} \rightarrow 0$ and $\frac{a}{n}$ also tends to zero

$\sin \frac{a}{n}$ should be written as $\frac{\sin \frac{a}{n}}{\frac{a}{n}}$ so that it looks like $\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta}$

The given limit = $\lim_{n \rightarrow \infty} \left(\frac{\sin \frac{a}{n}}{\frac{a}{n}} \right) \left(\frac{\frac{b}{n+1}}{\tan \frac{b}{n+1}} \right) \cdot \frac{a(n+1)}{n \cdot b}$

$$= \lim_{n \rightarrow \infty} \left(\frac{\sin \frac{a}{n}}{\frac{a}{n}} \right) \left(\frac{\frac{b}{n+1}}{\tan \frac{b}{n+1}} \right) \cdot \frac{a}{b} \left(1 + \frac{1}{n} \right) = 1 \times 1 \times \frac{a}{b} \times 1 = \frac{a}{b}$$

Do yourself - 5 :

(i) Evaluate :

(a) $\lim_{x \rightarrow 0} \frac{\sin \alpha x}{\tan \beta x}$

(b) $\lim_{x \rightarrow y} \frac{\sin^2 x - \sin^2 y}{x^2 - y^2}$

(c) $\lim_{h \rightarrow 0} \frac{(a+h)^2 \sin(a+h) - a^2 \sin a}{h}$

8. LIMIT OF EXPONENTIAL FUNCTIONS :

$$(a) \quad \lim_{x \rightarrow 0} \frac{a^x - 1}{x} = \ell \ln a (a > 0) \text{ In particular } \lim_{x \rightarrow 0} \frac{e^x - 1}{x} = 1.$$

$$\text{In general if } \lim_{x \rightarrow a} f(x) = 0, \text{ then } \lim_{x \rightarrow a} \frac{a^{f(x)} - 1}{f(x)} = \ell \ln a, a > 0$$

Illustration 11 : Evaluate : $\lim_{x \rightarrow 0} \frac{e^{\tan x} - e^x}{\tan x - x}$

Solution :

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{e^{\tan x} - e^x}{\tan x - x} &= \lim_{x \rightarrow 0} \frac{e^x \times e^{(\tan x - x)} - e^x}{\tan x - x} \\ &= \lim_{x \rightarrow 0} \frac{e^x (e^{\tan x - x} - 1)}{\tan x - x} = \lim_{x \rightarrow 0, y \rightarrow 0} \frac{e^x (e^y - 1)}{y} \text{ where } y = \tan x - x \text{ and } \lim_{y \rightarrow 0} \frac{e^y - 1}{y} = 1 \\ &= e^0 \times 1 \quad [\text{as } x \rightarrow 0, \tan x - x \rightarrow 0] \\ &= 1 \times 1 = 1 \end{aligned}$$

Do yourself - 6 :

(i) Evaluate : $\lim_{x \rightarrow a} \frac{e^x - e^a}{x - a}$

(ii) Evaluate : $\lim_{x \rightarrow 0} \frac{2^x - 1}{(1+x)^{1/2} - 1}$

(b) (i) $\lim_{x \rightarrow 0} (1+x)^{1/x} = e = \lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x$ (Note : The base and exponent depends on the same

variable.) **In general**, if $\lim_{x \rightarrow a} f(x) = 0$, then $\lim_{x \rightarrow a} (1 + f(x))^{1/f(x)} = e$

(ii) $\lim_{x \rightarrow 0} \frac{\ell n(1+x)}{x} = 1$

(iii) If $\lim_{x \rightarrow a} f(x) = 1$ and $\lim_{x \rightarrow a} \phi(x) = \infty$, then ; $\lim_{x \rightarrow a} [f(x)]^{\phi(x)} = e^k$

where $k = \lim_{x \rightarrow a} \phi(x) [f(x) - 1]$

Illustration 12 : Evaluate $\lim_{x \rightarrow 1} (\log_3 3x)^{\log_x 3}$

Solution : $\lim_{x \rightarrow 1} (\log_3 3x)^{\log_x 3} = \lim_{x \rightarrow 1} (\log_3 3 + \log_3 x)^{\log_x 3}$

$$= \lim_{x \rightarrow 1} (1 + \log_3 x)^{1/\log_3 x} = e$$

$$\because \log_b a = \frac{1}{\log_a b}$$

Illustration 13 : Evaluate : $\lim_{x \rightarrow 0} \frac{x \ln(1 + 2 \tan x)}{1 - \cos x}$

Solution : $\lim_{x \rightarrow 0} \frac{x \ln(1 + 2 \tan x)}{1 - \cos x} = \lim_{x \rightarrow 0} \frac{x \ln(1 + 2 \tan x)}{\frac{1 - \cos x}{x^2} \cdot x^2} \cdot \frac{2 \tan x}{2 \tan x} = 4$

Illustration 14 : Evaluate : $\lim_{x \rightarrow \infty} \left(\frac{2x^2 - 1}{2x^2 + 3} \right)^{4x^2 + 2}$

Solution : Since it is in the form of 1^∞

$$\lim_{x \rightarrow \infty} \left(\frac{2x^2 - 1}{2x^2 + 3} \right)^{4x^2 + 2} = e^{\lim_{x \rightarrow \infty} \left(\frac{2x^2 - 1 - 2x^2 - 3}{2x^2 + 3} \right) (4x^2 + 2)} = e^{-8}$$

Do yourself - 7 :

- (i) Evaluate : $\lim_{x \rightarrow \infty} x \{ \ln(x+a) - \ln x \}$ (ii) Evaluate : $\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n} \right)^{pn+q}$
- (iii) Evaluate : $\lim_{x \rightarrow 0} \left(1 + \tan^2 \sqrt{x} \right)^{\frac{1}{2x}}$ (iv) Evaluate : $\lim_{x \rightarrow \infty} \left(\frac{x+6}{x+1} \right)^{x+4}$

(c) If $\lim_{x \rightarrow a} f(x) = A > 0$ & $\lim_{x \rightarrow a} \phi(x) = B$, then $\lim_{x \rightarrow a} [f(x)]^{\phi(x)} = e^{B \ln A} = A^B$

Illustration 15 : Evaluate : $\lim_{x \rightarrow \infty} \left(\frac{7x^2 + 1}{5x^2 - 1} \right)^{\frac{x^5}{1-x^3}}$

Solution : Here $f(x) = \frac{7x^2 + 1}{5x^2 - 1}$, $\phi(x) = \frac{x^5}{1-x^3} = \frac{x^2 \cdot x^3}{1-x^3} = \frac{x^2}{\frac{1}{x^3} - 1}$

$$\therefore \lim_{x \rightarrow \infty} f(x) = \frac{7}{5} \quad \& \quad \lim_{x \rightarrow \infty} \phi(x) \rightarrow -\infty$$

$$\Rightarrow \lim_{x \rightarrow \infty} (f(x))^{\phi(x)} = \left(\frac{7}{5} \right)^{-\infty} = 0$$

Do yourself - 8 :

- (i) Evaluate : $\lim_{x \rightarrow \infty} \left(\frac{1 + 5x^2}{1 + 3x^2} \right)^{-x^2}$

9. LIMIT USING SERIES EXPANSION :

Expansion of function like binomial expansion, exponential & logarithmic expansion, expansion of $\sin x$, $\cos x$, $\tan x$ should be remembered by heart which are given below :

$$(a) \quad a^x = 1 + \frac{x \ln a}{1!} + \frac{x^2 \ln^2 a}{2!} + \frac{x^3 \ln^3 a}{3!} + \dots, x \in \mathbb{R}, a > 0, a \neq 1$$

$$(b) \quad e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots, x \in \mathbb{R}$$

$$(c) \quad \ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots \text{ for } -1 < x \leq 1$$

$$(d) \quad \sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots, x \in \mathbb{R}$$

$$(e) \quad \cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots, x \in \mathbb{R}$$

$$(f) \quad \tan x = x + \frac{x^3}{3} + \frac{2x^5}{15} + \dots, -\frac{\pi}{2} < x < \frac{\pi}{2}$$

$$(g) \quad \tan^{-1} x = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \dots, x \in (-1, 1)$$

$$(h) \quad \sin^{-1} x = x + \frac{1^2}{3!} x^3 + \frac{1^2 \cdot 3^2}{5!} x^5 + \frac{1^2 \cdot 3^2 \cdot 5^2}{7!} x^7 + \dots, x \in (-1, 1)$$

$$(i) \quad \sec^{-1} x = 1 + \frac{x^2}{2!} + \frac{5x^4}{4!} + \frac{61x^6}{6!} + \dots, x \in (-\infty, -1) \cup (1, \infty)$$

$$(j) \quad (1+x)^n = 1 + nx + \frac{n(n-1)}{2!} x^2 + \dots, n \in \mathbb{R}, x \in (-1, 1)$$

Illustration 16 : $\lim_{x \rightarrow 0} \frac{e^x - e^{-x} - 2x}{x - \sin x}$

Solution :

$$\lim_{x \rightarrow 0} \frac{e^x - e^{-x} - 2x}{x - \sin x} \Rightarrow \lim_{x \rightarrow 0} \frac{1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots - \left(1 - x + \frac{x^2}{2!} - \frac{x^3}{3!} + \dots\right) - 2x}{x - \left(x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots\right)}$$

$$\Rightarrow \lim_{x \rightarrow 0} \frac{2 \cdot \frac{x^3}{6} + 2 \cdot \frac{x^5}{5!} + \dots}{\frac{x^3}{6} + \frac{x^5}{5!} - \dots} \Rightarrow \lim_{x \rightarrow 0} \frac{x^3 \left(\frac{1}{3} + \frac{1}{60} x^2 + \dots\right)}{x^3 \left(\frac{1}{6} + \frac{1}{120} x^2 + \dots\right)} = \frac{1/3}{1/6} = 2$$

Do yourself - 9 :

(i) Evaluate : $\lim_{x \rightarrow 0} \frac{x - \sin x}{\sin(x^3)}$

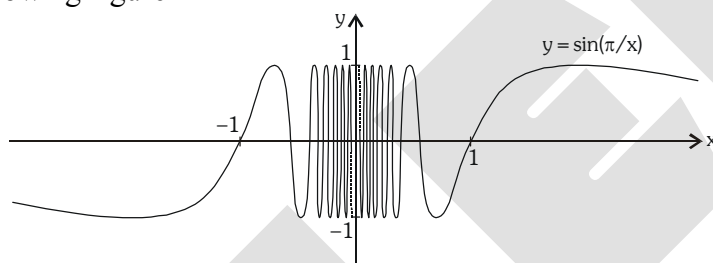
(ii) Evaluate : $\lim_{x \rightarrow 0} \frac{x - \tan^{-1} x}{x^3}$

Miscellaneous Illustrations :

Illustration 17: Evaluate $\lim_{x \rightarrow 0} \sin \frac{\pi}{x}$.

Solution : Again the function $f(x) = \sin(\pi/x)$ is undefined at 0. Evaluating the function for some small values of x , we get $f(1) = \sin \pi = 0$, $f\left(\frac{1}{2}\right) = \sin 2\pi = 0$,
 $f(0.1) = \sin 10\pi = 0$, $f(0.01) = \sin 100\pi = 0$.

On the basis of this information we might be tempted to guess that $\lim_{x \rightarrow 0} \sin \frac{\pi}{x} = 0$ but this time our guess is wrong. Note that although $f(1/n) = \sin n\pi = 0$ for any integer n , it is also true that $f(x) = 1$ for infinitely many values of x that approach 0. [In fact, $\sin(\pi/x) = 1$ when $\frac{\pi}{x} = \frac{\pi}{2} + 2n\pi$ and solving for x , we get $x = 2/(4n + 1)$]. The graph of f is given in following figure



The dashed line indicate that the values of $\sin(\pi/x)$ oscillate between 1 and -1 infinitely often as x approaches 0. Since the values of $f(x)$ do not approach a fixed number as x approaches 0,

$\Rightarrow \lim_{x \rightarrow 0} \sin \frac{\pi}{x}$ does not exist.

ANSWERS FOR DO YOURSELF

- | | | | | | | | | | | | |
|-----|-----|-----------------------------|-------|-----------------------|-------|--------------------------|-------|-------|-------|-------|-------|
| 1 : | (i) | (a) T | (b) F | (c) F | (d) T | (e) T | (f) T | (g) T | (h) T | (i) F | (j) T |
| 2 : | (i) | $-\frac{1}{3}$ | | | | | | | | | |
| 3 : | (i) | $\frac{\sqrt{q}}{\sqrt{p}}$ | (ii) | $\frac{2}{3\sqrt{3}}$ | (iii) | $\frac{1}{\sqrt{24}}$ | | | | | |
| 4 : | (i) | 1 | (ii) | $-\frac{1}{2}$ | | | | | | | |
| 5 : | (i) | (a) $\frac{\alpha}{\beta}$ | (b) | $\frac{\sin 2y}{2y}$ | (c) | $2a \sin a + a^2 \cos a$ | | | | | |
| 6 : | (i) | e^a | (ii) | $2 \ln 2$ | | | | | | | |
| 7 : | (i) | a | (ii) | e^p | (iii) | $e^{\frac{1}{2}}$ | (iv) | e^5 | | | |
| 8 : | (i) | 0 | | | | | | | | | |
| 9 : | (i) | $\frac{1}{6}$ | (ii) | $\frac{1}{3}$ | | | | | | | |

EXERCISE (O-1)

[SINGLE CORRECT CHOICE TYPE]

1. $\lim_{x \rightarrow 1} \left(\frac{1}{1-x} - \frac{3}{1-x^3} \right)$ is equal to
 (A) -1 (B) 0 (C) 1 (D) D.N.E.
2. $\lim_{x \rightarrow 0} \frac{\sqrt{1+x} - \sqrt{1-x}}{2x}$ is equal to
 (A) 0 (B) 1 (C) $\frac{1}{2}$ (D) $\frac{1}{4}$
3. $\lim_{x \rightarrow 2} \frac{\sqrt{1+\sqrt{2+x}} - \sqrt{3}}{x-2}$ is equal to
 (A) $\frac{1}{\sqrt{3}}$ (B) $\sqrt{3}$ (C) $\frac{1}{4\sqrt{3}}$ (D) $\frac{1}{8\sqrt{3}}$
4. $\lim_{x \rightarrow 1} \frac{\sqrt[n]{x} - 1}{\sqrt[m]{x} - 1}$ (m and n integers) is equal to
 (A) 0 (B) 1 (C) $\frac{m}{n}$ (D) $\frac{n}{m}$
5. If $\lim_{x \rightarrow a} \frac{2x - \sqrt{x^2 + 3a^2}}{\sqrt{x+a} - \sqrt{2a}} = \sqrt{2}$ (where $a \in \mathbb{R}^+$), then a is equal to -
 (A) $\frac{1}{3}$ (B) $\frac{1}{2\sqrt{2}}$ (C) $\frac{1}{3\sqrt{2}}$ (D) $\frac{1}{9}$
6. $\lim_{x \rightarrow 0} \frac{\ell n(\sin 3x)}{\ell n(\sin x)}$ is equal to
 (A) 0 (B) 1 (C) 2 (D) Non existent
7. $\lim_{x \rightarrow 0} \frac{\sqrt[3]{1+x^2} - \sqrt[4]{1-2x}}{x+x^2}$ is equal to
 (A) $\frac{1}{4}$ (B) $\frac{1}{2}$ (C) 1 (D) D.N.E.
8. $\lim_{x \rightarrow 1} \frac{\sqrt[3]{7+x^3} - \sqrt{3+x^2}}{x-1}$ is equal to
 (A) $\frac{1}{4}$ (B) $\frac{1}{6}$ (C) $-\frac{1}{4}$ (D) $-\frac{1}{6}$

9. $\lim_{n \rightarrow \infty} \frac{(n+1)^4 - (n-1)^4}{(n+1)^4 + (n-1)^4}$ is equal to
 (A) -1 (B) 0 (C) 1 (D) D.N.E.
10. $\lim_{x \rightarrow \infty} \frac{(x+1)^{10} + (x+2)^{10} + \dots + (x+100)^{10}}{x^{10} + 10^{10}}$ is equal to
 (A) 1 (B) 100 (C) 200 (D) 10
11. $\lim_{x \rightarrow -\infty} (\sqrt{x^2 - 2x - 1} - \sqrt{x^2 - 7x + 3})$ is equal to
 (A) $-\frac{5}{2}$ (B) $\frac{5}{2}$ (C) 0 (D) D.N.E.
12. If $\lim_{n \rightarrow \infty} (\sqrt{2n^2 + n} - \lambda \sqrt{2n^2 - n}) = \frac{1}{\sqrt{2}}$ (where λ is a real number), then-
 (A) $\lambda = 1$ (B) $\lambda = -1$ (C) $\lambda = \pm 1$ (D) $\lambda \in (-\infty, 1)$
13. Let $U_n = \frac{n!}{(n+2)!}$ where $n \in \mathbb{N}$. If $S_n = \sum_{n=1}^n U_n$ then $\lim_{n \rightarrow \infty} S_n$ equals
 (A) 2 (B) 1 (C) 1/2 (D) Non existent
14. For $n \in \mathbb{N}$, let $a_n = \sum_{k=1}^n 2k$ and $b_n = \sum_{k=1}^n (2k-1)$. Then $\lim_{n \rightarrow \infty} (\sqrt{a_n} - \sqrt{b_n})$ is equal to-
 (A) 1 (B) $\frac{1}{2}$ (C) 0 (D) 2
15. Let $P_n = \prod_{k=2}^n \left(1 - \frac{1}{k+1} C_2\right)$. If $\lim_{n \rightarrow \infty} P_n$ can be expressed as lowest rational in the form $\frac{a}{b}$, then value of $(a + b)$ is
 (A) 4 (B) 8 (C) 10 (D) 12
16. $\lim_{x \rightarrow -1} \frac{\cos 2 - \cos 2x}{x^2 - |x|}$ is equal to
 (A) 0 (B) $\cos 2$ (C) $2\sin 2$ (D) $\sin 1$
17. $\lim_{x \rightarrow 0} \left(\left[\frac{-5 \sin x}{x} \right] + \left[\frac{6 \sin x}{x} \right] \right)$ (where $[.]$ denotes greatest integer function) is equal to -
 (A) 0 (B) -12 (C) 1 (D) 2
18. Let $f(x) = \left[\frac{\sin x}{x} \right] + \left[\frac{2 \sin 2x}{x} \right] + \dots + \left[\frac{10 \sin 10x}{x} \right]$ (where $[y]$ is the largest integer $\leq y$). The value of $\lim_{x \rightarrow 0} f(x)$ equals
 (A) 55 (B) 164 (C) 165 (D) 375

19. Let $f(x) = \frac{\sin\{x\}}{x^2 + ax + b}$. If $f(5^+)$ & $f(3^+)$ exists finitely and are not zero, then the value of $(a + b)$ is (where $\{.\}$ represents fractional part function) -

(A) 7 (B) 10 (C) 11 (D) 20

20. $\lim_{x \rightarrow 0} \frac{|\cos(\sin(3x))| - 1}{x^2}$ equals

(A) $-\frac{9}{2}$ (B) $-\frac{3}{2}$ (C) $\frac{3}{2}$ (D) $\frac{9}{2}$

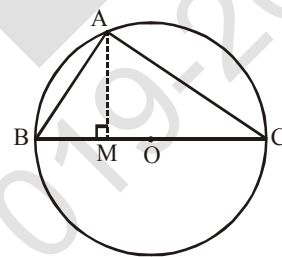
21. Let $a = \min\{x^2 + 2x + 3, x \in \mathbb{R}\}$ and $b = \lim_{\theta \rightarrow 0} \frac{1 - \cos \theta}{\theta^2}$. Then value of $\sum_{r=0}^n a^r \cdot b^{n-r}$ is :

(A) $\frac{2^{n+1} - 1}{3 \cdot 2^n}$ (B) $\frac{2^{n+1} + 1}{3 \cdot 2^n}$ (C) $\frac{4^{n+1} - 1}{3 \cdot 2^n}$ (D) N.O.T.

22. Let BC is diameter of a circle centred at O. Point A is a variable point, moving on the circumference of circle. If BC = 1 unit, then

$\lim_{A \rightarrow B} \frac{BM}{(\text{Area of sector OAB})^2}$ is equal to -

(A) 1 (B) 2
(C) 4 (D) 16



23. $\lim_{x \rightarrow \infty} \left(\frac{x^2 - 2x + 1}{x^2 - 4x + 2} \right)^x$ is equal to

(A) 1 (B) e (C) $\frac{1}{e^2}$ (D) e^2

24. $\lim_{x \rightarrow 0} (1 + \sin x)^{\cos x}$ is equal to

(A) 0 (B) e (C) 1 (D) $\frac{1}{e}$

25. $\lim_{x \rightarrow 0} (\cos x + a \sin bx)^{\frac{1}{x}}$ is equal to

(A) e^a (B) e^{ab} (C) e^b (D) $e^{a/b}$

26. $\lim_{x \rightarrow 0} \left(\tan \left(\frac{\pi}{4} + x \right) \right)^{1/x}$ is equal to

(A) e^{-2} (B) $\frac{1}{e}$ (C) e (D) e^2

27. $\lim_{n \rightarrow \infty} (4^n + 5^n)^{1/n}$ is equal to

(A) 5 (B) 4 (C) 0 (D) D.N.E.

28. $\lim_{x \rightarrow \infty} \left(\frac{1^{1/x} + 2^{1/x} + 3^{1/x} + \dots + n^{1/x}}{n} \right)^{nx}$, $n \in \mathbb{N}$ is equal to
- (A) $n!$ (B) 1 (C) $\frac{1}{n!}$ (D) 0
29. If $\lim_{x \rightarrow \lambda} \left(2 - \frac{\lambda}{x} \right)^{\lambda \tan\left(\frac{\pi x}{2\lambda}\right)} = \frac{1}{e}$, then λ is equal to -
- (A) $-\pi$ (B) π (C) $\frac{\pi}{2}$ (D) $-\frac{2}{\pi}$
30. If $\lim_{x \rightarrow 0} (1 + ax + bx^2)^{2/x} = e^3$, then
- (A) $a = \frac{3}{2}$ and $b \in \mathbb{R}$ (B) $a = \frac{3}{2}$ and $b \in \mathbb{R}^+$
(C) $a = 0$ and $b = 1$ (D) $a = 1$ and $b = 0$
31. If $f(x)$ is a polynomial of least degree, such that $\lim_{x \rightarrow 0} \left(1 + \frac{f(x) + x^2}{x^2} \right)^{1/x} = e^2$, then $f(2)$ is -
- (A) 2 (B) 8 (C) 10 (D) 12
32. Let $f(x) = \frac{\tan x}{x}$, then the value of $\lim_{x \rightarrow 0} \left([f(x)] + x^2 \right)^{\frac{1}{\{f(x)\}}}$ is equal to (where $[\cdot]$, $\{\cdot\}$ denotes greatest integer function and fractional part functions respectively) -
- (A) e^{-3} (B) e^3 (C) e^2 (D) non-existent
33. $\lim_{n \rightarrow \infty} \frac{e^n}{\left(1 + \frac{1}{n}\right)^{n^2}}$ equals -
- (A) 1 (B) $\frac{1}{2}$ (C) e (D) $\sqrt{e}$
34. If $f(x)$ is odd linear polynomial with $f(1) = 1$, then $\lim_{x \rightarrow 0} \frac{2^{f(\tan x)} - 2^{f(\sin x)}}{x^2 f(\sin x)}$ is
- (A) 1 (B) $\ln 2$ (C) $\frac{1}{2} \ln 2$ (D) $\cos 2$
35. $\lim_{x \rightarrow 0} \frac{x(1 + a \cos x) - b \sin x}{x^3} = 1$ then
- (A) $a = -5/2$ (B) $a = -3/2, b = -1/2$ (C) $a = -3/2, b = -5/2$ (D) $a = -5/2, b = -3/2$

[MULTIPLE CORRECT CHOICE TYPE]

36. Consider following statements and identify correct options

$$(i) \lim_{x \rightarrow 4} \left(\frac{2x}{x-4} - \frac{8}{x-4} \right) = \lim_{x \rightarrow 4} \frac{2x}{x-4} - \lim_{x \rightarrow 4} \frac{8}{x-4}$$

$$(ii) \lim_{x \rightarrow 1} \frac{x^2 + 6x - 7}{x^2 + 5x - 6} = \frac{\lim_{x \rightarrow 1} (x^2 + 6x - 7)}{\lim_{x \rightarrow 1} (x^2 + 5x - 6)}$$

$$(iii) \lim_{x \rightarrow 1} \frac{x-3}{x^2 + 2x - 4} = \frac{\lim_{x \rightarrow 1} (x-3)}{\lim_{x \rightarrow 1} (x^2 + 2x - 4)}$$

(iv) If $\lim_{x \rightarrow 5} f(x) = 2$ and $\lim_{x \rightarrow 5} g(x) = 0$, then $\lim_{x \rightarrow 5} \frac{f(x)}{g(x)}$ does not exist.

(v) If $\lim_{x \rightarrow 5} f(x) = 0$ and $\lim_{x \rightarrow 5} g(x) = 2$, then $\lim_{x \rightarrow 5} \frac{f(x)}{g(x)}$ does not exist.

(A) Only one is true.

(B) Only two are true.

(C) Only three are false.

(D) Only two are false.

37. Which of the following limits equal to $\frac{1}{2}$

$$(A) \lim_{n \rightarrow \infty} \left(\frac{1}{1.3} + \frac{1}{3.5} + \dots + \frac{1}{(2n-1)(2n+1)} \right)$$

$$(B) \lim_{x \rightarrow \infty} \left[\frac{3x^2}{2x+1} - \frac{(2x-1)(3x^2+x+2)}{4x^2} \right]$$

$$(C) \lim_{n \rightarrow \infty} \frac{1}{n^2} (1 + 2 + 3 + \dots + n)$$

$$(D) \lim_{n \rightarrow \infty} \frac{(n+2)! + (n+1)!}{(n+2)! - (n+1)!}$$

38. Let $f(x) = \begin{cases} \sin x; & \text{where } x = \text{integer} \\ 0; & \text{otherwise} \end{cases}$; $g(x) = \begin{cases} x^2 + 1; & x \neq 0, 2 \\ 4; & x = 0 \\ 5; & x = 2 \end{cases}$, then

$$(A) \lim_{x \rightarrow 0} g(f(x)) = 4$$

$$(B) \lim_{x \rightarrow 0} f(g(x)) = 0$$

$$(C) \lim_{x \rightarrow 1} f(g(x)) = 0$$

$$(D) \lim_{x \rightarrow 1} g(f(x)) = 5$$

39. If $f(x) = \begin{cases} x^2 & \text{if } x \text{ is rational} \\ 0 & \text{if } x \text{ is irrational} \end{cases}$, then

$$(A) \lim_{x \rightarrow 0} f(x) = 0$$

$$(B) \lim_{x \rightarrow 0} f(x) \text{ does not exist}$$

$$(C) \lim_{x \rightarrow 2} f(x) = 4$$

$$(D) \lim_{x \rightarrow 2} f(x) \text{ does not exist}$$

40. Let $f(\beta) = \lim_{\alpha \rightarrow \beta} \frac{\sin^2 \alpha - \sin^2 \beta}{\alpha^2 - \beta^2}$, then $f\left(\frac{\pi}{4}\right)$ is greater than-

$$(A) \lim_{x \rightarrow 0} \frac{1 - \cos^3 x}{x \sin 2x}$$

$$(B) \lim_{x \rightarrow \pi/2} \frac{\cot x - \cos x}{(\pi - 2x)^3}$$

$$(C) \lim_{x \rightarrow \infty} (\cos \sqrt{x+1} - \cos \sqrt{x})$$

$$(D) \lim_{x \rightarrow a} \frac{\sqrt{a+2x} - \sqrt{3x}}{\sqrt{3a+x} - 2\sqrt{x}} \text{ where } a > 0$$

41. If $\frac{\sin x + ae^x + be^{-x} + c \ln(1+x)}{x^3}$ has a finite limit L as $x \rightarrow 0$, then

(A) $a = -\frac{1}{2}$

(B) $b = \frac{1}{2}$

(C) $\mathbf{c} = 0$

(D) $L = -\frac{1}{3}$

42. Let $\ell = \lim_{x \rightarrow \infty} \frac{a^x - a^{-x}}{a^x + a^{-x}} (a > 0)$, then

$$(A) \ell = 1 \quad \forall \mathbf{a} > 0$$

(B) $\ell = -1 \ \forall \ a \in (0, 1)$ (C) $\ell = 0$, if $a = 1$

(D) $\ell = 1 \ \forall \mathbf{a} > 1$

[MATCH THE COLUMN TYPE]

43. For the function $g(t)$ whose graph is given, match the entries of column-I to column-II

Column-I

Column-II

(A) $\lim_{t \rightarrow 0^+} g(t) + \lim_{t \rightarrow 2^-} g(t)$

(P) $\lim_{t \rightarrow 2^+} g(t)$

(B) $\lim_{t \rightarrow 0^-} g(t) + g(2)$

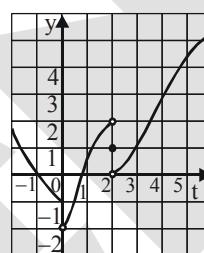
(Q) does not exist

(C) $\lim_{t \rightarrow 0} g(t)$

(R) 0

(D) $\lim_{t \rightarrow 2} g(t)$

(S) $\lim_{t \rightarrow 4} g(t)$



44. Column-I

Column-II

(A) $\lim_{n \rightarrow \infty} n \sin\left(\frac{\pi}{4n}\right) \cos\left(\frac{\pi}{4n}\right)$ is equal to

(P) 0

(B) $\lim_{x \rightarrow 0} \frac{\sin x^\circ}{x}$ is equal to

(Q) $\frac{1}{2}$

(C) $\lim_{x \rightarrow 0} \left(\frac{1}{\sin x} - \frac{1}{\tan x} \right)$ is equal to

$$(R) \quad \frac{\pi}{4}$$

(D) $\lim_{x \rightarrow \pi/2} \frac{1 + \cos 2x}{(\pi - 2x)^2}$ is equal to

$$(S) \quad \frac{\pi}{180}$$

45. Column-I

Column-II

(A) $\lim_{x \rightarrow \infty} \frac{a^x}{a^x + 1}$ ($a > 0$) can be equal to

(P) $\lim_{x \rightarrow \infty} x(e^{1/x} - 1)$

(B) $\lim_{x \rightarrow 2} \frac{\sin(e^{x-2} - 1)}{\log(x - 1)}$ is equal to

(Q) $\lim_{x \rightarrow 0} \frac{a^x + b^x + c^x - 3}{x} (a, b, c > 0 \& abc = 1)$

(C) $\lim_{x \rightarrow e} \frac{(\ell \ln x - 1)e}{x - e}$ is equal to

$$(R) \quad \lim_{x \rightarrow 0} \frac{e^{4x} - e^{3x}}{x}$$

(D) $\lim_{x \rightarrow 0} \frac{x(5^x - 1)}{(1 - \cos x)4 \ln 5}$ is equal to

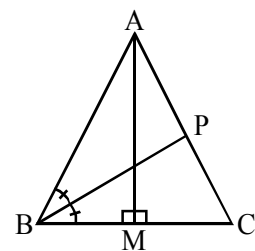
(S) $\frac{1}{2}$

(T) 0

EXERCISE (O-2)

[SINGLE CORRECT CHOICE TYPE]

- $\lim_{h \rightarrow 0} \frac{\sin(a+3h) - 3\sin(a+2h) + 3\sin(a+h) - \sin a}{h^3}$ is equal to
 (A) $\cos a$ (B) $-\cos a$ (C) $\sin a$ (D) $\sin a \cos a$
- $\lim_{x \rightarrow \frac{\pi}{2}} \tan^2 x (\sqrt{2\sin^2 x + 3\sin x + 4} - \sqrt{\sin^2 x + 6\sin x + 2})$ is equal to
 (A) $\frac{3}{4}$ (B) $\frac{1}{6}$ (C) $\frac{1}{12}$ (D) $\frac{5}{12}$
- $\lim_{x \rightarrow \infty} x \left(\arctan \frac{x+1}{x+2} - \arctan \frac{x}{x+2} \right)$ is equal to
 (A) $\frac{1}{2}$ (B) $-\frac{1}{2}$ (C) 1 (D) D.N.E
- $\lim_{h \rightarrow 0} \frac{\tan(a+2h) - 2\tan(a+h) + \tan a}{h^2}$ is equal to
 (A) $\tan a$ (B) $\tan^2 a$ (C) $\sec a$ (D) $2(\sec^2 a)(\tan a)$
- $\lim_{x \rightarrow 0} \left(2^{x-1} + \frac{1}{2} \right)^{1/x}$ equals
 (A) $\sqrt{2}$ (B) $\frac{1}{2} \ln 2$ (C) $\ln 2$ (D) 2
- If $\lim_{x \rightarrow 0} (\cos x + a^3 \sin(b^6 x))^{\frac{1}{x}} = e^{512}$, then the value of ab^2 is equal to
 (A) -512 (B) 512 (C) 8 (D) $8\sqrt{8}$
- The value of $\lim_{x \rightarrow 0} \frac{\sin(\sqrt[3]{x}) \ln(1+3x)}{(\tan^{-1} \sqrt{x})^2 (e^{5\sqrt[3]{x}} - 1)}$ is equal to
 (A) $\frac{1}{5}$ (B) $\frac{3}{5}$ (C) $\frac{2}{5}$ (D) $\frac{4}{5}$
- The figure shows an isosceles triangle ABC with $\angle B = \angle C$. The bisector of angle B intersects the side AC at the point P. Suppose that BC remains fixed but the altitude AM approaches 0, so that $A \rightarrow M$ (mid-point of BC). Limiting value of BP, is
 (A) $\frac{a}{3}$ (B) $\frac{a}{2}$ (C) $\frac{2a}{3}$ (D) $\frac{3a}{4}$
 where a is fixed side BC.
- The value of $\lim_{x \rightarrow 2} \frac{\sec^x \theta - \tan^x \theta - 1}{x - 2}$ is equal to
 (A) $\sec^2 \theta \cdot \ln \sec \theta + \tan^2 \theta \cdot \ln \tan \theta$ (B) $\sec^2 \theta \cdot \ln \tan \theta + \tan^2 \theta \cdot \ln \sec \theta$
 (C) $\sec^2 \theta \cdot \ln \tan \theta - \tan^2 \theta \cdot \ln \sec \theta$ (D) $\sec^2 \theta \cdot \ln \sec \theta - \tan^2 \theta \cdot \ln \tan \theta$



10. Consider the function $f(x) = \begin{cases} 1-x, & 0 \leq x \leq 1 \\ x+2, & 1 < x < 2 \\ 4-x, & 2 \leq x \leq 4 \end{cases}$. Let $\lim_{x \rightarrow 1} f(f(x)) = \ell$ and $\lim_{x \rightarrow 2} f(f(x)) = m$ then which one of the following hold good ?
 (A) ℓ exists but m does not. (B) m exists but ℓ does not.
 (C) Both ℓ and m exist (D) Neither ℓ nor m exist.
11. If $f(x) = e^x$, then $\lim_{x \rightarrow 0} f(f(x))^{\frac{1}{\{f(x)\}}}$ is equal to (where $\{x\}$ denotes fractional part of x).
 (A) $f(1)$ (B) $f(0)$ (C) 0 (D) does not exist
12. Let $f(x)$ be a quadratic function such that $f(0) = f(1) = 0$ & $f(2) = 1$, then $\lim_{x \rightarrow 0} \frac{\cos\left(\frac{\pi}{2} \cos^2 x\right)}{f^2(x)}$ is equal to
 (A) $\frac{\pi}{2}$ (B) π (C) 2π (D) 4π

[MULTIPLE CORRECT CHOICE TYPE]

13. If $\ell = \lim_{x \rightarrow a} \frac{\sqrt{3x^2 + a^2} - \sqrt{x^2 + 3a^2}}{(x-a)}$ then -
 (A) $\ell = 1 \forall a \in \mathbb{R}$ (B) $\ell = 1 \forall a > 0$ (C) $\ell = -1 \forall a < 0$ (D) $\ell = \text{D.N.E. if } a = 0$
14. Which of the following limits vanish ?
 (A) $\lim_{x \rightarrow \infty} \frac{\sin x}{x}$ (B) $\lim_{x \rightarrow \infty} \frac{\arctan x}{x}$ (C) $\lim_{x \rightarrow \infty} \frac{x + \sin x}{x + \cos x}$ (D) $\lim_{x \rightarrow 1} \frac{\arcsin x}{\tan \frac{\pi x}{2}}$
15. Which of the following statement are true for the function f defined for $-1 \leq x \leq 3$ in the figure shown.
 (A) $\lim_{x \rightarrow -1^+} f(x) = 1$
 (B) $\lim_{x \rightarrow 2} f(x)$ does not exist
 (C) $\lim_{x \rightarrow 1^-} f(x) = 2$
 (D) $\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^-} f(x)$
-
16. Let $f(x) = x + \sqrt{x^2 + 2x}$ and $g(x) = \sqrt{x^2 + 2x} - x$, then
 (A) $\lim_{x \rightarrow \infty} g(x) = 1$ (B) $\lim_{x \rightarrow \infty} f(x) = 1$ (C) $\lim_{x \rightarrow -\infty} f(x) = -1$ (D) $\lim_{x \rightarrow -\infty} g(x) = -1$
17. If $A = \lim_{x \rightarrow 0} \frac{\sin^{-1}(\sin x)}{\cos^{-1}(\cos x)}$ and $B = \lim_{x \rightarrow 0} \frac{[x]}{x}$, then (where $[.]$ denotes greatest integer function)-
 (A) $A = 1$ (B) A does not exist (C) $B = 0$ (D) $B = 1$

18. Which of the following limit tends to unity ?

(A) $\lim_{x \rightarrow 0} \frac{1 - \cos x + 2 \sin x - \sin^3 x - x^2 + 3x^4}{\tan^3 x - 6 \sin^2 x + x - 5x^3}$

(B) $\lim_{x \rightarrow \infty} \frac{x}{[x]}$

(C) $\lim_{x \rightarrow \infty} \frac{1}{(\sqrt{x} + \sqrt{x + \sqrt{x}} - \sqrt{x})}$

(D) $\lim_{x \rightarrow \infty} \left(\frac{\sqrt{x}}{\sqrt{x} + \sqrt{x + \sqrt{x}}} \right)$

19. Which of the following limits does not exist ?

(A) $\lim_{x \rightarrow 1^+} ([x])^{\frac{1}{x-1}}$

(B) $\lim_{x \rightarrow 3} \frac{(x^2 - 9 - \sqrt{x^2 - 6x + 9})}{|x - 1| - 2}$

(C) $\lim_{x \rightarrow 0^+} (x)^{\ln x}$

(D) $\lim_{x \rightarrow 0^+} \left(\frac{1 - \cos(\sin^2 x)}{x^2} \right)^{\frac{\ln(1-2x^2)}{\sin^2 x}}$

(where $[.]$ represents greatest integer function)

20. The value(s) of 'n' for which $\lim_{x \rightarrow 1} \frac{e^{x-1} - x}{(x-1)^n}$ exists is/are -

(A) 1

(B) 2

(C) 3

(D) 4

21. Let $f : \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \rightarrow \mathbb{R}$, $f(x) = \begin{cases} \lim_{n \rightarrow \infty} \left(\frac{(\tan x)^{2n} + x^2}{\sin^2 x + (\tan x)^{2n}} \right); & x \neq 0 \\ 1 & ; x = 0 \end{cases}$, $n \in \mathbb{N}$. Which of the following

holds good ?

(A) $f\left(-\frac{\pi}{4}\right) = f\left(\frac{\pi}{4}\right)$

(B) $f\left(-\frac{\pi}{4}\right) = f\left(-\frac{\pi^+}{4}\right)$

(C) $f\left(\frac{\pi^-}{4}\right) = f\left(\frac{\pi^+}{4}\right)$

(D) $f(0^+) = f(0) = f(0^-)$

22. Let $f(x) = \begin{cases} \frac{\tan^2 \{x\}}{x^2 - [x]^2} & \text{for } x > 0 \\ 1 & \text{for } x = 0 \\ \sqrt{\{x\} \cot \{x\}} & \text{for } x < 0 \end{cases}$ where $[x]$ is the step up function and $\{x\}$ is the fractional part

function of x , then-

(A) $\lim_{x \rightarrow 0^+} f(x) = 1$

(B) $\lim_{x \rightarrow 0^-} f(x) = 1$

(C) $\cot^{-1} \left(\lim_{x \rightarrow 0^-} f(x) \right)^2 = 1$

(D) None

23. $\lim_{x \rightarrow c} f(x)$ does not exist when (where $[x]$ is the step up function, $\{x\}$ is the fractional part function of x & $\text{sgn}(x)$ denotes signum function), then-
- (A) $f(x) = [[x]] - [2x - 1]$; $c = 3$ (B) $f(x) = [x] - x$, $c = 1$
- (C) $f(x) = \{x\}^2 - \{-x\}^2$, $c = 0$ (D) $f(x) = \frac{\tan(\text{sgn } x)}{\text{sgn } x}$, $c = 0$
24. Which of the following limits does not exist ?
- (A) $\lim_{x \rightarrow \infty} \text{cosec}^{-1}\left(\frac{x}{x+7}\right)$ (B) $\lim_{x \rightarrow 1} \sec^{-1}(\sin^{-1} x)$
- (C) $\lim_{x \rightarrow 0^+} x^{\frac{1}{x}}$ (D) $\lim_{x \rightarrow 0} \left(\tan\left(\frac{\pi}{8} + x\right) \right)^{\cot x}$
25. Which of the following statement(s) is (are) INCORRECT ?
- (A) If $\lim_{x \rightarrow c} f(x)$ and $\lim_{x \rightarrow c} g(x)$ both does not exist then $\lim_{x \rightarrow c} f(x)g(x)$ also does not exist.
- (B) If $\lim_{x \rightarrow c} f(x)$ and $\lim_{x \rightarrow c} g(x)$ both does not exist then $\lim_{x \rightarrow c} f(g(x))$ also does not exist.
- (C) If $\lim_{x \rightarrow c} f(x)$ exists and $\lim_{x \rightarrow c} g(x)$ does not exist then $\lim_{x \rightarrow c} g(f(x))$ does not exist.
- (D) If $\lim_{x \rightarrow c} f(x)$ and $\lim_{x \rightarrow c} g(x)$ both exist then $\lim_{x \rightarrow c} f(g(x))$ and $\lim_{x \rightarrow c} g(f(x))$ also exist.

EXERCISE (S-1)

1. $\lim_{x \rightarrow 1} \frac{x^2 - x \ln x + \ln x - 1}{x - 1}$ 2. $\lim_{x \rightarrow 1} \frac{\left[\sum_{k=1}^{100} x^k \right] - 100}{x - 1}$
3. $\lim_{x \rightarrow \frac{\pi}{4}} \frac{1 - \tan x}{1 - \sqrt{2} \sin x}$ 4. $\lim_{x \rightarrow 0} \frac{8}{x^8} \left[1 - \cos \frac{x^2}{2} - \cos \frac{x^2}{4} + \cos \frac{x^2}{2} \cos \frac{x^2}{4} \right]$
5. $\lim_{\theta \rightarrow \frac{\pi}{4}} \frac{\sqrt{2} - \cos \theta - \sin \theta}{(4\theta - \pi)^2}$
6. $\lim_{h \rightarrow 0} \frac{\sin\left(\frac{\pi}{3} + 4h\right) - 4\sin\left(\frac{\pi}{3} + 3h\right) + 6\sin\left(\frac{\pi}{3} + 2h\right) - 4\sin\left(\frac{\pi}{3} + h\right) + \sin \frac{\pi}{3}}{h^4}$
7. $\lim_{x \rightarrow \infty} x^2 \left(\sqrt{\frac{x+2}{x}} - \sqrt[3]{\frac{x+3}{x}} \right)$
8. $\lim_{x \rightarrow -\infty} \frac{(3x^4 + 2x^2) \sin \frac{1}{x} + |x|^3 + 5}{|x|^3 + |x|^2 + |x| + 1}$

9. If $\ell = \lim_{n \rightarrow \infty} \sum_{r=2}^n \left((r+1) \sin \frac{\pi}{r+1} - r \sin \frac{\pi}{r} \right)$ then find $\{\ell\}$. (where $\{\}$ denotes the fractional part function)
10. Find a & b if : (i) $\lim_{x \rightarrow \infty} \left[\frac{x^2+1}{x+1} - ax - b \right] = 0$ (ii) $\lim_{x \rightarrow -\infty} \left[\sqrt{x^2 - x + 1} - ax - b \right] = 0$
11. $\lim_{x \rightarrow 0} [\ell n(1 + \sin^2 x) \cdot \cot(\ell n^2(1 + x))]$
12. $\lim_{x \rightarrow 0} \frac{27^x - 9^x - 3^x + 1}{\sqrt{2} - \sqrt{1 + \cos x}}$
13. (a) $\lim_{x \rightarrow 0} \tan^{-1} \frac{a}{x^2}$, where $a \in \mathbb{R}$; (b) Plot the graph of the function $f(x) = \lim_{t \rightarrow 0} \left(\frac{2x}{\pi} \tan^{-1} \frac{x}{t^2} \right)$
14. Let $\{a_n\}$, $\{b_n\}$, $\{c_n\}$ be sequences such that
 (i) $a_n + b_n + c_n = 2n + 1$; (ii) $a_n b_n + b_n c_n + c_n a_n = 2n - 1$;
 (iii) $a_n b_n c_n = -1$; (iv) $a_n < b_n < c_n$
 Then find the value of $\lim_{n \rightarrow \infty} (na_n)$.
15. Let $f(x) = ax^3 + bx^2 + cx + d$ and $g(x) = x^2 + x - 2$.
 If $\lim_{x \rightarrow 1} \frac{f(x)}{g(x)} = 1$ and $\lim_{x \rightarrow -2} \frac{f(x)}{g(x)} = 4$, then find the value of $\frac{c^2 + d^2}{a^2 + b^2}$.
16. $\lim_{x \rightarrow \infty} \left[\frac{2x^2 + 3}{2x^2 + 5} \right]^{8x^2 + 3}$
17. $\lim_{x \rightarrow \infty} \left(\frac{x+c}{x-c} \right)^x = 4$ then find c
18. $\lim_{x \rightarrow 1} \left(\tan \frac{\pi x}{4} \right)^{\tan \frac{\pi x}{2}}$
19. $\lim_{x \rightarrow 0} \left(\frac{x-1+\cos x}{x} \right)^{\frac{1}{x}}$
20. If $n \in \mathbb{N}$ and $a_n = 2^2 + 4^2 + 6^2 + \dots + (2n)^2$ and $b_n = 1^2 + 3^2 + 5^2 + \dots + (2n-1)^2$.
 Find the value $\lim_{n \rightarrow \infty} \frac{\sqrt{a_n} - \sqrt{b_n}}{\sqrt{n}}$.

EXERCISE (S-2)

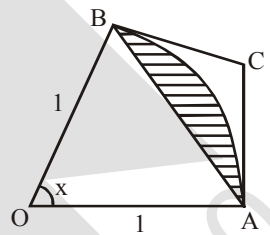
1. $\lim_{n \rightarrow \infty} \left(\frac{\sqrt{n^2 + n} - 1}{n} \right)^{2\sqrt{n^2 + n} - 1}$
2. $\lim_{x \rightarrow \infty} \left(\frac{a_1^{\frac{1}{x}} + a_2^{\frac{1}{x}} + a_3^{\frac{1}{x}} + \dots + a_n^{\frac{1}{x}}}{n} \right)^{nx}$, $n \in \mathbb{N}$, where $a_1, a_2, a_3, \dots, a_n > 0$
3. $\lim_{x \rightarrow 0} \left[\frac{(1+x)^{1/x}}{e} \right]^{1/x}$

4. If $\lim_{x \rightarrow \infty} \frac{a(2x^3 - x^2) + b(x^3 + 5x^2 - 1) - c(3x^3 + x^2)}{a(5x^4 - x) - bx^4 + c(4x^4 + 1) + 2x^2 + 5x} = 1$, then the value of $(a + b + c)$ can be expressed in the lowest form as $\frac{p}{q}$. Find the value of $(p + q)$.

5. $\lim_{x \rightarrow 0} \left[\frac{\ln(1+x)^{1+x}}{x^2} - \frac{1}{x} \right]$

6. Let $L = \prod_{n=3}^{\infty} \left(1 - \frac{4}{n^2} \right)$; $M = \prod_{n=2}^{\infty} \left(\frac{n^3 - 1}{n^3 + 1} \right)$ and $N = \prod_{n=1}^{\infty} \frac{(1 + n^{-1})^2}{1 + 2n^{-1}}$, then find the value of $L^{-1} + M^{-1} + N^{-1}$.

7. A circular arc of radius 1 subtends an angle of x radians, $0 < x < \frac{\pi}{2}$ as shown in the figure. The point C is the intersection of the two tangent lines at A & B . Let $T(x)$ be the area of triangle ABC & let $S(x)$ be the area of the shaded region. Compute :



- (a) $T(x)$ (b) $S(x)$ & (c) the limit of $\frac{T(x)}{S(x)}$ as $x \rightarrow 0$.

8. Let $f(x) = \lim_{n \rightarrow \infty} \sum_{n=1}^n 3^{n-1} \sin^3 \frac{x}{3^n}$ and $g(x) = x - 4f(x)$. Evaluate $\lim_{x \rightarrow 0} (1 + g(x))^{\cot x}$.

9. If $f(n, \theta) = \prod_{r=1}^n \left(1 - \tan^2 \frac{\theta}{2^r} \right)$, then compute $\lim_{n \rightarrow \infty} f(n, \theta)$

10. Evaluate $\lim_{x \rightarrow \infty} \left(\frac{x}{e} - x \left(\frac{x}{x+1} \right)^x \right)$

11. $f(x)$ is the function such that $\lim_{x \rightarrow 0} \frac{f(x)}{x} = 1$. If $\lim_{x \rightarrow 0} \frac{x(1 + a \cos x) - b \sin x}{(f(x))^3} = 1$, then find the value of a and b .

12. Through a point A on a circle, a chord AP is drawn & on the tangent at A a point T is taken such that $AT = AP$. If TP produced meet the diameter through A at Q , prove that the limiting value of AQ when P moves upto A is double the diameter of the circle.

13. At the end points A, B of the fixed segment of length L , lines are drawn meeting in C and making angles θ and 2θ respectively with the given segment. Let D be the foot of the altitude CD and let x represents the length of AD . Find the value of x as θ tends to zero i.e. $\lim_{\theta \rightarrow 0} x$.

14. Let $f(x) = \lim_{n \rightarrow \infty} \frac{2x^{2n} \sin \frac{1}{x} + x}{1 + x^{2n}}$, $n \in \mathbb{N}$, then find

- (a) $\lim_{x \rightarrow \infty} xf(x)$, (b) $\lim_{x \rightarrow 1} f(x)$, (c) $\lim_{x \rightarrow 0} f(x)$, (d) $\lim_{x \rightarrow -\infty} f(x)$

15. Using Sandwich theorem, evaluate

(a) $\lim_{n \rightarrow \infty} \left(\frac{1}{\sqrt{n^2}} + \frac{1}{\sqrt{n^2 + 1}} + \frac{1}{\sqrt{n^2 + 2}} + \dots + \frac{1}{\sqrt{n^2 + 2n}} \right)$

(b) $\lim_{n \rightarrow \infty} \frac{1}{1 + n^2} + \frac{2}{2 + n^2} + \dots + \frac{n}{n + n^2}$

EXERCISE (JM)

1. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a positive increasing function with $\lim_{x \rightarrow \infty} \frac{f(3x)}{f(x)} = 1$. Then $\lim_{x \rightarrow \infty} \frac{f(2x)}{f(x)} =$ [AIEEE-2010]

(1) 1 (2) $\frac{2}{3}$ (3) $\frac{3}{2}$ (4) 3

2. $\lim_{x \rightarrow 2} \left(\frac{\sqrt{1 - \cos\{2(x-2)\}}}{x-2} \right)$ [AIEEE-2011]

(1) equals $-\sqrt{2}$ (2) equals $\frac{1}{\sqrt{2}}$
(3) does not exist (4) equals $\sqrt{2}$

3. Let $f : \mathbb{R} \rightarrow [0, \infty)$ be such that $\lim_{x \rightarrow 5} f(x)$ exists and $\lim_{x \rightarrow 5} \frac{(f(x))^2 - 9}{\sqrt{|x-5|}} = 0$. Then $\lim_{x \rightarrow 5} f(x)$ equal - [AIEEE-2011]

(1) 3 (2) 0 (3) 1 (4) 2

4. $\lim_{x \rightarrow 0} \frac{\sin(\pi \cos^2 x)}{x^2}$ is equal to : [JEE Mains Offline-2014]

(1) $\frac{\pi}{2}$ (2) 1 (3) $-\pi$ (4) π

5. If $\lim_{x \rightarrow 2} \frac{\tan(x-2)\{x^2 + (k-2)x - 2k\}}{x^2 - 4x + 4} = 5$ then k is equal to [JEE Mains Online-2014]

(1) 3 (2) 1 (3) 0 (4) 2

6. Let $p = \lim_{x \rightarrow 0^+} (1 + \tan^2 \sqrt{x})^{\frac{1}{2x}}$ then $\log p$ is equal to - [JEE(Main)-2016]

(1) $\frac{1}{4}$ (2) 2 (3) 1 (4) $\frac{1}{2}$

7. $\lim_{x \rightarrow \frac{\pi}{2}} \frac{\cot x - \cos x}{(\pi - 2x)^3}$ equals :- [JEE(Main)-2017]

(1) $\frac{1}{4}$ (2) $\frac{1}{24}$ (3) $\frac{1}{16}$ (4) $\frac{1}{8}$

8. For each $t \in \mathbb{R}$, let $[t]$ be the greatest integer less than or equal to t . Then

$\lim_{x \rightarrow 0^+} x \left(\left[\frac{1}{x} \right] + \left[\frac{2}{x} \right] + \dots + \left[\frac{15}{x} \right] \right)$ [JEE(Main)-2018]

(1) is equal to 15. (2) is equal to 120.
(3) does not exist (in $\mathbb{R}$). (4) is equal to 0.

EXERCISE (JA)

1. Let $L = \lim_{x \rightarrow 0} \frac{a - \sqrt{a^2 - x^2} - \frac{x^2}{4}}{x^4}$, $a > 0$. If L is finite, then - [JEE 2009, 4]

(A) $a = 2$ (B) $a = 1$ (C) $L = \frac{1}{64}$ (D) $L = \frac{1}{32}$

2. If $\lim_{x \rightarrow 0} \left[1 + x \ln(1 + b^2) \right]^{\frac{1}{x}} = 2b \sin^2 \theta$, $b > 0$ and $\theta \in (-\pi, \pi]$, then the value of θ is-

(A) $\pm \frac{\pi}{4}$ (B) $\pm \frac{\pi}{3}$ (C) $\pm \frac{\pi}{6}$ (D) $\pm \frac{\pi}{2}$

[JEE 2011, 3M, -1M]

3. If $\lim_{x \rightarrow \infty} \left(\frac{x^2 + x + 1}{x + 1} - ax - b \right) = 4$, then - [JEE 2012, 3M, -1M]

(A) $a = 1$, $b = 4$ (B) $a = 1$, $b = -4$
(C) $a = 2$, $b = -3$ (D) $a = 2$, $b = 3$

4. Let $\alpha(a)$ and $\beta(a)$ be the roots of the equation $(\sqrt[3]{1+a} - 1)x^2 + (\sqrt{1+a} - 1)x + (\sqrt[5]{1+a} - 1) = 0$ where $a > -1$. Then $\lim_{a \rightarrow 0^+} \alpha(a)$ and $\lim_{a \rightarrow 0^+} \beta(a)$ are [JEE 2012, 3M, -1M]

(A) $-\frac{5}{2}$ and 1 (B) $-\frac{1}{2}$ and -1 (C) $-\frac{7}{2}$ and 2 (D) $-\frac{9}{2}$ and 3

5. The largest value of the non-negative integer a for which $\lim_{x \rightarrow 1} \left\{ \frac{-ax + \sin(x-1) + a}{x + \sin(x-1) - 1} \right\}^{\frac{1-x}{1-\sqrt{x}}} = \frac{1}{4}$ is

[JEE(Advanced)-2014, 3]

6. Let m and n be two positive integers greater than 1. If $\lim_{\alpha \rightarrow 0} \left(\frac{e^{\cos(\alpha^n)} - e}{\alpha^m} \right) = -\left(\frac{e}{2} \right)$ then the value of $\frac{m}{n}$ is [JEE 2015, 4M, -0M]

7. Let $\alpha, \beta \in \mathbb{R}$ be such that $\lim_{x \rightarrow 0} \frac{x^2 \sin(\beta x)}{\alpha x - \sin x} = 1$. Then $6(\alpha + \beta)$ equals

[JEE(Advanced)-2016, 3(0)]

8. Let $f(x) = \frac{1-x(1+|1-x|)}{|1-x|} \cos\left(\frac{1}{1-x}\right)$ for $x \neq 1$. Then [JEE(Advanced)-2017, 4]

- (A) $\lim_{x \rightarrow 1^+} f(x)$ does not exist (B) $\lim_{x \rightarrow 1^-} f(x)$ does not exist
 (C) $\lim_{x \rightarrow 1^-} f(x) = 0$ (D) $\lim_{x \rightarrow 1^+} f(x) = 0$

9. For any positive integer n , define $f_n : (0, \infty) \rightarrow \mathbb{R}$ as

$$f_n(x) = \sum_{j=1}^n \tan^{-1} \left(\frac{1}{1+(x+j)(x+j-1)} \right) \text{ for all } x \in (0, \infty).$$

(Here, the inverse trigonometric function $\tan^{-1}x$ assume values in $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$.)

Then, which of the following statement(s) is (are) TRUE ?

[JEE(Advanced)-2018]

- (A) $\sum_{j=1}^5 \tan^2(f_j(0)) = 55$
 (B) $\sum_{j=1}^{10} (1 + f'_j(0)) \sec^2(f_j(0)) = 10$
 (C) For any fixed positive integer n , $\lim_{x \rightarrow \infty} \tan(f_n(x)) = \frac{1}{n}$
 (D) For any fixed positive integer n , $\lim_{x \rightarrow \infty} \sec^2(f_n(x)) = 1$

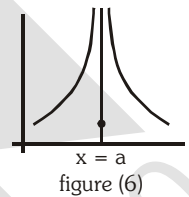
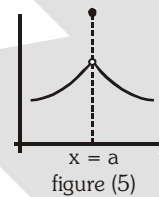
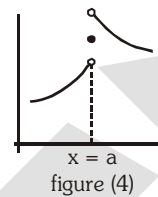
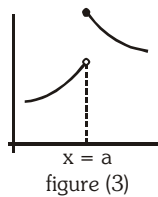
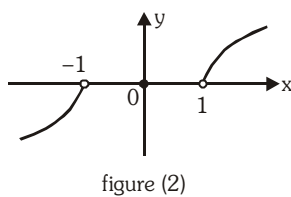
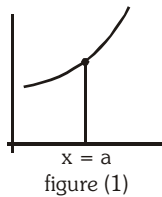
CONTINUITY

1. CONTINUOUS FUNCTIONS :

A function for which a small change in the independent variable causes only a small change and not a sudden jump in the dependent variable are called continuous functions. Naively, we may say that a function is continuous at a fixed point if we can draw the graph of the function around that point without lifting the pen from the plane of the paper.

A function $f(x)$ is said to be continuous at $x = a$, if $\lim_{x \rightarrow a} f(x)$ exists and is equal to $f(a)$. Symbolically $f(x)$ is continuous at $x = a$ if $\lim_{h \rightarrow 0} f(a - h) = \lim_{h \rightarrow 0} f(a + h) = f(a) = \text{finite quantity}$.

i.e. $\text{LHL}|_{x=a} = \text{RHL}|_{x=a} = \text{value of } f(x)|_{x=a} = \text{finite quantity. } (h > 0)$



In figure (1) and (2) $f(x)$ is continuous at $x = a$ and $x = 0$ respectively and in figure (3) to (6) $f(x)$ is discontinuous at $x = a$.

Note 1 : Continuity of a function must be discussed only at points which are in the domain of the function.

Note 2 : If $x = a$ is an isolated point of domain then $f(x)$ is always considered to be continuous at $x = a$.

Illustration 1 : If $f(x) = \begin{cases} \sin \frac{\pi x}{2}, & x < 1 \\ [x] & x \geq 1 \end{cases}$ then find whether $f(x)$ is continuous or not at $x = 1$, where $[]$ denotes greatest integer function.

Solution : $f(x) = \begin{cases} \sin \frac{\pi x}{2}, & x < 1 \\ [x], & x \geq 1 \end{cases}$
For continuity at $x = 1$, we determine, $f(1)$, $\lim_{x \rightarrow 1^-} f(x)$ and $\lim_{x \rightarrow 1^+} f(x)$.
Now, $f(1) = [1] = 1$

$$\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^-} \sin \frac{\pi x}{2} = \sin \frac{\pi}{2} = 1 \text{ and } \lim_{x \rightarrow 1^+} f(x) = \lim_{x \rightarrow 1^+} [x] = 1$$

so $f(1) = \lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^+} f(x)$
 $\therefore f(x)$ is continuous at $x = 1$

Illustration 2 : Let $f(x) = \begin{cases} \frac{a(1 - x \sin x) + b \cos x + 5}{x^2} & x < 0 \\ 3 & x = 0 \\ \left(1 + \left(\frac{cx + dx^3}{x^2}\right)\right)^{\frac{1}{x}} & x > 0 \end{cases}$

If f is continuous at $x = 0$, then find out the values of a , b , c and d .

Solution : Since $f(x)$ is continuous at $x = 0$, so at $x = 0$, both left and right limits must exist and both must be equal to 3.

$$\text{Now } \lim_{x \rightarrow 0^-} \frac{a(1 - x \sin x) + b \cos x + 5}{x^2} = \lim_{x \rightarrow 0^-} \frac{(a + b + 5) + \left(-a - \frac{b}{2}\right)x^2 + \dots}{x^2} = 3$$

(By the expansions of $\sin x$ and $\cos x$)

$$\text{If } \lim_{x \rightarrow 0^-} f(x) \text{ exists then } a + b + 5 = 0 \text{ and } -a - \frac{b}{2} = 3 \Rightarrow a = -1 \text{ and } b = -4$$

$$\text{since } \lim_{x \rightarrow 0^+} \left(1 + \left(\frac{cx + dx^3}{x^2}\right)\right)^{\frac{1}{x}} \text{ exists } \Rightarrow \lim_{x \rightarrow 0^+} \frac{cx + dx^3}{x^2} = 0 \Rightarrow c = 0$$

$$\text{Now } \lim_{x \rightarrow 0^+} (1 + dx)^{\frac{1}{x}} = \lim_{x \rightarrow 0^+} \left[(1 + dx)^{\frac{1}{dx}}\right]^d = e^d$$

$$\text{So } e^d = 3 \Rightarrow d = \ln 3,$$

$$\text{Hence } a = -1, b = -4, c = 0 \text{ and } d = \ln 3.$$

Do yourself - 1 :

(i) If $f(x) = \begin{cases} \cos x; x \geq 0 \\ x + k; x < 0 \end{cases}$ find the value of k if $f(x)$ is continuous at $x = 0$.

(ii) If $f(x) = \begin{cases} \frac{|x+2|}{\tan^{-1}(x+2)} & ; x \neq -2 \\ 2 & ; x = -2 \end{cases}$ then discuss the continuity of $f(x)$ at $x = -2$

2. CONTINUITY OF THE FUNCTION IN AN INTERVAL :

- (a) A function is said to be continuous in (a, b) if f is continuous at each & every point belonging to (a, b) .
- (b) A function is said to be continuous in a closed interval $[a, b]$ if :
- f is continuous in the open interval (a, b)
 - f is right continuous at 'a' i.e. $\lim_{x \rightarrow a^+} f(x) = f(a) = \text{a finite quantity}$
 - f is left continuous at 'b' i.e. $\lim_{x \rightarrow b^-} f(x) = f(b) = \text{a finite quantity}$

Note :

- All polynomials, trigonometrical functions, exponential & logarithmic functions are continuous in their domains.
- If $f(x)$ & $g(x)$ are two functions that are continuous at $x = c$ then the function defined by :
 $F_1(x) = f(x) \pm g(x)$; $F_2(x) = K f(x)$, where K is any real number ; $F_3(x) = f(x) \cdot g(x)$ are also continuous at $x = c$.

$$\text{Further, if } g(c) \text{ is not zero, then } F_4(x) = \frac{f(x)}{g(x)} \text{ is also continuous at } x = c.$$

Illustration 3 : Discuss the continuity of $f(x) = \begin{cases} |x+1| & , \quad x < -2 \\ 2x+3 & , \quad -2 \leq x < 0 \\ x^2+3 & , \quad 0 \leq x < 3 \\ x^3-15 & , \quad x \geq 3 \end{cases}$

Solution : We write $f(x)$ as $f(x) = \begin{cases} -x-1 & , \quad x < -2 \\ 2x+3 & , \quad -2 \leq x < 0 \\ x^2+3 & , \quad 0 \leq x < 3 \\ x^3-15 & , \quad x \geq 3 \end{cases}$

As we can see, $f(x)$ is defined as a polynomial function in each of intervals $(-\infty, -2)$, $(-2, 0)$, $(0, 3)$ and $(3, \infty)$. Therefore, it is continuous in each of these four open intervals. Thus we check the continuity at $x = -2, 0, 3$.

At the point $x = -2$

$$\lim_{x \rightarrow -2^-} f(x) = \lim_{x \rightarrow -2^-} (-x-1) = +2-1 = 1$$

$$\lim_{x \rightarrow -2^+} f(x) = \lim_{x \rightarrow -2^+} (2x+3) = 2 \cdot (-2) + 3 = -1$$

Therefore, $\lim_{x \rightarrow -2} f(x)$ does not exist and hence $f(x)$ is discontinuous at $x = -2$.

At the point $x = 0$

$$\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} (2x+3) = 3$$

$$\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} (x^2+3) = 3$$

$$f(0) = 0^2 + 3 = 3$$

Therefore $f(x)$ is continuous at $x = 0$.

At the point $x = 3$

$$\lim_{x \rightarrow 3^-} f(x) = \lim_{x \rightarrow 3^-} (x^2+3) = 3^2+3 = 12$$

$$\lim_{x \rightarrow 3^+} f(x) = \lim_{x \rightarrow 3^+} (x^3-15) = 3^3-15 = 12$$

$$f(3) = 3^3 - 15 = 12$$

Therefore, $f(x)$ is continuous at $x = 3$.

We find that $f(x)$ is continuous at all points in $\mathbb{R}$ except at $x = -2$

Do yourself -2 :

(i) If $f(x) = \begin{cases} \frac{x^2}{a} & ; 0 \leq x < 1 \\ -1 & ; 1 \leq x < \sqrt{2} \\ \frac{2b^2 - 4b}{x^2} & ; \sqrt{2} \leq x < \infty \end{cases}$ then find the value of a & b if $f(x)$ is continuous in $[0, \infty)$

(ii) Discuss the continuity of $f(x) = \begin{cases} |x-3| & ; 0 \leq x < 1 \\ \sin x & ; 1 \leq x \leq \frac{\pi}{2} \\ \log_{\frac{\pi}{2}} x & ; \frac{\pi}{2} < x < 3 \end{cases}$ in $[0, 3)$

3. TYPES OF DISCONTINUITIES :

Type-1 : (Removable type of discontinuities) :- In case $\lim_{x \rightarrow a} f(x)$ exists but is not equal to $f(a)$ ($f(a)$ is defined) then the function is said to have a removable discontinuity or discontinuity of the first kind. In this case we can redefine the function such that $\lim_{x \rightarrow a} f(x) = f(a)$ & make it continuous at $x = a$.

Illustration 4 : Examine the function, $f(x) = \begin{cases} x-1 & , x < 0 \\ 1/4 & , x = 0 \\ x^2 - 1 & , x > 0 \end{cases}$. Discuss the continuity, and if discontinuous

remove the discontinuity by redefining the function (if possible).

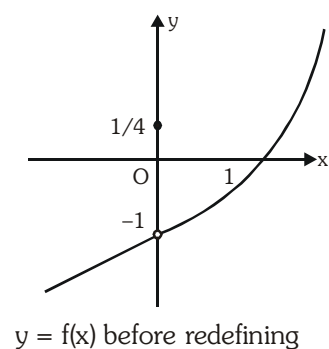
Solution :

Graph of $f(x)$ is shown, from graph it is seen that

$$\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^+} f(x) = -1, \text{ but } f(0) = 1/4$$

Thus, $f(x)$ has removable discontinuity and $f(x)$ could be made continuous by taking $f(0) = -1$

$$\Rightarrow f(x) = \begin{cases} x-1 & , x < 0 \\ -1 & , x = 0 \\ x^2 - 1 & , x > 0 \end{cases}$$



Do yourself -3 :

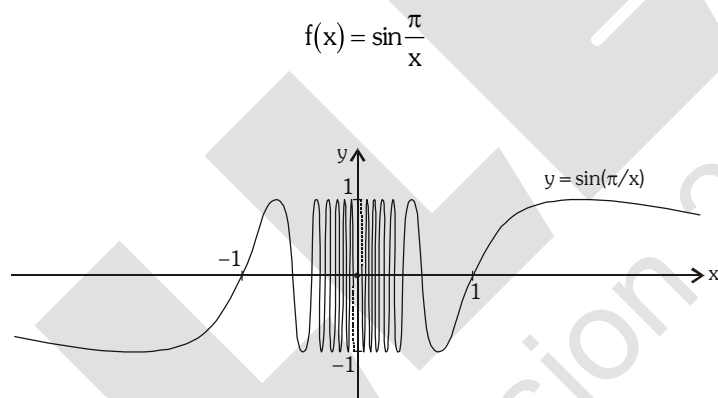
(i) If $f(x) = \begin{cases} \frac{1}{x-1} & ; 1 < x < 2 \\ x^2 - 3 & ; 2 \leq x < 4 \\ 5 & ; x = 4 \\ 14 - \frac{x^{1/2}}{2} & ; x > 4 \end{cases}$, then discuss the types of discontinuity for the function.

Type-2 : (Non-Removable type of discontinuities) :

In case $\lim_{x \rightarrow a} f(x)$ does not exist then it is not possible to make the function continuous by redefining it.

Such a discontinuity is known as non-removable discontinuity or discontinuity of the 2nd kind.

Example : $f(x) = \begin{cases} \sin \frac{\pi}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases}$ discuss continuity at $x = 0$



$f(x)$ has non removable type discontinuity at $x = 0$

Example : From the adjacent graph note that

- (i) f is continuous at $x = -1$
- (ii) f has removable discontinuity at $x = 1$
- (iii) f has non-removable discontinuity at $x = 0$

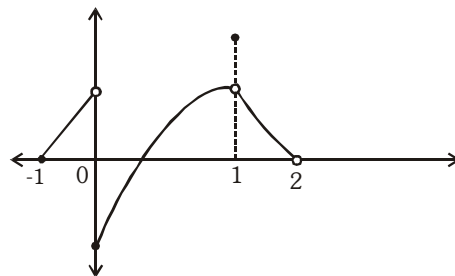


Illustration 5 : Show that the function, $f(x) = \begin{cases} \frac{e^{1/x} - 1}{e^{1/x} + 1} & ; \text{when } x \neq 0 \\ 0, & ; \text{when } x = 0 \end{cases}$ has non-removable discontinuity at $x = 0$.

Solution : We have, $f(x) = \begin{cases} \frac{e^{1/x} - 1}{e^{1/x} + 1} & ; \text{ when } x \neq 0 \\ 0, & ; \text{ when } x = 0 \end{cases}$

$$\Rightarrow \lim_{x \rightarrow 0^+} f(x) = \lim_{h \rightarrow 0} f(0 + h) = \lim_{h \rightarrow 0} \frac{e^{\frac{1}{h}} - 1}{e^{\frac{1}{h}} + 1} = \lim_{h \rightarrow 0} \frac{1 - \frac{1}{e^{1/h}}}{1 + \frac{1}{e^{1/h}}} = 1 \quad [\because e^{1/h} \rightarrow \infty]$$

$$\Rightarrow \lim_{x \rightarrow 0^-} f(x) = \lim_{h \rightarrow 0} \frac{e^{-1/h} - 1}{e^{-1/h} + 1} = \frac{0 - 1}{0 + 1} = -1 \quad [\because h \rightarrow 0 ; e^{-1/h} \rightarrow 0]$$

$$\lim_{x \rightarrow 0^-} f(x) = -1$$

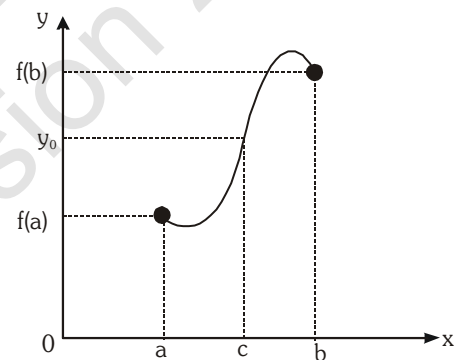
$\Rightarrow \lim_{x \rightarrow 0^+} f(x) \neq \lim_{x \rightarrow 0^-} f(x)$. Thus $f(x)$ has non-removable discontinuity.

Do yourself -4 :

(i) Discuss the type of discontinuity for $f(x) = \begin{cases} -1 & ; x \leq -1 \\ |x| & ; -1 < x < 1 \\ (x+1) & ; x \geq 1 \end{cases}$

4. THE INTERMEDIATE VALUE THEOREM :

Suppose $f(x)$ is continuous on an interval I , and a and b are any two points of I . Then if y_0 is a number between $f(a)$ and $f(b)$, there exists a number c between a and b such that $f(c) = y_0$.



The function f , being continuous on $[a, b]$ takes on every value between $f(a)$ and $f(b)$

Note that a function f which is continuous in $[a, b]$ possesses the following properties :

- (i) If $f(a)$ & $f(b)$ possess opposite signs, then there exists at least one root of the equation $f(x) = 0$ in the open interval (a, b) .
- (ii) If K is any real number between $f(a)$ & $f(b)$, then there exists at least one root of the equation $f(x) = K$ in the open interval (a, b) .

Note : In above cases the number of roots is always odd.

Illustration 6 : Show that the function, $f(x) = (x - a)^2(x - b)^2 + x$, takes the value $\frac{a+b}{2}$ for some

$$x_0 \in (a, b)$$

Solution :

$$f(x) = (x - a)^2(x - b)^2 + x$$

$$f(a) = a$$

$$f(b) = b$$

$$\& \frac{a+b}{2} \in (f(a), f(b))$$

$\therefore$ By intermediate value theorem, there is atleast one $x_0 \in (a, b)$ such that $f(x_0) = \frac{a+b}{2}$.

Illustration 7 : Let $f: [0, 1] \xrightarrow{\text{onto}} [0, 1]$ be a continuous function, then prove that $f(x) = x$ for atleast one $x \in [0, 1]$

Solution :

Consider $g(x) = f(x) - x$

$$g(0) = f(0) - 0 = f(0) \geq 0 \quad \{\because 0 \leq f(x) \leq 1\}$$

$$g(1) = f(1) - 1 \leq 0$$

$$\Rightarrow g(0) \cdot g(1) \leq 0$$

$$\Rightarrow g(x) = 0 \text{ has atleast one root in } [0, 1]$$

$$\Rightarrow f(x) = x \text{ for atleast one } x \in [0, 1]$$

Do yourself -5 :

(i) If $f(x)$ is continuous in $[a, b]$ such that $f(c) = \frac{2f(a) + 3f(b)}{5}$, then prove that $c \in (a, b)$

5. SOME IMPORTANT POINTS :

(a) If $f(x)$ is continuous & $g(x)$ is discontinuous at $x = a$ then the product function $\phi(x) = f(x) \cdot g(x)$ will **not necessarily be discontinuous at $x = a$** , e.g.

$$f(x) = x \text{ \& } g(x) = \begin{cases} \sin \frac{\pi}{x} & x \neq 0 \\ 0 & x = 0 \end{cases}$$

$f(x)$ is continuous at $x = 0$ & $g(x)$ is discontinuous at $x = 0$, but $f(x) \cdot g(x)$ is continuous at $x = 0$.

(b) If $f(x)$ and $g(x)$ both are discontinuous at $x = a$ then the product function $\phi(x) = f(x) \cdot g(x)$ **is not necessarily be discontinuous at $x = a$** , e.g.

$$f(x) = -g(x) = \begin{cases} 1 & x \geq 0 \\ -1 & x < 0 \end{cases}$$

$f(x)$ & $g(x)$ both are discontinuous at $x = 0$ but the product function $f \cdot g(x)$ is still continuous at $x = 0$

- (c) If $f(x)$ and $g(x)$ both are discontinuous at $x = a$ then $f(x) \pm g(x)$ is not necessarily be discontinuous at $x = a$
- (d) A continuous function whose domain is closed must have a range also in closed interval.
- (e) If f is continuous at $x = a$ & g is continuous at $x = f(a)$ then the composite $g[f(x)]$ is continuous at $x = a$. eg. $f(x) = \frac{x \sin x}{x^2 + 2}$ & $g(x) = |x|$ are continuous at $x = 0$, hence the composite

$$(g \circ f)(x) = \left| \frac{x \sin x}{x^2 + 2} \right| \text{ will also be continuous at } x = 0$$

Illustration 8: If $f(x) = \frac{x+1}{x-1}$ and $g(x) = \frac{1}{x-2}$, then discuss the continuity of $f(x)$, $g(x)$ and $f \circ g(x)$ in its domain.

Solution : $f(x) = \frac{x+1}{x-1}$

$f(x)$ is a rational function it must be continuous in its domain and f is not defined at $x = 1$.

$$g(x) = \frac{1}{x-2}$$

$g(x)$ is also a rational function. It must be continuous in its domain and g is not defined at $x = 2$.

Consider $g(x) = 1$

$$\frac{1}{x-2} = 1 \Rightarrow x = 3$$

$\therefore f \circ g(x)$ is continuous in its domain : $\mathbb{R} - \{2, 3\}$

Do yourself -6 :

- (i) Let $f(x) = [x]$ & $g(x) = \text{sgn}(x)$ (where $[.]$ denotes greatest integer function), then discuss the continuity of $f(x) \pm g(x)$, $f(x) \cdot g(x)$ & $\frac{f(x)}{g(x)}$ at $x = 0$.
- (ii) If $f(x) = \sin|x|$ & $g(x) = \tan|x|$ then discuss the continuity of $f(x) \pm g(x)$; $\frac{f(x)}{g(x)}$ & $f(x) \cdot g(x)$

6. CONTINUITY OVER COUNTABLE SET :

There are functions which are continuous over a countable set and else where discontinuous.

Illustration 9: If $f(x) = \begin{cases} x & \text{if } x \in \mathbb{Q} \\ -x & \text{if } x \notin \mathbb{Q} \end{cases}$, find the points where $f(x)$ is continuous

Solution : Let $x = a$ be the point at which $f(x)$ is continuous.

$$\Rightarrow \lim_{\substack{x \rightarrow a \\ \text{through rational}}} f(x) = \lim_{\substack{x \rightarrow a \\ \text{through irrational}}} f(x)$$

$$\Rightarrow a = -a$$

$$\Rightarrow a = 0 \Rightarrow \text{function is continuous at } x = 0.$$

Do yourself -7 :

(i) If $g(x) = \begin{cases} x & \text{if } x \in \mathbb{Q} \\ 0 & \text{if } x \notin \mathbb{Q} \end{cases}$, then find the points where function is continuous.

(ii) If $f(x) = \begin{cases} x^2 & ; x \in \mathbb{Q} \\ 1-x^2 & ; x \notin \mathbb{Q} \end{cases}$, then find the points where function is continuous.

ANSWERS FOR DO YOURSELF

1. (i) 1 (ii) discontinuous at $x = -2$
2. (i) $a = -1$ & $b = 1$ (ii) Discontinuous at $x = 1$ & continuous at $x = \frac{\pi}{2}$
3. (i) Removable discontinuity at $x = 4$.
4. (i) Non-removable discontinuity at $x = -1, 1$
6. (i) All are discontinuous at $x = 0$.
(ii) $f(x)g(x)$ & $f(x) \pm g(x)$ are continuous in $\mathbb{R} - \{x : x = (2n+1)\frac{\pi}{2}; n \in \mathbb{Z}\}$
 $\frac{f(x)}{g(x)}$ is continuous in $\mathbb{R} - \{x : x = \frac{n\pi}{2}; n \in \mathbb{Z}\}$
7. (i) $x = 0$ (ii) $x = \pm \frac{1}{\sqrt{2}}$

EXERCISE (O-1)

[SINGLE CORRECT CHOICE TYPE]

1. Let $f(x) = \begin{cases} ax+1 & \text{if } x < 1 \\ 3 & \text{if } x = 1 \\ bx^2+1 & \text{if } x > 1 \end{cases}$. If $f(x)$ is continuous at $x = 1$ then $(a - b)$ is equal to-
- (A) 0 (B) 1 (C) 2 (D) 4
2. For the function $f(x) = \begin{cases} \frac{1}{x+2^{\left(\frac{1}{x-2}\right)}}, & x \neq 2 \\ k, & x = 2 \end{cases}$ which of the following holds ?
- (A) $k = 1/2$ and f is continuous at $x = 2$ (B) $k \neq 0, 1/2$ and f is continuous at $x = 2$
 (C) f can not be continuous at $x = 2$ (D) $k = 0$ and f is continuous at $x = 2$.
3. The function $f(x) = \frac{4-x^2}{4x-x^3}$, is-
- (A) discontinuous at only one point in its domain.
 (B) discontinuous at two points in its domain.
 (C) discontinuous at three points in its domain.
 (D) continuous everywhere in its domain.
4. If $f(x) = \begin{cases} -4\sin x + \cos x & \text{for } x \leq -\frac{\pi}{2} \\ a\sin x + b & \text{for } -\frac{\pi}{2} < x < \frac{\pi}{2} \\ \cos x + 2 & \text{for } x \geq \frac{\pi}{2} \end{cases}$ is continuous then :
- (A) $a = -1, b = 3$ (B) $a = 1, b = -3$
 (C) $a = 1, b = 3$ (D) $a = -1, b = -3$
5. The function $f(x) = \begin{cases} \frac{1}{4}(3x^2+1) & -\infty < x \leq 1 \\ 5-4x & 1 < x < 4 \\ 4-x & 4 \leq x < \infty \end{cases}$ is -
- (A) continuous at $x = 1$ & $x = 4$
 (B) continuous at $x = 1$, discontinuous at $x = 4$
 (C) continuous at $x = 4$, discontinuous at $x = 1$
 (D) discontinuous at $x = 1$ & $x = 4$
6. If $f(x) = \frac{x^2 - bx + 25}{x^2 - 7x + 10}$ for $x \neq 5$ and f is continuous at $x = 5$, then $f(5)$ has the value equal to-
- (A) 0 (B) 5 (C) 10 (D) 25

7. If $f(x) = \frac{x - e^x + \cos 2x}{x^2}$, $x \neq 0$ is continuous at $x = 0$, then -

- (A) $f(0) = \frac{5}{2}$ (B) $[f(0)] = -2$ (C) $\{f(0)\} = -0.5$ (D) $[f(0)]. \{f(0)\} = -1.5$

where $[.]$ and $\{.\}$ denotes greatest integer and fractional part function

8. $y = f(x)$ is a continuous function such that its graph passes through $(a, 0)$. Then $\lim_{x \rightarrow a} \frac{\log_e (1 + 3f(x))}{2f(x)}$

is-

- (A) 1 (B) 0 (C) $\frac{3}{2}$ (D) $\frac{2}{3}$

9. In $[1, 3]$, the function $[x^2 + 1], [.]$ denoting the greatest integer function, is continuous -

- (A) for all x
(B) for all x except at nine points
(C) for all x except at seven points
(D) for all x except at eight points

10. Number of points of discontinuity of $f(x) = [2x^3 - 5]$ in $[1, 2)$, is equal to-

(where $[x]$ denotes greatest integer less than or equal to x)

- (A) 14 (B) 13 (C) 10 (D) 8

11. Given $f(x) = \begin{cases} |x+1| & \text{if } x < -2 \\ 2x+3 & \text{if } -2 \leq x < 0 \\ x^2+3 & \text{if } 0 \leq x < 3 \\ x^3-15 & \text{if } x \geq 3 \end{cases}$. Then number of point(s) of discontinuity of $f(x)$ is-

- (A) 0 (B) 1 (C) 2 (D) 3

12. If $f(x)$ is continuous and $f\left(\frac{9}{2}\right) = \frac{2}{9}$, then the value of $\lim_{x \rightarrow 0} f\left(\frac{1 - \cos 3x}{x^2}\right)$ is-

- (A) $\frac{2}{9}$ (B) $\frac{9}{2}$ (C) 0 (D) data insufficient

13. f is a continuous function on the real line. Given that $x^2 + (f(x) - 2)x - \sqrt{3} \cdot f(x) + 2\sqrt{3} - 3 = 0$. Then the value of $f(\sqrt{3})$

- (A) can not be determined (B) is $2(1 - \sqrt{3})$
(C) is zero (D) is $\frac{2(\sqrt{3} - 2)}{\sqrt{3}}$

14. The function $f(x) = [x]^2 - [x^2]$ (where $[y]$ is the greatest integer less than or equal to y), is discontinuous at :

- (A) all integers (B) all integers except 0 & 1
(C) all integers except 0 (D) all integers except 1

EXERCISE (O-2)

[SINGLE CORRECT CHOICE TYPE]

- Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function $\forall x \in \mathbb{R}$ and $f(x) = 5 \forall x \in \text{irrational}$. Then the value of $f(3)$ is -
 (A) 1 (B) 2 (C) 5 (D) cannot determine
- If $f(x) = \frac{1}{(x-1)(x-2)}$ and $g(x) = \frac{1}{x^2}$, then set of points in domain of $f \circ g(x)$ at which $f \circ g(x)$ is discontinuous.
 (A) $\left\{-1, 0, 1, \frac{1}{\sqrt{2}}\right\}$ (B) ϕ
 (C) $\{0, 1\}$ (D) $\left\{0, 1, \frac{1}{\sqrt{2}}\right\}$
- Consider the function $f(x) = \begin{cases} \frac{x}{[x]} & \text{if } 1 \leq x < 2 \\ 1 & \text{if } x = 2 \\ \sqrt{6-x} & \text{if } 2 < x \leq 3 \end{cases}$
 where $[x]$ denotes step up function then function -
 (A) has removable discontinuity at $x = 3$
 (B) has removable discontinuity at $x = 2$
 (C) has non removable discontinuity at $x = 2$
 (D) is continuous at $x = 2$
- Consider $f(x) = \begin{cases} x[x]^2 \log_{(1+x)} 2, & \text{for } -1 < x < 0 \\ k, & \text{for } x = 0 \\ \frac{\ln(e^{x^2} + 2\sqrt{\{x\}})}{\tan \sqrt{x}}, & \text{for } 0 < x < 1 \end{cases}$
 where $[*]$ & $\{*\}$ are the greatest integer function & fractional part function respectively, then :-
 (A) $k = \ln 2 \Rightarrow f$ is continuous at $x = 0$ (B) $k = 2 \Rightarrow f$ is continuous at $x = 0$
 (C) $k = e^2 \Rightarrow f$ is continuous at $x = 0$ (D) f has a non-removable discontinuity at $x = 0$
- The function $f(x) = [x] \cdot \cos \frac{2x-1}{2} \pi$, where $[\cdot]$ denotes the greatest integer function, is discontinuous at :-
 (A) all x (B) all integer points (C) no x (D) x which is not an integer
- Consider the function defined on $[0, 1] \rightarrow \mathbb{R}$, $f(x) = \frac{\sin x - x \cos x}{x^2}$ if $x \neq 0$ and $f(0) = 0$, then the function $f(x)$:-
 (A) has a removable discontinuity at $x = 0$ (B) has a non removable discontinuity at $x = 0$
 (C) limit doesnot exist at $x = 0$ (D) is continuous at $x = 0$

[MULTIPLE CORRECT CHOICE TYPE]

7. Which of the following function(s) is/are discontinuous at $x = 0$?

- (A) $f(x) = \sin \frac{\pi}{2x}$, $x \neq 0$ and $f(0) = 1$ (B) $g(x) = x \sin \left(\frac{\pi}{x} \right)$, $x \neq 0$ and $g(0) = \pi$
 (C) $h(x) = \frac{|x|}{x}$, $x \neq 0$ and $h(0) = 1$ (D) $k(x) = \frac{1}{1 + e^{\cot x}}$, $x \neq 0$ and $k(0) = 0$.

8. A function $f(x)$ is defined as $f(x) = \frac{A \sin x + \sin 2x}{x^3}$, ($x \neq 0$). If the function is continuous at $x = 0$, then -

- (A) $A = -2$ (B) $f(0) = -1$ (C) $A = 1$ (D) $f(0) = 1$

9. Which of the following function(s) can't be defined at $x = 0$ to make it continuous at $x = 0$?

- (A) $f(x) = \frac{1}{1 + 2^{\frac{1}{x}}}$ (B) $f(x) = \tan^{-1} \frac{1}{x}$ (C) $f(x) = \frac{e^{\frac{1}{x}} - 1}{e^{\frac{1}{x}} + 1}$ (D) $f(x) = \frac{1}{\ln |x|}$

10. Which of the following function(s) can be defined continuously at $x = 0$?

- (A) $f(x) = \frac{1}{1 + 2^{\cot x}}$ (B) $f(x) = \cos \left(\frac{|\sin x|}{x} \right)$
 (C) $f(x) = x \sin \frac{\pi}{x}$ (D) $f(x) = \frac{1}{\ln |x|}$

11. Let $f(x) = \begin{cases} \frac{e^x - 1 + ax}{x^2}, & x > 0 \\ b, & x = 0 \\ \frac{\sin \frac{x}{2}}{x}, & x < 0 \end{cases}$, then -

- (A) $f(x)$ is continuous at $x = 0$ if $a = -1$, $b = \frac{1}{2}$.
 (B) $f(x)$ is discontinuous at $x = 0$, if $b \neq \frac{1}{2}$.
 (C) $f(x)$ has non-removable discontinuity at $x = 0$ if $a \neq -1$.
 (D) $f(x)$ has removable discontinuity at $x = 0$ if $a = -1$, $b \neq \frac{1}{2}$.

12. Which of the following function(s) can be defined continuously at $x = 0$?

- (A) $f(x) = \frac{1 - \sec^2 2x}{4x^2}$ (B) $g(x) = \frac{\csc x - 1}{x \csc x}$ (where $\csc x = \operatorname{cosec} x$)
 (C) $h(x) = \frac{\sin 5x}{x}$ (D) $l(x) = (1 + 2x^2)^{\frac{1}{x^2}}$

13. If f is defined on an interval $[a, b]$. Which of the following statement(s) is/are INCORRECT ?
- (A) If $f(a)$ and $f(b)$, have opposite sign, then there must be a point $c \in (a, b)$ such that $f(c) = 0$.
- (B) If f is continuous on $[a, b]$, $f(a) < 0$ and $f(b) > 0$, then there must be a point $c \in (a, b)$ such that $f(c) = 0$.
- (C) If f is continuous on $[a, b]$ and there is a point c in (a, b) such that $f(c) = 0$, then $f(a)$ and $f(b)$ have opposite sign.
- (D) If f has no zeroes on $[a, b]$, then $f(a)$ and $f(b)$ have the same sign.
14. Which of the following functions can be defined at indicated point so that resulting function is continuous -
- (A) $f(x) = \frac{x^2 - 2x - 8}{x + 2}$ at $x = -2$
- (B) $f(x) = \frac{x - 7}{|x - 7|}$ at $x = 7$
- (C) $f(x) = \frac{x^3 + 64}{x + 4}$ at $x = -4$
- (D) $f(x) = \frac{3 - \sqrt{x}}{9 - x}$ at $x = 9$
15. In which of the following cases the given equations has atleast one root in the indicated interval ?
- (A) $x - \cos x = 0$ in $(0, \pi/2)$
- (B) $x + \sin x = 1$ in $(0, \pi/6)$
- (C) $\frac{a}{x-1} + \frac{b}{x-3} = 0$, $a, b > 0$ in $(1, 3)$
- (D) $f(x) - g(x) = 0$ in (a, b) where f and g are continuous on $[a, b]$ and $f(a) > g(a)$ and $f(b) < g(b)$.

[MATRIX TYPE]

16. Column-I

Column-II

- | | |
|---|-------------------|
| (A) $\lim_{x \rightarrow 1} \frac{x^3 - 1}{\ln x}$ is | (P) 2 |
| (B) $\lim_{x \rightarrow 0} \frac{x(\cos x - \cos 2x)}{2 \sin x - \sin 2x}$ is | (Q) 3 |
| (C) $\lim_{x \rightarrow 0} \frac{\tan x \sqrt{\tan x} - \sin x \sqrt{\sin x}}{x^3 \cdot \sqrt{x}}$ is | (R) $\frac{3}{2}$ |
| (D) If $f(x) = \cos(x \cos \frac{1}{x})$ and $g(x) = \frac{\ln(\sec^2 x)}{x \sin x}$ are both continuous at $x = 0$ then $f(0) + g(0)$ equals | (S) $\frac{3}{4}$ |

17. Match the function in Column-I with its behaviour at $x=0$ in column-II, where $[.]$ denotes greatest integer function & $\text{sgn}(x)$ denotes signum function.

Column-I

- (A) $f(x) = [x][1+x]$
 (B) $f(x) = [-x][1+x]$
 (C) $f(x) = (\text{sgn}(x))[2-x][1+|x|]$
 (D) $f(x) = [\cos x]$

Column-II

- (P) LHL exist at $x=0$
 (Q) RHL exist at $x=0$
 (R) Continuous at $x=0$
 (S) $\lim_{x \rightarrow 0} f(x)$ exists but function is discontinuous at $x=0$
 (T) $\lim_{x \rightarrow 0} f(x)$ does not exist

EXERCISE (S-1)

1. If the function $f(x) = \frac{3x^2 + ax + a + 3}{x^2 + x - 2}$, ($x \neq -2$) is continuous at $x = -2$. Find $f(-2)$.
 2. Find all possible values of a and b so that $f(x)$ is continuous for all $x \in \mathbb{R}$ if

$$f(x) = \begin{cases} |ax+3| & \text{if } x \leq -1 \\ |3x+a| & \text{if } -1 < x \leq 0 \\ \frac{b \sin 2x}{x} - 2b & \text{if } 0 < x < \pi \\ \cos^2 x - 3 & \text{if } x \geq \pi \end{cases}$$

3. The function $f(x) = \begin{cases} \left(\frac{6}{5}\right)^{\frac{\tan 6x}{\tan 5x}} & \text{if } 0 < x < \frac{\pi}{2} \\ b+2 & \text{if } x = \frac{\pi}{2} \\ (1+|\cos x|)^{\left(\frac{a|\tan x|}{b}\right)} & \text{if } \frac{\pi}{2} < x < \pi \end{cases}$

Determine the values of 'a' & 'b', if f is continuous at $x = \pi/2$.

4. Suppose that $f(x) = x^3 - 3x^2 - 4x + 12$ and $h(x) = \begin{cases} \frac{f(x)}{x-3} & , x \neq 3 \\ K & , x = 3 \end{cases}$ then
 (a) find all zeroes of $f(x)$.
 (b) find the value of K that makes h continuous at $x = 3$.
 (c) using the value of K found in (b), determine whether h is an even function.

5. Let $f(x) = \begin{cases} \frac{1 - \sin \pi x}{1 + \cos 2\pi x}, & x < \frac{1}{2} \\ p, & x = \frac{1}{2} \\ \frac{\sqrt{2x-1}}{\sqrt{4+\sqrt{2x-1}}-2}, & x > \frac{1}{2} \end{cases}$. Determine the value of p , if possible, so that the function is continuous at $x = 1/2$.

6. Given the function $g(x) = \sqrt{6-2x}$ and $h(x) = 2x^2 - 3x + a$. Then
 (a) evaluate $h(g(2))$ (b) If $f(x) = \begin{cases} g(x), & x \leq 1 \\ h(x), & x > 1 \end{cases}$, find 'a' so that f is continuous.
7. Let $f(x) = \begin{cases} 1+x, & 0 \leq x \leq 2 \\ 3-x, & 2 < x \leq 3 \end{cases}$. Determine the form of $g(x) = f[f(x)]$ & hence find the point of discontinuity of g, if any.
8. Let $f(x) = \begin{cases} \frac{\ln \cos x}{\sqrt[4]{1+x^2}-1} & \text{if } x > 0 \\ \frac{e^{\sin 4x} - 1}{\ln(1+\tan 2x)} & \text{if } x < 0 \end{cases}$

Is it possible to define $f(0)$ to make the function continuous at $x = 0$. If yes what is the value of $f(0)$.

9. Determine a & b so that f is continuous at $x = \frac{\pi}{2}$ where $f(x) = \begin{cases} \frac{1-\sin^3 x}{3\cos^2 x} & \text{if } x < \frac{\pi}{2} \\ a & \text{if } x = \frac{\pi}{2} \\ \frac{b(1-\sin x)}{(\pi-2x)^2} & \text{if } x > \frac{\pi}{2} \end{cases}$
10. Determine the values of a, b & c for which the function $f(x) = \begin{cases} \frac{\sin(a+1)x + \sin x}{x} & \text{for } x < 0 \\ c & \text{for } x = 0 \\ \frac{(x+bx^2)^{1/2} - x^{1/2}}{bx^{3/2}} & \text{for } x > 0 \end{cases}$ is continuous at $x = 0$.

EXERCISE (S-2)

1. If $f(x) = \frac{\sin 3x + A \sin 2x + B \sin x}{x^5}$ ($x \neq 0$) is cont. at $x = 0$. Find A & B. Also find $f(0)$.
2. Let $f(x) = \begin{cases} \frac{\left(\frac{\pi}{2} - \sin^{-1}(1-\{x\}^2)\right) \sin^{-1}(1-\{x\})}{\sqrt{2}(\{x\} - \{x\}^3)} & \text{for } x \neq 0 \\ \frac{\pi}{2} & \text{for } x = 0 \end{cases}$ where $\{x\}$ is the fractional part of x.

Consider another function $g(x)$; such that

$$g(x) = \begin{cases} f(x) & \text{for } x \geq 0 \\ 2\sqrt{2}f(x) & \text{for } x < 0 \end{cases}$$

Discuss the continuity of the functions $f(x)$ & $g(x)$ at $x = 0$.

3. If $f(x) = x + \{-x\} + [x]$, where $[x]$ is the integral part & $\{x\}$ is the fractional part of x. Discuss the continuity of f in $[-2, 2]$.

4. Find the locus of (a, b) for which the function $f(x) = \begin{cases} ax - b & \text{for } x \leq 1 \\ 3x & \text{for } 1 < x < 2 \\ bx^2 - a & \text{for } x \geq 2 \end{cases}$

is continuous at $x = 1$ but discontinuous at $x = 2$.

5. A function $f: \mathbb{R} \rightarrow \mathbb{R}$ is defined as $f(x) = \lim_{n \rightarrow \infty} \frac{ax^2 + bx + c + e^{nx}}{1 + c \cdot e^{nx}}$ where f is continuous on $\mathbb{R}$. Find the values of a, b and c .

6. Let $f(x) = \begin{cases} (\sin x + \cos x)^{\cos x} & ; \quad -\frac{\pi}{2} < x < 0 \\ a & ; \quad x = 0 \\ \frac{e^{1/x} + e^{2/x} + e^{3/|x|}}{ae^{2/x} + be^{3/|x|}} & ; \quad 0 < x < \frac{\pi}{2} \end{cases}$

If $f(x)$ is continuous at $x = 0$, find the value of $(a^2 + b^2)$.

7. Given $f(x) = \sum_{r=1}^n \tan\left(\frac{x}{2^r}\right) \sec\left(\frac{x}{2^{r-1}}\right); r, n \in \mathbb{N}$

$$g(x) = \lim_{n \rightarrow \infty} \frac{\ell n \left(f(x) + \tan \frac{x}{2^n} \right) - \left(f(x) + \tan \frac{x}{2^n} \right)^n \cdot \left[\sin \left(\tan \frac{x}{2} \right) \right]}{1 + \left(f(x) + \tan \frac{x}{2^n} \right)^n}$$

$$= k \text{ for } x = \frac{\pi}{4} \text{ and the domain of } g(x) \text{ is } (0, \pi/2).$$

where $[\]$ denotes the greatest integer function.

Find the value of k , if possible, so that $g(x)$ is continuous at $x = \pi/4$. Also state the points of discontinuity of $g(x)$ in $(0, \pi/4)$, if any.

8. Let $f(x) = x^3 - x^2 - 3x - 1$ and $h(x) = \frac{f(x)}{g(x)}$, where h is a rational function such that

(a) Domain of $h(x)$ is $\mathbb{R} - \{-1\}$

(b) $\lim_{x \rightarrow \infty} h(x) = \infty$ and (c) $\lim_{x \rightarrow -1} h(x) = \frac{1}{2}$.

Find $\lim_{x \rightarrow 0} (3h(x) + f(x) - 2g(x))$

9. (a) Let f be a real valued continuous function on $\mathbb{R}$ and satisfying $f(-x) - f(x) = 0 \quad \forall \quad x \in \mathbb{R}$. If $f(-5) = 5, f(-2) = 4, f(3) = -2$ and $f(0) = 0$ then find the minimum number of zero's of the equation $f(x) = 0$.

(b) Find the number of points of discontinuity of the function $f(x) = [5x] + \{3x\}$ in $[0, 5]$ where $[y]$ and $\{y\}$ denote largest integer less than or equal to y and fractional part of y respectively.

10. (a) If $g: [a, b] \rightarrow [a, b]$ is continuous & onto function, then show that there is some $c \in [a, b]$ such that $g(c) = c$.

(b) Let f be continuous on the interval $[0, 1]$ to $\mathbb{R}$ such that $f(0) = f(1)$. Prove that there exists a point c in $\left[0, \frac{1}{2}\right]$ such that $f(c) = f\left(c + \frac{1}{2}\right)$

EXERCISE (JM)

1. The function $f: \mathbb{R}/\{0\} \rightarrow \mathbb{R}$ given by $f(x) = \frac{1}{x} - \frac{2}{e^{2x}-1}$ can be made continuous at $x=0$ by defining $f(0)$ as- [AIEEE 2007]

(1) 2 (2) -1 (3) 0 (4) 1

2. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function defined by $f(x) = \frac{1}{e^x + 2e^{-x}}$

Statement-1 : $f(c) = \frac{1}{3}$, for some $c \in \mathbb{R}$.

Statement-2 : $0 < f(x) \leq \frac{1}{2\sqrt{2}}$, for all $x \in \mathbb{R}$.

[AIEEE-2010]

(1) Statement-1 is true, Statement-2 is true ; Statement-2 is a correct explanation for Statement-1.

(2) Statement-1 is true, Statement-2 is true ; Statement-2 is not a correct explanation for statement-1.

(3) Statement-1 is true, Statement-2 is false.

(4) Statement-1 is false, Statement-2 is true.

3. The values of p and q for which the function $f(x) = \begin{cases} \frac{\sin(p+1)x + \sin x}{x}, & x < 0 \\ q, & x = 0 \\ \frac{\sqrt{x+x^2} - \sqrt{x}}{x^2}, & x > 0 \end{cases}$ is continuous for all x in $\mathbb{R}$,

are :-

[AIEEE 2011]

(1) $p = -\frac{3}{2}$, $q = \frac{1}{2}$ (2) $p = \frac{1}{2}$, $q = \frac{3}{2}$ (3) $p = \frac{1}{2}$, $q = -\frac{3}{2}$ (4) $p = \frac{5}{2}$, $q = \frac{1}{2}$

4. Define $F(x)$ as the product of two real functions $f_1(x) = x$, $x \in \mathbb{R}$,

and $f_2(x) = \begin{cases} \sin \frac{1}{x}, & \text{if } x \neq 0 \\ 0, & \text{if } x = 0 \end{cases}$ as follows : $F(x) = \begin{cases} f_1(x) \cdot f_2(x) & \text{if } x \neq 0 \\ 0, & \text{if } x = 0 \end{cases}$

[AIEEE 2011]

Statement-1 : $F(x)$ is continuous on $\mathbb{R}$.

Statement-2 : $f_1(x)$ and $f_2(x)$ are continuous on $\mathbb{R}$.

(1) Statement-1 is false, statement-2 is true.

(2) Statement-1 is true, statement-2 is true; Statement-2 is correct explanation for statement 1.

(3) Statement-1 is true, statement-2 is true, statement-2 is not a correct explanation for statement 1

(4) Statement-1 is true, statement-2 is false

5. If $f(x)$ is continuous and $f(9/2) = 2/9$, then $\lim_{x \rightarrow 0} f\left(\frac{1 - \cos 3x}{x^2}\right)$ is equal to: [JEE Mains Offline-2014]

(1) 9/2 (2) 0 (3) 2/9 (4) 8/9

6. If the function $f(x) = \begin{cases} \frac{\sqrt{2+\cos x}-1}{(\pi-x)^2}, & x \neq \pi \\ k, & x = \pi \end{cases}$ is continuous at $x = \pi$, then k equals:-

- (1) $\frac{1}{4}$ (2) $\frac{1}{2}$ (3) 2 (4) 0

[JEE Mains Offline-2014]

EXERCISE (JA)

1. Discuss the continuity of the function $f(x) = \begin{cases} \frac{e^{1/(x-1)} - 2}{e^{1/(x-1)} + 2}, & x \neq 1 \\ 1, & x = 1 \end{cases}$

at $x = 1$.

[REE 2001 (Mains), 3 out of 100]

2. For every integer n , let a_n and b_n be real numbers. Let function $f : \mathbb{R} \rightarrow \mathbb{R}$ be given by

$$f(x) = \begin{cases} a_n + \sin \pi x, & \text{for } x \in [2n, 2n+1] \\ b_n + \cos \pi x, & \text{for } x \in (2n-1, 2n) \end{cases}, \text{ for all integers } n.$$

If f is continuous, then which of the following holds(s) for all n ?

[JEE 2012, 4]

- (A) $a_{n-1} - b_{n-1} = 0$ (B) $a_n - b_n = 1$ (C) $a_n - b_{n+1} = 1$ (D) $a_{n-1} - b_n = -1$

3. For every pair of continuous function $f, g : [0, 1] \rightarrow \mathbb{R}$ such that

$$\max\{f(x) : x \in [0, 1]\} = \max\{g(x) : x \in [0, 1]\},$$

the correct statement(s) is(are) :

- (A) $(f(c))^2 + 3f(c) = (g(c))^2 + 3g(c)$ for some $c \in [0, 1]$
 (B) $(f(c))^2 + f(c) = (g(c))^2 + 3g(c)$ for some $c \in [0, 1]$
 (C) $(f(c))^2 + 3f(c) = (g(c))^2 + g(c)$ for some $c \in [0, 1]$
 (D) $(f(c))^2 = (g(c))^2$ for some $c \in [0, 1]$

[JEE(Advanced)-2014, 3]

4. Let $[x]$ be the greatest integer less than or equal to x . Then, at which of the following point(s) the function $f(x) = x \cos(\pi(x + [x]))$ is discontinuous?

[JEE(Advanced)-2017, 4(-2)]

- (A) $x = -1$ (B) $x = 0$ (C) $x = 2$ (D) $x = 1$

DIFFERENTIABILITY

1. MEANING OF DERIVATIVE :

The instantaneous rate of change of a function with respect to the dependent variable is called derivative. Let 'f' be a given function of one variable and let Δx denote a number (positive or negative) to be added to the number x . Let Δf denote the corresponding change of 'f' then $\Delta f = f(x + \Delta x) - f(x)$

$$\Rightarrow \frac{\Delta f}{\Delta x} = \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

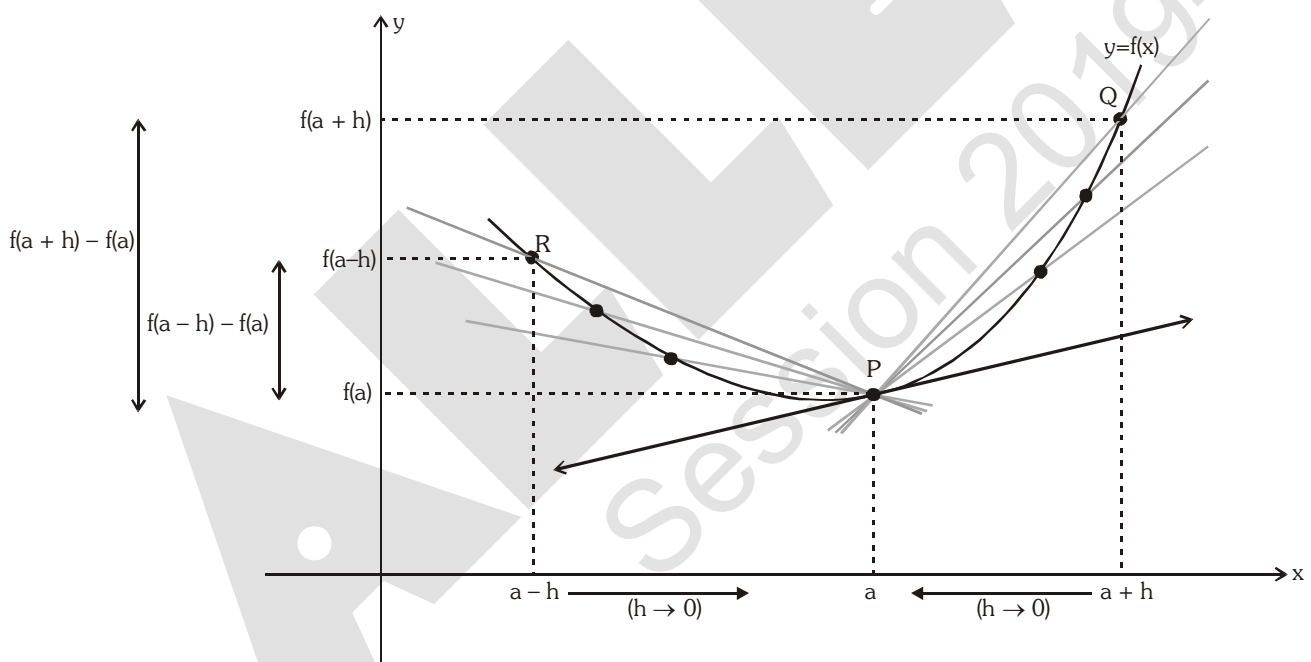
If $\Delta f/\Delta x$ approaches a limit as Δx approaches zero, this limit is the derivative of 'f' at the point x . The derivative of a function 'f' is a function ; this function is denoted by symbols such as

$$f'(x), \frac{df}{dx}, \frac{d}{dx}f(x) \text{ or } \frac{df(x)}{dx}$$

$$\Rightarrow \frac{df}{dx} = \lim_{\Delta x \rightarrow 0} \frac{\Delta f}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

The derivative evaluated at a point a , is written, $f'(a)$, $\left. \frac{df(x)}{dx} \right|_{x=a}$, $f'(x)_{x=a}$, etc.

2. EXISTENCE OF DERIVATIVE AT $x = a$:



(a) Right hand derivative :

The right hand derivative of $f(x)$ at $x = a$ denoted by $Rf'(a)$ is defined as :

$$Rf'(a) = \lim_{h \rightarrow 0^+} \frac{f(a+h) - f(a)}{h}, \text{ provided the limit exists \& is finite. } (h > 0)$$

(b) Left hand derivative :

The left hand derivative of $f(x)$ at $x = a$ denoted by $Lf'(a)$ is defined as :

$$Lf'(a) = \lim_{h \rightarrow 0^+} \frac{f(a-h) - f(a)}{-h}, \text{ provided the limit exists \& is finite. } (h > 0)$$

Hence $f(x)$ is said to be **derivable or differentiable at $x = a$** . If $Rf'(a) = Lf'(a) = \text{finite quantity}$

and it is denoted by $f'(a)$; where $f'(a) = Lf'(a) = Rf'(a)$ & it is called derivative or differential coefficient of $f(x)$ at $x = a$.

3. DIFFERENTIABILITY & CONTINUITY :

Theorem : If a function $f(x)$ is derivable at $x = a$, then $f(x)$ is continuous at $x = a$.

Proof: $f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$ exists.

$$\text{Also } f(a+h) - f(a) = \frac{f(a+h) - f(a)}{h} \cdot h \quad [h \neq 0]$$

$$\therefore \lim_{h \rightarrow 0} [f(a+h) - f(a)] = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} \cdot h = f'(a) \cdot 0 = 0$$

$$\Rightarrow \lim_{h \rightarrow 0} [f(a+h) - f(a)] = 0 \Rightarrow \lim_{h \rightarrow 0} f(a+h) = f(a) \Rightarrow f(x) \text{ is continuous at } x = a.$$

Note :

- (i) Differentiable $\Rightarrow$ Continuous ; Continuity $\nRightarrow$ Differentiable ; Not Differentiable $\nRightarrow$ Not Continuous
But Not Continuous $\Rightarrow$ Not Differentiable
- (ii) All polynomial, rational, trigonometric, logarithmic and exponential function are continuous and differentiable in their domains.
- (iii) If $f(x)$ & $g(x)$ are differentiable at $x = a$, then the functions $f(x) + g(x)$, $f(x) - g(x)$, $f(x) \cdot g(x)$ will also be differentiable at $x = a$ & if $g(a) \neq 0$ then the function $f(x)/g(x)$ will also be differentiable at $x = a$.

Illustration 1 : Let $f(x) = \begin{cases} \operatorname{sgn}(x) + x; & -\infty < x < 0 \\ -1 + \sin x; & 0 \leq x < \frac{\pi}{2} \\ \cos x; & \frac{\pi}{2} \leq x < \infty \end{cases}$. Discuss the continuity & differentiability at

$$x = 0 \text{ \& } \frac{\pi}{2}.$$

Solution :

$$f(x) = \begin{cases} -1 + x; & -\infty < x < 0 \\ -1 + \sin x; & 0 \leq x < \frac{\pi}{2} \\ \cos x; & \frac{\pi}{2} \leq x < \infty \end{cases}$$

To check the differentiability at $x = 0$

$$\text{LHD} = \lim_{h \rightarrow 0} \frac{f(0-h) - f(0)}{-h} = \lim_{h \rightarrow 0} \frac{-1 + 0 - h - (-1)}{-h} = 1$$

$$\text{RHD} = \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h} = \lim_{h \rightarrow 0} \frac{-1 + \sinh + 1}{h} = 1$$

$$\therefore \text{LHD} = \text{RHD}$$

$$\therefore \text{Differentiable at } x = 0.$$

$$\Rightarrow \text{Continuous at } x = 0.$$

To check the continuity at $x = \frac{\pi}{2}$

$$\text{LHL } \lim_{x \rightarrow \frac{\pi}{2}^-} f(x) = \lim_{x \rightarrow \frac{\pi}{2}^-} (-1 + \sin x) = 0$$

$$\text{RHL } \lim_{x \rightarrow \frac{\pi}{2}^+} f(x) = \lim_{x \rightarrow \frac{\pi}{2}^+} \cos x = 0$$

$$\therefore \text{LHL} = \text{RHL} = f\left(\frac{\pi}{2}\right) = 0$$

$$\therefore \text{Continuous at } x = \frac{\pi}{2}.$$

To check the differentiability at $x = \frac{\pi}{2}$

$$\text{LHD} = \lim_{h \rightarrow 0} \frac{f\left(\frac{\pi}{2} - h\right) - f\left(\frac{\pi}{2}\right)}{-h} = \lim_{h \rightarrow 0} \frac{-1 + \cosh - 0}{-h} = 0$$

$$\text{RHD} = \lim_{h \rightarrow 0} \frac{f\left(\frac{\pi}{2} + h\right) - f\left(\frac{\pi}{2}\right)}{h} = \lim_{h \rightarrow 0} \frac{-\sinh - 0}{h} = -1$$

$$\therefore \text{LHD} \neq \text{RHD}$$

$$\therefore \text{not differentiable at } x = \frac{\pi}{2}.$$

Illustration 2 : If $f(x) = \begin{cases} A + Bx^2 & x < 1 \\ 3Ax - B + 2 & x \geq 1 \end{cases}$

then find A and B so that $f(x)$ become differentiable at $x = 1$.

Solution :
$$\text{Rf}'(1) = \lim_{h \rightarrow 0} \frac{f(1+h) - f(1)}{h} = \lim_{h \rightarrow 0} \frac{3A(1+h) - B + 2 - 3A + B - 2}{h} = \lim_{h \rightarrow 0} \frac{3Ah}{h} = 3A$$

$$\text{Lf}'(1) = \lim_{h \rightarrow 0} \frac{f(1-h) - f(1)}{-h} = \lim_{h \rightarrow 0} \frac{A + B(1-h)^2 - 3A + B - 2}{-h} = \lim_{h \rightarrow 0} \frac{(-2A + 2B - 2) + Bh^2 - 2Bh}{-h}$$

hence for this limit to be defined

$$-2A + 2B - 2 = 0$$

$$B = A + 1$$

$$\text{Lf}'(1) = \lim_{h \rightarrow 0} -(Bh - 2B) = 2B$$

$$\therefore \text{Lf}'(1) = \text{Rf}'(1)$$

$$3A = 2B = 2(A + 1)$$

$$A = 2, B = 3$$

Illustration 3 : $f(x) = \begin{cases} [\cos \pi x] & x \leq 1 \\ 2\{x\} - 1 & x > 1 \end{cases}$ comment on the derivability at $x=1$, where $[]$ denotes greatest integer

function & $\{ \}$ denotes fractional part function.

Solution :

$$Lf'(1) = \lim_{h \rightarrow 0} \frac{f(1-h) - f(1)}{-h} = \lim_{h \rightarrow 0} \frac{[\cos(\pi - \pi h)] + 1}{-h} = \lim_{h \rightarrow 0} \frac{-1 + 1}{-h} = 0$$

$$Rf'(1) = \lim_{h \rightarrow 0} \frac{f(1+h) - f(1)}{h} = \lim_{h \rightarrow 0} \frac{2\{1+h\} - 1 + 1}{h} = \lim_{h \rightarrow 0} \frac{2h}{h} = 2$$

Hence $f(x)$ is not differentiable at $x = 1$.

Do yourself -1 :

(i) A function is defined as follows :

$$f(x) = \begin{cases} x^3 & ; \quad x^2 < 1 \\ x & ; \quad x^2 \geq 1 \end{cases} \text{ discuss the continuity and differentiability at } x = 1.$$

(ii) If $f(x) = \begin{cases} ax^3 + b, & \text{for } 0 \leq x \leq 1 \\ 2 \cos \pi x + \tan^{-1} x, & \text{for } 1 < x \leq 2 \end{cases}$ be the differentiable function in $[0, 2]$, then find a and b . (where $[.]$ denotes the greatest integer function)

4. IMPORTANT NOTE :

(a) Let $Rf'(a) = p$ & $Lf'(a) = q$ where p & q are finite then :

- (i) $p = q \Rightarrow f$ is differentiable at $x = a \Rightarrow f$ is continuous at $x = a$
- (ii) $p \neq q \Rightarrow f$ is not differentiable at $x = a$, but f is continuous at $x = a$.

Illustration 4 : Determine the values of x for which the following functions fails to be continuous or

$$\text{differentiable } f(x) = \begin{cases} (1-x), & x < 1 \\ (1-x)(2-x), & 1 \leq x \leq 2 \\ (3-x), & x > 2 \end{cases} \text{ , Justify your answer.}$$

Solution : By the given definition it is clear that the function f is continuous and differentiable at all points except possibly at $x = 1$ and $x = 2$.

Check the differentiability at $x = 1$

$$q = \text{LHD} = \lim_{h \rightarrow 0} \frac{f(1-h) - f(1)}{-h} = \lim_{h \rightarrow 0} \frac{1 - (1-h) - 0}{-h} = -1$$

$$p = \text{RHD} = \lim_{h \rightarrow 0} \frac{f(1+h) - f(1)}{h} = \lim_{h \rightarrow 0} \frac{\{1 - (1+h)\} \{2 - (1+h)\} - 0}{h} = -1$$

$\therefore q = p \therefore$ Differentiable at $x = 1. \Rightarrow$ Continuous at $x = 1$.

Check the differentiability at $x = 2$

$$q = \text{LHD} = \lim_{h \rightarrow 0} \frac{f(2-h) - f(2)}{-h} = \lim_{h \rightarrow 0} \frac{(1-2+h)(2-2+h) - 0}{-h} = 1 = \text{finite}$$

$$p = \text{RHD} = \lim_{h \rightarrow 0} \frac{f(2+h) - f(2)}{h} = \lim_{h \rightarrow 0} \frac{(3-2-h) - 0}{h} \rightarrow \infty \quad (\text{not finite})$$

$\therefore q \neq p \quad \therefore$ not differentiable at $x = 2$.

Now we have to check the continuity at $x = 2$

$$\text{LHL} = \lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^-} (1-x)(2-x) = \lim_{h \rightarrow 0} (1-(2-h))(2-(2-h)) = 0$$

$$\text{RHL} = \lim_{x \rightarrow 2^+} f(x) = \lim_{x \rightarrow 2^+} (3-x) = \lim_{h \rightarrow 0} (3-(2+h)) = 1$$

$\therefore \text{LHL} \neq \text{RHL}$

$\Rightarrow$ not continuous at $x = 2$.

Do yourself - 2 :

(i) Let $f(x) = (x-1)|x-1|$. Discuss the continuity and differentiability of $f(x)$ at $x = 1$.

(b) Vertical tangent :

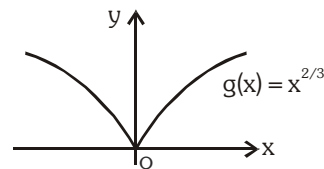
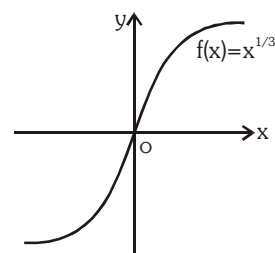
- (i) If $y = f(x)$ is continuous at $x = a$ and $\lim_{x \rightarrow a} |f'(x)|$ approaches to ∞ , then $y = f(x)$ has a vertical tangent at $x = a$. If a function has vertical tangent at $x = a$ then it is non differentiable at $x = a$.

e.g. (1) $f(x) = x^{1/3}$ has vertical tangent at $x = 0$

since $f'_+(0) \rightarrow \infty$ and $f'_-(0) \rightarrow \infty$ hence $f(x)$ is not differentiable at $x = 0$

(2) $g(x) = x^{2/3}$ have vertical tangent at $x = 0$

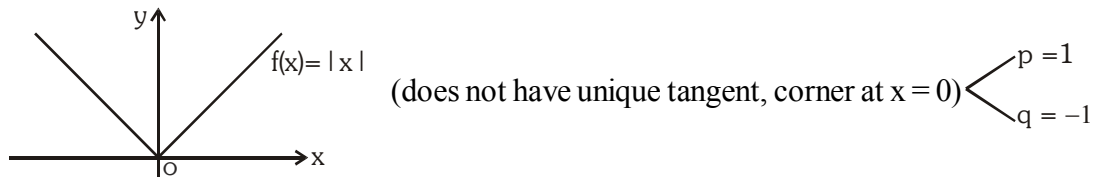
since $g'_+(0) \rightarrow \infty$ and $g'_-(0) \rightarrow -\infty$ hence $g(x)$ is not differentiable at $x = 0$.



(c) Geometrical interpretation of differentiability :

- (i) If the function $y = f(x)$ is differentiable at $x = a$, then a unique non vertical tangent can be drawn to the curve $y = f(x)$ at the point $P(a, f(a))$ & $f'(a)$ represent the slope of the tangent at point P .
- (ii) If a function $f(x)$ does not have a unique tangent (p & q are finite but unequal), then f is continuous at $x = a$, it geometrically implies a corner at $x = a$.

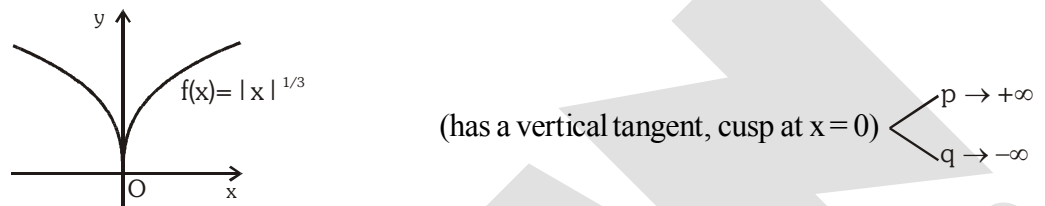
e.g. $f(x) = |x|$ is continuous but not differentiable at $x = 0$ & there is corner at $x = 0$.



(iii) If one of p & q tends to ∞ and other tends to $-\infty$, then there will be a cusp at $x = a$. Where

$$p = Rf'(a) \text{ and } q = Lf'(a)$$

e.g. (1) $f(x) = |x|^{1/3}$ is continuous but not differentiable at $x = 0$ & there is cusp at $x = 0$.



(2) $f(x) = x^{1/3}$ is continuous but not differentiable at $x = 0$ because $Rf'(0) \rightarrow \infty$ and $Lf'(0) \rightarrow \infty$.



Note : corner/cusp/vertical tangent $\Rightarrow$ non differentiable

non differentiable $\Rightarrow$ corner/cusp/vertical tangent

Illustration 5 : If $f(x) = \begin{cases} x-3 & x < 0 \\ x^2-3x+2 & x \geq 0 \end{cases}$. Draw the graph of the function & discuss the continuity and differentiability of $f(|x|)$ and $|f(x)|$.

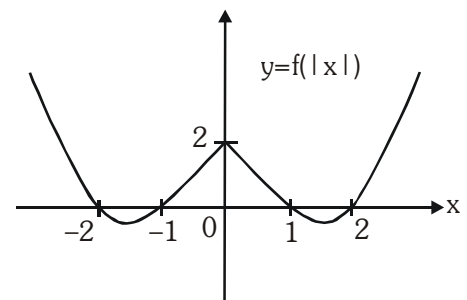
Solution :

$$f(|x|) = \begin{cases} |x|-3; & |x| < 0 \rightarrow \text{not possible} \\ |x|^2-3|x|+2; & |x| \geq 0 \end{cases}$$

$$f(|x|) = \begin{cases} x^2+3x+2, & x < 0 \\ x^2-3x+2, & x \geq 0 \end{cases}$$

at $x = 0$

$$q = \text{LHD} = \lim_{h \rightarrow 0} \frac{f(0-h) - f(0)}{-h} = \lim_{h \rightarrow 0} \frac{h^2 - 3h + 2 - 2}{-h} = 3$$



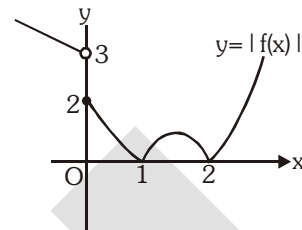
$$p = \text{RHD} = \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h} = \lim_{h \rightarrow 0} \frac{h^2 - 3h + 2 - 2}{h} = -3$$

$$\therefore q \neq p$$

$\therefore$ not differentiable at $x = 0$. but p & q are both are finite

$\Rightarrow$ continuous at $x = 0$

$$\text{Now, } |f(x)| = \begin{cases} 3-x, & x < 0 \\ (x^2 - 3x + 2), & 0 \leq x < 1 \\ -(x^2 - 3x + 2), & 1 \leq x \leq 2 \\ (x^2 - 3x + 2), & x > 2 \end{cases}$$



To check differentiability at $x = 0$

$$q = \text{LHD} = \lim_{h \rightarrow 0} \frac{f(0-h) - f(0)}{-h} = \lim_{h \rightarrow 0} \frac{3+h-2}{-h} = \lim_{h \rightarrow 0} \frac{(1+h)}{-h} \rightarrow -\infty$$

$$p = \text{RHD} = \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h} = \lim_{h \rightarrow 0} \frac{h^2 - 3h + 2 - 2}{h} = -3$$

$\Rightarrow$ not differentiable at $x = 0$.

Now to check continuity at $x = 0$

$$\text{LHL} = \lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} 3 - x = 3$$

$$\text{RHL} = \lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} x^2 - 3x + 2 = 2$$

$\Rightarrow$ not continuous at $x = 0$.

To check differentiability at $x = 1$

$$q = \text{LHD} = \lim_{h \rightarrow 0} \frac{f(1-h) - f(1)}{-h} = \lim_{h \rightarrow 0} \frac{(1-h)^2 - 3(1-h) + 2 - 0}{-h} = \lim_{h \rightarrow 0} \frac{h^2 + h}{-h} = -1$$

$$p = \text{RHD} = \lim_{h \rightarrow 0} \frac{f(1+h) - f(1)}{h} = \lim_{h \rightarrow 0} \frac{-(h^2 + 2h + 1 - 3 + 3h + 2) - 0}{h} = \lim_{h \rightarrow 0} \frac{-(h^2 - h)}{h} = 1$$

$\Rightarrow$ not differentiable at $x = 1$.

but $|f(x)|$ is continuous at $x = 1$, because $p \neq q$ and both are finite.

To check differentiability at $x = 2$

$$q = \text{LHD} = \lim_{h \rightarrow 0} \frac{f(2-h) - f(2)}{-h} = \lim_{h \rightarrow 0} \frac{-(4 + h^2 - 4h - 6 + 3h + 2) - 0}{-h} = \lim_{h \rightarrow 0} \frac{h^2 - h}{h} = -1$$

$$p = \text{RHD} = \lim_{h \rightarrow 0} \frac{f(2+h) - f(2)}{h} = \lim_{h \rightarrow 0} \frac{(h^2 + 4h + 4 - 6 - 3h + 2) - 0}{h} = \lim_{h \rightarrow 0} \frac{(h^2 + h)}{h} = 1$$

$\Rightarrow$ not differentiable at $x = 2$.

but $|f(x)|$ is continuous at $x = 2$, because $p \neq q$ and both are finite.

Do yourself - 3 :

(i) Let $f(x) = \begin{cases} -4 & ; -4 < x < 0 \\ x^2 - 4 & ; 0 \leq x < 4 \end{cases}$

Discuss the continuity and differentiability of $g(x) = |f(x)|$.

(ii) Let $f(x) = \min \{|x - 1|, |x + 1|, 1\}$. Find the number of points where it is not differentiable.

5. DIFFERENTIABILITY OVER AN INTERVAL :

- (a) $f(x)$ is said to be differentiable over an open interval (a, b) if it is differentiable at each & every point of the open interval (a, b) .
- (b) $f(x)$ is said to be differentiable over the closed interval $[a, b]$ if :
- (i) $f(x)$ is differentiable in (a, b) &
- (ii) for the points a and b , $f(a^+)$ & $f(b^-)$ exist.

Illustration 6 : If $f(x) = \begin{cases} e^{-|x|}, & -5 < x < 0 \\ -e^{-|x-1|} + e^{-1} + 1, & 0 \leq x < 2 \\ e^{-|x-2|}, & 2 \leq x < 4 \end{cases}$

Discuss the continuity and differentiability of $f(x)$ in the interval $(-5, 4)$.

Solution :

$$f(x) = \begin{cases} e^{+x} & -5 < x < 0 \\ -e^{x-1} + e^{-1} + 1 & 0 \leq x \leq 1 \\ -e^{-x+1} + e^{-1} + 1 & 1 < x < 2 \\ e^{-x+2} & 2 \leq x < 4 \end{cases}$$

Check the differentiability at $x = 0$

$$\text{LHD} = \lim_{h \rightarrow 0} \frac{f(0-h) - f(0)}{-h} = \lim_{h \rightarrow 0} \frac{e^{-h} - 1}{-h} = 1$$

$$\text{RHD} = \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h} = \lim_{h \rightarrow 0} \frac{-e^{h-1} + e^{-1} + 1 - 1}{h} = -e^{-1}$$

$$\therefore \text{LHD} \neq \text{RHD}$$

$\therefore$ Not differentiable at $x = 0$, but continuous at $x = 0$ since LHD and RHD both are finite.

Check the differentiability at $x = 1$

$$\text{LHD} = \lim_{h \rightarrow 0} \frac{f(1-h) - f(1)}{-h} = \lim_{h \rightarrow 0} \frac{-e^{1-h-1} + e^{-1} + 1 - e^{-1}}{-h} = -1$$

$$\text{RHD} = \lim_{h \rightarrow 0} \frac{f(1+h) - f(1)}{h} = \lim_{h \rightarrow 0} \frac{-e^{1-h-1} + e^{-1} + 1 - e^{-1}}{h} = 1$$

$$\therefore \text{LHD} \neq \text{RHD}$$

$\therefore$ Not differentiable at $x = 1$, but continuous at $x = 1$ since LHD and RHD both are finite.

6. DETERMINATION OF FUNCTION WHICH SATISFYING THE GIVEN FUNCTIONAL EQUATION :

Illustration 7: Let $f(x+y) = f(x) + f(y) - 2xy - 1$ for all x and y . If $f(0)$ exists and $f'(0) = -\sin\alpha$, then find $f\{f(0)\}$.

Solution :

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\{f(x) + f(h) - 2xh - 1\} - f(x)}{h} \quad (\text{Using the given relation})$$

$$= \lim_{h \rightarrow 0} -2x + \lim_{h \rightarrow 0} \frac{f(h) - 1}{h} = -2x + \lim_{h \rightarrow 0} \frac{f(h) - f(0)}{h}$$

[Putting $x = 0 = y$ in the given relation we find $f(0) = f(0) + f(0) + 0 - 1 \Rightarrow f(0) = 1$]

$$\therefore f(x) = -2x + f(0) = -2x - \sin\alpha$$

$$\Rightarrow f(x) = -x^2 - (\sin\alpha) \cdot x + c$$

$$f(0) = -0 - 0 + c \Rightarrow c = 1$$

$$\therefore f(x) = -x^2 - (\sin\alpha) \cdot x + 1$$

So, $f\{f(0)\} = f(-\sin\alpha) = -\sin^2\alpha + \sin^2\alpha + 1$

$$\therefore f\{f(0)\} = 1.$$

Do yourself - 5 :

- (i) A function $f: \mathbb{R} \rightarrow \mathbb{R}$ satisfies the equation $f(x+y) = f(x) \cdot f(y)$ for all $x, y \in \mathbb{R}$, $f(x) \neq 0$. Suppose that the function is differentiable everywhere and $f'(0) = 2$. Prove that $f(x) = 2f(x)$.

Miscellaneous Illustrations :

Illustration 8 : Discuss the continuity and differentiability of the function $y = f(x)$ defined parametrically; $x = 2t - |t - 1|$ and $y = 2t^2 + t|t|$.

Solution : Here $x = 2t - |t - 1|$ and $y = 2t^2 + t|t|$.

Now when $t < 0$;

$$x = 2t - \{-(t-1)\} = 3t - 1 \quad \text{and} \quad y = 2t^2 - t^2 = t^2 \Rightarrow y = \frac{1}{9}(x+1)^2$$

when $0 \leq t < 1$

$$x = 2t - (-(t-1)) = 3t - 1 \quad \text{and} \quad y = 2t^2 + t^2 = 3t^2 \Rightarrow y = \frac{1}{3}(x+1)^2$$

when $t \geq 1$;

$$x = 2t - (t-1) = t+1 \quad \text{and} \quad y = 2t^2 + t^2 = 3t^2 \Rightarrow y = 3(x-1)^2$$

$$\text{Thus, } y = f(x) = \begin{cases} \frac{1}{9}(x+1)^2, & x < -1 \\ \frac{1}{3}(x+1)^2, & -1 \leq x < 2 \\ 3(x-1)^2, & x \geq 2 \end{cases}$$

We have to check differentiability at $x = -1$ and 2 .

Differentiability at $x = -1$;

$$\text{LHD} = f'_-(-1) = \lim_{h \rightarrow 0} \frac{f(-1-h) - f(-1)}{-h} = \lim_{h \rightarrow 0} \frac{\frac{1}{9}(-1-h+1)^2 - 0}{-h} = 0$$

$$\text{RHD} = f'_+(-1) = \lim_{h \rightarrow 0} \frac{f(-1+h) - f(-1)}{h} = \lim_{h \rightarrow 0} \frac{\frac{1}{9}(-1+h+1)^2 - 0}{h} = 0$$

Hence $f(x)$ is differentiable at $x = -1$.

$\Rightarrow$ continuous at $x = -1$.

To check differentiability at $x = 2$;

$$\text{LHD} = f'_-(2) = \lim_{h \rightarrow 0} \frac{\frac{1}{3}(2-h+1)^2 - 3}{-h} = 2 \quad \& \quad \text{RHD} = f'_+(2) = \lim_{h \rightarrow 0} \frac{3(2+h-1)^2 - 3}{h} = 6$$

Hence $f(x)$ is not differentiable at $x = 2$.

But continuous at $x = 2$, because LHD & RHD both are finite.

$\therefore f(x)$ is continuous for all x and differentiable for all x , except $x = 2$.

ANSWERS FOR DO YOURSELF

1: (i) Continuous but not differentiable at $x = 1$

$$(ii) \quad a = \frac{1}{6}, b = \frac{\pi}{4} - \frac{13}{6}$$

2: (i) Continuous & differentiable at $x = 1$

3: (i) Continuous everywhere but not differentiable at $x = 2$ only

(ii) 5

4: (i) 3 (ii) (a) 1 & 2 (b) Not continuous at $x = 1$ & 2 and not differentiable at $x = 1$ & 2.

EXERCISE (O-1)

[SINGLE CORRECT CHOICE TYPE]

- Let $f(x) = [\tan^2 x]$, (where $[.]$ denotes greatest integer function). Then -
 (A) $\lim_{x \rightarrow 0} f(x)$ does not exist (B) $f(x)$ is continuous at $x = 0$.
 (C) $f(x)$ is not differentiable at $x = 0$ (D) $f'(0) = 1$
- The number of points where $f(x) = [\sin x + \cos x]$ (where $[.]$ denotes the greatest integer function), $x \in (0, 2\pi)$ is not continuous is -
 (A) 3 (B) 4 (C) 5 (D) 6
- If 6, 8 and 12 are l^{th} , m^{th} and n^{th} terms of an A.P. and $f(x) = nx^2 + 2/x - 2m$, then the equation $f(x) = 0$ has -
 (A) a root between 0 and 1 (B) both roots imaginary.
 (C) both roots negative. (D) both roots greater than 1.
- Let f be differentiable at $x = 0$ and $f'(0) = 1$. Then $\lim_{h \rightarrow 0} \frac{f(h) - f(-2h)}{h} =$
 (A) 3 (B) 2 (C) 1 (D) -1
- Let $g(x) = \begin{cases} 3x^2 - 4\sqrt{x} + 1 & \text{for } x < 1 \\ ax + b & \text{for } x \geq 1 \end{cases}$.
 If $g(x)$ is continuous and differentiable for all numbers in its domain then -
 (A) $a = b = 4$ (B) $a = b = -4$
 (C) $a = 4$ and $b = -4$ (D) $a = -4$ and $b = 4$
- If $f(x)f(y) + 2 = f(x) + f(y) + f(xy)$ and $f(1) = 2$, $f'(1) = 2$ then $\text{sgn } f(x)$ is equal to (where sgn denotes signum function) -
 (A) 0 (B) 1 (C) -1 (D) 4
- The function $g(x) = \begin{cases} x + b, & x < 0 \\ \cos x, & x \geq 0 \end{cases}$ can be made differentiable at $x = 0$ -
 (A) if b is equal to zero (B) if b is not equal to zero
 (C) if b takes any real value (D) for no value of b
- Which one of the following functions is continuous everywhere in its domain but has atleast one point where it is not differentiable ?
 (A) $f(x) = x^{1/3}$ (B) $f(x) = \frac{|x|}{x}$ (C) $f(x) = e^{-x}$ (D) $f(x) = \tan x$
- If the right hand derivative of $f(x) = [x] \tan \pi x$ at $x = 7$ is $k\pi$, then k is equal to ($[y]$ denotes greatest integer $\leq y$)
 (A) 6 (B) 7 (C) -7 (D) 49

10. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous onto function satisfying $f(x) + f(-x) = 0, \forall x \in \mathbb{R}$. If $f(-3) = 2$ and $f(5) = 4$ in $[-5, 5]$, then the equation $f(x) = 0$ has -
- (A) exactly three real roots (B) exactly two real roots
(C) atleast five real roots (D) atleast three real roots

11. Let $f(x) = \begin{cases} \lim_{n \rightarrow \infty} \frac{ax(x-1) \left(\cot \frac{\pi x}{4} \right)^n + (px^2 + 2)}{\left(\cot \frac{\pi x}{4} \right)^n + 1}, & x \in (0, 1) \cup (1, 2) \\ 0, & x = 1 \end{cases}$

If $f(x)$ is differentiable for all $x \in (0, 2)$ then $(a^2 + p^2)$ equals -

- (A) 18 (B) 20 (C) 22 (D) 24
12. If $2x + 3|y| = 4y$, then y as a function of x i.e. $y = f(x)$, is -
- (A) discontinuous at one point
(B) non differentiable at one point
(C) discontinuous & non differentiable at same point
(D) continuous & differentiable everywhere
13. If $f(x) = (x^5 + 1) |x^2 - 4x - 5| + \sin|x| + \cos(|x - 1|)$, then $f(x)$ is not differentiable at -
- (A) 2 points (B) 3 points (C) 4 points (D) zero points
14. Let $f(x) = \begin{cases} x^3 + 2x^2 & x \in \mathbb{Q} \\ -x^3 + 2x^2 + ax & x \notin \mathbb{Q} \end{cases}$, then the integral value of 'a' so that $f(x)$ is differentiable at $x = 1$, is
- (A) 2 (B) 6 (C) 7 (D) not possible
15. Let $\mathbb{R}$ be the set of real numbers and $f : \mathbb{R} \rightarrow \mathbb{R}$, be a differentiable function such that $|f(x) - f(y)| \leq |x - y|^3 \forall x, y \in \mathbb{R}$. If $f(10) = 100$, then the value of $f(20)$ is equal to -
- (A) 0 (B) 10 (C) 20 (D) 100
16. For what triplets of real numbers (a, b, c) with $a \neq 0$ the function

$$f(x) = \begin{cases} x & x \leq 1 \\ ax^2 + bx + c & \text{otherwise} \end{cases} \text{ is differentiable for all real } x?$$

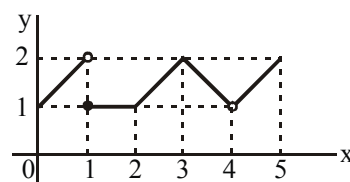
- (A) $\{(a, 1 - 2a, a) \mid a \in \mathbb{R}, a \neq 0\}$ (B) $\{(a, 1 - 2a, c) \mid a, c \in \mathbb{R}, a \neq 0\}$
(C) $\{(a, b, c) \mid a, b, c \in \mathbb{R}, a + b + c = 1\}$ (D) $\{(a, 1 - 2a, 0) \mid a \in \mathbb{R}, a \neq 0\}$
17. Number of points of non-differentiability of the function $g(x) = [x^2] \{\cos^2 4x\} + \{x^2\} [\cos^2 4x] + x^2 \sin^2 4x + [x^2] [\cos^2 4x] + \{x^2\} \{\cos^2 4x\}$ in $(-50, 50)$ where $[x]$ and $\{x\}$ denotes the greatest integer function and fractional part function of x respectively, is equal to :-
- (A) 98 (B) 99 (C) 100 (D) 0

18. Let $f(x) = [n + p \sin x]$, $x \in (0, \pi)$, $n \in \mathbb{Z}$ and p is a prime number. The number of points where $f(x)$ is not differentiable is :-
 (A) $p - 1$ (B) $p + 1$ (C) $2p + 1$ (D) $2p - 1$
 Here $[x]$ denotes greatest integer function.
19. The function $f(x) = (x^2 - 1) |x^2 - 3x + 2| + \cos(|x|)$ is NOT differentiable at :
 (A) -1 (B) 0 (C) 1 (D) 2

EXERCISE (O-2)

[MULTIPLE CORRECT CHOICE TYPE]

1. If $f(x) = x(\sqrt{x} - \sqrt{x+1})$, then -
 (A) $Rf'(0)$ exist (B) $Lf'(0)$ exist but $Rf'(0)$ does not exist
 (C) $\lim_{x \rightarrow 0^+} f(x)$ exist (D) $f(x)$ is differentiable at $x = 0$.
2. The function $f(x) = \begin{cases} |x-3|, & x \geq 1 \\ \left(\frac{x^2}{4}\right) - \left(\frac{3x}{2}\right) + \left(\frac{13}{4}\right), & x < 1 \end{cases}$ is -
 (A) continuous at $x = 1$ (B) differentiable at $x = 1$
 (C) continuous at $x = 3$ (D) differentiable at $x = 3$
3. Select the correct statements -
 (A) The function f defined by $f(x) = \begin{cases} 2x^2 + 3 & \text{for } x \leq 1 \\ 3x + 2 & \text{for } x > 1 \end{cases}$ is neither differentiable nor continuous at $x = 1$.
 (B) The function $f(x) = x^2|x|$ is twice differentiable at $x = 0$.
 (C) If f is continuous at $x = 5$ and $f(5) = 2$ then $\lim_{x \rightarrow 2} f(4x^2 - 11)$ exists
 (D) If $\lim_{x \rightarrow a} (f(x) + g(x)) = 2$ and $\lim_{x \rightarrow a} (f(x) - g(x)) = 1$ then $\lim_{x \rightarrow a} f(x) \cdot g(x)$ need not exist.
4. If $f(x) = \text{sgn}(x^5)$, then which of the following is/are **false** (where sgn denotes signum function) -
 (A) $f'_+(0) = 1$
 (B) $f'_-(0) = -1$
 (C) f is continuous but not differentiable at $x = 0$
 (D) f is discontinuous at $x = 0$
5. Graph of $f(x)$ is shown in adjacent figure, then in $[0, 5]$
 (A) $f(x)$ has non removable discontinuity at two points
 (B) $f(x)$ is non differentiable at three points in its domain
 (C) $\lim_{x \rightarrow 1} f(x) = 1$
 (D) Number of points of discontinuity = number of points of non-differentiability



6. Let S denotes the set of all points where $\sqrt[5]{x^2|x|^3} - \sqrt{x^2|x|} - 1$ is not differentiable then S is a subset of -
- (A) $\{0,1\}$ (B) $\{0,1,-1\}$ (C) $\{0,1\}$ (D) $\{0\}$
7. Which of the following statements is/are correct ?
- (A) There exist a function $f : [0,1] \rightarrow \mathbb{R}$ which is discontinuous at every point in $[0,1]$ & $|f(x)|$ is continuous at every point in $[0,1]$
- (B) Let $F(x) = f(x) \cdot g(x)$. If $f(x)$ is differentiable at $x = a$, $f(a) = 0$ and $g(x)$ is continuous at $x = a$ then $F(x)$ is always differentiable at $x = a$.
- (C) If $Rf'(a) = 2$ & $Lf'(a) = 3$, then $f(x)$ is non-differentiable at $x = a$ but will be always continuous at $x = a$
- (D) If $f(a)$ and $f(b)$ possess opposite signs then there must exist at least one solution of the equation $f(x) = 0$ in (a,b) provided f is continuous on $[a,b]$
8. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function. Define $g : \mathbb{R} \rightarrow \mathbb{R}$ by $g(x) = |f(x)|$ for all x . Then which of the following is/are not always true-
- (A) If f is continuous then g is also continuous (B) If f is one-one then g is also one-one
- (C) If f is onto then g is also onto (D) If f is differentiable then g is also differentiable
9. The function $\phi(x) = [x] - \sin|x|$ (where $[.]$ denotes greater integer function) is -
- (A) derivable at $x = 0$ (B) continuous at $x = 0$
- (C) $\lim_{x \rightarrow 0} \phi(x)$ does not exists (D) continuous and derivable at $x = 0$
10. Let $f(x) = \begin{cases} x^2 \cos \frac{1}{x}, & x < 0 \\ 0, & x = 0 \\ x^2 \sin \frac{1}{x}, & x > 0 \end{cases}$, then which of the following is (are) correct ?
- (A) $f(x)$ is continuous but not differentiable at $x = 0$
- (B) $f(x)$ is continuous and differentiable at $x = 0$
- (C) $f'(x)$ is continuous but not differentiable at $x = 0$
- (D) $f'(x)$ is discontinuous at $x = 0$

[MATCH THE COLUMN]

10. Column - I

Column - II

(A) If $f(x)$ is derivable at $x = 3$ & $f'(3) = 2$,

(P) 0

then $\lim_{h \rightarrow 0} \frac{f(3+h^2) - f(3-h^2)}{2h^2}$ equals

(B) Let $f(x)$ be a function satisfying the condition $f(-x) = f(x)$ for all real x . If $f'(0)$ exists, then its value is equal to

(Q) 1

(C) For the function $f(x) = \begin{cases} \frac{x}{1+e^{1/x}}, & x \neq 0 \\ 0, & x = 0 \end{cases}$

(R) 2

the derivative from the left $Lf'(0)$ equals

(D) The number of points at which the function $f(x) = \max. \{a-x, a+x, b\}$, $-\infty < x < \infty$, $0 < a < b$ cannot be differentiable is

(S) 3

EXERCISE (S-1)

1. Discuss the continuity & differentiability of the function $f(x) = \sin x + \sin |x|$, $x \in \mathbb{R}$. Draw a rough sketch of the graph of $f(x)$.

2. Examine the continuity and differentiability of $f(x) = |x| + |x-1| + |x-2|$, $x \in \mathbb{R}$. Also draw the graph of $f(x)$.

3. If the function $f(x)$ defined as $f(x) = \begin{cases} -\frac{x^2}{2} & \text{for } x \leq 0 \\ x^n \sin \frac{1}{x} & \text{for } x > 0 \end{cases}$ is continuous but not derivable at $x = 0$ then find the range of n .

4. A function f is defined as follows : $f(x) = \begin{cases} 1 & \text{for } -\infty < x < 0 \\ 1 + |\sin x| & \text{for } 0 \leq x < \frac{\pi}{2} \\ 2 + \left(x - \frac{\pi}{2}\right)^2 & \text{for } \frac{\pi}{2} \leq x < +\infty \end{cases}$

Discuss the continuity & differentiability at $x = 0$ & $x = \pi/2$.

5. Examine the origin for continuity & derivability in the case of the function f defined by $f(x) = x \tan^{-1} (1/x)$, $x \neq 0$ and $f(0) = 0$.

6. Let $f(0) = 0$ and $f'(0) = 1$. For a positive integer k , show that

$$\lim_{x \rightarrow 0} \frac{1}{x} \left(f(x) + f\left(\frac{x}{2}\right) + \dots + f\left(\frac{x}{k}\right) \right) = 1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{k}$$

7. Let $f(x) = xe^{-\left(\frac{1}{|x|} + \frac{1}{x}\right)}$; $x \neq 0$, $f(0) = 0$, test the continuity & differentiability at $x = 0$

8. If $f(x) = |x-1| \cdot ([x] - [-x])$, then find $Rf'(1)$ & $Lf'(1)$ where $[x]$ denotes greatest integer function.

9. If $f(x) = \begin{cases} ax^2 - b & \text{if } |x| < 1 \\ -\frac{1}{|x|} & \text{if } |x| \geq 1 \end{cases}$ is derivable at $x = 1$. Find the values of a & b .
10. Let $g(x) = \begin{cases} a\sqrt{x+2}, & 0 < x < 2 \\ bx+2, & 2 \leq x < 5 \end{cases}$. If $g(x)$ is derivable on $(0, 5)$, then find $(2a + b)$.

EXERCISE (S-2)

1. Let $f(x)$ be defined in the interval $[-2, 2]$ such that $f(x) = \begin{cases} -1, & -2 \leq x \leq 0 \\ x-1, & 0 < x \leq 2 \end{cases}$ & $g(x) = f(|x|) + |f(x)|$. Test the differentiability of $g(x)$ in $(-2, 2)$.
2. Discuss the continuity & the derivability in $[0, 2]$ of $f(x) = \begin{cases} 2x-3[x] & \text{for } x \geq 1 \\ \sin \frac{\pi x}{2} & \text{for } x < 1 \end{cases}$ where $[.]$ denotes the greatest integer function
3. Examine the function, $f(x) = x \cdot \frac{a^{1/x} - a^{-1/x}}{a^{1/x} + a^{-1/x}}$, $x \neq 0$ ($a > 0$) and $f(0) = 0$ for continuity and existence of the derivative at the origin.
4. For any real number x , let $[x]$ denote the largest integer less than or equal to x . Let f be a real valued function defined on the interval $[-3, 3]$ by $f(x) = \begin{cases} -x - [-x] & \text{if } [x] \text{ is even} \\ x - [x] & \text{if } [x] \text{ is odd} \end{cases}$. If L denotes the number of point of discontinuity and M denotes the number of points of non-derivability of $f(x)$, then find $(L + M)$.
5. $f(x) = \begin{cases} 1-x & , (0 \leq x \leq 1) \\ x+2 & , (1 < x < 2) \\ 4-x & , (2 \leq x \leq 4) \end{cases}$. Discuss the continuity & differentiability of $y = f[f(x)]$ for $0 \leq x \leq 4$.
6. A derivable function $f: \mathbb{R}^+ \rightarrow \mathbb{R}$ satisfies the condition $f(x) - f(y) \geq \ell \ln(x/y) + x - y$ for every $x, y \in \mathbb{R}^+$. If g denotes the derivative of f then compute the value of the sum $\sum_{n=1}^{100} g\left(\frac{1}{n}\right)$.
7. If $\lim_{x \rightarrow 0} \frac{1 - \cos\left(1 - \cos \frac{x}{2}\right)}{2^m x^n}$ is equal to the left hand derivative of $e^{-|x|}$ at $x = 0$, then find the value of $(n - 10m)$
8. If f is a differentiable function such that $f\left(\frac{x+y}{3}\right) = \frac{f(x) + f(y) + f(0)}{3}$, $\forall x, y \in \mathbb{R}$ and $f'(0) = 2$, find $f(x)$
9. If $\lim_{x \rightarrow 0} \frac{f(3 - \sin x) - f(3 + x)}{x} = 8$, then $|f'(3)|$ is
10. Let $f(x)$ be a differentiable function such that $2f(x+y) + f(x-y) = 3f(x) + 3f(y) + 2xy$ $\forall x, y \in \mathbb{R}$ & $f'(0) = 0$, then $f(10) + f'(10)$ is equal to

EXERCISE (JM)

1. Let $f(x) = x|x|$ and $g(x) = \sin x$. [AIEEE-2009]

Statement-1 : $g \circ f$ is differentiable at $x = 0$ and its derivative is continuous at that point.

Statement-2 : $g \circ f$ is twice differentiable at $x = 0$.

- (1) Statement-1 is true, Statement-2 is false.
 (2) Statement-1 is false, Statement-2 is true.
 (3) Statement-1 is true, Statement-2 is true; Statement-2 is a correct explanation for Statement-1.
 (4) Statement-1 is true, Statement-2 is true ; Statement-2 is not a correct explanation for statement-1.

2. If function $f(x)$ is differentiable at $x = a$ then $\lim_{x \rightarrow a} \frac{x^2 f(a) - a^2 f(x)}{x - a}$ [AIEEE-2011]

- (1) $2a f(a) + a^2 f'(a)$ (2) $-a^2 f'(a)$ (3) $af(a) - a^2 f'(a)$ (4) $2af(a) - a^2 f'(a)$

3. Consider the function,
 $f(x) = |x - 2| + |x - 5|$, $x \in \mathbb{R}$.

Statement-1 : $f'(4) = 0$.

Statement-2 : f is continuous in $[2, 5]$, differentiable in $(2, 5)$ and $f(2) = f(5)$. [AIEEE 2012]

- (1) Statement-1 is true, Statement-2 is false.
 (2) Statement-1 is false, Statement-2 is true.
 (3) Statement-1 is true, Statement-2 is true ; Statement-2 is a correct explanation for Statement 1.
 (4) Statement-1 is true, Statement-2 is true ; Statement-2 is not a correct explanation for Statement 1.

4. Let $f(x) = x|x|$, $g(x) = \sin x$ and $h(x) = (g \circ f)(x)$. Then [On-line 2014]

- (1) $h'(x)$ is differentiable at $x = 0$
 (2) $h'(x)$ is continuous at $x = 0$ but is not differentiable at $x = 0$
 (3) $h(x)$ is differentiable at $x = 0$ but $h'(x)$ is not continuous at $x = 0$
 (4) $h(x)$ is not differentiable at $x = 0$

5. Let $f, g: \mathbb{R} \rightarrow \mathbb{R}$ be two functions defined by $f(x) = \begin{cases} x \sin\left(\frac{1}{x}\right), & x \neq 0 \\ 0, & x = 0 \end{cases}$, and $g(x) = xf(x)$:-

Statement I : f is a continuous function at $x = 0$.

[On-line 2014]

Statement II : g is a differentiable function at $x = 0$.

- (1) Statement I is false and statement II is true
 (2) Statement I is true and statement II is false
 (3) Both statement I and II are true
 (4) Both statements I and II are false

6. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a function such that $|f(x)| \leq x^2$, for all $x \in \mathbb{R}$. Then, at $x = 0$, f is:

- (1) Neither continuous nor differentiable [On-line 2014]
 (2) differentiable but not continuous
 (3) continuous as well as differentiable
 (4) continuous but not differentiable

7. For $x \in \mathbb{R}$, $f(x) = |\log 2 - \sin x|$ and $g(x) = f(f(x))$, then :

[JEE(Main)-2016]

- (1) g is differentiable at $x = 0$ and $g'(0) = -\sin(\log 2)$
 (2) g is not differentiable at $x = 0$
 (3) $g'(0) = \cos(\log 2)$
 (4) $g'(0) = -\cos(\log 2)$

8. Let $S = \{t \in \mathbb{R} : f(x) = |x - \pi| \cdot (e^{|x|} - 1) \sin|x| \text{ is not differentiable at } t\}$. Then the set S is equal to:

[JEE(Main)-2018]

- (1) $\{0\}$ (2) $\{\pi\}$ (3) $\{0, \pi\}$ (4) ϕ (an empty set)

EXERCISE (JA)

1. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function such that

$$f(x + y) = f(x) + f(y), \forall x, y \in \mathbb{R}.$$

If $f(x)$ is differentiable at $x = 0$, then

- (A) $f(x)$ is differentiable only in a finite interval containing zero
 (B) $f(x)$ is continuous $\forall x \in \mathbb{R}$
 (C) $f'(x)$ is constant $\forall x \in \mathbb{R}$
 (D) $f(x)$ is differentiable except at finitely many points

[JEE 2011, 4M]

2. If $f(x) = \begin{cases} -x - \frac{\pi}{2}, & x \leq -\frac{\pi}{2} \\ -\cos x, & -\frac{\pi}{2} < x \leq 0 \\ x - 1, & 0 < x \leq 1 \\ \ln x, & x > 1 \end{cases}$ then -

[JEE 2011, 4M]

- (A) $f(x)$ is continuous at $x = -\frac{\pi}{2}$ (B) $f(x)$ is not differentiable at $x = 0$
 (C) $f(x)$ is differentiable at $x = 1$ (D) $f(x)$ is differentiable at $x = -\frac{3}{2}$

3. Let $f(x) = \begin{cases} x^2 \left| \cos \frac{\pi}{x} \right|, & x \neq 0 \\ 0, & x = 0 \end{cases}$, $x \in \mathbb{R}$, then f is -

[JEE 2012, 3M, -1M]

- (A) differentiable both at $x = 0$ and at $x = 2$
 (B) differentiable at $x = 0$ but not differentiable at $x = 2$
 (C) not differentiable at $x = 0$ but differentiable at $x = 2$
 (D) differentiable neither at $x = 0$ nor at $x = 2$

4. Let $f_1 : \mathbb{R} \rightarrow \mathbb{R}$, $f_2 : [0, \infty) \rightarrow \mathbb{R}$, $f_3 : \mathbb{R} \rightarrow \mathbb{R}$ and $f_4 : \mathbb{R} \rightarrow [0, \infty)$ be defined by

$$f_1(x) = \begin{cases} |x| & \text{if } x < 0, \\ e^x & \text{if } x \geq 0; \end{cases}$$

$$f_2(x) = x^2;$$

$$f_3(x) = \begin{cases} \sin x & \text{if } x < 0, \\ x & \text{if } x \geq 0 \end{cases}$$

$$\text{and } f_4(x) = \begin{cases} f_2(f_1(x)) & \text{if } x < 0, \\ f_2(f_1(x)) - 1 & \text{if } x \geq 0. \end{cases}$$

List-I

P. f_4 is

Q. f_3 is

R. $f_2 \circ f_1$ is

S. f_2 is

Codes :

	P	Q	R	S
(A)	3	1	4	2
(B)	1	3	4	2
(C)	3	1	2	4
(D)	1	3	2	4

List-II

1. onto but not one-one

2. neither continuous nor one-one

3. differentiable but not one-one

4. continuous and one-one

[JEE(Advanced)-2014, 3(-1)]

5. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ and $g : \mathbb{R} \rightarrow \mathbb{R}$ be respectively given by $f(x) = |x| + 1$ and $g(x) = x^2 + 1$. Define

$$h : \mathbb{R} \rightarrow \mathbb{R} \text{ by } h(x) = \begin{cases} \max\{f(x), g(x)\} & \text{if } x \leq 0, \\ \min\{f(x), g(x)\} & \text{if } x > 0. \end{cases}$$

The number of points at which $h(x)$ is not differentiable is

[JEE(Advanced)-2014, 3]

6. Let $a, b \in \mathbb{R}$ and $f : \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = a \cos(|x^3 - x|) + b|x| \sin(|x^3 + x|)$. Then f is -

(A) differentiable at $x = 0$ if $a = 0$ and $b = 1$

(B) differentiable at $x = 1$ if $a = 1$ and $b = 0$

(C) **NOT** differentiable at $x = 0$ if $a = 1$ and $b = 0$

(D) **NOT** differentiable at $x = 1$ if $a = 1$ and $b = 1$

[JEE(Advanced)-2016, 4(-2)]

7. Let $f : \left[-\frac{1}{2}, 2\right] \rightarrow \mathbb{R}$ and $g : \left[-\frac{1}{2}, 2\right] \rightarrow \mathbb{R}$ be function defined by $f(x) = [x^2 - 3]$ and $g(x) = |x| f(x) + |4x - 7| f(x)$, where $[y]$ denotes the greatest integer less than or equal to y for $y \in \mathbb{R}$. Then

(A) f is discontinuous exactly at three points in $\left[-\frac{1}{2}, 2\right]$

(B) f is discontinuous exactly at four points in $\left[-\frac{1}{2}, 2\right]$

(C) g is NOT differentiable exactly at four points in $\left(-\frac{1}{2}, 2\right)$

(D) g is NOT differentiable exactly at five points in $\left(-\frac{1}{2}, 2\right)$

[JEE(Advanced)-2016, 4(-2)]

8. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a differentiable function with $f(0) = 1$ and satisfying the equation

$$f(x+y) = f(x)f'(y) + f'(x)f(y) \text{ for all } x, y \in \mathbb{R}.$$

Then, then value of $\log_e(f(4))$ is _____.

[JEE(Advanced)-2018]

9. Let $f_1 : \mathbb{R} \rightarrow \mathbb{R}$, $f_2 : \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \rightarrow \mathbb{R}$, $f_3 : \left(-1, e^{\frac{\pi}{2}} - 2\right) \rightarrow \mathbb{R}$ and $f_4 : \mathbb{R} \rightarrow \mathbb{R}$ be functions defined by

$$(i) f_1(x) = \sin\left(\sqrt{1-e^{-x^2}}\right)$$

$$(ii) f_2(x) = \begin{cases} \frac{|\sin x|}{\tan^{-1} x} & \text{if } x \neq 0 \\ 1 & \text{if } x = 0 \end{cases}, \text{ where the inverse trigonometric function } \tan^{-1} x \text{ assumes values in } \left(-\frac{\pi}{2}, \frac{\pi}{2}\right),$$

$$(iii) f_3(x) = [\sin(\log_e(x+2))], \text{ where for } t \in \mathbb{R}, [t] \text{ denotes the greatest integer less than or equal to } t,$$

$$(iv) f_4(x) = \begin{cases} x^2 \sin\left(\frac{1}{x}\right) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

List-I

P. The function f_1 is

Q. The function f_2 is

R. The function f_3 is

S. The function f_4 is

List-II

1. NOT continuous at $x = 0$

2. continuous at $x = 0$ and NOT differentiable at $x = 0$

3. differentiable at $x = 0$ and its derivative is NOT continuous at $x = 0$

4. differentiable at $x = 0$ and its derivative is continuous at $x = 0$

The correct option is :

(A) P $\rightarrow$ 2; Q $\rightarrow$ 3; R $\rightarrow$ 1; S $\rightarrow$ 4

(B) P $\rightarrow$ 4; Q $\rightarrow$ 1; R $\rightarrow$ 2; S $\rightarrow$ 3

(C) P $\rightarrow$ 4; Q $\rightarrow$ 2; R $\rightarrow$ 1; S $\rightarrow$ 3

(D) P $\rightarrow$ 2; Q $\rightarrow$ 1; R $\rightarrow$ 4; S $\rightarrow$ 3

[JEE(Advanced)-2018]

ANSWER KEY

LIMIT

EXERCISE (O-1)

- | | | | | | | | |
|--|--------------------|------------------|---|----------------|------------------|----------------|--------------|
| 1. A | 2. C | 3. D | 4. C | 5. D | 6. B | 7. B | 8. C |
| 9. B | 10. B | 11. A | 12. A | 13. C | 14. B | 15. A | 16. C |
| 17. A | 18. D | 19. A | 20. A | 21. C | 22. D | 23. D | 24. C |
| 25. B | 26. D | 27. A | 28. A | 29. C | 30. A | 31. D | 32. B |
| 33. D | 34. C | 35. D | 36. B,C | 37. A,C | 38. A,B,C | 39. A,D | |
| 40. B,C,D | 41. A,B,C,D | 42. B,C,D | 43. (A)→(P,R); (B)→(P,R); (C)→(Q); (D)→(Q) | | | | |
| 44. (A)→(R);(B)→(S);(C)→(P);(D)→(Q) | | | 45. (A)→(P,Q,R,S,T);(B)→(P,R);(C)→(P,R); (D)→(S) | | | | |

EXERCISE (O-2)

- | | | | | | | | |
|--------------------|----------------|------------------|----------------|------------------|------------------|------------------|----------------|
| 1. B | 2. C | 3. A | 4. D | 5. A | 6. C | 7. B | 8. C |
| 9. D | 10. A | 11. D | 12. C | 13. B,C,D | 14. A,B,D | 15. A,C,D | 16. A,C |
| 17. B,C | 18. B,D | 19. B,C,D | 20. A,B | 21. A,D | 22. A,C | 23. B,C | 24. A,D |
| 25. A,B,C,D | | | | | | | |

EXERCISE (S-1)

1. 2 2. 5050 3. 2 4. $\frac{1}{32}$ 5. $\frac{1}{16\sqrt{2}}$ 6. $\frac{\sqrt{3}}{2}$ 7. $1/2$
8. -2 9. $\pi - 3$ 10. (i) $a = 1, b = -1$ (ii) $a = -1, b = \frac{1}{2}$ 11. 1
12. $8\sqrt{2}(\ln 3)^2$ 13. (a) $\pi/2$ if $a > 0$; 0 if $a = 0$ and $-\pi/2$ if $a < 0$; (b) $f(x) = |x|$
14. $-1/2$ 15. 16 16. e^{-8} 17. $c = \ln 2$ 18. e^{-1} 19. $e^{-1/2}$ 20. $\frac{\sqrt{3}}{2}$

EXERCISE (S-2)

1. e^{-1} 2. $(a_1 \cdot a_2 \cdot a_3 \dots a_n)$ 3. $e^{\frac{1}{2}}$ 4. 167 5. 1/2 6. 8
7. $T(x) = \frac{1}{2} \tan^2 \frac{x}{2} \cdot \sin x$ or $\tan \frac{x}{2} - \frac{\sin x}{2}$, $S(x) = \frac{1}{2}x - \frac{1}{2}\sin x$, limit = $\frac{3}{2}$ 8. $g(x) = \sin x$ and $\ell = e$
9. $\frac{\theta}{\tan \theta}$ 10. $-\frac{1}{2e}$ 11. $a = -5/2$, $b = -3/2$ 13. $\frac{2L}{3}$
14. (a) 2, (b) D.N.E., (c) 0, (d) 0 15. (a) 2; (b) 1/2

EXERCISE (JM)

- 1. 1 2. 3 3. 1 4. 4 5. 1 6. 4 7. 3 8. 2**

EXERCISE (JA)

- 1. A,C 2. D 3. B 4. B 5. 0 6. 2 7. 7 8. A,C 9. D**

CONTINUITY

EXERCISE (O-1)

1. A 2. C 3. D 4. A 5. B 6. A 7. D
 8. C 9. D 10. B 11. B 12. A 13. B 14. D

EXERCISE (O-2)

1. C 2. B 3. B 4. D 5. C 6. D 7. A,B,C,D
 8. A,B 9. A,B,C 10. B,C,D 11. A,B,C,D 12. A,C,D 13. A, C, D
 14. A, C, D 15. A, B, C, D 16. (A)→(Q); (B)→(R); (C)→(S); (D)→(P)
 17. (A)→(P,Q,R); (B)→(P,Q,T); (C)→(P,Q,T); (D)→(P,Q,S)

EXERCISE (S-1)

1. -1 2. $a = 0, b = 1$ 3. $a = 0; b = -1$ 4. (a) -2, 2, 3; (b) $K = 5$; (c) even
 5. P not possible. 6. (a) $4 - 3\sqrt{2} + a$, (b) $a = 3$
 7. $g(x) = 2 + x$ for $0 \leq x \leq 1$, $2 - x$ for $1 < x \leq 2$, $4 - x$ for $2 < x \leq 3$, g is discontinuous at $x = 1$ & $x = 2$
 8. $f(0^+) = -2$; $f(0^-) = 2$ hence $f(0)$ not possible to define 9. $a = 1/2, b = 4$
 10. $a = -3/2, b \neq 0, c = 1/2$

EXERCISE (S-2)

1. $A = -4, B = 5, f(0) = 1$
 2. $f(0^+) = \frac{\pi}{2}$; $f(0^-) = \frac{\pi}{4\sqrt{2}} \Rightarrow f$ is discontin. at $x = 0$; $g(0^+) = g(0^-) = g(0) = \pi/2 \Rightarrow g$ is cont. at $x = 0$
 3. discontinuous at all integral values in $[-2, 2]$
 4. locus $(a, b) \rightarrow x, y$ is $y = x - 3$ excluding the points where $y = 3$ intersects it.
 5. $c = 1, a, b \in \mathbb{R}$ 6. $e^2 + e^{-2}$
 7. $k = 0$; $g(x) = \begin{cases} \ln(\tan x) & \text{if } 0 < x < \frac{\pi}{4} \\ 0 & \text{if } \frac{\pi}{4} \leq x < \frac{\pi}{2} \end{cases}$. Hence $g(x)$ is continuous everywhere.
 8. $g(x) = 4(x + 1)$ and limit $= -\frac{39}{4}$ 9. (a) 5(b) 30

EXERCISE (JM)

1. 4 2. 1 3. 1 4. 4 5. 3 6. 1

EXERCISE (JA)

1. Discontinuous at $x = 1$; $f(1^+) = 1$ and $f(1^-) = -1$ 2. B,D 3. A,D 4. A,C,D

DIFFERENTIABILITY

EXERCISE (O-1)

1. B 2. C 3. A 4. A 5. C 6. B 7. D
8. A 9. B 10. D 11. B 12. B 13. A 14. D
15. D 16. A 17. D 18. D 19. D

EXERCISE (O-2)

1. A,C,D 2. A,B,C 3. B,C 4. A,B,C 5. B,C 6. A,B,C,D 7. A,B,C,D
8. B,C,D 9. A,B,D 10. B,D 11. (A) R, (B) P, (C) Q, (D) R

EXERCISE (S-1)

1. $f(x)$ is conti. but not derivable at $x = 0$
2. conti. $\forall x \in \mathbb{R}$, not diff. at $x = 0, 1$ & 2
3. $0 < n \leq 1$
4. conti. but not diff. at $x = 0$; diff. & conti. at $x = \pi/2$
5. conti. but not diff. at $x = 0$
7. f is conti. but not diff. at $x = 0$
8. $Rf'(1) = 3, Lf'(1) = -1$
9. $a = 1/2, b = 3/2$
10. 3

EXERCISE (S-2)

1. not derivable at $x = 0$ & $x = 1$
2. f is conti. at $x = 1, 3/2$ & disconti. at $x = 2$, f is not diff. at $x = 1, 3/2, 2$
3. If $a \in (0, 1)$ $Rf'(0) = -1$; $Lf'(0) = 1 \Rightarrow$ continuous but not derivable
 $a = 1$; $f(x) = 0$ which is constant $\Rightarrow$ continuous and derivable
 If $a > 1$, $Lf'(0) = -1$; $Rf'(0) = 1 \Rightarrow$ continuous but not derivable
4. 8 5. f is conti. but not diff. at $x = 1$, disconti. at $x = 2$ & $x = 3$. cont. & diff. at all other points
6. 5150 7. 74 8. $f(x) = 2x + c$ 9. 4 10. 120

EXERCISE (JM)

1. 1 2. 4 3. 4 4. 2 5. 3 6. 3 7. 3 8. 4

EXERCISE (JA)

1. B,C 2. A,B,C,D 3. B 4. D 5. 3 6. A,B 7. B,C 8. 2
9. D

CONTENTS

METHOD OF DIFFERENTIATION

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JEE (Main) Syllabus :

Differentiation of the sum, difference, product and quotient of two functions. Differentiation of trigonometric, inverse trigonometric, logarithmic, exponential, composite and implicit functions; derivatives of order upto two.

JEE (Advanced) Syllabus :

Derivative of a function, derivative of the sum, difference, product and quotient of two functions, chain rule, derivatives of polynomial, rational, trigonometric, inverse trigonometric, exponential and logarithmic functions. Derivatives of implicit functions, derivatives up to order two. L'Hospital rule of evaluation of limits of functions.

METHODS OF DIFFERENTIATION

The process of calculating derivative is called differentiation.

1. DERIVATIVE OF $f(x)$ FROM THE FIRST PRINCIPLE :

Obtaining the derivative using the definition $\lim_{\delta x \rightarrow 0} \frac{\delta y}{\delta x} = \lim_{\delta x \rightarrow 0} \frac{f(x + \delta x) - f(x)}{\delta x} = f'(x) = \frac{dy}{dx}$ is called calculating derivative using first principle or ab initio or delta method.

Note : $\frac{dy}{dx}$ can also be represented as y_1 or y' or Dy or $f'(x)$. $\frac{dy}{dx}$ represents instantaneous rate of change of y w.r.t. x .

Illustration 1 : Differentiate each of following functions by first principle :

(i) $f(x) = \tan x$ (ii) $f(x) = e^{\sin x}$

Solution : (i) $f(x) = \lim_{h \rightarrow 0} \frac{\tan(x+h) - \tan x}{h} = \lim_{h \rightarrow 0} \frac{\tan(x+h-x)[1 + \tan x \tan(x+h)]}{h}$

$$= \lim_{h \rightarrow 0} \frac{\tanh}{h} \cdot (1 + \tan^2 x) = \sec^2 x.$$

Ans.

(ii) $f(x) = \lim_{h \rightarrow 0} \frac{e^{\sin(x+h)} - e^{\sin x}}{h} = \lim_{h \rightarrow 0} e^{\sin x} \frac{[e^{\sin(x+h) - \sin x} - 1]}{\sin(x+h) - \sin x} \left(\frac{\sin(x+h) - \sin x}{h} \right)$

$$= e^{\sin x} \lim_{h \rightarrow 0} \frac{\sin(x+h) - \sin x}{h} = e^{\sin x} \cos x$$

Ans.

Do yourself - 1 :

(i) Differentiate each of following functions by first principle:

(a) $f(x) = \ln x$ (b) $f(x) = \frac{1}{x}$

2. DERIVATIVE OF STANDARD FUNCTIONS :

	$f(x)$	$f'(x)$		$f(x)$	$f'(x)$
(i)	x^n	nx^{n-1}	(ii)	e^x	e^x
(iii)	a^x	$a^x \ln a, a > 0$	(iv)	$\ln x$	$1/x$
(v)	$\log_a x$	$(1/x) \log_a e, a > 0, a \neq 1$	(vi)	$\sin x$	$\cos x$
(vii)	$\cos x$	$-\sin x$	(viii)	$\tan x$	$\sec^2 x$
(ix)	$\sec x$	$\sec x \tan x$	(x)	$\operatorname{cosec} x$	$-\operatorname{cosec} x \cdot \cot x$
(xi)	$\cot x$	$-\operatorname{cosec}^2 x$	(xii)	constant	0
(xiii)	$\sin^{-1} x$	$\frac{1}{\sqrt{1-x^2}}, -1 < x < 1$	(xiv)	$\cos^{-1} x$	$\frac{-1}{\sqrt{1-x^2}}, -1 < x < 1$
(xv)	$\tan^{-1} x$	$\frac{1}{1+x^2}, x \in \mathbb{R}$	(xvi)	$\sec^{-1} x$	$\frac{1}{ x \sqrt{x^2-1}}, x > 1$
(xvii)	$\operatorname{cosec}^{-1} x$	$\frac{-1}{ x \sqrt{x^2-1}}, x > 1$	(xviii)	$\cot^{-1} x$	$\frac{-1}{1+x^2}, x \in \mathbb{R}$

3. FUNDAMENTAL THEOREMS :

If f and g are derivable functions of x , then,

$$(a) \quad \frac{d}{dx}(f \pm g) = \frac{df}{dx} \pm \frac{dg}{dx}$$

$$(b) \quad \frac{d}{dx}(cf) = c \frac{df}{dx}, \text{ where } c \text{ is any constant}$$

$$(c) \quad \frac{d}{dx}(fg) = f \frac{dg}{dx} + g \frac{df}{dx} \text{ known as "PRODUCT RULE"}$$

$$(d) \quad \frac{d}{dx}\left(\frac{f}{g}\right) = \frac{g\left(\frac{df}{dx}\right) - f\left(\frac{dg}{dx}\right)}{g^2} \text{ where } g \neq 0 \text{ known as "QUOTIENT RULE"}$$

$$(e) \quad \text{If } y = f(u) \text{ \& } u = g(x) \text{ then } \frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} \text{ known as "CHAIN RULE"}$$

Note : In general if $y = f(u)$ then $\frac{dy}{dx} = f'(u) \cdot \frac{du}{dx}$.

Illustration 2 : If $y = e^x \tan x + x \log_e x$, find $\frac{dy}{dx}$.

Solution :

$$y = e^x \cdot \tan x + x \cdot \log_e x$$

On differentiating we get,

$$\frac{dy}{dx} = e^x \cdot \tan x + e^x \cdot \sec^2 x + 1 \cdot \log x + x \cdot \frac{1}{x}$$

$$\text{Hence, } \frac{dy}{dx} = e^x(\tan x + \sec^2 x) + (\log x + 1)$$

Ans.

Illustration 3 : If $y = \frac{\log x}{x} + e^x \sin 2x + \log_5 x$, find $\frac{dy}{dx}$.

Solution :

On differentiating we get,

$$\frac{dy}{dx} = \frac{d}{dx}\left(\frac{\log x}{x}\right) + \frac{d}{dx}(e^x \sin 2x) + \frac{d}{dx}(\log_5 x) = \frac{\frac{1}{x} \cdot x - \log x \cdot 1}{x^2} + e^x \sin 2x + 2e^x \cdot \cos 2x + \frac{1}{x \log_e 5}$$

$$\text{Hence, } \frac{dy}{dx} = \left(\frac{1 - \log x}{x^2}\right) + e^x(\sin 2x + 2\cos 2x) + \frac{1}{x \log_e 5}$$

Ans.

Illustration 4 : If $y = \log_e(\tan^{-1} \sqrt{1+x^2})$, find $\frac{dy}{dx}$.

Solution :

$$y = \log_e(\tan^{-1} \sqrt{1+x^2})$$

On differentiating we get,

$$= \frac{1}{\tan^{-1} \sqrt{1+x^2}} \cdot \frac{1}{1+(\sqrt{1+x^2})^2} \cdot \frac{1}{2\sqrt{1+x^2}} \cdot 2x$$

$$= \frac{x}{(\tan^{-1} \sqrt{1+x^2}) \{1+(\sqrt{1+x^2})^2\} \sqrt{1+x^2}} = \frac{x}{(\tan^{-1} \sqrt{1+x^2})(2+x^2)\sqrt{1+x^2}}$$

Ans.

Do yourself -2 :

(i) Find $\frac{dy}{dx}$ if -

(a) $y = (x+1)(x+2)(x+3)$ (b) $y = e^{5x} \tan(x^2+2)$

4. LOGARITHMIC DIFFERENTIATION :

To find the derivative of a function :

- (a) which is the product or quotient of a number of functions or
(b) of the form $[f(x)]^{g(x)}$ where f & g are both derivable functions.

It is convenient to take the logarithm of the function first & then differentiate.

Illustration 5 : If $y = (\sin x)^{\ln x}$, find $\frac{dy}{dx}$

Solution : $\ln y = \ln x \cdot \ln (\sin x)$
On differentiating we get,

$$\frac{1}{y} \frac{dy}{dx} = \frac{1}{x} \ln (\sin x) + \ln x \cdot \frac{\cos x}{\sin x} \Rightarrow \frac{dy}{dx} = (\sin x)^{\ln x} \left[\frac{\ln (\sin x)}{x} + \cot x \ln x \right] \quad \text{Ans.}$$

Illustration 6 : If $x = \exp \left(\tan^{-1} \left(\frac{y-x^2}{x^2} \right) \right)$, then $\frac{dy}{dx}$ equals -

- (A) $x [1 + \tan (\log x) + \sec^2 x]$ (B) $2x [1 + \tan (\log x)] + \sec^2 x$
(C) $2x [1 + \tan (\log x)] + \sec x$ (D) $2x + x[1 + \tan(\log x)]^2$

Solution : Taking log on both sides, we get

$$\log x = \tan^{-1} \left(\frac{y-x^2}{x^2} \right) \Rightarrow \tan (\log x) = (y-x^2) / x^2$$

$$\Rightarrow y = x^2 + x^2 \tan (\log x)$$

On differentiating, we get

$$\therefore \frac{dy}{dx} = 2x + 2x \tan (\log x) + x \sec^2 (\log x) \Rightarrow 2x [1 + \tan (\log x)] + x \sec^2 (\log x)$$

$$= 2x + x[1 + \tan(\log x)]^2 \quad \text{Ans. (D)}$$

Illustration 7 : If $y = \frac{x^{1/2}(1-2x)^{2/3}}{(2-3x)^{3/4}(3-4x)^{4/5}}$ find $\frac{dy}{dx}$

Solution : $\ln y = \frac{1}{2} \ln x + \frac{2}{3} \ln (1-2x) - \frac{3}{4} \ln (2-3x) - \frac{4}{5} \ln (3-4x)$

On differentiating we get,

$$\Rightarrow \frac{1}{y} \frac{dy}{dx} = \frac{1}{2x} - \frac{4}{3(1-2x)} + \frac{9}{4(2-3x)} + \frac{16}{5(3-4x)}$$

$$\frac{dy}{dx} = y \left(\frac{1}{2x} - \frac{4}{3(1-2x)} + \frac{9}{4(2-3x)} + \frac{16}{5(3-4x)} \right)$$

Ans.

Do yourself -3 :

(i) Find $\frac{dy}{dx}$ if $y = x^x$

(ii) Find $\frac{dy}{dx}$ if $y = e^x \cdot e^{x^2} \cdot e^{x^3} \cdot e^{x^4}$

5. PARAMETRIC DIFFERENTIATION :

If $y = f(\theta)$ & $x = g(\theta)$ where θ is a parameter, then $\frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta} = \frac{f'(\theta)}{g'(\theta)}$

Illustration 8: If $y = a \cos t$ and $x = a(t - \sin t)$ find the value of $\frac{dy}{dx}$ at $t = \frac{\pi}{2}$

Solution : $\frac{dy}{dx} = \frac{-a \sin t}{a(1 - \cos t)} \Rightarrow \frac{dy}{dx} \Big|_{t=\frac{\pi}{2}} = -1$

Ans.

Illustration 9: Prove that the function represented parametrically by the equations. $x = \frac{1+t}{t^3}$; $y = \frac{3}{2t^2} + \frac{2}{t}$

satisfies the relationship : $x(y')^3 = 1 + y'$ (where $y' = \frac{dy}{dx}$)

Solution : Here $x = \frac{1+t}{t^3} = \frac{1}{t^3} + \frac{1}{t^2}$
Differentiating w.r. to t

$$\frac{dx}{dt} = -\frac{3}{t^4} - \frac{2}{t^3}$$

$$y = \frac{3}{2t^2} + \frac{2}{t}$$

Differentiating w.r. to t

$$\frac{dy}{dt} = -\frac{3}{t^3} - \frac{2}{t^2}$$

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = t = y'$$

Since $x = \frac{1+t}{t^3} \Rightarrow x = \frac{1+y'}{(y')^3}$ or $x(y')^3 = 1 + y'$

Ans.**6. DERIVATIVE OF A FUNCTION W.R.T. ANOTHER FUNCTION :**

Let $y = f(x)$; $z = g(x)$ then $\frac{dy}{dz} = \frac{dy/dx}{dz/dx} = \frac{f'(x)}{g'(x)}$

Illustration 10: Differentiate $\log_e (\tan x)$ with respect to $\sin^{-1}(e^x)$.

Solution : $\frac{d(\log_e \tan x)}{d(\sin^{-1}(e^x))} = \frac{\frac{d}{dx}(\log_e \tan x)}{\frac{d}{dx} \sin^{-1}(e^x)} = \frac{\cot x \cdot \sec^2 x}{e^x \cdot 1/\sqrt{1-e^{2x}}} = \frac{e^{-x} \sqrt{1-e^{2x}}}{\sin x \cos x}$

Ans.

(i) Find $\frac{dy}{dx}$ at $t = \frac{\pi}{4}$ if $y = \cos^4 t$ & $x = \sin^4 t$.

(ii) Find the slope of the tangent at a point P(t) on the curve $x = at^2$, $y = 2at$.

(iii) Differentiate $x^{\ell_{\text{nx}}}$ with respect to ℓ_{nx} .

7. DIFFERENTIATION OF IMPLICIT FUNCTIONS : $\phi(x, y) = 0$

(a) To find dy/dx of implicit functions, we differentiate each term w.r.t. x regarding y as a function of x & then collect terms with dy/dx together on one side.

(b) Also $\frac{dy}{dx} = -\frac{\frac{\partial \phi}{\partial x}}{\frac{\partial \phi}{\partial y}}$, where $\frac{\partial \phi}{\partial x}$ = partial derivative of $\phi(x, y)$ w.r.t. x taking y as a constant and

$$\frac{\partial \phi}{\partial y} = \text{partial derivative of } \phi(x, y) \text{ w.r.t. } y \text{ taking } x \text{ as a constant.}$$

(c) In the case of implicit functions, generally, both x & y are present in answers of dy/dx .

Illustration 11: If $x^y + y^x = 2$, then find $\frac{dy}{dx}$.

Let $u = x^y$ and $v = y^x$

$$u + v = 2 \Rightarrow \frac{du}{dx} + \frac{dv}{dx} = 0$$

Now $u = x^y$ and $v = y^x$

$$\Rightarrow \ln u = y \ln x \quad \text{and} \quad \ln v = x \ln y$$

$$\Rightarrow \frac{1}{u} \frac{du}{dx} = \frac{y}{x} + \ell n x \frac{dy}{dx} \quad \text{and} \quad \frac{1}{v} \frac{dv}{dx} = \ell n y + \frac{x}{y} \frac{dy}{dx}$$

$$\Rightarrow \frac{du}{dx} = x^y \left(\frac{y}{x} + \ell n x \frac{dy}{dx} \right) \text{ and } \frac{dv}{dx} = y^x \left(\ell n y + \frac{x}{y} \frac{dy}{dx} \right)$$

$$\Rightarrow x^y \left(\frac{y}{x} + \ln x \frac{dy}{dx} \right) + y^x \left(\ln y + \frac{x}{y} \frac{dy}{dx} \right) = 0 \Rightarrow \frac{dy}{dx} = - \frac{\left(y^x \ln y + x^y \cdot \frac{y}{x} \right)}{\left(x^y \ln x + y^x \cdot \frac{x}{y} \right)}$$

Ans.

Aliter :

$$\phi(x, y) = x^y + y^x - 2 = 0$$

$$\frac{dy}{dx} = \frac{-\partial\phi/\partial x}{\partial\phi/\partial y} = \frac{yx^{y-1} + y^x \ln y}{x^y \ln x + xy^{x-1}}$$

Illustration 12 : If $y = \frac{\sin x}{1 + \frac{\cos x}{1 + \frac{\sin x}{1 + \cos x}}}$, prove that $\frac{dy}{dx} = \frac{(1+y)\cos x + y \sin x}{1 + 2y + \cos x - \sin x}$.

Solution : Given function is $y = \frac{\sin x}{1 + \frac{\cos x}{1 + y}} = \frac{(1+y)\sin x}{1 + y + \cos x}$

$$\text{or } y + y^2 + y \cos x = (1 + y) \sin x \quad \dots(i)$$

Differentiate both sides with respect to x ,

$$\frac{dy}{dx} + 2y \frac{dy}{dx} + \frac{dy}{dx} \cos x - y \sin x = (1 + y) \cos x + \frac{dy}{dx} \sin x$$

$$\frac{dy}{dx} (1 + 2y + \cos x - \sin x) = (1 + y) \cos x + y \sin x$$

$$\text{or } \frac{dy}{dx} = \frac{(1+y)\cos x + y \sin x}{1 + 2y + \cos x - \sin x}$$

Ans.

Aliter :

$$\text{From (i) } \phi(x, y) = (1 + y)\sin x - y - y^2 - y \cos x = 0$$

$$\frac{dy}{dx} = -\frac{\partial \phi / \partial x}{\partial \phi / \partial y} = -\frac{(1+y)\cos x + y \sin x}{\sin x - 1 - 2y - \cos x} = \frac{(1+y)\cos x + y \sin x}{1 + 2y + \cos x - \sin x}$$

Do yourself -5 :

(i) Find $\frac{dy}{dx}$, if $x + y = \sin(x - y)$

(ii) If $x^2 + xe^y + y = 0$, find y' , also find the value of y' at point $(0, 0)$.

8. DIFFERENTIATION BY TRIGONOMETRIC TRANSFORMATION :

Some Standard Substitutions :

Expression

Substitution

$$\sqrt{a^2 - x^2}$$

$$x = a \sin \theta \text{ or } a \cos \theta$$

$$\sqrt{a^2 + x^2}$$

$$x = \tan \theta \text{ or } \cot \theta$$

$$\sqrt{x^2 - a^2}$$

$$x = a \sec \theta \text{ or } a \operatorname{cosec} \theta$$

$$\sqrt{\frac{a+x}{a-x}} \text{ or } \sqrt{\frac{a-x}{a+x}}$$

$$x = a \cos \theta \text{ or } a \cos 2\theta$$

$$\sqrt{2ax - x^2}$$

$$x = a(1 - \cos \theta)$$

Illustration 13: If $f(x) = \sin^{-1}\left(\frac{2x}{1+x^2}\right)$ then find

- (i) $f(2)$ (ii) $f\left(\frac{1}{2}\right)$ (iii) $f(1)$

Solution : $x = \tan\theta$, where $-\frac{\pi}{2} < \theta < \frac{\pi}{2} \Rightarrow y = \sin^{-1}(\sin 2\theta)$

$$y = \begin{cases} \pi - 2\theta & \frac{\pi}{2} < 2\theta < \pi \\ 2\theta & -\frac{\pi}{2} \leq 2\theta \leq \frac{\pi}{2} \\ -(\pi + 2\theta) & -\pi < 2\theta < -\frac{\pi}{2} \end{cases} \Rightarrow f(x) = \begin{cases} \pi - 2 \tan^{-1} x & x > 1 \\ 2 \tan^{-1} x & -1 \leq x \leq 1 \\ -(\pi + 2 \tan^{-1} x) & x < -1 \end{cases}$$

$$\Rightarrow f(x) = \begin{cases} -\frac{2}{1+x^2} & x > 1 \\ \frac{2}{1+x^2} & -1 < x < 1 \\ \frac{-2}{1+x^2} & x < -1 \end{cases}$$

(i) $f(2) = -\frac{2}{5}$ (ii) $f\left(\frac{1}{2}\right) = \frac{8}{5}$ (iii) $f(1^+) = -1$ and $f(1^-) = +1 \Rightarrow f(1)$ does not exist **Ans.**

Illustration 14: $\frac{d}{dx} \left\{ \sin^2 \left(\cot^{-1} \sqrt{\frac{1+x}{1-x}} \right) \right\} =$

- (A) $-\frac{1}{2}$ (B) 0 (C) $\frac{1}{2}$ (D) -1

Solution : Let $y = \sin^2 \left(\cot^{-1} \sqrt{\frac{1+x}{1-x}} \right)$. Put $x = \cos 2\theta$ $\theta \in \left(0, \frac{\pi}{2} \right]$

$$\therefore y = \sin^2 \cot^{-1} \left(\sqrt{\frac{1+\cos 2\theta}{1-\cos 2\theta}} \right) = \sin^2 \cot^{-1} (\cot \theta)$$

$$\therefore y = \sin^2 \theta = \frac{1 - \cos 2\theta}{2} = \frac{1-x}{2} = \frac{1}{2} - \frac{x}{2}$$

$$\therefore \frac{dy}{dx} = -\frac{1}{2}.$$

Ans (A)

Illustration 15: Obtain differential coefficient of $\tan^{-1} \frac{\sqrt{1+x^2}-1}{x}$ with respect to $\cos^{-1} \sqrt{\frac{1+\sqrt{1+x^2}}{2\sqrt{1+x^2}}}$

Solution : Assume $u = \tan^{-1} \frac{\sqrt{1+x^2}-1}{x}$, $v = \cos^{-1} \sqrt{\frac{1+\sqrt{1+x^2}}{2\sqrt{1+x^2}}}$

The function needs simplification before differentiation Let $x = \tan \theta$; $\theta \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

$$\therefore u = \tan^{-1} \left(\frac{\sec \theta - 1}{\tan \theta} \right) = \tan^{-1} \left(\frac{1 - \cos \theta}{\sin \theta} \right) = \tan^{-1} \left(\tan \frac{\theta}{2} \right) = \frac{\theta}{2}$$

$$v = \cos^{-1} \sqrt{\frac{1 + \sec \theta}{2 \sec \theta}} = \cos^{-1} \sqrt{\frac{1 + \cos \theta}{2}} = \cos^{-1} \left(\cos \frac{\theta}{2} \right) = \frac{\theta}{2} \Rightarrow u = v$$

$$\therefore \frac{du}{dv} = 1.$$

Ans.

Do yourself : 6

(i) If $y = \cos^{-1}(4x^3 - 3x)$, then find :

(a) $f' \left(-\frac{\sqrt{3}}{2} \right)$, (b) $f'(0)$, (c) $f' \left(\frac{\sqrt{3}}{2} \right)$.

9. DERIVATIVE OF A FUNCTION AND ITS INVERSE FUNCTION :

If g is inverse of f , then

(a) $g\{f(x)\} = x$ (b) $f\{g(x)\} = x$
 $g'\{f(x)\}f'(x) = 1$ $f'\{g(x)\}g'(x) = 1$

Illustration 16 : If g is inverse of f and $f(x) = \frac{1}{1+x^n}$, then $g'(x)$ equals :-

(A) $1+x^n$ (B) $1+[f(x)]^n$ (C) $1+[g(x)]^n$ (D) none of these

Solution : Since g is the inverse of f . Therefore

$$f(g(x)) = x \quad \text{for all } x$$

$$\Rightarrow \frac{d}{dx} f(g(x)) = 1 \quad \text{for all } x$$

$$\Rightarrow f(g(x)) g'(x) = 1 \Rightarrow g'(x) = \frac{1}{f'(g(x))} = 1 + (g(x))^n \quad \text{Ans. (C)}$$

Do yourself -7 :

(i) If g is inverse of f and $f(x) = 2x + \sin x$; then $g'(x)$ equals:

(A) $-\frac{3}{x^2} + \frac{1}{\sqrt{1-x^2}}$ (B) $2 + \sin^{-1}x$ (C) $2 + \cos g(x)$ (D) $\frac{1}{2 + \cos(g(x))}$

10. HIGHER ORDER DERIVATIVES :

Let a function $y = f(x)$ be defined on an interval (a, b) . If $f(x)$ is differentiable function, then its derivative $f'(x)$ [or (dy/dx) or y'] is called the first derivative of y w.r.t. x . If $f'(x)$ is again differentiable function on (a, b) , then its derivative $f''(x)$ [or d^2y/dx^2 or y''] is called second derivative of y w.r.t. x .

Similarly, the 3rd order derivative of y w.r.to x , if it exists, is defined by $\frac{d^3y}{dx^3} = \frac{d}{dx} \left(\frac{d^2y}{dx^2} \right)$ and denoted by $f'''(x)$ or y''' and so on.

Note : If $x = f(\theta)$ and $y = g(\theta)$ where ' θ ' is a parameter then $\frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta}$ & $\frac{d^2y}{dx^2} = \frac{d}{d\theta} \left(\frac{dy}{dx} \right) / \frac{dx}{d\theta}$

$$\text{In general } \frac{d^n y}{dx^n} = \frac{d}{d\theta} \left(\frac{d^{n-1} y}{dx^{n-1}} \right) \bigg/ \frac{dx}{d\theta}.$$

11. ANALYSIS AND GRAPHS OF SOME INVERSE TRIGONOMETRIC FUNCTIONS :

$$\text{(a)} \quad y = f(x) = \sin^{-1}\left(\frac{2x}{1+x^2}\right) = \begin{cases} 2 \tan^{-1} x & |x| \leq 1 \\ \pi - 2 \tan^{-1} x & x > 1 \\ -(\pi + 2 \tan^{-1} x) & x < -1 \end{cases}$$

Important points :

- (i) Domain is $x \in \mathbb{R}$ & range is $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$
(ii) f is continuous for all x but not differentiable at $x=1, -1$

$$(iii) \frac{dy}{dx} = \begin{cases} \frac{2}{1+x^2} & \text{for } |x| < 1 \\ \text{non existent} & \text{for } |x| = 1 \\ \frac{-2}{1+x^2} & \text{for } |x| > 1 \end{cases}$$

- (iv) Increasing in $[-1, 1]$ & Decreasing in $(-\infty, -1], [1, \infty)$

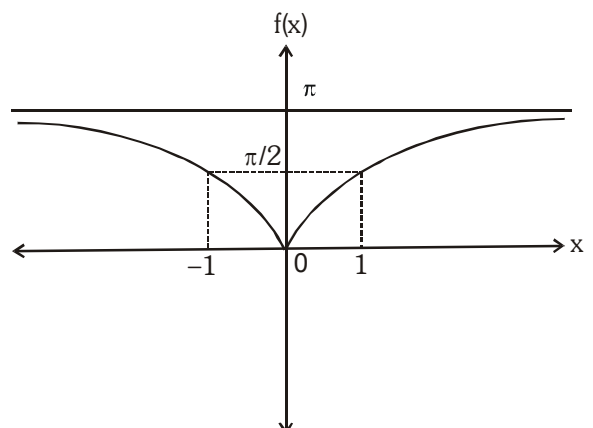
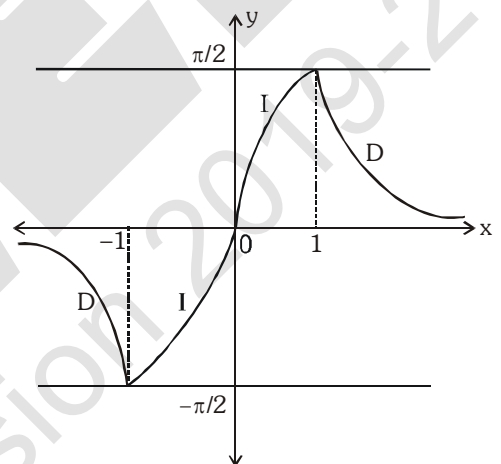
(b) Consider $y = f(x) = \cos^{-1} \left(\frac{1 - x^2}{1 + x^2} \right) = \begin{cases} 2 \tan^{-1} x & \text{if } x \geq 0 \\ -2 \tan^{-1} x & \text{if } x < 0 \end{cases}$

Important points :

- (i) Domain is $x \in \mathbb{R}$ & range is $[0, \pi)$
- (ii) Continuous for all x but not differentiable at $x = 0$

$$(iii) \frac{dy}{dx} = \begin{cases} \frac{2}{1+x^2} & \text{for } x > 0 \\ \text{non existent} & \text{for } x = 0 \\ -\frac{2}{1+x^2} & \text{for } x < 0 \end{cases}$$

- (iv) Increasing in $[0, \infty)$ & Decreasing in $(-\infty, 0]$



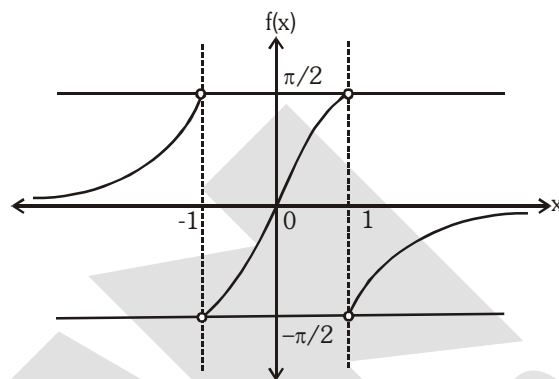
$$(c) \quad y = f(x) = \tan^{-1} \frac{2x}{1-x^2} = \begin{cases} 2 \tan^{-1} x & |x| < 1 \\ \pi + 2 \tan^{-1} x & x < -1 \\ -(\pi - 2 \tan^{-1} x) & x > 1 \end{cases}$$

Important points :

(i) Domain is $\mathbb{R} - \{1, -1\}$ & range is $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

$$(ii) \quad \frac{dy}{dx} = \begin{cases} \frac{2}{1+x^2} & |x| \neq 1 \\ \text{non existent} & |x| = 1 \end{cases}$$

(iii) It is bounded for all x



$$(d) \quad y = f(x) = \sin^{-1}(3x - 4x^3) = \begin{cases} -(\pi + 3 \sin^{-1} x) & \text{if } -1 \leq x < -\frac{1}{2} \\ 3 \sin^{-1} x & \text{if } -\frac{1}{2} \leq x \leq \frac{1}{2} \\ \pi - 3 \sin^{-1} x & \text{if } \frac{1}{2} < x \leq 1 \end{cases}$$

Important points :

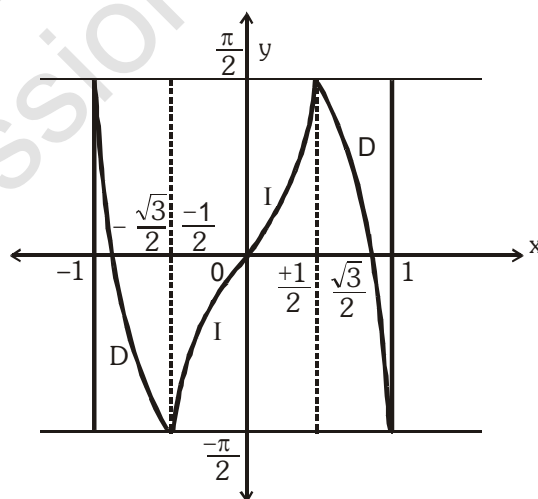
(i) Domain is $x \in [-1, 1]$ & range is $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

(ii) Continuous everywhere in its domain

(iii) Not derivable at $x = -\frac{1}{2}, \frac{1}{2}, 1, -1$

$$(iv) \quad \frac{dy}{dx} = \begin{cases} \frac{3}{\sqrt{1-x^2}} & \text{if } x \in \left(-\frac{1}{2}, \frac{1}{2}\right) \\ -\frac{3}{\sqrt{1-x^2}} & \text{if } x \in \left(-1, -\frac{1}{2}\right) \cup \left(\frac{1}{2}, 1\right) \end{cases}$$

(v) Increasing in $\left[-\frac{1}{2}, \frac{1}{2}\right]$ and Decreasing in $\left[-1, -\frac{1}{2}\right], \left[\frac{1}{2}, 1\right]$



$$(e) \quad y = f(x) = \cos^{-1} (4x^3 - 3x) = \begin{cases} 3 \cos^{-1} x - 2\pi & \text{if } -1 \leq x < -\frac{1}{2} \\ 2\pi - 3 \cos^{-1} x & \text{if } -\frac{1}{2} \leq x \leq \frac{1}{2} \\ 3 \cos^{-1} x & \text{if } \frac{1}{2} < x \leq 1 \end{cases}$$

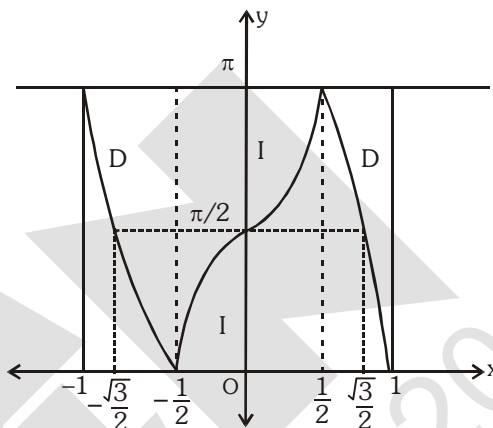
Important points :

- (i) Domain is $x \in [-1, 1]$ & range is $[0, \pi]$
- (ii) Continuous everywhere in its domain
- (iii) Not derivable at $x = -\frac{1}{2}, \frac{1}{2}, 1, -1$

$$(iv) \frac{dy}{dx} = \begin{cases} \frac{3}{\sqrt{1-x^2}} & \text{if } x \in \left(-\frac{1}{2}, \frac{1}{2}\right) \\ -\frac{3}{\sqrt{1-x^2}} & \text{if } x \in \left(-1, -\frac{1}{2}\right) \cup \left(\frac{1}{2}, 1\right) \end{cases}$$

- (v) Increasing in $\left[-\frac{1}{2}, \frac{1}{2}\right]$ &

Decreasing in $\left[-1, -\frac{1}{2}\right], \left[\frac{1}{2}, 1\right]$



GENERAL NOTE :

Concavity is decided by the sign of 2nd derivative as :

$$\frac{d^2y}{dx^2} \geq 0 \Rightarrow \text{Concave upwards} ; \quad \frac{d^2y}{dx^2} \leq 0 \Rightarrow \text{Concave downwards}$$

Illustration 17: Find the interval for which $f(x) = x^3 + x + 1$ is

- (i) concave upwards (ii) concave downwards.

Solution :

$$f(x) = x^3 + x + 1$$

$$f(x) = 3x^2 + 1$$

$$f'(x) = 6x$$

- (i) $f''(x) = 6x \geq 0 \Rightarrow$ Concave upwards
 $\Rightarrow x \in [0, \infty)$
- (ii) $f''(x) = 6x \leq 0 \Rightarrow$ Concave downwards
 $\Rightarrow x \in (-\infty, 0]$

Ans.

Illustration 18: If $x = a(t + \sin t)$ and $y = a(1 - \cos t)$, find $\frac{d^2y}{dx^2}$.

Solution : Here $x = a(t + \sin t)$ and $y = a(1 - \cos t)$
Differentiating both sides w.r.t. t , we get :

$$\frac{dx}{dt} = a(1 + \cos t) \quad \text{and} \quad \frac{dy}{dt} = a(\sin t)$$

$$\therefore \frac{dy}{dx} = \frac{a \sin t}{a(1 + \cos t)} = \frac{2 \sin \frac{t}{2} \cdot \cos \frac{t}{2}}{2 \cos^2 \frac{t}{2}} = \tan\left(\frac{t}{2}\right)$$

Again differentiating both sides, we get,

$$\frac{d^2y}{dx^2} = \sec^2\left(\frac{t}{2}\right) \cdot \frac{1}{2} \cdot \frac{dt}{dx} = \frac{1}{2} \sec^2\left(\frac{t}{2}\right) \cdot \frac{1}{a(1 + \cos t)} = \frac{1}{2a} \cdot \frac{\sec^2\left(\frac{t}{2}\right)}{2 \cos^2 \frac{t}{2}}$$

$$\text{Hence,} \quad \frac{d^2y}{dx^2} = \frac{1}{4a} \cdot \sec^4\left(\frac{t}{2}\right)$$

Ans.

Illustration 19: $y = f(x)$ and $x = g(y)$ are inverse functions of each other then express $g'(y)$ and $g''(y)$ in terms of derivative of $f(x)$.

Solution : $\frac{dy}{dx} = f'(x)$ and $\frac{dx}{dy} = g'(y)$

$$\Rightarrow g'(y) = \frac{1}{f'(x)} \quad \dots\dots\dots(i)$$

Again differentiating w.r.t. to y

$$g''(y) = \frac{d}{dy} \left(\frac{1}{f'(x)} \right) = \frac{d}{dx} \left(\frac{1}{f'(x)} \right) \cdot \frac{dx}{dy} = -\frac{f''(x)}{(f'(x))^2} \cdot \left(\frac{1}{f'(x)} \right)$$

$$\Rightarrow g''(y) = -\frac{f''(x)}{(f'(x))^3} \quad \dots\dots\dots(ii)$$

$$\text{Which can also be remembered as} \quad \frac{d^2x}{dy^2} = -\frac{\frac{d^2y}{dx^2}}{\left(\frac{dy}{dx}\right)^3}$$

Ans.

Do yourself : 8

- (i) If $y = xe^{x^2}$ then find y'' .
- (ii) Find y'' at $x = \pi/4$, if $y = x \tan x$.
- (iii) Prove that the function $y = e^x \sin x$ satisfies the relationship $y'' - 2y' + 2y = 0$.

12. DIFFERENTIATION OF DETERMINANTS :

If $F(x) = \begin{vmatrix} f(x) & g(x) & h(x) \\ l(x) & m(x) & n(x) \\ u(x) & v(x) & w(x) \end{vmatrix}$, where $f, g, h, l, m, n, u, v, w$ are differentiable functions of x then

$$F'(x) = \begin{vmatrix} f'(x) & g'(x) & h'(x) \\ l(x) & m(x) & n(x) \\ u(x) & v(x) & w(x) \end{vmatrix} + \begin{vmatrix} f(x) & g(x) & h(x) \\ l'(x) & m'(x) & n'(x) \\ u(x) & v(x) & w(x) \end{vmatrix} + \begin{vmatrix} f(x) & g(x) & h(x) \\ l(x) & m(x) & n(x) \\ u'(x) & v'(x) & w'(x) \end{vmatrix}$$

Note : Sometimes it is better to expand the determinant first & then differentiate.

Illustration 20: If $f(x) = \begin{vmatrix} x & x^2 & x^3 \\ 1 & 2x & 3x^2 \\ 0 & 2 & 6x \end{vmatrix}$, find $f'(x)$.

Solution : Here, $f(x) = \begin{vmatrix} x & x^2 & x^3 \\ 1 & 2x & 3x^2 \\ 0 & 2 & 6x \end{vmatrix}$

On differentiating, we get,

$$\Rightarrow f'(x) = \begin{vmatrix} \frac{d}{dx}(x) & \frac{d}{dx}(x^2) & \frac{d}{dx}(x^3) \\ 1 & 2x & 3x^2 \\ 0 & 2 & 6x \end{vmatrix} + \begin{vmatrix} x & x^2 & x^3 \\ \frac{d}{dx}(1) & \frac{d}{dx}(2x) & \frac{d}{dx}(3x^2) \\ 0 & 2 & 6x \end{vmatrix} + \begin{vmatrix} x & x^2 & x^3 \\ 1 & 2x & 3x^2 \\ \frac{d}{dx}(0) & \frac{d}{dx}(2) & \frac{d}{dx}(6x) \end{vmatrix}$$

$$\text{or } f'(x) = \begin{vmatrix} 1 & 2x & 3x^2 \\ 1 & 2x & 3x^2 \\ 0 & 2 & 6x \end{vmatrix} + \begin{vmatrix} x & x^2 & x^3 \\ 0 & 2 & 6x \\ 0 & 2 & 6x \end{vmatrix} + \begin{vmatrix} x & x^2 & x^3 \\ 1 & 2x & 3x^2 \\ 0 & 0 & 6 \end{vmatrix}$$

As we know if any two rows or columns are equal, then value of determinant is zero.

$$= 0 + 0 + \begin{vmatrix} x & x^2 & x^3 \\ 1 & 2x & 3x^2 \\ 0 & 0 & 6 \end{vmatrix} \quad \therefore f'(x) = 6(2x^2 - x^2)$$

Therefore, $f'(x) = 6x^2$

Ans.

Do yourself : 9

(i) If $f(x) = \begin{vmatrix} e^x & x^2 \\ \ln x & \sin x \end{vmatrix}$, then find $f'(1)$. (ii) If $f(x) = \begin{vmatrix} 2x & x^2 & x^3 \\ x^2 + 2x & 1 & 3x + 1 \\ 2x & 1 - 3x^2 & 5x \end{vmatrix}$, then find $f'(1)$.

13. L'HÔPITAL'S RULE :

- (a) This rule is applicable for the indeterminate forms of the type $\frac{0}{0}$, $\frac{\infty}{\infty}$. If the function $f(x)$ and $g(x)$ are differentiable in certain neighbourhood of the point 'a', except, may be, at the point 'a' itself and $g'(x) \neq 0$, and if

$$\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} g(x) = 0 \quad \text{or} \quad \lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} g(x) = \infty,$$

$$\text{then} \quad \lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$$

provided the limit $\lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$ exists (L'Hôpital's rule). The point 'a' may be either finite or improper ($+\infty$ or $-\infty$).

- (b) Indeterminate forms of the type $0 \cdot \infty$ or $\infty - \infty$ are reduced to forms of the type $\frac{0}{0}$ or $\frac{\infty}{\infty}$ by algebraic transformations.
- (c) Indeterminate forms of the type 1^∞ , ∞^0 or 0^0 are reduced to forms of the type $0 \times \infty$ by taking logarithms or by the transformation $[f(x)]^{\phi(x)} = e^{\phi(x) \cdot \ln f(x)}$.

Illustration 21: Evaluate $\lim_{x \rightarrow 0} |x|^{\sin x}$

Solution :

$$\begin{aligned} \lim_{x \rightarrow 0} |x|^{\sin x} &= \lim_{x \rightarrow 0} e^{\sin x \log_e |x|} = e^{\lim_{x \rightarrow 0} \frac{\log_e |x|}{\operatorname{cosec} x}} \\ &= e^{\lim_{x \rightarrow 0} \frac{1/x}{-\operatorname{cosec} x \cot x}} \quad (\text{applying L'Hôpital's rule}) \\ &= e^{\lim_{x \rightarrow 0} \frac{-\sin^2 x}{x \cos x}} = e^{\lim_{x \rightarrow 0} \left(\frac{\sin x}{x} \right)^2 \cdot \left(\frac{x}{\cos x} \right)} = e^{-(1)^2 \cdot (1)} = e^0 = 1 \end{aligned}$$

Ans.

Illustration 22: Solve $\lim_{x \rightarrow 0^+} \log_{\sin x} \sin 2x$.

Solution :

$$\begin{aligned} \text{Here } \lim_{x \rightarrow 0^+} \log_{\sin x} \sin 2x &= \lim_{x \rightarrow 0^+} \frac{\log \sin 2x}{\log \sin x} \quad \left(\frac{-\infty}{-\infty} \text{ form} \right) \\ &= \lim_{x \rightarrow 0^+} \frac{\frac{1}{\sin 2x} \cdot 2 \cos 2x}{\frac{1}{\sin x} \cdot \cos x} \quad \{\text{applying L'Hôpital's rule}\} \\ &= \lim_{x \rightarrow 0^+} \frac{\left(\frac{2x}{\sin(2x)} \right) \cos 2x}{\left(\frac{x}{\sin x} \right) \cos x} = \lim_{x \rightarrow 0^+} \frac{\cos 2x}{\cos x} = 1 \end{aligned}$$

Ans.

Illustration 23: Evaluate $\lim_{n \rightarrow \infty} \left(\frac{e^n}{\pi} \right)^{1/n}$.

Solution : Here, $A = \lim_{n \rightarrow \infty} \left(\frac{e^n}{\pi} \right)^{1/n}$ (∞^0 form)

$$\begin{aligned} \therefore \log A &= \lim_{n \rightarrow \infty} \frac{1}{n} \log \left(\frac{e^n}{\pi} \right) = \lim_{n \rightarrow \infty} \frac{n \log e - \log \pi}{n} \left(\frac{\infty}{\infty} \text{ form} \right) \\ &= \lim_{n \rightarrow \infty} \frac{\log e - 0}{1} \quad \{ \text{applying L'Hôpital's rule} \} \end{aligned}$$

$$\log A = 1 \Rightarrow A = e^1 \text{ or } \lim_{n \rightarrow \infty} \left(\frac{e^n}{\pi} \right)^{1/n} = e$$

Ans.

Do yourself : 10

- (i) Using L'Hôpital's rule find (a) $\lim_{x \rightarrow 0} \frac{\tan x - x}{x^3}$ (b) $\lim_{x \rightarrow 0} \frac{e^x - x - 1}{x^2}$
- (ii) Using L'Hôpital's rule verify that : (a) $\lim_{x \rightarrow 0} \frac{\sin x - \tan x}{x^3} = -\frac{1}{2}$ (b) $\lim_{x \rightarrow 0} \frac{\ln(1+x)}{x} = 1$

Miscellaneous Illustrations :

Illustration 24: Find second order derivative of $y = \sin x$ with respect to $z = e^x$.

Solution :

$$\begin{aligned} \frac{dy}{dz} &= \frac{dy/dx}{dz/dx} = \frac{\cos x}{e^x} \\ \Rightarrow \frac{d^2y}{dz^2} &= \frac{d}{dx} \left(\frac{\cos x}{e^x} \right) \cdot \frac{dx}{dz} = \frac{-e^x \sin x - \cos x e^x}{(e^x)^2} \cdot \frac{1}{e^x} \\ \Rightarrow \frac{d^2y}{dz^2} &= -\frac{(\sin x + \cos x)}{e^{2x}} \end{aligned}$$

Ans.

Illustration 25: Let a function f satisfies $f\left(\frac{x+y}{2}\right) = \frac{f(x)+f(y)}{2} \forall x, y \in \mathbb{R}$ and $f'(0) = a, f(0) = b$, then find $f(x)$ hence find $f''(x)$.

Solution :

$$f\left(\frac{x+y}{2}\right) = \frac{f(x)+f(y)}{2}$$

Diff. w.r.t. 'x'

$$f'\left(\frac{x+y}{2}\right) \cdot \frac{1}{2} = \frac{1}{2} f'(x) \quad \left(\because x \text{ \& } y \text{ are independent to each other, } \therefore \frac{dy}{dx} = 0 \right)$$

$$f'\left(\frac{x+y}{2}\right) = f'(x)$$

$$\text{Let } x = 0 \text{ \& } y = x \quad f'\left(\frac{x}{2}\right) = f'(0) = a$$

$$\Rightarrow f'(x) = a$$

$$\text{On integrating, we get } f(x) = ax + b \quad (\because f(0) = b)$$

$$\Rightarrow f''(x) = 0$$

Illustration 26: Prove that $\frac{1}{2^2} \sec^2 \frac{x}{2} + \frac{1}{2^4} \sec^2 \frac{x}{2^2} + \frac{1}{2^6} \sec^2 \frac{x}{2^3} + \dots \infty = \operatorname{cosec}^2 x - \frac{1}{x^2}$

Solution : Let $\cos \frac{x}{2} \cdot \cos \frac{x}{2^2} \cdot \cos \frac{x}{2^3} \dots \infty$

$$= \lim_{n \rightarrow \infty} \cos \frac{x}{2} \cdot \cos \frac{x}{2^2} \cdot \cos \frac{x}{2^3} \dots \cos \frac{x}{2^n}$$

$$\Rightarrow \lim_{n \rightarrow \infty} \frac{\sin x}{2^n \sin \frac{x}{2^n}} = \frac{\sin x}{x} \Rightarrow \cos \frac{x}{2} \cdot \cos \frac{x}{2^2} \cdot \cos \frac{x}{2^3} \dots \infty = \frac{\sin x}{x}$$

$$\Rightarrow \ln \left(\cos \frac{x}{2} \right) + \ln \left(\cos \frac{x}{2^2} \right) + \ln \left(\cos \frac{x}{2^3} \right) + \dots \infty = \ln \sin x - \ln x$$

Diff. w.r.t. x

$$-\left(\frac{1}{2} \tan \frac{x}{2} + \frac{1}{2^2} \tan \frac{x}{2^2} + \dots \infty \right) = \cot x - \frac{1}{x}$$

Diff. w.r.t. x again

$$\frac{1}{2^2} \sec^2 \frac{x}{2} + \frac{1}{2^4} \sec^2 \frac{x}{2^2} + \frac{1}{2^6} \sec^2 \frac{x}{2^3} + \dots \infty = \operatorname{cosec}^2 x - \frac{1}{x^2}$$

Hence proved

ANSWERS FOR DO YOURSELF

- | | |
|--|---|
| 1: (i) (a) $\frac{1}{x}$ | (b) $-\frac{1}{x^2}$ |
| 2: (i) (a) $3x^2 + 12x + 11$ | (b) $5e^{5x} \tan(x^2 + 2) + 2xe^{5x} \sec^2(x^2 + 2)$ |
| 3: (i) $x^x (\ln x + 1)$ | (ii) $y(1 + 2x + 3x^2 + 4x^3)$ |
| 4: (i) -1 | (ii) $\frac{1}{t}$ |
| | (iii) $2(x^{\ln x})(\ln x)$ |
| 5: (i) $\frac{\cos(x-y)-1}{\cos(x-y)+1}$ | (ii) $y' = -\left(\frac{2x + e^y}{xe^y + 1} \right), -1$ |
| 6: (i) (a) -6 | (b) 3 (c) -6 |
| 7: (i) D | |
| 8: (i) $y'' = 4y + 2xy'$ | (ii) $\pi + 4$ |
| 9: (i) $e(\sin 1 + \cos 1) - 1$ | (ii) 9 |
| 10: (i) (a) $\frac{1}{3}$ | (b) $\frac{1}{2}$ |

EXERCISE (O-1)

- If $y = \frac{1}{1+x^{\beta-\alpha} + x^{\gamma-\alpha}} + \frac{1}{1+x^{\alpha-\beta} + x^{\gamma-\beta}} + \frac{1}{1+x^{\alpha-\gamma} + x^{\beta-\gamma}}$, then $\frac{dy}{dx}$ is equal to-
 (A) 0 (B) 1
 (C) $(\alpha + \beta + \gamma)x^{\alpha+\beta+\gamma-1}$ (D) $\alpha\beta\gamma$
- If $f(x) = |\cos x|$, then $f\left(\frac{3\pi}{4}\right)$ is equal to -
 (A) $-\frac{1}{\sqrt{2}}$ (B) $\frac{1}{\sqrt{2}}$ (C) 1 (D) -1
- $\frac{d}{dx}(e^x \sin \sqrt{3} x)$ equals-
 (A) $e^x \sin(\sqrt{3} x + \pi/3)$ (B) $2e^x \sin(\sqrt{3} x + \pi/3)$
 (C) $\frac{1}{2} e^x \sin(\sqrt{3} x + \pi/3)$ (D) $\frac{1}{2} e^x \sin(\sqrt{3} x - \pi/3)$
- $\frac{d}{dx}(\ln \sin \sqrt{x})$ is equal to-
 (A) $\frac{\tan \sqrt{x}}{2\sqrt{x}}$ (B) $\frac{\cot \sqrt{x}}{\sqrt{x}}$ (C) $\frac{\cot \sqrt{x}}{2x}$ (D) $\frac{\cot \sqrt{x}}{2\sqrt{x}}$
- If $y = \sqrt{\frac{1-x}{1+x}}$, then $\frac{dy}{dx}$ equals-
 (A) $\frac{y}{1-x^2}$ (B) $\frac{y}{x^2-1}$ (C) $\frac{y}{1+x^2}$ (D) $\frac{y}{y^2-1}$
- If $y = \ln \left\{ \frac{x + \sqrt{a^2 + x^2}}{a} \right\}$, then the value of $\frac{dy}{dx}$ is-
 (A) $\sqrt{a^2 - x^2}$ (B) $a\sqrt{a^2 + x^2}$ (C) $\frac{1}{\sqrt{a^2 + x^2}}$ (D) $x\sqrt{a^2 + x^2}$
- If $x = y \ln(xy)$, then $\frac{dx}{dy}$ equals-
 (A) $\frac{y(x-y)}{x(x+y)}$ (B) $\frac{x(x+y)}{y(x-y)}$ (C) $\frac{y(x+y)}{x(x-y)}$ (D) $\frac{x(x-y)}{y(x+y)}$
- If $(\cos x)^y = (\sin y)^x$, then $\frac{dy}{dx}$ equals-
 (A) $\frac{\log \sin y - y \tan x}{\log \cos x + x \cot y}$ (B) $\frac{\log \sin y + y \tan x}{\log \cos x - x \cot y}$
 (C) $\frac{\log \sin y + y \tan x}{\log \cos x + x \cot y}$ (D) $\frac{\log \sin y + y \tan x}{\log \cos y - y \cot x}$

9. If $2^x + 2^y = 2^{x+y}$, then $\frac{dy}{dx}$ is equal to-

- (A) $\frac{2^x + 2^y}{2^x - 2^y}$ (B) $\frac{2^x + 2^y}{1 + 2^{x+y}}$ (C) $2^{x-y} \left(\frac{2^y - 1}{1 - 2^x} \right)$ (D) $\frac{2^x + y - 2^x}{2^y}$

10. If $x = a(t - \sin t)$, $y = a(1 + \cos t)$, then $\frac{dy}{dx}$ equals-

- (A) $-\tan \frac{t}{2}$ (B) $\cot \frac{t}{2}$ (C) $-\cot \frac{t}{2}$ (D) $\tan \frac{t}{2}$

11. The differential coefficient of $\sec^{-1} \left(\frac{1}{2x^2 - 1} \right)$ w.r.t. $\sqrt{1-x^2}$ is-

- (A) $1/x^2$ (B) $2/x^3$ (C) $x/2$ (D) $2/x$

12. If $x^3 - y^3 + 3xy^2 - 3x^2y + 1 = 0$, then at $(0, 1)$ $\frac{dy}{dx}$ equals-

- (A) 1 (B) -1 (C) 2 (D) 0

13. $\frac{d}{d\theta} \left(\tan^{-1} \left(\frac{1 - \cos \theta}{\sin \theta} \right) \right)$ equals, if $-\pi < \theta < \pi$ -

- (A) $1/2$ (B) 1 (C) $\sec \theta$ (D) $\operatorname{cosec} \theta$

14. $\frac{d}{dx} \cot^{-1} \left(\frac{1+x}{1-x} \right)$ is equal to, if $x > -1$

- (A) $\frac{1}{1+x^2}$ (B) $\frac{1}{1-x^2}$ (C) $-\frac{1}{1+x^2}$ (D) $\frac{-1}{1-x^2}$

15. If $y = \tan^{-1}(\cot x) + \cot^{-1}(\tan x)$, then $\frac{dy}{dx}$ is equal to-

- (A) 1 (B) 0 (C) -1 (D) -2

16. $\frac{d}{dx} \left(\tan^{-1} \left(\frac{\sqrt{x} - x}{1 + x^{3/2}} \right) \right)$ equals- ($x \geq 0$)

- (A) $\frac{1}{2\sqrt{x}(1+x)} - \frac{1}{1+x^2}$ (B) $\frac{1}{2\sqrt{x}(1+x)} + \frac{1}{1+x^2}$ (C) $\frac{1}{1+x} - \frac{1}{1+x^2}$ (D) $\frac{1}{1+x} + \frac{1}{1+x^2}$

17. If g is the inverse of f and $f'(x) = \frac{1}{1+x^3}$ then $g'(x)$ is equal to-

- (A) $1 + [g(x)]^3$ (B) $\frac{-1}{2(1+x^2)}$ (C) $\frac{1}{2(1+x^2)}$ (D) $\frac{1}{1 + [g(x)]^3}$

18. If $x^2 + y^2 = 1$, then-

- (A) $yy'' - 2(y')^2 + 1 = 0$ (B) $yy'' + (y')^2 + 1 = 0$
(C) $yy'' + (y')^2 - 1 = 0$ (D) $yy'' + 2(y')^2 + 1 = 0$

8. Suppose that $f(0) = 0$ and $f'(0) = 2$, and let $g(x) = f(-x + f(f(x)))$. The value of $g'(0)$ is equal to
 (A) 0 (B) 1 (C) 6 (D) 8
9. If f is differentiable in $(0, 6)$ & $f'(4) = 5$, then $\lim_{x \rightarrow 2} \frac{f(4) - f(x^2)}{2 - x} =$
 (A) 5 (B) $5/4$ (C) 10 (D) 20
10. If $y = x + e^x$ then $\frac{d^2x}{dy^2}$ is :
 (A) e^x (B) $-\frac{e^x}{(1+e^x)^3}$ (C) $-\frac{e^x}{(1+e^x)^2}$ (D) $\frac{-1}{(1+e^x)^3}$
11. If f is twice differentiable such that $f''(x) = -f(x)$, $f'(x) = g(x)$
 $h'(x) = [f(x)]^2 + [g(x)]^2$ and
 $h(0) = 2$, $h(1) = 4$
 then the equation $y = h(x)$ represents :
 (A) a curve of degree 2 (B) a curve passing through the origin
 (C) a straight line with slope 2 (D) a straight line with y intercept equal to -2 .
12. If $y = (A + Bx)e^{mx} + (m-1)^{-2}e^x$ then $\frac{d^2y}{dx^2} - 2m \frac{dy}{dx} + m^2y$ is equal to -
 (A) e^x (B) e^{mx} (C) e^{-mx} (D) $e^{(1-m)x}$
13. If $\frac{d^2x}{dy^2} \left(\frac{dy}{dx} \right)^3 + \frac{d^2y}{dx^2} = K$ then the value of K is equal to
 (A) 1 (B) -1 (C) 2 (D) 0
14. Let $f(x)$ be differentiable at $x = h$, then $\lim_{x \rightarrow h} \frac{(x+h)f(x) - 2hf(h)}{x-h}$ is equal to -
 (A) $f(h) + 2hf'(h)$ (B) $2f(h) + hf'(h)$ (C) $hf(h) + 2f'(h)$ (D) $hf(h) - 2f'(h)$
15. If $y = at^2 + 2bt + c$ and $t = ax^2 + 2bx + c$, then $\frac{d^3y}{dx^3}$ equals
 (A) $24a^2(at+b)$ (B) $24a(ax+b)^2$
 (C) $24a(at+b)^2$ (D) $24a^2(ax+b)$
16. Let $f(x) = x + \sin x$. Suppose g denotes the inverse function of f . The value of $g'\left(\frac{\pi}{4} + \frac{1}{\sqrt{2}}\right)$ has the value equal to
 (A) $\sqrt{2} - 1$ (B) $\frac{\sqrt{2} + 1}{\sqrt{2}}$
 (C) $2 - \sqrt{2}$ (D) $\sqrt{2} + 1$

Multiple objective type :

17. If $f(x) = (2x - 3\pi)^5 + \frac{4}{3}x + \cos x$ and g is the inverse function of f , then
 (A) $g'(2\pi) = \frac{7}{3}$ (B) $g'(2\pi) = \frac{3}{7}$ (C) $g''(2\pi) = \frac{7}{3}$ (D) $g''(2\pi) = 0$
18. If $f(x) = x \cdot |x|$, then its derivative is :
 (A) $2x$ (B) $-2x$ (C) $2|x|$ (D) $2x \operatorname{sgn} x$
19. If $y = \tan x \tan 2x \tan 3x$, ($\sin 12x \neq 0$) then $\frac{dy}{dx}$ has the value equal to
 (A) $3 \sec^2 3x \tan x \tan 2x + \sec^2 x \tan 2x \tan 3x + 2 \sec^2 2x \tan 3x \tan x$
 (B) $2y (\operatorname{cosec} 2x + 2 \operatorname{cosec} 4x + 3 \operatorname{cosec} 6x)$
 (C) $3 \sec^2 3x - 2 \sec^2 2x - \sec^2 x$
 (D) $\sec^2 x + 2 \sec^2 2x + 3 \sec^2 3x$
20. Which of the following statements are true ?
 (A) If $xe^{xy} = y + \sin^2 x$, then at $x = 0$, $(dy/dx) = 1$.
 (B) If $f(x) \equiv a_0 x^{2m+1} + a_1 x^{2m} + a_3 x^{2m-1} + \dots + a_{2m+1} = 0$ ($a_0 \neq 0$) is a polynomial equation with rational coefficients then the equation $f'(x) = 0$ must have a real root. ($m \in \mathbb{N}$).
 (C) If $(x - r)$ is a factor of the polynomial $f(x) = a_n x^n + a_{n-1} x^{n-1} + a_{n-2} x^{n-2} + \dots + a_0$ repeated m times where $1 \leq m \leq n$ then r is a root of the equation $f'(x) = 0$ repeated $(m-1)$ times.
 (D) If $y = \sin^{-1}(\cos \sin^{-1} x) + \cos^{-1}(\sin \cos^{-1} x)$ then $\frac{dy}{dx}$ is dependent on x .
21. Let $f(x) = \frac{\sqrt{x-2}\sqrt{x-1}}{\sqrt{x-1}-1} \cdot x$ then
 (A) $f'(10) = 1$ (B) $f'(3/2) = -1$ (C) domain of $f(x)$ is $x \geq 1$ (D) none
22. $\lim_{x \rightarrow \pi} \frac{x^\pi - \pi^x}{x^x - \pi^\pi}$ is equal to -
 (A) $\log_{\pi e} \left(\frac{e}{\pi} \right)$ (B) $\log_{\pi e} \left(\frac{\pi}{e} \right)$
 (C) $\tan(\cot^{-1}(\ln \pi) - \cot^{-1}(1))$ (D) $\tan(\tan^{-1}(1) - \tan^{-1}(\ln \pi))$
23. Let $P(x)$ be the polynomial $x^3 + ax^2 + bx + c$, where $a, b, c \in \mathbb{R}$. If $P(-3) = P(2) = 0$ and $P'(-3) < 0$, which of the following is a possible value of 'c' ?
 (A) -27 (B) -18 (C) -6 (D) -3
24. Two functions f & g have first & second derivatives at $x = 0$ & satisfy the relations,
 $f(0) = \frac{2}{g(0)}$, $f'(0) = 2g'(0) = 4g(0)$, $g''(0) = 5$ $f''(0) = 6f(0) = 3$ then -
 (A) if $h(x) = \frac{f(x)}{g(x)}$ then $h'(0) = \frac{15}{4}$ (B) if $k(x) = f(x) \cdot g(x) \sin x$ then $k'(0) = 2$
 (C) $\lim_{x \rightarrow 0} \frac{g'(x)}{f'(x)} = \frac{1}{2}$ (D) none

25. If $y = x^{(\ln x)^{\ln(\ln x)}}$, then $\frac{dy}{dx}$ is equal to :

(A) $\frac{y}{x} (\ln x^{\ln x - 1} + 2 \ln x \ln(\ln x))$

(B) $\frac{y}{x} (\ln x)^{\ln(\ln x)} (2 \ln(\ln x) + 1)$

(C) $\frac{y}{x \ln x} ((\ln x)^2 + 2 \ln(\ln x))$

(D) $\frac{y \ln y}{x \ln x} (2 \ln(\ln x) + 1)$

EXERCISE (S-1)

1. Let f, g and h are differentiable functions. If $f(0) = 1$; $g(0) = 2$; $h(0) = 3$ and the derivatives of their pair wise products at $x = 0$ are $(fg)'(0) = 6$; $(gh)'(0) = 4$ and $(hf)'(0) = 5$ then compute the value of $(fgh)'(0)$.

2. (a) If $y = (\cos x)^{\ln x} + (\ln x)^x$ find $\frac{dy}{dx}$

(b) If $y = e^{x^{e^x}} + e^{x^{+e}} + x^{e^{e^x}}$. Find $\frac{dy}{dx}$.

3. If $y = \frac{x^2}{2} + \frac{1}{2}x\sqrt{x^2+1} + \ln\sqrt{x+\sqrt{x^2+1}}$ prove that $2y = xy' + \ln y'$. Where ' denotes the derivative.

4. If $y = \ln(x^{e^x \cdot a^y})^{y^x}$ find $\frac{dy}{dx}$.

5. If $y = 1 + \frac{x_1}{x-x_1} + \frac{x_2 \cdot x}{(x-x_1)(x-x_2)} + \frac{x_3 \cdot x^2}{(x-x_1)(x-x_2)(x-x_3)} + \dots$ upto $(n+1)$ terms then prove

that $\frac{dy}{dx} = \frac{y}{x} \left[\frac{x_1}{x_1-x} + \frac{x_2}{x_2-x} + \frac{x_3}{x_3-x} + \dots + \frac{x_n}{x_n-x} \right]$

6. If $x = \operatorname{cosec} \theta - \sin \theta$; $y = \operatorname{cosec}^n \theta - \sin^n \theta$, then show that $(x^2 + 4) \left(\frac{dy}{dx} \right)^2 - n^2(y^2 + 4) = 0$.

7. If a curve is represented parametrically by the equations

$$x = \sin\left(t + \frac{7\pi}{12}\right) + \sin\left(t - \frac{\pi}{12}\right) + \sin\left(t + \frac{3\pi}{12}\right),$$

$$y = \cos\left(t + \frac{7\pi}{12}\right) + \cos\left(t - \frac{\pi}{12}\right) + \cos\left(t + \frac{3\pi}{12}\right)$$

then find the value of $\frac{d}{dt} \left(\frac{x}{y} - \frac{y}{x} \right)$ at $t = \frac{\pi}{8}$.

8. If $x = \frac{1 + \ln t}{t^2}$ and $y = \frac{3 + 2 \ln t}{t}$. Show that $y \frac{dy}{dx} = 2x \left(\frac{dy}{dx} \right)^2 + 1$.

9. Differentiate $\frac{\sqrt{1+x^2} + \sqrt{1-x^2}}{\sqrt{1+x^2} - \sqrt{1-x^2}}$ w.r.t. $\sqrt{1-x^4}$.

10. Let $g(x)$ be a polynomial, of degree one & $f(x)$ be defined by $f(x) = \begin{cases} g(x), & x \leq 0 \\ \left(\frac{1+x}{2+x} \right)^{1/x}, & x > 0 \end{cases}$

Find the continuous function $f(x)$ satisfying $f'(1) = f(-1)$.

11. If $\sqrt{1-x^6} + \sqrt{1-y^6} = a^3 \cdot (x^3 - y^3)$, prove that $\frac{dy}{dx} = \frac{x^2}{y^2} \sqrt{\frac{1-y^6}{1-x^6}}$.
12. If $y = x + \frac{1}{x + \frac{1}{x + \frac{1}{x + \dots}}}$, prove that $\frac{dy}{dx} = \frac{1}{2 - \frac{x}{x + \frac{1}{x + \dots}}}$.
13. Let $f(x) = x + \frac{1}{2x} + \frac{1}{2x} + \frac{1}{2x} + \dots \infty$. Compute the value of $f(100) \cdot f'(100)$.
14. Find the derivative with respect to x of the function :
 $(\log_{\cos x} \sin x) (\log_{\sin x} \cos x)^{-1} + \arcsin \frac{2x}{1+x^2}$ at $x = \frac{\pi}{4}$.
15. Suppose $f(x) = \tan(\sin^{-1}(2x))$
 (a) Find the domain and range of f .
 (b) Express $f(x)$ as an algebraic function of x .
 (c) Find $f'(1/4)$.
16. (a) Let $f(x) = x^2 - 4x - 3$, $x > 2$ and let g be the inverse of f . Find the value of g' where $f(x) = 2$.
 (b) Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined as $f(x) = x^3 + 3x^2 + 6x - 5 + 4e^{2x}$ and $g(x) = f^{-1}(x)$, then find $g'(-1)$.
 (c) Suppose f^{-1} is the inverse function of a differentiable function f and let $G(x) = \frac{1}{f^{-1}(x)}$.
 If $f(3) = 2$ and $f'(3) = \frac{1}{9}$, find $G'(2)$.
17. If $y = \tan^{-1} \frac{1}{x^2+x+1} + \tan^{-1} \frac{1}{x^2+3x+3} + \tan^{-1} \frac{1}{x^2+5x+7} + \tan^{-1} \frac{1}{x^2+7x+13} + \dots$ to n terms.
 Find dy/dx , expressing your answer in 2 terms.
18. If $y = \tan^{-1} \frac{u}{\sqrt{1-u^2}}$ & $x = \sec^{-1} \frac{1}{2u^2-1}$, $u \in \left(0, \frac{1}{\sqrt{2}}\right) \cup \left(\frac{1}{\sqrt{2}}, 1\right)$ prove that $2 \frac{dy}{dx} + 1 = 0$.
19. If $y = \cot^{-1} \frac{\sqrt{1+\sin x} + \sqrt{1-\sin x}}{\sqrt{1+\sin x} - \sqrt{1-\sin x}}$, find $\frac{dy}{dx}$ if $x \in \left(0, \frac{\pi}{2}\right) \cup \left(\frac{\pi}{2}, \pi\right)$.
20. If $f(x) = \begin{vmatrix} (x-a)^4 & (x-a)^3 & 1 \\ (x-b)^4 & (x-b)^3 & 1 \\ (x-c)^4 & (x-c)^3 & 1 \end{vmatrix}$, then $f'(x) = \lambda \cdot \begin{vmatrix} (x-a)^4 & (x-a)^2 & 1 \\ (x-b)^4 & (x-b)^2 & 1 \\ (x-c)^4 & (x-c)^2 & 1 \end{vmatrix}$. Find the value of λ .
21. (a) If $y = y(x)$ and it follows the relation $e^{xy} + y \cos x = 2$, then find (i) $y'(0)$ and (ii) $y''(0)$.
 (b) A twice differentiable function $f(x)$ is defined for all real numbers and satisfies the following conditions
 $f(0) = 2$; $f'(0) = -5$ and $f''(0) = 3$.
 The function $g(x)$ is defined by $g(x) = e^{ax} + f(x) \forall x \in \mathbb{R}$, where 'a' is any constant.
 If $g'(0) + g''(0) = 0$. Find the value(s) of 'a'.
22. If $x = 2\cos t - \cos 2t$ & $y = 2\sin t - \sin 2t$, find the value of (d^2y/dx^2) when $t = (\pi/2)$.
23. Find the value of the expression $y^3 \frac{d^2y}{dx^2}$ on the ellipse $3x^2 + 4y^2 = 12$.

24. If $f(x) = \begin{vmatrix} \cos(x+x^2) & \sin(x+x^2) & -\cos(x+x^2) \\ \sin(x-x^2) & \cos(x-x^2) & \sin(x-x^2) \\ \sin 2x & 0 & \sin 2x^2 \end{vmatrix}$ then find $f'(x)$.

25. Let $P(x)$ be a polynomial of degree 4 such that $P(1) = P(3) = P(5) = P'(7) = 0$. If the real number $x \neq 1, 3, 5$ is such that $P(x) = 0$ can be expressed as $x = p/q$ where 'p' and 'q' are relatively prime, then $(p + q)$ equals.

EXERCISE (S-2)

- Find a polynomial function $f(x)$ such that $f(2x) = f'(x) f''(x)$.
- If $2x = y^{\frac{1}{5}} + y^{-\frac{1}{5}}$ then $(x^2 - 1) \frac{d^2y}{dx^2} + x \frac{dy}{dx} = ky$, then find the value of 'k'.
- Let $y = x \sin kx$. Find the possible value of k for which the differential equation $\frac{d^2y}{dx^2} + y = 2k \cos kx$ holds true for all $x \in \mathbb{R}$.
- Prove that if $|a_1 \sin x + a_2 \sin 2x + \dots + a_n \sin nx| \leq |\sin x|$ for $x \in \mathbb{R}$, then $|a_1 + 2a_2 + 3a_3 + \dots + na_n| \leq 1$.
- The function $f: \mathbb{R} \rightarrow \mathbb{R}$ satisfies $f(x^2) \cdot f''(x) = f'(x) \cdot f'(x^2)$ for all real x . Given that $f(1) = 1$ and $f'''(1) = 8$, compute the value of $f'(1) + f''(1)$.
- Show that $R = \frac{[1 + (dy/dx)^2]^{\frac{3}{2}}}{d^2y/dx^2}$ can be reduced to the form $R^{2/3} = \frac{1}{(d^2y/dx^2)^{2/3}} + \frac{1}{(d^2x/dy^2)^{2/3}}$.
- Let $f(x) = \frac{\sin x}{x}$ if $x \neq 0$ and $f(0) = 1$. Define the function $f'(x)$ for all x and find $f''(0)$ if it exist.
- Suppose f and g are two functions such that $f, g: \mathbb{R} \rightarrow \mathbb{R}$,
 $f(x) = \ln(1 + \sqrt{1+x^2})$ and $g(x) = \ln(x + \sqrt{1+x^2})$
then find the value of $x e^{g(x)} \left(f\left(\frac{1}{x}\right) \right)' + g'(x)$ at $x = 1$.
- Let $\frac{f(x+y) - f(x)}{2} = \frac{f(y) - a}{2} + xy$ for all real x and y . If $f(x)$ is differentiable and $f'(0)$ exists for all real permissible values of 'a' and is equal to $\sqrt{5a-1-a^2}$. Prove that $f(x)$ is positive for all real x .
- If $y = \tan^{-1} \frac{x}{1+\sqrt{1-x^2}} + \sin \left(2 \tan^{-1} \sqrt{\frac{1-x}{1+x}} \right)$, then find $\frac{dy}{dx}$ for $x \in (-1, 1)$.
- If $f: \mathbb{R} \rightarrow \mathbb{R}$ is a function such that $f(x) = x^3 + x^2 f'(1) + x f''(2) + f'''(3)$ for all $x \in \mathbb{R}$, then prove that $f(2) = f(1) - f(0)$.
- If α be a repeated root of a quadratic equation $f(x) = 0$ & $A(x), B(x), C(x)$ be the polynomials of degree

3, 4 & 5 respectively, then show that $\begin{vmatrix} A(x) & B(x) & C(x) \\ A(\alpha) & B(\alpha) & C(\alpha) \\ A'(\alpha) & B'(\alpha) & C'(\alpha) \end{vmatrix}$ is divisible by $f(x)$, where dash denotes

the derivative.

EXERCISE (JM)

1. Let y be an implicit function of x defined by $x^{2x} - 2x^x \cot y - 1 = 0$. Then $y'(1)$ equals :-
[AIEEE-2009]
(1) $\log 2$ (2) $-\log 2$ (3) -1 (4) 1
2. Let $f: (-1, 1) \rightarrow \mathbb{R}$ be a differentiable function with $f(0) = -1$ and $f'(0) = 1$. Let $g(x) = [f(2f(x) + 2)]^2$. Then $g'(0)$:-
[AIEEE-2010]
(1) 4 (2) -4 (3) 0 (4) -2
3. $\frac{d^2x}{dy^2}$ equals :-
[AIEEE-2011]
(1) $\left(\frac{d^2y}{dx^2}\right)\left(\frac{dy}{dx}\right)^{-2}$ (2) $-\left(\frac{d^2y}{dx^2}\right)\left(\frac{dy}{dx}\right)^{-3}$ (3) $\left(\frac{d^2y}{dx^2}\right)^{-1}$ (4) $-\left(\frac{d^2y}{dx^2}\right)^{-1}\left(\frac{dy}{dx}\right)^{-3}$
4. If $y = \sec(\tan^{-1}x)$, then $\frac{dy}{dx}$ at $x = 1$ is equal to :
[JEE(Main)-2013]
(1) $\frac{1}{\sqrt{2}}$ (2) $\frac{1}{2}$ (3) 1 (4) $\sqrt{2}$
5. If g is the inverse of a function f and $f'(x) = \frac{1}{1+x^5}$, then $g'(x)$ is equal to :
[JEE(Main)-2014]
(1) $1 + x^5$ (2) $5x^4$ (3) $\frac{1}{1+\{g(x)\}^5}$ (4) $1+\{g(x)\}^5$
6. If for $x \in \left(0, \frac{1}{4}\right)$, the derivative of $\tan^{-1}\left(\frac{6x\sqrt{x}}{1-9x^3}\right)$ is $\sqrt{x} \cdot g(x)$, then $g(x)$ equals :-
[JEE(Main)-2017]
(1) $\frac{3}{1+9x^3}$ (2) $\frac{9}{1+9x^3}$ (3) $\frac{3x\sqrt{x}}{1-9x^3}$ (4) $\frac{3x}{1-9x^3}$

EXERCISE (JA)

1. Let $g(x) = \ln f(x)$ where $f(x)$ is a twice differentiable positive function on $(0, \infty)$ such that $f(x+1) = x f(x)$. Then for $N = 1, 2, 3, \dots$,

$$g''\left(N + \frac{1}{2}\right) - g''\left(\frac{1}{2}\right) =$$

[JEE 2008, 3]

(A) $-4 \left\{ 1 + \frac{1}{9} + \frac{1}{25} + \dots + \frac{1}{(2N-1)^2} \right\}$

(B) $4 \left\{ 1 + \frac{1}{9} + \frac{1}{25} + \dots + \frac{1}{(2N-1)^2} \right\}$

(C) $-4 \left\{ 1 + \frac{1}{9} + \frac{1}{25} + \dots + \frac{1}{(2N+1)^2} \right\}$

(D) $4 \left\{ 1 + \frac{1}{9} + \frac{1}{25} + \dots + \frac{1}{(2N+1)^2} \right\}$

2. Let f and g be real valued functions defined on interval $(-1, 1)$ such that $g''(x)$ is continuous, $g(0) \neq 0$, $g'(0) = 0$, $g''(0) \neq 0$, and $f(x) = g(x) \sin x$.

Statement-1 : $\lim_{x \rightarrow 0} [g(x) \cot x - g(0) \operatorname{cosec} x] = f''(0)$

and

Statement-2 : $f'(0) = g(0)$

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation of statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

[JEE 2008, 3]

3. If the function $f(x) = x^3 + e^{\frac{x}{2}}$ and $g(x) = f^{-1}(x)$, then the value of $g'(1)$ is

[JEE 2009, 4]

4. Let $f(\theta) = \sin \left(\tan^{-1} \left(\frac{\sin \theta}{\sqrt{\cos 2\theta}} \right) \right)$, where $-\frac{\pi}{4} < \theta < \frac{\pi}{4}$. Then the value of $\frac{d}{d(\tan \theta)}(f(\theta))$ is

[JEE 2011, 4]

5. The slope of the tangent to the curve $(y - x^5)^2 = x(1 + x^2)^2$ at the point (1,3) is

[JEE(Advanced)-2014, 3]

6. Let $f : \mathbb{R} \rightarrow \mathbb{R}$, $g : \mathbb{R} \rightarrow \mathbb{R}$ and $h : \mathbb{R} \rightarrow \mathbb{R}$ be differentiable functions such that $f(x) = x^3 + 3x + 2$, $g(f(x)) = x$ and $h(g(g(x))) = x$ for all $x \in \mathbb{R}$. Then-

[JEE(Advanced)-2016, 4(-2)]

(A) $g'(2) = \frac{1}{15}$

(B) $h'(1) = 666$

(C) $h(0) = 16$

(D) $h(g(3)) = 36$

ANSWER KEY

EXERCISE (O-1)

1. A 2. B 3. B 4. D 5. B 6. C 7. B 8. B
 9. C 10. C 11. D 12. A 13. A 14. C 15. D 16. A
 17. A 18. B 19. A 20. B

EXERCISE (O-2)

1. A 2. D 3. A 4. A 5. D 6. A 7. D 8. C
 9. D 10. B 11. C 12. A 13. D 14. A 15. D 16. C
 17. B,D 18. C,D 19. A,B,C 20. A,C 21. A,B 22. A,C,D 23. A
 24. A,B,C 25. B,D

EXERCISE (S-1)

1. 16 2. (a) $Dy = (\cos x)^{\ell nx} \left[\frac{\ell n(\cos x)}{x} - \tan x \ell nx \right] + (\ell nx)^x \left[\frac{1}{\ell nx} + \ell n(\ell nx) \right];$
 (b) $\frac{dy}{dx} = e^{x e^x} \cdot x^{e^x} \left[\frac{e^x}{x} + e^x \ell nx \right] + e^{x e^x} x^{e^x-1} x^{e^x} [1 + e \ell nx] + x^{e^x} e^{e^x} \left[\frac{1}{x} + e^x \ell nx \right]$
 4. $\frac{y}{x} \cdot \frac{x \ell nx + x \ell nx \cdot \ell ny + 1}{\ell nx(1-x-y \ell na)}$ 7. 8 9. $\frac{1+\sqrt{1-x^4}}{x^6}$ 10. $f(x) = \begin{cases} -\frac{2}{3} \left[\frac{1}{6} + \ell n \frac{3}{2} \right] x & \text{if } x \leq 0 \\ \left(\frac{1+x}{2+x} \right)^{1/x} & \text{if } x > 0 \end{cases}$
 13. 100 14. $\frac{32}{16+\pi^2} - \frac{8}{\ell n 2}$ 15. (a) $\left(-\frac{1}{2}, \frac{1}{2} \right), (-\infty, \infty);$ (b) $f(x) = \frac{2x}{\sqrt{1-4x^2}};$ (c) $\frac{16\sqrt{3}}{9}$
 16. (a) $1/6;$ (b) $\frac{1}{14};$ (c) -1 17. $\frac{1}{1+(x+n)^2} - \frac{1}{1+x^2}$ 19. $\frac{1}{2}$ or $-\frac{1}{2}$ 20. 3
 21. (a) (i) $y'(0) = -1;$ (ii) $y''(0) = 2;$ (b) $a = 1, -2$ 22. $\frac{-3}{2}$ 23. $\frac{-9}{4}$ 24. $2(1+2x) \cdot \cos 2(x+x^2)$ 25. 100

EXERCISE (S-2)

1. $\frac{4x^3}{9}$ 2. 25 3. $k=1, -1$ or 0 5. 6 7. $f'(x) = \begin{cases} \frac{x \cos x - \sin x}{x^2} & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}; f''(0) = -\frac{1}{3}$
 8. Zero 10. $\frac{1-2x}{2\sqrt{1-x^2}}$

EXERCISE (S-3)

1. $\frac{5}{6}$ 2. $\frac{1}{6}$ 3. $-\frac{1}{3}$ 4. $-\frac{1}{2}$ 5. 1000 6. 11
 7. f is cont. but not derivable at $x=0$ 8. (a) $a_1^{p_1} \cdot a_2^{p_2} \dots a_n^{p_n};$ (b) $a_1;$ (c) a_n 10. (A) S; (B) P; (C) R

EXERCISE (JM)

1. 3 2. 2 3. 2 4. 1 5. 4 6. 2

EXERCISE (JA)

1. A 2. A 3. 2 4. 1 5. 8 6. B,C

NURTURE COURSE
QUADRATIC EQUATION

CONTENTS

QUADRATIC EQUATION

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JEE(Advanced) Syllabus :

Quadratic equations with real coefficients, relations between roots and coefficients, formation of quadratic equations with given roots, symmetric functions of roots.

JEE(Main) Syllabus :

Quadratic equations in real and complex number system and their solutions.

Relation between roots and co-efficients, nature of roots, formation of quadratic equations with given roots.

QUADRATIC EQUATION

1. INTRODUCTION :

The algebraic expression of the form $ax^2 + bx + c$, $a \neq 0$ is called a quadratic expression, because the highest order term in it is of second degree. Quadratic equation means, $ax^2 + bx + c = 0$. In general whenever one says zeroes of the expression $ax^2 + bx + c$, it implies roots of the equation $ax^2 + bx + c = 0$, unless specified otherwise.

A quadratic equation has exactly two roots which may be real (equal or unequal) or imaginary.

2. SOLUTION OF QUADRATIC EQUATION & RELATION BETWEEN ROOTS & CO-EFFICIENTS :

(a) The general form of quadratic equation is $ax^2 + bx + c = 0$, $a \neq 0$.

The roots can be found in following manner :

$$a \left(x^2 + \frac{b}{a}x + \frac{c}{a} \right) = 0 \Rightarrow \left(x + \frac{b}{2a} \right)^2 + \frac{c}{a} - \frac{b^2}{4a^2} = 0$$

$$\left(x + \frac{b}{2a} \right)^2 = \frac{b^2}{4a^2} - \frac{c}{a} \Rightarrow \boxed{x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}}$$

This expression can be directly used to find the two roots of a quadratic equation.

(b) The expression $b^2 - 4ac \equiv D$ is called the discriminant of the quadratic equation.

(c) If α & β are the roots of the quadratic equation $ax^2 + bx + c = 0$, then :

$$(i) \alpha + \beta = -b/a \quad (ii) \alpha\beta = c/a \quad (iii) |\alpha - \beta| = \sqrt{D}/|a|$$

(d) A quadratic equation whose roots are α & β is $(x - \alpha)(x - \beta) = 0$ i.e.

$$x^2 - (\alpha + \beta)x + \alpha\beta = 0 \text{ i.e. } x^2 - (\text{sum of roots})x + \text{product of roots} = 0.$$

Illustration 1 : If α, β are the roots of a quadratic equation $x^2 - 3x + 5 = 0$, then the equation whose roots are $(\alpha^2 - 3\alpha + 7)$ and $(\beta^2 - 3\beta + 7)$ is -

$$(A) x^2 + 4x + 1 = 0 \quad (B) x^2 - 4x + 4 = 0 \quad (C) x^2 - 4x - 1 = 0 \quad (D) x^2 + 2x + 3 = 0$$

Solution : Since α, β are the roots of equation $x^2 - 3x + 5 = 0$

$$\text{So } \alpha^2 - 3\alpha + 5 = 0$$

$$\beta^2 - 3\beta + 5 = 0$$

$$\therefore \alpha^2 - 3\alpha = -5$$

$$\beta^2 - 3\beta = -5$$

$$\text{Putting in } (\alpha^2 - 3\alpha + 7) \text{ \& } (\beta^2 - 3\beta + 7) \quad \dots\dots\dots(i)$$

$$-5 + 7, -5 + 7$$

$$\therefore 2 \text{ and } 2 \text{ are the roots.}$$

$$\therefore \text{The required equation is}$$

$$x^2 - 4x + 4 = 0.$$

Ans. (B)

Illustration 2 : If α and β are the roots of $ax^2 + bx + c = 0$, find the value of $(a\alpha + b)^{-2} + (a\beta + b)^{-2}$.

Solution : We know that $\alpha + \beta = -\frac{b}{a}$ & $\alpha\beta = \frac{c}{a}$

$$\begin{aligned} (a\alpha + b)^{-2} + (a\beta + b)^{-2} &= \frac{1}{(a\alpha + b)^2} + \frac{1}{(a\beta + b)^2} \\ &= \frac{a^2\beta^2 + b^2 + 2ab\beta + a^2\alpha^2 + b^2 + 2ab\alpha}{(a^2\alpha\beta + ba\beta + ba\alpha + b^2)^2} = \frac{a^2(\alpha^2 + \beta^2) + 2ab(\alpha + \beta) + 2b^2}{(a^2\alpha\beta + ab(\alpha + \beta) + b^2)^2} \\ (\alpha^2 + \beta^2 \text{ can always be written as } (\alpha + \beta)^2 - 2\alpha\beta) \\ &= \frac{a^2[(\alpha + \beta)^2 - 2\alpha\beta] + 2ab(\alpha + \beta) + 2b^2}{(a^2\alpha\beta + ab(\alpha + \beta) + b^2)^2} = \frac{a^2\left[\frac{b^2 - 2ac}{a^2}\right] + 2ab\left(-\frac{b}{a}\right) + 2b^2}{\left(a^2\frac{c}{a} + ab\left(-\frac{b}{a}\right) + b^2\right)^2} = \frac{b^2 - 2ac}{a^2c^2} \end{aligned}$$

Alternatively :

If α & β are roots of $ax^2 + bx + c = 0$

then, $a\alpha^2 + b\alpha + c = 0$

$$\Rightarrow a\alpha + b = -\frac{c}{\alpha}$$

$$\text{same as } a\beta + b = -\frac{c}{\beta}$$

$$\begin{aligned} \therefore (a\alpha + b)^{-2} &= (\alpha\beta + b)^{-2} = \frac{\alpha^2}{c^2} + \frac{\beta^2}{c^2} \\ &= \frac{(\alpha + \beta)^2 - 2\alpha\beta}{c^2} \\ &= \frac{(-b/a)^2 - 2(c/a)}{c^2} \\ &= \frac{b^2 - 2ac}{a^2c^2} \end{aligned}$$

Do yourself - 1 :

(i) Find the roots of following equations :

(a) $x^2 + 3x + 2 = 0$

(b) $x^2 - 8x + 16 = 0$

(c) $x^2 - 2x - 1 = 0$

(ii) Find the roots of the equation $a(x^2 + 1) - (a^2 + 1)x = 0$, where $a \neq 0$.

(iii) Solve : $\frac{6-x}{x^2-4} = 2 + \frac{x}{x+2}$

(iv) If the roots of $4x^2 + 5k = (5k + 1)x$ differ by unity, then find the values of k .

3. NATURE OF ROOTS :

(a) Consider the quadratic equation $ax^2 + bx + c = 0$ where $a, b, c \in \mathbb{R}$ & $a \neq 0$ then ;

$$x = \frac{-b \pm \sqrt{D}}{2a}$$

- (i) $D > 0 \Leftrightarrow$ roots are real & distinct (unequal).
- (ii) $D = 0 \Leftrightarrow$ roots are real & coincident (equal)
- (iii) $D < 0 \Leftrightarrow$ roots are imaginary.
- (iv) If $p + iq$ is one root of a quadratic equation, then the other root must be the conjugate $p - iq$ & vice versa. ($p, q \in \mathbb{R}$ & $i = \sqrt{-1}$).

(b) Consider the quadratic equation $ax^2 + bx + c = 0$ where $a, b, c \in \mathbb{Q}$ & $a \neq 0$ then ;

- (i) If D is a perfect square, then roots are rational.
- (ii) If $\alpha = p + \sqrt{q}$ is one root in this case, (where p is rational & $\sqrt{q}$ is a surd) then other root will be $p - \sqrt{q}$.

Illustration 3 : If the coefficient of the quadratic equation are rational & the coefficient of x^2 is 1, then find the equation one of whose roots is $\tan \frac{\pi}{8}$.

Solution : We know that $\tan \frac{\pi}{8} = \sqrt{2} - 1$

Irrational roots always occur in conjugational pairs.

Hence if one root is $(-1 + \sqrt{2})$, the other root will be $(-1 - \sqrt{2})$. Equation is

$$(x - (-1 + \sqrt{2}))(x - (-1 - \sqrt{2})) = 0 \quad \Rightarrow \quad x^2 + 2x - 1 = 0$$

Illustration 4 : Find all the integral values of a for which the quadratic equation $(x - a)(x - 10) + 1 = 0$ has integral roots.

Solution : Here the equation is $x^2 - (a + 10)x + 10a + 1 = 0$. Since integral roots will always be rational it means D should be a perfect square.

$$\text{From (i) } D = a^2 - 20a + 96.$$

$$\Rightarrow D = (a - 10)^2 - 4 \Rightarrow 4 = (a - 10)^2 - D$$

If D is a perfect square it means we want difference of two perfect square as 4 which is possible only when $(a - 10)^2 = 4$ and $D = 0$.

$$\Rightarrow (a - 10) = \pm 2 \quad \Rightarrow \quad a = 12, 8$$

Ans.

Do yourself - 2 :

- (i) If $2 + \sqrt{3}$ is a root of the equation $x^2 + bx + c = 0$, where $b, c \in \mathbb{Q}$, find b, c .
- (ii) For the following equations, find the nature of the roots (real & distinct, real & coincident or imaginary).
- (a) $x^2 - 6x + 10 = 0$
- (b) $x^2 - (7 + \sqrt{3})x + 6(1 + \sqrt{3}) = 0$
- (c) $4x^2 + 28x + 49 = 0$
- (iii) If ℓ, m are real and $\ell \neq m$, then show that the roots of $(\ell - m)x^2 - 5(\ell + m)x - 2(\ell - m) = 0$ are real and unequal.

4. ROOTS UNDER PARTICULAR CASES :

Let the quadratic equation $ax^2 + bx + c = 0$ has real roots and

- (a) If $b = 0 \Rightarrow$ roots are equal in magnitude but opposite in sign
- (b) If $c = 0 \Rightarrow$ one root is zero other is $-b/a$
- (c) If $a = c \Rightarrow$ roots are reciprocal to each other
- (d) If $\begin{cases} a > 0, c < 0 \\ a < 0, c > 0 \end{cases} \Rightarrow$ roots are of opposite signs
- (e) If $\begin{cases} a > 0, b > 0, c > 0 \\ a < 0, b < 0, c < 0 \end{cases} \Rightarrow$ both roots are negative.
- (f) If $\begin{cases} a > 0, b < 0, c > 0 \\ a < 0, b > 0, c < 0 \end{cases} \Rightarrow$ both roots are positive.
- (g) If sign of $a = \text{sign of } b \neq \text{sign of } c \Rightarrow$ Greater root in magnitude is negative.
- (h) If sign of $b = \text{sign of } c \neq \text{sign of } a \Rightarrow$ Greater root in magnitude is positive.
- (i) If $a + b + c = 0 \Rightarrow$ one root is 1 and second root is c/a or $(-b-a)/a$.

Illustration 5 : If equation $\frac{x^2 - bx}{ax - c} = \frac{k-1}{k+1}$ has roots equal in magnitude & opposite in sign, then the value of k is -

- (A) $\frac{a+b}{a-b}$ (B) $\frac{a-b}{a+b}$ (C) $\frac{a}{b} + 1$ (D) $\frac{a}{b} - 1$

Solution : Let the roots are α & $-\alpha$.

given equation is

$$(x^2 - bx)(k+1) = (k-1)(ax - c) \quad \{\text{Considering, } x \neq c/a \text{ \& } k \neq -1\}$$

$$\Rightarrow x^2(k+1) - bx(k+1) = ax(k-1) - c(k-1)$$

$$\Rightarrow x^2(k+1) - bx(k+1) - ax(k-1) + c(k-1) = 0$$

$$\text{Now sum of roots} = 0 \quad (\because \alpha - \alpha = 0)$$

$$\therefore b(k+1) + a(k-1) = 0 \Rightarrow k = \frac{a-b}{a+b}$$

Ans. (B)

***Illustration 6 :** If roots of the equation $(a - b)x^2 + (c - a)x + (b - c) = 0$ are equal, then a, b, c are in

- (A) A.P. (B) H.P. (C) G.P. (D) none of these

Solution : $(a - b)x^2 + (c - a)x + (b - c) = 0$

As roots are equal so

$$B^2 - 4AC = 0 \Rightarrow (c - a)^2 - 4(a - b)(b - c) = 0 \Rightarrow (c - a)^2 - 4ab + 4b^2 + 4ac - 4bc = 0$$

$$\Rightarrow (c - a)^2 + 4ac - 4b(c + a) + 4b^2 = 0 \Rightarrow (c + a)^2 - 2 \cdot (2b)(c + a) + (2b)^2 = 0$$

$$\Rightarrow [c + a - 2b]^2 = 0 \Rightarrow c + a - 2b = 0 \Rightarrow c + a = 2b$$

Hence a, b, c are in A. P.

Alternative method :

$\therefore$ Sum of the coefficients = 0

Hence one root is 1 and other root is $\frac{b-c}{a-b}$.

Given that both roots are equal, so

$$1 = \frac{b-c}{a-b} \Rightarrow a - b = b - c \Rightarrow 2b = a + c$$

Hence a, b, c are in A.P.

Ans. (A)

Do yourself - 3 :

(i) Consider $f(x) = x^2 + bx + c$.

(a) Find c if $x = 0$ is a root of $f(x) = 0$.

(b) Find c if $\alpha, \frac{1}{\alpha}$ are roots of $f(x) = 0$.

(c) Comment on sign of b & c, if $\alpha < 0 < \beta$ & $|\beta| > |\alpha|$, where α, β are roots of $f(x) = 0$.

5. IDENTITY :

An equation which is true for every value of the variable within the domain is called an identity, for example : $5(a - 3) = 5a - 15$, $(a + b)^2 = a^2 + b^2 + 2ab$ for all $a, b \in \mathbb{R}$.

Note : A quadratic equation cannot have three or more roots & if it has, it becomes an identity.

If $ax^2 + bx + c = 0$ is an identity $\Leftrightarrow a = b = c = 0$

Illustration 7 : If the equation $(\lambda^2 - 5\lambda + 6)x^2 + (\lambda^2 - 3\lambda + 2)x + (\lambda^2 - 4) = 0$ has more than two roots, then find the value of λ ?

Solution : As the equation has more than two roots so it becomes an identity. Hence

$$\lambda^2 - 5\lambda + 6 = 0 \Rightarrow \lambda = 2, 3$$

$$\text{and } \lambda^2 - 3\lambda + 2 = 0 \Rightarrow \lambda = 1, 2$$

$$\text{and } \lambda^2 - 4 = 0 \Rightarrow \lambda = 2, -2$$

So $\lambda = 2$

Ans. $\lambda = 2$

6. COMMON ROOTS OF TWO QUADRATIC EQUATIONS :**(a)** Only one common root.Let α be the common root of $ax^2 + bx + c = 0$ & $a'x^2 + b'x + c' = 0$ then

$$a\alpha^2 + b\alpha + c = 0 \text{ \& \; } a'\alpha^2 + b'\alpha + c' = 0. \text{ By Cramer's Rule } \frac{\alpha^2}{bc' - b'c} = \frac{\alpha}{a'c - ac'} = \frac{1}{ab' - a'b}$$

$$\text{Therefore, } \alpha = \frac{ca' - c'a}{ab' - a'b} = \frac{bc' - b'c}{a'c - ac'}$$

So the condition for a common root is $(ca' - c'a)^2 = (ab' - a'b)(bc' - b'c)$.**(b)** If both roots are same then $\frac{a}{a'} = \frac{b}{b'} = \frac{c}{c'}$.**Illustration 8 :** Find p and q such that $px^2 + 5x + 2 = 0$ and $3x^2 + 10x + q = 0$ have both roots in common.**Solution :** $a_1 = p, b_1 = 5, c_1 = 2$

$$a_2 = 3, b_2 = 10, c_2 = q$$

We know that :

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2} \Rightarrow \frac{p}{3} = \frac{5}{10} = \frac{2}{q} \Rightarrow p = \frac{3}{2}; q = 4$$

***Illustration 9 :** Find the possible value(s) of a for which the equations $x^2 + ax + 1 = 0$ and $x^2 + x + a = 0$ have atleast one common root.**Solution :** Let α is a common root

$$\text{then } \alpha^2 + a\alpha + 1 = 0$$

$$\& \quad \alpha^2 + \alpha + a = 0$$

by cramer's rule

$$\frac{\alpha^2}{a^2 - 1} = \frac{\alpha}{1 - a} = \frac{1}{1 - a}$$

$$\Rightarrow (1 - a)^2 = (a^2 - 1)(1 - a)$$

$$\Rightarrow a = 1, -2$$

Do yourself - 4 :**(i)** If $x^2 + bx + c = 0$ & $2x^2 + 9x + 10 = 0$ have both roots in common, find b & c.**(ii)** If $x^2 - 7x + 10 = 0$ & $x^2 - 5x + c = 0$ have a common root, find c.**(iii)** Show that $x^2 + (a^2 - 2)x - 2a^2 = 0$ and $x^2 - 3x + 2 = 0$ have exactly one common root for all $a \in \mathbb{R}$.

7. REMAINDER THEOREM :

If we divide a polynomial $f(x)$ by $(x - \alpha)$ the remainder obtained is $f(\alpha)$. If $f(\alpha)$ is 0 then $(x - \alpha)$ is a factor of $f(x)$.

Consider $f(x) = x^3 - 9x^2 + 23x - 15$

If $f(1) = 0 \Rightarrow (x - 1)$ is a factor of $f(x)$.

If $f(x) = (x - 2)(x^2 - 7x + 9) + 3$. Hence $f(2) = 3$ is remainder when $f(x)$ is divided by $(x - 2)$.

Illustrations 10: A polynomial in x of degree greater than three, leaves remainders 2, 1 and -1 when divided, respectively, by $(x - 1)$, $(x + 2)$ and $(x + 1)$. What will be the remainder when it is divided by $(x - 1)(x + 2)(x + 1)$.

Solution : Let required polynomial be $f(x) = p(x)(x - 1)(x + 2)(x + 1) + a_0x^2 + a_1x + a_2$
By remainder theorem, $f(1) = 2$, $f(-2) = 1$, $f(-1) = -1$.

$$\begin{aligned}\Rightarrow a_0 + a_1 + a_2 &= 2 \\ 4a_0 - 2a_1 + a_2 &= 1 \\ a_0 - a_1 + a_2 &= -1\end{aligned}$$

$$\text{Solving we get, } a_0 = \frac{7}{6}, a_1 = \frac{3}{2}, a_2 = \frac{2}{3}$$

Remainder when $f(x)$ is divided by $(x - 1)(x + 2)(x + 1)$

$$\text{will be } \frac{7}{6}x^2 + \frac{3}{2}x + \frac{2}{3}.$$

8. SOLUTION OF RATIONAL INEQUALITIES :

Let $y = \frac{f(x)}{g(x)}$ be an expression in x where $f(x)$ & $g(x)$ are polynomials in x . Now, if it is given that

$y > 0$ (or < 0 or ≥ 0 or ≤ 0), this calls for all the values of x for which y satisfies the constraint. This solution set can be found by following steps :

Step I : Factorize $f(x)$ & $g(x)$ and generate the form :

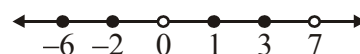
$$y = \frac{(x - a_1)^{n_1}(x - a_2)^{n_2} \dots (x - a_k)^{n_k}}{(x - b_1)^{m_1}(x - b_2)^{m_2} \dots (x - b_p)^{m_p}}$$

where $n_1, n_2, \dots, n_k, m_1, m_2, \dots, m_p$ are natural numbers and $a_1, a_2, \dots, a_k, b_1, b_2, \dots, b_p$ are real numbers. Clearly, here $a_1, a_2, \dots, a_k$ are roots of $f(x) = 0$ & $b_1, b_2, \dots, b_p$ are roots of $g(x) = 0$.

Step II : Here y vanishes (becomes zero) for $a_1, a_2, \dots, a_k$. These points are marked on the number line with a black dot. They are solution of $y = 0$.

If $g(x) = 0$, $y = \frac{f(x)}{g(x)}$ attains an undefined form, hence $b_1, b_2, \dots, b_k$ are excluded from the solution. These points are marked with white dots.

$$\text{e.g. } f(x) = \frac{(x - 1)^3(x + 2)^4(x - 3)^5(x + 6)}{x^2(x - 7)^3}$$



Step-III : Check the value of y for any real number greater than the right most marked number on the number line. If it is positive, then y is positive for all the real numbers greater than the right most marked number and vice versa.

Step-IV : If the exponent of a factor is odd, then the point is called simple point and if the exponent of a factor is even, then the point is called double point

$$\frac{(x-1)^3(x+2)^4(x-3)^5(x+6)}{x^2(x-7)^3}$$

Here 1, 3, -6 and 7 are simple points and -2 & 0 are double points.

From right to left, beginning above the number line (if y is positive in step 3 otherwise from below the line), a wavy curve should be drawn which passes through all the marked points so that when passing through a simple point, the curve intersects the number line and when passing through a double point, the curve remains on the same side of number line.

$$f(x) = \frac{(x-1)^3(x+2)^4(x-3)^5(x+6)}{x^2(x-7)^3}$$


As exponents of $(x+2)$ and x are even, the curve does not cross the number line. This method is called wavy curve method.

Step-V : The intervals where the curve is above number line, y will be positive and the intervals where the curve is below the number line, y will be negative. The appropriate intervals are chosen in accordance with the sign of inequality & their union represents the solution of inequality.

Note :

- (i) Points where denominator is zero will never be included in the answer.
- (ii) If you are asked to find the intervals where $f(x)$ is non-negative or non-positive then make the intervals closed corresponding to the roots of the numerator and let it remain open corresponding to the roots of denominator.
- (iii) Normally we cannot cross-multiply in inequalities. But we cross multiply if we are sure that quantity in denominator is always positive.
- (iv) Normally we cannot square in inequalities. But we can square if we are sure that both sides are non negative.
- (v) We can multiply both sides with a negative number by changing the sign of inequality.
- (vi) We can add or subtract equal quantity to both sides of inequalities without changing the sign of inequality.

Illustration 11: Find x such that $3x^2 - 7x + 6 < 0$

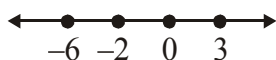
Solution : $D = 49 - 72 < 0$

As $D < 0$, $3x^2 - 7x + 6$ will always be positive. Hence $x \in \phi$.

Illustration 12: $(x^2 - x - 6)(x^2 + 6x) \geq 0$

Solution : $(x-3)(x+2)(x)(x+6) \geq 0$

Consider $E = x(x-3)(x+2)(x+6)$, $E = 0 \Rightarrow x = 0, 3, -2, -6$ (all are simple points)



For $x \geq 3$ $E = \underbrace{x}_{+ve} \underbrace{(x-3)}_{+ve} \underbrace{(x+2)}_{+ve} \underbrace{(x+6)}_{+ve}$
 $= \text{positive}$



Hence for $x(x-3)(x+2)(x+6) \geq 0$

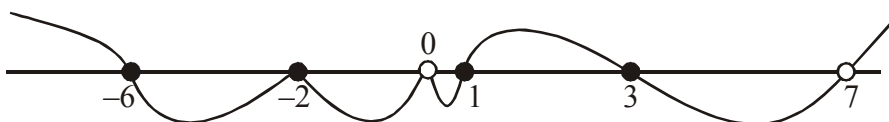
$$x \in (-\infty, -6] \cup [-2, 0] \cup [3, \infty)$$

Illustration 13: Let $f(x) = \frac{(x-1)^3(x+2)^4(x-3)^5(x+6)}{x^2(x-7)^3}$. Solve the following inequality

- (i) $f(x) > 0$ (ii) $f(x) \geq 0$ (iii) $f(x) < 0$ (iv) $f(x) \leq 0$

Solution :

We mark on the number line zeros of the function : 1, -2, 3 and -6 (with black circles) and the points of discontinuity 0 and 7 (with white circles), isolate the double points : -2 and 0 and draw the wavy curve :



From graph, we get

- (i) $x \in (-\infty, -6) \cup (1, 3) \cup (7, \infty)$
 (ii) $x \in (-\infty, -6] \cup \{-2\} \cup [1, 3] \cup (7, \infty)$
 (iii) $x \in (-6, -2) \cup (-2, 0) \cup (0, 1) \cup (3, 7)$
 (iv) $x \in [-6, 0) \cup (0, 1] \cup [3, 7)$

Do yourself - 5 :

(i) Find range of x such that

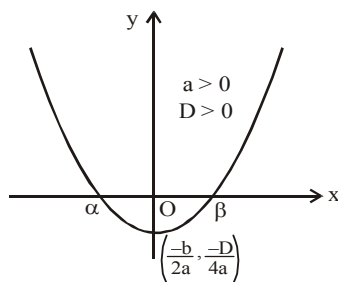
- | | |
|-------------------------------------|--|
| (a) $(x-2)(x+3) \geq 0$ | (b) $\frac{x}{x+1} > 2$ |
| (c) $\frac{3x-1}{4x+1} \leq 0$ | (d) $\frac{(2x-1)(x+3)(2-x)(1-x)^2}{x^4(x+6)(x-9)(2x^2+4x+9)} < 0$ |
| (e) $\frac{7x-17}{x^2-3x+4} \geq 1$ | (f) $x^2+2 \leq 3x \leq 2x^2-5$ |

9. QUADRATIC EXPRESSION AND ITS GRAPHS :

Consider the quadratic expression, $y = ax^2 + bx + c$, $a \neq 0$ & $a, b, c \in \mathbb{R}$ then ;

- (a) The graph between x, y is always a parabola. If $a > 0$ then the shape of the parabola is concave upwards & if $a < 0$ then the shape of the parabola is concave downwards.
- (b) The graph of $y = ax^2 + bx + c$ can be divided in 6 broad categories which are as follows :
(Let the real roots of quadratic equation $ax^2 + bx + c = 0$ be α & β where $\alpha \leq \beta$).

Fig. 1

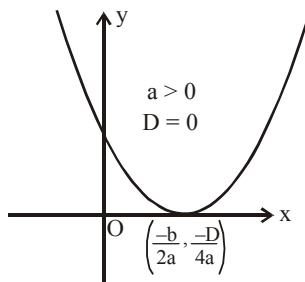


Roots are real & distinct

$$ax^2 + bx + c > 0 \quad \forall x \in (-\infty, \alpha) \cup (\beta, \infty)$$

$$ax^2 + bx + c < 0 \quad \forall x \in (\alpha, \beta)$$

Fig. 2

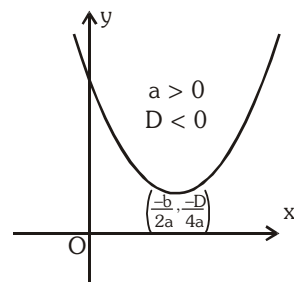


Roots are coincident

$$ax^2 + bx + c > 0 \quad \forall x \in \mathbb{R} - \{\alpha\}$$

$$ax^2 + bx + c = 0 \quad \text{for } x = \alpha = \beta$$

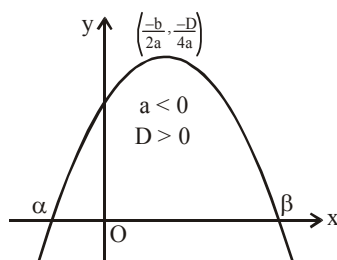
Fig. 3



Roots are complex conjugate

$$ax^2 + bx + c > 0 \quad \forall x \in \mathbb{R}$$

Fig. 4

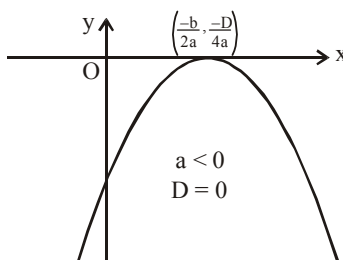


Roots are real & distinct

$$ax^2 + bx + c > 0 \quad \forall x \in (\alpha, \beta)$$

$$ax^2 + bx + c < 0 \quad \forall x \in (-\infty, \alpha) \cup (\beta, \infty)$$

Fig. 5

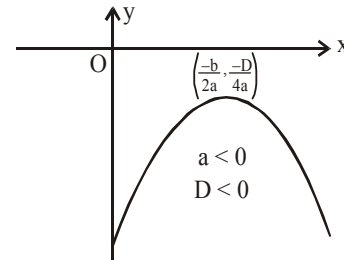


Roots are coincident

$$ax^2 + bx + c < 0 \quad \forall x \in \mathbb{R} - \{\alpha\}$$

$$ax^2 + bx + c = 0 \quad \text{for } x = \alpha = \beta$$

Fig. 6



Roots are complex conjugate

$$ax^2 + bx + c < 0 \quad \forall x \in \mathbb{R}$$

Important Note :

- (i) The quadratic expression $ax^2 + bx + c > 0$ for each $x \in \mathbb{R} \Rightarrow a > 0, D < 0$ & vice-versa (Fig. 3)
- (ii) The quadratic expression $ax^2 + bx + c < 0$ for each $x \in \mathbb{R} \Rightarrow a < 0, D < 0$ & vice-versa (Fig. 6)

10. MAXIMUM & MINIMUM VALUES OF QUADRATIC EXPRESSIONS : $y = ax^2 + bx + c$:

We know that $y = ax^2 + bx + c$ takes following form : $y = a \left[\left(x + \frac{b}{2a} \right)^2 - \frac{(b^2 - 4ac)}{4a^2} \right]$,

which is a parabola. $\therefore$ vertex = $\left(\frac{-b}{2a}, \frac{-D}{4a} \right)$

When $a > 0$, y will take a minimum value at vertex ; $x = \frac{-b}{2a}$; $y_{\min} = \frac{-D}{4a}$

When $a < 0$, y will take a maximum value at vertex ; $x = \frac{-b}{2a}$; $y_{\max} = \frac{-D}{4a}$.

If quadratic expression $ax^2 + bx + c$ is a perfect square, then $a > 0$ and $D = 0$

Illustration 14 : The value of the expression $x^2 + 2bx + c$ will be positive for all real x if -

- (A) $b^2 - 4c > 0$ (B) $b^2 - 4c < 0$ (C) $c^2 < b$ (D) $b^2 < c$

Solution : As $a > 0$, so this expression will be positive if $D < 0$

$$\text{so } 4b^2 - 4c < 0$$

$$b^2 < c$$



Ans. (D)

Illustration 15 : The minimum value of the expression $4x^2 + 2x + 1$ is -

- (A) $1/4$ (B) $1/2$ (C) $3/4$ (D) 1

Solution : Since $a = 4 > 0$

$$\text{therefore its minimum value} = -\frac{D}{4a} = \frac{4(4)(1) - (2)^2}{4(4)} = \frac{16 - 4}{16} = \frac{12}{16} = \frac{3}{4}$$

Ans. (C)

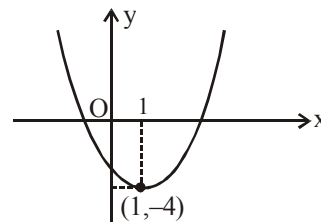
***Illustration 16 :** If $y = x^2 - 2x - 3$, then find the range of y when :

- (i) $x \in \mathbb{R}$ (ii) $x \in [0, 3]$ (iii) $x \in [-2, 0]$

Solution : We know that minimum value of y will occur at

$$x = -\frac{b}{2a} = -\frac{(-2)}{2 \times 1} = 1$$

$$y_{\min} = -\frac{D}{4a} = \frac{-(4 + 3 \times 4)}{4} = -4$$



- (i) $x \in \mathbb{R};$

$$y \in [-4, \infty)$$

Ans.

- (ii) $x \in [0, 3]$

$$f(0) = -3, f(1) = -4, \quad f(3) = 0$$

$$\therefore f(3) > f(0)$$

$\therefore y$ will take all the values from minimum to $f(3)$.

$$y \in [-4, 0]$$

- (iii) $x \in [-2, 0]$

This interval does not contain the minimum value of y for $x \in \mathbb{R}$.

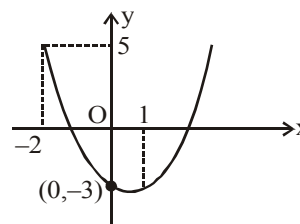
y will take values from $f(0)$ to $f(-2)$

$$f(0) = -3$$

$$f(-2) = 5$$

$$y \in [-3, 5]$$

Ans.



Ans.

Illustration 17 : If $ax^2 + bx + 10 = 0$ does not have real & distinct roots, find the minimum value of $5a - b$.

Solution : Either $f(x) \geq 0 \quad \forall x \in \mathbb{R}$ or $f(x) \leq 0 \quad \forall x \in \mathbb{R}$

$$\therefore f(0) = 10 > 0 \Rightarrow f(x) \geq 0 \quad \forall x \in \mathbb{R}$$

$$\Rightarrow f(-5) = 25a - 5b + 10 \geq 0$$

$$\Rightarrow 5a - b \geq -2$$

Ans.

Do yourself - 6

(i) Find the minimum value of :

(a) $y = x^2 + 2x + 2$

(b) $y = 4x^2 - 16x + 15$

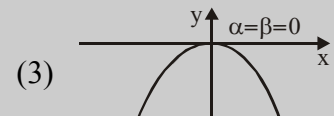
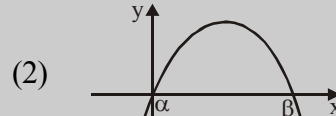
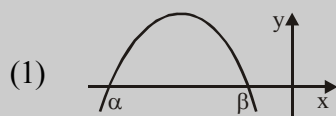
(ii) For following graphs of $y = ax^2 + bx + c$ with $a, b, c \in \mathbb{R}$, comment on the sign of :

(i) a

(ii) b

(iii) c

(iv) D

(v) $\alpha + \beta$ (vi) $\alpha\beta$ (iii) Given the roots of equation $ax^2 + bx + c = 0$ are real & distinct, where $a, b, c \in \mathbb{R}^+$, then the vertex of the graph will lie in which quadrant.

*(iv) Find the range of 'a' for which :

(a) $ax^2 + 3x + 4 > 0 \quad \forall x \in \mathbb{R}$

(b) $ax^2 + 4x - 2 < 0 \quad \forall x \in \mathbb{R}$

11. INEQUALITIES INVOLVING MODULUS FUNCTION :**Properties of modulus function :**(i) $|x| \geq a \Rightarrow x \geq a$ or $x \leq -a$, where a is positive.(ii) $|x| \leq a \Rightarrow x \in [-a, a]$, where a is positive(iii) $|x| > |y| \Rightarrow x^2 > y^2$ (iv) $||a| - |b|| \leq |a \pm b| \leq |a| + |b|$ (v) $|x + y| = |x| + |y| \Rightarrow xy \geq 0$ (vi) $|x - y| = |x| + |y| \Rightarrow xy \leq 0$ **Illustration 18:** If x satisfies $|x - 1| + |x - 2| + |x - 3| \geq 6$, then

(A) $0 \leq x \leq 4$

(B) $x \leq -2$ or $x \geq 4$

(C) $x \leq 0$ or $x \geq 4$

(D) none of these

Solution :**Case I :** $x \leq 1$, then

$$1 - x + 2 - x + 3 - x \geq 6 \Rightarrow x \leq 0$$

$$\text{Hence } x \leq 0 \quad \dots(i)$$

Case II : $1 < x \leq 2$, then

$$x - 1 + 2 - x + 3 - x \geq 6 \Rightarrow x \leq -2$$

$$\text{But } 1 < x \leq 2 \Rightarrow \text{No solution.} \quad \dots(ii)$$

Case III : $2 < x \leq 3$, then

$$x - 1 + x - 2 + 3 - x \geq 6 \Rightarrow x \geq 6$$

$$\text{But } 2 < x \leq 3 \Rightarrow \text{No solution.} \quad \dots(iii)$$

Case IV : $x > 3$, then

$$x - 1 + x - 2 + x - 3 \geq 6 \Rightarrow x \geq 4$$

$$\text{Hence } x \geq 4 \quad \dots(iv)$$

From (i), (ii), (iii) and (iv) the given inequality holds for $x \leq 0$ or $x \geq 4$.

Illustration 19 : Solve for x : (a) $\|x-1\|+2\leq 4$. (b) $\frac{x-4}{x+2}\leq \left|\frac{x-2}{x-1}\right|$

Solution : (a) $\|x-1\|+2\leq 4\Rightarrow -4\leq |x-1|+2\leq 4$
 $\Rightarrow -6\leq |x-1|\leq 2$
 $\Rightarrow |x-1|\leq 2\Rightarrow -2\leq x-1\leq 2$
 $\Rightarrow -1\leq x\leq 3\Rightarrow x\in[-1, 3]$
 (b) **Case 1 :** Given inequation will be statisfied for all x such that

$$\frac{x-4}{x+2}\leq 0\Rightarrow x\in(-2, 4]-\{1\}\quad \text{.....(i)}$$

(Note : $\{1\}$ is not in domain of RHS)

$$\text{Case 2 : } \frac{x-4}{x+2}>0\Rightarrow x\in(-\infty, -2)\cup(4, \infty)\quad \text{.....(ii)}$$

Given inequation becomes

$$\frac{x-2}{x-1}\geq \frac{x-4}{x+2}\quad \text{or}\quad \frac{x-2}{x-1}\leq -\frac{x-4}{x+2}$$

on solving we get

$$x\in(-2, 4/5)\cup(1, \infty)$$

taking intersection with (ii) we get

$$x\in(4, \infty)\quad \text{.....(iii)}$$

on solving we get

$$x\in(-2, 0]\cup(1, 5/2]$$

taking intersection with (ii) we get

$$x\in\phi$$

Hence, solution of the original inequation : $x\in(-2, \infty)-\{1\}$ (taking union of (i) & (iii))

Illustration 20 : The equation $|x|+\left|\frac{x}{x-1}\right|=\frac{x^2}{|x-1|}$ is always true for x belongs to

- (A) $\{0\}$ (B) $(1, \infty)$ (C) $(-1, 1)$ (D) $(-\infty, \infty)$

Solution : $\frac{x^2}{|x-1|}=\left|x+\frac{x}{x-1}\right|$

$$\therefore |x|+\left|\frac{x}{x-1}\right|=\left|x+\frac{x}{x-1}\right| \text{ is true only if } \left(x\cdot\frac{x}{x-1}\right)\geq 0\Rightarrow x\in\{0\}\cup(1, \infty). \text{ Ans (A,B)}$$

12. IRRATIONAL INEQUALITIES :

Illustration 21 : Solve for x , if $\sqrt{x^2-3x+2}>x-2$

Solution :
$$\left[\begin{array}{l} x^2-3x+2\geq 0 \\ x-2\geq 0 \\ (x^2-3x+2)>(x-2)^2 \end{array}\right] \Rightarrow \left[\begin{array}{l} (x-1)(x-2)\geq 0 \\ (x-2)\geq 0 \Rightarrow x>2 \\ x-2>0 \end{array}\right]$$

$$\text{or} \Rightarrow \left[\begin{array}{l} (x-1)(x-2)\geq 0 \\ x-2<0 \end{array}\right] \Rightarrow x\leq 1$$

Hence, solution set of the original inequation is $x\in\mathbb{R}-(1,2]$

Do yourself - 7 :

(i) Solve for x if $\frac{|x^2 - 4|}{x^2 + x - 2} > 1$

(ii) Solve for x if $\sqrt{x^2 - x} > (x - 1)$

13. LOGARITHMIC INEQUALITIES :

Points to remember :

(i) $\log_a x < \log_a y \Leftrightarrow \begin{cases} x < y & \text{if } a > 1 \\ x > y & \text{if } 0 < a < 1 \end{cases}$

(ii) If $a > 1$, then (a) $\log_a x < p \Rightarrow 0 < x < a^p$ (b) $\log_a x > p \Rightarrow x > a^p$

(iii) If $0 < a < 1$, then (a) $\log_a x < p \Rightarrow x > a^p$ (b) $\log_a x > p \Rightarrow 0 < x < a^p$

Illustration 22 : Solve for x : (a) $\log_{0.5}(x^2 - 5x + 6) \geq -1$ (b) $\log_{1/3}(\log_4(x^2 - 5)) > 0$

Solution : (a) $\log_{0.5}(x^2 - 5x + 6) \geq -1 \Rightarrow 0 < x^2 - 5x + 6 \leq (0.5)^{-1}$
 $\Rightarrow 0 < x^2 - 5x + 6 \leq 2$

$$\begin{cases} x^2 - 5x + 6 > 0 \\ x^2 - 5x + 6 \leq 2 \end{cases} \Rightarrow x \in [1, 2) \cup (3, 4]$$

Hence, solution set of original inequation : $x \in [1, 2) \cup (3, 4]$

(b) $\log_{1/3}(\log_4(x^2 - 5)) > 0 \Rightarrow 0 < \log_4(x^2 - 5) < 1$

$$\begin{cases} 0 < \log_4(x^2 - 5) \Rightarrow x^2 - 5 > 1 \\ \log_4(x^2 - 5) < 1 \Rightarrow 0 < x^2 - 5 < 4 \end{cases} \Rightarrow 1 < (x^2 - 5) < 4$$

$$\Rightarrow 6 < x^2 < 9 \Rightarrow x \in (-3, -\sqrt{6}) \cup (\sqrt{6}, 3)$$

Hence, solution set of original inequation : $x \in (-3, -\sqrt{6}) \cup (\sqrt{6}, 3)$ **Illustration 23 :** Solve for x : $\log_2 x \leq \frac{2}{\log_2 x - 1}$.

Solution : Let $\log_2 x = t$

$$t \leq \frac{2}{t-1} \Rightarrow t - \frac{2}{t-1} \leq 0$$

$$\Rightarrow \frac{t^2 - t - 2}{t-1} \leq 0 \Rightarrow \frac{(t-2)(t+1)}{(t-1)} \leq 0$$

$$\Rightarrow t \in (-\infty, -1] \cup (1, 2]$$

or $\log_2 x \in (-\infty, -1] \cup (1, 2]$

or $x \in \left(0, \frac{1}{2}\right] \cup (2, 4]$

Illustration 24 : Solve the inequation : $\log_{2x+3} x^2 < \log_{2x+3} (2x+3)$

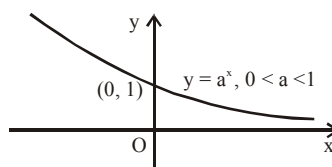
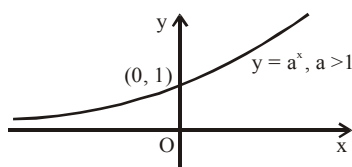
Solution : This inequation is equivalent to the collection of the systems

$$\left[\begin{array}{l} 2x+3 > 1 \\ 0 < x^2 < 2x+3 \end{array} \right] \Rightarrow \left[\begin{array}{l} x > -1 \\ (x-3)(x+1) < 0 \text{ \& } x \neq 0 \end{array} \right] \Rightarrow \left[\begin{array}{l} x > -1 \\ -1 < x < 3 \end{array} \right] \Rightarrow -1 < x < 3 \text{ \& } x \neq 0$$

$$\text{or} \quad \left[\begin{array}{l} 0 < 2x+3 < 1 \\ x^2 > 2x+3 > 0 \end{array} \right] \Rightarrow \left[\begin{array}{l} -\frac{3}{2} < x < -1 \\ (x-3)(x+1) > 0 \end{array} \right] \Rightarrow \left[\begin{array}{l} -\frac{3}{2} < x < -1 \\ x < -1 \text{ or } x > 3 \end{array} \right] \Rightarrow -\frac{3}{2} < x < -1$$

Hence, solution of the original inequation is $x \in \left(-\frac{3}{2}, -1\right) \cup (-1, 0) \cup (0, 3)$

14. EXPONENTIAL INEQUATIONS :



$$\text{If } a^{f(x)} > b \Rightarrow \begin{cases} f(x) > \log_a b \text{ when } a > 1 \\ f(x) < \log_a b \text{ when } 0 < a < 1 \end{cases}$$

Illustration 25 : Solve for x : $2^{x+2} > \left(\frac{1}{4}\right)^{\frac{1}{x}}$

Solution : We have $2^{x+2} > 2^{-2/x}$. Since the base $2 > 1$, we have $x+2 > -\frac{2}{x}$ (the sign of the inequality is retained).

$$\begin{aligned} \text{Now } x+2+\frac{2}{x} > 0 &\Rightarrow \frac{x^2+2x+2}{x} > 0 \\ \Rightarrow \frac{(x+1)^2+1}{x} > 0 &\Rightarrow x \in (0, \infty) \end{aligned}$$

Illustration 26 : Solve for x : $(1.25)^{1-x} < (0.64)^{2(1+\sqrt{x})}$

Solution : We have $\left(\frac{5}{4}\right)^{1-x} < \left(\frac{16}{25}\right)^{2(1+\sqrt{x})}$ or $\left(\frac{4}{5}\right)^{x-1} < \left(\frac{4}{5}\right)^{4(1+\sqrt{x})}$

Since the base $0 < \frac{4}{5} < 1$, the inequality is equivalent to the inequality $x-1 > 4(1+\sqrt{x})$

$$\Rightarrow \frac{x-5}{4} > \sqrt{x}$$

Now, R.H.S. is positive

$$\Rightarrow \frac{x-5}{4} > 0 \Rightarrow x > 5 \quad \dots\dots(i)$$

we have $\frac{x-5}{4} > \sqrt{x}$

both sides are positive, so squaring both sides

$$\Rightarrow \frac{(x-5)^2}{16} > x \quad \text{or} \quad \frac{(x-5)^2}{16} - x > 0$$

$$\text{or} \quad x^2 - 26x + 25 > 0 \quad \text{or} \quad (x-25)(x-1) > 0$$

$$\Rightarrow x \in (-\infty, 1) \cup (25, \infty) \quad \dots\dots(ii)$$

intersection (i) & (ii) gives $x \in (25, \infty)$

Do yourself-8 :

(i) Solve for x : (a) $\log_{0.3}(x^2 + 8) > \log_{0.3}(9x)$ (b) $\log_7\left(\frac{2x-6}{2x-1}\right) > 0$

(ii) Solve for x : $\left(\frac{2}{3}\right)^{\frac{|x|-1}{|x|+1}} > 1$

15. MAXIMUM & MINIMUM VALUES OF RATIONAL ALGEBRAIC EXPRESSIONS :

$$y = \frac{a_1x^2 + b_1x + c_1}{a_2x^2 + b_2x + c_2}, \frac{1}{ax^2 + bx + c}, \frac{a_1x + b_1}{a_2x^2 + b_2x + c_2}, \frac{a_1x^2 + b_1x + c_1}{a_2x + b_2} :$$

Sometime we have to find range of expression of form $\frac{a_1x^2 + b_1x + c_1}{a_2x^2 + b_2x + c_2}$.

The following procedure is used :

Step 1 : Equate the given expression to y i.e. $y = \frac{a_1x^2 + b_1x + c_1}{a_2x^2 + b_2x + c_2}$

Step 2 : By cross multiplying and simplifying, obtain a quadratic equation in x.

$$(a_1 - a_2y)x^2 + (b_1 - b_2y)x + (c_1 - c_2y) = 0$$

Step 3 : Put Discriminant ≥ 0 and solve the inequality for possible set of values of y.

Illustration 27 : For $x \in \mathbb{R}$, find the set of values attainable by $\frac{x^2 - 3x + 4}{x^2 + 3x + 4}$.

Solution : Let $y = \frac{x^2 - 3x + 4}{x^2 + 3x + 4}$
 $x^2(y-1) + 3x(y+1) + 4(y-1) = 0$

Case- I : $y \neq 1$

For $y \neq 1$ above equation is a quadratic equation.

So for $x \in \mathbb{R}$, $D \geq 0$

$$\Rightarrow 9(y+1)^2 - 16(y-1)^2 \geq 0 \quad \Rightarrow 7y^2 - 50y + 7 \leq 0$$

$$\Rightarrow (7y-1)(y-7) \leq 0 \quad \Rightarrow y \in \left[\frac{1}{7}, 7\right] - \{1\}$$

Case II : when $y = 1$

$$\Rightarrow 1 = \frac{x^2 - 3x + 4}{x^2 + 3x + 4}$$

$$\Rightarrow x^2 + 3x + 4 = x^2 - 3x + 4$$

$$\Rightarrow x = 0$$

Hence $y = 1$ for real value of x .

$$\text{so range of } y \text{ is } \left[\frac{1}{7}, 7 \right]$$

Illustration 28 : Find the values of a for which the expression $\frac{ax^2 + 3x - 4}{3x - 4x^2 + a}$ assumes all real values for real values of x .

Solution : Let $y = \frac{ax^2 + 3x - 4}{3x - 4x^2 + a}$

$$x^2(a + 4y) + 3(1 - y)x - (4 + ay) = 0$$

$$\text{If } x \in \mathbb{R}, D \geq 0$$

$$\Rightarrow 9(1 - y)^2 + 4(a + 4y)(4 + ay) \geq 0 \Rightarrow (9 + 16a)y^2 + (4a^2 + 46)y + (9 + 16a) \geq 0$$

$$\text{for all } y \in \mathbb{R}, (9 + 16a) > 0 \text{ \& } D \leq 0$$

$$\Rightarrow (4a^2 + 46)^2 - 4(9 + 16a)(9 + 16a) \leq 0 \Rightarrow 4(a^2 - 8a + 7)(a^2 + 8a + 16) \leq 0$$

$$\Rightarrow a^2 - 8a + 7 \leq 0 \Rightarrow 1 \leq a \leq 7$$

$$9 + 16a > 0 \text{ \& } 1 \leq a \leq 7$$

$$\text{Taking intersection, } a \in [1, 7]$$

Now, checking the boundary values of a

$$\text{For } a = 1$$

$$y = \frac{x^2 + 3x - 4}{3x - 4x^2 + 1} = -\frac{(x - 1)(x + 4)}{(x - 1)(4x + 1)}$$

$$\therefore x \neq 1 \Rightarrow y \neq -1$$

$$\Rightarrow a = 1 \text{ is not possible.}$$

$$\text{if } a = 7$$

$$y = \frac{7x^2 + 3x - 4}{3x - 4x^2 + 7} = \frac{(7x - 4)(x + 1)}{(7 - 4x)(x + 1)} \quad \therefore x \neq -1 \Rightarrow y \neq -1$$

So y will assume all real values for some real values of x .

$$\text{So } a \in (1, 7)$$

Do yourself - 9 :

(i) Prove that the expression $\frac{8x - 4}{x^2 + 2x - 1}$ cannot have values between 2 and 4, in its domain.

(ii) Find the range of $\frac{x^2 + 2x + 1}{x^2 + 2x + 7}$, where x is real

16. LOCATION OF ROOTS :

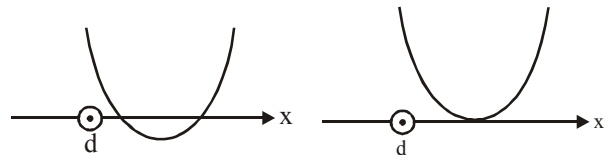
This article deals with an elegant approach of solving problems on quadratic equations when the roots are located / specified on the number line with variety of constraints :

Consider the quadratic equation $ax^2 + bx + c = 0$ with $a > 0$ and let $f(x) = ax^2 + bx + c$

Type-1 : Both roots of the quadratic equation are greater than a specific number (say d).

The necessary and sufficient condition for this are :

$$(i) D \geq 0 ; \quad (ii) f(d) > 0 ; \quad (iii) -\frac{b}{2a} > d$$



Note : When both roots of the quadratic equation are less than a specific number d then the necessary and sufficient condition will be :

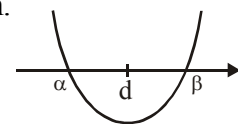
$$(i) D \geq 0 ; \quad (ii) f(d) > 0 ; \quad (iii) -\frac{b}{2a} < d$$

Type-2 :

Both roots lie on either side of a fixed number say (d) . Alternatively one root is greater than ' d ' and other root less than ' d ' or ' d ' lies between the roots of the given equation.

The necessary and sufficient condition for this are : $f(d) < 0$

Note : Consideration of discriminant is not needed.

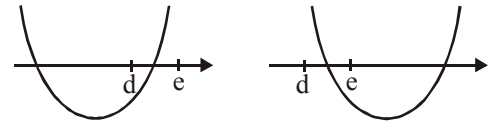


Type-3 :

Exactly one root lies in the interval (d, e) .

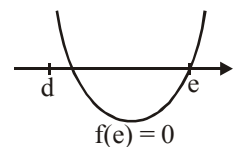
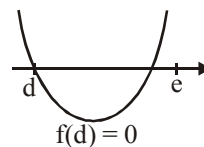
The necessary and sufficient condition for this are :

$$f(d) \cdot f(e) < 0$$



Note : The extremes of the intervals found by given conditions give ' d ' or ' e ' as the root of the equation.

Hence in this case also check for end points.



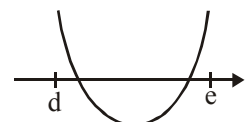
Type-4 :

When both roots are confined between the number d and e ($d < e$).

The necessary and sufficient condition for this are :

$$(i) D \geq 0 ; \quad (ii) f(d) > 0 ; \quad (iii) f(e) > 0$$

$$(iv) d < -\frac{b}{2a} < e$$



Type-5 :

One root is greater than e and the other roots is less than d ($d < e$).

The necessary and sufficient condition for this are : $f(d) < 0$ and $f(e) < 0$

Note : If $a < 0$ in the quadratic equation $ax^2 + bx + c = 0$ then we divide the whole equation by ' a '.

Now assume $x^2 + \frac{b}{a}x + \frac{c}{a}$ as $f(x)$. This makes the coefficient of x^2 positive and hence above cases are applicable.

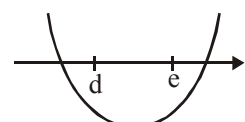


Illustration 29 : Find the values of the parameter 'a' for which the roots of the quadratic equation $x^2 + 2(a-1)x + a + 5 = 0$ are

- | | |
|--|--|
| (i) real and distinct | (ii) equal |
| (iii) opposite in sign | (iv) equal in magnitude but opposite in sign |
| (v) positive | (vi) negative |
| (vii) greater than 3 | (viii) smaller than 3 |
| (ix) such that both the roots lie in the interval (1, 3) | |

Solution :

Let $f(x) = x^2 + 2(a-1)x + a + 5 = Ax^2 + Bx + C$ (say)

$$\Rightarrow A = 1, B = 2(a-1), C = a + 5.$$

$$\text{Also } D = B^2 - 4AC = 4(a-1)^2 - 4(a+5) = 4(a+1)(a-4)$$

(i) $D > 0$

$$\Rightarrow (a+1)(a-4) > 0 \Rightarrow a \in (-\infty, -1) \cup (4, \infty).$$

(ii) $D = 0$

$$\Rightarrow (a+1)(a-4) = 0 \Rightarrow a = -1, 4.$$

(iii) This means that 0 lies between the roots of the given equation.

$$\Rightarrow f(0) < 0 \text{ and } D > 0 \text{ i.e. } a \in (-\infty, -1) \cup (4, \infty)$$

$$\Rightarrow a + 5 < 0 \Rightarrow a < -5 \Rightarrow a \in (-\infty, -5).$$

(iv) This means that the sum of the roots is zero

$$\Rightarrow -2(a-1) = 0 \text{ and } D > 0 \text{ i.e. } a \in (-\infty, -1) \cup (4, \infty) \Rightarrow a = 1$$

which does not belong to $(-\infty, -1) \cup (4, \infty)$

$$\Rightarrow a \in \phi.$$

(v) This implies that both the roots are greater than zero

$$\Rightarrow -\frac{B}{A} > 0, \frac{C}{A} > 0, D \geq 0 \Rightarrow -(a-1) > 0, a+5 > 0, a \in (-\infty, -1] \cup [4, \infty)$$

$$\Rightarrow a < 1, -5 < a, a \in (-\infty, -1] \cup [4, \infty) \Rightarrow a \in (-5, -1].$$

(vi) This implies that both the roots are less than zero

$$\Rightarrow -\frac{B}{A} < 0, \frac{C}{A} > 0, D \geq 0 \Rightarrow -(a-1) < 0, a+5 > 0, a \in (-\infty, -1] \cup [4, \infty)$$

$$\Rightarrow a > 1, a > -5, a \in (-\infty, -1] \cup [4, \infty) \Rightarrow a \in [4, \infty).$$

(vii) In this case

$$-\frac{B}{2a} > 3, A.f(3) > 0 \text{ and } D \geq 0.$$

$$\Rightarrow -(a-1) > 3, 7a+8 > 0 \text{ and } a \in (-\infty, -1] \cup [4, \infty)$$

$$\Rightarrow a < -2, a > -8/7 \text{ and } a \in (-\infty, -1] \cup [4, \infty)$$

Since no value of 'a' can satisfy these conditions simultaneously, there can be no value of a for which both the roots will be greater than 3.

(viii) In this case

$$-\frac{B}{2a} < 3, A.f(3) > 0 \text{ and } D \geq 0.$$

$$\Rightarrow a > -2, a > -8/7 \text{ and } a \in (-\infty, -1] \cup [4, \infty) \Rightarrow a \in (-8/7, -1] \cup [4, \infty)$$

(ix) In this case

$$1 < -\frac{B}{2A} < 3, A.f(1) > 0, A.f(3) > 0, D \geq 0.$$

$$\Rightarrow 1 < -1(a-1) < 3, 3a+4 > 0, 7a+8 > 0, a \in (-\infty, -1] \cup [4, \infty)$$

$$\Rightarrow -2 < a < 0, a > -4/3, a > -8/7, a \in (-\infty, -1] \cup [4, \infty) \Rightarrow a \in \left(-\frac{8}{7}, -1\right]$$

Illustration 30 : Find value of k for which one root of equation $x^2 - (k+1)x + k^2 + k - 8 = 0$ exceeds 2 & other is less than 2.

Solution : $4 - 2(k+1) + k^2 + k - 8 < 0 \Rightarrow k^2 - k - 6 < 0$

$$(k-3)(k+2) < 0 \Rightarrow -2 < k < 3$$

Taking intersection, $k \in (-2, 3)$.

Illustration 31 : Find all possible values of a for which exactly one root of $x^2 - (a+1)x + 2a = 0$ lies in interval $(0, 3)$.

Solution : $f(0) \cdot f(3) < 0$

$$\Rightarrow 2a(9 - 3(a+1) + 2a) < 0 \Rightarrow 2a(-a+6) < 0$$

$$\Rightarrow a(a-6) > 0 \Rightarrow a < 0 \text{ or } a > 6$$

Checking the extremes.

If $a = 0$, $x^2 - x = 0$

$$x = 0, 1$$

$$1 \in (0, 3)$$

If $a = 6$, $x^2 - 7x + 12 = 0$

$$x = 3, 4 \quad \text{But } 4 \notin (0, 3)$$

Hence solution set is

$$a \in (-\infty, 0] \cup (6, \infty)$$

Do yourself - 10 :

(i) If α, β are roots of $7x^2 + 9x - 2 = 0$, find their position with respect to following $(\alpha < \beta)$:

(a) -3 (b) 0 (c) 1

(ii) If $a > 1$, roots of the equation $(1-a)x^2 + 3ax - 1 = 0$ are -

(A) one positive one negative (B) both negative

(C) both positive (D) both non-real

(iii) Find the set of value of a for which the roots of the equation $x^2 - 2ax + a^2 + a - 3 = 0$ are less than 3.

(iv) If α, β are the roots of $x^2 - 3x + a = 0$, $a \in \mathbb{R}$ and $\alpha < 1 < \beta$, then find the values of a .

(v) If α, β are roots of $4x^2 - 16x + \lambda = 0$, $\lambda \in \mathbb{R}$ such that $1 < \alpha < 2$ and $2 < \beta < 3$, then find the range of λ .

17. GENERAL QUADRATIC EXPRESSION IN TWO VARIABLES :

$f(x, y) = ax^2 + 2hxy + by^2 + 2gx + 2fy + c$ may be resolved into two linear factors if ;

$$\Delta = abc + 2fgh - af^2 - bg^2 - ch^2 = 0 \quad \text{OR} \quad \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} = 0$$

Illustration 32 : If $x^2 + 2xy + 2x + my - 3$ have two linear factor then m is equal to -

- (A) 6, 2 (B) -6, 2 (C) 6, -2 (D) -6, -2

Solution : Here $a=1, h=1, b=0, g=1, f=m/2, c=-3$

$$\text{So } \Delta = 0 \Rightarrow \begin{vmatrix} 1 & 1 & 1 \\ 1 & 0 & m/2 \\ 1 & m/2 & -3 \end{vmatrix} = 0$$

$$\Rightarrow -\frac{m^2}{4} - (-3 - m/2) + m/2 = 0 \quad \Rightarrow -\frac{m^2}{4} + m + 3 = 0$$

$$\Rightarrow m^2 - 4m - 12 = 0 \quad \Rightarrow m = -2, 6$$

Ans. (C)

Do yourself - 11 :

- (i) Find the value of k for which the expression $x^2 + 2xy + ky^2 + 2x + k = 0$ can be resolved into two linear factors.

18. THEORY OF EQUATIONS :

Let $\alpha_1, \alpha_2, \alpha_3, \dots, \alpha_n$ are roots of the equation, $f(x) = a_0x^n + a_1x^{n-1} + a_2x^{n-2} + \dots + a_{n-1}x + a_n = 0$, where $a_0, a_1, \dots, a_n$ are constants and $a_0 \neq 0$.

$$f(x) = a_0(x - \alpha_1)(x - \alpha_2)(x - \alpha_3) \dots (x - \alpha_n)$$

$$\therefore a_0x^n + a_1x^{n-1} + \dots + a_{n-1}x + a_n = a_0(x - \alpha_1)(x - \alpha_2) \dots (x - \alpha_n)$$

Comparing the coefficients of like powers of x , we get

$$\sum \alpha_i = -\frac{a_1}{a_0} = S_1 \text{ (say)}$$

$$\text{or } S_1 = -\frac{\text{coefficient of } x^{n-1}}{\text{coefficient of } x^n}$$

$$S_2 = \sum_{i \neq j} \alpha_i \alpha_j = (-1)^2 \frac{a_2}{a_0}$$

$$S_3 = \sum_{i \neq j \neq k} \alpha_i \alpha_j \alpha_k = (-1)^3 \frac{a_3}{a_0}$$

⋮

$$S_n = \alpha_1 \alpha_2 \dots \alpha_n = (-1)^n \frac{a_n}{a_0} = (-1)^n \frac{\text{constant term}}{\text{coefficient of } x^n}$$

where S_k denotes the sum of the product of root taken k at a time.

Quadratic equation : If α, β are the roots of the quadratic equation $ax^2 + bx + c = 0$,

$$\text{then } \alpha + \beta = -\frac{b}{a} \text{ and } \alpha\beta = \frac{c}{a}$$

Cubic equation : If α, β, γ are roots of a cubic equation $ax^3 + bx^2 + cx + d = 0$, then

$$\alpha + \beta + \gamma = -\frac{b}{a}, \alpha\beta + \beta\gamma + \gamma\alpha = \frac{c}{a} \text{ and } \alpha\beta\gamma = -\frac{d}{a}$$

Note :

- (i) If α is a root of the equation $f(x) = 0$, then the polynomial $f(x)$ is exactly divisible by $(x - \alpha)$ or $(x - \alpha)$ is a factor of $f(x)$ and conversely.
- (ii) Every equation of n th degree ($n \geq 1$) has exactly n root & if the equation has more than n roots, it is an identity.
- (iii) If the coefficients of the equation $f(x) = 0$ are all real and $\alpha + i\beta$ is its root, then $\alpha - i\beta$ is also a root. i.e. **imaginary roots occur in conjugate pairs.**
- (iv) If the coefficients in the equation are all rational & $\alpha + \sqrt{\beta}$ is one of its roots, then $\alpha - \sqrt{\beta}$ is also a root where $\alpha, \beta \in \mathbb{Q}$ & β is not a perfect square.
- (v) If there be any two real numbers 'a' & 'b' such that $f(a)$ & $f(b)$ are of opposite signs, then $f(x) = 0$ must have atleast one real root between 'a' and 'b'.
- (vi) Every equation $f(x) = 0$ of degree odd has atleast one real root of a sign opposite to that of its last term.

Illustration 33 : If two roots are equal, find the roots of $4x^3 + 20x^2 - 23x + 6 = 0$.

Solution : Let roots be α, α and β

$$\therefore \alpha + \alpha + \beta = -\frac{20}{4} \Rightarrow 2\alpha + \beta = -5 \quad \dots\dots\dots (i)$$

$$\therefore \alpha \cdot \alpha + \alpha\beta + \alpha\beta = -\frac{23}{4} \Rightarrow \alpha^2 + 2\alpha\beta = -\frac{23}{4} \text{ \& } \alpha^2\beta = -\frac{6}{4}$$

from equation (i)

$$\alpha^2 + 2\alpha(-5 - 2\alpha) = -\frac{23}{4} \Rightarrow \alpha^2 - 10\alpha - 4\alpha^2 = -\frac{23}{4} \Rightarrow 12\alpha^2 + 40\alpha - 23 = 0$$

$$\therefore \alpha = 1/2, -\frac{23}{6}$$

$$\text{when } \alpha = \frac{1}{2}$$

$$\alpha^2\beta = \frac{1}{4}(-5 - 1) = -\frac{3}{2}$$

$$\text{when } \alpha = -\frac{23}{6} \Rightarrow \alpha^2\beta = \frac{23 \times 23}{36} \left(-5 - 2 \times \left(-\frac{23}{6} \right) \right) \neq -\frac{3}{2} \Rightarrow \alpha = \frac{1}{2} \quad \beta = -6$$

$$\text{Hence roots of equation} = \frac{1}{2}, \frac{1}{2}, -6$$

Ans.

Illustration 34 : If α, β, γ are the roots of $x^3 - px^2 + qx - r = 0$, find :

$$(i) \sum \alpha^3 \quad (ii) \alpha^2(\beta + \gamma) + \beta^2(\gamma + \alpha) + \gamma^2(\alpha + \beta)$$

Solution :

We know that $\alpha + \beta + \gamma = p$

$$\alpha\beta + \beta\gamma + \gamma\alpha = q$$

$$\alpha\beta\gamma = r$$

$$(i) \alpha^3 + \beta^3 + \gamma^3 = 3\alpha\beta\gamma + (\alpha + \beta + \gamma)\{(\alpha + \beta + \gamma)^2 - 3(\alpha\beta + \beta\gamma + \gamma\alpha)\} \\ = 3r + p\{p^2 - 3q\} = 3r + p^3 - 3pq$$

$$(ii) \alpha^2(\beta + \gamma) + \beta^2(\alpha + \gamma) + \gamma^2(\alpha + \beta) = \alpha^2(p - \alpha) + \beta^2(p - \beta) + \gamma^2(p - \gamma) \\ = p(\alpha^2 + \beta^2 + \gamma^2) - 3r - p^3 + 3pq = p(p^2 - 2q) - 3r - p^3 + 3pq = pq - 3r$$

Illustration 35 : If $b^2 < 2ac$ and $a, b, c, d \in \mathbb{R}$, then prove that $ax^3 + bx^2 + cx + d = 0$ has exactly one real root.

Solution :

Let α, β, γ be the roots of $ax^3 + bx^2 + cx + d = 0$

$$\text{Then } \alpha + \beta + \gamma = -\frac{b}{a}$$

$$\alpha\beta + \beta\gamma + \gamma\alpha = \frac{c}{a}$$

$$\alpha\beta\gamma = \frac{-d}{a}$$

$$\alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \gamma\alpha) = \frac{b^2}{a^2} - \frac{2c}{a} = \frac{b^2 - 2ac}{a^2}$$

$\Rightarrow \alpha^2 + \beta^2 + \gamma^2 < 0$, which is not possible if all α, β, γ are real. So atleast one root is non-real, but complex roots occurs in pair. Hence given cubic equation has two non-real and one real roots.

Do yourself - 12 :

- (i) Let α, β be two of the roots of the equation $x^3 - px^2 + qx - r = 0$. If $\alpha + \beta = 0$, then show that $pq = r$
- (ii) If two roots of $x^3 + 3x^2 - 9x + c = 0$ are equal, then find the value of c .
- (iii) If α, β, γ be the roots of $ax^3 + bx^2 + cx + d = 0$, then find the value of

$$(a) \sum \alpha^2 \quad (b) \sum \frac{1}{\alpha} \quad (c) \sum \alpha^2(\beta + \gamma)$$

19. TRANSFORMATION OF THE EQUATION :

Let $ax^2 + bx + c = 0$ be a quadratic equation with two roots α and β . If we have to find an equation whose roots are $f(\alpha)$ and $f(\beta)$, i.e. some expression in α & β , then this equation can be found by finding α in terms of y . Now as α satisfies given equation, put this α in terms of y directly in the equation.

$$y = f(\alpha)$$

By transformation, $\alpha = g(y)$

$$a(g(y))^2 + b(g(y)) + c = 0$$

This is the required equation in y .

Illustration 36 : If the roots of $ax^2 + bx + c = 0$ are α and β , then find the equation whose roots are :

- (a) $\frac{-2}{\alpha}, \frac{-2}{\beta}$ (b) $\frac{\alpha}{\alpha+1}, \frac{\beta}{\beta+1}$ (c) α^2, β^2

Solution :

(a) $\frac{-2}{\alpha}, \frac{-2}{\beta}$

$$\text{put, } y = \frac{-2}{\alpha} \Rightarrow \alpha = \frac{-2}{y}$$

$$a\left(\frac{-2}{y}\right)^2 + b\left(\frac{-2}{y}\right) + c = 0 \Rightarrow cy^2 - 2by + 4a = 0$$

Required equation is $cx^2 - 2bx + 4a = 0$

(b) $\frac{\alpha}{\alpha+1}, \frac{\beta}{\beta+1}$

$$\text{put, } y = \frac{\alpha}{\alpha+1} \Rightarrow \alpha = \frac{y}{1-y}$$

$$\Rightarrow a\left(\frac{y}{1-y}\right)^2 + b\left(\frac{y}{1-y}\right) + c = 0 \Rightarrow (a+c-b)y^2 + (-2c+b)y + c = 0$$

Required equation is $(a+c-b)x^2 + (b-2c)x + c = 0$

(c) α^2, β^2

$$\text{put } y = \alpha^2 \Rightarrow \alpha = \sqrt{y}$$

$$ay + b\sqrt{y} + c = 0$$

$$b^2y = a^2y^2 + c^2 + 2acy$$

$$\Rightarrow a^2y^2 + (2ac - b^2)y + c^2 = 0$$

Required equation is $a^2x^2 + (2ac - b^2)x + c^2 = 0$

Illustration 37 : If the roots of $ax^3 + bx^2 + cx + d = 0$ are α, β, γ then find equation whose roots are

$$\frac{1}{\alpha\beta}, \frac{1}{\beta\gamma}, \frac{1}{\gamma\alpha}.$$

Solution :

$$\text{Put } y = \frac{1}{\alpha\beta} = \frac{\gamma}{\alpha\beta\gamma} = -\frac{a\gamma}{d} \quad (\because \alpha\beta\gamma = -\frac{d}{a})$$

$$\text{Put } x = -\frac{dy}{a}$$

$$\Rightarrow a\left(-\frac{dy}{a}\right)^3 + b\left(-\frac{dy}{a}\right)^2 + c\left(-\frac{dy}{a}\right) + d = 0$$

Required equation is $d^2x^3 - bdx^2 + acx - a^2 = 0$

Do yourself - 13 :

(i) If α, β are the roots of $ax^2 + bx + c = 0$, then find the equation whose roots are

(a) $\frac{1}{\alpha^2}, \frac{1}{\beta^2}$ (b) $\frac{1}{a\alpha+b}, \frac{1}{a\beta+b}$ (c) $\alpha + \frac{1}{\beta}, \beta + \frac{1}{\alpha}$

(ii) If α, β are roots of $x^2 - px + q = 0$, then find the quadratic equation whose root are $(\alpha^2 - \beta^2)(\alpha^3 - \beta^3)$ and $\alpha^2\beta^3 + \alpha^3\beta^2$.

Miscellaneous Illustrations :

Illustrations 38: If α, β are the roots of $x^2 + px + q = 0$, and γ, δ are the roots of $x^2 + rx + s = 0$, evaluate $(\alpha - \gamma)(\alpha - \delta)(\beta - \gamma)(\beta - \delta)$ in terms of p, q, r and s . Deduce the condition that the equations have a common root.

Solution :

α, β are the roots of $x^2 + px + q = 0$

$$\therefore \alpha + \beta = -p, \alpha\beta = q \quad \dots\dots(1)$$

and γ, δ are the roots of $x^2 + rx + s = 0$

$$\therefore \gamma + \delta = -r, \gamma\delta = s \quad \dots\dots(2)$$

Now, $(\alpha - \gamma)(\alpha - \delta)(\beta - \gamma)(\beta - \delta)$

$$\begin{aligned} &= [\alpha^2 - \alpha(\gamma + \delta) + \gamma\delta] [\beta^2 - \beta(\gamma + \delta) + \gamma\delta] \\ &= (\alpha^2 + r\alpha + s) (\beta^2 + r\beta + s) \\ &= \alpha^2\beta^2 + r\alpha\beta(\alpha + \beta) + r^2\alpha\beta + s(\alpha^2 + \beta^2) + sr(\alpha + \beta) + s^2 \\ &= \alpha^2\beta^2 + r\alpha\beta(\alpha + \beta) + r^2\alpha\beta + s((\alpha + \beta)^2 - 2\alpha\beta) + sr(\alpha + \beta) + s^2 \\ &= q^2 - pqr + r^2q + s(p^2 - 2q) + sr(-p) + s^2 \\ &= (q - s)^2 - rpq + r^2q + sp^2 - prs \\ &= (q - s)^2 - rq(p - r) + sp(p - r) \\ &= (q - s)^2 + (p - r)(sp - rq) \end{aligned}$$

For a common root (Let $\alpha = \gamma$ or $\beta = \delta$) $\dots\dots(3)$

then $(\alpha - \gamma)(\alpha - \delta)(\beta - \gamma)(\beta - \delta) = 0 \quad \dots\dots(4)$

from (3) and (4), we get

$$(q - s)^2 + (p - r)(sp - rq) = 0$$

$\Rightarrow (q - s)^2 = (p - r)(rq - sp)$, which is the required condition.

Illustrations 39: If $(y^2 - 5y + 3)(x^2 + x + 1) < 2x$ for all $x \in \mathbb{R}$, then find the interval in which y lies.

Solution :

$$(y^2 - 5y + 3)(x^2 + x + 1) < 2x, \forall x \in \mathbb{R}$$

$$\Rightarrow y^2 - 5y + 3 < \frac{2x}{x^2 + x + 1}$$

$$\text{Let } \frac{2x}{x^2 + x + 1} = P$$

$$\Rightarrow px^2 + (p - 2)x + p = 0$$

$$(1) \text{ Since } x \text{ is real, } (p - 2)^2 - 4p^2 \geq 0$$

$$\Rightarrow -2 \leq p \leq \frac{2}{3}$$

$$(2) \text{ The minimum value of } 2x/(x^2 + x + 1) \text{ is } -2.$$

$$\text{So, } y^2 - 5y + 3 < -2 \Rightarrow y^2 - 5y + 5 < 0$$

$$\Rightarrow y \in \left(\frac{5 - \sqrt{5}}{2}, \frac{5 + \sqrt{5}}{2} \right)$$

ANSWERS FOR DO YOURSELF

- 1: (i) (a) $-1, -2$; (b) 4 ; (c) $1 \pm \sqrt{2}$; (ii) $a, \frac{1}{a}$; (iii) $\frac{7}{3}$ (iv) $3, -\frac{1}{5}$
- 2: (i) $b = -4, c = 1$; (ii) (a) imaginary; (b) real & distinct; (c) real & coincident
- 3: (i) (a) $c = 0$; (b) $c = 1$; (c) $b \rightarrow \text{negative}, c \rightarrow \text{negative}$
- 4: (i) $b = \frac{9}{2}, c = 5$; (ii) $c = 0, 6$
- 5: (i) (a) $x \in (-\infty, -3] \cup [2, \infty)$; (b) $x \in (-2, -1)$; (c) $\left(-\frac{1}{4}, \frac{1}{3}\right]$;
 (d) $x \in (-6, -3) \cup \left(\frac{1}{2}, 2\right) - \{1\} \cup (9, \infty)$; (e) $[3, 7]$; (f) ϕ
- 6: (i) (a) 1 (b) -1
 (ii) (1) (i) $a < 0$ (ii) $b < 0$ (iii) $c < 0$ (iv) $D > 0$ (v) $\alpha + \beta < 0$ (vi) $\alpha\beta > 0$
 (2) (i) $a < 0$ (ii) $b > 0$ (iii) $c = 0$ (iv) $D > 0$ (v) $\alpha + \beta > 0$ (vi) $\alpha\beta = 0$
 (3) (i) $a < 0$ (ii) $b = 0$ (iii) $c = 0$ (iv) $D = 0$ (v) $\alpha + \beta = 0$ (vi) $\alpha\beta = 0$
 (iii) Third quadrant
 (iv) (a) $a > 9/16$ (b) $a < -2$
- 7: (i) $x \in (-\infty, -2) \cup (1, 3/2)$ (ii) $x \in \mathbb{R} - (0, 1]$
- 8: (i) (a) $x \in (1, 8)$ (b) $x \in (-\infty, 1/2)$ (ii) $x \in (-1, 1)$
- 9: (ii) least value = 0, greatest value = 1.
- 10: (i) $-3 < \alpha < 0 < \beta < 1$; (ii) C ; (iii) $a < 2$; (iv) $a < 2$; (v) $12 < \lambda < 16$
- 11: (i) $0, 2$
- 12: (ii) $-27, 5$; (iii) (a) $\frac{1}{a^2}(b^2 - 2ac)$, (b) $-\frac{c}{d}$, (c) $\frac{1}{a^2}(3ad - bc)$
- 13: (i) (a) $c^2y^2 + y(2ac - b^2) + a^2 = 0$; (b) $acx^2 - bx + 1 = 0$; (c) $acx^2 + (a + c)bx + (a + c)^2 = 0$
 (ii) $x^2 - p(p^4 - 5p^2q + 5q^2)x + p^2q^2(p^2 - 4q)(p^2 - q) = 0$

EXERCISE (E-1)

- For what values of a does the equation $9x^2 - 2x + a = 6 - ax$ possess equal roots ?
- Find the value of k for which the equation $(k - 1)x^2 + (k + 4)x + k + 7 = 0$ has equal roots.
- Find the values of a for which the roots of the equation $(2a - 5)x^2 - 2(a - 1)x + 3 = 0$ are equal.
- Form a quadratic equation whose roots are the numbers $\frac{1}{10 - \sqrt{72}}$ and $\frac{1}{10 + 6\sqrt{2}}$.
- For what values of a is the sum of the roots of the equation $x^2 + (2 - a - a^2)x - a^2 = 0$ equal to zero ?
- For what values of a is the ratio of the roots of the equation $ax^2 - (a + 3)x + 3 = 0$ equal to 1.5 ?
- The roots x_1 and x_2 of the equation $x^2 + px + 12 = 0$ are such that $x_2 - x_1 = 1$. Find p .
- Find k in the equation $5x^2 - kx + 1 = 0$ such that the difference between the roots of the equation is unity.
- Find p in the equation $x^2 - 4x + p = 0$ if it is known that the sum of the squares of its roots is equal to 16.
- For what values of a is the difference between the roots of the equation $2x^2 - (a + 1)x + (a - 1) = 0$ equal to their product ?
- Express $x_1^3 + x_2^3$ in terms of the coefficients of the equation $x^2 + px + q = 0$, where x_1 and x_2 are the roots of the equation.
- Assume that x_1 and x_2 are roots of the equation $3x^2 - ax + 2a - 1 = 0$. Calculate $x_1^3 + x_2^3$.
- Without solving the equation $3x^2 - 5x - 2 = 0$, find the sum of the cubes of its roots.
- For what values of m does the equation $x^2 - x + m = 0$ possess no real roots ?
- For what values of c does the equation $(c - 2)x^2 + 2(c - 2)x + 2 = 0$ possess no real roots ?
- Find integral values of k for which the quadratic equation $(k - 12)x^2 + 2(k - 12)x + 2 = 0$ possess no real roots ?
- For what values of k is the inequality $x^2 - (k - 3)x - k + 6 > 0$ valid for all real x ?
- For what integral k is the inequality $x^2 - 2(4k - 1)x + 15k^2 - 2k - 7 > 0$ valid for any real x ?
- Find the least integral value of k for which the equation $x^2 - 2(k + 2)x + 12 + k^2 = 0$ has two different real roots.
- Find all values of a for which the inequality $(a + 4)x^2 - 2ax + 2a - 6 < 0$ is satisfied for all $x \in \mathbb{R}$.
- Find all values of a for which the inequality $(a - 3)x^2 - 2ax + 3a - 6 > 0$ is satisfied for all values of x .
- Find all values of a for which the inequality $(a - 1)x^2 - (a + 1)x + a + 1 > 0$ is satisfied for all real x .
- For what values of a do the equations $x^2 + ax + 1 = 0$ and $x^2 + x + a = 0$ have exactly one root in common ?
- The trinomial $ax^2 + bx + c$ has no real roots, $a + b + c < 0$. Find the sign of the number c .
- For what values of a do the graphs of the functions $y = 2ax + 1$ and $y = (a - 6)x^2 - 2$ not intersect?

EXERCISE (E-2)

- For what values of 'a' does the quadratic equation $x^2 + (2a\sqrt{a^2 - 3})x + 4 = 0$ possess equal roots ?
- Find the value of a for which one root of the equation $x^2 + (2a - 1)x + a^2 + 2 = 0$ is twice as large as the other.
- For what values of a is the ratio of the roots of the equation $x^2 + ax + a + 2 = 0$ equal to 2 ?
- For what values of a do the roots x_1 and x_2 of the equation $x^2 - (3a + 2)x + a^2 = 0$ satisfy the relation $x_1 = 9x_2$? Find the roots.
- Find a such that one of the roots of the equation $x^2 - \frac{15}{4}x + a = 0$ is the square of the other.
- For what value of a is the difference between the roots of the equation $(a - 2)x^2 - (a - 4)x - 2 = 0$ equal to 3 ?
- Find all the values of a for which the sum of the roots of the equation $x^2 - 2a(x - 1) - 1 = 0$ is equal to the sum of the squares of its roots.
- Find the coefficients of the equation $x^2 + px + q = 0$ such that its roots are equal to p and q.
- Calculate $\frac{1}{x_1^3} + \frac{1}{x_2^3}$, where x_1 and x_2 are roots of the equation $2x^2 - 3ax - 2 = 0$.
- For what values of m does the equation $x^2 - x + m^2 = 0$ possess no real roots ?
- For what values of m does the equation $mx^2 - (m + 1)x + 2m - 1 = 0$ possess no real roots ?
- For what values of a is the inequality $ax^2 + 2ax + 0.5 > 0$ valid throughout the entire number axis ?
- For what values of a does the equation $(2 - x)(x + 1) = a$ possess real and positive roots ?
- For what least integral k is the quadratic trinomial $(k - 2)x^2 + 8x + k + 4$ positive for all values of x ?
- Given two quadratic equations $x^2 - x + m = 0$ and $x^2 - x + 3m = 0$, $m \neq 0$. Find the value of m for which one of the roots of the second equation is equal to double the root of the first equation.
- For what values of p does the vertex of the parabola $y = x^2 + 2px + 13$ lie at a distance of 5 from the origin ?
- If the roots of the equation $x^2 - 5x + 16 = 0$ are α , β and the roots of the equation $x^2 + px + q = 0$ are

$(\alpha^2 + \beta^2)$ and $\frac{\alpha\beta}{2}$, then-

[AIEEE-2002]

(1) $p = 1$ and $q = 56$

(2) $p = 1$ and $q = -56$

(3) $p = -1$ and $q = 56$

(4) $p = -1$ and $q = -56$

- If α and β be the roots of the equation $(x - a)(x - b) = c$ and $c \neq 0$, then roots of the equation $(x - \alpha)(x - \beta) + c = 0$ are-

[AIEEE-2002]

(1) a and c

(2) b and c

(3) a and b

(4) a + b and b + c

- If $\alpha^2 = 5\alpha - 3$, $\beta^2 = 5\beta - 3$ then the value of $\frac{\alpha}{\beta} + \frac{\beta}{\alpha}$ is-

[AIEEE-2002]

(1) 19/3

(2) 25/3

(3) -19/3

(4) none of these

20. The value of a for which one roots of the quadratic equation $(a^2 - 5a + 3)x^2 + (3a - 1)x + 2 = 0$ is twice as large as the other is [AIEEE-2003]
 (1) $-2/3$ (2) $1/3$ (3) $-1/3$ (4) $2/3$
21. The number of real solutions of the equation $x^2 - 3|x| + 2 = 0$, is- [AIEEE-2003]
 (1) 4 (2) 1 (3) 3 (4) 2
22. If $(1 - p)$ is a root of quadratic equation $x^2 + px + (1 - p) = 0$ then its roots are-
 (1) $0, -1$ (2) $-1, 1$ (3) $0, 1$ (4) $-1, 2$ [AIEEE-2004]
23. If one root of the equation $x^2 + px + 12 = 0$ is 4, while the equation $x^2 + px + q = 0$ has equal roots, then the value of 'q' is- [AIEEE-2004]
 (1) 3 (2) 12 (3) $49/4$ (4) 4
24. If value of a for which the sum of the squares of the roots of the equation $x^2 - (a - 2)x - a - 1 = 0$ assume the least value is- [AIEEE-2005]
 (1) 2 (2) 3 (3) 0 (4) 1
25. If the roots of the equation $x^2 - bx + c = 0$ be two consecutive integers, then $b^2 - 4c$ equals- [AIEEE-2005]
 (1) 1 (2) 2 (3) 3 (4) -2
26. If both the roots of the quadratic equation $x^2 - 2kx + k^2 + k - 5 = 0$ are less than 5, then k lies in the interval- [AIEEE-2005]
 (1) $[4, 5]$ (2) $(-\infty, 4)$ (3) $(6, \infty)$ (4) $(5, 6)$
27. If $x^2 + 2ax + 10 - 3a > 0$ for all $x \in \mathbb{R}$, then [JEE 2004 (Screening)]
 (1) $-5 < a < 2$ (2) $a < -5$ (3) $a > 5$ (4) $2 < a < 5$

EXERCISE (S-1)

1. Solve following Inequalities over the set of real numbers -

- (i) $\frac{x^2 + 2x - 3}{x^2 + 1} < 0$ (ii) $\frac{(x-1)(x+2)^2}{-1-x} < 0$
- (iii) $x^4 - 2x^2 - 63 \leq 0$ (iv) $\frac{x+1}{(x-1)^2} < 1$
- (v) $\frac{x^2 - 7x + 12}{2x^2 + 4x + 5} > 0$ (vi) $\frac{x^2 + 6x - 7}{x^2 + 1} \leq 2$
- (vii) $\frac{x^4 + x^2 + 1}{x^2 - 4x - 5} < 0$ (viii) $\frac{x+7}{x-5} + \frac{3x+1}{2} \geq 0$
- (ix) $\frac{1}{x+2} < \frac{3}{x-3}$ (x) $\frac{14x}{x+1} - \frac{9x-30}{x-4} < 0$
- (xi) $\frac{x^2 - 5x + 12}{x^2 - 4x + 5} > 3$ (xii) $\frac{x^2 + 2}{x^2 - 1} < -2$
- (xiii) $\frac{(2-x^2)(x-3)^3}{(x+1)(x^2-3x-4)} \geq 0$ (xiv) $\frac{5-4x}{3x^2-x-4} < 4$

$$(xv) \frac{(x+2)(x^2-2x+1)}{4+3x-x^2} \geq 0$$

$$(xvi) \frac{x^4-3x^3+2x^2}{x^2-x-30} > 0$$

$$(xvii) \frac{2x}{x^2-9} \leq \frac{1}{x+2}$$

$$(xviii) \frac{1}{x-2} + \frac{1}{x-1} > \frac{1}{x}$$

$$(xix) \frac{20}{(x-3)(x-4)} + \frac{10}{x-4} + 1 > 0$$

$$(xx) \frac{(x-2)(x-4)(x-7)}{(x+2)(x+4)(x+7)} > 1$$

$$(xxi) (x^2-2x)(2x-2) - 9 \frac{2x-2}{x^2-2x} \leq 0$$

$$(xxii) \frac{|x+2|-x}{x} < 2$$

$$(xxiii) \left| \frac{x^2-5x+4}{x^2-4} \right| \leq 1$$

$$(xxiv) \frac{1}{x-2} - \frac{1}{x} \leq \frac{2}{x+2}$$

2. α, β are the roots of the equation $K(x^2-x) + x + 5 = 0$. If K_1 & K_2 are the two values of K for which the roots α, β are connected by the relation $(\alpha/\beta) + (\beta/\alpha) = 4/5$. Find the value of $(K_1/K_2) + (K_2/K_1)$.
3. Let a, b be arbitrary real numbers. Find the smallest natural number 'b' for which the equation $x^2 + 2(a+b)x + (a-b+8) = 0$ has unequal real roots for all $a \in \mathbb{R}$.
4. Find the range of values of a , such that $f(x) = \frac{ax^2 + 2(a+1)x + 9a+4}{x^2-8x+32}$ is always negative.
5. Let the quadratic equation $x^2 + 3x - k = 0$ has roots a, b and $x^2 + 3x - 10 = 0$ has roots c, d such that modulus of difference of the roots of the first equation is equal to twice the modulus of the difference of the roots of the second equation. If the value of 'k' can be expressed as rational number in the lowest form as m/n then find the value of $(m+n)$.
6. When $y^2 + my + 2$ is divided by $(y-1)$ then the quotient is $f(y)$ and the remainder is R_1 . When $y^2 + my + 2$ is divided by $(y+1)$ then quotient is $g(y)$ and the remainder is R_2 . If $R_1 = R_2$ then find the value of m .
7. If the quadratic equations, $x^2 + bx + c = 0$ and $bx^2 + cx + 1 = 0$ have a common root then prove that either $b+c+1=0$ or $b^2+c^2+1=bc+b+c$.
8. Find the value of m for which the quadratic equations $x^2 - 11x + m = 0$ and $x^2 - 14x + 2m = 0$ may have common root.
9. If α, β are the roots of $ax^2 + bx + c = 0$, ($a \neq 0$) and $\alpha + \delta, \beta + \delta$ are the roots of $Ax^2 + Bx + C = 0$, ($A \neq 0$) for some constant δ , then prove that, $\frac{b^2-4ac}{a^2} = \frac{B^2-4AC}{A^2}$.
10. If α, β are the roots of the quadratic equation $ax^2+bx+c=0$ then which of the following expressions in α, β will denote the symmetric functions of roots. Give proper reasoning.
 - (i) $f(\alpha, \beta) = \alpha^2 - \beta$
 - (ii) $f(\alpha, \beta) = \alpha^2\beta + \alpha\beta^2$
 - (iii) $f(\alpha, \beta) = \ln \frac{\alpha}{\beta}$
 - (iv) $f(\alpha, \beta) = \cos(\alpha - \beta)$

11. Let $P(x) = 4x^2 + 6x + 4$ and $Q(y) = 4y^2 - 12y + 25$. Find the unique pair of real numbers (x, y) that satisfy $P(x) \cdot Q(y) = 28$.
12. Find the product of the real roots of the equation, $x^2 + 18x + 30 = 2\sqrt{x^2 + 18x + 45}$
13. We call 'p' a good number if the inequality $\frac{2x^2 + 2x + 3}{x^2 + x + 1} \leq p$ is satisfied for any real x. Find the smallest integral good number.
14. Find the values of 'a' for which $-3 < [(x^2 + ax - 2)/(x^2 + x + 1)] < 2$ is valid for all real x.
15. Suppose $a, b, c \in I$ such that greatest common divisor of $x^2 + ax + b$ and $x^2 + bx + c$ is $(x + 1)$ and the least common multiple of $x^2 + ax + b$ and $x^2 + bx + c$ is $(x^3 - 4x^2 + x + 6)$. Find the value of $(a + b + c)$.
16. If roots of the equation $(x - \alpha)(x - 4 + \beta) + (x - 2 + \alpha)(x + 2 - \beta) = 0$ are p and q then find the absolute value of the sum of the roots of the equations $2(x - p)(x - q) - (x - \alpha)(x - 4 + \beta) = 0$ and $2(x - p)(x - q) - (x - 2 + \alpha)(x + 2 - \beta) = 0$.
17. If the roots of $x^2 - ax + b = 0$ are real & differ by a quantity which is less than c ($c > 0$), prove that b lies between $(1/4)(a^2 - c^2)$ & $(1/4)a^2$.
18. Find all values of p for which the roots of the equation $(p - 3)x^2 - 2px + 5p = 0$ are real and positive.
19. Find all the values of the parameter 'a' for which both roots of the quadratic equation $x^2 - ax + 2 = 0$ belong to the interval $(0, 3)$.
20. At what values of 'a' do all the zeroes of the function $f(x) = (a - 2)x^2 + 2ax + a + 3$ lie on the interval $(-2, 1)$?
21. Consider the quadratic polynomial $f(x) = x^2 - 4ax + 5a^2 - 6a$
 - (a) Find the smallest positive integral value of 'a' for which $f(x)$ positive for every real x.
 - (b) Find the largest distance between the roots of the equation $f(x) = 0$
 - (c) Find the set of values of 'a' for which range of $f(x)$ is $[-8, \infty)$
22. Solve the inequality. Where ever base is not given take it as 10.
 - (i) $(\log_2 x)^4 - \left(\log_{\frac{1}{2}} \frac{x^5}{4}\right)^2 - 20 \log_2 x + 148 < 0$
 - (ii) $(\log 100x)^2 + (\log 10x)^2 + \log x \leq 14$
 - (iii) $\log_{1/2}(x + 1) > \log_2(2 - x)$
 - (iv) $\log_x 2 \cdot \log_{2x} 2 \cdot \log_2 4x > 1$
 - (v) $\log_{1/5}(2x^2 + 5x + 1) < 0$
 - (vi) $\log_{1/2} x + \log_3 x > 1$
 - (vii) $\log_{x^2}(2 + x) < 1$
 - (viii) $(\log_{|x+6|} 2) \cdot \log_2(x^2 - x - 2) \geq 1$
 - (ix) $\log_x \frac{4x + 5}{6 - 5x} < -1$
 - (x) $\log_3 \frac{|x^2 - 4x| + 3}{x^2 + |x - 5|} \geq 0$

EXERCISE (S-2)

- Solve the following where $x \in \mathbb{R}$.
 - $(x-1)|x^2 - 4x + 3| + 2x^2 + 3x - 5 = 0$
 - $3|x^2 - 4x + 2| = 5x - 4$
 - $|x^3 + 1| + x^2 - x - 2 = 0$
 - $2^{|x+2|} - |2^{x+1} - 1| = 2^{x+1} + 1$
- Let α, β and γ are the roots of the cubic $x^3 - 3x^2 + 1 = 0$. Find a cubic whose roots are $\frac{\alpha}{\alpha-2}, \frac{\beta}{\beta-2}$ and $\frac{\gamma}{\gamma-2}$. Hence or otherwise find the value of $(\alpha-2)(\beta-2)(\gamma-2)$.
- Suppose a cubic polynomial $f(x) = x^3 + px^2 + qx + 72$ is divisible by both $x^2 + ax + b$ and $x^2 + bx + a$ (where a, b, p, q are constants and $a \neq b$). Find the sum of the squares of the roots of the cubic polynomial.
- Given $x, y \in \mathbb{R}$, $x^2 + y^2 > 0$. If the maximum and minimum value of the expression $E = \frac{x^2 + y^2}{x^2 + xy + 4y^2}$ are M and m , and A denotes the average value of M and m , compute (2016) A .
- Find the complete set of real values of 'a' for which both roots of the quadratic equation $(a^2 - 6a + 5)x^2 - \sqrt{a^2 + 2a}x + (6a - a^2 - 8) = 0$ lie on either side of the origin.
- Find the values of K so that the quadratic equation $x^2 + 2(K-1)x + K + 5 = 0$ has at least one positive root.
- Let $P(x) = x^2 + bx + c$, where b and c are integer. If $P(x)$ is a factor of both $x^4 + 6x^2 + 25$ and $3x^4 + 4x^2 + 28x + 5$, find the value of $P(1)$.
- Let α, β, γ be distinct real numbers such that

$$a\alpha^2 + b\alpha + c = (\sin\theta)\alpha^2 + (\cos\theta)\alpha$$

$$a\beta^2 + b\beta + c = (\sin\theta)\beta^2 + (\cos\theta)\beta$$

$$a\gamma^2 + b\gamma + c = (\sin\theta)\gamma^2 + (\cos\theta)\gamma$$
 (where $a, b, c \in \mathbb{R}$)
 - Find the maximum value of the expression $\frac{a^2 + b^2}{a^2 + 3ab + 5b^2}$
 - If $\vec{V}_1 = a\hat{i} + b\hat{j} + c\hat{k}$ makes an angle $\frac{\pi}{3}$ with $\vec{V}_2 = \hat{i} + \hat{j} + \sqrt{2}\hat{k}$, then find the number of values of $\theta \in [0, 2\pi]$
- Find the product of uncommon real roots of the two polynomials $P(x) = x^4 + 2x^3 - 8x^2 - 6x + 15$ and $Q(x) = x^3 + 4x^2 - x - 10$.
- If $ax^{17} + bx^{16} + 1$ is divisible by $x^2 - x - 1$, then find the integral value of a .

EXERCISE (JM)

- If the roots of the equation $bx^2 + cx + a = 0$ be imaginary, then for all real values of x , the expression $3b^2x^2 + 6bcx + 2c^2$ is :- [AIEEE-2009]
 - Greater than $-4ab$
 - Less than $-4ab$
 - Greater than $4ab$
 - Less than $4ab$
- Let for $a \neq a_1 \neq 0$, $f(x) = ax^2 + bx + c$, $g(x) = a_1x^2 + b_1x + c_1$ and $p(x) = f(x) - g(x)$. If $p(x) = 0$ only for $x = -1$ and $p(-2) = 2$, then the value of $p(2)$ is: [AIEEE-2011]
 - 18
 - 3
 - 9
 - 6

3. Sachin and Rahul attempted to solve a quadratic equation. Sachin made a mistake in writing down the constant term and ended up in roots (4, 3). Rahul made a mistake in writing down coefficient of x to get roots (3, 2). The correct roots of equation are: [AIEEE-2011]
 (1) $-4, -3$ (2) $6, 1$ (3) $4, 3$ (4) $-6, -1$
4. The equation $e^{\sin x} - e^{-\sin x} - 4 = 0$ has : [AIEEE-2012]
 (1) exactly four real roots. (2) infinite number of real roots.
 (3) no real roots. (4) exactly one real root.
5. If the equations $x^2 + 2x + 3 = 0$ and $ax^2 + bx + c = 0$, $a, b, c \in \mathbb{R}$, have a common root, then $a : b : c$ is : [JEE-MAIN-2013]
 (1) $1 : 2 : 3$ (2) $3 : 2 : 1$ (3) $1 : 3 : 2$ (4) $3 : 1 : 2$
6. Let α and β be the roots of equation $x^2 - 6x - 2 = 0$. If $a_n = \alpha^n - \beta^n$, for $n \geq 1$, then the value of $\frac{a_{10} - 2a_8}{2a_9}$ is equal to : [JEE-MAIN-2015]
 (1) 3 (2) -3 (3) 6 (4) -6
7. The sum of all real values of x satisfying the equation $(x^2 - 5x + 5)^{x^2 + 4x - 60} = 1$ is :- [JEE-MAIN-2016]
 (1) 5 (2) 3 (3) -4 (4) 6

EXERCISE (JA)

1. The smallest value of k , for which both the roots of the equation, $x^2 - 8kx + 16(k^2 - k + 1) = 0$ are real, distinct and have values at least 4, is [JEE 2009, 4 (-1)]
2. Let p and q be real numbers such that $p \neq 0$, $p^3 \neq q$ and $p^3 \neq -q$. If α and β are nonzero complex numbers satisfying $\alpha + \beta = -p$ and $\alpha^3 + \beta^3 = q$, then a quadratic equation having $\frac{\alpha}{\beta}$ and $\frac{\beta}{\alpha}$ as its roots is [JEE 2010, 3]
 (A) $(p^3 + q)x^2 - (p^3 + 2q)x + (p^3 + q) = 0$ (B) $(p^3 + q)x^2 - (p^3 - 2q)x + (p^3 + q) = 0$
 (C) $(p^3 - q)x^2 - (5p^3 - 2q)x + (p^3 - q) = 0$ (D) $(p^3 - q)x^2 - (5p^3 + 2q)x + (p^3 - q) = 0$
3. Let α and β be the roots of $x^2 - 6x - 2 = 0$, with $\alpha > \beta$. If $a_n = \alpha^n - \beta^n$ for $n \geq 1$, then the value of $\frac{a_{10} - 2a_8}{2a_9}$ is [JEE 2011]
 (A) 1 (B) 2 (C) 3 (D) 4
4. A value of b for which the equations

$$x^2 + bx - 1 = 0$$

$$x^2 + x + b = 0,$$
 have one root in common is - [JEE 2011]
 (A) $-\sqrt{2}$ (B) $-i\sqrt{3}$ (C) $i\sqrt{5}$ (D) $\sqrt{2}$

5. Let S be the set of all non-zero numbers α such that the quadratic equation $\alpha x^2 - x + \alpha = 0$ has two distinct real roots x_1 and x_2 satisfying the inequality $|x_1 - x_2| < 1$. Which of the following intervals is(are) a subset(s) of S ? [JEE 2015, 4M, -0M]

(A) $\left(-\frac{1}{2}, -\frac{1}{\sqrt{5}}\right)$ (B) $\left(-\frac{1}{\sqrt{5}}, 0\right)$ (C) $\left(0, \frac{1}{\sqrt{5}}\right)$ (D) $\left(\frac{1}{\sqrt{5}}, \frac{1}{2}\right)$

6. Let $-\frac{\pi}{6} < \theta < -\frac{\pi}{12}$. Suppose α_1 and β_1 are the roots of the equation $x^2 - 2x\sec\theta + 1 = 0$ and α_2 and β_2 are the roots of the equation $x^2 + 2x\tan\theta - 1 = 0$. If $\alpha_1 > \beta_1$ and $\alpha_2 > \beta_2$, then $\alpha_1 + \beta_2$ equals [JEE(Advanced)-2016, 3(-1)]

(A) $2(\sec\theta - \tan\theta)$ (B) $2\sec\theta$ (C) $-2\tan\theta$ (D) 0

PARAGRAPH

Let p, q be integers and let α, β be the roots of the equation, $x^2 - x - 1 = 0$, where $\alpha \neq \beta$. For $n = 0, 1, 2, \dots$, let $a_n = p\alpha^n + q\beta^n$.

FACT : If a and b are rational numbers and $a + b\sqrt{5} = 0$, then $a = 0 = b$.

7. If $a_4 = 28$, then $p + 2q =$ [JEE(Advanced)-2017, 3(-1)]
 (A) 14 (B) 7 (C) 12 (D) 21
8. $a_{12} =$ [JEE(Advanced)-2017, 3(-1)]
 (A) $2a_{11} + a_{10}$ (B) $a_{11} - a_{10}$ (C) $a_{11} + a_{10}$ (D) $a_{11} + 2a_{10}$

ANSWER KEY

EXERCISE (E-1)

1. $a = 20 \pm 6\sqrt{5}$
2. $k_1 = \frac{-22}{3}, k_2 = 2$
3. $a = 4$
4. $28x^2 - 20x + 1 = 0$
5. $a_1 = -2, a_2 = 1$
6. $a_1 = 2, a_2 = 9/2$
7. $p = \pm 7$
8. $k = \pm 3\sqrt{5}$
9. $p = 0$
10. $a = 2$
11. $x_1^3 + x_2^3 = 3pq - p^3$
12. $x_1^3 + x_2^3 = \frac{a(a^2 - 18a + 9)}{27}$
13. $\frac{215}{27}$
14. For all $m \in \left(\frac{1}{4}, +\infty\right)$
15. For all $c \in [2, 4)$
16. $k = 13$
17. $(-3, 5)$
18. $k = 3$
19. $k = 3$
20. For all $a \in (-\infty, -6)$
21. For all $a \in (6, +\infty)$
22. For all $a \in \left(\frac{5}{3}, +\infty\right)$
23. $a = -2$
24. $c < 0$
25. For all $a \in (-6, 3)$

EXERCISE (E-2)

1. $a = \pm 2$
2. $a = -4$
3. $a_1 = -\frac{3}{2}, a_2 = 6$
4. $\{2, 18\}$ for $a = 6, \left\{\frac{2}{19}, \frac{18}{19}\right\}$ for $a = -\frac{6}{19}$
5. $a_1 = -\frac{125}{8}, a_2 = \frac{27}{8}$
6. $a_1 = 3/2, a_2 = 3$
7. $a_1 = 1/2, a_2 = 1$
8. $p_1 = 0, q_1 = 0, p_2 = 1, q_2 = -2$
9. $\frac{1}{x_1^3} + \frac{1}{x_2^3} = -\frac{27a^3 + 36a}{8}$
10. For all $m \in \left(-\infty, -\frac{1}{2}\right) \cup \left(\frac{1}{2}, \infty\right)$
11. For all $m \in \left(-\infty, -\frac{1}{7}\right) \cup (1, +\infty)$
12. $\left[0, \frac{1}{2}\right)$
13. For all $a \in \left[2, \frac{9}{4}\right]$
14. $k = 5$
15. $m = -2$
16. $\{-4, -3, 3, 4\}$
17. 4
18. 3
19. 1
20. 4
21. 1
22. 1
23. 3
24. 4
25. 1
26. 2
27. 1

EXERCISE (S-1)

1. (i) $(-3, 1)$ (ii) $(-\infty, -2) \cup (-2, -1) \cup (1, +\infty)$ (iii) $[-3, 3]$ (iv) $(-\infty, 0) \cup (3, +\infty)$
- (v) $(-\infty, 3) \cup (4, +\infty)$ (vi) $(-\infty, +\infty)$ (vii) $(-1, 5)$ (viii) $[1, 3] \cup (5, +\infty)$
- (ix) $(-9/2, -2) \cup (3, +\infty)$ (x) $(-1, 1) \cup (4, 6)$ (xi) $(1/2, 3)$ (xii) $(-1, 0) \cup (0, 1)$
- (xiii) $[-\sqrt{2}, -1) \cup (-1, \sqrt{2}] \cup [3, 4)$ (xiv) $(-\infty, -\sqrt{7}/2) \cup (-1, \sqrt{7}/2) \cup (4/3, +\infty)$
- (xv) $(-\infty, -2] \cup (-1, 4)$ (xvi) $(-\infty, -5) \cup (1, 2) \cup (6, +\infty)$ (xvii) $(-\infty, -3) \cup (-2, 3)$
- (xviii) $(-\sqrt{2}, 0) \cup (1, \sqrt{2}) \cup (2, +\infty)$ (xix) $(-\infty, -2) \cup (-1, 3) \cup (4, +\infty)$
- (xx) $(-\infty, -7) \cup (-4, -2)$ (xxi) $(-\infty, -1] \cup (0, 1] \cup (2, 3]$ (xxii) $(-\infty, 0) \cup (1, +\infty)$

$$(xxiii) [0, 8/5] \cup [5/2, +\infty) \quad (xxiv) \left(-2, \frac{(3-\sqrt{17})}{2}\right] \cup (0, 2) \cup \left[\frac{(3+\sqrt{17})}{2}, +\infty\right)$$

$$2. 254 \quad 3. 5 \quad 4. a \in \left(-\infty, -\frac{1}{2}\right) \quad 5. 191 \quad 6. 0 \quad 8. 0 \text{ or } 24$$

$$9. 2x^2 + 2x \cos(A - B) - 2 \quad 10. (ii) \text{ and } (iv) \quad 11. \left(-\frac{3}{4}, \frac{3}{2}\right) \quad 12. 20 \quad 13. 4$$

$$14. -2 < a < 1 \quad 15. -6 \quad 16. 4 \quad 18. \text{ For all } p \in \left[3, \frac{15}{4}\right] \quad 19. 2\sqrt{2} \leq a < \frac{11}{3}$$

$$20. \left(-\infty, -\frac{1}{4}\right) \cup \{2\} \cup (5, 6] \quad 21. (a) 7, (b) 6, (c) 2 \text{ or } 4$$

$$22. (i) x \in \left(\frac{1}{16}, \frac{1}{8}\right) \cup (8, 16) \quad (ii) \frac{1}{\sqrt{10^9}} \leq x \leq 10 \quad (iii) -1 < x < \frac{1-\sqrt{5}}{2} \text{ or } \frac{1+\sqrt{5}}{2} < x < 2$$

$$(iv) 2^{-\sqrt{2}} < x < 2^{-1} ; 1 < x < 2^{\sqrt{2}} \quad (v) (-\infty, -2.5) \cup (0, \infty)$$

$$(vi) 0 < x < 3^{1/\log 3} \text{ (where base of log is 2)} \quad (vii) -2 < x < -1, -1 < x < 0, 0 < x < 1, x > 2$$

$$(viii) x < -7, -5 < x \leq -2, x \geq 4 \quad (ix) \frac{1}{2} < x < 1 \quad (x) x \leq -\frac{2}{3} ; \frac{1}{2} \leq x \leq 2$$

EXERCISE (S-2)

1. (a) $x = 1$; (b) $x = 2$ or 5 ; (c) $x = -1$ or 1 ; (d) $x \geq -1$ or $x = -3$
 2. $3y^3 - 9y^2 - 3y + 1 = 0$; $(\alpha - 2)(\beta - 2)(\gamma - 2) = 3$ 3. 146 4. 1344
 5. $(-\infty, -2] \cup [0, 1) \cup (2, 4) \cup (5, \infty)$ 6. $K \leq -1$ 7. $P(1) = 4$ 8. (a) 2, (b) 3
 9. 6 10. 987

EXERCISE (JM)

1. 1 2. 1 3. 2 4. 3 5. 1 6. 1 7. 2

EXERCISE (JA)

1. 2 2. B 3. C 4. B 5. A, D 6. C 7. C 8. C

Important Notes

NURTURE COURSE

CIRCLE

CONTENTS

CIRCLE

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JEE (Main) Syllabus :

Standard form of equation of a circle, general form of the equation of a circle, its radius and centre, equation of a circle when the end points of a diameter are given, points of intersection of a line and a circle with the centre at the origin and condition for a line to be tangent to a circle, equation of the tangent.

JEE (Advanced) Syllabus :

Equation of a circle in various forms, equations of tangent, normal and chord. Parametric equations of a circle, intersection of a circle with a straight line or a circle, equation of a circle through the points of intersection of two circles and those of a circle and a straight line.

CIRCLE

1.(A) DEFINITION :

A circle is the locus of a point which moves in a plane in such a way that its distance from a fixed point (in the same given plane) remains constant. The fixed point is called the centre of the circle and the constant distance is called the radius of the circle.

Equation of a circle :

The curve traced by the moving point is called its circumference i.e. the equation of any circle is satisfied by co-ordinates of all points on its circumference.

or

The equation of the circle means the equation of its circumference.

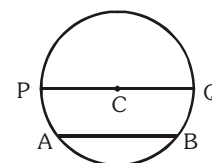
or

It is the set of all points lying on the circumference of the circle.

Chord and diameter - the line joining any two points on the circumference is called a chord. If any chord passing through its centre is called its diameter.

AB = chord, PQ = diameter

C = centre

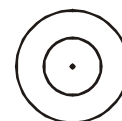


(B) BASIC THEOREMS & RESULTS OF CIRCLES :

(a) **Concentric circles** : Circles having same centre.

(b) **Congruent circles** : Iff their radii are equal.

(c) **Congruent arcs** : Iff they have same degree measure at the centre.



Theorem 1 :

(i) If two arcs of a circle (or of congruent circles) are congruent, the corresponding chords are equal.

Converse : If two chords of a circle are equal then their corresponding arcs are congruent.

(ii) Equal chords of a circle (or of congruent circles) subtend equal angles at the centre.

Converse : If the angle subtended by two chords of a circle (or of congruent circles) at the centre are equal, the chords are equal.

Theorem 2 :

(i) The perpendicular from the centre of a circle to a chord bisects the chord.

Converse : The line joining the mid point of a chord to the centre of a circle is perpendicular to the chord.

(ii) Perpendicular bisectors of two chords of a circle intersect at its centre.

Theorem 3 :

(i) There is one and only one circle passing through three non collinear points.

(ii) If two circles intersect in two points, then the line joining the centres is perpendicular bisector of common chords.

Theorem 4 :

(i) Equal chords of a circle (or of congruent circles) are equidistant from the centre.

Converse : Chords of a circle (or of congruent circles) which are equidistant from the centre are equal in length.

(ii) If two equal chords are drawn from a point on the circle, then the centre of circle will lie on angle bisector of these two chords.

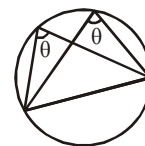
(iii) Of any two chords of a circle larger will be near to centre.

Theorem 5 :

(i) The degree measure of an arc or angle subtended by an arc at the centre is double the angle subtended by it at any point of alternate segment.

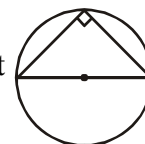


(ii) Angle in the same segment of a circle are equal.



(iii) The angle in a semi circle is right angle.

Converse : The arc of a circle subtending a right angle in alternate segment is semi circle.

**Theorem 6 :**

Any angle subtended by a minor arc in the alternate segment is acute and any angle subtended by a major arc in the alternate segment is obtuse.

Theorem 7 :

If a line segment joining two points subtends equal angles at two other points lying on the same side of the line segment, the four points are concyclic, i.e. lie on the same circle.

(d) Cyclic Quadrilaterals :

A quadrilateral is called a cyclic quadrilateral if its all vertices lie on a circle.

Theorem 1 :

The sum of either pair of opposite angles of a cyclic quadrilateral is 180°

OR

The opposite angles of a cyclic quadrilateral are supplementary.

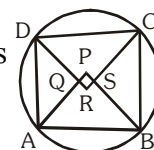
Converse : If the sum of any pair of opposite angle of a quadrilateral is 180° , then the quadrilateral is cyclic.

Theorem 2 :

If a side of a cyclic quadrilateral is produced, then the exterior angle is equal to the interior opposite angle.

Theorem 3 :

The internal angle bisectors of a cyclic quadrilateral form a quadrilateral which is also cyclic.

**Theorem 4 :**

If two sides of a cyclic quadrilateral are parallel then the remaining two sides are equal and the diagonals are also equal.

OR

A cyclic trapezium is isosceles and its diagonals are equal.

Converse : If two non-parallel sides of a trapezium are equal, then it is cyclic.

OR

An isosceles trapezium is always cyclic.

Theorem 5 :

When the opposite sides of cyclic quadrilateral (provided that they are not parallel) are produced, then the exterior angle bisectors intersect at right angle.

(C) TANGENTS TO A CIRCLE :

Theorem 1 :

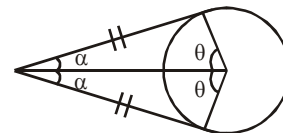
A tangent to a circle is perpendicular to the radius through the point of contact.

Converse : A line drawn through the end point of a radius and perpendicular to it is a tangent to the circle.

Theorem 2 :

If two tangents are drawn to a circle from an external point, then :

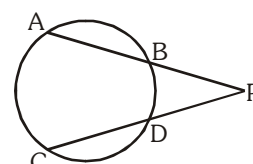
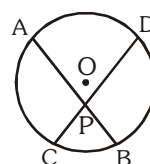
- they are equal.
- they subtend equal angles at the centre,
- they are equally inclined to the segment, joining the centre to that point.



Theorem 3 :

If two chords of a circle intersect inside or outside the circle when produced, the rectangle formed by the two segments of one chord is equal in area to the rectangle formed by the two segments of the other chord.

$$PA \times PB = PC \times PD$$

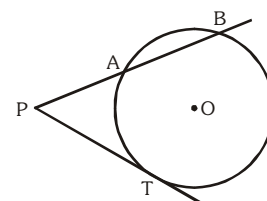


Theorem 4 :

If PAB is a secant to a circle intersecting the circle at A and B and PT is tangent segment, then $PA \times PB = PT^2$

OR

Area of the rectangle formed by the two segments of a chord is equal to the area of the square of side equal to the length of the tangent from the point on the circle.



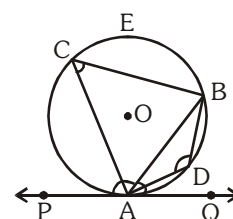
Theorem 5 :

If a chord is drawn through the point of contact of a tangent to a circle, then the angles which this chord makes with the given tangent are equal respectively to the angles formed in the corresponding alternate segments.

$$\angle BAQ = \angle ACB \text{ and } \angle BAP = \angle ADB$$

Converse :

If a line is drawn through an end point of a chord of a circle so that the angle formed with the chord is equal to the angle subtended by the chord in the alternate segment, then the line is a tangent to the circle.



2. STANDARD EQUATIONS OF THE CIRCLE :

(a) Central Form :

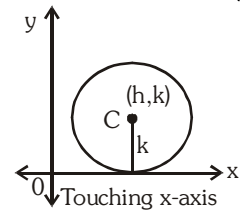
If (h, k) is the centre and r is the radius of the circle then its equation is

$$(x-h)^2 + (y-k)^2 = r^2$$

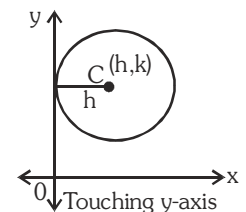
Special Cases :

- If centre is origin $(0,0)$ and radius is ' r ' then equation of circle is $x^2 + y^2 = r^2$ and this is called the standard form.
- If radius of circle is zero then equation of circle is $(x-h)^2 + (y-k)^2 = 0$. Such circle is called zero circle or **point circle**.

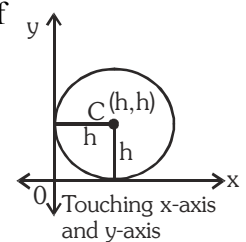
- (iii) When circle touches x-axis then equation of the circle is
 $(x-h)^2 + (y-k)^2 = k^2$.



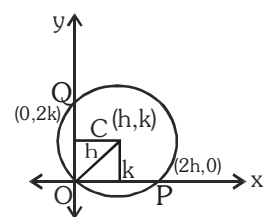
- (iv) When circle touches y-axis then equation of circle is
 $(x-h)^2 + (y-k)^2 = h^2$.



- (v) When circle touches both the axes (x-axis and y-axis) then equation of circle $(x-h)^2 + (y-h)^2 = h^2$.



- (vi) When circle passes through the origin and centre of the circle is (h,k) then radius $\sqrt{h^2 + k^2} = r$ and intercept cut on x-axis $OP = 2h$, and intercept cut on y-axis is $OQ = 2k$ and equation of circle is $(x-h)^2 + (y-k)^2 = h^2 + k^2$ or $x^2 + y^2 - 2hx - 2ky = 0$



Note : Centre of the circle may exist in any quadrant hence for general cases use $\pm$ sign before h & k .

(b) General equation of circle

$x^2 + y^2 + 2gx + 2fy + c = 0$. where g, f, c are constants and centre is $(-g, -f)$

i.e. $\left(-\frac{\text{coefficient of } x}{2}, -\frac{\text{coefficient of } y}{2} \right)$ and radius $r = \sqrt{g^2 + f^2 - c}$

Note :

- If $(g^2 + f^2 - c) > 0$, then r is real and positive and the circle is a real circle.
- If $(g^2 + f^2 - c) = 0$, then radius $r = 0$ and circle is a point circle.
- If $(g^2 + f^2 - c) < 0$, then r is imaginary then circle is also an imaginary circle with real centre.
- $x^2 + y^2 + 2gx + 2fy + c = 0$, has three constants and to get the equation of the circle at least three conditions should be known $\Rightarrow$ A unique circle passes through three non collinear points.
- The general second degree in x and y , $ax^2 + by^2 + 2hxy + 2gx + 2fy + c = 0$ represents a circle if :**
 - coefficient of $x^2 =$ coefficient of y^2 or $a = b \neq 0$
 - coefficient of $xy = 0$ or $h = 0$
 - $(g^2 + f^2 - c) \geq 0$ (for a real circle)

(c) **Intercepts cut by the circle on axes :**

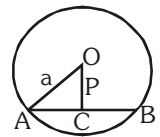
The intercepts cut by the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ on :

(i) x-axis = $2\sqrt{g^2 - c}$

(ii) y-axis = $2\sqrt{f^2 - c}$

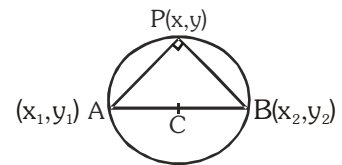
Note :

- (i) If the circle cuts the x-axis at two distinct point, then $g^2 - c > 0$
- (ii) If the circle cuts the y-axis at two distinct point, then $f^2 - c > 0$
- (iii) If circle touches x-axis then $g^2 = c$.
- (iv) If circle touches y-axis then $f^2 = c$.
- (v) Circle lies completely above or below the x-axis then $g^2 < c$.
- (vi) Circle lies completely to the right or left to the y-axis, then $f^2 < c$.
- (vii) Intercept cut by a line on the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ or length of chord of the circle = $2\sqrt{a^2 - P^2}$ where a is the radius and P is the length of perpendicular from the centre to the chord.



(d) **Equation of circle in diameter form :**

If $A(x_1, y_1)$ and $B(x_2, y_2)$ are the end points of the diameter of the circle and $P(x, y)$ is the point other than A and B on the circle then from geometry we know that $\angle APB = 90^\circ$.



$$\Rightarrow (\text{Slope of PA}) \times (\text{Slope of PB}) = -1$$

$$\Rightarrow \therefore \left(\frac{y - y_1}{x - x_1} \right) \left(\frac{y - y_2}{x - x_2} \right) = -1$$

$$\Rightarrow (x - x_1)(x - x_2) + (y - y_1)(y - y_2) = 0$$

Note : This will be the circle of least radius passing through (x_1, y_1) and (x_2, y_2)

(e) **Equation of circle in parametric forms :**

(i) The parametric equation of the circle $x^2 + y^2 = r^2$ are $x = r \cos \theta$, $y = r \sin \theta$; $\theta \in [0, 2\pi)$ and $(r \cos \theta, r \sin \theta)$ are called the parametric co-ordinates.

(ii) The parametric equation of the circle $(x - h)^2 + (y - k)^2 = r^2$ is $x = h + r \cos \theta$, $y = k + r \sin \theta$ where θ is parameter.

(iii) The parametric equation of the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ are

$$x = -g + \sqrt{g^2 + f^2 - c} \cos \theta,$$

$$y = -f + \sqrt{g^2 + f^2 - c} \sin \theta \text{ where } \theta \text{ is parameter.}$$

Note : Equation of a straight line joining two point α & β on the circle $x^2 + y^2 = a^2$ is

$$x \cos \frac{\alpha + \beta}{2} + y \sin \frac{\alpha + \beta}{2} = a \cos \frac{\alpha - \beta}{2}.$$

Illustration 1 : Find the centre and the radius of the circles

(a) $3x^2 + 3y^2 - 8x - 10y + 3 = 0$

(b) $x^2 + y^2 + 2x \sin \theta + 2y \cos \theta - 8 = 0$

(c) $2x^2 + \lambda xy + 2y^2 + (\lambda - 4)x + 6y - 5 = 0$, for some λ .

Solution :

(a) We rewrite the given equation as

$$x^2 + y^2 - \frac{8}{3}x - \frac{10}{3}y + 1 = 0 \Rightarrow g = -\frac{4}{3}, f = -\frac{5}{3}, c = 1$$

Hence the centre is $\left(\frac{4}{3}, \frac{5}{3}\right)$ and the radius is $\sqrt{\frac{16}{9} + \frac{25}{9} - 1} = \sqrt{\frac{32}{9}} = \frac{4\sqrt{2}}{3}$ units

(b) $x^2 + y^2 + 2x \sin\theta + 2y \cos\theta - 8 = 0$. Centre of this circle is $(-\sin\theta, -\cos\theta)$

$$\text{Radius} = \sqrt{\sin^2\theta + \cos^2\theta + 8} = \sqrt{1+8} = 3 \text{ units}$$

(c) $2x^2 + \lambda xy + 2y^2 + (\lambda - 4)x + 6y - 5 = 0$

We rewrite the equation as

$$x^2 + \frac{\lambda}{2}xy + y^2 + \left(\frac{\lambda-4}{2}\right)x + 3y - \frac{5}{2} = 0 \quad \dots\dots (i)$$

Since, there is no term of xy in the equation of circle $\Rightarrow \frac{\lambda}{2} = 0 \Rightarrow \lambda = 0$

$$\text{So, equation (i) reduces to } x^2 + y^2 - 2x + 3y - \frac{5}{2} = 0$$

$$\therefore \text{ centre is } \left(1, -\frac{3}{2}\right) \quad \text{Radius} = \sqrt{1 + \frac{9}{4} + \frac{5}{2}} = \frac{\sqrt{23}}{2} \text{ units.}$$

Illustration 2 : If the lines $3x - 4y + 4 = 0$ and $6x - 8y - 7 = 0$ are tangents to a circle, then the radius of the circle is -

- (A) $3/2$ (B) $3/4$ (C) $1/10$ (D) $1/20$

Solution :

The diameter of the circle is perpendicular distance between the parallel lines (tangents)

$$3x - 4y + 4 = 0 \text{ and } 3x - 4y - \frac{7}{2} = 0 \text{ and so it is equal to } \frac{4 + 7/2}{\sqrt{9+16}} = \frac{3}{2}.$$

$$\text{Hence radius is } \frac{3}{4}.$$

Ans. (B)**Illustration 3 :** If $y = 2x + m$ is a diameter to the circle $x^2 + y^2 + 3x + 4y - 1 = 0$, then find m **Solution :**Centre of circle = $(-3/2, -2)$. This lies on diameter $y = 2x + m$

$$\Rightarrow -2 = (-3/2) \times 2 + m \Rightarrow m = 1$$

Illustration 4 : The equation of a circle which passes through the point $(1, -2)$ and $(4, -3)$ and whose centre lies on the line $3x + 4y = 7$ is

- (A) $15(x^2 + y^2) - 94x + 18y - 55 = 0$ (B) $15(x^2 + y^2) - 94x + 18y + 55 = 0$
(C) $15(x^2 + y^2) + 94x - 18y + 55 = 0$ (D) none of these

Solution :

$$\text{Let the circle be } x^2 + y^2 + 2gx + 2fy + c = 0 \quad \dots\dots (i)$$

Hence, substituting the points, $(1, -2)$ and $(4, -3)$ in equation (i)

$$5 + 2g - 4f + c = 0 \quad \dots\dots (ii)$$

$$25 + 8g - 6f + c = 0 \quad \dots\dots (iii)$$

centre $(-g, -f)$ lies on line $3x + 4y = 7$

$$\text{Hence } -3g - 4f = 7$$

solving for g, f, c , we get

$$\text{Here } g = \frac{-47}{15}, f = \frac{9}{15}, c = \frac{55}{15}$$

$$\text{Hence the equation is } 15(x^2 + y^2) - 94x + 18y + 55 = 0$$

Ans. (B)

Illustration 5 : A circle has radius equal to 3 units and its centre lies on the line $y = x - 1$. Find the equation of the circle if it passes through $(7, 3)$.

Solution : Let the centre of the circle be (α, β) . It lies on the line $y = x - 1$
 $\Rightarrow \beta = \alpha - 1$. Hence the centre is $(\alpha, \alpha - 1)$.
 $\Rightarrow$ The equation of the circle is $(x - \alpha)^2 + (y - \alpha + 1)^2 = 9$
 It passes through $(7, 3) \Rightarrow (7 - \alpha)^2 + (4 - \alpha)^2 = 9$
 $\Rightarrow 2\alpha^2 - 22\alpha + 56 = 0 \Rightarrow \alpha^2 - 11\alpha + 28 = 0$
 $\Rightarrow (\alpha - 4)(\alpha - 7) = 0 \Rightarrow \alpha = 4, 7$

Hence the required equations are

$$x^2 + y^2 - 8x - 6y + 16 = 0 \text{ and } x^2 + y^2 - 14x - 12y + 76 = 0.$$

Ans.

Illustration 6 : Let L_1 be a straight line through the origin and L_2 be the straight line $x + y = 1$. If the intercepts made by the circle $x^2 + y^2 - x + 3y = 0$ on L_1 & L_2 are equal, then which of the following equations can represent L_1 ?

- (A) $x + y = 0$ (B) $x - y = 0$ (C) $x + 7y = 0$ (D) $x - 7y = 0$

Solution : Let L_1 be $y = mx$
 lines L_1 & L_2 will be at equal distances from centre of the circle centre of the circle is $\left(\frac{1}{2}, -\frac{3}{2}\right)$

$$\Rightarrow \frac{\left|\frac{1}{2}m + \frac{3}{2}\right|}{\sqrt{1+m^2}} = \frac{\left|\frac{1}{2} - \frac{3}{2} - 1\right|}{\sqrt{2}} \Rightarrow \frac{(m+3)^2}{(1+m^2)} = 8$$

$$\Rightarrow 7m^2 - 6m - 1 = 0 \Rightarrow (m-1)(7m+1) = 0$$

$$\Rightarrow m = 1, m = -\frac{1}{7} \Rightarrow y = x, 7y + x = 0$$

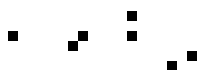
Ans. (B, C)

Do yourself - 1 :

- (i) Find the centre and radius of the circle $2x^2 + 2y^2 = 3x - 5y + 7$
- (ii) Find the equation of the circle whose centre is the point of intersection of the lines $2x - 3y + 4 = 0$ & $3x + 4y - 5 = 0$ and passes through the origin.
- (iii) Find the parametric form of the equation of the circle $x^2 + y^2 + px + py = 0$
- (iv) Find the equation of the circle the end points of whose diameter are the centres of the circles $x^2 + y^2 + 16x - 14y = 1$ & $x^2 + y^2 - 4x + 10y = 2$

3. POSITION OF A POINT W.R.T CIRCLE :

- (a) Let the circle is $x^2 + y^2 + 2gx + 2fy + c = 0$ and the point is (x_1, y_1) then -
 Point (x_1, y_1) lies out side the circle or on the circle or inside the circle according as
 $\Rightarrow x_1^2 + y_1^2 + 2gx_1 + 2fy_1 + c >, =, < 0$ or $S_1 >, =, < 0$
- (b) The greatest & the least distance of a point A from a circle with centre C & radius r is $AC + r$ & $|AC - r|$ respectively.



4. POWER OF A POINT W.R.T. CIRCLE :

Theorem : The power of point $P(x_1, y_1)$ w.r.t. the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ is S_1

where $S_1 = x_1^2 + y_1^2 + 2gx_1 + 2fy_1 + c$

Note : If P outside, inside or on the circle then power of point is positive, negative or zero respectively.

If from a point $P(x_1, y_1)$, inside or outside the circle, a secant be drawn intersecting the circle in two points A & B , then $PA \cdot PB = \text{constant}$.

The product $PA \cdot PB$ is called power of point $P(x_1, y_1)$ w.r.t. the circle

$S \equiv x^2 + y^2 + 2gx + 2fy + c = 0$, i.e. for number of secants $PA \cdot PB =$

$PA_1 \cdot PB_1 = PA_2 \cdot PB_2 = \dots = PT^2 = S_1$

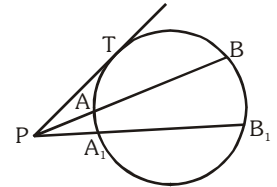


Illustration 7 : If $P(2, 8)$ is an interior point of a circle $x^2 + y^2 - 2x + 4y - p = 0$ which neither touches nor intersects the axes, then set for p is -

- (A) $p < -1$ (B) $p < -4$ (C) $p > 96$ (D) ϕ

Solution : For internal point $p(2, 8)$, $4 + 64 - 4 + 32 - p < 0 \Rightarrow p > 96$

and x intercept $= 2\sqrt{1+p}$ therefore $1 + p < 0$

$\Rightarrow p < -1$ and y intercept $= 2\sqrt{4+p} \Rightarrow p < -4$

Ans. (D)

Do yourself - 2 :

- (i) Find the position of the points $(1, 2)$ & $(6, 0)$ w.r.t. the circle $x^2 + y^2 - 4x + 2y - 11 = 0$
 (ii) Find the greatest and least distance of a point $P(7, 3)$ from circle $x^2 + y^2 - 8x - 6y + 16 = 0$.
 Also find the power of point P w.r.t. circle.

5. TANGENT LINE OF CIRCLE :

When a straight line meet a circle on two coincident points then it is called the tangent of the circle.

(a) Condition of Tangency :

The line $L = 0$ touches the circle $S = 0$ if P the length of the perpendicular from the centre to that line and radius of the circle r are equal i.e. $P = r$.

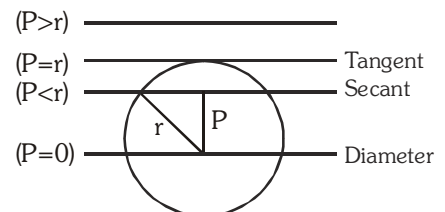


Illustration 8 : Find the range of parameter 'a' for which the variable line $y = 2x + a$ lies between the circles $x^2 + y^2 - 2x - 2y + 1 = 0$ and $x^2 + y^2 - 16x - 2y + 61 = 0$ without intersecting or touching either circle.

Solution : The given circles are $C_1 : (x - 1)^2 + (y - 1)^2 = 1$ and $C_2 : (x - 8)^2 + (y - 1)^2 = 4$
 The line $y - 2x - a = 0$ will lie between these circle if centre of the circles lie on opposite sides of the line, i.e. $(1 - 2 - a)(1 - 16 - a) < 0 \Rightarrow a \in (-15, -1)$

Line wouldn't touch or intersect the circles if, $\frac{|1 - 2 - a|}{\sqrt{5}} > 1$, $\frac{|1 - 16 - a|}{\sqrt{5}} > 2$

$\Rightarrow |1 + a| > \sqrt{5}$, $|15 + a| > 2\sqrt{5}$

$\Rightarrow a > \sqrt{5} - 1$ or $a < -\sqrt{5} - 1$, $a > 2\sqrt{5} - 15$ or $a < -2\sqrt{5} - 15$

Hence common values of 'a' are $(2\sqrt{5} - 15, -\sqrt{5} - 1)$.

Illustration 9 : The equation of a circle whose centre is $(3, -1)$ and which cuts off a chord of length 6 on the line $2x - 5y + 18 = 0$

(A) $(x - 3)^2 + (y + 1)^2 = 38$

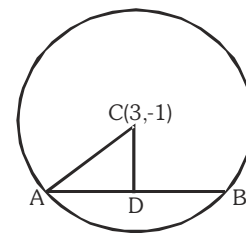
(B) $(x + 3)^2 + (y - 1)^2 = 38$

(C) $(x - 3)^2 + (y + 1)^2 = \sqrt{38}$

(D) none of these

Solution :

Let $AB (= 6)$ be the chord intercepted by the line $2x - 5y + 18 = 0$ from the circle and let CD be the perpendicular drawn from centre $(3, -1)$ to the chord AB .



$$\text{i.e., } AD = 3, CD = \frac{2 \cdot 3 - 5(-1) + 18}{\sqrt{2^2 + 5^2}} = \sqrt{29}$$

$$\text{Therefore, } CA^2 = 3^2 + (\sqrt{29})^2 = 38$$

$$\text{Hence required equation is } (x - 3)^2 + (y + 1)^2 = 38$$

Ans. (A)

Illustration 10 : The area of the triangle formed by line joining the origin to the points of intersection(s) of the line $x\sqrt{5} + 2y = 3\sqrt{5}$ and circle $x^2 + y^2 = 10$ is

(A) 3

(B) 4

(C) 5

(D) 6

Solution :

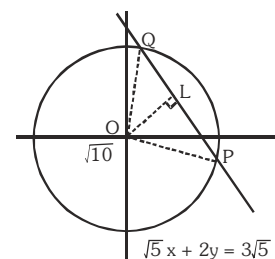
Length of perpendicular from origin to the line $x\sqrt{5} + 2y = 3\sqrt{5}$ is

$$OL = \frac{3\sqrt{5}}{\sqrt{(\sqrt{5})^2 + 2^2}} = \frac{3\sqrt{5}}{\sqrt{9}} = \sqrt{5}$$

$$\text{Radius of the given circle} = \sqrt{10} = OQ = OP$$

$$PQ = 2QL = 2\sqrt{OQ^2 - OL^2} = 2\sqrt{10 - 5} = 2\sqrt{5}$$

$$\text{Thus area of } \triangle OPQ = \frac{1}{2} \times PQ \times OL = \frac{1}{2} \times 2\sqrt{5} \times \sqrt{5} = 5$$



Ans. (C)

(b) Equation of the tangent ($T = 0$) :

(i) Tangent at the point (x_1, y_1) on the circle $x^2 + y^2 = a^2$ is $xx_1 + yy_1 = a^2$.

(ii) (1) The tangent at the point $(a \cos t, a \sin t)$ on the circle $x^2 + y^2 = a^2$ is $x \cos t + y \sin t = a$

(2) The point of intersection of the tangents at the points $P(\alpha)$ and $Q(\beta)$

$$\text{is } \left(\frac{a \cos \frac{\alpha + \beta}{2}}{\cos \frac{\alpha - \beta}{2}}, \frac{a \sin \frac{\alpha + \beta}{2}}{\cos \frac{\alpha - \beta}{2}} \right).$$

(iii) The equation of tangent at the point (x_1, y_1) on the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ is $xx_1 + yy_1 + g(x + x_1) + f(y + y_1) + c = 0$

(iv) If line $y = mx + c$ is a straight line touching the circle $x^2 + y^2 = a^2$, then $c = \pm a\sqrt{1 + m^2}$ and contact points are $\left(\mp \frac{am}{\sqrt{1 + m^2}}, \pm \frac{a}{\sqrt{1 + m^2}} \right)$ or $\left(\mp \frac{a^2 m}{c}, \pm \frac{a^2}{c} \right)$ and equation of tangent is $y = mx \pm a\sqrt{1 + m^2}$

(v) The equation of tangent with slope m of the circle $(x - h)^2 + (y - k)^2 = a^2$ is

$$(y - k) = m(x - h) \pm a\sqrt{1 + m^2}$$

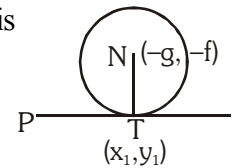
6. NORMAL OF CIRCLE :

Normal at a point is the straight line which is perpendicular to the tangent at the point of contact.

Note : Normal at point of the circle passes through the centre of the circle.

(a) Equation of normal at point (x_1, y_1) of circle $x^2 + y^2 + 2gx + 2fy + c = 0$ is

$$y - y_1 = \left(\frac{y_1 + f}{x_1 + g} \right) (x - x_1)$$



(b) The equation of normal on any point (x_1, y_1) of circle $x^2 + y^2 = a^2$ is $\frac{y}{x} = \frac{y_1}{x_1}$.

(c) If $x^2 + y^2 = a^2$ is the equation of the circle then at any point 't' of this circle $(a \cos t, a \sin t)$, the equation of normal is $x \sin t - y \cos t = 0$.

Illustration 12 : Find the equation of the normal to the circle $x^2 + y^2 - 5x + 2y - 48 = 0$ at the point $(5, 6)$.

Solution : Since normal to the circle always passes through the centre so equation of the normal will

be the line passing through $(5, 6)$ & $\left(\frac{5}{2}, -1\right)$

$$\text{i.e. } y + 1 = \frac{7}{5/2} \left(x - \frac{5}{2} \right) \Rightarrow 5y + 5 = 14x - 35$$

$$\Rightarrow 14x - 5y - 40 = 0$$

Ans.

Illustration 13 : If the straight line $ax + by = 2$; $a, b \neq 0$ touches the circle $x^2 + y^2 - 2x = 3$ and is normal to the circle $x^2 + y^2 - 4y = 6$, then the values of a and b are respectively

- (A) 1, -1 (B) 1, 2 (C) $-\frac{4}{3}, 1$ (D) 2, 1

Solution : Given $x^2 + y^2 - 2x = 3$

$\therefore$ centre is $(1, 0)$ and radius is 2

Given $x^2 + y^2 - 4y = 6$

$\therefore$ centre is $(0, 2)$ and radius is $\sqrt{10}$. Since line $ax + by = 2$ touches the first circle

$$\therefore \frac{|a(1) + b(0) - 2|}{\sqrt{a^2 + b^2}} = 2 \quad \text{or } |(a - 2)| = [2\sqrt{a^2 + b^2}] \quad \dots\dots\dots (i)$$

Also the given line is normal to the second circle. Hence it will pass through the centre of the second circle.

$$\therefore a(0) + b(2) = 2 \quad \text{or } 2b = 2 \quad \text{or } b = 1$$

$$\text{Putting this value in equation (i) we get } |a - 2| = 2\sqrt{a^2 + 1^2} \quad \text{or } (a - 2)^2 = 4(a^2 + 1)$$

$$\text{or } a^2 + 4 - 4a = 4a^2 + 4 \quad \text{or } 3a^2 + 4a = 0 \quad \text{or } a(3a + 4) = 0 \quad \text{or } a = 0, -\frac{4}{3} \quad (a \neq 0)$$

$$\therefore \text{ values of } a \text{ and } b \text{ are } \left(-\frac{4}{3}, 1 \right).$$

Ans. (C)

Illustration 14 : Find the equation of a circle having the lines $x^2 + 2xy + 3x + 6y = 0$ as its normal and having size just sufficient to contain the circle $x(x - 4) + y(y - 3) = 0$.

Solution :

Pair of normals are $(x + 2y)(x + 3) = 0$

$\therefore$ Normals are $x + 2y = 0$, $x + 3 = 0$.

Point of intersection of normals is the centre of required circle i.e. $C_1(-3, 3/2)$ and centre

of given circle is $C_2(2, 3/2)$ and radius $r_2 = \sqrt{4 + \frac{9}{4}} = \frac{5}{2}$

Let r_1 be the radius of required circle

$$\Rightarrow r_1 = C_1C_2 + r_2 = \sqrt{(-3-2)^2 + \left(\frac{3}{2} - \frac{3}{2}\right)^2} + \frac{5}{2} = \frac{15}{2}$$

Hence equation of required circle is $x^2 + y^2 + 6x - 3y - 45 = 0$

Do yourself - 4 :

- (i) Find the equation of the normal to the circle $x^2 + y^2 = 2x$, which is parallel to the line $x + 2y = 3$.

7. CHORD OF CONTACT (T = 0) :

A line joining the two points of contacts of two tangents drawn from a point outside the circle, is called chord of contact of that point.

If two tangents PT_1 & PT_2 are drawn from the point $P(x_1, y_1)$ to the circle

$S \equiv x^2 + y^2 + 2gx + 2fy + c = 0$, then the equation of the chord of contact T_1T_2 is :

$xx_1 + yy_1 + g(x + x_1) + f(y + y_1) + c = 0$ (i.e. $T = 0$ same as equation of tangent).

Remember :

(a) Length of chord of contact $T_1T_2 = \frac{2LR}{\sqrt{R^2 + L^2}}$.

(b) Area of the triangle formed by the pair of the tangents & its chord of contact $= \frac{RL^3}{R^2 + L^2}$,
where R is the radius of the circle & L is the length of the tangent from (x_1, y_1) on $S = 0$.

(c) Angle between the pair of tangents from $P(x_1, y_1) = \tan^{-1} \left(\frac{2RL}{L^2 - R^2} \right)$

(d) Equation of the circle circumscribing the triangle PT_1T_2 or quadrilateral CT_1PT_2 is :
 $(x - x_1)(x + g) + (y - y_1)(y + f) = 0$.

(e) The joint equation of a pair of tangents drawn from the point $A(x_1, y_1)$ to the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ is : $SS_1 = T^2$.

Where $S \equiv x^2 + y^2 + 2gx + 2fy + c$; $S_1 \equiv x_1^2 + y_1^2 + 2gx_1 + 2fy_1 + c$

$T \equiv xx_1 + yy_1 + g(x + x_1) + f(y + y_1) + c$.

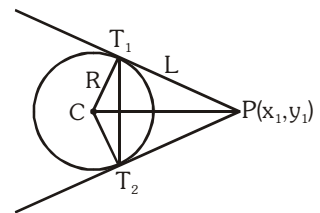


Illustration 15 : The chord of contact of tangents drawn from a point on the circle $x^2 + y^2 = a^2$ to the circle $x^2 + y^2 = b^2$ touches the circle $x^2 + y^2 = c^2$. Show that a, b, c are in GP.

Solution : Let $P(a\cos\theta, a\sin\theta)$ be a point on the circle $x^2 + y^2 = a^2$.

Then equation of chord of contact of tangents drawn from

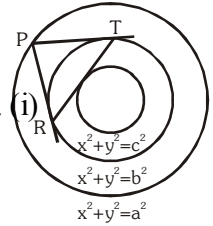
$P(a\cos\theta, a\sin\theta)$ to the circle $x^2 + y^2 = b^2$ is $ax\cos\theta + ay\sin\theta = b^2$ (i)

This touches the circle $x^2 + y^2 = c^2$ (ii)

$\therefore$ Length of perpendicular from $(0, 0)$ to (i) = radius of (ii)

$$\therefore \frac{|0 + 0 - b^2|}{\sqrt{(a^2 \cos^2 \theta + a^2 \sin^2 \theta)}} = c$$

or $b^2 = ac \Rightarrow a, b, c$ are in GP.



Do yourself - 5 :

- (i) Find the equation of the chord of contact of the point $(1, 2)$ with respect to the circle $x^2 + y^2 + 2x + 3y + 1 = 0$
- (ii) Tangents are drawn from the point $P(4, 6)$ to the circle $x^2 + y^2 = 25$. Find the area of the triangle formed by them and their chord of contact.

8. EQUATION OF THE CHORD WITH A GIVEN MIDDLE POINT ($T = S_1$) :

The equation of the chord of the circle $S \equiv x^2 + y^2 + 2gx + 2fy + c = 0$ in terms of its mid point M

(x_1, y_1) is $y - y_1 = -\frac{x_1 + g}{y_1 + f}(x - x_1)$. This on simplification can be put in the form

$xx_1 + yy_1 + g(x + x_1) + f(y + y_1) + c = x_1^2 + y_1^2 + 2gx_1 + 2fy_1 + c$ which is designated by $T = S_1$.

Note that : The shortest chord of a circle passing through a point 'M' inside the circle, is one chord whose middle point is M.

Illustration 16 : Find the locus of middle points of chords of the circle $x^2 + y^2 = a^2$, which subtend right angle at the point $(c, 0)$.

Solution : Let $N(h, k)$ be the middle point of any chord AB, which subtend a right angle at $P(c, 0)$.

Since $\angle APB = 90^\circ$

$$\therefore NA = NB = NP$$

(since distance of the vertices from middle point of the hypotenuse are equal)

$$\text{or } (NA)^2 = (NB)^2 = (h - c)^2 + (k - 0)^2 \dots (i)$$

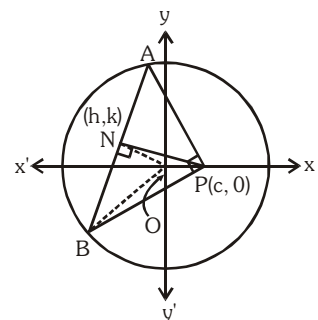
But also $\angle BNO = 90^\circ$

$$\therefore (OB)^2 = (ON)^2 + (NB)^2$$

$$\Rightarrow -(NB)^2 = (ON)^2 - (OB)^2 \Rightarrow -[(h - c)^2 + (k - 0)^2] = (h^2 + k^2) - a^2$$

$$\text{or } 2(h^2 + k^2) - 2ch + c^2 - a^2 = 0$$

$$\therefore \text{Locus of } N(h, k) \text{ is } 2(x^2 + y^2) - 2cx + c^2 - a^2 = 0$$



Ans.

Illustration 17 : Let a circle be given by $2x(x - a) + y(2y - b) = 0$ ($a \neq 0, b \neq 0$)

Find the condition on a and b if two chords, each bisected by the x -axis, can be drawn to the circle from $(a, b/2)$.

Solution : The given circle is $2x(x - a) + y(2y - b) = 0$

$$\text{or } x^2 + y^2 - ax - by/2 = 0$$

Let AB be the chord which is bisected by x -axis at a point M . Let its co-ordinates be $M(h, 0)$.

$$\text{and } S \equiv x^2 + y^2 - ax - by/2 = 0$$

$\therefore$ Equation of chord AB is $T = S_1$

$$hx + 0 - \frac{a}{2}(x + h) - \frac{b}{4}(y + 0) = h^2 + 0 - ah - 0$$

$$\text{Since it passes through } (a, b/2) \text{ we have } ah - \frac{a}{2}(a + h) - \frac{b^2}{8} = h^2 - ah \Rightarrow h^2 - \frac{3ah}{2} + \frac{a^2}{2} + \frac{b^2}{8} = 0$$

Now there are two chords bisected by the x -axis, so there must be two distinct real roots of h .

$$\therefore B^2 - 4AC > 0$$

$$\Rightarrow \left(\frac{-3a}{2} \right)^2 - 4.1. \left(\frac{a^2}{2} + \frac{b^2}{8} \right) > 0 \Rightarrow a^2 > 2b^2.$$

Ans.

Do yourself - 6 :

(i) Find the equation of the chord of $x^2 + y^2 - 6x + 10 - a = 0$ which is bisected at $(-2, 4)$.

(ii) Find the locus of mid point of chord of $x^2 + y^2 + 2gx + 2fy + c = 0$ that pass through the origin.

9. DIRECTOR CIRCLE :

The locus of point of intersection of two perpendicular tangents to a circle is called director circle. Let $P(h, k)$ is the point of intersection of two tangents drawn on the circle $x^2 + y^2 = a^2$. Then the equation of the pair of tangents is $SS_1 = T^2$

$$\text{i.e. } (x^2 + y^2 - a^2)(h^2 + k^2 - a^2) = (hx + ky - a^2)^2$$

As lines are perpendicular to each other then, coefficient of x^2 + coefficient of $y^2 = 0$

$$\Rightarrow [(h^2 + k^2 - a^2) - h^2] + [(h^2 + k^2 - a^2) - k^2] = 0$$

$$\Rightarrow h^2 + k^2 = 2a^2$$

$\therefore$ locus of (h, k) is $x^2 + y^2 = 2a^2$ which is the equation of the director circle.

$\therefore$ director circle is a concentric circle whose radius is $\sqrt{2}$ times the radius of the circle.

Note : The director circle of $x^2 + y^2 + 2gx + 2fy + c = 0$ is $x^2 + y^2 + 2gx + 2fy + 2c - g^2 - f^2 = 0$

Illustration 18 : Let P be any moving point on the circle $x^2 + y^2 - 2x = 1$, from this point chord of contact is drawn w.r.t. the circle $x^2 + y^2 - 2x = 0$. Find the locus of the circumcentre of the triangle CAB , C being centre of the circle and A, B are the points of contact.

Solution : The two circles are

$$(x - 1)^2 + y^2 = 1 \quad \dots\dots\dots (i)$$

$$(x - 1)^2 + y^2 = 2 \quad \dots\dots\dots (ii)$$

So the second circle is the director circle of the first. So $\angle APB = \pi/2$

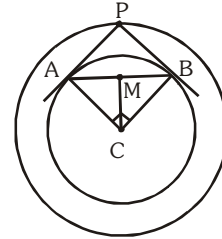
Also $\angle ACB = \pi/2$

Now circumcentre of the right angled triangle CAB would lie on the mid point of AB
So let the point be $M \equiv (h, k)$

$$\text{Now, } CM = CB \sin 45^\circ = \frac{1}{\sqrt{2}}$$

$$\text{So, } (h-1)^2 + k^2 = \left(\frac{1}{\sqrt{2}}\right)^2$$

$$\text{So, locus of M is } (x-1)^2 + y^2 = \frac{1}{2}.$$

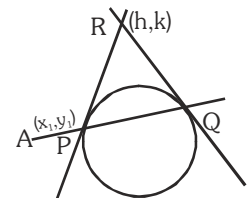


Do yourself - 7 :

- (i) Find the equation of the director circle of the circle $(x-h)^2 + (y-k)^2 = a^2$.
- (ii) If the angle between the tangents drawn to $x^2 + y^2 + 4x + 8y + c = 0$ from $(0, 0)$ is $\frac{\pi}{2}$, then find value of 'c'
- (iii) If two tangents are drawn from a point on the circle $x^2 + y^2 = 50$ to the circle $x^2 + y^2 = 25$, then find the angle between the tangents.

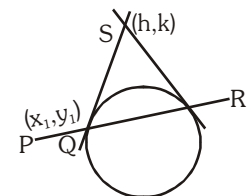
10. POLE AND POLAR :

Let any straight line through the given point $A(x_1, y_1)$ intersect the given circle $S=0$ in two points P and Q and if the tangent of the circle at P and Q meet at the point R then locus of point R is called polar of the point A and point A is called the pole, with respect to the given circle.



(a) The equation of the polar of point (x_1, y_1) w.r.t. circle $x^2 + y^2 = a^2$ ($T=0$).

Let PQR is a chord which passes through the point $P(x_1, y_1)$ which intersects the circle at points Q and R and the tangents are drawn at points Q and R meet at point $S(h, k)$ then equation of QR the chord of contact is $x_1h + y_1k = a^2 \therefore$ locus of point $S(h, k)$ is $xx_1 + yy_1 = a^2$ which is the equation of the polar.



Note :

- (i) The equation of the polar is the $T=0$, so the polar of point (x_1, y_1) w.r.t circle $x^2 + y^2 + 2gx + 2fy + c = 0$ is $xx_1 + yy_1 + g(x + x_1) + f(y + y_1) + c = 0$
- (ii) If point is outside the circle then equation of polar and chord of contact is same. So the chord of contact is polar.
- (iii) If point is inside the circle then chord of contact does not exist but polar exists.
- (iv) If point lies on the circle then polar, chord of contact and tangent on that point are same.
- (v) If the polar of P w.r.t. a circle passes through the point Q, then the polar of point Q will pass through P and hence P & Q are conjugate points of each other w.r.t. the given circle.
- (vi) If pole of a line w.r.t. a circle lies on second line. Then pole of second line lies on first line and hence both lines are conjugate lines of each other w.r.t. the given circle.
- (vii) If O be the centre of a circle and P be any point, then OP is perpendicular to the polar of P.

(viii) If O be the centre of a circle and P any point, then if OP (produce, if necessary) meet the polar of P in Q, then $OP \cdot OQ = (\text{radius})^2$

(b) Pole of a given line with respect to a circle

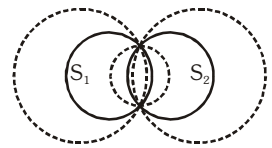
To find the pole of a line we assume the coordinates of the pole then from these coordinates we find the polar. This polar and given line represent the same line. Then by comparing the coefficients of similar terms we can get the coordinates of the pole. The pole of $\ell x + my + n = 0$

w.r.t. circle $x^2 + y^2 = a^2$ will be $\left(\frac{-\ell a^2}{n}, \frac{-ma^2}{n} \right)$

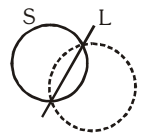
11. FAMILY OF CIRCLES :

- (a) The equation of the family of circles passing through the points of intersection of two circles

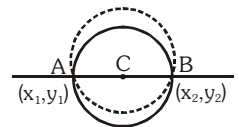
$$S_1 = 0 \text{ \& } S_2 = 0 \text{ is : } S_1 + K S_2 = 0 \quad (K \neq -1).$$



- (b) The equation of the family of circles passing through the point of intersection of a circle $S = 0$ & a line $L = 0$ is given by $S + KL = 0$.



- (c) The equation of a family of circles passing through two given points (x_1, y_1) & (x_2, y_2) can be written in the form :

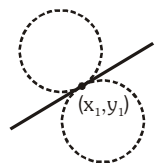


$$(x - x_1)(x - x_2) + (y - y_1)(y - y_2) + K \begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix} = 0 \text{ where } K \text{ is a parameter.}$$

- (d) The equation of a family of circles touching a fixed line $y - y_1 = m(x - x_1)$

at the fixed point (x_1, y_1) is $(x - x_1)^2 + (y - y_1)^2 + K[y - y_1 - m(x - x_1)] = 0$,

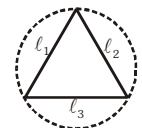
where K is a parameter.



- (e) Family of circles circumscribing a triangle whose sides are given by

$$L_1 = 0 ; L_2 = 0 \text{ \& } L_3 = 0 \text{ is given by ; } L_1 L_2 + \lambda L_2 L_3 + \mu L_3 L_1 = 0$$

provided coefficient of $xy = 0$ & coefficient of $x^2 = \text{coefficient of } y^2$.



- (f) Equation of circle circumscribing a quadrilateral whose sides in order are represented by the lines $L_1 = 0, L_2 = 0, L_3 = 0$ & $L_4 = 0$ is $L_1 L_3 + \lambda L_2 L_4 = 0$

provided coefficient of $x^2 = \text{coefficient of } y^2$ and coefficient of $xy = 0$.

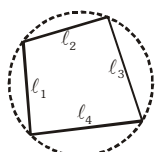


Illustration 19 : The equation of the circle through the points of intersection of $x^2 + y^2 - 1 = 0$, $x^2 + y^2 - 2x - 4y + 1 = 0$ and touching the line $x + 2y = 0$, is -

- (A) $x^2 + y^2 + x + 2y = 0$ (B) $x^2 + y^2 - x + 20 = 0$
(C) $x^2 + y^2 - x - 2y = 0$ (D) $2(x^2 + y^2) - x - 2y = 0$

Solution :

Family of circles is $x^2 + y^2 - 2x - 4y + 1 + \lambda(x^2 + y^2 - 1) = 0$

$$(1 + \lambda)x^2 + (1 + \lambda)y^2 - 2x - 4y + (1 - \lambda) = 0$$

$$x^2 + y^2 - \frac{2}{1+\lambda}x - \frac{4}{1+\lambda}y + \frac{1-\lambda}{1+\lambda} = 0$$

$$\text{Centre is } \left(\frac{1}{1+\lambda}, \frac{2}{1+\lambda} \right) \quad \text{and radius} = \sqrt{\left(\frac{1}{1+\lambda} \right)^2 + \left(\frac{2}{1+\lambda} \right)^2 - \frac{1-\lambda}{1+\lambda}} = \frac{\sqrt{4+\lambda^2}}{|1+\lambda|}$$

Since it touches the line $x + 2y = 0$, hence

Radius = Perpendicular distance from centre to the line.

$$\text{i.e., } \left| \frac{\frac{1}{1+\lambda} + 2 \cdot \frac{2}{1+\lambda}}{\sqrt{1^2 + 2^2}} \right| = \frac{\sqrt{4+\lambda^2}}{|1+\lambda|} \Rightarrow \sqrt{5} = \sqrt{4+\lambda^2} \Rightarrow \lambda = \pm 1$$

$\lambda = -1$ cannot be possible in case of circle. So $\lambda = 1$.

Thus, we get the equation of circle.

Ans. (C)

Do yourself - 8 :

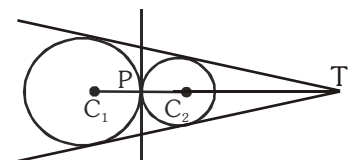
- (i) Prove that the polar of a given point with respect to any one of circles $x^2 + y^2 - 2kx + c^2 = 0$, where k is a variable, always passes through a fixed point, whatever be the value of k .
(ii) Find the equation of the circle passing through the points of intersection of the circle $x^2 + y^2 - 6x + 2y + 4 = 0$ & $x^2 + y^2 + 2x - 4y - 6 = 0$ and with its centre on the line $y = x$.
(iii) Find the equation of the circle through the points of intersection of the circles $x^2 + y^2 + 2x + 3y - 7 = 0$ and $x^2 + y^2 + 3x - 2y - 1 = 0$ and passing through the point $(1, 2)$.

12. DIRECT AND TRANSVERSE COMMON TANGENTS :

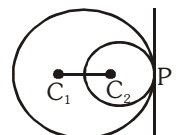
Let two circles having centre C_1 and C_2 and radii, r_1 and r_2 and C_1C_2 is the distance between their centres then :

(a) Both circles will touch :

- (i) **Externally** if $C_1C_2 = r_1 + r_2$ i.e. the distance between their centres is equal to sum of their radii and point P & T divides C_1C_2 in the ratio $r_1 : r_2$ (internally & externally respectively). In this case there are **three common tangents**.

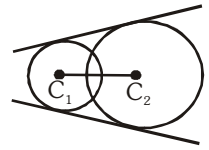


- (ii) **Internally** if $C_1C_2 = |r_1 - r_2|$ i.e. the distance between their centres is equal to difference between their radii and point P divides C_1C_2 in the ratio $r_1 : r_2$ **externally** and in this case there will be only **one common tangent**.

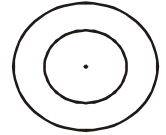


(b) The circles will intersect :

when $|r_1 - r_2| < C_1C_2 < r_1 + r_2$ in this case there are **two common tangents**.

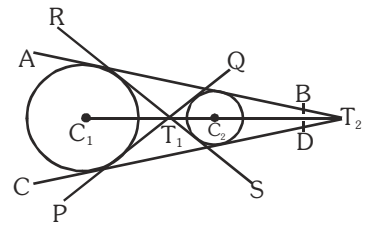
**(c) The circles will not intersect :**

(i) One circle will lie inside the other circle if $C_1C_2 < |r_1 - r_2|$. In this case there will be no common tangent.



(ii) When circles are apart from each other then $C_1C_2 > r_1 + r_2$ and in this case there

will be **four common tangents**. Lines PQ and RS are called **transverse** or **indirect** or **internal** common tangents and these lines meet line C_1C_2 on T_1 and T_1 divides the line C_1C_2 in the ratio $r_1 : r_2$ internally and lines AB & CD are called **direct** or **external** common tangents. These lines meet C_1C_2 produced on T_2 . Thus T_2 divides C_1C_2 externally in the ratio $r_1 : r_2$.



Note : Length of direct common tangent $= \sqrt{(C_1C_2)^2 - (r_1 - r_2)^2}$

Length of transverse common tangent $= \sqrt{(C_1C_2)^2 - (r_1 + r_2)^2}$

Illustration 20 : Prove that the circles $x^2 + y^2 + 2ax + c^2 = 0$ and $x^2 + y^2 + 2by + c^2 = 0$ touch each other,

$$\text{if } \frac{1}{a^2} + \frac{1}{b^2} = \frac{1}{c^2}.$$

Solution : Given circles are $x^2 + y^2 + 2ax + c^2 = 0$ (i)

and $x^2 + y^2 + 2by + c^2 = 0$ (ii)

Let C_1 and C_2 be the centres of circles (i) and (ii), respectively and r_1 and r_2 be their radii, then

$$C_1 = (-a, 0), C_2 = (0, -b), r_1 = \sqrt{a^2 - c^2}, r_2 = \sqrt{b^2 - c^2}$$

Here we find the two circles touch each other internally or externally.

$$\text{For touch, } |C_1C_2| = |r_1 \pm r_2|$$

$$\text{or } \sqrt{(a^2 + b^2)} = \left| \sqrt{(a^2 - c^2)} \pm \sqrt{(b^2 - c^2)} \right|$$

$$\text{On squaring } a^2 + b^2 = a^2 - c^2 + b^2 - c^2 \pm 2\sqrt{(a^2 - c^2)}\sqrt{(b^2 - c^2)}$$

$$\text{or } c^2 = \pm \sqrt{a^2b^2 - c^2(a^2 + b^2) + c^4}$$

$$\text{Again squaring, } c^4 = a^2b^2 - c^2(a^2 + b^2) + c^4$$

$$\text{or } c^2(a^2 + b^2) = a^2b^2$$

$$\text{or } \frac{1}{a^2} + \frac{1}{b^2} = \frac{1}{c^2}$$

Do yourself - 9 :

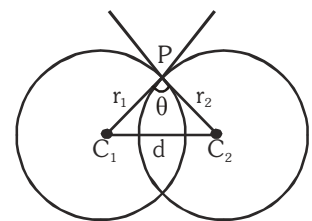
- (i) Two circles with radius 5 touches at the point (1, 2). If the equation of common tangent is $4x + 3y = 10$ and one of the circle is $x^2 + y^2 + 6x + 2y - 15 = 0$. Find the equation of other circle.
- (ii) Find the number of common tangents to the circles $x^2 + y^2 = 1$ and $x^2 + y^2 - 2x - 6y + 6 = 0$.

13. THE ANGLE OF INTERSECTION OF TWO CIRCLES :

Definition : The angle between the tangents of two circles at the point of intersection of the two circles is called angle of intersection of two circles. If two circles are $S_1 \equiv x^2 + y^2 + 2g_1x + 2f_1y + c_1 = 0$

$S_2 \equiv x^2 + y^2 + 2g_2x + 2f_2y + c_2 = 0$ and θ is the acute angle between them

$$\text{then } \cos \theta = \left| \frac{2g_1g_2 + 2f_1f_2 - c_1 - c_2}{2\sqrt{g_1^2 + f_1^2 - c_1} \sqrt{g_2^2 + f_2^2 - c_2}} \right| \quad \text{or} \quad \cos \theta = \left| \left(\frac{r_1^2 + r_2^2 - d^2}{2r_1r_2} \right) \right|$$



Here r_1 and r_2 are the radii of the circles and d is the distance between their centres.

If the angle of intersection of the two circles is a right angle then such circles are called "**Orthogonal circles**" and conditions for the circles to be orthogonal is -

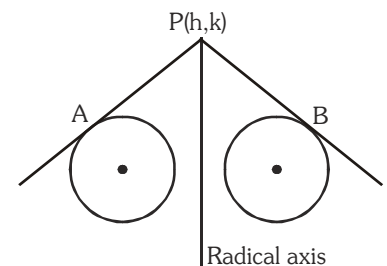
$$2g_1g_2 + 2f_1f_2 = c_1 + c_2$$

14. RADICAL AXIS OF THE TWO CIRCLES ($S_1 - S_2 = 0$) :

- (a) **Definition :** The locus of a point, which moves in such a way that the length of tangents drawn from it to the circles are equal and is called the radical axis. If two circles are -

$$S_1 \equiv x^2 + y^2 + 2g_1x + 2f_1y + c_1 = 0$$

$$S_2 \equiv x^2 + y^2 + 2g_2x + 2f_2y + c_2 = 0$$



Let $P(h,k)$ is a point and PA, PB are length of two tangents on the circles from point P , Then from definition -

$$\sqrt{h^2 + k^2 + 2g_1h + 2f_1k + c_1} = \sqrt{h^2 + k^2 + 2g_2h + 2f_2k + c_2} \quad \text{or} \quad 2(g_1 - g_2)h + 2(f_1 - f_2)k + c_1 - c_2 = 0$$

$\therefore$ locus of (h,k)

$$2x(g_1 - g_2) + 2y(f_1 - f_2) + c_1 - c_2 = 0$$

$$S_1 - S_2 = 0$$

which is the equation of radical axis.

Note :

- (i) To get the equation of the radical axis first of all make the coefficient of x^2 and $y^2=1$
- (ii) If circles touch each other then radical axis is the common tangent to both the circles.
- (iii) When the two circles intersect on real points then common chord is the radical axis of the two circles.
- (iv) The radical axis of the two circles is perpendicular to the line joining the centre of two circles but not always pass through mid point of it.
- (v) Radical axis (if exist) bisects common tangent to two circles.
- (vi) The radical axes of three circles (taking two at a time) meet at a point.
- (vii) If circles are concentric then the radical axis does not always exist but if circles are not concentric then radical axis always exists.
- (viii) If two circles are orthogonal to the third circle then radical axis of both circle passes through the centre of the third circle.
- (ix) A system of circle, every pair of which have the same radical axis, is called a **coaxial system of circles**.

(b) Radical centre :

The radical centre of three circles is the point from which length of tangents on three circles are equal i.e. the point of intersection of radical axis of the circles is the radical centre of the circles.

To get the radical axis of three circles $S_1=0$, $S_2=0$, $S_3=0$ we have to solve any two

$$S_1-S_2=0, S_2-S_3=0, S_3-S_1=0$$

Note :

- (i) The circle with centre as radical centre and radius equal to the length of tangent from radical centre to any of the circle, will cut the three circles orthogonally.
- (ii) If three circles are drawn on three sides of a triangle taking them as diameter then its orthocenter will be its radical centre.
- (iii) Locus of the centre of a variable circle orthogonal to two fixed circles is the radical axis between the two fixed circles.
- (iv) If two circles are orthogonal, then the polar of a point 'P' on first circle w.r.t. the second circle passes through the point Q which is the other end of the diameter through P. Hence locus of a point which moves such that its polars w.r.t. the circles $S_1=0$, $S_2=0$ & $S_3=0$ are concurrent is a circle which is orthogonal to all the three circles.

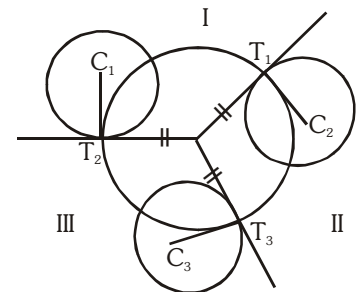


Illustration 21 : A and B are two fixed points and P moves such that $PA = nPB$ where $n \neq 1$. Show that locus of P is a circle and for different values of n all the circles have a common radical axis.

Solution : Let $A \equiv (a, 0)$, $B \equiv (-a, 0)$ and $P(h, k)$

$$\text{so } PA = nPB$$

$$\Rightarrow (h-a)^2 + k^2 = n^2[(h+a)^2 + k^2]$$

$$\Rightarrow (1 - n^2)h^2 + (1 - n^2)k^2 - 2ah(1 + n^2) + (1 - n^2)a^2 = 0$$

$$\Rightarrow h^2 + k^2 - 2ah\left(\frac{1+n^2}{1-n^2}\right) + a^2 = 0$$

Hence locus of P is

$$x^2 + y^2 - 2ax\left(\frac{1+n^2}{1-n^2}\right) + a^2 = 0, \text{ which is a circle of different values of } n.$$

Let n_1 and n_2 are two different values of n so their radical axis is $x = 0$ i.e. y -axis. Hence for different values of n the circles have a common radical axis.

Illustration 22 : Find the equation of the circle through the points of intersection of the circles $x^2 + y^2 - 4x - 6y - 12 = 0$ and $x^2 + y^2 + 6x + 4y - 12 = 0$ and cutting the circle $x^2 + y^2 - 2x - 4 = 0$ orthogonally.

Solution : The equation of the circle through the intersection of the given circles is $x^2 + y^2 - 4x - 6y - 12 + \lambda(-10x - 10y) = 0$ (i)

where $(-10x - 10y = 0)$ is the equation of radical axis for the circle

$$x^2 + y^2 - 4x - 6y - 12 = 0 \text{ and } x^2 + y^2 + 6x + 4y - 12 = 0.$$

Equation (i) can be re-arranged as

$$x^2 + y^2 - x(10\lambda + 4) - y(10\lambda + 6) - 12 = 0$$

It cuts the circle $x^2 + y^2 - 2x - 4 = 0$ orthogonally.

$$\text{Hence } 2gg_1 + 2ff_1 = c + c_1$$

$$\Rightarrow 2(5\lambda + 2)(1) + 2(5\lambda + 3)(0) = -12 - 4 \Rightarrow \lambda = -2$$

Hence the required circle is

$$x^2 + y^2 - 4x - 6y - 12 - 2(-10x - 10y) = 0$$

$$\text{i.e., } x^2 + y^2 + 16x + 14y - 12 = 0$$

Illustration 23 : Find the radical centre of circles $x^2 + y^2 + 3x + 2y + 1 = 0$, $x^2 + y^2 - x + 6y + 5 = 0$ and $x^2 + y^2 + 5x - 8y + 15 = 0$. Also find the equation of the circle cutting them orthogonally.

Solution : Given circles are $S_1 \equiv x^2 + y^2 + 3x + 2y + 1 = 0$

$$S_2 \equiv x^2 + y^2 - x + 6y + 5 = 0$$

$$S_3 \equiv x^2 + y^2 + 5x - 8y + 15 = 0$$

$$\text{Equations of two radical axes are } S_1 - S_2 \equiv 4x - 4y - 4 = 0 \text{ or } x - y - 1 = 0$$

$$\text{and } S_2 - S_3 \equiv -6x + 14y - 10 = 0 \text{ or } 3x - 7y + 5 = 0$$

Solving them the radical centre is $(3, 2)$. Also, if r is the length of the tangent drawn from the radical centre $(3, 2)$ to any one of the given circles, say S_1 , we have

$$r = \sqrt{S_1} = \sqrt{3^2 + 2^2 + 3.3 + 2.2 + 1} = \sqrt{27}$$

Hence $(3, 2)$ is the centre and $\sqrt{27}$ is the radius of the circle intersecting them orthogonally.

$$\therefore \text{ Its equation is } (x - 3)^2 + (y - 2)^2 = r^2 = 27 \Rightarrow x^2 + y^2 - 6x - 4y - 14 = 0$$

Alternative Method :

Let $x^2 + y^2 + 2gx + 2fy + c = 0$ be the equation of the circle cutting the given circles orthogonally.

$$\therefore 2g\left(\frac{3}{2}\right) + 2f(1) = c + 1 \quad \text{or} \quad 3g + 2f = c + 1 \quad \dots\dots (i)$$

$$2g\left(-\frac{1}{2}\right) + 2f(3) = c + 5 \quad \text{or} \quad -g + 6f = c + 5 \quad \dots\dots (ii)$$

$$\text{and} \quad 2g\left(\frac{5}{2}\right) + 2f(-4) = c + 15 \quad \text{or} \quad 5g - 8f = c + 15 \quad \dots\dots (iii)$$

Solving (i), (ii) and (iii) we get $g = -3$, $f = -2$ and $c = -14$

$$\therefore \text{equation of required circle is } x^2 + y^2 - 6x - 4y - 14 = 0$$

Ans.**Do yourself - 10 :**

- (i) Find the angle of intersection of two circles

$$S : x^2 + y^2 - 4x + 6y + 11 = 0 \text{ \& } S' : x^2 + y^2 - 2x + 8y + 13 = 0$$

- (ii) Find the equation of the radical axis of the circle $x^2 + y^2 - 3x - 4y + 5 = 0$ and $3x^2 + 3y^2 - 7x - 8y + 11 = 0$

- (iii) Find the radical centre of three circles described on the three sides $4x - 7y + 10 = 0$, $x + y - 5 = 0$ and $7x + 4y - 15 = 0$ of a triangle as diameters.

15. SOME IMPORTANT RESULTS TO REMEMBER :

- If the circle $S_1 = 0$, bisects the circumference of the circle $S_2 = 0$, then their common chord will be the diameter of the circle $S_2 = 0$.
- The radius of the director circle of a given circle is $\sqrt{2}$ times the radius of the given circle.
- The locus of the middle point of a chord of a circle subtend a right angle at a given point will be a circle.
- The length of side of an equilateral triangle inscribed in the circle $x^2 + y^2 = a^2$ is $\sqrt{3}a$
- If the lengths of tangents from the points A and B to a circle are ℓ_1 and ℓ_2 respectively, then if the points A and B are conjugate to each other, then $(AB)^2 = \ell_1^2 + \ell_2^2$.
- Length of transverse common tangent is less than the length of direct common tangent.

Do yourself - 11 :

- When the circles $x^2 + y^2 + 4x + 6y + 3 = 0$ and $2(x^2 + y^2) + 6x + 4y + c = 0$ intersect orthogonally, then find the value of c is
- Write the condition so that circles $x^2 + y^2 + 2ax + c = 0$ and $x^2 + y^2 + 2by + c = 0$ touch externally.

Miscellaneous Illustrations :

Illustration 24 : Find the equation of a circle which passes through the point (2, 0) and whose centre is the limit of the point of intersection of the lines $3x + 5y = 1$ and $(2 + c)x + 5c^2y = 1$ as $c \rightarrow 1$.

Solution : Solving the equations $(2 + c)x + 5c^2y = 1$ and $3x + 5y = 1$

$$\text{then } (2 + c)x + 5c^2\left(\frac{1-3x}{5}\right) = 1 \quad \text{or} \quad (2 + c)x + c^2(1 - 3x) = 1$$

$$\therefore x = \frac{1-c^2}{2+c-3c^2} \quad \text{or} \quad x = \frac{(1+c)(1-c)}{(3c+2)(1-c)} = \frac{1+c}{3c+2}$$

$$\therefore x = \lim_{c \rightarrow 1} \frac{1+c}{3c+2} \quad \text{or} \quad x = \frac{2}{5}$$

$$\therefore y = \frac{1-3x}{5} = \frac{1-\frac{6}{5}}{5} = -\frac{1}{25}$$

Therefore the centre of the required circle is $\left(\frac{2}{5}, -\frac{1}{25}\right)$ but circle passes through (2, 0)

$$\therefore \text{Radius of the required circle} = \sqrt{\left(\frac{2}{5} - 2\right)^2 + \left(-\frac{1}{25} - 0\right)^2} = \sqrt{\frac{64}{25} + \frac{1}{625}} = \sqrt{\frac{1601}{625}}$$

$$\text{Hence the required equation of the circle is } \left(x - \frac{2}{5}\right)^2 + \left(y + \frac{1}{25}\right)^2 = \frac{1601}{625}$$

$$\text{or } 25x^2 + 25y^2 - 20x + 2y - 60 = 0$$

Ans.

Illustration 25 : Two straight lines rotate about two fixed points. If they start from their position of coincidence such that one rotates at the rate double that of the other. Prove that the locus of their point of intersection is a circle.

Solution : Let $A \equiv (-a, 0)$ and $B \equiv (a, 0)$ be two fixed points.

Let one line which rotates about B an angle θ with the x-axis at any time t and at that time the second line which rotates about A make an angle 2θ with x-axis.

Now equation of line through B and A are respectively

$$y - 0 = \tan\theta(x - a) \quad \dots\dots (i)$$

$$\text{and } y - 0 = \tan 2\theta(x + a) \quad \dots\dots (ii)$$

$$\text{From (ii), } y = \frac{2 \tan \theta}{1 - \tan^2 \theta} (x + a)$$

$$= \left\{ \frac{\frac{2y}{(x-a)}}{1 - \frac{y^2}{(x-a)^2}} \right\} (x + a) \quad \text{(from (i))}$$

$$\Rightarrow y = \frac{2y(x-a)(x+a)}{(x-a)^2 - y^2} \Rightarrow (x-a)^2 - y^2 = 2(x^2 - a^2)$$

$$\text{or } x^2 + y^2 + 2ax - 3a^2 = 0 \text{ which is the required locus.}$$

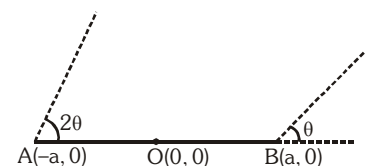


Illustration 26 : If the circle $x^2 + y^2 + 6x - 2y + k = 0$ bisects the circumference of the circle $x^2 + y^2 + 2x - 6y - 15 = 0$, then $k =$

- (A) 21 (B) -21 (C) 23 (D) -23

Solution : $2g_2(g_1 - g_2) + 2f_2(f_1 - f_2) = c_1 - c_2$
 $2(1)(3 - 1) + 2(-3)(-1 + 3) = k + 15$
 $4 - 12 = k + 15$ or $-8 = k + 15 \Rightarrow k = -23$

Ans. (D)

Illustration 27 : Find the equation of the circle of minimum radius which contains the three circles.

$$S_1 \equiv x^2 + y^2 - 4y - 5 = 0$$

$$S_2 \equiv x^2 + y^2 + 12x + 4y + 31 = 0$$

$$S_3 \equiv x^2 + y^2 + 6x + 12y + 36 = 0$$

Solution : For S_1 , centre = (0, 2) and radius = 3
 For S_2 , centre = (-6, -2) and radius = 3
 For S_3 , centre = (-3, -6) and radius = 3
 let $P(a, b)$ be the centre of the circle passing through the centres of the three given circles, then

$$(a - 0)^2 + (b - 2)^2 = (a + 6)^2 + (b + 2)^2$$

$$\Rightarrow (a + 6)^2 - a^2 = (b - 2)^2 - (b + 2)^2$$

$$(2a + 6)6 = 2b(-4)$$

$$b = \frac{2 \times 6(a + 3)}{-8} = -\frac{3}{2}(a + 3)$$

$$\text{again } (a - 0)^2 + (b - 2)^2 = (a + 3)^2 + (b + 6)^2$$

$$\Rightarrow (a + 3)^2 - a^2 = (b - 2)^2 - (b + 6)^2$$

$$(2a + 3)3 = (2b + 4)(-8)$$

$$(2a + 3)3 = -16 \left[-\frac{3}{2}(a + 3) + 2 \right]$$

$$6a + 9 = -8(-3a - 5)$$

$$6a + 9 = 24a + 40$$

$$18a = -31$$

$$a = -\frac{31}{18}, b = -\frac{23}{12}$$

$$\text{radius of the required circle} = 3 + \sqrt{\left(-\frac{31}{18}\right)^2 + \left(-\frac{23}{12} - 2\right)^2} = 3 + \frac{5}{36}\sqrt{949}$$

$$\therefore \text{equation of the required circle is } \left(x + \frac{31}{18}\right)^2 + \left(y + \frac{23}{12}\right)^2 = \left(3 + \frac{5}{36}\sqrt{949}\right)^2$$

Illustration 28 : Find the equation of the image of the circle $x^2 + y^2 + 16x - 24y + 183 = 0$ by the line mirror $4x + 7y + 13 = 0$.

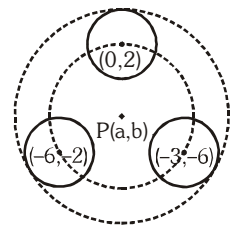
Solution : Centre of given circle = (-8, 12), radius = 5

the given line is $4x + 7y + 13 = 0$

let the centre of required circle is (h, k)

since radius will not change. so radius of required circle is 5.

Now (h, k) is the reflection of centre (-8, 12) in the line $4x + 7y + 13 = 0$



$$\text{Co-ordinates of A} = \left(\frac{-8+h}{2}, \frac{12+k}{2} \right)$$

$$\Rightarrow \frac{4(-8+h)}{2} + \frac{7(12+k)}{2} + 13 = 0$$

$$-32 + 4h + 84 + 7k + 26 = 0$$

$$4h + 7k + 78 = 0 \quad \dots\dots\dots(i)$$

$$\text{Also } \frac{k-12}{h+8} = \frac{7}{4}$$

$$4k - 48 = 7h + 56$$

$$4k = 7h + 104 \quad \dots\dots\dots(ii)$$

solving (i) & (ii)

$$h = -16, k = -2$$

$$\therefore \text{required circle is } (x + 16)^2 + (y + 2)^2 = 5^2$$

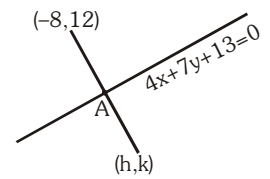


Illustration 29 : The circle $x^2 + y^2 - 6x - 10y + k = 0$ does not touch or intersect the coordinate axes and the point $(1, 4)$ is inside the circle. Find the range of the value of k .

Solution : Since $(1, 4)$ lies inside the circle

$$\Rightarrow S_1 < 0$$

$$\Rightarrow (1)^2 + (4)^2 - 6(1) - 10(4) + k < 0$$

$$\Rightarrow k < 29$$

Also centre of given circle is $(3, 5)$ and circle does not touch or intersect the coordinate axes

$$\Rightarrow r < CA \quad \& \quad r < CB$$

$$CA = 5$$

$$CB = 3$$

$$\Rightarrow r < 5 \quad \& \quad r < 3$$

$$\Rightarrow r < 3 \quad \text{or} \quad r^2 < 9$$

$$r^2 = 9 + 25 - k$$

$$r^2 = 34 - k \quad \Rightarrow \quad 34 - k < 9$$

$$k > 25$$

$$\Rightarrow k \in (25, 29)$$

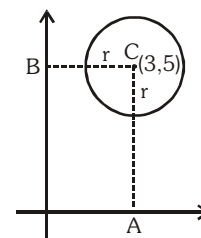


Illustration 30 : The circle $x^2 + y^2 - 4x - 8y + 16 = 0$ rolls up the tangent to it at $(2 + \sqrt{3}, 3)$ by 2 units, find the equation of the circle in the new position.

Solution : Given circle is $x^2 + y^2 - 4x - 8y + 16 = 0$

$$\text{let } P \equiv (2 + \sqrt{3}, 3)$$

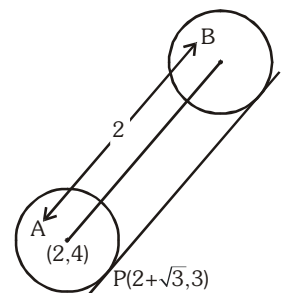
Equation of tangent to the circle at $P(2 + \sqrt{3}, 3)$ will be

$$(2 + \sqrt{3})x + 3y - 2(x + 2 + \sqrt{3}) - 4(y + 3) + 16 = 0$$

$$\text{or} \quad \sqrt{3}x - y - 2\sqrt{3} = 0$$

$$\text{slope} = \sqrt{3} \Rightarrow \tan\theta = \sqrt{3}$$

$$\theta = 60^\circ$$



line AB is parallel to the tangent at P

$\Rightarrow$ coordinates of point B = $(2 + 2\cos 60^\circ, 4 + 2\sin 60^\circ)$

thus B = $(3, 4 + \sqrt{3})$

radius of circle = $\sqrt{2^2 + 4^2 - 16} = 2$

$\therefore$ equation of required circle is $(x - 3)^2 + (y - 4 - \sqrt{3})^2 = 2^2$

Illustration 31 : A fixed circle is cut by a family of circles all of which, pass through two given points $A(x_1, y_1)$ and $B(x_2, y_2)$. Prove that the chord of intersection of the fixed circle with any circle of the family passes through a fixed point.

Solution :

Let $S = 0$ be the equation of fixed circle

let $S_1 = 0$ be the equation of any circle through A and B which intersect $S = 0$ in two points.

$L \equiv S - S_1 = 0$ is the equation of the chord of intersection of $S = 0$ and $S_1 = 0$

let $L_1 = 0$ be the equation of line AB

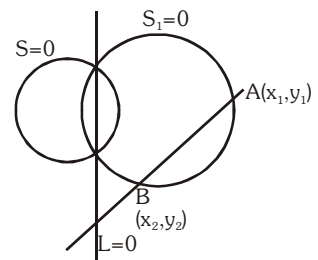
let S_2 be the equation of the circle whose diametrical ends are $A(x_1, y_1)$ & $B(x_2, y_2)$

then $S_1 \equiv S_2 - \lambda L_1 = 0$

$\Rightarrow L \equiv S - (S_2 - \lambda L_1) = 0$ or $L \equiv (S - S_2) + \lambda L_1 = 0$

or $L \equiv L' + \lambda L_1 = 0$ (i)

(i) implies each chord of intersection passes through the fixed point, which is the point of intersection of lines $L' = 0$ & $L_1 = 0$. Hence proved.



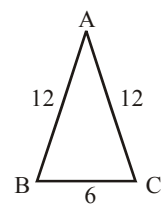
ANSWERS FOR DO YOURSELF

- 1: (i) Centre $\left(\frac{3}{4}, -\frac{5}{4}\right)$, Radius $\frac{3\sqrt{10}}{4}$ (ii) $17(x^2 + y^2) + 2x - 44y = 0$
- (iii) $x = \frac{p}{2}(-1 + \sqrt{2} \cos \theta)$; $y = \frac{p}{2}(-1 + \sqrt{2} \sin \theta)$ (iv) $x^2 + y^2 + 6x - 2y - 51 = 0$
- 2: (i) (1, 2) lie inside the circle and the point (6, 0) lies outside the circle
- (ii) min = 0, max = 6, power = 0
- 3: (i) $x \cos \alpha + y \sin \alpha = a(1 + \cos \alpha)$ (ii) $4x - 3y + 7 = 0$ & $4x - 3y - 43 = 0$
- (iii) $5x + 12y = \pm 26$; $\left(\mp \frac{10}{13}, \mp \frac{24}{13}\right)$ (iv) 1
- 4: (i) $x + 2y = 1$
- 5: (i) $4x + 7y + 10 = 0$ (ii) $\frac{405\sqrt{3}}{52}$ sq. units
- 6: (i) $5x - 4y + 26 = 0$ (ii) $x^2 + y^2 + gx + fy = 0$
- 7: (i) $(x - h)^2 + (y - k)^2 = 2a^2$ (ii) 10 (iii) angle between the tangents = 90°
- 8: (ii) $x^2 + y^2 - \frac{10x}{7} - \frac{10y}{7} - \frac{12}{7} = 0$ (iii) $x^2 + y^2 + 4x - 7y + 5 = 0$
- 9: (i) $(x - 5)^2 + (y - 5)^2 = 25$ (ii) 4
- 10: (i) 135° (ii) $x + 2y = 2$ (iii) (1, 2)
- 11: (i) 18 (ii) $a^{-2} + b^{-2} = c^{-1}$

EXERCISE (O-1)

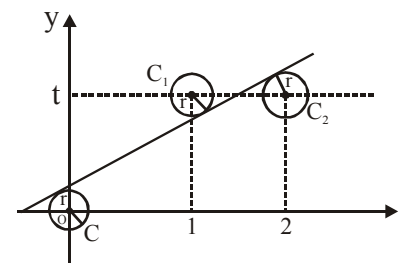
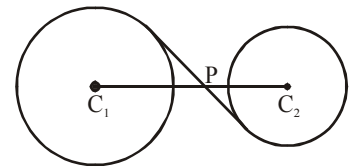
[SINGLE CORRECT]

- Centres of the three circles $x^2 + y^2 - 4x - 6y - 14 = 0$, $x^2 + y^2 + 2x + 4y - 5 = 0$ and $x^2 + y^2 - 10x - 16y + 7 = 0$
 (A) are the vertices of a right triangle
 (B) the vertices of an isosceles triangle which is not regular
 (C) vertices of a regular triangle
 (D) are collinear
- $y - 1 = m_1(x - 3)$ and $y - 3 = m_2(x - 1)$ are two family of straight lines, at right angled to each other. The locus of their point of intersection is
 (A) $x^2 + y^2 - 2x - 6y + 10 = 0$ (B) $x^2 + y^2 - 4x - 4y + 6 = 0$
 (C) $x^2 + y^2 - 2x - 6y + 6 = 0$ (D) $x^2 + y^2 - 4x - 4y - 6 = 0$
- Suppose that the equation of the circle having $(-3, 5)$ and $(5, -1)$ as end points of a diameter is $(x - a)^2 + (y - b)^2 = r^2$. Then $a + b + r$, ($r > 0$) is
 (A) 8 (B) 9 (C) 10 (D) 11
- B and C are fixed points having co-ordinates $(3, 0)$ and $(-3, 0)$ respectively. If the vertical angle BAC is 90° , then the locus of the centroid of the $\triangle ABC$ has the equation :
 (A) $x^2 + y^2 = 1$ (B) $x^2 + y^2 = 2$ (C) $9(x^2 + y^2) = 1$ (D) $9(x^2 + y^2) = 4$
- The area of an equilateral triangle inscribed in the circle $x^2 + y^2 - 2x = 0$ is
 (A) $\frac{3\sqrt{3}}{4}$ (B) $\frac{3\sqrt{3}}{2}$ (C) $\frac{3\sqrt{3}}{8}$ (D) none
- The equation of the image of the circle $x^2 + y^2 + 16x - 24y + 183 = 0$ by the line mirror $4x + 7y + 13 = 0$ is
 (A) $x^2 + y^2 + 32x - 4y + 235 = 0$ (B) $x^2 + y^2 + 32x + 4y - 235 = 0$
 (C) $x^2 + y^2 + 32x - 4y - 235 = 0$ (D) $x^2 + y^2 + 32x + 4y + 235 = 0$
- The radius of the circle passing through the vertices of the triangle ABC, is
 (A) $\frac{8\sqrt{15}}{5}$ (B) $\frac{3\sqrt{15}}{5}$
 (C) $3\sqrt{5}$ (D) $3\sqrt{2}$
- In the xy plane, the segment with end points $(3, 8)$ and $(-5, 2)$ is the diameter of the circle. The point $(k, 10)$ lies on the circle for
 (A) no value of k (B) exactly one integral k
 (C) exactly one non integral k (D) two real values of k
- $(6, 0)$, $(0, 6)$ and $(7, 7)$ are the vertices of a triangle. The circle inscribed in the triangle has the equation
 (A) $x^2 + y^2 - 9x + 9y + 36 = 0$ (B) $x^2 + y^2 - 9x - 9y + 36 = 0$
 (C) $x^2 + y^2 + 9x - 9y + 36 = 0$ (D) $x^2 + y^2 - 9x - 9y - 36 = 0$



10. The smallest distance between the circle $(x - 5)^2 + (y + 3)^2 = 1$ and the line $5x + 12y - 4 = 0$, is
 (A) $1/13$ (B) $2/13$ (C) $3/15$ (D) $4/15$
11. Consider the points $P(2, 1)$; $Q(0, 0)$; $P(4, -3)$ and the circle $S : x^2 + y^2 - 5x + 2y - 5 = 0$
 (A) exactly one point lies outside S (B) exactly two points lie outside S
 (C) all the three points lie outside S (D) none of the point lies outside S
12. If a circle of constant radius $3k$ passes through the origin 'O' and meets co-ordinate axes at A and B then the locus of the centroid of the triangle OAB is -
 (A) $x^2 + y^2 = (2k)^2$ (B) $x^2 + y^2 = (3k)^2$ (C) $x^2 + y^2 = (4k)^2$ (D) $x^2 + y^2 = (6k)^2$
13. The angle between the two tangents from the origin to the circle $(x - 7)^2 + (y + 1)^2 = 25$ equals
 (A) $\frac{\pi}{6}$ (B) $\frac{\pi}{3}$ (C) $\frac{\pi}{2}$ (D) $\frac{\pi}{4}$
14. Tangents are drawn from $(4, 4)$ to the circle $x^2 + y^2 - 2x - 2y - 7 = 0$ to meet the circle at A and B. The length of the chord AB is
 (A) $2\sqrt{3}$ (B) $3\sqrt{2}$ (C) $2\sqrt{6}$ (D) $6\sqrt{2}$
15. The area of the quadrilateral formed by the tangents from the point $(4, 5)$ to the circle $x^2 + y^2 - 4x - 2y - 11 = 0$ with the pair of radii through the points of contact of the tangents is :
 (A) 4 sq. units (B) 8 sq. units (C) 6 sq. units (D) none
16. The line joining $(5, 0)$ to $(10\cos\theta, 10\sin\theta)$ is divided internally in the ratio $2 : 3$ at P. If θ varies then the locus of P is :
 (A) a pair of straight lines (B) a circle
 (C) a straight line (D) a second degree curve which is not a circle
17. Combined equation to the pair of tangents drawn from the origin to the circle $x^2 + y^2 + 4x + 6y + 9 = 0$ is
 (A) $3(x^2 + y^2) = (x + 2y)^2$ (B) $2(x^2 + y^2) = (3x + y)^2$
 (C) $9(x^2 + y^2) = (2x + 3y)^2$ (D) $x^2 + y^2 = (2x + 3y)^2$
18. From $(3, 4)$ chords are drawn to the circle $x^2 + y^2 - 4x = 0$. The locus of the mid points of the chords is :
 (A) $x^2 + y^2 - 5x - 4y + 6 = 0$ (B) $x^2 + y^2 + 5x - 4y + 6 = 0$
 (C) $x^2 + y^2 - 5x + 4y + 6 = 0$ (D) $x^2 + y^2 - 5x - 4y - 6 = 0$
19. The locus of the center of the circles such that the point $(2, 3)$ is the mid point of the chord $5x + 2y = 16$ is
 (A) $2x - 5y + 11 = 0$ (B) $2x + 5y - 11 = 0$ (C) $2x + 5y + 11 = 0$ (D) none
20. In the xy-plane, the length of the shortest path from $(0, 0)$ to $(12, 16)$ that does not go inside the circle $(x - 6)^2 + (y - 8)^2 = 25$ is
 (A) $10\sqrt{3}$ (B) $10\sqrt{5}$ (C) $10\sqrt{3} + \frac{5\pi}{3}$ (D) $10 + 5\pi$
21. Tangents are drawn to a unit circle with centre at the origin from each point on the line $2x + y = 4$. Then the equation to the locus of the middle point of the chord of contact is -
 (A) $2(x^2 + y^2) = x + y$ (B) $2(x^2 + y^2) = x + 2y$ (C) $4(x^2 + y^2) = 2x + y$ (D) none
22. Chord AB of the circle $x^2 + y^2 = 100$ passes through the point $(7, 1)$ and subtends an angle of 60° at the circumference of the circle. If m_1 and m_2 are the slopes of two such chords then the value of $m_1 m_2$, is
 (A) -1 (B) 1 (C) $7/12$ (D) -3

23. Tangents PA and PB are drawn to the circle $x^2 + y^2 = 4$, then the locus of the point P if the triangle PAB is equilateral, is equal to-
- (A) $x^2 + y^2 = 16$ (B) $x^2 + y^2 = 8$ (C) $x^2 + y^2 = 64$ (D) $x^2 + y^2 = 32$
24. Sum of the abscissa and ordinate of the centre of the circle touching the line $3x + y + 2 = 0$ at the point $(-1, 1)$ and passing through the point $(3, 5)$ is-
- (A) 2 (B) 3 (C) 4 (D) 5
25. A circle of radius 5 is tangent to the line $4x - 3y = 18$ at $M(3, -2)$ and lies above the line. The equation of the circle, is-
- (A) $x^2 + y^2 - 6x + 4y - 12 = 0$ (B) $x^2 + y^2 + 2x - 2y - 3 = 0$
 (C) $x^2 + y^2 + 2x - 2y - 23 = 0$ (D) $x^2 + y^2 + 6x + 4y - 12 = 0$
26. If L_1 and L_2 are the length of the tangent from $(0, 5)$ to the circles $x^2 + y^2 + 2x - 4 = 0$ and $x^2 + y^2 - y + 1 = 0$ then
- (A) $L_1 = 2L_2$ (B) $L_2 = 2L_1$ (C) $L_1 = L_2$ (D) $L_1^2 = L_2$
27. In the figure given, two circles with centres C_1 and C_2 are 35 units apart, i.e. $C_1C_2 = 35$. The radii of the circles with centres C_1 and C_2 are 12 and 9 respectively. If P is the intersection of C_1C_2 and a common internal tangent to the circles, then $l(C_1P)$ equals-
- (A) 18 (B) 20 (C) 12 (D) 15
28. Circle K is inscribed in the first quadrant touching the circle $x^2 + y^2 = 36$ internally. The length of the radius of the circle K, is-
- (A) $\frac{6 - \sqrt{2}}{2}$ (B) $\frac{3\sqrt{2}}{2}$ (C) 3 (D) $6(\sqrt{2} - 1)$
29. Let C_1 and C_2 are circles defined by $x^2 + y^2 - 20x + 64 = 0$ and $x^2 + y^2 + 30x + 144 = 0$. The length of the shortest line segment PQ that is tangent to C_1 at P and to C_2 at Q is -
- (A) 15 (B) 18 (C) 20 (D) 24
30. As shown in the figure, three circles which have the same radius r , have centres at $(0, 0)$; $(1, 1)$ and $(2, 1)$. If they have a common tangent line, as shown then, their radius 'r' is -
- (A) $\frac{\sqrt{5} - 1}{2}$ (B) $\frac{\sqrt{5}}{10}$
 (C) $\frac{1}{2}$ (D) $\frac{\sqrt{3} - 1}{2}$
31. Two congruent circles with centres at $(2, 3)$ and $(5, 6)$ which intersect at right angles has radius equal to-
- (A) $2\sqrt{2}$ (B) 3 (C) 4 (D) none
32. The equation of a circle which touches the line $x + y = 5$ at $N(-2, 7)$ and cuts the circle $x^2 + y^2 + 4x - 6y + 9 = 0$ orthogonally, is -
- (A) $x^2 + y^2 + 7x - 11y + 38 = 0$ (B) $x^2 + y^2 = 53$
 (C) $x^2 + y^2 + x - y - 44 = 0$ (D) $x^2 + y^2 - x + y - 62 = 0$



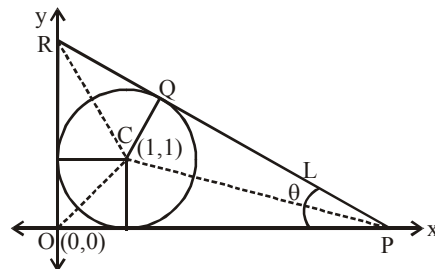
33. The angle at which the circle $(x-1)^2 + y^2 = 10$ and $x^2 + (y-2)^2 = 5$ intersect is -
 (A) $\frac{\pi}{6}$ (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{3}$ (D) $\frac{\pi}{2}$
34. Two circles whose radii are equal to 4 and 8 intersect at right angles. The length of their common chord is -
 (A) $\frac{16}{\sqrt{5}}$ (B) 8 (C) $4\sqrt{6}$ (D) $\frac{8\sqrt{5}}{5}$
35. The points (x_1, y_1) , (x_2, y_2) , (x_1, y_2) and (x_2, y_1) are always
 (A) collinear (B) concyclic
 (C) vertices of a square (D) vertices of a rhombus
36. Locus of all point P(x, y) satisfying $x^3 + y^3 + 3xy = 1$ consists of union of
 (A) a line and an isolated point (B) a line pair and an isolated point
 (C) a line and a circle (D) a circle and a isolated point.
37. Two circles of radii r_1 and r_2 are both touching the coordinate axes and intersecting each other orthogonally. The value of r_1/r_2 (where $r_1 > r_2$) equals -
 (A) $2 + \sqrt{3}$ (B) $\sqrt{3} + 1$ (C) $2 - \sqrt{3}$ (D) $2 + \sqrt{5}$

EXERCISE (O-2)

[COMPREHENSION]

Paragraph for Question nos. 1 to 3

In the diagram as shown, a circle is drawn with centre C(1, 1) and radius 1 and a line L. The line L is tangential to the circle at Q. Further L meet the y-axis at R and the x-axis at P is such a way that the angle OPQ equals θ where $0 < \theta < \frac{\pi}{2}$.



- The coordinates of Q are
 (A) $(1 + \cos\theta, 1 + \sin\theta)$ (B) $(\sin\theta, \cos\theta)$
 (C) $(1 + \sin\theta, \cos\theta)$ (D) $(1 + \sin\theta, 1 + \cos\theta)$
- Equation of the line PR is
 (A) $x\cos\theta + y\sin\theta = \sin\theta + \cos\theta + 1$ (B) $x\sin\theta + y\cos\theta = \cos\theta + \sin\theta - 1$
 (C) $x\sin\theta + y\cos\theta = \cos\theta + \sin\theta + 1$ (D) $x\tan\theta + y = 1 + \cot\left(\frac{\theta}{2}\right)$
- If the area bounded by the circle, the x-axis and PQ is $A(\theta)$, then $A\left(\frac{\pi}{4}\right)$ equals
 (A) $\sqrt{2} + 1 - \frac{3\pi}{8}$ (B) $\sqrt{2} - 1 + \frac{3\pi}{8}$ (C) $\sqrt{2} + 1 + \frac{\pi}{8}$ (D) $\sqrt{2} - 1 + \frac{\pi}{8}$

Paragraph for question Nos. 4 to 7

Consider the circle $S : x^2 + y^2 - 4x - 1 = 0$ and the line $L : y = 3x - 1$. If the line L cuts the circle at A & B .

4. Length of the chord AB equal -
 (A) $2\sqrt{5}$ (B) $\sqrt{5}$ (C) $5\sqrt{2}$ (D) $\sqrt{10}$
5. The angle subtended by the chord AB in the minor arc of S is-
 (A) $\frac{3\pi}{4}$ (B) $\frac{5\pi}{6}$ (C) $\frac{2\pi}{3}$ (D) $\frac{\pi}{4}$
6. Acute angle between the line L and the circle S is -
 (A) $\frac{\pi}{2}$ (B) $\frac{\pi}{3}$ (C) $\frac{\pi}{4}$ (D) $\frac{\pi}{6}$
7. If the equation of the circle on AB as diameter is of the form $x^2 + y^2 + ax + by + c = 0$ then the magnitude of the vector $\vec{V} = a\hat{i} + b\hat{j} + c\hat{k}$ has the value equal to-
 (A) $\sqrt{8}$ (B) $\sqrt{6}$ (C) $\sqrt{9}$ (D) $\sqrt{10}$

[MULTIPLE CHOICE]

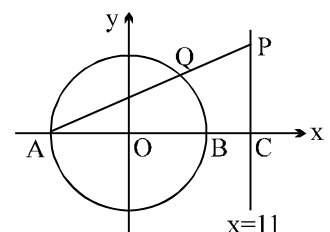
8. Which of the following lines have the intercepts of equal lengths on the circle, $x^2 + y^2 - 2x + 4y = 0$?
 (A) $3x - y = 0$ (B) $x + 3y = 0$ (C) $x + 3y + 10 = 0$ (D) $3x - y - 10 = 0$
9. A family of linear functions is given by $f(x) = 1 + c(x + 3)$ where $c \in \mathbb{R}$. If a member of this family meets a unit circle centred at origin in two coincident points then 'c' can be equal to
 (A) $-3/4$ (B) 0 (C) $3/4$ (D) 1
10. $\frac{x - x_1}{\cos \theta} = \frac{y - y_1}{\sin \theta} = r$, represents :
 (A) equation of a straight line, if θ is constant and r is variable
 (B) equation of a circle, if r is constant and θ is a variable
 (C) a straight line passing through a fixed point and having a known slope
 (D) a circle with a known centre and a given radius.
11. The equations of the tangents drawn from the origin to the circle, $x^2 + y^2 - 2rx - 2hy + h^2 = 0$ are :
 (A) $x = 0$ (B) $y = 0$
 (C) $(h^2 - r^2)x - 2rhy = 0$ (D) $(h^2 - r^2)x + 2rhy = 0$
12. Tangents PA and PB are drawn to the circle $S \equiv x^2 + y^2 - 2y - 3 = 0$ from the point $P(3,4)$. Which of the following alternative(s) is/are correct ?
 (A) The power of point $P(3,4)$ with respect to circle $S = 0$ is 14.
 (B) The angle between tangents from $P(3,4)$ to the circle $S = 0$ is $\frac{\pi}{3}$
 (C) The equation of circumcircle of ΔPAB is $x^2 + y^2 - 3x - 5y + 4 = 0$
 (D) The area of quadrilateral $PACB$ is $3\sqrt{7}$ square units where C is the centre of circle $S = 0$.

13. Consider the circles $C_1 : x^2 + y^2 = 16$ and $C_2 : x^2 + y^2 - 12x + 32 = 0$. Which of the following statement is/are correct ?
 (A) Number of common tangent to these circles is 3.
 (B) The point P with coordinates (4,1) lies outside the circle C_1 and inside the circle C_2 .
 (C) Their direct common tangent intersect at (12,0).
 (D) Slope of their radical axis is not defined.
14. Which of the following is/are True ?
 The circles $x^2 + y^2 - 6x - 6y + 9 = 0$ and $x^2 + y^2 + 6x + 6y + 9 = 0$ are such that -
 (A) they do not intersect
 (B) they touch each other
 (C) their exterior common tangents are parallel.
 (D) their interior common tangents are perpendicular.
15. Consider the circles $S_1 : x^2 + y^2 = 4$ and $S_2 : x^2 + y^2 - 2x - 4y + 4 = 0$ which of the following statements are correct ?
 (A) Number of common tangents to these circles is 2.
 (B) If the power of a variable point P w.r.t. these two circles is same then P moves on the line $x + 2y - 4 = 0$
 (C) Sum of the y-intercepts of both the circles is 6.
 (D) The circles S_1 and S_2 are orthogonal.
16. Two circles $x^2 + y^2 + px + py - 7 = 0$ and $x^2 + y^2 - 10x + 2py + 1 = 0$ intersect each other orthogonally then the value of p is -
 (A) 1 (B) 2 (C) 3 (D) 5

EXERCISE (S-1)

- Find the equation to the circle
 (i) Whose radius is 10 and whose centre is $(-5, -6)$.
 (ii) Whose radius is $a + b$ and whose centre is $(a, -b)$.
- Find the coordinates of the centres and the radii of the circles whose equations are :
 (i) $x^2 + y^2 - 4x - 8y = 41$ (ii) $\sqrt{1+m^2} (x^2 + y^2) - 2cx - 2mcy = 0$
- Find the equation to the circles which pass through the points :
 (i) $(0, 0)$, $(a, 0)$ and $(0, b)$ (ii) $(1, 2)$, $(3, -4)$ and $(5, -6)$
 (iii) $(1, 1)$, $(2, -1)$ and $(3, 2)$
- Find the equation to the circle which goes through the origin and cuts off intercepts equal to h and k from the positive parts of the axes.
- Find the equation to the circle which touches the axis of :
 (a) x at a distance +3 from the origin and intercepts a distance 6 on the axis of y.
 (b) x, pass through the point $(1, 1)$ and have line $x + y = 3$ as diameter.
- (a) Find the shortest distance from the point $M(-7, 2)$ to the circle $x^2 + y^2 - 10x - 14y - 151 = 0$.
 (b) Find the co-ordinate of the point on the circle $x^2 + y^2 - 12x - 4y + 30 = 0$, which is farthest from the origin.
- If the points $(\lambda, -\lambda)$ lies inside the circle $x^2 + y^2 - 4x + 2y - 8 = 0$, then find the range of λ .
- Show that the line $3x - 4y - c = 0$ will meet the circle having centre at $(2, 4)$ and the radius 5 in real and distinct points if $-35 < c < 15$.

9. (i) Write down the equation of the tangent to the circle $x^2 + y^2 - 3x + 10y = 15$ at the point $(4, -11)$
 (ii) Find the condition that the straight line $3x + 4y = k$ may touch the circle $x^2 + y^2 = 10x$.
10. Find the equation of the tangent to the circle
 (a) $x^2 + y^2 - 6x + 4y = 12$, which are parallel to the straight line $4x + 3y + 5 = 0$.
 (b) $x^2 + y^2 - 22x - 4y + 25 = 0$, which are perpendicular to the straight line $5x + 12y + 9 = 0$
 (c) $x^2 + y^2 = 25$, which are inclined at 30° to the axis of x .
11. Given that $x^2 + y^2 = 14x + 6y + 6$, find the largest possible value of the expression $E = 3x + 4y$.
12. The straight line $x - 2y + 1 = 0$ intersects the circle $x^2 + y^2 = 25$ in points T and T' , find the co-ordinates of a point of intersection of tangents drawn at T and T' to the circle.
13. Find the co-ordinates of the middle point of the chord which the circle $x^2 + y^2 - 2x + 2y - 2 = 0$ cuts off on the line $y = x - 1$.
 Find also the equation of the locus of the middle point of all chords of the circle which are parallel to the line $y = x - 1$.
14. Determine the nature of the quadrilateral formed by four lines $3x + 4y - 5 = 0$; $4x - 3y - 5 = 0$; $3x + 4y + 5 = 0$ and $4x - 3y + 5 = 0$. Find the equation of the circle inscribed and circumscribing this quadrilateral.
15. A circle $S = 0$ is drawn with its centre at $(-1, 1)$ so as to touch the circle $x^2 + y^2 - 4x + 6y - 3 = 0$ externally. Find the intercept made by the circle $S = 0$ on the coordinate axes.
16. The line $lx + my + n = 0$ intersects the curve $ax^2 + 2hxy + by^2 = 1$ at the point P and Q . The circle on PQ as diameter passes through the origin. Prove that $n^2(a + b) = l^2 + m^2$.
17. One of the diameters of the circle circumscribing the rectangle $ABCD$ is $4y = x + 7$. If A & B are the points $(-3, 4)$ & $(5, 4)$ respectively, then find the area of the rectangle.
18. Let L_1 be a straight line through the origin and L_2 be the straight line $x + y = 1$. If the intercepts made by the circle $x^2 + y^2 - x + 3y = 0$ on L_1 & L_2 are equal, then find the equation(s) which represent L_1 .
19. A circle passes through the points $(-1, 1)$, $(0, 6)$ and $(5, 5)$. Find the points on the circle the tangents at which are parallel to the straight line joining origin to the centre.
20. Find the equations of straight lines which pass through the intersection of the lines $x - 2y - 5 = 0$, $7x + y = 50$ & divide the circumference of the circle $x^2 + y^2 = 100$ into two arcs whose lengths are in the ratio $2 : 1$.
21. A line with gradient 2 is passing through the point $P(1, 7)$ and touches the circle $x^2 + y^2 + 16x + 12y + c = 0$ at the point Q . If (a, b) are the coordinates of the point Q , then find the value of $(7a + 7b + c)$.
22. In the given figure, the circle $x^2 + y^2 = 25$ intersects the x -axis at the point A and B . The line $x = 11$ intersects the x -axis at the point C . Point P moves along the line $x = 11$ above the x -axis and AP intersects the circle at Q . Find
 (i) The coordinates of the point P if the triangle AQB has the maximum area.
 (ii) The coordinates of the point P if Q is the middle point of AP .
 (iii) The coordinates of P if the area of the triangle AQB is $(1/4)^{\text{th}}$ of the area of the triangle APC .



23. A point moving around circle $(x + 4)^2 + (y + 2)^2 = 25$ with centre C broke away from it either at the point A or point B on the circle and moved along a tangent to the circle passing through the point D $(3, -3)$. Find the following.
- Equation of the tangents at A and B.
 - Coordinates of the points A and B.
 - Angle ADB and the maximum and minimum distances of the point D from the circle.
 - Area of quadrilateral ADBC and the ΔDAB .
 - Equation of the circle circumscribing the ΔDAB and also the length of the intercepts made by this circle on the coordinate axes.
24. Find the equation of a line with gradient 1 such that the two circles $x^2 + y^2 = 4$ and $x^2 + y^2 - 10x - 14y + 65 = 0$ intercept equal length on it.
25. Find the locus of the middle points of portions of the tangents to the circle $x^2 + y^2 = a^2$ terminated by the coordinate axes.
26. Tangents OP and OQ are drawn from the origin O to the circle $x^2 + y^2 + 2gx + 2fy + c = 0$. Find the equation of the circumcircle of the triangle OPQ.
27. If M and m are the maximum and minimum values of $\frac{y}{x}$ for pair of real number (x, y) which satisfy the equation $(x - 3)^2 + (y - 3)^2 = 6$, then find the value of $(M + m)$.
28. Tangents are drawn to the concentric circles $x^2 + y^2 = a^2$ and $x^2 + y^2 = b^2$ at right angle to one another. Show that the locus of their point of intersection is a 3rd concentric circle. Find its radius.
29. Find the equation of the circle passing through the point of intersection of the circles $x^2 + y^2 - 6x + 2y + 4 = 0$, $x^2 + y^2 + 2x - 4y - 6 = 0$ and with its centre on the line $y = x$.
30. Find the equation of the circle passing through the points of intersection of the circles $x^2 + y^2 - 2x - 4y - 4 = 0$ and $x^2 + y^2 - 10x - 12y + 40 = 0$ and whose radius is 4.
31. Find the equation of the circle through points of intersection of the circle $x^2 + y^2 - 2x - 4y + 4 = 0$ and the line $x + 2y = 4$ which touches the line $x + 2y = 0$.
32. Find the equations of the circles which pass through the common points of the following pair of circles.
- $x^2 + y^2 + 2x + 3y - 7 = 0$ and $x^2 + y^2 + 3x - 2y - 1 = 0$ through the point $(1, 2)$
 - $x^2 + y^2 + 4x - 6y - 12 = 0$ and $x^2 + y^2 - 5x + 17y = 19$ and having its centre on $x + y = 0$.
33. Find the radical centre of the following set of circles
 $x^2 + y^2 - 3x - 6y + 14 = 0$; $x^2 + y^2 - x - 4y + 8 = 0$; $x^2 + y^2 + 2x - 6y + 9 = 0$
34. Find the equation to the circle orthogonal to the two circles
 $x^2 + y^2 - 4x - 6y + 11 = 0$; $x^2 + y^2 - 10x - 4y + 21 = 0$ and has $2x + 3y = 7$ as diameter.
35. Find the equation to the circle, cutting orthogonally each of the following circles :
 $x^2 + y^2 - 2x + 3y - 7 = 0$; $x^2 + y^2 + 5x - 5y + 9 = 0$; $x^2 + y^2 + 7x - 9y + 29 = 0$.
36. Find the equation of the circle through the points of intersection of circles $x^2 + y^2 - 4x - 6y - 12 = 0$ and $x^2 + y^2 + 6x + 4y - 12 = 0$ & cutting the circle $x^2 + y^2 - 2x - 4 = 0$ orthogonally.

37. A variable circle passes through the point $A(a, b)$ & touches the x -axis. Show that the locus of the other end of the diameter through A is $(x - a)^2 = 4by$.
38. Circles C_1 and C_2 are externally tangent and they are both internally tangent to the circle C_3 . The radii of C_1 and C_2 are 4 and 10, respectively and the centres of the three circles are collinear. A chord of C_3 is also a common internal tangent of C_1 and C_2 . Given that the length of the chord is $\frac{m\sqrt{n}}{p}$ where m , n and p are positive integers, m and p are relatively prime and n is not divisible by the square of any prime, find the value of $(m + n + p)$.
39. The line $2x - 3y + 1 = 0$ is tangent to a circle $S = 0$ at $(1, 1)$. If the radius of the circle is $\sqrt{13}$. Find the equation of the circle S .
40. Find the equation of the circle which passes through the point $(1, 1)$ & which touches the circle $x^2 + y^2 + 4x - 6y - 3 = 0$ at the point $(2, 3)$ on it.
41. Find the equation of the circle whose radius is 3 and which touches the circle $x^2 + y^2 - 4x - 6y - 12 = 0$ internally at the point $(-1, -1)$.
42. The centre of the circle $S = 0$ lie on the line $2x - 2y + 9 = 0$ & $S = 0$ cuts orthogonally the circle $x^2 + y^2 = 4$. Show that circle $S = 0$ passes through two fixed points & find their coordinates.
43. Consider a circle S with centre at the origin and radius 4. Four circles A, B, C and D each with radius unity and centres $(-3, 0), (-1, 0), (1, 0)$ and $(3, 0)$ respectively are drawn. A chord PQ of the circle S touches the circle B and passes through the centre of the circle C . If the length of this chord can be expressed as $\sqrt{x}$, find x .

EXERCISE (S-2)

1. If the circle $x^2 + y^2 + 4x + 22y + a = 0$ bisects the circumference of the circle $x^2 + y^2 - 2x + 8y - b = 0$ (where $a, b > 0$), then find the maximum value of (ab) .
2. Real number x, y satisfies $x^2 + y^2 = 1$. If the maximum and minimum value of the expression $z = \frac{4-y}{7-x}$ are M and m respectively, then find the value $(2M + 6m)$.
3. Consider a family of circles passing through two fixed points $A(3, 7)$ & $B(6, 5)$. The chords in which the circle $x^2 + y^2 - 4x - 6y - 3 = 0$ cuts the members of the family are concurrent at a point. Find the coordinates of this point.
4. (a) Find the equation of a circle passing through the origin if the line pair, $xy - 3x + 2y - 6 = 0$ is orthogonal to it. If this circle is orthogonal to the circle $x^2 + y^2 - kx + 2ky - 8 = 0$ then find the value of k .
- (b) Find the equation of the circle which cuts the circle $x^2 + y^2 - 14x - 8y + 64 = 0$ and the coordinate axes orthogonally.

5. Find the equation of a circle which touches the line $x + y = 5$ at the point $(-2, 7)$ and cuts the circle $x^2 + y^2 + 4x - 6y + 9 = 0$ orthogonally.
6. A circle is drawn with its centre on the line $x + y = 2$ to touch the line $4x - 3y + 4 = 0$ and pass through the point $(0, 1)$. Find its equation.
7. Through a given point $P(5, 2)$, secants are drawn to cut the circle $x^2 + y^2 = 25$ at points $A_1(B_1)$, $A_2(B_2)$, $A_3(B_3)$, $A_4(B_4)$ and $A_5(B_5)$ such that $PA_1 + PB_1 = 5$, $PA_2 + PB_2 = 6$, $PA_3 + PB_3 = 7$, $PA_4 + PB_4 = 8$ and $PA_5 + PB_5 = 9$. Find the value of $\sum_{i=1}^5 PA_i^2 + \sum_{i=1}^5 PB_i^2$.
- [Note : $A_r(B_r)$ denotes that the line passing through $P(5, 2)$ meets the circle $x^2 + y^2 = 25$ at two points A_r and B_r .]
8. Find the equation of a circle which is co-axial with circles $2x^2 + 2y^2 - 2x + 6y - 3 = 0$ & $x^2 + y^2 + 4x + 2y + 1 = 0$. It is given that the centre of the circle to be determined lies on the radical axis of these two circles.
9. Find the locus of the mid point of all chords of the circle $x^2 + y^2 - 2x - 2y = 0$ such that the pair of lines joining $(0, 0)$ & the point of intersection of the chords with the circles make equal angle with axis of x .
10. A circle with center in the first quadrant is tangent to $y = x + 10$, $y = x - 6$, and the y -axis. Let (h, k) be the center of the circle. If the value of $(h + k) = a + b\sqrt{a}$ where $\sqrt{a}$ is a surd, find the value of $a + b$.
11. Let $S_1 = 0$ and $S_2 = 0$ be two circles intersecting at $P(6, 4)$ and both are tangent to x -axis and line $y = mx$ (where $m > 0$). If product of radii of the circles $S_1 = 0$ and $S_2 = 0$ is $\frac{52}{3}$, then find the value of m .
12. Consider two circles C_1 of radius 'a' and C_2 of radius 'b' ($b > a$) both lying in the first quadrant and touching the coordinate axes. In each of the conditions listed in **column-I**, the ratio of b/a is given in **column-II**.

Column-I

- (A) C_1 and C_2 touch each other
- (B) C_1 and C_2 are orthogonal
- (C) C_1 and C_2 intersect so that the common chord is longest
- (D) C_2 passes through the centre of C_1

Column-II

- (P) $2 + \sqrt{2}$
- (Q) 3
- (R) $2 + \sqrt{3}$
- (S) $3 + 2\sqrt{2}$
- (T) $3 - 2\sqrt{2}$

EXERCISE (JM)

- Three distinct points A, B and C are given in the 2-dimensional coordinate plane such that the ratio of the distance of any one of them from the point (1, 0) to the distance from the point (-1, 0) is equal to $\frac{1}{3}$. Then the circumcentre of the triangle ABC is at the point :- **[AIEEE-2009]**
 (1) $\left(\frac{5}{2}, 0\right)$ (2) $\left(\frac{5}{3}, 0\right)$ (3) (0, 0) (4) $\left(\frac{5}{4}, 0\right)$
- If P and Q are the points of intersection of the circles $x^2 + y^2 + 3x + 7y + 2p - 5 = 0$ and $x^2 + y^2 + 2x + 2y - p^2 = 0$, then there is a circle passing through P, Q and (1, 1) for :- **[AIEEE-2009]**
 (1) All except two values of p (2) Exactly one value of p
 (3) All values of p (4) All except one value of p
- For a regular polygon, let r and R be the radii of the inscribed and the circumscribed circles. A false statement among the following is :- **[AIEEE-2010]**
 (1) There is a regular polygon with $\frac{r}{R} = \frac{1}{2}$ (2) There is a regular polygon with $\frac{r}{R} = \frac{1}{\sqrt{2}}$
 (3) There is a regular polygon with $\frac{r}{R} = \frac{2}{3}$ (4) There is a regular polygon with $\frac{r}{R} = \frac{\sqrt{3}}{2}$
- The circle $x^2 + y^2 = 4x + 8y + 5$ intersects the line $3x - 4y = m$ at two distinct points if :- **[AIEEE-2010]**
 (1) $-85 < m < -35$ (2) $-35 < m < 15$ (3) $15 < m < 65$ (4) $35 < m < 85$
- The two circles $x^2 + y^2 = ax$ and $x^2 + y^2 = c^2$ ($c > 0$) touch each other if :- **[AIEEE-2011]**
 (1) $a = 2c$ (2) $|a| = 2c$ (3) $2|a| = c$ (4) $|a| = c$
- The equation of the circle passing through the points (1, 0) and (0, 1) and having the smallest radius is - **[AIEEE-2011]**
 (1) $x^2 + y^2 + x + y - 2 = 0$ (2) $x^2 + y^2 - 2x - 2y + 1 = 0$
 (3) $x^2 + y^2 - x - y = 0$ (4) $x^2 + y^2 + 2x + 2y - 7 = 0$
- The length of the diameter of the circle which touches the x-axis at the point (1, 0) and passes through the point (2, 3) is : **[AIEEE-2012]**
 (1) $5/3$ (2) $10/3$ (3) $3/5$ (4) $6/5$
- The circle passing through (1, -2) and touching the axis of x at (3, 0) also passes through the point : **[JEE (Main)-2013]**
 (1) (-5, 2) (2) (2, -5) (3) (5, -2) (4) (-2, 5)
- If a circle C passing through (4, 0) touches the circle $x^2 + y^2 + 4x - 6y - 12 = 0$ externally at a point (1, -1), then the radius of the circle C is :- **[JEE-Main (on line)-2013]**
 (1) $\sqrt{57}$ (2) $2\sqrt{5}$ (3) 4 (4) 5
- If the circle $x^2 + y^2 - 6x - 8y + (25 - a^2) = 0$ touches the axis of x, then a equals :- **[JEE-Main (on line)-2013]**
 (1) ± 4 (2) ± 3 (3) 0 (4) ± 2

11. Statement I : The only circle having radius $\sqrt{10}$ and a diameter along line $2x + y = 5$ is $x^2 + y^2 - 6x + 2y = 0$.
Statement II : $2x + y = 5$ is a normal to the circle $x^2 + y^2 - 6x + 2y = 0$. [JEE-Main (on line)-2013]
(1) Statement I is false, Statement II is true
(2) Statement I is true ; Statement II is false.
(3) Statement I is true, Statement II is true, Statement II is not a correct explanation for Statement I.
(4) Statement I is true : Statement II is true ; Statement II is a correct explanation for Statement I.
12. Let C be the circle with centre at (1, 1) and radius = 1. If T is the circle centred at (0, y), passing through origin and touching the circle C externally, then the radius of T is equal to : [JEE(Main)-2014]
(1) $\frac{\sqrt{3}}{\sqrt{2}}$ (2) $\frac{\sqrt{3}}{2}$ (3) $\frac{1}{2}$ (4) $\frac{1}{4}$
13. The number of common tangents to the circle $x^2 + y^2 - 4x - 6y - 12 = 0$ and $x^2 + y^2 + 6x + 18y + 26 = 0$, is : [JEE(Main)-2015]
(1) 3 (2) 4 (3) 1 (4) 2
14. If one of the diameters of the circle, given by the equation, $x^2 + y^2 - 4x + 6y - 12 = 0$, is a chord of a circle S, whose centre is at $(-3, 2)$, then the radius of S is :- [JEE(Main)-2016]
(1) 10 (2) $5\sqrt{2}$ (3) $5\sqrt{3}$ (4) 5
15. The centres of those circles which touch the circle, $x^2 + y^2 - 8x - 8y - 4 = 0$, externally and also touch the x-axis, lie on :- [JEE(Main)-2016]
(1) A parabola (2) A circle
(3) An ellipse which is not a circle (4) A hyperbola

EXERCISE (JA)

1. Tangents drawn from the point P(1, 8) to the circle $x^2 + y^2 - 6x - 4y - 11 = 0$ touch the circle at the points A and B. The equation of the circumcircle of the triangle PAB is
(A) $x^2 + y^2 + 4x - 6y + 19 = 0$ (B) $x^2 + y^2 - 4x - 10y + 19 = 0$
(C) $x^2 + y^2 - 2x + 6y - 29 = 0$ (D) $x^2 + y^2 - 6x - 4y + 19 = 0$ [JEE 2009, 3]
2. The centres of two circles C_1 and C_2 each of unit radius are at a distance of 6 units from each other. Let P be the mid point of the line segment joining the centres of C_1 and C_2 and C be a circle touching circles C_1 and C_2 externally. If a common tangent to C_1 and C passing through P is also a common tangent to C_2 and C, then the radius of the circle C is [JEE 2009, 4]
3. Two parallel chords of a circle of radius 2 are at a distance $\sqrt{3} + 1$ apart. If the chords subtend at the center, angles of $\frac{\pi}{k}$ and $\frac{2\pi}{k}$, where $k > 0$, then the value of [k] is [JEE 10, 3M]
[Note : [k] denotes the largest integer less than or equal to k]
4. The circle passing through the point $(-1, 0)$ and touching the y-axis at $(0, 2)$ also passes through the point -
(A) $\left(-\frac{3}{2}, 0\right)$ (B) $\left(-\frac{5}{2}, 2\right)$ (C) $\left(-\frac{3}{2}, \frac{5}{2}\right)$ (D) $(-4, 0)$

[JEE 2011, 3M, -1M]

5. The straight line $2x - 3y = 1$ divides the circular region $x^2 + y^2 \leq 6$ into two parts. If

$$S = \left\{ \left(2, \frac{3}{4} \right), \left(\frac{5}{2}, \frac{3}{4} \right), \left(\frac{1}{4}, -\frac{1}{4} \right), \left(\frac{1}{8}, \frac{1}{4} \right) \right\},$$

then the number of point(s) in S lying inside the smaller part is

[JEE 2011, 4M]

6. The locus of the mid-point of the chord of contact of tangents drawn from points lying on the straight line $4x - 5y = 20$ to the circle $x^2 + y^2 = 9$ is-

[JEE 2012, 3M, -1M]

- (A) $20(x^2 + y^2) - 36x + 45y = 0$ (B) $20(x^2 + y^2) + 36x - 45y = 0$
(C) $36(x^2 + y^2) - 20x + 45y = 0$ (D) $36(x^2 + y^2) + 20x - 45y = 0$

Paragraph for Question 7 and 8

A tangent PT is drawn to the circle $x^2 + y^2 = 4$ at the point $P(\sqrt{3}, 1)$. A straight line L , perpendicular to PT is a tangent to the circle $(x - 3)^2 + y^2 = 1$.

7. A common tangent of the two circles is

[JEE 2012, 3M, -1M]

- (A) $x = 4$ (B) $y = 2$ (C) $x + \sqrt{3}y = 4$ (D) $x + 2\sqrt{2}y = 6$

8. A possible equation of L is

[JEE 2012, 3M, -1M]

- (A) $x - \sqrt{3}y = 1$ (B) $x + \sqrt{3}y = 1$ (C) $x - \sqrt{3}y = -1$ (D) $x + \sqrt{3}y = 5$

9. Circle(s) touching x -axis at a distance 3 from the origin and having an intercept of length $2\sqrt{7}$ or y -axis is (are)

[JEE(Advanced) 2013, 3, (-1)]

- (A) $x^2 + y^2 - 6x + 8y + 9 = 0$ (B) $x^2 + y^2 - 6x + 7y + 9 = 0$
(C) $x^2 + y^2 - 6x - 8y + 9 = 0$ (D) $x^2 + y^2 - 6x - 7y + 9 = 0$

10. A circle S passes through the point $(0, 1)$ and is orthogonal to the circles $(x - 1)^2 + y^2 = 16$ and $x^2 + y^2 = 1$. Then :-

[JEE(Advanced)-2014, 3]

- (1) radius of S is 8 (B) radius of S is 7
(3) centre of S is $(-7, 1)$ (D) centre of S is $(-8, 1)$

11. Let RS be the diameter of the circle $x^2 + y^2 = 1$, where S is the point $(1, 0)$. Let P be a variable point (other than R and S) on the circle and tangents to the circle at S and P meet at the point Q . The normal to the circle at P intersects a line drawn through Q parallel to RS at point E . then the locus of E passes through the point(s)-

[JEE(Advanced)-2016, 4(-2)]

- (A) $\left(\frac{1}{3}, \frac{1}{\sqrt{3}} \right)$ (B) $\left(\frac{1}{4}, \frac{1}{2} \right)$ (C) $\left(\frac{1}{3}, -\frac{1}{\sqrt{3}} \right)$ (D) $\left(\frac{1}{4}, -\frac{1}{2} \right)$

12. For how many values of p , the circle $x^2 + y^2 + 2x + 4y - p = 0$ and the coordinate axes have exactly three common points ?

[JEE(Advanced)-2017, 3]

Paragraph "X"

Let S be the circle in the xy-plane defined by the equation $x^2 + y^2 = 4$.

(There are two question based on Paragraph "X", the question given below is one of them)

13. Let E_1E_2 and F_1F_2 be the chord of S passing through the point $P_0(1, 1)$ and parallel to the x-axis and the y-axis, respectively. Let G_1G_2 be the chord of S passing through P_0 and having slope -1 . Let the tangents to S at E_1 and E_2 meet at E_3 , the tangents of S at F_1 and F_2 meet at F_3 , and the tangents to S at G_1 and G_2 meet at G_3 . Then, the points E_3 , F_3 and G_3 lie on the curve [JEE(Advanced)-2018, 3(-1)]
- (A) $x + y = 4$ (B) $(x - 4)^2 + (y - 4)^2 = 16$
 (C) $(x - 4)(y - 4) = 4$ (D) $xy = 4$

Paragraph "X"

Let S be the circle in the xy-plane defined by the equation $x^2 + y^2 = 4$.

(There are two question based on Paragraph "X", the question given below is one of them)

14. Let P be a point on the circle S with both coordinates being positive. Let the tangent to S at P intersect the coordinate axes at the points M and N. Then, the mid-point of the line segment MN must lie on the curve - [JEE(Advanced)-2018, 3(-1)]
- (A) $(x + y)^2 = 3xy$ (B) $x^{2/3} + y^{2/3} = 2^{4/3}$ (C) $x^2 + y^2 = 2xy$ (D) $x^2 + y^2 = x^2y^2$
15. Let T be the line passing through the points $P(-2, 7)$ and $Q(2, -5)$. Let F_1 be the set of all pairs of circles (S_1, S_2) such that T is tangents to S_1 at P and tangent to S_2 at Q, and also such that S_1 and S_2 touch each other at a point, say, M. Let E_1 be the set representing the locus of M as the pair (S_1, S_2) varies in F_1 . Let the set of all straight line segments joining a pair of distinct points of E_1 and passing through the point $R(1, 1)$ be F_2 . Let E_2 be the set of the mid-points of the line segments in the set F_2 . Then, which of the following statement(s) is (are) TRUE ? [JEE(Advanced)-2018, 4(-2)]

- (A) The point $(-2, 7)$ lies in E_1 (B) The point $\left(\frac{4}{5}, \frac{7}{5}\right)$ does NOT lie in E_2
 (C) The point $\left(\frac{1}{2}, 1\right)$ lies in E_2 (D) The point $\left(0, \frac{3}{2}\right)$ does NOT lie in E_1

ANSWER KEY

EXERCISE (O-1)

- | | | | | | | | |
|-------|-------|-------|-------|-------|-------|-------|-------|
| 1. D | 2. B | 3. A | 4. A | 5. A | 6. D | 7. A | 8. B |
| 9. B | 10. B | 11. D | 12. A | 13. C | 14. B | 15. B | 16. B |
| 17. C | 18. A | 19. A | 20. C | 21. C | 22. A | 23. A | 24. C |
| 25. C | 26. C | 27. B | 28. D | 29. C | 30. B | 31. B | 32. A |
| 33. B | 34. A | 35. B | 36. A | 37. A | | | |

EXERCISE (O-2)

- | | | | | | | |
|------------|-----------|-------------|---------|---------|-----------|------|
| 1. D | 2. C | 3. A | 4. D | 5. A | 6. C | 7. B |
| 8. A,B,C,D | 9. A,B | 10. A,B,C,D | 11. A,C | 12. A,C | 13. A,C,D | |
| 14. A,C,D | 15. A,B,D | 16. B,C | | | | |

EXERCISE (S-1)

1. (i) $x^2 + y^2 + 10x + 12y = 39$; (ii) $x^2 + y^2 - 2ax + 2by = 2ab$
2. (i) $(2, 4)$; $\sqrt{61}$; (ii) $\left(\frac{c}{\sqrt{1+m^2}}, \frac{mc}{\sqrt{1+m^2}}\right)$; c
3. (i) $x^2 + y^2 - ax - by = 0$; (ii) $x^2 + y^2 - 22x - 4y + 25 = 0$; (iii) $x^2 + y^2 - 5x - y + 4 = 0$
4. $x^2 + y^2 - hx - ky = 0$
5. (a) $x^2 + y^2 - 6x \pm 6\sqrt{2}y + 9 = 0$; (b) $x^2 + y^2 + 4x - 10y + 4 = 0$; $x^2 + y^2 - 4x - 2y + 4 = 0$
6. (a) 2; (b) (9, 3) 7. $\lambda \in (-1, 4)$ 9. (i) $5x - 12y = 152$ (ii) $k = 40$ or -10
10. (a) $4x + 3y + 19 = 0$ and $4x + 3y - 31 = 0$; (b) $12x - 5y + 8 = 0$ and $12x - 5y - 252 = 0$
(c) $x - \sqrt{3}y \pm 10 = 0$
11. 73 12. $(-25, 50)$ 13. $\left(\frac{1}{2}, -\frac{1}{2}\right)$, $x + y = 0$
14. square of side 2; $x^2 + y^2 = 1$; $x^2 + y^2 = 2$ 15. zero, zero 17. 32 sq. unit
18. $x - y = 0$; $x + 7y = 0$ 19. (5, 1) & (-1, 5) 20. $4x - 3y - 25 = 0$ OR $3x + 4y - 25 = 0$
21. 4 22. (i) (11, 16), (ii) (11, 8), (iii) (11, 12)
23. (i) $3x - 4y = 21$; $4x + 3y = 3$; (ii) A(0, 1) and B(-1, -6); (iii) 90° , $5(\sqrt{2} \pm 1)$ units
(iv) 25 sq. units, 12.5 sq. units; (v) $x^2 + y^2 + x + 5y - 6$, x-intercept = 5; y-intercept = 7
24. $2x - 2y - 3 = 0$ 25. $a^2(x^2 + y^2) = 4x^2y^2$ 26. $x^2 + y^2 + gx + fy = 0$ 27. 6
28. $x^2 + y^2 = a^2 + b^2$; $r = \sqrt{a^2 + b^2}$ 29. $7x^2 + 7y^2 - 10x - 10y - 12 = 0$
30. $2x^2 + 2y^2 - 18x - 22y + 69 = 0$ and $x^2 + y^2 - 2y - 15 = 0$ 31. $x^2 + y^2 - x - 2y = 0$
32. (a) $x^2 + y^2 + 4x - 7y + 5 = 0$, (b) $2(x^2 + y^2) - x + y - 31 = 0$ 33. (1, 2)
34. $x^2 + y^2 - 4x - 2y + 3 = 0$ 35. $x^2 + y^2 - 16x - 18y - 4 = 0$
36. $x^2 + y^2 + 16x + 14y - 12 = 0$ 38. 19
39. $x^2 + y^2 - 6x + 4y = 0$ OR $x^2 + y^2 + 2x - 8y + 4 = 0$ 40. $x^2 + y^2 + x - 6y + 3 = 0$
41. $5x^2 + 5y^2 - 8x - 14y - 32 = 0$ 42. $(-4, 4)$; $(-1/2, 1/2)$ 43. 63

EXERCISE (S-2)

1. 625 2. 4 3. $\left(2, \frac{23}{3}\right)$ 4. (a) $x^2 + y^2 + 4x - 6y = 0$; $k = 1$; (b) $x^2 + y^2 = 64$
 5. $x^2 + y^2 + 7x - 11y + 38 = 0$ 6. $x^2 + y^2 - 2x - 2y + 1 = 0$ OR $x^2 + y^2 - 42x + 38y - 39 = 0$
 7. 215 8. $4x^2 + 4y^2 + 6x + 10y - 1 = 0$ 9. $x + y = 2$ 10. 10 11. $\sqrt{3}$
 12. (A) S; (B) R; (C) Q; (D) P

EXERCISE (JM)

1. 4 2. 4 3. 3 4. 2 5. 4 6. 3 7. 2 8. 3
 9. 4 10. 1 11. 1 12. 4 13. 1 14. 3 15. 1

EXERCISE (JA)

1. B 2. 8 3. 3 4. D 5. 2 6. A 7. D 8. A
 9. A,C 10. B,C 11. A,C 12. 2 13. A 14. D 15. B,D

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JEE (Main) Syllabus :

Sections of cones, equations of conic sections (parabola, ellipse and hyperbola) in standard forms, condition for $y = mx + c$ to be a tangent and point (s) of tangency.

JEE (Advanced) Syllabus :

Equations of a parabola, ellipse and hyperbola in standard form, their foci, directrices and eccentricity, parametric equations, equations of tangent and normal. Locus Problems.

PARABOLA

1. CONIC SECTIONS :

A conic section, or conic is the locus of a point which moves in a plane so that its distance from a fixed point is in a constant ratio to its perpendicular distance from a fixed straight line.

- (a) The fixed point is called the **focus**.
- (b) The fixed straight line is called the **directrix**.
- (c) The constant ratio is called the **eccentricity** denoted by e .
- (d) The line passing through the focus & perpendicular to the directrix is called the **axis**.
- (e) A point of intersection of a conic with its axis is called a **vertex**.

2. GENERAL EQUATION OF A CONIC : FOCAL DIRECTRIX PROPERTY :

The general equation of a conic with focus (p, q) & directrix $lx + my + n = 0$ is :

$$(l^2 + m^2) [(x - p)^2 + (y - q)^2] = e^2 (lx + my + n)^2 \equiv ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$$

3. DISTINGUISHING BETWEEN THE CONIC :

The nature of the conic section depends upon the position of the focus S w.r.t. the directrix & also upon the value of the eccentricity e . Two different cases arise.

Case (i) When the focus lies on the directrix :

In this case $D \equiv abc + 2fgh - af^2 - bg^2 - ch^2 = 0$ & the general equation of a conic represents a pair of straight lines and if :

$e > 1$ the lines will be real & distinct intersecting at S .

$e = 1$ the lines will be coincident.

$e < 1$ the lines will be imaginary.

Case (ii) When the focus does not lie on the directrix :

The conic represents:

a parabola	an ellipse	a hyperbola	a rectangular hyperbola
$e = 1 ; D \neq 0$ $h^2 = ab$	$0 < e < 1 ; D \neq 0$ $h^2 < ab$	$D \neq 0 ; e > 1 ;$ $h^2 > ab$	$e > 1 ; D \neq 0$ $h^2 > ab ; a + b = 0$

4. PARABOLA :

A parabola is the locus of a point which moves in a plane, such that its distance from a fixed point (focus) is equal to its perpendicular distance from a fixed straight line (directrix).

Standard equation of a parabola is $y^2 = 4ax$. For this parabola :

- (i) Vertex is $(0, 0)$
- (ii) Focus is $(a, 0)$
- (iii) Axis is $y = 0$
- (iv) Directrix is $x + a = 0$

(a) Focal distance :

The distance of a point on the parabola from the focus is called the **focal distance of the point**.

(b) Focal chord :

A chord of the parabola, which passes through the focus is called a **focal chord**.

(c) Double ordinate :

A chord of the parabola perpendicular to the axis of the symmetry is called a **double ordinate**.

(d) Latus rectum :

A double ordinate passing through the focus or a focal chord perpendicular to the axis of parabola is called the **latus rectum**. For $y^2 = 4ax$.

- Length of the latus rectum = $4a$.
- Length of the semi latus rectum = $2a$.
- Ends of the latus rectum are $L(a, 2a)$ & $L'(a, -2a)$

Note that :

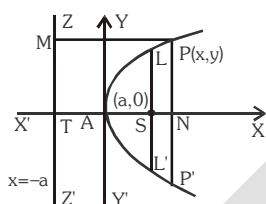
- Perpendicular distance from focus on directrix = half the latus rectum.
- Vertex is middle point of the focus & the point of intersection of directrix & axis.
- Two parabolas are said to be equal if they have the same latus rectum.

5. PARAMETRIC REPRESENTATION :

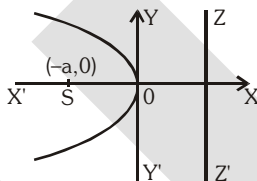
The simplest & the best form of representing the co-ordinates of a point on the parabola is $(at^2, 2at)$. The equation $x = at^2$ & $y = 2at$ together represents the parabola $y^2 = 4ax$, t being the parameter.

6. TYPE OF PARABOLA :

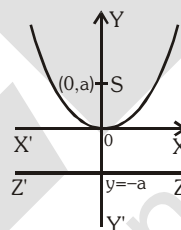
Four standard forms of the parabola are $y^2 = 4ax$; $y^2 = -4ax$; $x^2 = 4ay$; $x^2 = -4ay$



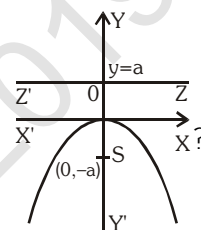
$$y^2 = 4ax$$



$$y^2 = -4ax$$



$$x^2 = 4ay$$



$$x^2 = -4ay$$

Parabola	Vertex	Focus	Axis	Directrix	Length of Latus rectum	Ends of Latus rectum	Parametric equation	Focal length
$y^2 = 4ax$	(0,0)	(a,0)	$y=0$	$x=-a$	$4a$	$(a, \pm 2a)$	$(at^2, 2at)$	$x+a$
$y^2 = -4ax$	(0,0)	(-a,0)	$y=0$	$x=a$	$4a$	$(-a, \pm 2a)$	$(-at^2, 2at)$	$x-a$
$x^2 = +4ay$	(0,0)	(0,a)	$x=0$	$y=-a$	$4a$	$(\pm 2a, a)$	$(2at, at^2)$	$y+a$
$x^2 = -4ay$	(0,0)	(0,-a)	$x=0$	$y=a$	$4a$	$(\pm 2a, -a)$	$(2at, -at^2)$	$y-a$
$(y-k)^2 = 4a(x-h)$	(h,k)	(h+a,k)	$y=k$	$x+a-h=0$	$4a$	$(h+a, k \pm 2a)$	$(h+at^2, k+2at)$	$x-h+a$
$(x-p)^2 = 4b(y-q)$	(p,q)	(p, b+q)	$x=p$	$y+b-q=0$	$4b$	$(p \pm 2a, q+a)$	$(p+2at, q+at^2)$	$y-q+b$

Illustration 1 : Find the vertex, axis, directrix, focus, latus rectum and the tangent at vertex for the parabola $9y^2 - 16x - 12y - 57 = 0$.

Solution : The given equation can be rewritten as $\left(y - \frac{2}{3}\right)^2 = \frac{16}{9}\left(x + \frac{61}{16}\right)$ which is of the form

$$Y^2 = 4AX. \text{ Hence the vertex is } \left(-\frac{61}{16}, \frac{2}{3}\right)$$

The axis is $y - \frac{2}{3} = 0 \Rightarrow y = \frac{2}{3}$

The directrix is $X + A = 0 \Rightarrow x + \frac{61}{16} + \frac{4}{9} = 0 \Rightarrow x = -\frac{613}{144}$

The focus is $X = A$ and $Y = 0 \Rightarrow x + \frac{61}{16} = \frac{4}{9}$ and $y - \frac{2}{3} = 0$

$\Rightarrow \text{focus} = \left(-\frac{485}{144}, \frac{2}{3}\right)$

Length of the latus rectum $= 4A = \frac{16}{9}$

The tangent at the vertex is $X = 0 \Rightarrow x = -\frac{61}{16}$.

Ans.

Illustration 2 : The length of latus rectum of a parabola, whose focus is (2, 3) and directrix is the line $x - 4y + 3 = 0$ is -

- (A) $\frac{7}{\sqrt{17}}$ (B) $\frac{14}{\sqrt{21}}$ (C) $\frac{7}{\sqrt{21}}$ (D) $\frac{14}{\sqrt{17}}$

Solution : The length of latus rectum $= 2 \times \text{perp. from focus to the directrix}$

$= 2 \times \left| \frac{2 - 4(3) + 3}{\sqrt{(1)^2 + (-4)^2}} \right| = \frac{14}{\sqrt{17}}$

Ans. (D)

Illustration 3 : Find the equation of the parabola whose focus is (-6, -6) and vertex (-2, 2).

Solution : Let S(-6, -6) be the focus and A(-2, 2) is vertex of the parabola. On SA take a point K(x₁, y₁) such that SA = AK. Draw KM perpendicular on SK. Then KM is the directrix

of the parabola. Since A bisects SK, $\left(\frac{-6 + x_1}{2}, \frac{-6 + y_1}{2}\right) = (-2, 2)$

$\Rightarrow -6 + x_1 = -4$ and $-6 + y_1 = 4$ or $(x_1, y_1) = (2, 10)$

Hence the equation of the directrix KM is

$y - 10 = m(x - 2)$ (i)

Also gradient of SK $= \frac{10 - (-6)}{2 - (-6)} = \frac{16}{8} = 2; \Rightarrow m = -\frac{1}{2}$

$y - 10 = -\frac{1}{2}(x - 2)$ (from (i))

$\Rightarrow x + 2y - 22 = 0$ is the directrix

Next, let PM be a perpendicular on the directrix KM from any point P(x, y) on the

parabola. From SP = PM, the equation of the parabola is $\sqrt{(x+6)^2 + (y+6)^2} = \frac{|x + 2y - 22|}{\sqrt{1^2 + 2^2}}$

or $5(x^2 + y^2 + 12x + 12y + 72) = (x + 2y - 22)^2$

or $4x^2 + y^2 - 4xy + 104x + 148y - 124 = 0$ or $(2x - y)^2 + 104x + 148y - 124 = 0$.

Ans.

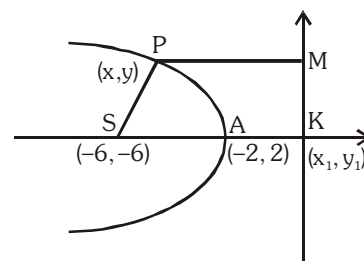


Illustration 4 : The extreme points of the latus rectum of a parabola are (7, 5) and (7, 3). Find the equation of the parabola.

Solution : Focus of the parabola is the mid-point of the latus rectum.

$\Rightarrow$ S is (7, 4). Also axis of the parabola is perpendicular to the latus rectum and passes through the focus. Its equation is

$$y - 4 = \frac{0}{5 - 3}(x - 7) \Rightarrow y = 4$$

Length of the latus rectum = (5 - 3) = 2

Hence the vertex of the parabola is at a distance $2/4 = 0.5$ from the focus. We have two parabolas, one concave rightwards and the other concave leftwards.

The vertex of the first parabola is (6.5, 4) and its equation is $(y - 4)^2 = 2(x - 6.5)$ and it meets the x-axis at (14.5, 0). The equation of the second parabola is $(y - 4)^2 = -2(x - 7.5)$. It meets the x-axis at (-0.5, 0).

Ans.

Do yourself - 1 :

- (i) Name the conic represented by the equation $\sqrt{ax} + \sqrt{by} = 1$, where $a, b \in \mathbb{R}$, $a, b > 0$.
- (ii) Find the vertex, axis, focus, directrix, latus rectum of the parabola $4y^2 + 12x - 20y + 67 = 0$.
- (iii) Find the equation of the parabola whose focus is (1, -1) and whose vertex is (2, 1). Also find its axis and latus rectum.
- (iv) Find the equation of the parabola whose latus rectum is 4 units, axis is the line $3x + 4y = 4$ and the tangent at the vertex is the line $4x - 3y + 7 = 0$.

7. POSITION OF A POINT RELATIVE TO A PARABOLA :

The point (x_1, y_1) lies **outside**, **on** or **inside** the parabola $y^2 = 4ax$ according as the expression $y_1^2 - 4ax_1$ is **positive**, **zero** or **negative**.

Illustration 5 : Find the value of α for which the point $(\alpha - 1, \alpha)$ lies inside the parabola $y^2 = 4x$.

Solution : $\because$ Point $(\alpha - 1, \alpha)$ lies inside the parabola $y^2 = 4x$

$$\therefore y_1^2 - 4ax_1 < 0$$

$$\Rightarrow \alpha^2 - 4(\alpha - 1) < 0$$

$$\Rightarrow \alpha^2 - 4\alpha + 4 < 0$$

$$(\alpha - 2)^2 < 0 \Rightarrow \alpha \in \phi$$

Ans.

8. CHORD JOINING TWO POINTS :

The equation of a chord of the parabola $y^2 = 4ax$ joining its two points $P(t_1)$ and $Q(t_2)$ is
 $y(t_1 + t_2) = 2x + 2at_1t_2$

Note :

(i) If PQ is focal chord then $t_1t_2 = -1$.

(ii) Extremities of focal chord can be taken as $(at^2, 2at)$ & $\left(\frac{a}{t^2}, \frac{-2a}{t}\right)$

Illustration 6 : Through the vertex O of a parabola $y^2 = 4x$ chords OP and OQ are drawn at right angles to one another. Show that for all position of P, PQ cuts the axis of the parabola at a fixed point.

Solution :

The given parabola is $y^2 = 4x$

..... (i)

Let $P \equiv (t_1^2, 2t_1)$, $Q \equiv (t_2^2, 2t_2)$

Slope of OP = $\frac{2t_1}{t_1^2} = \frac{2}{t_1}$ and slope of OQ = $\frac{2}{t_2}$

Since $OP \perp OQ$, $\frac{4}{t_1t_2} = -1$ or $t_1t_2 = -4$

..... (ii)

The equation of PQ is $y(t_1 + t_2) = 2(x + t_1t_2)$

$\Rightarrow y\left(t_1 - \frac{4}{t_1}\right) = 2(x - 4)$ [from (ii)]

$\Rightarrow 2(x - 4) - y\left(t_1 - \frac{4}{t_1}\right) = 0 \Rightarrow L_1 + \lambda L_2 = 0$

$\therefore$ variable line PQ passes through a fixed point which is point of intersection of $L_1 = 0$ & $L_2 = 0$
 i.e. (4, 0)

Ans.

9. LINE & A PARABOLA :

(a) The line $y = mx + c$ meets the parabola $y^2 = 4ax$ in two points real, coincident or imaginary according as $a > = < cm \Rightarrow$ condition of tangency is, $c = \frac{a}{m}$.

Note : Line $y = mx + c$ will be tangent to parabola $x^2 = 4ay$ if $c = -am^2$.

(b) Length of the chord intercepted by the parabola $y^2 = 4ax$ on the line $y = mx + c$ is :
 $\left(\frac{4}{m^2}\right)\sqrt{a(1+m^2)(a-mc)}$.

Note : Length of the focal chord making an angle α with the x - axis is $4a \operatorname{cosec}^2 \alpha$.

Illustration 7 : If the line $y = 3x + \lambda$ intersect the parabola $y^2 = 4x$ at two distinct points then set of values of λ is -

(A) (3, ∞)

(B) $(-\infty, 1/3)$

(C) (1/3, 3)

(D) none of these

Solution :Putting value of y from the line in the parabola -

$$(3x + \lambda)^2 = 4x$$

$$\Rightarrow 9x^2 + (6\lambda - 4)x + \lambda^2 = 0$$

 $\therefore$ line cuts the parabola at two distinct points

$$\therefore D > 0$$

$$\Rightarrow 4(3\lambda - 2)^2 - 4.9\lambda^2 > 0$$

$$\Rightarrow 9\lambda^2 - 12\lambda + 4 - 9\lambda^2 > 0$$

$$\Rightarrow \lambda < 1/3$$

$$\text{Hence, } \lambda \in (-\infty, 1/3)$$

Ans.(B)**Do yourself - 2 :**

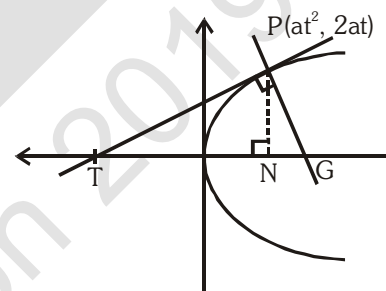
- (i) Find the value of 'a' for which the point $(a^2 - 1, a)$ lies inside the parabola $y^2 = 8x$.
- (ii) The focal distance of a point on the parabola $(x - 1)^2 = 16(y - 4)$ is 8. Find the co-ordinates.
- (iii) Show that the focal chord of parabola $y^2 = 4ax$ makes an angle α with x-axis is of length $4a \operatorname{cosec}^2 \alpha$.
- (iv) Find the condition that the straight line $ax + by + c = 0$ touches the parabola $y^2 = 4kx$.
- (v) Find the length of the chord of the parabola $y^2 = 8x$, whose equation is $x + y = 1$.

10. LENGTH OF SUBTANGENT & SUBNORMAL :

PT and PG are the tangent and normal respectively at the point P to the parabola $y^2 = 4ax$. Then

TN = length of subtangent = twice the abscissa of the point P
(Subtangent is always bisected by the vertex)

NG = length of subnormal which is constant for all points on the parabola & equal to its semilatus rectum $(2a)$.

**11. TANGENT TO THE PARABOLA $y^2 = 4ax$:****(a) Point form :**

Equation of tangent to the given parabola at its point (x_1, y_1) is

$$yy_1 = 2a(x + x_1)$$

(b) Slope form :

Equation of tangent to the given parabola whose slope is 'm', is

$$y = mx + \frac{a}{m}, (m \neq 0)$$

$$\text{Point of contact is } \left(\frac{a}{m^2}, \frac{2a}{m} \right)$$

(c) Parametric form :

Equation of tangent to the given parabola at its point $P(t)$, is $ty = x + at^2$

Note : Point of intersection of the tangents at the point t_1 & t_2 is $[at_1t_2, a(t_1 + t_2)]$.

Illustration 8 : A tangent to the parabola $y^2 = 8x$ makes an angle of 45° with the straight line $y = 3x + 5$. Find its equation and its point of contact.

Solution : Let the slope of the tangent be m

$$\therefore \tan 45^\circ = \left| \frac{3-m}{1+3m} \right| \Rightarrow 1+3m = \pm(3-m)$$

$$\therefore m = -2 \text{ or } \frac{1}{2}$$

As we know that equation of tangent of slope m to the parabola $y^2 = 4ax$ is $y = mx + \frac{a}{m}$

and point of contact is $\left(\frac{a}{m^2}, \frac{2a}{m} \right)$

for $m = -2$, equation of tangent is $y = -2x - 1$ and point of contact is $\left(\frac{1}{2}, -2 \right)$

for $m = \frac{1}{2}$, equation of tangent is $y = \frac{1}{2}x + 4$ and point of contact is $(8, 8)$

Ans.

Illustration 9 : Find the equation of the tangents to the parabola $y^2 = 9x$ which go through the point $(4, 10)$.
Solution : Equation of tangent to parabola $y^2 = 9x$ is

$$y = mx + \frac{9}{4m}$$

Since it passes through $(4, 10)$

$$\therefore 10 = 4m + \frac{9}{4m} \Rightarrow 16m^2 - 40m + 9 = 0$$

$$m = \frac{1}{4}, \frac{9}{4}$$

$$\therefore \text{equation of tangent's are } y = \frac{x}{4} + 9 \text{ \& } y = \frac{9}{4}x + 1$$

Ans.

Illustration 10: Find the locus of the point P from which tangents are drawn to the parabola $y^2 = 4ax$ having slopes m_1 and m_2 such that -

$$(i) m_1^2 + m_2^2 = \lambda \text{ (constant)} \quad (ii) \theta_1 - \theta_2 = \theta_0 \text{ (constant)}$$

where θ_1 and θ_2 are the inclinations of the tangents from positive x -axis.

Solution : Equation of tangent to $y^2 = 4ax$ is $y = mx + \frac{a}{m}$

Let it passes through $P(h, k)$

$$\therefore m^2h - mk + a = 0$$

$$(i) m_1^2 + m_2^2 = \lambda$$

$$(m_1 + m_2)^2 - 2m_1m_2 = \lambda$$

$$\frac{k^2}{h^2} - 2 \cdot \frac{a}{h} = \lambda$$

$$\therefore \text{locus of } P(h, k) \text{ is } y^2 - 2ax = \lambda x^2$$

$$(ii) \quad \theta_1 - \theta_2 = \theta_0$$

$$\tan(\theta_1 - \theta_2) = \tan \theta_0$$

$$\frac{m_1 - m_2}{1 + m_1 m_2} = \tan \theta_0$$

$$(m_1 + m_2)^2 - 4m_1 m_2 = \tan^2 \theta_0 (1 + m_1 m_2)^2$$

$$\frac{k^2}{h^2} - \frac{4a}{h} = \tan^2 \theta_0 \left(1 + \frac{a}{h}\right)^2$$

$$k^2 - 4ah = (h + a)^2 \tan^2 \theta_0$$

$$\therefore \text{locus of } P(h, k) \text{ is } y^2 - 4ax = (x + a)^2 \tan^2 \theta_0$$

Ans.

Do yourself - 3 :

- (i) Find the equation of the tangent to the parabola $y^2 = 12x$, which passes through the point (2, 5). Find also the co-ordinates of their points of contact.
- (ii) Find the equation of the tangents to the parabola $y^2 = 16x$, which are parallel and perpendicular respectively to the line $2x - y + 5 = 0$. Find also the co-ordinates of their points of contact.
- (iii) Prove that the locus of the point of intersection of tangents to the parabola $y^2 = 4ax$ which meet at an angle θ is $(x + a)^2 \tan^2 \theta = y^2 - 4ax$.

12. NORMAL TO THE PARABOLA $y^2 = 4ax$:**(a) Point form :**Equation of normal to the given parabola at its point (x_1, y_1) is

$$y - y_1 = -\frac{y_1}{2a}(x - x_1)$$

(b) Slope form :

Equation of normal to the given parabola whose slope is 'm', is

$$y = mx - 2am - am^3$$

foot of the normal is $(am^2, -2am)$ **(c) Parametric form :**Equation of normal to the given parabola at its point $P(t)$, is

$$y + tx = 2at + at^3$$

Note :(i) Point of intersection of normals at t_1 & t_2 is $(a(t_1^2 + t_2^2 + t_1 t_2 + 2), -at_1 t_2(t_1 + t_2))$.(ii) If the normal to the parabola $y^2 = 4ax$ at the point t_1 , meets the parabola again at the point

$$t_2, \text{ then } t_2 = -\left(t_1 + \frac{2}{t_1}\right).$$

(iii) If the normals to the parabola $y^2 = 4ax$ at the points t_1 & t_2 intersect again on the parabola at the point ' t_3 ' then $t_1 t_2 = 2$; $t_3 = -(t_1 + t_2)$ and the line joining t_1 & t_2 passes through a fixed point $(-2a, 0)$.

- (iv) If normal drawn to a parabola passes through a point $P(h, k)$ then $k = mh - 2am - am^3$,
i.e. $am^3 + m(2a - h) + k = 0$.

This gives $m_1 + m_2 + m_3 = 0$; $m_1m_2 + m_2m_3 + m_3m_1 = \frac{2a - h}{a}$; $m_1m_2m_3 = \frac{-k}{a}$

where m_1, m_2 , & m_3 are the slopes of the three concurrent normals :

- Algebraic sum of slopes of the three concurrent normals is zero.
- Algebraic sum of ordinates of the three co-normal points on the parabola is zero.
- Centroid of the Δ formed by three co-normal points lies on the axis of parabola (x-axis).

Illustration 11: Prove that the normal chord to a parabola $y^2 = 4ax$ at the point whose ordinate is equal to abscissa subtends a right angle at the focus.

Solution :

Let the normal at $P(at_1^2, 2at_1)$ meet the curve at $Q(at_2^2, 2at_2)$

$\therefore$ PQ is a normal chord.

and $t_2 = -t_1 - \frac{2}{t_1}$ (i)

By given condition $2at_1 = at_1^2$

$\therefore t_1 = 2$ from equation (i), $t_2 = -3$

then $P(4a, 4a)$ and $Q(9a, -6a)$

but focus $S(a, 0)$

$\therefore$ Slope of SP = $\frac{4a - 0}{4a - a} = \frac{4a}{3a} = \frac{4}{3}$

and Slope of SQ = $\frac{-6a - 0}{9a - a} = \frac{-6a}{8a} = -\frac{3}{4}$

$\therefore$ Slope of SP $\times$ Slope of SQ = $\frac{4}{3} \times -\frac{3}{4} = -1$

$\therefore \angle PSQ = \pi/2$

i.e. PQ subtends a right angle at the focus S.

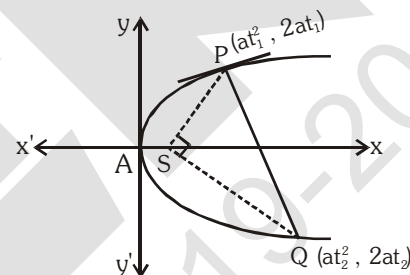


Illustration 12: If two normals drawn from any point to the parabola $y^2 = 4ax$ make angle α and β with the axis such that $\tan \alpha \cdot \tan \beta = 2$, then find the locus of this point.

Solution :

Let the point is (h, k) . The equation of any normal to the parabola $y^2 = 4ax$ is

$y = mx - 2am - am^3$

passes through (h, k)

$k = mh - 2am - am^3$

$am^3 + m(2a - h) + k = 0$ (i)

m_1, m_2, m_3 are roots of the equation, then $m_1 \cdot m_2 \cdot m_3 = -\frac{k}{a}$

but $m_1m_2 = 2$, $m_3 = -\frac{k}{2a}$

m_3 is root of (i) $\therefore a\left(-\frac{k}{2a}\right)^3 - \frac{k}{2a}(2a - h) + k = 0 \Rightarrow k^2 = 4ah$

Thus locus is $y^2 = 4ax$.

Ans.

Illustration 13 : Three normals are drawn from the point (14, 7) to the curve $y^2 - 16x - 8y = 0$. Find the coordinates of the feet of the normals.

Solution : The given parabola is $y^2 - 16x - 8y = 0$ (i)

Let the co-ordinates of the feet of the normal from (14, 7) be $P(\alpha, \beta)$. Now the equation of the tangent at $P(\alpha, \beta)$ to parabola (i) is

$$y\beta - 8(x + \alpha) - 4(y + \beta) = 0$$

$$\text{or } (\beta - 4)y = 8x + 8\alpha + 4\beta \quad \text{..... (ii)}$$

$$\text{Its slope} = \frac{8}{\beta - 4}$$

$$\text{Equation of the normal to parabola (i) at } (\alpha, \beta) \text{ is } y - \beta = \frac{4 - \beta}{8} (x - \alpha)$$

It passes through (14, 7)

$$\Rightarrow 7 - \beta = \frac{4 - \beta}{8} (14 - \alpha) \Rightarrow \alpha = \frac{6\beta}{\beta - 4} \quad \text{..... (iii)}$$

$$\text{Also } (\alpha, \beta) \text{ lies on parabola (i) i.e. } \beta^2 - 16\alpha - 8\beta = 0 \quad \text{..... (iv)}$$

$$\text{Putting the value of } \alpha \text{ from (iii) in (iv), we get } \beta^2 - \frac{96\beta}{\beta - 4} - 8\beta = 0$$

$$\Rightarrow \beta^2(\beta - 4) - 96\beta - 8\beta(\beta - 4) = 0 \Rightarrow \beta(\beta^2 - 4\beta - 96 - 8\beta + 32) = 0$$

$$\Rightarrow \beta(\beta^2 - 12\beta - 64) = 0 \Rightarrow \beta(\beta - 16)(\beta + 4) = 0$$

$$\Rightarrow \beta = 0, 16, -4$$

from (iii), $\alpha = 0$ when $\beta = 0$; $\alpha = 8$, when $\beta = 16$; $\alpha = 3$ when $\beta = -4$

Hence the feet of the normals are (0, 0), (8, 16) and (3, -4)

Ans.

Do yourself - 4 :

(i) If three distinct and real normals can be drawn to $y^2 = 8x$ from the point (a, 0), then -

(A) $a > 2$

(B) $a \in (2, 4)$

(C) $a > 4$

(D) none of these

(ii) Find the number of distinct normal that can be drawn from (-2, 1) to the parabola

$$y^2 - 4x - 2y - 3 = 0.$$

(iii) If $2x + y + k = 0$ is a normal to the parabola $y^2 = -16x$, then find the value of k.

(iv) Three normals are drawn from the point (7, 14) to the parabola $x^2 - 8x - 16y = 0$. Find the co-ordinates of the feet of the normals.

13. AN IMPORTANT CONCEPT :

If a family of straight lines can be represented by an equation $\lambda^2 P + \lambda Q + R = 0$ where λ is a parameter and P, Q, R are linear functions of x and y then the family of lines will be tangent to the curve $Q^2 = 4PR$.

Illustration 14 : If the equation $m^2(x + 1) + m(y - 2) + 1 = 0$ represents a family of lines, where 'm' is parameter then find the equation of the curve to which these lines will always be tangents.

Solution : $m^2(x + 1) + m(y - 2) + 1 = 0$

The equation of the curve to which above lines will always be tangents can be obtained by equating its discriminant to zero.

$$\begin{aligned}\therefore (y - 2)^2 - 4(x + 1) &= 0 \\ y^2 - 4y + 4 - 4x - 4 &= 0 \\ y^2 &= 4(x + y)\end{aligned}$$

Ans.

14. PAIR OF TANGENTS :

The equation of the pair of tangents which can be drawn from any point $P(x_1, y_1)$ outside the parabola to the parabola $y^2 = 4ax$ is given by : $SS_1 = T^2$ where :

$$S \equiv y^2 - 4ax ; \quad S_1 \equiv y_1^2 - 4ax_1 ; \quad T \equiv yy_1 - 2a(x + x_1).$$

15. DIRECTOR CIRCLE :

Locus of the point of intersection of the perpendicular tangents to the parabola $y^2 = 4ax$ is called the **director circle**. It's equation is $x + a = 0$ which is parabola's own directrix.

Illustration 15 : The angle between the tangents drawn from a point $(-a, 2a)$ to $y^2 = 4ax$ is -

- (A) $\pi/4$ (B) $\pi/2$ (C) $\pi/3$ (D) $\pi/6$

Solution : The given point $(-a, 2a)$ lies on the directrix $x = -a$ of the parabola $y^2 = 4ax$. Thus, the tangents are at right angle. **Ans.(B)**

Illustration 16 : The circle drawn with variable chord $x + ay - 5 = 0$ (a being a parameter) of the parabola $y^2 = 20x$ as diameter will always touch the line -

- (A) $x + 5 = 0$ (B) $y + 5 = 0$ (C) $x + y + 5 = 0$ (D) $x - y + 5 = 0$

Solution : Clearly $x + ay - 5 = 0$ will always pass through the focus of $y^2 = 20x$ i.e. $(5, 0)$. Thus the drawn circle will always touch the directrix of the parabola i.e. the line $x + 5 = 0$. **Ans.(A)**

Do yourself - 5 :

- (i) If the equation $\lambda^2x + \lambda y - \lambda^2 + 2\lambda + 7 = 0$ represents a family of lines, where ' λ ' is parameter, then find the equation of the curve to which these lines will always be tangents.
(ii) Find the angle between the tangents drawn from the origin to the parabola, $y^2 = 4a(x - a)$.

16. CHORD OF CONTACT :

Equation of the chord of contact of tangents drawn from a point $P(x_1, y_1)$ is $yy_1 = 2a(x + x_1)$

Note : The area of the triangle formed by the tangents from the point (x_1, y_1) & the chord of contact is

$$\frac{(y_1^2 - 4ax_1)^{3/2}}{2a} \text{ i.e. } \frac{(S_1)^{3/2}}{2a}, \text{ also note that the chord of contact exists only if the point P is not inside.}$$

Illustration 17 : If the line $x - y - 1 = 0$ intersect the parabola $y^2 = 8x$ at P & Q, then find the point of intersection of tangents at P & Q.

Solution : Let (h, k) be point of intersection of tangents then chord of contact is

$$yk = 4(x + h)$$

$$4x - yk + 4h = 0 \quad \dots\dots (i)$$

But given line is

$$x - y - 1 = 0 \quad \dots\dots (ii)$$

Comparing (i) and (ii)

$$\therefore \frac{4}{1} = \frac{-k}{-1} = \frac{4h}{-1} \quad \Rightarrow \quad h = -1, k = 4$$

$$\therefore \text{point} = (-1, 4)$$

Ans.

Illustration 18 : Find the locus of point whose chord of contact w.r.t. to the parabola $y^2 = 4bx$ is the tangent of the parabola $y^2 = 4ax$.

Solution : Equation of tangent to $y^2 = 4ax$ is $y = mx + \frac{a}{m}$ (i)

Let it is chord of contact for parabola $y^2 = 4bx$ w.r.t. the point P(h, k)

$\therefore$ Equation of chord of contact is $yk = 2b(x + h)$

$$y = \frac{2b}{k}x + \frac{2bh}{k} \quad \dots\dots (ii)$$

From (i) & (ii)

$$m = \frac{2b}{k}, \frac{a}{m} = \frac{2bh}{k} \Rightarrow a = \frac{4b^2h}{k^2}$$

$$\text{locus of P is } y^2 = \frac{4b^2}{a}x.$$

Ans.

17. CHORD WITH A GIVEN MIDDLE POINT :

Equation of the chord of the parabola $y^2 = 4ax$ whose middle point is (x_1, y_1) is $y - y_1 = \frac{2a}{y_1}(x - x_1)$.

This reduced to $T = S_1$, where $T \equiv yy_1 - 2a(x + x_1)$ & $S_1 \equiv y_1^2 - 4ax_1$.

Illustration 19 : Find the locus of middle point of the chord of the parabola $y^2 = 4ax$ which pass through a given (p, q).

Solution : Let P(h, k) be the mid point of chord of the parabola $y^2 = 4ax$, so equation of chord is $yk - 2a(x + h) = k^2 - 4ah$.

Since it passes through (p, q)

$$\therefore qk - 2a(p + h) = k^2 - 4ah$$

$$\therefore \text{Required locus is } y^2 - 2ax - qy + 2ap = 0.$$

Illustration 20 : Find the locus of the middle point of a chord of a parabola $y^2 = 4ax$ which subtends a right angle at the vertex.

Solution : The equation of the chord of the parabola whose middle point is (α, β) is

$$y\beta - 2a(x + \alpha) = \beta^2 - 4a\alpha$$

$$\Rightarrow y\beta - 2ax = \beta^2 - 2a\alpha$$

$$\text{or } \frac{y\beta - 2ax}{\beta^2 - 2a\alpha} = 1 \quad \dots\dots\dots (i)$$

Now, the equation of the pair of the lines OP and OQ joining the origin O i.e. the vertex to the points of intersection P and Q of the chord with the parabola $y^2 = 4ax$ is obtained by making the equation homogeneous by means of (i). Thus the equation of lines OP and

$$\text{OQ is } y^2 = \frac{4ax(y\beta - 2ax)}{\beta^2 - 2a\alpha}$$

$$\Rightarrow y^2(\beta^2 - 2a\alpha) - 4a\beta xy + 8a^2x^2 = 0$$

If the lines OP and OQ are at right angles, then the coefficient of x^2 + the coefficient of y^2 = 0

$$\text{Therefore, } \beta^2 - 2a\alpha + 8a^2 = 0 \Rightarrow \beta^2 = 2a(\alpha - 4a)$$

$$\text{Hence the locus of } (\alpha, \beta) \text{ is } y^2 = 2a(x - 4a)$$

Do yourself - 6 :

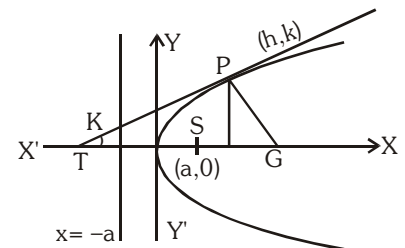
- (i) Find the equation of the chord of contacts of tangents drawn from a point (2, 1) to the parabola $x^2 = 2y$.
- (ii) Find the co-ordinates of the middle point of the chord of the parabola $y^2 = 16x$, the equation of which is $2x - 3y + 8 = 0$
- (iii) Find the locus of the mid-point of the chords of the parabola $y^2 = 4ax$ such that tangent at the extremities of the chords are perpendicular.

18. DIAMETER :

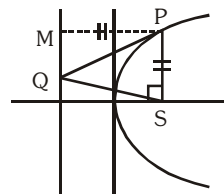
The locus of the middle points of a system of parallel chords of a Parabola is called a DIAMETER. Equation to the diameter of a parabola is $y = 2a/m$, where m = slope of parallel chords.

19. IMPORTANT HIGHLIGHTS :

- (a) If the tangent & normal at any point 'P' of the parabola intersect the axis at T & G then $ST = SG = SP$ where 'S' is the focus. In other words the tangent and the normal at a point P on the parabola are the bisectors of the angle between the focal radius SP & the perpendicular from P on the directrix. From this we conclude that all rays emanating from S will become parallel to the axis of the parabola after reflection.



- (b) The portion of a tangent to a parabola cut off between the directrix & the curve subtends a right angle at the **focus**.



- (c) The tangents at the extremities of a focal chord intersect at right angles on the **directrix**, and a circle on any focal chord as diameter touches the directrix. Also a circle on any focal radii of a point P ($at^2, 2at$) as diameter touches the tangent at the vertex and intercepts a chord of length $a\sqrt{1+t^2}$ on a normal at the point P.

- (d) Any tangent to a parabola & the perpendicular on it from the focus meet on the tangent at the vertex.

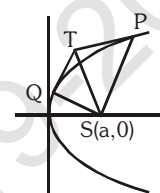
- (e) Semi latus rectum of the parabola $y^2 = 4ax$, is the harmonic mean between segments of any focal chord of the parabola is ; $2a = \frac{2bc}{b+c}$ i.e. $\frac{1}{b} + \frac{1}{c} = \frac{1}{a}$.

- (f) If the tangents at P and Q meet in T, then :

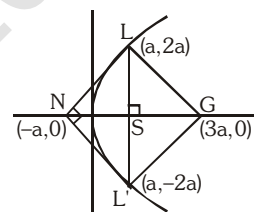
(i) TP and TQ subtend equal angles at the focus S.

(ii) $ST^2 = SP \cdot SQ$ &

(iii) The triangles SPT and STQ are similar.



- (g) Tangents and Normals at the extremities of the latus rectum of a parabola $y^2 = 4ax$ constitute a square, their points of intersection being $(-a, 0)$ & $(3a, 0)$.



Note :

- (i) The two tangents at the extremities of focal chord meet on the foot of the directrix.
- (ii) Figure LNL'G is square of side $2\sqrt{2}a$
- (h) The circle circumscribing the triangle formed by any three tangents to a parabola passes through the focus.

Do yourself - 7 :

- (i) The parabola $y^2 = 4x$ and $x^2 = 4y$ divide the square region bounded by the line $x = 4$, $y = 4$ and the co-ordinates axes. If S_1, S_2, S_3 are respectively the areas of these parts numbered from top to bottom; then find $S_1 : S_2 : S_3$.
- (ii) Let P be the point (1, 0) and Q a point on the parabola $y^2 = 8x$, then find the locus of the mid point of PQ.

Miscellaneous Illustrations :

Illustration 21 : The common tangent of the parabola $y^2 = 8ax$ and the circle $x^2 + y^2 = 2a^2$ is -
 (A) $y = x + a$ (B) $x + y + a = 0$ (C) $x + y + 2a = 0$ (D) $y = x + 2a$

Solution : Any tangent to parabola is $y = mx + \frac{2a}{m}$

$$\text{Solving with the circle } x^2 + (mx + \frac{2a}{m})^2 = 2a^2 \Rightarrow x^2(1 + m^2) + 4ax + \frac{4a^2}{m^2} - 2a^2 = 0$$

$$B^2 - 4AC = 0 \text{ gives } m = \pm 1$$

$$\text{Tangent } y = \pm x \pm 2a$$

Ans. (C,D)

Illustration 22 : If the tangent to the parabola $y^2 = 4ax$ meets the axis in T and tangent at the vertex A in Y and the rectangle TAYG is completed, show that the locus of G is $y^2 + ax = 0$.

Solution : Let $P(at^2, 2at)$ be any point on the parabola $y^2 = 4ax$.

$$\text{Then tangent at } P(at^2, 2at) \text{ is } ty = x + at^2$$

Since tangent meet the axis of parabola in T and tangent at the vertex in Y.

$\therefore$ Co-ordinates of T and Y are $(-at^2, 0)$ and $(0, at)$ respectively.

Let co-ordinates of G be (x_1, y_1) .

Since TAYG is rectangle.

$\therefore$ Mid-points of diagonals TY and GA is same

$$\Rightarrow \frac{x_1 + 0}{2} = \frac{-at^2 + 0}{2} \Rightarrow x_1 = -at^2 \quad \dots\dots\dots (i)$$

$$\text{and } \frac{y_1 + 0}{2} = \frac{0 + at}{2} \Rightarrow y_1 = at \quad \dots\dots\dots (ii)$$

$$\text{Eliminating } t \text{ from (i) and (ii) then we get } x_1 = -a \left(\frac{y_1}{a} \right)^2$$

$$\text{or } y_1^2 = -ax_1 \quad \text{or } y_1^2 + ax_1 = 0$$

$\therefore$ The locus of $G(x_1, y_1)$ is $y^2 + ax = 0$

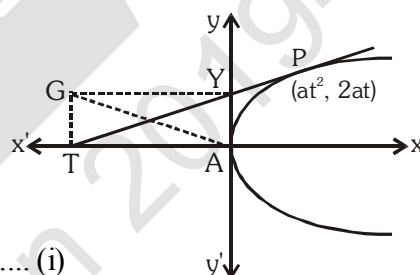


Illustration 23 : If $P(-3, 2)$ is one end of the focal chord PQ of the parabola $y^2 + 4x + 4y = 0$, then the slope of the normal at Q is -

- (A) $-1/2$ (B) 2 (C) $1/2$ (D) -2

Solution : The equation of the tangent at $(-3, 2)$ to the parabola $y^2 + 4x + 4y = 0$ is

$$2y + 2(x - 3) + 2(y + 2) = 0$$

$$\text{or } 2x + 4y - 2 = 0 \Rightarrow x + 2y - 1 = 0$$

Since the tangent at one end of the focal chord is parallel to the normal at the other end, the

slope of the normal at the other end of the focal chord is $-\frac{1}{2}$.

Ans. (A)

Illustration 24 : Prove that the two parabolas $y^2 = 4ax$ and $y^2 = 4c(x - b)$ cannot have common normal, other than the axis unless $b/(a - c) > 2$.

Solution : Given parabolas $y^2 = 4ax$ and $y^2 = 4c(x - b)$ have common normals. Then equation of normals in terms of slopes are $y = mx - 2am - am^3$ and $y = m(x - b) - 2cm - cm^3$ respectively then normals must be identical, compare the co-efficients

$$1 = \frac{2am + am^3}{mb + 2cm + cm^3}$$

$$\Rightarrow m[(c - a)m^2 + (b + 2c - 2a)] = 0, m \neq 0 \quad (\because \text{other than axis})$$

$$\text{and } m^2 = \frac{2a - 2c - b}{c - a}, m = \pm \sqrt{\frac{2(a - c) - b}{c - a}}$$

$$\text{or } m = \pm \sqrt{\left(-2 - \frac{b}{c - a}\right)}$$

$$\therefore -2 - \frac{b}{c - a} > 0$$

$$\text{or } -2 + \frac{b}{a - c} > 0 \Rightarrow \frac{b}{a - c} > 2$$

Illustration 25 : If r_1, r_2 be the length of the perpendicular chords of the parabola $y^2 = 4ax$ drawn through the vertex, then show that $(r_1 r_2)^{4/3} = 16a^2 (r_1^{2/3} + r_2^{2/3})$.

Solution : Since chord are perpendicular, therefore if one makes an angle θ then the other will make an angle $(90^\circ - \theta)$ with x-axis

Let $AP = r_1$ and $AQ = r_2$

If $\angle PAX = \theta$

then $\angle QAX = 90^\circ - \theta$

$\therefore$ Co-ordinates of P and Q are $(r_1 \cos \theta, r_1 \sin \theta)$

and $(r_2 \sin \theta, -r_2 \cos \theta)$ respectively.

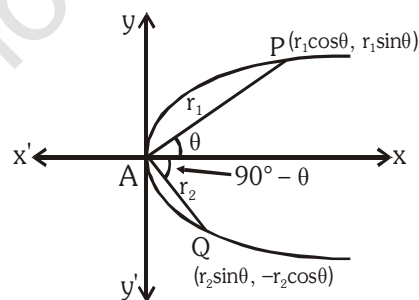
Since P and Q lies on $y^2 = 4ax$

$$\therefore r_1^2 \sin^2 \theta = 4ar_1 \cos \theta \text{ and } r_2^2 \cos^2 \theta = 4ar_2 \sin \theta$$

$$\Rightarrow r_1 = \frac{4a \cos \theta}{\sin^2 \theta} \text{ and } r_2 = \frac{4a \sin \theta}{\cos^2 \theta}$$

$$\therefore (r_1 r_2)^{4/3} = \left(\frac{4a \cos \theta}{\sin^2 \theta} \cdot \frac{4a \sin \theta}{\cos^2 \theta} \right)^{4/3} = \left(\frac{16a^2}{\sin \theta \cos \theta} \right)^{4/3} \quad \dots\dots (i)$$

$$\text{and } 16a^2 \cdot (r_1^{2/3} + r_2^{2/3}) = 16a^2 \left\{ \left(\frac{4a \cos \theta}{\sin^2 \theta} \right)^{2/3} + \left(\frac{4a \sin \theta}{\cos^2 \theta} \right)^{2/3} \right\}$$



$$= 16a^2 \cdot (4a)^{2/3} \left\{ \frac{(\cos \theta)^{2/3}}{(\sin \theta)^{4/3}} + \frac{(\sin \theta)^{2/3}}{(\cos \theta)^{4/3}} \right\} = 16a^2 \cdot (4a)^{2/3} \left\{ \frac{\cos^2 \theta + \sin^2 \theta}{(\sin \theta)^{4/3} (\cos \theta)^{4/3}} \right\}$$

$$= \frac{16a^2 \cdot (4a)^{2/3}}{(\sin \theta \cos \theta)^{4/3}} = \left(\frac{16a^2}{\cos \theta \cos \theta} \right)^{4/3} = (r_1 r_2)^{4/3} \quad \{\text{from (i)}\}$$

Illustration 26 : The area of the triangle formed by three points on a parabola is twice the area of the triangle formed by the tangents at these points.

Solution : Let the three points on the parabola be

$$(at_1^2, 2at_1), (at_2^2, 2at_2) \text{ and } (at_3^2, 2at_3)$$

The area of the triangle formed by these points

$$\Delta_1 = \frac{1}{2} [at_1^2 (2at_2 - 2at_3) + at_2^2 (2at_3 - 2at_1) + at_3^2 (2at_1 - 2at_2)]$$

$$= -a^2(t_2 - t_3)(t_3 - t_1)(t_1 - t_2).$$

The points of intersection of the tangents at these points are

$$(at_2 t_3, a(t_2 + t_3)), (at_3 t_1, a(t_3 + t_1)) \text{ and } (at_1 t_2, a(t_1 + t_2))$$

The area of the triangle formed by these three points

$$\Delta_2 = \frac{1}{2} \{at_2 t_3 (at_3 - at_2) + at_3 t_1 (at_1 - at_3) + at_1 t_2 (at_2 - at_1)\}$$

$$= \frac{1}{2} a^2 (t_2 - t_3)(t_3 - t_1)(t_1 - t_2)$$

$$\text{Hence } \Delta_1 = 2\Delta_2$$

Illustration 27: Prove that the orthocentre of any triangle formed by three tangents to a parabola lies on the directrix.

Solution : Let the equations of the three tangents be

$$t_1 y = x + at_1^2 \quad \dots\dots\dots \text{(i)}$$

$$t_2 y = x + at_2^2 \quad \dots\dots\dots \text{(ii)}$$

$$\text{and } t_3 y = x + at_3^2 \quad \dots\dots\dots \text{(iii)}$$

The point of intersection of (ii) and (iii) is found, by solving them, to be $(at_2 t_3, a(t_2 + t_3))$

The equation of the straight line through this point & perpendicular to (i) is

$$y - a(t_2 + t_3) = -t_1(x - at_2 t_3)$$

$$\text{i.e. } y + t_1 x = a(t_2 + t_3 + t_1 t_2 t_3) \quad \dots\dots\dots \text{(iv)}$$

Similarly, the equation of the straight line through the point of intersection of (iii) and (i) & perpendicular to (ii) is

$$y + t_2x = a(t_3 + t_1 + t_1t_2t_3) \quad \dots\dots\dots(v)$$

and the equation of the straight line through the point of intersection of (i) and (ii) & perpendicular to (iii) is

$$y + t_1x = a(t_1 + t_2 + t_1t_2t_3) \quad \dots\dots\dots(vi)$$

The point which is common to the straight lines (iv), (v) and (vi)

i.e. the orthocentre of the triangle, is easily seen to be the point whose coordinates are

$$x = -a, y = a(t_1 + t_2 + t_3 + t_1t_2t_3)$$

and this point lies on the directrix.

ANSWERS FOR DO YOURSELF

- 1: (i) Parabola (ii) Vertex : $\left(-\frac{7}{2}, \frac{5}{2}\right)$, Axis : $y = \frac{5}{2}$, Focus : $\left(-\frac{17}{4}, \frac{5}{2}\right)$, Directrix : $x = -\frac{11}{4}$; LR = 3
 (iii) $4x^2 + y^2 - 4xy + 8x + 46y - 71 = 0$; Axis : $2x - y = 3$; LR = $4\sqrt{5}$ unit
 (iv) $(3x + 4y - 4)^2 = 20(4x - 3y + 7)$
- 2: (i) $\left(-\infty, -\sqrt{\frac{8}{7}}\right) \cup \left(\sqrt{\frac{8}{7}}, \infty\right)$ (ii) $(-7, 8), (9, 8)$ (iv) $kb^2 = ac$ (v) $8\sqrt{3}$
- 3: (i) $x - y + 3 = 0, (3, 6); 3x - 2y + 4 = 0, \left(\frac{4}{3}, 4\right)$
 (ii) $2x - y + 2 = 0, (1, 4); x + 2y + 16 = 0, (16, -16)$
- 4: (i) C (ii) 1 (iii) 48 (iv) $(0, 0), (-4, 3)$ and $(16, 8)$
- 5: (i) $(y + 2)^2 = 28(x - 1)$ (ii) $\pi/2$
- 6: (i) $2x = y + 1$ (ii) $(14, 12)$ (iii) $y^2 = 2a(x - a)$
- 7: (i) $1 : 1 : 1$ (ii) $y^2 - 4x + 2 = 0$

EXERCISE (O-1)

[STRAIGHT OBJECTIVE TYPE]

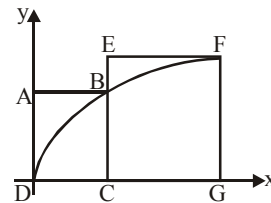
- The equation of the directrix of the parabola, $y^2 + 4y + 4x + 2 = 0$ is -
(A) $x = -1$ (B) $x = 1$ (C) $x = -3/2$ (D) $x = 3/2$
- Length of the latus rectum of the parabola $25[(x - 2)^2 + (y - 3)^2] = (3x - 4y + 7)^2$ is -
(A) 4 (B) 2 (C) $1/5$ (D) $2/5$
- If the line $x - 1 = 0$ is the directrix of the parabola $y^2 - kx + 8 = 0$ then one of the values of 'k' is -
(A) $1/8$ (B) 8 (C) 4 (D) $1/4$
- The length of the intercept on y - axis cut off by the parabola, $y^2 - 5y = 3x - 6$ is
(A) 1 (B) 2 (C) 3 (D) 5
- Maximum number of common chords of a parabola and a circle can be equal to
(A) 2 (B) 4 (C) 6 (D) 8
- A variable circle is drawn to touch the line $3x - 4y = 10$ and also the circle $x^2 + y^2 = 1$ externally then the locus of its centre is -
(A) straight line (B) circle
(C) pair of real, distinct straight lines (D) parabola
- The locus of the point of trisection of all the double ordinates of the parabola $y^2 = \ell x$ is a parabola whose latus rectum is -
(A) $\frac{\ell}{9}$ (B) $\frac{2\ell}{9}$ (C) $\frac{4\ell}{9}$ (D) $\frac{\ell}{36}$
- The straight line $y = m(x - a)$ will meet the parabola $y^2 = 4ax$ in two distinct real points if
(A) $m \in \mathbb{R}$ (B) $m \in [-1, 1]$
(C) $m \in (-\infty, 1] \cup [1, \infty)$ (D) $m \in \mathbb{R} - \{0\}$
- The vertex A of the parabola $y^2 = 4ax$ is joined to any point P on it and PQ is drawn at right angles to AP to meet the axis in Q. Projection of PQ on the axis is equal to
(A) twice the length of latus rectum (B) the latus length of rectum
(C) half the length of latus rectum (D) one fourth of the length of latus rectum
- If on a given base, a triangle be described such that the sum of the tangents of the base angles is a constant, then the locus of the vertex is :
(A) a circle (B) a parabola (C) an ellipse (D) a hyperbola
- The equation of the circle drawn with the focus of the parabola $(x - 1)^2 - 8y = 0$ as its centre and touching the parabola at its vertex is :
(A) $x^2 + y^2 - 4y = 0$ (B) $x^2 + y^2 - 4y + 1 = 0$
(C) $x^2 + y^2 - 2x - 4y = 0$ (D) $x^2 + y^2 - 2x - 4y + 1 = 0$
- Which one of the following equations represented parametrically, represents equation to a parabolic profile?
(A) $x = 3 \cos t$; $y = 4 \sin t$ (B) $x^2 - 2 = -2 \cos t$; $y = 4 \cos^2 \frac{t}{2}$
(C) $\sqrt{x} = \tan t$; $\sqrt{y} = \sec t$ (D) $x = \sqrt{1 - \sin t}$; $y = \sin \frac{t}{2} + \cos \frac{t}{2}$

13. Angle between the parabolas $y^2 = 4(x - 1)$ and $x^2 + 4(y - 3) = 0$ at the common end of their latus rectum, is -
 (A) $\tan^{-1}(1)$ (B) $\tan^{-1}1 + \cot^{-1}2 + \cot^{-1}3$
 (C) $\tan^{-1}(\sqrt{3})$ (D) $\tan^{-1}(2) + \tan^{-1}(3)$
14. If a focal chord of $y^2 = 4x$ makes an angle $\alpha, \alpha \in \left(0, \frac{\pi}{4}\right]$ with the positive direction of x-axis, then minimum length of this focal chord is -
 (A) $2\sqrt{2}$ (B) $4\sqrt{2}$ (C) 8 (D) 16
15. A parabola $y = ax^2 + bx + c$ crosses the x-axis at $(\alpha, 0)$ $(\beta, 0)$ both to the right of the origin. A circle also passes through these two points. The length of a tangent from the origin to the circle is :
 (A) $\sqrt{\frac{bc}{a}}$ (B) ac^2 (C) $\frac{b}{a}$ (D) $\sqrt{\frac{c}{a}}$
16. If $(2, -8)$ is one end of a focal chord of the parabola $y^2 = 32x$, then the other end of the focal chord, is -
 (A) $(32, 32)$ (B) $(32, -32)$ (C) $(-2, 8)$ (D) $(2, 8)$
17. Minimum distance between the curves $y^2 = x - 1$ and $x^2 = y - 1$ is equal to
 (A) $\frac{3\sqrt{2}}{4}$ (B) $\frac{5\sqrt{2}}{4}$ (C) $\frac{7\sqrt{2}}{4}$ (D) $\frac{\sqrt{2}}{4}$
18. The length of a focal chord of the parabola $y^2 = 4ax$ at a distance b from the vertex is c , then
 (A) $2a^2 = bc$ (B) $a^3 = b^2c$ (C) $ac = b^2$ (D) $b^2c = 4a^3$
19. Locus of trisection point of any arbitrary double ordinate of the parabola $x^2 = 4by$, is -
 (A) $9x^2 = by$ (B) $3x^2 = 2by$ (C) $9x^2 = 4by$ (D) $9x^2 = 2by$
20. Consider the graphs of $y = Ax^2$ and $y^2 + 3 = x^2 + 4y$, where A is a positive constant and $x, y \in \mathbb{R}$. Number of points in which the two graphs intersect, is -
 (A) exactly 4
 (B) exactly 2
 (C) at least 2 but the number of points varies for different positive values of A .
 (D) zero for atleast one positive A .
21. From an external point P , pair of tangent lines are drawn to the parabola, $y^2 = 4x$. If θ_1 & θ_2 are the inclinations of these tangents with the axis of x such that, $\theta_1 + \theta_2 = \frac{\pi}{4}$, then the locus of P is :
 (A) $x - y + 1 = 0$ (B) $x + y - 1 = 0$ (C) $x - y - 1 = 0$ (D) $x + y + 1 = 0$
22. y-intercept of the common tangent to the parabola $y^2 = 32x$ and $x^2 = 108y$ is
 (A) -18 (B) -12 (C) -9 (D) -6
23. The points of contact Q and R of tangent from the point $P(2, 3)$ on the parabola $y^2 = 4x$ are
 (A) $(9, 6)$ and $(1, 2)$ (B) $(1, 2)$ and $(4, 4)$ (C) $(4, 4)$ and $(9, 6)$ (D) $(9, 6)$ and $(\frac{1}{4}, 1)$
24. If the lines $(y - b) = m_1(x + a)$ and $(y - b) = m_2(x + a)$ are the tangents to the parabola $y^2 = 4ax$, then
 (A) $m_1 + m_2 = 0$ (B) $m_1 m_2 = 1$ (C) $m_1 m_2 = -1$ (D) $m_1 + m_2 = 1$

25. The equation of a straight line passing through the point (3,6) and cutting the curve $y = \sqrt{x}$ orthogonally is-
 (A) $4x + y - 18 = 0$ (B) $x + y - 9 = 0$ (C) $4x - y - 6 = 0$ (D) none
26. The equation of the common tangent touching the circle $(x - 3)^2 + y^2 = 9$ and the parabola $y^2 = 4x$ above the x-axis is -
 (A) $\sqrt{3}y = 3x + 1$ (B) $\sqrt{3}y = -(x + 3)$ (C) $\sqrt{3}y = x + 3$ (D) $\sqrt{3}y = -(3x + 1)$
27. Let BC be the latus rectum of the parabola $y^2 = 4x$ with vertex A. Minimum length of the projection of BC on a tangent drawn in the portion BAC is -
 (A) 2 (B) $2\sqrt{3}$ (C) $2\sqrt{2}$ (D) $2 + \sqrt{2}$
28. If $x + y = k$ is normal to $y^2 = 12x$, then 'k' is-
 (A) 3 (B) 9 (C) -9 (D) -3
29. Equation of the other normal to the parabola $y^2 = 4x$ which passes through the intersection of those at (4, -4) and (9a, -6a) is -
 (A) $5x - y + 115 = 0$ (B) $5x + y - 135 = 0$ (C) $5x - y - 115 = 0$ (D) $5x + y + 115 = 0$
30. Length of the normal chord of the parabola, $y^2 = 4x$, which makes an angle of $\frac{\pi}{4}$ with the axis of x is:
 (A) 8 (B) $8\sqrt{2}$ (C) 4 (D) $4\sqrt{2}$
31. Suppose that three points on the parabola $y = x^2$ have the property that their normal lines intersect at a common point (a,b). The sum of their x-coordinates is -
 (A) 0 (B) $\frac{2b-1}{2}$ (C) $\frac{a}{2}$ (D) $a + b$
32. The normal chord of a parabola $y^2 = 4ax$ at the point whose ordinate is equal to the abscissa, then angle subtended by normal chord at the focus is :
 (A) $\frac{\pi}{4}$ (B) $\tan^{-1}\sqrt{2}$ (C) $\tan^{-1}2$ (D) $\frac{\pi}{2}$
33. Which one of the following lines cannot be the normals to $x^2 = 4y$?
 (A) $x - y + 3 = 0$ (B) $x + y - 3 = 0$ (C) $x - 2y + 12 = 0$ (D) $x + 2y + 12 = 0$
34. Tangents are drawn from the points on the line $x - y + 3 = 0$ to parabola $y^2 = 8x$. Then the variable chords of contact pass through a fixed point whose coordinates are :
 (A) (3,2) (B) (2,4) (C) (3,4) (D) (4,1)
35. Consider two curves $C_1 : (y - \sqrt{3})^2 = 4(x - \sqrt{2})$ and $C_2 : x^2 + y^2 = (6 + 2\sqrt{2})x + 2\sqrt{3}y - 6(1 + \sqrt{2})$, then-
 (A) C_1 and C_2 touch each other only at one point.
 (B) C_1 and C_2 touch each other exactly at two points.
 (C) C_1 and C_2 intersect (but do not touch) at exactly two points.
 (D) C_1 and C_2 neither intersect nor touch each other.

36. ABCD and EFGC are squares and the curve $y = k\sqrt{x}$ passes through the origin D and the points B and F. The ratio $\frac{FG}{BC}$ is -

- (A) $\frac{\sqrt{5}+1}{2}$ (B) $\frac{\sqrt{3}+1}{2}$
 (C) $\frac{\sqrt{5}+1}{4}$ (D) $\frac{\sqrt{3}+1}{4}$



37. C is the centre of the circle with centre (0,1) and radius unity. P is parabola $y = ax^2$. The set of values of 'a' for which they meet at a point other than the origin, is-
- (A) $a > 0$ (B) $a \in \left(0, \frac{1}{2}\right)$ (C) $\left(\frac{1}{4}, \frac{1}{2}\right)$ (D) $\left(\frac{1}{2}, \infty\right)$
38. Tangents are drawn from the point $(-1, 2)$ on the parabola $y^2 = 4x$. The length, these tangents will intercept on the line $x = 2$ is :
- (A) 6 (B) $6\sqrt{2}$ (C) $2\sqrt{6}$ (D) none of these
39. If the locus of the middle points of the chords of the parabola $y^2 = 2x$ which touches the circle $x^2 + y^2 - 2x - 4 = 0$ is given by $(y^2 + 1 - x)^2 = \lambda(1 + y^2)$, then the value of λ is equal to-
- (A) 3 (B) 4 (C) 5 (D) 6

[MULTIPLE OBJECTIVE TYPE]

40. A variable circle is described to pass through the point (1,0) and tangent to the curve $y = \tan(\tan^{-1}x)$. The locus of the centre of the circle is a parabola whose -
- (A) length of the latus rectum is $2\sqrt{2}$ (B) axis of symmetry has the equation $x + y = 1$
 (C) vertex has the co-ordinates $(3/4, 1/4)$ (D) directrix is $x - y = 0$
41. The locus of the mid point of the focal radii of a variable point moving on the parabola, $y^2 = 8x$ is a parabola whose-
- (A) Latus rectum is half the latus rectum of the original parabola
 (B) Vertex is (1,0)
 (C) Directrix is y-axis
 (D) Focus has the co-ordinates (2,0)
42. Consider a circle with its centre lying on the focus of the parabola, $y^2 = 2px$ such that it touches the directrix of the parabola. Then a point of intersection of the circle & the parabola is
- (A) $\left(\frac{p}{2}, p\right)$ (B) $\left(\frac{p}{2}, -p\right)$ (C) $\left(-\frac{p}{2}, p\right)$ (D) $\left(-\frac{p}{2}, -p\right)$
43. Let $y^2 = 4ax$ be a parabola and $x^2 + y^2 + 2bx = 0$ be a circle. If parabola and circle touch each other externally then :
- (A) $a > 0, b > 0$ (B) $a > 0, b < 0$ (C) $a < 0, b > 0$ (D) $a < 0, b < 0$
44. The focus of the parabola is (1,1) and the tangent at the vertex has the equation $x + y = 1$. Then :
- (A) equation of the parabola is $(x - y)^2 = 2(x + y - 1)$
 (B) equation of the parabola is $(x - y)^2 = 4(x + y - 1)$
 (C) the co-ordinates of the vertex are $\left(\frac{1}{2}, \frac{1}{2}\right)$
 (D) length of the latus rectum is $2\sqrt{2}$

45. The straight line $y + x = 1$ touches the parabola
 (A) $x^2 + 4y = 0$ (B) $x^2 - x + y = 0$ (C) $4x^2 - 3x + y = 0$ (D) $x^2 - 2x + 2y = 0$
46. The parabola $x = y^2 + ay + b$ intersect the parabola $x^2 = y$ at $(1,1)$ at right angle. Which of the following is/are correct ?
 (A) $a = 4, b = -4$
 (B) $a = 2, b = -2$
 (C) Equation of the director circle for the parabola $x = y^2 + ay + b$ is $4x + 1 = 0$.
 (D) Area enclosed by the parabola $x = y^2 + ay + b$ and its latus rectum is $\frac{1}{6}$.

[ASSERTION AND REASON]

47. Consider a curve $C : y^2 - 8x - 2y - 15 = 0$ in which two tangents T_1 and T_2 are drawn from $P(-4,1)$.
Statement-1 : T_1 and T_2 are mutually perpendicular tangents.
Statement-2 : Point P lies on the axis of curve C .
 (A) Statement-1 is True, Statement-2 is True ; Statement-2 is a correct explanation for Statement-1.
 (B) Statement-1 is True, Statement-2 is True ; Statement-2 is NOT a correct explanation for Statement-1.
 (C) Statement-1 is True, Statement-2 is False.
 (D) Statement-1 is False, Statement-2 is True.

[COMPREHENSION TYPE]

Paragraph for question nos. 48 & 49

Consider the parabola $y^2 = 8x$

48. Area of the figure formed by the tangents and normals drawn at the extremities of its latus rectum is
 (A) 8 (B) 16 (C) 32 (D) 64
49. Distance between the tangent to the parabola and a parallel normal inclined at 30° with the x -axis, is
 (A) $\frac{16}{3}$ (B) $\frac{16\sqrt{3}}{9}$ (C) $\frac{2}{3}$ (D) $\frac{16}{\sqrt{3}}$

[MATRIX MATCH TYPE]

50. Identify the conic whose equations are given in column-I.

Column-I

(Equation of a conic)

- (A) $xy + a^2 = a(x + y)$
 (B) $2x^2 - 72xy + 23y^2 - 4x - 28y - 48 = 0$
 (C) $6x^2 - 5xy - 6y^2 + 14x + 5y + 4 = 0$
 (D) $14x^2 - 4xy + 11y^2 - 44x - 58y + 71 = 0$
 (E) $4x^2 - 4xy + y^2 - 12x + 6y + 9 = 0$

Column-II

(Nature of conic)

- (P) Ellipse
 (Q) Hyperbola
 (R) Parabola
 (S) line pair

EXERCISE (O-2)

[STRAIGHT OBJECTIVE TYPE]

- Two unequal parabolas have the same common axis which is the x-axis and have the same vertex which is the origin with their concavities in opposite direction. If a variable line parallel to the common axis meet the parabolas in P and P' the locus of the middle point of PP' is
(A) a parabola (B) a circle (C) an ellipse (D) a hyperbola
- PN is an ordinate of the parabola $y^2 = 4ax$ (P on $y^2 = 4ax$ and N on x-axis). A straight line is drawn parallel to the axis to bisect NP and meets the curve in Q. NQ meets the tangent at the vertex in a point T such that $AT = kNP$, then the value of k is (where A is the vertex)
(A) $3/2$ (B) $2/3$ (C) 1 (D) none
- Let A and B be two points on a parabola $y^2 = x$ with vertex V such that VA is perpendicular to VB and θ is the angle between the chord VA and the axis of the parabola. The value of $\frac{|VA|}{|VB|}$ is
(A) $\tan \theta$ (B) $\tan^3 \theta$ (C) $\cot^2 \theta$ (D) $\cot^3 \theta$
- Locus of the feet of the perpendiculars drawn from vertex of the parabola $y^2 = 4ax$ upon all such chords of the parabola which subtend a right angle at the vertex is
(A) $x^2 + y^2 - 4ax = 0$ (B) $x^2 + y^2 - 2ax = 0$
(C) $x^2 + y^2 + 2ax = 0$ (D) $x^2 + y^2 + 4ax = 0$
- The triangle PQR of area 'A' is inscribed in the parabola $y^2 = 4ax$ such that the vertex P lies at the vertex of the parabola and the base QR is a focal chord. The modulus of the difference of the ordinates of the point Q and R is :
(A) $\frac{A}{2a}$ (B) $\frac{A}{a}$ (C) $\frac{2A}{a}$ (D) $\frac{4A}{a}$
- Through the focus of the parabola $y^2 = 2px$ ($p > 0$) a line is drawn which intersects the curve at $A(x_1, y_1)$ and $B(x_2, y_2)$. The ratio $\frac{y_1 y_2}{x_1 x_2}$ equals-
(A) 2 (B) -1 (C) -4 (D) some function of p
- A circle with radius unity has its centre on the positive y-axis. If this circle touches the parabola $y = 2x^2$ tangentially at the point P and Q then the sum of the ordinates of P and Q, is-
(A) $15/4$ (B) $15/8$ (C) $2\sqrt{15}$ (D) 5
- The straight line joining any point P on the parabola $y^2 = 4ax$ to the vertex and perpendicular from the focus to the tangent at P, intersect at R, then the equation of the locus of R is
(A) $x^2 + 2y^2 - ax = 0$ (B) $2x^2 + y^2 - 2ax = 0$
(C) $2x^2 + 2y^2 - ay = 0$ (D) $2x^2 + y^2 - 2ay = 0$
- If two normals to a parabola $y^2 = 4ax$ intersect at right angles then the chord joining their feet pass through a fixed point whose co-ordinates are :
(A) $(-2a, 0)$ (B) $(a, 0)$ (C) $(2a, 0)$ (D) none

10. If the normal to a parabola $y^2 = 4ax$ at P meets the curve again in Q and if PQ and the normal at Q makes angles α and β respectively with the x-axis then $\tan \alpha (\tan \alpha + \tan \beta)$ has the value equal to
- (A) 0 (B) -2 (C) $-\frac{1}{2}$ (D) -1
11. If a normal to a parabola $y^2 = 4ax$ makes an angle ϕ with its axis, then it will cut the curve again at an angle
- (A) $\tan^{-1}(2 \tan \phi)$ (B) $\tan^{-1}\left(\frac{1}{2} \tan \phi\right)$ (C) $\cot^{-1}\left(\frac{1}{2} \tan \phi\right)$ (D) none
12. The tangent and normal at P(t), for all real positive t, to the parabola $y^2 = 4ax$ meet the axis of the parabola in T and G respectively, then the angle at which the tangent at P to the parabola is inclined to the tangent at P to the circle passing through the points P, T and G is
- (A) $\cot^{-1}t$ (B) $\cot^{-1}t^2$ (C) $\tan^{-1}t$ (D) $\tan^{-1}t^2$
13. Normal to the parabola $y^2 = 8x$ at the point P (2, 4) meets the parabola again at the point Q. If C is the centre of the circle described on PQ as diameter then the coordinates of the image of the point C in the line $y = x$ are
- (A) (-4, 10) (B) (-3, 8) (C) (4, -10) (D) (-3, 10)
14. PQ is a normal chord of the parabola $y^2 = 4ax$ at P, A being the vertex of the parabola. Through P a line is drawn parallel to AQ meeting the x-axis in R. Then the length of AR is -
- (A) equal to the length of the latus rectum.
(B) equal to the focal distance of the point P.
(C) equal to twice the focal distance of the point P.
(D) equal to the distance of the point P from the directrix.
15. Normals are drawn at points A, B, and C on the parabola $y^2 = 4x$ which intersect at P(h, k). The locus of the point P if the slope of the line joining the feet of two of them is 2, is
- (A) $x + y = 1$ (B) $x - y = 3$ (C) $y^2 = 2(x - 1)$ (D) $y^2 = 2\left(x - \frac{1}{2}\right)$
16. TP & TQ are tangents to the parabola, $y^2 = 4ax$ at P & Q. If the chord PQ passes through the fixed point (-a, b) then the locus of T is :
- (A) $ay = 2b(x - b)$ (B) $bx = 2a(y - a)$ (C) $by = 2a(x - a)$ (D) $ax = 2b(y - b)$
17. Through the vertex O of the parabola, $y^2 = 4ax$ two chords OP & OQ are drawn and the circles on OP & OQ as diameters intersect in R. If θ_1, θ_2 & ϕ are the angles made with the axis by the tangents at P & Q on the parabola & by OR then the value of, $\cot \theta_1 + \cot \theta_2 =$
- (A) $-2 \tan \phi$ (B) $-2 \tan(\pi - \phi)$ (C) 0 (D) $2 \cot \phi$
18. Length of the intercept on the normal at the point P($at^2, 2at$) of the parabola $y^2 = 4ax$ made by the circle described on the focal distance of the point P as diameter is :
- (A) $a\sqrt{2+t^2}$ (B) $\frac{a}{2}\sqrt{1+t^2}$ (C) $2a\sqrt{1+t^2}$ (D) $a\sqrt{1+t^2}$

[MULTIPLE OBJECTIVE TYPE]

19. P is a point on the parabola $y^2 = 4ax$ ($a > 0$) whose vertex is A. PA is produced to meet the directrix in D and M is the foot of the perpendicular from P on the directrix. If a circle is described on MD as a diameter then it intersects the x-axis at a point whose co-ordinates are :
- (A) $(-3a, 0)$ (B) $(-a, 0)$ (C) $(-2a, 0)$ (D) $(a, 0)$
20. If from the vertex of a parabola $y^2 = 4x$ a pair of chords be drawn at right angles to one another and with these chords as adjacent sides a rectangle be made, then the locus of the further end of the rectangle is -
- (A) an equal parabola (B) a parabola with focus at $(9, 0)$
 (C) a parabola with directrix as $x - 7 = 0$ (D) a parabola having tangent at its vertex $x = 8$
21. A circle 'S' is described on the focal chord of the parabola $y^2 = 4x$ as diameter. If the focal chord is inclined at an angle of 45° with axis of x, then which of the following is/are true ?
- (A) Radius of the circle is 4.
 (B) Centre of the circle is $(3, 2)$
 (C) The line $x + 1 = 0$ touches the circle
 (D) The circle $x^2 + y^2 + 2x - 6y + 3 = 0$ is orthogonal to 'S'.
22. PQ is a double ordinate of the parabola $y^2 = 4ax$. If the normal at P intersect the line passing through Q and parallel to axis of x at G, then locus of G is a parabola with -
- (A) length of latus rectum equal to $4a$ (B) vertex at $(4a, 0)$
 (C) directrix as the line $x - 3a = 0$ (D) focus at $(5a, 0)$
23. TP and TQ are tangents to parabola $y^2 = 4x$ and normals at P and Q intersect at a point R on the curve. The locus of the centre of the circle circumscribing ΔTPQ is a parabola whose
- (A) vertex is $(1, 0)$. (B) foot of directrix is $\left(\frac{7}{8}, 0\right)$.
 (C) length of latus-rectum is $\frac{1}{4}$. (D) focus is $\left(\frac{9}{8}, 0\right)$.
24. The locus of the point of intersection of those normals to the parabola $x^2 = 8y$ which are at right angles to each other, is a parabola. Which of the following hold(s) good in respect of the locus ?
- (A) Length of the latus rectum is 2. (B) Coordinates of focus are $\left(0, \frac{11}{2}\right)$
 (C) Equation of a director circle is $2y - 11 = 0$ (D) Equation of axis of symmetry is $y = 0$
25. Consider the parabola whose equation is $y = x^2 - 4x$ and the line $y = 2x - b$. Then which of the following is/are correct ?
- (A) For $b = 9$ the line is a tangent to the parabola.
 (B) For $b = 7$ the line cuts the parabola in A and B such that the $\angle AOB$ is a right angle when 'O' is the origin.
 (C) For some $b \in \mathbb{R}$ the line cuts the parabola in C and D such that x-axis bisects the $\angle COD$.
 (D) For $b > 9$ the line passes outside the parabola.

26. Through a point P (– 2, 0), tangents PQ and PR are drawn to the parabola $y^2 = 8x$. Two circles each passing through the focus of the parabola and one touching at Q and other at R are drawn. Which of the following point(s) with respect to the triangle PQR lie(s) on the radical axis of the two circles?
- (A) centroid (B) orthocenter (C) incentre (D) circumcenter

[COMPREHENSION TYPE]

Paragraph for question nos. 27 to 29

Tangents are drawn to the parabola $y^2 = 4x$ from the point P(6, 5) to touch the parabola at Q and R. C_1 is a circle which touches the parabola at Q and C_2 is a circle which touches the parabola at R. Both the circles C_1 and C_2 pass through the focus of the parabola.

27. Area of the ΔPQR equals
- (A) $\frac{1}{2}$ (B) 1 (C) 2 (D) $\frac{1}{4}$
28. Radius of the circle C_2 is
- (A) $5\sqrt{5}$ (B) $5\sqrt{10}$ (C) $10\sqrt{2}$ (D) $\sqrt{210}$
29. The common chord of the circles C_1 and C_2 passes through the
- (A) incentre of the ΔPQR (B) circumcenter of the ΔPQR
(C) centroid of the ΔPQR (D) orthocenter of the ΔPQR

[MATRIX MATCH TYPE]

30. Consider the parabola $y^2 = 12x$

Column-I

- (A) Tangent and normal at the extremities of the latus rectum intersect the x axis at T and G respectively. The coordinates of the middle point of T and G are
- (B) Variable chords of the parabola passing through a fixed point K on the axis, such that sum of the squares of the reciprocals of the two parts of the chords through K, is a constant. The coordinate of the point K are
- (C) All variable chords of the parabola subtending a right angle at the origin are concurrent at the point
- (D) AB and CD are the chords of the parabola which intersect at a point E on the axis. The radical axis of the two circles described on AB and CD as diameter always passes through

Column-II

- (P) (0, 0)
- (Q) (3, 0)
- (R) (6, 0)
- (S) (12, 0)

EXERCISE (S-1)

- 'O' is the vertex of the parabola $y^2 = 4ax$ & L is the upper end of the latus rectum. If LH is drawn perpendicular to OL meeting OX in H, prove that the length of the double ordinate through H is $4a\sqrt{5}$.
- A point P on a parabola $y^2 = 4x$, the foot of the perpendicular from it upon the directrix, and the focus are the vertices of an equilateral triangle, find the area of the equilateral triangle.
- Through the vertex O of a parabola $y^2 = 4x$, chords OP & OQ are drawn at right angles to one another. Show that for all positions of P, PQ cuts the axis of the parabola at a fixed point. Also find the locus of the middle point of PQ.
- Find the equations of the tangents to the parabola $y^2 = 16x$, which are parallel & perpendicular respectively to the line $2x - y + 5 = 0$. Find also the coordinates of their points of contact.
- Find the equations of the tangents of the parabola $y^2 = 12x$, which passes through the point (2,5). Also find the point of contact.
- Through the vertex O of the parabola $y^2 = 4ax$, a perpendicular is drawn to any tangent meeting it at P & the parabola at Q. Show that $OP \cdot OQ = \text{constant}$.
- In the parabola $y^2 = 4ax$, the tangent at the point P, whose abscissa is equal to the latus rectum meets the axis in T & the normal at P cuts the parabola again in Q. Prove that $PT : PQ = 4 : 5$.
- Show that the normals at the points $(4a, 4a)$ & at the upper end of the latus rectum of the parabola $y^2 = 4ax$ intersect on the same parabola.
- Three normals to $y^2 = 4x$ pass through the point (15, 12). Show that if one of the normals is given by $y = x - 3$ & find the equations of the others.
- The normal at a point P to the parabola $y^2 = 4ax$ meets its axis at G. Q is another point on the parabola such that QG is perpendicular to the axis of the parabola. Prove that $QG^2 - PG^2 = \text{constant}$.
- Prove that the locus of the middle point of portion of a normal to $y^2 = 4ax$ intercepted between the curve & the axis is another parabola. Find the vertex & the latus rectum of the second parabola.
- If the normal at P(18, 12) to the parabola $y^2 = 8x$ cuts it again at Q, show that $9PQ = 80\sqrt{10}$.
- Prove that, the normal to $y^2 = 12x$ at (3, 6) meets the parabola again in (27, -18) & circle on this normal chord as diameter is $x^2 + y^2 - 30x + 12y - 27 = 0$.
- From the point P(h, k) three normals are drawn to the parabola $x^2 = 8y$ and m_1, m_2 and m_3 are the slopes of three normals
 - Find the algebraic sum of the slopes of these three normals.
 - If two of the three normals are at right angles then the locus of point P is a conic, find the latus rectum of conic.
 - If the two normals from P are such that they make complementary angles with the axis then the locus of point P is a conic, find a directrix of conic.
- Show that the normals at two suitable distinct real points on the parabola $y^2 = 4ax$ ($a > 0$) intersect at a point on the parabola whose abscissa $> 8a$.
- The normal to the parabola $y^2 = 4x$ at the point P, Q & R are concurrent at the point (15, 12). Find
 - the equation of the circle circumscribing the triangle PQR
 - the co-ordinates of the centroid of the triangle PQR.
- From the point (-1, 2) tangent lines are drawn to the parabola $y^2 = 4x$. Find the equation of the chord of contact. Also find the area of the triangle formed by the chord of contact & the tangents.
- Find the equation of the circle which passes through the focus of the parabola $x^2 = 4y$ & touches it at the point (6, 9).
- Find the equations of the chords of the parabola $y^2 = 4ax$ which pass through the point $(-6a, 0)$ and which subtends an angle of 45° at the vertex.

EXERCISE (S-2)

1. PC is the normal at P to the parabola $y^2 = 4ax$, C being on the axis. CP is produced outwards to Q so that $PQ = CP$; show that the locus of Q is a parabola.
2. A quadrilateral is inscribed in a parabola $y^2 = 4ax$ and three of its sides pass through fixed points on the axis. Show that the fourth side also passes through fixed point on the axis of the parabola.
3. Tangents are drawn to the parabola $y^2 = 12x$ at the points A, B and C such that the three tangents form a triangle PQR. If θ_1, θ_2 and θ_3 be the inclinations of these tangents with the axis of x such that their cotangents form an arithmetical progression (in the same order) with common difference 2. Find the area of the triangle PQR.
4. Two straight lines one being a tangent to $y^2 = 4ax$ and the other to $x^2 = 4by$ are right angles. Find the locus of their point of intersection.
5. Let P(a,b) and Q(c,d) are the two points on the parabola $y^2 = 8x$ such that the normals at them meet in (18,12). Find the product (abcd).
6. A variable circle passes through the point A(2,1) and touches the x-axis. Locus of the other end of the diameter through A is a parabola.
 - (a) Find the length of the latus rectum of the parabola.
 - (b) Find the co-ordinates of the foot of the directrix of the parabola.
 - (c) The two tangents and two normals at the extremities of the latus rectum of the parabola constitutes a quadrilateral. Find area of quadrilateral.
7. Find the condition on 'a' & 'b' so that the two tangents drawn to the parabola $y^2 = 4ax$ from a point are normals to the parabola $x^2 = 4by$.
8. Two tangents to the parabola $y^2 = 8x$ meet the tangent at its vertex in the points P & Q. If $PQ = 4$ units, prove that the locus of the point of the intersection of the two tangents is $y^2 = 8(x+2)$.
9. Let a variable point A be lying on the directrix of parabola $y^2 = 4x$. Tangents AB & AC are drawn to the curve where B & C are points of contact of tangent. If the locus of centroid of ΔABC is a conic then find the length of its latus rectum.
10. A variable chords of the parabola $y^2 = 8x$ touches the parabola $y^2 = 2x$. The locus of the point of intersection of the tangent at the end of the chord is a parabola. Find its latus rectum.
11. Show that the circle through three points the normals at which to the parabola $y^2 = 4ax$ are concurrent at the point (h,k) is $2(x^2 + y^2) - 2(h + 2a)x - ky = 0$.
12. Let $L_1 : x + y = 0$ and $L_2 : x - y = 0$ are tangent to a parabola whose focus is S(1, 2).
If the length of latus-rectum of the parabola can be expressed as $\frac{m}{\sqrt{n}}$ (where m and n are coprime) then find the value of (m + n).
13. PQ, a variable chord of the parabola $y^2 = 4x$ subtends a right angle at the vertex. The tangents at P and Q meet at T and the normals at those points meet at N. If the locus of the mid point of TN is a parabola, then find its latus rectum.

EXERCISE (JM)

1. If two tangents drawn from a point P to the parabola $y^2 = 4x$ are at right angles then the locus of P is :-
[AIEEE-2010]

(1) $x = 1$ (2) $2x + 1 = 0$ (3) $x = -1$ (4) $2x - 1 = 0$

2. Given : A circle, $2x^2 + 2y^2 = 5$ and a parabola, $y^2 = 4\sqrt{5}x$.

Statement-I : An equation of a common tangent to these curves is $y = x + \sqrt{5}$.

Statement-II : If the line, $y = mx + \frac{\sqrt{5}}{m}$ ($m \neq 0$) is their common tangent, then m satisfies $m^4 - 3m^2 + 2 = 0$.
[JEE (Main)-2013]

- (1) Statement-I is true, Statement-II is true; statement-II is a **correct** explanation for Statement-I.
(2) Statement-I is true, Statement-II is true; statement-II is **not** a correct explanation for Statement-I.
(3) Statement-I is true, Statement-II is false.
(4) Statement-I is false, Statement-II is true.

3. Statement 1 : The slope of the tangent at any point P on a parabola, whose axis is the axis of x and vertex is at the origin, is inversely proportional to the ordinate of the point P.

Statement 2 : The system of parabolas $y^2 = 4ax$ satisfies a differential equation of degree 1 and order 1.

- (1) Statement 1 is True Statement 2 is True, Statement 2 is a correct explanation for Statement 1.
(2) Statement 1 is True, Statement 2 is False.
(3) Statement 1 is True, Statement 2 is True statement 2 is not a correct explanation for statement 1.
(4) Statement 1 is False, Statement 2 is True
[JEE-Main (On line)-2013]

4. **Statement 1** : The line $x - 2y = 2$ meets the parabola, $y^2 + 2x = 0$ only at the point $(-2, -2)$

Statement 2 : The line $y = mx - \frac{1}{2m}$ ($m \neq 0$) is tangent to the parabola, $y^2 = -2x$ at the point $\left(-\frac{1}{2m^2}, -\frac{1}{m}\right)$.

- (1) Statement 1 is false; Statement 2 is true.
(2) Statement 1 is true; Statement 2 is true; Statement 2 is not a correct explanation for Statement 1.
(3) Statement 1 is true; Statement 2 is false.
(4) Statement 1 is true; Statement 2 is true; Statement 2 is a correct explanation for Statement 1.
[JEE-Main (On line)-2013]

5. The point of intersection of the normals to the parabola $y^2 = 4x$ at the ends of its latus rectum is :

[JEE-Main (On line)-2013]

(1) $(0, 3)$ (2) $(2, 0)$ (3) $(3, 0)$ (4) $(0, 2)$

6. The slope of the line touching both, the parabolas $y^2 = 4x$ and $x^2 = -32y$ is : [JEE(Main)-2014]

(1) $\frac{1}{2}$ (2) $\frac{3}{2}$ (3) $\frac{1}{8}$ (4) $\frac{2}{3}$

7. Let O be the vertex and Q be any point on the parabola, $x^2 = 8y$. If the point P divides the line segment OQ internally in the ratio 1 : 3, then the locus of P is :-
[JEE(Main)-2015]

(1) $y^2 = 2x$ (2) $x^2 = 2y$ (3) $x^2 = y$ (4) $y^2 = x$

8. Let P be the point on the parabola, $y^2 = 8x$ which is at a minimum distance from the centre C of the circle, $x^2 + (y + 6)^2 = 1$. Then the equation of the circle, passing through C and having its centre at P is : [JEE(Main)-2016]
- (1) $x^2 + y^2 - 4x + 9y + 18 = 0$ (2) $x^2 + y^2 - 4x + 8y + 12 = 0$
- (3) $x^2 + y^2 - x + 4y - 12 = 0$ (4) $x^2 + y^2 - \frac{x}{4} + 2y - 24 = 0$
9. The radius of a circle, having minimum area, which touches the curve $y = 4 - x^2$ and the lines, $y = |x|$ is :- [JEE-Main 2017]
- (1) $4(\sqrt{2} + 1)$ (2) $2(\sqrt{2} + 1)$ (3) $2(\sqrt{2} - 1)$ (4) $4(\sqrt{2} - 1)$
10. Tangent and normal are drawn at P(16, 16) on the parabola $y^2 = 16x$, which intersect the axis of the parabola at A and B, respectively. If C is the centre of the circle through the points P, A and B and $\angle CPB = \theta$, then a value of $\tan \theta$ is - [JEE-Main 2018]
- (1) 2 (2) 3 (3) $\frac{4}{3}$ (4) $\frac{1}{2}$
11. If the tangent at (1, 7) to the curve $x^2 = y - 6$ touches the circle $x^2 + y^2 + 16x + 12y + c = 0$ then the value of c is : [JEE-Main 2018]
- (1) 185 (2) 85 (3) 95 (4) 195
12. Let A(4, -4) and B(9, 6) be points on the parabola, $y^2 = 4x$. Let C be chosen on the arc AOB of the parabola, where O is the origin, such that the area of $\triangle ACB$ is maximum. Then, the area (in sq. units) of $\triangle ACB$, is: [JEE (Main)-Jan 19]
- (1) $31\frac{3}{4}$ (2) 32 (3) $30\frac{1}{2}$ (4) $31\frac{1}{4}$
13. If the parabolas $y^2 = 4b(x - c)$ and $y^2 = 8ax$ have a common normal, then which one of the following is a valid choice for the ordered triad (a, b, c) [JEE (Main)-Jan 19]
- (1) (1, 1, 0) (2) $\left(\frac{1}{2}, 2, 3\right)$ (3) $\left(\frac{1}{2}, 2, 0\right)$ (4) (1, 1, 3)
14. If the tangent to the parabola $y^2 = x$ at a point (α, β) , ($\beta > 0$) is also a tangent to the ellipse, $x^2 + 2y^2 = 1$, then α is equal to : [JEE (Main)-Apr 19]
- (1) $2\sqrt{2} + 1$ (2) $\sqrt{2} - 1$ (3) $\sqrt{2} + 1$ (4) $2\sqrt{2} - 1$
15. The area (in sq. units) of the smaller of the two circles that touch the parabola, $y^2 = 4x$ at the point (1, 2) and the x-axis is :- [JEE (Main)-Apr 19]
- (1) $4\pi(2 - \sqrt{2})$ (2) $8\pi(3 - 2\sqrt{2})$ (3) $4\pi(3 + \sqrt{2})$ (4) $8\pi(2 - \sqrt{2})$
16. The tangents to the curve $y = (x - 2)^2 - 1$ at its points of intersection with the line $x - y = 3$, intersect at the point : [JEE (Main)-Apr 19]
- (1) $\left(-\frac{5}{2}, -1\right)$ (2) $\left(-\frac{5}{2}, 1\right)$ (3) $\left(\frac{5}{2}, -1\right)$ (4) $\left(\frac{5}{2}, 1\right)$
17. The equation of a common tangent to the curves, $y^2 = 16x$ and $xy = -4$ is : [JEE (Main)-Apr 19]
- (1) $x + y + 4 = 0$ (2) $x - 2y + 16 = 0$ (3) $2x - y + 2 = 0$ (4) $x - y + 4 = 0$

EXERCISE (JA)

1. The tangent PT and the normal PN to the parabola $y^2 = 4ax$ at a point P on it meet its axis at points T and N, respectively. The locus of the centroid of the triangle PTN is a parabola whose
- (A) vertex is $\left(\frac{2a}{3}, 0\right)$ (B) directrix is $x = 0$ [JEE 2009, 4]
- (C) latus rectum is $\frac{2a}{3}$ (D) focus is $(a, 0)$
2. Let A and B be two distinct point on the parabola $y^2 = 4x$. If the axis of the parabola touches a circle of radius r having AB as its diameter, then the slope of the line joining A and B can be -
- (A) $-\frac{1}{r}$ (B) $\frac{1}{r}$ (C) $\frac{2}{r}$ (D) $-\frac{2}{r}$ [JEE 2010, 3]
3. Consider the parabola $y^2 = 8x$. Let Δ_1 be the area of the triangle formed by the end points of its latus rectum and the point $P\left(\frac{1}{2}, 2\right)$ on the parabola, and Δ_2 be the area of the triangle formed by drawing tangents at P and at the end points of the latus rectum. Then $\frac{\Delta_1}{\Delta_2}$ is [JEE 2011, 4]
4. Let (x, y) be any point on the parabola $y^2 = 4x$. Let P be the point that divides the line segment from $(0, 0)$ to (x, y) in the ratio 1 : 3. Then the locus of P is- [JEE 2011, 3]
- (A) $x^2 = y$ (B) $y^2 = 2x$ (C) $y^2 = x$ (D) $x^2 = 2y$
5. Let L be a normal to the parabola $y^2 = 4x$. If L passes through the point $(9, 6)$, then L is given by - [JEE 2011, 4]
- (A) $y - x + 3 = 0$ (B) $y + 3x - 33 = 0$ (C) $y + x - 15 = 0$ (D) $y - 2x + 12 = 0$
6. Let S be the focus of the parabola $y^2 = 8x$ & let PQ be the common chord of the circle $x^2 + y^2 - 2x - 4y = 0$ and the given parabola. The area of the triangle PQS is [JEE 2012, 4M]

Paragraph for Question 7 and 8

Let PQ be a focal chord of the parabolas $y^2 = 4ax$. The tangents to the parabola at P and Q meet at a point lying on the line $y = 2x + a$, $a > 0$.

7. If chord PQ subtends an angle θ at the vertex of $y^2 = 4ax$, then $\tan \theta =$ [JEE(Advanced) 2013, 3, (-1)]
- (A) $\frac{2}{3}\sqrt{7}$ (B) $\frac{-2}{3}\sqrt{7}$ (C) $\frac{2}{3}\sqrt{5}$ (D) $\frac{-2}{3}\sqrt{5}$
8. Length of chord PQ is [JEE(Advanced) 2013, 3, (-1)]
- (A) $7a$ (B) $5a$ (C) $2a$ (D) $3a$

9. A line $L : y = mx + 3$ meets y -axis at $E(0,3)$ and the arc of the parabola $y^2 = 16x$, $0 \leq y \leq 6$ at the point $F(x_0, y_0)$. The tangent to the parabola at $F(x_0, y_0)$ intersects the y -axis at $G(0, y_1)$. The slope m of the line L is chosen such that the area of the triangle EFG has a local maximum.

Match List-I with List-II and select the correct answer using the code given below the lists.

List-I

- P. $m =$
Q. Maximum area of $\triangle EFG$ is
R. $y_0 =$
S. $y_1 =$

List-II

1. $\frac{1}{2}$
2. 4
3. 2
4. 1

Codes :

- | | P | Q | R | S |
|-----|---|---|---|---|
| (A) | 4 | 1 | 2 | 3 |
| (B) | 3 | 4 | 1 | 2 |
| (C) | 1 | 3 | 2 | 4 |
| (D) | 1 | 3 | 4 | 2 |

[JEE(Advanced) 2013, 3, (-1)]

10. The common tangents to the circle $x^2 + y^2 = 2$ and the parabola $y^2 = 8x$ touch the circle at the point P, Q and the parabola at the points R, S. Then the area of the quadrilateral PQRS is -
(A) 3 (B) 6 (C) 9 (D) 15

[JEE(Advanced)-2014, 3(-1)]

Paragraph For Questions 11 and 12

Let a, r, s, t be nonzero real numbers. Let $P(at^2, 2at)$, Q , $R(ar^2, 2ar)$ and $S(as^2, 2as)$ be distinct points on the parabola $y^2 = 4ax$. Suppose that PQ is the focal chord and lines QR and PK are parallel, where K is the point $(2a, 0)$.

11. The value of r is- [JEE(Advanced)-2014, 3(-1)]
(A) $-\frac{1}{t}$ (B) $\frac{t^2+1}{t}$ (C) $\frac{1}{t}$ (D) $\frac{t^2-1}{t}$

12. If $st = 1$, then the tangent at P and the normal at S to the parabola meet at a point whose ordinate is- [JEE(Advanced)-2014, 3(-1)]

- (A) $\frac{(t^2+1)^2}{2t^3}$ (B) $\frac{a(t^2+1)^2}{2t^3}$ (C) $\frac{a(t^2+1)^2}{t^3}$ (D) $\frac{a(t^2+2)^2}{t^3}$

13. If the normals of the parabola $y^2 = 4x$ drawn at the end points of its latus rectum are tangents to the circle $(x-3)^2 + (y+2)^2 = r^2$, then the value of r^2 is [JEE 2015, 4M, -0M]

14. Let the curve C be the mirror image of the parabola $y^2 = 4x$ with respect to the line $x + y + 4 = 0$. If A and B are the points of intersection of C with the line $y = -5$, then the distance between A and B is [JEE 2015, 4M, -0M]

15. Let P and Q be distinct points on the parabola $y^2 = 2x$ such that a circle with PQ as diameter passes through the vertex O of the parabola. If P lies in the first quadrant and the area of the triangle ΔOPQ is $3\sqrt{2}$, then which of the following is(are) the coordinates of P ? [JEE 2015, 4M, -2M]

(A) $(4, 2\sqrt{2})$ (B) $(9, 3\sqrt{2})$ (C) $(\frac{1}{4}, \frac{1}{\sqrt{2}})$ (D) $(1, \sqrt{2})$

16. The circle $C_1: x^2 + y^2 = 3$, with centre at O, intersects the parabola $x^2 = 2y$ at the point P in the first quadrant. Let the tangent to the circle C_1 at P touches other two circles C_2 and C_3 at R_2 and R_3 , respectively. Suppose C_2 and C_3 have equal radii $2\sqrt{3}$ and centres Q_2 and Q_3 , respectively. If Q_2 and Q_3 lie on the y-axis, then - [JEE(Advanced)-2016, 4(-2)]

(A) $Q_2Q_3 = 12$ (B) $R_2R_3 = 4\sqrt{6}$
(C) area of the triangle OR_2R_3 is $6\sqrt{2}$ (D) area of the triangle PQ_2Q_3 is $4\sqrt{2}$

17. Let P be the point on the parabola $y^2 = 4x$ which is at the shortest distance from the center S of the circle $x^2 + y^2 - 4x - 16y + 64 = 0$. Let Q be the point on the circle dividing the line segment SP internally. Then-

(A) $SP = 2\sqrt{5}$
(B) $SQ : QP = (\sqrt{5} + 1) : 2$
(C) the x-intercept of the normal to the parabola at P is 6
(D) the slope of the tangent to the circle at Q is $\frac{1}{2}$ [JEE(Advanced)-2016, 4(-2)]

18. If a chord, which is not a tangent, of the parabola $y^2 = 16x$ has the equation $2x + y = p$, and midpoint (h, k), then which of the following is(are) possible value(s) of p, h and k ? [JEE(Advanced)-2017, 4(-2)]

(A) $p = 5, h = 4, k = -3$ (B) $p = -1, h = 1, k = -3$
(C) $p = -2, h = 2, k = -4$ (D) $p = 2, h = 3, k = -4$

ANSWER KEY

PARABOLA

EXERCISE (O-1)

1. D 2. D 3. C 4. A 5. C 6. D 7. A 8. D
9. B 10. B 11. D 12. B 13. B 14. C 15. D 16. A
17. A 18. D 19. C 20. A 21. C 22. B 23. B 24. C
25. A 26. C 27. C 28. B 29. B 30. B 31. A 32. D
33. D 34. C 35. B 36. A 37. D 38. B 39. C
40. B,C,D 41. A,B,C,D 42. A,B 43. A,D 44. B,C,D 45. A,B,C 46. C,D
47. B 48. C 49. A 50. (A) S, (B) Q, (C) S, (D) P, (E) S

EXERCISE (O-2)

1. A 2. B 3. D 4. A 5. C 6. C 7. A 8. B
9. B 10. B 11. B 12. C 13. A 14. C 15. B 16. C
17. A 18. D 19. A,D 20. A,B,C,D 21. A,B,C 22. A,B,C,D
23. A,B,D 24. A,C 25. A,B,D 26. A,B,C,D 27. A 28. B 29. C
30. (A) Q, (B) R, (C) S, (D) P

EXERCISE (S-1)

2. $4\sqrt{3}$ 3. $(4, 0); y^2 = 2a(x - 4a)$ 4. $2x - y + 2 = 0, (1, 4); x + 2y + 16 = 0, (16, -16)$
5. $3x - 2y + 4 = 0; x - y + 3 = 0$ 9. $y = -4x + 72, y = 3x - 33$ 11. $(a, 0); a$
14. $(a) \frac{k-4}{h}; (b) 2; (c) 2y - 3 = 0$ 16. $(a) x^2 + y^2 - 17x - 6y = 0; (b) (26/3, 0)$
17. $x - y = 1; 8\sqrt{2}$ sq. units 18. $x^2 + y^2 + 18x - 28y + 27 = 0$ 19. $7y \pm 2(x + 6a) = 0$

EXERCISE (S-2)

3. 72 4. $(ax + by)(x^2 + y^2) + (bx - ay)^2 = 0$ 5. 512 6. $(a) 4, (b) (2, -1), (c) 8$ sq. units
7. $a^2 > 8b^2$ 9. 3 10. 32 12. 11 13. 25/2

EXERCISE (JM)

1. 3 2. 2 3. 3 4. 4 5. 3 6. 1 7. 2
8. 2 9. Bonus 10. 1 11. 3 12. 4 13. 1,2,3,4 14. 3
15. 2 16. 3 17. 4

EXERCISE (JA)

1. A,D 2. C,D 3. 2 4. C 5. A,B,D 6. 4 7. D 8. B 9. A
10. D 11. D 12. B 13. 2 14. 4 15. A,D 16. A,B,C 17. A,C,D 18. D

ELLIPSE

1. STANDARD EQUATION & DEFINITION :

Standard equation of an ellipse referred to its principal axes along the co-ordinate axes is $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

where $a > b$ & $b^2 = a^2(1 - e^2) \Rightarrow a^2 - b^2 = a^2 e^2$.

where e = eccentricity ($0 < e < 1$).

FOCI : $S \equiv (ae, 0)$ & $S' \equiv (-ae, 0)$.

(a) **Equation of directrices :**

$$x = \frac{a}{e} \quad \& \quad x = -\frac{a}{e}.$$

(b) **Vertices :**

$$A' \equiv (-a, 0) \quad \& \quad A \equiv (a, 0).$$

(c) **Major axis :** The line segment $A'A$ in which the foci S' & S lie is of length $2a$ & is called the **major axis** ($a > b$) of the ellipse. Point of intersection of major axis with directrix is called the

foot of the directrix (z) $\left(\pm \frac{a}{e}, 0\right)$.

(d) **Minor Axis :** The y-axis intersects the ellipse in the points $B' \equiv (0, -b)$ & $B \equiv (0, b)$. The line segment $B'B$ of length $2b$ ($b < a$) is called the **Minor Axis** of the ellipse.

(e) **Principal Axes :** The major & minor axis together are called **Principal Axes** of the ellipse.

(f) **Centre :** The point which bisects every chord of the conic drawn through it is called the **centre**

of the conic. $C \equiv (0, 0)$ the origin is the centre of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

(g) **Diameter :** A chord of the conic which passes through the centre is called a **diameter** of the conic.

(h) **Focal Chord :** A chord which passes through a focus is called a **focal chord**.

(i) **Double Ordinate :** A chord perpendicular to the major axis is called a **double ordinate**.

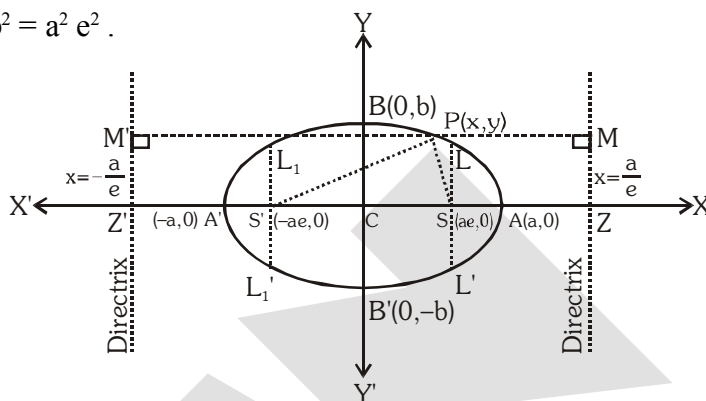
(j) **Latus Rectum :** The focal chord perpendicular to the major axis is called the **latus rectum**.

(i) Length of latus rectum $(LL') = \frac{2b^2}{a} = \frac{(\text{minor axis})^2}{\text{major axis}} = 2a(1 - e^2)$

(ii) Equation of latus rectum : $x = \pm ae$.

(iii) Ends of the latus rectum are $L\left(ae, \frac{b^2}{a}\right), L'\left(ae, -\frac{b^2}{a}\right), L_1\left(-ae, \frac{b^2}{a}\right)$

and $L_1'\left(-ae, -\frac{b^2}{a}\right)$.



(k) Focal radii : $SP = a - ex$ & $S'P = a + ex \Rightarrow SP + S'P = 2a = \text{Major axis.}$

(l) Eccentricity : $e = \sqrt{1 - \frac{b^2}{a^2}}$

Note :

(i) The sum of the focal distances of any point on the ellipse is equal to the major Axis. Hence distance of focus from the extremity of a minor axis is equal to semi major axis. **i.e. $BS = CA$.**

(ii) If the equation of the ellipse is given as $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ & nothing is mentioned, then the rule is to assume that $a > b$.

Illustration 1 : If LR of an ellipse is half of its minor axis, then its eccentricity is -

- (A) $\frac{3}{2}$ (B) $\frac{2}{3}$ (C) $\frac{\sqrt{3}}{2}$ (D) $\frac{\sqrt{2}}{3}$

Solution : As given $\frac{2b^2}{a} = b \Rightarrow 2b = a \Rightarrow 4b^2 = a^2$
 $\Rightarrow 4a^2(1 - e^2) = a^2 \Rightarrow 1 - e^2 = 1/4$
 $\therefore e = \sqrt{3}/2$

Ans. (C)

Illustration 2 : Find the equation of the ellipse whose foci are $(2, 3)$, $(-2, 3)$ and whose semi minor axis is of length $\sqrt{5}$.

Solution : Here S is $(2, 3)$ & S' is $(-2, 3)$ and $b = \sqrt{5} \Rightarrow SS' = 4 = 2ae \Rightarrow ae = 2$
 but $b^2 = a^2(1 - e^2) \Rightarrow 5 = a^2 - 4 \Rightarrow a = 3$.
 Hence the equation to major axis is $y = 3$
 Centre of ellipse is midpoint of SS' i.e. $(0, 3)$

$\therefore$ Equation to ellipse is $\frac{x^2}{a^2} + \frac{(y-3)^2}{b^2} = 1$ or $\frac{x^2}{9} + \frac{(y-3)^2}{5} = 1$ **Ans.**

Illustration 3 : Find the equation of the ellipse having centre at $(1, 2)$, one focus at $(6, 2)$ and passing through the point $(4, 6)$.

Solution : With centre at $(1, 2)$, the equation of the ellipse is $\frac{(x-1)^2}{a^2} + \frac{(y-2)^2}{b^2} = 1$. It passes through the point $(4, 6)$

$$\Rightarrow \frac{9}{a^2} + \frac{16}{b^2} = 1 \quad \dots\dots\dots (i)$$

Distance between the focus and the centre $= (6 - 1) = 5 = ae$

$$\Rightarrow b^2 = a^2 - a^2e^2 = a^2 - 25 \quad \dots\dots\dots (ii)$$

Solving for a^2 and b^2 from the equations (i) and (ii), we get $a^2 = 45$ and $b^2 = 20$.

Hence the equation of the ellipse is $\frac{(x-1)^2}{45} + \frac{(y-2)^2}{20} = 1$ **Ans.**

Do yourself - 1 :

- (i) If LR of an ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, ($a < b$) is half of its major axis, then find its eccentricity.
- (ii) Find the equation of the ellipse whose foci are (4, 6) & (16, 6) and whose semi-minor axis is 4.
- (iii) Find the eccentricity, foci and the length of the latus-rectum of the ellipse $x^2 + 4y^2 + 8y - 2x + 1 = 0$.

2. ANOTHER FORM OF ELLIPSE : $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, ($a < b$)

(a) $AA' = \text{Minor axis} = 2a$

(b) $BB' = \text{Major axis} = 2b$

(c) $a^2 = b^2 (1 - e^2)$

(d) Latus rectum $LL' = L_1L_1' = \frac{2a^2}{b}$, equation $y = \pm be$

(e) Ends of the latus rectum are :

$$L\left(\frac{a^2}{b}, be\right), L'\left(-\frac{a^2}{b}, be\right), L_1\left(\frac{a^2}{b}, -be\right), L_1'\left(-\frac{a^2}{b}, -be\right)$$

(f) Equation of directrix $y = \pm b/e$

(g) Eccentricity : $e = \sqrt{1 - \frac{a^2}{b^2}}$

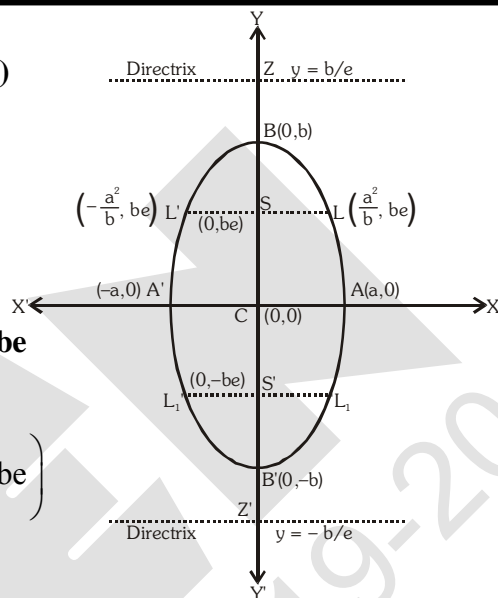


Illustration 4 : The equation of the ellipse with respect to coordinate axes whose minor axis is equal to the distance between its foci and whose LR = 10, will be-

- (A) $2x^2 + y^2 = 100$ (B) $x^2 + 2y^2 = 100$ (C) $2x^2 + 3y^2 = 80$ (D) none of these

Solution :

When $a > b$

As given $2b = 2ae \Rightarrow b = ae$ (i)

Also $\frac{2b^2}{a} = 10 \Rightarrow b^2 = 5a$ (ii)

Now since $b^2 = a^2 - a^2e^2 \Rightarrow b^2 = a^2 - b^2$ [From (i)]

$\Rightarrow 2b^2 = a^2$ (iii)

(ii), (iii) $\Rightarrow a^2 = 100, b^2 = 50$

Hence equation of the ellipse will be $\frac{x^2}{100} + \frac{y^2}{50} = 1 \Rightarrow x^2 + 2y^2 = 100$

Similarly when $a < b$ then required ellipse is $2x^2 + y^2 = 100$

Ans. (A, B)

Do yourself - 2 :

(i) The foci of an ellipse are $(0, \pm 2)$ and its eccentricity is $\frac{1}{\sqrt{2}}$. Find its equation

(ii) Find the centre, the length of the axes, eccentricity and the foci of ellipse

$$12x^2 + 4y^2 + 24x - 16y + 25 = 0$$

(iii) The equation $\frac{x^2}{8-t} + \frac{y^2}{t-4} = 1$, will represent an ellipse if

- (A) $t \in (1, 5)$ (B) $t \in (2, 8)$ (C) $t \in (4, 8) - \{6\}$ (D) $t \in (4, 10) - \{6\}$

Hence $y = mx + c$ is tangent to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ if $c^2 = a^2m^2 + b^2$.

The equation to the chord of the ellipse joining two points with eccentric angles α & β is given by

$$\frac{x}{a} \cos \frac{\alpha + \beta}{2} + \frac{y}{b} \sin \frac{\alpha + \beta}{2} = \cos \frac{\alpha - \beta}{2}.$$

Illustration 5 : For what value of λ does the line $y = x + \lambda$ touches the ellipse $9x^2 + 16y^2 = 144$.

Solution : $\therefore$ Equation of ellipse is $9x^2 + 16y^2 = 144$ or $\frac{x^2}{16} + \frac{y^2}{9} = 1$

Comparing this with $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ then we get $a^2 = 16$ and $b^2 = 9$

and comparing the line $y = x + \lambda$ with $y = mx + c$ $\therefore m = 1$ and $c = \lambda$

If the line $y = x + \lambda$ touches the ellipse $9x^2 + 16y^2 = 144$, then $c^2 = a^2m^2 + b^2$

$$\Rightarrow \lambda^2 = 16 \times 1^2 + 9 \Rightarrow \lambda^2 = 25 \quad \therefore \lambda = \pm 5 \quad \text{Ans.}$$

Illustration 6 : If α, β are eccentric angles of end points of a focal chord of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, then $\tan \alpha/2 \cdot \tan \beta/2$ is equal to -

- (A) $\frac{e-1}{e+1}$ (B) $\frac{1-e}{1+e}$ (C) $\frac{e+1}{e-1}$ (D) $\frac{e-1}{e+1}$

Solution : Equation of line joining points ' α ' and ' β ' is $\frac{x}{a} \cos \frac{\alpha + \beta}{2} + \frac{y}{b} \sin \frac{\alpha + \beta}{2} = \cos \frac{\alpha - \beta}{2}$

If it is a focal chord, then it passes through focus $(ae, 0)$, so $e \cos \frac{\alpha + \beta}{2} = \cos \frac{\alpha - \beta}{2}$

$$\Rightarrow \frac{\cos \frac{\alpha - \beta}{2}}{\cos \frac{\alpha + \beta}{2}} = \frac{e}{1} \Rightarrow \frac{\cos \frac{\alpha - \beta}{2} - \cos \frac{\alpha + \beta}{2}}{\cos \frac{\alpha - \beta}{2} + \cos \frac{\alpha + \beta}{2}} = \frac{e-1}{e+1}$$

$$\Rightarrow \frac{2 \sin \alpha/2 \sin \beta/2}{2 \cos \alpha/2 \cos \beta/2} = \frac{e-1}{e+1} \Rightarrow \tan \frac{\alpha}{2} \tan \frac{\beta}{2} = \frac{e-1}{e+1}$$

$$\text{using } (-ae, 0), \text{ we get } \tan \frac{\alpha}{2} \tan \frac{\beta}{2} = \frac{e+1}{e-1}$$

Ans. (A,C)

Do yourself - 3 :

(i) Find the position of the point $(4, 3)$ relative to the ellipse $2x^2 + 9y^2 = 113$.

(ii) A tangent to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, ($a > b$) having slope -1 intersects the axis of x & y in point A & B respectively. If O is the origin then find the area of triangle OAB.

(iii) Find the condition for the line $x \cos \theta + y \sin \theta = P$ to be a tangent to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

8. TANGENT TO THE ELLIPSE $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$:

- (a) **Point form :** Equation of tangent to the given ellipse at its point (x_1, y_1) is $\frac{xx_1}{a^2} + \frac{yy_1}{b^2} = 1$

Note : For general ellipse replace x^2 by (xx_1) , y^2 by (yy_1) , $2x$ by $(x + x_1)$, $2y$ by $(y + y_1)$, $2xy$ by $(xy_1 + yx_1)$ & c by (c) .

- (b) **Slope form :** Equation of tangent to the given ellipse whose slope is 'm', is $y = mx \pm \sqrt{a^2 m^2 + b^2}$

Point of contact are $\left(\frac{\mp a^2 m}{\sqrt{a^2 m^2 + b^2}}, \frac{\pm b^2}{\sqrt{a^2 m^2 + b^2}} \right)$

Note that there are two tangents to the ellipse having the same m , i.e. there are two tangents parallel to any given direction.

- (c) **Parametric form :** Equation of tangent to the given ellipse at its point $(a \cos \theta, b \sin \theta)$, is

$$\frac{x \cos \theta}{a} + \frac{y \sin \theta}{b} = 1$$

Note :

- (i) The eccentric angles of point of contact of two parallel tangents differ by π .

- (ii) Point of intersection of the tangents at the point α & β is $\left(a \frac{\cos \frac{\alpha+\beta}{2}}{\cos \frac{\alpha-\beta}{2}}, b \frac{\sin \frac{\alpha+\beta}{2}}{\cos \frac{\alpha-\beta}{2}} \right)$

Illustration 7: Find the equations of the tangents to the ellipse $3x^2 + 4y^2 = 12$ which are perpendicular to the line $y + 2x = 4$.

Solution : Let m be the slope of the tangent, since the tangent is perpendicular to the line $y + 2x = 4$.

$$\therefore mx - 2 = -1 \Rightarrow m = \frac{1}{2}$$

Since $3x^2 + 4y^2 = 12$ or $\frac{x^2}{4} + \frac{y^2}{3} = 1$

Comparing this with $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

$$\therefore a^2 = 4 \text{ and } b^2 = 3$$

So the equation of the tangent are $y = \frac{1}{2}x \pm \sqrt{4 \times \frac{1}{4} + 3}$

$$\Rightarrow y = \frac{1}{2}x \pm 2 \text{ or } x - 2y \pm 4 = 0.$$

Ans.

Illustration 8 : The tangent at a point P on an ellipse intersects the major axis in T and N is the foot of the perpendicular from P to the same axis. Show that the circle drawn on NT as diameter intersects the auxiliary circle orthogonally.

Solution : Let the equation of the ellipse be $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. Let $P(a\cos\theta, b\sin\theta)$ be a point on the ellipse.

The equation of the tangent at P is $\frac{x\cos\theta}{a} + \frac{y\sin\theta}{b} = 1$. It meets the major axis at $T \equiv (a\sec\theta, 0)$.

The coordinates of N are $(a\cos\theta, 0)$. The equation of the circle with NT as its diameter is $(x - a\sec\theta)(x - a\cos\theta) + y^2 = 0$.

$$\Rightarrow x^2 + y^2 - ax(\sec\theta + \cos\theta) + a^2 = 0$$

It cuts the auxiliary circle $x^2 + y^2 - a^2 = 0$ orthogonally if

$$2g \cdot 0 + 2f \cdot 0 = a^2 - a^2 = 0, \text{ which is true.}$$

Ans.**Do yourself - 4 :**

- (i) Find the equation of the tangents to the ellipse $9x^2 + 16y^2 = 144$ which are parallel to the line $x + 3y + k = 0$.
- (ii) Find the equation of the tangent to the ellipse $7x^2 + 8y^2 = 100$ at the point $(2, -3)$.

9. NORMAL TO THE ELLIPSE $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$:

- (a) **Point form :** Equation of the normal to the given ellipse at (x_1, y_1) is

$$\frac{a^2x}{x_1} - \frac{b^2y}{y_1} = a^2 - b^2 = a^2e^2.$$

- (b) **Slope form :** Equation of a normal to the given ellipse whose slope is 'm' is

$$y = mx \mp \frac{(a^2 - b^2)m}{\sqrt{a^2 + b^2m^2}}.$$

- (c) **Parametric form :** Equation of the normal to the given ellipse at the point $(a\cos\theta, b\sin\theta)$ is $ax\sec\theta - by\csc\theta = (a^2 - b^2)$.

Illustration 9 : Find the condition that the line $\ell x + my = n$ may be a normal to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

Solution : Equation of normal to the given ellipse at $(a\cos\theta, b\sin\theta)$ is $\frac{ax}{\cos\theta} - \frac{by}{\sin\theta} = a^2 - b^2$... (i)

If the line $\ell x + my = n$ is also normal to the ellipse then there must be a value of θ for which line (i) and line $\ell x + my = n$ are identical. For that value of θ we have

$$\left(\frac{\ell}{a}\right) \cos\theta = \left(\frac{m}{b}\right) \sin\theta = \frac{n}{(a^2 - b^2)} \quad \text{or} \quad \cos\theta = \frac{an}{\ell(a^2 - b^2)} \quad \dots (iii)$$

$$\text{and} \quad \sin\theta = \frac{-bn}{m(a^2 - b^2)} \quad \dots (iv)$$

Squaring and adding (iii) and (iv), we get $1 = \frac{n^2}{(a^2 - b^2)^2} \left(\frac{a^2}{\ell^2} + \frac{b^2}{m^2} \right)$ which is the required condition.

Illustration 10 : If the normal at an end of a latus-rectum of an ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ passes through one

extremity of the minor axis, show that the eccentricity of the ellipse is given by $e = \sqrt{\frac{\sqrt{5}-1}{2}}$

Solution :

The co-ordinates of an end of the latus-rectum are $(ae, b^2/a)$.

The equation of normal at $P(ae, b^2/a)$ is

$$\frac{a^2x}{ae} - \frac{b^2(y)}{b^2/a} = a^2 - b^2 \quad \text{or} \quad \frac{ax}{e} - ay = a^2 - b^2$$

It passes through one extremity of the minor axis

whose co-ordinates are $(0, -b)$

$$\therefore 0 + ab = a^2 - b^2 \Rightarrow (a^2b^2) = (a^2 - b^2)^2$$

$$\Rightarrow a^2 \cdot a^2(1 - e^2) = (a^2 e^2)^2 \Rightarrow 1 - e^2 = e^4$$

$$\Rightarrow e^4 + e^2 - 1 = 0 \Rightarrow (e^2)^2 + e^2 - 1 = 0$$

$$\therefore e^2 = \frac{-1 \pm \sqrt{1+4}}{2} \Rightarrow e = \sqrt{\frac{\sqrt{5}-1}{2}} \quad (\text{taking positive sign})$$

Ans.

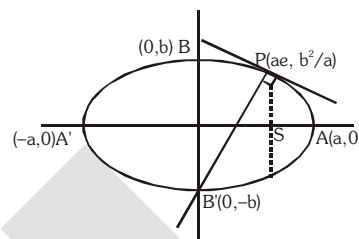


Illustration 11 : P and Q are corresponding points on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ and the auxiliary circles respectively. The normal at P to the ellipse meets CQ in R, where C is the centre of the ellipse. Prove that $CR = a + b$

Solution :

Let $P \equiv (a \cos \theta, b \sin \theta)$

$\therefore Q \equiv (a \cos \theta, a \sin \theta)$

Equation of normal at P is

$$(a \sec \theta)x - (b \csc \theta)y = a^2 - b^2 \quad \dots\dots\dots (i)$$

$$\text{equation of CQ is } y = \tan \theta \cdot x \quad \dots\dots\dots (ii)$$

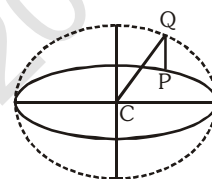
Solving equation (i) & (ii), we get $(a - b)x = (a^2 - b^2)\cos \theta$

$$x = (a + b) \cos \theta, \text{ \& } y = (a + b) \sin \theta$$

$$\therefore R \equiv ((a + b)\cos \theta, (a + b)\sin \theta)$$

$$\therefore CR = a + b$$

Ans.



Do yourself - 5 :

- (i) Find the equation of the normal to the ellipse $9x^2 + 16y^2 = 288$ at the point $(4, 3)$
- (ii) Let P be a variable point on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ with foci F_1 and F_2 . If A is the area of the triangle PF_1F_2 , then find maximum value of A.
- (iii) If the normal at the point $P(\theta)$ to the ellipse $\frac{x^2}{3} + \frac{y^2}{2} = 1$ intersects it again at the point $Q(2\theta)$, then find $\cos \theta$.
- (iv) Show that for all real values of 't' the line $2tx + y\sqrt{1-t^2} = 1$ touches a fixed ellipse. Find the eccentricity of the ellipse.

10. CHORD OF CONTACT :

If PA and PB be the tangents from point $P(x_1, y_1)$ to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

The equation of the chord of contact AB is $\frac{xx_1}{a^2} + \frac{yy_1}{b^2} = 1$ or **T = 0 (at x_1, y_1)**.

Illustration 12 : If tangents to the parabola $y^2 = 4ax$ intersect the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ at A and B, then find the locus of point of intersection of tangents at A and B.

Solution : Let $P \equiv (h, k)$ be the point of intersection of tangents at A & B

$$\therefore \text{equation of chord of contact AB is } \frac{xh}{a^2} + \frac{yk}{b^2} = 1 \quad \dots\dots\dots (i)$$

which touches the parabola.

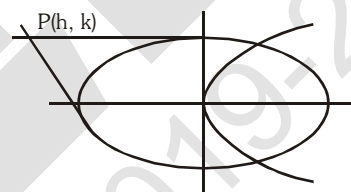
$$\text{Equation of tangent to parabola } y^2 = 4ax \text{ is } y = mx + \frac{a}{m}$$

$$\Rightarrow mx - y = -\frac{a}{m} \quad \dots\dots\dots (ii)$$

equation (i) & (ii) as must be same

$$\therefore \frac{\frac{m}{\left(\frac{h}{a^2}\right)}}{\left(\frac{k}{b^2}\right)} = \frac{-1}{1} = \frac{-a}{m} \Rightarrow m = -\frac{h b^2}{k a^2} \text{ \& } m = \frac{ak}{b^2}$$

$$\therefore -\frac{hb^2}{ka^2} = \frac{ak}{b^2} \Rightarrow \text{locus of P is } y^2 = -\frac{b^4}{a^3} \cdot x$$



Ans.

Do yourself - 6 :

(i) Find the equation of chord of contact to the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$ at the point (1, 3).

(ii) If the chord of contact of tangents from two points (x_1, y_1) and (x_2, y_2) to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ are at right angles, then find $\frac{x_1 x_2}{y_1 y_2}$.

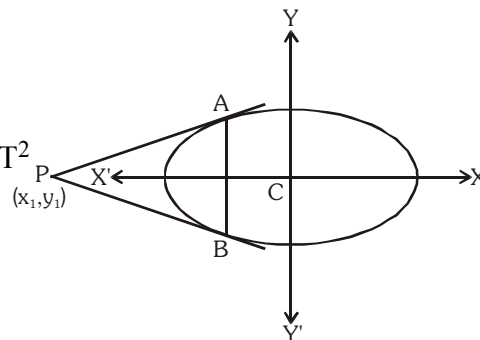
(iii) If a line $3x - y = 2$ intersects ellipse $\frac{x^2}{8} + \frac{y^2}{4} = 1$ at points A & B, then find co-ordinates of point of intersection of tangents at points A & B.

11. PAIR OF TANGENTS :

If $P(x_1, y_1)$ be any point lies outside the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, and a pair of tangents PA, PB can be drawn to it from P. Then the equation of pair of tangents of PA and PB is $SS_1 = T^2$

$$\text{where } S_1 = \frac{x_1^2}{a^2} + \frac{y_1^2}{b^2} - 1, \quad T = \frac{xx_1}{a^2} + \frac{yy_1}{b^2} - 1$$

$$\text{i.e. } \left(\frac{x^2}{a^2} + \frac{y^2}{b^2} - 1 \right) \left(\frac{x_1^2}{a^2} + \frac{y_1^2}{b^2} - 1 \right) = \left(\frac{xx_1}{a^2} + \frac{yy_1}{b^2} - 1 \right)^2$$



12. DIRECTOR CIRCLE :

Locus of the point of intersection of the tangents which meet at right angles is called the **Director Circle**. The equation to this locus is $x^2 + y^2 = a^2 + b^2$ i.e. a circle whose centre is the centre of the ellipse & whose radius is the length of the line joining the ends of the major & minor axis.

Illustration 13 : A tangent to the ellipse $x^2 + 4y^2 = 4$ meets the ellipse $x^2 + 2y^2 = 6$ at P and Q. Prove that the tangents at P and Q of the ellipse $x^2 + 2y^2 = 6$ are at right angles.

Solution : Given ellipse are $\frac{x^2}{4} + \frac{y^2}{1} = 1$ (i)

and, $\frac{x^2}{6} + \frac{y^2}{3} = 1$ (ii)

any tangent to (i) is $\frac{x \cos \theta}{2} + \frac{y \sin \theta}{1} = 1$ (iii)

It cuts (ii) at P and Q, and suppose tangent at P and Q meet at (h, k) Then equation of chord of contact of (h, k) with respect to ellipse (ii) is $\frac{hx}{6} + \frac{ky}{3} = 1$ (iv)

comparing (iii) and (iv), we get $\frac{\cos \theta}{h/3} = \frac{\sin \theta}{k/3} = 1$

$\Rightarrow \cos \theta = \frac{h}{3}$ and $\sin \theta = \frac{k}{3} \Rightarrow h^2 + k^2 = 9$

locus of the point (h, k) is $x^2 + y^2 = 9 \Rightarrow x^2 + y^2 = 6 + 3 = a^2 + b^2$
i.e. director circle of second ellipse. Hence the tangents are at right angles.

13. EQUATION OF CHORD WITH MID POINT (x_1, y_1) :

The equation of the chord of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, whose mid-point be (x_1, y_1) is $T = S_1$

where $T = \frac{xx_1}{a^2} + \frac{yy_1}{b^2} - 1$, $S_1 = \frac{x_1^2}{a^2} + \frac{y_1^2}{b^2} - 1$, i.e. $\left(\frac{xx_1}{a^2} + \frac{yy_1}{b^2} - 1\right) = \left(\frac{x_1^2}{a^2} + \frac{y_1^2}{b^2} - 1\right)$

Illustration 14 : Find the locus of the mid-point of focal chords of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

Solution : Let P $\equiv$ (h, k) be the mid-point

$\therefore$ equation of chord whose mid-point is given $\frac{xh}{a^2} + \frac{yk}{b^2} - 1 = \frac{h^2}{a^2} + \frac{k^2}{b^2} - 1$

since it is a focal chord,

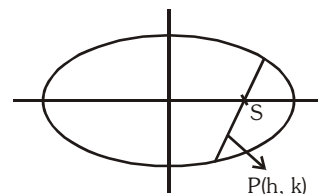
$\therefore$ It passes through focus, either (ae, 0) or (-ae, 0)

If it passes through (ae, 0)

$\therefore$ locus is $\frac{ex}{a} = \frac{x^2}{a^2} + \frac{y^2}{b^2}$

If it passes through (-ae, 0)

$\therefore$ locus is $-\frac{ex}{a} = \frac{x^2}{a^2} + \frac{y^2}{b^2}$



Ans.

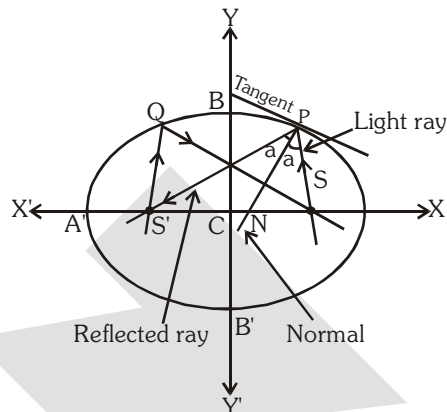
Do yourself - 7 :

- (i) Find the equation of chord of the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$ whose mid point be $(-1, 1)$.

14. IMPORTANT POINTS :

Referring to an ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

- (a) If P be any point on the ellipse with S & S' as its foci then $\ell(SP) + \ell(S'P) = 2a$.
- (b) The tangent & normal at a point P on the ellipse bisect the external & internal angles between the focal distances of P. This refers to the well known reflection property of the ellipse which states that rays from one focus are reflected through other focus & vice versa.
- (c) **The product of the length's of the perpendicular segments from the foci on any tangent to the ellipse is b^2** and the feet of these perpendiculars lie on its auxiliary circle and the tangents at these feet to the auxiliary circle meet on the ordinate of P and that the locus of their point of intersection is a similar ellipse as that of the original one.
- (d) The portion of the tangent to an ellipse between the point of contact & the directrix subtends a **right angle** at the corresponding focus.
- (e) If the normal at any point P on the ellipse with centre C meet the major & minor axes in G & g respectively, & if CF be perpendicular upon this normal, then
- (i) $PF \cdot PG = b^2$
 - (ii) $PF \cdot Pg = a^2$
 - (iii) $PG \cdot Pg = SP \cdot S'P$
 - (iv) $CG \cdot CT = CS^2$
 - (v) locus of the mid point of Gg is another ellipse having the same eccentricity as that of the original ellipse.



[where S and S' are the foci of the ellipse and T is the point where tangent at P meet the major axis]

- (f) Atmost four normals & two tangents can be drawn from any point to an ellipse.
- (g) The circle on any focal distance as diameter touches the auxiliary circle.
- (h) Perpendiculars from the centre upon all chords which join the ends of any perpendicular diameters of the ellipse are of constant length.
- (i) If the tangent at the point P of a standard ellipse meets the axes in T and t and CY is the perpendicular on it from the centre then,
- (i) $Tt \cdot PY = a^2 - b^2$ and
 - (ii) least value of Tt is $a + b$.

Do yourself - 8 :

- (i) A man running round a racecourse note that the sum of the distance of two flag-posts from him is always 20 meters and distance between the flag-posts is 16 meters. Find the area of the path he encloses in square meters
- (ii) If chord of contact of the tangent drawn from the point (α, β) to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ touches the circle $x^2 + y^2 = k^2$, then find the locus of the point (α, β) .

Miscellaneous Illustration :

Illustration 15 : A point moves so that the sum of the squares of its distances from two intersecting straight lines is constant. Prove that its locus is an ellipse.

Solution : Let two intersecting lines OA and OB, intersect at origin O and let both lines OA and OB makes equal angles with x axis.

i.e., $\angle XO A = \angle XO B = \theta$.

$\therefore$ Equations of straight lines OA and OB are

$y = x \tan \theta$ and $y = -x \tan \theta$

or $x \sin \theta - y \cos \theta = 0$ (i)

and $x \sin \theta + y \cos \theta = 0$ (ii)

Let $P(\alpha, \beta)$ is the point whose locus is to be determine.

According to the example $(PM)^2 + (PN)^2 = 2\lambda^2$ (say)

$$\therefore (\alpha \sin \theta + \beta \cos \theta)^2 + (\alpha \sin \theta - \beta \cos \theta)^2 = 2\lambda^2 \Rightarrow 2\alpha^2 \sin^2 \theta + 2\beta^2 \cos^2 \theta = 2\lambda^2$$

$$\text{or } \alpha^2 \sin^2 \theta + \beta^2 \cos^2 \theta = \lambda^2 \Rightarrow \frac{\alpha^2}{\lambda^2 \operatorname{cosec}^2 \theta} + \frac{\beta^2}{\lambda^2 \sec^2 \theta} = 1 \Rightarrow \frac{\alpha^2}{(\lambda \operatorname{cosec} \theta)^2} + \frac{\beta^2}{(\lambda \sec \theta)^2} = 1$$

$$\text{Hence required locus is } \frac{x^2}{(\lambda \operatorname{cosec} \theta)^2} + \frac{y^2}{(\lambda \sec \theta)^2} = 1$$

Ans.

Illustration 16 : Find the condition on 'a' and 'b' for which two distinct chords of the ellipse $\frac{x^2}{2a^2} + \frac{y^2}{2b^2} = 1$ passing through $(a, -b)$ are bisected by the line $x + y = b$.

Solution : Let $(t, b - t)$ be a point on the line $x + y = b$.

Then equation of chord whose mid point $(t, b - t)$ is

$$\frac{tx}{2a^2} + \frac{y(b-t)}{2b^2} - 1 = \frac{t^2}{2a^2} + \frac{(b-t)^2}{2b^2} - 1 \quad \text{..... (i)}$$

$$(a, -b) \text{ lies on (i) then } \frac{ta}{2a^2} - \frac{b(b-t)}{2b^2} = \frac{t^2}{2a^2} + \frac{(b-t)^2}{2b^2} \Rightarrow t^2(a^2 + b^2) - ab(3a + b)t + 2a^2b^2 = 0$$

$$\text{Since } t \text{ is real } B^2 - 4AC \geq 0 \Rightarrow a^2b^2(3a + b)^2 - 4(a^2 + b^2)2a^2b^2 \geq 0$$

$$\Rightarrow a^2 + 6ab - 7b^2 \geq 0 \Rightarrow a^2 + 6ab \geq 7b^2, \text{ which is the required condition.}$$

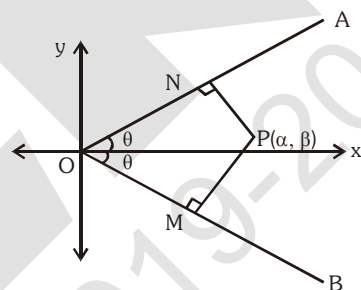


Illustration 17 : Any tangent to an ellipse is cut by the tangents at the ends of the major axis in T and T'. Prove that circle on TT' as diameter passes through foci.

Solution : Let ellipse be $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

and let $P(a\cos\phi, b\sin\phi)$ be any point on this ellipse

$\therefore$ Equation of tangent at $P(a\cos\phi, b\sin\phi)$ is

$$\frac{x}{a}\cos\phi + \frac{y}{b}\sin\phi = 1 \quad \dots(i)$$

The two tangents drawn at the ends of the major axis are $x = a$ and $x = -a$

$$\text{Solving (i) and } x = a \text{ we get } T = \left\{ a, \frac{b(1 - \cos\phi)}{\sin\phi} \right\} \equiv \left\{ a, b \tan\left(\frac{\phi}{2}\right) \right\}$$

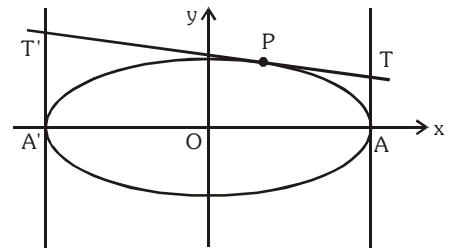
$$\text{and solving (i) and } x = -a \text{ we get } T' = \left\{ -a, \frac{b(1 + \cos\phi)}{\sin\phi} \right\} \equiv \left\{ -a, b \cot\left(\frac{\phi}{2}\right) \right\}$$

Equation of circle on TT' as diameter is $(x-a)(x+a) + (y-b\tan(\phi/2))(y-b\cot(\phi/2)) = 0$
or $x^2 + y^2 - by(\tan(\phi/2) + \cot(\phi/2)) - a^2 + b^2 = 0 \quad \dots(ii)$

Now put $x = \pm ae$ and $y = 0$ in LHS of (ii), we get

$$a^2e^2 + 0 - 0 - a^2 + b^2 = a^2 - b^2 - a^2 + b^2 = 0 = \text{RHS}$$

Hence foci lie on this circle



ANSWERS FOR DO YOURSELF

1: (i) $e = \frac{1}{\sqrt{2}}$ (ii) $\frac{(x-10)^2}{52} + \frac{(y-6)^2}{16} = 1$ (iii) $e = \frac{\sqrt{3}}{2}$; foci = $(1 \pm \sqrt{3}, -1)$; LR = 1

2: (i) $\frac{x^2}{4} + \frac{y^2}{8} = 1$

(ii) $C \equiv (-1, 2)$, length of major axis = $2b = \sqrt{3}$, length of minor axis = $2a = 1$; $e = \sqrt{\frac{2}{3}}$;

$$f\left(-1, 2 \pm \frac{1}{\sqrt{2}}\right)$$

(iii) C

3: (i) On the ellipse (ii) $\frac{1}{2}(a^2 + b^2)$ (iii) $P^2 = a^2\cos^2\theta + b^2\sin^2\theta$

4: (i) $3y + x \pm \sqrt{97} = 0$ (ii) $7x - 12y = 50$

5: (i) $4x - 3y = 7$ (ii) abe (iii) -1 (iv) $\frac{\sqrt{3}}{2}$

6: (i) $\frac{x}{16} + \frac{y}{3} = 1$ (ii) $-\frac{a^4}{b^4}$ (iii) $(12, -2)$

7: (i) $-9x + 16y = 25$ 8: (i) 60π (ii) $\frac{x^2}{a^4} + \frac{y^2}{b^4} = \frac{1}{k^2}$

EXERCISE (O-1)

[STRAIGHT OBJECTIVE TYPE]

- Let 'E' be the ellipse $\frac{x^2}{9} + \frac{y^2}{4} = 1$ & 'C' be the circle $x^2 + y^2 = 9$. Let P & Q be the points (1, 2) and (2, 1) respectively. Then :
 (A) Q lies inside C but outside E
 (B) Q lies outside both C & E
 (C) P lies inside both C & E
 (D) P lies inside C but outside E.
- The eccentricity of the ellipse $(x - 3)^2 + (y - 4)^2 = \frac{y^2}{9}$ is
 (A) $\frac{\sqrt{3}}{2}$
 (B) $\frac{1}{3}$
 (C) $\frac{1}{3\sqrt{2}}$
 (D) $\frac{1}{\sqrt{3}}$
- The equation, $2x^2 + 3y^2 - 8x - 18y + 35 = K$ represents
 (A) no locus if $K > 0$
 (B) an ellipse if $K < 0$
 (C) a point if $K = 0$
 (D) a hyperbola if $K > 0$
- If the ellipse $\frac{(x-h)^2}{M} + \frac{(y-k)^2}{N} = 1$ has major axis on the line $y = 2$, minor axis on the line $x = -1$, major axis has length 10 and minor axis has length 4. The number h,k,M,N (in this order only) are -
 (A) -1,2,5,2
 (B) -1,2,10,4
 (C) 1,-2,25,4
 (D) -1,2,25,4
- The y-axis is the directrix of the ellipse with eccentricity $e = 1/2$ and the corresponding focus is at (3, 0), equation to its auxiliary circle is
 (A) $x^2 + y^2 - 8x + 12 = 0$
 (B) $x^2 + y^2 - 8x - 12 = 0$
 (C) $x^2 + y^2 - 8x + 9 = 0$
 (D) $x^2 + y^2 = 4$
- Imagine that you have two thumbtacks placed at two points, A and B. If the ends of a fixed length of string are fastened to the thumbtacks and the string is drawn taut with a pencil, the path traced by the pencil will be an ellipse. The best way to maximise the area surrounded by the ellipse with a fixed length of string occurs when
 I the two points A and B have the maximum distance between them.
 II two points A and B coincide.
 III A and B are placed vertically.
 IV The area is always same regardless of the location of A and B .
 (A) I
 (B) II
 (C) III
 (D) IV
- The latus rectum of a conic section is the width of the function through the focus. The positive difference between the length of the latus rectum of $3y = x^2 + 4x - 9$ and $x^2 + 4y^2 - 6x + 16y = 24$ is -
 (A) $\frac{1}{2}$
 (B) 2
 (C) $\frac{3}{2}$
 (D) $\frac{5}{2}$
- Let S(5,12) and S'(-12,5) are the foci of an ellipse passing through the origin. The eccentricity of ellipse equals -
 (A) $\frac{1}{2}$
 (B) $\frac{1}{\sqrt{3}}$
 (C) $\frac{1}{\sqrt{2}}$
 (D) $\frac{2}{3}$

9. A circle has the same centre as an ellipse & passes through the foci F_1 & F_2 of the ellipse, such that the two curves intersect in 4 points. Let 'P' be any one of their point of intersection. If the major axis of the ellipse is 17 & the area of the triangle PF_1F_2 is 30, then the distance between the foci is :
 (A) 11 (B) 12 (C) 13 (D) none
10. An ellipse is inscribed in a circle and a point within the circle is chosen at random. If the probability that this point lies outside the ellipse is $\frac{2}{3}$ then the eccentricity of the ellipse is :
 (A) $\frac{2\sqrt{2}}{3}$ (B) $\frac{\sqrt{5}}{3}$ (C) $\frac{8}{9}$ (D) $\frac{2}{3}$
11. (a) Which of the following is an equation of the ellipse with centre $(-2, 1)$, major axis running from $(-2, 6)$ to $(-2, -4)$ and focus at $(-2, 5)$?
 (A) $\frac{(x-2)^2}{25} + \frac{(y+1)^2}{16} = 1$ (B) $\frac{(x+2)^2}{25} + \frac{(y-1)^2}{9} = 1$
 (C) $\frac{(x-2)^2}{9} + \frac{(y+1)^2}{25} = 1$ (D) $\frac{(x+2)^2}{9} + \frac{(y-1)^2}{25} = 1$
 (b) Which of the following statement(s) is/are correct for the ellipse of 11(a) ?
 (A) auxiliary circle is $(x+2)^2 + (y-1)^2 = 25$
 (B) director circle is $(x+2)^2 + (y-1)^2 = 34$
 (C) Latus rectum = $\frac{18}{5}$
 (D) eccentricity = $\frac{4}{5}$
12. $x - 2y + 4 = 0$ is a common tangent to $y^2 = 4x$ & $\frac{x^2}{4} + \frac{y^2}{b^2} = 1$. Then the value of b and the other common tangent are given by :
 (A) $b = \sqrt{3}$; $x + 2y + 4 = 0$ (B) $b = 3$; $x + 2y + 4 = 0$
 (C) $b = \sqrt{3}$; $x + 2y - 4 = 0$ (D) $b = \sqrt{3}$; $x - 2y - 4 = 0$
13. Consider the particle travelling clockwise on the elliptical path $\frac{x^2}{100} + \frac{y^2}{25} = 1$. The particle leaves the orbit at the point $(-8, 3)$ and travels in a straight line tangent to the ellipse. At what point will the particle cross the y-axis?
 (A) $\left(0, \frac{25}{3}\right)$ (B) $\left(0, \frac{23}{3}\right)$ (C) $(0, 9)$ (D) $\left(0, \frac{26}{3}\right)$

[MULTIPLE OBJECTIVE TYPE]

14. Consider the ellipse $\frac{x^2}{\tan^2 \alpha} + \frac{y^2}{\sec^2 \alpha} = 1$ where $\alpha \in (0, \pi/2)$.
 Which of the following quantities would vary as α varies ?
 (A) degree of flatness (B) ordinate of the vertex
 (C) coordinates of the foci (D) length of the latus rectum
15. The equation of the common tangents of the parabola $y^2 = 4x$ and an ellipse $\frac{x^2}{4} + \frac{y^2}{3} = 1$ are -
 (A) $x - 2y + 4 = 0$ (B) $x + 2y + 4 = 0$ (C) $2x - y + 1 = 0$ (D) $2x + y + 1 = 0$

16. Consider an ellipse $\frac{x^2}{a} + \frac{y^2}{b} = 1$. If A_1 and A_2 denotes area of ellipse and its director circle respectively, then-
- (A) $A_2 \geq 2A_1$ (B) $A_2 < 2A_1$
- (C) if $a = 4b$, then $A_1 = \frac{2}{5}A_2$ (D) if $a = 4b$, then $A_2 = \frac{2}{5}A_1$
17. If length of perpendicular drawn from origin to any normal of the ellipse $\frac{x^2}{16} + \frac{y^2}{25} = 1$ is ℓ , then ℓ cannot be -
- (A) 4 (B) $\frac{5}{2}$ (C) $\frac{1}{2}$ (D) $\frac{2}{3}$

[REASONING TYPE]

- 18. Statement-1 :** The ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$ and $\frac{x^2}{9} + \frac{y^2}{16} = 1$ are congruent.

and

Statement-2 : The ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$ and $\frac{x^2}{9} + \frac{y^2}{16} = 1$ have the same eccentricity.

- (A) Statement-1 is True, Statement-2 is True ; Statement-2 is a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True ; Statement-2 is NOT a correct explanation for Statement-1.
(C) Statement-1 is True, Statement-2 is False.
(D) Statement-1 is False, Statement-2 is True.

- 19.** Tangents are drawn from the point $P(-\sqrt{3}, \sqrt{2})$ to an ellipse $4x^2 + y^2 = 4$.

Statement-1 : The tangents are mutually perpendicular.

and

Statement-2 : The locus of the points from which mutually perpendicular tangents can be drawn to given ellipse is $x^2 + y^2 = 5$.

- (A) Statement-1 is True, Statement-2 is True ; Statement-2 is a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True ; Statement-2 is NOT a correct explanation for Statement-1.
(C) Statement-1 is True, Statement-2 is False.
(D) Statement-1 is False, Statement-2 is True.

[COMPREHENSION TYPE]

Paragraph for question nos. 20 to 22

Consider the ellipse $\frac{x^2}{9} + \frac{y^2}{4} = 1$ and the parabola $y^2 = 2x$. They intersect at P and Q in the first and fourth quadrants respectively. Tangents to the ellipse at P and Q intersect the x-axis at R and tangents to the parabola at P and Q intersect the x-axis at S.

- 20.** The ratio of the areas of the triangles PQS and PQR, is

- (A) 1:3 (B) 1:2 (C) 2:3 (D) 3:4

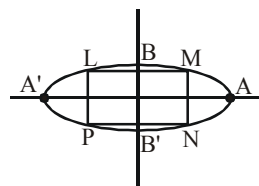
- 21.** The area of quadrilateral PRQS, is

- (A) $\frac{3\sqrt{15}}{2}$ (B) $\frac{15\sqrt{3}}{2}$ (C) $\frac{5\sqrt{3}}{2}$ (D) $\frac{5\sqrt{15}}{2}$

4. The normal at a variable point P on an ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ of eccentricity e meets the axes of the ellipse in Q and R then the locus of the mid-point of QR is a conic with an eccentricity e' such that :
 (A) e' is independent of e (B) $e' = 1$
 (C) $e' = e$ (D) $e' = 1/e$
5. The area of the rectangle formed by the perpendiculars from the centre of the standard ellipse to the tangent and normal at its point whose eccentric angle is $\pi/4$, is :
 (A) $\frac{(a^2 - b^2)ab}{a^2 + b^2}$ (B) $\frac{(a^2 - b^2)}{(a^2 + b^2)ab}$ (C) $\frac{(a^2 - b^2)}{ab(a^2 + b^2)}$ (D) $\frac{a^2 + b^2}{(a^2 - b^2)ab}$
6. The locus of the middle point of chords of an ellipse $\frac{x^2}{16} + \frac{y^2}{25} = 1$ passing through P(0, 5) is another ellipse E. The coordinates of the foci of the ellipse E, is
 (A) $\left(0, \frac{3}{5}\right)$ and $\left(0, \frac{-3}{5}\right)$ (B) (0, -4) and (0, 1)
 (C) (0, 4) and (0, 1) (D) $\left(0, \frac{11}{2}\right)$ and $\left(0, \frac{-1}{2}\right)$

[MULTIPLE OBJECTIVE TYPE]

7. Extremities of the latus rectum of the ellipses $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ ($a > b$) having a given major axis $2a$ lies on-
 (A) $x^2 = a(a - y)$ (B) $x^2 = a(a + y)$ (C) $y^2 = a(a + x)$ (D) $y^2 = a(a - x)$
8. If a number of ellipse (whose axes are x & y axes) be described having the same major axis $2a$ but a variable minor axis then the tangents at the ends of their latus rectum pass through fixed points which can be -
 (A) (0, a) (B) (0, 0) (C) (0, -a) (D) (a, a)
9. Tangents are drawn from any point on the ellipse $\frac{x^2}{9} + \frac{y^2}{4} = 1$ to the circle $x^2 + y^2 = 1$ and respective chord of contact always touches a conic 'C', then -
 (A) minimum distance between 'C' & ellipse is $\frac{3}{2}$
 (B) maximum distance between 'C' & ellipse is $\frac{10}{3}$
 (C) eccentricity of 'C' is $\frac{\sqrt{5}}{3}$
 (D) product of eccentricity of 'C' & ellipse is 1
10. Two lines are drawn from point P(α , β) which touches $y^2 = 8x$ at A, B and touches $\frac{x^2}{4} + \frac{y^2}{6} = 1$ at C, D, then -
 (A) $\alpha + \beta = -4$ (B) $\alpha\beta = 4$
 (C) Area of triangle PAB is $128\sqrt{2}$ (D) Area of triangle PAB is $32\sqrt{2}$



11. Let LMNP be a rectangle (which is not a square) inscribed in the ellipse as shown in the figure. Let ' λ ' denotes the number of ways in which we can select four points out of A, A', B, B', L, M, P and N such that normal at these four points are concurrent, then ' λ ' is less than or equal to-
- (A) 5 (B) 6 (C) 7 (D) 4
12. Let P(r, s), ($r < 0$) is a variable point on ellipse $C_1 : \frac{x^2}{16} + \frac{y^2}{9} = 1$. PA and PB are two tangents of parabola $C_2 : y^2 - 4x = 0$, where A and B are point of contacts, then -
- (A) number of common tangents of curves C_1 and C_2 is 4.
 (B) minimum integral value of $r + s$ is -5
 (C) mid point of chord AB lie on curve $\frac{y^2}{9} + \frac{(y^2 - 2x)^2}{64} = 1$
 (D) mid point of chord AB lie on curve $\frac{x^2}{9} + \frac{(y^2 - 2x)^2}{64} = 1$

[COMPREHENSION TYPE]

Paragraph for question nos. 13 to 16

Let the two foci of an ellipse be $(-1, 0)$ and $(3, 4)$ and the foot of perpendicular from the focus $(3, 4)$ upon a tangent to the ellipse be $(4, 6)$.

13. The foot of perpendicular from the focus $(-1, 0)$ upon the same tangent to the ellipse is
- (A) $\left(\frac{12}{5}, \frac{34}{5}\right)$ (B) $\left(\frac{7}{3}, \frac{11}{3}\right)$ (C) $\left(2, \frac{17}{4}\right)$ (D) $(-1, 2)$
14. The equation of auxiliary circle of the ellipse is
- (A) $x^2 + y^2 - 2x - 4y - 5 = 0$ (B) $x^2 + y^2 - 2x - 4y - 20 = 0$
 (C) $x^2 + y^2 + 2x + 4y - 20 = 0$ (D) $x^2 + y^2 + 2x + 4y - 5 = 0$
15. The length of semi-minor axis of the ellipse is
- (A) 1 (B) $2\sqrt{2}$ (C) $\sqrt{17}$ (D) $\sqrt{19}$
16. The equations of directrices of the ellipse are
- (A) $x - y + 2 = 0$, $x - y - 5 = 0$ (B) $x + y - \frac{21}{2} = 0$, $x + y + \frac{17}{2} = 0$
 (C) $x - y + \frac{3}{2} = 0$, $x - y - \frac{5}{2} = 0$ (D) $x + y - \frac{31}{2} = 0$, $x + y + \frac{19}{2} = 0$

EXERCISE (S-1)

1. (a) Find the equation of the ellipse with its centre $(1, 2)$, focus at $(6, 2)$ and passing through the point $(4, 6)$.
 (b) An ellipse passes through the points $(-3, 1)$ & $(2, -2)$ & its principal axis are along the coordinate axes in order. Find its equation.
2. If any two chords be drawn through two points on the major axis of an ellipse equidistant from the centre, show that $\tan \frac{\alpha}{2} \cdot \tan \frac{\beta}{2} \cdot \tan \frac{\gamma}{2} \cdot \tan \frac{\delta}{2} = 1$, where $\alpha, \beta, \gamma, \delta$ are the eccentric angles of the extremities of the chords.

15. Consider the family of circles, $x^2 + y^2 = r^2$, $2 < r < 5$. If in the first quadrant, the common tangent to a circle of the family and the ellipse $4x^2 + 25y^2 = 100$ meets the co-ordinate axes at A & B, then find the equation of the locus of the mid-point of AB.
16. If the normal at the point P(θ) to the ellipse $\frac{x^2}{14} + \frac{y^2}{5} = 1$, intersects it again at the point Q(2 θ), show that $\cos \theta = -(2/3)$.
17. Find the equation of the largest circle with centre (1, 0) that can be inscribed in the ellipse $x^2 + 4y^2 = 16$.
18. Let P_i and P_i' be the feet of the perpendiculars drawn from foci S, S' on a tangent T_i to an ellipse whose length of semi-major axis is 20. If $\sum_{i=1}^{10} (SP_i)(S'P_i') = 2560$, then find the value of $100e$.
19. A ray emanating from the point (-4, 0) is incident on the ellipse $9x^2 + 25y^2 = 225$ at the point P with abscissa 3. Find the equation of the reflected ray after first reflection.
20. If s, s' are the length of the perpendicular on a tangent from the foci, a, a' are those from the vertices, c is that from the centre and e is the eccentricity of the ellipse, $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, then prove that $\frac{ss'-c^2}{aa'-c^2} = e^2$.
21. An ellipse has foci at $F_1(9, 20)$ and $F_2(49, 55)$ in the xy-plane and is tangent to the x-axis. Find the length of its major axis.
22. Prove that, in an ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, the perpendicular from a focus upon any tangent and the line joining the centre of the ellipse to the point of contact meet on the corresponding directrix.

EXERCISE (S-2)

1. Rectangle ABCD has area 200. An ellipse with area 200π passes through A and C and has foci at B and D. Find the perimeter of the rectangle.
2. A tangent to the ellipse $x^2 + 4y^2 = 4$ meets the ellipse $x^2 + 2y^2 = 6$ at P & Q. Prove that the tangents at P & Q of the ellipse $x^2 + 2y^2 = 6$ are at right angles.
3. A straight line AB touches the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ & the circle $x^2 + y^2 = r^2$; where $a > r > b$. A focal chord of the ellipse, parallel to AB intersects the circle in P & Q, find the length of the perpendicular drawn from the centre of the ellipse to PQ. Hence show that $PQ = 2b$.
4. If the tangent at any point of an ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ makes an angle α with the major axis and an angle β with the focal radius of the point of contact then show that the eccentricity 'e' of the ellipse is given by the absolute value of $\frac{\cos \beta}{\cos \alpha}$.
5. A normal inclined at 45° to the axis of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is drawn. It meets the x-axis & the y-axis in P & Q respectively. If C is the centre of the ellipse, show that the area of triangle CPQ is $\frac{(a^2 - b^2)^2}{2(a^2 + b^2)}$ sq. units.

6. PG is the normal to a standard ellipse at P, G being on the major axis. GP is produced outwards to Q so that $PQ = GP$. Show that the locus of Q is an ellipse whose eccentricity is $\frac{a^2 - b^2}{a^2 + b^2}$.
7. Consider the parabola $y^2 = 4x$ and the ellipse $2x^2 + y^2 = 6$, intersecting at P and Q.
 (a) Prove that the two curves are orthogonal.
 (b) Find the area enclosed by the parabola and the common chord of the ellipse and parabola.
 (c) If tangent and normal at the point P on the ellipse intersect the x-axis at T and G respectively then find the area of the triangle PTG.
8. The tangents from (x_1, y_1) to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ intersect at right angles. Show that the normals at the points of contact meet on the line $\frac{y}{y_1} = \frac{x}{x_1}$.
9. Find the number of integral values of parameter 'a' for which three chords of the ellipse $\frac{x^2}{2a^2} + \frac{y^2}{a^2} = 1$ (other than its diameter) passing through the point $P\left(11a, -\frac{a^2}{4}\right)$ are bisected by the parabola $y^2 = 4ax$.
10. Prove that the equation to the circle, having double contact with the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ (with eccentricity e) at the ends of a latus rectum, is $x^2 + y^2 - 2ae^3x = a^2(1 - e^2 - e^4)$.
11. Consider the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ with centre 'O' where $a > b > 0$. Tangent at any point P on the ellipse meets the coordinate axes at X and Y and N is the foot of the perpendicular from the origin on the tangent at P. Minimum length of XY is 36 and maximum length of PN is 4.
 (a) Find the eccentricity of the ellipse.
 (b) Find the maximum area of an isosceles triangle inscribed in the ellipse if one of its vertex coincides with one end of the major axis of the ellipse.
 (c) Find the maximum area of the triangle OPN.
12. Consider an ellipse $\frac{x^2}{25} + \frac{y^2}{9} = 1$ with centre C and a point P on it with eccentric angle $\frac{\pi}{4}$. Normal drawn at P intersects the major and minor axes in A and B respectively. N_1 and N_2 are the feet of the perpendiculars from the foci S_1 and S_2 respectively on the tangent at P and N is the foot of the perpendicular from the centre of the ellipse on the normal at P. Tangent at P intersects the axis of x at T.

Match the entries of **Column-I** with the entries of **Column-II**.

Column-I	Column-II
(A) (CA)(CT) is equal to	(P) 9
(B) (PN)(PB) is equal to	(Q) 16
(C) $(S_1N_1)(S_2N_2)$ is equal to	(R) 17
(D) $(S_1P)(S_2P)$ is equal to	(S) 25

EXERCISE (JM)

- The ellipse $x^2 + 4y^2 = 4$ is inscribed in a rectangle aligned with the coordinate axes, which in turn is inscribed in another ellipse that passes through the point $(4, 0)$. Then the equation of the ellipse is :- [AIEEE-2009]
 (1) $4x^2 + 48y^2 = 48$ (2) $4x^2 + 64y^2 = 48$ (3) $x^2 + 16y^2 = 16$ (4) $x^2 + 12y^2 = 16$
- Equation of the ellipse whose axes are the axes of coordinates and which passes through the point $(-3, 1)$ and has eccentricity $\sqrt{2/5}$ is :- [AIEEE-2011]
 (1) $3x^2 + 5y^2 - 15 = 0$ (2) $5x^2 + 3y^2 - 32 = 0$
 (3) $3x^2 + 5y^2 - 32 = 0$ (4) $5x^2 + 3y^2 - 48 = 0$
- An ellipse is drawn by taking a diameter of the circle $(x - 1)^2 + y^2 = 1$ as its semi-minor axis and a diameter of the circle $x^2 + (y - 2)^2 = 4$ as its semi-major axis. If the centre of the ellipse is at the origin and its axes are the coordinate axes, then the equation of the ellipse is : [AIEEE-2012]
 (1) $x^2 + 4y^2 = 16$ (2) $4x^2 + y^2 = 4$ (3) $x^2 + 4y^2 = 8$ (4) $4x^2 + y^2 = 8$
- Statement-1 :** An equation of a common tangent to the parabola $y^2 = 16\sqrt{3}x$ and the ellipse $2x^2 + y^2 = 4$ is $y = 2x + 2\sqrt{3}$.
Statement-2 : If the line $y = mx + \frac{4\sqrt{3}}{m}$, ($m \neq 0$) is a common tangent to the parabola $y^2 = 16\sqrt{3}x$ and the ellipse $2x^2 + y^2 = 4$, then m satisfies $m^4 + 2m^2 = 24$. [AIEEE-2012]
 (1) Statement-1 is true, Statement-2 is false.
 (2) Statement-1 is false, Statement-2 is true.
 (3) Statement-1 is true, Statement-2 is true ; Statement-2 is a correct explanation for Statement-1.
 (4) Statement-1 is true, Statement-2 is true ; Statement-2 is not a correct explanation for Statement-1.
- The equation of the circle passing through the foci of the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$ and having centre at $(0, 3)$ is : [JEE (Main)-2013]
 (1) $x^2 + y^2 - 6y - 7 = 0$ (2) $x^2 + y^2 - 6y + 7 = 0$
 (3) $x^2 + y^2 - 6y - 5 = 0$ (4) $x^2 + y^2 - 6y + 5 = 0$
- If a and c are positive real number and the ellipse $\frac{x^2}{4c^2} + \frac{y^2}{c^2} = 1$ has four distinct points in common with the circle $x^2 + y^2 = 9a^2$, then [JEE-Main (On line)-2013]
 (1) $6ac + 9a^2 - 2c^2 > 0$ (2) $6ac + 9a^2 - 2c^2 < 0$
 (3) $9ac - 9a^2 - 2c^2 < 0$ (4) $9ac - 9a^2 - 2c^2 > 0$
- Equation of the line passing through the points of intersection of the parabola $x^2 = 8y$ and the ellipse $\frac{x^2}{3} + y^2 = 1$ is - [JEE-Main (On line)-2013]
 (1) $y + 3 = 0$ (2) $3y + 1 = 0$ (3) $3y - 1 = 0$ (4) $y - 3 = 0$
- Let the equations of two ellipses be $E_1 : \frac{x^2}{3} + \frac{y^2}{2} = 1$ and $E_2 : \frac{x^2}{16} + \frac{y^2}{b^2} = 1$. If the product of their eccentricities is $\frac{1}{2}$, then the length of the minor axis of ellipse E_2 is :- [JEE-Main (On line)-2013]
 (1) 9 (2) 8 (3) 2 (4) 4

9. If the curves $\frac{x^2}{\alpha} + \frac{y^2}{4} = 1$ and $y^3 = 16x$ intersect at right angles, then a value of α is :
[JEE-Main (On line)-2013]
- (1) $\frac{4}{3}$ (2) $\frac{3}{4}$ (3) $\frac{1}{2}$ (4) 2
10. A point on the ellipse, $4x^2 + 9y^2 = 36$, where the normal is parallel to the line, $4x - 2y - 5 = 0$, is :
[JEE-Main (On line)-2013]
- (1) $\left(\frac{8}{5}, -\frac{9}{5}\right)$ (2) $\left(-\frac{9}{5}, \frac{8}{5}\right)$ (3) $\left(\frac{8}{5}, \frac{9}{5}\right)$ (4) $\left(\frac{9}{5}, \frac{8}{5}\right)$
11. The locus of the foot of perpendicular drawn from the centre of the ellipse $x^2 + 3y^2 = 6$ on any tangent to it is :
[JEE(Main)-2014]
- (1) $(x^2 - y^2)^2 = 6x^2 + 2y^2$ (2) $(x^2 - y^2)^2 = 6x^2 - 2y^2$
(3) $(x^2 + y^2)^2 = 6x^2 + 2y^2$ (4) $(x^2 + y^2)^2 = 6x^2 - 2y^2$
12. The area (in sq. units) of the quadrilateral formed by the tangents at the end points of the latera recta to the ellipse $\frac{x^2}{9} + \frac{y^2}{5} = 1$ is :
[JEE (Main)-2015]
- (1) $\frac{27}{2}$ (2) 27 (3) $\frac{27}{4}$ (4) 18
13. The eccentricity of an ellipse whose centre is at the origin is $\frac{1}{2}$. If one of its directrices is $x = -4$, then the equation of the normal to it at $\left(1, \frac{3}{2}\right)$ is :-
[JEE(Main) 2017]
- (1) $x + 2y = 4$ (2) $2y - x = 2$ (3) $4x - 2y = 1$ (4) $4x + 2y = 7$
14. If tangents are drawn to the ellipse $x^2 + 2y^2 = 2$ at all points on the ellipse other than its four vertices then the mid points of the tangents intercepted between the coordinate axes lie on the curve :
[JEE (Main)-Jan 19]
- (1) $\frac{x^2}{2} + \frac{y^2}{4} = 1$ (2) $\frac{x^2}{4} + \frac{y^2}{2} = 1$ (3) $\frac{1}{2x^2} + \frac{1}{4y^2} = 1$ (4) $\frac{1}{4x^2} + \frac{1}{2y^2} = 1$
15. Let S and S' be the foci of the ellipse and B be any one of the extremities of its minor axis. If $\Delta S'BS$ is a right angled triangle with right angle at B and area $(\Delta S'BS) = 8$ sq. units, then the length of a latus rectum of the ellipse is :
[JEE (Main)-Jan 19]
- (1) $2\sqrt{2}$ (2) 2 (3) 4 (4) $4\sqrt{2}$
16. If the line $x - 2y = 12$ is tangent to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ at the point $\left(3, -\frac{9}{2}\right)$, then the length of the latus rectum of the ellipse is :
[JEE (Main)-Apr 19]
- (1) 9 (2) $8\sqrt{3}$ (3) $12\sqrt{2}$ (4) 5
17. If the normal to the ellipse $3x^2 + 4y^2 = 12$ at a point P on it is parallel to the line, $2x + y = 4$ and the tangent to the ellipse at P passes through Q(4, 4) then PQ is equal to :
[JEE (Main)-Apr 19]
- (1) $\frac{\sqrt{221}}{2}$ (2) $\frac{\sqrt{157}}{2}$ (3) $\frac{\sqrt{61}}{2}$ (4) $\frac{5\sqrt{5}}{2}$

EXERCISE (JA)

PARAGRAPH :

Tangents are drawn from the point $P(3,4)$ to the ellipse $\frac{x^2}{9} + \frac{y^2}{4} = 1$ touching the ellipse at points A and B.

[JEE 2010, 3+3+3]

1. The coordinates of A and B are

(A) $(3,0)$ and $(0,2)$ (B) $\left(-\frac{8}{5}, \frac{2\sqrt{261}}{15}\right)$ and $\left(-\frac{9}{5}, \frac{8}{5}\right)$

(C) $\left(-\frac{8}{5}, \frac{2\sqrt{161}}{15}\right)$ and $(0,2)$ (D) $(3,0)$ and $\left(-\frac{9}{5}, \frac{8}{5}\right)$

2. The orthocenter of the triangle PAB is

(A) $\left(5, \frac{8}{7}\right)$ (B) $\left(\frac{7}{5}, \frac{25}{8}\right)$

(C) $\left(\frac{11}{5}, \frac{8}{5}\right)$ (D) $\left(\frac{8}{25}, \frac{7}{5}\right)$

3. The equation of the locus of the point whose distances from the point P and the line AB are equal, is -

(A) $9x^2 + y^2 - 6xy - 54x - 62y + 241 = 0$ (B) $x^2 + 9y^2 + 6xy - 54x + 62y - 241 = 0$

(C) $9x^2 + 9y^2 - 6xy - 54x - 62y - 241 = 0$ (D) $x^2 + y^2 - 2xy + 27x + 31y - 120 = 0$

4. The ellipse $E_1 : \frac{x^2}{9} + \frac{y^2}{4} = 1$ is inscribed in a rectangle R whose sides are parallel to the coordinate axes.

Another ellipse E_2 passing through the point $(0,4)$ circumscribes the rectangle R. The eccentricity of the ellipse E_2 is -

[JEE 2012, 3M, -1M]

(A) $\frac{\sqrt{2}}{2}$ (B) $\frac{\sqrt{3}}{2}$ (C) $\frac{1}{2}$ (D) $\frac{3}{4}$

5. A vertical line passing through the point $(h,0)$ intersects the ellipse $\frac{x^2}{4} + \frac{y^2}{3} = 1$ at the points P and Q. Let the tangents to the ellipse at P and Q meet at the point R. If $\Delta(h)$ = area of the triangle PQR,

$\Delta_1 = \max_{1/2 \leq h \leq 1} \Delta(h)$ and $\Delta_2 = \min_{1/2 \leq h \leq 1} \Delta(h)$, then $\frac{8}{\sqrt{5}} \Delta_1 - 8 \Delta_2 =$ [JEE-Advanced 2013, 4, (-1)]

6. List-I

List-II

P. Let $y(x) = \cos(3 \cos^{-1} x)$, $x \in [-1, 1]$, $x \neq \pm \frac{\sqrt{3}}{2}$.

1. 1

Then $\frac{1}{y(x)} \left\{ (x^2 - 1) \frac{d^2 y(x)}{dx^2} + x \frac{dy(x)}{dx} \right\}$ equals

Q. Let $A_1, A_2, \dots, A_n$ ($n > 2$) be the vertices of a regular polygon of n sides with its centre at the origin. Let $\vec{a}_k$ be the position vector of the point A_k , $k = 1, 2, \dots, n$.

2. 2

If $\left| \sum_{k=1}^{n-1} (\vec{a}_k \times \vec{a}_{k+1}) \right| = \left| \sum_{k=1}^{n-1} (\vec{a}_k \cdot \vec{a}_{k+1}) \right|$, then the minimum value of n is

R. If the normal from the point $P(h, 1)$ on the ellipse $\frac{x^2}{6} + \frac{y^2}{3} = 1$ is perpendicular to the line $x + y = 8$, then the value of h is

3. 8

S. Number of positive solutions satisfying the equation

4. 9

$$\tan^{-1} \left(\frac{1}{2x+1} \right) + \tan^{-1} \left(\frac{1}{4x+1} \right) = \tan^{-1} \left(\frac{2}{x^2} \right) \text{ is}$$

Codes :

	P	Q	R	S
(A)	4	3	2	1
(B)	2	4	3	1
(C)	4	3	1	2
(D)	2	4	1	3

[JEE(Advanced)-2014, 3(-1)]

7. Suppose that the foci of the ellipse $\frac{x^2}{9} + \frac{y^2}{5} = 1$ are $(f_1, 0)$ and $(f_2, 0)$ where $f_1 > 0$ and $f_2 < 0$. Let P_1 and P_2 be two parabolas with a common vertex at $(0, 0)$ and with foci at $(f_1, 0)$ and $(2f_2, 0)$, respectively. Let T_1 be a tangent to P_1 which passes through $(2f_2, 0)$ and T_2 be a tangent to P_2 which passes through $(f_1, 0)$. If m_1 is the slope of T_1 and m_2 is the slope of T_2 , then the value of $\left(\frac{1}{m_1^2} + m_2^2 \right)$ is

[JEE 2015, 4M, -0M]

8. Let E_1 and E_2 be two ellipses whose centers are at the origin. The major axes of E_1 and E_2 lie along the x -axis and the y -axis, respectively. Let S be the circle $x^2 + (y - 1)^2 = 2$. The straight line $x + y = 3$ touches the curves S , E_1 and E_2 at P, Q and R , respectively. Suppose that $PQ = PR = \frac{2\sqrt{2}}{3}$. If e_1 and e_2 are the eccentricities of E_1 and E_2 , respectively, then the correct expression(s) is(are)

[JEE 2015, 4M, -0M]

(A) $e_1^2 + e_2^2 = \frac{43}{40}$ (B) $e_1 e_2 = \frac{\sqrt{7}}{2\sqrt{10}}$ (C) $|e_1^2 - e_2^2| = \frac{5}{8}$ (D) $e_1 e_2 = \frac{\sqrt{3}}{4}$

PARAGRAPH :

Let $F_1(x_1, 0)$ and $F_2(x_2, 0)$ for $x_1 < 0$ and $x_2 > 0$, be the foci of the ellipse $\frac{x^2}{9} + \frac{y^2}{8} = 1$. Suppose a parabola having vertex at the origin and focus at F_2 intersects the ellipse at point M in the first quadrant and at point N in the fourth quadrant.

9. The orthocentre of the triangle F_1MN is-

[JEE(Advanced)-2016, 4(-2)]

- (A) $\left(-\frac{9}{10}, 0\right)$ (B) $\left(\frac{2}{3}, 0\right)$ (C) $\left(\frac{9}{10}, 0\right)$ (D) $\left(\frac{2}{3}, \sqrt{6}\right)$

10. If the tangents to the ellipse at M and N meet at R and the normal to the parabola at M meets the x-axis at Q, then the ratio of area of the triangle MQR to area of the quadrilateral MF_1NF_2 is -

[JEE(Advanced)-2016, 3(0)]

- (A) 3 : 4 (B) 4 : 5 (C) 5 : 8 (D) 2 : 3

11. Consider two straight lines, each of which is tangent to both the circle $x^2 + y^2 = \frac{1}{2}$ and the parabola $y^2 = 4x$. Let these lines intersect at the point Q. Consider the ellipse whose center is at the origin $O(0, 0)$ and whose semi-major axis is OQ. If the length of the minor axis of this ellipse is $\sqrt{2}$, then the which of the following statement(s) is (are) TRUE ? [JEE(Advanced)-2018, 4(-2)]

- (A) For the ellipse, the eccentricity is $\frac{1}{\sqrt{2}}$ and the length of the latus rectum is 1
 (B) For the ellipse, the eccentricity is $\frac{1}{2}$ and the length of the latus rectum is $\frac{1}{2}$
 (C) The area of the region bounded by the ellipse between the lines $x = \frac{1}{\sqrt{2}}$ and $x = 1$ is $\frac{1}{4\sqrt{2}}(\pi - 2)$
 (D) The area of the region bounded by the ellipse between the lines $x = \frac{1}{\sqrt{2}}$ and $x = 1$ is $\frac{1}{16}(\pi - 2)$

12. Define the collections $\{E_1, E_2, E_3, \dots\}$ of ellipses and $\{R_1, R_2, R_3, \dots\}$ of rectangles as follows :

$$E_1 : \frac{x^2}{9} + \frac{y^2}{4} = 1 ;$$

R_1 : rectangle of largest area, with sides parallel to the axes, inscribed in E_1 ;

E_n : ellipse $\frac{x^2}{a_n^2} + \frac{y^2}{b_n^2} = 1$ of largest area inscribed in R_{n-1} , $n > 1$;

R_n : rectangle of largest area, with sides parallel to the axes, inscribed in E_n , $n > 1$.

Then which of the following options is/are correct ?

[JEE(Advanced)-2019, 4(-1)]

- (1) The eccentricities of E_{18} and E_{19} are NOT equal
 (2) The distance of a focus from the centre in E_9 is $\frac{\sqrt{5}}{32}$
 (3) The length of latus rectum of E_9 is $\frac{1}{6}$
 (4) $\sum_{n=1}^N (\text{area of } R_n) < 24$, for each positive integer N

ANSWER KEY

ELLIPSE

EXERCISE (O-1)

1. D 2. B 3. C 4. D 5. A 6. B 7. A 8. C
9. C 10. A 11. (a) D; (b) A,B,C,D 12. A 13. A 14. A,B,D 15. A,B
16. A,C 17. A,B 18. B 19. A 20. C 21. B 22. D
23. (A) S, (B) R, (C) P, (D) Q

EXERCISE (0-2)

1. D 2. A 3. B 4. C 5. A 6. C 7. A,B 8. A,C
9. A,B,C 10. A,D 11. A,B,C 12. B,C 13. A 14. B 15. C 16. D

EXERCISE (S-1)

1. (a) $20x^2 + 45y^2 - 40x - 180y - 700 = 0$; (b) $3x^2 + 5y^2 = 32$ 3. 16 4. 65
5. $\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}$ 7. $a^2p^2 + b^2q^2 = r^2 \sec^2 \frac{\pi}{8} = (4 - 2\sqrt{2})r^2$ 8. $\frac{18a}{17}$
10. $x + y - 5 = 0, x + y + 5 = 0$ 11. 24 sq. units 12. 54 14. $bx + a\sqrt{3}y = 2ab$
15. $25y^2 + 4x^2 = 4x^2y^2$ 17. $(x - 1)^2 + y^2 = \frac{11}{3}$ 18. 60 19. $12x + 5y = 48; 12x - 5y = 48$
21. 85

EXERCISE (S-2)

1. 80 3. $\sqrt{r^2 - b^2}$ 7. (b) $8/3$, (c) 4 9. 2 11. (a) $\frac{3}{5}$; (b) $240\sqrt{3}$; (c) 36
12. (A) Q; (B) S; (C) P; (D) R

EXERCISE (JM)

- | | | | | | | | |
|-------|-------|-------|-------|-------|-------|-------|-------|
| 1. 4 | 2. 3 | 3. 1 | 4. 3 | 5. 1 | 6. 4 | 7. 3 | 8. 4 |
| 9. 1 | 10. 4 | 11. 3 | 12. 2 | 13. 3 | 14. 3 | 15. 3 | 16. 1 |
| 17. 4 | | | | | | | |

EXERCISE (JA)

1. D 2. C 3. A 4. C 5. 9 6. A 7. 4 8. A,B
9. A 10. C 11. A,C 12. 3,4

HYPERBOLA

The **Hyperbola** is a conic whose eccentricity is greater than unity. ($e > 1$).

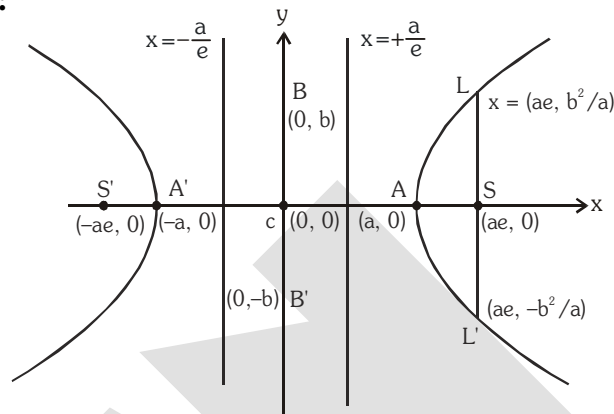
1. STANDARD EQUATION & DEFINITION(S):

Standard equation of the hyperbola is

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1, \text{ where } b^2 = a^2(e^2 - 1)$$

$$\text{or } a^2 e^2 = a^2 + b^2 \text{ i.e. } e^2 = 1 + \frac{b^2}{a^2}$$

$$= 1 + \left(\frac{\text{Conjugate Axis}}{\text{Transverse Axis}} \right)^2$$



(a) Foci :

$$S \equiv (ae, 0) \quad \& \quad S' \equiv (-ae, 0).$$

(b) Equations of directrices :

$$x = \frac{a}{e} \quad \& \quad x = -\frac{a}{e}.$$

(c) Vertices :

$$A \equiv (a, 0) \quad \& \quad A' \equiv (-a, 0).$$

(d) Latus rectum :

(i) Equation : $x = \pm ae$

(ii) Length = $\frac{2b^2}{a} = \frac{(\text{Conjugate Axis})^2}{(\text{Transverse Axis})} = 2a(e^2 - 1) = 2e$ (distance from focus to directrix)

(iii) Ends : $\left(ae, \frac{b^2}{a} \right), \left(ae, -\frac{b^2}{a} \right); \left(-ae, \frac{b^2}{a} \right), \left(-ae, -\frac{b^2}{a} \right)$

(e) (i) Transverse Axis :

The line segment A'A of length 2a in which the foci S' & S both lie is called the **Transverse Axis of the Hyperbola**.

(ii) Conjugate Axis :

The line segment B'B between the two points B' $\equiv (0, -b)$ & B $\equiv (0, b)$ is called as the **Conjugate Axis of the Hyperbola**.

The Transverse Axis & the Conjugate Axis of the hyperbola are together called the **Principal axes of the hyperbola**.

(f) Focal Property :

The difference of the focal distances of any point on the hyperbola is constant and equal to transverse axis i.e. $||\mathbf{PS}| - |\mathbf{PS}'|| = 2\mathbf{a}$. The distance $SS' =$ focal length.

(g) Focal distance :

Distance of any point P(x, y) on Hyperbola from foci **PS = ex - a** & **PS' = ex + a**.

Illustration 1 : Find the equation of the hyperbola whose directrix is $2x + y = 1$, focus $(1, 2)$ and eccentricity $\sqrt{3}$.

Solution : Let P(x, y) be any point on the hyperbola and PM is perpendicular from P on the directrix.
Then by definition $SP = e \text{ PM}$

$$\begin{aligned} \Rightarrow (\text{SP})^2 &= e^2 (\text{PM})^2 \Rightarrow (x-1)^2 + (y-2)^2 = 3 \left\{ \frac{2x+y-1}{\sqrt{4+1}} \right\}^2 \\ \Rightarrow 5(x^2 + y^2 - 2x - 4y + 5) &= 3(4x^2 + y^2 + 1 + 4xy - 2y - 4x) \\ \Rightarrow 7x^2 - 2y^2 + 12xy - 2x + 14y - 22 &= 0 \end{aligned}$$

which is the required hyperbola.

Illustration 2: The eccentricity of the hyperbola $4x^2 - 9y^2 - 8x = 32$ is -

- (A) $\frac{\sqrt{5}}{3}$ (B) $\frac{\sqrt{13}}{3}$ (C) $\frac{\sqrt{13}}{2}$ (D) $\frac{3}{2}$

Solution : $4x^2 - 9y^2 - 8x = 32 \Rightarrow 4(x-1)^2 - 9y^2 = 36 \Rightarrow \frac{(x-1)^2}{9} - \frac{y^2}{4} = 1$

Here $a^2 = 9$, $b^2 = 4$

$$\therefore \text{eccentricity } e = \sqrt{1 + \frac{b^2}{a^2}} = \sqrt{1 + \frac{4}{9}} = \frac{\sqrt{13}}{3} \quad \text{Ans.(B)}$$

Illustration 3 : If foci of a hyperbola are foci of the ellipse $\frac{x^2}{25} + \frac{y^2}{9} = 1$. If the eccentricity of the hyperbola be 2, then its equation is -

- (A) $\frac{x^2}{4} - \frac{y^2}{12} = 1$ (B) $\frac{x^2}{12} - \frac{y^2}{4} = 1$ (C) $\frac{x^2}{12} + \frac{y^2}{4} = 1$ (D) none of these

Solution : For ellipse $e = \frac{4}{5}$, so foci = $(\pm 4, 0)$

For hyperbola $e = 2$, so $a = \frac{ae}{e} = \frac{4}{2} = 2$, $b = 2\sqrt{4-1} = 2\sqrt{3}$

Hence equation of the hyperbola is $\frac{x^2}{4} - \frac{y^2}{12} = 1$ **Ans.(A)**

Illustration 4 : Find the coordinates of foci, the eccentricity and latus-rectum, equations of directrices for the hyperbola $9x^2 - 16y^2 - 72x + 96y - 144 = 0$.

Solution : Equation can be rewritten as $\frac{(x-4)^2}{4^2} - \frac{(y-3)^2}{3^2} = 1$ so $a = 4$, $b = 3$

$$b^2 = a^2(e^2 - 1) \text{ given } e = \frac{5}{4}$$

Foci : $X = \pm ae$, $Y = 0$ gives the foci as $(9, 3)$, $(-1, 3)$

Centre : $X = 0$, $Y = 0$ i.e. $(4, 3)$

Directrices : $X = \pm \frac{a}{e}$ i.e. $x - 4 = \pm \frac{16}{5}$ $\therefore$ directrices are $5x - 36 = 0$; $5x - 4 = 0$

$$\text{Latus-rectum} = \frac{2b^2}{a} = 2 \cdot \frac{9}{4} = \frac{9}{2}$$

Do yourself - 1 :

- (i) Find the eccentricity of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ which passes through $(4, 0)$ & $(3\sqrt{2}, 2)$
- (ii) Find the equation to the hyperbola, whose eccentricity is $\frac{5}{4}$, focus is $(a, 0)$ and whose directrix is $4x - 3y = a$.
- (iii) In the hyperbola $4x^2 - 9y^2 = 36$, find length of the axes, the co-ordinates of the foci, the eccentricity, and the latus rectum.
- (iv) Find the equation to the hyperbola, the distance between whose foci is 16 and whose eccentricity is $\sqrt{2}$.

2. CONJUGATE HYPERBOLA :

Two hyperbolas such that transverse & conjugate axes of one hyperbola are respectively the conjugate

& the transverse axes of the other are called **Conjugate Hyperbolas** of each other. eg. $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

& $-\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ are conjugate hyperbolas of each other.

Note that :

- (i) If e_1 & e_2 are the eccentricities of the hyperbola & its conjugate then $e_1^{-2} + e_2^{-2} = 1$.
- (ii) The foci of a **hyperbola** and its **conjugate** are **concylic and form the vertices of a square**.
- (iii) Two hyperbolas are said to be **similar** if they have the **same eccentricity**.

Illustration 5 : The eccentricity of the conjugate hyperbola to the hyperbola $x^2 - 3y^2 = 1$ is-

- (A) 2 (B) $2/\sqrt{3}$ (C) 4 (D) $4/3$

Solution : Equation of the conjugate hyperbola to the hyperbola $x^2 - 3y^2 = 1$ is

$$-x^2 + 3y^2 = 1 \Rightarrow -\frac{x^2}{1} + \frac{y^2}{1/3} = 1$$

Here $a^2 = 1$, $b^2 = 1/3$

$$\therefore \text{eccentricity } e = \sqrt{1 + a^2/b^2} = \sqrt{1 + 3} = 2$$

Ans. (A)

Do yourself - 2 :

- (i) Find eccentricity of conjugate hyperbola of hyperbola $4x^2 - 16y^2 = 64$, also find area of quadrilateral formed by foci of hyperbola & its conjugate hyperbola

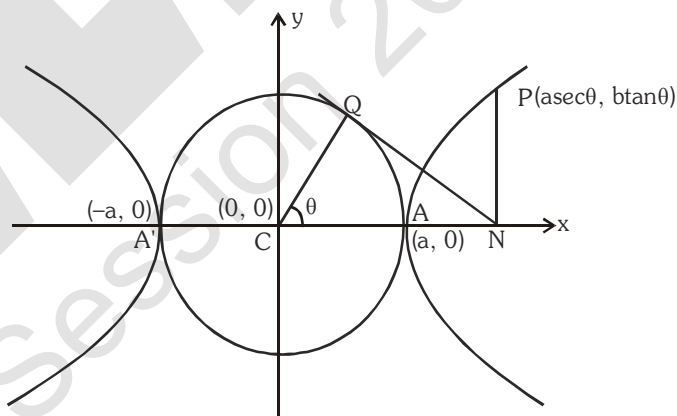
3. RECTANGULAR OR EQUILATERAL HYPERBOLA :

The particular kind of hyperbola in which the lengths of the transverse & conjugate axis are equal is called an **Equilateral Hyperbola**. Note that the eccentricity of the rectangular hyperbola is $\sqrt{2}$ and the length of it's latus rectum is equal to it's transverse or conjugate axis.

4. AUXILIARY CIRCLE :

A circle drawn with centre C & T.A. as a diameter is called the **Auxiliary Circle** of the hyperbola. Equation of the auxiliary circle is $x^2 + y^2 = a^2$.

Note from the figure that P & Q are called the "**Corresponding Points**" on the hyperbola & the auxiliary circle. 'θ' is called the **eccentric angle** of the point 'P' on the hyperbola. ($0 \leq \theta < 2\pi$).



Parametric Equation :

The equations $x = a \sec \theta$ & $y = b \tan \theta$ together represents the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ where

θ is a parameter. The parametric equations ; $x = a \cosh \phi$, $y = b \sinh \phi$ also represents the same hyperbola.

General Note :

Since the fundamental equation to the hyperbola only differs from that to the ellipse in having $-b^2$ instead of b^2 it will be found that many propositions for the hyperbola are derived from those for the ellipse by simply changing the sign of b^2 .

5. POSITION OF A POINT 'P' w.r.t. A HYPERBOLA :

The quantity $\frac{x_1^2}{a^2} - \frac{y_1^2}{b^2} = 1$ is **positive**, **zero** or **negative** according as the point (x_1, y_1) lies within, upon or outside the curve.

6. LINE AND A HYPERBOLA :

The straight line $y = mx + c$ is a secant, a tangent or passes outside the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ according as : $c^2 > = < a^2 m^2 - b^2$.

Equation of a chord of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ joining its two points $P(\alpha)$ & $Q(\beta)$ is

$$\frac{x}{a} \cos \frac{\alpha - \beta}{2} - \frac{y}{b} \sin \frac{\alpha + \beta}{2} = \cos \frac{\alpha + \beta}{2}$$

Illustration 6 : Show that the line $x \cos \alpha + y \sin \alpha = p$ touches the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ if $a^2 \cos^2 \alpha - b^2 \sin^2 \alpha = p^2$.

Solution : The given line is $x \cos \alpha + y \sin \alpha = p \Rightarrow y \sin \alpha = -x \cos \alpha + p$
 $\Rightarrow y = -x \cot \alpha + p \operatorname{cosec} \alpha$

Comparing this line with $y = mx + c$

$$m = -\cot \alpha, c = p \operatorname{cosec} \alpha$$

Since the given line touches the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ then

$$c^2 = a^2 m^2 - b^2 \Rightarrow p^2 \operatorname{cosec}^2 \alpha = a^2 \cot^2 \alpha - b^2 \text{ or } p^2 = a^2 \cos^2 \alpha - b^2 \sin^2 \alpha$$

Illustration 7 : If $(a \sec \theta, b \tan \theta)$ and $(a \sec \phi, b \tan \phi)$ are the ends of a focal chord of $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, then

$\tan \frac{\theta}{2} \tan \frac{\phi}{2}$ equal to -

$$(A) \frac{e-1}{e+1} \quad (B) \frac{1-e}{1+e} \quad (C) \frac{1+e}{1-e} \quad (D) \frac{e+1}{e-1}$$

Solution : Equation of chord connecting the points $(a \sec \theta, b \tan \theta)$ and $(a \sec \phi, b \tan \phi)$ is

$$\frac{x}{a} \cos \left(\frac{\theta - \phi}{2} \right) - \frac{y}{b} \sin \left(\frac{\theta + \phi}{2} \right) = \cos \left(\frac{\theta + \phi}{2} \right) \quad \dots\dots (i)$$

If it passes through $(ae, 0)$; we have, $e \cos\left(\frac{\theta - \phi}{2}\right) = \cos\left(\frac{\theta + \phi}{2}\right)$

$$\Rightarrow e = \frac{\cos\left(\frac{\theta + \phi}{2}\right)}{\cos\left(\frac{\theta - \phi}{2}\right)} = \frac{1 - \tan\frac{\theta}{2} \cdot \tan\frac{\phi}{2}}{1 + \tan\frac{\theta}{2} \cdot \tan\frac{\phi}{2}} \Rightarrow \tan\frac{\theta}{2} \cdot \tan\frac{\phi}{2} = \frac{1 - e}{1 + e}$$

Similarly if (i) passes through $(-ae, 0)$, $\tan\frac{\theta}{2} \cdot \tan\frac{\phi}{2} = \frac{1 + e}{1 - e}$

Ans. (B, C)

Do yourself - 3 :

(i) Find the condition for the line $\ell x + my + n = 0$ to touch the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

(ii) If the line $y = 5x + 1$ touch the hyperbola $\frac{x^2}{4} - \frac{y^2}{b^2} = 1$ $\{b > 4\}$, then -

(A) $b^2 = \frac{1}{5}$

(B) $b^2 = 99$

(C) $b^2 = 4$

(D) $b^2 = 100$

7. TANGENT TO THE HYPERBOLA $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$:

(a) **Point form** : Equation of the tangent to the given hyperbola at the point (x_1, y_1) is

$$\frac{xx_1}{a^2} - \frac{yy_1}{b^2} = 1.$$

Note : In general two tangents can be drawn from an external point (x_1, y_1) to the hyperbola and they are $y - y_1 = m_1(x - x_1)$ & $y - y_1 = m_2(x - x_1)$, where m_1 & m_2 are roots of the equation $(x_1^2 - a^2)m^2 - 2x_1y_1m + y_1^2 + b^2 = 0$. If $D < 0$, then **no tangent** can be drawn from (x_1, y_1) to the hyperbola.

(b) **Slope form** : The equation of tangents of slope m to the given hyperbola is $y = mx$

$$\pm \sqrt{a^2m^2 - b^2}. \text{ Point of contact are } \left(\mp \frac{a^2m}{\sqrt{a^2m^2 - b^2}}, \frac{\mp b^2}{\sqrt{a^2m^2 - b^2}} \right)$$

Note that there are **two parallel tangents having the same slope m** .

(c) **Parametric form** : Equation of the tangent to the given hyperbola at the point $(a \sec \theta, b \tan \theta)$ is

$$\frac{x \sec \theta}{a} - \frac{y \tan \theta}{b} = 1.$$

Note : Point of intersection of the tangents at θ_1 & θ_2 is $x = a \frac{\cos\left(\frac{\theta_1 - \theta_2}{2}\right)}{\cos\left(\frac{\theta_1 + \theta_2}{2}\right)}$, $y = b \tan\left(\frac{\theta_1 + \theta_2}{2}\right)$

Illustration 8 : Find the equation of the tangent to the hyperbola $x^2 - 4y^2 = 36$ which is perpendicular to the line $x - y + 4 = 0$.

Solution : Let m be the slope of the tangent. Since the tangent is perpendicular to the line $x - y = 0$
 $\therefore m \times 1 = -1 \Rightarrow m = -1$

$$\text{Since } x^2 - 4y^2 = 36 \quad \text{or} \quad \frac{x^2}{36} - \frac{y^2}{9} = 1$$

$$\text{Comparing this with } \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

$$\therefore a^2 = 36 \text{ and } b^2 = 9$$

$$\text{So the equation of tangents are } y = (-1)x \pm \sqrt{36 \times (-1)^2 - 9}$$

$$y = -x \pm \sqrt{27} \Rightarrow x + y \pm 3\sqrt{3} = 0$$

Ans.

Illustration 9 : The locus of the point of intersection of two tangents of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ if the product of their slopes is c^2 , will be -

$$(A) y^2 - b^2 = c^2(x^2 + a^2)$$

$$(B) y^2 + b^2 = c^2(x^2 - a^2)$$

$$(C) y^2 + a^2 = c^2(x^2 - b^2)$$

$$(D) y^2 - a^2 = c^2(x^2 + b^2)$$

Solution : Equation of any tangent of the hyperbola with slope m is $y = mx \pm \sqrt{a^2 m^2 - b^2}$

If it passes through (x_1, y_1) then

$$(y_1 - mx_1)^2 = a^2 m^2 - b^2 \Rightarrow (x_1^2 - a^2) m^2 - 2x_1 y_1 m + (y_1^2 + b^2) = 0$$

$$\text{If } m = m_1, m_2 \text{ then as given } m_1 m_2 = c^2 \Rightarrow \frac{y_1^2 + b^2}{x_1^2 - a^2} = c^2$$

$$\text{Hence required locus will be : } y^2 + b^2 = c^2(x^2 - a^2)$$

Ans.(B)

Illustration 10 : A common tangent to $9x^2 - 16y^2 = 144$ and $x^2 + y^2 = 9$ is -

$$(A) y = 3\sqrt{\frac{2}{7}}x - \frac{15}{\sqrt{7}} \quad (B) y = 3\sqrt{\frac{2}{7}}x + \frac{15}{\sqrt{7}} \quad (C) y = -3\sqrt{\frac{2}{7}}x + \frac{15}{\sqrt{7}} \quad (D) y = -3\sqrt{\frac{2}{7}}x - \frac{15}{\sqrt{7}}$$

Solution : $\frac{x^2}{16} - \frac{y^2}{9} = 1, x^2 + y^2 = 9$

$$\text{Equation of tangent } y = mx + \sqrt{16m^2 - 9} \quad (\text{for hyperbola})$$

$$\text{Equation of tangent } y = m'x + 3\sqrt{1+m'^2} \quad (\text{circle})$$

$$\text{For common tangent } m = m' \text{ and } 3\sqrt{1+m'^2} = \sqrt{16m^2 - 9}$$

$$\text{or } 9 + 9m^2 = 16m^2 - 9$$

$$\text{or } 7m^2 = 18 \Rightarrow m = \pm 3\sqrt{\frac{2}{7}}$$

$$\therefore \text{ required equation is } y = \pm 3\sqrt{\frac{2}{7}}x \pm 3\sqrt{1 + \frac{18}{7}}$$

$$\text{or } y = \pm 3\sqrt{\frac{2}{7}}x \pm \frac{15}{\sqrt{7}}$$

Ans. (A,B,C,D)

Solution : Equation of normal at any point Q is $ax \cos \theta + by \cot \theta = a^2 + b^2$

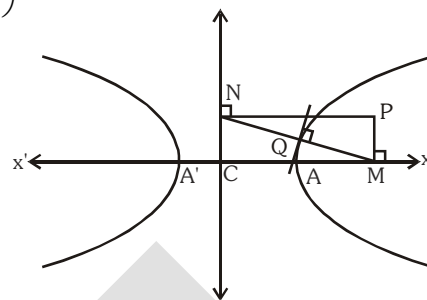
$$\therefore M \equiv \left(\frac{a^2 + b^2}{a} \sec \theta, 0 \right), N \equiv \left(0, \frac{a^2 + b^2}{b} \tan \theta \right)$$

$$\therefore \text{Let } P \equiv (h, k)$$

$$\Rightarrow h = \frac{a^2 + b^2}{a} \sec \theta, \quad k = \frac{a^2 + b^2}{b} \tan \theta$$

$$\Rightarrow \frac{a^2 h^2}{(a^2 + b^2)^2} - \frac{b^2 k^2}{(a^2 + b^2)^2} = \sec^2 \theta - \tan^2 \theta = 1$$

$$\therefore \text{locus of P is } (a^2 x^2 - b^2 y^2) = (a^2 + b^2).$$

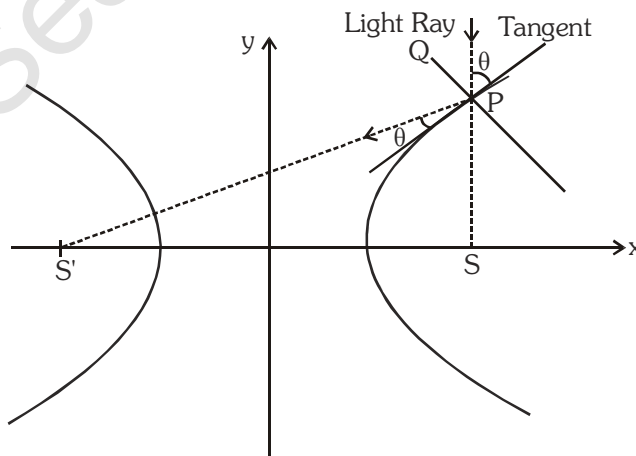


Do yourself - 5 :

- (i) Find the equation of normal to the hyperbola $\frac{x^2}{25} - \frac{y^2}{16} = 1$ at $(5, 0)$.
- (ii) Find the equation of normal to the hyperbola $\frac{x^2}{16} - \frac{y^2}{9} = 1$ at the point $\left(6, \frac{3}{2}\sqrt{5}\right)$.
- (iii) Find the condition for the line $\ell x + my + n = 0$ is normal to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$.

9. HIGHLIGHTS ON TANGENT AND NORMAL :

- (a) Locus of the feet of the perpendicular drawn from focus of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ upon any tangent is its auxiliary circle i.e. $x^2 + y^2 = a^2$ & the product of lengths to these perpendiculars is b^2 (**semi Conjugate Axis**)²
- (b) The portion of the tangent between the **point of contact** & the **directrix** subtends a **right angle** at the corresponding **focus**.
- (c) The tangent & normal at any point of a hyperbola **bisect** the angle between the **focal radii**. This spells the **reflection property of the hyperbola** as "An incoming light ray" aimed towards one **focus** is **reflected from the outer surface of the hyperbola towards the other focus**. It follows that if an ellipse and a hyperbola have the same foci, they cut at **right angles** at any of their **common point**.



Note that the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ & the hyperbola $\frac{x^2}{a^2 - k^2} - \frac{y^2}{k^2 - b^2} = 1$ ($a > k > b > 0$) are confocal and therefore orthogonal.

- (d) The foci of the hyperbola and the points P and Q in which any tangent meets the tangents at the vertices are concyclic with PQ as diameter of the circle.

10. DIRECTOR CIRCLE :

The locus of the intersection of tangents which are at **right angles** is known as the **Director Circle** of the hyperbola. The equation to the **director circle** is : $x^2 + y^2 = a^2 - b^2$.

If $b^2 < a^2$, this circle is **real**; if $b^2 = a^2$ the **radius** of the **circle** is **zero** & it reduces to a **point circle at the origin**. In this case the **centre** is the **only point** from which the **tangents at right angles** can be drawn to the **curve**.

If $b^2 > a^2$, the **radius** of the **circle** is **imaginary**, so that there is **no such circle** & so **no tangents at right angle** can be drawn to the curve.

Note : Equations of chord of contact, chord with a given middle point, pair of tangents from an external point are to be interpreted in the similar way as in ellipse.

11. ASYMPTOTES :

Definition : If the length of the perpendicular let fall from a point on a hyperbola to a straight line tends to zero as the point on the hyperbola moves to infinity along the hyperbola, then the straight line is called the **Asymptote of the Hyperbola**.

To find the asymptote of the hyperbola :

Let $y = mx + c$ is the **asymptote** of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$.

Solving these two we get the quadratic as $(b^2 - a^2m^2)x^2 - 2a^2mcx - a^2(b^2 + c^2) = 0$ (1)

In order that $y = mx + c$ be an asymptote,

both **roots** of equation (1) must approach

infinity, the conditions for which are :

coefficient of $x^2 = 0$ & coefficient of

$x = 0$.

$$\Rightarrow b^2 - a^2m^2 = 0 \text{ or } m = \pm \frac{b}{a} \text{ \& } a^2mc = 0 \Rightarrow c = 0.$$

$$\therefore \text{ equations of asymptote are } \frac{x}{a} + \frac{y}{b} = 0$$

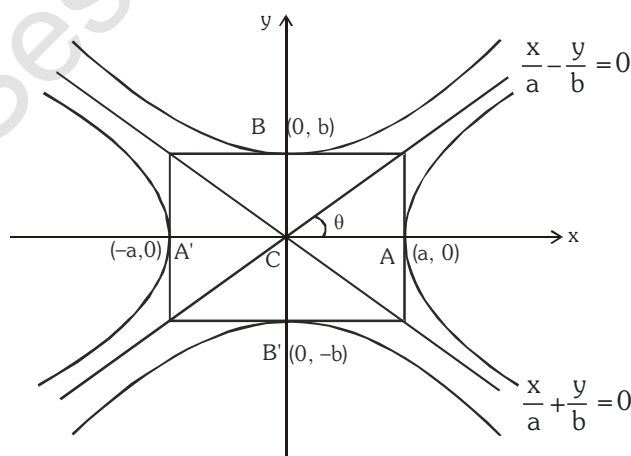
$$\text{and } \frac{x}{a} - \frac{y}{b} = 0.$$

$$\text{combined equation to the asymptotes } \frac{x^2}{a^2} - \frac{y^2}{b^2} = 0.$$

Particular Case :

When $b = a$ the asymptotes of the rectangular hyperbola.

$$x^2 - y^2 = a^2 \text{ are } y = \pm x \text{ which are at right angles.}$$



Note :

- (i) **Equilateral hyperbola** $\Leftrightarrow$ **rectangular hyperbola**.
- (ii) If a **hyperbola is equilateral** then the **conjugate hyperbola is also equilateral**.
- (iii) A **hyperbola** and its **conjugate** have the **same asymptote**.
- (iv) The equation of the **pair of asymptotes** differ the **hyperbola & the conjugate hyperbola** by the **same constant** only.
- (v) The asymptotes pass through the **centre of the hyperbola** & the bisectors of the angles between the asymptotes are the axes of the hyperbola.
- (vi) The asymptotes of a hyperbola are the diagonals of the rectangle formed by the lines drawn through the extremities of each axis parallel to the other axis.
- (vii) Asymptotes are the tangent to the hyperbola from the centre.
- (viii) A simple method to find the co-ordinates of the centre of the hyperbola expressed as a general equation of degree 2 should be remembered as : Let $f(x, y) = 0$ represents a hyperbola.

Find $\frac{\partial f}{\partial x}$ & $\frac{\partial f}{\partial y}$. Then the point of intersection of $\frac{\partial f}{\partial x} = 0$ & $\frac{\partial f}{\partial y} = 0$ gives the **centre of the hyperbola**.

Illustration 13 : Find the asymptotes of the hyperbola $2x^2 + 5xy + 2y^2 + 4x + 5y = 0$. Find also the general equation of all the hyperbolas having the same set of asymptotes.

Solution : Let $2x^2 + 5xy + 2y^2 + 4x + 5y + \lambda = 0$ be asymptotes. This will represent two straight line

$$\text{so } 4\lambda + 25 - \frac{25}{2} - 8 - \frac{25}{4}\lambda = 0$$

$$\Rightarrow \lambda = 2$$

$$\Rightarrow 2x^2 + 5xy + 2y^2 + 4x + 5y + 2 = 0 \text{ are asymptotes}$$

$$\Rightarrow (2x + y + 2) = 0 \text{ and } (x + 2y + 1) = 0 \text{ are asymptotes}$$

$$\text{and } 2x^2 + 5xy + 2y^2 + 4x + 5y + c = 0 \text{ is general equation of hyperbola.}$$

Illustration 14 : Find the hyperbola whose asymptotes are $2x - y = 3$ and $3x + y - 7 = 0$ and which passes through the point $(1, 1)$.

Solution : The equation of the hyperbola differs from the equation of the asymptotes by a constant

$$\Rightarrow \text{The equation of the hyperbola with asymptotes } 3x + y - 7 = 0 \text{ and } 2x - y = 3 \text{ is } (3x + y - 7)(2x - y - 3) + k = 0$$

$$\text{It passes through } (1, 1)$$

$$\Rightarrow k = -6.$$

$$\text{Hence the equation of the hyperbola is } (2x - y - 3)(3x + y - 7) = 6.$$

Do yourself - 6 :

- (i) Find the equation to the chords of the hyperbola $x^2 - y^2 = 9$ which is bisected at $(5, -3)$
- (ii) If m_1 and m_2 are the slopes of the tangents to the hyperbola $\frac{x^2}{25} - \frac{y^2}{16} = 1$ which pass through the point $(6, 2)$, then find the value of $11m_1m_2$ and $11(m_1 + m_2)$.
- (iii) Find the locus of the mid points of the chords of the circle $x^2 + y^2 = 16$ which are tangents to the hyperbola $9x^2 - 16y^2 = 144$.
- (iv) The asymptotes of a hyperbola are parallel to lines $2x + 3y = 0$ and $3x + 2y = 0$. The hyperbola has its centre at $(1, 2)$ and it passes through $(5, 3)$. Find its equation.

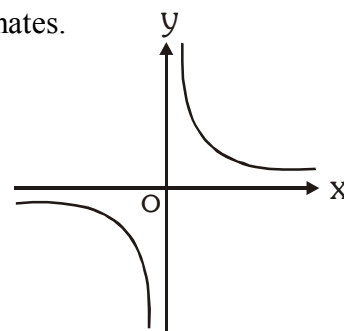
12. HIGHLIGHTS ON ASYMPTOTES

- (a) If from any point on the asymptote a straight line be drawn perpendicular to the transverse axis, the product of the segments of this line, intercepted between the point & the curve is always equal to the square of the semi conjugate axis.
- (b) Perpendicular from the foci on either asymptote meet it in the same points as the corresponding directrix & the common points of intersection lie on the auxiliary circle.
- (c) The tangent at any point P on a hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ with centre C, meets the asymptotes in Q and R and cuts off a ΔCQR of constant area equal to ab from the asymptotes & the portion of the tangent intercepted between the asymptote is bisected at the point of contact. This implies that locus of the centre of the circle circumscribing the ΔCQR in case of a rectangular hyperbola is the **hyperbola** itself.
- (d) If the angle between the asymptote of a hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is 2θ then the eccentricity of the **hyperbola** is $\sec\theta$.

13. RECTANGULAR HYPERBOLA :

Rectangular hyperbola referred to its asymptotes as axis of coordinates.

- (a) Equation is $\mathbf{xy} = \mathbf{c}^2$ with parametric representation $\mathbf{x} = \mathbf{ct}, \mathbf{y} = \mathbf{c/t}, \mathbf{t} \in \mathbf{R} - \{\mathbf{0}\}$.
- (b) Equation of a chord joining the points $\mathbf{P(t_1)} \& \mathbf{Q(t_2)}$ is $\mathbf{x + t_1 t_2 y = c(t_1 + t_2)}$ with slope, $\mathbf{m = \frac{-1}{t_1 t_2}}$
- (c) Equation of the tangent at $\mathbf{P(x_1, y_1)}$ is $\mathbf{\frac{x}{x_1} + \frac{y}{y_1} = 2}$
& at $\mathbf{P(t)}$ is $\mathbf{\frac{x}{t} + ty = 2c}$.
- (d) Equation of normal is $\mathbf{y - \frac{c}{t} = t^2(x - ct)}$
- (e) Chord with a given middle point as $\mathbf{(h, k)}$ is $\mathbf{kx + hy = 2hk}$.



Note :

For the hyperbola, $xy = c^2$

(i) Vertices : (c, c) & $(-c, -c)$.

(ii) Foci : $(\sqrt{2}c, \sqrt{2}c)$ & $(-\sqrt{2}c, -\sqrt{2}c)$

(iii) Directrices : $x + y = \pm\sqrt{2}c$

(iv) Latus rectum : $\ell = 2\sqrt{2}c = T \cdot A = C \cdot A$

Illustration 15 : A triangle has its vertices on a rectangular hyperbola. Prove that the orthocentre of the triangle also lies on the same hyperbola.

Solution : Let t_1, t_2 and t_3 are the vertices of the triangle ABC, described on the rectangular hyperbola $xy = c^2$.

$\therefore$ co-ordinates of A, B and C are $\left(ct_1, \frac{c}{t_1}\right)$, $\left(ct_2, \frac{c}{t_2}\right)$ and $\left(ct_3, \frac{c}{t_3}\right)$ respectively

Now slope of BC is $\frac{\frac{c}{t_3} - \frac{c}{t_2}}{ct_3 - ct_2} = -\frac{1}{t_2 t_3}$

$\therefore$ Slope of AD is $t_2 t_3$

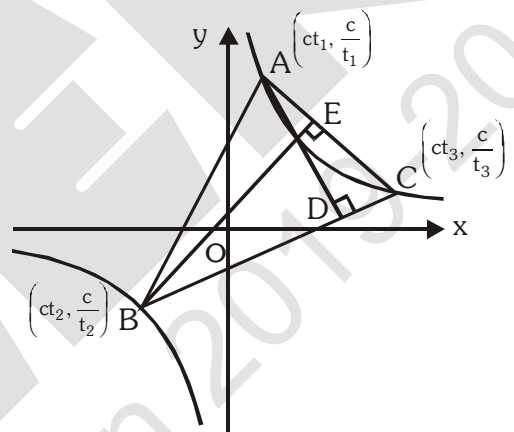
Equation of altitude AD is $y - \frac{c}{t_1} = t_2 t_3 (x - ct_1)$

or $t_1 y - c = x t_1 t_2 t_3 - c t_1^2 t_2 t_3$ (i)

Similarly equation of altitude BE is

$t_2 y - c = x t_1 t_2 t_3 - c t_1 t_2^2 t_3$ (ii)

Solving (i) and (ii), we get the orthocentre $\left(-\frac{c}{t_1 t_2 t_3}, -c t_1 t_2 t_3\right)$ which lies on $xy = c^2$.



Do yourself - 7 :

- (i) If equation $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ represents a rectangular hyperbola then write required conditions.
- (ii) Find the equation of tangent at the point $(1, 2)$ to the rectangular hyperbola $xy = 2$.
- (iii) Prove that the locus of point, tangents from where to hyperbola $x^2 - y^2 = a^2$ inclined at an angle α & β with x-axis such that $\tan \alpha \tan \beta = 2$ is also a hyperbola. Find the eccentricity of this hyperbola.

Miscellaneous Illustrations :

Illustration 16: Chords of the circle $x^2 + y^2 = a^2$ touch the hyperbola $x^2/a^2 - y^2/b^2 = 1$. Prove that locus of their middle point is the curve $(x^2 + y^2)^2 = a^2x^2 - b^2y^2$.

Solution : Let (h, k) be the mid-point of the chord of the circle $x^2 + y^2 = a^2$,
so that its equation by T = S₁ is $hx + ky = h^2 + k^2$

or $y = -\frac{h}{k}x + \frac{h^2 + k^2}{k}$ i.e. of the form $y = mx + c$

It will touch the hyperbola if $c^2 = a^2m^2 - b^2$

$$\therefore \left(\frac{h^2 + k^2}{k} \right)^2 = a^2 \left(-\frac{h}{k} \right)^2 - b^2 \quad \text{or} \quad (h^2 + k^2)^2 = a^2 h^2 - b^2 k^2$$

Generalising, the locus of mid-point (h, k) is $(x^2 + y^2)^2 = a^2x^2 - b^2y^2$

Illustration 17: C is the centre of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$. The tangent at any point P on this hyperbola meets the straight lines $bx - ay = 0$ and $bx + ay = 0$ in the points Q and R respectively. Show that $CQ \cdot CR = a^2 + b^2$.

Solution : P is $(a \sec \theta, b \tan \theta)$

Tangent at P is $\frac{x \sec \theta}{a} - \frac{y \tan \theta}{b} = 1$

It meets $bx - ay = 0$ i.e. $\frac{x}{a} = \frac{y}{b}$ in Q

$$\therefore Q \text{ is } \left(\frac{a}{\sec \theta - \tan \theta}, \frac{b}{\sec \theta - \tan \theta} \right)$$

It meets $bx + ay = 0$ i.e. $\frac{x}{a} = -\frac{y}{b}$ in R .

$$\therefore R \text{ is } \left(\frac{a}{\sec \theta + \tan \theta}, \frac{-b}{\sec \theta + \tan \theta} \right)$$

$$\therefore \text{CQ.CR} = \frac{\sqrt{(a^2 + b^2)}}{\sec \theta - \tan \theta} \cdot \frac{\sqrt{(a^2 + b^2)}}{\sec \theta + \tan \theta} = a^2 + b^2 \quad (\because \sec^2 \theta - \tan^2 \theta = 1) \quad \text{Ans.}$$

Illustration 18: A circle of variable radius cuts the rectangular hyperbola $x^2 - y^2 = 9a^2$ in points P, Q, R and S. Determine the equation of the locus of the centroid of triangle PQR.

Solution : Let the circle be $(x - h)^2 + (y - k)^2 = r^2$ where r is variable. Its intersection with $x^2 - y^2 = 9a^2$ is obtained by putting $y^2 = x^2 - 9a^2$.

$$x^2 + x^2 - 9a^2 - 2hx + h^2 + k^2 - r^2 = 2k\sqrt{(x^2 - 9a^2)}$$

$$\text{or } [2x^2 - 2hx + (h^2 + k^2 - r^2 - 9a^2)]^2 = 4k^2(x^2 - 9a^2)$$

$$\text{or } 4x^4 - 8hx^3 + \dots = 0$$

$\therefore$ Above gives the abscissas of the four points of intersection.

$$\therefore \Sigma x_1 = \frac{8h}{4} = 2h$$

$$x_1 + x_2 + x_3 + x_4 = 2h$$

$$\text{Similarly } y_1 + y_2 + y_3 + y_4 = 2k.$$

Now if (α, β) be the centroid of ΔPQR , then $3\alpha = x_1 + x_2 + x_3$, $3\beta = y_1 + y_2 + y_3$

$$\therefore x_4 = 2h - 3\alpha, y_4 = 2k - 3\beta$$

But (x_4, y_4) lies on $x^2 - y^2 = 9a^2$

$$\therefore (2h - 3\alpha)^2 + (2k - 3\beta)^2 = 9a^2$$

Hence the locus of centroid (α, β) is $(2h - 3x)^2 + (2k - 3y)^2 = 9a^2$

$$\text{or } \left(x - \frac{2h}{3}\right)^2 + \left(y - \frac{2k}{3}\right)^2 = a^2$$

Illustration 19 : If a circle cuts a rectangular hyperbola $xy = c^2$ in A, B, C, D and the parameters of these four points be t_1, t_2, t_3 and t_4 respectively, then prove that :

- $t_1 t_2 t_3 t_4 = 1$
- The centre of mean position of the four points bisects the distance between the centres of the two curves.

Solution :

- Let the equation of the hyperbola referred to rectangular asymptotes as axes be $xy = c^2$ or its parametric equation be

$$x = ct, y = c/t \quad \dots\dots\dots (i)$$

and that of the circle be

$$x^2 + y^2 + 2gx + 2fy + k = 0 \quad \dots\dots\dots (ii)$$

Solving (i) and (ii), we get

$$c^2 t^2 + \frac{c^2}{t^2} + 2gct + 2f \frac{c}{t} + k = 0$$

$$\text{or } c^2 t^4 + 2gct^3 + kt^2 + 2fct + c^2 = 0 \quad \dots\dots\dots (iii)$$

Above equation being of fourth degree in t gives us the four parameters t_1, t_2, t_3, t_4 of the points of intersection.

$$\therefore t_1 + t_2 + t_3 + t_4 = -\frac{2gc}{c^2} = -\frac{2g}{c} \quad \dots\dots\dots (iv)$$

$$\begin{aligned} t_1 t_2 t_3 + t_1 t_2 t_4 + t_3 t_4 t_1 + t_3 t_4 t_2 \\ = -\frac{2fc}{c^2} = -\frac{2f}{c} \quad \dots\dots\dots (v) \end{aligned}$$

$$t_1 t_2 t_3 t_4 = \frac{c^2}{c^2} = 1. \text{ It proves (a) } \dots\dots\dots \text{ (vi)}$$

Dividing (v) by (vi), we get

$$\frac{1}{t_1} + \frac{1}{t_2} + \frac{1}{t_3} + \frac{1}{t_4} = -\frac{2f}{c} \dots\dots\dots \text{(vii)}$$

(b) The centre of mean position of the four points of intersection is

$$\left[\frac{c}{4}(t_1 + t_2 + t_3 + t_4), \frac{c}{4} \left(\frac{1}{t_1} + \frac{1}{t_2} + \frac{1}{t_3} + \frac{1}{t_4} \right) \right] = \left[\frac{c}{4} \left(-\frac{2g}{c} \right), \frac{c}{4} \left(-\frac{2f}{c} \right) \right], \text{ by (iv) and}$$

(vii)

$$= (-g/2, -f/2)$$

Above is clearly the mid-point of $(0, 0)$ and $(-g, -f)$ i.e. the join of the centres of the two curves.

ANSWERS FOR DO YOURSELF

1: (i) $\sqrt{3}$

(ii) $7y^2 + 24xy - 24ax - 6ay + 15a^2 = 0$

(iii) $6, 4; (\pm\sqrt{13}, 0); \sqrt{13}/3; 8/3$

(iv) $x^2 - y^2 = 32$

2: (i) $\sqrt{5}$ & 40 sq. units

3: (i) $n^2 = a^2 \ell^2 - b^2 m^2$

(ii) B

4: (i) $24y = 30x \pm \sqrt{161}$

(ii) $5x - 3y = 9$

(iii) $y = \pm x \pm \sqrt{7}$

5: (i) $y = 0$;

(ii) $8\sqrt{5}x + 18y = 75\sqrt{5}$

$$\text{(iii)} \quad \frac{a^2}{\ell^2} - \frac{b^2}{m^2} = \frac{(a^2 + b^2)^2}{n^2}$$

6 : (i) $5x + 3y = 16$

(ii) 20 & 24

(iii) $(x^2 + y^2)^2 = 16x^2 - 9y^2$

(iv) $(2x + 3y - 8)(3x + 2y - 7) = 154$

7: (i) $\Delta \neq 0, h^2 > ab, a + b = 0$

(ii) $2x + y = 4$

(iii) $e = \sqrt{3}$

EXERCISE (O-1)

1. Consider the hyperbola $9x^2 - 16y^2 + 72x - 32y - 16 = 0$. Find the following:
 (a) centre (b) eccentricity (c) foci (d) equation of directrix
 (e) length of the latus rectum (f) equation of auxiliary circle
 (g) equation of director circle

[STRAIGHT OBJECTIVE TYPE]

2. The area of the quadrilateral with its vertices at the foci of the conics
 $9x^2 - 16y^2 - 18x + 32y - 23 = 0$ and
 $25x^2 + 9y^2 - 50x - 18y + 33 = 0$, is
 (A) $5/6$ (B) $8/9$ (C) $5/3$ (D) $16/9$
3. Eccentricity of the hyperbola conjugate to the hyperbola $\frac{x^2}{4} - \frac{y^2}{12} = 1$ is
 (A) $\frac{2}{\sqrt{3}}$ (B) 2 (C) $\sqrt{3}$ (D) $\frac{4}{3}$
4. The foci of a hyperbola coincide with the foci of the ellipse $\frac{x^2}{25} + \frac{y^2}{9} = 1$. Then the equation of the hyperbola with eccentricity 2 is
 (A) $\frac{x^2}{12} - \frac{y^2}{4} = 1$ (B) $\frac{x^2}{4} - \frac{y^2}{12} = 1$ (C) $3x^2 - y^2 + 12 = 0$ (D) $9x^2 - 25y^2 - 225 = 0$
5. If the eccentricity of the hyperbola $x^2 - y^2 \sec^2 \alpha = 5$ is $\sqrt{3}$ times the eccentricity of the ellipse $x^2 \sec^2 \alpha + y^2 = 25$, then a value of α is :
 (A) $\pi/6$ (B) $\pi/4$ (C) $\pi/3$ (D) $\pi/2$
6. The foci of the ellipse $\frac{x^2}{16} + \frac{y^2}{b^2} = 1$ and the hyperbola $\frac{x^2}{144} - \frac{y^2}{81} = \frac{1}{25}$ coincide. Then the value of b^2 is-
 (A) 5 (B) 7 (C) 9 (D) 4
7. The graph of the equation $x + y = x^3 + y^3$ is the union of -
 (A) line and an ellipse (B) line and a parabola (C) line and hyperbola (D) line and a point
8. The focal length of the hyperbola $x^2 - 3y^2 - 4x - 6y - 11 = 0$, is-
 (A) 4 (B) 6 (C) 8 (D) 10
9. The equation $\frac{x^2}{29-p} + \frac{y^2}{4-p} = 1$ ($p \neq 4, 29$) represents -
 (A) an ellipse if p is any constant greater than 4
 (B) a hyperbola if p is any constant between 4 and 29.
 (C) a rectangular hyperbola if p is any constant greater than 29.
 (D) no real curve if p is less than 29.

10. If $\frac{x^2}{\cos^2 \alpha} - \frac{y^2}{\sin^2 \alpha} = 1$ represents family of hyperbolas where ' α ' varies then -
(A) distance between the foci is constant
(B) distance between the two directrices is constant
(C) distance between the vertices is constant
(D) distances between focus and the corresponding directrix is constant
11. The locus of the point of intersection of the lines $\sqrt{3}x - y - 4\sqrt{3}t = 0$ & $\sqrt{3}tx + ty - 4\sqrt{3} = 0$ (where t is a parameter) is a hyperbola whose eccentricity is
(A) $\sqrt{3}$ (B) 2 (C) $\frac{2}{\sqrt{3}}$ (D) $\frac{4}{3}$
12. Latus rectum of the conic satisfying the differential equation, $x dy + y dx = 0$ and passing through the point (2, 8) is :
(A) $4\sqrt{2}$ (B) 8 (C) $8\sqrt{2}$ (D) 16
13. The magnitude of the gradient of the tangent at an extremity of latera recta of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is equal to (where e is the eccentricity of the hyperbola)
(A) be (B) e (C) ab (D) ae
14. The number of possible tangents which can be drawn to the curve $4x^2 - 9y^2 = 36$, which are perpendicular to the straight line $5x + 2y - 10 = 0$ is :
(A) zero (B) 1 (C) 2 (D) 4
15. Locus of the point of intersection of the tangents at the points with eccentric angles ϕ and $\frac{\pi}{2} - \phi$ on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is :
(A) $x = a$ (B) $y = b$ (C) $x = ab$ (D) $y = ab$
16. Locus of the feet of the perpendiculars drawn from either foci on a variable tangent to the hyperbola $16y^2 - 9x^2 = 1$ is
(A) $x^2 + y^2 = 9$ (B) $x^2 + y^2 = 1/9$ (C) $x^2 + y^2 = 7/144$ (D) $x^2 + y^2 = 1/16$
17. A tangent to the ellipse $\frac{x^2}{9} + \frac{y^2}{4} = 1$ with centre C meets its director circle at P and Q . Then the product of the slopes of CP and CQ , is -
(A) $\frac{9}{4}$ (B) $-\frac{4}{9}$ (C) $\frac{2}{9}$ (D) $-\frac{1}{4}$
18. The asymptote of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ form with any tangent to the hyperbola a triangle whose area is $a^2 \tan \lambda$ in magnitude then its eccentricity is :
(A) $\sec \lambda$ (B) $\operatorname{cosec} \lambda$ (C) $\sec^2 \lambda$ (D) $\operatorname{cosec}^2 \lambda$

19. In which of the following cases maximum number of normals can be drawn from a point P lying in the same plane
 (A) circle (B) parabola (C) ellipse (D) hyperbola
20. PQ is a double ordinate of the ellipse $x^2 + 9y^2 = 9$, the normal at P meets the diameter through Q at R, then the locus of the mid point of PR is
 (A) a circle (B) a parabola (C) an ellipse (D) a hyperbola
21. With one focus of the hyperbola $\frac{x^2}{9} - \frac{y^2}{16} = 1$ as the centre, a circle is drawn which is tangent to the hyperbola with no part of the circle being outside the hyperbola. The radius of the circle is
 (A) less than 2 (B) 2 (C) $\frac{11}{3}$ (D) none
22. If the normal to the rectangular hyperbola $xy = c^2$ at the point 't' meets the curve again at 't₁', then $t^3 t_1$ has the value equal to
 (A) 1 (B) -1 (C) 0 (D) none
23. Number of common tangent with finite slope to the curves $xy = c^2$ & $y^2 = 4ax$ is :
 (A) 0 (B) 1 (C) 2 (D) 4
24. Locus of the middle points of the parallel chords with gradient m of the rectangular hyperbola $xy = c^2$ is
 (A) $y + mx = 0$ (B) $y - mx = 0$ (C) $my - x = 0$ (D) $my + x = 0$

[MULTIPLE OBJECTIVE TYPE]

25. Let p and q be non-zero real numbers. Then the equation $(px^2 + qy^2 + r)(4x^2 + 4y^2 - 8x - 4) = 0$ represents
 (A) two straight lines and a circle, when $r = 0$ and p, q are of the opposite sign.
 (B) two circles, when $p = q$ and r is of sign opposite to that of p.
 (C) a hyperbola and a circle, when p and q are of opposite sign and $r \neq 0$.
 (D) a circle and an ellipse, when p and q are unequal but of same sign and r is of sign opposite to that of p.
26. Solutions of the differential equation $(1 - x^2) \frac{dy}{dx} + xy = ax$ where $a \in \mathbb{R}$, is
 (A) a conic which is an ellipse or a hyperbola with principal axes parallel to coordinates axes.
 (B) centre of the conic is (0, a)
 (C) length of one of the principal axes is 1.
 (D) length of one of the principal axes is equal to 2.
27. If θ is eliminated from the equations $a \sec \theta - x \tan \theta = y$ and $b \sec \theta + y \tan \theta = x$ (a and b are constant), then the eliminant denotes the equation of
 (A) the director circle of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$
 (B) auxiliary circle of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$
 (C) Director circle of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$
 (D) Director circle of the circle $x^2 + y^2 = \frac{a^2 + b^2}{2}$.

28. The tangent to the hyperbola, $x^2 - 3y^2 = 3$ at the point $(\sqrt{3}, 0)$ when associated with two asymptotes constitutes -
 (A) isosceles triangle which is not equilateral (B) an equilateral triangle
 (C) a triangles whose area is $\sqrt{3}$ sq. units (D) a right isosceles triangle.
29. If latus rectum of a hyperbola subtends a right angle at other focus of hyperbola, then eccentricity is equal to-
 (A) $1 - \sqrt{2}$ (B) $\tan \frac{\pi}{8}$ (C) $\cot \frac{\pi}{8}$ (D) $\left(\frac{1}{\sqrt{2}-1} \right)$
30. If the circle $x^2 + y^2 = a^2$ intersects the hyperbola $xy = c^2$ in four points $P(x_1, y_1)$, $Q(x_2, y_2)$, $R(x_3, y_3)$, $S(x_4, y_4)$, then -
 (A) $x_1 + x_2 + x_3 + x_4 = 0$ (B) $y_1 + y_2 + y_3 + y_4 = 0$
 (C) $x_1 x_2 x_3 x_4 = c^4$ (D) $y_1 y_2 y_3 y_4 = c^4$

[COMPREHENSION TYPE]

Paragraph for question nos. 31 to 33

The graph of the conic $x^2 - (y - 1)^2 = 1$ has one tangent line with positive slope that passes through the origin. the point of tangency being (a, b). Then

31. The value of $\sin^{-1} \left(\frac{a}{b} \right)$ is
 (A) $\frac{5\pi}{12}$ (B) $\frac{\pi}{6}$ (C) $\frac{\pi}{3}$ (D) $\frac{\pi}{4}$
32. Length of the latus rectum of the conic is
 (A) 1 (B) $\sqrt{2}$ (C) 2 (D) none
33. Eccentricity of the conic is
 (A) $\frac{4}{3}$ (B) $\sqrt{3}$ (C) 2 (D) none

Paragraph for question nos. 34 to 36

Consider the conic C : $\frac{x^2}{16} + \frac{y^2}{12} = 1$

34. Equation of circle touching C at one extremity of latus-rectum and passing through centre of C is/are-
 (A) $8x^2 + 8y^2 - 19x - 22y = 0$ (B) $8x^2 + 8y^2 - 19x + 22y = 0$
 (C) $8x^2 + 8y^2 + 19x - 22y = 0$ (D) $8x^2 + 8y^2 + 19x + 22y = 0$
35. The equation of parabolas with same latus-rectum as conic C, is/are-
 (A) $y^2 - 6x + 3 = 0$ (B) $y^2 + 6x - 21 = 0$
 (C) $y^2 - 6x - 21 = 0$ (D) $y^2 + 6x + 3 = 0$
36. If a hyperbola passes through extremities of minor axis of conic C and its transverse and conjugate axis coincides with the minor and major axis of conic C respectively, and the product of eccentricity of hyperbola and conic C is 1, then -
 (A) equation of hyperbola is $\frac{x^2}{36} - \frac{y^2}{12} = -1$. (B) equation of hyperbola is $\frac{x^2}{9} - \frac{y^2}{3} = -1$.
 (C) focus of hyperbola is $(0, 2\sqrt{3})$ (D) focus of hyperbola is $(0, -4\sqrt{3})$

[REASONING TYPE]

37. **Statement-1 :** Consider two hyperbola $S \equiv 2x^2 - 4y^2 - 8 = 0$ and $S' \equiv 2x^2 - 4y^2 + 8 = 0$. S and S' are conjugate of each other.

and

Statement-2 : Length of transverse axis and conjugate axis of one of the given hyperbolas are respectively equals to length of conjugate axis and transverse axis of other hyperbola.

- (A) Statement-1 is True, Statement-2 is True ; Statement-2 is a correct explanation for Statement-1.
 (B) Statement-1 is True, Statement-2 is True ; Statement-2 is NOT a correct explanation for Statement-1.
 (C) Statement-1 is True, Statement-2 is False.
 (D) Statement-1 is False, Statement-2 is True.

38. **Statement-1:** The points of intersection of the tangents at three distinct points A, B, C on the parabola $y^2 = 4x$ can be collinear.

Statement-2: If a line L does not intersect the parabola $y^2 = 4x$, then from every point of the line two tangents can be drawn to the parabola.

- (A) Statement-1 is True, Statement-2 is True ; Statement-2 is a correct explanation for Statement-1
 (B) Statement-1 is True, Statement-2 is True ; Statement-2 is NOT a correct explanation for Statement-1
 (C) Statement -1 is True, Statement -2 is False
 (D) Statement -1 is False, Statement -2 is True
39. **Statement-1:** The latus rectum is the shortest focal chord in a parabola of length $4a$.

because

Statement-2: As the length of a focal chord of the parabola $y^2 = 4ax$ is $a\left(t + \frac{1}{t}\right)^2$, which is minimum when $t = 1$.

- (A) Statement-1 is True, Statement-2 is True ; Statement-2 is a correct explanation for Statement-1
 (B) Statement-1 is True, Statement-2 is True ; Statement-2 is NOT a correct explanation for Statement-1
 (C) Statement -1 is True, Statement -2 is False
 (D) Statement -1 is False, Statement -2 is True
40. **Statement-1:** The quadrilateral formed by the pair of tangents drawn from the point $(0, 2)$ to the parabola $y^2 - 2y + 4x + 5 = 0$ and the normals at the point of contact of tangents is a square.

Statement-2: The angle between tangents drawn from the given point to the parabola is 90° .

- (A) Statement-1 is True, Statement-2 is True ; Statement-2 is a correct explanation for Statement-1
 (B) Statement-1 is True, Statement-2 is True ; Statement-2 is NOT a correct explanation for Statement-1
 (C) Statement -1 is True, Statement -2 is False
 (D) Statement -1 is False, Statement -2 is True

[MATRIX MATCH TYPE]

41. Match the properties given in **column-I** with the corresponding curves given in the **column-II**.

Column-I	Column-II
(A) The curve such that product of the distances of any of its tangent from two given points is constant, can be	(P) Circle
(B) A curve for which the length of the subnormal at any of its point is equal to 2 and the curve passes through (1, 2), can be	(Q) Parabola
(C) A curve passes through (1, 4) and is such that the segment joining any point P on the curve and the point of intersection of the normal at P with the x-axis is bisected by the y-axis. The curve can be	(R) Ellipse
(D) A curve passes through (1, 2) is such that the length of the normal at any of its point is equal to 2. The curve can be	(S) Hyperbola

EXERCISE (O-2)

[STRAIGHT OBJECTIVE TYPE]

- Let F_1, F_2 are the foci of the hyperbola $\frac{x^2}{16} - \frac{y^2}{9} = 1$ and F_3, F_4 are the foci of its conjugate hyperbola. If e_H and e_C are their eccentricities respectively then the statement which holds true is
 (A) Their equations of the asymptotes are different.
 (B) $e_H > e_C$
 (C) Area of the quadrilateral formed by their foci is 50 sq. units.
 (D) Their auxiliary circles will have the same equation.
- For each positive integer n , consider the point P with abscissa n on the curve $y^2 - x^2 = 1$. If d_n represents the shortest distance from the point P to the line $y = x$ then $\lim_{n \rightarrow \infty} (n \cdot d_n)$ has the value equal to-
 (A) $\frac{1}{2\sqrt{2}}$ (B) $\frac{1}{2}$ (C) $\frac{1}{\sqrt{2}}$ (D) 0
- AB is a double ordinate of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ such that $\triangle AOB$ (where 'O' is the origin) is an equilateral triangle, then the eccentricity e of the hyperbola satisfies
 (A) $e > \sqrt{3}$ (B) $1 < e < \frac{2}{\sqrt{3}}$ (C) $e = \frac{2}{\sqrt{3}}$ (D) $e > \frac{2}{\sqrt{3}}$
- P is a point on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, N is the foot of the perpendicular from P on the transverse axis. The tangent to the hyperbola at P meets the transverse axis at T. If O is the centre of the hyperbola, then $OT \cdot ON$ is equal to :
 (A) e^2 (B) a^2 (C) b^2 (D) b^2/a^2
- Let the major axis of a standard ellipse equals the transverse axis of a standard hyperbola and their director circles have radius equal to $2R$ and R respectively. If e_1 and e_2 are the eccentricities of the ellipse and hyperbola then the correct relation is
 (A) $4e_1^2 - e_2^2 = 6$ (B) $e_1^2 - 4e_2^2 = 2$ (C) $4e_2^2 - e_1^2 = 6$ (D) $2e_1^2 - e_2^2 = 4$

6. The tangent to the hyperbola $xy = c^2$ at the point P intersects the x-axis at T and the y-axis at T'. The normal to the hyperbola at P intersects the x-axis at N and the y-axis at N'. The areas of the triangles PNT and PN'T' are Δ and Δ' respectively, then $\frac{1}{\Delta} + \frac{1}{\Delta'}$ is
 (A) equal to 1 (B) depends on t (C) depends on c (D) equal to 2
7. The chord PQ of the rectangular hyperbola $xy = a^2$ meets the axis of x at A; C is the mid point of PQ & 'O' is the origin. Then the ΔACO is :
 (A) equilateral (B) isosceles (C) right angled (D) right isosceles.
8. The locus of the foot of the perpendicular from the centre of the hyperbola $xy = c^2$ on a variable tangent is :
 (A) $(x^2 - y^2)^2 = 4c^2 xy$ (B) $(x^2 + y^2)^2 = 2c^2 xy$
 (C) $(x^2 + y^2) = 4c^2 xy$ (D) $(x^2 + y^2)^2 = 4c^2 xy$
9. The equation to the chord joining two points (x_1, y_1) and (x_2, y_2) on the rectangular hyperbola $xy = c^2$ is :
 (A) $\frac{x}{x_1 + x_2} + \frac{y}{y_1 + y_2} = 1$ (B) $\frac{x}{x_1 - x_2} + \frac{y}{y_1 - y_2} = 1$
 (C) $\frac{x}{y_1 + y_2} + \frac{y}{x_1 + x_2} = 1$ (D) $\frac{x}{y_1 - y_2} + \frac{y}{x_1 - x_2} = 1$
10. At the point of intersection of the rectangular hyperbola $xy = c^2$ and the parabola $y^2 = 4ax$ tangents to the rectangular hyperbola and the parabola make an angle θ and ϕ respectively with the axis of X, then
 (A) $\theta = \tan^{-1}(-2 \tan \phi)$ (B) $\phi = \tan^{-1}(-2 \tan \theta)$
 (C) $\theta = \frac{1}{2} \tan^{-1}(-\tan \phi)$ (D) $\phi = \frac{1}{2} \tan^{-1}(-\tan \theta)$

[MULTIPLE OBJECTIVE TYPE]

11. Which of the following equations in parametric form can represent a hyperbolic profile, where 't' is a parameter.
 (A) $x = \frac{a}{2} \left(t + \frac{1}{t} \right)$ & $y = \frac{b}{2} \left(t - \frac{1}{t} \right)$ (B) $\frac{tx}{a} - \frac{y}{b} + t = 0$ & $\frac{x}{a} + \frac{ty}{b} - 1 = 0$
 (C) $x = e^t + e^{-t}$ & $y = e^t - e^{-t}$ (D) $x^2 - 6 = 2 \cos t$ & $y^2 + 2 = 4 \cos^2 \frac{t}{2}$
12. Let A(-1,0) and B(2,0) be two points on the x - axis . A point M is moving in xy-plane (other than x - axis) in such a way that $\angle MBA = 2\angle MAB$, then the point M moves along a conic whose
 (A) coordinate of vertices are $(\pm 3, 0)$.
 (B) length of latus-rectum equals 6.
 (C) eccentricity equals 2.
 (D) equation of directrices are $x = \pm \frac{1}{2}$.

13. Hyperbola $\frac{x^2}{a^2} - \frac{y^2}{3} = 1$ of eccentricity e is confocal with the ellipse $\frac{x^2}{8} + \frac{y^2}{4} = 1$. Let A, B, C & D are points of intersection of hyperbola & ellipse, then-
- (A) $e = \frac{5}{2}$
 (B) $e = 2$
 (C) A, B, C, D are concyclic points
 (D) Number of common tangents of hyperbola & ellipse is 2
14. If the ellipse $4x^2 + 9y^2 + 12x + 12y + 5 = 0$ is confocal with a hyperbola having same principal axes, then -
- (A) angle between normals at their each point of intersection is 90° .
 (B) centre of the ellipse is $\left(-\frac{3}{2}, -\frac{2}{3}\right)$
 (C) distance between foci of the hyperbola is $\frac{2\sqrt{10}}{3}$
 (D) ellipse and hyperbola has same length of latus rectum
15. If the eccentricity of the ellipse $\frac{x^2}{(\log a)^2} + \frac{y^2}{(\log b)^2} = 1$ ($a > b > 0$, $a, b \neq 1$) is $\frac{1}{\sqrt{2}}$ and 'e' be the eccentricity of the hyperbola $\frac{x^2}{(\log_b a)^2} - y^2 = 1$, then e^2 is greater than (where $\log x = \ln x$)-
- (A) $\frac{3}{2}$ (B) $\frac{1}{2}$ (C) $\frac{2}{3}$ (D) $\frac{5}{4}$

[COMPREHENSION TYPE]

Paragraph for question nos. 16 to 18

From a point 'P' three normals are drawn to the parabola $y^2 = 4x$ such that two of them make angles with the abscissa axis, the product of whose tangents is 2. Suppose the locus of the point 'P' is a part of a conic 'C'. Now a circle $S = 0$ is described on the chord of the conic 'C' as diameter passing through the point (1, 0) and with gradient unity. Suppose (a, b) are the coordinates of the centre of this circle. If L_1 and L_2 are the two asymptotes of the hyperbola with length of its transverse axis $2a$ and conjugate axis $2b$ (principal axes of the hyperbola along the coordinate axes) then answer the following questions.

16. Locus of P is a
- (A) circle (B) parabola (C) ellipse (D) hyperbola
17. Radius of the circle $S = 0$ is
- (A) 4 (B) 5 (C) $\sqrt{17}$ (D) $\sqrt{23}$
18. The angle $\alpha \in (0, \pi/2)$ between the two asymptotes of the hyperbola lies in the interval
- (A) $(0, 15^\circ)$ (B) $(30^\circ, 45^\circ)$ (C) $(45^\circ, 60^\circ)$ (D) $(60^\circ, 75^\circ)$

Paragraph for question nos. 19 to 21

A conic C passes through the point $(2, 4)$ and is such that the segment of any of its tangents at any point contained between the co-ordinate axes is bisected at the point of tangency. Let S denotes circle described on the foci F_1 and F_2 of the conic C as diameter.

19. Vertex of the conic C is

(A) $(2, 2), (-2, -2)$

(B) $(2\sqrt{2}, 2\sqrt{2}), (-2\sqrt{2}, -2\sqrt{2})$

(C) $(4, 4), (-4, -4)$

(D) $(\sqrt{2}, \sqrt{2}), (-\sqrt{2}, -\sqrt{2})$

20. Director circle of the conic is

(A) $x^2 + y^2 = 4$

(B) $x^2 + y^2 = 8$

(C) $x^2 + y^2 = 2$

(D) None

21. Equation of the circle S is

(A) $x^2 + y^2 = 16$

(B) $x^2 + y^2 = 8$

(C) $x^2 + y^2 = 32$

(D) $x^2 + y^2 = 4$

[REASONING TYPE]

22. **Statement-1:** Diagonals of any parallelogram inscribed in an ellipse always intersect at the centre of the ellipse.

Statement-2: Centre of the ellipse is the only point at which two chords can bisect each other and every chord passing through the centre of the ellipse gets bisected at the centre.

(A) Statement-1 is True, Statement-2 is True ; Statement-2 is a correct explanation for Statement-1

(B) Statement-1 is True, Statement-2 is True ; Statement-2 is NOT a correct explanation for Statement-1

(C) Statement -1 is True, Statement -2 is False

(D) Statement -1 is False, Statement -2 is True

23. **Statement-1:** If $P(2a, 0)$ be any point on the axis of parabola $y^2 = 4ax$, then the chord QPR , satisfy

$$\frac{1}{(PQ)^2} + \frac{1}{(PR)^2} = \frac{1}{4a^2}.$$

Statement-2 : There exists a point P on the axis of the parabola $y^2 = 4ax$ (other than vertex), such that

$$\frac{1}{(PQ)^2} + \frac{1}{(PR)^2} = \text{constant for all chord } QPR \text{ of the parabola.}$$

(A) Statement-1 is True, Statement-2 is True ; Statement-2 is a correct explanation for Statement-1

(B) Statement-1 is True, Statement-2 is True ; Statement-2 is NOT a correct explanation for Statement-1

(C) Statement -1 is True, Statement -2 is False

(D) Statement -1 is False, Statement -2 is True

EXERCISE (S-1)

- Find the equation to the hyperbola whose directrix is $2x + y = 1$, focus $(1, 1)$ & eccentricity $\sqrt{3}$. Find also the length of its latus rectum.
- The hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ passes through the point of intersection of the lines, $7x + 13y - 87 = 0$ and $5x - 8y + 7 = 0$ & the latus rectum is $32\sqrt{2}/5$. Find 'a' & 'b'.
- For the hyperbola $\frac{x^2}{100} - \frac{y^2}{25} = 1$, prove that
 - eccentricity $= \sqrt{5}/2$
 - SA. $S'A = 25$, where S & S' are the foci & A is the vertex.
- Find the centre, the foci, the directrices, the length of the latus rectum, the length & the equations of the axes of the hyperbola $16x^2 - 9y^2 + 32x + 36y - 164 = 0$.
- If θ_1 & θ_2 are the parameters of the extremities of a chord through $(ae, 0)$ of a hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, then show that $\tan \frac{\theta_1}{2} \cdot \tan \frac{\theta_2}{2} + \frac{e-1}{e+1} = 0$.
- Find the equation of the tangent to the hyperbola $x^2 - 4y^2 = 36$ which is perpendicular to the line $x - y + 4 = 0$.
- Tangents are drawn to the hyperbola $3x^2 - 2y^2 = 25$ from the point $(0, 5/2)$. Find their equations.
- A conic C satisfies the differential equation, $(1 + y^2)dx - xy dy = 0$ and passes through the point $(1, 0)$. An ellipse E which is confocal with C having its eccentricity equal to $\sqrt{2/3}$.
 - Find the length of the latus rectum of the conic C
 - Find the equation of the ellipse E.
 - Find the locus of the point of intersection of the perpendicular tangents to the ellipse E.
- A hyperbola has one focus at the origin and its eccentricity $= \sqrt{2}$ and one of its directrix is $x + y + 1 = 0$. Find the equation to its asymptotes.
- If the lines $x + y + 1 = 0$ and $2x - y + 2 = 0$ are the asymptotes of a hyperbola. If the line $x - 2 = 0$ touches the hyperbola then the equation of the hyperbola is $4(x + y + 1)(2x - y + 2) = \lambda$. Find the value of λ .
- If C is the centre of a hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, S, S' its foci and P a point on it. Prove that $SP \cdot S'P = CP^2 - a^2 + b^2$.
- Chords of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ are tangents to the circle drawn on the line joining the foci as diameter. Find the locus of the point of intersection of tangents at the extremities of the chords.
- Let 'p' be the perpendicular distance from the centre C of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ to the tangent drawn at a point R on the hyperbola. If S & S' are the two foci of the hyperbola, then show that $(RS + RS')^2 = 4a^2 \left(1 + \frac{b^2}{p^2}\right)$.

14. If two points P & Q on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ whose centre is C be such that CP is perpendicular to CQ & $a < b$, then prove that $\frac{1}{CP^2} + \frac{1}{CQ^2} = \frac{1}{a^2} - \frac{1}{b^2}$.
15. Locus of the feet of the perpendicular from centre of the hyperbola $x^2 - 4y^2 = 4$ upon a variable normal to it has the equation, $(x^2 + y^2)^2 (4y^2 - x^2) = \lambda x^2 y^2$, find λ .
16. Let P ($a \sec \theta$, $b \tan \theta$) and Q ($a \sec \phi$, $b \tan \phi$), where $\theta + \phi = \frac{\pi}{2}$, be two points on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$. If (h, k) is the point of intersection of the normals at P & Q, then find k.

EXERCISE (S-2)

- Tangent and normal are drawn at the upper end (x_1, y_1) of the latus rectum P with $x_1 > 0$ and $y_1 > 0$, of the hyperbola $\frac{x^2}{4} - \frac{y^2}{12} = 1$, intersecting the transverse axis at T and G respectively. Find the area of the triangle PTG.
- Find the equations of the tangents to the hyperbola $x^2 - 9y^2 = 9$ that are drawn from (3, 2). Find the area of the triangle that these tangents form with their chord of contact.
- The normal to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ drawn at an extremity of its latus rectum is parallel to an asymptote. Show that the eccentricity is equal to the square root of $(1 + \sqrt{5})/2$.
- An ellipse and a hyperbola have their principal axes along the coordinate axes and have a common foci separated by a distance $2\sqrt{13}$, the difference of their focal semi axes is equal to 4. If the ratio of their eccentricities is $3/7$. Find the equation of these curves.
- From the centre C of the hyperbola $x^2 - y^2 = 9$, CM is drawn perpendicular to the tangent at any point of the curve, meeting the tangent at M and the curve at N. Find the value of the product (CM)(CN).
- Prove that the part of the tangent at any point of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ intercepted between the point of contact and the transverse axis is a harmonic mean between the lengths of the perpendicular drawn from the foci on the normal at the same point.
- Ascertain the co-ordinates of the two points Q & R, where the tangent to the hyperbola $\frac{x^2}{45} - \frac{y^2}{20} = 1$ at the point P(9, 4) intersects the two asymptotes. Finally prove that P is the middle point of QR. Also compute the area of the triangle CQR where C is the centre of the hyperbola.
- The tangent at P on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ meets one of the asymptote in Q. If the locus of the mid point of PQ is a hyperbola, $\frac{x^2}{a^2} - \frac{y^2}{b^2} = \lambda$, find the value of 4λ .

9. Through any point P of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ a line QPR is drawn with a fixed gradient m, meeting the asymptotes in Q & R. Show that the product, $(QP)(PR) = \frac{a^2 b^2 (1 + m^2)}{b^2 - a^2 m^2}$.
10. Tangents are drawn from the point (α, β) to the hyperbola $3x^2 - 2y^2 = 6$ and are inclined at angles θ and ϕ to the x-axis. If $\tan \theta \cdot \tan \phi = 2$, prove that $\beta^2 = 2\alpha^2 - 7$.

EXERCISE (JM)

1. The equation of the hyperbola whose foci are $(-2, 0)$ and $(2, 0)$ and eccentricity is 2 is given by : **[AIEEE-2011]**
 (1) $-3x^2 + y^2 = 3$ (2) $x^2 - 3y^2 = 3$ (3) $3x^2 - y^2 = 3$ (4) $-x^2 + 3y^2 = 3$
2. A tangent to the hyperbola $\frac{x^2}{4} - \frac{y^2}{2} = 1$ meets x-axis at P and y-axis at Q. Lines PR and QR are drawn such that OPRQ is a rectangle (where O is the origin). Then R lies on : **[JEE-Main (On line)-2013]**
 (1) $\frac{2}{x^2} - \frac{4}{y^2} = 1$ (2) $\frac{4}{x^2} - \frac{2}{y^2} = 1$ (3) $\frac{4}{x^2} + \frac{2}{y^2} = 1$ (4) $\frac{2}{x^2} + \frac{4}{y^2} = 1$
3. A common tangent to the conics $x^2 = 6y$ and $2x^2 - 4y^2 = 9$ is : **[JEE-Main (On line)-2013]**
 (1) $x + y = \frac{9}{2}$ (2) $x + y = 1$ (3) $x - y = \frac{3}{2}$ (4) $x - y = 1$
4. The eccentricity of the hyperbola whose length of the latus rectum is equal to 8 and the length of its conjugate axis is equal to half of the distance between its foci, is : **[JEE (Main) 2016]**
 (1) $\sqrt{3}$ (2) $\frac{4}{3}$ (3) $\frac{4}{\sqrt{3}}$ (4) $\frac{2}{\sqrt{3}}$
5. A hyperbola passes through the point $P(\sqrt{2}, \sqrt{3})$ and has foci at $(\pm 2, 0)$. Then the tangent to this hyperbola at P also passes through the point : **[JEE (Main) 2017]**
 (1) $(-\sqrt{2}, -\sqrt{3})$ (2) $(3\sqrt{2}, 2\sqrt{3})$ (3) $(2\sqrt{2}, 3\sqrt{3})$ (4) $(\sqrt{3}, \sqrt{2})$
6. Tangents are drawn to the hyperbola $4x^2 - y^2 = 36$ at the point P and Q. If these tangents intersect at the point T(0, 3) then the area (in sq. units) of ΔPTQ is - **[JEE (Main) 2018]**
 (1) $54\sqrt{3}$ (2) $60\sqrt{3}$ (3) $36\sqrt{5}$ (4) $45\sqrt{5}$
7. Let $S = \left\{ (x, y) \in \mathbb{R}^2 : \frac{y^2}{1+r} - \frac{x^2}{1-r} = 1 \right\}$, where $r \neq \pm 1$. Then S represents : **[JEE (Main)-Jan 19]**
 (1) A hyperbola whose eccentricity is $\frac{2}{\sqrt{r+1}}$, where $0 < r < 1$.
 (2) An ellipse whose eccentricity is $\frac{1}{\sqrt{r+1}}$, where $r > 1$
 (3) A hyperbola whose eccentricity is $\frac{2}{\sqrt{1-r}}$, when $0 < r < 1$.
 (4) An ellipse whose eccentricity is $\sqrt{\frac{2}{r+1}}$, when $r > 1$

8. Equation of a common tangent to the parabola $y^2 = 4x$ and the hyperbola $xy = 2$ is :

[JEE (Main)-Jan 19]

(1) $x + 2y + 4 = 0$

(2) $x - 2y + 4 = 0$

(3) $x + y + 1 = 0$

(4) $4x + 2y + 1 = 0$

9. If a hyperbola has length of its conjugate axis equal to 5 and the distance between its foci is 13, then the eccentricity of the hyperbola is :-

[JEE (Main)-Jan 19]

(1) 2

(2) $\frac{13}{6}$

(3) $\frac{13}{8}$

(4) $\frac{13}{12}$

10. If the line $y = mx + 7\sqrt{3}$ is normal to the hyperbola $\frac{x^2}{24} - \frac{y^2}{18} = 1$, then a value of m is

[JEE (Main)-Apr 19]

(1) $\frac{\sqrt{5}}{2}$

(2) $\frac{3}{\sqrt{5}}$

(3) $\frac{2}{\sqrt{5}}$

(4) $\frac{\sqrt{15}}{2}$

11. If a directrix of a hyperbola centred at the origin and passing through the point $(4, -2\sqrt{3})$ is $5x = 4\sqrt{5}$ and its eccentricity is e , then :

[JEE (Main)-Apr 19]

(1) $4e^4 - 24e^2 + 35 = 0$

(2) $4e^4 + 8e^2 - 35 = 0$

(3) $4e^4 - 12e^2 - 27 = 0$

(4) $4e^4 - 24e^2 + 27 = 0$

12. Let P be the point of intersection of the common tangents to the parabola $y^2 = 12x$ and the hyperbola $8x^2 - y^2 = 8$. If S and S' denote the foci of the hyperbola where S lies on the positive x -axis then P divides SS' in a ratio :

[JEE (Main)-Apr 19]

(1) 5:4

(2) 14:13

(3) 2:1

(4) 13:11

EXERCISE (JA)

1. Consider a branch of the hyperbola, $x^2 - 2y^2 - 2\sqrt{2}x - 4\sqrt{2}y - 6 = 0$ with vertex at the point A . Let B be one of the end points of its latus rectum. If C is the focus of the hyperbola nearest to the point A , then the area of the triangle ABC is

[JEE 2008, 3]

(A) $1 - \sqrt{\frac{2}{3}}$

(B) $\sqrt{\frac{3}{2}} - 1$

(C) $1 + \sqrt{\frac{2}{3}}$

(D) $\sqrt{\frac{3}{2}} + 1$

2. The locus of the orthocentre of the triangle formed by the lines $(1 + p)x - py + p(1 + p) = 0$, $(1 + q)x - qy + q(1 + q) = 0$ and $y = 0$, where $p \neq q$, is

[JEE 2009, 3]

(A) a hyperbola

(B) a parabola

(C) an ellipse

(D) a straight line

3. An ellipse intersects the hyperbola $2x^2 - 2y^2 = 1$ orthogonally. The eccentricity of the ellipse is reciprocal of that of the hyperbola. If the axes of the ellipse are along the coordinate axes, then

[JEE 2009, 4]

(A) Equation of ellipse is $x^2 + 2y^2 = 2$

(B) The foci of ellipse are $(\pm 1, 0)$

(C) Equation of ellipse is $x^2 + 2y^2 = 4$

(D) The foci of ellipse are $(\pm \sqrt{2}, 0)$

4. Match the conics in **Column I** with the statements/expressions in **Column II**. [JEE 2009, 8]

Column I	Column II
(A) Circle	(p) The locus of the point (h, k) for which the line $hx + ky = 1$ touches the circle $x^2 + y^2 = 4$
(B) Parabola	(q) Points z in the complex plane satisfying $ z + 2 - z - 2 = \pm 3$
(C) Ellipse	(r) Points of the conic have parametric representation $x = \sqrt{3} \left(\frac{1-t^2}{1+t^2} \right), y = \frac{2t}{1+t^2}$
(D) Hyperbola	(s) The eccentricity of the conic lies in the interval $1 \leq e < \infty$
	(t) Points z in the complex plane satisfying $\operatorname{Re}(z + 1)^2 = z ^2 + 1$

Comprehension :

[JEE 2010, 3+3]

The circle $x^2 + y^2 - 8x = 0$ and hyperbola $\frac{x^2}{9} - \frac{y^2}{4} = 1$ intersect at the points A and B.

5. Equation of a common tangent with positive slope to the circle as well as to the hyperbola is -

- (A) $2x - \sqrt{5}y - 20 = 0$ (B) $2x - \sqrt{5}y + 4 = 0$
(C) $3x - 4y + 8 = 0$ (D) $4x - 3y + 4 = 0$

6. Equation of the circle with AB as its diameter is -

- (A) $x^2 + y^2 - 12x + 24 = 0$ (B) $x^2 + y^2 + 12x + 24 = 0$
(C) $x^2 + y^2 + 24x - 12 = 0$ (D) $x^2 + y^2 - 24x - 12 = 0$

7. The line $2x + y = 1$ is tangent to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$. If this line passes through the point of intersection of the nearest directrix and the x-axis, then the eccentricity of the hyperbola is

[JEE 2010, 3]

8. Let the eccentricity of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ be reciprocal to that of the ellipse $x^2 + 4y^2 = 4$. If the hyperbola passes through a focus of the ellipse, then - **[JEE 2011, 4]**

[JEE 2011, 4]

- (A) the equation of the hyperbola is $\frac{x^2}{3} - \frac{y^2}{2} = 1$

- (B) a focus of the hyperbola is $(2,0)$

- (C) the eccentricity of the hyperbola is $\sqrt{\frac{5}{3}}$

- (D) the equation of the hyperbola is $x^2 - 3y^2 = 3$

9. Let P(6, 3) be a point on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$. If the normal at the point P intersects the x-axis at (9,0), then the eccentricity of the hyperbola is - **[JEE 2011, 3]**

[JEE 2011, 3]

- (A) $\sqrt{\frac{5}{2}}$ (B) $\sqrt{\frac{3}{2}}$ (C) $\sqrt{2}$ (D) $\sqrt{3}$

10. Tangents are drawn to the hyperbola $\frac{x^2}{9} - \frac{y^2}{4} = 1$, parallel to the straight line $2x - y = 1$. The points of contact of the tangents on the hyperbola are [JEE 2012, 4M]

(A) $\left(\frac{9}{2\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$ (B) $\left(-\frac{9}{2\sqrt{2}}, -\frac{1}{\sqrt{2}}\right)$
 (C) $(3\sqrt{3}, -2\sqrt{2})$ (D) $(-3\sqrt{3}, 2\sqrt{2})$

11. Consider the hyperbola $H : x^2 - y^2 = 1$ and a circle S with center $N(x_2, 0)$. Suppose that H and S touch each other at a point $P(x_1, y_1)$ with $x_1 > 1$ and $y_1 > 0$. The common tangent to H and S at P intersects the x -axis at point M . If (l, m) is the centroid of the triangle ΔPMN , then the correct expression(s) is(are) [JEE 2015, 4M, -0M]

(A) $\frac{dl}{dx_1} = 1 - \frac{1}{3x_1^2}$ for $x_1 > 1$ (B) $\frac{dm}{dx_1} = \frac{x_1}{3(\sqrt{x_1^2 - 1})}$ for $x_1 > 1$
 (C) $\frac{dl}{dx_1} = 1 + \frac{1}{3x_1^2}$ for $x_1 > 1$ (D) $\frac{dm}{dy_1} = \frac{1}{3}$ for $y_1 > 0$

12. If $2x - y + 1 = 0$ is tangent to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{16} = 1$, then which of the following CANNOT be sides of a right angled triangle ? [JEE(Advanced)-2017, 4(-2)]

(A) 2a, 4, 1 (B) 2a, 8, 1 (C) a, 4, 1 (D) a, 4, 2

Column 1, 2 and 3 contain conics, equation of tangents to the conics and points of contact, respectively.

Column 1

Column 2

Column 3

(I) $x^2 + y^2 = a^2$

(i) $my = m^2x + a$

(P) $\left(\frac{a}{m^2}, \frac{2a}{m}\right)$

(II) $x^2 + a^2y^2 = a^2$

(ii) $y = mx + a\sqrt{m^2 + 1}$

(Q) $\left(\frac{-ma}{\sqrt{m^2 + 1}}, \frac{a}{\sqrt{m^2 + 1}}\right)$

(III) $y^2 = 4ax$

(iii) $y = mx + \sqrt{a^2m^2 - 1}$

(R) $\left(\frac{-a^2m}{\sqrt{a^2m^2 + 1}}, \frac{1}{\sqrt{a^2m^2 + 1}}\right)$

(IV) $x^2 - a^2y^2 = a^2$

(iv) $y = mx + \sqrt{a^2m^2 + 1}$

(S) $\left(\frac{-a^2m}{\sqrt{a^2m^2 - 1}}, \frac{-1}{\sqrt{a^2m^2 - 1}}\right)$

13. The tangent to a suitable conic (Column 1) at $\left(\sqrt{3}, \frac{1}{2}\right)$ is found to be $\sqrt{3}x + 2y = 4$, then which of the following options is the only **CORRECT** combination ? [JEE(Advanced)-2017, 3(-1)]

(A) (II) (iii) (R) (B) (IV) (iv) (S) (C) (IV) (iii) (S) (D) (II) (iv) (R)

14. If a tangent to a suitable conic (Column 1) is found to be $y = x + 8$ and its point of contact is $(8, 16)$, then which of the following options is the only **CORRECT** combination ?

[JEE(Advanced)-2017, 3(-1)]

- (A) (III) (i) (P) (B) (III) (ii) (Q) (C) (II) (iv) (R) (D) (I) (ii) (Q)

15. For $a = \sqrt{2}$, if a tangent is drawn to a suitable conic (Column 1) at the point of contact $(-1, 1)$, then which of the following options is the only **CORRECT** combination for obtaining its equation ?

[JEE(Advanced)-2017, 3(-1)]

- (A) (II) (ii) (Q) (B) (III) (i) (P) (C) (I) (i) (P) (D) (I) (ii) (Q)

16. Let $H : \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, where $a > b > 0$, be a hyperbola in the xy -plane whose conjugate axis LM subtends an angle of 60° at one of its vertices N . Let the area of the triangle LMN be $4\sqrt{3}$.

LIST-I

P. The length of the conjugate axis of H is

Q. The eccentricity of H is

R. The distance between the foci of H is

S. The length of the latus rectum of H is

The correct option is :

(A) $P \rightarrow 4$; $Q \rightarrow 2$; $R \rightarrow 1$; $S \rightarrow 3$

(C) $P \rightarrow 4$; $Q \rightarrow 1$; $R \rightarrow 3$; $S \rightarrow 2$

(B) $P \rightarrow 4$; $Q \rightarrow 3$; $R \rightarrow 1$; $S \rightarrow 2$

(D) $P \rightarrow 3$; $Q \rightarrow 4$; $R \rightarrow 2$; $S \rightarrow 1$

[JEE(Advanced)-2018, 3(-1)]

LIST-II

1. 8

2. $\frac{4}{\sqrt{3}}$

3. $\frac{2}{\sqrt{3}}$

4. 4

ANSWER KEY HYPERBOLA EXERCISE (O-1)

1. (a) $(-4, -1)$; (b) $\frac{5}{4}$; (c) $(1, -1), (-9, -1)$; (d) $5x + 4 = 0, 5x + 36 = 0$; (e) $\frac{9}{2}$; (f) $(x + 4)^2 + (y + 1)^2 = 16$; (g) $(x + 4)^2 + (y + 1)^2 = 7$
2. B 3. A 4. B 5. B 6. B 7. A 8. C 9. B
 10. A 11. B 12. C 13. B 14. A 15. B 16. D 17. B
 18. A 19. A 20. C 21. B 22. B 23. B 24. A
 25. A, B, C, D 26. A, B, D 27. C, D 28. B, C 29. C, D 30. A, B, C, D
 31. D 32. C 33. D 34. A, B, C, D 35. A, B, C, D 36. A, D
 37. B 38. D 39. A 40. D 41. (A) R, S, (B) Q, (C) R, (D) P

EXERCISE (O-2)

1. C 2. A 3. D 4. B 5. C 6. C 7. B 8. D
 9. A 10. A 11. A, C, D 12. B, C, D 13. B, C 14. A, B, C 15. B, C, D 16. B
 17. A 18. D 19. B 20. D 21. C 22. A 23. A

EXERCISE (S-1)

1. $\sqrt{\frac{48}{5}}$ 2. $a^2 = 25/2$; $b^2 = 16$ 4. $(-1, 2)$; $(4, 2)$ & $(-6, 2)$; $5x - 4 = 0$ & $5x + 14 = 0$; $\frac{32}{3}$; 6; 8;
 $y - 2 = 0$; $x + 1 = 0$; $4x - 3y + 10 = 0$; $4x + 3y - 2 = 0$.
6. $x + y \pm 3\sqrt{3} = 0$ 7. $3x + 2y - 5 = 0$; $3x - 2y + 5 = 0$ 8. (a) 2; (b) $\frac{x^2}{3} + \frac{y^2}{1} = 1$; (c) $x^2 + y^2 = 4$
9. $x + 1 = 0$ only $y + 1 = 0$ 10. 81 12. $\frac{x^2}{a^4} + \frac{y^2}{b^4} = \frac{1}{a^2 + b^2}$ 15. 25 16. $-\left(\frac{a^2 + b^2}{b}\right)$

EXERCISE (S-2)

1. 45 2. $y = \frac{5}{12}x + \frac{3}{4}$; $x - 3 = 0$; 8 sq. unit 4. $\frac{x^2}{49} + \frac{y^2}{36} = 1$; $\frac{x^2}{9} - \frac{y^2}{4} = 1$
 5. 9 7. $(15, 10)$ and $(3, -2)$ and 30 sq. units 8. 3

EXERCISE (JM)

1. 3 2. 2 3. 3 4. 4 5. 3 6. 4 7. 4 8. 1
 9. 4 10. 3 11. 1 12. 1

EXERCISE (JA)

1. B 2. D 3. A, B 4. (A) p, (B) s, t; (C) r; (D) q, s 5. B
 6. A 7. 2 8. B, D 9. B 10. A, B 11. A, B, D 12. B, C, D
 13. D 14. A 15. D 16. B

NURTURE COURSE
BINOMIAL THEOREM

CONTENTS

BINOMIAL THEOREM

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JEE (Main) Syllabus :

Binomial theorem for a positive integral index, general term and middle term, properties of Binomial coefficients and simple applications.

JEE (Advanced) Syllabus :

Binomial theorem for a positive integral index, properties of binomial coefficients.

BINOMIAL THEOREM

1. BINOMIAL EXPRESSION :

Any algebraic expression which contains two dissimilar terms is called binomial expression.

For example : $x - y$, $xy + \frac{1}{x}$, $\frac{1}{z} - 1$, $\frac{1}{(x-y)^{1/3}} + 3$ etc.

2. BINOMIAL THEOREM :

The formula by which any positive integral power of a binomial expression can be expanded in the form of a series is known as **BINOMIAL THEOREM**.

If $x, y \in \mathbb{R}$ and $n \in \mathbb{N}$, then :

$$(x + y)^n = {}^nC_0 x^n + {}^nC_1 x^{n-1} y + {}^nC_2 x^{n-2} y^2 + \dots + {}^nC_r x^{n-r} y^r + \dots + {}^nC_n y^n = \sum_{r=0}^n {}^nC_r x^{n-r} y^r$$

This theorem can be proved by induction.

Observations :

- The number of terms in the expansion is $(n+1)$ i.e. one more than the index.
- The sum of the indices of x & y in each term is n .
- The binomial coefficients of the terms $({}^nC_0, {}^nC_1, \dots)$ equidistant from the beginning and the end are equal. i.e. ${}^nC_r = {}^nC_{n-r}$
- Symbol nC_r can also be denoted by $\binom{n}{r}$, $C(n, r)$ or A_r^n .

Some important expansions :

- $(1 + x)^n = {}^nC_0 + {}^nC_1 x + {}^nC_2 x^2 + \dots + {}^nC_n x^n$.
- $(1 - x)^n = {}^nC_0 - {}^nC_1 x + {}^nC_2 x^2 + \dots + (-1)^n \cdot {}^nC_n x^n$.

Note : The coefficient of x^r in $(1 + x)^n = {}^nC_r$ & that in $(1 - x)^n = (-1)^r \cdot {}^nC_r$

Illustration 1 : Expand : $(y + 2)^6$.

Solution : ${}^6C_0 y^6 + {}^6C_1 y^5 \cdot 2 + {}^6C_2 y^4 \cdot 2^2 + {}^6C_3 y^3 \cdot 2^3 + {}^6C_4 y^2 \cdot 2^4 + {}^6C_5 y^1 \cdot 2^5 + {}^6C_6 \cdot 2^6$
 $= y^6 + 12y^5 + 60y^4 + 160y^3 + 240y^2 + 192y + 64$.

Illustration 2 : Write first 4 terms of $\left(1 - \frac{2y^2}{5}\right)^7$

Solution : ${}^7C_0, {}^7C_1 \left(-\frac{2y^2}{5}\right), {}^7C_2 \left(-\frac{2y^2}{5}\right)^2, {}^7C_3 \left(-\frac{2y^2}{5}\right)^3$

Illustration 3 : If in the expansion of $(1 + x)^m (1 - x)^n$, the coefficients of x and x^2 are 3 and -6 respectively then m is - [JEE 99]

- (A) 6 (B) 9 (C) 12 (D) 24

Solution : $(1 + x)^m (1 - x)^n = \left[1 + mx + \frac{(m)(m-1) \cdot x^2}{2} + \dots\right] \left[1 - nx + \frac{n(n-1)}{2} x^2 + \dots\right]$

$$\text{Coefficient of } x = m - n = 3 \quad \dots\dots(i)$$

$$\text{Coefficient of } x^2 = -mn + \frac{n(n+1)}{2} + \frac{m(m-1)}{2} = -6 \quad \dots\dots(ii)$$

Solving (i) and (ii), we get

$$m = 12 \text{ and } n = 9.$$

Do yourself - 1 :

(i) Expand $\left(3x^2 - \frac{x}{2}\right)^5$

(ii) Expand $(y + x)^n$

Pascal's triangle :

$(x+y)^0$	1	1
$(x+y)^1$	$x + y$	1 1
$(x+y)^2$	$x^2 + 2xy + y^2$	1 2 1
$(x+y)^3$	$x^3 + 3x^2y + 3xy^2 + y^3$	1 3 3 1
$(x+y)^4$	$x^4 + 4x^3y + 6x^2y^2 + 4xy^3 + y^4$	1 4 6 4 1

Pascal's triangle

- (i) **Pascal's triangle** - A triangular arrangement of numbers as shown. The numbers give the binomial coefficients for the expansion of $(x + y)^n$. The first row is for $n = 0$, the second for $n = 1$, etc. Each row has 1 as its first and last number. Other numbers are generated by adding the two numbers immediately to the left and right in the row above.
- (ii) Pascal triangle is formed by binomial coefficient.
- (iii) The number of terms in the expansion of $(x+y)^n$ is $(n + 1)$ i.e. one more than the index.
- (iv) The sum of the indices of x & y in each term is n .
- (v) Power of first variable (x) decreases while of second variable (y) increases.
- (vi) Binomial coefficients are also called **combinatorial coefficients**.
- (vii) Binomial coefficients of the terms equidistant from the beginning and end are equal.
- (viii) r^{th} term from the beginning in the expansion of $(x + y)^n$ is same as r^{th} term from end in the expansion of $(y + x)^n$.
- (ix) r^{th} term from the end in $(x + y)^n$ is $(n - r + 2)^{\text{th}}$ term from the beginning.

3. IMPORTANT TERMS IN THE BINOMIAL EXPANSION :

- (a) **General term:** The general term or the $(r + 1)^{\text{th}}$ term in the expansion of $(x + y)^n$ is given by

$$T_{r+1} = {}^nC_r x^{n-r} y^r$$

Illustration 4 : Find : (a) The coefficient of x^7 in the expansion of $\left(ax^2 + \frac{1}{bx}\right)^{11}$

(b) The coefficient of x^{-7} in the expansion of $\left(ax^2 + \frac{1}{bx}\right)^{11}$

Also, find the relation between a and b , so that these coefficients are equal.

Solution :

(a) In the expansion of $\left(ax^2 + \frac{1}{bx}\right)^{11}$, the general term is :

$$T_{r+1} = {}^{11}C_r (ax^2)^{11-r} \left(\frac{1}{bx}\right)^r = {}^{11}C_r \cdot \frac{a^{11-r}}{b^r} \cdot x^{22-3r}$$

putting $22 - 3r = 7$

$$\therefore 3r = 15 \Rightarrow r = 5$$

$$\therefore T_6 = {}^{11}C_5 \frac{a^6}{b^5} \cdot x^7$$

Hence the coefficient of x^7 in $\left(ax^2 + \frac{1}{bx}\right)^{11}$ is ${}^{11}C_5 a^6 b^{-5}$.

Ans.

Note that binomial coefficient of sixth term is ${}^{11}C_5$.

(b) In the expansion of $\left(ax - \frac{1}{bx^2}\right)^{11}$, general term is :

$$T_{r+1} = {}^{11}C_r (ax)^{11-r} \left(-\frac{1}{bx^2}\right)^r = (-1)^r {}^{11}C_r \frac{a^{11-r}}{b^r} \cdot x^{11-3r}$$

putting $11 - 3r = -7$

$$\therefore 3r = 18 \Rightarrow r = 6$$

$$\therefore T_7 = (-1)^6 \cdot {}^{11}C_6 \frac{a^5}{b^6} \cdot x^{-7}$$

Hence the coefficient of x^{-7} in $\left(ax - \frac{1}{bx^2}\right)^{11}$ is ${}^{11}C_6 a^5 b^{-6}$.

Ans.

Also given :

$$\text{Coefficient of } x^7 \text{ in } \left(ax^2 + \frac{1}{bx}\right)^{11} = \text{coefficient of } x^{-7} \text{ in } \left(ax - \frac{1}{bx^2}\right)^{11}$$

$$\Rightarrow {}^{11}C_5 a^6 b^{-5} = {}^{11}C_6 a^5 b^{-6}$$

$$\Rightarrow ab = 1 \quad (\because {}^{11}C_5 = {}^{11}C_6)$$

which is the required relation between a and b.

Ans.

Illustration 5 : Find the number of rational terms in the expansion of $(9^{1/4} + 8^{1/6})^{1000}$.

Solution : The general term in the expansion of $(9^{1/4} + 8^{1/6})^{1000}$ is

$$T_{r+1} = {}^{1000}C_r \left(9^{1/4}\right)^{1000-r} \left(8^{1/6}\right)^r = {}^{1000}C_r 3^{\frac{1000-r}{2}} 2^{\frac{r}{2}}$$

The above term will be rational if exponents of 3 and 2 are integers

It means $\frac{1000-r}{2}$ and $\frac{r}{2}$ must be integers

The possible set of values of r is $\{0, 2, 4, \dots, 1000\}$

Hence, number of rational terms is 501

Ans.

(b) Middle term :

The middle term(s) in the expansion of $(x + y)^n$ is (are) :

(i) If n is even, there is only one middle term which is given by $T_{(n+2)/2} = {}^nC_{n/2} \cdot x^{n/2} \cdot y^{n/2}$

(ii) If n is odd, there are two middle terms which are $T_{(n+1)/2}$ & $T_{[(n+1)/2]+1}$

Important Note :

Middle term has greatest binomial coefficient and if there are 2 middle terms their coefficients will be equal.

$$\Rightarrow {}^nC_r \text{ will be maximum } \begin{cases} \text{When } r = \frac{n}{2} \text{ if } n \text{ is even} \\ \text{When } r = \frac{n-1}{2} \text{ or } \frac{n+1}{2} \text{ if } n \text{ is odd} \end{cases}$$

$\Rightarrow$ The term containing greatest binomial coefficient will be middle term in the expansion of $(1+x)^n$

Illustration 6 : Find the middle term in the expansion of $\left(3x - \frac{x^3}{6}\right)^9$

Solution : The number of terms in the expansion of $\left(3x - \frac{x^3}{6}\right)^9$ is 10 (even). So there are two middle terms.

i.e. $\left(\frac{9+1}{2}\right)^{\text{th}}$ and $\left(\frac{9+3}{2}\right)^{\text{th}}$ are two middle terms. They are given by T_5 and T_6

$$\therefore T_5 = T_{4+1} = {}^9C_4 (3x)^5 \left(-\frac{x^3}{6}\right)^4 = {}^9C_4 3^5 x^5 \cdot \frac{x^{12}}{6^4} = \frac{9 \cdot 8 \cdot 7 \cdot 6}{1 \cdot 2 \cdot 3 \cdot 4} \cdot \frac{3^5}{2^4 \cdot 3^4} x^{17} = \frac{189}{8} x^{17}$$

$$\text{and } T_6 = T_{5+1} = {}^9C_5 (3x)^4 \left(-\frac{x^3}{6}\right)^5 = -{}^9C_4 3^4 \cdot x^4 \cdot \frac{x^{15}}{6^5} = \frac{-9 \cdot 8 \cdot 7 \cdot 6}{1 \cdot 2 \cdot 3 \cdot 4} \cdot \frac{3^4}{2^5 \cdot 3^5} x^{19} = -\frac{21}{16} x^{19} \quad \text{Ans.}$$

(c) Term independent of x :

Term independent of x does not contain x ; Hence find the value of r for which the exponent of x is zero.

Illustration 7 : The term independent of x in $\left[\sqrt{\frac{x}{3}} + \sqrt{\left(\frac{3}{2x^2}\right)}\right]^{10}$ is -

- (A) 1 (B) $\frac{5}{12}$ (C) ${}^{10}C_1$ (D) none of these

Solution : General term in the expansion is

$${}^{10}C_r \left(\frac{x}{3}\right)^{\frac{r}{2}} \left(\frac{3}{2x^2}\right)^{\frac{10-r}{2}} = {}^{10}C_r x^{\frac{3r}{2}-10} \cdot \frac{3^{5-r}}{2^{\frac{10-r}{2}}} \quad \text{For constant term, } \frac{3r}{2} = 10 \Rightarrow r = \frac{20}{3}$$

which is not an integer. Therefore, there will be no constant term.

Ans. (D)

Do yourself - 2 :

(i) Find the 7th term of $\left(3x^2 - \frac{1}{3}\right)^{10}$

(ii) Find the term independent of x in the expansion : $\left(2x^2 - \frac{3}{x^3}\right)^{25}$

(iii) Find the middle term in the expansion of: (a) $\left(\frac{2x}{3} - \frac{3}{2x}\right)^6$ (b) $\left(2x^2 - \frac{1}{x}\right)^7$

(d) Numerically greatest term :

Let numerically greatest term in the expansion of $(a + b)^n$ be T_{r+1} .

$$\Rightarrow \begin{cases} |T_{r+1}| \geq |T_r| \\ |T_{r+1}| \geq |T_{r+2}| \end{cases} \text{ where } T_{r+1} = {}^nC_r a^{n-r} b^r$$

Solving above inequalities we get $\frac{n+1}{1+\left|\frac{a}{b}\right|} - 1 \leq r \leq \frac{n+1}{1+\left|\frac{a}{b}\right|}$

Case I : When $\frac{n+1}{1+\left|\frac{a}{b}\right|}$ is an integer equal to m , then T_m and T_{m+1} will be numerically greatest term.

Case II : When $\frac{n+1}{1+\left|\frac{a}{b}\right|}$ is not an integer and its integral part is m , then T_{m+1} will be the numerically greatest term.

Illustration 8 : Find numerically greatest term in the expansion of $(3 - 5x)^{11}$ when $x = \frac{1}{5}$

Solution : Using $\frac{n+1}{1+\left|\frac{a}{b}\right|} - 1 \leq r \leq \frac{n+1}{1+\left|\frac{a}{b}\right|}$

$$\frac{11+1}{1+\left|\frac{3}{-5x}\right|} - 1 \leq r \leq \frac{11+1}{1+\left|\frac{3}{-5x}\right|}$$

solving we get $2 \leq r \leq 3$

$$\therefore r = 2, 3$$

so, the greatest terms are T_{2+1} and T_{3+1} .

$\therefore$ Greatest term (when $r = 2$)

$$T_3 = {}^{11}C_2 \cdot 3^9 \cdot (-5x)^2 = 55 \cdot 3^9 = T_4$$

From above we say that the value of both greatest terms are equal.

Ans.

Illustration 9 : Given T_3 in the expansion of $(1 - 3x)^6$ has maximum numerical value. Find the range of 'x'.

Solution : Using $\frac{n+1}{1+\left|\frac{a}{b}\right|} - 1 \leq r \leq \frac{n+1}{1+\left|\frac{a}{b}\right|}$

$$\frac{6+1}{1+\left|\frac{1}{-3x}\right|} - 1 \leq 2 \leq \frac{7}{1+\left|\frac{1}{-3x}\right|}$$

Let $|x| = t$

$$\frac{21t}{3t+1} - 1 \leq 2 \leq \frac{21t}{3t+1}$$

$$\begin{cases} \frac{21t}{3t+1} \leq 3 \\ \frac{21t}{3t+1} \geq 2 \end{cases} \Rightarrow \begin{cases} \frac{4t-1}{3t+1} \leq 0 \Rightarrow t \in \left[-\frac{1}{3}, \frac{1}{4}\right] \\ \frac{15t-2}{3t+1} \geq 0 \Rightarrow t \in \left(-\infty, -\frac{1}{3}\right] \cup \left[\frac{2}{15}, \infty\right) \end{cases}$$

$$\text{Common solution } t \in \left[\frac{2}{15}, \frac{1}{4}\right] \Rightarrow x \in \left[-\frac{1}{4}, -\frac{2}{15}\right] \cup \left[\frac{2}{15}, \frac{1}{4}\right]$$

Do yourself -3 :

- (i) Find the numerically greatest term in the expansion of $(3 - 2x)^9$, when $x = 1$.
- (ii) In the expansion of $\left(\frac{1}{2} + \frac{2x}{3}\right)^n$ when $x = -\frac{1}{2}$, it is known that 3rd term is the greatest term. Find the possible integral values of n .

4. PROPERTIES OF BINOMIAL COEFFICIENTS :

$$(1+x)^n = C_0 + C_1x + C_2x^2 + C_3x^3 + \dots + C_nx^n = \sum_{r=0}^n {}^nC_r x^r; n \in \mathbb{N} \quad \dots(i)$$

where $C_0, C_1, C_2, \dots, C_n$ are called combinatorial (binomial) coefficients.

- (a) The sum of all the binomial coefficients is 2^n .

Put $x = 1$, in (i) we get

$$C_0 + C_1 + C_2 + \dots + C_n = 2^n \Rightarrow \sum_{r=0}^n {}^nC_r = 2^n \quad \dots(ii)$$

- (b) Put $x = -1$ in (i) we get

$$C_0 - C_1 + C_2 - C_3 + \dots + C_n = 0 \Rightarrow \sum_{r=0}^n (-1)^r {}^nC_r = 0 \quad \dots(iii)$$

- (c) The sum of the binomial coefficients at odd position is equal to the sum of the binomial coefficients at even position and each is equal to 2^{n-1} .

$$\text{From (ii) \& (iii), } C_0 + C_2 + C_4 + \dots = C_1 + C_3 + C_5 + \dots = 2^{n-1}$$

- (d) ${}^nC_r + {}^nC_{r-1} = {}^{n+1}C_r$

$$(e) \frac{{}^nC_r}{{}^nC_{r-1}} = \frac{n-r+1}{r}$$

$$(f) {}^nC_r = \frac{n}{r} \cdot {}^{n-1}C_{r-1} = \frac{n}{r} \cdot \frac{n-1}{r-1} \cdot {}^{n-2}C_{r-2} = \dots = \frac{n(n-1)(n-2)\dots(n-r+1)}{r(r-1)(r-2)\dots 1}$$

$$(g) {}^nC_r = \frac{r+1}{n+1} \cdot {}^{n+1}C_{r+1}$$

Illustration 10 : Prove that : ${}^{25}C_{10} + {}^{24}C_{10} + \dots + {}^{10}C_{10} = {}^{26}C_{11}$

Solution : LHS = ${}^{10}C_{10} + {}^{11}C_{10} + {}^{12}C_{10} + \dots + {}^{25}C_{10}$
 $\Rightarrow {}^{11}C_{11} + {}^{11}C_{10} + {}^{12}C_{10} + \dots + {}^{25}C_{10}$
 $\Rightarrow {}^{12}C_{11} + {}^{12}C_{10} + \dots + {}^{25}C_{10}$
 $\Rightarrow {}^{13}C_{11} + {}^{13}C_{10} + \dots + {}^{25}C_{10}$
 and so on. $\therefore$ LHS = ${}^{26}C_{11}$

Aliter :

LHS = coefficient of x^{10} in $\{(1+x)^{10} + (1+x)^{11} + \dots + (1+x)^{25}\}$

$$\Rightarrow \text{coefficient of } x^{10} \text{ in } \left[(1+x)^{10} \frac{\{1+x\}^{16} - 1}{1+x-1} \right]$$

$$\Rightarrow \text{coefficient of } x^{10} \text{ in } \frac{[(1+x)^{26} - (1+x)^{10}]}{x}$$

$$\Rightarrow \text{coefficient of } x^{11} \text{ in } [(1+x)^{26} - (1+x)^{10}] = {}^{26}C_{11} - 0 = {}^{26}C_{11}$$

Illustration 11 : A student is allowed to select at most n books from a collection of $(2n+1)$ books. If the total number of ways in which he can select books is 63, find the value of n .

Solution : Given student selects at most n books from a collection of $(2n+1)$ books. It means that he selects one book or two books or three books or or n books. Hence, by the given condition-

$${}^{2n+1}C_1 + {}^{2n+1}C_2 + {}^{2n+1}C_3 + \dots + {}^{2n+1}C_n = 63 \quad \dots(i)$$

But we know that

$${}^{2n+1}C_0 + {}^{2n+1}C_1 + {}^{2n+1}C_2 + {}^{2n+1}C_3 + \dots + {}^{2n+1}C_{2n+1} = 2^{2n+1} \quad \dots(ii)$$

Since ${}^{2n+1}C_0 = {}^{2n+1}C_{2n+1} = 1$, equation (ii) can also be written as

$$2 + ({}^{2n+1}C_1 + {}^{2n+1}C_2 + {}^{2n+1}C_3 + \dots + {}^{2n+1}C_n) + ({}^{2n+1}C_{n+1} + {}^{2n+1}C_{n+2} + {}^{2n+1}C_{n+3} + \dots + {}^{2n+1}C_{2n-1} + {}^{2n+1}C_{2n}) = 2^{2n+1}$$

$$\Rightarrow 2 + ({}^{2n+1}C_1 + {}^{2n+1}C_2 + {}^{2n+1}C_3 + \dots + {}^{2n+1}C_n) + ({}^{2n+1}C_n + {}^{2n+1}C_{n-1} + \dots + {}^{2n+1}C_2 + {}^{2n+1}C_1) = 2^{2n+1}$$

$$(\because {}^{2n+1}C_r = {}^{2n+1}C_{2n+1-r})$$

$$\Rightarrow 2 + 2({}^{2n+1}C_1 + {}^{2n+1}C_2 + {}^{2n+1}C_3 + \dots + {}^{2n+1}C_n) = 2^{2n+1} \quad [\text{from (i)}]$$

$$\Rightarrow 2 + 2.63 = 2^{2n+1} \quad \Rightarrow 1 + 63 = 2^{2n}$$

$$\Rightarrow 64 = 2^{2n} \Rightarrow 2^6 = 2^{2n} \quad \therefore 2n = 6$$

Hence, $n = 3$.

Ans.

Illustration 12 : Prove that :

$$(i) \quad C_1 + 2C_2 + 3C_3 + \dots + nC_n = n \cdot 2^{n-1}$$

$$(ii) \quad C_0 + \frac{C_1}{2} + \frac{C_2}{3} + \dots + \frac{C_n}{n+1} = \frac{2^{n+1} - 1}{n+1}$$

Solution :

$$(i) \quad \text{L.H.S.} = \sum_{r=1}^n r \cdot {}^nC_r = \sum_{r=1}^n r \cdot \frac{n}{r} \cdot {}^{n-1}C_{r-1}$$

$$= n \sum_{r=1}^n {}^{n-1}C_{r-1} = n \cdot [{}^{n-1}C_0 + {}^{n-1}C_1 + \dots + {}^{n-1}C_{n-1}]$$

$$= n \cdot 2^{n-1}$$

Aliter : (Using method of differentiation)

$$(1+x)^n = {}^nC_0 + {}^nC_1x + {}^nC_2x^2 + \dots + {}^nC_nx^n \quad \text{.....(A)}$$

Differentiating (A), we get

$$n(1+x)^{n-1} = C_1 + 2C_2x + 3C_3x^2 + \dots + n.C_nx^{n-1}.$$

Put $x = 1$,

$$C_1 + 2C_2 + 3C_3 + \dots + n.C_n = n.2^{n-1}$$

$$\begin{aligned} \text{(ii) L.H.S.} &= \sum_{r=0}^n \frac{C_r}{r+1} = \frac{1}{n+1} \sum_{r=0}^n \frac{n+1}{r+1} {}^nC_r \\ &= \frac{1}{n+1} \sum_{r=0}^n {}^{n+1}C_{r+1} = \frac{1}{n+1} [{}^{n+1}C_1 + {}^{n+1}C_2 + \dots + {}^{n+1}C_{n+1}] = \frac{1}{n+1} [2^{n+1} - 1] \end{aligned}$$

Aliter : (Using method of integration)

Integrating (A), we get

$$\frac{(1+x)^{n+1}}{n+1} + C = C_0x + \frac{C_1x^2}{2} + \frac{C_2x^3}{3} + \dots + \frac{C_nx^{n+1}}{n+1} \quad (\text{where } C \text{ is a constant})$$

$$\text{Put } x = 0, \text{ we get, } C = -\frac{1}{n+1}$$

$$\therefore \frac{(1+x)^{n+1} - 1}{n+1} = C_0x + \frac{C_1x^2}{2} + \frac{C_2x^3}{3} + \dots + \frac{C_nx^{n+1}}{n+1}$$

$$\text{Put } x = 1, \text{ we get } C_0 + \frac{C_1}{2} + \frac{C_2}{3} + \dots + \frac{C_n}{n+1} = \frac{2^{n+1} - 1}{n+1}$$

$$\text{Put } x = -1, \text{ we get } C_0 - \frac{C_1}{2} + \frac{C_2}{3} - \dots = \frac{1}{n+1}$$

Illustration 13 : If $(1+x)^n = \sum_{r=0}^n {}^nC_r x^r$, then prove that $C_1^2 + 2.C_2^2 + 3.C_3^2 + \dots + n.C_n^2 = \frac{(2n-1)!}{((n-1)!)^2}$

$$\text{Solution : } (1+x)^n = C_0 + C_1x + C_2x^2 + C_3x^3 + \dots + C_nx^n \quad \text{.....(i)}$$

Differentiating both the sides, w.r.t. x , we get

$$n(1+x)^{n-1} = C_1 + 2C_2x + 3C_3x^2 + \dots + n.C_nx^{n-1} \quad \text{.....(ii)}$$

also, we have

$$(x+1)^n = C_0x^n + C_1x^{n-1} + C_2x^{n-2} + \dots + C_n \quad \text{.....(iii)}$$

Multiplying (ii) & (iii), we get

$$(C_1 + 2C_2x + 3C_3x^2 + \dots + C_nx^{n-1})(C_0x^n + C_1x^{n-1} + C_2x^{n-2} + \dots + C_n) = n(1+x)^{2n-1}$$

Equating the coefficients of x^{n-1} , we get

$$C_1^2 + 2C_2^2 + 3C_3^2 + \dots + n.C_n^2 = n.2^{n-1}C_{n-1} = \frac{(2n-1)!}{((n-1)!)^2}$$

Ans.

Illustration 14 : Prove that : $C_0 - 3C_1 + 5C_2 - \dots + (-1)^n(2n+1)C_n = 0$

$$\text{Solution : } T_r = (-1)^r(2r+1)C_r = 2(-1)^r.r.C_r + (-1)^r.C_r$$

$$\Sigma T_r = 2 \sum_{r=1}^n (-1)^r.r.C_r + \sum_{r=0}^n (-1)^r.C_r = 2 \sum_{r=1}^n (-1)^r.r.C_r + \sum_{r=0}^n (-1)^r.C_r$$

$$= 2 \left[{}^{n-1}C_0 - {}^{n-1}C_1 + \dots \right] + \left[{}^nC_0 - {}^nC_1 + \dots \right] = 0$$

Illustration 15 : Prove that $({}^{2n}C_0)^2 - ({}^{2n}C_1)^2 + ({}^{2n}C_2)^2 - \dots + (-1)^n ({}^{2n}C_{2n})^2 = (-1)^n \cdot {}^{2n}C_n$

Solution : $(1 - x)^{2n} = {}^{2n}C_0 - {}^{2n}C_1x + {}^{2n}C_2x^2 - \dots + (-1)^n {}^{2n}C_{2n}x^{2n}$ (i)

and $(x + 1)^{2n} = {}^{2n}C_0x^{2n} + {}^{2n}C_1x^{2n-1} + {}^{2n}C_2x^{2n-2} + \dots + {}^{2n}C_{2n}$ (ii)

Multiplying (i) and (ii), we get

$(x^2 - 1)^{2n} = ({}^{2n}C_0 - {}^{2n}C_1x + \dots + (-1)^n {}^{2n}C_{2n}x^{2n}) \times ({}^{2n}C_0x^{2n} + {}^{2n}C_1x^{2n-1} + \dots + {}^{2n}C_{2n})$ (iii)

Now, coefficient of x^{2n} in R.H.S.

$= ({}^{2n}C_0)^2 - ({}^{2n}C_1)^2 + ({}^{2n}C_2)^2 - \dots + (-1)^n ({}^{2n}C_{2n})^2$

$\therefore$ General term in L.H.S., $T_{r+1} = {}^{2n}C_r(x^2)^{2n-r}(-1)^r$

Putting $2(2n - r) = 2n$

$\therefore r = n$

$\therefore T_{n+1} = {}^{2n}C_n x^{2n} (-1)^n$

Hence coefficient of x^{2n} in L.H.S. $= (-1)^n \cdot {}^{2n}C_n$

But (iii) is an identity, therefore coefficient of x^{2n} in R.H.S. = coefficient of x^{2n} in L.H.S.

$\Rightarrow ({}^{2n}C_0)^2 - ({}^{2n}C_1)^2 + ({}^{2n}C_2)^2 - \dots + (-1)^n ({}^{2n}C_{2n})^2 = (-1)^n \cdot {}^{2n}C_n$

Illustration 16 : Prove that : ${}^nC_0 \cdot {}^{2n}C_n - {}^nC_1 \cdot {}^{2n-2}C_n + {}^nC_2 \cdot {}^{2n-4}C_n + \dots = 2^n$

Solution : L.H.S. = Coefficient of x^n in $[{}^nC_0(1 + x)^{2n} - {}^nC_1(1 + x)^{2n-2} + \dots]$

$=$ Coefficient of x^n in $[(1 + x)^2 - 1]^n$

$=$ Coefficient of x^n in $x^n(x + 2)^n = 2^n$

Illustration 17 : If $(1 + x)^n = C_0 + C_1x + C_2x^2 + \dots + C_nx^n$ then show that the sum of the products of

the C_i 's taken two at a time represented by : $\sum_{0 \leq i < j \leq n} C_i C_j$ is equal to $2^{2n-1} - \frac{2n!}{2.n!n!}$

Solution : Since $(C_0 + C_1 + C_2 + \dots + C_{n-1} + C_n)^2$

$= C_0^2 + C_1^2 + C_2^2 + \dots + C_{n-1}^2 + C_n^2 + 2(C_0C_1 + C_0C_2 + C_0C_3 + \dots + C_0C_n + C_1C_2 + C_1C_3 + \dots + C_1C_n + C_2C_3 + C_2C_4 + \dots + C_2C_n + \dots + C_{n-1}C_n)$

$(2^n)^2 = {}^{2n}C_n + 2 \sum_{0 \leq i < j \leq n} C_i C_j$

Hence $\sum_{0 \leq i < j \leq n} C_i C_j = 2^{2n-1} - \frac{2n!}{2.n!n!}$ **Ans.**

Illustration 18 : If $(1 + x)^n = C_0 + C_1x + C_2x^2 + \dots + C_nx^n$ then prove that $\sum_{0 \leq i < j \leq n} (C_i + C_j)^2 = (n-1) {}^{2n}C_n + 2^{2n}$

Solution : L.H.S. $\sum_{0 \leq i < j \leq n} (C_i + C_j)^2$
 $= (C_0 + C_1)^2 + (C_0 + C_2)^2 + \dots + (C_0 + C_n)^2 + (C_1 + C_2)^2 + (C_1 + C_3)^2 + \dots + (C_1 + C_n)^2 + (C_2 + C_3)^2 + (C_2 + C_4)^2 + \dots + (C_2 + C_n)^2 + \dots + (C_{n-1} + C_n)^2$

$= n(C_0^2 + C_1^2 + C_2^2 + \dots + C_n^2) + 2 \sum_{0 \leq i < j \leq n} C_i C_j$

$= n \cdot {}^{2n}C_n + 2 \cdot \left\{ 2^{2n-1} - \frac{2n!}{2.n!n!} \right\}$ {from Illustration 17}

$= n \cdot {}^{2n}C_n + 2^{2n} - {}^{2n}C_n = (n-1) \cdot {}^{2n}C_n + 2^{2n} = \text{R.H.S.}$

Do yourself - 4 :

- (i) $\binom{n}{0} + \binom{n}{1} + \binom{n}{2} + \dots + \binom{n}{n} =$
 (A) 2^{n-1} (B) $2^n C_n$ (C) 2^n (D) 2^{n+1}
- (ii) If $(1+x)^n = C_0 + C_1x + C_2x^2 + \dots + C_nx^n$, $n \in \mathbb{N}$. Prove that
- (a) $3C_0 - 8C_1 + 13C_2 - 18C_3 + \dots$ upto $(n+1)$ terms $= 0$, if $n \geq 2$.
- (b) $2C_0 + 2^2 \frac{C_1}{2} + 2^3 \frac{C_2}{3} + 2^4 \frac{C_3}{4} + \dots + 2^{n+1} \frac{C_n}{n+1} = \frac{3^{n+1} - 1}{n+1}$
- (c) $C_0^2 + \frac{C_1^2}{2} + \frac{C_2^2}{3} + \dots + \frac{C_n^2}{n+1} = \frac{(2n+1)!}{((n+1)!)^2}$

5. MULTINOMIAL THEOREM :

Using binomial theorem, we have $(x+a)^n = \sum_{r=0}^n {}^nC_r x^{n-r} a^r$, $n \in \mathbb{N}$

$$= \sum_{r=0}^n \frac{n!}{(n-r)!r!} x^{n-r} a^r = \sum_{r+s=n} \frac{n!}{r!s!} x^s a^r, \text{ where } s+r=n$$

This result can be generalized in the following form.

$$(x_1 + x_2 + \dots + x_k)^n = \sum_{r_1+r_2+\dots+r_k=n} \frac{n!}{r_1!r_2!\dots r_k!} x_1^{r_1} x_2^{r_2} \dots x_k^{r_k}$$

The general term in the above expansion $\frac{n!}{r_1!r_2!r_3!\dots r_k!} \cdot x_1^{r_1} x_2^{r_2} x_3^{r_3} \dots x_k^{r_k}$

The number of terms in the above expansion is equal to the number of non-negative integral solution of the equation $r_1 + r_2 + \dots + r_k = n$ because each solution of this equation gives a term in the above expansion. The number of such solutions is ${}^{n+k-1}C_{k-1}$

Particular cases :

(i) $(x+y+z)^n = \sum_{r+s+t=n} \frac{n!}{r!s!t!} x^r y^s z^t$

The above expansion has ${}^{n+3-1}C_{3-1} = {}^{n+2}C_2$ terms

(ii) $(x+y+z+u)^n = \sum_{p+q+r+s=n} \frac{n!}{p!q!r!s!} x^p y^q z^r u^s$

There are ${}^{n+4-1}C_{4-1} = {}^{n+3}C_3$ terms in the above expansion.

Illustration 19 : Find the coefficient of $x^2 y^3 z^4 w$ in the expansion of $(x-y-z+w)^{10}$

Solution : $(x-y-z+w)^{10} = \sum_{p+q+r+s=10} \frac{10!}{p!q!r!s!} (x)^p (-y)^q (-z)^r (w)^s$

We want to get $x^2 y^3 z^4 w$ this implies that $p=2, q=3, r=4, s=1$

$\therefore$ Coefficient of $x^2 y^3 z^4 w$ is $\frac{10!}{2!3!4!1!} (-1)^3 (-1)^4 = -12600$

Ans.

Illustration 20 : Find the total number of terms in the expansion of $(1 + x + y)^{10}$ and coefficient of x^2y^3 .

Solution : Total number of terms = $^{10+3-1}C_{3-1} = {}^{12}C_2 = 66$

$$\text{Coefficient of } x^2y^3 = \frac{10!}{2! \times 3! \times 5!} = 2520$$

Ans.

Illustration 21 : Find the coefficient of x^5 in the expansion of $(2 - x + 3x^2)^6$.

Solution : The general term in the expansion of $(2 - x + 3x^2)^6 = \frac{6!}{r!s!t!} 2^r (-x)^s (3x^2)^t$,

where $r + s + t = 6$.

$$= \frac{6!}{r!s!t!} 2^r \times (-1)^s \times (3)^t \times x^{s+2t}$$

For the coefficient of x^5 , we must have $s + 2t = 5$.

But, $r + s + t = 6$,

$\therefore s = 5 - 2t$ and $r = 1 + t$, where $0 \leq r, s, t \leq 6$.

Now $t = 0 \Rightarrow r = 1, s = 5$.

$t = 1 \Rightarrow r = 2, s = 3$.

$t = 2 \Rightarrow r = 3, s = 1$.

Thus, there are three terms containing x^5 and coefficient of x^5

$$= \frac{6!}{1! 5! 0!} \times 2^1 \times (-1)^5 \times 3^0 + \frac{6!}{2! 3! 1!} \times 2^2 \times (-1)^3 \times 3^1 + \frac{6!}{3! 1! 2!} \times 2^3 \times (-1)^1 \times 3^2$$

$$= -12 - 720 - 4320 = -5052.$$

Ans.

Illustration 22 : If $(1+x+x^2)^n = \sum_{r=0}^{2n} a_r x^r$, then prove that (a) $a_r = a_{2n-r}$ (b) $\sum_{r=0}^{n-1} a_r = \frac{1}{2}(3^n - a_n)$

Solution : (a) We have

$$(1 + x + x^2)^n = \sum_{r=0}^{2n} a_r x^r \quad \dots(A)$$

Replace x by $\frac{1}{x}$

$$\therefore \left(1 + \frac{1}{x} + \frac{1}{x^2}\right)^n = \sum_{r=0}^{2n} a_r \left(\frac{1}{x}\right)^r$$

$$\Rightarrow (x^2 + x + 1)^n = \sum_{r=0}^{2n} a_r x^{2n-r}$$

$$\Rightarrow \sum_{r=0}^{2n} a_r x^r = \sum_{r=0}^{2n} a_r x^{2n-r} \quad \{\text{Using (A)}\}$$

Equating the coefficient of x^{2n-r} on both sides, we get

$$a_{2n-r} = a_r \text{ for } 0 \leq r \leq 2n.$$

Hence $a_r = a_{2n-r}$.

(b) Putting $x=1$ in given series, then

$$\begin{aligned} a_0 + a_1 + a_2 + \dots + a_{2n} &= (1+1+1)^n \\ a_0 + a_1 + a_2 + \dots + a_{2n} &= 3^n \quad \dots(1) \end{aligned}$$

But $a_r = a_{2n-r}$ for $0 \leq r \leq 2n$

$\therefore$ series (1) reduces to

$$2(a_0 + a_1 + a_2 + \dots + a_{n-1}) + a_n = 3^n.$$

$$\therefore a_0 + a_1 + a_2 + \dots + a_{n-1} = \frac{1}{2}(3^n - a_n)$$

Do yourself - 5 :

(i) Find the coefficient of x^2y^5 in the expansion of $(3 + 2x - y)^{10}$.

6. APPLICATION OF BINOMIAL THEOREM :

Illustration 23 : If $(6\sqrt{6} + 14)^{2n+1} = [N] + F$ and $F = N - [N]$; where $[.]$ denotes greatest integer function, then NF is equal to

- (A) 20^{2n+1} (B) an even integer (C) odd integer (D) 40^{2n+1}

Solution : Since $(6\sqrt{6} + 14)^{2n+1} = [N] + F$

Let us assume that $f = (6\sqrt{6} - 14)^{2n+1}$; where $0 \leq f < 1$.

$$\begin{aligned} \text{Now, } [N] + F - f &= (6\sqrt{6} + 14)^{2n+1} - (6\sqrt{6} - 14)^{2n+1} \\ &= 2 \left[{}^{2n+1}C_1 (6\sqrt{6})^{2n} (14) + {}^{2n+1}C_3 (6\sqrt{6})^{2n-2} (14)^3 + \dots \right] \end{aligned}$$

$$\Rightarrow [N] + F - f = \text{even integer.}$$

Now $0 < F < 1$ and $0 < f < 1$

so $-1 < F - f < 1$ and $F - f$ is an integer so it can only be zero

$$\text{Thus } NF = (6\sqrt{6} + 14)^{2n+1} (6\sqrt{6} - 14)^{2n+1} = 20^{2n+1}.$$

Ans. (A,B)

Illustration 24 : Find the last three digits in 11^{50} .

Solution : Expansion of $(10 + 1)^{50} = {}^{50}C_0 10^{50} + {}^{50}C_1 10^{49} + \dots + {}^{50}C_{48} 10^2 + {}^{50}C_{49} 10 + {}^{50}C_{50}$

$$= \underbrace{{}^{50}C_0 10^{50} + {}^{50}C_1 10^{49} + \dots + {}^{50}C_{47} 10^3}_{1000K} + 49 \times 25 \times 100 + 500 + 1$$

$$\Rightarrow 1000K + 123001$$

$$\Rightarrow \text{Last 3 digits are 001.}$$

Illustration 25 : Prove that $2222^{5555} + 5555^{2222}$ is divisible by 7.

Solution : When 2222 is divided by 7 it leaves a remainder 3.

So adding & subtracting 3^{5555} , we get :

$$E = \underbrace{2222^{5555} - 3^{5555}}_{E_1} + \underbrace{3^{5555} + 5555^{2222}}_{E_2}$$

For E_1 : Now since $2222-3 = 2219$ is divisible by 7, therefore E_1 is divisible by 7

($\because x^n - a^n$ is divisible by $x - a$)

For E_2 : 5555 when divided by 7 leaves remainder 4.

So adding and subtracting 4^{2222} , we get :

$$E_2 = 3^{5555} + 4^{2222} + 5555^{2222} - 4^{2222}$$

$$= (243)^{1111} + (16)^{1111} + (5555)^{2222} - 4^{2222}$$

Again $(243)^{1111} + 16^{1111}$ and $(5555)^{2222} - 4^{2222}$ are divisible by 7

($\because x^n + a^n$ is divisible by $x + a$ when n is odd)

Hence $2222^{5555} + 5555^{2222}$ is divisible by 7.

Do yourself - 6 :

- (i) Prove that $5^{25} - 3^{25}$ is divisible by 2.
- (ii) Find the remainder when the number 9^{100} is divided by 8.
- (iii) Find last three digits in 19^{100} .
- (iv) Let $R = (8 + 3\sqrt{7})^{20}$ and $[.]$ denotes greatest integer function, then prove that :
 - (a) $[R]$ is odd
 - (b) $R - [R] = 1 - \frac{1}{(8 + 3\sqrt{7})^{20}}$
- (v) Find the digit at unit's place in the number $17^{1995} + 11^{1995} - 7^{1995}$.

7. BINOMIAL THEOREM FOR NEGATIVE OR FRACTIONAL INDICES :

If $n \in \mathbb{Q}$, then $(1 + x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \dots \infty$ provided $|x| < 1$.

Note :

- (i) When the index n is a positive integer the number of terms in the expansion of $(1 + x)^n$ is finite i.e. $(n+1)$ & the coefficient of successive terms are : ${}^nC_0, {}^nC_1, {}^nC_2, \dots, {}^nC_n$
- (ii) When the index is other than a positive integer such as negative integer or fraction, the number of terms in the expansion of $(1 + x)^n$ is infinite and the symbol nC_r cannot be used to denote the coefficient of the general term.
- (iii) Following expansion should be remembered ($|x| < 1$).
 - (a) $(1 + x)^{-1} = 1 - x + x^2 - x^3 + x^4 - \dots \infty$
 - (b) $(1 - x)^{-1} = 1 + x + x^2 + x^3 + x^4 + \dots \infty$
 - (c) $(1 + x)^{-2} = 1 - 2x + 3x^2 - 4x^3 + \dots \infty$
 - (d) $(1 - x)^{-2} = 1 + 2x + 3x^2 + 4x^3 + \dots \infty$
 - (e) $(1 + x)^{-3} = 1 - 3x + 6x^2 - 10x^3 + \dots + \frac{(-1)^r(r+1)(r+2)}{2!}x^r + \dots$
 - (f) $(1 - x)^{-3} = 1 + 3x + 6x^2 + 10x^3 + \dots + \frac{(r+1)(r+2)}{2!}x^r + \dots$
- (iv) The expansions in ascending powers of x are only valid if x is 'small'. If x is large i.e. $|x| > 1$ then we may find it convenient to expand in powers of $1/x$, which then will be small.

8. APPROXIMATIONS :

$$(1+x)^n = 1 + nx + \frac{n(n-1)}{1.2}x^2 + \frac{n(n-1)(n-2)}{1.2.3}x^3 + \dots$$

If $x < 1$, the terms of the above expansion go on decreasing and if x be very small, a stage may be reached when we may neglect the terms containing higher powers of x in the expansion. Thus, if x be so small that its square and higher powers may be neglected then $(1+x)^n = 1 + nx$, approximately.

This is an approximate value of $(1+x)^n$

Illustration 26 : If x is so small such that its square and higher powers may be neglected then find the approximate value of $\frac{(1-3x)^{1/2} + (1-x)^{5/3}}{(4+x)^{1/2}}$

Solution :

$$\frac{(1-3x)^{1/2} + (1-x)^{5/3}}{(4+x)^{1/2}} = \frac{1 - \frac{3}{2}x + 1 - \frac{5x}{3}}{2\left(1 + \frac{x}{4}\right)^{1/2}} = \frac{1}{2}\left(2 - \frac{19}{6}x\right)\left(1 + \frac{x}{4}\right)^{-1/2} = \frac{1}{2}\left(2 - \frac{19}{6}x\right)\left(1 - \frac{x}{8}\right)$$

$$= \frac{1}{2}\left(2 - \frac{x}{4} - \frac{19}{6}x\right) = 1 - \frac{x}{8} - \frac{19}{12}x = 1 - \frac{41}{24}x$$

Ans.

Illustration 27 : The value of cube root of 1001 upto five decimal places is –

- (A) 10.03333 (B) 10.00333 (C) 10.00033 (D) none of these

Solution :

$$(1001)^{1/3} = (1000+1)^{1/3} = 10\left(1 + \frac{1}{1000}\right)^{1/3} = 10\left\{1 + \frac{1}{3} \cdot \frac{1}{1000} + \frac{1/3(1/3-1)}{2!} \frac{1}{1000^2} + \dots\right\}$$

$$= 10\{1 + 0.0003333 - 0.00000011 + \dots\} = 10.00333$$

Ans. (B)

Illustration 28 : The sum of $1 + \frac{1}{4} + \frac{1.3}{4.8} + \frac{1.3.5}{4.8.12} + \dots \infty$ is –

- (A) $\sqrt{2}$ (B) $\frac{1}{\sqrt{2}}$ (C) $\sqrt{3}$ (D) $2^{3/2}$

Solution : Comparing with $1 + nx + \frac{n(n-1)}{2!}x^2 + \dots$

$$nx = 1/4 \quad \dots\dots(i)$$

$$\text{and } \frac{n(n-1)x^2}{2!} = \frac{1.3}{4.8}$$

$$\text{or } \frac{nx(nx-x)}{2!} = \frac{3}{32} \Rightarrow \frac{1}{4}\left(\frac{1}{4} - x\right) = \frac{3}{16} \quad (\text{by (i)})$$

$$\Rightarrow \left(\frac{1}{4} - x\right) = \frac{3}{4} \Rightarrow x = \frac{1}{4} - \frac{3}{4} = -\frac{1}{2} \quad \dots\dots(ii)$$

putting the value of x in (i)

$$n(-1/2) = 1/4 \Rightarrow n = -1/2$$

$$\therefore \text{sum of series} = (1+x)^n = (1-1/2)^{-1/2} = (1/2)^{-1/2} = \sqrt{2}$$

Ans. (A)

9. EXPONENTIAL SERIES :

- (a) e is an irrational number lying between 2.7 & 2.8. Its value correct upto 10 places of decimal is 2.7182818284.
- (b) Logarithms to the base ' e ' are known as the Napierian system, so named after Napier, their inventor. They are also called **Natural Logarithm**.
- (c) $e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots \infty$; where x may be any real or complex number & $e = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n$
- (d) $a^x = 1 + \frac{x}{1!} \ln a + \frac{x^2}{2!} \ln^2 a + \frac{x^3}{3!} \ln^3 a + \dots \infty$, where $a > 0$
- (e) $e = 1 + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \dots \infty$

10. LOGARITHMIC SERIES :

- (a) $\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots \infty$, where $-1 < x \leq 1$
- (b) $\ln(1-x) = -x - \frac{x^2}{2} - \frac{x^3}{3} - \frac{x^4}{4} + \dots \infty$, where $-1 \leq x < 1$

- Remember :** (i) $1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots \infty = \ln 2$ (ii) $e^{\ln x} = x$; for all $x > 0$
- (iii) $\ln 2 = 0.693$ (iv) $\ln 10 = 2.303$

ANSWERS FOR DO YOURSELF

1. (i) ${}^5C_0 x (3x^2)^5 + {}^5C_1 (3x^2)^4 \left(-\frac{x}{2}\right) + {}^5C_2 (3x^2)^3 \left(-\frac{x}{2}\right)^2 + {}^5C_3 (3x^2)^2 \left(-\frac{x}{2}\right)^3 + {}^5C_4 (3x^2)^1 \left(-\frac{x}{2}\right)^4 + {}^5C_5 \left(-\frac{x}{2}\right)^5$
- (ii) ${}^nC_0 y^n + {}^nC_1 y^{n-1} \cdot x + {}^nC_2 y^{n-2} \cdot x^2 + \dots + {}^nC_n \cdot x^n$
- 2: (i) $\frac{70}{3} x^8$; (ii) $\frac{25!}{10! 5!} 2^{15} 3^{10}$; (iii) (a) -20; (b) $-560x^5, 280x^2$
3. (i) 4^{th} & 5^{th} i.e. 489888 (ii) $n = 4, 5, 6$
4. (i) C
5. (i) -272160 or $-{}^{10}C_5 \times {}^5C_2 \times 108$
6. (ii) 1 (iii) 801 (v) 1

EXERCISE (O-1)

[SINGLE CORRECT CHOICE TYPE]

- If the coefficients of x^7 & x^8 in the expansion of $\left[2 + \frac{x}{3}\right]^n$ are equal, then the value of n is :
 (A) 15 (B) 45 (C) 55 (D) 56
- If the constant term of the binomial expansion $\left(2x - \frac{1}{x}\right)^n$ is -160 , then n is equal to -
 (A) 4 (B) 6 (C) 8 (D) 10
- The coefficient of x^{49} in the expansion of $(x-1)\left(x - \frac{1}{2}\right)\left(x - \frac{1}{2^2}\right) \dots \left(x - \frac{1}{2^{49}}\right)$ is equal to -
 (A) $-2\left(1 - \frac{1}{2^{50}}\right)$ (B) +ve coefficient of x
 (C) -ve coefficient of x (D) $-2\left(1 - \frac{1}{2^{49}}\right)$
- Set of value of r for which, ${}^{18}C_{r-2} + 2 \cdot {}^{18}C_{r-1} + {}^{18}C_r \geq {}^{20}C_{13}$ contains :
 (A) 4 element (B) 5 elements (C) 7 elements (D) 10 elements
- Number of rational terms in the expansion of $(\sqrt{2} + \sqrt[4]{3})^{100}$ is :
 (A) 25 (B) 26 (C) 27 (D) 28
- The largest real value for x such that $\sum_{k=0}^4 \left(\frac{5^{4-k}}{(4-k)!}\right) \left(\frac{x^k}{k!}\right) = \frac{8}{3}$ is -
 (A) $2\sqrt{2} - 5$ (B) $2\sqrt{2} + 5$ (C) $-2\sqrt{2} - 5$ (D) $-2\sqrt{2} + 5$
- The expression $[x + (x^3 - 1)^{1/2}]^5 + [x - (x^3 - 1)^{1/2}]^5$ is a polynomial of degree
 (A) 5 (B) 6 (C) 7 (D) 8
- Given $(1 - 2x + 5x^2 - 10x^3)(1 + x)^n = 1 + a_1x + a_2x^2 + \dots$ and that $a_1^2 = 2a_2$ then the value of n is-
 (A) 6 (B) 2 (C) 5 (D) 3
- The sum of the co-efficients of all the even powers of x in the expansion of $(2x^2 - 3x + 1)^{11}$ is -
 (A) 2.6^{10} (B) 3.6^{10} (C) 6^{11} (D) none
- Co-efficient of α^t in the expansion of,
 $(\alpha + p)^{m-1} + (\alpha + p)^{m-2}(\alpha + q) + (\alpha + p)^{m-3}(\alpha + q)^2 + \dots (\alpha + q)^{m-1}$ where $\alpha \neq -q$ and $p \neq q$ is :
 (A) $\frac{{}^mC_t(p^t - q^t)}{p - q}$ (B) $\frac{{}^mC_t(p^{m-t} - q^{m-t})}{p - q}$ (C) $\frac{{}^mC_t(p^t + q^t)}{p - q}$ (D) $\frac{{}^mC_t(p^{m-t} + q^{m-t})}{p - q}$

11. If $n \in \mathbb{N}$ & n is even, then $\frac{1}{1.(n-1)!} + \frac{1}{3!(n-3)!} + \frac{1}{5!(n-5)!} + \dots + \frac{1}{(n-1)!1!} =$
 (A) 2^n (B) $\frac{2^{n-1}}{n!}$ (C) $2^n n!$ (D) none of these
12. Let $\binom{n}{k}$ represents the combination of 'n' things taken 'k' at a time, then the value of the sum $\binom{99}{97} + \binom{98}{96} + \binom{97}{95} + \dots + \binom{3}{1} + \binom{2}{0}$ equals -
 (A) $\binom{99}{97}$ (B) $\binom{100}{98}$ (C) $\binom{99}{98}$ (D) $\binom{100}{97}$

[COMPREHENSION TYPE]

Paragraph for question nos. 13 to 15

If $n \in \mathbb{N}$ and if $(1 + 4x + 4x^2)^n = \sum_{r=0}^{2n} a_r x^r$, where $a_0, a_1, a_2, \dots, a_{2n}$ are real numbers.

13. The value of $2 \sum_{r=0}^n a_{2r}$, is
 (A) $9^n - 1$ (B) $9^n + 1$ (C) $9^n - 2$ (D) $9^n + 2$
14. The value of $2 \sum_{r=1}^n a_{2r-1}$, is-
 (A) $9^n - 1$ (B) $9^n + 1$ (C) $9^n - 2$ (D) $9^n + 2$
15. The value of a_{2n-1} is -
 (A) 2^{2n} (B) $n \cdot 2^{2n}$ (C) $(n-1)2^{2n}$ (D) $(n+1)2^{2n}$

EXERCISE (O-2)

[SINGLE CORRECT CHOICE TYPE]

1. Let $(5 + 2\sqrt{6})^n = p + f$ where $n, p \in \mathbb{N}$ and $0 < f < 1$ then the value of $f^2 - f + pf - p$ is -
 (A) a natural number (B) a negative integer (C) a prime number (D) are irrational number
2. Greatest term in the binomial expansion of $(a + 2x)^9$ when $a = 1$ & $x = \frac{1}{3}$ is :
 (A) 3rd & 4th (B) 4th & 5th (C) only 4th (D) only 5th
3. If $\sum_{r=1}^{10} r(r-1) {}^{10}C_r = k \cdot 2^9$, then k is equal to-
 (A) 10 (B) 45 (C) 90 (D) 100
4. The sum $\frac{\binom{11}{0}}{1} + \frac{\binom{11}{1}}{2} + \frac{\binom{11}{2}}{3} + \dots + \frac{\binom{11}{11}}{12}$ equals $\left(\text{where } \binom{n}{r} \text{ denotes } {}^nC_r \right)$
 (A) $\frac{2^{11}}{12}$ (B) $\frac{2^{12}}{12}$ (C) $\frac{2^{11}-1}{12}$ (D) $\frac{2^{12}-1}{12}$

5. **Statement-1** : The sum of the series ${}^nC_0 \cdot {}^mC_r + {}^nC_1 \cdot {}^mC_{r-1} + {}^nC_2 \cdot {}^mC_{r-2} + \dots + {}^nC_r \cdot {}^mC_0$ is equal to ${}^{n+m}C_r$, where nC_r 's and mC_r 's denotes the combinatorial coefficients in the expansion of $(1+x)^n$ and $(1+x)^m$ respectively.

Statement-2 : Number of ways in which r children can be selected out of $(n+m)$ children consisting of n boys and m girls if each selection may consist of any number of boys and girls is equal to ${}^{n+m}C_r$.

- (A) Statement-1 is true, statement-2 is true ; statement-2 is a correct explanation for statement-1.
 (B) Statement-1 is true, statement-2 is true ; statement-2 is NOT a correct explanation for statement-1.
 (C) Statement-1 is true, statement-2 is false.
 (D) Statement-1 is false, statement-2 is true.

[MULTIPLE CORRECT CHOICE TYPE]

6. In the expansion of $\left(\frac{x+1}{x^{2/3} - x^{1/3} + 1} - \frac{x-1}{x - x^{1/2}} \right)^{10}$, the term which does not contain x is-
- (A) ${}^{11}C_4 - {}^{10}C_3$ (B) ${}^{10}C_7$ (C) ${}^{10}C_4$ (D) ${}^{11}C_5 - {}^{10}C_5$
7. Let $(1+x^2)^2(1+x)^n = A_0 + A_1x + A_2x^2 + \dots$. If A_0, A_1, A_2 are in A.P. then the value of n is-
- (A) 2 (B) 3 (C) 5 (D) 7
8. In the expansion of $\left(x^3 + 3 \cdot 2^{-\log_{\sqrt{2}} \sqrt{x^3}} \right)^{11}$
- (A) there appears a term with the power x^2 (B) there does not appear a term with the power x^2
 (C) there appears a term with the power x^{-3} (D) the ratio of the co-efficient of x^3 to that of x^{-3} is $1/3$
9. If it is known that the third term of the binomial expansion $(x + x^{\log_{10} x})^5$ is 10^6 then x is equal to-
- (A) 10 (B) $10^{-5/2}$ (C) 100 (D) 5
10. Which of the following statement(s) is/are correct ?
- (A) $1 + \frac{2}{2} + \frac{3}{2^2} + \frac{4}{2^3} + \dots + \infty = 4$
- (B) Integral part of $(9 + 4\sqrt{5})^n$, $n \in \mathbb{N}$ is even.
- (C) $({}^nC_0 + {}^nC_1 + {}^nC_2 + \dots + {}^nC_n)^2 = 1 + {}^{2n}C_1 + {}^{2n}C_2 + \dots + {}^{2n}C_{2n}$
- (D) $\frac{1}{(3+2x)^2}$ can be expanded as infinite series in ascending powers of x only if $|x| < \frac{2}{3}$.
11. If $(9 + \sqrt{80})^n = I + f$ where I, n are integers and $0 < f < 1$, then -
- (A) I is an odd integer (B) I is an even integer
 (C) $(I + f)(1 - f) = 1$ (D) $1 - f = (9 - \sqrt{80})^n$
12. If for $n \in \mathbb{I}$, $n > 10$; $1 + (1+x) + (1+x)^2 + \dots + (1+x)^n = \sum_{k=0}^n a_k \cdot x^k$, $x \neq 0$ then
- (A) $\sum_{k=0}^n a_k = 2^{n+1}$ (B) $a_{n-2} = \frac{n(n+1)}{2}$
- (C) $a_p > a_{p-1}$ for $p < \frac{n}{2}$, $p \in \mathbb{N}$ (D) $(a_9)^2 - (a_8)^2 = {}^{n+2}C_{10} ({}^{n+1}C_{10} - {}^{n+1}C_9)$

13. Let $P(n) = \sum_{r=0}^n \frac{(-1)^r r}{r+1} {}^nC_r$. Now which of the following holds good ?
- (A) $|P_{10}|$ is harmonic mean of $|P_9|$ & $|P_{11}|$ (B) $\sum_{r=5}^{10} P(r)P(r-1) = -\frac{6}{55}$
- (C) $|P_{10}|$ is arithmetic mean of $|P_9|$ & $|P_{11}|$ (D) $\sum_{r=5}^{10} P(r)P(r-1) = \frac{6}{55}$
14. Let $(1+x)^m = C_0 + C_1x + C_2x^2 + C_3x^3 + \dots + C_mx^m$, where $C_r = {}^mC_r$ and $A = C_1C_3 + C_2C_4 + C_3C_5 + C_4C_6 + \dots + C_{m-2}C_m$, then -
- (A) $A \geq {}^{2m}C_{m-2}$ (B) $A < {}^{2m}C_{m-2}$
- (C) $A > C_0^2 + C_1^2 + C_2^2 + \dots + C_m^2$ (D) $A < C_0^2 + C_1^2 + C_2^2 + \dots + C_m^2$
15. Consider $E = \left(\sqrt[3]{x} + \sqrt[5]{y}\right)^z = I + f, 0 \leq f < 1$
- (A) If $x = 5, y = 2, z = 100$, then number of irrational terms in expansion of E is 98
- (B) If $x = 5, y = 2, z = 100$, then number of rational terms in expansion of E is 4
- (C) If $x = 16, y = 1$ & $z = 6$, then $I = 197$
- (D) If $x = 16, y = 1$ & $z = 6$, then $f = (\sqrt{2} - 1)^6$

EXERCISE (S-1)

1. (a) If the coefficients of $(2r+4)^{\text{th}}, (r-2)^{\text{th}}$ terms in the expansion of $(1+x)^{18}$ are equal, find r .
- (b) If the coefficients of the $r^{\text{th}}, (r+1)^{\text{th}}$ & $(r+2)^{\text{th}}$ terms in the expansion of $(1+x)^{14}$ are in AP, find r .
- (c) If the coefficients of $2^{\text{nd}}, 3^{\text{rd}}$ & 4^{th} terms in the expansion of $(1+x)^{2n}$ are in AP, show that $2n^2 - 9n + 7 = 0$.
2. Find the term independent of x in the expansion of (i) $\left[\sqrt{\frac{x}{3}} + \frac{\sqrt{3}}{2x^2}\right]^{10}$ (ii) $\left[\frac{1}{2}x^{1/3} + x^{-1/5}\right]^8$
3. Prove that the ratio of the coefficient of x^{10} in $(1-x^2)^{10}$ & the term independent of x in $\left(x - \frac{2}{x}\right)^{10}$ is $1 : 32$.
4. Find the sum of the series $\sum_{r=0}^n (-1)^r \cdot {}^nC_r \left[\frac{1}{2^r} + \frac{3^r}{2^{2r}} + \frac{7^r}{2^{3r}} + \frac{15^r}{2^{4r}} + \dots \text{up to } m \text{ terms} \right]$
5. Find numerically greatest term in the expansion of :
- (i) $(2+3x)^9$ when $x = \frac{3}{2}$ (ii) $(3-5x)^{15}$ when $x = \frac{1}{5}$
6. Find the term independent of x in the expansion of $(1+x+2x^3)\left(\frac{3x^2}{2} - \frac{1}{3x}\right)^9$.
7. Let $(1+x^2)^2 \cdot (1+x)^n = \sum_{k=0}^{n+4} a_k \cdot x^k$. If a_1, a_2 & a_3 are in AP, find n .

8. Let $f(x) = 1 - x + x^2 - x^3 + \dots + x^{16} - x^{17} = a_0 + a_1(1+x) + a_2(1+x)^2 + \dots + a_{17}(1+x)^{17}$, find the value of a_2 .
9. Let $N = {}^{2000}C_1 + 2 \cdot {}^{2000}C_2 + 3 \cdot {}^{2000}C_3 + \dots + 2000 \cdot {}^{2000}C_{2000}$. Prove that N is divisible by 2^{2003} .
10. Find the coefficient of
 (a) $x^2 y^3 z^4$ in the expansion of $(ax - by + cz)^9$.
 (b) $a^2 b^3 c^4 d$ in the expansion of $(a - b - c + d)^{10}$.
11. Find the coefficient of
 (a) x^4 in the expansion of $(1 + x + x^2 + x^3)^{11}$
 (b) x^4 in the expansion of $(2 - x + 3x^2)^6$
12. Find the coefficient of x^r in the expression :
 $(x+3)^{n-1} + (x+3)^{n-2}(x+2) + (x+3)^{n-3}(x+2)^2 + \dots + (x+2)^{n-1}$
13. (a) Show that the integral part in each of the following is odd. $n \in \mathbb{N}$
 (A) $(5 + 2\sqrt{6})^n$ (B) $(8 + 3\sqrt{7})^n$
 (b) Show that the integral part in each of the following is even. $n \in \mathbb{N}$
 (A) $(3\sqrt{3} + 5)^{2n+1}$ (B) $(5\sqrt{5} + 11)^{2n+1}$
14. Given that $(1 + x + x^2)^n = a_0 + a_1x + a_2x^2 + \dots + a_{2n}x^{2n}$, find the values of :
 (i) $a_0 + a_1 + a_2 + \dots + a_{2n}$;
 (ii) $a_0 - a_1 + a_2 - a_3 + \dots + a_{2n}$; (iii) $a_0^2 - a_1^2 + a_2^2 - a_3^2 + \dots + a_{2n}^2$
15. Prove the following identities using the theory of permutation where $C_0, C_1, C_2, \dots, C_n$ are the combinatorial coefficients in the expansion of $(1+x)^n$, $n \in \mathbb{N}$:
 (a) $C_0^2 + C_1^2 + C_2^2 + \dots + C_n^2 = \frac{(2n)!}{n!n!}$
 (b) $C_0 C_1 + C_1 C_2 + C_2 C_3 + \dots + C_{n-1} C_n = \frac{(2n)!}{(n+1)!(n-1)!}$
 (c) $C_0 C_r + C_1 C_{r+1} + C_2 C_{r+2} + \dots + C_{n-r} C_n = \frac{2n!}{(n-r)!(n+r)!}$
 (d) $\sum_{r=0}^{n-2} ({}^n C_r \cdot {}^n C_{r+2}) = \frac{(2n)!}{(n-2)!(n+2)!}$
 (e) ${}^{100}C_{10} + 5 \cdot {}^{100}C_{11} + 10 \cdot {}^{100}C_{12} + 10 \cdot {}^{100}C_{13} + 5 \cdot {}^{100}C_{14} + {}^{100}C_{15} = {}^{105}C_{90}$
16. If $C_0, C_1, C_2, \dots, C_n$ are the combinatorial coefficients in the expansion of $(1+x)^n$, $n \in \mathbb{N}$, then prove the following :
 (a) $C_1 + 2C_2 + 3C_3 + \dots + n \cdot C_n = n \cdot 2^{n-1}$
 (b) $C_0 + 2C_1 + 3C_2 + \dots + (n+1)C_n = (n+2)2^{n-1}$
 (c) $C_0 + 3C_1 + 5C_2 + \dots + (2n+1)C_n = (n+1)2^n$
 (d) $(C_0+C_1)(C_1+C_2)(C_2+C_3) \dots (C_{n-1}+C_n) = \frac{C_0 \cdot C_1 \cdot C_2 \dots C_{n-1} (n+1)^n}{n!}$
 (e) $1 \cdot C_0^2 + 3 \cdot C_1^2 + 5 \cdot C_2^2 + \dots + (2n+1) C_n^2 = \frac{(n+1)(2n)!}{n!n!}$

17. Prove that

$$(a) \quad \frac{C_1}{C_0} + \frac{2C_2}{C_1} + \frac{3C_3}{C_2} + \dots + \frac{n \cdot C_n}{C_{n-1}} = \frac{n(n+1)}{2} \quad (b) \quad C_0 + \frac{C_1}{2} + \frac{C_2}{3} + \dots + \frac{C_n}{n+1} = \frac{2^{n+1}-1}{n+1}$$

$$(c) \quad 2 \cdot C_0 + \frac{2^2 \cdot C_1}{2} + \frac{2^3 \cdot C_2}{3} + \frac{2^4 \cdot C_3}{4} + \dots + \frac{2^{n+1} \cdot C_n}{n+1} = \frac{3^{n+1}-1}{n+1}$$

$$(d) \quad C_0 - \frac{C_1}{2} + \frac{C_2}{3} - \dots + (-1)^n \frac{C_n}{n+1} = \frac{1}{n+1}$$

EXERCISE (S-2)

1. If $(7 + 4\sqrt{3})^n = p + \beta$ where n & p are positive integers and β is a proper fraction show that $(1 - \beta)(p + \beta) = 1$.

2. Let $P = (2 + \sqrt{3})^5$ and $f = P - [P]$, where $[P]$ denotes the greatest integer function.

Find the value of $\left(\frac{f^2}{1-f}\right)$.

3. For which positive values of x is the fourth term in the expansion of $(5 + 3x)^{10}$ is the greatest.

4. Prove that $\sum_{K=0}^n {}^nC_K \sin Kx \cdot \cos(n-K)x = 2^{n-1} \sin nx$.

5. Let $a = (4^{1/401} - 1)$ and let $b_n = {}^nC_1 + {}^nC_2 \cdot a + {}^nC_3 \cdot a^2 + \dots + {}^nC_n \cdot a^{n-1}$.

Find the value of $(b_{2006} - b_{2005})$

6. Let a and b be the coefficient of x^3 in $(1 + x + 2x^2 + 3x^3)^4$ and $(1 + x + 2x^2 + 3x^3 + 4x^4)^4$ respectively. Find the value of $(a - b)$.

7. Find the sum of the roots (real or complex) of the equation $x^{2001} + \left(\frac{1}{2} - x\right)^{2001} = 0$.

8. Find the index n of the binomial $\left(\frac{x}{5} + \frac{2}{5}\right)^n$ if the 9th term of the expansion has numerically the greatest coefficient ($n \in \mathbb{N}$).

9. Find the coefficient of x^{49} in the polynomial

$$\left(x - \frac{C_1}{C_0}\right) \left(x - 2^2 \cdot \frac{C_2}{C_1}\right) \left(x - 3^2 \cdot \frac{C_3}{C_2}\right) \dots \left(x - 50^2 \cdot \frac{C_{50}}{C_{49}}\right), \text{ where } C_r = {}^{50}C_r.$$

10. If $\binom{n}{r}$ denotes nC_r , then

$$(a) \quad \text{Evaluate : } 2^{15} \binom{30}{0} \binom{30}{15} - 2^{14} \binom{30}{1} \binom{29}{14} + 2^{13} \binom{30}{2} \binom{28}{13} - \dots - \binom{30}{15} \binom{15}{0}$$

$$(b) \quad \text{Prove that : } \sum_{r=1}^n \binom{n-1}{n-r} \binom{n}{r} = \binom{2n-1}{n-1} \quad (c) \quad \text{Prove that : } \binom{n}{r} \binom{r}{k} = \binom{n}{k} \binom{n-k}{r-k}$$

EXERCISE (JM)

1. The remainder left out when $8^{2n} - (62)^{2n+1}$ is divided by 9 is :- [AIEEE 2009]
 (1) 7 (2) 8 (3) 0 (4) 2

2. Let $S_1 = \sum_{j=1}^{10} j(j-1)^{10} C_j$, $S_2 = \sum_{j=1}^{10} j^{10} C_j$ and $S_3 = \sum_{j=1}^{10} j^{210} C_j$. [AIEEE-2010]

Statement-1 : $S_3 = 55 \times 2^9$.

Statement-2 : $S_1 = 90 \times 2^8$ and $S_2 = 10 \times 2^8$.

- (1) Statement-1 is true, Statement-2 is true ; Statement-2 is a correct explanation for Statement-1.
 (2) Statement-1 is true, Statement-2 is true ; Statement-2 is not a correct explanation for Statement-1.
 (3) Statement-1 is true, Statement-2 is false.
 (4) Statement-1 is false, Statement-2 is true.
3. The coefficient of x^7 in the expansion of $(1 - x - x^2 + x^3)^6$ is :- [AIEEE 2011]
 (1) -144 (2) 132 (3) 144 (4) -132
4. If n is a positive integer, then $(\sqrt{3} + 1)^{2n} - (\sqrt{3} - 1)^{2n}$ is : [AIEEE 2012]
 (1) a rational number other than positive integers (2) an irrational number
 (3) an odd positive integer (4) an even positive integer

5. The term independent of x in expansion of $\left(\frac{x+1}{x^{2/3} - x^{1/3} + 1} - \frac{x-1}{x - x^{1/2}}\right)^{10}$ is : [JEE-Main 2013]
 (1) 4 (2) 120 (3) 210 (4) 310

6. If the coefficients of x^3 and x^4 in the expansion of $(1 + ax + bx^2)(1 - 2x)^{18}$ in powers of x are both zero, then (a, b) is equal to :- [JEE(Main)-2014]

- (1) $\left(16, \frac{251}{3}\right)$ (2) $\left(14, \frac{251}{3}\right)$ (3) $\left(14, \frac{272}{3}\right)$ (4) $\left(16, \frac{272}{3}\right)$

7. The sum of coefficients of integral powers of x in the binomial expansion of $(1 - 2\sqrt{x})^{50}$ is : [JEE(Main)-2015]

- (1) $\frac{1}{2}(3^{50} - 1)$ (2) $\frac{1}{2}(2^{50} + 1)$ (3) $\frac{1}{2}(3^{50} + 1)$ (4) $\frac{1}{2}(3^{50})$

8. If the number of terms in the expansion of $\left(1 - \frac{2}{x} + \frac{4}{x^2}\right)^n$, $x \neq 0$, is 28, then the sum of the coefficients of all the terms in this expansion, is :- [JEE(Main)-2016]

- (1) 729 (2) 64 (3) 2187 (4) 243

9. The value of $({}^{21}C_1 - {}^{10}C_1) + ({}^{21}C_2 - {}^{10}C_2) + ({}^{21}C_3 - {}^{10}C_3) + ({}^{21}C_4 - {}^{10}C_4) + \dots + ({}^{21}C_{10} - {}^{10}C_{10})$ is :- [JEE(Main)-2017]

- (1) $2^{20} - 2^{10}$ (2) $2^{21} - 2^{11}$ (3) $2^{21} - 2^{10}$ (4) $2^{20} - 2^9$

10. The sum of the co-efficients of all odd degree terms in the expansion of $(x + \sqrt{x^3 - 1})^5 + (x - \sqrt{x^3 - 1})^5$, $(x > 1)$ is - [JEE(Main)-2018]

- (1) 0 (2) 1 (3) 2 (4) -1

EXERCISE (JA)

- For $r = 0, 1, \dots, 10$, let A_r, B_r and C_r denote, respectively, the coefficient of x^r in the expansions of $(1+x)^{10}$, $(1+x)^{20}$ and $(1+x)^{30}$. Then $\sum_{r=1}^{10} A_r(B_{10}B_r - C_{10}A_r)$ is equal to -
 (A) $B_{10} - C_{10}$ (B) $A_{10}(B_{10}^2 - C_{10}A_{10})$ (C) 0 (D) $C_{10} - B_{10}$
[JEE 2010, 5]
- The coefficients of three consecutive terms of $(1+x)^{n+5}$ are in the ratio 5 : 10 : 14. Then $n =$
[JEE (Advanced) 2013, 4M, -1M]
- Coefficient of x^{11} in the expansion of $(1+x^2)^4(1+x^3)^7(1+x^4)^{12}$ is -
 (A) 1051 (B) 1106 (C) 1113 (D) 1120
[JEE(Advanced)-2014, 3(-1)]
- The coefficient of x^9 in the expansion of $(1+x)(1+x^2)(1+x^3)\dots(1+x^{100})$ is **[JEE 2015, 4M, -0M]**
- Let m be the smallest positive integer such that the coefficient of x^2 in the expansion of $(1+x)^2 + (1+x)^3 + \dots + (1+x)^{49} + (1+mx)^{50}$ is $(3n+1)^{51}C_3$ for some positive integer n . Then the value of n is **[JEE(Advanced)-2016, 3(0)]**
- Let $X = \binom{10}{C_1}^2 + 2\binom{10}{C_2}^2 + 3\binom{10}{C_3}^2 + \dots + 10\binom{10}{C_{10}}^2$, where $\binom{10}{C_r}, r \in \{1, 2, \dots, 10\}$ denote binomial coefficients. Then, the value of $\frac{1}{1430}X$ is _____. **[JEE(Advanced)-2018, 3(0)]**

ANSWER KEY

EXERCISE (O-1)

1. C 2. B 3. A 4. C 5. B 6. A 7. C
 8. A 9. B 10. B 11. B 12. D 13. B 14. A 15. B

EXERCISE (O-2)

1. B 2. B 3. B 4. D 5. A 6. A,C,D 7. A,B
 8. B,C,D 9. A,B 10. A,C 11. A,C,D 12. B,C,D 13. A,D 14. B,D 15. A,C

EXERCISE (S-1)

1. (a) $r = 6$ (b) $r = 5$ or 9 2. (i) $\frac{5}{12}$ (ii) $T_6 = 7$ 4. $\frac{(2^{mn} - 1)}{(2^n - 1)(2^m)}$
 5. (i) $T_7 = \frac{7 \cdot 3^{13}}{2}$ (ii) 455×3^{12} 6. $\frac{17}{54}$ 7. $n = 2$ or 3 or 4 8. 816
 10. (a) $-1260 \cdot a^2 b^3 c^4$; (b) -12600 11. (a) 990 (b) 3660
 12. ${}^nC_r(3^{n-r} - 2^{n-r})$ 14. (i) 3^n (ii) 1 , (iii) a_n

EXERCISE (S-2)

2. 722 3. $\frac{5}{8} < x < \frac{20}{21}$ 5. 2^{10} 6. 0 7. 500
 8. $n = 12$ 9. -22100 10. (a) $\binom{30}{15}$

EXERCISE (JM)

1. 4 2. 3 3. 1 4. 2 5. 3 6. 4 7. 3
 8. Bonus 9. 1 10. 3

EXERCISE (JA)

1. D 2. 6 3. C 4. 8 5. 5 6. 646

[illegible]

NURTURE COURSE

PERMUTATION & COMBINATION

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PERMUTATION & COMBINATION

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IIT-JEE Syllabus :

Permutations and Combinations

PERMUTATION & COMBINATION

1. FUNDAMENTAL PRINCIPLE OF COUNTING (counting without actual counting):

If an event A can occur in 'm' different ways and another event B can occur in 'n' different ways, then the total number of different ways of -

- (a) simultaneous occurrence of both events in a definite order is $m \times n$. This can be extended to any number of events (known as multiplication principle).
- (b) happening exactly one of the events is $m + n$ (known as addition principle).

Example : There are 15 IITs in India and let each IIT has 10 branches, then the IITJEE topper can select the IIT and branch in $15 \times 10 = 150$ number of ways.

Example : There are 15 IITs & 20 NITs in India, then a student who cleared both IITJEE & AIEEE exams can select an institute in $(15 + 20) = 35$ number of ways.

Illustration 1 : A college offers 6 courses in the morning and 4 in the evening. The possible number of choices with the student if he wants to study one course in the morning and one in the evening is-

- (A) 24 (B) 2 (C) 12 (D) 10

Solution : The student has 6 choices from the morning courses out of which he can select one course in 6 ways.

For the evening course, he has 4 choices out of which he can select one in 4 ways.

Hence the total number of ways $6 \times 4 = 24$. **Ans.(A)**

Illustration 2 : A college offers 6 courses in the morning and 4 in the evening. The number of ways a student can select exactly one course, either in the morning or in the evening-

- (A) 6 (B) 4 (C) 10 (D) 24

Solution : The student has 6 choices from the morning courses out of which he can select one course in 6 ways.

For the evening course, he has 4 choices out of which he can select one in 4 ways.

Hence the total number of ways $6 + 4 = 10$. **Ans. (C)**

Do yourself - 1 :

- (i) There are 3 ways to go from A to B, 2 ways to go from B to C and 1 way to go from A to C. In how many ways can a person travel from A to C ?
- (ii) There are 2 red balls and 3 green balls. All balls are identical except colour. In how many ways can a person select two balls ?

2. FACTORIAL NOTATION :

- (i) A Useful Notation : $n!$ (factorial n) = $n.(n-1).(n-2).....3.2.1$; $n! = n.(n-1)!$ where $n \in \mathbb{N}$
- (ii) $0! = 1! = 1$
- (iii) Factorials of negative integers are not defined.
- (iv) $n!$ is also denoted by $\lfloor n$
- (v) $(2n)! = 2^n \cdot n! [1.3.5.7.....(2n-1)]$
- (vi) Prime factorisation of $n!$: Let p be a prime number and n be a positive integer, then exponent of p in $n!$ is denoted by $E_p(n!)$ and is given by

$$E_p(n!) = \left\lfloor \frac{n}{p} \right\rfloor + \left\lfloor \frac{n}{p^2} \right\rfloor + \left\lfloor \frac{n}{p^3} \right\rfloor + \dots + \left\lfloor \frac{n}{p^k} \right\rfloor$$

where, $p^k \leq n < p^{k+1}$ and $[x]$ denotes the integral part of x .

If we isolate the power of each prime contained in any number n , then n can be written as $n = 2^{\alpha_1} \cdot 3^{\alpha_2} \cdot 5^{\alpha_3} \cdot 7^{\alpha_4} \dots$, where α_i are whole numbers.

Illustration 3 : Find the exponent of 6 in $50!$

Solution: $E_2(50!) = \left\lfloor \frac{50}{2} \right\rfloor + \left\lfloor \frac{50}{4} \right\rfloor + \left\lfloor \frac{50}{8} \right\rfloor + \left\lfloor \frac{50}{16} \right\rfloor + \left\lfloor \frac{50}{32} \right\rfloor + \left\lfloor \frac{50}{64} \right\rfloor$ (where $[]$ denotes integral part)

$$E_2(50!) = 25 + 12 + 6 + 3 + 1 + 0 = 47$$

$$E_3(50!) = \left\lfloor \frac{50}{3} \right\rfloor + \left\lfloor \frac{50}{9} \right\rfloor + \left\lfloor \frac{50}{27} \right\rfloor + \left\lfloor \frac{50}{81} \right\rfloor$$

$$E_3(50!) = 16 + 5 + 1 + 0 = 22$$

$$\Rightarrow 50! \text{ can be written as } 50! = 2^{47} \cdot 3^{22} \dots$$

$$\text{Therefore exponent of 6 in } 50! = 22$$

Ans.

3. PERMUTATION & COMBINATION :

- (a) **Permutation :** Each of the arrangements in a definite order which can be made by taking some or all of the things at a time is called a PERMUTATION. In permutation, order of appearance of things is taken into account; when the order is changed, a different permutation is obtained. Generally, it involves the problems of arrangements (standing in a line, seated in a row), problems on digit, problems on letters from a word etc.

${}^n P_r$ denotes the number of permutations of n **different** things, taken r at a time ($n \in \mathbb{N}$, $r \in \mathbb{W}$, $r \leq n$)

$${}^n P_r = n(n-1)(n-2) \dots (n-r+1) = \frac{n!}{(n-r)!}$$

Note :

- (i) ${}^n P_n = n!$, ${}^n P_0 = 1$, ${}^n P_1 = n$
- (ii) Number of arrangements of n **distinct** things taken all at a time = $n!$
- (iii) ${}^n P_r$ is also denoted by A_r^n or $P(n,r)$.

(b) **Combination :**

Each of the groups or selections which can be made by taking some or all of the things without considering the order of the things in each group is called a COMBINATION.

Generally, involves the problem of selections, choosing, distributed groups formation, committee formation, geometrical problems etc.

${}^n C_r$ denotes the number of combinations of n different things taken r at a time ($n \in \mathbb{N}$, $r \in \mathbb{W}$, $r \leq n$)

$${}^n C_r = \frac{n!}{r!(n-r)!}$$

Note :

(i) ${}^n C_r$ is also denoted by $\binom{n}{r}$ or $C(n, r)$.

(ii) ${}^n P_r = {}^n C_r \cdot r!$

Illustration 4 : If a denotes the number of permutations of $(x+2)$ things taken all at a time, b the number of permutations of x things taken 11 at a time and c the number of permutations of $(x-11)$ things taken all at a time such that $a = 182bc$, then the value of x is
(A) 15 (B) 12 (C) 10 (D) 18

Solution :

$${}^{x+2} P_{x+2} = a \Rightarrow a = (x+2)!$$

$${}^x P_{11} = b \Rightarrow b = \frac{x!}{(x-11)!}$$

$$\text{and } {}^{x-11} P_{x-11} = c \Rightarrow c = (x-11)!$$

$$\therefore a = 182bc$$

$$(x+2)! = 182 \frac{x!}{(x-11)!} (x-11)! \Rightarrow (x+2)(x+1) = 182 = 14 \times 13$$

$$\therefore x+1 = 13 \Rightarrow x = 12$$

Ans. (B)

Illustration 5 : A box contains 5 different red and 6 different white balls. In how many ways can 6 balls be drawn so that there are atleast two balls of each colour ?

Solution :

The selections of 6 balls, consisting of atleast two balls of each colour from 5 red and 6 white balls, can be made in the following ways

Red balls (5)	White balls (6)	Number of ways
2	4	${}^5 C_2 \times {}^6 C_4 = 150$
3	3	${}^5 C_3 \times {}^6 C_3 = 200$
4	2	${}^5 C_4 \times {}^6 C_2 = 75$

Therefore total number of ways = 425

Ans.

Illustration 6 : How many 4 letter words can be formed from the letters of the word 'ANSWER' ? How many of these words start with a vowel ?

Solution : Number of ways of arranging 4 different letters from 6 different letters are

$${}^6C_4 4! = \frac{6!}{2!} = 360.$$

There are two vowels (A & E) in the word 'ANSWER'.

$$\text{Total number of 4 letter words starting with A : } A_ _ _ = {}^5C_3 3! = \frac{5!}{2!} = 60$$

$$\text{Total number of 4 letter words starting with E : } E_ _ _ = {}^5C_3 3! = \frac{5!}{2!} = 60$$

$\therefore$ Total number of 4 letter words starting with a vowel = $60 + 60 = 120$. **Ans.**

Illustration 7 : If all the letters of the word 'RAPID' are arranged in all possible manner as they are in a dictionary, then find the rank of the word 'RAPID'.

Solution : First of all, arrange all letters of given word alphabetically : 'ADIPR'

$$\text{Total number of words starting with A } _ _ _ _ = 4! = 24$$

$$\text{Total number of words starting with D } _ _ _ _ = 4! = 24$$

$$\text{Total number of words starting with I } _ _ _ _ = 4! = 24$$

$$\text{Total number of words starting with P } _ _ _ _ = 4! = 24$$

$$\text{Total number of words starting with RAD } _ _ = 2! = 2$$

$$\text{Total number of words starting with RAI } _ _ = 2! = 2$$

$$\text{Total number of words starting with RAPD } _ = 1$$

$$\text{Total number of words starting with RAPI } _ = 1$$

$\therefore$ Rank of the word RAPID = $24 + 24 + 24 + 24 + 2 + 2 + 1 + 1 = 102$ **Ans.**

Do yourself -2 :

- (i) Find the exponent of 10 in ${}^{75}C_{25}$.
- (ii) If ${}^{10}P_r = 5040$, then find the value of r .
- (iii) Find the number of ways of selecting 4 even numbers from the set of first 100 natural numbers.
- (iv) If all letters of the word 'RANK' are arranged in all possible manner as they are in a dictionary, then find the rank of the word 'RANK'.
- (v) How many words can be formed using all letters of the word 'LEARN'? In how many of these words vowels are together?

4. PROPERTIES OF nP_r and nC_r :

- (a) The number of permutation of n different objects taken r at a time, when p particular objects are always to be included is $r! \cdot {}^{n-p}C_{r-p}$ ($p \leq r \leq n$)
- (b) The number of permutations of n different objects taken r at a time, when repetition is allowed any number of times is n^r .
- (c) Following properties of nC_r should be remembered :
 - (i) ${}^nC_r = {}^nC_{n-r}$; ${}^nC_0 = {}^nC_n = 1$
 - (ii) ${}^nC_x = {}^nC_y \Rightarrow x = y$ or $x + y = n$
 - (iii) ${}^nC_r + {}^nC_{r-1} = {}^{n+1}C_r$
 - (iv) ${}^nC_0 + {}^nC_1 + {}^nC_2 + \dots + {}^nC_n = 2^n$
 - (v) ${}^nC_r = \frac{n}{r} {}^{n-1}C_{r-1}$
 - (vi) nC_r is maximum when $r = \frac{n}{2}$ if n is even & $r = \frac{n-1}{2}$ or $r = \frac{n+1}{2}$, if n is odd.

- (d) The number of combinations of n different things taking r at a time,
 (i) when p particular things are always to be included $= {}^{n-p}P_{r-p}$
 (ii) when p particular things are always to be excluded $= {}^{n-p}P_r$
 (iii) when p particular things are always to be included and q particular things are to be excluded
 $= {}^{n-p-q}P_{r-p}$

Illustration 8 : There are 6 pockets in the coat of a person. In how many ways can he put 4 pens in these pockets?

- (A) 360 (B) 1296 (C) 4096 (D) none of these

Solution : First pen can be put in 6 ways.

Similarly each of second, third and fourth pen can be put in 6 ways.

Hence total number of ways $= 6 \times 6 \times 6 \times 6 = 1296$

Ans.(B)

Illustration 9 : A delegation of four students is to be selected from a total of 12 students. In how many ways can the delegation be selected, if-

- (a) all the students are equally willing ?
 (b) two particular students have to be included in the delegation ?
 (c) two particular students do not wish to be together in the delegation ?
 (d) two particular students wish to be included together only ?
 (e) two particular students refuse to be together and two other particular students wish to be together only in the delegation ?

Solution : (a) Formation of delegation means selection of 4 out of 12.

Hence the number of ways $= {}^{12}C_4 = 495$.

- (b) If two particular students are already selected. Here we need to select only 2 out of the remaining 10. Hence the number of ways $= {}^{10}C_2 = 45$.
 (c) The number of ways in which both are selected $= 45$. Hence the number of ways in which the two are not included together $= 495 - 45 = 450$
 (d) There are two possible cases

(i) Either both are selected. In this case, the number of ways in which the selection can be made $= 45$.

(ii) Or both are not selected. In this case all the four students are selected from the remaining ten students. This can be done in ${}^{10}C_4 = 210$ ways.

Hence the total number of ways of selection $= 45 + 210 = 255$

- (e) We assume that students A and B wish to be selected together and students C and D do not wish to be together. Now there are following 6 cases.

- | | |
|------------------------------|------------------------|
| (i) (A, B, C) selected, | (D) not selected |
| (ii) (A, B, D) selected, | (C) not selected |
| (iii) (A, B) selected, | (C, D) not selected |
| (iv) (C) selected, | (A, B, D) not selected |
| (v) (D) selected, | (A, B, C) not selected |
| (vi) A, B, C, D not selected | |

Do yourself-3 :

- (i) Find the number of ways of selecting 5 members from a committee of 5 men & 2 women such that all women are always included.
- (ii) Out of first 20 natural numbers, 3 numbers are selected such that there is exactly one even number. How many different selections can be made ?
- (iii) How many four letter words can be made from the letters of the word 'PROBLEM'. How many of these start as well as end with a vowel ?

5. FORMATION OF GROUPS :

- (a) (i) The number of ways in which $(m + n)$ different things can be divided into two groups such that one of them contains m things and other has n things, is $\frac{(m+n)!}{m! n!}$ ($m \neq n$).

- (ii) If $m = n$, it means the groups are equal & in this case the number of divisions is $\frac{(2n)!}{n! n! 2!}$.

As in any one way it is possible to interchange the two groups without obtaining a new distribution.

- (iii) If $2n$ things are to be divided equally between two persons then the number of ways :

$$\frac{(2n)!}{n! n! (2!)} \times 2!$$

- (b) (i) Number of ways in which $(m + n + p)$ different things can be divided into three groups containing m , n & p things respectively is : $\frac{(m+n+p)!}{m! n! p!}$, $m \neq n \neq p$.

- (ii) If $m = n = p$ then the number of groups = $\frac{(3n)!}{n! n! n! 3!}$.

- (iii) If $3n$ things are to be divided equally among three people then the number of ways in which it can be done is $\frac{(3n)!}{(n!)^3}$.

- (c) In general, the number of ways of dividing n distinct objects into ℓ groups containing p objects each and m groups containing q objects each is equal to $\frac{n!(\ell+m)!}{(p!)^\ell (q!)^m \ell! m!}$

Here $\ell p + m q = n$

Illustration 13 : In how many ways can 15 students be divided into 3 groups of 5 students each such that 2 particular students are always together ? Also find the number of ways if these groups are to be sent to three different colleges.

Solution : Here first we separate those two particular students and make 3 groups of 5, 5 and 3 of the remaining 13 so that these two particular students always go with the group of 3 students.

$$\therefore \text{Number of ways} = \frac{13!}{5!5!3!} \cdot \frac{1}{2!}$$

Now if these groups are to be sent to three different colleges, total number of

$$\text{ways} = \frac{13!}{5!5!3!} \cdot \frac{1}{2!} \cdot 3!$$

Ans.

Illustration 14 : Find the number of ways of dividing 52 cards among 4 players equally such that each gets exactly one Ace.

Solution : Total number of ways of dividing 48 cards (Excluding 4 Aces) in 4 groups = $\frac{48!}{(12!)^4 4!}$

Now, distribute exactly one Ace to each group of 12 cards. Total number of ways

$$= \frac{48!}{(12!)^4 4!} \times 4!$$

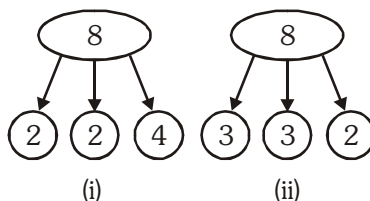
Now, distribute these groups of cards among four players

$$= \frac{48!}{(12!)^4 4!} \times 4! 4! = \frac{48!}{(12!)^4} \times 4!$$

Ans.

Illustration 15 : In how many ways can 8 different books be distributed among 3 students if each receives at least 2 books ?

Solution : If each receives at least two books, then the division trees would be as shown below :



The number of ways of division for tree in figure (i) is $\left[\frac{8!}{(2!)^2 4! 2!} \right]$.

The number of ways of division for tree in figure (ii) is $\left[\frac{8!}{(3!)^2 2! 2!} \right]$.

The total number of ways of distribution of these groups among 3 students

$$\text{is } \left[\frac{8!}{(2!)^2 4! 2!} + \frac{8!}{(3!)^2 2! 2!} \right] \times 3!.$$

Ans.

Do yourself-4 :

- (i) Find the number of ways in which 16 constables can be assigned to patrol 8 villages, 2 for each.
- (ii) In how many ways can 6 different books be distributed among 3 students such that none gets equal number of books and each gets atleast one book ?
- (iii) n different toys are to be distributed among n children. Find the number of ways in which these toys can be distributed so that exactly one child gets no toy.

6. PRINCIPLE OF INCLUSION AND EXCLUSION :

In the Venn's diagram (i), we get

$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

$$n(A' \cap B') = n(U) - n(A \cup B)$$

In the Venn's diagram (ii), we get

$$n(A \cup B \cup C)$$

$$= n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(A \cap C) + n(A \cap B \cap C)$$

$$n(A' \cap B' \cap C') = n(U) - n(A \cup B \cup C)$$

In general, we have $n(A_1 \cup A_2 \cup \dots \cup A_n)$

$$= \sum n(A_i) - \sum_{i \neq j} n(A_i \cap A_j) + \sum_{i \neq j \neq k} n(A_i \cap A_j \cap A_k) + \dots + (-1)^{n+1} \sum n(A_1 \cap A_2 \cap \dots \cap A_n)$$

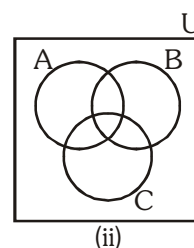
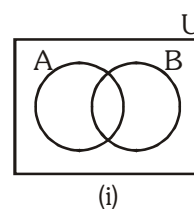
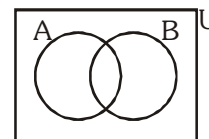


Illustration 16 : Find the number of permutations of letters a,b,c,d,e,f,g taken all at a time if neither 'beg' nor 'cad' pattern appear.



Solution :

The total number of permutations without any restrictions; $n(U) = 7!$

(b e g) a c d f

Let A be the set of all possible permutations in which 'beg' pattern always appears : $n(A) = 5!$

(c a d) b e f g

Let B be the set of all possible permutations in which 'cad' pattern always appears : $n(B) = 5!$

(c a d) (b e g) f

$n(A \cap B)$: Number of all possible permutations when both 'beg' and 'cad' patterns appear.

$$n(A \cap B) = 3!$$

Therefore, the total number of permutations in which 'beg' and 'cad' patterns do not appear

$$n(A' \cap B') = n(U) - n(A \cap B) = n(U) - n(A) - n(B) + n(A \cap B)$$

$$= 7! - 5! - 5! + 3!$$

Ans.

Do yourself-5 :

- (i) Find the number of n digit numbers formed using first 5 natural numbers, which contain the digits 2 & 4 essentially.

7. PERMUTATIONS OF ALIKE OBJECTS :

Case-I : Taken all at a time -

The number of permutations of n things taken all at a time : when p of them are similar of one type, q of them are similar of second type, r of them are similar of third type and the remaining $n - (p + q + r)$

are all different is : $\frac{n!}{p! q! r!}$.

Illustration 17 : In how many ways the letters of the word "ARRANGE" can be arranged without altering the relative position of vowels & consonants.

Solution : The consonants in their positions can be arranged in $\frac{4!}{2!} = 12$ ways.

The vowels in their positions can be arranged in $\frac{3!}{2!} = 3$ ways

$\therefore$ Total number of arrangements = $12 \times 3 = 36$

Ans.

Illustration 18 : How many numbers can be formed with the digits 1, 2, 3, 4, 3, 2, 1 so that the odd digits always occupy the odd places?

(A) 17 (B) 18 (C) 19 (D) 20

Solution : There are 4 odd digits (1, 1, 3, 3) and 4 odd places (first, third, fifth and seventh). At these places the odd digits can be arranged in $\frac{4!}{2!2!} = 6$ ways

Then at the remaining 3 places, the remaining three digits (2, 2, 4) can be arranged in $\frac{3!}{2!} = 3$ ways

$\therefore$ The required number of numbers = $6 \times 3 = 18$.

Ans. (B)

Illustration 19 : (a) How many permutations can be made by using all the letters of the word HINDUSTAN ?

(b) How many of these permutations begin and end with a vowel ?

(c) In how many of these permutations, all the vowels come together ?

(d) In how many of these permutations, none of the vowels come together ?

(e) In how many of these permutations, do the vowels and the consonants occupy the same relative positions as in HINDUSTAN ?

Solution : (a) The total number of permutations = Arrangements of nine letters taken all at a time
 $= \frac{9!}{2!} = 181440$.

(b) We have 3 vowels and 6 consonants, in which 2 consonants are alike. The first place can be filled in 3 ways and the last in 2 ways. The rest of the places can be filled in $\frac{7!}{2!}$ ways.

Hence the total number of permutations = $3 \times 2 \times \frac{7!}{2!} = 15120$.

(c) Assume the vowels (I, U, A) as a single letter. The letters (IUA), H, D, S, T, N, N can be arranged in $\frac{7!}{2!}$ ways. Also IUA can be arranged among themselves in $3! = 6$ ways.

Hence the total number of permutations = $\frac{7!}{2!} \times 6 = 15120$.

(d) Let us divide the task into two parts. In the first, we arrange the 6 consonants as shown below in $\frac{6!}{2!}$ ways.

$\times C \times C \times C \times C \times C \times C \times$ (Here C stands for a consonant and $\times$ stands for a gap between two consonants)

Now 3 vowels can be placed in 7 places (gaps between the consonants) in ${}^7C_3 \cdot 3! = 210$ ways.

Hence the total number of permutations = $\frac{6!}{2!} \times 210 = 75600$.

- (e) In this case, the vowels can be arranged among themselves in $3! = 6$ ways.

Also, the consonants can be arranged among themselves in $\frac{6!}{2!}$ ways.

Hence the total number of permutations = $\frac{6!}{2!} \times 6 = 2160$.

Ans.

Illustration 20 : If all the letters of the word 'PROPER' are arranged in all possible manner as they are in a dictionary, then find the rank of the word 'PROPER' .

Solution : First of all, arrange all letters of given word alphabetically : EOPRRR.

Total number of words starting with-

$$E_{\text{-----}} = \frac{5!}{2!2!} = 30$$

$$O_{-----} = \frac{5!}{2!2!} = 30$$

$$\text{PE} \text{ --- } = \frac{4!}{2!} = 12$$

PO $\frac{4!}{2!} = 12$

$$\text{PP} \quad \text{---} \quad \text{---} \quad \text{---} \quad \text{---} = \frac{4!}{2!} = 12$$

PRE = 3! = 6

$$\text{PROE} = 2! = 2$$

$$\overline{\text{PROPER}} = 1 = 1$$

Rank of the word PROPER = 105

Ans.

Case-II : Taken some at a time

Illustration 21: Find the total number of 4 letter words formed using four letters from the word "PARALLELOPIPED".

Solution : Given letters are PPP, LLL, AA, EE, R, O, I, D.

Cases	No. of ways of selection	No. of ways of arrangements	Total
All distinct	8C_4	${}^8C_4 \times 4!$	1680
2 alike, 2 distinct	${}^4C_1 \times {}^7C_2$	${}^4C_1 \times {}^7C_2 \times \frac{4!}{2!}$	1008
2 alike, 2 other alike	4C_2	${}^4C_2 \times \frac{4!}{2!2!}$	36
3 alike, 1 distinct	${}^2C_1 \times {}^7C_1$	${}^2C_1 \times {}^7C_1 \times \frac{4!}{3!}$	56
		Total	2780

Ans.

Illustration 22 : Find the number of all 6 digit numbers such that all the digits of each number are selected from the set $\{1,2,3,4,5\}$ and any digit that appears in the number appears at least twice.

Solution :

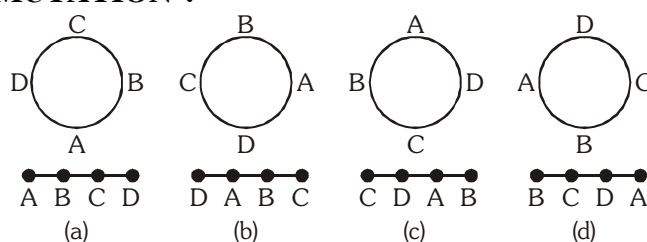
Cases	No. of ways of selection	No. of ways of arrangements	Total
All alike	5C_1	${}^5C_1 \times 1$	5
4 alike + 2 other alike	${}^5C_2 \times 2!$	${}^5C_2 \times 2 \times \frac{6!}{2!4!}$	300
3 alike + 3 other alike	5C_2	${}^5C_2 \times \frac{6!}{3!3!}$	200
2 alike + 2 other alike + 2 other alike	5C_3	${}^5C_3 \times \frac{6!}{2!2!2!}$	900
		Total	1405

Ans.

Do yourself-6 :

- In how many ways can the letters of the word 'ALLEN' be arranged ? Also find its rank if all these words are arranged as they are in dictionary.
- How many numbers greater than 1000 can be formed from the digits 1, 1, 2, 2, 3 ?

8. CIRCULAR PERMUTATION :



Let us consider that persons A,B,C,D are sitting around a round table. If all of them (A,B,C,D) are shifted by one place in anticlockwise order, then we will get Fig.(b) from Fig.(a). Now, if we shift A,B,C,D in anticlockwise order, we will get Fig.(c). Again, if we shift them, we will get Fig.(d) and in the next time, Fig.(a).

Thus, we see that if 4 persons are sitting at a round table, they can be shifted four times and the four different arrangements, thus obtained will be the same, because anticlockwise order of A,B,C,D does not change.

But if A,B,C,D are sitting in a row and they are shifted in such an order that the last occupies the place of first, then the four arrangements will be different.

Thus, if there are 4 things, then for each circular arrangement number of linear arrangements is 4.

Similarly, if n different things are arranged along a circle, for each circular arrangement number of linear arrangements is n .

Therefore, the number of linear arrangements of n different things is $n \times$ (number of circular arrangements of n different things). Hence, the number of circular arrangements of n different things is -

$$\frac{1}{n} \times (\text{number of linear arrangements of } n \text{ different things}) = \frac{n!}{n} = (n-1)!$$

Therefore note that :

- (i) The number of circular permutations of n different things taken all at a time is : $(n-1)!$.

If clockwise & anti-clockwise circular permutations are considered to be same, then it is : $\frac{(n-1)!}{2}$.

- (ii) The number of circular permutations of n different things taking r at a time distinguishing clockwise & anticlockwise arrangements is : $\frac{{}^n P_r}{r}$

Illustration 23 : In how many ways can 5 boys and 5 girls be seated at a round table so that no two girls are together?

- (A) $5! \times 5!$ (B) $5! \times 4!$ (C) $\frac{1}{2}(5!)^2$ (D) $\frac{1}{2}(5! \times 4!)$

Solution : Leaving one seat vacant between two boys, 5 boys may be seated in $4!$ ways. Then at remaining 5 seats, 5 girls sit in $5!$ ways. Hence the required number of ways = $4! \times 5!$

Ans. (B)

Illustration 24 : The number of ways in which 7 girls can stand in a circle so that they do not have same neighbours in any two arrangements ?

- (A) 720 (B) 380 (C) 360 (D) none of these

Solution : Seven girls can stand in a circle by $\frac{(7-1)!}{2!}$ number of ways, because there is no difference in anticlockwise and clockwise order of their standing in a circle.

$$\therefore \frac{(7-1)!}{2!} = 360$$

Ans. (C)

Illustration 25 : The number of ways in which 20 different pearls of two colours can be set alternately on a necklace, there being 10 pearls of each colour, is

- (A) $9! \times 10!$ (B) $5(9!)^2$ (C) $(9!)^2$ (D) none of these

Solution : Ten pearls of one colour can be arranged in $\frac{1}{2} \cdot (10-1)!$ ways. The number of arrangements of 10 pearls of the other colour in 10 places between the pearls of the first colour = $10!$

$$\therefore \text{The required number of ways} = \frac{1}{2} \times 9! \times 10! = 5(9!)^2$$

Ans. (B)

Illustration 26 : A person invites a group of 10 friends at dinner. They sit

- (i) 5 on one round table and 5 on other round table,
(ii) 4 on one round table and 6 on other round table.

Find the number of ways in each case in which he can arrange the guests.

Solution : (i) The number of ways of selection of 5 friends for first table is ${}^{10}C_5$. Remaining 5 friends are left for second table.

The total number of permutations of 5 guests at a round table is $4!$. Hence, the total

$$\text{number of arrangements is } {}^{10}C_5 \times 4! \times 4! = \frac{10!4!4!}{5!5!} = \frac{10!}{25}$$

(ii) The number of ways of selection of 6 guests is ${}^{10}C_6$.

The number of ways of permutations of 6 guests on round table is $5!$. The number of permutations of 4 guests on round table is $3!$

$$\text{Therefore, total number of arrangements is : } {}^{10}C_6 \cdot 5! \times 3! = \frac{(10)!}{6!4!} \cdot 5! \cdot 3! = \frac{(10)!}{24}$$

Ans. (B)

Do yourself-7 :

- (i) In how many ways can 3 men and 3 women be seated around a round table such that all men are always together ?
- (ii) Find the number of ways in which 10 different diamonds can be arranged to make a necklace.
- (iii) Find the number of ways in which 6 persons out of 5 men & 5 women can be seated at a round table such that 2 men are never together.
- (iv) In how many ways can 8 persons be seated on two round tables of capacity 5 & 3.

9. TOTAL NUMBER OF COMBINATIONS :

- (a) Given n different objects, the number of ways of selecting atleast one of them is,
 ${}^nC_1 + {}^nC_2 + {}^nC_3 + \dots + {}^nC_n = 2^n - 1$. This can also be stated as the total number of combinations of n distinct things.
- (b) (i) Total number of ways in which it is possible to make a selection by taking some or all out of $p + q + r + \dots$ things, where p are alike of one kind, q alike of a second kind, r alike of third kind & so on is given by : $(p + 1)(q + 1)(r + 1)\dots - 1$.
- (ii) The total number of ways of selecting one or more things from p identical things of one kind, q identical things of second kind, r identical things of third kind and n different things is given by :
 $(p + 1)(q + 1)(r + 1)2^n - 1$.

Illustration 27 : A is a set containing n elements. A subset P of A is chosen. The set A is reconstructed by replacing the elements of P . A subset Q of A is again chosen. The number of ways of choosing P and Q so that $P \cap Q = \phi$ is :-

- (A) $2^{2n} - 2^n C_n$ (B) 2^n (C) $2^n - 1$ (D) 3^n

Solution : Let $A = \{a_1, a_2, a_3, \dots, a_n\}$. For $a_i \in A$, we have the following choices :

- (i) $a_i \in P$ and $a_i \in Q$ (ii) $a_i \in P$ and $a_i \notin Q$
- (iii) $a_i \notin P$ and $a_i \in Q$ (iv) $a_i \notin P$ and $a_i \notin Q$

Out of these only (ii), (iii) and (iv) imply $a_i \notin P \cap Q$. Therefore, the number of ways in which none of $a_1, a_2, \dots, a_n$ belong to $P \cap Q$ is 3^n . **Ans. (D)**

Illustration 28 : There are 3 books of mathematics, 4 of science and 5 of english. How many different collections can be made such that each collection consists of-

- (i) one book of each subject ?
- (ii) at least one book of each subject ?
- (iii) at least one book of english ?

Solution :

- (i) ${}^3C_1 \times {}^4C_1 \times {}^5C_1 = 60$
- (ii) $(2^3 - 1)(2^4 - 1)(2^5 - 1) = 7 \times 15 \times 31 = 3255$
- (iii) $(2^5 - 1)(2^3)(2^4) = 31 \times 128 = 3968$

Ans.

Illustration 29 : Find the number of groups that can be made from 5 red balls, 3 green balls and 4 black balls, if at least one ball of all colours is always to be included. Given that all balls are identical except colours.

Solution : After selecting one ball of each colour, we have to find total number of combinations that can be made from 4 red, 2 green and 3 black balls. These will be $(4 + 1)(2 + 1)(3 + 1)$
 $= 60$ **Ans.**

Do yourself-8 :

- (i) There are p copies each of n different books. Find the number of ways in which atleast one book can be selected ?
- (ii) There are 10 questions in an examination. In how many ways can a candidate answer the questions, if he attempts atleast one question.

10. DIVISORS :

Let $N = p^a \cdot q^b \cdot r^c \dots$ where $p, q, r, \dots$ are distinct primes & $a, b, c, \dots$ are natural numbers then :

- (a) The total numbers of divisors of N including 1 & N is $= (a + 1)(b + 1)(c + 1) \dots$
- (b) The sum of these divisors is

$$= (p^0 + p^1 + p^2 + \dots + p^a)(q^0 + q^1 + q^2 + \dots + q^b)(r^0 + r^1 + r^2 + \dots + r^c) \dots$$
- (c) Number of ways in which N can be resolved as a product of two factor is =

$$\frac{1}{2} (a + 1)(b + 1)(c + 1) \dots$$
 if N is not a perfect square

$$\frac{1}{2} [(a + 1)(b + 1)(c + 1) \dots + 1]$$
 if N is a perfect square
- (d) Number of ways in which a composite number N can be resolved into two factors which are relatively prime (or coprime) to each other is equal to 2^{n-1} where n is the number of different prime factors in N .

Note :

- (i) Every natural number except 1 has atleast 2 divisors. If it has exactly two divisors then it is called a prime. System of prime numbers begin with 2. All primes except 2 are odd.
- (ii) A number having more than 2 divisors is called composite. 2 is the only even number which is not composite.
- (iii) Two natural numbers are said to be relatively prime or coprime if their HCF is one. For two natural numbers to be relatively prime, it is not necessary that one or both should be prime. It is possible that they both are composite but still coprime, eg. 4 and 25.
- (iv) 1 is neither prime nor composite however it is co-prime with every other natural number.
- (v) Two prime numbers are said to be twin prime numbers if their non-negative difference is 2 (e.g. 5 & 7, 19 & 17 etc).
- (vi) All divisors except 1 and the number itself are called proper divisors.

Illustration 30: Find the number of proper divisors of the number 38808. Also find the sum of these divisors.

Solution :(i) The number $38808 = 2^3 \cdot 3^2 \cdot 7^2 \cdot 11$

Hence the total number of divisors (excluding 1 and itself i.e. 38808)

$$= (3 + 1)(2 + 1)(2 + 1)(1 + 1) - 2 = 70$$

(ii) The sum of these divisors

$$= (2^0 + 2^1 + 2^2 + 2^3)(3^0 + 3^1 + 3^2)(7^0 + 7^1 + 7^2)(11^0 + 11^1) - 1 - 38808$$

$$= (15)(13)(57)(12) - 1 - 38808 = 133380 - 1 - 38808 = 94571. \quad \text{Ans.}$$

Illustration 31: In how many ways the number 18900 can be split in two factors which are relative prime (or coprime) ?

Solution:

$$\text{Here } N = 18900 = 2^2 \cdot 3^3 \cdot 5^2 \cdot 7^1$$

Number of different prime factors in 18900 = $n = 4$ Hence number of ways in which 18900 can be resolved into two factors which are relative prime (or coprime) = $2^{4-1} = 2^3 = 8$. **Ans.**

Illustration 32: Find the total number of proper factors of the number 35700. Also find

- (i) sum of all these factors,
- (ii) sum of the odd proper divisors,
- (iii) the number of proper divisors divisible by 10 and the sum of these divisors.

Solution:

$$35700 = 5^2 \times 2^2 \times 3^1 \times 7^1 \times 17^1$$

The total number of factors is equal to the total number of selections from (5,5), (2,2), (3), (7) and (17), which is given by $3 \times 3 \times 2 \times 2 \times 2 = 72$.These include 1 and 35700. Therefore, the number of proper divisors (excluding 1 and 35700) is $72 - 2 = 70$

(i) Sum of all these factors (proper) is :

$$(5^0 + 5^1 + 5^2)(2^0 + 2^1 + 2^2)(3^0 + 3^1)(7^0 + 7^1)(17^0 + 17^1) - 1 - 35700$$

$$= 31 \times 7 \times 4 \times 8 \times 18 - 1 - 35700 = 89291$$

(ii) The sum of odd proper divisors is :

$$(5^0 + 5^1 + 5^2)(3^0 + 3^1)(7^0 + 7^1)(17^0 + 17^1) - 1$$

$$= 31 \times 4 \times 8 \times 18 - 1 = 17856 - 1 = 17855$$

(iii) The number of proper divisors divisible by 10 is equal to number of selections from (5,5), (2,2), (3), (7), (17) consisting of at least one 5 and at least one 2 and 35700 is to be excluded and is given by $2 \times 2 \times 2 \times 2 \times 2 - 1 = 31$.

Sum of these divisors is :

$$(5^1 + 5^2)(2^1 + 2^2)(3^0 + 3^1)(7^0 + 7^1)(17^0 + 17^1) - 35700$$

$$= 30 \times 6 \times 4 \times 8 \times 18 - 35700 = 67980$$

Ans.**Do yourself-9 :**

- (i) Find the number of ways in which the number 94864 can be resolved as a product of two factors.
- (ii) Find the number of different sets of solution of $xy = 1440$.

11. TOTAL DISTRIBUTION :

- (a) **Distribution of distinct objects :** Number of ways in which n distinct things can be distributed to p persons if there is no restriction to the number of things received by them is given by : p^n
- (b) **Distribution of alike objects :** Number of ways to distribute n alike things among p persons so that each may get none, one or more thing(s) is given by ${}^{n+p-1}C_{p-1}$.

Illustration 33: In how many ways can 5 different mangoes, 4 different oranges & 3 different apples be distributed among 3 children such that each gets atleast one mango ?

Solution : 5 different mangoes can be distributed by following ways among 3 children such that each gets atleast 1 :

$$\begin{matrix} 3 & 1 & 1 \\ 2 & 2 & 1 \end{matrix}$$

$$\text{Total number of ways : } \left(\frac{5!}{3!1!1!2!} + \frac{5!}{2!2!2!} \right) \times 3!$$

Now, the number of ways of distributing remaining fruits (i.e. 4 oranges + 3 apples) among 3 children = 3^7 (as each fruit has 3 options).

$$\therefore \text{Total number of ways} = \left(\frac{5!}{3!2!} + \frac{5!}{(2!)^3} \right) \times 3! \times 3^7 \quad \text{Ans.}$$

Illustration 34: In how many ways can 12 identical apples be distributed among four children if each gets atleast 1 apple and not more than 4 apples.

Solution : Let x, y, z & w be the number of apples given to the children.

$$\Rightarrow x + y + z + w = 12$$

Giving one-one apple to each

$$\text{Now, } x + y + z + w = 8 \quad \dots\dots(i)$$

$$\text{Here, } 0 \leq x \leq 3, 0 \leq y \leq 3, 0 \leq z \leq 3, 0 \leq w \leq 3$$

$$x = 3 - t_1, y = 3 - t_2, z = 3 - t_3, w = 3 - t_4.$$

Putting value of x, y, z, w in equation (i)

$$\text{Put } 12 - 8 = t_1 + t_2 + t_3 + t_4$$

$$\Rightarrow t_1 + t_2 + t_3 + t_4 = 4$$

(Here max. value that t_1, t_2, t_3 & t_4 can attain is 3, so we have to remove those cases when any of t_i getting value 4)

$$= {}^7C_3 - (\text{all cases when atleast one is 4})$$

$$= {}^7C_3 - 4 = 35 - 4 = 31 \quad \text{Ans.}$$

Illustration 35: Find the number of non negative integral solutions of the inequation $x + y + z \leq 20$.

Solution : Let w be any number ($0 \leq w \leq 20$), then we can write the equation as :

$$x + y + z + w = 20 \quad (\text{here } x, y, z, w \geq 0)$$

$$\text{Total ways} = {}^{23}C_3$$

Ans.

Illustration 36: Find the number of integral solutions of $x + y + z + w < 25$, where $x > -2, y > 1, z \geq 2, w \geq 0$.

Solution : Given $x + y + z + w < 25$

$$x + y + z + w + v = 25 \quad \dots\dots(i)$$

Let $x = -1 + t_1, y = 2 + t_2, z = 2 + t_3, w = t_4, v = 1 + t_5$ where ($t_1, t_2, t_3, t_4 \geq 0$)

Putting value of x, y, z, w, v in equation (i)

$$\Rightarrow t_1 + t_2 + t_3 + t_4 + t_5 = 21.$$

$$\text{Number of solutions} = {}^{25}C_4$$

Ans.

Illustration 37: Find the number of positive integral solutions of the inequation $x + y + z \geq 150$, where $0 < x \leq 60$, $0 < y \leq 60$, $0 < z \leq 60$.

Solution : Let $x = 60 - t_1$, $y = 60 - t_2$, $z = 60 - t_3$ (where $0 \leq t_1 \leq 59$, $0 \leq t_2 \leq 59$, $0 \leq t_3 \leq 59$)

Given $x + y + z \geq 150$

or $x + y + z - w = 150$ (where $0 \leq w \leq 147$)(i)

Putting values of x, y, z in equation (i)

$$60 - t_1 + 60 - t_2 + 60 - t_3 - w = 150$$

$$30 = t_1 + t_2 + t_3 + w$$

$$\text{Total solutions} = {}^{33}C_3$$

Ans.

Illustration 38: Find the number of positive integral solutions of $xy = 12$

Solution : $xy = 12$

$$xy = 2^2 \times 3^1$$

(i) 3 has 2 ways either 3 can go to x or y

(ii) 2^2 can be distributed between x & y as distributing 2 identical things between 2 persons

(where each person can get 0, 1 or 2 things). Let two person be ℓ_1 & ℓ_2

$$\Rightarrow \ell_1 + \ell_2 = 2$$

$$\Rightarrow {}^{2+1}C_1 = {}^3C_1 = 3$$

$$\text{So total ways} = 2 \times 3 = 6.$$

Alternatively :

$$xy = 12 = 2^2 \times 3^1$$

$$x = 2^{a_1} 3^{a_2} \quad 0 \leq a_1 \leq 2$$

$$0 \leq a_2 \leq 1$$

$$y = 2^{b_1} 3^{b_2} \quad 0 \leq b_1 \leq 2$$

$$0 \leq b_2 \leq 1$$

$$2^{a_1+b_1} 3^{a_2+b_2} = 2^2 3^1$$

$$\Rightarrow a_1 + b_1 = 2 \rightarrow {}^3C_1 \text{ ways}$$

$$a_2 + b_2 = 1 \rightarrow {}^2C_1 \text{ ways}$$

$$\text{Number of solutions} = {}^3C_1 \times {}^2C_1 = 3 \times 2 = 6$$

Ans.

Illustration 39 : Find the number of solutions of the equation $xyz = 360$ when (i) $x, y, z \in \mathbb{N}$ (ii) $x, y, z \in \mathbb{I}$

Solution : (i) $xyz = 360 = 2^3 \times 3^2 \times 5$ ($x, y, z \in \mathbb{N}$)

$$x = 2^{a_1} 3^{a_2} 5^{a_3} \quad (\text{where } 0 \leq a_1 \leq 3, 0 \leq a_2 \leq 2, 0 \leq a_3 \leq 1)$$

$$y = 2^{b_1} 3^{b_2} 5^{b_3} \quad (\text{where } 0 \leq b_1 \leq 3, 0 \leq b_2 \leq 2, 0 \leq b_3 \leq 1)$$

$$z = 2^{c_1} 3^{c_2} 5^{c_3} \quad (\text{where } 0 \leq c_1 \leq 3, 0 \leq c_2 \leq 2, 0 \leq c_3 \leq 1)$$

$$\Rightarrow 2^{a_1} 3^{a_2} 5^{a_3} \cdot 2^{b_1} 3^{b_2} 5^{b_3} \cdot 2^{c_1} 3^{c_2} 5^{c_3} = 2^3 \times 3^2 \times 5^1$$

$$\Rightarrow 2^{a_1+b_1+c_1} \cdot 3^{a_2+b_2+c_2} \cdot 5^{a_3+b_3+c_3} = 2^3 \times 3^2 \times 5^1$$

$$\Rightarrow a_1 + b_1 + c_1 = 3 \rightarrow {}^5C_2 = 10$$

$$a_2 + b_2 + c_2 = 2 \rightarrow {}^4C_2 = 6$$

$$a_3 + b_3 + c_3 = 1 \rightarrow {}^3C_2 = 3$$

$$\text{Total solutions} = 10 \times 6 \times 3 = 180.$$

- (ii) If $x, y, z \in I$ then, (a) all positive (b) 1 positive and 2 negative.

$$\text{Total number of ways} = 180 + {}^3C_2 \times 180 = 720$$

Ans.

Do yourself -10 :

- (i) In how many ways can 12 identical apples be distributed among 4 boys. (a) If each boy receives any number of apples. (b) If each boy receives atleast 2 apples.
- (ii) Find the number of non-negative integral solutions of the equation $x + y + z = 10$.
- (iii) Find the number of integral solutions of $x + y + z = 20$, if $x \geq -4$, $y \geq 1$, $z \geq 2$

12. DEARRANGEMENT :

There are n letters and n corresponding envelopes. The number of ways in which letters can be placed in the envelopes (one letter in each envelope) so that no letter is placed in correct envelope is

$$n! \left[1 - \frac{1}{1!} + \frac{1}{2!} - \dots + \frac{(-1)^n}{n!} \right]$$

Proof : n letters are denoted by $1, 2, 3, \dots, n$. Let A_i denote the set of distribution of letters in envelopes (one letter in each envelope) so that the i^{th} letter is placed in the corresponding envelope. Then, $n(A_i) = 1 \times (n-1)!$ [since the remaining $n-1$ letters can be placed in $n-1$ envelopes in $(n-1)!$ ways]

Then, $n(A_i \cap A_j)$ represents the number of ways where letters i and j can be placed in their corresponding envelopes. Then,

$$n(A_i \cap A_j) = 1 \times 1 \times (n-2)!$$

$$\text{Also } n(A_i \cap A_j \cap A_k) = 1 \times 1 \times 1 \times (n-3)!$$

Hence, the required number is

$$n(A_1' \cup A_2' \cup \dots \cup A_n') = n! - n(A_1 \cup A_2 \cup \dots \cup A_n)$$

$$= n! - \left[\sum n(A_i) - \sum n(A_i \cap A_j) + \sum n(A_i \cap A_j \cap A_k) - \dots + (-1)^n \sum n(A_i \cap A_2 \dots \cap A_n) \right]$$

$$= n! - [{}^nC_1(n-1)! - {}^nC_2(n-2)! + {}^nC_3(n-3)! + \dots + (-1)^{n-1} \times {}^nC_n 1]$$

$$= n! - \left[\frac{n!}{1!(n-1)!} (n-1)! - \frac{n!}{2!(n-2)!} (n-2)! + \dots + (-1)^{n-1} \right] = n! \left[1 - \frac{1}{1!} + \frac{1}{2!} - \dots + \frac{(-1)^n}{n!} \right]$$

Illustration 40: A person writes letters to six friends and addresses the corresponding envelopes. In how many ways can the letters be placed in the envelopes so that

- (i) all the letters are in the wrong envelopes.
- (ii) at least two of them are in the wrong envelopes.

Solution :

- (i) The number of ways in which all letters be placed in wrong envelopes

$$= 6! \left(1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \frac{1}{5!} + \frac{1}{6!} \right) = 720 \left(\frac{1}{2} - \frac{1}{6} + \frac{1}{24} - \frac{1}{120} + \frac{1}{720} \right)$$

$$= 360 - 120 + 30 - 6 + 1 = 265.$$

- (ii) The number of ways in which at least two of them in the wrong envelopes

$$= {}^6C_4 \cdot 2! \left(1 - \frac{1}{1!} + \frac{1}{2!} \right) + {}^6C_3 \cdot 3! \left(1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} \right) + {}^6C_2 \cdot 4! \left(1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} \right)$$

$$+ {}^6C_1 \cdot 5! \left(1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \frac{1}{5!} \right) + {}^6C_0 \cdot 6! \left(1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \frac{1}{5!} + \frac{1}{6!} \right)$$

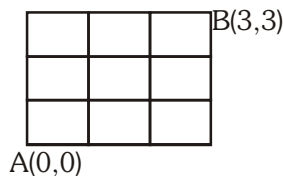
$$= 15 + 40 + 135 + 264 + 265 = 719.$$

Ans.**Do yourself - 11 :**

- (i) There are four balls of different colours and four boxes of colours same as those of the balls. Find the number of ways in which the balls, one in each box, could be placed in such a way that a ball does not go to box of its own colour.

Miscellaneous Illustrations :

Illustration 41: In how many ways can a person go from point A to point B if he can travel only to the right or upward along the lines (Grid Problem) ?

**Solution :**

To reach the point B from point A, a person has to travel along 3 horizontal and 3 vertical strips. Therefore, we have to arrange 3H and 3V in a row. Total number of ways =

$$\frac{6!}{3!3!} = 20 \text{ ways } \mathbf{Ans.}$$

Illustration 42: Find sum of all numbers formed using the digits 2,4,6,8 taken all at a time and no digit being repeated.

Solution :

All possible numbers = $4! = 24$

If 2 occupies the unit's place then total numbers = 6

Hence, 2 comes at unit's place 6 times.

Sum of all the digits occurring at unit's place

$$= 6 \times (2 + 4 + 6 + 8)$$

Same summation will occur for ten's, hundred's & thousand's place. Hence required sum

$$= 6 \times (2 + 4 + 6 + 8) \times (1 + 10 + 100 + 1000) = 133320$$

Ans.

Illustration 43: Find the sum of all the numbers greater than 1000 using the digits 0,1,2,2.

Solution :

(i) When 1 is at thousand's place, total numbers formed will be $= \frac{3!}{2!} = 3$

(ii) When 2 is at thousand's place, total numbers formed will be $= 3! = 6$

(iii) When 1 is at hundred's, ten's or unit's place then total numbers formed will be-
Thousand's place is fixed i.e. only the digit 2 will come here, remaining two places can be filled in $2!$ ways.
So total numbers $= 2!$

(iv) When 2 is at hundred's, ten's or unit's place then total numbers formed will be-
Thousand's place has 2 options and other two places can be filled in 2 ways.
So total numbers $= 2 \times 2 = 4$

Sum $= 10^3 (1 \times 3 + 2 \times 6) + 10^2 (1 \times 2 + 2 \times 4) + 10^1 (1 \times 2 + 2 \times 4) + (1 \times 2 + 2 \times 4)$
 $= 15 \times 10^3 + 10^3 + 10^2 + 10$
 $= 16110$

Ans.

Illustration 44 : Find the number of positive integral solutions of $x + y + z = 20$, if $x \neq y \neq z$.

Solution :

$x \geq 1$

$y = x + t_1 \quad t_1 \geq 1$
 $z = y + t_2 \quad t_2 \geq 1$
 $x + x + t_1 + x + t_1 + t_2 = 20$
 $3x + 2t_1 + t_2 = 20$

(i) $x = 1 \quad 2t_1 + t_2 = 17$
 $t_1 = 1, 2, \dots, 8 \Rightarrow 8 \text{ ways}$

(ii) $x = 2 \quad 2t_1 + t_2 = 14$
 $t_1 = 1, 2, \dots, 6 \Rightarrow 6 \text{ ways}$

(iii) $x = 3 \quad 2t_1 + t_2 = 11$
 $t_1 = 1, 2, \dots, 5 \Rightarrow 5 \text{ ways}$

(vi) $x = 4 \quad 2t_1 + t_2 = 8$
 $t_1 = 1, 2, 3 \Rightarrow 3 \text{ ways}$

(v) $x = 5 \quad 2t_1 + t_2 = 5$
 $t_1 = 1, 2 \Rightarrow 2 \text{ ways}$

Total $= 8 + 6 + 5 + 3 + 2 = 24$

But each solution can be arranged by $3!$ ways.

So total solutions $= 24 \times 3! = 144$.

Ans.

Illustration 45: A regular polygon of 15 sides is constructed. In how many ways can a triangle be formed using the vertices of the polygon such that no side of triangle is same as that of polygon?

Solution : Select one point out of 15 point, therefore total number of ways = ${}^{15}C_1$
Suppose we select point P_1 . Now we have to choose 2 more point which are not consecutive.

since we can not select P_2 & P_{15} .

Total points left are 12.

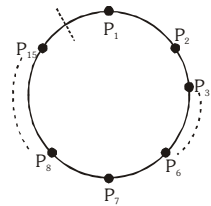
Now we have to select 2 points out of 12 points

which are not consecutive

$$\text{Total ways} = {}^{12-2+1}C_2 = {}^{11}C_2$$

Every select triangle will be repeated 3 times.

$$\text{So total number of ways} = \frac{{}^{15}C_1 \times {}^{11}C_2}{3} = 275$$



Alternative :

First of all let us cut the polygon between points P_1 & P_{15} . Now there are 15 points on a straight line and we have to select 3 points out of these, such that the selected points are not consecutive.

x O y O z O w

Here bubbles represents the selected points,

x represents the number of points before first selected point,

y represents the number of points between Ist & IInd selected point,

z represents the number of points between IInd & IIIrd selected point

and w represents the number of points after IIIrd selected point.

$$x + y + z + w = 15 - 3 = 12$$

$$\text{here } x \geq 0, y \geq 1, z \geq 1, w \geq 0$$

$$\text{Put } y = 1 + y' \text{ \& } z = 1 + z' \text{ (} y' \geq 0, z' \geq 0 \text{)}$$

$$\Rightarrow x + y' + z' + w = 10$$

$$\text{Total number of ways} = {}^{13}C_3$$

These selections include the cases when both the points P_1 & P_{15} are selected. We have to remove those cases. Here a represents number of points between P_1 & 3rd selected point & b represents number of points between 3rd selected point and P_{15}

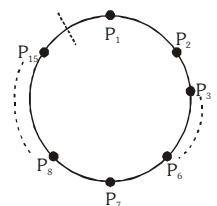
$$\Rightarrow a + b = 15 - 3 = 12 \quad (a \geq 1, b \geq 1)$$

$$\text{put } a = 1 + t_1 \text{ \& } b = 1 + t_2$$

$$t_1 + t_2 = 10$$

$$\text{Total number of ways} = {}^{11}C_1 = 11$$

$$\text{Therefore required number of ways} = {}^{13}C_3 - {}^{11}C_1 = 286 - 11 = 275$$



Ans.

Illustration 46: Find the number of ways in which three numbers can be selected from the set $\{5^1, 5^2, 5^3, \dots, 5^{11}\}$ so that they form a G.P.

Solution : Any three selected numbers which are in G.P. have their powers in A.P.

Set of powers is $= \{1, 2, \dots, 6, 7, \dots, 11\}$

By selecting any two numbers from $\{1, 3, 5, 7, 9, 11\}$, the middle number is automatically fixed.

Total number of ways $= {}^6C_2$

Now select any two numbers from $\{2, 4, 6, 8, 10\}$ and again middle number is automatically fixed. Total number of ways $= {}^5C_2$

$\therefore$ Total number of ways are $= {}^6C_2 + {}^5C_2 = 15 + 10 = 25$

Ans.

ANSWERS FOR DO YOURSELF

- | | | | | |
|---|------------------------------|--------------------------|-----------|-------------|
| 1 : (i) 7 | (ii) 3 | | | |
| 2 : (i) 0 | (ii) $r = 4$ | (iii) ${}^{50}C_4$ | (iv) 20 | (v) 120, 48 |
| 3 : (i) 10 | (ii) 450 | (iii) 840, 40 | | |
| 4 : (i) $\frac{16!}{(2!)^8 8!} \times 8!$ | (ii) 360 | (iii) ${}^nC_2 \cdot n!$ | | |
| 5 : (i) $5^n - 4^n - 4^n + 3^n$ | | | | |
| 6 : (i) 60, 6 th | (ii) 60 | | | |
| 7 : (i) 36 | (ii) $\frac{9!}{2} = 181440$ | (iii) 5400 | (iv) 2688 | |
| 8 : (i) $(p+1)^n - 1$ | (ii) $2^{10} - 1$ | | | |
| 9 : (i) 23 | (ii) 36 | | | |
| 10 : (i) (a) ${}^{15}C_3$ (b) 7C_3 | (ii) ${}^{12}C_2$ | (iii) ${}^{23}C_2$ | | |
| 11 : (i) 9 | | | | |

EXERCISE (O-1)

ONLY ONE CORRECT :

1. Number of natural numbers between 100 and 1000 such that at least one of their digits is 7, is
(A) 225 (B) 243 (C) 252 (D) none
2. Number of 4 digit numbers of the form $N = abcd$ which satisfy following three conditions :
(i) $4000 \leq N < 6000$ (ii) N is multiple of 5 (iii) $3 \leq b < c \leq 6$
is equal to
(A) 12 (B) 18 (C) 24 (D) 48
3. How many of the 900 three digit numbers have at least one even digit?
(A) 775 (B) 875 (C) 450 (D) 750
4. The number of different seven digit numbers that can be written using only three digits 1, 2 & 3 under the condition that the digit 2 occurs exactly twice in each number is
(A) 672 (B) 640 (C) 512 (D) none
5. Out of seven consonants and four vowels, the number of words of six letters, formed by taking four consonants and two vowels is (Assume that each ordered group of letter is a word):
(A) 210 (B) 462 (C) 151200 (D) 332640
6. Number of 5 digit numbers which are divisible by 5 and each number containing the digit 5, digits being all different is equal to $k(4!)$, the value of k is
(A) 84 (B) 168 (C) 188 (D) 208
7. The number of six digit numbers that can be formed from the digits 1, 2, 3, 4, 5, 6 & 7 so that digits do not repeat and the terminal digits are even is :
(A) 144 (B) 72 (C) 288 (D) 720
8. If the letters of the word "VARUN" are written in all possible ways and then are arranged as in a dictionary, then the rank of the word VARUN is :
(A) 98 (B) 99 (C) 100 (D) 101
9. A new flag is to be designed with six vertical strips using some or all of the colours yellow, green, blue and red. Then, the number of ways this can be done such that no two adjacent strips have the same colour is -
(A) 12×81 (B) 16×192 (C) 20×125 (D) 24×216
10. Consider the five points comprising of the vertices of a square and the intersection point of its diagonals. How many triangles can be formed using these points?
(A) 4 (B) 6 (C) 8 (D) 10
11. A 5 digit number divisible by 3 is to be formed using the numerals 0, 1, 2, 3, 4 & 5 without repetition. The total number of ways this can be done is :
(A) 3125 (B) 600 (C) 240 (D) 216

12. Number of permutations of 1, 2, 3, 4, 5, 6, 7, 8 and 9 taken all at a time, such that the digit
1 appearing somewhere to the left of 2
3 appearing to the left of 4 and
5 somewhere to the left of 6, is
(e.g. 815723946 would be one such permutation)
(A) $9 \cdot 7!$ (B) $8!$ (C) $5! \cdot 4!$ (D) $8! \cdot 4!$
13. The number of ways in which 5 different books can be distributed among 10 people if each person can get at most one book is :
(A) 252 (B) 10^5 (C) 5^{10} (D) ${}^{10}C_5 \cdot 5!$
14. 5 Indian & 5 American couples meet at a party & shake hands. If no wife shakes hands with her own husband & no Indian wife shakes hands with a male, then the number of hand shakes that takes place in the party is :
(A) 95 (B) 110 (C) 135 (D) 150
15. A student has to answer 10 out of 13 questions in an examination. The number of ways in which he can answer if he must answer atleast 3 of the first five questions is :
(A) 276 (B) 267 (C) 80 (D) 1200
16. The number of n digit numbers which consists of the digits 1 & 2 only if each digit is to be used atleast once, is equal to 510 then n is equal to
(A) 7 (B) 8 (C) 9 (D) 10
17. There are counters available in x different colours. The counters are all alike except for the colour. The total number of arrangements consisting of y counters, assuming sufficient number of counters of each colour, if no arrangement consists of all counters of the same colour is :
(A) $x^y - x$ (B) $x^y - y$ (C) $y^x - x$ (D) $y^x - y$
18. A question paper on mathematics consists of twelve questions divided into three parts A, B and C, each containing four questions. In how many ways can an examinee answer five questions, selecting atleast one from each part.
(A) 624 (B) 208 (C) 1248 (D) 2304
19. If m denotes the number of 5 digit numbers if each successive digits are in their descending order of magnitude and n is the corresponding figure, when the digits are in their ascending order of magnitude then $(m - n)$ has the value
(A) ${}^{10}C_4$ (B) 9C_5 (C) ${}^{10}C_3$ (D) 9C_3
20. A rack has 5 different pairs of shoes. The number of ways in which 4 shoes can be chosen from it, so that there will be no complete pair is :
(A) 1920 (B) 200 (C) 110 (D) 80
21. Number of ways in which 8 people can be arranged in a line if A and B must be next each other and C must be somewhere behind D, is equal to
(A) 10080 (B) 5040 (C) 5050 (D) 10100
22. The kindergarten teacher has 25 kids in her class. She takes 5 of them at a time, to zoological garden as often as she can, without taking the same 5 kids more than once. Then the number of visits, the teacher makes to the garden exceeds that of a kid by :
(A) ${}^{25}C_5 - {}^{24}C_5$ (B) ${}^{24}C_5$ (C) ${}^{24}C_4$ (D) none

23. Seven different coins are to be divided amongst three persons. If no two of the persons receive the same number of coins but each receives atleast one coin & none is left over, then the number of ways in which the division may be made is
 (A) 420 (B) 630 (C) 710 (D) none
24. The number of ways in which 8 distinguishable apples can be distributed among 3 boys such that every boy should get atleast 1 apple & atmost 4 apples is $K \cdot {}^7P_3$ where K has the value equal to
 (A) 14 (B) 66 (C) 44 (D) 22
25. An old man while dialing a 7 digit telephone number remembers that the first four digits consists of one 1's, one 2's and two 3's. He also remembers that the fifth digit is either a 4 or 5 while has no memorising of the sixth digit, he remembers that the seventh digit is 9 minus the sixth digit. Maximum number of distinct trials he has to try to make sure that he dials the correct telephone number, is
 (A) 360 (B) 240 (C) 216 (D) none
26. Number of ways in which 9 different toys be distributed among 4 children belonging to different age groups in such a way that distribution among the 3 elder children is even and the youngest one is to receive one toy more, is :
 (A) $\frac{(5!)^2}{8}$ (B) $\frac{9!}{2}$ (C) $\frac{9!}{3! (2!)^3}$ (D) none
27. Let P_n denotes the number of ways in which three people can be selected out of 'n' people sitting in a row, if no two of them are consecutive. If, $P_{n+1} - P_n = 15$ then the value of 'n' is :
 (A) 7 (B) 8 (C) 9 (D) 10
28. There are six periods in each working day of a school. Number of ways in which 5 subjects can be arranged if each subject is allotted at least one period and no period remains vacant is
 (A) 210 (B) 1800 (C) 360 (D) 3600
29. Number of ways in which 7 green bottles and 8 blue bottles can be arranged in a row if exactly 1 pair of green bottles is side by side, is (Assume all bottles to be alike except for the colour).
 (A) 84 (B) 360 (C) 504 (D) none
30. There are 10 red balls of different shades & 9 green balls of identical shades. Then the number of arranging them in a row so that no two green balls are together is
 (A) $(10!) \cdot {}^{11}P_9$ (B) $(10!) \cdot {}^{11}C_9$
 (C) $10!$ (D) $10! \cdot 9!$
31. A gentleman invites a party of $m + n$ ($m \neq n$) friends to a dinner & places m at one table T_1 and n at another table T_2 , the table being round. If not all people shall have the same neighbour in any two arrangement, then the number of ways in which he can arrange the guests, is
 (A) $\frac{(m+n)!}{4mn}$ (B) $\frac{1}{2} \frac{(m+n)!}{mn}$ (C) $2 \frac{(m+n)!}{mn}$ (D) none
32. A team of 8 students goes on an excursion, in two cars, of which one can seat 5 and the other only 4. If internal arrangement inside the car does not matter then the number of ways in which they can travel, is
 (A) 91 (B) 182 (C) 126 (D) 3920

33. One hundred management students who read at least one of the three business magazines are surveyed to study the readership pattern. It is found that 80 read Business India, 50 read Business world, and 30 read Business Today. Five students read all the three magazines. How many read exactly two magazines?
- (A) 50 (B) 10 (C) 95 (D) 65
34. Number of cyphers at the end of $^{2002}C_{1001}$ is
- (A) 0 (B) 1 (C) 2 (D) 200
35. Three vertices of a convex n sided polygon are selected. If the number of triangles that can be constructed such that none of the sides of the triangle is also the side of the polygon is 30, then the polygon is a
- (A) Heptagon (B) Octagon (C) Nonagon (D) Decagon
36. Number of 5 digit numbers divisible by 25 that can be formed using only the digits 1, 2, 3, 4, 5 & 0 taken five at a time is
- (A) 2 (B) 32 (C) 42 (D) 52
37. Let P_n denotes the number of ways of selecting 3 people out of 'n' sitting in a row, if no two of them are consecutive and Q_n is the corresponding figure when they are in a circle. If $P_n - Q_n = 6$, then 'n' is equal to :
- (A) 8 (B) 9 (C) 10 (D) 12
38. Let m denote the number of ways in which 4 different books are distributed among 10 persons, each receiving none or one only and let n denote the number of ways of distribution if the books are all alike. Then :
- (A) $m = 4n$ (B) $n = 4m$ (C) $m = 24n$ (D) none
39. Number of 7 digit numbers the sum of whose digits is 61 is :
- (A) 12 (B) 24 (C) 28 (D) none
40. There are 2 identical white balls, 3 identical red balls and 4 green balls of different shades. The number of ways in which they can be arranged in a row so that atleast one ball is separated from the balls of the same colour, is :
- (A) $6(7! - 4!)$ (B) $7(6! - 4!)$ (C) $8! - 5!$ (D) none
41. Number of ways in which four different toys and five indistinguishable marbles can be distributed between Amar, Akbar and Anthony, if each child receives atleast one toy and one marble, is
- (A) 42 (B) 100 (C) 150 (D) 216
42. In a unique hockey series between India & Pakistan, they decide to play on till a team wins 5 matches. The number of ways in which the series can be won by India, if no match ends in a draw is :
- (A) 126 (B) 252 (C) 225 (D) none
43. There are 100 different books in a shelf. Number of ways in which 3 books can be selected so that no two of which are neighbours is
- (A) $^{100}C_3 - 98$ (B) $^{97}C_3$ (C) $^{96}C_3$ (D) $^{98}C_3$

44. Number of positive integral solutions satisfying the equation $(x_1 + x_2 + x_3)(y_1 + y_2) = 77$, is
 (A) 150 (B) 270 (C) 420 (D) 1024
45. There are counters available in 3 different colours (atleast four of each colour). Counters are all alike except for the colour. If 'm' denotes the number of arrangements of four counters if no arrangement consists of counters of same colour and 'n' denotes the corresponding figure when every arrangement consists of counters of each colour, then :
 (A) $m = 2n$ (B) $6m = 13n$ (C) $3m = 5n$ (D) $5m = 3n$
46. There are 12 books on Algebra and Calculus in our library, the books of the same subject being different. If the number of selections each of which consists of 3 books on each topic is greatest then the number of books of Algebra and Calculus in the library are respectively:
 (A) 3 and 9 (B) 4 and 8 (C) 5 and 7 (D) 6 and 6
47. There are $(p + q)$ different books on different topics in Mathematics. ($p \neq q$)
 If L = The number of ways in which these books are distributed between two students X and Y such that X get p books and Y gets q books.
 M = The number of ways in which these books are distributed between two students X and Y such that one of them gets p books and another gets q books.
 N = The number of ways in which these books are divided into two groups of p books and q books then,
 (A) $L = M = N$ (B) $L = 2M = 2N$ (C) $2L = M = 2N$ (D) $L = M = 2N$
48. Number of ways in which 5 A's and 6 B's can be arranged in a row which reads the same backwards and forwards, is
 (A) 6 (B) 8 (C) 10 (D) 12
49. A person writes letters to his 5 friends and addresses the corresponding envelopes. Number of ways in which the letters can be placed in the envelope, so that atleast two of them are in the wrong envelopes, is,
 (A) 1 (B) 2 (C) 118 (D) 119

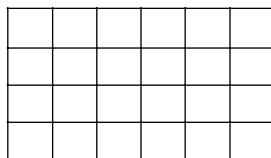
MATCH THE COLUMN :

- | 50. Column-I | Column-II |
|---|----------------|
| (A) Number of increasing permutations of m symbols are there from the n set numbers $\{a_1, a_2, \dots, a_n\}$ where the order among the numbers is given by $a_1 < a_2 < a_3 < \dots < a_{n-1} < a_n$ is | (P) n^m |
| (B) There are m men and n monkeys. Number of ways in which every monkey has a master, if a man can have any number of monkeys | (Q) ${}^m C_n$ |
| (C) Number of ways in which n red balls and $(m - 1)$ green balls can be arranged in a line, so that no two red balls are together, is (balls of the same colour are alike) | (R) ${}^n C_m$ |
| (D) Number of ways in which ' m ' different toys can be distributed in ' n ' children if every child may receive any number of toys, is | (S) m^n |

EXERCISE (O-2)

ONLY ONE CORRECT :

- In a certain strange language, words are written with letters from the following six-letter alphabet : A, G, K, N, R, U. Each word consists of six letters and none of the letters repeat. Each combination of these six letters is a word in this language. The word "KANGUR" remains in the dictionary at,
(A) 248th (B) 247th (C) 246th (D) 253rd
- All possible three digits even numbers which can be formed with the condition that if 5 is one of the digit, then 7 is the next digit is :
(A) 5 (B) 325 (C) 345 (D) 365
- Number of 3 digit numbers in which the digit at hundredth's place is greater than the other two digit is
(A) 285 (B) 281 (C) 240 (D) 204
- The number of three digit numbers having only two consecutive digits identical is :
(A) 153 (B) 162 (C) 180 (D) 161
- A committee of 5 is to be chosen from a group of 9 people. Number of ways in which it can be formed if two particular persons either serve together or not at all and two other particular persons refuse to serve with each other, is
(A) 41 (B) 36 (C) 47 (D) 76
- There are m points on a straight line AB & n points on the line AC none of them being the point A. Triangles are formed with these points as vertices, when
(i) A is excluded (ii) A is included. The ratio of number of triangles in the two cases is:
(A) $\frac{m+n-2}{m+n}$ (B) $\frac{m+n-2}{m+n-1}$ (C) $\frac{m+n-2}{m+n+2}$ (D) $\frac{m(n-1)}{(m+1)(n+1)}$
- There are 10 straight lines in a plane, such that no 3 are concurrent and no 2 are parallel to each other. If points of intersection of above lines are joined, then maximum number of lines thus formed are (including old lines) -
(A) 610 (B) 620 (C) 630 (D) 640
- Number of rectangles in the grid shown which are not squares is



- (A) 160 (B) 162 (C) 170 (D) 185
- Six people are going to sit in a row on a bench. A and B are adjacent, C does not want to sit adjacent to D. E and F can sit anywhere. Number of ways in which these six people can be seated, is
(A) 200 (B) 144 (C) 120 (D) 56
- Given 11 points, of which 5 lie on one circle, other than these 5, no 4 lie on one circle. Then the maximum number of circles that can be drawn so that each contains atleast three of the given points is :
(A) 216 (B) 156 (C) 172 (D) none

11. Six married couple are sitting in a room. Find the number of ways in which 4 people can be selected so that
- (a) they do not form a couple (b) they form exactly one couple
(c) they form at least one couple (d) they form atmost one couple
12. The number of ways of choosing a committee of 2 women & 3 men from 5 women & 6 men, if Mr. A refuses to serve on the committee if Mr. B is a member & Mr. B can only serve, if Miss C is the member of the committee, is
- (A) 60 (B) 84 (C) 124 (D) none
13. Product of all the even divisors of $N = 1000$, is
- (A) $32 \cdot 10^2$ (B) $64 \cdot 2^{14}$ (C) $64 \cdot 10^{18}$ (D) $128 \cdot 10^6$
14. Two classrooms A and B having capacity of 25 and $(n-25)$ seats respectively. A_n denotes the number of possible seating arrangements of room 'A', when 'n' students are to be seated in these rooms, starting from room 'A' which is to be filled up full to its capacity. If $A_n - A_{n-1} = 25! \cdot ({}^{49}C_{25})$ then 'n' equals -
- (A) 50 (B) 48 (C) 49 (D) 51
15. A guardian with 6 wards wishes everyone of them to study either Law or Medicine or Engineering. Number of ways in which he can make up his mind with regard to the education of his wards if every one of them be fit for any of the branches to study, and atleast one child is to be sent in each discipline is :
- (A) 120 (B) 216 (C) 729 (D) 540
16. Consider the word $W = \text{MISSISSIPPI}$
- (a) If N denotes the number of different selections of 5 letters from the word $W = \text{MISSISSIPPI}$ then N belongs to the set
- (A) $\{15, 16, 17, 18, 19\}$ (B) $\{20, 21, 22, 23, 24\}$
(C) $\{25, 26, 27, 28, 29\}$ (D) $\{30, 31, 32, 33, 34\}$
- (b) Number of ways in which the letters of the word W can be arranged if atleast one vowel is separated from rest of the vowels
- (A) $\frac{8! \cdot 16!}{4! \cdot 4! \cdot 2!}$ (B) $\frac{8! \cdot 16!}{4 \cdot 4! \cdot 2!}$ (C) $\frac{8! \cdot 16!}{4! \cdot 2!}$ (D) $\frac{8!}{4! \cdot 2!} \cdot \frac{16!}{4!}$
- (c) If the number of arrangements of the letters of the word W if all the S's and P's are separated is
- (K) $\left(\frac{10!}{4! \cdot 4!} \right)$, then K equals -
- (A) $\frac{6}{5}$ (B) 1 (C) $\frac{4}{3}$ (D) $\frac{3}{2}$

Paragraph for Question Nos. 17 to 19

16 players $P_1, P_2, P_3, \dots, P_{16}$ take part in a tennis tournament. Lower suffix player is better than any higher suffix player. These players are to be divided into 4 groups each comprising of 4 players and the best from each group is selected for semifinals.

17. Number of ways in which 16 players can be divided into four equal groups, is
- (A) $\frac{35}{27} \prod_{r=1}^8 (2r-1)$ (B) $\frac{35}{24} \prod_{r=1}^8 (2r-1)$ (C) $\frac{35}{52} \prod_{r=1}^8 (2r-1)$ (D) $\frac{35}{6} \prod_{r=1}^8 (2r-1)$

18. Number of ways in which they can be divided into 4 equal groups if the players P_1, P_2, P_3 and P_4 are in different groups, is :

(A) $\frac{(11)!}{36}$ (B) $\frac{(11)!}{72}$ (C) $\frac{(11)!}{108}$ (D) $\frac{(11)!}{216}$

19. Number of ways in which these 16 players can be divided into four equal groups, such that when the best player is selected from each group, P_6 is one among them, is $(k) \frac{12!}{(4!)^3}$. The value of k is :

(A) 36 (B) 24 (C) 18 (D) 20

MORE THAN ONE ARE CORRECT :

20. Lines $y = x + i$ & $y = -x + j$ are drawn in $x - y$ plane such that $i \in \{1, 2, 3, 4\}$ & $j \in \{1, 2, 3, 4, 5, 6\}$. If m represents the total number of squares formed by the lines and n represents the total number of triangles formed by the given lines & x -axis, then correct option/s is/are-

(A) $m + n = 50$ (B) $m - n = 2$ (C) $m + n = 48$ (D) $m - n = 4$

21. The combinatorial coefficient $C(n, r)$ is equal to

- (A) number of possible subsets of r members from a set of n distinct members.
(B) number of possible binary messages of length n with exactly r 1's.
(C) number of non decreasing 2-D paths from the lattice point $(0, 0)$ to (r, n) .
(D) number of ways of selecting r things out of n different things when a particular thing is always included plus the number of ways of selecting ' r ' things out of n , when a particular thing is always excluded.

22. There are 10 questions, each question is either True or False. Number of different sequences of incorrect answers is also equal to

- (A) Number of ways in which a normal coin tossed 10 times would fall in a definite order if both Heads and Tails are present.
(B) Number of ways in which a multiple choice question containing 10 alternatives with one or more than one correct alternatives, can be answered.
(C) Number of ways in which it is possible to draw a sum of money with 10 coins of different denominations taken some or all at a time.
(D) Number of different selections of 10 indistinguishable things taken some or all at a time.

23. The maximum number of permutations of $2n$ letters in which there are only a's & b's, taken all at a time is given by :

(A) ${}^{2n}C_n$ (B) $\frac{2}{1} \cdot \frac{6}{2} \cdot \frac{10}{3} \dots \frac{4n-6}{n-1} \cdot \frac{4n-2}{n}$
(C) $\frac{n+1}{1} \cdot \frac{n+2}{2} \cdot \frac{n+3}{3} \cdot \frac{n+4}{4} \dots \frac{2n-1}{n-1} \cdot \frac{2n}{n}$ (D) $\frac{2^n \cdot [1 \cdot 3 \cdot 5 \dots (2n-3)(2n-1)]}{n!}$

24. Number of ways in which 3 numbers in A.P. can be selected from $1, 2, 3, \dots, n$ is :

(A) $\left(\frac{n-1}{2}\right)^2$ if n is even (B) $\frac{n(n-2)}{4}$ if n is odd
(C) $\frac{(n-1)^2}{4}$ if n is odd (D) $\frac{n(n-2)}{4}$ if n is even

25. The combinatorial coefficient ${}^{n-1}C_p$ denotes

- (A) the number of ways in which n things of which p are alike and rest different can be arranged in a circle.
(B) the number of ways in which p different things can be selected out of n different thing if a particular thing is always excluded.

- (C) number of ways in which n alike balls can be distributed in p different boxes so that no box remains empty and each box can hold any number of balls.
- (D) the number of ways in which $(n - 2)$ white balls and p black balls can be arranged in a line if black balls are separated, balls are all alike except for the colour.
26. Which of the following statements are correct?
- (A) Number of words that can be formed with 6 only of the letters of the word "CENTRIFUGAL" if each word must contain all the vowels is $3 \cdot 7!$
- (B) There are 15 balls of which some are white and the rest black. If the number of ways in which the balls can be arranged in a row, is maximum then the number of white balls must be equal to 7 or 8. Assume balls of the same colour to be alike.
- (C) There are 12 things, 4 alike of one kind, 5 alike and of another kind and the rest are all different. The total number of combinations is 240.
- (D) Number of selections that can be made of 6 letters from the word "COMMITTEE" is 35.
27. Number of ways in which the letters of the word 'B U L B U L' can be arranged in a line in a definite order is also equal to the
- (A) number of ways in which 2 alike Apples and 4 alike Mangoes can be distributed in 3 children so that each child receives any number of fruits.
- (B) Number of ways in which 6 different books can be tied up into 3 bundles, if each bundle is to have equal number of books.
- (C) coefficient of $x^2y^2z^2$ in the expansion of $(x + y + z)^6$.
- (D) number of ways in which 6 different prizes can be distributed equally in three children.

MATCH THE COLUMN:

28.	Column-I	Column-II
(A)	Four different movies are running in a town. Ten students go to watch these four movies. The number of ways in which every movie is watched by atleast one student, is (Assume each way differs only by number of students watching a movie)	(P) 11 (Q) 36
(B)	Consider 8 vertices of a regular octagon and its centre. If T denotes the number of triangles and S denotes the number of straight lines that can be formed with these 9 points then the value of $(T - S)$ equals	
(C)	In an examination, 5 children were found to have their mobiles in their pocket. The Invigilator fired them and took their mobiles in his possession. Towards the end of the test, Invigilator randomly returned their mobiles. The number of ways in which at most two children did not get their own mobiles is	(R) 52 (S) 60
(D)	The product of the digits of 3214 is 24. The number of 4 digit natural numbers such that the product of their digits is 12, is	
(E)	The number of ways in which a mixed double tennis game can be arranged from amongst 5 married couple if no husband & wife plays in the same game, is	(T) 84

EXERCISE (S-1)

1. Four visitors A, B, C & D arrive at a town which has 5 hotels. In how many ways can they disperse themselves among 5 hotels, if 4 hotels are used to accommodate them.
2. There are 6 roads between A & B and 4 roads between B & C.
 - (i) In how many ways can one drive from A to C by way of B ?
 - (ii) In how many ways can one drive from A to C and back to A, passing through B on both trips ?
 - (iii) In how many ways can one drive the circular trip described in (ii) without using the same road more than once.
3.
 - (i) Find the number of four letter word that can be formed from the letters of the word HISTORY. (each letter to be used atmost once)
 - (ii) How many of them contain only consonants?
 - (iii) How many of them begin & end in a consonant?
 - (iv) How many of them begin with a vowel?
 - (v) How many contain the letters Y?
 - (vi) How many begin with T & end in a vowel?
 - (vii) How many begin with T & also contain S?
 - (viii) How many contain both vowels?
4. If repetitions are not permitted
 - (i) How many 3 digit numbers can be formed from the six digits 2, 3, 5, 6, 7 & 9 ?
 - (ii) How many of these are less than 400 ?
 - (iii) How many are even ?
 - (iv) How many are odd ?
 - (v) How many are multiples of 5 ?
5. How many two digit numbers are there in which the tens digit and the units digit are different and odd ?
6. Every telephone number consists of 7 digits. How many telephone numbers are there which do not include any other digits but 2, 3, 5 & 7 ?
7.
 - (a) In how many ways can four passengers be accommodated in three railway carriages, if each carriage can accommodate any number of passengers.
 - (b) In how many ways four persons can be accommodated in 3 different chairs if each person can occupy only one chair.
8. How many odd numbers of five distinct digits can be formed with the digits 0,1,2,3,4 ?
9. Number of ways in which 7 different colours in a rainbow can be arranged if green is always in the middle.
10. Find the number of ways in which the letters of the word "MIRACLE" can be arranged if vowels always occupy the odd places.
11. A letter lock consists of three rings each marked with 10 different letters. Find the number of ways in which it is possible to make an unsuccessful attempts to open the lock.

12. Find the number of ways in which letters of the word VALEDICTORY be arranged so that the vowels may never be separated.
13. (i) Prove that : ${}^nP_r = {}^{n-1}P_r + r \cdot {}^{n-1}P_{r-1}$
(ii) If ${}^{20}C_{r+2} = {}^{20}C_{2r-3}$ find ${}^{12}C_r$
(iii) Prove that ${}^{n-1}C_3 + {}^{n-1}C_4 > {}^nC_3$ if $n > 7$.
(iv) Find r if ${}^{15}C_{3r} = {}^{15}C_{r+3}$
14. There are 2 women participating in a chess tournament. Every participant played 2 games with the other participants. The number of games that the men played between themselves exceeded by 66 as compared to the number of games that the men played with the women. Find the number of participants & the total numbers of games played in the tournament.
15. 5 boys & 4 girls sit in a straight line. Find the number of ways in which they can be seated if 2 girls are together & the other 2 are also together but separate from the first 2.
16. An examination paper consists of 12 questions divided into parts A & B.
Part-A contains 7 questions & Part-B contains 5 questions. A candidate is required to attempt 8 questions selecting atleast 3 from each part. In how many maximum ways can the candidate select the questions ?
17. In how many ways can a team of 6 horses be selected out of a stud of 16, so that there shall always be 3 out of A B C A' B' C', but never A A', B B' or C C' together.
18. During a draw of lottery, tickets bearing numbers 1, 2, 3,....., 40, 6 tickets are drawn out & then arranged in the descending order of their numbers. In how many ways, it is possible to have 4th ticket bearing number 25.
19. Find the number of distinct natural numbers upto a maximum of 4 digits and divisible by 5, which can be formed with the digits 0, 1, 2, 3, 4, 5, 6, 7, 8, 9 each digit not occurring more than once in each number.
20. In how many ways can you divide a pack of 52 cards equally among 4 players. In how many ways the cards can be divided in 4 sets, 3 of them having 17 cards each & the 4th with 1 card.
21. A firm of Chartered Accountants in Bombay has to send 10 clerks to 5 different companies, two clerks in each. Two of the companies are in Bombay and the others are outside. Two of the clerks prefer to work in Bombay while three others prefer to work outside. In how many ways can the assignment be made if the preferences are to be satisfied.
22. In a certain algebraical exercise book there are 4 examples on arithmetical progressions, 5 examples on permutation-combination and 6 examples on binomial theorem. Number of ways a teacher can select for his pupils atleast one but not more than 2 examples from each of these sets, is _____.
23. Find the number of ways in which two squares can be selected from an 8 by 8 chess board of size 1×1 so that they are not in the same row and in the same column.
24. There are 10 seats in a double decker bus, 6 in the lower deck and 4 on the upper deck. Ten passengers board the bus, of them 3 refuse to go to the upper deck and 2 insist on going up. The number of ways in which the passengers can be accommodated is _____. (Assume all seats to be duly numbered)
25. Find the number of permutations of the word "AUROBIND" in which vowels appear in an alphabetical order.

26. In how many different ways a grandfather along with two of his grandsons and four grand daughters can be seated in a line for a photograph so that he is always in the middle and the two grandsons are never adjacent to each other.
27. Define a 'good word' as a sequence of letters that consists only of the letters A, B and C and in which A never immediately followed by B, B is never immediately followed by C, and C is never immediately followed by A. If the number of n-letter good words are 384, find the value of n.
28. In how many other ways can the letters of the word **MULTIPLE** be arranged;
- without changing the order of the vowels
 - keeping the position of each vowel fixed &
 - without changing the relative order/position of vowels & consonants.
29. Each of 3 committees has 1 vacancy which is to be filled from a group of 6 people. Find the number of ways the 3 vacancies can be filled if ;
- Each person can serve on atmost 1 committee.
 - There is no restriction on the number of committees on which a person can serve.
 - Each person can serve on atmost 2 committees.
30. How many 6 digits odd numbers greater than 60,0000 can be formed from the digits 5, 6, 7, 8, 9, 0 if
- repetitions are not allowed
 - repetitions are allowed.
31. Find the number of ways in which the letters of the word 'KUTKUT' can be arranged so that no two alike letters are together.
32. If as many more words as possible be formed out of the letters of the word "DOGMATIC" then find the number of words in which the relative order of vowels and consonants remain unchanged.
33. Find the number of ways in which 3 distinct numbers can be selected from the set $\{3^1, 3^2, 3^3, \dots, 3^{100}, 3^{101}\}$ so that they form a G.P.

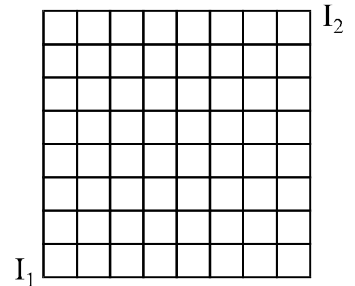
Paragraph for Question 34 & 35

Consider the number $N = 2910600$.

On the basis of above information, answer the following questions :

34. Total number of divisors of N, which are divisible by 15 but not by 36 are-
- (A) 92 (B) 94 (C) 96 (D) 98
35. Total number of ways, in which the given number can be split into two factors such that their highest common factor is a prime number is equal to-
- (A) 16 (B) 32 (C) 48 (D) 64
36. Determine the number of paths from the origin to the point (9, 9) in the cartesian plane which never pass through (5, 5) in paths consisting only of steps going 1 unit North and 1 unit East.
37. There are 20 books on Algebra & Calculus in our library. Prove that the greatest number of selections each of which consists of 5 books on each topic is possible only when there are 10 books on each topic in the library.

38. (a) How many divisors are there of the number $x = 21600$. Find also the sum of these divisors.
 (b) In how many ways the number 7056 can be resolved as a product of 2 factors.
 (c) Find the number of ways in which the number 300300 can be split into 2 factors which are relatively prime.
 (d) Find the number of positive integers that are divisors of atleast one of the numbers 10^{10} ; 15^7 ; 18^{11} .
39. There are 10 different books in a shelf. Find the number of ways in which 3 books can be selected so that exactly two of them are consecutive.
40. On the normal chess board as shown, I_1 & I_2 are two insects which starts moving towards each other. Each insect moving with the same constant speed. Insect I_1 can move only to the right or upward along the lines while the insect I_2 can move only to the left or downward along the lines of the chess board. Find the total number of ways the two insects can meet at same point during their trip.
41. A committee of 10 members is to be formed with members chosen from the faculties of Arts, Economics, Education, Engineering, Medicine and Science. Number of possible ways in which the faculties representation be distributed on this committee, is _____.
 (Assume every department contains more than 10 members).
42. If x_1, x_2, x_3 are the whole numbers and gives remainders 0, 1, 2 respectively, when divided by 3 then total number of different solutions of the equation $x_1 + x_2 + x_3 = 33$ are k, then $\frac{k}{11}$ is equal to

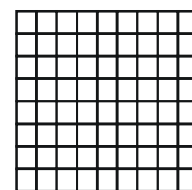


EXERCISE (S-2)

1. The straight lines l_1, l_2 & l_3 are parallel & lie in the same plane. A total of m points are taken on the line l_1 , n points on l_2 & k points on l_3 . How many maximum number of triangles are there whose vertices are at these points?
2. (a) How many five digits numbers divisible by 3 can be formed using the digits 0, 1, 2, 3, 4, 7 and 8 if each digit is to be used atmost once.
 (b) Find the number of 4 digit positive integers if the product of their digits is divisible by 3.
3. Find the number of words each consisting of 3 consonants & 3 vowels that can be formed from the letters of the word “Circumference”. In how many of these c’s will be together.
4. Find the number of three elements sets of positive integers $\{a, b, c\}$ such that $a \times b \times c = 2310$.

Paragraph for Question 5 & 6

If 10 vertical equispaced (1 cm) lines and 9 horizontal equispaced lines (1 cm) are drawn in a plane as shown in the given figure.



On the basis of above information, answer the following questions :

5. Total number of rectangles with one side odd & one side even are given by-
 (A) 600 (B) 700 (C) 800 (D) 900
6. If squares of odd side length are selected from the above grid, then sum of their areas is equal to-
 (A) $\sum_{r=1}^4 (11-2r)(10-2r)(2r-1)^2 \text{ cm}^2$ (B) $\sum_{r=1}^8 (9-2r)(7-2r)(2r+1)^2 \text{ cm}^2$
 (C) $\sum_{r=1}^8 (11-2r)(9-2r)(2r+1)^2 \text{ cm}^2$ (D) $\sum_{r=1}^5 (11+2r)(9+2r)(2r-1)^2 \text{ cm}^2$
7. How many 4 digit numbers are there which contains not more than 2 different digits?

Instruction for question nos. 8 to 10 :

2 American men; 2 British men; 2 Chinese men and one each of Dutch, Egyptian, French and German persons are to be seated for a round table conference.

8. If the number of ways in which they can be seated if exactly two pairs of persons of same nationality are together is $p(6!)$, then find p .
9. If the number of ways in which only American pair is adjacent is equal to $q(6!)$, then find q .
10. If the number of ways in which no two people of the same nationality are together given by $r(6!)$, find r .
11. For each positive integer k , let S_k denote the increasing arithmetic sequence of integers whose first term is 1 and whose common difference is k . For example, S_3 is the sequence 1, 4, 7, 10..... Find the number of values of k for which S_k contain the term 361.
12. A shop sells 6 different flavours of ice-cream. In how many ways can a customer choose 4 ice-cream cones if
 (i) they are all of different flavours
 (ii) they are non necessarily of different flavours
 (iii) they contain only 3 different flavours
 (iv) they contain only 2 or 3 different flavours?
13. How many different ways can 15 Candy bars be distributed between Ram, Shyam, Ghanshyam and Balram, if Ram can not have more than 5 candy bars and Shyam must have at least two. Assume all Candy bars to be alike.
14. Find the sum of all numbers greater than 10000 formed by using the digits 0, 1, 2, 4, 5 no digit being repeated in any number.
15. There are 3 cars of different make available to transport 3 girls and 5 boys on a field trip. Each car can hold up to 3 children. Find
 (a) the number of ways in which they can be accommodated.
 (b) the numbers of ways in which they can be accommodated if 2 or 3 girls are assigned to one of the cars.
- In both the cases internal arrangement of children inside the car is considered to be immaterial.

EXERCISE (JM)

- From 6 different novels and 3 different dictionaries, 4 novels and 1 dictionary are to be selected and arranged in a row on a shelf so that the dictionary is always in the middle. Then the number of such arrangements is - [AIEEE 2009]
 (1) At least 750 but less than 1000 (2) At least 1000
 (3) Less than 500 (4) At least 500 but less than 750
- There are two urns. Urn A has 3 distinct red balls and urn B has 9 distinct blue balls. From each urn two balls are taken out at random and then transferred to the other. The number of ways in which this can be done is - [AIEEE-2010]
 (1) 3 (2) 36 (3) 66 (4) 108
- Statement - 1 :** The number of ways of distributing 10 identical balls in 4 distinct boxes such that no box is empty is 9C_3 . [AIEEE-2011]
Statement - 2 : The number of ways of choosing any 3 places from 9 different places is 9C_3 .
 (1) Statement-1 is true, Statement-2 is false.
 (2) Statement-1 is false, Statement-2 is true
 (3) Statement-1 is true, Statement-2 is true; Statement-2 is a correct explanation for Statement-1
 (4) Statement-1 is true, Statement-2 is true; Statement-2 is **not** a correct explanation for Statement-1.
- There are 10 points in a plane, out of these 6 are collinear. If N is the number of triangles formed by joining these points, then : [AIEEE-2011]
 (1) $N > 190$ (2) $N \leq 100$ (3) $100 < N \leq 140$ (4) $140 < N \leq 190$
- Assuming the balls to be identical except for difference in colours, the number of ways in which one or more balls can be selected from 10 white, 9 green and 7 black balls is - [AIEEE-2012]
 (1) 879 (2) 880 (3) 629 (4) 630
- Let A and B be two sets containing 2 elements and 4 elements respectively. The number of subsets of $A \times B$ having 3 or more elements is [JEE (Main)-2013]
 (1) 256 (2) 220 (3) 219 (4) 211
- Let T_n be the number of all possible triangles formed by joining vertices of an n-sided regular polygon. If $T_{n+1} - T_n = 10$, then the value of n is : [JEE (Main)-2013]
 (1) 7 (2) 5 (3) 10 (4) 8
- The number of points, having both co-ordinates as integers, that lie in the interior of the triangle with vertices (0, 0), (0, 41) and (41, 0) is : [JEE (Main)-2015]
 (1) 820 (2) 780 (3) 901 (4) 861
- Let A and B be two sets containing four and two elements respectively. Then the number of subsets of the set $A \times B$, each having at least three elements is : [JEE (Main)-2015]
 (1) 275 (2) 510 (3) 219 (4) 256
- The number of integers greater than 6000 that can be formed, using the digits 3, 5, 6, 7 and 8 without repetition, is : [JEE (Main)-2015]
 (1) 120 (2) 72 (3) 216 (4) 192
- If all the words (with or without meaning) having five letters, formed using the letters of the word SMALL and arranged as in a dictionary; then the position of the word SMALL is : [JEE (Main)-2016]
 (1) 58th (2) 46th (3) 59th (4) 52nd

12. A man X has 7 friends, 4 of them are ladies and 3 are men. His wife Y also has 7 friends, 3 of them are ladies and 4 are men. Assume X and Y have no common friends. Then the total number of ways in which X and Y together can throw a party inviting 3 ladies and 3 men, so that 3 friends of each of X and Y are in this party, is : [JEE (Main)-2017]
 (1) 484 (2) 485 (3) 468 (4) 469
13. From 6 different novels and 3 different dictionaries, 4 novels and 1 dictionary are to be selected and arranged in a row on a shelf so that the dictionary is always in the middle. The number of such arrangements is- [JEE(Main)-2018]
 (1) less than 500 (2) at least 500 but less than 750
 (3) at least 750 but less than 1000 (4) at least 1000

EXERCISE (JA)

1. The number of seven digit integers, with sum of the digits equal to 10 and formed by using the digits 1, 2 and 3 only, is
 (A) 55 (B) 66 (C) 77 (D) 88 [JEE 2009, 3]
2. Let $S = \{1, 2, 3, 4\}$. The total number of unordered pairs of disjoint subsets of S is equal to - [JEE 10, 5M, -2M]
 (A) 25 (B) 34 (C) 42 (D) 41
3. The total number of ways in which 5 balls of different colours can be distributed among 3 persons so that each person gets at least one ball is - [JEE 2012, 3M, -1M]
 (A) 75 (B) 150 (C) 210 (D) 243

Paragraph for Question 4 and 5 :

Let a_n denotes the number of all n-digit positive integers formed by the digits 0, 1 or both such that no consecutive digits in them are 0. Let b_n = the number of such n-digit integers ending with digit 1 and c_n = the number of such n-digit integers ending with digit 0.

4. The value of b_6 is [JEE 2012, 3M, -1M]
 (A) 7 (B) 8 (C) 9 (D) 11
5. Which of the following is correct ? [JEE 2012, 3M, -1M]
 (A) $a_{17} = a_{16} + a_{15}$ (B) $c_{17} \neq c_{16} + c_{15}$ (C) $b_{17} \neq b_{16} + c_{16}$ (D) $a_{17} = c_{17} + b_{16}$
6. Let $n_1 < n_2 < n_3 < n_4 < n_5$ be positive integers such that $n_1 + n_2 + n_3 + n_4 + n_5 = 20$. The number of such distinct arrangements $(n_1, n_2, n_3, n_4, n_5)$ is [JEE(Advanced)-2014, 3]
7. Let $n \geq 2$ be an integer. Take n distinct points on a circle and join each pair of points by a line segment. Colour the line segment joining every pair of adjacent points by blue and the rest by red. If the number of red and blue line segments are equal, then the value of n is [JEE(Advanced)-2014, 3]
8. Six cards and six envelopes are numbered 1, 2, 3, 4, 5, 6 and cards are to be placed in envelopes so that each envelope contains exactly one card and no card is placed in the envelope bearing the same number and moreover the card numbered 1 is always placed in envelope numbered 2. Then the number of ways it can be done is - [JEE(Advanced)-2014, 3(-1)]
 (A) 264 (B) 265 (C) 53 (D) 67

9. Let n be the number of ways in which 5 boys and 5 girls can stand in a queue in such a way that all the girls stand consecutively in the queue. Let m be the number of ways in which 5 boys and 5 girls can stand in a queue in such a way that exactly four girls stand consecutively in the queue. Then the value of $\frac{m}{n}$ is [JEE (Advanced) 2015, 4M, -0M]
10. A debate club consists of 6 girls and 4 boys. A team of 4 members is to be selected from this club including the selection of a captain (from among these 4 members) for the team. If the team has to include at most one boy, then the number of ways of selecting the team is [JEE(Advanced)-2016, 3(-1)]
- (A) 380 (B) 320 (C) 260 (D) 95
11. Words of length 10 are formed using the letters A, B, C, D, E, F, G, H, I, J. Let x be the number of such words where no letter is repeated; and let y be the number of such words where exactly one letter is repeated twice and no other letter is repeated. Then $\frac{y}{9x} =$ [JEE(Advanced)-2017, 3]
12. Let $S = \{1, 2, 3, \dots, 9\}$. For $k = 1, 2, \dots, 5$, let N_k be the number of subsets of S , each containing five elements out of which exactly k are odd. Then $N_1 + N_2 + N_3 + N_4 + N_5 =$ [JEE(Advanced)-2017, 3(-1)]
- (A) 125 (B) 252 (C) 210 (D) 126
13. The number of 5 digit numbers which are divisible by 4, with digits from the set $\{1, 2, 3, 4, 5\}$ and the repetition of digits is allowed, is [JEE(Advanced)-2018, 3(0)]
14. In a high school, a committee has to be formed from a group of 6 boys $M_1, M_2, M_3, M_4, M_5, M_6$ and 5 girls G_1, G_2, G_3, G_4, G_5 .
- (i) Let α_1 be the total number of ways in which the committee can be formed such that the committee has 5 members, having exactly 3 boys and 2 girls.
- (ii) Let α_2 be the total number of ways in which the committee can be formed such that the committee has at least 2 members, and having an equal number of boys and girls.
- (iii) Let α_3 be the total number of ways in which the committee can be formed such that the committee has 5 members, at least 2 of them being girls.
- (iv) Let α_4 be the total number of ways in which the committee can be formed such that the committee has 4 members, having at least 2 girls and such that both M_1 and G_1 are NOT in the committee together.

LIST-I

- P. The value of α_1 is
 Q. The value of α_2 is
 R. The value of α_3 is
 S. The value of α_4 is

LIST-II

1. 136
 2. 189
 3. 192
 4. 200
 5. 381
 6. 461

The correct option is :-

- (A) $P \rightarrow 4; Q \rightarrow 6; R \rightarrow 2; S \rightarrow 1$ (B) $P \rightarrow 1; Q \rightarrow 4; R \rightarrow 2; S \rightarrow 3$
 (C) $P \rightarrow 4; Q \rightarrow 6; R \rightarrow 5; S \rightarrow 2$ (D) $P \rightarrow 4; Q \rightarrow 2; R \rightarrow 3; S \rightarrow 1$

[JEE(Advanced)-2018, 3(-1)]

ANSWER KEY

EXERCISE (O-1)

- | | | | | | | | |
|-------|--------------------------------|-------|-------|-------|-------|-------|-------|
| 1. C | 2. C | 3. A | 4. A | 5. C | 6. B | 7. D | 8. C |
| 9. A | 10. C | 11. D | 12. A | 13. D | 14. C | 15. A | 16. C |
| 17. A | 18. A | 19. B | 20. D | 21. B | 22. B | 23. B | 24. D |
| 25. B | 26. C | 27. B | 28. B | 29. C | 30. B | 31. A | 32. C |
| 33. A | 34. B | 35. C | 36. C | 37. C | 38. C | 39. C | 40. A |
| 41. D | 42. A | 43. D | 44. C | 45. B | 46. D | 47. C | 48. C |
| 49. D | 50. (A) R; (B) S; (C) Q; (D) P | | | | | | |

EXERCISE (O-2)

- | | | | | | | | |
|---------------------------------------|---------|------------------------|-----------|-----------|-----------|---------|------|
| 1. A | 2. D | 3. A | 4. B | 5. A | 6. A | 7. D | 8. A |
| 9. B | 10. B | 11. 240, 240, 255, 480 | 12. C | 13. C | 14. A | 15. D | |
| 16. (a) C ; (b) B ; (c) B | 17. A | 18. C | 19. D | 20. A,B | 21. A,B,D | 22. B,C | |
| 23. A,B,C,D | 24. C,D | 25. B,D | 26. A,B,D | 27. A,C,D | | | |
| 28. (A) T; (B) R; (C) P; (D) Q; (E) S | | | | | | | |

EXERCISE (S-1)

- | | | | | | | | |
|---|----------------------------------|----------------------------------|-----------------------|-----------------------------------|-----------------------------|--|--|
| 1. 120 | 2. (i) 24 ; (ii) 576 ; (iii) 360 | | | | | | |
| 3. (i) 840; (ii) 120; (iii) 400; (iv) 240; (v) 480; (vi) 40; (vii) 60; (viii) 240 | | | | | | | |
| 4. (i) 120 ; (ii) 40 ; (iii) 40 ; (iv) 80 ; (v) 20 | 5. 20 | 6. 4^7 | 7. (a) 3^4 ; (b) 24 | | | | |
| 8. 36 | 9. 720 | 10. 576 | 11. 999 | 12. 967680 | 13. (ii) 792 ; (iv) $r = 3$ | | |
| 14. 13, 156 | 15. 43200 | 16. 420 | 17. 960 | 18. ${}^{24}C_2 \cdot {}^{15}C_3$ | 19. 1106 | | |
| 20. $\frac{52!}{(13!)^4} ; \frac{52!}{3!(17!)^3}$ | 21. 5400 | 22. 3150 | 23. 1568 | 24. 172800 | 25. ${}^8C_4 \cdot 4!$ | | |
| 26. 528 | 27. $n = 8$ | 28. (i) 3359; (ii) 59; (iii) 359 | 29. 120, 216, 210 | 30. 240, 15552 | | | |
| 31. 30 | 32. 719 | 33. 2500 | 34. C | 35. C | 36. 30980 | | |
| 38. (a) 72; 78120; (b) 23; (c) 32 ; (d) 435 | 39. 56 | 40. 12870 | 41. 3003 | 42. 6 | | | |

EXERCISE (S-2)

- | | | | |
|--|----------------------|------------------------|--|
| 1. ${}^{m+n+k}C_3 - ({}^mC_3 + {}^nC_3 + {}^kC_3)$ | 2. (a) 744; (b) 7704 | 3. 22100, 52 | 4. 40 |
| 5. C | 6. A | 7. 576 | 8. 60 |
| 9. 64 | 10. 244 | 11. 24 | 12. (i) 15, (ii) 126, (iii) 60, (iv) 105 |
| 13. 440 | 14. 3119976 | 15. (a) 1680; (b) 1140 | |

EXERCISE (JM)

- | | | | | | | | |
|------|-------|-------|-------|-------|------|------|------|
| 1. 2 | 2. 4 | 3. 3 | 4. 2 | 5. 1 | 6. 3 | 7. 2 | 8. 2 |
| 9. 3 | 10. 4 | 11. 1 | 12. 2 | 13. 4 | | | |

EXERCISE (JA)

- | | | | | | | | |
|------|-------|-------|-------|---------|-------|------|------|
| 1. C | 2. D | 3. B | 4. B | 5. A | 6. 7 | 7. 5 | 8. C |
| 9. 5 | 10. A | 11. 5 | 12. D | 13. 625 | 14. C | | |

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DEFINITE INTEGRATION

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JEE (Main) Syllabus :

Integral as an anti derivative. Fundamental integrals involving algebraic, trigonometric, exponential and logarithmic functions. Integration by substitution, by parts and by partial fractions. Integration using trigonometric identities. Evaluation of simple integrals of the type $\int_a^b f(x) dx$. Fundamental Theorem of Calculus. Properties of definite integrals. Evaluation of definite integrals.

JEE (Advanced) Syllabus :

Integration as the inverse process of differentiation, indefinite integrals of standard functions, definite integrals and their properties, Fundamental Theorem of Integral Calculus. Integration by parts, integration by the methods of substitution and partial fractions.

INDEFINITE INTEGRATION

If f & F are function of x such that $F'(x) = f(x)$ then the function F is called a **PRIMITIVE OR ANTIDERIVATIVE OR INTEGRAL** of $f(x)$ w.r.t. x and is written symbolically as

$$\int f(x) dx = F(x) + C \Leftrightarrow \frac{d}{dx} \{F(x) + C\} = f(x), \text{ where } C \text{ is called the constant of integration.}$$

1. GEOMETRICAL INTERPRETATION OF INDEFINITE INTEGRAL :

$\int f(x) dx = F(x) + C = y$ (say), represents a family of curves. The different values of C will correspond to different members of this family and these members can be obtained by shifting any one of the curves parallel to itself. This is the geometrical interpretation of indefinite integral.

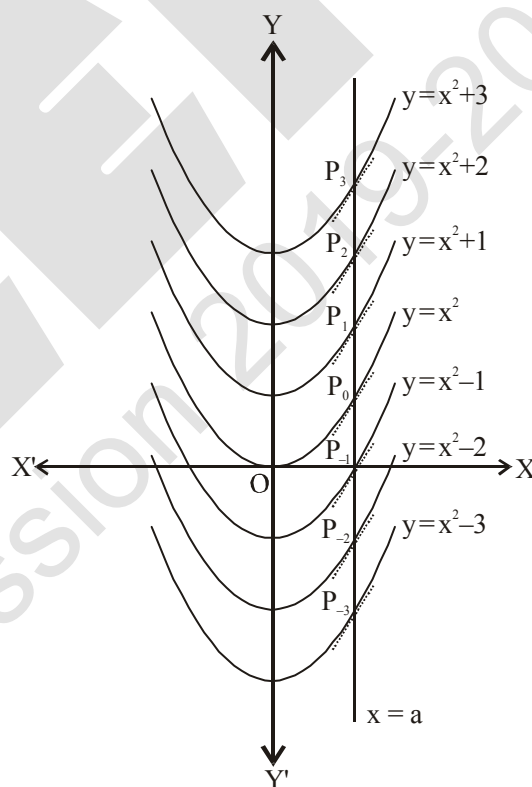
Let $f(x) = 2x$. Then $\int f(x) dx = x^2 + C$. For different values of C , we get different integrals. But these integrals are very similar geometrically.

Thus, $y = x^2 + C$, where C is arbitrary constant, represents a family of integrals. By assigning different values to C , we get different members of the family. These together constitute the indefinite integral. In this case, each integral represents a parabola with its axis along y -axis.

If the line $x = a$ intersects the parabolas $y = x^2$, $y = x^2 + 1$, $y = x^2 + 2$, $y = x^2 - 1$, $y = x^2 - 2$ at $P_0, P_1, P_2, P_{-1}, P_{-2}$ etc., respectively, then $\frac{dy}{dx}$ at these points

equals $2a$. This indicates that the tangents to the curves at these points are parallel. Thus,

$\int 2x dx = x^2 + C = f(x) + C$ (say), implies that the tangents to all the curves $f(x) + C, C \in \mathbb{R}$, at the points of intersection of the curves by the line $x = a, (a \in \mathbb{R})$, are parallel.



2. STANDARD FORMULAE :

$$(i) \int (ax + b)^n dx = \frac{(ax + b)^{n+1}}{a(n+1)} + C; n \neq -1$$

$$(ii) \int \frac{dx}{ax + b} = \frac{1}{a} \ell n |ax + b| + C$$

$$(iii) \int e^{ax+b} dx = \frac{1}{a} e^{ax+b} + C$$

$$(iv) \int a^{px+q} dx = \frac{1}{p} \frac{a^{px+q}}{\ell n a} + C, (a > 0)$$

$$(v) \int \sin(ax + b) dx = -\frac{1}{a} \cos(ax + b) + C$$

$$(vi) \int \cos(ax + b) dx = \frac{1}{a} \sin(ax + b) + C$$

$$(vii) \int \tan(ax+b)dx = \frac{1}{a} \ell n |\sec(ax+b)| + C \quad (viii) \int \cot(ax+b)dx = \frac{1}{a} \ell n |\sin(ax+b)| + C$$

$$(ix) \int \sec^2(ax+b)dx = \frac{1}{a} \tan(ax+b) + C$$

$$(x) \int \operatorname{cosec}^2(ax+b)dx = -\frac{1}{a} \cot(ax+b) + C$$

$$(xi) \int \operatorname{cosec}(ax+b) \cdot \cot(ax+b)dx = -\frac{1}{a} \operatorname{cosec}(ax+b) + C$$

$$(xii) \int \sec(ax+b) \cdot \tan(ax+b)dx = \frac{1}{a} \sec(ax+b) + C$$

$$(xiii) \int \sec x dx = \ell n |\sec x + \tan x| + C \quad \ell n \left| \tan \left(\frac{\pi}{4} + \frac{x}{2} \right) \right| + C$$

$$(xiv) \int \operatorname{cosec} x dx = \ell n |\operatorname{cosec} x - \cot x| + C = \ell n \left| \tan \frac{x}{2} \right| + C = -\ell n |\operatorname{cosec} x + \cot x| + C$$

$$(xv) \int \frac{dx}{\sqrt{a^2 - x^2}} = \sin^{-1} \frac{x}{a} + C$$

$$(xvi) \int \frac{dx}{a^2 + x^2} = \frac{1}{a} \tan^{-1} \frac{x}{a} + C$$

$$(xvii) \int \frac{dx}{x\sqrt{x^2 - a^2}} = \frac{1}{a} \sec^{-1} \frac{x}{a} + C$$

$$(xviii) \int \frac{dx}{\sqrt{x^2 + a^2}} = \ell n \left[x + \sqrt{x^2 + a^2} \right] + C$$

$$(xix) \int \frac{dx}{\sqrt{x^2 - a^2}} = \ell n \left[x + \sqrt{x^2 - a^2} \right] + C$$

$$(xx) \int \frac{dx}{a^2 - x^2} = \frac{1}{2a} \ell n \left| \frac{a+x}{a-x} \right| + C$$

$$(xxi) \int \frac{dx}{x^2 - a^2} = \frac{1}{2a} \ell n \left| \frac{x-a}{x+a} \right| + C$$

$$(xxii) \int \sqrt{a^2 - x^2} dx = \frac{x}{2} \sqrt{a^2 - x^2} + \frac{a^2}{2} \sin^{-1} \frac{x}{a} + C$$

$$(xxiii) \int \sqrt{x^2 + a^2} dx = \frac{x}{2} \sqrt{x^2 + a^2} + \frac{a^2}{2} \ell n \left(x + \sqrt{x^2 + a^2} \right) + C$$

$$(xxiv) \int \sqrt{x^2 - a^2} dx = \frac{x}{2} \sqrt{x^2 - a^2} - \frac{a^2}{2} \ell n \left(x + \sqrt{x^2 - a^2} \right) + C$$

$$(xxv) \int e^{ax} \cdot \sin bx dx = \frac{e^{ax}}{a^2 + b^2} (a \sin bx - b \cos bx) + C = \frac{e^{ax}}{\sqrt{a^2 + b^2}} \sin \left(bx - \tan^{-1} \frac{b}{a} \right) + C$$

$$(xxvi) \int e^{ax} \cdot \cos bx dx = \frac{e^{ax}}{a^2 + b^2} (a \cos bx + b \sin bx) + C = \frac{e^{ax}}{\sqrt{a^2 + b^2}} \cos \left(bx - \tan^{-1} \frac{b}{a} \right) + C$$

3. TECHNIQUES OF INTEGRATION :

(a) Substitution or change of independent variable :

If $\phi(x)$ is a continuous differentiable function, then to evaluate integrals of the form

$\int f(\phi(x))\phi'(x)dx$, we substitute $\phi(x) = t$ and $\phi'(x)dx = dt$.

Hence $I = \int f(\phi(x))\phi'(x)dx$ reduces to $\int f(t)dt$.

(i) Fundamental deductions of method of substitution :

$$\int [f(x)]^n f'(x)dx \quad \text{OR} \quad \int \frac{f'(x)}{[f(x)]^n} dx \quad \text{put } f(x) = t \text{ \& proceed.}$$

Illustration 1 : Evaluate $\int \frac{\cos^3 x}{\sin^2 x + \sin x} dx$

Solution : $I = \int \frac{(1 - \sin^2 x) \cos x}{\sin x(1 + \sin x)} dx = \int \frac{1 - \sin x}{\sin x} \cos x dx$

Put $\sin x = t \Rightarrow \cos x dx = dt$

$$\Rightarrow I = \int \frac{1-t}{t} dt = \ell n |t| - t + C = \ell n |\sin x| - \sin x + C$$

Ans.

Illustration 2 : Evaluate $\int \frac{(x^2 - 1) dx}{(x^4 + 3x^2 + 1) \tan^{-1} \left(x + \frac{1}{x} \right)}$

Solution : The given integral can be written as

$$I = \int \frac{\left(1 - \frac{1}{x^2}\right) dx}{\left[\left(x + \frac{1}{x}\right)^2 + 1\right] \tan^{-1} \left(x + \frac{1}{x}\right)}$$

Let $\left(x + \frac{1}{x}\right) = t$. Differentiating we get $\left(1 - \frac{1}{x^2}\right) dx = dt$

$$\text{Hence } I = \int \frac{dt}{(t^2 + 1) \tan^{-1} t}$$

Now make one more substitution $\tan^{-1} t = u$. Then $\frac{dt}{t^2 + 1} = du$ and $I = \int \frac{du}{u} = \ell n |u| + C$

Returning to t , and then to x , we have

$$I = \ell n |\tan^{-1} t| + C = \ell n \left| \tan^{-1} \left(x + \frac{1}{x} \right) \right| + C$$

Ans.

Do yourself -1 :

(i) Evaluate : $\int \frac{x^2}{9 + 16x^6} dx$

(ii) Evaluate : $\int \cos^3 x dx$

(ii) Standard substitutions :

$$\int \frac{dx}{\sqrt{a^2 + x^2}} \text{ or } \int \sqrt{a^2 + x^2} dx ; \text{ put } x = a \tan \theta \text{ or } x = a \cot \theta$$

$$\int \frac{dx}{\sqrt{a^2 - x^2}} \text{ or } \int \sqrt{a^2 - x^2} dx ; \text{ put } x = a \sin \theta \text{ or } x = a \cos \theta$$

$$\int \frac{dx}{\sqrt{x^2 - a^2}} \text{ or } \int \sqrt{x^2 - a^2} dx ; \text{ put } x = a \sec \theta \text{ or } x = a \csc \theta$$

$$\int \sqrt{\frac{a-x}{a+x}} dx ; \text{ put } x = a \cos 2\theta$$

$$\int \sqrt{\frac{x-\alpha}{\beta-x}} dx \text{ or } \int \sqrt{(x-\alpha)(\beta-x)} ; \text{ put } x = \alpha \cos^2 \theta + \beta \sin^2 \theta$$

$$\int \sqrt{\frac{x-\alpha}{x-\beta}} dx \text{ or } \int \sqrt{(x-\alpha)(x-\beta)} ; \text{ put } x = \alpha \sec^2 \theta - \beta \tan^2 \theta$$

$$\int \frac{dx}{\sqrt{(x-\alpha)(x-\beta)}} ; \text{ put } x - \alpha = t^2 \text{ or } x - \beta = t^2.$$

Illustration 3 : Evaluate $\int \frac{dx}{\sqrt{(x-a)(b-x)}}$

Solution : Put $x = a \cos^2 \theta + b \sin^2 \theta$, the given integral becomes

$$\begin{aligned} I &= \int \frac{2(b-a) \sin \theta \cos \theta d\theta}{\{(a \cos^2 \theta + b \sin^2 \theta - a)(b - a \cos^2 \theta - b \sin^2 \theta)\}^{\frac{1}{2}}} \\ &= \int \frac{2(b-a) \sin \theta \cos \theta d\theta}{(b-a) \sin \theta \cos \theta} = \left(\frac{b-a}{b-a}\right) \int 2 d\theta = 2\theta + C = 2 \sin^{-1} \sqrt{\frac{x-a}{b-a}} + C \end{aligned}$$

Ans.

Illustration 4 : Evaluate $\int \sqrt{\frac{1-\sqrt{x}}{1+\sqrt{x}}} \cdot \frac{1}{x} dx$

Solution : Put $x = \cos^2 \theta \Rightarrow dx = -2 \sin \theta \cos \theta d\theta$

$$\begin{aligned} \Rightarrow I &= \int \sqrt{\frac{1-\cos \theta}{1+\cos \theta}} \cdot \frac{1}{\cos^2 \theta} (-2 \sin \theta \cos \theta) d\theta = -\int 2 \tan \frac{\theta}{2} \tan \theta d\theta \\ &= -4 \int \frac{\sin^2(\theta/2)}{\cos \theta} d\theta = -2 \int \frac{1-\cos \theta}{\cos \theta} d\theta = -2 \ln |\sec \theta + \tan \theta| + 2\theta + C \\ &= -2 \ln \left| \frac{1+\sqrt{1-x}}{\sqrt{x}} \right| + 2 \cos^{-1} \sqrt{x} + C \end{aligned}$$

Do yourself -2 :

(i) Evaluate : $\int \sqrt{\frac{x-3}{2-x}} dx$

(ii) Evaluate : $\int \frac{dx}{x\sqrt{x^2+4}}$

(b) **Integration by part :** $\int u \cdot v dx = u \int v dx - \int \left[\frac{du}{dx} \cdot \int v dx \right] dx$ where u & v are differentiable functions and are commonly designated as first & second function respectively.

Note : While using integration by parts, choose u & v such that

(i) $\int v dx$ & (ii) $\int \left[\frac{du}{dx} \cdot \int v dx \right] dx$ are simple to integrate.

This is generally obtained by choosing first function as the function which comes first in the word **ILATE**, where; I-Inverse function, L-Logarithmic function, A-Algebraic function, T-Trigonometric function & E-Exponential function.

Illustration 5 : Evaluate : $\int \cos \sqrt{x} dx$

Solution : Consider $I = \int \cos \sqrt{x} dx$

Let $\sqrt{x} = t$ then $\frac{1}{2\sqrt{x}} dx = dt$

i.e. $dx = 2\sqrt{x} dt$ or $dx = 2t dt$

so $I = \int \cos t \cdot 2t dt$

Taking t as first function, integrate it by part

$$\Rightarrow I = 2 \left[t \int \cos t dt - \int \left\{ \frac{dt}{dt} \int \cos t dt \right\} dt \right]$$

$$I = 2 \left[t \sin t - \int 1 \cdot \sin t dt \right] = 2 [t \sin t + \cos t] + C$$

$$I = 2 [\sqrt{x} \sin \sqrt{x} + \cos \sqrt{x}] + C$$

Ans.

Illustration 6 : Evaluate : $\int \frac{x}{1+\sin x} dx$

Solution : Let $I = \int \frac{x}{1+\sin x} dx = \int \frac{x(1-\sin x)}{(1+\sin x)(1-\sin x)} dx$

$$= \int \frac{x(1-\sin x)}{1-\sin^2 x} dx = \int \frac{x(1-\sin x)}{\cos^2 x} dx = \int x \sec^2 x dx - \int x \sec x \tan x dx$$

$$= \left[x \int \sec^2 x dx - \int \left\{ \frac{dx}{dx} \int \sec^2 x dx \right\} dx \right]$$

$$- \left[x \int \sec x \tan x dx - \int \left\{ \frac{dx}{dx} \int \sec x \tan x dx \right\} dx \right]$$

$$\begin{aligned}
&= \left[x \tan x - \int \tan x dx \right] - \left[x \sec x - \int \sec x dx \right] \\
&= \left[x \tan x - \ln |\sec x| \right] - \left[x \sec x - \ln |\sec x + \tan x| \right] + C \\
&= x(\tan x - \sec x) + \ln \left| \frac{(\sec x + \tan x)}{\sec x} \right| + C \\
&= \frac{-x(1 - \sin x)}{\cos x} + \ln |1 + \sin x| + C
\end{aligned}$$

Ans.

Do yourself - 3 :

(i) Evaluate : $\int x e^x dx$

(ii) Evaluate : $\int x^3 \sin(x^2) dx$

Two classic integrands :

(i) $\int e^x [f(x) + f'(x)] dx = e^x \cdot f(x) + C$

Illustration 7 : Evaluate $\int e^x \left(\frac{1-x}{1+x^2} \right)^2 dx$

Solution :

$$\begin{aligned}
\int e^x \left(\frac{1-x}{1+x^2} \right)^2 dx &= \int e^x \frac{(1-2x+x^2)}{(1+x^2)^2} dx \\
&= \int e^x \left(\frac{1}{(1+x^2)} - \frac{2x}{(1+x^2)^2} \right) dx = \frac{e^x}{1+x^2} + C
\end{aligned}$$

Ans.

Illustration 8 : The value of $\int e^x \left(\frac{x^4+2}{(1+x^2)^{5/2}} \right) dx$ is equal to -

(A) $\frac{e^x(x+1)}{(1+x^2)^{3/2}} + C$ (B) $\frac{e^x(1-x+x^2)}{(1+x^2)^{3/2}} + C$ (C) $\frac{e^x(1-x)}{(1+x^2)^{3/2}} + C$ (D) none of these

Solution :

$$\begin{aligned}
\text{Let } I &= \int e^x \left(\frac{x^4+2}{(1+x^2)^{5/2}} \right) dx = \int e^x \left(\frac{1}{(1+x^2)^{1/2}} + \frac{1-2x^2}{(1+x^2)^{5/2}} \right) dx \\
&= \int e^x \left(\frac{1}{(1+x^2)^{1/2}} - \frac{x}{(1+x^2)^{3/2}} + \frac{x}{(1+x^2)^{3/2}} + \frac{1-2x^2}{(1+x^2)^{5/2}} \right) dx \\
&= \frac{e^x}{(1+x^2)^{1/2}} + \frac{x e^x}{(1+x^2)^{3/2}} + C = \frac{e^x \{1+x^2+x\}}{(1+x^2)^{3/2}} + C
\end{aligned}$$

Ans. (D)

Do yourself - 4 :

(i) Evaluate : $\int e^x \left(\tan^{-1} x + \frac{1}{1+x^2} \right) dx$

(ii) Evaluate : $\int x e^{x^2} (\sin x^2 + \cos x^2) dx$

(ii) $\int [f(x) + x f'(x)] dx = x f(x) + C$

Illustration 9: Evaluate $\int \frac{x + \sin x}{1 + \cos x} dx$

Solution :
$$I = \int \frac{x + \sin x}{1 + \cos x} dx = \int \left(\frac{x + \sin x}{2 \cos^2 \frac{x}{2}} \right) dx = \int \left(x \frac{1}{2} \sec^2 \frac{x}{2} + \tan \frac{x}{2} \right) dx = x \tan \frac{x}{2} + C$$
 Ans.

Do yourself -5 :

(i) Evaluate : $\int (\tan(e^x) + xe^x \sec^2(e^x)) dx$ (ii) Evaluate : $\int (\ln x + 1) dx$

(c) Integration of trigonometric functions :

(i) $\int \sin^m x \cos^n x dx$

Case-I : When m & $n \in$ natural numbers.

- * If one of them is odd, then substitute for the term of even power.
- * If both are odd, substitute either of the term.
- * If both are even, use trigonometric identities to convert integrand into cosines of multiple angles.

Case-II : $m + n$ is a negative even integer.

- * In this case the best substitution is $\tan x = t$.

Illustration 10 : Evaluate $\int \sin^3 x \cos^5 x \, dx$

Solution : Put $\cos x = t$; $-\sin x \, dx = dt$.

$$\begin{aligned}\text{so that } I &= -\int (1-t^2).t^5 dt \\ &= \int (t^7 - t^5) dt = \frac{t^8}{8} - \frac{t^6}{6} = \frac{\cos^8 x}{8} - \frac{\cos^6 x}{6} + C\end{aligned}$$

Alternate :

Put $\sin x = t$; $\cos x \, dx = dt$

$$\begin{aligned}\text{so that } I &= \int t^3(1-t^2)^2 dt = \int (t^3 - 2t^5 + t^7) dt \\ &= \frac{\sin^4 x}{4} - \frac{2\sin^6 x}{6} + \frac{\sin^8 x}{8} + C\end{aligned}$$

Note : This problem can also be handled by successive reduction or by trigonometric identities.

Illustration 11: Evaluate $\int \sin^2 x \cos^4 x \, dx$

Solution :

$$\int \sin^2 x \cos^4 x dx = \int \left(\frac{1 - \cos 2x}{2} \right) \left(\frac{\cos 2x + 1}{2} \right)^2 dx$$
$$= \int \frac{1}{8} (1 - \cos 2x) (\cos^2 2x + 2 \cos 2x + 1) dx$$

$$\begin{aligned}
&= \frac{1}{8} \int (\cos^2 2x + 2 \cos 2x + 1 - \cos^3 2x - 2 \cos^2 2x - \cos 2x) dx \\
&= \frac{1}{8} \int (-\cos^3 2x - \cos^2 2x + \cos 2x + 1) dx \\
&= -\frac{1}{8} \int \left(\frac{\cos 6x + 3 \cos 2x}{4} + \frac{1 + \cos 4x}{2} - \cos 2x - 1 \right) dx \\
&= -\frac{1}{32} \left[\frac{\sin 6x}{6} + \frac{3 \sin 2x}{2} \right] - \frac{1}{16} x - \frac{\sin 4x}{64} + \frac{\sin 2x}{16} + \frac{x}{8} + C \\
&= -\frac{\sin 6x}{192} - \frac{\sin 4x}{64} + \frac{1}{64} \sin 2x + \frac{x}{16} + C
\end{aligned}$$

Illustration 12 : Evaluate $\int \frac{\sqrt{\sin x}}{\cos^{9/2} x} dx$

Solution : Let $I = \int \frac{\sin^{1/2} x}{\cos^{9/2} x} dx = \int \frac{dx}{\sin^{-1/2} x \cos^{9/2} x}$

Here $m + n = \frac{1}{2} - \frac{9}{2} = -4$ (negative even integer).

Divide Numerator & Denominator by $\cos^4 x$.

$$\begin{aligned}
I &= \int \sqrt{\tan x} \sec^4 x dx = \int \sqrt{\tan x} (1 + \tan^2 x) \sec^2 x dx \\
&= \int \sqrt{t} (1 + t^2) dt \quad (\text{using } \tan x = t) \\
&= \frac{2}{3} t^{3/2} + \frac{2}{7} t^{7/2} + C = \frac{2}{3} \tan^{3/2} x + \frac{2}{7} \tan^{7/2} x + C
\end{aligned}$$

Do yourself -6 :

(i) Evaluate : $\int \frac{\sin^2 x}{\cos^4 x} dx$ (ii) Evaluate : $\int \frac{\sqrt{\sin x} dx}{\cos^{5/2} x}$ (iii) Evaluate : $\int \sin^2 x \cos^5 x dx$

(ii) $\int \frac{dx}{a + b \sin^2 x}$ OR $\int \frac{dx}{a + b \cos^2 x}$ OR $\int \frac{dx}{a \sin^2 x + b \sin x \cos x + c \cos^2 x}$

Divide N^r & D^r by $\cos^2 x$ & put $\tan x = t$.

Illustration 13 : Evaluate : $\int \frac{dx}{2 + \sin^2 x}$

Solution : Divide numerator and denominator by $\cos^2 x$

$$I = \int \frac{\sec^2 x dx}{2 \sec^2 x + \tan^2 x} = \int \frac{\sec^2 x dx}{2 + 3 \tan^2 x}$$

Let $\sqrt{3} \tan x = t \quad \therefore \sqrt{3} \sec^2 x dx = dt$

So $I = \frac{1}{\sqrt{3}} \int \frac{dt}{2 + t^2} = \frac{1}{\sqrt{3}} \cdot \frac{1}{\sqrt{2}} \tan^{-1} \frac{t}{\sqrt{2}} + C = \frac{1}{\sqrt{6}} \tan^{-1} \left(\frac{\sqrt{3} \tan x}{\sqrt{2}} \right) + C$

Ans.

Illustration 14: Evaluate: $\int \frac{dx}{(2 \sin x + 3 \cos x)^2}$

Solution : Divide numerator and denominator by $\cos^2 x$

$$\therefore I = \int \frac{\sec^2 x dx}{(2 \tan x + 3)^2}$$

$$\text{Let } 2 \tan x + 3 = t, \quad \therefore 2 \sec^2 x dx = dt$$

$$I = \frac{1}{2} \int \frac{dt}{t^2} = -\frac{1}{2t} + C = -\frac{1}{2(2 \tan x + 3)} + C$$

Ans.

Do yourself -7 :

(i) Evaluate: $\int \frac{dx}{1 + 4 \sin^2 x}$

(ii) Evaluate: $\int \frac{dx}{3 \sin^2 x + \sin x \cos x + 1}$

(iii) $\int \frac{dx}{a + b \sin x}$ OR $\int \frac{dx}{a + b \cos x}$ OR $\int \frac{dx}{a + b \sin x + c \cos x}$

Convert sines & cosines into their respective tangents of half the angles

& put $\tan \frac{x}{2} = t$

In this case $\sin x = \frac{2t}{1+t^2}$, $\cos x = \frac{1-t^2}{1+t^2}$, $x = 2 \tan^{-1} t$; $dx = \frac{2dt}{1+t^2}$

Illustration 15: Evaluate: $\int \frac{dx}{3 \sin x + 4 \cos x}$

Solution :

$$I = \int \frac{dx}{3 \sin x + 4 \cos x} = \int \frac{\sec^2 \frac{x}{2} dx}{3 \left\{ \frac{2 \tan \frac{x}{2}}{1 + \tan^2 \frac{x}{2}} \right\} + 4 \left\{ \frac{1 - \tan^2 \frac{x}{2}}{1 + \tan^2 \frac{x}{2}} \right\}} = \int \frac{\sec^2 \frac{x}{2} dx}{4 + 6 \tan \frac{x}{2} - 4 \tan^2 \frac{x}{2}}$$

let $\tan \frac{x}{2} = t$, $\therefore \frac{1}{2} \sec^2 \frac{x}{2} dx = dt$

so $I = \int \frac{2dt}{4 + 6t - 4t^2} = \frac{1}{2} \int \frac{dt}{1 - \left(t^2 - \frac{3}{2}t\right)} = \frac{1}{2} \int \frac{dt}{\frac{25}{16} - \left(t - \frac{3}{4}\right)^2}$

$$= \frac{1}{2} \cdot \frac{1}{2 \left(\frac{5}{4}\right)} \ln \left| \frac{\frac{5}{4} + \left(t - \frac{3}{4}\right)}{\frac{5}{4} - \left(t - \frac{3}{4}\right)} \right| + C = \frac{1}{5} \ln \left| \frac{1 + 2 \tan \frac{x}{2}}{4 - 2 \tan \frac{x}{2}} \right| + C$$

Ans.

Do yourself-8 :

(i) Evaluate : $\int \frac{dx}{3 + \sin x}$

(ii) Evaluate : $\int \frac{dx}{1 + 4 \sin x + 3 \cos x}$

(iv) $\int \frac{a \cos x + b \sin x + c}{p \cos x + q \sin x + r} dx$

Express Numerator (N^r) = $\ell(D^r) + m \frac{d}{dx}(D^r) + n$ & proceed.

Illustration 16 : Evaluate : $\int \frac{2 + 3 \cos \theta}{\sin \theta + 2 \cos \theta + 3} d\theta$

Solution :

Write the Numerator = $\ell(\text{denominator}) + m(\text{d.c. of denominator}) + n$

$$\Rightarrow 2 + 3 \cos \theta = \ell(\sin \theta + 2 \cos \theta + 3) + m(\cos \theta - 2 \sin \theta) + n.$$

Comparing the coefficients of $\sin \theta$, $\cos \theta$ and constant terms,

$$\text{we get } 3\ell + n = 2, \quad 2\ell + m = 3, \quad \ell - 2m = 0 \Rightarrow \ell = 6/5, m = 3/5 \text{ and } n = -8/5$$

$$\text{Hence } I = \int \frac{6}{5} d\theta + \frac{3}{5} \int \frac{\cos \theta - 2 \sin \theta}{\sin \theta + 2 \cos \theta + 3} d\theta - \frac{8}{5} \int \frac{d\theta}{\sin \theta + 2 \cos \theta + 3}$$

$$= \frac{6}{5} \theta + \frac{3}{5} \ell n |\sin \theta + 2 \cos \theta + 3| - \frac{8}{5} I_3 \text{ where } I_3 = \int \frac{d\theta}{\sin \theta + 2 \cos \theta + 3}$$

$$\text{In } I_3, \text{ put } \tan \frac{\theta}{2} = t \Rightarrow \sec^2 \frac{\theta}{2} d\theta = 2dt$$

$$I_3 = 2 \int \frac{dt}{t^2 + 2t + 5} = 2 \int \frac{dt}{(t+1)^2 + 2^2} = 2 \cdot \frac{1}{2} \tan^{-1} \left(\frac{t+1}{2} \right) = \tan^{-1} \left(\frac{\tan \theta / 2 + 1}{2} \right)$$

$$\text{Hence } I = \frac{6\theta}{5} + \frac{3}{5} \ell n |\sin \theta + 2 \cos \theta + 3| - \frac{8}{5} \tan^{-1} \left(\frac{\tan \theta / 2 + 1}{2} \right) + C$$

Ans.**Do yourself -9 :**

(i) Evaluate : $\int \frac{\sin x}{\sin x + \cos x} dx$

(ii) Evaluate : $\int \frac{3 \sin x + 2 \cos x}{3 \cos x + 2 \sin x} dx$

(d) Integration of rational function :

- (i) Rational function is defined as the ratio of two polynomials in the form $\frac{P(x)}{Q(x)}$, where $P(x)$ and $Q(x)$ are polynomials in x and $Q(x) \neq 0$. If the degree of $P(x)$ is less than the degree of $Q(x)$, then the rational function is called proper, otherwise, it is called improper. The improper rational function can be reduced to the proper rational functions by long division process. Thus, if $\frac{P(x)}{Q(x)}$ is improper, then $\frac{P(x)}{Q(x)} = T(x) + \frac{P_1(x)}{Q(x)}$, where $T(x)$ is a polynomial in x and $\frac{P_1(x)}{Q(x)}$ is proper rational function. It is always possible to write the integrand as a sum of simpler rational functions by a method called partial fraction decomposition. After this, the integration can be carried out easily using the already known methods.

S. No.	Form of the rational function	Form of the partial fraction
1.	$\frac{px^2 + qx + r}{(x-a)(x-b)(x-c)}$	$\frac{A}{x-a} + \frac{B}{x-b} + \frac{C}{x-c}$
2.	$\frac{px^2 + qx + r}{(x-a)^2(x-b)}$	$\frac{A}{x-a} + \frac{B}{(x-a)^2} + \frac{C}{x-b}$
3.	$\frac{px^2 + qx + r}{(x-a)(x^2 + bx + c)}$ where $x^2 + bx + c$ cannot be factorised further	$\frac{A}{x-a} + \frac{Bx + C}{x^2 + bx + c}$

Illustration 17: Evaluate : $\int \frac{x}{(x-2)(x+5)} dx$

Solution : $\frac{x}{(x-2)(x+5)} = \frac{A}{x-2} + \frac{B}{x+5}$
or $x = A(x+5) + B(x-2)$.
by comparing the coefficients, we get
 $A = 2/7$ and $B = 5/7$ so that

$$\int \frac{x}{(x-2)(x+5)} dx = \frac{2}{7} \int \frac{dx}{x-2} + \frac{5}{7} \int \frac{dx}{x+5} = \frac{2}{7} \ln|(x-2)| + \frac{5}{7} \ln|(x+5)| + C \quad \text{Ans.}$$

Illustration 18: Evaluate $\int \frac{x^4}{(x+2)(x^2+1)} dx$

Solution : $\frac{x^4}{(x+2)(x^2+1)} = (x-2) + \frac{3x^2+4}{(x+2)(x^2+1)}$

$$\text{Now, } \frac{3x^2+4}{(x+2)(x^2+1)} = \frac{16}{5(x+2)} + \frac{-\frac{1}{5}x + \frac{2}{5}}{x^2+1}$$

$$\text{So, } \frac{x^4}{(x+2)(x^2+1)} = x-2 + \frac{16}{5(x+2)} + \frac{-\frac{1}{5}x + \frac{2}{5}}{x^2+1}$$

$$\text{Now, } \int \left((x-2) + \frac{16}{5(x+2)} + \frac{-\frac{1}{5}x + \frac{2}{5}}{x^2+1} \right) dx$$

$$= \frac{x^2}{2} - 2x + \frac{16}{5} \tan^{-1} x + \frac{1}{10} \ln|x+2| - \frac{1}{10} \ln(x^2+1) + C \quad \text{Ans.}$$

Do yourself - 10 :

(i) Evaluate : $\int \frac{3x+2}{(x+1)(x+3)} dx$

(ii) Evaluate : $\int \frac{x^2-1}{(x+1)(x+2)^2} dx$

$$(ii) \int \frac{dx}{ax^2 + bx + c}, \int \frac{dx}{\sqrt{ax^2 + bx + c}}, \int \sqrt{ax^2 + bx + c} dx$$

Express $ax^2 + bx + c$ in the form of perfect square & then apply the standard results.

$$(iii) \int \frac{px + q}{ax^2 + bx + c} dx, \int \frac{px + q}{\sqrt{ax^2 + bx + c}} dx$$

Express $px + q = \ell$ (differential coefficient of denominator) + m.

Illustration 19 : Evaluate $\int \frac{dx}{2x^2 + x - 1}$

Solution :

$$\begin{aligned}
 I &= \int \frac{dx}{2x^2 + x - 1} = \frac{1}{2} \int \frac{dx}{x^2 + \frac{x}{2} - \frac{1}{2}} = \frac{1}{2} \int \frac{dx}{x^2 + \frac{x}{2} + \frac{1}{16} - \frac{1}{16} - \frac{1}{2}} \\
 &= \frac{1}{2} \int \frac{dx}{(x + 1/4)^2 - 9/16} = \frac{1}{2} \int \frac{dx}{(x + 1/4)^2 - (3/4)^2} \\
 &= \frac{1}{2} \cdot \frac{1}{2(3/4)} \log \left| \frac{x + 1/4 - 3/4}{x + 1/4 + 3/4} \right| + C \quad \left\{ \text{using, } \int \frac{dx}{x^2 - a^2} = \frac{1}{2a} \log \left| \frac{x - a}{x + a} \right| + C \right\} \\
 &= \frac{1}{3} \log \left| \frac{x - 1/2}{x + 1} \right| + C = \frac{1}{3} \log \left| \frac{2x - 1}{2(x + 1)} \right| + C
 \end{aligned}$$

Ans.

Illustration 20 : Evaluate $\int \frac{3x + 2}{4x^2 + 4x + 5} dx$

Solution : Express $3x + 2 = \ell(\text{d.c. of } 4x^2 + 4x + 5) + m$
or, $3x + 2 = \ell(8x + 4) + m$
Comparing the coefficients, we get
 $8\ell = 3$ and $4\ell + m = 2 \Rightarrow \ell = 3/8$ and $m = 2 - 4\ell = 1/2$

$$\begin{aligned}
 \Rightarrow I &= \frac{3}{8} \int \frac{8x + 4}{4x^2 + 4x + 5} dx + \frac{1}{2} \int \frac{dx}{4x^2 + 4x + 5} \\
 &= \frac{3}{8} \log |4x^2 + 4x + 5| + \frac{1}{8} \int \frac{dx}{x^2 + x + \frac{5}{4}} \\
 &= \frac{3}{8} \log |4x^2 + 4x + 5| + \frac{1}{8} \tan^{-1} \left(x + \frac{1}{2} \right) + C
 \end{aligned}$$

Ans.

Do yourself -11 :

(i) Evaluate : $\int \frac{dx}{x^2 + x + 1}$

(ii) Evaluate : $\int \frac{5x + 4}{\sqrt{x^2 + 4x + 1}} dx$

(iv) Integrals of the form $\int \frac{x^2+1}{x^4+Kx^2+1} dx$ OR $\int \frac{x^2-1}{x^4+Kx^2+1} dx$, where K is any constant.

Divide N^r & D^r by x^2 & proceed.

Note : Sometimes it is useful to write the integral as a sum of two related integrals, which can be evaluated by making suitable substitutions e.g.

$$* \quad \int \frac{2x^2}{x^4+1} dx = \int \frac{x^2+1}{x^4+1} dx + \int \frac{x^2-1}{x^4+1} dx \qquad * \quad \int \frac{2}{x^4+1} dx = \int \frac{x^2+1}{x^4+1} dx - \int \frac{x^2-1}{x^4+1} dx$$

These integrals can be called as **Algebraic Twins**.

Illustration 21 : Evaluate : $\int \frac{4}{\sin^4 x + \cos^4 x} dx$

Solution :

$$\begin{aligned} I &= 4 \int \frac{1}{\sin^4 x + \cos^4 x} dx = 4 \int \frac{\sec^4 x}{1 + \tan^4 x} dx \\ &= 4 \int \frac{(\tan^2 x + 1) \sec^2 x}{(\tan^4 x + 1)} dx \end{aligned}$$

Now, put $\tan x = t \Rightarrow \sec^2 x \, dx = dt$

$$\Rightarrow I = 4 \int \frac{1+t^2}{1+t^4} dt = 4 \int \frac{1/t^2 + 1}{t^2 + 1/t^2} dt$$

Now, put $t - 1/t = z \Rightarrow \left(1 + \frac{1}{t^2}\right) dt = dz$

$$\Rightarrow I = 4 \int \frac{dz}{z^2 + 2} = \frac{4}{\sqrt{2}} \tan^{-1} \frac{z}{\sqrt{2}} + C = 2\sqrt{2} \tan^{-1} \frac{t-1/t}{\sqrt{2}} + C$$

$$= 2\sqrt{2} \tan^{-1} \left(\frac{\tan x - 1/\tan x}{\sqrt{2}} \right) + C$$

Ans.

Illustration 22 : Evaluate : $\int \frac{1}{x^4 + 5x^2 + 1} dx$

Solution :

$$\begin{aligned} I &= \frac{1}{2} \int \frac{2}{x^4 + 5x^2 + 1} dx \\ \Rightarrow I &= \frac{1}{2} \int \frac{1+x^2}{x^4 + 5x^2 + 1} dx + \frac{1}{2} \int \frac{1-x^2}{x^4 + 5x^2 + 1} dx \\ &= \frac{1}{2} \int \frac{1+1/x^2}{x^2 + 5 + 1/x^2} dx - \frac{1}{2} \int \frac{1-1/x^2}{x^2 + 5 + 1/x^2} dx \end{aligned}$$

{dividing N^r and D^r by x^2 }

$$= \frac{1}{2} \int \frac{(1+1/x^2)}{(x-1/x)^2+7} dx - \frac{1}{2} \int \frac{(1-1/x^2)dx}{(x+1/x)^2+3}$$

$$= \frac{1}{2} \int \frac{dt}{t^2+(\sqrt{7})^2} - \frac{1}{2} \int \frac{du}{u^2+(\sqrt{3})^2}$$

where $t = x - \frac{1}{x}$ and $u = x + \frac{1}{x}$

$$I = \frac{1}{2} \cdot \frac{1}{\sqrt{7}} \left(\tan^{-1} \frac{t}{\sqrt{7}} \right) - \frac{1}{2} \cdot \frac{1}{\sqrt{3}} \left(\tan^{-1} \frac{u}{\sqrt{3}} \right) + C$$

$$= \frac{1}{2} \left[\frac{1}{\sqrt{7}} \tan^{-1} \left(\frac{x-1/x}{\sqrt{7}} \right) - \frac{1}{\sqrt{3}} \tan^{-1} \left(\frac{x+1/x}{\sqrt{3}} \right) \right] + C$$

Ans.

Do yourself -12 :

(i) Evaluate : $\int \frac{x^2+1}{x^4-x^2+1} dx$

(ii) Evaluate : $\int \frac{1}{1+x^4} dx$

(e) Integration of Irrational functions :

(i) $\int \frac{dx}{(ax+b)\sqrt{px+q}}$ & $\int \frac{dx}{(ax^2+bx+c)\sqrt{px+q}}$; put $px+q = t^2$

(ii) $\int \frac{dx}{(ax+b)\sqrt{px^2+qx+r}}$, put $ax+b = \frac{1}{t}$; $\int \frac{dx}{(ax^2+bx+c)\sqrt{px^2+q}}$, put $x = \frac{1}{t}$

Illustration 23 : Evaluate $\int \frac{x+2}{(x^2+3x+3)\sqrt{x+1}} dx$

Solution : Let, $I = \int \frac{x+2}{(x^2+3x+3)\sqrt{x+1}} dx$ Put $x+1 = t^2 \Rightarrow dx = 2t dt$

$$\therefore I = \int \frac{(t^2-1)+2}{\{(t^2-1)^2+3(t^2-1)+3\}\sqrt{t^2}} \cdot (2t) dt = 2 \int \frac{t^2+1}{t^4+t^2+1} dt = 2 \int \frac{1+1/t^2}{t^2+1+1/t^2} dt$$

$$= 2 \int \frac{1+1/t^2}{(t-1/t)^2+(\sqrt{3})^2} dt = 2 \int \frac{du}{u^2+(\sqrt{3})^2} \quad \left\{ \text{where } u = t - \frac{1}{t} \right\}$$

$$= \frac{2}{\sqrt{3}} \tan^{-1} \left(\frac{u}{\sqrt{3}} \right) + C = \frac{2}{\sqrt{3}} \tan^{-1} \left(\frac{t^2-1}{\sqrt{3}t} \right) + C = \frac{2}{\sqrt{3}} \tan^{-1} \left(\frac{x}{\sqrt{3}(x+1)} \right) + C \quad \text{Ans.}$$

Illustration 24: Evaluate $\int \frac{dx}{(x-1)\sqrt{x^2+x+1}}$

Solution : Let, $I = \int \frac{dx}{(x-1)\sqrt{x^2+x+1}}$ put $x-1 = \frac{1}{t} \Rightarrow dx = -1/t^2 dt$

$$I = \int \frac{-1/t^2 dt}{1/t \sqrt{\left(\frac{1}{t}+1\right)^2 + \left(\frac{1}{t}+1\right)+1}} = -\int \frac{dt}{\sqrt{3t^2+3t+1}}$$

$$= -\frac{1}{\sqrt{3}} \int \frac{dt}{\sqrt{\left(t+\frac{1}{2}\right)^2 + 1/12}} = -\frac{1}{\sqrt{3}} \log \left| \left(t+\frac{1}{2}\right) + \sqrt{\left(t+\frac{1}{2}\right)^2 + 1/12} \right| + C$$

$$= -\frac{1}{\sqrt{3}} \log \left| \left(\frac{1}{x-1} + \frac{1}{2}\right) + \sqrt{\frac{12\left(\frac{1}{x-1} + \frac{1}{2}\right)^2 + 1}{12}} \right| + C$$

Ans.

Illustration 25: Evaluate : $\int \frac{dx}{(1+x^2)\sqrt{1-x^2}}$

Solution : Let, $I = \int \frac{dx}{(1+x^2)\sqrt{1-x^2}}$ Put $x = \frac{1}{t}$, So that $dx = \frac{-1}{t^2} dt$

$$\therefore I = \int \frac{-1/t^2 dt}{(1+1/t^2)\sqrt{1-1/t^2}} = -\int \frac{tdt}{(t^2+1)\sqrt{t^2-1}}$$

Again let, $t^2 = u$. So that $2t dt = du$.

$$= -\frac{1}{2} \int \frac{du}{(u+1)\sqrt{u-1}} \text{ which reduces to the form } \int \frac{dx}{P\sqrt{Q}} \text{ where both P and Q are linear}$$

so that we put $u-1 = z^2$ so that $du = 2z dz$

$$\therefore I = -\frac{1}{2} \int \frac{2zdz}{(z^2+1+1)\sqrt{z^2}} = -\int \frac{dz}{(z^2+2)}$$

$$I = -\frac{1}{\sqrt{2}} \tan^{-1} \left(\frac{z}{\sqrt{2}} \right) + C$$

$$I = -\frac{1}{\sqrt{2}} \tan^{-1} \left(\frac{\sqrt{u-1}}{\sqrt{2}} \right) + C = -\frac{1}{\sqrt{2}} \tan^{-1} \left(\frac{\sqrt{t^2-1}}{\sqrt{2}} \right) + C$$

$$= -\frac{1}{\sqrt{2}} \tan^{-1} \left(\frac{\sqrt{1-x^2}}{\sqrt{2}x} \right) + C$$

Ans.

Do yourself -13 :

(i) Evaluate : $\int \frac{x}{(x-3)\sqrt{x+1}} dx$

(ii) Evaluate : $\int \frac{dx}{x^2\sqrt{1+x^2}}$

(f) Manipulating integrands :

(i) $\int \frac{dx}{x(x^n + 1)}$, $n \in \mathbb{N}$, take x^n common & put $1 + x^{-n} = t$.

(ii) $\int \frac{dx}{x^2(x^n + 1)^{\frac{(n-1)}{n}}}$, $n \in \mathbb{N}$, take x^n common & put $1 + x^{-n} = t^n$

(iii) $\int \frac{dx}{x^n(1 + x^n)^{1/n}}$, take x^n common and put $1 + x^{-n} = t^n$.

Illustration 26 : Evaluate : $\int \frac{dx}{x^n(1 + x^n)^{1/n}}$

Solution : Let $I = \int \frac{dx}{x^n(1 + x^n)^{1/n}} = \int \frac{dx}{x^{n+1} \left(1 + \frac{1}{x^n}\right)^{1/n}}$

Put $1 + \frac{1}{x^n} = t^n$, then $\frac{1}{x^{n+1}} dx = -t^{n-1} dt$

$I = - \int \frac{t^{n-1} dt}{t} = - \int t^{n-2} dt = -\frac{t^{n-1}}{n-1} + C = \frac{-1}{n-1} \left(1 + \frac{1}{x^n}\right)^{\frac{n-1}{n}} + C$

Ans.**Do yourself -14 :**

(i) Evaluate : $\int \frac{dx}{x(x^2 + 1)}$

(ii) Evaluate : $\int \frac{dx}{x^2(x^3 + 1)^{2/3}}$

(iii) Evaluate : $\int \frac{dx}{x^3(x^3 + 1)^{1/3}}$

Miscellaneous Illustrations :

Illustration 27 : Evaluate $\int \frac{\cos^4 x dx}{\sin^3 x \{\sin^5 x + \cos^5 x\}^{\frac{3}{5}}}$

Solution : $I = \int \frac{\cos^4 x}{\sin^3 x \{\sin^5 x + \cos^5 x\}^{\frac{3}{5}}} dx = \int \frac{\cos^4 x}{\sin^6 x \{1 + \cot^5 x\}^{\frac{3}{5}}} dx = \int \frac{\cot^4 x \operatorname{cosec}^2 x dx}{(1 + \cot^5 x)^{3/5}}$

Put $1 + \cot^5 x = t$

$5 \cot^4 x \operatorname{cosec}^2 x dx = -dt$

$= -\frac{1}{5} \int \frac{dt}{t^{3/5}} = -\frac{1}{2} t^{2/5} + C = -\frac{1}{2} (1 + \cot^5 x)^{2/5} + C$

Ans.

Illustration 28 : $\int \frac{dx}{\cos^6 x + \sin^6 x}$ is equal to -

- (A) $\ln|\tan x - \cot x| + C$ (B) $\ln|\cot x - \tan x| + C$
 (C) $\tan^{-1}(\tan x - \cot x) + C$ (D) $\tan^{-1}(-2\cot 2x) + C$

Solution : Let $I = \int \frac{dx}{\cos^6 x + \sin^6 x} = \int \frac{\sec^6 x}{1 + \tan^6 x} dx = \int \frac{(1 + \tan^2 x)^2 \sec^2 x dx}{1 + \tan^6 x}$

If $\tan x = p$, then $\sec^2 x dx = dp$

$$\begin{aligned} \Rightarrow I &= \int \frac{(1+p^2)^2 dp}{1+p^6} = \int \frac{(1+p^2)}{p^4 - p^2 + 1} dp = \int \frac{p^2 \left(1 + \frac{1}{p^2}\right)}{p^2 \left(p^2 + \frac{1}{p^2} - 1\right)} dp \\ &= \int \frac{dk}{k^2 + 1} = \tan^{-1}(k) + C \quad \left(\text{where } p - \frac{1}{p} = k, \left(1 + \frac{1}{p^2}\right) dp = dk \right) \\ &= \tan^{-1}\left(p - \frac{1}{p}\right) + C = \tan^{-1}(\tan x - \cot x) + C = \tan^{-1}(-2\cot 2x) + C \quad \text{Ans. (C,D)} \end{aligned}$$

Illustration 29 : Evaluate: $\int \frac{2 \sin 2x - \cos x}{6 - \cos^2 x - 4 \sin x} dx$

Solution : $I = \int \frac{2 \sin 2x - \cos x}{6 - \cos^2 x - 4 \sin x} dx = \int \frac{(4 \sin x - 1) \cos x}{6 - (1 - \sin^2 x) - 4 \sin x} dx = \int \frac{(4 \sin x - 1) \cos x}{\sin^2 x - 4 \sin x + 5} dx$

Put $\sin x = t$, so that $\cos x dx = dt$.

$$\therefore I = \int \frac{(4t - 1)dt}{(t^2 - 4t + 5)} \quad \dots (i)$$

Now, let $(4t - 1) = \lambda(2t - 4) + \mu$

Comparing coefficients of like powers of t , we get

$$\begin{aligned} 2\lambda &= 4, -4\lambda + \mu = -1 \\ \lambda &= 2, \mu = 7 \end{aligned} \quad \dots (ii)$$

$$\begin{aligned} \therefore I &= \int \frac{2(2t - 4) + 7}{t^2 - 4t + 5} dt \quad \{\text{using (i) and (ii)}\} \\ &= 2 \int \frac{2t - 4}{t^2 - 4t + 5} dt + 7 \int \frac{dt}{t^2 - 4t + 5} = 2 \log|t^2 - 4t + 5| + 7 \int \frac{dt}{t^2 - 4t + 4 - 4 + 5} \\ &= 2 \log|t^2 - 4t + 5| + 7 \int \frac{dt}{(t - 2)^2 + (1)^2} = 2 \log|t^2 - 4t + 5| + 7 \tan^{-1}(t - 2) + C \\ &= 2 \log|\sin^2 x - 4 \sin x + 5| + 7 \tan^{-1}(\sin x - 2) + C. \end{aligned}$$

Ans.

Illustration 30 : The value of $\int \sqrt{\frac{3-x}{3+x}} \cdot \sin^{-1}\left(\frac{1}{\sqrt{6}}\sqrt{3-x}\right) dx$, is equal to -

(A) $\frac{1}{4} \left\{ -3 \left(\cos^{-1}\left(\frac{x}{3}\right) \right)^2 + 2\sqrt{9-x^2} \cdot \cos^{-1}\left(\frac{x}{3}\right) + 2x \right\} + C$

(B) $\frac{1}{4} \left\{ -3 \left(\cos^{-1}\left(\frac{x}{3}\right) \right)^2 + 2\sqrt{9-x^2} \cdot \sin^{-1}\left(\frac{x}{3}\right) + 2x \right\} + C$

(C) $\frac{1}{4} \left\{ -3 \left(\sin^{-1}\left(\frac{x}{3}\right) \right)^2 + 2\sqrt{9-x^2} \cdot \sin^{-1}\left(\frac{x}{3}\right) + 2x \right\} + C$

(D) none of these

Solution : Here, $I = \int \sqrt{\frac{3-x}{3+x}} \cdot \sin^{-1}\left(\frac{1}{\sqrt{6}}\sqrt{3-x}\right) dx$

Put $x = 3\cos 2\theta \Rightarrow dx = -6\sin 2\theta d\theta$

$$= \int \sqrt{\frac{3-3\cos 2\theta}{3+3\cos 2\theta}} \cdot \sin^{-1}\left(\frac{1}{\sqrt{6}}\sqrt{3-3\cos 2\theta}\right) (-6 \sin 2\theta) d\theta$$

$$= \int \frac{\sin \theta}{\cos \theta} \cdot \sin^{-1}(\sin \theta) \cdot (-6 \sin 2\theta) d\theta = -6 \int \theta \cdot (2 \sin^2 \theta) d\theta$$

$$= -6 \int \theta (1 - \cos 2\theta) d\theta = -6 \left\{ \frac{\theta^2}{2} - \int \theta \cos 2\theta d\theta \right\} + C$$

$$= -6 \left\{ \frac{\theta^2}{2} - \left(\theta \frac{\sin 2\theta}{2} - \int 1 \cdot \left(\frac{\sin 2\theta}{2} \right) d\theta \right) \right\} + C = -3\theta^2 + 6 \left\{ \theta \frac{\sin 2\theta}{2} + \frac{\cos 2\theta}{4} \right\} + C$$

$$= \frac{1}{4} \left\{ -3 \left(\cos^{-1}\left(\frac{x}{3}\right) \right)^2 + 2\sqrt{9-x^2} \cdot \cos^{-1}\left(\frac{x}{3}\right) + 2x \right\} + C$$

Ans. (A)

Illustration 31 : Evaluate : $\int \frac{\tan\left(\frac{\pi}{4} - x\right)}{\cos^2 x \sqrt{\tan^3 x + \tan^2 x + \tan x}} dx$

Solution : $I = \int \frac{\tan\left(\frac{\pi}{4} - x\right)}{\cos^2 x \sqrt{\tan^3 x + \tan^2 x + \tan x}} dx = \int \frac{(1 - \tan^2 x) dx}{(1 + \tan x)^2 \cos^2 x \sqrt{\tan^3 x + \tan^2 x + \tan x}}$

$$I = \int \frac{-\left(1 - \frac{1}{\tan^2 x}\right) \sec^2 x dx}{\left(\tan x + 2 + \frac{1}{\tan x}\right) \sqrt{\tan x + 1 + \frac{1}{\tan x}}}$$

$$\text{let, } y = \sqrt{\tan x + 1 + \frac{1}{\tan x}} \Rightarrow 2y \, dy = \left(\sec^2 x - \frac{1}{\tan^2 x} \cdot \sec^2 x \right) dx$$

$$\therefore I = \int \frac{-2y \, dy}{(y^2 + 1) \cdot y} = -2 \int \frac{dy}{1 + y^2}$$

$$= -2 \tan^{-1} y + c = -2 \tan^{-1} \left(\sqrt{\tan x + 1 + \frac{1}{\tan x}} \right) + C$$

Ans.

ANSWERS FOR DO YOURSELF

1: (i) $\frac{1}{36} \tan^{-1} \left(\frac{4x^3}{3} \right) + C$

(ii) $\sin x - \frac{1}{3} \sin^3 x + C$

2: (i) $\sqrt{(x-2)(3-x)} - \sin^{-1} \sqrt{3-x} + C$

(ii) $\frac{1}{2} \ln \left[\frac{\sqrt{x^2+4}-2}{x} \right] + C$

3: (i) $xe^x - e^x + C$

(ii) $\frac{1}{2} [-x^2 \cos x^2 + \sin x^2] + C$

4: (i) $e^x \tan^{-1} x + C$

(ii) $\frac{1}{2} e^{x^2} \sin(x^2) + C$

5: (i) $x \tan(e^x) + C$

(ii) $x \ln x + C$

6: (i) $\frac{1}{3} \tan^3 x + C$

(ii) $\frac{2}{3} \tan^{3/2} x + C$ (iii) $\frac{1}{3} \sin^3 x - \frac{2}{5} \sin^5 x + \frac{1}{7} \sin^7 x + C$

7: (i) $\frac{1}{\sqrt{5}} \tan^{-1} (\sqrt{5} \tan x) + C$

(ii) $\frac{2}{\sqrt{15}} \tan^{-1} \left(\frac{8 \tan x + 1}{\sqrt{15}} \right) + C$

8: (i) $\frac{1}{\sqrt{2}} \tan^{-1} \left(\frac{3 \tan x / 2 + 1}{2\sqrt{2}} \right) + C$

(ii) $\frac{1}{2\sqrt{6}} \ln \left| \frac{\sqrt{6} + \tan x / 2 - 2}{\sqrt{6} - \tan x / 2 + 2} \right| + C$

9: (i) $\frac{1}{2} x - \frac{1}{2} \ln |\sin x + \cos x| + C$

(ii) $\frac{12}{13} x - \frac{5}{13} \ln |3 \cos x + 2 \sin x| + C$

10: (i) $-\frac{1}{2} \ln |x+1| + \frac{7}{2} \ln |x+3| + C$

(ii) $\ln |x+2| + \frac{3}{x+2} + C$

11: (i) $\frac{2}{\sqrt{3}} \tan^{-1} \left(\frac{2x+1}{\sqrt{3}} \right) + C$

(ii) $5\sqrt{x^2+4x+1} - 6 \ln [(x+2) + \sqrt{x^2+4x+1}] + C$

12: (i) $\tan^{-1} \left(\frac{x^2-1}{x} \right) + C$

(ii) $\frac{1}{2\sqrt{2}} \left[\tan^{-1} \left(\frac{x^2-1}{\sqrt{2}x} \right) - \frac{1}{2} \ln \frac{x^2 - \sqrt{2}x + 1}{x^2 + \sqrt{2}x + 1} \right]$

13: (i) $2\sqrt{x+1} + \frac{3}{2} \ln \left| \frac{\sqrt{x+1}-2}{\sqrt{x+1}+2} \right| + C$

(ii) $-\frac{1}{x} \sqrt{1+x^2} + C$

14: (i) $-\frac{1}{2} \ln \left(\frac{x^2+1}{x^2} \right) + C$

(ii) $-\left(1 + \frac{1}{x^3}\right)^{1/3} + C$ (iii) $-\frac{1}{2} \left(1 + \frac{1}{x^3}\right)^{2/3} + C$

EXERCISE (O-1)

1. $\int \frac{1-x^7}{x(1+x^7)} dx$ equals -
- (A) $\ln x + \frac{2}{7} \ln(1+x^7) + c$ (B) $\ln x - \frac{2}{7} \ln(1-x^7) + c$
 (C) $\ln x - \frac{2}{7} \ln(1+x^7) + c$ (D) $\ln x + \frac{2}{7} \ln(1-x^7) + c$
2. Primitive of $\frac{3x^4-1}{(x^4+x+1)^2}$ w.r.t. x is -
- (A) $\frac{x}{x^4+x+1} + c$ (B) $-\frac{x}{x^4+x+1} + c$ (C) $\frac{x+1}{x^4+x+1} + c$ (D) $-\frac{x+1}{x^4+x+1} + c$
3. If $\int \frac{\cos x - \sin x + 1 - x}{e^x + \sin x + x} dx = \ln(f(x)) + g(x) + C$ where C is the constant of integration and $f(x)$ is positive, then $f(x) + g(x)$ has the value equal to
- (A) $e^x + \sin x + 2x$ (B) $e^x + \sin x$ (C) $e^x - \sin x$ (D) $e^x + \sin x + x$
4. Integral of $\sqrt{1+2\cot x(\cot x + \operatorname{cosec} x)}$ w.r.t. x is
- (A) $2 \ln \cos \frac{x}{2} + c$ (B) $2 \ln \sin \frac{x}{2} + c$
 (C) $\frac{1}{2} \ln \cos \frac{x}{2} + c$ (D) $\ln \sin x - \ln(\operatorname{cosec} x - \cot x) + c$
5. $\int x \cdot \frac{\ln(x+\sqrt{1+x^2})}{\sqrt{1+x^2}} dx$ equals -
- (A) $\sqrt{1+x^2} \ln(x+\sqrt{1+x^2}) - x + c$ (B) $\frac{x}{2} \cdot \ln^2(x+\sqrt{1+x^2}) - \frac{x}{\sqrt{1+x^2}} + c$
 (C) $\frac{x}{2} \cdot \ln^2(x+\sqrt{1+x^2}) + \frac{x}{\sqrt{1+x^2}} + c$ (D) $\sqrt{1+x^2} \ln(x+\sqrt{1+x^2}) + x + c$
6. Let $g(x)$ be an antiderivative for $f(x)$. Then $\ln(1+(g(x))^2)$ is an antiderivative for
- (A) $\frac{2f(x)g(x)}{1+(f(x))^2}$ (B) $\frac{2f(x)g(x)}{1+(g(x))^2}$ (C) $\frac{2f(x)}{1+(f(x))^2}$ (D) none
7. A function $y = f(x)$ satisfies $f''(x) = -\frac{1}{x^2} - \pi^2 \sin(\pi x)$; $f'(2) = \pi + \frac{1}{2}$ and $f(1) = 0$. The value of $f\left(\frac{1}{2}\right)$ is
- (A) $\ln 2$ (B) 1 (C) $\frac{\pi}{2} - \ln 2$ (D) $1 - \ln 2$

8. Consider $f(x) = \frac{x^2}{1+x^3}$; $g(t) = \int f(t)dt$. If $g(1) = 0$ then $g(x)$ equals-
- (A) $\frac{1}{3} \ln(1+x^3)$ (B) $\frac{1}{3} \ln\left(\frac{1+x^3}{2}\right)$ (C) $\frac{1}{2} \ln\left(\frac{1+x^3}{3}\right)$ (D) $\frac{1}{3} \ln\left(\frac{1+x^3}{3}\right)$
9. $\int \frac{e^{\sqrt{x}}}{\sqrt{x}} (x + \sqrt{x}) dx$
- (A) $2e^{\sqrt{x}} [x - \sqrt{x} + 1] + C$ (B) $e^{\sqrt{x}} [x - 2\sqrt{x} + 1]$
(C) $e^{\sqrt{x}} (x + \sqrt{x}) + C$ (D) $e^{\sqrt{x}} (x + \sqrt{x} + 1) + C$
10. $\int \frac{dx}{\sqrt[3]{x^{5/2}(x+1)^{7/2}}}$
- (A) $-\left(\frac{x+1}{x}\right)^{1/6} + C$ (B) $6\left(\frac{x+1}{x}\right)^{-1/6} + C$ (C) $\left(\frac{x}{x+1}\right)^{5/6} + C$ (D) $-\left(\frac{x}{x+1}\right)^{5/6} + C$
11. Let $f(x) = \frac{2\sin^2 x - 1}{\cos x} + \frac{\cos x(2\sin x + 1)}{1 + \sin x}$ then $\int e^x (f(x) + f'(x)) dx$ where c is the constant of integration
- (A) $e^x \tan x + c$ (B) $e^x \cot x + c$ (C) $e^x \operatorname{cosec}^2 x + c$ (D) $e^x \sec^2 x + c$
12. $\int \frac{x^2(1 - \ln x)}{\ln^4 x - x^4} dx$ equals
- (A) $\frac{1}{2} \ln\left(\frac{x}{\ln x}\right) - \frac{1}{4} \ln(\ln^2 x - x^2) + C$ (B) $\frac{1}{4} \ln\left(\frac{\ln x - x}{\ln x + x}\right) - \frac{1}{2} \tan^{-1}\left(\frac{\ln x}{x}\right) + C$
(C) $\frac{1}{4} \ln\left(\frac{\ln x + x}{\ln x - x}\right) + \frac{1}{2} \tan^{-1}\left(\frac{\ln x}{x}\right) + C$ (D) $\frac{1}{4} \left(\ln\left(\frac{\ln x - x}{\ln x + x}\right) + \tan^{-1}\left(\frac{\ln x}{x}\right) \right) + C$
13. If $\int \frac{(2x+3)dx}{x(x+1)(x+2)(x+3)+1} = C - \frac{1}{f(x)}$, where $f(x)$ is of the form of $ax^2 + bx + c$ then $(a + b + c)$ equals
- (A) 4 (B) 5 (C) 6 (D) none
14. $\int e^x \left(\frac{x^2 - 3}{(x-1)^2} \right) dx$ is equal to -
- (A) $e^x \frac{(x+3)}{(x-1)} + C$ (B) $e^x \left(\frac{x-3}{x-1} \right) + C$ (C) $e^x \left(\frac{x+1}{x-1} \right) + C$ (D) $e^x \left(\frac{1}{x-1} \right)^2 + C$
(where C is constant of integration)

15. $\int \frac{x^3}{(2x^2+1)^3} dx$ is equal to-

(A) $\frac{1}{4} \left(2 + \frac{1}{x^2} \right)^{-2} + C$

(B) $-\frac{1}{4} \left(2 + \frac{1}{x^2} \right)^{-2} + C$

(C) $\frac{1}{2} \left(2 + \frac{1}{x^2} \right)^{-2} + C$

(D) $\frac{1}{4} \left(2 + \frac{1}{x^2} \right)^2 + C$

(where 'C' is integration constant)

EXERCISE (O-2)

1. $\int \frac{\cot^{-1}(e^x)}{e^x} dx$ is equal to -

(A) $\frac{1}{2} \ln(e^{2x} + 1) - \frac{\cot^{-1}(e^x)}{e^x} + x + c$

(B) $\frac{1}{2} \ln(e^{2x} + 1) + \frac{\cot^{-1}(e^x)}{e^x} + x + c$

(C) $\frac{1}{2} \ln(e^{2x} + 1) - \frac{\cot^{-1}(e^x)}{e^x} - x + c$

(D) $\frac{1}{2} \ln(e^{2x} + 1) + \frac{\cot^{-1}(e^x)}{e^x} - x + c$

2. $\int (\sin(101x) \cdot \sin^{99} x) dx$ equals

(A) $\frac{\sin(100x)(\sin x)^{100}}{100} + C$

(B) $\frac{\cos(100x)(\sin x)^{100}}{100} + C$

(C) $\frac{\cos(100x)(\cos x)^{100}}{100} + C$

(D) $\frac{\sin(100x)(\sin x)^{101}}{101} + C$

3. The evaluation of $\int \frac{p x^{p+2q-1} - q x^{q-1}}{x^{2p+2q} + 2x^{p+q} + 1} dx$ is

(A) $-\frac{x^p}{x^{p+q} + 1} + C$

(B) $\frac{x^q}{x^{p+q} + 1} + C$

(C) $-\frac{x^q}{x^{p+q} + 1} + C$

(D) $\frac{x^p}{x^{p+q} + 1} + C$

4. $\int \sec^2 \theta (\sec \theta + \tan \theta)^2 d\theta$

(A) $\frac{(\sec \theta + \tan \theta)}{2} [2 + \tan \theta (\sec \theta + \tan \theta)] + C$

(B) $\frac{(\sec \theta + \tan \theta)}{3} [2 + 4 \tan \theta (\sec \theta + \tan \theta)] + C$

(C) $\frac{(\sec \theta + \tan \theta)}{3} [2 + \tan \theta (\sec \theta + \tan \theta)] + C$

(D) $\frac{3(\sec \theta + \tan \theta)}{2} [2 + \tan \theta (\sec \theta + \tan \theta)] + C$

5. The integral $\int \sqrt{\cot x} e^{\sqrt{\sin x}} \sqrt{\cos x} dx$ equals

(A) $\frac{\sqrt{\tan x} e^{\sqrt{\sin x}}}{\sqrt{\cos x}} + C$

(B) $2e^{\sqrt{\sin x}} + C$

(C) $-\frac{1}{2} e^{\sqrt{\sin x}} + C$

(D) $\frac{\sqrt{\cot x} e^{\sqrt{\sin x}}}{2\sqrt{\cos x}} + C$

Paragraph for Question Nos. 6 to 8

A curve is represented parametrically by the equations $x = e^t \cos t$ and $y = e^t \sin t$ where t is a parameter. Then

6. The relation between the parameter 't' and the angle α between the tangent to the given curve and the x-axis is given by, 't' equals

$$(A) \frac{\pi}{2} - \alpha$$

(B) $\frac{\pi}{4} + \alpha$

$$(C) \propto -\frac{\pi}{4}$$

(D) $\frac{\pi}{4} - \alpha$

7. The value of $\frac{d^2y}{dx^2}$ at the point where $t = 0$ is

(A) 1

(B) 2

(C) - 2

(D) 3

8. If $F(t) = \int (x + y) dt$ then the value of $F\left(\frac{\pi}{2}\right) - F(0)$ is

(A) 1

(B) – 1

(C) $e^{\pi/2}$

(D) 0

Multiple Correct :

9. Which one of the following is **FALSE** ?

$$(A) \quad x \cdot \int \frac{dx}{x} = x \ln |x| + C$$

(B) $x \cdot \int \frac{dx}{x} = x \ln |x| + Cx$

$$(C) \frac{1}{\cos x} \cdot \int \cos x \, dx = \tan x + C$$

(D) $\frac{1}{\cos x} \cdot \int \cos x \, dx = x + C$

- 10.** If $I_n = \int (\sin x)^n dx$ $n \in \mathbb{N}$, then $5I_4 - 6I_6$ is equal to-

(A) $\sin x.(\cos x)^5 + C$

(B) $\cos x.(\sin x)^5 + C$

$$(C) \frac{\sin 2x}{8} [\cos^2 2x + 1 - 2 \cos 2x] + C$$

(D) $\frac{\sin 2x}{8} [\cos^2 2x + 1 + 2 \cos 2x] + C$

- 11.** Let $f(x) = \sin^3 x + \sin^3\left(x + \frac{2\pi}{3}\right) + \sin^3\left(x + \frac{4\pi}{3}\right)$ then the primitive of $f(x)$ w.r.t. x is

(A) $-\frac{3 \sin 3x}{4} + C$

(B) $\frac{1}{2} \cos^2 \left(\frac{3x}{2} \right) + C$

(C) $\frac{\sin 3x}{4} + C$

(D) $\frac{\cos 3x}{4} + C$

where C is an arbitrary constant.

- 12.** Suppose $J = \int \frac{\sin^2 x + \sin x}{1 + \sin x + \cos x} dx$ and $K = \int \frac{\cos^2 x + \cos x}{1 + \sin x + \cos x} dx$. If C is an arbitrary constant of integration then which of the following is/are correct?

(A) $J = \frac{1}{2} (x - \sin x + \cos x) + C$

(B) $J = K - (\sin x + \cos x) + C$

(C) $J = X - K + C$

(D) $K = \frac{1}{2}(x - \sin x + \cos x) + C$

Match the column :

13. Column-I

Column-II

(A) Let $f(x) = \int x^{\sin x} (1 + x \cos x \cdot \ln x + \sin x) dx$ and $f\left(\frac{\pi}{2}\right) = \frac{\pi^2}{4}$

(P) rational

then the value of $f(\pi)$ is

(B) Let $g(x) = \int \frac{1 + 2 \cos x}{(\cos x + 2)^2} dx$ and $g(0) = 0$

(Q) irrational

then the value of $g\left(\frac{\pi}{2}\right)$ is

(R) integral

(C) If real numbers x and y satisfy $(x + 5)^2 + (y - 12)^2 = (14)^2$ then

the minimum value of $\sqrt{(x^2 + y^2)}$ is

(S) prime

(D) Let $k(x) = \int \frac{(x^2 + 1) dx}{\sqrt[3]{x^3 + 3x + 6}}$ and $k(-1) = \frac{1}{\sqrt[3]{2}}$ then the value

of $k(-2)$ is

14. Column-I

Column-II

(A) $\int \frac{x^2(x^6 + x^5 - 1) dx}{(2x^6 + 3x^5 + 2)^2}$

(P) $\frac{1}{6} \left(\frac{1}{x^{-3}} + \frac{3}{x^{-2}} \right)^{\frac{1}{2}} + C$

(B) $\int \frac{(x^5 + x^4 + x^2) dx}{\sqrt{4x^7 + 5x^6 + 10x^4}}$

(Q) $\frac{1}{2} (1 + x^{-2} + x^{-5})^{-2} + C$

(C) $\int \frac{(2x^{12} + 5x^9) dx}{(x^5 + x^3 + 1)^3}$

(R) $-\frac{1}{6} (2x^3 + 3x^2 + 2x^{-3})^{-1} + C$

(where C is the constant of integration.)

(S) $x \left(\frac{x^3}{25} + \frac{x^2}{20} + \frac{1}{10} \right)^{\frac{1}{2}} + C$

EXERCISE (S-1)

1. $\int \frac{(\sqrt{x}+1)(x^2-\sqrt{x})}{x\sqrt{x}+x+\sqrt{x}} dx$
2. A function g defined for all positive real numbers, satisfies $g'(x^2) = x^3$ for all $x > 0$ and $g(1) = 1$. Compute $g(4)$.
3. $\int \left[\sin \alpha \sin(x-\alpha) + \sin^2 \left(\frac{x}{2} - \alpha \right) \right] dx$
4. $\int \frac{x^2+3}{x^6(x^2+1)} dx$
5. $\int \frac{dx}{\cot \frac{x}{2} \cdot \cot \frac{x}{3} \cdot \cot \frac{x}{6}}$
6. $\int \sqrt{\frac{\operatorname{cosec} x - \cot x}{\operatorname{cosec} x + \cot x}} \cdot \frac{\sec x}{\sqrt{1+2 \sec x}} dx$
7. $\int \frac{\ln \left(\ln \left(\frac{1+x}{1-x} \right) \right)}{1-x^2} dx$
8. $\int \left[\left(\frac{x}{e} \right)^x + \left(\frac{e}{x} \right)^x \right] \ell n x dx$
9. $\int \frac{x^5+3x^4-x^3+8x^2-x+8}{x^2+1} dx$
10. $\int \frac{(\sqrt{x}+1)dx}{\sqrt{x}(\sqrt[3]{x}+1)}$
11. $\int \sin^{-1} \sqrt{\frac{x}{a+x}} dx$
12. $\int \frac{x \ell n x}{(x^2-1)^{3/2}} dx$
13. $\int \left[\frac{\sqrt{x^2+1} \left[\ell n(x^2+1) - 2 \ell n x \right]}{x^4} \right] dx$
14. $\int \frac{\tan 2\theta}{\sqrt{\cos^6 \theta + \sin^6 \theta}} d\theta$
15. $\int \frac{3x^2+1}{(x^2-1)^3} dx$
16. $\int \frac{(ax^2-b) dx}{x\sqrt{c^2x^2-(ax^2+b)^2}}$
17. $\int \frac{(e^{\sqrt{x}} - e^{-\sqrt{x}}) \cos \left(e^{\sqrt{x}} + e^{-\sqrt{x}} + \frac{\pi}{4} \right) + (e^{\sqrt{x}} + e^{-\sqrt{x}}) \cos \left(e^{\sqrt{x}} - e^{-\sqrt{x}} + \frac{\pi}{4} \right)}{\sqrt{x}} dx$
18. $\int \frac{x^2+x}{(e^x+x+1)^2} dx$
19. $\int \frac{e^{\cos x} (x \sin^3 x + \cos x)}{\sin^2 x} dx$
20. $\int \frac{5x^4+4x^5}{(x^5+x+1)^2} dx$
21. $\int (\sin x)^{-1/3} (\cos x)^{-1/3} dx$
22. $\int \frac{dx}{\sin x + \sec x}$
23. $\int \frac{4x^5-7x^4+8x^3-2x^2+4x-7}{x^2(x^2+1)^2} dx$
24. Let $\int \frac{f'(x)g(x)-g'(x)f(x)}{(f(x)+g(x))\sqrt{f(x)g(x)-g^2(x)}} dx = \sqrt{m} \tan^{-1} \left(\sqrt{\frac{f(x)-g(x)}{ng(x)}} \right) + C$,

where $m, n \in \mathbb{N}$ and 'C' is constant of integration ($g(x) > 0$). Find the value of $(m^2 + n^2)$.

25. If the value $\int \frac{1 - (\cot x)^{2008}}{\tan x + (\cot x)^{2009}} dx = \frac{1}{k} \ln |\sin^k x + \cos^k x| + C$, then find k.
26. $\int \frac{dx}{(x - \alpha)\sqrt{(x - \alpha)(x - \beta)}}$
27. Suppose $\int \frac{1 - 7\cos^2 x}{\sin^7 x \cos^2 x} dx = \frac{g(x)}{\sin^7 x} + C$, where C is arbitrary constant of integration. Then find the value of $g'(0) + g''\left(\frac{\pi}{4}\right)$

EXERCISE (S-2)

1. $\int \frac{\tan(\ln x) \tan\left(\ln \frac{x}{2}\right) \tan(\ln 2)}{x} dx$
2. $\int \frac{e^x (2 - x^2)}{(1 - x)\sqrt{1 - x^2}} dx$
3. $\int \frac{dx}{\left(x + \sqrt{x(1+x)}\right)^2}$
4. $\int \frac{dx}{\sqrt{\sin^3 x \sin(x + \alpha)}}$
5. $\int \frac{x}{(7x - 10 - x^2)^{3/2}} dx$
6. $\int \frac{(1 + x^2) dx}{1 - 2x^2 \cos \alpha + x^4} \alpha \in (0, \pi)$
7. $\int \sqrt{\frac{\sin(x - a)}{\sin(x + a)}} dx$
8. $\int \frac{\cos^2 x}{1 + \tan x} dx$
9. Let $f(x)$ is a quadratic function such that $f(0) = 1$ and $\int \frac{f(x) dx}{x^2(x+1)^3}$ is a rational function, find the value of $f'(0)$
10. $\int \frac{\cos x - \sin x}{7 - 9\sin 2x} dx$
11. $\int \cos 2\theta \cdot \ln \frac{\cos \theta + \sin \theta}{\cos \theta - \sin \theta} d\theta$

Match of Column :

12. $I_1 = \int \tan x \tan(ax + b) dx$ and $I_2 = \int \cot x \cot(ax + b) dx$

Column-I

- (A) value of I_1 for $a = 1$ is
- (B) value of I_2 for $a = 1$ is
- (C) value of I_1 for $a = -1$ is
- (D) value of I_2 for $a = -1$ is

Column-II

- (P) $x - \cot b \ln \frac{\cos(x - b)}{\cos x} + C$
- (Q) $\cot b \ln \frac{\sin x}{\sin(x + b)} - x + C$
- (R) $\cot b \ln \left(\frac{\cos x}{\cos(x + b)} \right) - x + C$
- (S) $x + \cot b \ln \left(\frac{\sin x}{\sin(b - x)} \right) + C$

EXERCISE (JM)

1. If the integral $\int \frac{5 \tan x}{\tan x - 2} dx = x + a \ln|\sin x - 2 \cos x| + k$ then a is equal to : [AIEEE-2012]

- (1) 2 (2) -1 (3) -2 (4) 1

2. If $\int f(x)dx = \Psi(x)$, then $\int x^5 f(x^3)dx$ is equal to : [JEE-MAIN-2013]

- (1) $\frac{1}{3} [x^3 \Psi(x^3) - \int x^2 \Psi(x^3) dx] + C$ (2) $\frac{1}{3} x^3 \Psi(x^3) - 3 \int x^3 \Psi(x^3) dx + C$
(3) $\frac{1}{3} x^3 \Psi(x^3) - \int x^2 \Psi(x^3) dx + C$ (4) $\frac{1}{3} [x^3 \Psi(x^3) - \int x^3 \Psi(x^3) dx] + C$

3. The integral $\int \left(1 + x - \frac{1}{x}\right) e^{x + \frac{1}{x}} dx$ is equal to : [JEE-MAIN-2014]

- (1) $(x-1)e^{x + \frac{1}{x}} + c$ (2) $xe^{x + \frac{1}{x}} + c$ (3) $(x+1)e^{x + \frac{1}{x}} + c$ (4) $-xe^{x + \frac{1}{x}} + c$

4. The integral $\int \frac{dx}{x^2(x^4 + 1)^{\frac{3}{4}}}$ equals : [JEE-MAIN-2015]

- (1) $-(x^4 + 1)^{\frac{1}{4}} + c$ (2) $-\left(\frac{x^4 + 1}{x^4}\right)^{\frac{1}{4}} + c$ (3) $\left(\frac{x^4 + 1}{x^4}\right)^{\frac{1}{4}} + c$ (4) $(x^4 + 1)^{\frac{1}{4}} + c$

5. The integral $\int \frac{2x^{12} + 5x^9}{(x^5 + x^3 + 1)^3} dx$ is equal to :- [JEE-MAIN 2016]

- (1) $\frac{-x^{10}}{2(x^5 + x^3 + 1)^2} + C$ (2) $\frac{-x^5}{(x^5 + x^3 + 1)^2} + C$ (3) $\frac{x^{10}}{2(x^5 + x^3 + 1)^2} + C$ (4) $\frac{x^5}{2(x^5 + x^3 + 1)^2} + C$

where C is an arbitrary constant.

6. Let $I_n = \int \tan^n x dx, (n > 1)$. $I_4 + I_6 = a \tan^5 x + bx^5 + C$, where C is a constant of integration, then the ordered pair (a, b) is equal to :- [JEE-MAIN 2017]

- (1) $\left(-\frac{1}{5}, 0\right)$ (2) $\left(-\frac{1}{5}, 1\right)$ (3) $\left(\frac{1}{5}, 0\right)$ (4) $\left(\frac{1}{5}, -1\right)$

7. The integral $\int \frac{\sin^2 x \cos^2 x}{(\sin^5 x + \cos^3 x \sin^2 x + \sin^3 x \cos^2 x + \cos^5 x)^2} dx$ is equal to [JEE-MAIN 2018]

- (1) $\frac{-1}{3(1 + \tan^3 x)} + C$ (2) $\frac{1}{1 + \cot^3 x} + C$ (3) $\frac{-1}{1 + \cot^3 x} + C$ (4) $\frac{1}{3(1 + \tan^3 x)} + C$

(where C is a constant of integration)

8. For $x^2 \neq n\pi + 1$, $n \in \mathbb{N}$ (the set of natural numbers), the integral $\int x \sqrt{\frac{2\sin(x^2 - 1) - \sin 2(x^2 - 1)}{2\sin(x^2 - 1) + \sin 2(x^2 - 1)}} dx$ is

equal to :

(where c is a constant of integration)

(1) $\log_e \left| \sec \left(\frac{x^2 - 1}{2} \right) \right| + c$

(2) $\log_e \left| \frac{1}{2} \sec^2(x^2 - 1) \right| + c$

(3) $\frac{1}{2} \log_e \left| \sec^2 \left(\frac{x^2 - 1}{2} \right) \right| + c$

(4) $\frac{1}{2} \log_e |\sec(x^2 - 1)| + c$ [JEE-MAIN 2019]

9. If $\int \frac{\sqrt{1-x^2}}{x^4} dx = A(x) (\sqrt{1-x^2})^m + C$, for a suitable chosen integer m and a function $A(x)$, where C

is a constant of integration then $(A(x))^m$ equals :

[JEE-MAIN 2019]

(1) $\frac{-1}{3x^3}$

(2) $\frac{-1}{27x^9}$

(3) $\frac{1}{9x^4}$

(4) $\frac{1}{27x^6}$

10. The integral $\int \cos(\log_e x) dx$ is equal to : (where C is a constant of integration) [JEE-MAIN 2019]

(1) $\frac{x}{2} [\sin(\log_e x) - \cos(\log_e x)] + C$

(2) $\frac{x}{2} [\cos(\log_e x) + \sin(\log_e x)] + C$

(3) $x [\cos(\log_e x) + \sin(\log_e x)] + C$

(4) $x [\cos(\log_e x) - \sin(\log_e x)] + C$

EXERCISE (JA)

1. $\int \frac{x^2 - 1}{x^3 \sqrt{2x^4 - 2x^2 + 1}} dx$ is equal to -

[JEE 2006, (3M, -1M)]

(A) $\frac{\sqrt{2x^4 - 2x^2 + 1}}{x^2} + c$

(B) $\frac{\sqrt{2x^4 - 2x^2 + 1}}{x^3} + c$

(C) $\frac{\sqrt{2x^4 - 2x^2 + 1}}{x} + c$

(D) $\frac{\sqrt{2x^4 - 2x^2 + 1}}{2x^2} + c$

2. Let $f(x) = \frac{x}{(1+x^n)^{1/n}}$ for $n \geq 2$ and $g(x) = \underbrace{(f \circ f \circ \dots \circ f)}_{f \text{ occurs } n \text{ times}}(x)$. Then $\int x^{n-2} g(x) dx$ equals.

(A) $\frac{1}{n(n-1)} (1+nx^n)^{1-\frac{1}{n}} + K$

(B) $\frac{1}{n-1} (1+nx^n)^{1-\frac{1}{n}} + K$

[JEE 2007, 3]

(C) $\frac{1}{n(n+1)} (1+nx^n)^{1+\frac{1}{n}} + K$

(D) $\frac{1}{n+1} (1+nx^n)^{1+\frac{1}{n}} + K$

3. Let $F(x)$ be an indefinite integral of $\sin^2 x$.

[JEE 2007, 3]

Statement-1 : The function $F(x)$ satisfies $F(x + \pi) = F(x)$ for all real x .

because

Statement-2 : $\sin^2(x + \pi) = \sin^2 x$ for all real x .

(A) Statement-1 is True, Statement-2 is True ; Statement-2 is a correct explanation for Statement-1.

(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.

(C) Statement-1 is True, Statement-2 is False.

(D) Statement-1 is False, Statement-2 is True.

4. Let $I = \int \frac{e^x}{e^{4x} + e^{2x} + 1} dx$, $J = \int \frac{e^{-x}}{e^{-4x} + e^{-2x} + 1} dx$. Then, for an arbitrary constant c , the value of $J - I$ equals [JEE 2008, 3 (-1)]

(A) $\frac{1}{2} \ln \left(\frac{e^{4x} - e^{2x} + 1}{e^{4x} + e^{2x} + 1} \right) + c$

(B) $\frac{1}{2} \ln \left(\frac{e^{2x} + e^x + 1}{e^{2x} - e^x + 1} \right) + c$

(C) $\frac{1}{2} \ln \left(\frac{e^{2x} - e^x + 1}{e^{2x} + e^x + 1} \right) + c$

(D) $\frac{1}{2} \ln \left(\frac{e^{4x} + e^{2x} + 1}{e^{4x} - e^{2x} + 1} \right) + c$

5. The integral $\int \frac{\sec^2 x}{(\sec x + \tan x)^{9/2}} dx$ equals (for some arbitrary constant K) [JEE 2012, 3M, -1M]

(A) $-\frac{1}{(\sec x + \tan x)^{11/2}} \left\{ \frac{1}{11} - \frac{1}{7} (\sec x + \tan x)^2 \right\} + K$

(B) $\frac{1}{(\sec x + \tan x)^{11/2}} \left\{ \frac{1}{11} - \frac{1}{7} (\sec x + \tan x)^2 \right\} + K$

(C) $-\frac{1}{(\sec x + \tan x)^{11/2}} \left\{ \frac{1}{11} + \frac{1}{7} (\sec x + \tan x)^2 \right\} + K$

(D) $\frac{1}{(\sec x + \tan x)^{11/2}} \left\{ \frac{1}{11} + \frac{1}{7} (\sec x + \tan x)^2 \right\} + K$

ANSWER KEY

INDEFINITE INTEGRATION

EXERCISE (O-1)

1. C 2. B 3. B 4. B 5. A 6. B 7. D 8. B
9. A 10. B 11. A 12. B 13. B 14. C 15. A

EXERCISE (O-2)

1. C 2. A 3. C 4. C 5. B 6. C 7. B 8. C
9. A,C,D 10. B,C 11. B,D 12. B,C 13. (A) Q ; (B) P ; (C) P,R ; (D) P,R,S
14. (A) R ; (B) S ; (C) Q

EXERCISE (S-1)

1. $\frac{x^2}{2} - x + C$ 2. $\frac{67}{5}$ 3. $\frac{1}{2}(x - \sin x) + C$ 4. $C - \frac{2}{x} + \frac{2}{3x^3} - \frac{3}{5x^5} - 2 \tan^{-1} x$
5. $2 \ln \left(\sec \frac{x}{2} \right) - 3 \ln \left(\sec \frac{x}{3} \right) - 6 \ln \left(\sec \frac{x}{6} \right) + C$ 6. $\sin^{-1} \left(\frac{1}{2} \sec^2 \frac{x}{2} \right) + C$
7. $\frac{1}{2} \left[\ln \left(\frac{1+x}{1-x} \right) \cdot \ln \left(\ln \frac{1+x}{1-x} \right) - \ln \left(\frac{1+x}{1-x} \right) \right] + C$ 8. $\left(\frac{x}{e} \right)^x - \left(\frac{e}{x} \right)^x + C$
9. $\frac{x^4}{4} + x^3 - x^2 + 5x + \frac{1}{2} \ln(x^2 + 1) + 3 \tan^{-1} x + C$ 10. $6 \left[\frac{t^4}{4} - \frac{t^2}{2} + t + \frac{1}{2} \ln(1+t^2) - \tan^{-1} t \right] + C$ where $t = x^{1/6}$
11. $(a+x) \arctan \sqrt{\frac{x}{a}} - \sqrt{ax} + C$ 12. $\operatorname{arcsec} x - \frac{\ln x}{\sqrt{x^2-1}} + C$ 13. $\frac{(x^2+1)\sqrt{x^2+1}}{9x^3} \left[2 - 3 \ln \left(1 + \frac{1}{x^2} \right) \right]$
14. $\ln \left(\frac{1 + \sqrt{1 + 3 \cos^2 2\theta}}{\cos 2\theta} \right) + C$ 15. $C - \frac{x}{(x^2-1)^2}$ 16. $\sin^{-1} \left(\frac{ax^2+b}{cx} \right) + k$
17. $2\sqrt{2} \left(\cos(e^{-\sqrt{x}}) \right) \left(\sin(e^{\sqrt{x}}) + \cos(e^{\sqrt{x}}) \right) + C$ 18. $C - \ln(1 + (x+1)e^{-x}) - \frac{1}{1 + (x+1)e^{-x}}$
19. $C - e^{\cos x}(x + \operatorname{cosec} x)$ 20. $C - \frac{x+1}{x^5+x+1}$ or $C + \frac{x^5}{x^5+x+1}$ 21. $-\frac{3(1+4 \tan^2 x)}{8(\tan x)^{8/3}} + C$
22. $\frac{1}{2\sqrt{3}} \ln \frac{\sqrt{3} + \sin x - \cos x}{\sqrt{3} - \sin x + \cos x} + \arctan(\sin x + \cos x) + C$ 23. $4 \ln x + \frac{7}{x} + 6 \tan^{-1}(x) + \frac{6x}{1+x^2} + C$
24. 8 25. 2010 26. $\frac{-2}{\alpha-\beta} \sqrt{\frac{x-\beta}{x-\alpha}} + C$ 27. 5

EXERCISE (S-2)

1. $\ln\left(\frac{\sec(\ln x)}{\sec(\ln x/2)x^{\tan(\ln 2)}}\right) + C$
2. $e^x \sqrt{\frac{1+x}{1-x}} + c$
3. $2\ln\frac{t}{2t+1} + \frac{1}{2t+1} + C$, when $t = x + \sqrt{x^2 + x}$
4. $C - \frac{2}{\sin \alpha} \sqrt{\frac{\sin(x + \alpha)}{\sin x}}$
5. $\frac{2(7x - 20)}{9\sqrt{7x - 10 - x^2}} + C$
6. $\frac{1}{2} \left(\operatorname{cosec} \frac{\alpha}{2} \right) \tan^{-1} \left(\left(\frac{x^2 - 1}{2x} \right) \operatorname{cosec} \frac{\alpha}{2} \right)$
7. $\cos a \cdot \arccos \left(\frac{\cos x}{\cos a} \right) - \sin a \cdot \ln \left(\sin x + \sqrt{\sin^2 x - \sin^2 a} \right) + C$
8. $\frac{1}{4} \ln(\cos x + \sin x) + \frac{x}{2} + \frac{1}{8} (\sin 2x + \cos 2x) + C$
9. 3
10. $\frac{1}{24} \ln \left| \frac{(4 + 3 \sin x + 3 \cos x)}{(4 - 3 \sin x - 3 \cos x)} \right| + C$
11. $\frac{1}{2} (\sin 2\theta) \ln \left(\frac{\cos \theta + \sin \theta}{\cos \theta - \sin \theta} \right) - \frac{1}{2} \ln(\sec 2\theta) + C$
12. (A) R ; (B) Q; (C) P ; (D) S

EXERCISE (JM)

1. 1
2. 3
3. 2
4. 2
5. 3
6. 3
7. 1
8. 1 or 3
9. 2
10. 2

EXERCISE (JA)

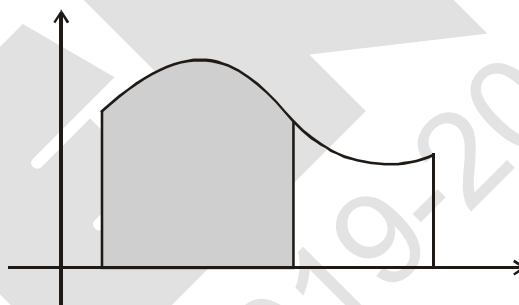
1. D
2. A
3. D
4. C
5. C

DEFINITE INTEGRATION

A definite integral is denoted by $\int_a^b f(x) dx$ which represent the algebraic area bounded by the curve $y = f(x)$, the ordinates $x = a$, $x = b$ and the x axis.

1. THE FUNDAMENTAL THEOREM OF CALCULUS :

The Fundamental Theorem of Calculus is appropriately named because it establishes a connection between the two branches of calculus : differential calculus and integral calculus. Differential calculus arose from the tangent problem, whereas integral calculus arose from a seemingly unrelated problem, the area problem. Newton's teacher at Cambridge, Isaac Barrow (1630-1677), discovered that these two problems are actually closely related. In fact, he realized that differentiation and integration are inverse processes. The Fundamental Theorem of Calculus gives the precise inverse relationship between the derivative and the integral. It was Newton and Leibnitz who exploited this relationship and used it to develop calculus into a systematic mathematical method. In particular, they saw that the Fundamental Theorem enabled them to compute areas and integrals very easily without having to compute them as limits of sums.



The Fundamental Theorem of Calculus, Part 1

If f is continuous on $[a, b]$, then the function g defined by

$$g(x) = \int_a^x f(t) dt \quad a \leq x \leq b$$

is continuous on $[a, b]$ and differentiable on (a, b) , and $g'(x) = f(x)$.

The Fundamental Theorem of Calculus, Part 2

If f is continuous on $[a, b]$, then

$$\int_a^b f(x) dx = F(b) - F(a)$$

where F is any antiderivative of f , that is, a function such that $F' = f$.

Note : If $\int_a^b f(x) dx = 0 \Rightarrow$ then the equation $f(x) = 0$ has atleast one root lying in (a, b) provided f is a continuous function in (a, b) .

2. PROPERTIES OF DEFINITE INTEGRAL :

(a) $\int_a^b f(x) dx = \int_a^b f(t) dt$ provided f is same

$$(b) \int_a^b f(x) dx = - \int_b^a f(x) dx$$

(c) $\int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx$, where c may lie inside or outside the interval $[a, b]$. This property is to be used when f is piecewise continuous in (a, b) .

Illustration 1 : If $f(x) = \begin{cases} x^2, & 0 < x < 2 \\ 3x - 4, & 2 \leq x < 3 \end{cases}$ then evaluate $\int_0^3 f(x) dx$

Solution :

$$\begin{aligned} \int_0^3 f(x) dx &= \int_0^2 f(x) dx + \int_2^3 f(x) dx = \int_0^2 x^2 dx + \int_2^3 (3x - 4) dx \\ &= \left(\frac{x^3}{3} \right)_0^2 + \left(\frac{3x^2}{2} - 4x \right)_2^3 = \frac{8}{3} + \frac{27}{2} - 12 - 6 + 8 = 37/6 \end{aligned}$$

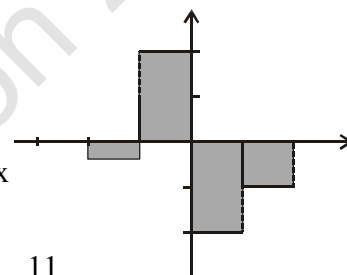
Ans.

Illustration 2 : If $f(x) = \begin{cases} 3[x] - 5 \frac{|x|}{x}, & x \neq 0 \\ 2, & x = 0 \end{cases}$ then $\int_{-3/2}^2 f(x) dx$ is equal to ($[.]$ denotes the greatest integer function)

- (A) $-\frac{11}{2}$ (B) $-\frac{7}{2}$ (C) -6 (D) $-\frac{17}{2}$

Solution :

$$\begin{aligned} 3[x] - 5 \frac{|x|}{x} &= 3[x] - 5, \text{ if } x > 0 \\ &= 3[x] + 5, \text{ if } x < 0 \\ \Rightarrow \int_{-3/2}^2 f(x) dx &= \int_{-3/2}^{-1} (-1) dx + \int_{-1}^0 (2) dx + \int_0^1 (-5) dx + \int_1^2 (-2) dx \\ &= -1 \left(-1 + \frac{3}{2} \right) + 2(1) + 1(-5) + (-2) = -\frac{1}{2} + 2 - 5 - 2 = -\frac{11}{2} \end{aligned}$$



Ans. (A)

Illustration 3 : The value of $\int_1^2 (x^{[x^2]} + [x^2]^x) dx$, where $[.]$ denotes the greatest integer function, is equal to -

- (A) $\frac{5}{4} + \sqrt{3} + (2^{\sqrt{3}} - 2^{\sqrt{2}}) + \frac{1}{\log 3} (9 - 3^{\sqrt{3}})$
 (B) $\frac{5}{4} + \sqrt{3} + \frac{\sqrt{2}}{3} + \frac{1}{\log 2} (2^{\sqrt{3}} - 2^{\sqrt{2}}) + \frac{1}{\log 3} (9 - 3^{\sqrt{3}})$
 (C) $\frac{5}{4} + \frac{\sqrt{2}}{3} + \frac{1}{\log 2} (2^{\sqrt{3}} - 2^{\sqrt{2}}) + \frac{1}{\log 3} (9 - 3^{\sqrt{3}})$
 (D) none of these

Solution : We have, $I = \int_1^2 (x^{[x^2]} + [x^2]^x) dx = \int_1^{\sqrt{2}} (x+1) dx + \int_{\sqrt{2}}^{\sqrt{3}} (x^2 + 2^x) dx + \int_{\sqrt{3}}^2 (x^3 + 3^x) dx$

$$= \left(\frac{x^2}{2} + x \right)_1^{\sqrt{2}} + \left(\frac{x^3}{3} + \frac{2^x}{\log 2} \right)_{\sqrt{2}}^{\sqrt{3}} + \left(\frac{x^4}{4} + \frac{3^x}{\log 3} \right)_{\sqrt{3}}^2$$

$$= \frac{5}{4} + \sqrt{3} + \frac{\sqrt{2}}{3} + \frac{1}{\log 2} (2^{\sqrt{3}} - 2^{\sqrt{2}}) + \frac{1}{\log 3} (3^2 - 3^{\sqrt{3}})$$

Ans. (B)

Illustration 4 : Evaluate : $\int_{-10}^{20} [\cot^{-1} x] dx$. Here $[.]$ is the greatest integer function.

Solution : $I = \int_{-10}^{20} [\cot^{-1} x] dx$, we know $\cot^{-1} x \in (0, \pi) \forall x \in \mathbb{R}$

$$\text{Thus } [\cot^{-1} x] = \begin{cases} 3, & x \in (-\infty, \cot 3) \\ 2, & x \in (\cot 3, \cot 2) \\ 1, & x \in (\cot 2, \cot 1) \\ 0 & x \in (\cot 1, \infty) \end{cases}$$

$$\text{Hence } I = \int_{-10}^{\cot 3} 3 dx + \int_{\cot 3}^{\cot 2} 2 dx + \int_{\cot 2}^{\cot 1} 1 dx + \int_{\cot 1}^{20} 0 dx = 30 + \cot 1 + \cot 2 + \cot 3$$

Ans.**Do yourself -1 :****Evaluate :**

(i) $\int_0^3 |x^2 - x - 2| dx$

(ii) $\int_0^4 \{x\} dx$, where $\{.\}$ denotes fractional part of x .

(iii) $\int_0^{\pi/2} |\sin x - \cos x| dx$

(iv) If $f(x) = \begin{cases} 2 & 0 \leq x \leq 1 \\ x + [x] & 1 \leq x < 3 \end{cases}$, where $[.]$ denotes the greatest integer function. Evaluate $\int_0^2 f(x) dx$

(d) $\int_{-a}^a f(x) dx = \int_0^a [f(x) + f(-x)] dx = \begin{cases} 0 & \text{; if } f(x) \text{ is an odd function} \\ 2 \int_0^a f(x) dx & \text{; if } f(x) \text{ is an even function} \end{cases}$

Illustration 5 : Evaluate $\int_{-1/2}^{1/2} \cos x \ln \left(\frac{1+x}{1-x} \right) dx$

Solution : $f(-x) = \cos(-x) \ln \left(\frac{1-x}{1+x} \right) = -\cos x \ln \left(\frac{1+x}{1-x} \right) = -f(x)$

$\Rightarrow f(x)$ is odd

Hence, the value of the given integral = 0.

Ans.

Illustration 6 : If $f(x) = \begin{vmatrix} \cos x & e^{x^2} & 2x \cos^2 x / 2 \\ x^2 & \sec x & \sin x + x^3 \\ 1 & 2 & x + \tan x \end{vmatrix}$, then the value of $\int_{-\pi/2}^{\pi/2} (x^2 + 1)(f(x) + f''(x))dx$

(A) 1

(B) -1

(C) 2

(D) none of these

Solution : As, $f(x) = \begin{vmatrix} \cos x & e^{x^2} & 2x \cos^2 x / 2 \\ x^2 & \sec x & \sin x + x^3 \\ 1 & 2 & x + \tan x \end{vmatrix}$

$$\Rightarrow f(-x) = -f(x) \Rightarrow f(x) \text{ is odd}$$

$$\Rightarrow f(x) \text{ is even} \Rightarrow f'(x) \text{ is odd}$$

Thus, $f(x) + f''(x)$ is odd function let,

$$\phi(x) = (x^2 + 1) \cdot \{f(x) + f''(x)\}$$

$$\Rightarrow \phi(-x) = -\phi(x)$$

i.e. $\phi(x)$ is odd

$$\therefore \int_{-\pi/2}^{\pi/2} \phi(x)dx = 0$$

Ans. (D)

Do yourself -2 :

Evaluate :

(i) $\int_{-\pi/2}^{\pi/2} (x^2 \sin^3 x + \cos x)dx$

(ii) $\int_{-\pi/2}^{\pi/2} \ln \left[2 \left(\frac{4 - \sin \theta}{4 + \sin \theta} \right) \right] d\theta$

(e) $\int_a^b f(x)dx = \int_a^b f(a+b-x)dx$, In particular $\int_0^a f(x)dx = \int_0^a f(a-x)dx$

Illustration 7 : If f, g, h be continuous functions on $[0, a]$ such that $f(a-x) = -f(x)$, $g(a-x) = g(x)$ and $3h(x) - 4h(a-x) = 5$, then prove that $\int_0^a f(x)g(x)h(x)dx = 0$

Solution : $I = \int_0^a f(x)g(x)h(x)dx = \int_0^a f(a-x)g(a-x)h(a-x)dx = -\int_0^a f(x)g(x)h(a-x)dx$

$$7I = 3I + 4I$$

$$= \int_0^a f(x)g(x)\{3h(x) - 4h(a-x)\}dx = 5 \int_0^a f(x)g(x)dx = 0$$

$$(\text{since } f(a-x)g(a-x) = -f(x)g(x))$$

$$\Rightarrow I = 0$$

Ans.

Illustration 8 : Evaluate $\int_{-\pi}^{\pi} \frac{x \sin x}{e^x + 1} dx$

Solution : $I = \int_{-\pi}^0 \frac{x \sin x}{e^x + 1} dx + \int_0^{\pi} \frac{x \sin x}{e^x + 1} dx = I_1 + I_2$

where $I_1 = \int_{-\pi}^0 \frac{x \sin x}{e^x + 1} dx$

Put $x = -t \Rightarrow dx = -dt$

$$\Rightarrow I_1 = \int_{\pi}^0 \frac{(-t) \sin(-t)(-dt)}{e^{-t} + 1} = \int_0^{\pi} \frac{t \sin t dt}{e^{-t} + 1} = \int_0^{\pi} \frac{e^t t \sin t dt}{e^t + 1} = \int_0^{\pi} \frac{e^x x \sin x dx}{e^x + 1}$$

Hence $I = I_1 + I_2 = \int_0^{\pi} \frac{e^x x \sin x}{e^x + 1} dx + \int_0^{\pi} \frac{x \sin x}{e^x + 1} dx$

$$I = \int_0^{\pi} x \sin x dx = \int_0^{\pi} (\pi - x) \sin(\pi - x) dx = \pi \int_0^{\pi} \sin x dx - I$$

$$\Rightarrow 2I = \pi \int_0^{\pi} \sin x dx = \pi [-\cos x]_0^{\pi} = 2\pi \Rightarrow I = \pi$$

Ans.

Illustration 9 : Evaluate $\int_0^2 \frac{dx}{(17 + 8x - 4x^2)[e^{6(1-x)} + 1]}$

Solution : Let $I = \int_0^2 \frac{dx}{(17 + 8x - 4x^2)[e^{6(1-x)} + 1]}$

Also $I = \int_0^2 \frac{dx}{(17 + 8x - 4x^2)[e^{-6(1-x)} + 1]} \left[\because \int_0^a f(x) dx = \int_0^a f(a-x) dx \right]$

Adding, we get

$$2I = \int_0^2 \frac{1}{17 + 8x - 4x^2} \left(\frac{1}{e^{6(1-x)} + 1} + \frac{1}{e^{-6(1-x)} + 1} \right) dx$$

$$= \int_0^2 \frac{1}{17 + 8x - 4x^2} dx = -\frac{1}{4} \int_0^2 \frac{dx}{x^2 - 2x - 17/4}$$

$$= -\frac{1}{4} \int_0^2 \frac{dx}{(x-1)^2 - 21/4} = -\frac{1}{4} \times \frac{1}{2 \times \frac{\sqrt{21}}{2}} \left[\log \left| \frac{x-1 - \frac{\sqrt{21}}{2}}{x-1 + \frac{\sqrt{21}}{2}} \right| \right]_0^2$$

$$= -\frac{1}{4\sqrt{21}} \left[\log \left| \frac{2x-2-\sqrt{21}}{2x-2+\sqrt{21}} \right| \right]_0^2 \Rightarrow I = -\frac{1}{8\sqrt{21}} \left[\log \left| \frac{2-\sqrt{21}}{2+\sqrt{21}} \right| - \log \left| \frac{2+\sqrt{21}}{\sqrt{21}-2} \right| \right]$$

$$= -\frac{1}{4\sqrt{21}} \left[\log \left| \frac{\sqrt{21}-2}{2+\sqrt{21}} \right| \right]$$

Ans.

Illustration 10: $\int_0^1 \cot^{-1}(1-x+x^2) dx$ equals -

- (A) $\frac{\pi}{2} + \log 2$ (B) $\frac{\pi}{2} - \log 2$ (C) $\pi - \log 2$ (D) none of these

Solution :
$$I = \int_0^1 \tan^{-1} \left(\frac{1}{1-x+x^2} \right) dx = \int_0^1 \tan^{-1} \left(\frac{x+(1-x)}{1-x(1-x)} \right) dx$$
$$= \int_0^1 [\tan^{-1} x + \tan^{-1}(1-x)] dx = \int_0^1 \tan^{-1} x dx + \int_0^1 \tan^{-1}(1-x) dx$$
$$= 2 \int_0^1 \tan^{-1} x dx = 2 \left[x \tan^{-1} x - \frac{1}{2} \log(1+x^2) \right]_0^1 = 2 \frac{\pi}{4} - \log 2 = \frac{\pi}{2} - \log 2$$
 Ans. (B)

Illustration 11 : $\int_0^{\pi/2} \frac{a \sin x + b \cos x}{\sin x + \cos x} dx$

Solution :
$$I = \int_0^{\pi/2} \frac{a \sin x + b \cos x}{\sin x + \cos x} dx \quad \dots(i)$$
$$I = \int_0^{\pi/2} \frac{a \sin(\pi/2-x) + b \cos(\pi/2-x)}{\sin(\pi/2-x) + \cos(\pi/2-x)} dx = \int_0^{\pi/2} \frac{a \cos x + b \sin x}{\sin x + \cos x} dx \quad \dots(ii)$$
$$\therefore 2I = \int_0^{\pi/2} \frac{(a+b)(\sin x + \cos x)}{\sin x + \cos x} dx = \int_0^{\pi/2} (a+b) dx = (a+b)\pi/2 \Rightarrow I = (a+b)\pi/4$$
 Ans.

Illustration 12 : $\int_0^{\pi/2} \frac{2^{\sin x}}{2^{\sin x} + 2^{\cos x}} dx$ equals -

- (A) 2 (B) π (C) $\frac{\pi}{4}$ (D) $\frac{\pi}{2}$

Solution :
$$I = \int_0^{\pi/2} \frac{2^{\sin x}}{2^{\sin x} + 2^{\cos x}} dx = \int_0^{\pi/2} \frac{2^{\sin(\pi/2-x)}}{2^{\sin(\pi/2-x)} + 2^{\cos(\pi/2-x)}} dx = \int_0^{\pi/2} \frac{2^{\cos x}}{2^{\cos x} + 2^{\sin x}} dx$$
$$2I = \int_0^{\pi/2} dx = \frac{\pi}{2} \Rightarrow I = \frac{\pi}{4}$$
 Ans. (C)

Do yourself - 3 :

Evaluate :

(i) $\int_1^5 \frac{\sqrt{x}}{\sqrt{6-x} + \sqrt{x}} dx$

(ii) $\int_{\pi/6}^{\pi/3} \frac{dx}{1 + \tan^5 x}$

(f) $\int_0^{2a} f(x) dx = \int_0^a f(x) dx + \int_0^a f(2a-x) dx = \begin{cases} 2 \int_0^a f(x) dx & ; \text{ if } f(2a-x) = f(x) \\ 0 & ; \text{ if } f(2a-x) = -f(x) \end{cases}$

Illustration 13: Evaluate $\int_0^{\pi} \frac{x dx}{1 + \cos^2 x}$

Solution : Let $I = \int_0^{\pi} \frac{x dx}{1 + \cos^2 x} = \int_0^{\pi} \frac{(\pi - x) dx}{1 + \cos^2(\pi - x)} = \int_0^{\pi} \frac{\pi dx}{1 + \cos^2 x} - I$

$$\Rightarrow 2I = \int_0^{\pi} \frac{\pi dx}{1 + \cos^2 x} = 2\pi \int_0^{\pi/2} \frac{dx}{1 + \cos^2 x} = 2\pi \int_0^{\pi/2} \frac{\sec^2 x dx}{2 + \tan^2 x}$$

Let $\tan x = t$ so that for $x \rightarrow 0$, $t \rightarrow 0$ and for $x \rightarrow \pi/2$, $t \rightarrow \infty$. Hence we can write,

$$I = \pi \int_0^{\infty} \frac{dt}{2 + t^2} = \pi \frac{1}{\sqrt{2}} \left[\tan^{-1} \frac{t}{\sqrt{2}} \right]_0^{\infty} = \frac{\pi^2}{2\sqrt{2}}$$

Ans.

Illustration 14: Prove that $\int_0^{\pi/2} \log(\sin x) dx = \int_0^{\pi/2} \log(\cos x) dx = -\frac{\pi}{2} \log 2$

Solution : Let $I = \int_0^{\pi/2} \log(\sin x) dx$ (i)

$$\text{then } I = \int_0^{\pi/2} \log \sin \left(\frac{\pi}{2} - x \right) dx = \int_0^{\pi/2} \log(\cos x) dx \quad \text{..... (ii)}$$

adding (i) and (ii), we get

$$2I = \int_0^{\pi/2} \log \sin x dx + \int_0^{\pi/2} \log \cos x dx = \int_0^{\pi/2} (\log \sin x + \log \cos x) dx$$

$$\begin{aligned} \Rightarrow 2I &= \int_0^{\pi/2} \log(\sin x \cos x) dx = \int_0^{\pi/2} \log \left(\frac{2 \sin x \cos x}{2} \right) dx \\ &= \int_0^{\pi/2} \log \left(\frac{\sin 2x}{2} \right) dx = \int_0^{\pi/2} \log(\sin 2x) dx - \int_0^{\pi/2} (\log 2) dx = \int_0^{\pi/2} \log \sin 2x \cdot dx - (\log 2)(x)_0^{\pi/2} \end{aligned}$$

$$\Rightarrow 2I = \int_0^{\pi/2} \log(\sin 2x) dx - \frac{\pi}{2} \log 2 \quad \text{..... (iii)}$$

Let $I_1 = \int_0^{\pi/2} \log(\sin 2x) dx$, putting $2x = t$, we get

$$I_1 = \int_0^{\pi} \log(\sin t) \frac{dt}{2} = \frac{1}{2} \int_0^{\pi} \log(\sin t) dt = \frac{1}{2} \cdot 2 \int_0^{\pi/2} \log(\sin t) dt$$

$$I_1 = \int_0^{\pi/2} \log(\sin x) dx$$

$$\therefore \text{ (iii) becomes ; } 2I = I - \frac{\pi}{2} \log 2$$

$$\text{Hence } \int_0^{\pi/2} \log \sin x dx = -\frac{\pi}{2} \log 2$$

Ans.

Illustration 15 : $\int_0^{\pi/2} (2 \log \sin x - \log \sin 2x) dx$ equals -

- (A) $\pi \log 2$ (B) $-\pi \log 2$ (C) $(\pi/2) \log 2$ (D) $-(\pi/2) \log 2$

Solution :
$$I = \int_0^{\pi/2} (2 \log \sin x - \log 2 \sin x \cos x) dx = \int_0^{\pi/2} (2 \log \sin x - \log 2 - \log \sin x - \log \cos x) dx$$
$$= \int_0^{\pi/2} \log \sin x dx - \int_0^{\pi/2} \log 2 dx - \int_0^{\pi/2} \log \cos x dx = -(\pi/2) \log 2$$
 Ans. (D)

Do yourself -4 :

Evaluate :

(i) $\int_{-\sqrt{3}}^{\sqrt{3}} \frac{dx}{(1+e^x)(1+x^2)}$

(ii) $\int_0^{\pi/2} \ln(\sin^2 x \cos x) dx$

(iii) $\int_0^{\pi/2} \frac{\sin x - \cos x}{1 + \sin x \cos x} dx$

(iv) $\int_{-\pi/2}^{\pi/2} \sqrt{\cos x - \cos^3 x} dx$

(g) $\int_0^{nT} f(x) dx = n \int_0^T f(x) dx$, ($n \in \mathbb{I}$); where 'T' is the period of the function i.e. $f(T+x) = f(x)$

Note that : $\int_x^{T+x} f(t) dt$ will be independent of x and equal to $\int_0^T f(t) dt$

(h) $\int_{a+nT}^{b+nT} f(x) dx = \int_a^b f(x) dx$ where f(x) is periodic with period T & $n \in \mathbb{I}$.

(i) $\int_{mT}^{nT} f(x) dx = (n-m) \int_0^T f(x) dx$, ($n, m \in \mathbb{I}$) if f(x) is periodic with period 'T'.

Illustration 16 : Evaluate $\int_0^{4\pi} |\cos x| dx$

Solution : Note that $|\cos x|$ is a periodic function with period π . Hence the given integral.

$$I = 4 \int_0^{\pi} |\cos x| dx = 4 \left[\int_0^{\pi/2} \cos x dx - \int_{\pi/2}^{\pi} \cos x dx \right] = 4 \left[[\sin x]_0^{\pi/2} - [\sin x]_{\pi/2}^{\pi} \right] = 4[1+1] = 8 \quad \text{Ans.}$$

Illustration 17: Evaluate $\int_0^{16\pi/3} |\sin x| dx$

Solution :
$$\int_0^{16\pi/3} |\sin x| dx = \int_0^{5\pi} |\sin x| dx + \int_{5\pi}^{5\pi+\pi/3} |\sin x| dx = 5 \int_0^{\pi} |\sin x| dx + \int_0^{\pi/3} |\sin x| dx$$

$$= 5[-\cos x]_0^{\pi} + [-\cos x]_0^{\pi/3} = 10 + \left(-\frac{1}{2} + 1\right) = \frac{21}{2}$$
 Ans.

Illustration 18: Evaluate : $\int_0^{2n\pi} [\sin x + \cos x] dx$. Here $[.]$ is the greatest integer function.

Solution : Let $I = \int_0^{2n\pi} [\sin x + \cos x] dx = n \int_0^{2\pi} [\sin x + \cos x] dx$

($\because$ $[\sin x + \cos x]$ is periodic function with period 2π)

$$[\sin x + \cos x] = \begin{cases} 1, & 0 \leq x \leq \frac{\pi}{2} \\ 0, & \frac{\pi}{2} \leq x \leq \frac{3\pi}{4} \\ -1, & \frac{3\pi}{4} < x \leq \pi \\ -2, & \pi < x \leq \frac{5\pi}{4} \\ -1, & \frac{5\pi}{4} < x \leq \frac{3\pi}{2} \\ 0, & \frac{3\pi}{2} < x \leq 2\pi \end{cases}$$

$$\text{Hence } I = n \left[\int_0^{\pi/2} 1 dx + \int_{\pi/2}^{3\pi/4} 0 dx + \int_{3\pi/4}^{\pi} -1 dx + \int_{\pi}^{5\pi/4} -2 dx + \int_{5\pi/4}^{3\pi/2} -1 dx + \int_{3\pi/2}^{2\pi} 0 dx \right]$$

$$I = n \left[\frac{\pi}{2} + 0 - \pi + \frac{3\pi}{4} - 3\pi + 2\pi - \frac{7\pi}{4} + \frac{3\pi}{2} + 0 \right] = -n\pi$$
 Ans.

Do yourself -5 :

Evaluate :

(i) $\int_{-1.5}^{10} \{2x\} dx$, where $\{.\}$ denotes fractional part of x .

(ii) $\int_{20\pi+\frac{\pi}{6}}^{20\pi+\frac{\pi}{3}} (\sin x + \cos x) dx$

3. WALLI'S FORMULA :

If $m, n \in \mathbb{N}$ & $m, n \geq 2$, then

$$(a) \int_0^{\pi/2} \sin^n x \, dx = \int_0^{\pi/2} \cos^n x \, dx = \frac{(n-1)(n-3)\dots(1 \text{ or } 2)}{n(n-2)\dots(1 \text{ or } 2)} K$$

$$\text{where } K = \begin{cases} \pi/2 & \text{if } n \text{ is even} \\ 1 & \text{if } n \text{ is odd} \end{cases}$$

$$(b) \int \sin^n x \cdot \cos^m x \, dx = \frac{[(n-1)(n-3)(n-5)\dots 1 \text{ or } 2][(m-1)(m-3)\dots 1 \text{ or } 2]}{(m+n)(m+n-2)(m+n-4)\dots 1 \text{ or } 2} K$$

$$\text{Where } K = \begin{cases} \frac{\pi}{2} & \text{if both } m \text{ and } n \text{ are even} \\ 1 & \text{otherwise} \end{cases}$$

Illustration 19: $\int_{-\pi/2}^{\pi/2} \sin^4 x \cos^6 x \, dx =$

(A) $\frac{3\pi}{64}$ (B) $\frac{3\pi}{572}$ (C) $\frac{3\pi}{256}$ (D) $\frac{3\pi}{128}$

Solution : $I = \int_{-\pi/2}^{\pi/2} \sin^4 x \cos^6 x \, dx = 2 \int_0^{\pi/2} \sin^4 x \cos^6 x \, dx = 2 \frac{(3.1)(5.3.1)}{10.8.6.4.2} \cdot \frac{\pi}{2} = \frac{3\pi}{256}$ **Ans. (C)**

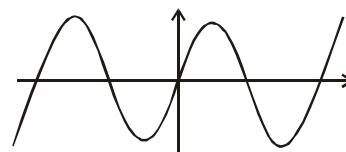
4. DERIVATIVE OF ANTIDERIVATIVE FUNCTION (Newton-Leibnitz Formula) :

If $h(x)$ & $g(x)$ are differentiable functions of x then, $\frac{d}{dx} \int_{g(x)}^{h(x)} f(t) dt = f[h(x)] \cdot h'(x) - f[g(x)] \cdot g'(x)$

Illustration 20: Find the points of maxima/minima of $\int \frac{-}{+}$

Solution : Let $f(x) = \int_0^{x^2} \frac{t^2 - 5t + 4}{2 + e^t} dt$

$$f'(x) = \frac{x^4 - 5x^2 + 4}{2 + e^{x^2}} \cdot 2x - 0 = \frac{(x-1)(x+1)(x-2)(x+2)2x}{2 + e^{x^2}}$$



From the wavy curve, it is clear that $f'(x)$ changes its sign at $x = \pm 2, \pm 1, 0$ and hence the points of maxima are $-1, 1$ and of the minima are $-2, 0, 2$.

Illustration 21 : Evaluate $\frac{d}{dt} \int_2^{t^3} \frac{1}{\log x} dx$

Solution : $\frac{d}{dt} \int_2^{t^3} \frac{1}{\log x} dx = \frac{1}{\log t^3} \cdot \frac{d}{dt}(t^3) - \frac{1}{\log 2} \cdot \frac{d}{dt}(2) = \frac{3t^2}{3 \log t} - \frac{2t}{2 \log t} = \frac{t(t-1)}{\log t}$ **Ans.**

Do yourself - 6 :

(i) If $f(x) = \int_{1/x}^{\sqrt{x}} \sin t dt$, then find $f'(1)$.

(ii) $\int_{\pi/3}^x \sqrt{3 - \sin^2 t} dt + \int_0^y \cos t dt = 0$, then evaluate $\frac{dy}{dx}$.

5. DEFINITE INTEGRAL AS LIMIT OF A SUM :

An alternative way of describing $\int_a^b f(x) dx$ is that the definite integral $\int_a^b f(x) dx$ is a limiting case of the summation of an infinite series, provided $f(x)$ is continuous on $[a, b]$

i.e. $\int_a^b f(x) dx = \lim_{n \rightarrow \infty} h \sum_{r=0}^{n-1} f(a + rh)$ where $h = \frac{b-a}{n}$. The converse is also true i.e., if we have an infinite series of the above form, it can be expressed as a definite integral.

Step I : Express the given series in the form $\sum \frac{1}{n} f\left(\frac{r}{n}\right)$

Step II : Then the limit is its sum when $n \rightarrow \infty$, i.e. $\lim_{n \rightarrow \infty} \frac{1}{n} f\left(\frac{r}{n}\right)$

Step III : Replace $\frac{r}{n}$ by x and $\frac{1}{n}$ by dx and $\lim_{n \rightarrow \infty} \sum$ by the sign of $\int$

Step IV : The lower and the upper limit of integration are the limiting values of – for the first and the last term of r respectively.

Illustration 22 : Evaluate $\lim_{n \rightarrow \infty} \left(\frac{1}{2n+1} + \frac{1}{2n+2} + \dots + \frac{1}{6n} \right)$

Solution : Let $S_n = \frac{1}{2n+1} + \frac{1}{2n+2} + \dots + \frac{1}{6n} = \sum_{r=1}^{4n} \frac{1}{2n+r} = \sum_{r=1}^{4n} \frac{1}{n} \cdot \frac{1}{2 + \left(\frac{r}{n}\right)}$

$$\Rightarrow S = \lim_{n \rightarrow \infty} S_n = \int_0^4 \frac{dx}{2+x} = [\ln |2+x|]_0^4 = \ln 6 - \ln 2 = \ln 3$$

Ans.

Illustration 23 : Evaluate $\lim_{n \rightarrow \infty} \left[\frac{\sqrt{n}}{(3+4\sqrt{n})^2} + \frac{\sqrt{n}}{\sqrt{2}(3\sqrt{2}+4\sqrt{n})^2} + \frac{\sqrt{n}}{\sqrt{3}(3\sqrt{3}+4\sqrt{n})^2} + \dots + \frac{1}{49n} \right]$

Solution : Let $p = \lim_{n \rightarrow \infty} \left[\frac{\sqrt{n}}{(3+4\sqrt{n})^2} + \frac{\sqrt{n}}{\sqrt{2}(3\sqrt{2}+4\sqrt{n})^2} + \dots + \frac{\sqrt{n}}{\sqrt{n}(3\sqrt{n}+4\sqrt{n})^2} \right]$

Analyzing the expression with the view of increasing integral value we get the expression in terms of r as

$$= \lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{\sqrt{n}}{\sqrt{r}(3\sqrt{r}+4\sqrt{n})^2} = \lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{1}{n\sqrt{\frac{r}{n}} \left(3\sqrt{\frac{r}{n}} + 4 \right)^2} = \int_0^1 \frac{dx}{\sqrt{x}(3\sqrt{x}+4)^2}$$

Put $3\sqrt{x} + 4 = t$, $\therefore \frac{3}{2\sqrt{x}} dx = dt$

Hence $p = \frac{2}{3} \int_4^7 \frac{dt}{t^2} = \frac{2}{3} \left[-\frac{1}{t} \right]_4^7 = \frac{2}{3} \left(-\frac{1}{7} + \frac{1}{4} \right) = \frac{1}{14}$

Ans.

Do yourself - 7 :

Evaluate :

(i) $\lim_{n \rightarrow \infty} \left[\frac{1}{n+2.1} + \frac{1}{n+2.2} + \frac{1}{n+2.3} + \dots + \frac{1}{3n} \right]$

(ii) $\lim_{n \rightarrow \infty} \sum_{r=0}^{n-1} \frac{1}{\sqrt{n^2 - r^2}}$

6. ESTIMATION OF DEFINITE INTEGRAL :

(a) If $f(x)$ is continuous in $[a, b]$ and its range in this interval is $[m, M]$, then

$$m(b-a) \leq \int_a^b f(x) dx \leq M(b-a)$$

Illustration 24 : Prove that $4 \leq \int_1^3 \sqrt{3+x^3} dx \leq 2\sqrt{30}$

Solution : Since the function $f(x) = \sqrt{3+x^3}$ increases monotonically on the interval $[1, 3]$, $m = 2$, $M = \sqrt{30}$, $b - a = 2$.

Hence, $2.2 \leq \int_1^3 \sqrt{3+x^3} dx \leq 2\sqrt{30} \Rightarrow 4 \leq \int_1^3 \sqrt{3+x^3} dx \leq 2\sqrt{30}$

Ans.

(b) If $f(x) \leq \phi(x)$ for $a \leq x \leq b$ then $\int_a^b f(x) dx \leq \int_a^b \phi(x) dx$

Illustration 25 : Prove that $\frac{\pi}{6} \leq \int_0^1 \frac{dx}{\sqrt{4-x^2-x^3}} \leq \frac{\pi}{4\sqrt{2}}$

Solution : Since $4-x^2 \geq 4-x^2-x^3 \geq 4-2x^2 > 0 \forall x \in [0, 1]$

$$\sqrt{4-x^2} \geq \sqrt{4-x^2-x^3} \geq \sqrt{4-2x^2} > 0 \forall x \in [0, 1]$$

$$\Rightarrow 0 < \frac{1}{\sqrt{4-x^2}} \leq \frac{1}{\sqrt{4-x^2-x^3}} \leq \frac{1}{\sqrt{4-2x^2}} \forall x \in [0, 1]$$

$$\Rightarrow \int_0^1 \frac{dx}{\sqrt{4-x^2}} \leq \int_0^1 \frac{dx}{\sqrt{4-x^2-x^3}} \leq \int_0^1 \frac{dx}{\sqrt{4-2x^2}} \forall x \in [0, 1]$$

$$\Rightarrow \left[\sin^{-1} \frac{x}{2} \right]_0^1 \leq \int_0^1 \frac{dx}{\sqrt{4-x^2-x^3}} \leq \frac{1}{\sqrt{2}} \left[\sin^{-1} \frac{x}{\sqrt{2}} \right]_0^1 \Rightarrow \frac{\pi}{6} \leq \int_0^1 \frac{dx}{\sqrt{4-x^2-x^3}} \leq \frac{\pi}{4\sqrt{2}} \text{ Ans.}$$

(c) $\left| \int_a^b f(x) dx \right| \leq \int_a^b |f(x)| dx.$

Illustration 26 : Prove that $\left| \int_{10}^{19} \frac{\sin x}{1+x^8} dx \right| < 10^{-7}$

Solution : To find $I = \left| \int_{10}^{19} \frac{\sin x}{1+x^8} dx \right| \leq \int_{10}^{19} \left| \frac{\sin x}{1+x^8} \right| dx$ (i)

Since $|\sin x| \leq 1$ for $x \geq 10$

The inequality $\left| \frac{\sin x}{1+x^8} \right| \leq \frac{1}{|1+x^8|}$ (ii)

also, $10 \leq x \leq 19$

$$\Rightarrow 1+x^8 > 10^8$$

$$\Rightarrow \frac{1}{1+x^8} < \frac{1}{10^8} \text{ or } \frac{1}{|1+x^8|} < 10^{-8} \text{ (iii)}$$

from (ii) and (iii) ;

$$\left| \frac{\sin x}{1+x^8} \right| < 10^{-8}$$

$$\left| \int_{10}^{19} \frac{\sin x}{1+x^8} dx \right| < \int_{10}^{19} 10^{-8} dx$$

$$\therefore \left| \int_{10}^{19} \frac{\sin x}{1+x^8} dx \right| < (19-10) \cdot 10^{-8} < 10^{-7}$$

Ans.

Illustration 27: If $f(x)$ is integrable function such that $|f(x) - f(y)| \leq |x^2 - y^2|$, $\forall x, y \in [a, b]$ then prove

$$\text{that } \left| \int_a^b \frac{f(x) - f(a)}{x + a} dx \right| \leq \frac{(a - b)^2}{2}.$$

Solution : Given, $\left| \int_a^b \frac{f(x) - f(a)}{x + a} dx \right| \leq \int_a^b \left| \frac{f(x) - f(a)}{x + a} \right| dx$

$$\leq \int_a^b \left| \frac{x^2 - a^2}{x + a} \right| dx = \int_a^b |x - a| dx = \int_a^b (x - a) dx = \frac{(a - b)^2}{2}$$

(d) If $f(x) \geq 0$ on the interval $[a, b]$, then $\int_a^b f(x) dx \geq 0$.

Illustration 28: If $f(x)$ is a continuous function such that $f(x) \geq 0 \forall x \in [2, 10]$ and $\int_4^8 f(x) dx = 0$, then find $f(6)$.

Solution : $f(x)$ is above the x-axis or on the x-axis for all $x \in [2, 10]$. If $f(x)$ is greater than zero for any sub interval of $[4, 8]$, then $\int_4^8 f(x) dx$ must be greater than zero.

$$\text{But } \int_4^8 f(x) dx = 0 \Rightarrow f(x) = 0 \forall x \in [4, 8] \\ \Rightarrow f(6) = 0.$$

Do yourself - 8 :

(i) Prove that $4 \leq \int_1^3 \sqrt{3 + x^2} dx \leq 4\sqrt{3}$

(ii) Prove that $\frac{\pi}{4} \leq \int_0^{2\pi} \frac{dx}{5 + 3 \sin x} \leq \pi$.

(iii) Show that $\frac{3}{5}(2^{1/3} - 1) \leq \int_0^1 \frac{x^4}{(1 + x^6)^{2/3}} dx \leq 1$

Miscellaneous Illustrations :

Illustration 29: Evaluate : $\int_0^\pi \frac{x^3 \cos^4 x \sin^2 x}{(\pi^2 - 3\pi x + 3x^2)} dx$

Solution : Let $I = \int_0^\pi \frac{x^3 \cos^4 x \sin^2 x}{(\pi^2 - 3\pi x + 3x^2)} dx$ (i)

$$= \int_0^\pi \frac{(\pi - x)^3 \cos^4(\pi - x) \sin^2(\pi - x) dx}{\pi^2 - 3\pi(\pi - x) + 3(\pi - x)^2} \quad (\text{By Prop.})$$

$$= \int_0^{\pi} \frac{(\pi^3 - x^3 - 3\pi^2 x + 3\pi x^2) \cos^4 x \sin^2 x}{(\pi^2 - 3\pi x + 3x^2)} dx \quad \dots\dots\dots (ii)$$

Adding (i) and (ii) we have

$$2I = \int_0^{\pi} \frac{(\pi^3 - 3\pi^2 x + 3\pi x^2) \cos^4 x \sin^2 x}{(\pi^2 - 3\pi x + 3x^2)} dx$$

$$\Rightarrow 2I = \pi \int_0^{\pi} \cos^4 x \sin^2 x dx \Rightarrow 2I = 2\pi \int_0^{\pi/2} \cos^4 x \sin^2 x dx$$

$$\therefore I = \pi \int_0^{\pi/2} \cos^4 x \sin^2 x dx$$

Using walli's formula, we get $I = \pi \frac{(3.1)(1)}{6.4.2} \frac{\pi}{2} = \frac{\pi^2}{32}$ **Ans.**

Illustration 30 : Let f be an injective function such that $f(x)f(y) + 2 = f(x) + f(y) + f(xy)$ for all non negative real x and y with $f(0) = 1$ and $f(1) = 2$ find $f(x)$ and show that $3 \int f(x) dx - x(f(x) + 2)$ is a constant.

Solution :

We have $f(x)f(y) + 2 = f(x) + f(y) + f(xy)$

Putting $x = 1$ & $y = 1$

then $f(1)f(1) + 2 = 3f(1)$

we get $f(1) = 1, 2$

$f(1) \neq 1$ ($\because f(0) = 1$ & function is injective)

then $f(1) = 2$

Replacing y by $\frac{1}{x}$ in (1) then

$$f(x)f\left(\frac{1}{x}\right) + 2 = f(x) + f\left(\frac{1}{x}\right) + f(1) \Rightarrow f(x)f\left(\frac{1}{x}\right) = f(x) + f\left(\frac{1}{x}\right)$$

Hence $f(x)$ is of the type

$$f(x) = 1 \pm x^n$$

$$\therefore f(1) = 2$$

$$\therefore f(x) = 1 + x^n$$

$$\text{and } f(x) = nx^{n-1} \Rightarrow f(1) = n = 2$$

$$f(x) = 1 + x^2$$

$$\therefore 3 \int f(x) dx - x(f(x) + 2) = 3 \int (1 + x^2) dx - x(1 + x^2 + 2)$$

$$= 3 \left(x + \frac{x^3}{3} \right) - x(3 + x^2) + c = c = \text{constant}$$

Illustration 31 : Evaluate : $\int_{-1}^1 [x[1 + \sin \pi x] + 1] dx$, $[.]$ is the greatest integer function.

Solution : Let $I = \int_{-1}^1 [x[1 + \sin \pi x] + 1] dx = \int_{-1}^0 [x[1 + \sin \pi x] + 1] dx + \int_0^1 [x[1 + \sin \pi x] + 1] dx$

Now $[1 + \sin \pi x] = 0$ if $-1 < x < 0$

$[1 + \sin \pi x] = 1$ if $0 < x < 1$

$$\therefore I = \int_{-1}^0 1 \cdot dx + \int_0^1 [x + 1] dx = 1 + 1 \int_0^1 dx = 1 + 1 = 2.$$

Ans.

Illustration 32 : Find the limit, when $n \rightarrow \infty$ of

$$\frac{1}{\sqrt{(2n-1)^2}} + \frac{1}{\sqrt{(4n-2)^2}} + \frac{1}{\sqrt{(6n-3)^2}} + \dots + \frac{1}{n}$$

Solution :

$$\begin{aligned} \text{Let } P &= \lim_{n \rightarrow \infty} \left[\frac{1}{\sqrt{2n-1^2}} + \frac{1}{\sqrt{4n-2^2}} + \frac{1}{\sqrt{6n-3^2}} + \dots + \frac{1}{n} \right] \\ &= \lim_{n \rightarrow \infty} \left[\frac{1}{\sqrt{1(2n)-1^2}} + \frac{1}{\sqrt{2(2n)-2^2}} + \frac{1}{\sqrt{3(2n)-3^2}} + \dots + \frac{1}{\sqrt{n(2n)-n^2}} \right] \\ &= \lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{1}{\sqrt{r(2n)-r^2}} = \lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{1}{n \cdot \sqrt{2 \frac{r}{n} - \left(\frac{r}{n}\right)^2}} = \int_0^1 \frac{dx}{\sqrt{(2x-x^2)}} \end{aligned}$$

$$\text{Put } x = t^2 \Rightarrow dx = 2t dt$$

$$\therefore P = \int_0^1 \frac{2t dt}{t \sqrt{2-t^2}} = \left[2 \sin^{-1} \left(\frac{t}{\sqrt{2}} \right) \right]_0^1 = 2 \sin^{-1} \left(\frac{1}{\sqrt{2}} \right) = 2 \left(\frac{\pi}{4} \right)$$

$$\text{Hence } P = \pi/2.$$

Ans.

Illustration 33 : If $f(x) = \begin{cases} 1-|x|, & |x| \leq 1 \\ |x|-1, & |x| > 1 \end{cases}$, and $g(x) = f(x-1) + f(x+1)$. Find the value of $\int_{-3}^5 g(x) dx$.

Solution :

Given,

$$f(x) = \begin{cases} -x-1, & x < -1 \\ 1+x, & -1 \leq x < 0 \\ 1-x, & 0 \leq x \leq 1 \\ x-1, & x > 1 \end{cases}; \quad f(x-1) = \begin{cases} -x, & x-1 < -1 \Rightarrow x < 0 \\ x, & -1 \leq x-1 < 0 \Rightarrow 0 \leq x < 1 \\ 2-x, & 0 \leq x-1 \leq 1 \Rightarrow 1 \leq x \leq 2 \\ x-2, & x-1 > 1 \Rightarrow x > 2 \end{cases}$$

Similarly

$$f(x+1) = \begin{cases} -x-2, & x+1 < -1 \Rightarrow x < -2 \\ x+2, & -1 \leq x+1 < 0 \Rightarrow -2 \leq x < -1 \\ -x, & 0 \leq x+1 \leq 1 \Rightarrow -1 \leq x \leq 0 \\ x, & x+1 > 1 \Rightarrow x > 0 \end{cases}$$

$$\Rightarrow g(x) = f(x-1) + f(x+1) = \begin{cases} -2x-2, & x < -2 \\ 2, & -2 \leq x < -1 \\ -2x, & -1 \leq x \leq 0 \\ 2x, & 0 < x < 1 \\ 2, & 1 < x \leq 2 \\ 2x-2, & 2 < x \end{cases}$$

Clearly $g(x)$ is even,

$$\begin{aligned} \text{Now } \int_{-3}^5 g(x) dx &= 2 \int_0^3 g(x) dx + \int_3^5 g(x) dx \\ &= 2 \left(\int_0^1 2x dx + \int_1^2 2 dx + \int_2^3 (2x-2) dx \right) + \int_3^5 (2x-2) dx = 24 \end{aligned}$$

ANSWERS FOR DO YOURSELF

- | | | | |
|-----------------------------|--|-----------------------|--------------------|
| 1: (i) $\frac{31}{6}$ | (ii) 2 | (iii) $2(\sqrt{2}-1)$ | (iv) $\frac{9}{2}$ |
| 2: (i) 2 | (ii) $\pi \ln 2$ | | |
| 3: (i) 2 | (ii) $\pi/12$ | | |
| 4: (i) $\frac{\pi}{3}$ | (ii) $-\left(\frac{3\pi}{2}\right) \ln 2$ | (iii) 0 | (iv) $\frac{4}{3}$ |
| 5: (i) $\frac{23}{4}$ | (ii) $(\sqrt{3}-1)$ | | |
| 6: (i) $\frac{3}{2} \sin 1$ | (ii) $\frac{dy}{dx} = \frac{-\sqrt{3-\sin^2 x}}{\cos y}$ | | |
| 7: (i) $\frac{1}{2} \ln 3$ | (ii) $\frac{\pi}{2}$ | | |

EXERCISE (O-1)

1. If $g(x) = \int_0^x \cos^4 t \, dt$, then $g(x + \pi)$ equals
 (A) $g(x) + g(\pi)$ (B) $g(x) - g(\pi)$ (C) $g(x)g(\pi)$ (D) $[g(x)/g(\pi)]$
2. $\int_0^1 \frac{\tan^{-1} x}{x} dx =$
 (A) $\int_0^{\pi/4} \frac{\sin x}{x} dx$ (B) $\int_0^{\pi/2} \frac{x}{\sin x} dx$ (C) $\frac{1}{2} \int_0^{\pi/2} \frac{x}{\sin x} dx$ (D) $\frac{1}{2} \int_0^{\pi/4} \frac{x}{\sin x} dx$
3. The value of the defined integral $\int_0^{\pi/2} (\sin x + \cos x) \cdot \sqrt{\frac{e^x}{\sin x}} dx$ equals
 (A) $2\sqrt{e^{\pi/2}}$ (B) $\sqrt{e^{\pi/2}}$ (C) $2\sqrt{e^{\pi/2}} \cdot \cos 1$ (D) $\frac{1}{2} e^{\pi/4}$
4. Variable x and y are related by equation $x = \int_0^y \frac{dt}{\sqrt{1+t^2}}$. The value of $\frac{d^2 y}{dx^2}$ is equal to
 (A) $\frac{y}{\sqrt{1+y^2}}$ (B) y (C) $\frac{2y}{\sqrt{1+y^2}}$ (D) $4y$
5. If $\int_0^x f(t) dt = x + \int_x^1 t^2 \cdot f(t) dt + \frac{\pi}{4} - 1$, then the value of the integral $\int_{-1}^1 f(x) dx$ is equal to
 (A) 0 (B) $\pi/4$ (C) $\pi/2$ (D) π
6. If $I = \int_0^{\pi/2} \ln(\sin x) dx$ then $\int_{-\pi/4}^{\pi/4} \ln(\sin x + \cos x) dx$
 (A) $\frac{I}{2}$ (B) $\frac{I}{4}$ (C) $\frac{I}{\sqrt{2}}$ (D) I
7. If $f(x) = x \sin x^2$; $g(x) = x \cos x^2$ for $x \in [-1, 2]$
 $A = \int_{-1}^2 f(x) dx$; $B = \int_{-1}^2 g(x) dx$, then
 (A) $A > 0$; $B < 0$ (B) $A < 0$; $B > 0$ (C) $A > 0$; $B > 0$ (D) $A < 0$; $B < 0$
8. The value of the definite integral $\int_0^{\pi/2} \sin x \sin 2x \sin 3x dx$ is equal to :
 (A) $\frac{1}{3}$ (B) $-\frac{2}{3}$ (C) $-\frac{1}{3}$ (D) $\frac{1}{6}$
9. Value of the definite integral $\int_{-1/2}^{1/2} (\sin^{-1}(3x - 4x^3) - \cos^{-1}(4x^3 - 3x)) dx$
 (A) 0 (B) $-\frac{\pi}{2}$ (C) $\frac{7\pi}{2}$ (D) $\frac{\pi}{2}$

10. $\lim_{x \rightarrow \infty} \left(x^3 \int_{-1/x}^{1/x} \frac{\ln(1+t^2)}{1+e^t} dt \right)$ equals
 (A) $1/3$ (B) $2/3$ (C) 1 (D) 0
11. If $\frac{C_0}{1} + \frac{C_1}{2} + \frac{C_2}{3} = 0$, where C_0, C_1, C_2 are all real, the equation $C_2x^2 + C_1x + C_0 = 0$ has :
 (A) atleast one root is $(0,1)$
 (B) one root is $(1,2)$ & other is $(3,4)$
 (C) one root is $(-1,1)$ & the other is $(-5,-2)$
 (D) both roots imaginary
12. The value of the definite integral $\int_1^e ((x+1)e^x \cdot \ln x) dx$ is -
 (A) e (B) e^{e+1} (C) $e^e(e-1)$ (D) $e^e(e-1) + e$
13. $\lim_{n \rightarrow \infty} \frac{1}{\sqrt{n}\sqrt{n+1}} + \frac{1}{\sqrt{n}\sqrt{n+2}} + \dots + \frac{1}{\sqrt{n}\sqrt{4n}}$ is equal to -
 (A) 2 (B) 4 (C) $2(\sqrt{2}-1)$ (D) $2\sqrt{2}-1$
14. The value of the definite integral $\int_{19}^{37} (\{x\}^2 + 3(\sin 2\pi x)) dx$ where $\{x\}$ denotes the fractional part function.
 (A) 0 (B) 6 (C) 9 (D) can't be determined
15. $\lim_{x \rightarrow 0} \frac{\int_0^x \sin t^2 dt}{x(1 - \cos x)}$ equals -
 (A) $\frac{1}{3}$ (B) 2 (C) $\frac{1}{2}$ (D) $\frac{2}{3}$
16. If $g(x)$ is the inverse of $f(x)$ and $f(x)$ has domain $x \in [1, 5]$, where $f(1) = 2$ and $f(5) = 10$ then the values of $\int_1^5 f(x) dx + \int_2^{10} g(y) dy$ equals -
 (A) 48 (B) 64 (C) 71 (D) 52
17. The value of the definite integral $\int_2^4 (x(3-x)(4+x)(6-x)(10-x) + \sin x) dx$ equals-
 (A) $\cos 2 + \cos 4$ (B) $\cos 2 - \cos 4$ (C) $\sin 2 + \sin 4$ (D) $\sin 2 - \sin 4$
18. The value of $\int_{-1}^1 \frac{dx}{\sqrt{|x|}}$ is -
 (A) $\frac{1}{2}$ (B) 2 (C) 4 (D) undefined

19. $\int_{-1}^1 \frac{x^3 + |x| + 1}{x^2 + 2|x| + 1} dx = a \ln 2 + b$ then -

- (A) $a = 2; b = 1$ (B) $a = 2; b = 0$ (C) $a = 3; b = -2$ (D) $a = 4; b = -1$

20. The true solution set of the inequality, $\sqrt{5x - 6 - x^2} + \left(\frac{\pi}{2} \int_0^x dz \right) > x \int_0^{\pi} \sin^2 x dx$ is :

- (A) $\mathbb{R}$ (B) $(1, 6)$ (C) $(-6, 1)$ (D) $(2, 3)$

21. Let $I_1 = \int_0^{\pi/2} e^{-x^2} \sin(x) dx$; $I_2 = \int_0^{\pi/2} e^{-x^2} dx$; $I_3 = \int_0^{\pi/2} e^{-x^2} (1+x) dx$

and consider the statements

- I $I_1 < I_2$ II $I_2 < I_3$ III $I_1 = I_3$

Which of the following is(are) true ?

- (A) I only (B) II only
(C) Neither I nor II nor III (D) Both I and II

22. Let $u = \int_0^1 \frac{\ln(x+1)}{x^2+1} dx$ and $v = \int_0^{\pi/2} \ln(\sin 2x) dx$ then -

- (A) $u = 4v$ (B) $4u + v = 0$ (C) $u + 4v = 0$ (D) $2u + v = 0$

23. $\int_{\frac{1}{2}}^{\frac{3}{2}} \left\{ \frac{1}{2} (|x-3| + |1-x| - 4) \right\} dx$ equals-

- (A) $-\frac{3}{2}$ (B) $\frac{9}{8}$ (C) $\frac{1}{4}$ (D) $\frac{3}{2}$

Where $\{.\}$ denotes the fraction part function.

24. Suppose that $F(x)$ is an antiderivative of $f(x) = \frac{\sin x}{x}$, $x > 0$ then $\int_1^3 \frac{\sin 2x}{x} dx$ can be expressed as -

- (A) $F(6) - F(2)$ (B) $\frac{1}{2}(F(6) - F(2))$ (C) $\frac{1}{2}(F(3) - F(1))$ (D) $2(F(6) - F(2))$

25. Let f be a positive function. Let $I_1 = \int_{1-k}^k x f(x(1-x)) dx$; $I_2 = \int_{1-k}^k f(x(1-x)) dx$, where $2k - 1 > 0$.

Then $\frac{I_2}{I_1}$ is -

- (A) k (B) $1/2$ (C) 1 (D) 2

26. $\lim_{h \rightarrow 0} \frac{\int_a^{x+h} \ln^2 t dt - \int_a^x \ln^2 t dt}{h} =$

- (A) 0 (B) $\ln^2 x$ (C) $\frac{2 \ln x}{x}$ (D) does not exist

27. $\int_0^{\infty} f\left(x + \frac{1}{x}\right) \cdot \frac{\ln x}{x} dx$
 (A) is equal to zero (B) is equal to one (C) is equal to $\frac{1}{2}$ (D) can not be evaluated
28. $\int_0^{\pi/2} \frac{\cos \theta - \sin \theta}{(1 + \cos \theta)(1 + \sin \theta)} d\theta$ equals -
 (A) $\cos^{-1}\left(\frac{1}{\sqrt{2}}\right)$ (B) $\cos^{-1}(0)$ (C) $\cos^{-1}(1)$ (D) $\cos^{-1}(-1)$
29. Let $f(x)$ be a continuous function on $[0, 4]$ satisfying $f(x) f(4 - x) = 1$.
 The value of the definite integral $\int_0^4 \frac{1}{1 + f(x)} dx$ equals -
 (A) 0 (B) 1 (C) 2 (D) 4
30. If $g(x) = \int_1^x e^{t^2} dt$ then the value of $\int_1^{x^3} e^{t^2} dt$ equals
 (A) $g(x^3) - g(3)$ (B) $g(x^3) + g(3)$ (C) $g(x^3) - 3$ (D) $g(x^3) - 3g(x)$

EXERCISE (O-2)

1. Let $f(x) = \int_0^{g(x)} \frac{dt}{\sqrt{1+t^2}}$ where $g(x) = \int_0^{\cos x} (1 + \sin t^2) dt$. Also $h(x) = e^{-|x|}$ and $l(x) = x^2 \sin \frac{1}{x}$ if $x \neq 0$ and $l(0) = 0$ then $f'\left(\frac{\pi}{2}\right)$ equals
 (A) $l'(0)$ (B) $h'(0^-)$ (C) $h'(0^+)$ (D) $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x \sin x}$
2. For positive integers $k = 1, 2, 3, \dots, n$, let S_k denotes the area of ΔAOB_k (where 'O' is origin) such that that $\angle AOB_k = \frac{k\pi}{2n}$, $OA = 1$ and $OB_k = k$. The value of the $\lim_{n \rightarrow \infty} \frac{1}{n^2} \sum_{k=1}^n S_k$ is
 (A) $\frac{2}{\pi^2}$ (B) $\frac{4}{\pi^2}$ (C) $\frac{8}{\pi^2}$ (D) $\frac{1}{2\pi^2}$
3. The value of the definite integral $\int_0^{\pi/2} \sin |2x - \alpha| dx$ where $\alpha \in [0, \pi]$
 (A) 1 (B) $\cos \alpha$ (C) $\frac{1 + \cos \alpha}{2}$ (D) $\frac{1 - \cos \alpha}{2}$
4. The absolute value of $\frac{\int_0^{\pi/2} (x \cos x + 1) e^{\sin x} dx}{\int_0^{\pi/2} (x \sin x - 1) e^{\cos x} dx}$ is equal to -
 (A) e (B) πe (C) $e/2$ (D) π/e

5. Let f be a continuous function satisfying $f'(\ln x) = \begin{cases} 1 & \text{for } 0 < x \leq 1 \\ x & \text{for } x > 1 \end{cases}$ and $f(0) = 0$ then $f(x)$ can be defined as
- (A) $f(x) = \begin{cases} 1 & \text{if } x \leq 0 \\ 1 - e^x & \text{if } x > 0 \end{cases}$ (B) $f(x) = \begin{cases} 1 & \text{if } x \leq 0 \\ e^x - 1 & \text{if } x > 0 \end{cases}$
- (C) $f(x) = \begin{cases} x & \text{if } x < 0 \\ e^x & \text{if } x > 0 \end{cases}$ (D) $f(x) = \begin{cases} x & \text{if } x \leq 0 \\ e^x - 1 & \text{if } x > 0 \end{cases}$
6. Suppose f is continuous and satisfies $f(x) + f(-x) = x^2$ then the integral $\int_{-1}^1 f(x) dx$ has the value equal to
- (A) $\frac{2}{3}$ (B) $\frac{1}{3}$ (C) $\frac{4}{3}$ (D) zero
7. The true set of values of 'a' for which the inequality $\int_a^0 (3^{-2x} - 2 \cdot 3^{-x}) dx \geq 0$ is true is -
- (A) $[0, 1]$ (B) $(-\infty, -1]$ (C) $[0, \infty)$ (D) $(-\infty, -1] \cup [0, \infty)$
8. If the value of the integral $\int_1^2 e^{x^2} dx$ is α , then the value of $\int_e^{e^4} \sqrt{\ln x} dx$ is -
- (A) $e^4 - e - \alpha$ (B) $2e^4 - e - \alpha$ (C) $2(e^4 - e) - \alpha$ (D) $2e^4 - 1 - \alpha$
9. If $\int_0^1 \frac{e^t dt}{1+t} = A$ then the value of $\int_0^1 \frac{e^t dt}{(1+t)^2}$ is -
- (A) $A + \frac{e}{2} - 1$ (B) $A - \frac{e}{2} + 1$ (C) $A - \frac{e}{2} - 1$ (D) $A + \frac{e}{2} + 1$
10. Which one of the following functions is not continuous on $(0, \pi)$?
- (A) $f(x) = \cot x$ (B) $g(x) = \int_0^x t \sin \frac{1}{t} dt$
- (C) $h(x) = \begin{cases} 1 & 0 < x \leq \frac{3\pi}{4} \\ 2 \sin \frac{2}{9} x & \frac{3\pi}{4} < x < \pi \end{cases}$ (D) $l(x) = \begin{cases} x \sin x, & 0 < x \leq \frac{\pi}{2} \\ \frac{\pi}{2} \sin(x + \pi), & \frac{\pi}{2} < x < \pi \end{cases}$
11. $\int_0^{\pi/2} \frac{\sqrt{\sec x - \tan x}}{\sqrt{\sec x + \tan x}} \cdot \frac{\operatorname{cosec} x}{\sqrt{1 + 2 \operatorname{cosec} x}} dx$ has the value -
- (A) $\pi/6$ (B) $\pi/4$ (C) $\pi/3$ (D) $2\pi/3$

12. The value of $\lim_{n \rightarrow \infty} \sum_{r=1}^{r=4n} \frac{\sqrt{n}}{\sqrt{r}(3\sqrt{r} + 4\sqrt{n})^2}$ is equal to
 (A) $\frac{1}{35}$ (B) $\frac{1}{14}$ (C) $\frac{1}{10}$ (D) $\frac{1}{5}$
13. The value of $\int_{\pi}^{2\pi} [2 \cos x] dx$ where $[.]$ represents the greatest integer function, is -
 (A) $-\frac{5\pi}{6}$ (B) $-\frac{\pi}{2}$ (C) $-\pi$ (D) none
14. $\lim_{k \rightarrow 0} \frac{1}{k} \int_0^k (1 + \sin 2x)^{\frac{1}{x}} dx$ -
 (A) 2 (B) 1 (C) e^2 (D) non existent
15. The value of the definite integral $\int_0^{\pi/3} \ln(1 + \sqrt{3} \tan x) dx$ equals -
 (A) $\frac{\pi}{3} \ln 2$ (B) $\frac{\pi}{3}$ (C) $\frac{\pi^2}{6} \ln 2$ (D) $\frac{\pi}{2} \ln 2$

Paragraph for Question Nos. 16 to 18

Let the function f satisfies

$$f(x) \cdot f'(-x) = f(-x) \cdot f'(x) \text{ for all } x \text{ and } f(0) = 3.$$

16. The value of $f(x) \cdot f(-x)$ for all x , is
 (A) 4 (B) 9 (C) 12 (D) 16
17. $\int_{-51}^{51} \frac{dx}{3 + f(x)}$ has the value equal to
 (A) 17 (B) 34 (C) 102 (D) 0
18. Number of roots of $f(x) = 0$ in $[-2, 2]$ is
 (A) 0 (B) 1 (C) 2 (D) 4

Multiple Correct :

19. $\int_0^{\infty} \frac{x}{(1+x)(1+x^2)} dx =$
 (A) $\frac{\pi}{4}$ (B) $\frac{\pi}{2}$
 (C) is same as $\int_0^{\infty} \frac{dx}{(1+x)(1+x^2)}$ (D) cannot be evaluated
20. Let $u = \int_0^{\infty} \frac{dx}{x^4 + 7x^2 + 1}$ & $v = \int_0^{\infty} \frac{x^2 dx}{x^4 + 7x^2 + 1}$ then -
 (A) $v > u$ (B) $6v = \pi$ (C) $3u + 2v = 5\pi/6$ (D) $u + v = \pi/3$

21. Which of the following pair(s) of functions are the primitive of one and the same function ?

(A) $f(x) = \ln ax$; $g(x) = \ln x$

(B) $f(x) = 2 \sin^2 x$; $g(x) = -\cos 2x$

(C) $f(x) = (e^x + e^{-x})^2$; $g(x) = (e^x - e^{-x})^2$

(D) $f(x) = \frac{e^x(e^x + e^{-x})}{2}$; $g(x) = \frac{e^x(e^x - e^{-x})}{2}$

22. Let $S_n = \sum_{k=1}^n \frac{k^2 + n^2}{n^3}$ and $T_n = \sum_{k=0}^{n-1} \frac{k^2 + n^2}{n^3}$ for $n = 1, 2, 3, \dots$. Then -

(A) $S_n < \frac{4}{3}$

(B) $T_n > \frac{4}{3}$

(C) $S_n > \frac{4}{3}$

(D) $T_n < \frac{4}{3}$

23. Which of the following statement(s) is/are **TRUE** ?

(A) $\int_0^1 \ell n x dx = -1$

(B) $\lim_{n \rightarrow \infty} \frac{1}{n} \ln \left(\left(1 + \frac{1}{n}\right) \left(1 + \frac{2}{n}\right) \dots \left(1 + \frac{n}{n}\right) \right) = 1 + 2 \ln 2$.

(C) Let f be a continuous and non-negative function defined on $[a, b]$.

If $\int_a^b f(x) dx = 0$ then $f(x) = 0 \forall x \in [a, b]$

(D) Let f be a continuous function defined on $[a, b]$ such that $\int_a^b f(x) dx = 0$, then there exists atleast one

$c \in (a, b)$ for which $f(c) = 0$.

24. Let $f(x) = \begin{cases} x+1, & 0 \leq x \leq 1 \\ 2x^2 - 6x + 6, & 1 < x \leq 2 \end{cases}$ and $g(t) = \int_{t-1}^t f(x) dx$ for $t \in [1, 2]$

Which of the following hold(s) good ?

(A) $f(x)$ is continuous and differentiable in $[0, 2]$

(B) $g'(t)$ vanishes for $t = 3/2$ and 2

(C) $g(t)$ is maximum at $t = 3/2$

(D) $g(t)$ is minimum at $t = 1$

EXERCISE (S-1)

1. Evaluate : (i) $\int_0^1 e^{\ln \tan^{-1} x} \cdot \sin^{-1}(\cos x) \cdot dx$ (ii) $\int_{1/3}^3 \frac{\sin^{-1} \frac{x}{\sqrt{1+x^2}}}{x} dx$
2. Evaluate : $\int_0^1 \frac{\sin^{-1} \sqrt{x}}{\sqrt{x(1-x)}} dx$
3. Evaluate : $\int_0^{\pi/2} \frac{1+2\cos x}{(2+\cos x)^2} dx$
4. Evaluate : $\int_0^{\pi/2} e^x \left\{ \cos(\sin x) \cos^2 \frac{x}{2} + \sin(\sin x) \sin^2 \frac{x}{2} \right\} dx$
5. Evaluate : $\int_1^e \left\{ (1+x)e^x + (1-x)e^{-x} \right\} \ln x dx$
6. If $P = \int_0^\infty \frac{x^2}{1+x^4} dx$; $Q = \int_0^\infty \frac{x dx}{1+x^4}$ and $R = \int_0^\infty \frac{dx}{1+x^4}$, then prove that :
 (a) $Q = \frac{\pi}{4}$, (b) $P = R$, (c) $P - \sqrt{2}Q + R = \frac{\pi}{2\sqrt{2}}$
7. $\int_1^2 \frac{(x^2-1)dx}{x^3 \cdot \sqrt{2x^4-2x^2+1}} = \frac{u}{v}$ where u and v are in their lowest form. Find the value of $\frac{(1000)u}{v}$.
8. Let $h(x) = (f \circ g)(x) + K$ where K is any constant. If $\frac{d}{dx}(h(x)) = -\frac{\sin x}{\cos^2(\cos x)}$ then compute the value of $j(0)$ where $j(x) = \int_{g(x)}^{f(x)} \frac{f(t)}{g(t)} dt$, where f and g are trigonometric functions.
9. Evaluate : $\int_0^{\pi/2} \sqrt{\frac{1-\sin 2x}{1+\sin 2x}} dx$
10. If a_1, a_2 and a_3 are the three values of a which satisfy the equation $\int_0^{\pi/2} (\sin x + a \cos x)^3 dx - \frac{4a}{\pi-2} \int_0^{\pi/2} x \cos x dx = 2$ then find the value of $1000(a_1^2 + a_2^2 + a_3^2)$.
11. Evaluate : $\int_{-\sqrt{2}}^{\sqrt{2}} \frac{2x^7 + 3x^6 - 10x^5 - 7x^3 - 12x^2 + x + 1}{x^2 + 2} dx$
12. Evaluate : $\int_{-2}^2 \frac{x^2 - x}{\sqrt{x^2 + 4}} dx$
13. Let $u = \int_0^{\pi/4} \left(\frac{\cos x}{\sin x + \cos x} \right)^2 dx$ and $v = \int_0^{\pi/4} \left(\frac{\sin x + \cos x}{\cos x} \right)^2 dx$. Find the value of $\frac{v}{u}$.

14. Evaluate : $\int_0^{\pi/4} \frac{x \, dx}{\cos x (\cos x + \sin x)}$
15. Evaluate : $\int_0^1 \frac{\sin^{-1} \sqrt{x}}{x^2 - x + 1} dx$
16. Evaluate : $\int_1^{1+\sqrt{5}} \frac{x^2 + 1}{x^4 - x^2 + 1} \ln \left(1 + x - \frac{1}{x} \right) dx$
17. $\lim_{n \rightarrow \infty} n^2 \int_{-1/n}^{1/n} (2010 \sin x + 2012 \cos x) |x| \, dx$
18. If $\int_0^{\pi} \sqrt{(\cos x + \cos 2x + \cos 3x)^2 + (\sin x + \sin 2x + \sin 3x)^2} \, dx$ has the value equal to $\left(\frac{\pi}{k} + \sqrt{w} \right)$ where k and w are positive integers, find the value of $(k^2 + w^2)$.
19. Evaluate : $\int_0^{\pi/2} \frac{a \sin x + b \cos x}{\sin \left(\frac{\pi}{4} + x \right)} dx$
20. A continuous real function f satisfies $f(2x) = 3 f(x) \, \forall x \in \mathbb{R}$
If $\int_0^1 f(x) dx = 1$, then compute the value of definite integral $\int_1^2 f(x) dx$
21. Evaluate : $\int_0^{\pi} \frac{(ax + b) \sec x \tan x}{4 + \tan^2 x} dx (a, b > 0)$
22. Evaluate : $\int_0^{\pi} \frac{(2x + 3) \sin x}{(1 + \cos^2 x)} dx$
23. Evaluate : $\int_0^3 \sqrt{\frac{x}{3-x}} \, dx$
24. Let $I_n = \int_{-n}^n \left(\{x+1\} \cdot \{x^2+2\} + \{x^2+2\} \{x^3+4\} \right) dx$, where $\{.\}$ denotes the fractional part of x . Find I_1 .
25. Evaluate : $\int_0^a \frac{\ln(1+ax)}{1+x^2} dx, a \in \mathbb{N}$
26. Evaluate : $\int_0^{\ln 3} \frac{e^x + 1}{e^{2x} + 1} dx$

27. Let $I = \int_0^1 \frac{2+3x+4x^2}{2\sqrt{1+x+x^2}} dx$. Find the value of I^2 .
28. Evaluate : $\int_{-1}^1 \frac{(2x^{332} + x^{998} + 4x^{1668} \cdot \sin x^{691})}{1+x^{666}} dx$
29. Show that $\int_0^{p+q\pi} |\cos x| dx = 2q + \sin p$ where $q \in \mathbb{N}$ & $-\frac{\pi}{2} < p < \frac{\pi}{2}$.
30. Evaluate : $\int_0^\pi \frac{x^2 \sin 2x \cdot \sin\left(\frac{\pi}{2} \cdot \cos x\right)}{2x - \pi} dx$
31. If $\phi(x) = \cos x - \int_0^x (x-t)\phi(t)dt$. Then find the value of $\phi''(x) + \phi(x)$.
32. (a) Let $g(x) = x^c \cdot e^{2x}$ & let $f(x) = \int_0^x e^{2t} \cdot (3t^2 + 1)^{1/2} dt$. For a certain value of 'c', the limit of $\frac{f'(x)}{g'(x)}$ as $x \rightarrow \infty$ is finite and non-zero. Determine the value of 'c' and the limit.
- (b) Find the constants 'a' ($a > 0$) and 'b' such that $\lim_{x \rightarrow 0} \frac{\int_0^x \frac{t^2 dt}{\sqrt{a+t}}}{bx - \sin x} = 1$.
33. Evaluate : $\lim_{x \rightarrow +\infty} \frac{d}{dx} \int_{2\sin \frac{1}{x}}^{3\sqrt{x}} \frac{3t^4 + 1}{(t-3)(t^2+3)} dt$
34. Evaluate (a) $\lim_{n \rightarrow \infty} \left[\left(1 + \frac{1}{n^2}\right) \left(1 + \frac{2^2}{n^2}\right) \left(1 + \frac{3^2}{n^2}\right) \dots \left(1 + \frac{n^2}{n^2}\right) \right]^{1/n}$ (b) $\lim_{n \rightarrow \infty} \frac{1}{n} \left[\frac{1}{n+1} + \frac{2}{n+2} + \dots + \frac{3n}{4n} \right]$
35. Find a positive real valued continuously differentiable functions f on the real line such that for all x
- $$f^2(x) = \int_0^x \left((f(t))^2 + (f'(t))^2 \right) dt + e^2$$
36. Let $f(x)$ be a function defined on $\mathbb{R}$ such that $f'(x) = f'(3-x) \forall x \in [0,3]$ with $f(0) = -32$ and $f(3) = 46$. Then find the value of $\int_0^3 f(x) dx$.
37. Let f and g be function that are differentiable for all real numbers x and that have the following properties:
- (i) $f'(x) = f(x) - g(x)$; (ii) $g'(x) = g(x) - f(x)$; (iii) $f(0) = 5$; (iv) $g(0) = 1$
- (a) Prove that $f(x) + g(x) = 6$ for all x . (b) Find $f(x)$ and $g(x)$

38. Consider a function $f(n) = \frac{1}{1+n^2}$. Let $\alpha_n = \frac{1}{n} \sum_{r=1}^n f\left(\frac{r}{n}\right)$ and $\beta_n = \frac{1}{n} \sum_{r=0}^{n-1} f\left(\frac{r}{n}\right)$ for $n = 1, 2, 3, \dots$

Also $\alpha = \lim_{n \rightarrow \infty} \alpha_n$ & $\beta = \lim_{n \rightarrow \infty} \beta_n$. Then prove (a) $\alpha_n < \beta_n$ (b) $\alpha = \beta$ (c) $\alpha_n < \frac{\pi}{4} < \beta_n$

39. Let $U_{10} = \int_0^{\frac{\pi}{2}} x \sin^{10} x dx$, then find the value of $\left(\frac{100U_{10} - 1}{U_8} \right)$.

40. Prove the inequalities :

$$(a) \frac{\pi}{6} < \int_0^1 \frac{dx}{\sqrt{4-x^2-x^3}} < \frac{\pi\sqrt{2}}{8}$$

$$(b) 2e^{-1/4} < \int_0^2 e^{x^2-x} dx < 2e^2.$$

EXERCISE (S-2)

1. Evaluate : $\int_0^1 x (\tan^{-1} x)^2 dx$

2. Evaluate : $\int_0^1 x^5 \sqrt{\frac{1+x^2}{1-x^2}} dx$

3. Let $A = \int_{3/4}^{4/3} \frac{2x^2 + x + 1}{x^3 + x^2 + x + 1} dx$, then find the value of e^A .

4. Evaluate : $\int_0^1 \frac{2-x^2}{(1+x)\sqrt{1-x^2}} dx$

5. Suppose f is continuous, $f(0) = 0$, $f(1) = 1$, $f'(x) > 0$ and $\int_0^1 f(x) dx = \frac{1}{3}$. Find the value of the definite integral $\int_0^1 f^{-1}(y) dy$.

6. Prove that :

$$(a) \int_{\alpha}^{\beta} \sqrt{(x-\alpha)(\beta-x)} dx = \frac{(\beta-\alpha)^2 \pi}{8}$$

$$(b) \int_{\alpha}^{\beta} \sqrt{\frac{x-\alpha}{\beta-x}} dx = (\beta-\alpha) \frac{\pi}{2}$$

$$(c) \int_{\alpha}^{\beta} \frac{dx}{x \sqrt{(x-\alpha)(\beta-x)}} = \frac{\pi}{\sqrt{\alpha\beta}} \text{ where } \alpha, \beta > 0$$

$$(d) \int_{\alpha}^{\beta} \frac{x \cdot dx}{\sqrt{(x-\alpha)(\beta-x)}} = (\alpha+\beta) \frac{\pi}{2}, \text{ where } \alpha < \beta$$

7. (a) Let $\beta(n) = \int_0^{n\pi} \sqrt{1-\sin t} dt$. Find the value of $\beta(2) - \beta(1)$.

(b) Determine a positive integer $n \leq 5$, such that $\int_0^1 e^x (x-1)^n dx = 16 - 6e$.

8. Evaluate : $\int_0^1 \frac{1-x}{1+x} \frac{dx}{\sqrt{x+x^2+x^3}}$

9. Evaluate : $\int_0^{\sqrt{3}} \sin^{-1} \frac{2x}{1+x^2} dx$
10. Evaluate : $\int_1^{16} \tan^{-1} \sqrt{\sqrt{x}-1} dx$
11. Evaluate : $\int_{\frac{\sqrt{3a^2+b^2}}{2}}^{\frac{\sqrt{a^2+b^2}}{2}} \frac{x \cdot dx}{\sqrt{(x^2-a^2)(b^2-x^2)}}$
12. (a) Show that $\int_0^\infty \frac{dx}{x^2+2x\cos\theta+1} = 2 \int_0^1 \frac{dx}{x^2+2x\cos\theta+1}$
- (b) Evaluate : $f(\theta) = \int_0^\infty \frac{\tan^{-1} x}{x^2+2x\cos\theta+1} dx, \theta \in (0, \pi)$.
13. Evaluate : $\lim_{n \rightarrow \infty} \frac{1}{n^2} \sum_{k=0}^{n-1} \left[k \int_k^{k+1} \sqrt{(x-k)(k+1-x)} dx \right]$
14. Let $y = f(x)$ be a quadratic function with $f'(2) = 1$. Find the value of the integral $\int_{-\pi}^{2+\pi} f(x) \cdot \sin\left(\frac{x-2}{2}\right) dx$.
15. Find the range of the function, $f(x) = \int_{-1}^1 \frac{\sin xdt}{1-2t\cos x+t^2}$
16. Let $f(x) = \begin{cases} -1 & \text{if } -2 \leq x \leq 0 \\ |x-1| & \text{if } 0 < x \leq 2 \end{cases}$ and $g(x) = \int_{-2}^x f(t)dt$. Define $g(x)$ as a function of x and test the continuity and differentiability of $g(x)$ in $(-2, 2)$.
17. If $y = x^{\int_1^x \ln t dt}$, find $\frac{dy}{dx}$ at $x = e$.
18. A curve C_1 is defined by : $\frac{dy}{dx} = e^x \cos x$ for $x \in [0, 2\pi]$ and passes through the origin. Prove that the roots of the function $y = 0$ (other than zero) occurs in the ranges $\frac{\pi}{2} < x < \pi$ and $\frac{3\pi}{2} < x < 2\pi$.
19. Determine a pair of number a and b for which $\int_0^1 \frac{ax+b}{(x^2+3x+2)^2} dx = \frac{5}{2}$.
20. Let $F(x) = \int_{-1}^x \sqrt{4+t^2} dt$ and $G(x) = \int_x^1 \sqrt{4+t^2} dt$ then compute the value of $(FG)'(0)$ where dash denotes the derivative.

21. A student forgot the product rule for differentiation and made the mistake of thinking that $(f \cdot g)' = f'g'$. However he was lucky to get the correct answer. The function f that he used was $f(x) = e^{x^2}$.

If the domain of $g(x)$ was the interval $\left(\frac{1}{2}, \infty\right)$ with $g(1) = e$. Find the value of $g(5)$.

22. (a) $\lim_{n \rightarrow \infty} \left[\frac{n!}{n^n} \right]^{1/n}$ (b) Let $P_n = \sqrt[n]{\frac{(3n)!}{(2n)!}}$ ($n = 1, 2, 3, \dots$), then find $\lim_{n \rightarrow \infty} \frac{P_n}{n}$.

23. If $f(x) = x + \sin x$ and I denotes the value of integral $\int_{\pi}^{2\pi} (f^{-1}(x) + \sin x) dx$ then the value of $\left[\frac{2I}{3} \right]$

(where $[.]$ denotes greatest integer function)

24. Prove the inequalities :

(a) $\frac{1}{3} < \int_0^1 x^{(\sin x + \cos x)^2} dx < \frac{1}{2}$ (b) $\frac{1}{2} \leq \int_0^2 \frac{dx}{2+x^2} \leq \frac{5}{6}$

25. For $a \geq 2$, if the value of the definite integral $\int_0^{\infty} \frac{dx}{a^2 + (x - (1/x))^2}$ equals $\frac{\pi}{5050}$. Find the value of a .

26. Find the value of the definite integral $\int_0^{\pi} |\sqrt{2} \sin x + 2 \cos x| dx$.

27. Evaluate : $\int_0^{2\pi} \frac{dx}{2 + \sin 2x}$

28. Evaluate : $\int_0^{\pi/2} \tan^{-1} \left[\frac{\sqrt{1+\sin x} + \sqrt{1-\sin x}}{\sqrt{1+\sin x} - \sqrt{1-\sin x}} \right] dx$

29. Let $I = \int_0^{\pi/2} \frac{\cos x + 4}{3 \sin x + 4 \cos x + 25} dx$ and $J = \int_0^{\pi/2} \frac{\sin x + 3}{3 \sin x + 4 \cos x + 25} dx$.

If $25I = a\pi + b/n \frac{c}{d}$ where a, b, c and $d \in \mathbb{N}$ and $\frac{c}{d}$ is not a perfect square of a rational then find the value of $(a + b + c + d)$.

30. Comment upon the nature of roots of the quadratic equation $x^2 + 2x = k + \int_0^1 |t + k| dt$ depending on the value of $k \in \mathbb{R}$.

31. If $U_n = \int_0^{\pi/2} \frac{\sin^2 nx}{\sin^2 x} dx$, then show that $U_1, U_2, U_3, \dots, U_n$ constitute an AP.

Hence or otherwise find the value of U_n .

EXERCISE (JM)

1. $\int_0^{\pi} [\cot x] dx$, where $[\cdot]$ denotes the greatest integer function, is equal to - [AIEEE-2009]

(1) -1 (2) $-\frac{\pi}{2}$ (3) $\frac{\pi}{2}$ (4) 1

2. Let $p(x)$ be a function defined on R such that $p'(x) = p'(1-x)$, for all $x \in [0, 1]$, $p(0) = 1$ and $p(1) = 41$. Then $\int_0^1 p(x) dx$ equals :- [AIEEE-2010]

(1) $\sqrt{41}$ (2) 21 (3) 41 (4) 42

3. The value of $\int_0^1 \frac{8 \log(1+x)}{1+x^2} dx$ is :- [AIEEE-2011]

(1) $\frac{\pi}{2} \log 2$ (2) $\log 2$ (3) $\pi \log 2$ (4) $\frac{\pi}{8} \log 2$

4. Let $[\cdot]$ denote the greatest integer function then the value of $\int_0^{1.5} x[x^2] dx$ is :- [AIEEE-2011]

(1) $\frac{5}{4}$ (2) 0 (3) $\frac{3}{2}$ (4) $\frac{3}{4}$

5. If $g(x) = \int_0^x \cos 4t dt$, then $g(x + \pi)$ equals : [AIEEE-2012]

(1) $g(x) \cdot g(\pi)$ (2) $\frac{g(x)}{g(\pi)}$ (3) $g(x) + g(\pi)$ (4) $g(x) - g(\pi)$

6. **Statement-I** : The value of the integral $\int_{\pi/6}^{\pi/3} \frac{dx}{1 + \sqrt{\tan x}}$ is equal to $\frac{\pi}{6}$. [JEE-MAIN-2013]

Statement-II : $\int_a^b f(x) dx = \int_a^b f(a+b-x) dx$.

- (1) Statement-I is true, Statement-II is true; Statement-II is a **correct** explanation for Statement-I.
 (2) Statement-I is true, Statement-II is true; Statement-II is **not** a correct explanation for Statement-I.
 (3) Statement-I is true, Statement-II is false.
 (4) Statement-I is false, Statement-II is true.

7. The integral $\int_0^{\pi} \sqrt{1 + 4 \sin^2 \frac{x}{2}} - 4 \sin \frac{x}{2} dx$ equals : [JEE-MAIN-2014]

(1) $\pi - 4$ (2) $\frac{2\pi}{3} - 4 - 4\sqrt{3}$ (3) $4\sqrt{3} - 4$ (4) $4\sqrt{3} - 4 - \frac{\pi}{3}$

8. The integral $\int_2^4 \frac{\log x^2}{\log x^2 + \log(36 - 12x + x^2)} dx$ is equal to : [JEE-MAIN-2015]

- (1) 1 (2) 6 (3) 2 (4) 4

9. The integral $\int_{\frac{\pi}{4}}^{\frac{3\pi}{4}} \frac{dx}{1 + \cos x}$ is equal to :- [JEE-MAIN-2017]

- (1) -1 (2) -2 (3) 2 (4) 4

10. The value of $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\sin^2 x}{1 + 2^x} dx$ is : [JEE-MAIN-2018]

- (1) $\frac{\pi}{2}$ (2) 4π (3) $\frac{\pi}{4}$ (4) $\frac{\pi}{8}$

11. The value of $\int_{-\pi/2}^{\pi/2} \frac{dx}{[x] + [\sin x] + 4}$, where $[t]$ denotes the greatest integer less than or equal to t , is : [JEE-MAIN 2019]

- (1) $\frac{1}{12}(7\pi + 5)$ (2) $\frac{3}{10}(4\pi - 3)$ (3) $\frac{1}{12}(7\pi - 5)$ (4) $\frac{3}{20}(4\pi - 3)$

12. If $\int_0^x f(t) dt = x^2 + \int_x^1 t^2 f(t) dt$, then $f(1/2)$ is : [JEE-MAIN 2019]

- (1) $\frac{6}{25}$ (2) $\frac{24}{25}$ (3) $\frac{18}{25}$ (4) $\frac{4}{5}$

13. Let $I = \int_a^b (x^4 - 2x^2) dx$. If I is minimum then the ordered pair (a, b) is : [JEE-MAIN 2019]

- (1) $(-\sqrt{2}, 0)$ (2) $(-\sqrt{2}, \sqrt{2})$ (3) $(0, \sqrt{2})$ (4) $(\sqrt{2}, -\sqrt{2})$

14. $\lim_{n \rightarrow \infty} \left(\frac{n}{n^2 + 1^2} + \frac{n}{n^2 + 2^2} + \frac{n}{n^2 + 3^2} + \dots + \frac{1}{5n} \right)$ is equal to : [JEE-MAIN 2019]

- (1) $\frac{\pi}{4}$ (2) $\tan^{-1}(2)$ (3) $\tan^{-1}(3)$ (4) $\frac{\pi}{2}$

EXERCISE (JA)

1. Let f be a non-negative function defined on the interval $[0, 1]$. If $\int_0^x \sqrt{1 - (f'(t))^2} dt = \int_0^x f(t) dt$, $0 \leq x \leq 1$, and $f(0) = 0$, then - [JEE 2009, 3]

(A) $f\left(\frac{1}{2}\right) < \frac{1}{2}$ and $f\left(\frac{1}{3}\right) > \frac{1}{3}$ (B) $f\left(\frac{1}{2}\right) > \frac{1}{2}$ and $f\left(\frac{1}{3}\right) > \frac{1}{3}$
 (C) $f\left(\frac{1}{2}\right) < \frac{1}{2}$ and $f\left(\frac{1}{3}\right) < \frac{1}{3}$ (D) $f\left(\frac{1}{2}\right) > \frac{1}{2}$ and $f\left(\frac{1}{3}\right) < \frac{1}{3}$

2. If $I_n = \int_{-\pi}^{\pi} \frac{\sin nx}{(1 + \pi^x) \sin x} dx$, $n = 0, 1, 2, \dots$, then - [JEE 2009, 4]

(A) $I_n = I_{n+2}$ (B) $\sum_{m=1}^{10} I_{2m+1} = 10\pi$ (C) $\sum_{m=1}^{10} I_{2m} = 0$ (D) $I_n = I_{n+1}$

3. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function which satisfies $f(x) = \int_0^x f(t) dt$. Then the value of $f(\ln 5)$ is..... [JEE 2009, 4]

4. The value of $\lim_{x \rightarrow 0} \frac{1}{x^3} \int_0^x \frac{t \ln(1+t)}{t^4 + 4} dt$ is [JEE 2010, 3 (-1)]

(A) 0 (B) $\frac{1}{12}$ (C) $\frac{1}{24}$ (D) $\frac{1}{64}$

5. The value(s) of $\int_0^1 \frac{x^4(1-x)^4}{1+x^2} dx$ is (are) [JEE 2010, 3]

(A) $\frac{22}{7} - \pi$ (B) $\frac{2}{105}$ (C) 0 (D) $\frac{71}{15} - \frac{3\pi}{2}$

6. Let f be a real-valued function defined on the interval $(-1, 1)$ such that $e^{-x} f(x) = 2 + \int_0^x \sqrt{t^4 + 1} dt$, for all $x \in (-1, 1)$, and let f^{-1} be the inverse function of f . Then $(f^{-1})'(2)$ is equal to- [JEE2010, 5 (-2)]

(A) 1 (B) $\frac{1}{3}$ (C) $\frac{1}{2}$ (D) $\frac{1}{e}$

7. For any real number x , let $[x]$ denote the largest integer less than or equal to x . Let f be a real valued function defined on the interval $[-10, 10]$ by $f(x) = \begin{cases} x - [x] & \text{if } [x] \text{ is odd,} \\ 1 + [x] - x & \text{if } [x] \text{ is even} \end{cases}$

Then the value of $\frac{\pi^2}{10} \int_{-10}^{10} f(x) \cos \pi x dx$ is

[JEE 2010, 3]

8. The value of $\int_{\sqrt{\ln 2}}^{\sqrt{\ln 3}} \frac{x \sin x^2}{\sin x^2 + \sin(\ln 6 - x^2)} dx$ is [JEE 2011, 3 (-1)]

(A) $\frac{1}{4} \ln \frac{3}{2}$ (B) $\frac{1}{2} \ln \frac{3}{2}$ (C) $\ln \frac{3}{2}$ (D) $\frac{1}{6} \ln \frac{3}{2}$

9. Let S be the area of the region enclosed by $y = e^{-x^2}$, $y = 0$, $x = 0$, and $x = 1$. Then - [JEE 2012, 4]

(A) $S \geq \frac{1}{e}$ (B) $S \geq 1 - \frac{1}{e}$ (C) $S \leq \frac{1}{4} \left(1 + \frac{1}{\sqrt{e}} \right)$ (D) $S \leq \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{e}} \left(1 - \frac{1}{\sqrt{2}} \right)$

10. The value of the integral $\int_{-\pi/2}^{\pi/2} \left(x^2 + \ln \frac{\pi+x}{\pi-x} \right) \cos x dx$ is [JEE 2012, 3, (-1)]

(A) 0 (B) $\frac{\pi^2}{2} - 4$ (C) $\frac{\pi^2}{2} + 4$ (D) $\frac{\pi^2}{2}$

11. For $a \in \mathbb{R}$ (the set of all real numbers), $a \neq -1$.

$$\lim_{n \rightarrow \infty} \frac{(1^a + 2^a + \dots + n^a)}{(n+1)^{a-1} [(na+1) + (na+2) + \dots + (na+n)]} = \frac{1}{60}$$

Then $a =$

[JEE(Advanced) 2013, 3, (-1)]

(A) 5 (B) 7 (C) $-\frac{15}{2}$ (D) $-\frac{17}{2}$

12. Let $f : [a, b] \rightarrow [1, \infty)$ be a continuous function and let $g : \mathbb{R} \rightarrow \mathbb{R}$ be defined as

$$g(x) = \begin{cases} 0 & \text{if } x < a \\ \int_a^x f(t) dt & \text{if } a \leq x \leq b \\ \int_a^b f(t) dt & \text{if } x > b \end{cases}$$

Then

[JEE(Advanced)-2014, 3]

- (A) $g(x)$ is continuous but not differentiable at a
 (B) $g(x)$ is differentiable on $\mathbb{R}$
 (C) $g(x)$ is continuous but not differentiable at b
 (D) $g(x)$ is continuous and differentiable at either a or b but not both.

13. The value of $\int_0^1 4x^3 \left\{ \frac{d^2}{dx^2} (1-x^2)^5 \right\} dx$ is [JEE(Advanced)-2014, 3]

14. The following integral $\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} (2 \operatorname{cosec} x)^{17} dx$ is equal to - [JEE(Advanced)-2014, 3(-1)]

(A) $\int_0^{\log(1+\sqrt{2})} 2(e^u + e^{-u})^{16} du$ (B) $\int_0^{\log(1+\sqrt{2})} (e^u + e^{-u})^{17} du$
 (C) $\int_0^{\log(1+\sqrt{2})} (e^u - e^{-u})^{17} du$ (D) $\int_0^{\log(1+\sqrt{2})} 2(e^u - e^{-u})^{16} du$

15. Let $f : [0, 2] \rightarrow \mathbb{R}$ be a function which is continuous on $[0, 2]$ and is differentiable on $(0, 2)$ with

$f(0) = 1$. Let $F(x) = \int_0^x f(\sqrt{t}) dt$ for $x \in [0, 2]$. If $F'(x) = f'(x)$ for all $x \in (0, 2)$, then $F(2)$ equals -

- (A) $e^2 - 1$ (B) $e^4 - 1$ (C) $e - 1$ (D) e^4

[JEE(Advanced)-2014, 3(-1)]

Paragraph For Questions 16 and 17

Given that for each $a \in (0, 1)$, $\lim_{h \rightarrow 0^+} \int_h^{1-h} t^{-a} (1-t)^{a-1} dt$ exists. Let this limit be $g(a)$. In addition, it is given that the function $g(a)$ is differentiable on $(0, 1)$.

16. The value of $g\left(\frac{1}{2}\right)$ is -

[JEE(Advanced)-2014, 3(-1)]

- (A) π (B) 2π (C) $\frac{\pi}{2}$ (D) $\frac{\pi}{4}$

17. The value of $g'\left(\frac{1}{2}\right)$ is-

[JEE(Advanced)-2014, 3(-1)]

- (A) $\frac{\pi}{2}$ (B) π (C) $-\frac{\pi}{2}$ (D) 0

18. **List-I**

P. The number of polynomials $f(x)$ with non-negative integer coefficients of degree ≤ 2 , satisfying

$f(0) = 0$ and $\int_0^1 f(x) dx = 1$, is

Q. The number of points in the interval $[-\sqrt{13}, \sqrt{13}]$

at which $f(x) = \sin(x^2) + \cos(x^2)$ attains its maximum value, is

R. $\int_{-2}^2 \frac{3x^2}{(1+e^x)} dx$ equals

S. $\frac{\left(\int_{-\frac{1}{2}}^{\frac{1}{2}} \cos 2x \log\left(\frac{1+x}{1-x}\right) dx\right)}{\left(\int_0^{\frac{1}{2}} \cos 2x \log\left(\frac{1+x}{1-x}\right) dx\right)}$ equals

List-II

1. 8

2. 2

3. 4

4. 0

Codes :

	P	Q	R	S
(A)	3	2	4	1
(B)	2	3	4	1
(C)	3	2	1	4
(D)	2	3	1	4

[JEE(Advanced)-2014, 3(-1)]

19. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function defined by $f(x) = \begin{cases} [x] & , x \leq 2 \\ 0 & , x > 2 \end{cases}$,

where $[x]$ is the greatest integer less than or equal to x . If $I = \int_{-1}^2 \frac{x f(x^2)}{2 + f(x+1)} dx$, then the value of

$(4I - 1)$ is

[JEE 2015, 4M, -0M]

20. If $\alpha = \int_0^1 \left(e^{9x+3\tan^{-1}x} \right) \left(\frac{12+9x^2}{1+x^2} \right) dx$, where $\tan^{-1}x$ takes only principal values, then the value of $\left(\log_e |1+\alpha| - \frac{3\pi}{4} \right)$ is

[JEE 2015, 4M, -0M]

21. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous odd function, which vanishes exactly at one point and $f(1) = \frac{1}{2}$. Suppose

that $F(x) = \int_{-1}^x f(t) dt$ for all $x \in [-1, 2]$ and $G(x) = \int_{-1}^x t |f(f(t))| dt$ for all $x \in [-1, 2]$. If $\lim_{x \rightarrow 1} \frac{F(x)}{G(x)} = \frac{1}{14}$,

then the value of $f\left(\frac{1}{2}\right)$ is

[JEE 2015, 4M, -0M]

22. The option(s) with the values of a and L that satisfy the following equation is(are)

$$\frac{\int_0^{4\pi} e^t (\sin^6 at + \cos^4 at) dt}{\int_0^{\pi} e^t (\sin^6 at + \cos^4 at) dt} = L ?$$

[JEE 2015, 4M, -0M]

- (A) $a = 2, L = \frac{e^{4\pi} - 1}{e^{\pi} - 1}$ (B) $a = 2, L = \frac{e^{4\pi} + 1}{e^{\pi} + 1}$ (C) $a = 4, L = \frac{e^{4\pi} - 1}{e^{\pi} - 1}$ (D) $a = 4, L = \frac{e^{4\pi} + 1}{e^{\pi} + 1}$

23. Let $f(x) = 7\tan^8 x + 7\tan^6 x - 3\tan^4 x - 3\tan^2 x$ for all $x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$. Then the correct expression(s) is(are)

[JEE 2015, 4M, -0M]

- (A) $\int_0^{\pi/4} x f(x) dx = \frac{1}{12}$ (B) $\int_0^{\pi/4} f(x) dx = 0$ (C) $\int_0^{\pi/4} x f(x) dx = \frac{1}{6}$ (D) $\int_0^{\pi/4} f(x) dx = 1$

24. Let $f'(x) = \frac{192x^3}{2 + \sin^4 \pi x}$ for all $x \in \mathbb{R}$ with $f\left(\frac{1}{2}\right) = 0$. If $m \leq \int_{1/2}^1 f(x) dx \leq M$, then the possible values of m and M are

[JEE 2015, 4M, -0M]

- (A) $m = 13, M = 24$ (B) $m = \frac{1}{4}, M = \frac{1}{2}$ (C) $m = -11, M = 0$ (D) $m = 1, M = 12$

Paragraph For Questions 25 and 26

Let $F : \mathbb{R} \rightarrow \mathbb{R}$ be a thrice differentiable function. Suppose that $F(1) = 0, F(3) = -4, F'(x) < 0$ for all $x \in (1/2, 3)$. Let $f(x) = xF(x)$ for all $x \in \mathbb{R}$.

25. The correct statement(s) is(are)

[JEE 2015, 4M, -0M]

- (A) $f'(1) < 0$ (B) $f(2) < 0$
(C) $f'(x) \neq 0$ for any $x \in (1, 3)$ (D) $f'(x) = 0$ for some $x \in (1, 3)$

26. If $\int_1^3 x^2 F'(x) dx = -12$ and $\int_1^3 x^3 F''(x) dx = 40$, then the correct expression(s) is(are) [JEE 2015, 4M, -0M]

(A) $9f'(3) + f'(1) - 32 = 0$ (B) $\int_1^3 f(x) dx = 12$
 (C) $9f'(3) - f'(1) + 32 = 0$ (D) $\int_1^3 f(x) dx = -12$

27. The value of $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{x^2 \cos x}{1 + e^x} dx$ is equal to

[JEE(Advanced)-2016, 3(-1)]

(A) $\frac{\pi^2}{4} - 2$ (B) $\frac{\pi^2}{4} + 2$ (C) $\pi^2 - e^{\frac{\pi}{2}}$ (D) $\pi^2 + e^{\frac{\pi}{2}}$

28. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a differentiable function such that $f(0) = 0$, $f\left(\frac{\pi}{2}\right) = 3$ and $f(0) = 1$. If

$$g(x) = \int_x^{\frac{\pi}{2}} [f'(t) \operatorname{cosec} t - \cot t \operatorname{cosec} t f(t)] dt$$

for $x \in \left(0, \frac{\pi}{2}\right]$, then $\lim_{x \rightarrow 0} g(x) =$

[JEE(Advanced)-2017, 3]

29. If $I = \sum_{k=1}^{98} \int_k^{k+1} \frac{k+1}{x(x+1)} dx$, then

[JEE(Advanced)-2017, 4]

(A) $I < \frac{49}{50}$ (B) $I < \log_e 99$ (C) $I > \frac{49}{50}$ (D) $I > \log_e 99$

30. If $g(x) = \int_{\sin x}^{\sin(2x)} \sin^{-1}(t) dt$, then

[JEE(Advanced)-2017, 4]

(A) $g'\left(\frac{\pi}{2}\right) = -2\pi$ (B) $g'\left(-\frac{\pi}{2}\right) = 2\pi$ (C) $g'\left(\frac{\pi}{2}\right) = 2\pi$ (D) $g'\left(-\frac{\pi}{2}\right) = -2\pi$

31. For each positive integer n , let

$$y_n = \frac{1}{n} (n+1)(n+2) \dots (n+n)^{1/n}$$

For $x \in \mathbb{R}$, let $[x]$ be the greatest integer less than or equal to x . If $\lim_{n \rightarrow \infty} y_n = L$, then the value of

$[L]$ is _____

[JEE(Advanced)-2018, 3(0)]

32. The value of the integral

$$\int_0^{\frac{1}{2}} \frac{1 + \sqrt{3}}{\left((x+1)^2 (1-x)^6\right)^{\frac{1}{4}}} dx$$

is _____.

[JEE(Advanced)-2018, 3(0)]

ANSWER KEY

DEFINITE INTEGRATION

EXERCISE (O-1)

- | | | | | | | | |
|-------|-------|-------|-------|-------|-------|-------|-------|
| 1. A | 2. C | 3. A | 4. B | 5. C | 6. A | 7. A | 8. D |
| 9. B | 10. A | 11. A | 12. D | 13. A | 14. B | 15. D | 16. A |
| 17. B | 18. C | 19. B | 20. D | 21. D | 22. B | 23. C | 24. A |
| 25. D | 26. B | 27. A | 28. C | 29. C | 30. A | | |

EXERCISE (O-2)

- | | | | | | | | |
|-----------|-------|---------|-----------|-------------|---------|-----------|-------|
| 1. C | 2. A | 3. A | 4. A | 5. D | 6. B | 7. D | 8. B |
| 9. B | 10. D | 11. C | 12. C | 13. B | 14. C | 15. A | 16. B |
| 17. A | 18. A | 19. A,C | 20. B,C,D | 21. A,B,C,D | 22. C,D | 23. A,C,D | |
| 24. B,C,D | | | | | | | |

EXERCISE (S-1)

1. (i) $\frac{\pi^2}{8} - \frac{\pi}{4}(1 + \ln 2) + \frac{1}{2}$; (ii) $\frac{\pi \ln 3}{2}$
2. $\frac{\pi^2}{4}$
3. $\frac{1}{2}$
4. $\frac{1}{2}[e^{\pi/2}(\cos 1 + \sin 1) - 1]$
5. $e^{1+e} + e^{1-e} + e^{-e} - e^e + e - e^{-1}$
7. 125
8. $1 - \sec(1)$
9. $\ln 2$
10. 5250
11. $\frac{\pi}{2\sqrt{2}} - \frac{16\sqrt{2}}{5}$
12. $4\sqrt{2} - 4\ln(\sqrt{2} + 1)$
13. 4
14. $\frac{\pi}{8}\ln 2$
15. $\frac{\pi^2}{6\sqrt{3}}$
16. $\frac{\pi}{8}\ln 2$
17. 2012
18. 153
19. $\frac{\pi(a+b)}{2\sqrt{2}}$
20. 5
21. $\frac{(a\pi + 2b)\pi}{3\sqrt{3}}$
22. $\frac{\pi(\pi + 3)}{2}$
23. $\frac{3\pi}{2}$
24. $2/3$
25. $\tan^{-1}(a) \cdot \ln \sqrt{1+a^2}$
26. $\frac{1}{2}\left[\frac{\pi}{6} + \ln 3 - \ln 2\right]$
27. 3
28. $\frac{\pi + 4}{666}$
30. $\frac{8}{\pi}$
31. $-\cos x$
32. (a) $c = 1$ and Limit will be $\frac{\sqrt{3}}{2}$; (b) $a = 4$ and $b = 1$
33. 13.5
34. (a) $2e^{(1/2)(\pi-4)}$; (b) $3 - \ln 4$
35. $f(x) = e^{x+1}$
36. 21
37. (b) $f(x) = 3 + 2e^{2x}$; $g(x) = 3 - 2e^{2x}$
39. 90

EXERCISE (S-2)

1. $\frac{\pi}{4} \left(\frac{\pi}{4} - 1 \right) + \frac{1}{2} \ln 2$ 2. $\frac{3\pi + 8}{24}$ 3. $\frac{16}{9}$ 4. $\frac{\pi}{2}$ 5. $2/3$ 7. (a) 4; (b) 3
8. $\frac{\pi}{3}$ 9. $\frac{\pi\sqrt{3}}{3}$ 10. $\frac{16\pi}{3} - 2\sqrt{3}$ 11. $\frac{\pi}{12}$ 12. (b) $\frac{\pi\theta}{4\sin\theta}$ 13. $\frac{\pi}{16}$
14. I = 8 as $\int_0^{\pi/2} y \sin y dy = 1$ 15. $\left\{ -\frac{\pi}{2}, \frac{\pi}{2} \right\}$

$$16. \quad g(x) \text{ is cont. in } (-2, 2); g(x) \text{ is der. at } x = 1 \text{ \& not der. at } x = 0. \text{ Note that : } g(x) = \begin{cases} -(x+2) & \text{for } -2 \leq x \leq 0 \\ -2 + x - \frac{x^2}{2} & \text{for } 0 < x < 1 \\ \frac{x^2}{2} - x - 1 & \text{for } 1 \leq x \leq 2 \end{cases}$$

$$17. 1 + e \quad 19. a = 15, b = \frac{45}{2} \quad 20. 0 \quad 21. g(5) = 3e^5 \quad 22. (a) \frac{1}{e}; (b) \frac{27}{4e}$$

$$23. 9 \quad 25. 2525 \quad 26. 2\sqrt{6} \quad 27. \frac{2\pi}{\sqrt{3}} \quad 28. \frac{3\pi^2}{16} \quad 29. 62$$

$$30. \text{ real \& distinct } \forall k \in \mathbb{R} \quad 31. U_n = \frac{n\pi}{2}$$

EXERCISE (JM)

$$\begin{array}{lllllll} 1. 2 & 2. 2 & 3. 3 & 4. 4 & 5. 3, 4 & 6. 4 & 7. 4 & 8. 1 \\ 9. 3 & 10. 3 & 11. 4 & 12. 2 & 13. 2 & 14. 2 & & \end{array}$$

EXERCISE (JA)

$$\begin{array}{lllllll} 1. C & 2. A, B, C & 3. 0 & 4. B & 5. A & 6. B & 7. 4 & 8. A \\ 9. A, B, D & 10. B & 11. B & 12. A, C & 13. 2 & 14. A & 15. B & 16. A \\ 17. D & 18. D & 19. 0 & 20. 9 & 21. 7 & 22. A, C & 23. A, B & 24. D \\ 25. A, B, C & 26. C, D & 27. A & 28. 2 & 29. B, C & 30. Bonus & 22. 1 & 23. 2 \end{array}$$

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AREA UNDER THE CURVE

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JEE (Main) Syllabus :

Ordinary differential equations, their order and degree. Formation of differential equations. Solution of differential equations by the method of separation of variables, solution of homogeneous and linear differential equations of the type : $dy + p(x)y = q(x)dx$.

Determining areas of the regions bounded by simple curves in standard form.

JEE (Advanced) Syllabus :

Formation of ordinary differential equations, solution of homogeneous differential equations, separation of variables method, linear first order differential equations.

Application of definite integrals to the determination of areas involving simple curves.

DIFFERENTIAL EQUATION

1. DIFFERENTIAL EQUATION :

An equation that involves independent and dependent variables and the derivatives of the dependent variables is called a **differential equation**.

A differential equation is said to be **ordinary**, if the differential coefficients have reference to a

single independent variable only e.g. $\frac{d^2y}{dx^2} - \frac{2dy}{dx} + \cos x = 0$ and it is said to be **partial** if there are

two or more independent variables. e.g. $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = 0$ is a partial differential equation. We are concerned with ordinary differential equations only.

2. ORDER OF DIFFERENTIAL EQUATION :

The order of a differential equation is the order of the highest differential coefficient occurring in it.

3. DEGREE OF DIFFERENTIAL EQUATION :

The exponent of the highest order differential coefficient, when the differential equation is expressed as a polynomial in all the differential coefficient.

Thus the differential equation :

$$f(x, y) \left[\frac{d^m y}{dx^m} \right]^p + \phi(x, y) \left[\frac{d^{m-1} (y)}{dx^{m-1}} \right]^q + \dots = 0 \text{ is of order } m \text{ \& degree } p.$$

Note :

- (i) The exponents of all the differential coefficient should be free from radicals and fraction.
- (ii) The degree is always positive natural number.
- (iii) The degree of differential equation may or may not exist.

Illustration 1: Find the order and degree of the following differential equation :

(i) $\sqrt{\frac{d^2y}{dx^2}} = \sqrt[3]{\frac{dy}{dx}} + 3$ (ii) $\frac{d^2y}{dx^2} = \sin\left(\frac{dy}{dx}\right)$ (iii) $\frac{dy}{dx} = \sqrt{3x+5}$

Solution :

(i) The given differential equation can be re-written as $\left(\frac{d^2y}{dx^2}\right)^3 = \left(\frac{dy}{dx} + 3\right)^2$

Hence order is 2 and degree is 3.

- (ii) The given differential equation has the order 2. Since the given differential equation cannot be written as a polynomial in the differential coefficients, the degree of the equation is not defined.
- (iii) Its order is 1 and degree 1. **Ans.**

Ans.

Illustration 2 : The order and degree of the differential equation $\left(\frac{d^2s}{dt^2}\right)^2 + 3\left(\frac{ds}{dt}\right)^3 + 4 = 0$ are -

- (A) 2, 2 (B) 2, 3 (C) 3, 2 (D) none of these

Solution : Clearly order is 2 and degree is 2 (from the definition of order and degree of differential equations). **Ans. (A)**

Do yourself - 1 :

Find the order and degree of following differential equations

- (i) $[1 + (y')^2]^{1/2} = x^2 + y$ (ii) $(1 + y')^{1/2} = y''$ (iii) $y' = \sin y$

4. FORMATION OF A DIFFERENTIAL EQUATION :

In order to obtain a differential equation whose solution is

$$f(x_1, y_1, c_1, c_2, c_3, \dots, c_n) = 0$$

where $c_1, c_2, \dots, c_n$ are 'n' arbitrary constants, we have to eliminate the 'n' constants for which we require (n+1) equations.

A differential equation is obtained as follows :

- Differentiate the given equation w.r.t the independent variable (say x) as many times as the number of independent arbitrary constants in it.
- Eliminate the arbitrary constants.
- The eliminant is the required differential equation.

Note :

- A differential equation represents a family of curves all satisfying some common properties. This can be considered as the geometrical interpretation of the differential equation.
- For there being n differentiation, the resulting equation must contain a derivative of n^{th} order i.e. equal to number of independent arbitrary constant.

Illustration 3 : Find the differential equation of all parabolas whose axes is parallel to the x-axis and having latus rectum a.

Solution : Equation of parabola whose axes is parallel to x-axis and having latus rectum 'a' is $(y - \beta)^2 = a(x - \alpha)$

Differentiating both sides, we get $2(y - \beta) \frac{dy}{dx} = a$

Again differentiating, we get

$$\Rightarrow 2(y - \beta) \frac{d^2y}{dx^2} + 2\left(\frac{dy}{dx}\right)^2 = 0 \quad \Rightarrow \quad a \frac{d^2y}{dx^2} + 2\left(\frac{dy}{dx}\right)^3 = 0.$$

Ans.

Illustration 4: Find the differential equation whose solution represents the family : $c(y + c)^2 = x^3$

Solution : $c(y + c)^2 = x^3 \quad \dots(i)$

Differentiating, we get, c. $[2(y + c)] \frac{dy}{dx} = 3x^2$

Writing the value of c from (i), we have

$$\frac{2x^3}{(y+c)^2} (y+c) \frac{dy}{dx} = 3x^2 \quad \Rightarrow \quad \frac{2x^3}{y+c} \frac{2x^3}{y+c} = 3x^2$$

$$\text{i.e. } \frac{2x}{y+c} \frac{dy}{dx} = 3 \Rightarrow \frac{2x}{3} \left[\frac{dy}{dx} \right] = y + c$$

Hence $c = \frac{2x}{3} \left[\frac{dy}{dx} \right] - y$

Substituting value of c in equation (i), we get $\left[\frac{2x}{3} \left(\frac{dy}{dx} \right) - y \right] \left[\frac{2x}{3} \frac{dy}{dx} \right]^2 = x^3$,

which is the required differential equation.

Ans.

Illustration 5 : Find the differential equation whose solution represents the family : $y = a \cos \theta x + b \sin \theta x$, where $\theta = \text{fixed constant}$

Solution : $y = a \cos \theta x + b \sin \theta x$, $\theta = \text{fixed constant}$ (i)

Differentiating, we get $\frac{dy}{dx} = -\theta a \sin \theta x + \theta b \cos \theta x$

Again differentiating, we get $\frac{d^2y}{dx^2} = -\theta^2 a \cos \theta x - \theta^2 b \sin \theta x$

using equation (i), we get $\frac{d^2y}{dx^2} = -\theta^2 y$

Ans.

Do yourself - 2

Eliminate the arbitrary constants and obtain the differential equation satisfied by it.

(i) $y = 2x + ce^x$ (ii) $y = \left(\frac{a}{x^2}\right) + bx$ (iii) $y = ae^{2x} + be^{-2x} + c$

5. SOLUTION OF DIFFERENTIAL EQUATION :

The solution of the differential equation is a relation between the variables of the equation not containing the derivatives, but satisfying the given differential equation (i.e., from which the given differential equation can be derived).

Thus, the solution of $\frac{dy}{dx} = e^x$ could be obtained by simply integrating both sides, i.e., $y = e^x + c$ and

that of, $\frac{dy}{dx} = px + q$ is $y = \frac{px^2}{2} + qx + c$, where c is arbitrary constant.

- (i) **A general solution** or an integral of a differential equation is a relation between the variables (not involving the derivatives) which contains the same number of the arbitrary constants as the order of the differential equation.

For example, a general solution of the differential equation $\frac{d^2x}{dt^2} = -4x$ is $x = A \cos 2t + B \sin 2t$

where A and B are the arbitrary constants.

- (ii) **Particular solution or particular integral** is that solution of the differential equation which is obtained from the general solution by assigning particular values to the arbitrary constant in the general solution.

For example, $x = 10 \cos 2t + 5 \sin 2t$ is a particular solution of differential equation $\frac{d^2x}{dt^2} = -4x$.

Note :

- (i) The general solution of a differential equation can be expressed in different (but equivalent) forms. For example

$$\log x - \log (y + 2) = k \quad \dots(i)$$

where k is an arbitrary constant is the general solution of the differential equation $xy' = y + 2$.

The solution given by equation (i) can also be re-written as

$$\log \left(\frac{x}{y+2} \right) = k \text{ or } \frac{x}{y+2} = e^k = c_1 \quad \dots(ii)$$

$$\text{or } x = c_1(y + 2) \quad \dots(iii)$$

where $c_1 = e^k$ is another arbitrary constant. The solution (iii) can also be written as

$$y + 2 = c_2x$$

where $c_2 = 1/c_1$ is another arbitrary constant.

- (ii) All differential equations that we come across have unique solutions or a family of solutions.

For example, the differential equation $\left| \frac{dy}{dx} \right| + |y| = 0$ has only the trivial solution, i.e. $y = 0$.

The differential equation $\left| \frac{dy}{dx} \right| + |y| + c = 0$, $c > 0$ has no solution.

6. ELEMENTARY TYPES OF FIRST ORDER & FIRST DEGREE DIFFERENTIAL EQUATIONS :

- (a) **Separation of Variables :**

Some differential equations can be solved by the method of separation of variables (or “variable separable”). This method is only possible, if we can express the differential equation in the form

$$A(x)dx + B(y) dy = 0$$

where A(x) is a function of 'x' only and B(y) is a function of 'y' only.

A general solution of this is given by,

$$\int A(x) dx + \int B(y)dy = c$$

where 'c' is the arbitrary constant.

Illustration 6 : Solve the differential equation $xy \frac{dy}{dx} = \frac{1+y^2}{1+x^2} (1+x+x^2)$.

Solution : Differential equation can be rewritten as

$$xy \frac{dy}{dx} = (1+y^2) \left(1 + \frac{x}{1+x^2}\right) \Rightarrow \frac{y}{1+y^2} dy = \left(\frac{1}{x} + \frac{1}{1+x^2}\right) dx$$

Integrating, we get

$$\frac{1}{2} \ln(1+y^2) = \ln x + \tan^{-1} x + \ln c \Rightarrow \sqrt{1+y^2} = cxe^{\tan^{-1} x}.$$

Ans.

Illustration 7 : Solve the differential equation $(x^3 - y^2 x^3) \frac{dy}{dx} + y^3 + x^2 y^3 = 0$.

Solution : The given equation $(x^3 - y^2 x^3) \frac{dy}{dx} + y^3 + x^2 y^3 = 0$

Case-I : $y = 0$ is one solution of differential equation

Case-II : If $y \neq 0$, then

$$\Rightarrow \frac{1-y^2}{y^3} dy + \frac{1+x^2}{x^3} dx = 0 \Rightarrow \int \left(\frac{1}{y^3} - \frac{1}{y}\right) dy + \int \left(\frac{1}{x^3} + \frac{1}{x}\right) dx = 0$$

$$\Rightarrow \log\left(\frac{x}{y}\right) = \frac{1}{2} \left(\frac{1}{y^2} + \frac{1}{x^2}\right) + c$$

$$y = 0 \text{ or } \log\left(\frac{x}{y}\right) = \frac{1}{2} \left(\frac{1}{y^2} + \frac{1}{x^2}\right) + c$$

Ans.

Illustration 8 : Solve : $\frac{dy}{dx} = (x-3)(y+1)^{2/3}$

Solution : Case-I : $y = -1$ is one solution of differential equation

Case-II : If $y \neq -1$, then

$$\frac{dy}{dx} = (x-3)(y+1)^{2/3}$$

$$\int \frac{dy}{(y+1)^{2/3}} = \int (x-3) dx$$

$$\text{Integrate and solve for } y : 3(y+1)^{1/3} = \frac{1}{2}(x-3)^2 + C$$

$$(y+1)^{1/3} = \frac{1}{6}(x-3)^2 + C_0 \Rightarrow y+1 = \left(\frac{1}{6}(x-3)^2 + C_0\right)^3$$

$$y = -1 \text{ or } y = \left(\frac{1}{6}(x-3)^2 + C_0\right)^3 - 1$$

Ans.

Do yourself - 3 :

Solve the following differential equations :

(i) $\frac{2dy}{dx} = \frac{y(x+1)}{x}$ (ii) $\sqrt{1+4x^2} dy = y^3 x dx$ (iii) $(\tan y) \frac{dy}{dx} = \sin(x+y) + \sin(x-y)$

(i) Equation of the form :

$$\Rightarrow y' = f(ax + by + c), b \neq 0$$

To solve this, substitute $t = ax + by + c$. Then the equation reduces to separable type in the variable t and x which can be solved.

Illustration 9 : Solve $\frac{dy}{dx} = \cos(x+y) - \sin(x+y)$.

Solution : $\frac{dy}{dx} = \cos(x+y) - \sin(x+y)$

Substituting, $x+y=t$, we get $\frac{dy}{dx} = \frac{dt}{dx} - 1$

Therefore $\frac{dt}{dx} - 1 = \cos t - \sin t$

$$\Rightarrow \int \frac{dt}{1 + \cos t - \sin t} = \int dx \Rightarrow \int \frac{\sec^2 \frac{t}{2} dt}{2 \left(1 - \tan \frac{t}{2} \right)} = \int dx \Rightarrow -\ln \left| 1 - \tan \frac{x+y}{2} \right| = x + c.$$

Ans.

Illustration 10: Solve : $y' = (x+y+1)^2$

Solution : $y' = (x+y+1)^2$ (i)

Let $t = x + y + 1$

$$\frac{dt}{dx} = 1 + \frac{dy}{dx}$$

Substituting in equation (i) we get

$$\frac{dt}{dx} = t^2 + 1 \Rightarrow \int \frac{dt}{1+t^2} = \int dx \Rightarrow \tan^{-1} t = x + C \Rightarrow t = \tan(x+C)$$

$$x + y + 1 = \tan(x+C) \Rightarrow y = \tan(x+C) - x - 1$$

Ans.**Do yourself - 4 :**

Solve the following differential equations :

(i) $\frac{dy}{dx} = (y-4x)^2$ (ii) $\tan^2(x+y)dx - dy = 0$

(ii) Equation of the form : $\Rightarrow \frac{dy}{dx} = \frac{a_1x + b_1y + c_1}{a_2x + b_2y + c_2}$

Case I : If $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$ then

$$\text{Let } \frac{a_1}{a_2} = \frac{b_1}{b_2} = \lambda \quad \text{then } a_1 = \lambda a_2 \quad \dots(i); \quad b_1 = \lambda b_2 \quad \dots(ii)$$

from (i) and (ii), differential equation becomes

$$\frac{dy}{dx} = \frac{\lambda a_2x + \lambda b_2y + c_1}{a_2x + b_2y + c_2} \Rightarrow \frac{dy}{dx} = \frac{\lambda(a_2x + b_2y) + c_1}{a_2x + b_2y + c_2}$$

$$\text{or we can say, } \frac{dy}{dx} = f(a_2x + b_2y)$$

which can be solved by substituting $t = a_2x + b_2y$

Illustration 11 : Solve : $(x + y)dx + (3x + 3y - 4) dy = 0$

Solution :

$$\text{Let } t = x + y$$

$$\Rightarrow dy = dt - dx$$

$$\text{So we get, } tdx + (3t - 4)(dt - dx) = 0$$

$$2dx + \left(\frac{3t-4}{2-t}\right)dt = 0 \Rightarrow 2dx - 3dt + \frac{2}{2-t}dt = 0$$

Integrating and replacing t by $x + y$, we get

$$2x - 3t - 2[\ln|(2 - t)|] = c_1$$

$$\Rightarrow 2x - 3(x + y) - 2[\ln|(2 - x - y)|] = c_1$$

$$\Rightarrow x + 3y + 2\ln|(2 - x - y)| = c$$

Ans.

Case II : If $a_2 + b_1 = 0$, then a simple cross multiplication and substituting $d(xy)$ for $xdy + ydx$ and integrating term by term, yield the results easily.

Illustration 12 : Solve $\frac{dy}{dx} = \frac{x - 2y + 1}{2x + 2y + 3}$

Solution :

$$\frac{dy}{dx} = \frac{x - 2y + 1}{2x + 2y + 3}$$

$$\Rightarrow 2xdy + 2y dy + 3dy = xdx - 2y dx + dx$$

$$\Rightarrow (2y + 3) dy = (x + 1) dx - 2(xdy + ydx)$$

On integrating, we get

$$\Rightarrow \int (2y + 3)dy = \int (x + 1)dx - \int 2d(xy)$$

$$\text{Solving : } y^2 + 3y = \frac{x^2}{2} + x - 2xy + c$$

Ans.

Do yourself - 5 :

Solve the following differential equations :

(i) $\frac{dy}{dx} = \frac{2x - y + 2}{2y - 4x + 1}$

(ii) $\frac{dy}{dx} = \frac{3x - 5y}{5x + y + 3}$

(iii) Equation of the form :

$$\Rightarrow yf(xy)dx + xg(xy)dy = 0 \quad \dots\dots\dots (i)$$

The substitution $xy = z$, reduces differential equation of this form to the form in which the variables are separable.

Let $xy = z \quad \dots\dots\dots (ii)$

$$dy = \left[\frac{xdz - zdx}{x^2} \right] \quad \dots\dots\dots (iii)$$

using equation (ii) & (iii), equation (i) becomes $\frac{z}{x}f(z)dx + xg(z)\left[\frac{xdz - zdx}{x^2}\right] = 0$

$$\Rightarrow \frac{z}{x}f(z)dx + g(z)dz - \frac{z}{x}g(z)dx = 0 \quad \Rightarrow \quad \frac{z}{x}\{f(z) - g(z)\}dx + g(z)dz = 0$$

$$\Rightarrow \frac{1}{x}dx + \frac{g(z)dz}{z\{f(z) - g(z)\}} = 0$$

Illustration 13 : Solve $y(xy + 1)dx + x(1 + xy + x^2y^2)dy = 0$ **Solution :** Let $xy = v$

$$y = \frac{v}{x} \Rightarrow dy = \frac{xdv - vdx}{x^2}$$

Now, differential equation becomes $\Rightarrow \frac{v}{x}(v + 1)dx + x(1 + v + v^2)\left(\frac{xdv - vdx}{x^2}\right) = 0$

On solving, we get

$$v^3dx - x(1 + v + v^2)dv = 0$$

separating the variables & integrating we get

$$\Rightarrow \int \frac{dx}{x} - \int \left(\frac{1}{v^3} + \frac{1}{v^2} + \frac{1}{v} \right) dv = 0 \quad \Rightarrow \quad \ln x + \frac{1}{2v^2} + \frac{1}{v} - \ln v = c$$

$$\Rightarrow 2v^2 \ln \left(\frac{v}{x} \right) - 2v - 1 = -2cv^2 \quad \Rightarrow \quad 2x^2y^2 \ln y - 2xy - 1 = Kx^2y^2 \quad \text{where } K = -2c$$

Do yourself - 6 :

Solve the following differential equations :

(i) $(y - xy^2)dx - (x + x^2y)dy = 0$

(ii) $y(1 + 2xy)dx + x(1 - xy)dy = 0$

(iv) Transformation to polar-co-ordinates :

Sometimes conversion of cartesian co-ordinates into polar coordinates helps us in separating the variables.

(1) $x = r \cos \theta, y = r \sin \theta$

then $x^2 + y^2 = r^2$

$x dx + y dy = r dr$

$x dy - y dx = r^2 d\theta$

(2) $x = r \sec \theta, y = r \tan \theta$

then $x^2 - y^2 = r^2$

$x dx - y dy = r dr$

$x dy - y dx = r^2 \sec \theta d\theta$

Illustration 14 : Solve : $\frac{x+y \frac{dy}{dx}}{y-x \frac{dy}{dx}} = x^2 + 2y^2 + \frac{y^4}{x^2}$

Solution : The given equation can be reduced to

$$\frac{x dx + y dy}{y dx - x dy} = \frac{(x^2 + y^2)^2}{x^2}$$

Substituting $x = r \cos \theta$

$y = r \sin \theta$

we get, $\frac{r dr}{r^2 d\theta} = \frac{-(r^2)^2}{r^2 \cos^2 \theta} \Rightarrow \int \frac{dr}{r^3} = -\int \sec^2 \theta d\theta \Rightarrow -\frac{1}{2r^2} = -\tan \theta + c$

Substituting, $\frac{1}{2(x^2 + y^2)} = \frac{y}{x} + K$

Ans.

Do yourself - 7 :

Solve the following differential equations :

(i) $x dx + y dy = x dy - y dx$

(ii) $y dx - x dy = xy dy - x^2 dx$

(b) **Homogeneous equations :**

A function $f(x, y)$ is said to be a homogeneous function of degree n , if the substitution $x = \lambda x, y = \lambda y, \lambda > 0$ produces the equality

$f(\lambda x, \lambda y) = \lambda^n f(x, y)$

The degree of homogeneity 'n' can be any real number.

Illustration 15 : Find the degree of homogeneity of function

$$(i) \quad f(x, y) = x^2 + y^2 \quad (ii) \quad f(x, y) = (x^{3/2} + y^{3/2})/(x + y) \quad (iii) \quad f(x, y) = \sin\left(\frac{x}{y}\right)$$

Solution :

$$(i) \quad f(\lambda x, \lambda y) = \lambda^2 x^2 + \lambda^2 y^2 = \lambda^2 (x^2 + y^2) = \lambda^2 f(x, y)$$

degree of homogeneity $\rightarrow 2$

$$(ii) \quad f(\lambda x, \lambda y) = \frac{\lambda^{3/2} x^{3/2} + \lambda^{3/2} y^{3/2}}{\lambda x + \lambda y}$$

$$f(\lambda x, \lambda y) = \lambda^{1/2} f(x, y)$$

degree of homogeneity $\rightarrow 1/2$

$$(iii) \quad f(\lambda x, \lambda y) = \sin\left(\frac{\lambda x}{\lambda y}\right) = \lambda^0 \sin\left(\frac{x}{y}\right) = \lambda^0 f(x, y)$$

degree of homogeneity $\rightarrow 0$

Illustration 16 : Determine whether or not each of the following functions is homogeneous.

$$(i) \quad f(x, y) = x^2 - xy \quad (ii) \quad f(x, y) = \frac{xy}{x + y^2} \quad (iii) \quad f(x, y) = \sin xy$$

Solution :

$$(i) \quad f(\lambda x, \lambda y) = \lambda^2 x^2 - \lambda^2 xy = \lambda^2 (x^2 - xy) = \lambda^2 f(x, y) \quad \text{homogeneous.}$$

$$(ii) \quad f(\lambda x, \lambda y) = \frac{\lambda^2 xy}{\lambda x + \lambda^2 y^2} \neq \lambda^n f(x, y) \quad \text{not homogeneous.}$$

$$(iii) \quad f(\lambda x, \lambda y) = \sin(\lambda^2 xy) \neq \lambda^n f(x, y) \quad \text{not homogeneous.}$$

Do yourself - 8 :

(i) Find the degree of homogeneity of function $f(x, y) = x^3 \ln\left[\frac{\sqrt{x+y}}{\sqrt{x-y}}\right]$

(ii) Find the degree of homogeneity of function $f(x, y) = ax^{2/3} + bx^{1/3}y^{1/3} + cy^{2/3}$

(iii) Determine whether or not each of the following functions is homogeneous.

$$(a) \quad f(x, y) = \sqrt{x^2 + 2xy + 3y^2} \quad (b) \quad f(x, y) = x + y \cos \frac{y}{x} \quad (c) \quad f(x, y) = x \sin y + y \sin x.$$

(i) Homogeneous first order differential equation

A differential equation of the form $\frac{dy}{dx} = \frac{f(x, y)}{g(x, y)}$

where $f(x, y)$ and $g(x, y)$ are homogeneous functions of x, y and of the same degree, is said to be homogeneous. Such equations can be solved by substituting

$$y = vx,$$

so that the dependent variable y is changed to another variable v .

Since $f(x, y)$ and $g(x, y)$ are homogeneous functions of the same degree say, n , they can be written as

$$f(x, y) = x^n f_1\left(\frac{y}{x}\right) \quad \text{and} \quad g(x, y) = x^n g_1\left(\frac{y}{x}\right).$$

As $y = vx$, we have $\frac{dy}{dx} = v + x \frac{dv}{dx}$.

The given differential equation, therefore, becomes

$$v + x \frac{dv}{dx} = \frac{f_1(v)}{g_1(v)} \Rightarrow \frac{g_1(v)dv}{f_1(v) - v g_1(v)} = \frac{dx}{x},$$

so that the variables v and x are now separable.

Note : Sometimes homogeneous equation can be solved by substituting $x = vy$ or by using polar coordinate substitution.

Illustration 17: The solution of the differential equation $\frac{dy}{dx} = \frac{\sin y + x}{\sin 2y - x \cos y}$ is -

(A) $\sin^2 y = x \sin y + \frac{x^2}{2} + c$

(B) $\sin^2 y = x \sin y - \frac{x^2}{2} + c$

$$(C) \sin^2 y = x + \sin y + \frac{x^2}{2} + c$$

(D) $\sin^2 y = x - \sin y + \frac{x^2}{2} + c$

Solution :

Here, $\frac{dy}{dx} = \frac{\sin y + x}{\sin 2y - x \cos y}$

$$\Rightarrow \cos y \frac{dy}{dx} = \frac{\sin y + x}{2 \sin y - x}, \quad (\text{put } \sin y = t)$$

$$\Rightarrow \frac{dt}{dx} = \frac{t+x}{2t-x} \quad (\text{put } t = vx)$$

$$\frac{x dv}{dx} + v = \frac{vx + x}{2vx - x} = \frac{v + 1}{2v - 1}$$

$$\therefore x \frac{dv}{dx} = \frac{v+1}{2v-1} - v = \frac{v+1-2v^2+v}{2v-1}$$

or $\frac{2v-1}{-2v^2+2v+1} dv = \frac{dx}{x}$ on solving, we get

$$\sin^2 y = x \sin y + \frac{x^2}{2} + c.$$

Ans. (A)

Illustration 18: Solve the differential equation $(1 + 2e^{x/y}) dx + 2e^{x/y} (1 - x/y) dy = 0$.

Solution :

The equation is homogeneous of degree 0.

Put $x = vy$, $dx = v dy + y dv$,

Then, differential equation becomes

$$(1 + 2e^v)(v \, dy + y \, dv) + 2e^v(1 - v) \, dy = 0 \Rightarrow (v + 2e^v) \, dy + y(1 + 2e^v) \, dv = 0$$

$$\frac{dy}{y} + \frac{1+2e^v}{v+2e^v} dv = 0$$

Integrating and replacing v by x/y , we get $\ln y + \ln (v + 2e^v) = \ln c$ and $x + 2ye^{x/y} = c$

Ans.

Do yourself - 9 :

Solve the following differential equations :

(i) $y' = \frac{3x-y}{x+y}$

(ii) $(x - y \ln y + y \ln x) dx + x(\ln y - \ln x) dy = 0$

(iii) $(3xy + y^2)dx + (x^2 + xy)dy = 0, y(1) = 1$

(ii) Equations reducible to homogeneous formThe equation of the form $\frac{dy}{dx} = \frac{a_1x + b_1y + c_1}{a_2x + b_2y + c_2}$ where $\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$ can be reduced to homogeneous form by changing the variable x, y to u, v as

$$x = u + h, y = v + k$$

where h, k are the constants to be chosen so as to make the given equation homogeneous.

We have

$$\frac{dy}{dx} = \frac{dv}{du}$$

$$\therefore \text{The equation becomes, } \frac{dv}{du} = \frac{a_1u + b_1v + (a_1h + b_1k + c_1)}{a_2u + b_2v + (a_2h + b_2k + c_2)}$$

Let h and k be chosen so as to satisfy the equation

$$a_1h + b_1k + c_1 = 0 \quad \dots(i)$$

$$a_2h + b_2k + c_2 = 0 \quad \dots(ii)$$

Solve for h and k from (i) and (ii)

$$\text{Now } \frac{du}{dv} = \frac{a_1u + b_1v}{a_2u + b_2v}$$

is a homogeneous equation and can be solved by substituting $v = ut$.**Illustration 19 :** Solve $\frac{dy}{dx} = \frac{x+2y+3}{2x+3y+4}$ **Solution :** Put $x = X + h, y = Y + k$

$$\text{We have } \frac{dY}{dX} = \frac{X+2Y+(h+2k+3)}{2X+3Y+(2h+3k+4)}$$

To determine h and k , we write

$$h + 2k + 3 = 0, 2h + 3k + 4 = 0 \Rightarrow h = 1, k = -2$$

$$\text{So that } \frac{dY}{dX} = \frac{X+2Y}{2X+3Y}$$

Putting $Y = VX$, we get

$$V + X \frac{dV}{dX} = \frac{1+2V}{2+3V} \Rightarrow \frac{2+3V}{3V^2-1} dV = -\frac{dX}{X}$$

$$\Rightarrow \left[\frac{2+\sqrt{3}}{2(\sqrt{3}V-1)} - \frac{2-\sqrt{3}}{2(\sqrt{3}V+1)} \right] dV = -\frac{dX}{X}$$

$$\Rightarrow \frac{2+\sqrt{3}}{2\sqrt{3}} \log(\sqrt{3}V-1) - \frac{2-\sqrt{3}}{2\sqrt{3}} \log(\sqrt{3}V+1) = (-\log X + c)$$

$$\frac{2+\sqrt{3}}{2\sqrt{3}} \log(\sqrt{3}Y-X) - \frac{2-\sqrt{3}}{2\sqrt{3}} \log(\sqrt{3}Y+X) = A \text{ where } A \text{ is another constant}$$

and $X = x - 1$, $Y = y + 2$. **Ans.**

Do yourself - 10 :

(i) Solve the differential equation : $\frac{dy}{dx} = \frac{x+2y-5}{2x+y-4}$

(c) Linear differential equations :

A differential equation is said to be linear if the dependent variable & its differential coefficients occur in the first degree only and are not multiplied together.

The n^{th} order linear differential equation is of the form ;

$a_0(x) \frac{d^n y}{dx^n} + a_1(x) \frac{d^{n-1} y}{dx^{n-1}} + \dots + a_n(x) y = \phi(x)$, where $a_0(x)$, $a_1(x)$ $a_n(x)$ are called the coefficients of the differential equation.

Note that a linear differential equation is always of the first degree but every differential equation of the first degree need not be linear . e.g. the differential equation $\frac{d^2 y}{dx^2} + \left(\frac{dy}{dx}\right)^3 + y^2 = 0$ is not linear, though its degree is 1.

Illustration 20: Which of the following equation is linear ?

(A) $\frac{dy}{dx} + xy^2 = 1$ (B) $x^2 \frac{dy}{dx} + y = e^x$ (C) $\frac{dy}{dx} + 3y = xy^2$ (D) $x \frac{dy}{dx} + y^2 = \sin x$

Solution : Clearly answer is (B)

Illustration 21 : Which of the following equation is non-linear ?

(A) $\frac{dy}{dx} = \cos x$ (B) $\frac{d^2 y}{dx^2} + y = 0$ (C) $dx + dy = 0$ (D) $x \left(\frac{dy}{dx}\right) + \frac{3}{\left(\frac{dy}{dx}\right)} = y^2$

Solution : Clearly answer is (D)

(i) Linear differential equations of first order :

The most general form of a linear differential equation of first order is $\frac{dy}{dx} + Py = Q$, where P & Q are functions of x.

To solve such an equation multiply both sides by $e^{\int P dx}$.

$$\text{So that we get } e^{\int P dx} \left[\frac{dy}{dx} + Py \right] = Q e^{\int P dx} \quad \dots(i)$$

$$\frac{d}{dx} \left(e^{\int P dx} \cdot y \right) = Q e^{\int P dx} \quad \dots(ii)$$

On integrating equation (ii), we get

$$y e^{\int P dx} = \int Q e^{\int P dx} dx + c$$

This is the required general solution.

Note :

- (i) The factor $e^{\int P dx}$ on multiplying by which the left hand side of the differential equation becomes the differential coefficient of some function of x & y, is called integrating factor of the differential equation popularly abbreviated as I.F.
- (ii) Sometimes a given differential equation becomes linear if we take y as the independent variable and x as the dependent variable. e.g. the equation ;

$(x + y + 1) \frac{dy}{dx} = y^2 + 3$ can be written as $(y^2 + 3) \frac{dx}{dy} = x + y + 1$ which is a linear differential equation.

Illustration 22 : Solve $(1 + y^2) + (x - e^{\tan^{-1} y}) \frac{dy}{dx} = 0$.

Solution : Differential equation can be rewritten as $(1 + y^2) \frac{dx}{dy} + x = e^{\tan^{-1} y}$

$$\text{or } \frac{dx}{dy} + \frac{1}{1+y^2} \cdot x = \frac{e^{\tan^{-1} y}}{1+y^2} \quad \dots(i)$$

$$\text{I. F} = e^{\int \frac{1}{1+y^2} dy} = e^{\tan^{-1} y}$$

$$\text{so solution is } x e^{\tan^{-1} y} = \int \frac{e^{\tan^{-1} y} e^{\tan^{-1} y}}{1+y^2} dy$$

$$\text{Let } e^{\tan^{-1} y} = t \Rightarrow \frac{e^{\tan^{-1} y}}{1+y^2} dy = dt$$

$$x e^{\tan^{-1} y} = \int t dt \quad [\text{Putting } e^{\tan^{-1} y} = t]$$

$$\text{or } x e^{\tan^{-1} y} = \frac{t^2}{2} + \frac{c}{2} \Rightarrow 2x e^{\tan^{-1} y} = e^{2 \tan^{-1} y} + c.$$

Ans.

Illustration 23 : The solution of differential equation $(x^2 - 1) \frac{dy}{dx} + 2xy = \frac{1}{x^2 - 1}$ is -

(A) $y(x^2 - 1) = \frac{1}{2} \log \left| \frac{x-1}{x+1} \right| + C$

(B) $y(x^2 + 1) = \frac{1}{2} \log \left| \frac{x-1}{x+1} \right| - C$

(C) $y(x^2 - 1) = \frac{5}{2} \log \left| \frac{x-1}{x+1} \right| + C$

(D) none of these

Solution : The given differential equation is

$$(x^2 - 1) \frac{dy}{dx} + 2xy = \frac{1}{x^2 - 1} \Rightarrow \frac{dy}{dx} + \frac{2x}{x^2 - 1} y = \frac{1}{(x^2 - 1)^2} \quad \dots(i)$$

This is linear differential equation of the form

$$\frac{dy}{dx} + Py = Q, \text{ where } P = \frac{2x}{x^2 - 1} \text{ and } Q = \frac{1}{(x^2 - 1)^2}$$

$$\therefore \text{I.F.} = e^{\int P dx} = e^{\int \frac{2x}{x^2 - 1} dx} = e^{\log(x^2 - 1)} = (x^2 - 1)$$

multiplying both sides of (i) by I.F. = $(x^2 - 1)$, we get

$$(x^2 - 1) \frac{dy}{dx} + 2xy = \frac{1}{x^2 - 1}$$

integrating both sides we get

$$y(x^2 - 1) = \int \frac{1}{x^2 - 1} dx + C$$

$$[\text{Using : } y(\text{I.F.}) = \int Q(\text{I.F.}) dx + C]$$

$$\Rightarrow y(x^2 - 1) = \frac{1}{2} \log \left| \frac{x-1}{x+1} \right| + C.$$

This is the required solution.

Ans.(A)

Do yourself - 11 :

Solve the following differential equations :

(i) $\frac{xdy}{dx} = 2y + x^4 + 6x^2 + 2x, x \neq 0$

(ii) $(x - a) \frac{dy}{dx} + 3y = 12(x - a)^3, x > a > 0$

(iii) $y \ln y dx + (x - \ln y) dy = 0$

(ii) Equation reducible to linear form :

The equation of the form $\frac{dy}{dx} + Py = Qy^n$, where P and Q are functions of x,

is called **Bernoulli's equation**.

On dividing by y^n , we get $y^{-n} \frac{dy}{dx} + Py^{-n+1} = Q$

Let $y^{-n+1} = t$, so that $(-n+1)y^{-n} \frac{dy}{dx} = \frac{dt}{dx}$

then equation becomes $\frac{dt}{dx} + P(1-n)t = Q(1-n)$

which is linear with t as a dependent variable.

Illustration 24: Solve the differential equation $x \frac{dy}{dx} + y = x^3 y^6$.

Solution : The given differential equation can be written as

$$\frac{1}{y^6} \frac{dy}{dx} + \frac{1}{xy^5} = x^2$$

Putting $y^{-5} = v$ so that

$$-5 y^{-6} \frac{dy}{dx} = \frac{dv}{dx} \text{ or } y^{-6} \frac{dy}{dx} = -\frac{1}{5} \frac{dv}{dx} \text{ we get}$$

$$-\frac{1}{5} \frac{dv}{dx} + \frac{1}{x} v = x^2 \Rightarrow \frac{dv}{dx} - \frac{5}{x} v = -5x^2 \quad \dots\dots(i)$$

This is the standard form of the linear differential equation having integrating factor

$$\text{I.F.} = e^{\int -\frac{5}{x} dx} = e^{-5 \log x} = \frac{1}{x^5}$$

Multiplying both sides of (i) by I.F. and integrating w.r.t. x

$$\text{We get } v \cdot \frac{1}{x^5} = \int -5x^2 \cdot \frac{1}{x^5} dx$$

$$\Rightarrow \frac{v}{x^5} = \frac{5}{2} x^{-2} + c$$

$$\Rightarrow y^{-5} x^{-5} = \frac{5}{2} x^{-2} + c \text{ which is the required solution.}$$

Ans.

Illustration 25 : Find the solution of differential equation $\frac{dy}{dx} - y \tan x = -y^2 \sec x$.

Solution : $\frac{1}{y^2} \frac{dy}{dx} - \frac{1}{y} \tan x = -\sec x$

$$\frac{1}{y} = v ; \frac{-1}{y^2} \frac{dy}{dx} = \frac{dv}{dx}$$

$$\therefore \frac{-dv}{dx} - v \tan x = -\sec x$$

$$\frac{dv}{dx} + v \tan x = \sec x,$$

Here $P = \tan x$, $Q = \sec x$

$$\text{I.F.} = e^{\int \tan x dx} = |\sec x|$$

$$v |\sec x| = \int \sec^2 x dx + c$$

Hence the solution is $y^{-1} |\sec x| = \tan x + c$

Ans.

Do yourself - 12 :

Solve the following differential equations :

(i) $y' + 3y = e^{3x} y^2$ (ii) $x dy - \{y + xy^3 (1 + \ln x)\} dx = 0$ (iii) $\frac{dy}{dx} + y = y^2 (\cos x - \sin x)$

7. TRAJECTORIES :

A curve which cuts every member of a given family of curves according to a given law is called a Trajectory of the given family.

The trajectory will be called **Orthogonal** if each trajectory cuts every member of given family at right angle.

Working rule for finding orthogonal trajectory

1. Form the differential equation of family of curves
2. Write $-\frac{1}{dy/dx}$ for $\frac{dy}{dx}$ or $-\frac{r^2 d\theta}{dr}$ for $\frac{dr}{d\theta}$ if differential equation is in the polar form.
3. Solve the new differential equation to get the equation of orthogonal trajectories.

Note: A family of curves is self-orthogonal if it is its own orthogonal family.

Illustration 26: Find the value of k such that the family of parabolas $y = cx^2 + k$ is the orthogonal trajectory of the family of ellipses $x^2 + 2y^2 - y = c$.

Solution : Differentiate both sides of $x^2 + 2y^2 - y = c$ w.r.t. x , We get

$$2x + 4y \frac{dy}{dx} - \frac{dy}{dx} = 0$$

$$\text{or } 2x + (4y - 1) \frac{dy}{dx} = 0, \text{ is the differential equation of the given family of curves.}$$

Replacing $\frac{dy}{dx}$ by $-\frac{dx}{dy}$ to obtain the differential equation of the orthogonal trajectories, we get

$$2x + \frac{(1-4y)}{\frac{dy}{dx}} = 0 \Rightarrow \frac{dy}{dx} = \frac{4y-1}{2x}$$

$$\Rightarrow \int \frac{dy}{4y-1} = \int \frac{dx}{2x} \Rightarrow \frac{1}{4} \ln(4y-1) = \frac{1}{2} \ln x + \frac{1}{2} \ln a, \text{ where } a \text{ is any constant.}$$

$$\Rightarrow \ln(4y-1) = 2 \ln x + 2 \ln a \text{ or } 4y-1 = a^2 x^2$$

or, $y = \frac{1}{4} a^2 x^2 + \frac{1}{4}$, is the required orthogonal trajectory, which is of the form

$$y = cx^2 + k \text{ where } c = \frac{a^2}{4}, k = \frac{1}{4}.$$

Ans.

Illustration 27 : Prove that $\frac{x^2}{1+\lambda} + \frac{y^2}{4+\lambda} = 1$ are self orthogonal family of curves.

Solution :
$$\frac{x^2}{1+\lambda} + \frac{y^2}{4+\lambda} = 1 \quad \dots(i)$$

Differentiating (i) with respect to x, we have

$$\frac{x}{1+\lambda} + \frac{y}{4+\lambda} \frac{dy}{dx} = 0 \quad \dots(ii)$$

From (i) and (ii), we have to eliminate λ .

Now, (ii) gives

$$\lambda = -\frac{\left[4x + y \frac{dy}{dx}\right]}{x + y \frac{dy}{dx}}$$

$$\Rightarrow 1 + \lambda = \frac{(1-4)x}{x + y(dy/dx)}, \quad 4 + \lambda = \frac{-(1-4)y(dy/dx)}{x + y(dy/dx)}$$

Substituting these values in (i), we get

$$\left(x + y \frac{dy}{dx}\right) \left(x - y \frac{dx}{dy}\right) = -3 \quad \dots(iii)$$

as the differential equation of the given family.

Changing dy/dx to $-dx/dy$ in (iii), we obtain

$$\left(x - y \frac{dx}{dy}\right) \left(x + y \frac{dy}{dx}\right) = -3 \quad \dots(iv)$$

which is the same as (iii). Thus we see that the family (i) is self-orthogonal, i.e., every member of the family (i) cuts every other member of the same family orthogonally.

Do yourself - 13 :

(i) Find the orthogonal trajectories of the following families of curves :

(a) $x + 2y = C$

(b) $y = Ce^{-2x}$

Note :

Following exact differentials must be remembered :

(i) $x dy + y dx = d(xy)$	(ii) $\frac{x dy - y dx}{x^2} = d\left(\frac{y}{x}\right)$	(iii) $\frac{y dx - x dy}{y^2} = d\left(\frac{x}{y}\right)$
(iv) $\frac{x dy + y dx}{xy} = d(\ln xy)$	(v) $\frac{dx + dy}{x + y} = d(\ln(x + y))$	(vi) $\frac{x dy - y dx}{xy} = d\left(\ln \frac{y}{x}\right)$
(vii) $\frac{y dx - x dy}{xy} = d\left(\ln \frac{x}{y}\right)$	(viii) $\frac{x dy - y dx}{x^2 + y^2} = d\left(\tan^{-1} \frac{y}{x}\right)$	(ix) $\frac{y dx - x dy}{x^2 + y^2} = d\left(\tan^{-1} \frac{x}{y}\right)$
(x) $\frac{x dx + y dy}{x^2 + y^2} = d\left[\ln \sqrt{x^2 + y^2}\right]$	(xi) $d\left(-\frac{1}{xy}\right) = \frac{x dy + y dx}{x^2 y^2}$	(xii) $d\left(\frac{e^x}{y}\right) = \frac{y e^x dx - e^x dy}{y^2}$
(xiii) $d\left(\frac{e^y}{x}\right) = \frac{x e^y dy - e^y dx}{x^2}$		

Illustration 28: Solve $\frac{y + \sin x \cos^2(xy)}{\cos^2(xy)} dx + \left(\frac{x}{\cos^2(xy)} + \sin y \right) dy = 0$.

Solution : The given differential equation can be written as;

$$\begin{aligned} & \frac{y \, dx + x \, dy}{\cos^2(xy)} + \sin x \, dx + \sin y \, dy = 0. \\ \Rightarrow & \sec^2(xy) \, d(xy) + \sin x \, dx + \sin y \, dy = 0 \\ \Rightarrow & d(\tan(xy)) + d(-\cos x) + d(-\cos y) = 0 \\ \Rightarrow & \tan(xy) - \cos x - \cos y = c. \end{aligned}$$

Ans.

Do yourself - 14 :

Solve the following differential equations :

(i) $x \, dx + y \, dy + 4y^3(x^2 + y^2)dy = 0.$

(ii) $x \, dy - y \, dx - (1 - x^2) \, dx = 0$.

8. APPLICATION OF DIFFERENTIAL EQUATIONS :

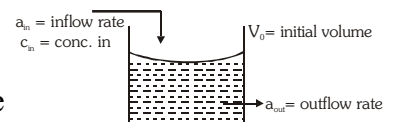
(a) Mixing Problems

A chemical in a liquid solution with given concentration c_{in} gm/lit. (or dispersed in a gas) runs into a container with a rate of a_{in} lit/min. holding the liquid (or the gas) with, possibly, a specified amount of the chemical dissolved as well. The mixture is kept uniform by stirring and flows out of the container at a known rate (a_{out} litre/min.). In this process it is often important to know the concentration of the chemical in the container at any given time. The differential equation describing the process is based on the formula.

$$\begin{array}{l} \text{Rate of change} \\ \text{of amount} \\ \text{in container} \end{array} = \begin{pmatrix} \text{rate at which} \\ \text{chemical} \\ \text{arrives} \end{pmatrix} - \begin{pmatrix} \text{rate at which} \\ \text{chemical} \\ \text{departs} \end{pmatrix} \quad \text{.....(i)}$$

$$\text{Arrival rate} = (\text{conc. in}) \times (\text{inflow rate}) = c_{\text{in}} \times a_{\text{in}}$$

If $y(t)$ denotes the amount of substance in the tank at time t & $V(t)$ denotes the amount of mixture in tank at that time



$$\begin{aligned} \text{Departure rate} &= \left(\frac{\text{concentration in}}{\text{container at time t}} \right) \cdot (\text{outflow rate}) \\ &= \frac{y(t)}{V(t)} \cdot (a_{\text{out}}) \end{aligned}$$

where volume of mixture at time t, $V(t) = \text{initial volume} + (\text{inflow rate} - \text{outflow rate}) \times t$
 $= V_0 + (a_{in} - a_{out})t$

Accordingly, Equation (i) becomes

$$\frac{dy(t)}{dt} = (\text{chemical's given arrival rate}) - \frac{y(t)}{V(t)} \cdot (\text{out flow rate}) \quad \text{.....(ii)}$$

$$\frac{d(y(t))}{dt} = c_{in} a_{in} - \frac{y(t)}{V_0 + (a_{in} - a_{out})t} \cdot a_{out}$$

This leads to a first order linear D.E. which can be solved to obtain $y(t)$ i.e. amount of chemical at time 't'.

Illustration 29 : A tank contains 20 kg of salt dissolved in 5000 L of water. Brine that contains 0.03 kg of salt per liter of water enters the tank at a rate of 25 L/min. The solution is kept thoroughly mixed and drains from the tank at the same rate. How much salt remains in the tank after half an hour ?

Solution : Let $y(t)$ be the amount of salt after t min.

Given $y(0) = 20$ kg

$$\text{rate in} = \left(\frac{0.03 \text{ kg}}{\text{L}} \right) \left(\frac{25 \text{ L}}{\text{min.}} \right) = \frac{0.75 \text{ kg}}{\text{min.}}$$

As $a_{\text{in}} = a_{\text{out}}$, so the tank always contains 5000 L of liquid so the conc. at time 't' is

$$\left(\frac{y(t)}{5000} \right) \frac{\text{kg}}{\text{L}}$$

$$\text{so rate out} = \left(\frac{y(t) \text{ kg}}{5000 \text{ L}} \right) \left(\frac{25 \text{ L}}{\text{min}} \right) = \frac{y(t) \text{ kg}}{200 \text{ min}}$$

$$\frac{dy(t)}{dt} = 0.75 - \frac{y(t)}{200}$$

by solving as linear D.E. or variable separable and using initial condition, we get

$$y(t) = 150 - 130 e^{-t/200}$$

The amount of salt after 30 min is

$$y(30) = 150 - 130 e^{-30/100} = 38.1 \text{ kg}$$

Do yourself - 15 :

- (i) A tank initially holds 10 lit. of fresh water. At $t = 0$, a brine solution containing $\frac{1}{2}$ kg of salt per lit. is poured into the tank at a rate of 2 lit/min. while the well-stirred mixture leaves the tank at the same rate. Find
- the amount and
 - the concentration of salt in the tank at any time t .

(b) Exponential Growth and Decay :

In general, if $y(t)$ is the value of quantity y at time t and if the rate of change of y with respect to t is proportional to its value $y(t)$ at that time, then

$$\frac{dy(t)}{dt} = ky(t), \text{ where } k \text{ is a constant} \quad \dots(i)$$

$$\int \frac{dy(t)}{y(t)} = \int k dt$$

Solving, we get $y(t) = Ae^{kt}$

equation (i) is sometimes called the law of natural growth (if $k > 0$) or law of natural decay (if $k < 0$).

In the context of population growth, we can write

$$\frac{dP}{dt} = kP \quad \text{or} \quad \frac{1}{P} \frac{dP}{dt} = k$$

where k is growth rate divided by the population size; it is called **the relative growth rate**.

Illustration 30 : A certain radioactive material is known to decay at a rate proportional to the amount present. If initially there is 50 kg of the material present and after two hours it is observed that the material has lost 10 percent of its original mass, find (a) an expression for the mass of the material remaining at any time t , (b) the mass of the material after four hours, and (c) the time at which the material has decayed to one half of its initial mass.

Solution : (a) Let N denote the amount of material present at time t .

$$\text{So, } \frac{dN}{dt} - kN = 0$$

This differential equation is separable and linear, its solution is

$$N = ce^{kt} \quad \dots(i)$$

At $t=0$, we are given that $N = 50$. Therefore, from (i), $50 = ce^{k(0)}$ or $c = 50$. Thus,

$$N = 50e^{kt} \quad \dots(ii)$$

At $t = 2$, 10 percent of the original mass of 50kg or 5kg has decayed. Hence, at $t = 2$, $N = 50 - 5 = 45$. Substituting these values into (ii) and solving for k , we have

$$45 = 50e^{2k} \text{ or } k = \frac{1}{2} \ln \frac{45}{50}$$

Substituting this value into (ii), we obtain the amount of mass present at any time t as

$$N = 50e^{\frac{1}{2}(\ln 0.9)t} \quad \dots(iii)$$

where t is measured in hours.

(b) We require N at $t = 4$. Substituting $t = 4$ into (iii) and then solving for N , we find $N = 50e^{-2 \ln(0.9)} \text{ kg}$

(c) We require when $N = 50/2 = 25$. Substituting $N = 25$ into (iii) and solving for t , we find

$$25 = 50e^{\frac{1}{2}(\ln 0.9)t} \Rightarrow t = \ln\left(\frac{1}{2}\right) / \left[\frac{1}{2} \ln(0.9)\right] \text{ hours}$$

(c) **Geometrical applications :**

Let $P(x_1, y_1)$ be any point on the curve $y = f(x)$, then slope of the tangent at point P is $\left(\frac{dy}{dx}\right)_{(x_1, y_1)}$

(i) The equation of the tangent at P is $y - y_1 = \frac{dy}{dx}(x - x_1)$

$$x\text{-intercept of the tangent} = x_1 - y_1 \left(\frac{dx}{dy} \right)$$

$$y\text{-intercept of the tangent} = y_1 - x_1 \left(\frac{dy}{dx} \right)$$

(ii) The equation of normal at P is $y - y_1 = -\frac{1}{(dy/dx)}(x - x_1)$

$$x \text{ and } y\text{-intercepts of normal are ; } x_1 + y_1 \frac{dy}{dx} \text{ and } y_1 + x_1 \frac{dx}{dy}$$

(iii) Length of tangent = PT = $|y_1| \sqrt{1 + (dx/dy)^2}_{(x_1, y_1)}$

(iv) Length of normal = PN = $|y_1| \sqrt{1 + (dy/dx)^2}_{(x_1, y_1)}$

(v) Length of sub-tangent = ST = $\left| y_1 \left(\frac{dx}{dy} \right)_{(x_1, y_1)} \right|$

(vi) Length of sub-normal = SN = $\left| y_1 \left(\frac{dy}{dx} \right)_{(x_1, y_1)} \right|$

(vii) Length of radius vector = $\sqrt{x_1^2 + y_1^2}$

Do yourself - 16 :

- (i) At each point (x,y) of a curve the intercept of the tangent on the y-axis is equal to $2xy^2$. Find the curve.
- (ii) Find the equation of the curve for which the normal at any point (x,y) passes through the origin.

Miscellaneous Illustrations :

Illustration 31 : Solve $(y \log x - 1) y dx = x dy$.

Solution : The given differential equation can be written as

$$x \frac{dy}{dx} + y = y^2 \log x \quad \dots(i)$$

Divide by xy^2 . Hence $\frac{1}{y^2} \frac{dy}{dx} + \frac{1}{xy} = \frac{1}{x} \log x$

Let $\frac{1}{y} = v \Rightarrow -\frac{1}{y^2} \frac{dy}{dx} = \frac{dv}{dx}$ so that $\frac{dv}{dx} - \frac{1}{x} v = -\frac{1}{x} \log x \quad \dots(ii)$

(ii) is the standard linear differential equation with $P = -\frac{1}{x}$, $Q = -\frac{1}{x} \log x$

$$\text{I.F.} = e^{\int p dx} = e^{\int -1/x dx} = 1/x$$

The solution is given by

$$v \cdot \frac{1}{x} = \int \frac{1}{x} \left(-\frac{1}{x} \log x \right) dx = -\int \frac{\log x}{x^2} dx = \frac{\log x}{x} - \int \frac{1}{x} \cdot \frac{1}{x} dx = \frac{\log x}{x} + \frac{1}{x} + C$$

$$\Rightarrow v = 1 + \log x + cx = \log ex + cx$$

$$\text{or } \frac{1}{y} = \log ex + cx \text{ or } y (\log ex + cx) = 1.$$

Ans.

Illustration 32: For a certain curve $y = f(x)$ satisfying $\frac{d^2y}{dx^2} = 6x - 4$, $f(x)$ has a local minimum value 5 when $x = 1$. Find the equation of the curve and also the global maximum and global minimum values of $f(x)$ given that $0 \leq x \leq 2$.

Solution : Integrating $\frac{d^2y}{dx^2} = 6x - 4$, we get $\frac{dy}{dx} = 3x^2 - 4x + A$

When $x = 1$, $\frac{dy}{dx} = 0$, so that $A = 1$. Hence

$$\frac{dy}{dx} = 3x^2 - 4x + 1 \quad \dots(i)$$

Integrating, we get $y = x^3 - 2x^2 + x + B$

When $x = 1$, $y = 5$, so that $B = 5$.

Thus we have $y = x^3 - 2x^2 + x + 5$.

From (i), we get the critical points $x = 1/3$, $x = 1$

At the critical point $x = \frac{1}{3}$, $\frac{d^2y}{dx^2}$ is negative.

Therefore at $x = 1/3$, y has a local maximum.

At $x = 1$, $\frac{d^2y}{dx^2}$ is positive.

Therefore at $x = 1$, y has a local minimum.

$$\text{Also } f(1) = 5, f\left(\frac{1}{3}\right) = \frac{139}{27}, f(0) = 5, f(2) = 7$$

Hence the global maximum value = 7, and the global minimum value = 5.

Ans.

Illustration 33 : Solve $\frac{dy}{dx} = \tan y \cot x - \sec y \cos x$.

Solution : $\frac{dy}{dx} = \tan y \cot x - \sec y \cos x$.

Rearrange it :

$$(\sin x - \sin y) \cos x \, dx + \sin x \cos y \, dy = 0.$$

Put $u = \sin y$, So, $du = \cos y \, dy$:

Substituting, we get

$$(\sin x - u) \cos x \, dx + \sin x \, du = 0, \quad \frac{du}{dx} - u \frac{\cos x}{\sin x} = -\cos x$$

The equation is first-order linear in u .

The integrating factor is

$$I = \exp \int -\frac{\cos x}{\sin x} \, dx = \exp \{-\ln(\sin x)\} = \frac{1}{\sin x}.$$

$$\text{Hence, } u \frac{1}{\sin x} = -\int \frac{\cos x}{\sin x} \, dx = -\ln |\sin x| + C.$$

Solve for u : $u = -\sin x \ln |\sin x| + C \sin x$.

Put y back : $\sin y = -\sin x \ln |\sin x| + C \sin x$.

Ans.

Illustration 34 : Solve the equation $x \int_0^x y(t) \, dt = (x+1) \int_0^x t y(t) \, dt, x > 0$

Solution : Differentiating the equation with respect to x , we get

$$xy(x) + 1 \cdot \int_0^x y(t) \, dt = (x+1)xy(x) + 1 \cdot \int_0^x ty(t) \, dt$$

$$\text{i.e., } \int_0^x y(t) \, dt = x^2 y(x) + \int_0^x ty(t) \, dt$$

Differentiating again with respect to x , we get $y(x) = x^2 y'(x) + 2xy(x) + xy(x)$

$$\text{i.e., } (1 - 3x)y(x) = \frac{x^2 dy(x)}{dx}$$

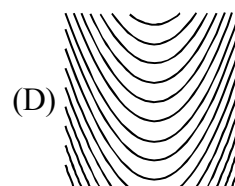
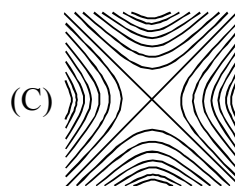
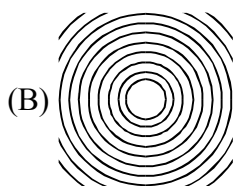
$$\text{i.e., } \frac{(1 - 3x)dx}{x^2} = \frac{dy(x)}{y(x)}, \text{ integrating we get}$$

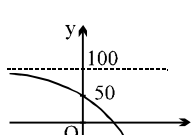
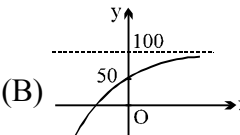
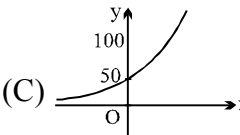
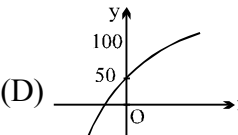
$$\text{i.e., } y = \frac{c}{x^3} e^{-1/x}$$

Ans.

EXERCISE (O-1)

- Number of values of $m \in \mathbb{N}$ for which $y = e^{mx}$ is a solution of the differential equation $D^3y - 3D^2y - 4Dy + 12y = 0$, is
(A) 0 (B) 1 (C) 2 (D) more than 2
- Number of straight lines which satisfy the differential equation $\frac{dy}{dx} + x \left(\frac{dy}{dx} \right)^2 - y = 0$ is:
(A) 1 (B) 2 (C) 3 (D) 4
- The value of the constant 'm' and 'c' for which $y = mx + c$ is a solution of the differential equation $D^2y - 3Dy - 4y = -4x$.
(A) is $m = -1$; $c = 3/4$ (B) is $m = 1$; $c = -3/4$ (C) no such real m, c (D) is $m = 1$; $c = 3/4$
- Consider the two statements
Statement-1: $y = \sin kt$ satisfies the differential equation $y'' + 9y = 0$.
Statement-2: $y = e^{kt}$ satisfy the differential equation $y'' + y' - 6y = 0$
The value of k for which both the statements are correct is
(A) -3 (B) 0 (C) 2 (D) 3
- If $y = \frac{x}{\ln|cx|}$ (where c is an arbitrary constant) is the general solution of the differential equation $\frac{dy}{dx} = \frac{y}{x} + \phi\left(\frac{x}{y}\right)$ then the function $\phi\left(\frac{x}{y}\right)$ is :
(A) $\frac{x^2}{y^2}$ (B) $-\frac{x^2}{y^2}$ (C) $\frac{y^2}{x^2}$ (D) $-\frac{y^2}{x^2}$
- The differential equation corresponding to the family of curves $y = e^x(ax + b)$ is
(A) $\frac{d^2y}{dx^2} + 2\frac{dy}{dx} - y = 0$ (B) $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + y = 0$ (C) $\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + y = 0$ (D) $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} - y = 0$
- Water is drained from a vertical cylindrical tank by opening a valve at the base of the tank. It is known that the rate at which the water level drops is proportional to the square root of water depth y, where the constant of proportionality $k > 0$ depends on the acceleration due to gravity and the geometry of the hole.
If t is measured in minutes and $k = \frac{1}{15}$ then the time to drain the tank if the water is 4 meter deep to start with is
(A) 30 min (B) 45 min (C) 60 min (D) 80 min
- The general solution of the differential equation $\frac{dy}{dx} = \frac{1-x}{y}$ is a family of curves which looks most like which of the following?



9. Spherical rain drop evaporates at a rate proportional to its surface area. The differential equation corresponding to the rate of change of the radius of the rain drop if the constant of proportionality is $K > 0$, is
- (A) $\frac{dr}{dt} + K = 0$ (B) $\frac{dr}{dt} - K = 0$ (C) $\frac{dr}{dt} = Kr$ (D) none
10. The x-intercept of the tangent to a curve is equal to the ordinate of the point of contact. The equation of the curve through the point (1, 1) is
- (A) $y e^{\frac{x}{y}} = e$ (B) $x e^{\frac{x}{y}} = e$ (C) $x e^{\frac{y}{x}} = e$ (D) $y e^{\frac{y}{x}} = e$
11. A function $f(x)$ satisfying $\int_0^1 f(tx) dt = n f(x)$, where $x > 0$, is
- (A) $f(x) = c \cdot x^{\frac{1-n}{n}}$ (B) $f(x) = c \cdot x^{\frac{n}{n-1}}$ (C) $f(x) = c \cdot x^{\frac{1}{n}}$ (D) $f(x) = c \cdot x^{(1-n)}$
12. Which one of the following curves represents the solution of the initial value problem $Dy = 100 - y$, where $y(0) = 50$
- (A)  (B)  (C)  (D) 
13. The real value of m for which the substitution, $y = u^m$ will transform the differential equation, $2x^4 y \frac{dy}{dx} + y^4 = 4x^6$ into a homogeneous equation is :
- (A) $m = 0$ (B) $m = 1$ (C) $m = 3/2$ (D) no value of m
14. A curve C passes through origin and has the property that at each point (x, y) on it the normal line at that point passes through $(1, 0)$. The equation of a common tangent to the curve C and the parabola $y^2 = 4x$ is
- (A) $x = 0$ (B) $y = 0$ (C) $y = x + 1$ (D) $x + y + 1 = 0$
15. A function $y = f(x)$ satisfies $(x+1) \cdot f'(x) - 2(x^2+x)f(x) = \frac{e^{x^2}}{(x+1)}$, $\forall x > -1$
If $f(0) = 5$, then $f(x)$ is
- (A) $\left(\frac{3x+5}{x+1}\right) \cdot e^{x^2}$ (B) $\left(\frac{6x+5}{x+1}\right) \cdot e^{x^2}$ (C) $\left(\frac{6x+5}{(x+1)^2}\right) \cdot e^{x^2}$ (D) $\left(\frac{5-6x}{x+1}\right) \cdot e^{x^2}$
16. The equation to the orthogonal trajectories of the system of parabolas $y = ax^2$ is
- (A) $\frac{x^2}{2} + y^2 = c$ (B) $x^2 + \frac{y^2}{2} = c$ (C) $\frac{x^2}{2} - y^2 = c$ (D) $x^2 - \frac{y^2}{2} = c$
17. If $\int_a^x t y(t) dt = x^2 + y(x)$ then y as a function of x is
- (A) $y = 2 - (2 + a^2) e^{\frac{x^2-a^2}{2}}$ (B) $y = 1 - (2 + a^2) e^{\frac{x^2-a^2}{2}}$
(C) $y = 2 - (1 + a^2) e^{\frac{x^2-a^2}{2}}$ (D) none
18. If $f(x)$ and $g(x)$ are two differentiable functions on $\mathbb{R}^+$ such that $xf'(x) + g(x) = 0$ and $xg'(x) + f(x) = 0$ for all $x \in \mathbb{R}^+$ and $f(1) + g(1) = 4$, then the value of $f''(2) \cdot g''(2)$ is -
- (A) $\frac{1}{2}$ (B) $\frac{1}{4}$ (C) $\frac{1}{8}$ (D) $\frac{1}{16}$

EXERCISE (O-2)

One or more than one correct

- The differential equation, $x \frac{dy}{dx} + \frac{3}{\frac{dy}{dx}} = y^2$:
 (A) is of order 1 (B) is of degree 2 (C) is linear (D) is non linear
- Which of the following pair of curves is/are orthogonal?
 (A) $16x^2 + y^2 = c$ and $y^{16} = kx$ (B) $y = x + ce^{-x}$ and $x + 2 = y + ke^{-y}$
 (C) $y = cx^2$ and $x^2 + 2y^2 = k$ (D) $x^2 - y^2 = c$ and $xy = k$
 where c and k arbitrary constant.
- Consider $g(x) = \begin{cases} \sin x & 0 \leq x < \frac{\pi}{2} \\ \cos x & x \geq \frac{\pi}{2} \end{cases}$ and a continuous function $y = f(x)$ satisfies $\frac{5dy}{dx} + 5y = g(x)$, $f(0) = 0$, then-
 (A) $f\left(\frac{\pi}{4}\right) = \frac{e^{-\pi/4}}{10}$ (B) $f\left(\frac{\pi}{4}\right) = \frac{e^{-\pi/4} - 1}{10}$
 (C) $f\left(\frac{\pi}{2}\right) = \frac{e^{-\pi/2} + 1}{10}$ (D) $f\left(\frac{\pi}{2}\right) = e^{-\pi/2}$
- A curve $y = f(x)$ has the property that the perpendicular distance of the origin from the normal at any point P of the curve is equal to the distance of the point P from the x-axis. Then the differential equation of the curve
 (A) is homogeneous.
 (B) can be converted into linear differential equation with some suitable substitution.
 (C) is the family of circles touching the x-axis at the origin.
 (D) the family of circles touching the y-axis at the origin.
- The function $f(x)$ satisfying the equation, $f^2(x) + 4f'(x) \cdot f(x) + [f'(x)]^2 = 0$.
 (A) $f(x) = c \cdot e^{(2-\sqrt{3})x}$ (B) $f(x) = c \cdot e^{(2+\sqrt{3})x}$
 (C) $f(x) = c \cdot e^{(\sqrt{3}-2)x}$ (D) $f(x) = c \cdot e^{-(2+\sqrt{3})x}$
 where c is an arbitrary constant.
- A function $y = f(x)$ satisfying the differential equation $\frac{dy}{dx} \cdot \sin x - y \cos x + \frac{\sin^2 x}{x^2} = 0$ is such that, $y \rightarrow 0$ as $x \rightarrow \infty$ then the statement which is correct is
 (A) $\lim_{x \rightarrow 0} f(x) = 1$ (B) $\int_0^{\pi/2} f(x) dx$ is less than $\frac{\pi}{2}$
 (C) $\int_0^{\pi/2} f(x) dx$ is greater than unity (D) $f(x)$ is an odd function

7. If a function $y = f(x)$ satisfies the differential equation $f(x) \cdot \sin 2x - \cos x + (1 + \sin^2 x) f'(x) = 0$ with initial condition $y(0) = 0$. Then-
- (A) Range of $f(x)$ is $\left[-\frac{1}{2}, \frac{1}{2}\right]$ (B) $f(1) < f(2)$
- (C) $f(1) > f(2)$ (D) $f(x)$ is odd function
8. A function $y = f(x)$ satisfies the condition $f'(x) \sin x + f(x) \cos x = 1$, $f(x)$ being bounded when $x \rightarrow 0$. If $I = \int_0^{\pi/2} f(x) dx$ then
- (A) $\frac{\pi}{2} < I < \frac{\pi^2}{4}$ (B) $\frac{\pi}{4} < I < \frac{\pi^2}{2}$ (C) $f\left(\frac{\pi}{4}\right) < f\left(\frac{\pi}{3}\right)$ (D) $f\left(\frac{\pi}{4}\right) > f\left(\frac{\pi}{3}\right)$
9. If $y = f(x)$ is solution of the differential equation, $x^2 \frac{dy}{dx} \cdot \cos \frac{1}{x} - y \sin \frac{1}{x} = -1$, where $y \rightarrow -1$ as $x \rightarrow \infty$, then
- (A) $f(x) = \sin \frac{1}{x} - \cos \frac{1}{x}$ (B) $\lim_{x \rightarrow \infty} (-f(x))^x = \frac{1}{e}$
- (C) $\lim_{x \rightarrow \infty} (-f(x))^x = e$ (D) $f'(x) + 2xf'(x) + x^2 f''(x) = -\frac{2}{x^2} \sin \frac{1}{x}$
10. Consider the differential equation $\frac{dy}{dx} + y \tan x = x \tan x + 1$. Then
- (A) The integral curves satisfying the differential equation and given by $y = x + c \cos x$.
- (B) The angle at which the integral curves cut the y-axis is $\frac{\pi}{4}$.
- (C) Tangents to all the integral curves at their point of intersection with y-axis are parallel.
- (D) none of these

EXERCISE (S-1)

(E-1)

[FORMATION & VARIABLES SEPARABLE]

1. State the order and degree of the following differential equations:
- (i) $\left[\frac{d^2x}{dt^2}\right]^3 + \left[\frac{dx}{dt}\right]^4 - xt = 0$ (ii) $\frac{d^2y}{dx^2} = \left[1 + \left(\frac{dy}{dx}\right)^2\right]^{3/2}$
2. (a) Form a differential equation for the family of curves represented by $ax^2 + by^2 = 1$, where a & b are arbitrary constants.
- (b) Obtain the differential equation of the family of circles $x^2 + y^2 + 2gx + 2fy + c = 0$; where g , f & c are arbitrary constants.
- (c) Obtain the differential equation associated with the primitive, $y = c_1 e^{3x} + c_2 e^{2x} + c_3 e^x$, where c_1 , c_2 , c_3 are arbitrary constants.

Solve the following differential equation for Q.3 to Q.9.

3. $\frac{\ln(\sec x + \tan x)}{\cos x} dx = \frac{\ln(\sec y + \tan y)}{\cos y} dy$

4. $(1 - x^2)(1 - y) dx = xy(1 + y) dy$

5. $\frac{dy}{dx} + \frac{\sqrt{(x^2 - 1)(y^2 - 1)}}{xy} = 0$

6. $y - x \frac{dy}{dx} = a \left(y^2 + \frac{dy}{dx} \right)$

7. $\frac{dy}{dx} = \sin(x + y) + \cos(x + y)$

8. $\frac{dy}{dx} = \frac{x(2 \ln x + 1)}{\sin y + y \cos y}$

9. (a) $\frac{dy}{dx} + \sin \frac{x + y}{2} = \sin \frac{x - y}{2}$

(b) $\sin x \cdot \frac{dy}{dx} = y \cdot \ln y$ if $y = e$, when $x = \frac{\pi}{2}$

10. The population P of a town decreases at a rate proportional to the number by which the population exceeds 1000, proportionality constant being $k > 0$. Find

(a) Population at any time t , given initial population of the town being 2500.

(b) If 10 years later the population has fallen to 1900, find the time when the population will be 1500.

(c) Predict about the population of the town in the long run.

11. It is known that the decay rate of radium is directly proportional to its quantity at each given instant. Find the law of variation of a mass of radium as a function of time if at $t = 0$, the mass of the radius was m_0 and during time $t_0 \propto \%$ of the original mass of radium decay.

12. A normal is drawn at a point $P(x, y)$ of a curve. It meets the x -axis at Q . If PQ is of constant length k , then show that the differential equation describing such curves is, $y \frac{dy}{dx} = \pm \sqrt{k^2 - y^2}$. Find the equation of such a curve passing through $(0, k)$.

13. A curve is such that the length of the polar radius of any point on the curve is equal to the length of the tangent drawn at this point. Form the differential equation and solve it to find the equation of the curve.

14. Given $y(0) = 2000$ and $\frac{dy}{dx} = 32000 - 20y^2$, then find the value of $\lim_{x \rightarrow \infty} y(x)$.

15. Let $f(x)$ is a continuous function which takes positive values for $x \geq 0$ and satisfy $\int_0^x f(t) dt = x\sqrt{f(x)}$

with $f(1) = \frac{1}{2}$. Find the value of $f(\sqrt{2} + 1)$.

16. $\frac{dy}{dx} = y + \int_0^1 y dx$ given $y = 1$, where $x = 0$

(E-2)

[HOMOGENEOUS]

1. $\frac{dy}{dx} = \frac{x^2 + xy}{x^2 + y^2}$
2. Find the equation of a curve such that the projection of its ordinate upon the normal is equal to its abscissa.
3. The perpendicular from the origin to the tangent at any point on a curve is equal to the abscissa of the point of contact. Find the equation of the curve satisfying the above condition and which passes through (1, 1).
4. Find the curve for which any tangent intersects the y-axis at the point equidistant from the point of tangency and the origin.
5. $(x - y) dy = (x + y + 1) dx$
6. $\frac{dy}{dx} = \frac{x + 2y - 3}{2x + y - 3}$
7. $\frac{dy}{dx} = \frac{y - x + 1}{y + x + 5}$
8. $\frac{dy}{dx} = \frac{x + y + 1}{2x + 2y + 3}$
9. $\frac{dy}{dx} = \frac{2(y + 2)^2}{(x + y - 1)^2}$

(E-3)

[LINEAR]

1. If $y = y(x)$ is solution of differential equation $\frac{dy}{dx} - y = 1 - e^{-x}$ and $y(0) = y_0$ has a finite value, when $x \rightarrow \infty$, then find y_0 .
2. $\frac{dy}{dx} + \frac{x}{1 + x^2} y = \frac{1}{2x(1 + x^2)}$
3. $(1 - x^2) \frac{dy}{dx} + 2xy = x(1 - x^2)^{1/2}$
4. (a) Find the curve such that the area of the trapezium formed by the co-ordinate axes, ordinate of an arbitrary point & the tangent at this point equals half the square of its abscissa.
(b) A curve in the first quadrant is such that the area of the triangle formed in the first quadrant by the x-axis, a tangent to the curve at any of its point P and radius vector of the point P is 2sq. units. If the curve passes through (2, 1), find the equation of the curve.
5. $x(x - 1) \frac{dy}{dx} - (x - 2)y = x^3(2x - 1)$
6. $\sin x \frac{dy}{dx} + 3y = \cos x$
7. $x(x^2 + 1) \frac{dy}{dx} = y(1 - x^2) + x^3 \cdot \ln x$
8. $x \frac{dy}{dx} - y = 2x^2 \operatorname{cosec} 2x$

9. $(1 + y^2) dx = (\tan^{-1} y - x) dy$
10. Let the function $\ln f(x)$ is defined where $f(x)$ exists for $x \geq 2$ & k is fixed positive real number, prove that if $\frac{d}{dx} (x \cdot f(x)) \leq -k f(x)$ then $f(x) \leq A x^{-1-k}$ where A is independent of x .
11. Find the differentiable function which satisfies the equation $f(x) = -\int_0^x f(t) \tan t dt + \int_0^x \tan(t-x) dt$ where $x \in (-\pi/2, \pi/2)$
12. $y - x Dy = b(1 + x^2 Dy)$
13. Find all functions $f(x)$ defined on $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ with real values and has a primitive $F(x)$ such that
$$f(x) + \cos x \cdot F(x) = \frac{\sin 2x}{(1 + \sin x)^2}.$$
14. $2 \frac{dy}{dx} - y \sec x = y^3 \tan x$
15. $x^2 y - x^3 \frac{dy}{dx} = y^4 \cos x$
16. $y(2xy + e^x) dx - e^x dy = 0$
17. A tank contains 100 litres of fresh water. A solution containing 1 gm/litre of soluble lawn fertilizer runs into the tank at the rate of 1 lit/min and the mixture is pumped out of the tank at the rate of 3 litres/min. Find the time when the amount of fertilizer in the tank is maximum.

(E-4)**(GENERAL – CHANGE OF VARIABLE BY A SUITABLE SUBSTITUTION)**

1. $(x - y^2) dx + 2xy dy = 0$
2. $(x^3 + y^2 + 2) dx + 2y dy = 0$
3. $x \frac{dy}{dx} + y \ln y = xye^x$
4. $\frac{dy}{dx} - \frac{\tan y}{1+x} = (1+x) e^x \sec y$
5. $\frac{dy}{dx} = \frac{e^y}{x^2} - \frac{1}{x}$
6. $\left(\frac{dy}{dx}\right)^2 - (x+y) \frac{dy}{dx} + xy = 0$
7. $\frac{dy}{dx} = \frac{y^2 - x}{2y(x+1)}$
8. $(1 - xy + x^2 y^2) dx = x^2 dy$
9. $\frac{dy}{dx} = e^{x-y} (e^x - e^y)$
10. $(x^2 + y) \frac{dy}{dx} = 6x$

EXERCISE (S-2)

(MISCELLANEOUS)

1. $\frac{dy}{dx} - y \ln 2 = 2^{\sin x} \cdot (\cos x - 1) \ln 2$, y being bounded when $x \rightarrow +\infty$.
2. The light rays emanating from a point source situated at origin when reflected from the mirror of a search light are reflected as beam parallel to the x -axis. Show that the surface is parabolic, by first forming the differential equation and then solving it.
3. $\frac{ydx - xdy}{(x-y)^2} = \frac{dx}{2\sqrt{1-x^2}}$, given that $y = 2$ when $x = 1$.
4. Consider the differential equation, $\frac{dy}{dx} + P(x)y = Q(x)$
 - (i) If two particular solutions of given equation $u(x)$ and $v(x)$ are known, find the general solution of the same equation in terms of $u(x)$ and $v(x)$.
 - (ii) If α and β are constants such that the linear combinations $\alpha \cdot u(x) + \beta \cdot v(x)$ is a solution of the given equation, find the relation between α and β .
 - (iii) If $w(x)$ is the third particular solution different from $u(x)$ and $v(x)$ then find the ratio $\frac{v(x) - u(x)}{w(x) - u(x)}$.
5. Find the integral curve of the differential equation, $x(1-x \ln y) \cdot \frac{dy}{dx} + y = 0$ which passes through $\left(1, \frac{1}{e}\right)$.
6. A tank consists of 50 litres of fresh water. Two litres of brine each litre containing 5 gms of dissolved salt are run into tank per minute; the mixture is kept uniform by stirring, and runs out at the rate of one litre per minute. If 'm' grams of salt are present in the tank after t minute, express 'm' in terms of t and find the amount of salt present after 10 minutes.
7. Find the orthogonal trajectories for the given family of curves when 'a' is the parameter.
 - (i) $y = ax^2$ (ii) $\cos y = a e^{-x}$ (iii) $x^k + y^k = a^k$
 - (iv) Find the isogonal trajectories for the family of rectangular hyperbolas $x^2 - y^2 = a^2$ which makes with it an angle of 45° .
8. Let $f(x, y, c_1) = 0$ and $f(x, y, c_2) = 0$ define two integral curves of a homogeneous first order differential equation. If P_1 and P_2 are respectively the points of intersection of these curves with an arbitrary line, $y = mx$ then prove that the slopes of these two curves at P_1 and P_2 are equal.
9. Let $f(x)$ be a differentiable function and satisfy $f(0) = 2$, $f'(0) = 3$ and $f''(x) = f(x)$. Find
 - (a) the range of the function $f(x)$
 - (b) the value of the function when $x = \ln 2$
 - (c) the area enclosed by $y = f(x)$ in the 2nd quadrant
10. Show that the curve such that the distance between the origin and the tangent at an arbitrary point is equal to the distance between the origin and the normal at the same point, $\sqrt{x^2 + y^2} = c e^{\pm \tan^{-1} \frac{y}{x}}$.
11. Find the curve possessing the property that the intercept, the tangent at any point of a curve cuts off on the y -axis is equal to the square of the abscissa of the point of tangency.

EXERCISE (JM)

1. If $\frac{dy}{dx} = y + 3 > 0$ and $y(0) = 2$, then $y(\ln 2)$ is equal to :- [AIEEE-2011]

(1) 13 (2) -2 (3) 7 (4) 5

2. Let I be the purchase value of an equipment and $V(t)$ be the value after it has been used for t years.

The value $V(t)$ depreciates at a rate given by differential equation $\frac{dV(t)}{dt} = -k(T - t)$, where $k > 0$ is a constant and T is the total life in years of the equipment. Then the scrap value $V(T)$ of the equipment is :- [AIEEE-2011]

(1) $I - \frac{k(T-t)^2}{2}$ (2) e^{-kT} (3) $T^2 - \frac{I}{k}$ (4) $I - \frac{kT^2}{2}$

3. The curve that passes through the point $(2, 3)$, and has the property that the segment of any tangent to it lying between the coordinate axes is bisected by the point of contact, is given by : [AIEEE-2011]

(1) $\left(\frac{x}{2}\right)^2 + \left(\frac{y}{3}\right)^2 = 2$ (2) $2y - 3x = 0$ (3) $y = \frac{6}{x}$ (4) $x^2 + y^2 = 13$

4. Consider the differential equation $y^2 dx + \left(x - \frac{1}{y}\right) dy = 0$. If $y(1) = 1$, then x is given by : [AIEEE-2011]

(1) $1 - \frac{1}{y} + \frac{e^y}{e}$ (2) $4 - \frac{2}{y} - \frac{e^y}{e}$ (3) $3 - \frac{1}{y} + \frac{e^y}{e}$ (4) $1 + \frac{1}{y} - \frac{e^y}{e}$

5. The population $p(t)$ at time t of a certain mouse species satisfies the differential equation $\frac{dp(t)}{dt} = 0.5 p(t) - 450$. If $p(0) = 850$, then the time at which the population becomes zero is :

[AIEEE-2012]

(1) $\ln 18$ (2) $2 \ln 18$ (3) $\ln 9$ (4) $\frac{1}{2} \ln 18$

6. At present a firm is manufacturing 2000 items. It is estimated that the rate of change of production

P w.r.t. additional number of workers x is given by $\frac{dP}{dx} = 100 - 12\sqrt{x}$. If the firm employs 25 more

workers, then the new level of production of items is :

[JEE (Main)-2013]

(1) 2500 (2) 3000 (3) 3500 (4) 4500

7. If the surface area of a sphere of radius r is increasing uniformly at the rate $8\text{cm}^2/\text{s}$, then the rate of change of its volume is :

[JEE-Main (On line)-2013]

(1) proportional to r^2 (2) constant (3) proportional to r (4) proportional to $\sqrt{r}$

8. Consider the differential equation $\frac{dy}{dx} = \frac{y^3}{2(xy^2 - x^2)}$:

Statement 1 : The substitution $z = y^2$ transforms the above equation into a first order homogenous differential equation.

Statement 2 : The solution of this differential equation is $y^2 e^{\frac{y^2}{x}} = C$. [JEE-Main (On line)-2013]

(1) Statement 1 is false and statement 2 is true. (2) Both statements are true.
(3) Statement 1 is true and statement 2 is false. (4) Both statements are false.

9. If a curve passes through the point $\left(2, \frac{7}{2}\right)$ and has slope $\left(1 - \frac{1}{x^2}\right)$ at any point (x, y) on it, then the ordinate of the point on the curve whose abscissa is -2 is : **[JEE-Main (On line)-2013]**
 (1) $-\frac{5}{2}$ (2) $\frac{5}{2}$ (3) $-\frac{3}{2}$ (4) $\frac{3}{2}$
10. The equation of the curve passing through the origin and satisfying the differential equation $(1 + x^2) \frac{dy}{dx} + 2xy = 4x^2$ is : **[JEE-Main (On line)-2013]**
 (1) $(1 + x^2)y = x^3$ (2) $3(1 + x^2)y = 4x^3$ (3) $3(1 + x^2)y = 2x^3$ (4) $(1 + x^2)y = 3x^3$
11. Let the population of rabbits surviving at a time t be governed by the differential equation $\frac{dp(t)}{dt} = \frac{1}{2}p(t) - 200$. If $p(0) = 100$, then $p(t)$ equals : **[JEE(Main)-2014]**
 (1) $400 - 300e^{t/2}$ (2) $300 - 200e^{-t/2}$ (3) $600 - 500e^{t/2}$ (4) $400 - 300e^{-t/2}$
12. Let $y(x)$ be the solution of the differential equation $(x \log x) \frac{dy}{dx} + y = 2x \log x$, $(x \geq 1)$. Then $y(e)$ is equal to : **[JEE(Main)-2015]**
 (1) 2 (2) $2e$ (3) e (4) 0
13. If a curve $y = f(x)$ passes through the point $(1, -1)$ and satisfies the differential equation, $y(1 + xy) dx = x dy$, then $f\left(-\frac{1}{2}\right)$ is equal to : **[JEE(Main)-2016]**
 (1) $\frac{4}{5}$ (2) $-\frac{2}{5}$ (3) $-\frac{4}{5}$ (4) $\frac{2}{5}$
14. If $(2 + \sin x) \frac{dy}{dx} + (y + 1) \cos x = 0$ and $y(0) = 1$, then $y\left(\frac{\pi}{2}\right)$ is equal to :- **[JEE(Main)-2017]**
 (1) $\frac{4}{3}$ (2) $\frac{1}{3}$ (3) $-\frac{2}{3}$ (4) $-\frac{1}{3}$
15. Let $y = y(x)$ be the solution of the differential equation $\sin x \frac{dy}{dx} + y \cos x = 4x$, $x \in (0, \pi)$. If $y\left(\frac{\pi}{2}\right) = 0$, then $y\left(\frac{\pi}{6}\right)$ is equal to : **[JEE(Main)-2018]**
 (1) $\frac{-8}{9\sqrt{3}}\pi^2$ (2) $-\frac{8}{9}\pi^2$ (3) $-\frac{4}{9}\pi^2$ (4) $\frac{4}{9\sqrt{3}}\pi^2$
16. Let $f: [0, 1] \rightarrow \mathbb{R}$ be such that $f(xy) = f(x)f(y)$ for all $x, y \in [0, 1]$, and $f(0) \neq 0$. If $y = y(x)$ satisfies the differential equation, $\frac{dy}{dx} = f(x)$ with $y(0) = 1$, then $y\left(\frac{1}{4}\right) + y\left(\frac{3}{4}\right)$ is equal to **[JEE(Main)-2019]**
 (1) 4 (2) 3 (3) 5 (4) 2
17. If $\frac{dy}{dx} + \frac{3}{\cos^2 x} y = \frac{1}{\cos^2 x}$, $x \in \left(-\frac{\pi}{3}, \frac{\pi}{3}\right)$ and $y\left(\frac{\pi}{4}\right) = \frac{4}{3}$, then $y\left(-\frac{\pi}{4}\right)$ equals : **[JEE(Main)-2019]**
 (1) $\frac{1}{3} + e^6$ (2) $\frac{1}{3}$ (3) $-\frac{4}{3}$ (4) $\frac{1}{3} + e^3$

18. The curve amongst the family of curves, represented by the differential equation, $(x^2 - y^2)dx + 2xy dy = 0$ which passes through (1,1) is : [JEE(Main)-2019]
- (1) A circle with centre on the y-axis
 - (2) A circle with centre on the x-axis
 - (3) An ellipse with major axis along the y-axis
 - (4) A hyperbola with transverse axis along the x-axis
19. The solution of the differential equation $\frac{dy}{dx} = (x - y)^2$, when $y(1) = 1$, is :- [JEE(Main)-2019]
- (1) $\log_e \left| \frac{2-y}{2-x} \right| = 2(y-1)$
 - (2) $\log_e \left| \frac{2-x}{2-y} \right| = x-y$
 - (3) $-\log_e \left| \frac{1+x-y}{1-x+y} \right| = x+y-2$
 - (4) $-\log_e \left| \frac{1-x+y}{1+x-y} \right| = 2(x-1)$
20. If a curve passes through the point (1, -2) and has slope of the tangent at any point (x, y) on it as $\frac{x^2 - 2y}{x}$, then the curve also passes through the point : [JEE(Main)-2019]
- (1) $(-\sqrt{2}, 1)$
 - (2) $(\sqrt{3}, 0)$
 - (3) $(-1, 2)$
 - (4) $(3, 0)$

EXERCISE (JA)

1. Match the statements/expressions in Column-I with the open intervals in Column-II.

Column-I

Column-II

(A) Interval contained in the domain of definition of non-zero solutions of the differential equation $(x-3)^2 y' + y = 0$

(p) $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

(B) Interval containing the value of the integral

(q) $\left(0, \frac{\pi}{2}\right)$

$$\int_1^5 (x-1)(x-2)(x-3)(x-4)(x-5)dx$$

(C) Interval in which at least one of the points of local maximum of $\cos^2 x + \sin x$ lies

(r) $\left(\frac{\pi}{8}, \frac{5\pi}{4}\right)$

(D) Interval in which $\tan^{-1}(\sin x + \cos x)$ is increasing

(s) $\left(0, \frac{\pi}{8}\right)$

(t) $(-\pi, \pi)$

[JEE 2009, 2+2+2+2]

2. Match the statements/expressions in Column-I with the values given in Column-II.

Column-I

Column-II

- (A) The number of solutions of the equation $x e^{\sin x} - \cos x = 0$ in the interval $\left(0, \frac{\pi}{2}\right)$ (p) 1
- (B) Value(s) of k for which the planes $kx + 4y + z = 0$, $4x + ky + 2z = 0$ and $2x + 2y + z = 0$ intersect in a straight line (q) 2
- (C) Value(s) of k for which $|x-1| + |x-2| + |x+1| + |x+2| = 4k$ has integer solution(s) (r) 3
- (D) If $y' = y + 1$ and $y(0) = 1$, then value(s) of $y(\ln 2)$ (s) 4
- (E) If $y' = y + 1$ and $y(0) = 1$, then value(s) of $y(\ln 2)$ (t) 5

[JEE 2009, 2+2+2+2]

3. Let f be a real valued differentiable function on $\mathbb{R}$ (the set of all real numbers) such that $f(1) = 1$. If the y -intercept of the tangent at any point $P(x, y)$ on the curve $y = f(x)$ is equal to the cube of the abscissa of P , then the value of $f(-3)$ is equal to [JEE 2010, 3]

4. (a) Let $f : [1, \infty) \rightarrow [2, \infty)$ be a differentiable function such that $f(1) = 2$. If $6 \int_1^x f(t) dt = 3x f(x) - x^3$ for all $x \geq 1$, then the value of $f(2)$ is [JEE 2011, 4]

- (b) Let $y'(x) + y(x)g(x) = g(x)g'(x)$, $y(0) = 0$, $x \in \mathbb{R}$, where $f'(x)$ denotes $\frac{df(x)}{dx}$ and $g(x)$ is a given non-constant differentiable function on $\mathbb{R}$ with $g(0) = g(2) = 0$. Then the value of $y(2)$ is [JEE 2011, 4]

5. If $y(x)$ satisfies the differential equation $y' - y \tan x = 2x \sec x$ and $y(0) = 0$, then [JEE 2012, 4M]

(A) $y\left(\frac{\pi}{4}\right) = \frac{\pi^2}{8\sqrt{2}}$ (B) $y'\left(\frac{\pi}{4}\right) = \frac{\pi^2}{18}$ (C) $y\left(\frac{\pi}{4}\right) = \frac{\pi^2}{9}$ (D) $y'\left(\frac{\pi}{3}\right) = \frac{4\pi}{3} + \frac{2\pi^2}{3\sqrt{3}}$

6. Let $f: \left[\frac{1}{2}, 1\right] \rightarrow \mathbb{R}$ (the set of all real numbers) be a positive, non-constant and differentiable function such that $f'(x) < 2f(x)$ and $f\left(\frac{1}{2}\right) = 1$. Then the value of $\int_{\frac{1}{2}}^1 f(x) dx$ lies in the interval

[JEE(Advanced) 2013, 2M]

(A) $(2e - 1, 2e)$ (B) $(e - 1, 2e - 1)$ (C) $\left(\frac{e-1}{2}, e-1\right)$ (D) $\left(0, \frac{e-1}{2}\right)$

7. A curve passes through the point $\left(1, \frac{\pi}{6}\right)$. Let the slope of the curve at each point (x, y) be

$\frac{y}{x} + \sec\left(\frac{y}{x}\right)$, $x > 0$. Then the equation of the curve is [JEE(Advanced) 2013, 2M]

- (A) $\sin\left(\frac{y}{x}\right) = \log x + \frac{1}{2}$ (B) $\operatorname{cosec}\left(\frac{y}{x}\right) = \log x + 2$
 (C) $\sec\left(\frac{2y}{x}\right) = \log x + 2$ (D) $\cos\left(\frac{2y}{x}\right) = \log x + \frac{1}{2}$

Paragraph for Question 8 and 9

Let $f : [0, 1] \rightarrow \mathbb{R}$ (the set of all real numbers) be a function. Suppose the function f is twice differentiable, $f(0) = f(1) = 0$ and satisfies $f''(x) - 2f'(x) + f(x) \geq e^x$, $x \in [0, 1]$.

8. If the function $e^{-x}f(x)$ assumes its minimum in the interval $[0, 1]$ at $x = \frac{1}{4}$, which of the following is true ? [JEE(Advanced) 2013, 3, (-1)]

- (A) $f'(x) < f(x)$, $\frac{1}{4} < x < \frac{3}{4}$ (B) $f'(x) > f(x)$, $0 < x < \frac{1}{4}$
 (C) $f'(x) < f(x)$, $0 < x < \frac{1}{4}$ (D) $f'(x) < f(x)$, $\frac{3}{4} < x < 1$

9. Which of the following is true for $0 < x < 1$? [JEE(Advanced) 2013, 3, (-1)]

- (A) $0 < f(x) < \infty$ (B) $-\frac{1}{2} < f(x) < \frac{1}{2}$ (C) $-\frac{1}{4} < f(x) < 1$ (D) $-\infty < f(x) < 0$

10. The function $y = f(x)$ is the solution of the differential equation $\frac{dy}{dx} + \frac{xy}{x^2 - 1} = \frac{x^4 + 2x}{\sqrt{1 - x^2}}$ in $(-1, 1)$

satisfying $f(0) = 0$. Then $\int_{\frac{\sqrt{3}}{2}}^{\frac{\sqrt{3}}{2}} f(x) dx$ is [JEE(Advanced)-2014, 3(-1)]

- (A) $\frac{\pi}{3} - \frac{\sqrt{3}}{2}$ (B) $\frac{\pi}{3} - \frac{\sqrt{3}}{4}$ (C) $\frac{\pi}{6} - \frac{\sqrt{3}}{4}$ (D) $\frac{\pi}{6} - \frac{\sqrt{3}}{2}$

11. Let $y(x)$ be a solution of the differential equation $(1 + e^x)y' + ye^x = 1$. If $y(0) = 2$, then which of the following statements is(are) true ? [JEE 2015, 4M, -2M]

- (A) $y(-4) = 0$
 (B) $y(-2) = 0$
 (C) $y(x)$ has a critical point in the interval $(-1, 0)$
 (D) $y(x)$ has no critical point in the interval $(-1, 0)$

12. Consider the family of all circles whose centers lie on the straight line $y = x$. If this family of circles is represented

by the differential equation $Py'' + Qy' + 1 = 0$, where P, Q are functions of x, y and y' (here $y' = \frac{dy}{dx}$, $y'' = \frac{d^2y}{dx^2}$),

then which of the following statements is (are) true?

[JEE 2015, 4M, -2M]

(A) $P = y + x$

(B) $P = y - x$

(C) $P + Q = 1 - x + y + y' + (y')^2$

(D) $P - Q = x + y - y' - (y')^2$

13. A solution curve of the differential equation $(x^2 + xy + 4x + 2y + 4) \frac{dy}{dx} - y^2 = 0$, $x > 0$, passes through the point $(1, 3)$. The the solution curve-

(A) intersects $y = x + 2$ exactly at one point

(B) intersects $y = x + 2$ exactly at two points

(C) intersects $y = (x + 2)^2$

(D) does NOT intersect $y = (x + 3)^2$

[JEE(Advanced)-2016, 4(-2)]

14. Let $f : (0, \infty) \rightarrow \mathbb{R}$ be a differentiable function such that $f'(x) = 2 - \frac{f(x)}{x}$ for all $x \in (0, \infty)$ and

$f(1) \neq 1$. Then

[JEE(Advanced)-2016, 4(-2)]

(A) $\lim_{x \rightarrow 0^+} f'\left(\frac{1}{x}\right) = 1$

(B) $\lim_{x \rightarrow 0^+} xf\left(\frac{1}{x}\right) = 2$

(C) $\lim_{x \rightarrow 0^+} x^2 f'(x) = 0$

(D) $|f(x)| \leq 2$ for all $x \in (0, 2)$

15. If $y = y(x)$ satisfies the differential equation $8\sqrt{x}(\sqrt{9+\sqrt{x}})dy = (\sqrt{4+\sqrt{9+\sqrt{x}}})^{-1} dx$, $x > 0$ and

$y(0) = \sqrt{7}$, then $y(256) =$

[JEE(Advanced)-2017, 3(-1)]

(A) 80

(B) 3

(C) 16

(D) 9

16. If $f : \mathbb{R} \rightarrow \mathbb{R}$ is a differentiable function such that $f'(x) > 2f(x)$ for all $x \in \mathbb{R}$, and $f(0) = 1$, then

[JEE(Advanced)-2017, 3(-2)]

(A) $f(x) > e^{2x}$ in $(0, \infty)$

(B) $f(x)$ is decreasing in $(0, \infty)$

(C) $f(x)$ is increasing in $(0, \infty)$

(D) $f'(x) < e^{2x}$ in $(0, \infty)$

17. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ and $g : \mathbb{R} \rightarrow \mathbb{R}$ be two non-constant differentiable functions. If $f'(x) = (e^{(f(x)-g(x))})g'(x)$ for all $x \in \mathbb{R}$, and $f(1) = g(2) = 1$, then which of the following statement(s) is (are) TRUE ?

[JEE(Advanced)-2018, 4(-2)]

(A) $f(2) < 1 - \log_e 2$

(B) $f(2) > 1 - \log_e 2$

(C) $g(1) > 1 - \log_e 2$

(D) $g(1) < 1 - \log_e 2$

18. Let $f : (0, \pi) \rightarrow \mathbb{R}$ be a twice differentiable function such that

$$\lim_{t \rightarrow x} \frac{f(x) \sin t - f(t) \sin x}{t - x} = \sin^2 x \text{ for all } x \in (0, \pi).$$

If $f\left(\frac{\pi}{6}\right) = -\frac{\pi}{12}$, then which of the following statement(s) is (are) TRUE ?

[JEE(Advanced)-2018, 4(-2)]

(A) $f\left(\frac{\pi}{4}\right) = \frac{\pi}{4\sqrt{2}}$

(B) $f(x) < \frac{x^4}{6} - x^2$ for all $x \in (0, \pi)$

(C) There exists $\alpha \in (0, \pi)$ such that $f'(\alpha) = 0$

(D) $f''\left(\frac{\pi}{2}\right) + f\left(\frac{\pi}{2}\right) = 0$

19. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a differentiable function with $f(0) = 0$. If $y = f(x)$ satisfies the differential equation $\frac{dy}{dx} = (2 + 5y)(5y - 2)$,

then the value of $\lim_{x \rightarrow -\infty} f(x)$ is _____.

[JEE(Advanced)-2018, 3(0)]

ANSWER KEY

DIFFERENTIAL EQUATION

EXERCISE (O-1)

1. C 2. B 3. B 4. A 5. D 6. B 7. C 8. B 9. A
10. A 11. A 12. B 13. C 14. A 15. B 16. A 17. A 18. B

EXERCISE (O-2)

1. A,B,D 2. A,B,C,D 3. A,C 4. A,B,D 5. C,D
6. A,B,C 7. A,B,D 8. A,B,C 9. A,B,D 10. A,B,C

EXERCISE (S-1)

(E-1)

[FORMATION & VARIABLES SEPARABLE]

1. (i) order 2 & degree 3 (ii) order 2 & degree 2
2. (a) $xy \frac{d^2y}{dx^2} + x \left(\frac{dy}{dx} \right)^2 - y \frac{dy}{dx} ;$ (b) $[1 + (y')^2] \cdot y''' - 3y'(y'')^2 = 0$ (c) $\frac{d^3y}{dx^3} - 6 \frac{d^2y}{dx^2} + 11 \frac{dy}{dx} - 6y = 0$
3. $\ln^2(\sec x + \tan x) - \ln^2(\sec y + \tan y) = c$ 4. $\ln x (1 - y)^2 = c - \frac{1}{2} y^2 - 2y + \frac{1}{2} x^2$
5. $\sqrt{x^2 - 1} - \sec^{-1} x + \sqrt{y^2 - 1} = c$ 6. $y = c (1 - ay) (x + a)$
7. $\ln \left[1 + \tan \frac{x+y}{2} \right] = x + c$ 8. $y \sin y = x^2 \ln x + c$
9. (a) $\ln \left| \tan \frac{y}{4} \right| = c - 2 \sin \frac{x}{2},$ (b) $y = e^{\tan(x/2)}$
10. (a) $P = 1000 + 1500e^{-kt}$ where $k = \frac{1}{10} \ln \left(\frac{5}{3} \right);$ (b) $T = 10 \log_{5/3}(3) ;$ (c) $P = 1000$ as $t \rightarrow \infty$
11. $m = m_0 e^{-kt}$ where $k = -\frac{1}{t_0} \ln \left(1 - \frac{\alpha}{100} \right)$ 12. $x^2 + y^2 = k^2$ 13. $y = kx$ or $xy = c$
14. 40 15. $1/4$ 16. $y = \frac{1}{3-e} (2e^x - e + 1)$

(E-2)

[HOMOGENEOUS]

1. $c(x-y)^{2/3} (x^2+xy+y^2)^{1/6} = \exp \left[\frac{1}{\sqrt{3}} \tan^{-1} \frac{x+2y}{x\sqrt{3}} \right], \text{ where } \exp x \equiv e^x$
2. $\frac{y^2 \pm y\sqrt{y^2-x^2}}{x^2} = \ell n \left| \left(y \pm \sqrt{y^2-x^2} \right) \cdot \frac{c^2}{x^3} \right|, \text{ where same sign has to be taken.}$
3. $x^2 + y^2 - 2x = 0$
4. $x^2 + y^2 = cx$
5. $\arctan \frac{2y+1}{2x+1} = \ln c \sqrt{x^2 + y^2 + x + y + \frac{1}{2}}$
6. $(x+y-2) = c(y-x)^3$
7. $\tan^{-1} \frac{y+3}{x+2} + \ln c \sqrt{(y+3)^2 + (x+2)^2} = 0$
8. $x+y+\frac{4}{3} = ce^{3(x-2y)}$
9. $e^{-2 \tan^{-1} \frac{y+2}{x-3}} = c.(y+2)$

(E-3)

[LINEAR]

1. $-\frac{1}{2}$
2. $y\sqrt{1+x^2} = c + \frac{1}{2} \ln \left[\tan \frac{1}{2} \arctan x \right]$. Another form is $y\sqrt{1+x^2} = c + \frac{1}{2} \ln \frac{\sqrt{1+x^2}-1}{x}$
3. $y = c(1-x^2) + \sqrt{1-x^2}$
4. (a) $y = cx^2 \pm x$; (b) $xy = 2$
5. $y(x-1) = x^2(x^2-x+c)$
6. $\left(\frac{1}{3} + y\right) \tan^3 \frac{x}{2} = c + 2 \tan \frac{x}{2} - x$
7. $4(x^2+1)y + x^3(1-2\ln x) = cx$
8. $y = cx + x \ln \tan x$
9. $x = ce^{-\arctan y} + \arctan y - 1$
11. $\cos x - 1$
12. $y(1+bx) = b + cx$
13. $f(x) = -\frac{2 \cos x}{(1+\sin x)^2} - Ce^{-\sin x} \cdot \cos x$
14. $\frac{1}{y^2} = -1 + (c+x) \cot\left(\frac{x}{2} + \frac{\pi}{4}\right)$
15. $x^3 y^3 = 3 \sin x + c$
16. $y^{-1} e^x = c - x^2$
17. $27\frac{7}{9}$ minutes

(E-4)

1. $y^2 + x \ln ax = 0$
2. $y^2 = 3x^2 - 6x - x^3 + ce^{-x} + 4$
3. $x \ln y = e^x(x-1) + c$
4. $\sin y = (e^x + c)(1+x)$
5. $cx^2 + 2xe^{-y} = 1$
6. $y = ce^x$; $y = c + \frac{x^2}{2}$
7. $y^2 = -1 + (x+1) \ln \frac{c}{x+1}$ or $x + (x+1) \ln \frac{c}{x+1}$
8. $y = \frac{1}{x} \tan(\ln|cx|)$
9. $e^y = c \cdot \exp(-e^x) + e^x - 1$
10. $y = 3 \ln(x^2 + y + 3) + C$

EXERCISE (S-2)

(MISCELLANEOUS)

1. $y = 2^{\sin x}$
3. $\frac{\sin^{-1} x}{2} + \frac{y}{x-y} = \frac{\pi}{4} - 2$
4. (i) $y = u(x) + K(u(x) - v(x))$, where K is any constant; (ii) $\alpha + \beta = 1$; (iii) constant
5. $x(ey + \ln y + 1) = 1$
6. $y = 5t \left(1 + \frac{50}{50+t}\right)$ gms; $91\frac{2}{3}$ gms
7. (i) $x^2 + 2y^2 = c$, (ii) $\sin y = ce^{-x}$, (iii) $y = cx$ if $k = 2$ and $\frac{1}{x^{k-2}} - \frac{1}{y^{k-2}} = \frac{1}{c^{k-2}}$ if $k \neq 2$
- (iv) $x^2 - y^2 + 2xy = c$; $x^2 - y^2 - 2xy = c$
9. (a) $(-\infty, \infty)$ (b) $\frac{19}{4}$ (c) $3 - \sqrt{5}$
11. $y = cx - x^2$

EXERCISE (JM)

- | | | | | | | | | |
|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| 1. 3 | 2. 4 | 3. 3 | 4. 4 | 5. 2 | 6. 3 | 7. 3 | 8. 2 | 9. 3 |
| 10. 2 | 11. 1 | 12. 1 | 13. 1 | 14. 2 | 15. 2 | 16. 2 | 17. 1 | 18. 2 |
| 19. 4 | 20. 2 | | | | | | | |

EXERCISE (JA)

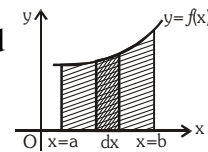
1. (A) p,q,s; (B) p,t; (C) p,q,r,t; (D) s
2. (A) p; (B) q,s; (C) q,r,s,t; (D) r
3. 9
4. (a) Bonus; (b) 0
5. A,D
6. D
7. A
8. C
9. D
10. B
11. A,C
12. B,C
13. A,D
14. A
15. B
16. A,C
17. B,C
18. B,C,D
19. 0.4

AREA UNDER THE CURVE

1. AREA UNDER THE CURVES :

- (a) Area bounded by the curve $y = f(x)$, the x-axis and the ordinates at $x = a$ and

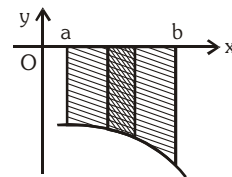
$x = b$ is given by $A = \int_a^b y \, dx$, where $y = f(x)$ lies above the x-axis



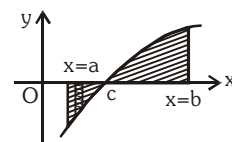
and $b > a$. Here vertical strip of thickness dx is considered at distance x .

- (b) If $y = f(x)$ lies completely below the x-axis then A is negative and we consider

the magnitude only, i.e. $A = \left| \int_a^b y \, dx \right|$



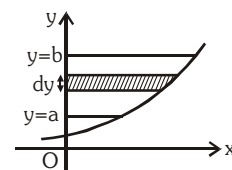
- (c) If curve crosses the x-axis at $x = c$, then $A = \left| \int_a^c y \, dx \right| + \int_c^b y \, dx$



- (d) Sometimes integration w.r.t. y is very useful (horizontal strip) :

Area bounded by the curve, y-axis and the two abscissae at

$y = a$ & $y = b$ is written as $A = \int_a^b x \, dy$.



Note : If the curve is symmetric and suppose it has 'n' symmetric portions, then total area = n (Area of one symmetric portion).

Illustration 1 : Find the area bounded by $y = \sec^2 x$, $x = \frac{\pi}{6}$, $x = \frac{\pi}{3}$ & x-axis

Solution : Area bounded $= \int_{\pi/6}^{\pi/3} y \, dx = \int_{\pi/6}^{\pi/3} \sec^2 x \, dx = [\tan x]_{\pi/6}^{\pi/3} = \tan \frac{\pi}{3} - \tan \frac{\pi}{6} = \sqrt{3} - \frac{1}{\sqrt{3}} = \frac{2}{\sqrt{3}}$ sq.units.

Illustration 2 : Find the area in the first quadrant bounded by $y = 4x^2$, $x = 0$, $y = 1$ and $y = 4$.

Solution : Required area $= \int_1^4 x \, dy = \int_1^4 \frac{\sqrt{y}}{2} \, dy = \frac{1}{2} \left[\frac{2}{3} y^{3/2} \right]_1^4$
 $= \frac{1}{3} [4^{3/2} - 1] = \frac{1}{3} [8 - 1]$
 $= \frac{7}{3} = 2\frac{1}{3}$ sq.units.

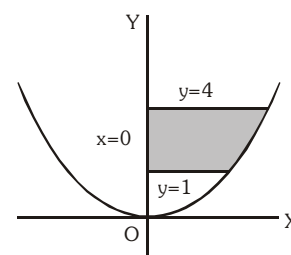
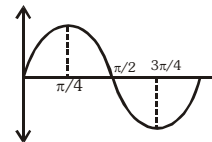


Illustration 3 : Find the area bounded by the curve $y = \sin 2x$, x-axis and the lines $x = \frac{\pi}{4}$ and $x = \frac{3\pi}{4}$

Solution : Required area $= \int_{\pi/4}^{\pi/2} \sin 2x dx + \left| \int_{\pi/2}^{3\pi/4} \sin 2x dx \right| = \left(-\frac{\cos 2x}{2} \right) \Big|_{\pi/4}^{\pi/2} + \left| \left(-\frac{\cos 2x}{2} \right) \Big|_{\pi/2}^{3\pi/4} \right|$

$$= -\frac{1}{2}[-1-0] + \left| \frac{1}{2}(0+(-1)) \right| = 1 \text{ sq. unit}$$



Do yourself - 1 :

- Find the area bounded by $y = x^2 + 2$ above x-axis between $x = 2$ & $x = 3$.
- Using integration, find the area of the curve $y = \sqrt{1-x^2}$ with co-ordinate axes bounded in first quadrant.
- Find the area bounded by the curve $y = 2\cos x$ and the x-axis from $x = 0$ to $x = 2\pi$.
- Find the area bounded by the curve $y = x|x|$, x-axis and the ordinates $x = -\frac{1}{2}$ and $x=1$.

2. AREA ENCLOSED BETWEEN TWO CURVES :

- (a) Area bounded by two curves $y = f(x)$ & $y = g(x)$ such that $f(x) > g(x)$ is

$$A = \int_{x_1}^{x_2} (y_1 - y_2) dy$$

$$A = \int_{x_1}^{x_2} [f(x) - g(x)] dx$$

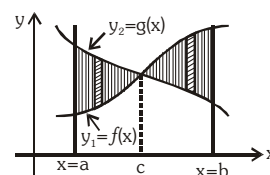
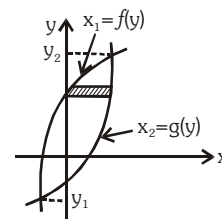
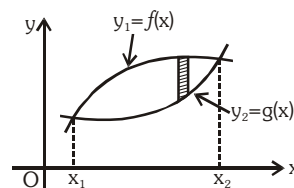
- (b) In case horizontal strip is taken we have

$$A = \int_{y_1}^{y_2} (x_1 - x_2) dy$$

$$A = \int_{y_1}^{y_2} [f(y) - g(y)] dy$$

- (c) If the curves $y_1 = f(x)$ and $y_2 = g(x)$ intersect at $x = c$, then required area

$$A = \int_a^c (g(x) - f(x)) dx + \int_c^b (f(x) - g(x)) dx = \int_a^b |f(x) - g(x)| dx$$



Note : Required area must have all the boundaries indicated in the problem.

Illustration 4 : Find the area bounded by the curve $y = (x-1)(x-2)(x-3)$ lying between the ordinates $x = 0$ and $x = 3$ and x -axis

Solution : To determine the sign, we follow the usual rule of change of sign.

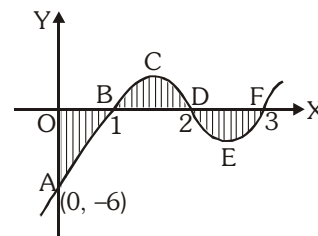
$$y = +ve \quad \text{for } x > 3$$

$$y = -ve \quad \text{for } 2 < x < 3$$

$$y = +ve \quad \text{for } 1 < x < 2$$

$$y = -ve \quad \text{for } x < 1.$$

$$\begin{aligned} \int_0^3 |y| dx &= \int_0^1 |y| dx + \int_1^2 |y| dx + \int_2^3 |y| dx \\ &= \int_0^1 -y dx + \int_1^2 y dx + \int_2^3 -y dx \end{aligned}$$



$$\text{Now let } F(x) = \int (x-1)(x-2)(x-3) dx = \int (x^3 - 6x^2 + 11x - 6) dx = \frac{1}{4}x^4 - 2x^3 + \frac{11}{2}x^2 - 6x.$$

$$\therefore F(0) = 0, F(1) = -\frac{9}{4}, F(2) = -2, F(3) = -\frac{9}{4}.$$

$$\text{Hence required Area} = -[F(1) - F(0)] + [F(2) - F(1)] - [F(3) - F(2)] = 2\frac{3}{4} \text{ sq.units.}$$

Illustration 5 : Compute the area of the figure bounded by the straight lines $x = 0$, $x = 2$ and the curves $y = 2^x$, $y = 2x - x^2$.

Solution : The required area $= \int_0^2 (y_1 - y_2) dx$

$$\text{where } y_1 = 2^x \text{ and } y_2 = 2x - x^2 = \int_0^2 (2^x - 2x + x^2) dx$$

$$= \left[\frac{2^x}{\ln 2} - x^2 + \frac{1}{3}x^3 \right]_0^2 = \left(\frac{4}{\ln 2} - 4 + \frac{8}{3} \right) = \frac{1}{\ln 2} = \frac{3}{\ln 2} - \frac{4}{3} \text{ sq.units.}$$

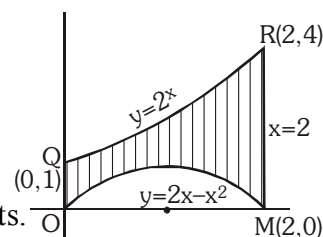


Illustration 6 : Compute the area of the figure bounded by the parabolas $x = -2y^2$, $x = 1 - 3y^2$.

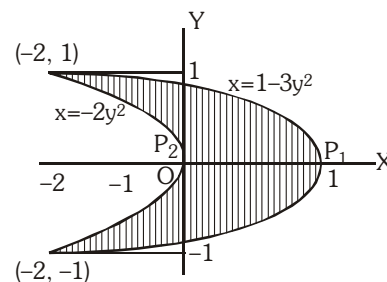
Solution : Solving the equations $x = -2y^2$, $x = 1 - 3y^2$, we find that ordinates of the points of intersection of the two curves as $y_1 = -1$, $y_2 = 1$.

The points are $(-2, -1)$ and $(-2, 1)$.

The required area

$$2 \int_0^1 (x_1 - x_2) dy = 2 \int_0^1 [(1 - 3y^2) - (-2y^2)] dy$$

$$= 2 \int_0^1 (1 - y^2) dy = 2 \left[y - \frac{y^3}{3} \right]_0^1 = \frac{4}{3} \text{ sq.units.}$$



Do yourself - 2 :

- Find the area bounded by $y = \sqrt{x}$ and $y = x$.
- Find the area bounded by the curves $x = y^2$ and $x = 3 - 2y^2$.
- Find the area of the region bounded by the curves $x = \frac{1}{2}$, $x = 2$, $y = \log x$ and $y = 2^x$.

3. CURVE TRACING :

The following procedure is to be applied in sketching the graph of a function $y = f(x)$ which in turn will be extremely useful to quickly and correctly evaluate the area under the curves.

- (a) Symmetry : The symmetry of the curve is judged as follows :
- (i) If all the powers of y in the equation are even then the curve is symmetrical about the axis of x .
 - (ii) If all the powers of x are even, the curve is symmetrical about the axis of y .
 - (iii) If powers of x & y both are even, the curve is symmetrical about the axis of x as well as y .
 - (iv) If the equation of the curve remains unchanged on interchanging x and y , then the curve is symmetrical about $y = x$.
 - (v) If on interchanging the signs of x & y both, the equation of the curve is unaltered then there is symmetry in opposite quadrants.
- (b) Find dy/dx & equate it to zero to find the points on the curve where you have horizontal tangents.
- (c) Find the points where the curve crosses the x -axis & also the y -axis.
- (d) Examine if possible the intervals when $f(x)$ is increasing or decreasing. Examine what happens to ' y ' when $x \rightarrow \infty$ or $-\infty$.

Illustration 7 : Find the area of a loop as well as the whole area of the curve $a^2y^2 = x^2(a^2 - x^2)$.

Solution : The curve is symmetrical about both the axes. It cuts x -axis at $(0, 0)$, $(-a, 0)$, $(a, 0)$

$$\text{Area of a loop} = 2 \int_0^a y \, dx = 2 \int_0^a \frac{x}{a} \sqrt{a^2 - x^2} \, dx$$

$$= -\frac{1}{a} \int_0^a \sqrt{a^2 - x^2} (-2x) \, dx = \frac{1}{a} \left[\frac{2}{3} (a^2 - x^2)^{3/2} \right]_0^a = \frac{2}{3} a^2$$

$$\text{Total area} = 2 \times \frac{2}{3} a^2 = \frac{4}{3} a^2 \text{ sq. units.}$$

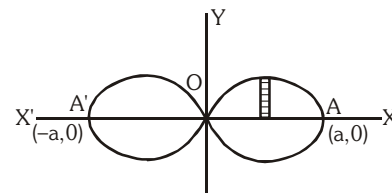


Illustration 8 : Find the whole area included between the curve $x^2y^2 = a^2(y^2 - x^2)$ and its asymptotes.

- Solution :**
- (i) The curve is symmetric about both the axes (even powers of x & y)
 - (ii) Asymptotes are $x = \pm a$

$$\begin{aligned} A &= 4 \int_0^a y \, dx \\ &= 4 \int_0^a \frac{ax}{\sqrt{a^2 - x^2}} \, dx \\ &= 4a \left| -\sqrt{a^2 - x^2} \right|_0^a \\ &= 4a^2 \end{aligned}$$

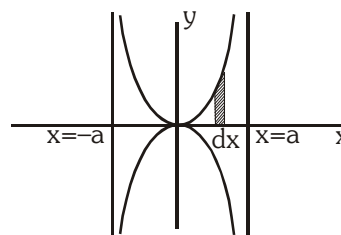
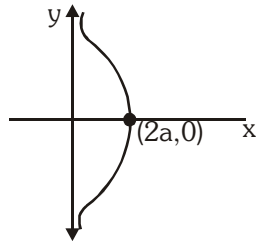


Illustration 9 : Find the area bounded by the curve $xy^2 = 4a^2(2a-x)$ and its asymptote.

- Solution :**
- (i) The curve is symmetrical about the x-axis as it contains even powers of y.
 - (ii) It passes through (2a,0).
 - (iii) Its asymptote is $x = 0$, i.e., y-axis.



$$A = 2 \int_0^{2a} y dx = 2 \int_0^{2a} 2a \sqrt{\frac{2a-x}{x}} dx$$

Put $x = 2a \sin^2 \theta$

$$A = 16a^2 \int_0^{\pi/2} \cos^2 \theta d\theta$$

$$= 4\pi a^2$$

4. IMPORTANT POINTS :

- (a) Since area remains invariant even if the co-ordinate axes are shifted, hence shifting of origin in many cases proves to be very convenient in computing the area.

Illustration 10: Find the area enclosed by $|x - 1| + |y + 1| = 1$.

Solution : Shift the origin to $(1, -1)$.

$$X = x - 1 \qquad Y = y + 1$$

$$|\mathbf{X}| + |\mathbf{Y}| = 1$$

$$\text{Area} = \sqrt{2} \times \sqrt{2} = 2 \text{ sq. units}$$

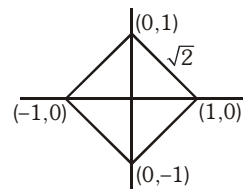


Illustration 11: Find the area of the region common to the circle $x^2 + y^2 + 4x + 6y - 3 = 0$ and the parabola $x^2 + 4x = 6y + 14$.

Solution : Circle is $x^2 + y^2 + 4x + 6y - 3 = 0$

$$\Rightarrow (x + 2)^2 + (y + 3)^2 = 16$$

Shifting origin to $(-2,-3)$.

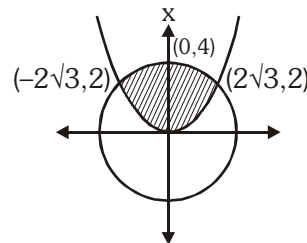
$$X^2 + Y^2 = 16$$

equation of parabola $\rightarrow (x + 2)^2 = 6(y + 3)$

$$\Rightarrow X^2 = 6Y$$

Solving circle & parabola, we get $X = \pm 2\sqrt{3}$

Hence they intersect at $(-2\sqrt{3}, 2)$ & $(2\sqrt{3}, 2)$



$$A = 2 \left[\int_0^2 \sqrt{6Y} \, dY + \int_2^4 \sqrt{16 - Y^2} \, dY \right]$$

$$= 2 \left[\frac{2}{3} \sqrt{6} \left[Y^{3/2} \right]_0^2 + \left[\frac{1}{2} Y \sqrt{16 - Y^2} + \frac{16}{2} \sin^{-1} \frac{Y}{4} \right]_2^4 \right] = \left(\frac{4\sqrt{3}}{3} + \frac{16\pi}{3} \right) \text{sq. units}$$

Do yourself : 3

- (i) Find the area inside the circle $x^2 - 2x + y^2 - 4y + 1 = 0$ and outside the ellipse $x^2 - 2x + 4y^2 - 16y + 13 = 0$

- (b) If $y = f(x)$ is a monotonic function in (a, b) , then the area bounded by the ordinates at $x = a$, $x = b$, $y = f(x)$ and $y = f(c)$ [where $c \in (a, b)$] is minimum when $c = \frac{a+b}{2}$.

Proof : Let the function $y = f(x)$ be monotonically increasing.

$$\text{Required area } A = \int_a^c [f(c) - f(x)] dx + \int_c^b [f(x) - f(c)] dx$$

$$\text{For minimum area, } \frac{dA}{dc} = 0$$

$$\Rightarrow [f'(c) \cdot c + f(c) - f'(c)a - f(c)] + [-f(c) - f'(c)b + f'(c)c + f(c)] = 0$$

$$\Rightarrow f'(c) \left\{ c - \frac{a+b}{2} \right\} = 0$$

$$\Rightarrow c = \frac{a+b}{2} \quad (\because f'(c) \neq 0)$$

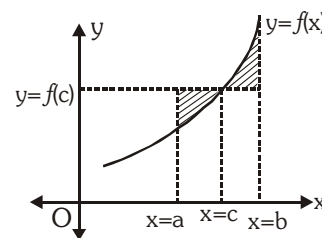


Illustration 12 : Find the value of 'a' for which area bounded by $x = 1$, $x = 2$, $y = 6x^2$ and $y = f(a)$ is minimum.

Solution : Let $b = f(a)$.

$$A = \int_1^a (b - 6x^2) dx + \int_a^2 (6x^2 - b) dx = \left[bx - 2x^3 \right]_1^a + \left[2x^3 - bx \right]_a^2$$

$$= 8a^3 - 18a^2 + 18$$

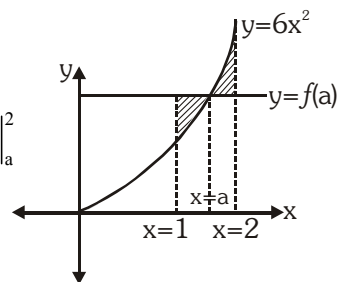
$$\text{For minimum area } \frac{dA}{da} = 0$$

$$\Rightarrow 24a^2 - 36a = 0 \Rightarrow a = 1.5$$

$$\text{Alternatively, } y = 6x^2 \Rightarrow \frac{dy}{dx} = 12x$$

Hence $y = f(x)$ is monotonically increasing. Hence bounded area is minimum when

$$a = \left(\frac{1+2}{2} \right) = 1.5$$



Do yourself - 4 :

- (i) Find the value of 'a' ($0 < a < \frac{\pi}{2}$) for which the area bounded by the curve $f(x) = \sin^3 x + \sin x$, $y = f(a)$ between $x = 0$ & $x = \pi$ is minimum.
- (d) The area bounded by a curve & an axis is equal to the area bounded by the inverse of that curve & the other axis, i.e., the area bounded by $y = f(x)$ and x-axis (say) is equal to the area bounded by $y = f^{-1}(x)$ and y-axis.

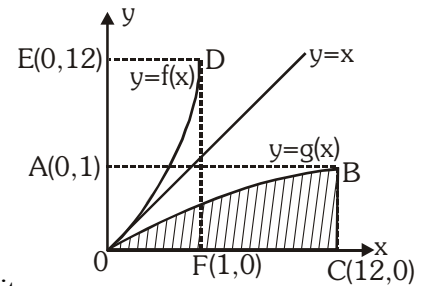
Illustration 13: If $y = g(x)$ is the inverse of a bijective mapping $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = 6x^5 + 4x^3 + 2x$, find the area bounded by $g(x)$, the x -axis and the ordinate at $x = 12$.

Solution :

$$f(\mathbf{x}) = 12$$

$$\Rightarrow 6x^5 + 4x^3 + 2x = 12 \Rightarrow x = 1$$

$$\begin{aligned} \int_0^{12} g(x) dx &= \text{area of rectangle OEDF} - \int_0^1 f(x) dx \\ &= 1 \times 12 - \int_0^1 (6x^5 + 4x^3 + 2x) dx = 12 - 3 = 9 \text{ sq. units.} \end{aligned}$$



Do yourself - 5 :

- (i) Find the area bounded by the inverse of bijective function $f(x) = 4x^3 + 6x$, the x-axis and the ordinates $x = 0$ & $x = 44$.

5. USEFUL RESULTS :

- (a) Whole area of the ellipse, $x^2/a^2 + y^2/b^2 = 1$ is πab sq.units.
- (b) Area enclosed between the parabolas $y^2 = 4ax$ & $x^2 = 4by$ is $16ab/3$ sq.units.
- (c) Area included between the parabola $y^2 = 4ax$ & the line $y = mx$ is $8a^2/3 m^3$ sq.units.
- (d) The area of the region bounded by one arch of $\sin ax$ (or $\cos ax$) and x-axis is $2/a$ sq.units.
- (e) Average value of a function $y = f(x)$ over an interval $a \leq x \leq b$ is defined as : $y(av) = \frac{1}{b-a} \int_a^b f(x)dx$.

Miscellaneous Illustration :

Illustration 14: Find the smaller of the areas bounded by the parabola $4y^2 - 3x - 8y + 7 = 0$ and the ellipse $x^2 + 4y^2 - 2x - 8y + 1 = 0$.

Solution :

$$C_1 \text{ is } 4(y^2 - 2y) = 3x - 7$$

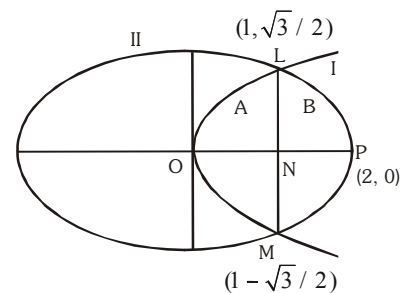
or $4(y-1)^2 = 3x - 3 = 3(x-1)$ (i)

Above is parabola with vertex at $(1, 1)$

$$C_2 \text{ is } (x^2 - 2x) + 4(y^2 - 2y) = -1$$

or $(x - 1)^2 + 4(y - 1)^2 = -1 + 1 + 4$

$$\text{or } \frac{(x-1)^2}{2^2} + \frac{(y-1)^2}{1^2} = 1 \quad \dots\dots(\text{ii})$$



Above represents an ellipse with centre at (1, 1). Shift the origin to (1, 1) and this will not affect the magnitude of required area but will make the calculation simpler.

Thus the two curves are

$$4Y^2 = 3X \text{ and } \frac{X^2}{2^2} + \frac{Y^2}{1} = 1 \text{ They meet at } \left(1, \pm \frac{\sqrt{3}}{2}\right)$$

$$\begin{aligned}\text{Required area} &= 2(A + B) = 2 \left[\int Y_1 dX + \int Y_2 dX \right] \\ &= 2 \left[\frac{\sqrt{3}}{2} \int_0^1 \sqrt{X} dX + \int_1^2 \frac{\sqrt{4-X^2}}{2} dX \right] = \left[\frac{\sqrt{3}}{6} + \frac{2\pi}{3} \right] \text{sq. units.}\end{aligned}$$

Illustration 15 : Find the area bounded by the regions $y \geq \sqrt{x}$, $x > -\sqrt{y}$ & curve $x^2 + y^2 = 2$.

Solution :

Common region is given by the diagram

If area of region OAB = λ

then area of OCD = λ

Because $y = \sqrt{x}$ & $x = -\sqrt{y}$

will bound same area with x & y axes respectively.

$$y = \sqrt{x} \Rightarrow y^2 = x$$

$x = -\sqrt{y} \Rightarrow x^2 = y$ and hence both the curves are symmetric with respect to the line $y = x$

$$\text{Area of first quadrant OBC} = \frac{\pi r^2}{4} = \frac{\pi}{2} \quad (\because r = \sqrt{2})$$

$$\text{Area of region OCA} = \frac{\pi}{2} - \lambda$$

$$\text{Area of shaded region} = \left(\frac{\pi}{2} - \lambda \right) + \lambda = \frac{\pi}{2} \text{ sq. units.}$$

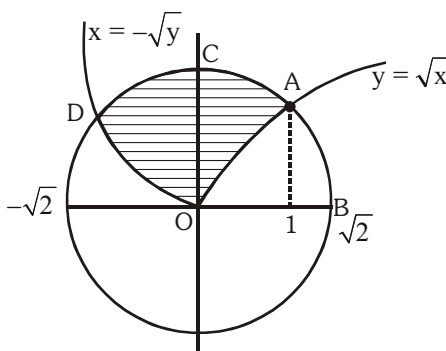


Illustration 16 : Find the equation of line passing through the origin & dividing the curvilinear triangle with vertex at the origin, bounded by the curves $y = 2x - x^2$, $y = 0$ & $x = 1$ in two parts of equal areas.

Solution :

$$\text{Area of region OBA} = \int_0^1 (2x - x^2) dx$$

$$= \left[x^2 - \frac{x^3}{3} \right]_0^1 = \frac{2}{3}$$

$$\frac{2}{3} = A_1 + A_1 \Rightarrow A_1 = \frac{1}{3}$$

Let pt. C has coordinates $(1, y)$

$$\text{Area of } \triangle OCB = \frac{1}{2} \times 1 \times y = \frac{1}{3}$$

$$y = \frac{2}{3}$$

$$\text{C has coordinates } \left(1, \frac{2}{3} \right)$$

$$\text{Line OC has slope } m = \frac{\frac{2}{3} - 0}{1 - 0} = \frac{2}{3}$$

$$\text{Equation of line OC is } y = mx \Rightarrow y = \frac{2}{3}x.$$

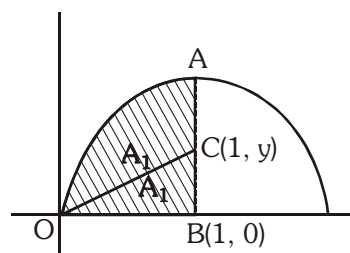


Illustration 17: The area bounded by $y = x^2 + 1$ and the tangents to it drawn from the origin is :-

- (A) $\frac{8}{3}$ sq. units (B) $\frac{1}{3}$ sq. units (C) $\frac{2}{3}$ sq. units (D) none of these

Solution :

The parabola is even function & let the equation of tangent is $y = mx$.

Now we calculate the point of intersection of parabola & tangent

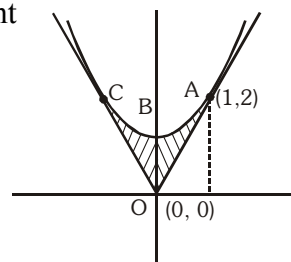
$$\mathbf{m}\mathbf{x} = \mathbf{x}^2 + 1$$

$$x^2 - mx + 1 = 0 \Rightarrow D = 0$$

$$\Rightarrow m^2 - 4 = 0 \Rightarrow m = \pm 2$$

Two tangents are possible $y = 2x$ & $y = -2x$

Intersection of $y = x^2 + 1$ & $y = 2x$ is $x = 1$ & $y = 2$



$$\text{Area of shaded region OAB} = \int_0^1 (y_2 - y_1) dx = \int_0^1 ((x^2 + 1) - 2x) dx = \frac{1}{3} \text{ sq. units}$$

$$\text{Area of total shaded region} = 2\left(\frac{1}{3}\right) = \frac{2}{3} \text{ sq. units}$$

Illustration 18: Find the area bounded by x-axis and the curve given by $x = a \sin t$, $y = a \cos t$ for $0 \leq t \leq \pi$.

Solution :

$$\text{Area} = \int_0^\pi y \frac{dx}{dt} \cdot dt = \int_0^\pi a \cos t (a \cos t) dt = \frac{a^2}{2} \int_0^\pi (1 + \cos 2t) dt = \frac{a^2}{2} \left[t + \frac{\sin 2t}{2} \right]_0^\pi = \frac{a^2}{2} |\pi| = \frac{\pi a^2}{2}$$

Alternatively,

$$\text{Area} = \int_0^\pi x \frac{dy}{dt} \cdot dt = \left| \int_0^\pi a \sin t (-a \sin t) dt \right| = \frac{a^2}{2} \left| \int_0^\pi (\cos 2t - 1) dt \right| = \frac{a^2}{2} \left| -\frac{\sin 2t}{2} - t \right|_0^\pi = \frac{\pi a^2}{2}$$

ANSWERS FOR DO YOURSELF

- | | | | | | | | | |
|------------|------------|--------------------------|-------------|----------------------------|--------------|--|-------------|-------------------------|
| 1 : | (i) | $\frac{25}{3}$ sq. units | (ii) | $\frac{\pi}{4}$ sq. units. | (iii) | 8 sq. units. | (iv) | $\frac{3}{8}$ sq. units |
| 2 : | (i) | $\frac{1}{6}$ sq. units | (ii) | 4 sq. units | (iii) | $\frac{4-\sqrt{2}}{\log 2} - \frac{5}{2} \log 2 + \frac{3}{2}$ sq. units | | |
| 3 : | (i) | 2π sq. units | | | | | | |
| 4 : | (i) | $\frac{\pi}{4}$ | | | | | | |
| 5 : | (i) | 60 sq. units. | | | | | | |

EXERCISE (O-1)

- The area bounded in the first quadrant by the normal at $(1, 2)$ on the curve $y^2 = 4x$, x -axis & the curve is given by :
 (A) $\frac{10}{3}$ (B) $\frac{7}{3}$ (C) $\frac{4}{3}$ (D) $\frac{9}{2}$
- Suppose $y = f(x)$ and $y = g(x)$ are two functions whose graphs intersect at the three points $(0, 4)$, $(2, 2)$ and $(4, 0)$ with $f(x) > g(x)$ for $0 < x < 2$ and $f(x) < g(x)$ for $2 < x < 4$. If $\int_0^4 [f(x) - g(x)] dx = 10$ and $\int_2^4 [g(x) - f(x)] dx = 5$, the area between two curves for $0 < x < 2$, is
 (A) 5 (B) 10 (C) 15 (D) 20
- Let 'a' be a positive constant number. Consider two curves $C_1: y = e^x$, $C_2: y = e^{a-x}$. Let S be the area of the part surrounding by C_1 , C_2 and the y -axis, then $\lim_{a \rightarrow 0} \frac{S}{a^2}$ equals
 (A) 4 (B) $1/2$ (C) 0 (D) $1/4$
- The area of the region(s) enclosed by the curves $y = x^2$ and $y = \sqrt{|x|}$ is
 (A) $1/3$ (B) $2/3$ (C) $1/6$ (D) 1
- Area enclosed by the graph of the function $y = \ln^2 x - 1$ lying in the 4th quadrant is
 (A) $\frac{2}{e}$ (B) $\frac{4}{e}$ (C) $2\left(e + \frac{1}{e}\right)$ (D) $4\left(e - \frac{1}{e}\right)$
- The area bounded by the curve $y = f(x)$ (where $f(x) \geq 0$), the co-ordinate axes & the line $x = x_1$ is given by $x_1 \cdot e^{x_1}$. Therefore $f(x)$ equals :
 (A) e^x (B) $x e^x$ (C) $x e^x - e^x$ (D) $x e^x + e^x$
- The slope of the tangent to a curve $y = f(x)$ at $(x, f(x))$ is $2x + 1$. If the curve passes through the point $(1, 2)$ then the area of the region bounded by the curve, the x -axis and the line $x = 1$ is
 (A) $\frac{5}{6}$ (B) $\frac{6}{5}$ (C) $\frac{1}{6}$ (D) 1
- The area bounded by the curves $y = x(x - 3)^2$ and $y = x$ is (in sq. units) :
 (A) 28 (B) 32 (C) 4 (D) 8
- Area of the region enclosed between the curves $x = y^2 - 1$ and $x = |y| \sqrt{1 - y^2}$ is
 (A) 1 (B) $4/3$ (C) $2/3$ (D) 2
- The curve $y = ax^2 + bx + c$ passes through the point $(1, 2)$ and its tangent at origin is the line $y = x$. The area bounded by the curve, the ordinate of the curve at minima and the tangent line is
 (A) $\frac{1}{24}$ (B) $\frac{1}{12}$ (C) $\frac{1}{8}$ (D) $\frac{1}{6}$

7. Which of the following statement(s) is/are True for the function

$$f(x) = (x-1)^2(x-2) + 1 \text{ defined on } [0, 2]?$$

(A) Range of f is $\left[\frac{23}{27}, 1\right]$.

(B) The coordinates of the turning point of the graph of $y = f(x)$ occur at $(1, 1)$ and $\left(\frac{5}{3}, \frac{23}{27}\right)$.

(C) The value of p for which the equation $f(x) = p$ has 3 distinct solutions lies in interval $\left(\frac{23}{27}, 1\right)$.

(D) The area enclosed by $y = f(x)$, the lines $x = 0$ and $y = 1$ as x varies from 0 to 1 is $\frac{7}{12}$.

8. If any curve passing through $(-2, 0)$ whose length of sub normal at any point is 4 unit, then -

(A) there are two possible such curves those touches each other

(B) area bounded by any possible curve and y -axis is $\frac{32}{3}$

(C) there will be a unique such curve

(D) line $x + 2 = 0$ is tangent to the curve

9. Let $f(x) = 2 - |x - 1|$ & $g(x) = (x - 1)^2$, then -

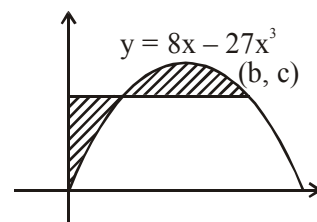
(A) area bounded by $f(x)$ & $g(x)$ is $\frac{7}{6}$ (B) area bounded by $f(x)$ & $g(x)$ is $\frac{7}{3}$

(C) area bounded by $f(x)$, $g(x)$ & x -axis is $\frac{5}{3}$ (D) area bounded by $f(x)$, $g(x)$ & x -axis is $\frac{5}{6}$

10. The figure shows a horizontal line $y = c$ passing through (b, c) intersecting the curve $y = 8x - 27x^3$. If the shaded areas are equal, then

(A) $b = \frac{1}{9}$ (B) $b = \frac{4}{9}$

(C) $c = \frac{32}{27}$ (D) $c = \frac{23}{27}$



11. If A_1 denotes area of the region bounded by the curves $C_1 : y = (x - 1)e^x$, tangent to C_1 at $(1, 0)$ & y -axis and A_2 denotes the area of the region bounded by C_1 and co-ordinate axes in fourth quadrant, then -

(A) $A_1 > A_2$ (B) $A_1 < A_2$ (C) $2A_1 + A_2 = 2$ (D) $A_1 + 2A_2 = 4$

12. Area bounded by $y = \sec^{-1} x$, $y = \cot^{-1} x$ and line $x = 1$ is given by-

(A) $\int_1^{\sqrt{\frac{1+\sqrt{5}}{2}}} (\cot^{-1} x - \sec^{-1} x) dx$

(B) $\int_0^{\alpha} \sec x dx + \int_{\alpha}^{\pi/4} \cot x dx - \frac{\pi}{4}$, where $\sin \alpha = \cos^2 \alpha$

(C) $\int_0^{\alpha} \sec x dx + \int_{\alpha}^{\pi/4} \cot x dx - \frac{\pi}{4} + 1$, where $\sin \alpha = \cos^2 \alpha$

(D) $\int_1^{\frac{1+\sqrt{5}}{2}} (\cot^{-1} x - \sec^{-1} x) dx$

13. Area bounded by the curve $y = \cot x$, $x = \frac{\pi}{4}$ and $y = 0$ is-

(A) $\int_0^{\pi/4} \tan\left(\frac{\pi}{4} - x\right) dx$ (B) $\frac{\pi}{4} - \int_0^1 \tan^{-1} x dx$ (C) $1 - \int_0^1 \tan^{-1} x dx$ (D) $\int_0^{\pi/4} \tan^{-1} x dx$

14. If $y = f(x)$ is the solution of equation $y dx + dy = -e^x y^2 dy$, $f(0) = 1$ and area bounded by the curve $y = f(x)$, $y = e^x$ and $x = 1$ is A, then -

(A) curve $y = f(x)$ is passing through $(-2, e)$ (B) curve $y = f(x)$ is passing through $\left(1, \frac{1}{e}\right)$

(C) $A = e - \frac{2}{\sqrt{e}} + 3$ (D) $A = e + \frac{2}{\sqrt{e}} - 3$

15. The function $f(x) = x^{1/3}(x-1)$

(A) has 2 inflection points.

(B) is strictly increasing for $x > 1/4$ and strictly decreasing for $x < 1/4$.

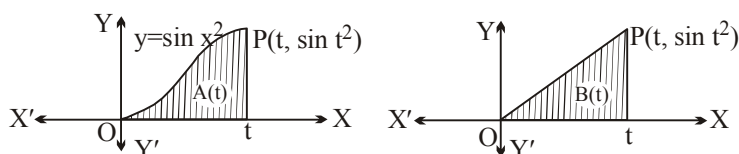
(C) is concave down in $(-1/2, 0)$.

(D) Area enclosed by the curve lying in the fourth quadrant is $9/28$.

EXERCISE (S-1)

- Find the area bounded on the right by the line $x + y = 2$, on the left by the parabola $y = x^2$ and below by the x -axis.
- Find the area of the region $\{(x, y) : 0 \leq y \leq x^2 + 1, 0 \leq y \leq x + 1, 0 \leq x \leq 2\}$.
- Find the area of the region bounded by curves $f(x) = (x-4)^2$, $g(x) = 16 - x^2$ and the x -axis.
- A figure is bounded by the curves $y = \left| \sqrt{2} \sin \frac{\pi x}{4} \right|$, $y = 0$, $x = 2$ & $x = 4$. At what angles to the positive x -axis straight lines must be drawn through $(4, 0)$ so that these lines partition the figure into three parts of the same area.
- Find the area bounded by the curves $y = \sqrt{1-x^2}$ and $y = x^3 - x$. Also find the ratio in which the y -axis divided this area.
- If the area enclosed by the parabolas $y = a - x^2$ and $y = x^2$ is $18\sqrt{2}$ sq. units. Find the value of 'a'.
- The line $3x + 2y = 13$ divides the area enclosed by the curve, $9x^2 + 4y^2 - 18x - 16y - 11 = 0$ into two parts. Find the ratio of the larger area to the smaller area.

8. Consider two curves $C_1 : y = \frac{1}{x}$ and $C_2 : y = \ln x$ on the xy plane. Let D_1 denotes the region surrounded by C_1 , C_2 and the line $x = 1$ and D_2 denotes the region surrounded by C_1 , C_2 and the line $x = a$. If $D_1 = D_2$. Find the value of 'a'.
9. Find the area enclosed between the curves : $y = \log_e(x + e)$, $x = \log_e(1/y)$ & the x -axis.
10. For what value of 'a' is the area bounded by the curve $y = a^2x^2 + ax + 1$ and the straight line $y = 0$, $x = 0$ & $x = 1$ the least ?
11. Find the positive value of 'a' for which the parabola $y = x^2 + 1$ bisects the area of the rectangle with vertices $(0, 0)$, $(a, 0)$, $(0, a^2 + 1)$ and $(a, a^2 + 1)$.
12. Compute the area of the curvilinear triangle bounded by the y -axis & the curve, $y = \tan x$ & $y = (2/3)\cos x$.
13. Find the value of 'c' for which the area of the figure bounded by the curve, $y = 8x^2 - x^5$, the straight lines $x = 1$ & $x = c$ & the abscissa axis is equal to $16/3$.
14. Find the area bounded by the curve $y = x e^{-x^2}$, the x -axis, and the line $x = c$ where $y(c)$ is maximum.
15. A polynomial function $f(x)$ satisfies the condition $f(x + 1) = f(x) + 2x + 1$. Find $f(x)$ if $f(0) = 1$. Find also the equations of the pair of tangents from the origin on the curve $y = f(x)$ and compute the area enclosed by the curve and the pair of tangents.
16. The figure shows two regions in the first quadrant.



$A(t)$ is the area under the curve $y = \sin x^2$ from 0 to t and $B(t)$ is the area of the triangle with vertices O , P and $M(t, 0)$. Find $\lim_{t \rightarrow 0} \frac{A(t)}{B(t)}$.

EXERCISE (S-2)

1. Compute the area of the region bounded by the curves $y = e^x$, $\ln x$ & $y = \ln x/(e^x)$ where $\ln e = 1$.
2. Find the values of m ($m > 0$) for which the area bounded by the line $y = mx + 2$ and $x = 2y - y^2$ is, (i) $9/2$ square units & (ii) minimum. Also find the minimum area.
3. Let A_n be the area bounded by the curve $y = (\tan x)^n$ & the lines $x = 0$, $y = 0$ & $x = \pi/4$. Prove that for $n > 2$, $A_n + A_{n-2} = 1/(n-1)$ & deduce that $1/(2n+2) < A_n < 1/(2n-2)$.
4. Find the area bounded by the curve $y = x e^{-x}$; $xy = 0$ and $x = c$ where c is the x -coordinate of the curve's inflection point.
5. Consider the curve $y = x^n$ where $n > 1$ in the 1st quadrant. If the area bounded by the curve, the x -axis and the tangent line to the graph of $y = x^n$ at the point $(1, 1)$ is maximum then find the value of n .

EXERCISE (JM)

1. The area of the region enclosed by the curves $y = x$, $x = e$, $y = \frac{1}{x}$ and the positive x-axis is:-
[AIEEE-2011]
(1) $\frac{3}{2}$ square units (2) $\frac{5}{2}$ square units (3) $\frac{1}{2}$ square units (4) 1 square units
2. The area bounded by the curves $y^2 = 4x$ and $x^2 = 4y$ is :-
[AIEEE-2011]
(1) 0 (2) $\frac{32}{3}$ (3) $\frac{16}{3}$ (4) $\frac{8}{3}$
3. The area bounded between the parabolas $x^2 = \frac{y}{4}$ and $x^2 = 9y$, and the straight line $y = 2$ is :
[AIEEE-2012]
(1) $10\sqrt{2}$ (2) $20\sqrt{2}$ (3) $\frac{10\sqrt{2}}{3}$ (4) $\frac{20\sqrt{2}}{3}$
4. The area (in square units) bounded by the curves $y = \sqrt{x}$, $2y - x + 3 = 0$, x-axis and lying in the first quadrant is :
[JEE (Main)-2013]
(1) 9 (2) 36 (3) 18 (4) $\frac{27}{4}$
5. The area bounded by the curve $y = \ln(x)$ and the lines $y = 0$, $y = \ln(3)$ and $x = 0$ is equal to :
[JEE-Main (On line)-2013]
(1) $3 \ln(3) - 2$ (2) 3 (3) 2 (4) $3 \ln(3) + 2$
6. The area of the region (in sq. units), in the first quadrant, bounded by the parabola $y = 9x^2$ and the lines $x = 0$, $y = 1$ and $y = 4$, is :-
[JEE-Main (On line)-2013]
(1) $\frac{7}{9}$ (2) $\frac{14}{3}$ (3) $\frac{14}{9}$ (4) $\frac{7}{3}$
7. The area under the curve $y = |\cos x - \sin x|$, $0 \leq x \leq \frac{\pi}{2}$, and above x-axis is :
[JEE-Main (On line)-2013]
(1) $2\sqrt{2}$ (2) $2\sqrt{2} + 2$ (3) 0 (4) $2\sqrt{2} - 2$
8. Let $f : [-2, 3] \rightarrow [0, \infty)$ be a continuous function such that $f(1-x) = f(x)$ for all $x \in [-2, 3]$. If R_1 is the numerical value of the area of the region bounded by $y = f(x)$, $x = -2$, $x = 3$ and the axis of x and $R_2 = \int_{-2}^3 x f(x) dx$, then :
[JEE-Main (On line)-2013]
(1) $2R_1 = 3R_2$ (2) $R_1 = R_2$ (3) $3R_1 = 2R_2$ (4) $R_1 = 2R_2$
9. The area of the region described by $A = \{(x, y) : x^2 + y^2 \leq 1 \text{ and } y^2 \leq 1 - x\}$ is :
[JEE(Main)-2014]
(1) $\frac{\pi}{2} + \frac{4}{3}$ (2) $\frac{\pi}{2} - \frac{4}{3}$ (3) $\frac{\pi}{2} - \frac{2}{3}$ (4) $\frac{\pi}{2} + \frac{2}{3}$

10. The area (in sq. units) of the region $\{(x, y) : y^2 \geq 2x \text{ and } x^2 + y^2 \leq 4x, x \geq 0, y \geq 0\}$ is :-
[JEE(Main)-2016]
- (1) $\frac{\pi}{2} - \frac{2\sqrt{2}}{3}$ (2) $\pi - \frac{4}{3}$ (3) $\pi - \frac{8}{3}$ (4) $\pi - \frac{4\sqrt{2}}{3}$
11. The area (in sq. units) of the region $\{(x, y) : x \geq 0, x + y \leq 3, x^2 \leq 4y \text{ and } y \leq 1 + \sqrt{x}\}$ is :
[JEE(Main)-2017]
- (1) $\frac{5}{2}$ (2) $\frac{59}{12}$ (3) $\frac{3}{2}$ (4) $\frac{7}{3}$
12. Let $g(x) = \cos x^2$, $f(x) = \sqrt{x}$ and α, β ($\alpha < \beta$) be the roots of the quadratic equation $18x^2 - 9\pi x + \pi^2 = 0$. Then the area (in sq. units) bounded by the curve $y = (g \circ f)(x)$ and the lines $x = \alpha$, $x = \beta$ and $y = 0$ is-
[JEE(Main)-2018]
- (1) $\frac{1}{2}(\sqrt{3} + 1)$ (2) $\frac{1}{2}(\sqrt{3} - \sqrt{2})$ (3) $\frac{1}{2}(\sqrt{2} - 1)$ (4) $\frac{1}{2}(\sqrt{3} - 1)$
13. The area (in sq. units) bounded by the parabola $y = x^2 - 1$, the tangent at the point $(2, 3)$ to it and the y -axis is :
[JEE(Main)-2019]
- (1) $\frac{14}{3}$ (2) $\frac{56}{3}$ (3) $\frac{8}{3}$ (4) $\frac{32}{3}$
14. The area of the region $A = \{(x, y) : 0 \leq y \leq x|x| + 1 \text{ and } -1 \leq x \leq 1\}$ in sq. units, is : [JEE(Main)-2019]
- (1) $\frac{2}{3}$ (2) $\frac{1}{3}$ (3) 2 (4) $\frac{4}{3}$
15. The area (in sq. units) of the region $A = \{(x, y) \in \mathbb{R} \times \mathbb{R} | 0 \leq x \leq 3, 0 \leq y \leq 4, y \leq x^2 + 3x\}$ is :
[JEE(Main)-2019]
- (1) $\frac{53}{6}$ (2) $\frac{59}{6}$ (3) 8 (4) $\frac{26}{3}$
16. The area (in sq. units) of the region bounded by the curves $y = 2^x$ and $y = |x + 1|$, in the first quadrant is :
[JEE(Main)-2019]
- (1) $\frac{3}{2} - \frac{1}{\log_e 2}$ (2) $\frac{1}{2}$ (3) $\log_e 2 + \frac{3}{2}$ (4) $\frac{3}{2}$
17. If the area (in sq. units) of the region $\{(x, y) : y^2 \leq 4x, x + y \leq 1, x \geq 0, y \geq 0\}$ is $a\sqrt{2} + b$, then $a - b$ is equal to :
[JEE(Main)-2019]
- (1) $\frac{8}{3}$ (2) $\frac{10}{3}$ (3) 6 (4) $-\frac{2}{3}$

EXERCISE (JA)

1. Area of the region bounded by the curve $y = e^x$ and lines $x = 0$ and $y = e$ is [JEE 2009, 4]

(A) $e - 1$ (B) $\int_1^e \ln(e+1-y)dy$ (C) $e - \int_0^1 e^x dx$ (D) $\int_1^e \ln y dy$

2. Comprehension (3 questions together) :

Consider the polynomial $f(x) = 1 + 2x + 3x^2 + 4x^3$. Let s be the sum of all distinct real roots of $f(x)$ and let $t = |s|$.

- (i) The real number s lies in the interval

(A) $\left(-\frac{1}{4}, 0\right)$ (B) $\left(-11, -\frac{3}{4}\right)$ (C) $\left(-\frac{3}{4}, -\frac{1}{2}\right)$ (D) $\left(0, \frac{1}{4}\right)$

- (ii) The area bounded by the curve $y = f(x)$ and the lines $x = 0$, $y = 0$ and $x = t$, lies in the interval

(A) $\left(\frac{3}{4}, 3\right)$ (B) $\left(\frac{21}{64}, \frac{11}{16}\right)$ (C) $(9, 10)$ (D) $\left(0, \frac{21}{64}\right)$

- (iii) The function $f'(x)$ is

(A) increasing in $\left(-t, -\frac{1}{4}\right)$ and decreasing in $\left(-\frac{1}{4}, t\right)$

(B) decreasing in $\left(-t, -\frac{1}{4}\right)$ and increasing in $\left(-\frac{1}{4}, t\right)$

(C) increasing in $(-t, t)$

(D) decreasing in $(-t, t)$

[JEE 2010, 3+3+3]

3. (a) Let the straight line $x = b$ divide the area enclosed by $y = (1-x)^2$, $y = 0$ and $x = 0$ into two parts $R_1(0 \leq x \leq b)$ and $R_2(b \leq x \leq 1)$ such that $R_1 - R_2 = \frac{1}{4}$. Then b equals

(A) $\frac{3}{4}$ (B) $\frac{1}{2}$ (C) $\frac{1}{3}$ (D) $\frac{1}{4}$

- (b) Let $f: [-1, 2] \rightarrow [0, \infty)$ be a continuous function such that $f(x) = f(1-x)$ for all $x \in [-1, 2]$.

Let $R_1 = \int_{-1}^2 x f(x) dx$, and R_2 be the area of the region bounded by $y = f(x)$, $x = -1$, $x = 2$, and the x -axis. Then -

(A) $R_1 = 2R_2$ (B) $R_1 = 3R_2$ (C) $2R_1 = R_2$ (D) $3R_1 = R_2$

[JEE 2011, 3+3]

4. The area enclosed by the curve $y = \sin x + \cos x$ and $y = |\cos x - \sin x|$ over the interval $\left[0, \frac{\pi}{2}\right]$ is

[JEE(Advanced) 2013, 2M]

(A) $4(\sqrt{2}-1)$ (B) $2\sqrt{2}(\sqrt{2}-1)$ (C) $2(\sqrt{2}+1)$ (D) $2\sqrt{2}(\sqrt{2}+1)$

ANSWER KEY

AREA UNDER THE CURVE

EXERCISE (O-1)

1. A 2. C 3. D 4. B 5. B 6. D 7. A 8. D 9. D
10. A

EXERCISE (O-2)

1. A 2. A 3. C 4. A,C 5. A,C,D 6. A,D
7. B,C,D 8. A,B,D 9. B,C 10. B,C 11. B,C 12. A,B
13. A,B 14. A,D 15. A,B,C,D

EXERCISE (S-1)

1. $5/6$ sq. units 2. $23/6$ sq. units 3. 64 4. $\pi - \tan^{-1} \frac{2\sqrt{2}}{3\pi}; \pi - \tan^{-1} \frac{4\sqrt{2}}{3\pi}$
5. $\frac{\pi}{2}; \frac{\pi-1}{\pi+1}$ 6. $a = 9$ 7. $\frac{3\pi+2}{\pi-2}$ 8. e 9. 2 sq. units
10. $a = -3/4$ 11. $\sqrt{3}$ 12. $\frac{1}{3} + \ell n\left(\frac{\sqrt{3}}{2}\right)$ sq. units
13. $C = -1$ or $(8 - \sqrt{17})^{1/3}$ 14. $\frac{1}{2}(1 - e^{-1/2})$
15. $f(x) = x^2 + 1; y = \pm 2x; A = \frac{2}{3}$ sq. units 16. $2/3$

EXERCISE (S-2)

1. $(e^2 - 5)/4$ e sq. units 2. (i) $m = 1$, (ii) $m = \infty; A_{\min} = 4/3$ 4. $1 - 3e^{-2}$
5. $\sqrt{2} + 1$ 6. $a = 1 + e^2, b = 1 + e^{-2}$ 7. $1/2$
8. $a = 8$ or $\frac{2}{5}(6 - \sqrt{21})$ 9. $\frac{(\pi-1)}{2}$ 10. 104 11. $a = \frac{2}{3}$
12. $\alpha = \pi/3$, ratio = $2 : \sqrt{3}$
13. $a = 1/2$ gives minima, $A\left(\frac{1}{2}\right) = \frac{3\sqrt{3} - \pi}{12}$; $a = 0$ gives local maxima $A(0) = 1 - \frac{\pi}{4}$;
 $a = 1$ gives maximum value, $A(1) = \pi/4$
14. $(16/9)x^2$

EXERCISE (JM)

1. 1 2. 3 3. 4 4. 1 5. 3 6. 3 7. 4 8. 4 9. 1
10. 3 11. 1 12. 4 13. 3 14. 3 15. 2 16. 1 17. 3

EXERCISE (JA)

1. B, C, D 2. (i) C; (ii) A; (iii) B 3. (a) B; (b) C 4. B 5. 3 6. A,D
7. B,C 8. 4

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MATHEMATICAL REASONING

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JEE (Main) Syllabus :

MATHEMATICAL REASONING: Statements, logical operations and, or, implies, implied by, if and only if. Understanding of tautology, contradiction, converse and contrapositive

STATISTICS : Measures of Dispersion: Calculation of mean, median, mode of grouped and ungrouped data calculation of standard deviation, variance and mean deviation for grouped and ungrouped data.

Ex.2 Find the mean of the following freq. dist.

x_i	5	15	25	35	45	55
f_i	12	18	27	20	17	6

Sol. Let assumed mean $a = 35$, $h = 10$

$$\text{here } N = \sum f_i = 100, \quad u_i = \frac{(x_i - 35)}{10}$$

$$\therefore \sum f_i u_i = (12 \times -3) + (18 \times -2) + (27 \times -1) + (20 \times 0) + (17 \times 1) + (6 \times 2) = -70$$

$$\therefore \bar{x} = a + \left(\frac{\sum f_i u_i}{N} \right) h = 35 + \frac{(-70)}{100} \times 10 = 28$$

(v) Weighted mean : If $w_1, w_2, \dots, w_n$ are the weights assigned to the values $x_1, x_2, \dots, x_n$ respectively then their weighted mean is defined as

$$\text{Weighted mean} = \frac{w_1 x_1 + w_2 x_2 + \dots + w_n x_n}{w_1 + \dots + w_n} = \frac{\sum_{i=1}^n w_i x_i}{\sum_{i=1}^n w_i}$$

Ex.3 Find the weighted mean of first n natural numbers when their weights are equal to their squares respectively

$$\text{Sol. Weighted Mean} = \frac{1 \cdot 1^2 + 2 \cdot 2^2 + \dots + n \cdot n^2}{1^2 + 2^2 + \dots + n^2} = \frac{1^3 + 2^3 + \dots + n^3}{1^2 + 2^2 + \dots + n^2} = \frac{[n(n+1)/2]^2}{[n(n+1)(2n+1)/6]} = \frac{3n(n+1)}{2(2n+1)}$$

(vi) Combined mean : If $\bar{x}_1$ and $\bar{x}_2$ be the means of two groups having n_1 and n_2 terms respectively then the mean (combined mean) of their composite group is given by

$$\text{combined mean} = \frac{n_1 \bar{x}_1 + n_2 \bar{x}_2}{n_1 + n_2}$$

$$\text{If there are more than two groups then, combined mean} = \frac{n_1 \bar{x}_1 + n_2 \bar{x}_2 + n_3 \bar{x}_3 + \dots}{n_1 + n_2 + n_3 + \dots}$$

Ex.4 The mean income of a group of persons is Rs. 400 and another group of persons is Rs. 480. If the mean income of all the persons of these two groups is Rs. 430 then find the ratio of the number of persons in the groups.

Sol. Here $\bar{x}_1 = 400$, $\bar{x}_2 = 480$, $\bar{x} = 430$

$$\therefore \bar{x} = \frac{n_1 \bar{x}_1 + n_2 \bar{x}_2}{n_1 + n_2} \Rightarrow 430 = \frac{400n_1 + 480n_2}{n_1 + n_2}$$

$$\Rightarrow \frac{n_1}{n_2} = \frac{5}{3}$$

(vii) Properties of Arithmetic mean :

- Sum of deviations of variate from their A.M. is always zero i.e. $\sum (x_i - \bar{x}) = 0$, $\sum f_i (x_i - \bar{x}) = 0$
- Sum of square of deviations of variate from their A.M. is minimum i.e. $\sum (x_i - \bar{x})^2$ is minimum
- If $\bar{x}$ is the mean of variate x_i then
 - A.M. of $(x_i + \lambda) = \bar{x} + \lambda$
 - A.M. of $(\lambda x_i) = \lambda \bar{x}$
 - A.M. of $(ax_i + b) = a\bar{x} + b$ (where λ, a, b are constant)
- A.M. is independent of change of assumed mean i.e. it is not effected by any change in assumed mean.

2. MEDIAN :

The median of a series is the value of middle term of the series when the values are written in ascending order. Therefore median, divided an arranged series into two equal parts.

Formulae of median :

(i) **For ungrouped distribution :** Let n be the number of variate in a series then

$$\text{Median} = \begin{cases} \left(\frac{n+1}{2}\right)^{\text{th}} \text{ term, (when } n \text{ is odd)} \\ \text{Mean of } \left(\frac{n}{2}\right)^{\text{th}} \text{ and } \left(\frac{n}{2}+1\right)^{\text{th}} \text{ terms, (when } n \text{ is even)} \end{cases}$$

(ii) **For ungrouped freq. dist. :** First we prepare the cumulative frequency (c.f.) column and Find value of N then

$$\text{Median} = \begin{cases} \left(\frac{N+1}{2}\right)^{\text{th}} \text{ term, (when } N \text{ is odd)} \\ \text{Mean of } \left(\frac{N}{2}\right)^{\text{th}} \text{ and } \left(\frac{N}{2}+1\right)^{\text{th}} \text{ terms, (when } N \text{ is even)} \end{cases}$$

(iii) **For grouped freq. dist :** Prepare c.f. column and find value of $\frac{N}{2}$ then find the class which contain value of c.f. is equal or just greater to $N/2$, this is median class

$$\therefore \text{Median} = \ell + \frac{\left(\frac{N}{2} - F\right)}{f} \times h$$

where

ℓ — lower limit of median class

f — freq. of median class

F — c.f. of the class preceeding median class

h — Class interval of median class

Ex.5 Find the median of following freq. dist.

class	0-10	10-20	20-30	30-40	40-50
f	8	30	40	12	10

class	f_i	c.f.
0-10	8	8
10-20	30	38
20-30	40	78
30-40	12	90
40-50	10	100

Sol.

Here $\frac{N}{2} = \frac{100}{2} = 50$ which lies in the value 78 of c.f. hence corresponding class of this c.f. is 20-30 is the median class, so $\ell = 20$, $f = 40$, $F = 38$, $h = 10$

$$\therefore \text{Median} = \ell + \frac{\left(\frac{N}{2} - F\right)}{f} \times h = 20 + \frac{(50-38)}{40} \times 10 = 23$$

3. **MODE :**

In a frequency distribution the mode is the value of that variate which have the maximum frequency

Method for determining mode :

(i) **For ungrouped dist. :** The value of that variate which is repeated maximum number of times

(ii) **For ungrouped freq. dist. :** The value of that variate which have maximum frequency.

(iii) **For grouped freq. dist. :** First we find the class which have maximum frequency, this is model class

$$\therefore \text{Mode} = \ell + \frac{f_0 - f_1}{2f_0 - f_1 - f_2} \times h$$

where

ℓ — lower limit of model class

f_0 — freq. of the model class

f_1 — freq. of the class preceeding model class

f_2 — freq. of the class succeeding model class

h — class interval of model class

Ex. 6 Find the mode of the following frequency dist

class	0–10	10–20	20–30	30–40	40–50	50–60	60–70	70–80
f_i	2	18	30	45	35	20	6	3

Sol. Here the class 30–40 has maximum freq. so this is the model class

$$\ell = 30, f_0 = 45, f_1 = 30, f_2 = 35, h = 10$$

$$\therefore \text{Mode} = \ell + \frac{f_0 - f_1}{2f_0 - f_1 - f_2} \times h = 30 + \frac{45 - 30}{2 \times 45 - 30 - 35} \times 10 = 36$$

4. **RELATION BETWEEN MEAN, MEDIAN AND MODE :**

In a moderately asymmetric distribution following relation between mean, median and mode of a distribution. It is known as imprical formula.

$$\text{Mode} = 3 \text{ Median} - 2 \text{ Mean}$$

Note (i) Median always lies between mean and mode

(ii) For a symmetric distribution the mean, median and mode are coincide.

5. **MEASURES OF DISPERSION :**

The dispersion of a statistical distribution is the measure of deviation of its values about the their average (central) value.

It gives an idea of scatteredness of different values from the average value.

Generally the following measures of dispersion are commonly used.

(i) Range (ii) Mean deviation (iii) Variance and standard deviation

(i) **Range :** The difference between the greatest and least values of variate of a distribution, are called the range of that distribution.

If the distribution is grouped distribution, then its range is the difference between upper limit of the maximum class and lower limit of the minimum class.

$$\text{Also, coefficient of range} = \frac{\text{difference of extreme values}}{\text{sum of extreme values}}$$

Ex.7 Find the range of following numbers 10, 8, 12, 11, 14, 9, 6

Sol. Here greatest value and least value of the distribution are 14 and 6 resp. therefore

$$\text{Range} = 14 - 6 = 8$$

(ii) Mean deviation (M.D.) : The mean deviation of a distribution is, the mean of absolute value of deviations of variate from their statistical average (Mean, Median, Mode).

If A is any statistical average of a distribution then mean deviation about A is defined as

$$\text{Mean deviation} = \frac{\sum_{i=1}^n |x_i - A|}{n} \quad (\text{for ungrouped dist.})$$

$$\text{Mean deviation} = \frac{\sum_{i=1}^n f_i |x_i - A|}{N} \quad (\text{for freq. dist.})$$

Note :- is minimum when it taken about the median

$$\text{Coefficient of Mean deviation} = \frac{\text{Mean deviation}}{A}$$

(where A is the central tendency about which Mean deviation is taken)

Ex.8 Find the mean deviation of number 3, 4, 5, 6, 7

Sol. Here $n = 5$, $\bar{x} = 5$

$$\begin{aligned} \therefore \text{Mean deviation} &= \frac{\sum |x_i - \bar{x}|}{n} \\ &= \frac{1}{5} [|3 - 5| + |4 - 5| + |5 - 5| + |6 - 5| + |7 - 5|] \\ &= \frac{1}{5} [2 + 1 + 0 + 1 + 2] = \frac{6}{5} = 1.2 \end{aligned}$$

Ex.9 Find the mean deviation about mean from the following data

x_i	3	9	17	23	27
f_i	8	10	12	9	5

Sol.

x_i	f_i	$f_i x_i$	$ x_i - \bar{x} $	$f_i x_i - \bar{x} $
3	8	24	12	96
9	10	90	6	60
17	12	204	2	24
23	9	207	8	72
27	5	135	12	60
	$N = 44$	$\sum f_i x_i = 660$		$\sum f_i x_i - \bar{x} = 312$

$$\text{Mean } (\bar{x}) = \frac{\sum f_i x_i}{N} = \frac{660}{44} = 15$$

$$\text{Mean deviation} = \frac{\sum f_i |x_i - \bar{x}|}{N} = \frac{312}{44} = 7.09$$

(iii) **Variance and standard deviation :** The variance of a distribution is, the mean of squares of deviation of variate from their mean. It is denoted by σ^2 or $\text{var}(x)$.

The positive square root of the variance are called the standard deviation. It is denoted by σ or S.D.

Hence standard deviation = $+\sqrt{\text{variance}}$

Formulae for variance :

(i) **for ungrouped dist. :**

$$\sigma_x^2 = \frac{\sum (x_i - \bar{x})^2}{n}$$

$$\sigma_x^2 = \frac{\sum x_i^2}{n} - \bar{x}^2 = \frac{\sum x_i^2}{n} - \left(\frac{\sum x_i}{n} \right)^2$$

$$\sigma_d^2 = \frac{\sum d_i^2}{n} - \left(\frac{\sum d_i}{n} \right)^2, \text{ where } d_i = x_i - a$$

(ii) **For freq. dist. :**

$$\sigma_x^2 = \frac{\sum f_i (x_i - \bar{x})^2}{N}$$

$$\sigma_x^2 = \frac{\sum f_i x_i^2}{N} - (\bar{x})^2 = \frac{\sum f_i x_i^2}{N} - \left(\frac{\sum f_i x_i}{N} \right)^2$$

$$\sigma_d^2 = \frac{\sum f_i d_i^2}{N} - \left(\frac{\sum f_i d_i}{N} \right)^2$$

$$\sigma_u^2 = h^2 \left[\frac{\sum f_i u_i^2}{N} - \left(\frac{\sum f_i u_i}{N} \right)^2 \right] \text{ where } u_i = \frac{d_i}{h}$$

(iii) Coefficient of S.D. = $\frac{\sigma}{\bar{x}}$

Coefficient of variation = $\frac{\sigma}{\bar{x}} \times 100$ (in percentage)

Note :- $\sigma^2 = \sigma_x^2 = \sigma_d^2 = h^2 \sigma_u^2$

Ex.10 Find the variance of first n natural numbers

Sol. $\sigma^2 = \frac{\sum x_i^2}{n} - \left(\frac{\sum x_i}{n} \right)^2 = \frac{\sum n^2}{n} - \left(\frac{\sum n}{n} \right)^2 = \frac{n(n+1)(2n+1)}{6n} - \left\{ \frac{n(n+1)}{2n} \right\}^2 = \frac{n^2-1}{12}$

Ex.11 If $\sum_{i=1}^{18} (x_i - 8) = 9$ and $\sum_{i=1}^{18} (x_i - 8)^2 = 45$, then find the standard deviation of $x_1, x_2, \dots, x_{18}$

Sol. Let $(x_i - 8) = d_i$

$$\therefore \sigma_x = \sigma_d = \sqrt{\frac{\sum d_i^2}{n} - \left(\frac{\sum d_i}{n} \right)^2} = \sqrt{\frac{45}{18} - \left(\frac{9}{18} \right)^2} = \sqrt{\frac{5}{2} - \frac{1}{4}} = \frac{3}{2}$$

Ex.12 Find the coefficient of variation of first n natural numbers

Sol. For first n natural numbers.

$$\text{Mean } (\bar{x}) = \frac{n+1}{2}, \text{ S.D. } (\sigma) = \sqrt{\frac{n^2-1}{12}}$$

$$\therefore \text{ coefficient of variance} = \frac{\sigma}{\bar{x}} \times 100 = \sqrt{\frac{n^2-1}{12}} \times \frac{1}{\left(\frac{n+1}{2}\right)} \times 100 = \sqrt{\frac{(n-1)}{3(n+1)}} \times 100$$

6. MEAN SQUARE DEVIATION :

The mean square deviation of a distribution is the mean of the square of deviations of variate from assumed mean. It is denoted by S^2

Hence
$$S^2 = \frac{\sum (x_i - a)^2}{n} = \frac{\sum d_i^2}{n} \quad (\text{for ungrouped dist.})$$

$$S^2 = \frac{\sum f_i (x_i - a)^2}{N} = \frac{\sum f_i d_i^2}{N} \quad (\text{for freq. dist.}), \quad \text{where } d_i = (x_i - a)$$

7. RELATION BETWEEN VARIANCE AND MEAN SQUARE DEVIATION :

$$\therefore \sigma^2 = \frac{\sum f_i d_i^2}{N} - \left(\frac{\sum f_i d_i}{N} \right)^2$$

$$\Rightarrow \sigma^2 = s^2 - d^2, \quad \text{where } d = \bar{x} - a = \frac{\sum f_i d_i}{N}$$

$$\Rightarrow s^2 = \sigma^2 + d^2 \Rightarrow s^2 \geq \sigma^2$$

Hence the variance is the minimum value of mean square deviation of a distribution

Ex.13 Determine the variance of the following frequency dist.

class	0-2	2-4	4-6	6-8	8-10	10-12
f_i	2	7	12	19	9	1

Sol. Let $a = 7, h = 2$

class	x_i	f_i	$u_i = \frac{x_i - a}{h}$	$f_i u_i$	$f_i u_i^2$
0-2	1	2	-3	-6	18
2-4	3	7	-2	-14	28
4-6	5	12	-1	-12	12
6-8	7	19	0	0	0
8-10	9	9	1	9	9
10-12	11	1	2	2	4
		$N = 50$		$\sum f_i u_i = -21$	$\sum f_i u_i^2 = 71$

$$\therefore \sigma^2 = h^2 \left[\frac{\sum f_i u_i^2}{N} - \left(\frac{\sum f_i u_i}{N} \right)^2 \right] = 4 \left[\frac{71}{50} - \left(\frac{-21}{50} \right)^2 \right] = 4[1.42 - 0.1764] = 4.97$$

8. MATHEMATICAL PROPERTIES OF VARIANCE :

- $\text{Var.}(x_i + \lambda) = \text{Var.}(x_i)$

$$\text{Var.}(\lambda x_i) = \lambda^2 \cdot \text{Var.}(x_i)$$

$$\text{Var.}(ax_i + b) = a^2 \cdot \text{Var.}(x_i)$$

where λ, a, b , are constant

- If means of two series containing n_1, n_2 terms are $\bar{x}_1, \bar{x}_2$ and their variance's are σ_1^2, σ_2^2 respectively and their combined mean is $\bar{x}$ then the variance σ^2 of their combined series is given by following formula

$$\sigma^2 = \frac{n_1(\sigma_1^2 + d_1^2) + n_2(\sigma_2^2 + d_2^2)}{(n_1 + n_2)} \quad \text{where } d_1 = \bar{x}_1 - \bar{x}, d_2 = \bar{x}_2 - \bar{x}$$

i.e.
$$\sigma^2 = \frac{n_1\sigma_1^2 + n_2\sigma_2^2}{n_1 + n_2} + \frac{n_1n_2}{(n_1 + n_2)^2} (\bar{x}_1 - \bar{x}_2)^2$$

SOLVED EXAMPLES

Ex.1 If in an examination different weights are assigned to different subjects Physics (2), Chemistry (1), English (1), Mathematics (2) A student scores 60 in Physics, 70 in Chemistry, 70 in English and 80 in Mathematics, then weighted mean is-

- (1) 60 (2) 70 (3) 80 (4) 85

Sol.(2) Weighted mean =
$$\frac{\sum_{i=1}^n w_i X_i}{\sum_{i=1}^n w_i} = \frac{2 \times 60 + 1 \times 70 + 1 \times 70 + 2 \times 80}{6} = 70$$

Ex.2 The mean of two groups of sizes 200 and 300 are 25 and 10 respectively. Their standard deviation are 3 and 4 respectively. The variance of combined sample of size 500 is-

- (1) 64 (2) 65.2 (3) 67.2 (4) 64.2

Sol.(3) Combined mean $\bar{x} = \frac{n_1 \bar{x}_1 + n_2 \bar{x}_2}{n_1 + n_2} = \frac{200 \times 25 + 300 \times 10}{500} = 16$

Here $d_1 = \bar{x}_1 - \bar{x} = 25 - 16 = 9$ and $d_2 = \bar{x}_2 - \bar{x} = 10 - 16 = -6$

We know that
$$\sigma^2 = \frac{n_1(\sigma_1^2 + d_1^2) + n_2(\sigma_2^2 + d_2^2)}{n_1 + n_2} = \frac{200(9 + 81) + 300(16 + 36)}{500} = \frac{33600}{500} = 67.2$$

Ex.3 If the mean of the series $x_1, x_2, \dots, x_n$ is $\bar{x}$, then the mean of the series $x_i + 2i, i = 1, 2, \dots, n$ will be-

- (1) $\bar{x} + n$ (2) $\bar{x} + n + 1$ (3) $\bar{x} + 2$ (4) $\bar{x} + 2n$

Sol.(2) As given $\bar{x} = \frac{x_1 + x_2 + \dots + x_n}{n}$ (1)

If the mean of the series $x_i + 2i, i = 1, 2, \dots, n$ be $\bar{X}$, then

$$\begin{aligned} \bar{X} &= \frac{(x_1 + 2) + (x_2 + 2.2) + (x_3 + 2.3) + \dots + (x_n + 2.n)}{n} \\ &= \frac{x_1 + x_2 + \dots + x_n}{n} + \frac{2(1 + 2 + 3 + \dots + n)}{n} \\ &= \bar{x} + \frac{2n(n+1)}{2n} \quad \text{from (1)} \\ &= \bar{x} + n + 1 \end{aligned}$$

Ex.4 The variance of first 20-natural numbers is-

- (1) $\frac{133}{4}$ (2) $\frac{379}{12}$ (3) $\frac{133}{2}$ (4) $\frac{399}{4}$

$$\begin{aligned}
 \text{Sol.(1)} \quad \therefore \sigma^2 &= \frac{\sum x_i^2}{n} - \left(\frac{\sum x_i}{n} \right)^2 \\
 &= \frac{1}{20} [1^2 + 2^2 + \dots + 20^2] - \left[\frac{1}{20} (1 + 2 + \dots + 20) \right]^2 \\
 &= \frac{1}{20} \frac{20 \times 21 (2 \times 20 + 1)}{6} - \left[\frac{1}{20} \frac{20 \times 21}{2} \right]^2 = \frac{7 \times 41}{2} - \frac{441}{4} = \frac{133}{4}.
 \end{aligned}$$

In fact, the variance of first n -natural numbers is $\frac{n^2 - 1}{12}$

Ex.5 The mean of the following freq. table is 50 and $\Sigma f = 120$

class	0-20	20-40	40-60	60-80	80-100
f	17	f_1	32	f_2	19

the missing frequencies are-

- (1) 28, 24 (2) 24, 36 (3) 36, 28 (4) None of these

$$\text{Sol.(1)} \quad \Sigma f = 120 = 17 + f_1 + 32 + f_2 + 19$$

$$\Rightarrow f_1 + f_2 = 52 \quad \dots (1)$$

$$\text{and } \Sigma fx = (10 \times 17) + (30 \times f_1) + (50 \times 32) + (70 \times f_2) + (90 \times 19) = 30f_1 + 70f_2 + 3480$$

$$\therefore \bar{x} = \frac{\Sigma fx}{\Sigma f} \Rightarrow 50 = \frac{30f_1 + 70f_2 + 3480}{120}$$

$$\Rightarrow 30f_1 + 70f_2 = 2520 \Rightarrow 3f_1 + 7f_2 = 252 \quad \dots (2)$$

by (1) and (2) we get $f_1 = 28, f_2 = 24$

Ex.6 A student obtained 75%, 80%, 85% marks in three subjects. If the marks of another subject are added then his average marks can not be less than-

- (1) 60% (2) 65% (3) 80% (4) 90%

$$\text{Sol.(1)} \quad \text{Total marks obtained from three subjects out of } 300 = 75 + 80 + 85 = 240$$

if the marks of another subject is added then total marks obtained out of 400 is greater than 240

if marks obtained in fourth subject is 0 then

$$\text{minimum average marks} = \frac{240}{400} \times 100 = 60\%$$

Ex.7 The mean and variance of a series containing 5 terms are 8 and 24 respectively. The mean and variance of another series containing 3 terms are also 8 and 24 respectively. The variance of their combined series will be-

- (1) 20 (2) 24 (3) 25 (4) 42

$$\text{Sol.(2)} \quad \text{Using } \sigma^2 = \frac{n_1\sigma_1^2 + n_2\sigma_2^2}{n_1 + n_2} + \frac{n_1n_2}{(n_1 + n_2)^2} (\bar{x}_1 - \bar{x}_2)^2 \Rightarrow \sigma^2 = \frac{5(24) + 3(24)}{5 + 3} + \frac{5(3)}{(5 + 3)^2} (8 - 8)^2 = 24$$

Ex.8 The mean deviation about median from the following data 340, 150, 210, 240, 300, 310, 320, is-
 (1) 52.4 (2) 52.5 (3) 52.8 (4) none of these

Sol.(3) Arranging the observations in ascending order of magnitude, we have 150, 210, 240, 300, 310, 320, 340. Clearly, the middle observation is 300. So, median = 300

Calculation of Mean deviation

x_i	$ x_i - 300 $
340	40
150	150
210	90
240	60
300	0
310	10
320	20
Total	$\sum x_i - 300 = 370$

$$\text{Mean deviation from median} = \frac{1}{7} \sum |x_i - 300| = \frac{370}{7} = 52.8$$

Ex.9 Variance of the data given below is

Size of item	3.5	4.5	5.5	6.5	7.5	8.5	9.5
Frequency	3	7	22	60	85	32	8

(1) 1.29 (2) 2.19 (3) 1.32 (4) none of these

Sol.(3) Let the assumed mean be $a = 6.5$

Calculation of variance

x_i	f_i	$d_i = x_i - 6.5$	$f_i d_i$	$f_i d_i^2$
3.5	3	-3	-9	27
4.5	7	-2	-14	28
5.5	22	-1	-22	22
6.5	60	0	0	0
7.5	85	1	85	85
8.5	32	2	64	128
9.5	8	3	24	72
$N = \sum f_i = 217$		$\sum f_i d_i = 128$		$\sum f_i d_i^2 = 362$

Here $N = 217$, $\sum f_i d_i = 128$ and $\sum f_i d_i^2 = 362$

$$\therefore \text{Var}(X) = \left(\frac{1}{N} \sum f_i d_i^2 \right) - \left(\frac{1}{N} \sum f_i d_i \right)^2 = \frac{362}{217} - \left(\frac{128}{217} \right)^2 = 1.668 - 0.347 = 1.321$$

Ex.10 If a variable takes the value 0, 1, 2, ..., n with frequencies proportional to the binomial coefficients ${}^nC_0, {}^nC_1, \dots, {}^nC_n$ then the mean of the distribution is-

- (1) $\frac{n(n+1)}{4}$ (2) $\frac{n}{2}$ (3) $\frac{n(n-1)}{2}$ (4) $\frac{n(n+1)}{2}$

Sol.(2) $N = \sum f_i = k [{}^nC_0 + {}^nC_1 + \dots + {}^nC_n] = k2^n$

$$\sum f_i x_i = k [1 \cdot {}^nC_1 + 2 \cdot {}^nC_2 + \dots + n \cdot {}^nC_n] = k \sum_{r=1}^n r \cdot {}^nC_r = kn \sum_{r=1}^n {}^{n-1}C_{r-1} = kn2^{n-1}$$

$$\text{Thus } \bar{x} = \frac{1}{2^n} (n \cdot 2^{n-1}) = \frac{n}{2}.$$

Ex.11 The mean and variance of 5 observations of an experiment are 4 and 5.2 respectively. If from these observations three are 1, 2 and 6, then the remaining will be-

- (1) 2, 9 (2) 5, 6 (3) 4, 7 (4) 3, 8

Sol.(3) As given $\bar{x} = 4$, $n = 5$ and $\sigma^2 = 5.2$. If the remaining observations are x_1, x_2 then

$$\sigma^2 = \frac{\sum (x_i - \bar{x})^2}{n} = 5.2$$

$$\Rightarrow \frac{(x_1 - 4)^2 + (x_2 - 4)^2 + (1 - 4)^2 + (2 - 4)^2 + (6 - 4)^2}{5} = 5.2$$

$$\Rightarrow (x_1 - 4)^2 + (x_2 - 4)^2 = 9 \quad \dots(1)$$

$$\text{Also } \bar{x} = 4 \Rightarrow \frac{x_1 + x_2 + 1 + 2 + 6}{5} = 4 \Rightarrow x_1 + x_2 = 11 \quad \dots(2)$$

from eq.(1), (2) $x_1, x_2 = 4, 7$

Ex.12 The mean deviation of the series $a, a + d, a + 2d, \dots, a + 2nd$ from its mean is-

- (1) $\frac{n+1}{2n+1} |d|$ (2) $\frac{n(n+1)}{2n+1} |d|$ (3) $\frac{n(n-1)}{2n+1} |d|$ (4) none of these

Sol.(2) Number of terms in the series $= 2n + 1$

$$\therefore \text{mean } \bar{x} = \frac{a + (a + d) + (a + 2d) + \dots + (a + 2nd)}{2n + 1} = \frac{1}{2n + 1} \left[\frac{2n + 1}{2} (a + a + 2nd) \right] = a + nd$$

$$\text{Also } \sum |x_i - \bar{x}| = |-nd| + |(1-n)d| + \dots + |-d| + 0 + |d| + \dots + |nd|$$

$$= 2|d| [n + (n-1) + \dots + 1] = 2|d| \frac{n(n+1)}{2} = n(n+1) |d|$$

$$\therefore \text{mean deviation from mean} = \frac{\sum |x_i - \bar{x}|}{N} = \frac{n(n+1)}{2n+1} |d|$$

Ex.13 Let $x_1, x_2, \dots, x_n$ be values taken by a variable X and $y_1, y_2, \dots, y_n$ be the values taken by a variable Y such that $y_i = ax_i + b$, $i = 1, 2, \dots, n$. Then-

- (1) $\text{Var}(Y) = a^2 \text{Var}(X)$ (2) $\text{Var}(Y) = a^2 \text{Var}(X) + b$ (3) $\text{Var}(Y) = \text{Var}(X) + b$ (4) None of these

Sol.(1) We have,

$$\text{Var}(Y) = \frac{1}{n} \sum_{i=1}^n (y_i - \bar{y})^2$$

$$[\because y_i = ax_i + b; i = 1, 2, \dots, n \Rightarrow \bar{Y} = a\bar{X} + b]$$

$$\Rightarrow \text{Var}(Y) = \frac{1}{n} \sum_{i=1}^n a^2 (x_i - \bar{X})^2$$

$$\Rightarrow \text{Var}(Y) = a^2 \left\{ \frac{1}{n} \sum_{i=1}^n (x_i - \bar{X})^2 \right\} = a^2 \text{Var}(X)$$

Ex.14 The mean square deviation of a set of n observations $x_1, x_2, \dots, x_n$ about a point c is defined as

$$\frac{1}{n} \sum_{i=1}^n (x_i - c)^2$$

The mean square deviation about -2 and 2 are 18 and 10 respectively, then standard deviation of this set of observations is-

- (1) 3 (2) 2 (3) 1 (4) None of these

Sol.(1) $\because \frac{1}{n} \sum (x_i + 2)^2 = 18$ and $\frac{1}{n} \sum (x_i - 2)^2 = 10$

$$\Rightarrow \sum (x_i + 2)^2 = 18n \text{ and } \sum (x_i - 2)^2 = 10n$$

$$\Rightarrow \sum (x_i + 2)^2 + \sum (x_i - 2)^2 = 28n \text{ and } \sum (x_i + 2)^2 - \sum (x_i - 2)^2 = 8n$$

$$\Rightarrow 2\sum x_i^2 + 8n = 28n \text{ and } 8\sum x_i = 8n$$

$$\Rightarrow \sum x_i^2 = 10n \text{ and } \sum x_i = n$$

$$\Rightarrow \frac{\sum x_i^2}{n} = 10 \text{ and } \frac{\sum x_i}{n} = 1$$

$$\therefore \sigma = \sqrt{\frac{\sum x_i^2}{n} - \left(\frac{\sum x_i}{n}\right)^2} = \sqrt{10 - (1)^2} = 3$$

CHECK YOUR GRASP

STATISTICS

EXERCISE-I

Arithmetic mean, weighted mean, Combined mean

- Mean of the first n terms of the A.P. $a, (a + d), (a + 2d), \dots$ is-
 (1) $a + \frac{nd}{2}$ (2) $a + \frac{(n-1)d}{2}$
 (3) $a + (n-1)d$ (4) $a + nd$
- The A.M. of first n even natural number is -
 (1) $n(n+1)$ (2) $\frac{n+1}{2}$ (3) $\frac{n}{2}$ (4) $n+1$
- The A.M. of ${}^nC_0, {}^nC_1, {}^nC_2, \dots, {}^nC_n$ is -
 (1) $\frac{2^n}{n}$ (2) $\frac{2^{n+1}}{n}$ (3) $\frac{2^n}{n+1}$ (4) $\frac{2^{n+1}}{n+1}$
- If the mean of numbers 27, 31, 89, 107, 156 is 82, then the mean of numbers 130, 126, 68, 50, 1 will be-
 (1) 80 (2) 82 (3) 75 (4) 157
- If the mean of n observations $x_1, x_2, \dots, x_n$ is $\bar{x}$, then the sum of deviations of observations from mean is :-
 (1) 0 (2) $n\bar{x}$
 (3) $\frac{\bar{x}}{n}$ (4) None of these
- The mean of 9 terms is 15. if one new term is added and mean become 16, then the value of new term is :-
 (1) 23 (2) 25 (3) 27 (4) 30
- If the mean of first n natural numbers is equal to $\frac{n+7}{3}$, then n is equal to-
 (1) 10 (2) 11
 (3) 12 (4) none of these
- The mean of first three terms is 14 and mean of next two terms is 18. The mean of all the five terms is-
 (1) 15.5 (2) 15.0 (3) 15.2 (4) 15.6
- If the mean of five observations $x, x+2, x+4, x+6$ and $x+8$ is 11, then the mean of last three observations is-
 (1) 11 (2) 13 (3) 15 (4) 17
- The mean of a set of numbers is $\bar{x}$. If each number is decreased by λ , the mean of the new set is-
 (1) $\bar{x}$ (2) $\bar{x} + \lambda$ (3) $\lambda - \bar{x}$ (4) $\bar{x} - \lambda$
- The mean of 50 observations is 36. If its two observations 30 and 42 are deleted, then the mean of the remaining observations is-
 (1) 48 (2) 36
 (3) 38 (4) none of these
- In a frequency dist., if d_i is deviation of variates from a number ℓ and mean $= \ell + \frac{\sum f_i d_i}{\sum f_i}$, then ℓ is :-
 (1) Lower limit
 (2) Assumed mean
 (3) Number of observation
 (4) Class interval
- The A.M. of n observation is $\bar{x}$. If the sum of $n - 4$ observations is K , then the mean of remaining observations is-
 (1) $\frac{\bar{x} - K}{4}$ (2) $\frac{n\bar{x} - K}{n - 4}$
 (3) $\frac{n\bar{x} - K}{4}$ (4) $\frac{n\bar{x} - (n-4)K}{4}$
- The mean of values $1, \frac{1}{2}, \frac{1}{3}, \dots, \frac{1}{n}$ which have frequencies 1, 2, 3, $\dots, n$ resp., is :-
 (1) $\frac{2n+1}{3}$ (2) $\frac{2}{n}$ (3) $\frac{n+1}{2}$ (4) $\frac{2}{n+1}$
- The sum of squares of deviation of variates from their A.M. is always :-
 (1) Zero
 (2) Minimum
 (3) Maximum
 (4) Nothing can be said
- If the mean of following freq. dist. is 2.6, then the value of f is :-

x_i	1	2	3	4	5
f_i	5	4	f	2	3

 (1) 1 (2) 3
 (3) 8 (4) None of these

17. The weighted mean (W.M.) is computed by the formula ?

$$(1) W.M. = \frac{\sum x_i}{\sum w_i} \quad (2) W.M. = \frac{\sum w_i}{\sum x_i}$$

$$(3) W.M. = \frac{\sum w_i x_i}{\sum x_i} \quad (4) W.M. = \frac{\sum w_i x_i}{\sum w_i}$$

18. The weighted mean of first n natural numbers when their weights are equal to corresponding natural number, is :-

$$(1) \frac{n+1}{2} \quad (2) \frac{2n+1}{3}$$

$$(3) \frac{(n+1)(2n+1)}{6} \quad (4) \text{None of these}$$

19. The average income of a group of persons is $\bar{x}$ and that of another group is $\bar{y}$. If the number of persons of both group are in the ratio 4 : 3, then average income of combined group is :-

$$(1) \frac{\bar{x} + \bar{y}}{7} \quad (2) \frac{3\bar{x} + 4\bar{y}}{7}$$

$$(3) \frac{4\bar{x} + 3\bar{y}}{7} \quad (4) \text{None of these}$$

20. In a group of students, the mean weight of boys is 65 kg. and mean weight of girls is 55 kg. If the mean weight of all students of group is 61 kg, then the ratio of the number of boys and girls in the group is :-

$$(1) 2 : 3 \quad (2) 3 : 1 \quad (3) 3 : 2 \quad (4) 4 : 3$$

Median, Mode

21. The median of an arranged series of n even observations, will be :-

$$(1) \left(\frac{n+1}{2} \right) \text{th term}$$

$$(2) \left(\frac{n}{2} \right) \text{th term}$$

$$(3) \left(\frac{n}{2} + 1 \right) \text{th term}$$

$$(4) \text{Mean of } \left(\frac{n}{2} \right) \text{th and } \left(\frac{n}{2} + 1 \right) \text{th terms}$$

22. The median of the numbers 6, 14, 12, 8, 10, 9, 11, is :-

$$(1) 8 \quad (2) 10 \quad (3) 10.5 \quad (4) 11$$

23. Median of the following freq. dist.

x_i	3	6	10	12	7	15
f_i	3	4	2	8	13	10

$$(1) 7 \quad (2) 10$$

$$(3) 8.5 \quad (4) \text{None of these}$$

24. Median is independent of change of :-

$$(1) \text{only Origin}$$

$$(2) \text{only Scale}$$

$$(3) \text{Origin and scale both}$$

$$(4) \text{Neither origin nor scale}$$

25. A series which have numbers three 4's, four 5's, five 6's, eight 7's, seven 8's and six 9's then the mode of numbers is :-

$$(1) 9 \quad (2) 8 \quad (3) 7 \quad (4) 6$$

26. Mode of the following frequency distribution

$x :$	4	5	6	7	8
$f :$	6	7	10	8	3

$$(1) 5 \quad (2) 6 \quad (3) 8 \quad (4) 10$$

27. The mode of the following freq. dist is :-

Class	1-10	11-20	21-30	31-40	41-50
f_i	5	7	8	6	4

$$(1) 24 \quad (2) 23.83$$

$$(3) 27.16 \quad (4) \text{None of these}$$

Symmetric and asymmetric distribution, Range

28. For a normal dist :-

$$(1) \text{mean} = \text{median}$$

$$(2) \text{median} = \text{mode}$$

$$(3) \text{mean} = \text{mode}$$

$$(4) \text{mean} = \text{median} = \text{mode}$$

29. The relationship between mean, median and mode for a moderately skewed distribution is-

$$(1) \text{mode} = \text{median} - 2 \text{ mean}$$

$$(2) \text{mode} = 2 \text{ median} - \text{mean}$$

$$(3) \text{mode} = 2 \text{ median} - 3 \text{ mean}$$

$$(4) \text{mode} = 3 \text{ median} - 2 \text{ mean}$$

30. The range of observations 2, 3, 5, 9, 8, 7, 6, 5, 7, 4, 3 is :-
 (1) 6 (2) 7 (3) 5.5 (4) 11

Mean Deviation

31. The mean deviation of a frequency dist. is equal to :-

(1) $\frac{\sum d_i}{\sum f_i}$ (2) $\frac{\sum |d_i|}{\sum f_i}$
 (3) $\frac{\sum f_i d_i}{\sum f_i}$ (4) $\frac{\sum f_i |d_i|}{\sum f_i}$

32. Mean deviation from the mean for the observation -1, 0, 4 is-

(1) $\sqrt{\frac{14}{3}}$ (2) $\frac{2}{3}$
 (3) 2 (4) none of these

33. Mean deviation of the observations 70, 42, 63, 34, 44, 54, 55, 46, 38, 48 from median is :-

(1) 7.8 (2) 8.6
 (3) 7.6 (4) 8.8

34. Mean deviation of 5 observations from their mean 3 is 1.2, then coefficient of mean deviation is :-

(1) 0.24 (2) 0.4
 (3) 2.5 (4) None of these

35. The mean deviation from median is

- (1) greater than the mean deviation from any other central value
 (2) less than the mean deviation from any other central value
 (3) equal to the mean deviation from any other central value
 (4) maximum if all values are positive

Variance and Standard Deviation

36. The variate x and u are related by $u = \frac{x-a}{h}$ then correct relation between σ_x and σ_u is :-

(1) $\sigma_x = h\sigma_u$ (2) $\sigma_x = h + \sigma_u$
 (3) $\sigma_u = h\sigma_x$ (4) $\sigma_u = h + \sigma_x$

37. The S.D. of the first n natural numbers is-

(1) $\sqrt{\frac{n^2-1}{2}}$ (2) $\sqrt{\frac{n^2-1}{3}}$
 (3) $\sqrt{\frac{n^2-1}{4}}$ (4) $\sqrt{\frac{n^2-1}{12}}$

38. The variance of observations 112, 116, 120, 125, 132 is :-

(1) 58.8 (2) 48.8
 (3) 61.8 (4) None of these

39. If $\sum_{i=1}^{10} (x_i - 15) = 12$ and $\sum_{i=1}^{10} (x_i - 15)^2 = 18$ then the S.D. of observations $x_1, x_2, \dots, x_{10}$ is :-

(1) $\frac{2}{5}$ (2) $\frac{3}{5}$
 (3) $\frac{4}{5}$ (4) None of these

40. The S.D. of 7 scored 1, 2, 3, 4, 5, 6, 7 is-

(1) 4 (2) 2
 (3) $\sqrt{7}$ (4) none of these

41. The variance of series a, a + d, a + 2d,, a + 2nd is :-

(1) $\frac{n(n+1)}{2}d^2$ (2) $\frac{n(n+1)}{3}d^2$
 (3) $\frac{n(n+1)}{6}d^2$ (4) $\frac{n(n+1)}{12}d^2$

42. Variance is independent of change of-

- (1) only origin
 (2) only scale
 (3) origin and scale both
 (4) none of these

43. If the coefficient of variation and standard deviation of a distribution are 50% and 20 respectively, then its mean is-
- (1) 40 (2) 30
(3) 20 (4) None of these
44. If each observation of a dist. whose S.D. is σ , is increased by λ , then the variance of the new observations is -
- (1) σ (2) $\sigma + \lambda$ (3) σ^2 (4) $\sigma^2 + \lambda$
45. The variance of 2, 4, 6, 8, 10 is-
- (1) 8 (2) $\sqrt{8}$
(3) 6 (4) none of these

46. If each observation of a dist., whose variance is σ^2 , is multiplied by λ , then the S.D. of the new new observations is-
- (1) σ (2) $\lambda\sigma$
(3) $|\lambda|\sigma$ (4) $\lambda^2\sigma$
47. The standard deviation of variate x_i is σ . Then standard deviation of the variate $\frac{ax_i + b}{c}$, where a, b, c are constants is-
- (1) $\left(\frac{a}{c}\right)\sigma$ (2) $\left|\frac{a}{c}\right|\sigma$
(3) $\left(\frac{a^2}{c^2}\right)\sigma$ (4) None of these

CHECK YOUR GRASP

ANSWER-KEY

EXERCISE-I

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Ans.	2	4	3	3	1	2	2	4	2	4	2	2	3	4	2	1	4	2	3	3
Que.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
Ans.	4	2	3	4	3	2	2	4	4	2	4	3	2	2	2	1	4	2	2	2
Que.	41	42	43	44	45	46	47													
Ans.	2	1	1	3	1	3	2													

BRAIN TEASERS

STATISTICS

EXERCISE-II

- The A.M. of the series 1, 2, 4, 8, 16,, 2^n is-
 (1) $\frac{2^n - 1}{n}$ (2) $\frac{2^{n+1} - 1}{n + 1}$
 (3) $\frac{2^n - 1}{n + 1}$ (4) $\frac{2^{n+1} - 1}{n}$
- If the mean of n observations $1^2, 2^2, 3^2, \dots$ n^2 is $\frac{46n}{11}$, then n is equal to-
 (1) 11 (2) 12
 (3) 23 (4) 22
- The weighted mean of first n natural numbers whose weights are equal, is :-
 (1) $\frac{n+1}{2}$ (2) $\frac{2n+1}{2}$
 (3) $\frac{2n+1}{3}$ (4) $\frac{(2n+1)(n+1)}{6}$
- The average age of a group of men and women is 30 years. If average age of men is 32 and that of women is 27, then the percentage of women in the group is-
 (1) 60 (2) 50
 (3) 40 (4) 30
- Mean and median of four numbers a, b, c and d ($b < a < d < c$) is 35 and 25 respectively then the value of $b + c - a - d$ will be :-
 (1) 90 (2) 115 (3) 40 (4) 10
- Variance of the group $\alpha, \alpha + 2, \alpha + 4, \alpha + 6, \dots$ upto n terms ($\alpha \neq 0$) is :-
 (1) $\frac{n^2 - 1}{12} + 2n + \alpha$ (2) $\frac{n^2 - 1}{3} + \alpha$
 (3) $\frac{n^2 - 1}{3}$ (4) None
- Product of n positive numbers is unit. The sum of these numbers can not be less than-
 (1) 1 (2) n
 (3) n^2 (4) none of these
- The A.M. of first n terms of the series 1.3.5, 3.5.7, 5.7.9,, is-
 (1) $3n^3 + 6n^2 + 7n - 1$ (2) $n^3 + 8n^2 + 7n - 1$
 (3) $2n^3 + 8n^2 - 7n - 2$ (4) $2n^3 + 8n^2 + 7n - 2$
- The observations 29, 32, 48, 50, $x, x + 2, 72, 78, 84, 95$ are arranged in ascending order and their median is 63 then the value of x is :-
 (1) 61 (2) 62 (3) 62.5 (4) 63
- If the mode of a distribution is 18 and the mean is 24, then median is-
 (1) 18 (2) 24 (3) 22 (4) 21
- If the mean and S.D. of n observations $x_1, x_2, \dots, x_n$ are $\bar{x}$ and σ resp, then the sum of squares of observations is :-
 (1) $n(\sigma^2 + \bar{x}^2)$ (2) $n(\sigma^2 - \bar{x}^2)$
 (3) $n(\bar{x}^2 - \sigma^2)$ (4) None of these
- The variance of observations 8, 12, 13, 15, 22, is :-
 (1) 21 (2) 21.2
 (3) 21.4 (4) None of these
- If the mean of a set of observations $x_1, x_2, \dots, x_{10}$ is 20, then the mean of $x_1 + 4, x_2 + 8, x_3 + 12, \dots, x_{10} + 40$ is-
 (1) 34 (2) 42 (3) 38 (4) 40
- The mean of values 0, 1, 2,, n when their weights are $1, {}^nC_1, {}^nC_2, \dots, {}^nC_n$, resp., is
 (1) $\frac{2^n}{n+1}$ (2) $\frac{n+1}{2}$
 (3) $\frac{2^{n+1}}{n(n+1)}$ (4) $\frac{n}{2}$

15. For 15 observations of x , mean and median were found to be 12 and 20 respectively. Later an observation which was 25 found to be wrong then replaced by its correct value 55, then new mean and median will be :-
 (1) 14 and 50 respectively
 (2) 12 and 20 respectively
 (3) 14 and 20 respectively
 (4) Mean is 14 but median can't be determined.
16. If a variable takes the discrete values $\alpha + 4$, $\alpha - \frac{7}{2}$, $\alpha - \frac{5}{2}$, $\alpha - 3$, $\alpha - 2$, $\alpha + \frac{1}{2}$, $\alpha - \frac{1}{2}$, $\alpha + 5$ ($\alpha > 0$), then the median of these values-
 (1) $\alpha - \frac{5}{4}$ (2) $\alpha - \frac{1}{2}$
 (3) $\alpha - 2$ (4) $\alpha + \frac{5}{4}$
17. The S.D. of first n odd natural numbers is :-
 (1) $\sqrt{\frac{n^2 - 1}{2}}$ (2) $\sqrt{\frac{n^2 - 1}{3}}$
 (3) $\sqrt{\frac{n^2 - 1}{6}}$ (4) $\sqrt{\frac{n^2 - 1}{12}}$
18. If the sum and sum of squares of 10 observations are 12 and 18 resp., then, The S.D. of observations is :-
 (1) $\frac{1}{5}$ (2) $\frac{2}{5}$ (3) $\frac{3}{5}$ (4) $\frac{4}{5}$
19. The mean of n values of a distribution is $\bar{x}$. If its first value is increased by 1, second by 2, then the mean of new values will be-
 (1) $\bar{x} + n$ (2) $\bar{x} + n/2$
 (3) $\bar{x} + \left(\frac{n+1}{2}\right)$ (4) None of these
20. The mean of the series $x_1, x_2, \dots, x_n$ is $\bar{X}$. If x_2 is replaced by λ , then the new mean is-
 (1) $\frac{\bar{X} - x_2 + \lambda}{n}$ (2) $\frac{n\bar{X} + x_2 - \lambda}{n}$
 (3) $\frac{(n-1)\bar{X} + \lambda}{n}$ (4) $\frac{n\bar{X} - x_2 + \lambda}{n}$
21. The mean square deviation about -1 and $+1$ of a set of observations are 7 and 3 respectively then standard deviation of the set is :-
 (1) $\sqrt{2}$ (2) $\sqrt{3}$ (3) 2 (4) None
22. The mean deviation of the numbers 1, 2, 3, 4, 5 is-
 (1) 0 (2) 1.2
 (3) 2 (4) 1.4
23. If mean = (3 median - mode) $\times$, then the value of x is-
 (1) 1 (2) 2 (3) $1/2$ (4) $3/2$
24. A man spends equal ammount on purchasing three kinds of pens at the rate 5 Rs/pen, 10 Rs/pen, 20 Rs/pen, then average cost of one pen is :-
 (1) 10 Rs (2) $\frac{35}{3}$ Rs
 (3) $\frac{60}{7}$ Rs (4) None of these
25. The median of 21 observation is 40. if each observations greater than the median are increased by 6, then the median of the observations will be-
 (1) 40 (2) 46
 (3) $46 + 40/21$ (4) $46 - 40/21$
26. The coefficient of range of the following distribution 10, 14, 11, 9, 8, 12, 6
 (1) 0.4 (2) 2.5
 (3) 8 (4) 0.9

27. The S.D. of the following freq. dist. :-

Class	0 – 10	10 – 20	20 – 30	30 – 40
f_i	1	3	4	2

- (1) 7.8 (2) 9
(3) 8.1 (4) 0.9
28. The mean of a dist. is 4. if its coefficient of variation is 58%. Then the S.D. of the dist. is:-
(1) 2.23 (2) 3.23
(3) 2.32 (4) None of these
29. The mean of a set of observations is $\bar{x}$. If each observation is divided by α , ($\alpha \neq 0$) and then is increased by 10, then the mean of the new set is
(1) $\frac{\bar{x}}{\alpha}$ (2) $\frac{\bar{x} + 10}{\alpha}$
(3) $\frac{\bar{x} + 10\alpha}{\alpha}$ (4) $\frac{\alpha\bar{x} + 10}{\alpha}$
30. The average age of a teacher and three students is 20 years. If all students are of equal age and the difference between the age of the teacher and that of a student is 20 years, then the age of the teacher is-
(1) 25 years (2) 30 years
(3) 35 years (4) 45 years
31. Median of 5 observations i.e.
 $3^{\log_9 4}, 5^{\log_{1/2} 8}, e^{2\ln 3}, \ln\left(\frac{1}{e}2\right) + 3, e^{2\ln 3 + \frac{1}{\log_4 e}}$:-
(1) 1 (2) 2 (3) 9 (4) 36
32. Median of ${}^{2n}C_0, {}^{2n}C_1, {}^{2n}C_2, \dots, {}^{2n}C_n$ (when n is even) is-
(1) ${}^{2n}C_{\frac{n-1}{2}}$ (2) ${}^{2n}C_{\frac{n}{2}}$
(3) ${}^{2n}C_{\frac{n+1}{2}}$ (4) None of these

33. The mean deviation from mean of observations 5, 10, 15, 20,85 is :-

(1) 43.71 (2) 21.17
(3) 38.7 (4) None of these

34. If standard deviation of variate x_i is 10, then variance of the variate ($50 + 5x_i$) will be-

(1) 50 (2) 250
(3) 500 (4) 2500

35. The S.D. of the numbers 31, 32, 33, 47 is-

(1) $2\sqrt{6}$ (2) $4\sqrt{3}$

(3) $\sqrt{\frac{47^2 - 1}{12}}$ (4) None of these

36. The sum of the squares of deviation of 10 observations from their mean 50 is 250, then coefficient of variation is-

(1) 10% (2) 40%
(3) 50% (4) None of these

37. The median and standard deviation (S.D.) of a distribution will be, If each term is increased by 2 -

(1) median and S.D. will increased by 2
(2) median will increased by 2 but S.D. will remain same
(3) median will remain same but S.D. will increased by 2
(4) median and S.D. will remain same

38. If $\bar{X}_1$ and $\bar{X}_2$ are the means of two series such that $\bar{X}_1 < \bar{X}_2$ and $\bar{X}$ is the mean of the combined series, then-

(1) $\bar{X} < \bar{X}_1$ (2) $\bar{X} > \bar{X}_2$
(3) $\bar{X}_1 < \bar{X} < \bar{X}_2$ (4) $\bar{X} = \frac{\bar{X}_1 + \bar{X}_2}{2}$

39. The median of 19 observations of a group is 30. If two observations with values 8 and 32 are further included, then the median of the new group of 21 observation will be
 (1) 28 (2) 30
 (3) 32 (4) 34
40. The coefficient of mean deviation from median of observations 40, 62, 54, 90, 68, 76 is :-
 (1) 2.16 (2) 0.2
 (3) 5 (4) None of these
41. A group of 10 observations has mean 5 and S.D. $2\sqrt{6}$. another group of 20 observations has mean 5 and S.D. $3\sqrt{2}$, then the S.D. of combined group of 30 observations is :-
 (1) $\sqrt{5}$ (2) $2\sqrt{5}$
 (3) $3\sqrt{5}$ (4) None of these
42. For the values $x_1, x_2, \dots, x_{101}$ of a distribution $x_1 < x_2 < x_3 < \dots < x_{100} < x_{101}$. The mean deviation of this distribution with respect to a number k will be minimum when k is equal to -
 (1) x_1
 (2) x_{51}
 (3) x_{50}
 (4) $\frac{x_1 + x_2 + \dots + x_{101}}{101}$
43. In any discrete series (when all the value are not same) the relationship between M.D. about mean and S.D. is-
 (1) M.D. = S.D. (2) M.D. > S.D.
 (3) M.D. < S.D. (4) M.D. $\leq$ S.D.
44. Median of observations x_i such that $(x_i^2 - 7x_i + 12)(x_i^3 - x_i^2 - 4x_i + 4) = 0$ will be :-
 (1) 1 (2) 2 (3) 3 (4) None

BRAIN TEASERS								ANSWER-KEY					EXERCISE-II							
Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Ans.	2	1	1	3	3	3	2	4	2	3	1	2	2	4	3	1	2	3	3	4
Que.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
Ans.	2	2	3	3	1	1	2	3	3	3	3	2	2	4	1	1	2	3	2	2
Que.	41	42	43	44																
Ans.	2	2	3	2																

PREVIOUS YEAR QUESTIONS

STATISTICS

EXERCISE-III

1. All the students of a class performed poorly in Mathematics. The teacher decided to give grace marks of 10 to each of the students. Which of the following statistical measures will not change even after the grace marks were given? **[JEE(Main)-2013]**
 - (1) mean (2) median
 - (3) mode (4) variance
2. The variance of first 50 even natural numbers is :- **[JEE(Main)-2014]**
 - (1) $\frac{833}{4}$ (2) 833
 - (3) 437 (4) $\frac{437}{4}$
3. The mean of the data set comprising of 16 observations is 16. If one of the observation valued 16 is deleted and three new observations valued 3, 4 and 5 are added to the data, then the mean of the resultant data, is : **[JEE(Main)-2015]**
 - (1) 15.8 (2) 14.0
 - (3) 16.8 (4) 16.0
4. If the standard deviation of the numbers 2, 3, a and 11 is 3.5, then which of the following is true ? **[JEE(Main)-2016]**
 - (1) $3a^2 - 23a + 44 = 0$
 - (2) $3a^2 - 26a + 55 = 0$
 - (3) $3a^2 - 32a + 84 = 0$
 - (4) $3a^2 - 34a + 91 = 0$
5. If $\sum_{i=1}^9 (x_i - 5) = 9$ and $\sum_{i=1}^9 (x_i - 5)^2 = 45$, then the standard deviation of the 9 items $x_1, x_2, \dots, x_9$ is - **[JEE(Main)-2018]**
 - (1) 4 (2) 2 (3) 3 (4) 9
6. 5 students of a class have an average height 150 cm and variance 18 cm^2 . A new student, whose height is 156 cm, joined them. The variance (in cm^2) of the height of these six students is : **[JEE(Main)-19]**
 - (1) 22 (2) 20 (3) 16 (4) 18
7. A data consists of n observations : $x_1, x_2, \dots, x_n$. If $\sum_{i=1}^n (x_i + 1)^2 = 9n$ and $\sum_{i=1}^n (x_i - 1)^2 = 5n$, then the standard deviation of this data is : **[JEE(Main)-19]**
 - (1) 5 (2) $\sqrt{5}$
 - (3) $\sqrt{7}$ (4) 2
8. The mean of five observations is 5 and their variance is 9.20. If three of the given five observations are 1, 3 and 8, then a ratio of other two observations is : **[JEE(Main)-19]**
 - (1) 4 : 9 (2) 6 : 7
 - (3) 5 : 8 (4) 10 : 3
9. The outcome of each of 30 items was observed; 10 items gave an outcome $\frac{1}{2} - d$ each, 10 items gave outcome $\frac{1}{2}$ each and the remaining 10 items gave outcome $\frac{1}{2} + d$ each. If the variance of this outcome data is $\frac{4}{3}$ then $|d|$ equals :- **[JEE(Main)-19]**
 - (1) 2 (2) $\frac{\sqrt{5}}{2}$
 - (3) $\frac{2}{3}$ (4) $\sqrt{2}$
10. The mean and the variance of five observation are 4 and 5.20, respectively. If three of the observations are 3, 4 and 4; then the absolute value of the difference of the other two observations, is : **[JEE(Main)-19]**
 - (1) 1 (2) 3
 - (3) 7 (4) 5

- [JEE(Main)-19]**

- (3) $4\sqrt{\frac{5}{3}}$

(4) $\sqrt{6}$

- [JEE(Main)-19]**

Marks	2	3	5	7
Frequency	$(x+1)^2$	$2x-5$	x^2-3x	x

[JEE(Main)-19]

- (1) 525 (2) 380
(3) 480 (4) 400

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14
Ans.	4	2	2	3	2	2	2	1	4	3	3	2	1	4

MATHEMATICAL REASONING

1. STATEMENT :

A sentence which is either true or false but cannot be both are called a statement. A sentence which is an exclamatory or a wish or an imperative or an interrogative can not be a statement.

If a statement is true then its truth value is T and if it is false then its truth value is F

For ex.

(i) "New Delhi is the capital of India", a true statement

(ii) " $3 + 2 = 6$ ", a false statement

(iii) "Where are you going ?" not a statement because it cannot be defined as true or false

Note : A statement cannot be both true and false at a time

2. SIMPLE STATEMENT :

Any statement whose truth value does not depend on other statement are called simple statement

For ex. (i) " $\sqrt{2}$ is an irrational number" (ii) "The set of real number is an infinite set"

3. COMPOUND STATEMENT :

A statement which is a combination of two or more simple statements are called compound statement

Here the simple statements which form a compound statement are known as its sub statements

For ex.

(i) "If x is divisible by 2 then x is even number"

(ii) " $\triangle ABC$ is equilateral if and only if its three sides are equal"

4. LOGICAL CONNECTIVES :

The words or phrases which combined simple statements to form a compound statement are called logical connectives.

In the following table some possible connectives, their symbols and the nature of the compound statement formed by them

S.N.	Connectives	symbol	use	operation
1.	and	$\wedge$	$p \wedge q$	conjunction
2.	or	$\vee$	$p \vee q$	disjunction
3.	not	$\sim$ or $'$	$\sim p$ or p'	negation
4.	If then	$\Rightarrow$ or $\rightarrow$	$p \Rightarrow q$ or $p \rightarrow q$	Implication or conditional
5.	If and only if (iff)	$\Leftrightarrow$ or $\leftrightarrow$	$p \Leftrightarrow q$ or $p \leftrightarrow q$	Equivalence or Bi-conditional

Explanation :

(i) $p \wedge q \equiv$ statement p and q

($p \wedge q$ is true only when p and q both are true otherwise it is false)

(ii) $p \vee q \equiv$ statement p or q

($p \vee q$ is true if at least one from p and q is true i.e. $p \vee q$ is false only when p and q both are false)

(iii) $\sim p \equiv$ not statement p

($\sim p$ is true when p is false and $\sim p$ is false when p is true)

(iv) $p \Rightarrow q \equiv$ statement p then statement q

($p \Rightarrow q$ is false only when p is true and q is false otherwise it is true for all other cases)

(v) $p \Leftrightarrow q \equiv$ statement p if and only if statement q

($p \Leftrightarrow q$ is true only when p and q both are true or false otherwise it is false)

5. TRUTH TABLE :

A table which shows the relationship between the truth value of compound statement $S(p, q, r, \dots)$ and the truth values of its sub statements $p, q, r, \dots$ is said to be truth table of compound statement S

If p and q are two simple statements then truth table for basic logical connectives are given below

Conjunction

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

Disjunction

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

Negation

p	$(\sim p)$
T	F
F	T

Conditional

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

Biconditional

p	q	$p \rightarrow q$	$q \rightarrow p$	$(p \rightarrow q) \wedge (q \rightarrow p)$ or $p \Leftrightarrow q$
T	T	T	T	T
T	F	F	T	F
F	T	T	F	F
F	F	T	T	T

Note : If the compound statement contain n sub statements then its truth table will contain 2^n rows.

6. LOGICAL EQUIVALENCE :

Two compound statements $S_1(p, q, r, \dots)$ and $S_2(p, q, r, \dots)$ are said to be logically equivalent or simply equivalent if they have same truth values for all logically possibilities

Two statements S_1 and S_2 are equivalent if they have identical truth table i.e. the entries in the last column of their truth table are same. If statements S_1 and S_2 are equivalent then we write $S_1 \equiv S_2$

For ex. The truth table for $(p \rightarrow q)$ and $(\sim p \vee q)$ given as below

p	q	$(\sim p)$	$p \rightarrow q$	$\sim p \vee q$
T	T	F	T	T
T	F	F	F	F
F	T	T	T	T
F	F	T	T	T

We observe that last two columns of the above truth table are identical hence compound statements

$(p \rightarrow q)$ and $(\sim p \vee q)$ are equivalent

i.e.

$$p \rightarrow q \equiv \sim p \vee q$$

7. TAUTOLOGY AND CONTRADICTION :

(i) **Tautology** : A statement is said to be a tautology if it is true for all logical possibilities i.e. its truth value always T. It is denoted by t.

For ex. the statement $p \vee \sim(p \wedge q)$ is a tautology

p	q	$p \wedge q$	$\sim(p \wedge q)$	$p \vee \sim(p \wedge q)$
T	T	T	F	T
T	F	F	T	T
F	T	F	T	T
F	F	F	T	T

Clearly, The truth value of $p \vee \sim(p \wedge q)$ is T for all values of p and q. so $p \vee \sim(p \wedge q)$ is a tautology

(ii) **Contradiction** : A statement is a contradiction if it is false for all logical possibilities.

i.e. its truth value always F. It is denoted by c.

For ex. The statement $(p \vee q) \wedge (\sim p \wedge \sim q)$ is a contradiction

p	q	$\sim p$	$\sim q$	$p \vee q$	$(\sim p \wedge \sim q)$	$(p \vee q) \wedge (\sim p \wedge \sim q)$
T	T	F	F	T	F	F
T	F	F	T	T	F	F
F	T	T	F	T	F	F
F	F	T	T	F	T	F

Clearly, then truth value of $(p \vee q) \wedge (\sim p \wedge \sim q)$ is F for all value of p and q. So $(p \vee q) \wedge (\sim p \wedge \sim q)$ is a contradiction.

Note : The negation of a tautology is a contradiction and negation of a contradiction is a tautology

8. DUALITY :

Two compound statements S_1 and S_2 are said to be duals of each other if one can be obtained from the other by replacing $\wedge$ by $\vee$ and $\vee$ by $\wedge$

If a compound statement contains the special variable t (tautology) and c (contradiction) then obtain its dual we replaced t by c and c by t in addition to replacing $\wedge$ by $\vee$ and $\vee$ by $\wedge$.

Note :

- (i) the connectives $\wedge$ and $\vee$ are also called dual of each other.
- (ii) If $S^*(p, q)$ is the dual of the compound statement $S(p, q)$ then
 - (a) $S^*(\sim p, \sim q) \equiv \sim S(p, q)$ (ii) $\sim S^*(p, q) \equiv S(\sim p, \sim q)$

For ex. The duals of the following statements

- (i) $(p \wedge q) \vee (r \vee s)$ (ii) $(p \vee t) \wedge (p \vee c)$
- (iii) $\sim(p \wedge q) \vee [p \wedge \sim(q \vee \sim s)]$

are as given below

- (i) $(p \vee q) \wedge (r \wedge s)$
- (ii) $(p \wedge c) \vee (p \wedge t)$
- (iii) $\sim(p \vee q) \wedge [p \vee \sim(q \wedge \sim s)]$

9. CONVERSE, INVERSE AND CONTRAPOSITIVE OF THE CONDITIONAL STATEMENT ($p \rightarrow q$):

- (i) **Converse** : The converse of the conditional statement $p \rightarrow q$ is defined as $q \rightarrow p$
 (ii) **Inverse** : The inverse of the conditional statement $p \rightarrow q$ is defined as $\sim p \rightarrow \sim q$
 (iii) **Contrapositive** : The contrapositive of conditional statement $p \rightarrow q$ is defined as $\sim q \rightarrow \sim p$

10. NEGATION OF COMPOUND STATEMENTS :

If p and q are two statements then

- (i) **Negation of conjunction** : $\sim(p \wedge q) \equiv \sim p \vee \sim q$

p	q	$\sim p$	$\sim q$	$(p \wedge q)$	$\sim(p \wedge q)$	$(\sim p \vee \sim q)$
T	T	F	F	T	F	F
T	F	F	T	F	T	T
F	T	T	F	F	T	T
F	F	T	T	F	T	T

- (ii) **Negation of disjunction** : $\sim(p \vee q) \equiv \sim p \wedge \sim q$

p	q	$\sim p$	$\sim q$	$(p \vee q)$	$\sim(p \vee q)$	$(\sim p \wedge \sim q)$
T	T	F	F	T	F	F
T	F	F	T	T	F	F
F	T	T	F	T	F	F
F	F	T	T	F	T	T

- (iii) **Negation of conditional** : $\sim(p \rightarrow q) \equiv p \wedge \sim q$

p	q	$\sim q$	$(p \rightarrow q)$	$\sim(p \rightarrow q)$	$(p \wedge \sim q)$
T	T	F	T	F	F
T	F	T	F	T	T
F	T	F	T	F	F
F	F	T	T	F	F

- (iv) **Negation of biconditional** : $\sim(p \leftrightarrow q) \equiv (p \wedge \sim q) \vee (q \wedge \sim p)$

we know that $p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$

$$\therefore \sim(p \leftrightarrow q) \equiv \sim[(p \rightarrow q) \wedge (q \rightarrow p)]$$

$$\equiv \sim(p \rightarrow q) \vee \sim(q \rightarrow p)$$

$$\equiv (p \wedge \sim q) \vee (q \wedge \sim p)$$

Note : The above result also can be proved by preparing truth table for $\sim(p \leftrightarrow q)$ and $(p \wedge \sim q) \vee (q \wedge \sim p)$

11. ALGEBRA OF STATEMENTS :

If p, q, r are any three statements then the some law of algebra of statements are as follow

(i) Idempotent Laws :

$$(a) p \wedge p \equiv p \quad (b) p \vee p \equiv p$$

$$\text{i.e. } p \wedge p \equiv p \equiv p \vee p$$

p	$(p \wedge p)$	$(p \vee p)$
T	T	T
F	F	F

(ii) Comutative laws :

$$(a) p \wedge q \equiv q \wedge p \quad (b) p \vee q \equiv q \vee p$$

p	q	$(p \wedge q)$	$(q \wedge p)$	$(p \vee q)$	$(q \vee p)$
T	T	T	T	T	T
T	F	F	F	T	T
F	T	F	F	T	T
F	F	F	F	F	F

(iii) Associative laws :

$$(a) (p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$$

$$(b) (p \vee q) \vee r \equiv p \vee (q \vee r)$$

p	q	r	$(p \wedge q)$	$(q \wedge r)$	$(p \wedge q) \wedge r$	$p \wedge (q \wedge r)$
T	T	T	T	T	T	T
T	T	F	T	F	F	F
T	F	T	F	F	F	F
T	F	F	F	F	F	F
F	T	T	F	T	F	F
F	T	F	F	F	F	F
F	F	T	F	F	F	F
F	F	F	F	F	F	F

Similarly we can proved result (b)

$$(iv) \text{Distributive laws : } (a) p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r) \quad (c) p \wedge (q \wedge r) \equiv (p \wedge q) \wedge (p \wedge r)$$

$$(b) p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r) \quad (d) p \vee (q \vee r) \equiv (p \vee q) \vee (p \vee r)$$

p	q	r	$(q \vee r)$	$(p \wedge q)$	$(p \wedge r)$	$p \wedge (q \vee r)$	$(p \wedge q) \vee (p \wedge r)$
T	T	T	T	T	T	T	T
T	T	F	T	T	F	T	T
T	F	T	T	F	T	T	T
T	F	F	F	F	F	F	F
F	T	T	T	F	F	F	F
F	T	F	T	F	F	F	F
F	F	T	T	F	F	F	F
F	F	F	F	F	F	F	F

Similarly we can prove result (b), (c), (d)

(v) **De Morgan Laws :** (a) $\sim (p \wedge q) \equiv \sim p \vee \sim q$

(b) $\sim (p \vee q) \equiv \sim p \wedge \sim q$

p	q	$\sim p$	$\sim q$	$(p \wedge q)$	$\sim (p \wedge q)$	$(\sim p \vee \sim q)$
T	T	F	F	T	F	F
T	F	F	T	F	T	T
F	T	T	F	F	T	T
F	F	T	T	F	T	T

Similarly we can prove result (b)

(vi) **Involution laws (or Double negation laws) :** $\sim(\sim p) \equiv p$

p	$\sim p$	$\sim(\sim p)$
T	F	T
F	T	F

(vii) **Identity Laws :** If p is a statement and t and c are tautology and contradiction respectively then

(a) $p \wedge t \equiv p$

(b) $p \vee t \equiv t$

(c) $p \wedge c \equiv c$

(d) $p \vee c \equiv p$

p	t	c	$(p \wedge t)$	$(p \vee t)$	$(p \wedge c)$	$(p \vee c)$
T	T	F	T	T	F	T
F	T	F	F	T	F	F

(viii) **Complement Laws :**

(a) $p \wedge (\sim p) \equiv c$

(b) $p \vee (\sim p) \equiv t$

(c) $(\sim \sim) \equiv c$

(d) $(\sim \sim) \equiv t$

p	$\sim p$	$(p \wedge \sim p)$	$(p \vee \sim p)$
T	F	F	T
F	T	F	T

(ix) **Contrapositive laws :** $p \rightarrow q \equiv \sim q \rightarrow \sim p$

p	q	$\sim p$	$\sim q$	$p \rightarrow q$	$\sim q \rightarrow \sim p$
T	T	F	F	T	T
T	F	F	T	F	F
F	T	T	F	T	T
F	F	T	T	T	T

12. QUANTIFIED STATEMENTS AND QUANTIFIERS :

The words or phrases "All", "Some", "None", "There exists a" are examples of quantifiers.

A statement containing one or more of these words (or phrases) is a quantified statement.

E.g. (1) All dogs are poodles

(2) Some books have hard covers

(3) There exists an odd number which is prime.

Note : Phrases "There exists a" and "Atleast one" and the word "some" have the same meaning.

NEGATION OF QUANTIFIED STATEMENTS :

(1) '**None**' is the negation of '**at least one**' or '**some**' or '**few**'

Statement : Some dogs are poodles.

Negation : No dogs are poodles.

Similarly negation of '**some**' is '**none**'

(2) The negation of "**some A are B**" or "**There exist A which is B**" is "**No A are (is) B**" or "**There does not exist any A which is B**".

Statement-1 : Some boys in the class are smart

Statement-2 : There exists a boy in the class who is smart

Statement-3 : Atleast one boy in the class is smart

All the three above statements have same meaning as they all indicate "**existence**" of smart boy in the class.

Negation of these statements is

No boy in the class is smart.

or

There does not exist any boy in the class who is smart.

(3) Negation of "**All A are B**" is "**Some A are not B**".

Statement : All boys in the class are smart.

Negation : Some boys in the class are not smart.

or

There exists a boy in the class who is not smart.

SOLVED EXAMPLES

Ex.1 Which of the following is correct for the statements p and q ?

- (1) $p \wedge q$ is true when at least one from p and q is true
- (2) $p \rightarrow q$ is true when p is true and q is false
- (3) $p \leftrightarrow q$ is true only when both p and q are true
- (4) $\sim(p \vee q)$ is true only when both p and q are false

Sol.(4) We know that $p \wedge q$ is true only when both p and q are true so option (1) is not correct
 we know that $p \rightarrow q$ is false only when p is true and q is false so option (2) is not correct
 we know that $p \leftrightarrow q$ is true only when either p and q both are true or both are false
 so option (3) is not correct
 we know that $\sim(p \vee q)$ is true only when $(p \vee q)$ is false
 i.e. p and q both are false
 So option (4) is correct

Ex.2 $\sim(p \vee q) \vee (\sim p \wedge q)$ is equivalent to-

- (1) p
- (2) $\sim p$
- (3) q
- (4) $\sim q$

Sol.(2) $\because \sim(p \vee q) \vee (\sim p \wedge q) \equiv (\sim p \wedge \sim q) \vee (\sim p \wedge q)$ (By Demorgan Law)
 $\equiv \sim p \wedge (\sim q \vee q)$ (By distributive laws)
 $\equiv \sim p \wedge t$ (By complement laws)
 $\equiv \sim p$ (By Identity Laws)

Ex.3 Which of the following is logically equivalent to $(p \wedge q)$?

- (1) $p \rightarrow \sim q$
- (2) $\sim p \vee \sim q$
- (3) $\sim(p \rightarrow \sim q)$
- (4) $\sim(\sim p \wedge \sim q)$

Sol.(3) $\because p \rightarrow \sim q \equiv \sim p \vee \sim q \equiv \sim(p \wedge q)$ ($\because p \rightarrow q \equiv \sim p \vee q$)
 i.e. $\sim(p \rightarrow \sim q) \equiv p \wedge q$
 $\because \sim p \vee \sim q \equiv \sim(p \wedge q)$
 and $\sim(\sim p \wedge \sim q) \equiv p \vee q$

Ex.4 If $p \rightarrow (q \vee r)$ is false, then the truth values of p, q, r respectively are-

- (1) T, F, F
- (2) F, F, F
- (3) F, T, T
- (4) T, T, F

Sol.(1) We know $p \rightarrow (q \vee r)$ is false only when p is true and $(q \vee r)$ is false. but $(q \vee r)$ is false only when q and r both are false
 Hence truth values of p, q, r are respectively T, F, F

Ex.5 Statement $(p \wedge \sim q) \wedge (\sim p \vee q)$ is

- (1) a tautology
- (2) a contradiction
- (3) neither a tautology nor a contradiction
- (4) None of these

Sol.(2) $\because (p \wedge \sim q) \wedge (\sim p \vee q)$
 $\equiv (p \wedge \sim q) \wedge \sim(p \wedge \sim q)$ (By Demorgan Laws)
 $\equiv c$, where c is contradiction (By complement laws)

Ex.6 Negation of the statement $p \rightarrow (q \wedge r)$ is-

- (1) $\sim p \rightarrow \sim(q \wedge r)$
- (2) $\sim p \vee (q \wedge r)$
- (3) $(q \wedge r) \rightarrow p$
- (4) $p \wedge (\sim q \vee \sim r)$

Sol.(4) $\sim(p \rightarrow (q \wedge r)) \equiv p \wedge \sim(q \wedge r)$ ($\because \sim(p \rightarrow q) \equiv p \wedge \sim q$)
 $\equiv p \wedge (\sim q \vee \sim r)$

Ex.7 If $x = 5$ and $y = -2$ then $x - 2y = 9$. The contrapositive of this statement is-

- (1) If $x - 2y \neq 9$ then $x \neq 5$ or $y \neq -2$ (2) If $x - 2y \neq 9$ then $x \neq 5$ and $y \neq -2$
 (3) If $x - 2y = 9$ then $x = 5$ and $y = -2$ (4) None of these

Sol.(1) Let p, q, r be the three statements such that

$$p : x = 5, \quad q : y = -2 \quad \text{and} \quad r : x - 2y = 9$$

Here given statement is $(p \wedge q) \rightarrow r$ and its contrapositive is $\sim r \rightarrow \sim(p \wedge q)$

$$\text{i.e. } \sim r \rightarrow (\sim p \vee \sim q)$$

$$\text{i.e. if } x - 2y \neq 9 \text{ then } x \neq 5 \text{ or } y \neq -2$$

Ex.8 Which of the following is wrong ?

- (1) $p \rightarrow q$ is logically equivalent to $\sim p \vee q$
 (2) If the $(p \vee q) \wedge (q \vee r)$ is true then truth values of p, q, r are T, F, T respectively
 (3) $\sim(p \wedge (q \vee r)) \equiv (\sim p \vee \sim q) \wedge (\sim p \vee \sim r)$
 (4) The truth value of $p \wedge \sim(p \vee q)$ is always T

Sol.(4) We know that $p \rightarrow q \equiv \sim p \vee q$

If $(p \vee q) \wedge (q \vee r)$ is true then

$(p \vee q)$ and $(q \vee r)$ both are true.

i.e. truth values of p, q, r may be T, F, T respectively

$$\therefore \sim(p \wedge (q \vee r)) \equiv \sim((p \wedge q) \vee (p \wedge r)) \equiv \sim(p \wedge q) \wedge \sim(p \wedge r) \equiv (\sim p \vee \sim q) \wedge (\sim p \vee \sim r)$$

If p is true and q is false then $\sim(p \vee q)$ is false i.e. $p \wedge \sim(p \vee q)$ is false

Ex.9 If $S^*(p, q, r)$ is the dual of the compound statement $S(p, q, r)$ and $S(p, q, r) = \sim p \wedge [(q \vee r)]$ then $S^*(\sim p, \sim q, \sim r)$ is equivalent to-

- (1) $S(p, q, r)$ (2) $\sim S(\sim p, \sim q, \sim r)$ (3) $\sim S(p, q, r)$ (4) $S^*(p, q, r)$

Sol.(3) $\therefore S(p, q, r) = \sim p \wedge [(q \vee r)]$

$$\text{So } S(\sim p, \sim q, \sim r) \equiv \sim(\sim p) \wedge [\sim(\sim q \vee \sim r)] \equiv p \wedge (q \wedge r)$$

$$S^*(p, q, r) \equiv \sim p \vee [\sim(q \wedge r)]$$

$$S^*(\sim p, \sim q, \sim r) \equiv p \vee (q \vee r)$$

Clearly $S^*(\sim p, \sim q, \sim r) \equiv \sim S(p, q, r)$

Ex.10 The negation of the statement "If a quadrilateral is a square then it is a rhombus"

- (1) If a quadrilateral is not a square then it is a rhombus
 (2) If a quadrilateral is a square then it is not a rhombus
 (3) a quadrilateral is a square and it is not a rhombus
 (4) a quadrilateral is not a square and it is a rhombus

Sol.(3) Let p and q be the statements as given below

p : a quadrilateral is a square

q : a quadrilateral is a rhombus

the given statement is $p \rightarrow q$

$$\therefore \sim(p \rightarrow q) \equiv p \wedge \sim q$$

Therefore the negation of the given statement is a quadrilateral is a square and it is not a rhombus

CHECK YOUR GRASP

MATHEMATICAL REASONING

EXERCISE-I

1. The inverse of the statement $(p \wedge \sim q) \rightarrow r$ is-
 (1) $\sim(p \vee \sim q) \rightarrow \sim r$ (2) $(\sim p \wedge q) \rightarrow \sim r$
 (3) $(\sim p \vee q) \rightarrow \sim r$ (4) None of these
 2. $(\sim p \vee \sim q)$ is logically equivalent to-
 (1) $p \wedge q$ (2) $\sim p \rightarrow q$
 (3) $p \rightarrow \sim q$ (4) $\sim p \rightarrow \sim q$
 3. The equivalent statement of $(p \leftrightarrow q)$ is-
 (1) $(p \wedge q) \vee (p \vee q)$
 (2) $(p \rightarrow q) \vee (q \rightarrow p)$
 (3) $(\sim p \vee q) \vee (p \vee \sim q)$
 (4) $(\sim p \vee q) \wedge (p \vee \sim q)$
 4. If the compound statement $p \rightarrow (\sim p \vee q)$ is false then the truth value of p and q are respectively-
 (1) T, T (2) T, F (3) F, T (4) F, F
 5. The statement $(p \rightarrow \sim p) \wedge (\sim p \rightarrow p)$ is-
 (1) a tautology
 (2) a contradiction
 (3) neither a tautology nor a contradiction
 (4) None of these
 6. Negation of the statement $(p \wedge r) \rightarrow (r \vee q)$ is-
 (1) $\sim(p \wedge r) \rightarrow \sim(r \vee q)$ (2) $(\sim p \vee \sim r) \vee (r \vee q)$
 (3) $(p \wedge r) \wedge (r \wedge q)$ (4) $(p \wedge r) \wedge (\sim r \wedge \sim q)$
 7. The dual of the statement $\sim p \wedge [\sim q \wedge (p \vee q) \wedge \sim r]$ is-
 (1) $\sim p \vee [\sim q \vee (p \vee q) \vee \sim r]$
 (2) $p \vee [q \vee (\sim p \wedge \sim q) \vee r]$
 (3) $\sim p \vee [\sim q \vee (p \wedge q) \vee \sim r]$
 (4) $\sim p \vee [\sim q \wedge (p \wedge q) \wedge \sim r]$
 8. Which of the following is correct-
 (1) $(\sim p \vee \sim q) \equiv (p \wedge q)$
 (2) $(p \rightarrow q) \equiv (\sim q \rightarrow \sim p)$
 (3) $\sim(p \rightarrow \sim q) \equiv (p \wedge \sim q)$
 (4) $\sim(p \leftrightarrow q) \equiv (p \rightarrow q) \vee (q \rightarrow p)$
 9. The contrapositive of $p \rightarrow (\sim q \rightarrow \sim r)$ is-
 (1) $(\sim q \wedge r) \rightarrow \sim p$ (2) $(q \rightarrow r) \rightarrow \sim p$
 (3) $(q \vee \sim r) \rightarrow \sim p$ (4) None of these
 10. The converse of $p \rightarrow (q \rightarrow r)$ is-
 (1) $(q \wedge \sim r) \vee p$ (2) $(\sim q \vee r) \vee p$
 (3) $(q \wedge \sim r) \wedge \sim p$ (4) $(q \wedge \sim r) \wedge p$
 11. If p and q are two statement then $(p \leftrightarrow \sim q)$ is true when-
 (1) p and q both are true
 (2) p and q both are false
 (3) p is false and q is true
 (4) None of these
 12. Statement $(p \wedge q) \rightarrow p$ is-
 (1) a tautology (2) a contradiction
 (3) neither (1) nor (2) (4) None of these
 13. If statements p, q, r have truth values T, F, T respectively then which of the following statement is true-
 (1) $(p \rightarrow q) \wedge r$ (2) $(p \rightarrow q) \vee \sim r$
 (3) $(p \wedge q) \vee (q \wedge r)$ (4) $(p \rightarrow q) \rightarrow r$
 14. If statement $p \rightarrow (q \vee r)$ is true then the truth values of statements p, q, r respectively-
 (1) T, F, T (2) F, T, F
 (3) F, F, F (4) All of these
 15. Which of the following statement is a contradiction-
 (1) $(p \wedge q) \wedge (\sim(p \vee q))$ (2) $p \vee (\sim p \wedge q)$
 (3) $(p \rightarrow q) \rightarrow p$ (4) $\sim p \vee \sim q$
 16. The negative of the statement "If a number is divisible by 15 then it is divisible by 5 or 3"
 (1) If a number is divisible by 15 then it is not divisible by 5 and 3
 (2) A number is divisible by 15 and it is not divisible by 5 or 3
 (3) A number is divisible by 15 or it is not divisible by 5 and 3
 (4) A number is divisible by 15 and it is not divisible by 5 and 3
 17. If $x = 5$ and $y = -2$ then $x - 2y = 9$. The contrapositive of this statement is-
 (1) If $x - 2y \neq 9$ then $x \neq 5$ or $y \neq -2$
 (2) If $x - 2y \neq 9$ then $x \neq 5$ and $y \neq -2$
 (3) If $x - 2y = 9$ then $x = 5$ and $y = -2$
 (4) None of these

18. The negation of the statement " $2 + 3 = 5$ and $8 < 10$ " is-
 (1) $2 + 3 \neq 5$ and $8 \not< 10$
 (2) $2 + 3 \neq 5$ or $8 > 10$
 (3) $2 + 3 \neq 5$ or $8 \geq 10$
 (4) None of these
19. For any three simple statement p, q, r the statement $(p \wedge q) \vee (q \wedge r)$ is true when-
 (1) p and r true and q is false
 (2) p and r false and q is true
 (3) p, q, r all are false
 (4) q and r true and p is false
20. Which of the following statement is a tautology-
 (1) $(\sim p \vee \sim q) \vee (p \vee \sim q)$
 (2) $(\sim p \vee \sim q) \wedge (p \vee \sim q)$
 (3) $\sim p \wedge (\sim p \vee \sim q)$
 (4) $\sim q \wedge (\sim p \vee \sim q)$
21. Which of the following statement is a contradiction-
 (1) $(\sim p \vee \sim q) \vee (p \vee \sim q)$
 (2) $(p \rightarrow q) \vee (p \wedge \sim q)$
 (3) $(\sim p \wedge q) \wedge (\sim q)$
 (4) $(\sim p \wedge q) \vee (\sim q)$
22. The negation of the statement $q \vee (p \wedge \sim r)$ is equivalent to-
 (1) $\sim q \wedge (p \rightarrow r)$ (2) $\sim q \wedge \sim(p \rightarrow r)$
 (3) $\sim q \wedge (\sim p \wedge r)$ (4) None of these
23. Let Q be a non empty subset of N . and q is a statement as given below :-
 q : There exists an even number $a \in Q$.
 Negation of the statement q will be :-
 (1) There is no even number in the set Q .
 (2) Every $a \in Q$ is an odd number.
 (3) (1) and (2) both
 (4) None of these
24. The statement $\sim(p \rightarrow q) \leftrightarrow (\sim p \vee \sim q)$ is-
 (1) a tautology
 (2) a contradiction
 (3) neither a tautology nor a contradiction
 (4) None of these
25. Which of the following is equivalent to $(p \wedge q)$
 (1) $p \rightarrow \sim q$ (2) $\sim(\sim p \wedge \sim q)$
 (3) $\sim(p \rightarrow \sim q)$ (4) None of these
26. The dual of the following statement "Reena is healthy and Meena is beautiful" is-
 (1) Reena is beautiful and Meena is healthy
 (2) Reena is beautiful or Meena is healthy
 (3) Reena is healthy or Meena is beautiful
 (4) None of these
27. If p is any statement, t and c are a tautology and a contradiction respectively then which of the following is not correct-
 (1) $p \wedge t \equiv p$ (2) $p \wedge c \equiv c$
 (3) $p \vee t \equiv c$ (4) $p \vee c \equiv p$
28. If $S^*(p, q)$ is the dual of the compound statement $S(p, q)$ then $S^*(\sim p, \sim q)$ is equivalent to-
 (1) $S(\sim p, \sim q)$ (2) $\sim S(p, q)$
 (3) $\sim S^*(p, q)$ (4) None of these
29. If p is any statement, t is a tautology and c is a contradiction then which of the following is not correct-
 (1) $p \wedge (\sim c) \equiv p$
 (2) $p \vee (\sim t) \equiv p$
 (3) $t \vee c \equiv p \vee t$
 (4) $(p \wedge t) \vee (p \vee c) \equiv (t \wedge c)$
30. If p, q, r are simple statement with truth values T, F, T respectively then the truth value of $((\sim p \vee q) \wedge \sim r) \rightarrow p$ is-
 (1) True
 (2) False
 (3) True if r is false
 (4) True if q is true
31. Which of the following is wrong-
 (1) $p \vee \sim p$ is a tautology
 (2) $\sim(\sim p) \leftrightarrow p$ is a tautology
 (3) $p \wedge \sim p$ is a contradiction
 (4) $((p \wedge p) \rightarrow q) \rightarrow p$ is a tautology

- 32.** The statement "If $2^2 = 5$ then I get first class" is logically equivalent to-

 - (1) $2^2 = 5$ and I do not get first class
 - (2) $2^2 = 5$ or I do not get first class
 - (3) $2^2 \neq 5$ or I get first class
 - (4) None of these

33. If statement $(p \vee \sim r) \rightarrow (q \wedge r)$ is false and statement q is true then statement p is-

 - (1) true
 - (2) false
 - (3) may be true or false
 - (4) None of these

34. Which of the following statement are not logically equivalent-

 - (1) $\sim(p \vee \sim q)$ and $(\sim p \wedge q)$
 - (2) $\sim(p \rightarrow q)$ and $(p \wedge \sim q)$
 - (3) $(p \rightarrow q)$ and $(\sim q \rightarrow \sim p)$
 - (4) $(p \rightarrow q)$ and $(\sim p \wedge q)$

35. Consider the following statements
 p : Virat kohli plays cricket.
 q : Virat kohli is good at maths
 r : Virat kohli is successful.
 then negation of the statement "If virat kohli plays cricket and is not good at maths then he is successful" will be :-

 - (1) $\sim p \wedge (q \wedge r)$
 - (2) $(\sim p \vee q) \wedge r$
 - (3) $p \wedge (\sim q \wedge \sim r)$
 - (4) None of these

36. Let p statement "If $2 > 5$ then earth will not rotate" and q be the statement " $2 \not> 5$ or earth will not rotate".

Statement-1 : p and q are equivalent.

Statement-2 : $m \rightarrow n$ and $\sim m \vee n$ are equivalent.

 - (1) Statement-1 is true, Statement-2 is true; Statement-2 is not the correct explanation of Statement-1.
 - (2) Statement-1 is false, Statement-2 is true.
 - (3) Statement-1 is true, Statement-2 is false.
 - (4) Statement-1 is true, Statement-2 is true; Statement-2 is the correct explanation of Statement-1.

37. Which of the following is a tautology :-

 - (1) $[(\sim p \wedge p) \rightarrow q] \longrightarrow (p \wedge p)$
 - (2) $[(\sim p \wedge p) \rightarrow q] \longrightarrow (\sim p \rightarrow p)$
 - (3) $[(\sim p \wedge p) \rightarrow q] \longrightarrow (p \rightarrow p)$
 - (4) None of these

38. Negation of the statement "No one in the class is fond of music" is :-

 - (1) everyone in the class is fond of music.
 - (2) Some of the students in the class are fond of music.
 - (3) There exists a student in the class who is fond of music.
 - (4) (2) and (3) both

CHECK YOUR GRASP								ANSWER-KEY					EXERCISE-I							
Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Ans.	3	3	4	2	2	4	3	2	1	1	3	1	4	4	1	4	1	3	4	1
Que.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38		
Ans.	3	1	3	3	3	3	3	2	4	1	4	3	3	4	3	4	3	4		

PREVIOUS YEAR QUESTIONS

STATISTICS

EXERCISE-II

- The negation of the statement
[JEE(Main)-2012]
"If I become a teacher, then I will open a school", is :
(1) I will not become a teacher or I will open a school.
(2) I will become a teacher and I will not open a school.
(3) Either I will not become a teacher or I will not open a school.
(4) Neither I will become a teacher nor I will open a school.
- Consider :
Statement-I : $(p \wedge \sim q) \wedge (\sim p \wedge q)$ is a fallacy.
Statement-II : $(p \rightarrow q) \leftrightarrow (\sim q \rightarrow \sim p)$ is a tautology. [JEE(Main)-2013]
(1) Statement-I is true, Statement-II is true; statement-II is a **correct** explanation for Statement-I.
(2) Statement-I is true, Statement-II is true; statement-II is **not** a correct explanation for Statement-I.
(3) Statement-I is true, Statement-II is false.
(4) Statement-I is false, Statement-II is true.
- The statement $\sim(p \leftrightarrow \sim q)$ is : [JEE(Main)-2014]
(1) equivalent to $p \leftrightarrow q$
(2) equivalent to $\sim p \leftrightarrow q$
(3) a tautology
(4) a fallacy
- The negation of $\sim s \vee (\sim r \wedge s)$ is equivalent to : [JEE(Main)-2015]
(1) $s \vee (r \vee \sim s)$ (2) $s \wedge r$
(3) $s \wedge \sim r$ (4) $s \wedge (r \wedge \sim s)$
- The Boolean Expression $(p \wedge \sim q) \vee q \vee (\sim p \wedge q)$ is equivalent to :- [JEE(Main)-2016]
(1) $p \vee \sim q$ (2) $\sim p \wedge q$
(3) $p \wedge q$ (4) $p \vee q$

- The following statement
 $(p \rightarrow q) \rightarrow [(\sim p \rightarrow q) \rightarrow q]$ is : [JEE(Main)-2017]
(1) a fallacy
(2) a tautology
(3) equivalent to $\sim p \rightarrow q$
(4) equivalent to $p \rightarrow \sim q$
- The Boolean expression $\sim(p \vee q) \vee (\sim p \wedge q)$ is equivalent to : [JEE(Main)-2018]
(1) p (2) q
(3) $\sim q$ (4) $\sim p$
- If the Boolean expression $(p \oplus q) \wedge (\sim p \odot q)$ is equivalent to $p \wedge q$, where $\oplus, \odot \in \{\wedge, \vee\}$, then the ordered pair $(\oplus, \odot)$ is: [JEE(Main)-19]
(1) $(\wedge, \vee)$ (2) $(\vee, \vee)$
(3) $(\wedge, \wedge)$ (4) $(\vee, \wedge)$
- The logical statement
$$[\sim(\sim p \vee q) \vee (p \wedge r) \wedge (\sim q \wedge r)]$$
is equivalent to : [JEE(Main)-19]
(1) $(p \wedge r) \wedge \sim q$ (2) $(\sim p \wedge \sim q) \wedge r$
(3) $\sim p \vee r$ (4) $(p \wedge \sim q) \vee r$
- Consider the following three statements :
P : 5 is a prime number.
Q : 7 is a factor of 192.
R : L.C.M. of 5 and 7 is 35.
Then the truth value of which one of the following statements is true ? [JEE(Main)-19]
(1) $(P \wedge Q) \vee (\sim R)$
(2) $(\sim P) \wedge (\sim Q \wedge R)$
(3) $(\sim P) \vee (Q \wedge R)$
(4) $P \vee (\sim Q \wedge R)$

11. If q is false and $p \wedge q \leftrightarrow r$ is true, then which one of the following statements is a tautology?

[JEE(Main)-19]

- (1) $(p \vee r) \rightarrow (p \wedge r)$
- (2) $p \vee r$
- (3) $p \wedge r$
- (4) $(p \wedge r) \rightarrow (p \vee r)$

12. Contrapositive of the statement

"If two numbers are not equal, then their squares are not equal." is :- [JEE(Main)-19]

- (1) If the squares of two numbers are equal, then the numbers are equal.
- (2) If the squares of two numbers are equal, then the numbers are not equal.
- (3) If the squares of two numbers are not equal, then the numbers are equal.
- (4) If the squares of two numbers are not equal, then the numbers are not equal.

13. The contrapositive of the statement "If you are born in India, then you are a citizen of India", is :

[JEE(Main)-19]

- (1) If you are born in India, then you are not a citizen of India.
- (2) If you are not a citizen of India, then you are not born in India.
- (3) If you are a citizen of India, then you are born in India.
- (4) If you are not born in India, then you are not a citizen of India.

14. For any two statements p and q , the negation of the expression $p \vee (\sim p \wedge q)$ is

[JEE(Main)-19]

- (1) $p \wedge q$
- (2) $p \leftrightarrow q$
- (3) $\sim p \vee \sim q$
- (4) $\sim p \wedge \sim q$

15. If the truth value of the statement $P \rightarrow (\sim p \vee r)$ is false(F), then the truth values of the statements p, q, r are respectively :

[JEE(Main)-19]

- (1) F, T, T
- (2) T, F, F
- (3) T, T, F
- (4) T, F, T

16. The Boolean expression $\sim(p \Rightarrow (\sim q))$ is equivalent to :

[JEE(Main)-19]

- (1) $(\sim p) \Rightarrow q$
- (2) $p \vee q$
- (3) $q \Rightarrow \sim p$
- (4) $p \wedge q$

PREVIOUS YEARS QUESTIONS

ANSWER-KEY

EXERCISE-II

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Ans.	2	2	1	2	4	2	4	1	1	4	4	1	2	4	3	4

NURTURE COURSE
SEQUENCE & SERIES

CONTENTS

SEQUENCE & PROGRESSION

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JEE-Advanced Syllabus :

Arithmetic, geometric and harmonic progressions, arithmetic, geometric and harmonic means, sums of finite arithmetic and geometric progressions, infinite geometric series, sums of squares and cubes of the first n natural numbers.

JEE-Mains Syllabus :

Arithmetic and Geometric progressions, insertion of arithmetic, geometric means between two given numbers. Relation between A.M. and G.M. Sum upto n terms of special series: S_n , S_n^2 , S_n^3 . Arithmetico – Geometric progression.

SEQUENCE & SERIES

1. DEFINITION :

Sequence :

A succession of terms $a_1, a_2, a_3, a_4, \dots$ formed according to some rule or law.

Examples are : 1, 4, 9, 16, 25

$-1, 1, -1, 1, \dots$

$\frac{x}{1!}, \frac{x^2}{2!}, \frac{x^3}{3!}, \frac{x^4}{4!}, \dots$

A finite sequence has a finite (i.e. limited) number of terms, as in the first example above. An infinite sequence has an unlimited number of terms, i.e. there is no last term, as in the second and third examples.

Series :

The indicated sum of the terms of a sequence. In the case of a finite sequence $a_1, a_2, a_3, \dots, a_n$

the corresponding series is $a_1 + a_2 + a_3 + \dots + a_n = \sum_{k=1}^n a_k$. This series has a finite or limited number of terms and is called a finite series.

2. ARITHMETIC PROGRESSION (A.P.) :

A.P. is a sequence whose terms differ by a fixed number. This fixed number is called the common difference. If a is the first term & d the common difference, then A.P. can be written as

$a, a + d, a + 2d, \dots, a + (n - 1)d, \dots$

(a) n^{th} term of AP $T_n = a + (n - 1)d$, where $d = t_n - t_{n-1}$

(b) The sum of the first n terms : $S_n = \frac{n}{2}[a + \ell] = \frac{n}{2}[2a + (n - 1)d]$

where ℓ is n^{th} term.

Note :

(i) n^{th} term of an A.P. is of the form $An + B$ i.e. a linear expression in ' n ', in such a case the coefficient of n is the common difference of the A.P. i.e. A .

(ii) Sum of first ' n ' terms of an A.P. is of the form $An^2 + Bn$ i.e. a quadratic expression in ' n ', in such case the common difference is twice the coefficient of n^2 . i.e. $2A$

(iii) Also n^{th} term $T_n = S_n - S_{n-1}$

Illustration 1 :

If $(x + 1)$, $3x$ and $(4x + 2)$ are first three terms of an A.P. then its 5^{th} term is -

(A) 14

(B) 19

(C) 24

(D) 28

Solution : $(x + 1), 3x, (4x + 2)$ are in AP

$$\Rightarrow 3x - (x + 1) = (4x + 2) - 3x \quad \Rightarrow \quad x = 3$$

$$\therefore a = 4, d = 9 - 4 = 5 \quad \Rightarrow \quad T_5 = 4 + (4)5 = 24$$

Ans. (C)**Illustration 2 :**

The sum of first four terms of an A.P. is 56 and the sum of its last four terms is 112. If its first term is 11 then find the number of terms in the A.P.

Solution :

$$a + a + d + a + 2d + a + 3d = 56$$

$$4a + 6d = 56$$

$$44 + 6d = 56 \quad (\text{as } a = 11)$$

$$6d = 12 \quad \text{hence } d = 2$$

Let total number of terms = n

Now sum of last four terms.

$$a + (n - 1)d + a + (n - 2)d + a + (n - 3)d + a + (n - 4)d = 112$$

$$\Rightarrow 4a + (4n - 10)d = 112 \quad \Rightarrow \quad 44 + (4n - 10)2 = 112$$

$$\Rightarrow 4n - 10 = 34$$

$$\Rightarrow n = 11$$

Ans.**Illustration 3 :**

The sum of first n terms of two A.P.s. are in ratio $\frac{7n+1}{4n+27}$. Find the ratio of their 11th terms.

Solution :

Let a_1 and a_2 be the first terms and d_1 and d_2 be the common differences of two A.P.s respectively then

$$\frac{\frac{n}{2}[2a_1 + (n-1)d_1]}{\frac{n}{2}[2a_2 + (n-1)d_2]} = \frac{7n+1}{4n+27} \Rightarrow \frac{a_1 + \left(\frac{n-1}{2}\right)d_1}{a_2 + \left(\frac{n-1}{2}\right)d_2} = \frac{7n+1}{4n+27}$$

For ratio of 11th terms

$$\frac{n-1}{2} = 10 \Rightarrow n = 21$$

$$\text{so ratio of 11th terms is } \frac{7(21)+1}{4(21)+27} = \frac{148}{111} = \frac{4}{3}$$

Ans.**Do yourself - 1 :**

(i) Write down the sequence whose n^{th} terms is : (a) $\frac{2^n}{n}$ (b) $\frac{3+(-1)^n}{3^n}$

(ii) For an A.P, show that $t_m + t_{2n+m} = 2t_{m+n}$

(iii) If the sum of p terms of an A.P. is q and the sum of its q terms is p , then find the sum of its $(p+q)$ term.

3. PROPERTIES OF A.P. :

- (a) If each term of an A.P. is increased, decreased, multiplied or divided by the some nonzero number, then the resulting sequence is also an A.P.
- (b) Three numbers in A.P. : $a - d, a, a + d$
 Four numbers in A.P. : $a - 3d, a - d, a + d, a + 3d$
 Five numbers in A.P. : $a - 2d, a - d, a, a + d, a + 2d$
 Six numbers in A.P. : $a - 5d, a - 3d, a - d, a + d, a + 3d, a + 5d$ etc.
- (c) The common difference can be zero, positive or negative.
- (d) k^{th} term from the last = $(n - k + 1)^{\text{th}}$ term from the beginning (If total number of terms = n).
- (e) The sum of the two terms of an AP equidistant from the beginning & end is constant and equal to the sum of first & last terms. $\Rightarrow T_k + T_{n-k+1} = \text{constant} = a + \ell$.
- (f) Any term of an AP (except the first) is equal to half the sum of terms which are equidistant from it. $a_n = (1/2)(a_{n-k} + a_{n+k})$, $k < n$
 For $k = 1$, $a_n = (1/2)(a_{n-1} + a_{n+1})$; For $k = 2$, $a_n = (1/2)(a_{n-2} + a_{n+2})$ and so on.
- (g) If a, b, c are in AP, then $2b = a + c$.

Illustration 4 :

Four numbers are in A.P. If their sum is 20 and the sum of their squares is 120, then the middle terms are -

- (A) 2, 4 (B) 4, 6 (C) 6, 8 (D) 8, 10

Solution :

Let the numbers are $a - 3d, a - d, a + d, a + 3d$
 given, $a - 3d + a - d + a + d + a + 3d = 20 \Rightarrow 4a = 20 \Rightarrow a = 5$
 and $(a - 3d)^2 + (a - d)^2 + (a + d)^2 + (a + 3d)^2 = 120 \Rightarrow 4a^2 + 20d^2 = 120$
 $\Rightarrow 4 \times 5^2 + 20d^2 = 120 \Rightarrow d^2 = 1 \Rightarrow d = \pm 1$
 Hence numbers are 2, 4, 6, 8 or 8, 6, 4, 2

Ans. (B)

Illustration 5 :

If $a_1, a_2, a_3, \dots, a_n$ are in A.P. where $a_i > 0$ for all i , show that :

$$\frac{1}{\sqrt{a_1} + \sqrt{a_2}} + \frac{1}{\sqrt{a_2} + \sqrt{a_3}} + \dots + \frac{1}{\sqrt{a_{n-1}} + \sqrt{a_n}} = \frac{(n-1)}{\sqrt{a_1} + \sqrt{a_n}}$$

Solution :

$$\begin{aligned} \text{L.H.S.} &= \frac{1}{\sqrt{a_1} + \sqrt{a_2}} + \frac{1}{\sqrt{a_2} + \sqrt{a_3}} + \dots + \frac{1}{\sqrt{a_{n-1}} + \sqrt{a_n}} \\ &= \frac{1}{\sqrt{a_2} + \sqrt{a_1}} + \frac{1}{\sqrt{a_3} + \sqrt{a_2}} + \dots + \frac{1}{\sqrt{a_n} + \sqrt{a_{n-1}}} \\ &= \frac{\sqrt{a_2} - \sqrt{a_1}}{(a_2 - a_1)} + \frac{\sqrt{a_3} - \sqrt{a_2}}{(a_3 - a_2)} + \dots + \frac{\sqrt{a_n} - \sqrt{a_{n-1}}}{a_n - a_{n-1}} \end{aligned}$$

Let 'd' is the common difference of this A.P.

$$\text{then } a_2 - a_1 = a_3 - a_2 = \dots = a_n - a_{n-1} = d$$

Now L.H.S.

$$\begin{aligned} &= \frac{1}{d} \{ \sqrt{a_2} - \sqrt{a_1} + \sqrt{a_3} - \sqrt{a_2} + \dots + \sqrt{a_{n-1}} - \sqrt{a_{n-2}} + \sqrt{a_n} - \sqrt{a_{n-1}} \} = \frac{1}{d} \{ \sqrt{a_n} - \sqrt{a_1} \} \\ &= \frac{a_n - a_1}{d(\sqrt{a_n} + \sqrt{a_1})} = \frac{a_1 + (n-1)d - a_1}{d(\sqrt{a_n} + \sqrt{a_1})} = \frac{1}{d} \frac{(n-1)d}{(\sqrt{a_n} + \sqrt{a_1})} = \frac{n-1}{\sqrt{a_n} + \sqrt{a_1}} = \text{R.H.S.} \end{aligned}$$

Do yourself - 2 :

(i) Find the sum of first 24 terms of the A.P. $a_1, a_2, a_3, \dots$, if it is known that

$$a_1 + a_5 + a_{10} + a_{15} + a_{20} + a_{24} = 225.$$

(ii) Find the number of terms common to the two A.P.'s 3, 7, 11, 407 and 2, 9, 16,, 709

4. GEOMETRIC PROGRESSION (G.P.) :

G.P. is a sequence of non zero numbers each of the succeeding term is equal to the preceding term multiplied by a constant. Thus in a GP the ratio of successive terms is constant. This constant factor is called the COMMON RATIO of the sequence & is obtained by dividing any term by the immediately previous term. Therefore $a, ar, ar^2, ar^3, ar^4, \dots$ is a GP with 'a' as the first term & 'r' as common ratio.

(a) n^{th} term ; $T_n = a r^{n-1}$

(b) Sum of the first n terms; $S_n = \frac{a(r^n - 1)}{r - 1}$, if $r \neq 1$

(c) Sum of infinite G.P. , $S_\infty = \frac{a}{1 - r}$; $0 < |r| < 1$

5. PROPERTIES OF GP :

(a) If each term of a G.P. be multiplied or divided by the some non-zero quantity, then the resulting sequence is also a G.P.

(b) Three consecutive terms of a GP : $a/r, a, ar$;

Four consecutive terms of a GP : $a/r^3, a/r, ar, ar^3$ & so on.

(c) If a, b, c are in G.P. then $b^2 = ac$.

(d) If in a G.P, the product of two terms which are equidistant from the first and the last term, is constant and is equal to the product of first and last term. $\Rightarrow T_k \cdot T_{n-k+1} = \text{constant} = a \cdot \ell$

(e) If each term of a G.P. be raised to the same power, then resulting sequence is also a G.P.

(f) In a G.P., $T_r^2 = T_{r-k} \cdot T_{r+k}$, $k < r, r \neq 1$

- (g) If the terms of a given G.P. are chosen at regular intervals, then the new sequence is also a G.P.
- (h) If $a_1, a_2, a_3, \dots, a_n$ is a G.P. of positive terms, then $\log a_1, \log a_2, \dots, \log a_n$ is an A.P. and vice-versa.
- (i) If $a_1, a_2, a_3, \dots$ and $b_1, b_2, b_3, \dots$ are two G.P.'s then $a_1 b_1, a_2 b_2, a_3 b_3, \dots$ & $\frac{a_1}{b_1}, \frac{a_2}{b_2}, \frac{a_3}{b_3}, \dots$ is also in G.P.

Illustration 6 :

If a, b, c, d and p are distinct real numbers such that

$$(a^2 + b^2 + c^2)p^2 - 2p(ab + bc + cd) + (b^2 + c^2 + d^2) \leq 0 \text{ then } a, b, c, d \text{ are in}$$

- (A) A.P. (B) G.P. (C) H.P. (D) none of these

Solution :

Here, the given condition $(a^2 + b^2 + c^2)p^2 - 2p(ab + bc + ca) + b^2 + c^2 + d^2 \leq 0$

$$\Rightarrow (ap - b)^2 + (bp - c)^2 + (cp - d)^2 \leq 0$$

$\therefore$ a square can not be negative

$$\therefore ap - b = 0, bp - c = 0, cp - d = 0 \Rightarrow p = \frac{b}{a} = \frac{c}{b} = \frac{d}{c} \Rightarrow a, b, c, d \text{ are in G.P.}$$

Ans. (B)

Illustration 7 :

If a, b, c are in G.P., then the equations $ax^2 + 2bx + c = 0$ and $dx^2 + 2ex + f = 0$ have a common root

if $\frac{d}{a}, \frac{e}{b}, \frac{f}{c}$ are in -

- (A) A.P. (B) G.P. (C) H.P. (D) none of these

Solution :

$$a, b, c \text{ are in G.P.} \Rightarrow b^2 = ac$$

Now the equation $ax^2 + 2bx + c = 0$ can be rewritten as $ax^2 + 2\sqrt{ac}x + c = 0$

$$\Rightarrow (\sqrt{ax} + \sqrt{c})^2 = 0 \Rightarrow x = -\sqrt{\frac{c}{a}}, -\sqrt{\frac{c}{a}}$$

If the two given equations have a common root, then this root must be $-\sqrt{\frac{c}{a}}$.

$$\text{Thus } d\frac{c}{a} - 2e\sqrt{\frac{c}{a}} + f = 0 \Rightarrow \frac{d}{a} + \frac{f}{c} = \frac{2e}{c}\sqrt{\frac{c}{a}} = \frac{2e}{\sqrt{ac}} = \frac{2e}{b} \Rightarrow \frac{d}{a}, \frac{e}{b}, \frac{f}{c} \text{ are in A.P.} \quad \text{Ans. (A)}$$

Illustration 8 :

A number consists of three digits which are in G.P. the sum of the right hand and left hand digits exceeds twice the middle digit by 1 and the sum of the left hand and middle digits is two third of the sum of the middle and right hand digits. Find the numbers.

Solution :Let the three digits be a , ar and ar^2 then number is

$$100a + 10ar + ar^2 \quad \dots(i)$$

Given, $a + ar^2 = 2ar + 1$

or $a(r^2 - 2r + 1) = 1$

or $a(r - 1)^2 = 1 \quad \dots(ii)$

Also given $a + ar = \frac{2}{3}(ar + ar^2)$

$$\Rightarrow 3 + 3r = 2r + 2r^2 \Rightarrow 2r^2 - r - 3 = 0 \Rightarrow (r + 1)(2r - 3) = 0$$

$$\therefore r = -1, 3/2$$

for $r = -1$, $a = \frac{1}{(r-1)^2} = \frac{1}{4} \notin \mathbb{I} \quad \therefore r \neq -1$

for $r = 3/2$, $a = \frac{1}{\left(\frac{3}{2}-1\right)^2} = 4 \quad \{\text{from (ii)}\}$

From (i), number is $400 + 10 \cdot 4 \cdot \frac{3}{2} + 4 \cdot \frac{9}{4} = 469$

Ans.**Illustration 9 :**Find the value of $0.32\overline{58}$ **Solution :**

Let $R = 0.32\overline{58} \Rightarrow R = 0.32585858\dots \dots\dots (i)$

Here number of figures which are not recurring is 2 and number of figures which are recurring is also 2.

then $100R = 32.585858\dots \dots\dots (ii)$

and $10000R = 3258.5858\dots \dots\dots (iii)$

Subtracting (ii) from (iii), we get

$$9900R = 3226 \Rightarrow R = \frac{1613}{4950}$$

Aliter Method : $R = .32 + .0058 + .0058 + .000058 + \dots\dots\dots$

$$= .32 + \frac{58}{10^4} \left(1 + \frac{1}{10^2} + \frac{1}{10^4} + \dots\dots\dots \infty \right)$$

$$= .32 + \frac{58}{10^4} \left(\frac{1}{1 - \frac{1}{100}} \right)$$

$$= \frac{32}{100} + \frac{58}{9900} = \frac{3168 + 58}{9900} = \frac{3226}{9900} = \frac{1613}{4950}$$

Do yourself - 3 :

- (i) Find a three digit number whose consecutive digits form a G.P. If we subtract 792 from this number, we get a number consisting of the same digits written in the reverse order. Now, if we increase the second digit of the required number by 2, then the resulting digits will form an A.P.
- (ii) If the third term of G.P. is 4, then find the product of first five terms.
- (iii) If a, b, c are respectively the p^{th} , q^{th} and r^{th} terms of the given G.P., then show that $(q - r) \log a + (r - p) \log b + (p - q) \log c = 0$, where a, b, c > 0.
- (iv) Find three numbers in G.P., whose sum is 52 and the sum of whose products in pairs is 624.
- (v) The rational number which equals the number $2.\overline{357}$ with recurring decimal is -

(A) $\frac{2357}{999}$

(B) $\frac{2379}{997}$

(C) $\frac{785}{333}$

(D) $\frac{2355}{1001}$

6. HARMONIC PROGRESSION (H.P.) :

A sequence is said to be in H.P. if the reciprocal of its terms are in AP.

If the sequence $a_1, a_2, a_3, \dots, a_n$ is an HP then $1/a_1, 1/a_2, \dots, 1/a_n$ is an AP. Here we do not have the formula for the sum of the n terms of an HP. The general form of a harmonic progression

is $\frac{1}{a}, \frac{1}{a+d}, \frac{1}{a+2d}, \dots, \frac{1}{a+(n-1)d}$

Note : No term of any H.P. can be zero.

(i) If a, b, c are in HP, then $b = \frac{2ac}{a+c}$ or $\frac{a}{c} = \frac{a-b}{b-c}$

Illustration 10 :

The sum of three numbers are in H.P. is 37 and the sum of their reciprocals is $1/4$. Find the numbers.

Solution :

Three numbers are in H.P. can be taken as

$$\frac{1}{a-d}, \frac{1}{a}, \frac{1}{a+d}$$

then $\frac{1}{a-d} + \frac{1}{a} + \frac{1}{a+d} = 37 \dots\dots(i)$

and $\frac{1}{a-d} + \frac{1}{a} + \frac{1}{a+d} = \frac{1}{4} \Rightarrow a = \frac{1}{12}$

from (i), $\frac{12}{1-12d} + 12 + \frac{12}{1+12d} = 37 \Rightarrow \frac{12}{1-12d} + \frac{12}{1+12d} = 25$

$\Rightarrow \frac{24}{1-144d^2} = 25 \Rightarrow 1-144d^2 = \frac{24}{25} \Rightarrow d^2 = \frac{1}{25 \times 144}$

$\therefore d = \pm \frac{1}{60}$

$\therefore a-d, a, a+d$ are $\frac{1}{15}, \frac{1}{12}, \frac{1}{10}$ or $\frac{1}{10}, \frac{1}{12}, \frac{1}{15}$

Hence, three numbers in H.P. are 15, 12, 10 or 10, 12, 15

Ans.

Illustration 11 :

Suppose a is a fixed real number such that $\frac{a-x}{px} = \frac{a-y}{qy} = \frac{a-z}{rz}$

If p, q, r are in A.P., then prove that x, y, z are in H.P.

Solution :

$\because$ p, q, r are in A.P.

$\therefore$ $q - p = r - q$ (i)

$\Rightarrow$ $p - q = q - r = k$ (let)

$$\text{given } \frac{a-x}{px} = \frac{a-y}{qy} = \frac{a-z}{rz} \Rightarrow \frac{\frac{a}{x} - 1}{p} = \frac{\frac{a}{y} - 1}{q} = \frac{\frac{a}{z} - 1}{r}$$

$$\Rightarrow \frac{\left(\frac{a}{x} - 1\right) - \left(\frac{a}{y} - 1\right)}{p - q} = \frac{\left(\frac{a}{y} - 1\right) - \left(\frac{a}{z} - 1\right)}{q - r} \quad (\text{by law of proportion})$$

$$\Rightarrow \frac{\frac{a}{x} - \frac{a}{y}}{k} = \frac{\frac{a}{y} - \frac{a}{z}}{k} \quad \{\text{from (i)}\}$$

$$\Rightarrow a \left(\frac{1}{x} - \frac{1}{y} \right) = a \left(\frac{1}{y} - \frac{1}{z} \right) \Rightarrow \frac{1}{x} - \frac{1}{y} = \frac{1}{y} - \frac{1}{z}$$

$$\therefore \frac{2}{y} = \frac{1}{x} + \frac{1}{z}$$

$$\therefore \frac{1}{x}, \frac{1}{y}, \frac{1}{z} \text{ are in A.P.}$$

Hence x, y, z are in H.P.

Do yourself - 4 :

(i) If the 7th term of a H.P. is 8 and the 8th term is 7. Then find the 28th term.

(ii) In a H.P., if 5th term is 6 and 3rd term is 10. Find the 2nd term.

(iii) If the $p^{\text{th}}, q^{\text{th}}$ and r^{th} terms of a H.P. are a, b, c respectively, then prove that $\frac{q-r}{a} + \frac{r-p}{b} + \frac{p-q}{c} = 0$.

7. MEANS**(a) ARITHMETIC MEAN :**

If three terms are in A.P. then the middle term is called the A.M. between the other two, so if

a, b, c are in A.P., b is A.M. of a & c . So A.M. of a and $c = \frac{a+c}{2} = b$.

n-ARITHMETIC MEANS BETWEEN TWO NUMBERS :

If a, b be any two given numbers & $a, A_1, A_2, \dots, A_n, b$ are in AP, then $A_1, A_2, \dots, A_n$ are the 'n' A.M's between a & b then. $A_1 = a + d, A_2 = a + 2d, \dots, A_n = a + nd$ or $b - d$, where

$$d = \frac{b-a}{n+1}$$

$$\Rightarrow A_1 = a + \frac{b-a}{n+1}, A_2 = a + \frac{2(b-a)}{n+1}, \dots$$

Note : Sum of n A.M's inserted between a & b is equal to n times the single A.M. between a & b i.e. $\sum_{r=1}^n A_r = nA$ where A is the single A.M. between a & b .

(b) GEOMETRIC MEAN :

If a, b, c are in G.P., then b is the G.M. between a & c , $b^2 = ac$. So G.M. of a and $c = \sqrt{ac} = b$

n-GEOMETRIC MEANS BETWEEN TWO NUMBERS :

If a, b are two given positive numbers & $a, G_1, G_2, \dots, G_n, b$ are in G.P. Then $G_1, G_2, G_3, \dots, G_n$ are 'n' G.Ms between a & b . where $b = ar^{n+1} \Rightarrow r = (b/a)^{1/(n+1)}$

$$G_1 = a(b/a)^{1/(n+1)}, \quad G_2 = a(b/a)^{2/(n+1)}, \dots, \quad G_n = a(b/a)^{n/(n+1)}$$

$$= ar, \quad = ar^2, \dots, \quad = ar^n = b/r$$

Note : The product of n G.Ms between a & b is equal to n^{th} power of the single G.M. between a & b i.e. $\prod_{r=1}^n G_r = (G)^n$ where G is the single G.M. between a & b

(c) HARMONIC MEAN :

If a, b, c are in H.P., then b is H.M. between a & c . So H.M. of a and $c = \frac{2ac}{a+c} = b$.

Insertion of 'n' HM's between a and b :

$a, H_1, H_2, H_3, \dots, H_n, b \rightarrow$ H.P

$$\frac{1}{a}, \frac{1}{H_1}, \frac{1}{H_2}, \frac{1}{H_3}, \dots, \frac{1}{H_n}, \frac{1}{b} \rightarrow \text{A.P.}$$

$$\frac{1}{b} = \frac{1}{a} + (n+1)D \Rightarrow D = \frac{\frac{1}{b} - \frac{1}{a}}{n+1}$$

$$\frac{1}{H_n} = \frac{1}{a} + n \left(\frac{\frac{1}{b} - \frac{1}{a}}{n+1} \right)$$

Important note :

- (i) If A, G, H, are respectively A.M., G.M., H.M. between two positive number a & b then
 (a) $G^2 = AH$ (A, G, H constitute a GP) (b) $A \geq G \geq H$ (c) $A = G = H \Leftrightarrow a = b$
 (ii) Let $a_1, a_2, \dots, a_n$ be n positive real numbers, then we define their arithmetic mean (A), geometric mean (G) and harmonic mean (H) as $A = \frac{a_1 + a_2 + \dots + a_n}{n}$

$$G = (a_1 a_2 \dots a_n)^{1/n} \text{ and } H = \frac{n}{\left(\frac{1}{a_1} + \frac{1}{a_2} + \frac{1}{a_3} + \dots + \frac{1}{a_n} \right)}$$

It can be shown that $A \geq G \geq H$. Moreover equality holds at either place if and only if $a_1 = a_2 = \dots = a_n$

Illustration 12 :

If $2x^3 + ax^2 + bx + 4 = 0$ (a and b are positive real numbers) has 3 real roots, then prove that $a + b \geq 6(2^{1/3} + 4^{1/3})$.

Solution :

Let α, β, γ be the roots of $2x^3 + ax^2 + bx + 4 = 0$. Given that all the coefficients are positive, so all the roots will be negative.

$$\text{Let } \alpha_1 = -\alpha, \alpha_2 = -\beta, \alpha_3 = -\gamma \quad \Rightarrow \quad \alpha_1 + \alpha_2 + \alpha_3 = \frac{a}{2}$$

$$\alpha_1 \alpha_2 + \alpha_2 \alpha_3 + \alpha_3 \alpha_1 = \frac{b}{2}$$

$$\alpha_1 \alpha_2 \alpha_3 = 2$$

Applying AM $\geq$ GM, we have

$$\frac{\alpha_1 + \alpha_2 + \alpha_3}{3} \geq (\alpha_1 \alpha_2 \alpha_3)^{1/3} \Rightarrow a \geq 6 \times 2^{1/3}$$

$$\text{Also } \frac{\alpha_1 \alpha_2 + \alpha_2 \alpha_3 + \alpha_1 \alpha_3}{3} > (\alpha_1 \alpha_2 \alpha_3)^{2/3} \Rightarrow b \geq 6 \times 4^{1/3}$$

Therefore $a + b \geq 6(2^{1/3} + 4^{1/3})$.

Illustration 13 :

If $a_i > 0 \forall i \in \mathbb{N}$ such that $\prod_{i=1}^n a_i = 1$, then prove that $(1 + a_1)(1 + a_2)(1 + a_3) \dots (1 + a_n) \geq 2^n$

Solution :

Using A.M. $\geq$ G.M.

$$1 + a_1 \geq 2\sqrt{a_1}$$

$$1 + a_2 \geq 2\sqrt{a_2}$$

$\vdots$

$$1 + a_n \geq 2\sqrt{a_n} \Rightarrow (1 + a_1)(1 + a_2) \dots (1 + a_n) \geq 2^n (a_1 a_2 a_3 \dots a_n)^{1/2}$$

$$\text{As } a_1 a_2 a_3 \dots a_n = 1$$

Hence $(1 + a_1)(1 + a_2) \dots (1 + a_n) \geq 2^n$.

Illustration 14 :

If a, b, x, y are positive natural numbers such that $\frac{1}{x} + \frac{1}{y} = 1$ then prove that $\frac{a^x}{x} + \frac{b^y}{y} \geq ab$.

Solution :

Consider the positive numbers $a^x, a^x, \dots, y$ times and $b^y, b^y, \dots, x$ times

For all these numbers,

$$AM = \frac{\{a^x + a^x + \dots, y \text{ times}\} + \{b^y + b^y + \dots, x \text{ times}\}}{x + y} = \frac{ya^x + xa^y}{(x + y)}$$

$$GM = \left\{ (a^x \cdot a^x \dots, y \text{ times})(b^y \cdot b^y \dots, x \text{ times}) \right\}^{\frac{1}{(x+y)}} = \left[(a^{xy}) \cdot (b^{xy}) \right]^{\frac{1}{(x+y)}} = (ab)^{\frac{xy}{(x+y)}}$$

$$\text{As } \frac{1}{x} + \frac{1}{y} = 1, \frac{x+y}{xy} = 1, \text{ i.e., } x + y = xy$$

$$\text{So using } AM \geq GM \quad \frac{ya^x + xa^y}{x + y} \geq (ab)^{\frac{xy}{x+y}}$$

$$\therefore \frac{ya^x + xa^y}{xy} \geq ab \quad \text{or} \quad \frac{a^x}{x} + \frac{a^y}{y} \geq ab.$$

Do yourself - 5 :

(i) If $\frac{a^n + b^n}{a^{n-1} + b^{n-1}}$ is the G.M. between a & b then find the value of 'n'.

(ii) If b is the harmonic mean between a and c , then prove that $\frac{1}{b-a} + \frac{1}{b-c} = \frac{1}{a} + \frac{1}{c}$.

8. ARITHMETICO - GEOMETRIC SERIES :

A series, each term of which is formed by multiplying the corresponding term of an A.P. & G.P. is called the Arithmetico-Geometric Series, e.g. $1 + 3x + 5x^2 + 7x^3 + \dots$

Here $1, 3, 5, \dots$ are in A.P. & $1, x, x^2, x^3, \dots$ are in G.P.

(a) SUM OF N TERMS OF AN ARITHMETICO-GEOMETRIC SERIES :

$$\text{Let } S_n = a + (a+d)r + (a+2d)r^2 + \dots + [a + (n-1)d]r^{n-1}$$

$$\text{then } S_n = \frac{a}{1-r} + \frac{dr(1-r^{n-1})}{(1-r)^2} - \frac{[a + (n-1)d]r^n}{1-r}, \quad r \neq 1$$

(b) SUM TO INFINITY :

$$\text{If } 0 < |r| < 1 \text{ \& } n \rightarrow \infty, \text{ then } \lim_{n \rightarrow \infty} r^n = 0, S_\infty = \frac{a}{1-r} + \frac{dr}{(1-r)^2}$$

Illustration 15 :

Find the sum of series $4 - 9x + 16x^2 - 25x^3 + 36x^4 - 49x^5 + \dots \infty$.

Solution :

$$\text{Let } S = 4 - 9x + 16x^2 - 25x^3 + 36x^4 - 49x^5 + \dots \infty$$

$$-Sx = -4x + 9x^2 - 16x^3 + 25x^4 - 36x^5 + \dots \infty$$

On subtraction, we get

$$S(1+x) = 4 - 5x + 7x^2 - 9x^3 + 11x^4 - 13x^5 + \dots \infty$$

$$-S(1+x)x = -4x + 5x^2 - 7x^3 + 9x^4 - 11x^5 + \dots \infty$$

On subtraction, we get

$$S(1+x)^2 = 4 - x + 2x^2 - 2x^3 + 2x^4 - 2x^5 + \dots \infty$$

$$= 4 - x + 2x^2(1 - x + x^2 - \dots \infty) = 4 - x + \frac{2x^2}{1+x} = \frac{4+3x+x^2}{1+x}$$

$$S = \frac{4+3x+x^2}{(1+x)^3} \quad \text{Ans.}$$

Illustration 16 :

Find the sum of series upto n terms $\left(\frac{2n+1}{2n-1}\right) + 3\left(\frac{2n+1}{2n-1}\right)^2 + 5\left(\frac{2n+1}{2n-1}\right)^3 + \dots$

Solution :

For $x \neq 1$, let

$$S = x + 3x^2 + 5x^3 + \dots + (2n-3)x^{n-1} + (2n-1)x^n \quad \dots (i)$$

$$\Rightarrow xS = x^2 + 3x^3 + \dots + (2n-5)x^{n-1} + (2n-3)x^n + (2n-1)x^{n+1} \quad \dots (ii)$$

Subtracting (ii) from (i), we get

$$\begin{aligned} (1-x)S &= x + 2x^2 + 2x^3 + \dots + 2x^{n-1} + 2x^n - (2n-1)x^{n+1} = x + \frac{2x^2(1-x^{n-1})}{1-x} - (2n-1)x^{n+1} \\ &= \frac{x}{1-x} [1 - x + 2x - 2x^n - (2n-1)x^n + (2n-1)x^{n+1}] \end{aligned}$$

$$\Rightarrow S = \frac{x}{(1-x)^2} [(2n-1)x^{n+1} - (2n+1)x^n + 1 + x]$$

$$\text{Thus } \left(\frac{2n+1}{2n-1}\right) + 3\left(\frac{2n+1}{2n-1}\right)^2 + \dots + (2n-1)\left(\frac{2n+1}{2n-1}\right)^n$$

$$= \left(\frac{2n+1}{2n-1}\right) \left(\frac{2n-1}{2}\right)^2 \left[(2n-1)\left(\frac{2n+1}{2n-1}\right)^{n+1} - (2n+1)\left(\frac{2n+1}{2n-1}\right)^n + 1 + \frac{2n+1}{2n-1} \right]$$

$$= \frac{4n^2-1}{4} \cdot \frac{4n}{2n-1} = n(2n+1)$$

Ans.

Do yourself - 6 :

(i) Find sum to n terms of the series $3 + 5 \times \frac{1}{4} + 7 \times \frac{1}{4^2} + \dots$

(ii) If the sum to the infinity of the series $3 + 5r + 7r^2 + \dots$ is $\frac{44}{9}$, then find the value of r.

(iii) If the sum to infinity of the series $3 + (3+d) \cdot \frac{1}{4} + (3+2d) \cdot \frac{1}{4^2} + \dots$ is $\frac{44}{9}$ then find d.

9. SIGMA NOTATIONS (Σ)

THEOREMS :

$$(a) \sum_{r=1}^n (a_r \pm b_r) = \sum_{r=1}^n a_r \pm \sum_{r=1}^n b_r \quad (b) \sum_{r=1}^n k a_r = k \sum_{r=1}^n a_r \quad (c) \sum_{r=1}^n k = nk \text{ ; where } k \text{ is a constant.}$$

10. RESULTS

$$(a) \sum_{r=1}^n r = \frac{n(n+1)}{2} \text{ (sum of the first } n \text{ natural numbers)}$$

$$(b) \sum_{r=1}^n r^2 = \frac{n(n+1)(2n+1)}{6} \text{ (sum of the squares of the first } n \text{ natural numbers)}$$

$$(c) \sum_{r=1}^n r^3 = \frac{n^2(n+1)^2}{4} = \left[\sum_{r=1}^n r \right]^2 \text{ (sum of the cubes of the first } n \text{ natural numbers)}$$

$$(d) \sum_{r=1}^n r^4 = \frac{n}{30} (n+1)(2n+1)(3n^2+3n-1)$$

$$(e) \sum_{r=1}^n (2r-1) = n^2 \text{ (sum of first } n \text{ odd natural numbers)}$$

$$(f) \sum_{r=1}^n 2r = n(n+1) \text{ (sum of first } n \text{ even natural numbers)}$$

Note :

If n^{th} term of a sequence is given by $T_n = an^3 + bn^2 + cn + d$ where a, b, c, d are constants,
then sum of n terms $S_n = \Sigma T_n = a\Sigma n^3 + b\Sigma n^2 + c\Sigma n + \Sigma d$

This can be evaluated using the above results.

Illustration 17 :

Sum up to 16 terms of the series $\frac{1^3}{1} + \frac{1^3+2^3}{1+3} + \frac{1^3+2^3+3^3}{1+3+5} + \dots$ is

- (A) 450 (B) 456 (C) 446 (D) none of these

Solution :

$$t_n = \frac{1^3 + 2^3 + 3^3 + \dots + n^3}{1 + 3 + 5 + \dots (2n-1)} = \frac{\left\{ \frac{n(n+1)}{2} \right\}^2}{\frac{n}{2} \{2 + 2(n-1)\}} = \frac{\frac{n^2(n+1)^2}{4}}{n^2} = \frac{(n+1)^2}{4} = \frac{n^2}{4} + \frac{n}{2} + \frac{1}{4}$$

$$\therefore S_n = \Sigma t_n = \frac{1}{4} \Sigma n^2 + \frac{1}{2} \Sigma n + \frac{1}{4} \Sigma 1 = \frac{1}{4} \cdot \frac{n(n+1)(2n+1)}{6} + \frac{1}{2} \cdot \frac{n(n+1)}{2} + \frac{1}{4} \cdot n$$

$$\therefore S_{16} = \frac{16 \cdot 17 \cdot 33}{24} + \frac{16 \cdot 17}{4} + \frac{16}{4} = 446$$

Ans. (C)

11. METHOD OF DIFFERENCE :

Some times the n^{th} term of a sequence or a series can not be determined by the method, we have discussed earlier. So we compute the difference between the successive terms of given sequence for obtained the n^{th} terms.

If $T_1, T_2, T_3, \dots, T_n$ are the terms of a sequence then some times the terms $T_2 - T_1, T_3 - T_2, \dots$ constitute an AP/GP. n^{th} term of the series is determined & the sum to n terms of the sequence can easily be obtained.

Case 1 :

- (a) If difference series are in A.P., then

Let $T_n = an^2 + bn + c$, where a, b, c are constant

- (b) If difference of difference series are in A.P.

Let $T_n = an^3 + bn^2 + cn + d$, where a, b, c, d are constant

Case 2 :

- (a) If difference are in G.P., then

Let $T_n = ar^n + b$, where r is common ratio & a, b are constant

- (b) If difference of difference are in G.P., then

Let $T_n = ar^n + bn + c$, where r is common ratio & a, b, c are constant

Determine constant by putting $n = 1, 2, 3, \dots, n$ and putting the value of $T_1, T_2, T_3, \dots$

and sum of series $(S_n) = \sum T_n$

Do yourself - 7 :

- (i) Find the sum of the series upto n terms $1 + \frac{1+2}{2} + \frac{1+2+3}{3} + \frac{1+2+3+4}{4} + \dots$
- (ii) Find the sum of ' n ' terms of the series whose n^{th} term is $t_n = 3n^2 + 2n$.

Miscellaneous Illustration :**Illustration 18 :**

If $\sum_{r=1}^n T_r = \frac{n}{8}(n+1)(n+2)(n+3)$, then find $\sum_{r=1}^n \frac{1}{T_r}$.

Solution : $\therefore T_n = S_n - S_{n-1}$

$$= \sum_{r=1}^n T_r - \sum_{r=1}^{n-1} T_r = \frac{n(n+1)(n+2)(n+3)}{8} - \frac{(n-1)n(n+1)(n+2)}{8} = \frac{n(n+1)(n+2)}{8} [(n+3) - (n-1)]$$

$$T_n = \frac{n(n+1)(n+2)}{8} (4) = \frac{n(n+1)(n+2)}{2}$$

$$\Rightarrow \frac{1}{T_n} = \frac{2}{n(n+1)(n+2)} = \frac{(n+2) - n}{n(n+1)(n+2)} = \frac{1}{n(n+1)} - \frac{1}{(n+1)(n+2)} \quad \dots\dots\dots (i)$$

$$\text{Let } V_n = \frac{1}{n(n+1)}$$

$$\therefore \frac{1}{T_n} = V_n - V_{n+1}$$

Putting $n = 1, 2, 3, \dots, n$

$$\Rightarrow \frac{1}{T_1} + \frac{1}{T_2} + \frac{1}{T_3} + \dots + \frac{1}{T_n} = (V_1 - V_{n+1}) \Rightarrow \sum_{r=1}^n \frac{1}{T_r} = \frac{n^2 + 3n}{2(n+1)(n+2)}$$

Illustration 19 :

Find the sum of n terms of the series $1 \cdot 3 \cdot 5 + 3 \cdot 5 \cdot 7 + 5 \cdot 7 \cdot 9 + \dots$

Solution :

The n^{th} term is $(2n-1)(2n+1)(2n+3)$

$$T_n = (2n-1)(2n+1)(2n+3)$$

$$T_n = \frac{1}{8}(2n-1)(2n+1)(2n+3)\{(2n+5) - (2n-3)\}$$

$$= \frac{1}{8}(V_n - V_{n-1}) \quad [\text{Let } V_n = (2n-1)(2n+1)(2n+3)(2n+5)]$$

$$S_n = \sum T_n = \frac{1}{8}[V_n - V_0]$$

$$\therefore S_n = \frac{(2n-1)(2n+1)(2n+3)(2n+5)}{8} + \frac{15}{8} = n(2n^3 + 8n^2 + 7n - 2)$$

Ans.

Illustration 20 :

Find the sum of n terms of the series $3 + 7 + 14 + 24 + 37 + \dots$

Solution :

Clearly here the differences between the successive terms are

$7-3, 14-7, 24-14, \dots$ i.e. $4, 7, 10, 13, \dots$, which are in A.P.

$$\text{Let } S = 3 + 7 + 14 + 24 + \dots + T_n$$

$$S = 3 + 7 + 14 + \dots + T_{n-1} + T_n$$

Subtracting, we get

$$0 = 3 + [4 + 7 + 10 + 13 + \dots (n-1) \text{ terms}] - T_n$$

$$\therefore T_n = 3 + S_{n-1} \text{ of an A.P. whose } a = 4 \text{ and } d = 3.$$

$$\therefore T_n = 3 + \left(\frac{n-1}{2}\right)(2 \cdot 4 + (n-2)3) = \frac{6 + (n-1)(3n+2)}{4} \text{ or, } T_n = \frac{1}{2}(3n^2 - n + 4)$$

Now putting $n = 1, 2, 3, \dots, n$ and adding

$$\therefore S_n = \frac{1}{2} [3 \sum n^2 - \sum n + 4n] = \frac{1}{2} \left[3 \frac{n(n+1)(2n+1)}{6} - \frac{n(n+1)}{2} + 4n \right] = \frac{n}{2} (n^2 + n + 4) \quad \text{Ans.}$$

Aliter Method :

$$\text{Let } T_n = an^2 + bn + c$$

$$\text{Now, } T_1 = 3 = a + b + c \quad \dots(i)$$

$$T_2 = 7 = 4a + 2b + c \quad \dots(ii)$$

$$T_3 = 14 = 9a + 3b + c \quad \dots(iii)$$

Solving (i), (ii) & (iii) we get

$$a = \frac{3}{2}, b = -\frac{1}{2} \text{ \& } c = 2$$

$$\therefore T_n = \frac{1}{2} (3n^2 - n + 4)$$

$$\Rightarrow S_n = \sum T_n = \frac{1}{2} [3 \sum n^2 - \sum n + 4n] = \frac{1}{2} \left[3 \frac{n(n+1)(2n+1)}{6} - \frac{n(n+1)}{2} + 4n \right] = \frac{n}{2} (n^2 + n + 4) \quad \text{Ans.}$$

Illustration 21 :

Find the sum of n -terms of the series $1 + 4 + 10 + 22 + \dots$

Solution :

$$\text{Let } S = 1 + 4 + 10 + 22 + \dots + T_n \quad \dots(i)$$

$$S = 1 + 4 + 10 + \dots + T_{n-1} + T_n \quad \dots(ii)$$

$$(i) - (ii) \Rightarrow T_n = 1 + (3 + 6 + 12 + \dots + T_n - T_{n-1})$$

$$T_n = 1 + 3 \left(\frac{2^{n-1} - 1}{2 - 1} \right)$$

$$T_n = 3 \cdot 2^{n-1} - 2$$

$$\text{So } S_n = \sum T_n = 3 \sum 2^{n-1} - \sum 2$$

$$= 3 \left(\frac{2^n - 1}{2 - 1} \right) - 2n = 3 \cdot 2^n - 2n - 3$$

Ans.

Aliter Method :

$$\text{Let } T_n = ar^n + b, \text{ where } r = 2$$

$$\text{Now } T_1 = 1 = ar + b \quad \dots(i)$$

$$T_2 = 4 = ar^2 + b \quad \dots(ii)$$

Solving (i) & (ii), we get

$$a = \frac{3}{2}, b = -2$$

$$\therefore T_n = 3 \cdot 2^{n-1} - 2$$

$$\Rightarrow S_n = \sum T_n = 3 \sum 2^{n-1} - \sum 2$$

$$= 3 \left(\frac{2^n - 1}{2 - 1} \right) - 2n = 3 \cdot 2^n - 2n - 3$$

Ans.

Illustration 22 :

The series of natural numbers is divided into groups (1), (2, 3, 4), (5, 6, 7, 8, 9) and so on. Show that the sum of the numbers in n^{th} group is $n^3 + (n - 1)^3$

Solution :

The groups are (1), (2, 3, 4), (5, 6, 7, 8, 9)

The number of terms in the groups are 1, 3, 5.....

$\therefore$ The number of terms in the n^{th} group $= (2n - 1)$

the last term of the n^{th} group is n^2

If we count from last term common difference should be -1

$$\text{So the sum of numbers in the } n^{\text{th}} \text{ group} = \left(\frac{2n-1}{2} \right) \{ 2n^2 + (2n-2)(-1) \}$$

$$= (2n-1)(n^2 - n + 1) = 2n^3 - 3n^2 + 3n - 1 = n^3 + (n-1)^3$$

Illustration 23 :

Find the natural number 'a' for which $\sum_{k=1}^n f(a+k) = 16(2^n - 1)$, where the function f satisfied

$f(x+y) = f(x) \cdot f(y)$ for all natural number x, y and further $f(1) = 2$.

Solution :

It is given that

$$f(x+y) = f(x) \cdot f(y) \text{ and } f(1) = 2$$

$$f(1+1) = f(1) \cdot f(1) \Rightarrow f(2) = 2^2, f(1+2) = f(1) \cdot f(2) \Rightarrow f(3) = 2^3, f(2+2) = f(2) \cdot f(2) \Rightarrow f(4) = 2^4$$

Similarly $f(k) = 2^k$ and $f(a) = 2^a$

$$\text{Hence, } \sum_{k=1}^n f(a+k) = \sum_{k=1}^n f(a) \cdot f(k) = f(a) \sum_{k=1}^n f(k) = 2^a \sum_{k=1}^n 2^k = 2^a \{ 2^1 + 2^2 + \dots + 2^n \}$$

$$= 2^a \left\{ \frac{2(2^n - 1)}{2 - 1} \right\} = 2^{a+1}(2^n - 1)$$

But $\sum_{k=1}^n f(a+k) = 16(2^n - 1)$

$$2^{a+1}(2^n - 1) = 16(2^n - 1)$$

$$\therefore 2^{a+1} = 2^4$$

$$\therefore a+1 = 4 \Rightarrow a = 3$$

Ans.

ANSWERS FOR DO YOURSELF

1: (i) (a) $\frac{2}{1}, \frac{4}{2}, \frac{8}{3}, \frac{16}{4}, \dots$, (b) $\frac{2}{3}, \frac{4}{9}, \frac{2}{27}, \frac{4}{81}, \dots$; (iii) $-(p+q)$

2: (i) 900 (ii) 14

3: (i) 931 (ii) 4^5 (iv) 4, 12, 36 (v) C

4: (i) 2 (ii) 15

5: (i) $\frac{1}{2}$

6: (i) $4 + \frac{8}{9} \left(1 - \frac{1}{4^{n-1}} \right) - \left(\frac{2n+1}{3 \times 4^{n-1}} \right)$ (ii) $\frac{1}{4}$ (iii) 2

7: (i) $\frac{n(n+3)}{4}$ (ii) $\frac{n(n+1)(2n+3)}{2}$

EXERCISE (O-1)

- If $a_1, a_2, a_3, \dots, a_n, \dots$ are in A.P. such that $a_4 - a_7 + a_{10} = m$, then the sum of first 13 terms of this A.P., is : [JEE-MAINS Online 2013]
 (A) 15 m (B) 10 m (C) 12 m (D) 13 m
- Let $a_1, a_2, a_3, \dots$ be an A.P. such that $\frac{a_1 + a_2 + \dots + a_p}{a_1 + a_2 + a_3 + \dots + a_q} = \frac{p^3}{q^3}$; $p \neq q$. Then $\frac{a_6}{a_{21}}$ is equal to : [JEE-MAINS Online 2013]
 (A) $\frac{121}{1861}$ (B) $\frac{11}{41}$ (C) $\frac{121}{1681}$ (D) $\frac{41}{11}$
- Given sum of the first n terms of an A. P. is $2n + 3n^2$. Another A. P. is formed with the same first term and double of the common difference, the sum of n terms of the new A. P. is :- [JEE-MAINS Online 2013]
 (A) $n + 4n^2$ (B) $n^2 + 4n$ (C) $3n + 2n^2$ (D) $6n^2 - n$
- If a, b, c are in AP, then $(a - c)^2$ equals
 (A) $4(b^2 - ac)$ (B) $4(b^2 + ac)$ (C) $4b^2 - ac$ (D) $b^2 - 4ac$
- If the sum of n terms of an AP is $Pn + Qn^2$, where P, Q are constants, then its common difference is
 (A) $2Q$ (B) $P + Q$ (C) $2P$ (D) $P - Q$
- Given a sequence of 4 numbers, first three of which are in G.P. and the last three are in A.P. with common difference six. If first and last terms of this sequence are equal, then the last term is : [JEE-MAINS Online 2013]
 (A) 8 (B) 16 (C) 2 (D) 4
- The first term of an infinite G.P. is 1 and every term is equals to the sum of the successive terms, then its fourth term will be-
 (A) $\frac{1}{2}$ (B) $\frac{1}{8}$ (C) $\frac{1}{4}$ (D) $\frac{1}{16}$
- If G be the GM between x and y , then the value of $\frac{1}{G^2 - x^2} + \frac{1}{G^2 - y^2}$ is equal to
 (A) G^2 (B) $\frac{2}{G^2}$ (C) $\frac{1}{G^2}$ (D) $3G^2$
- $2 + 4 + 7 + 11 + 16 + \dots$ to n terms =
 (A) $\frac{1}{6}(n^2 + 3n + 8)$ (B) $\frac{n}{6}(n^2 + 3n + 8)$ (C) $\frac{1}{6}(n^2 - 3n + 8)$ (D) $\frac{n}{6}(n^2 - 3n + 8)$

10. If a, b, c are in HP, then $\frac{a-b}{b-c}$ is equal to
- (A) $\frac{a}{b}$ (B) $\frac{b}{a}$ (C) $\frac{a}{c}$ (D) $\frac{c}{b}$
11. If a, b and c are positive real numbers then $\frac{a}{b} + \frac{b}{c} + \frac{c}{a}$ is greater than or equal to
- (A) 3 (B) 6 (C) 27 (D) 5
12. If $a_1, a_2, a_3, \dots, a_n \in \mathbb{R}^+$ and $a_1 \cdot a_2 \cdot a_3 \cdot \dots \cdot a_n = 1$, then minimum value of $(1+a_1+a_1^2)(1+a_2+a_2^2) \dots (1+a_n+a_n^2)$ is equal to :-
- (A) 3^{n+1} (B) 3^n (C) 3^{n-1} (D) none of these
13. The value of $1^2 + 3^2 + 5^2 + \dots + 25^2$ is : [JEE-MAINS Online 2013]
- (A) 1728 (B) 1456 (C) 2925 (D) 1469
14. The sum of the series : $1 + \frac{1}{1+2} + \frac{1}{1+2+3} + \dots$ up to 10 terms, is:
- [JEE-MAINS Online 2013]
- (A) $\frac{22}{13}$ (B) $\frac{18}{11}$ (C) $\frac{20}{11}$ (D) $\frac{16}{9}$
15. If $a = \sum_{n=0}^{\infty} x^n, b = \sum_{n=0}^{\infty} y^n, c = \sum_{n=0}^{\infty} (xy)^n$ where $|x|, |y| < 1$; then-
- (A) $abc = a + b + c$ (B) $ab + bc = ac + b$ (C) $ac + bc = ab + c$ (D) $ab + ac = bc + a$
16. If $r > 1$ and $x = a + \frac{a}{r} + \frac{a}{r^2} + \dots$ to ∞ , $y = b - \frac{b}{r} + \frac{b}{r^2} - \dots$ to ∞ and $z = c + \frac{c}{r^2} + \frac{c}{r^4} + \dots$ to ∞ , then
- $\frac{xy}{z} =$
- (A) $\frac{ab}{c}$ (B) $\frac{ac}{b}$ (C) $\frac{bc}{a}$ (D) None of these
17. If a, b, c are positive real numbers such that $ab^2c^3 = 64$ then minimum value of $\left(\frac{1}{a} + \frac{2}{b} + \frac{3}{c}\right)$ is equal to:-
- (A) 6 (B) 2 (C) 3 (D) None of these
18. In a GP, first term is 1. If $4T_2 + 5T_3$ is minimum, then its common ratio is
- (A) $\frac{2}{5}$ (B) $-\frac{2}{5}$ (C) $\frac{3}{5}$ (D) $-\frac{3}{5}$

19. The sum $\frac{3}{1^2} + \frac{5}{1^2+2^2} + \frac{7}{1^2+2^2+3^2} + \dots$ upto 11-terms is :- [JEE-MAINS Online 2013]
- (A) $\frac{11}{4}$ (B) $\frac{60}{11}$ (C) $\frac{7}{2}$ (D) $\frac{11}{2}$
20. The sum of the series : $(2)^2 + 2(4)^2 + 3(6)^2 + \dots$ upto 10 terms is : [JEE-MAINS Online 2013]
- (A) 11300 (B) 12100 (C) 12300 (D) 11200

EXERCISE (O-2)

1. If $x = \sum_{n=0}^{\infty} a^n$, $y = \sum_{n=0}^{\infty} b^n$, $z = \sum_{n=0}^{\infty} c^n$ where a, b, c are in A.P. and $|a| < 1$, $|b| < 1$, $|c| < 1$, then x, y, z are in- [AIEEE 2005]
- (A) HP (B) Arithmetic - Geometric Progression
(C) AP (D) GP
2. If a, b, c are distinct positive real in H.P., then the value of the expression, $\frac{b+a}{b-a} + \frac{b+c}{b-c}$ is equal to
- (A) 1 (B) 2 (C) 3 (D) 4
3. Along a road lies an odd number of stones placed at intervals of 10 m. These stones have to be assembled around the middle stone. A person can carry only one stone at a time. A man carried out the job starting with the stone in the middle, carrying stones in succession, thereby covering a distance of 4.8 km. Then the number of stones is
- (A) 15 (B) 29 (C) 31 (D) 35
4. If $S = 1^2 + 3^2 + 5^2 + \dots + (99)^2$ then the value of the sum $2^2 + 4^2 + 6^2 + \dots + (100)^2$ is
- (A) $S + 2550$ (B) $2S$ (C) $4S$ (D) $S + 5050$
5. In an A.P. with first term 'a' and the common difference d ($a, d \neq 0$), the ratio ' ρ ' of the sum of the first n terms to sum of n terms succeeding them does not depend on n . Then the ratio $\frac{a}{d}$ and the ratio ' ρ ', respectively are
- (A) $\frac{1}{2}, \frac{1}{4}$ (B) $2, \frac{1}{3}$ (C) $\frac{1}{2}, \frac{1}{3}$ (D) $\frac{1}{2}, 2$
6. The arithmetic mean of the nine numbers in the given set $\{9, 99, 999, \dots, 999999999\}$ is a 9 digit number N , all whose digits are distinct. The number N does not contain the digit
- (A) 0 (B) 2 (C) 5 (D) 9
7. If for an A.P. $a_1, a_2, a_3, \dots, a_n, \dots$
 $a_1 + a_3 + a_5 = -12$ and $a_1 a_2 a_3 = 8$, then the value of $a_2 + a_4 + a_6$ equals
- (A) -12 (B) -16 (C) -18 (D) -21

8. An H.M. is inserted between the number $1/3$ and an unknown number. If we diminish the reciprocal of the inserted number by 6, it is the G.M. of the reciprocal of $1/3$ and that of the unknown number. If all the terms of the respective H.P. are distinct then
 (A) the unknown number is 27 (B) the unknown number is $1/27$
 (C) the H.M. is 15 (D) the G.M. is 21
9. If $x \in \mathbb{R}$, the numbers $(5^{1+x} + 5^{1-x})$, $a/2$, $(25^x + 25^{-x})$ form an A.P. then 'a' must lie in the interval
 (A) $[1, 5]$ (B) $[2, 5]$ (C) $[5, 12]$ (D) $[12, \infty)$
10. If the sum of the first 11 terms of an arithmetical progression equals that of the first 19 terms, then the sum of its first 30 terms, is
 (A) equal to 0 (B) equal to -1 (C) equal to 1 (D) non unique
11. Let $s_1, s_2, s_3, \dots$ and $t_1, t_2, t_3, \dots$ are two arithmetic sequences such that $s_1 = t_1 \neq 0$; $s_2 = 2t_2$ and
 $\sum_{i=1}^{10} s_i = \sum_{i=1}^{15} t_i$. Then the value of $\frac{s_2 - s_1}{t_2 - t_1}$ is
 (A) $8/3$ (B) $3/2$ (C) $19/8$ (D) 2
12. If $\frac{1+3+5+\dots \text{upto } n \text{ terms}}{4+7+10+\dots \text{upto } n \text{ terms}} = \frac{20}{7 \log_{10} x}$ and
 $n = \log_{10} x + \log_{10} x^{\frac{1}{2}} + \log_{10} x^{\frac{1}{4}} + \log_{10} x^{\frac{1}{8}} + \dots + \infty$, then x is equal to
 (A) 10^3 (B) 10^5 (C) 10^6 (D) 10^7
13. Let $a_n, n \in \mathbb{N}$ is an A.P. with common difference 'd' and all whose terms are non-zero. If n approaches infinity, then the sum $\frac{1}{a_1 a_2} + \frac{1}{a_2 a_3} + \dots + \frac{1}{a_n a_{n+1}}$ will approach
 (A) $\frac{1}{a_1 d}$ (B) $\frac{2}{a_1 d}$ (C) $\frac{1}{2a_1 d}$ (D) $a_1 d$
14. The sum of the first three terms of an increasing G.P. is 21 and the sum of their squares is 189. Then the sum of its first n terms is
 (A) $3(2^n - 1)$ (B) $12\left(1 - \frac{1}{2^n}\right)$ (C) $6\left(1 - \frac{1}{2^n}\right)$ (D) $6(2^n - 1)$
15. If $a \neq 1$ and $\ln a^2 + (\ln a^2)^2 + (\ln a^2)^3 + \dots = 3(\ln a + (\ln a)^2 + (\ln a)^3 + (\ln a)^4 + \dots)$ then 'a' is equal to
 (A) $e^{1/5}$ (B) $\sqrt{e}$ (C) $\sqrt[3]{e}$ (D) $\sqrt[4]{e}$
16. If $abcd = 1$ where a, b, c, d are positive reals then the minimum value of
 $a^2 + b^2 + c^2 + d^2 + ab + ac + ad + bc + bd + cd$ is
 (A) 6 (B) 10 (C) 12 (D) 20

17. For which positive integers n is the ratio, $\frac{\sum_{k=1}^n k^2}{\sum_{k=1}^n k}$ an integer?
- (A) odd n only (B) even n only
(C) $n = 1 + 6k$ only, where $k \geq 0$ and $k \in \mathbb{I}$ (D) $n = 1 + 3k$, integer $k \geq 0$
18. **Statement-1:** If $27abc \geq (a+b+c)^3$ and $3a+4b+5c=12$ then $\frac{1}{a^2} + \frac{1}{b^3} + \frac{1}{c^5} = 10$; where a, b, c are positive real numbers.
- Statement-2:** For positive real numbers $A.M. \geq G.M.$
- (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.

MULTIPLE

19. Let $a_1, a_2, a_3, \dots$ and $b_1, b_2, b_3, \dots$ be arithmetic progressions such that $a_1 = 25, b_1 = 75$ and $a_{100} + b_{100} = 100$. Then
- (A) the difference between successive terms in progression 'a' is opposite of the difference in progression 'b'.
(B) $a_n + b_n = 100$ for any n .
(C) $(a_1 + b_1), (a_2 + b_2), (a_3 + b_3), \dots$ are in A.P.
(D) $\sum_{r=1}^{100} (a_r + b_r) = 10000$
20. If $\sin(x-y), \sin x$ and $\sin(x+y)$ are in H.P., then $\sin x \cdot \sec \frac{y}{2}$ can be
- (A) 2 (B) $\sqrt{2}$ (C) $-\sqrt{2}$ (D) -2

EXERCISE (S-1)

- The sum of n terms of two arithmetic series are in the ratio of $(7n+1) : (4n+27)$. Find the ratio of their n^{th} term.
- In an AP of which 'a' is the 1st term, if the sum of the 1st p terms is equal to zero, show that the sum of the next q terms is $-\left(\frac{aq(p+q)}{p-1}\right)$
- The interior angles of a convex polygon form an arithmetic progression with a common difference of 4° . Determine the number of sides of the polygon if its largest interior angle is 172° .

4. There are n AM's between 1 & 31 such that 7th mean : $(n-1)^{\text{th}}$ mean = 5 : 9, then find the value of n .
5. Find the value of the sum $\sum_{k=0}^{359} k \cdot \cos k^\circ$.
6. The first term of an arithmetic progression is 1 and the sum of the first nine terms equal to 369. The first and the ninth term of a geometric progression coincide with the first and the ninth term of the arithmetic progression. Find the seventh term of the geometric progression.
7. The sequence $a_1, a_2, a_3, \dots, a_{98}$ satisfies the relation $a_{n+1} = a_n + 1$ for $n = 1, 2, 3, \dots, 97$ and has the sum equal to 4949. Evaluate $\sum_{k=1}^{49} a_{2k}$.
8. For an increasing G.P. $a_1, a_2, a_3, \dots, a_n$, if $a_6 = 4a_4$, $a_9 - a_7 = 192$, then the value of $\sum_{i=1}^{\infty} \frac{1}{a_i}$ is
9. In a set of four numbers, the first three are in GP & the last three are in A.P. with common difference 6. If the first number is the same as the fourth, find the four numbers.
10. Find three numbers a, b, c between 2 & 18 such that ;
 (i) their sum is 25
 (ii) the numbers 2, a, b are consecutive terms of an AP &
 (iii) the numbers $b, c, 18$ are consecutive terms of a G.P.
11. If the 10th term of an HP is 21 and 21st term of the same HP is 10, then find the 210th term.
12. The p^{th} term T_p of H.P. is $q(p+q)$ and q^{th} term T_q is $p(p+q)$ when $p > 2, q > 2$. Prove that
 (a) $T_{p+q} = pq$; (b) $T_{pq} = p+q$; (c) $T_{p+q} > T_{pq}$
13. (a) The harmonic mean of two numbers is 4. The arithmetic mean A & the geometric mean G satisfy the relation $2A + G^2 = 27$. Find the two numbers.
 (b) The AM of two numbers exceeds their GM by 15 & HM by 27. Find the numbers.
14. If $A_1, A_2, A_3, \dots, A_{51}$ are arithmetic means inserted between the numbers a and b , then find the value of $\left(\frac{b + A_{51}}{b - A_{51}} \right) - \left(\frac{A_1 + a}{A_1 - a} \right)$.
15. If $a > 0$, then minimum value of $a + 2a^2 + a^3 + 15 + a^{-1} + a^{-3} + a^{-4}$ is
16. If $a, b, c, d > 0$ such that $a + 2b + 3c + 4d = 50$, then find the maximum value of $\left(\frac{a^2 b^4 c^3 d}{16} \right)^{1/10}$
17. If number of coins earned in n^{th} game is $n2^{n+2} - 2^n$ and total number of coins earned in first 10 games is $10(B \cdot 2^{10} + 1)$, where $B \in \mathbb{N}$, then the value of B is
18. Find the n^{th} term and the sum to n terms of the sequence :
 (i) $1 + 5 + 13 + 29 + 61 + \dots$ (ii) $6 + 13 + 22 + 33 + \dots$
19. Sum the following series to n terms and to infinity :

(i) $\frac{1}{1.4.7} + \frac{1}{4.7.10} + \frac{1}{7.10.13} + \dots$

(ii) $\sum_{r=1}^n r(r+1)(r+2)(r+3)$

(iii) $\sum_{r=1}^n \frac{1}{4r^2 - 1}$

EXERCISE (S-2)

1. If the first 3 consecutive terms of a geometrical progression are the real roots of the equation $2x^3 - 19x^2 + 57x - 54 = 0$ find the sum to infinite number of terms of G.P.
2. Find the sum of the infinite series $\frac{1.3}{2} + \frac{3.5}{2^2} + \frac{5.7}{2^3} + \frac{7.9}{2^4} + \dots \infty$.
3. Let $S = \sum_{n=1}^{99} \frac{5^{100}}{(25)^n + 5^{100}}$. Find [S].
Where [y] denotes largest integer less than or equal to y.
4. If $3^{2\sin 2x-1}$, 14, $3^{4-2\sin 2x}$ form first three terms of an A.P., then find the sum $1 + \sin 2x + \sin^2 2x + \dots \infty$.
5. Given that the cubic $ax^3 - ax^2 + 9bx - b = 0$ ($a \neq 0$) has all three positive roots. Find the harmonic mean of the roots independent of a and b, hence deduce that the root are all equal. Find also the minimum value of $(a + b)$ if a and b $\in \mathbb{N}$.
6. If $\tan\left(\frac{\pi}{12} - x\right)$, $\tan\frac{\pi}{12}$, $\tan\left(\frac{\pi}{12} + x\right)$ in order are three consecutive terms of a G.P. then sum of all the solutions in $[0, 314]$ is $k\pi$. Find the value of k.
7. If the roots of $10x^3 - cx^2 - 54x - 27 = 0$ are in harmonic progression, then find c and all the roots.
8. In a GP the ratio of the sum of the first eleven terms to the sum of the last eleven terms is $1/8$ and the ratio of the sum of all the terms without the first nine to the sum of all the terms without the last nine is 2. Find the number of terms in the GP.
9. In the quadratic equation $A(\sqrt{3} - \sqrt{2})x^2 + \frac{B}{(\sqrt{3} + \sqrt{2})}x + C = 0$ with α, β as its roots.

If $A = (49 + 20\sqrt{6})^{1/4}$; $B = \text{sum of the infinite G.P. as } 8\sqrt{3} + \frac{8\sqrt{6}}{\sqrt{3}} + \frac{16}{\sqrt{3}} + \dots \infty$

and $|\alpha - \beta| = (6\sqrt{6})^k$ where $k = \log_6 10 - 2 \log_6 \sqrt{5} + \log_6 \sqrt{(\log_6 18 + \log_6 72)}$,

then find the value of C.

10. If a,b,c,d,e be 5 numbers such that a,b,c are in AP ; b,c,d are in GP & c,d,e are in HP then :
 - (i) Prove that a,c,e are in GP.
 - (ii) Prove that $e = (2b - a)^2/a$
 - (iii) If $a = 2$ & $e = 18$, find all possible values of b,c,d.
11. Prove that the average of the numbers $n \sin n^\circ$, $n = 2, 4, 6, \dots, 180$, is $\cot 1^\circ$.
12. If one AM 'a' and two GM's p and q be inserted between any two given numbers then show that $p^3 + q^3 = 2apq$.
13. Find the sum of the n terms of the sequence $\frac{1}{1+1^2+1^4} + \frac{2}{1+2^2+2^4} + \frac{3}{1+3^2+3^4} + \dots$

EXERCISE (JM)

1. The sum to infinity of the series $1 + \frac{2}{3} + \frac{6}{3^2} + \frac{10}{3^3} + \frac{14}{3^4} + \dots$ is :- [AIEEE-2009]
 (1) 4 (2) 6 (3) 2 (4) 3
2. A person is to count 4500 currency notes. Let a_n denote the number of notes he counts in the n^{th} minute. If $a_1 = a_2 = \dots = a_{10} = 150$ and $a_{10}, a_{11}, \dots$ are in an AP with common difference -2 , then the time taken by him to count all notes is :- [AIEEE-2010]
 (1) 24 minutes (2) 34 minutes (3) 125 minutes (4) 135 minutes
3. A man saves Rs. 200 in each of the first three months of his service. In each of the subsequent months his saving increases by Rs. 40 more than the saving of immediately previous month. His total saving from the start of service will be Rs. 11040 after :- [AIEEE-2011]
 (1) 20 months (2) 21 months (3) 18 months (4) 19 months
4. Let a_n be the n^{th} term of an A.P. If $\sum_{r=1}^{100} a_{2r} = \alpha$ and $\sum_{r=1}^{100} a_{2r-1} = \beta$, then the common difference of the A.P. is : [AIEEE-2011]
 (1) $\frac{\alpha - \beta}{200}$ (2) $\alpha - \beta$ (3) $\frac{\alpha - \beta}{100}$ (4) $\beta - \alpha$
5. **Statement-1 :** The sum of the series $1 + (1 + 2 + 4) + (4 + 6 + 9) + (9 + 12 + 16) + \dots + (361 + 380 + 400)$ is 8000.
Statement-2 : $\sum_{k=1}^n (k^3 - (k-1)^3) = n^3$, for any natural number n . [AIEEE-2012]
 (1) Statement-1 is true, Statement-2 is false.
 (2) Statement-1 is false, Statement-2 is true.
 (3) Statement-1 is true, Statement-2 is true ; Statement-2 is a correct explanation for Statement-1.
 (4) Statement-1 is true, Statement-2 is true ; Statement-2 is not a correct explanation for Statement-1.
6. If 100 times the 100th term of an A.P. with non-zero common difference equals the 50 times its 50th term, then the 150th term of this A.P. is : [AIEEE-2012]
 (1) zero (2) -150
 (3) 150 times its 50th term (4) 150
7. The sum of first 20 terms of the sequence 0.7, 0.77, 0.777,, is : [JEE(Main)-2013]
 (1) $\frac{7}{81}(179 - 10^{-20})$ (2) $\frac{7}{9}(99 - 10^{-20})$ (3) $\frac{7}{81}(179 + 10^{-20})$ (4) $\frac{7}{9}(99 - 10^{-20})$
8. Let α and β be the roots of equation $px^2 + qx + r = 0$, $p \neq 0$. If p, q, r are in A.P. and $\frac{1}{\alpha} + \frac{1}{\beta} = 4$, then the value of $|\alpha - \beta|$ is: [JEE(Main)-2014]
 (1) $\frac{\sqrt{61}}{9}$ (2) $\frac{2\sqrt{17}}{9}$ (3) $\frac{\sqrt{34}}{9}$ (4) $\frac{2\sqrt{13}}{9}$

9. Three positive numbers form an increasing G.P. If the middle term in this G.P. is doubled, the new numbers are in A.P. Then the common ratio of the G.P. is : [JEE(Main)-2014]
 (1) $\sqrt{2} + \sqrt{3}$ (2) $3 + \sqrt{2}$ (3) $2 - \sqrt{3}$ (4) $2 + \sqrt{3}$
10. If $(10)^9 + 2(11)^1 (10)^8 + 3(11)^2 (10)^7 + \dots + 10 (11)^9 = k (10)^9$, then k is equal to : [JEE(Main)-2014]
 (1) $\frac{121}{10}$ (2) $\frac{441}{100}$ (3) 100 (4) 110
11. If m is the A.M. of two distinct real numbers l and n ($l, n > 1$) and G_1, G_2 and G_3 are three geometric means between l and n , then $G_1^4 + 2G_2^4 + G_3^4$ equals - [JEE(Main)-2015]
 (1) $4lmn^2$ (2) $4l^2m^2n^2$ (3) $4l^2mn$ (4) $4lm^2n$
12. If the 2nd, 5th and 9th terms of a non-constant A.P. are in G.P., then the common ratio of this G.P. is:- [JEE(Main)-2016]
 (1) $\frac{7}{4}$ (2) $\frac{8}{5}$ (3) $\frac{4}{3}$ (4) 1
13. If the sum of the first ten terms of the series $\left(1\frac{3}{5}\right)^2 + \left(2\frac{2}{5}\right)^2 + \left(3\frac{1}{5}\right)^2 + 4^2 + \left(4\frac{4}{5}\right)^2 + \dots$, is $\frac{16}{5}m$, then m is equal to :- [JEE(Main)-2016]
 (1) 99 (2) 102 (3) 101 (4) 100
14. For any three positive real numbers a, b and c , $9(25a^2 + b^2) + 25(c^2 - 3ac) = 15b(3a + c)$. Then : [JEE(Main)-2017]
 (1) a, b and c are in G.P. (2) b, c and a are in G.P.
 (3) b, c and a are in A.P. (4) a, b and c are in A.P.
15. Let $a_1, a_2, a_3, \dots, a_{49}$ be in A.P. such that $\sum_{k=0}^{12} a_{4k+1} = 416$ and $a_9 + a_{43} = 66$.
 If $a_1^2 + a_2^2 + \dots + a_{17}^2 = 140m$, then m is equal to- [JEE(Main)-2018]
 (1) 68 (2) 34 (3) 33 (4) 66
16. Let A be the sum of the first 20 terms and B be the sum of the first 40 terms of the series $1^2 + 2 \cdot 2^2 + 3^2 + 2 \cdot 4^2 + 5^2 + 2 \cdot 6^2 + \dots$. If $B - 2A = 100\lambda$, then λ is equal to : [JEE(Main)-2018]
 (1) 248 (2) 464 (3) 496 (4) 232

EXERCISE (JA)

1. If the sum of first n terms of an A.P. is cn^2 , then the sum of squares of these n terms is
 (A) $\frac{n(4n^2 - 1)c^2}{6}$ (B) $\frac{n(4n^2 + 1)c^2}{3}$ (C) $\frac{n(4n^2 - 1)c^2}{3}$ (D) $\frac{n(4n^2 + 1)c^2}{6}$ [JEE 2009, 3 (-1)]
2. Let $a_1, a_2, a_3, \dots, a_{11}$ be real numbers satisfying
 $a_1 = 15, 27 - 2a_2 > 0$ and $a_k = 2a_{k-1} - a_{k-2}$ for $k = 3, 4, \dots, 11$.
 If $\frac{a_1^2 + a_2^2 + \dots + a_{11}^2}{11} = 90$, then the value of $\frac{a_1 + a_2 + \dots + a_{11}}{11}$ is equal to [JEE 2010, 3+3]

3. The minimum value of the sum of real numbers a^{-5} , a^{-4} , $3a^{-3}$, 1 , a^8 and a^{10} with $a > 0$ is
[JEE 2011, 4]
4. Let $a_1, a_2, a_3, \dots, a_{100}$ be an arithmetic progression with $a_1 = 3$ and $S_p = \sum_{i=1}^p a_i$, $1 \leq p \leq 100$. For any integer n with $1 \leq n \leq 20$, let $m = 5n$. If $\frac{S_m}{S_n}$ does not depend on n , then a_2 is [JEE 2011, 4]
5. Let $a_1, a_2, a_3, \dots$ be in harmonic progression with $a_1 = 5$ and $a_{20} = 25$. The least positive integer n for which $a_n < 0$ is [JEE 2012, 3 (-1)]
(A) 22 (B) 23 (C) 24 (D) 25
6. Let $S_n = \sum_{k=1}^{4n} (-1)^{\frac{k(k+1)}{2}} k^2$. Then S_n can take value(s) [JEE-Advanced 2013, 4, (-1)]
(A) 1056 (B) 1088 (C) 1120 (D) 1332
7. A pack contains n cards numbered from 1 to n . Two consecutive numbered cards are removed from the pack and the sum of the numbers on the remaining cards is 1224. If the smaller to the numbers on the removed cards is k , then $k - 20 =$ [JEE-Advanced 2013, 4, (-1)]
8. Let a, b, c be positive integers such that $\frac{b}{a}$ is an integer. If a, b, c are in geometric progression and the arithmetic mean of a, b, c is $b + 2$, then the value of $\frac{a^2 + a - 14}{a + 1}$ is [JEE(Advanced)-2014, 3]
9. Suppose that all the terms of an arithmetic progression (A.P.) are natural numbers. If the ratio of the sum of the first seven terms to the sum of the first eleven terms is $6 : 11$ and the seventh term lies in between 130 and 140, then the common difference of this A.P. is [JEE 2015, 4M, -0M]
10. Let $b_i > 1$ for $i = 1, 2, \dots, 101$. Suppose $\log_e b_1, \log_e b_2, \dots, \log_e b_{101}$ are in Arithmetic Progression (A.P.) with the common difference $\log_e 2$. Suppose $a_1, a_2, \dots, a_{101}$ are in A.P. such that $a_1 = b_1$ and $a_{51} = b_{51}$. If $t = b_1 + b_2 + \dots + b_{51}$ and $s = a_1 + a_2 + \dots + a_{51}$ then [JEE(Advanced)-2016, 3(-1)]
(A) $s > t$ and $a_{101} > b_{101}$ (B) $s > t$ and $a_{101} < b_{101}$
(C) $s < t$ and $a_{101} > b_{101}$ (D) $s < t$ and $a_{101} < b_{101}$
11. The sides of the right angled triangle are in arithmetic progression. If the triangle has area 24, then what is the length of its smallest side ? [JEE(Advanced)-2017, 3]
12. Let X be the set consisting of the first 2018 terms of the arithmetic progression 1, 6, 11, $\dots$, and Y be the set consisting of the first 2018 terms of the arithmetic progression 9, 16, 23, $\dots$. Then, the number of elements in the set $X \cup Y$ is [JEE(Advanced)-2018, 3]

ANSWER KEY

EXERCISE (O-1)

- | | | | | | | |
|-------|-------|-------|-------|-------|-------|-------|
| 1. D | 2. C | 3. D | 4. A | 5. A | 6. A | 7. B |
| 8. C | 9. B | 10. C | 11. A | 12. B | 13. C | 14. C |
| 15. C | 16. A | 17. C | 18. B | 19. D | 20. B | |

EXERCISE (O-2)

- | | | | | | | |
|-------|-------|-------|-------|-------------|---------|-------|
| 1. A | 2. B | 3. C | 4. D | 5. C | 6. A | 7. D |
| 8. B | 9. D | 10. A | 11. C | 12. B | 13. A | 14. A |
| 15. D | 16. B | 17. D | 18. D | 19. A,B,C,D | 20. B,C | |

EXERCISE (S-1)

1. $(14n - 6)/(8n + 23)$ 3. 12 4. $n = 14$ 5. -180 6. 27 7. 2499
 8. 2 9. $(8, -4, 2, 8)$ 10. $a = 5, b = 8, c = 12$ 11. 1
 13. (a) 6, 3 (b) 120, 30 14. 102 15. 22 16. 5 17. 7
 18. (i) $2^{n+1} - 3; 2^{n+2} - 4 - 3n$ (ii) $n^2 + 4n + 1; (1/6)n(n+1)(2n+13) + n$
 19. (i) $s_n = (1/24) - [1/\{6(3n+1)(3n+4)\}]$; $s_\infty = 1/24$ (ii) $(1/5)n(n+1)(n+2)(n+3)(n+4)$
 (iii) $n/(2n+1)$

EXERCISE (S-2)

1. $\frac{27}{2}$ 2. 23 3. 49 4. 2 5. 28 6. 4950
 7. $C = 9; (3, -3/2, -3/5)$ 8. $n = 38$ 9. 128
 10. (iii) $b = 4, c = 6, d = 9$ or $b = -2, c = -6, d = -18$ 13. $\frac{n(n+1)}{2(n^2+n+1)}$

EXERCISE (JM)

- | | | | | | | |
|-------|-------|-------|-------|-------|-------|-------|
| 1. 4 | 2. 2 | 3. 2 | 4. 3 | 5. 3 | 6. 1 | 7. 3 |
| 8. 4 | 9. 4 | 10. 3 | 11. 4 | 12. 3 | 13. 3 | 14. 3 |
| 15. 2 | 16. 1 | | | | | |

EXERCISE (JA)

- | | | | | | | |
|------|------|-------|-----------|----------|--------|------|
| 1. C | 2. 0 | 3. 8 | 4. 9 or 3 | 5. D | 6. A,D | 7. 5 |
| 8. 4 | 9. 9 | 10. B | 11. 6 | 12. 3748 | | |

NURTURE COURSE

**Height and Distance
Principle of Mathematical Induction,
Sets, Relations
(FOR JEE-MAIN)**

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SETS

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RELATIONS

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JEE (Main) Syllabus :

Heights and Distances

MATHEMATICAL INDUCTION

Principle of Mathematical Induction and its simple applications.

SETS, RELATIONS

Sets and their representation; Union, intersection and complement of sets and their algebraic properties; Power set; Relation, Types of relations, equivalence relations.

HEIGHT AND DISTANCE

1. INTRODUCTION :

One of the important application of trigonometry is in finding the height and distance of the point which are not directly measurable. This is done with the help of trigonometric ratios.

2. ANGLES OF ELEVATION AND DEPRESSION :

Let OP be a horizontal line in the vertical plane in which an object R is given and let OR be joined.

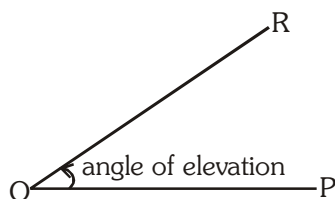


Fig. (a)

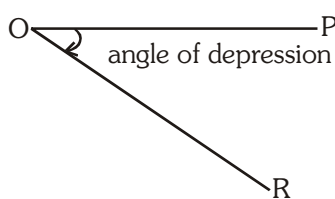


Fig. (b)

In Fig. (a), where the object R is above the horizontal line OP, the angle POR is called the angle of elevation of the object R as seen from the point O. In Fig. (b) where the object R is below the horizontal line OP, the angle POR is called the angle of depression of the object R as seen from the point O.

Remark :

Unless stated to the contrary, it is assumed that the height of the observer is neglected, and that the angles of elevation are measured from the ground.

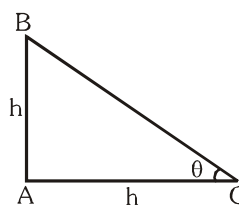
Ex.1 Find the angle of elevation of the sun when the length of shadow of a vertical pole is equal to its height.

Sol. Let height of the pole AB = h and

length of the shadow of the Pole (AC) = h

$$\text{In } \triangle ABC \tan \theta = \frac{AB}{AC} = \frac{h}{h} = 1 \Rightarrow \tan \theta = 1$$

$$\Rightarrow \tan \theta = \tan 45^\circ \Rightarrow \theta = 45^\circ$$



Ex.2 The shadow of the tower standing on a level ground is found to be 60 metres longer when the sun's altitude is 30° than when it is 45° . The height of the tower is-

(1) 60 m

(2) $30(\sqrt{3} - 1)$ m

(3) $60\sqrt{3}$ m

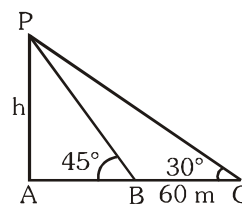
(4) $30(\sqrt{3} + 1)$ m.

Sol.(4) $AC = h \cot 30^\circ = \sqrt{3} h$

$$AB = h \cot 45^\circ = h$$

$$\therefore BC = AC - AB = h(\sqrt{3} - 1) \Rightarrow 60 = h(\sqrt{3} - 1)$$

$$\therefore h = \frac{60}{\sqrt{3} - 1} = \frac{60(\sqrt{3} + 1)}{3 - 1} = 30(\sqrt{3} + 1)$$



- Ex.3** The angle of elevation of the tower observed from each of the three point A,B,C on the ground, forming a triangle is the same angle α . If R is the circum - radius of the triangle ABC, then the height of the tower is -
 (1) $R \sin \alpha$ (2) $R \cos \alpha$ (3) $R \cot \alpha$ (4) $R \tan \alpha$

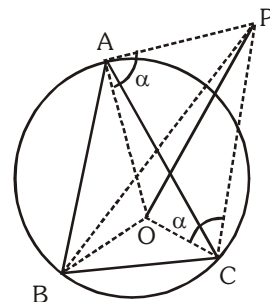
Sol.(4) The tower makes equal angles at the vertices of the triangle, therefore foot of the tower is at the circumcentre.

From ΔOCP , OP is perpendicular to OC .

$$\angle OCP = \alpha$$

$$\text{so } \tan \alpha = \frac{OP}{OA} \Rightarrow OP = OA \tan \alpha$$

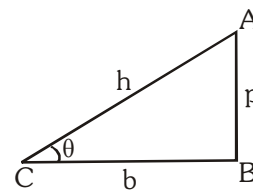
$$OP = R \tan \alpha$$



3. SOME USEFUL RESULTS :

- In a triangle ABC,

$$\sin \theta = \frac{p}{h}, \quad \cos \theta = \frac{b}{h}, \quad \tan \theta = \frac{p}{b}$$

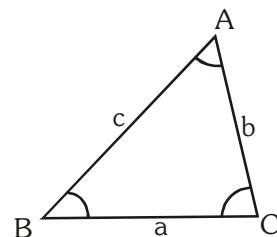


- In any triangle ABC,

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} \quad [\text{By sine rule}]$$

or cosine formula

$$\text{i.e. } \cos A = \frac{b^2 + c^2 - a^2}{2bc}; \cos B = \frac{a^2 + c^2 - b^2}{2ac}, \cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

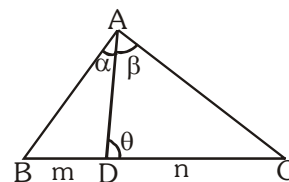


- In any triangle ABC

if $BD : DC = m : n$ and $\angle BAD = \alpha$

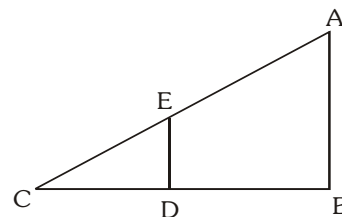
$\angle CAD = \beta$ and $\angle ADC = \theta$,

then $(m+n) \cot \theta = m \cot \alpha + n \cot \beta$



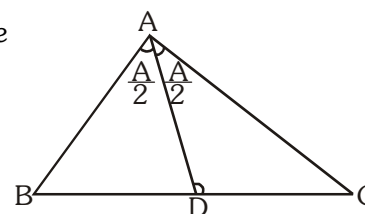
- In a triangle ABC, if $DE \parallel AB$

$$\text{then, } \frac{AB}{DE} = \frac{BC}{DC}$$



- In a triangle the internal bisector of an angle divides the opposite side in the ratio of the arms of the angle

$$\therefore \frac{BD}{DC} = \frac{AB}{AC}$$



- In an isosceles triangle the median is perpendicular to the base

SOLVED EXAMPLES

Ex.1 A tower subtends an angle of 30° at a point on the same level as its foot, and at a second point h m above the first, the depression of the foot of tower is 60° . The height of the tower is.

- (1) h m (2) $3h$ m (3) $\sqrt{3} h$ m (4) $\frac{h}{3}$ m.

Sol.(4) Let OP be the tower of height x , A the point on the same level as the foot O of the tower and B be the point h m above A (see Fig.) Then $\angle AOB = 60^\circ$ and $\angle PAO = 30^\circ$. From right-angled triangle AOP , we have

$$OA = x \cot 30^\circ$$

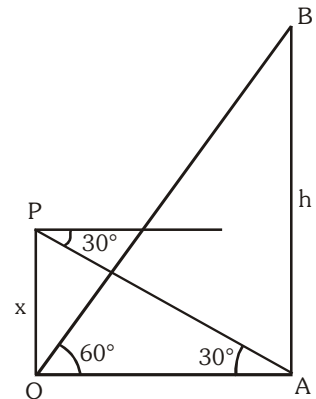
and from right-angled triangle OAB , we have

$$OA = h \cot 60^\circ$$

Therefore, from (1) and (2), we get

$$x \cot 30^\circ = h \cot 60^\circ$$

$$\sqrt{3} x = \frac{1}{\sqrt{3}} h \Rightarrow x = \frac{1}{3} h$$



Ex.2 At a point on level ground, the angle of elevation of a vertical tower is found to be such that its tangent is $\frac{5}{12}$.

On walking 192 metres towards the tower, the tangent of the angle of elevation is $\frac{3}{4}$. Find the height of the tower.

Sol. Let AB be the tower and let the angle of elevation of its top at C be α . Let D be a point at a distance of 192 metres from C such that the angle of elevation of the top of the tower at D be β .

Let h be the height of the tower and $AD = x$,

It is given that $\tan \alpha = \frac{5}{12}$ and $\tan \beta = \frac{3}{4}$.

In $\triangle ABC$, we have

$$\tan \alpha = \frac{AB}{AC} \Rightarrow \tan \alpha = \frac{h}{192 + x} \Rightarrow \frac{5}{12} = \frac{h}{192 + x} \quad \dots (i)$$

In $\triangle ABD$, we have

$$\tan \beta = \frac{AB}{AD} \Rightarrow \tan \beta = \frac{h}{x} \Rightarrow \frac{3}{4} = \frac{h}{x} \quad \dots (ii)$$

We have to find h . This means that we have to eliminate x from (i) and (ii).

$$\text{From (ii), we have } 3x = 4h \Rightarrow x = \frac{4h}{3}$$

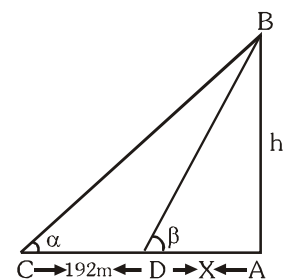
Substituting this value of x in (i), we get

$$\frac{5}{12} = \frac{h}{192 + 4h/3} \Rightarrow 5 \left(192 + \frac{4h}{3} \right) = 12h$$

$$\Rightarrow 5(576 + 4h) = 36h \Rightarrow 2880 + 20h = 36h$$

$$\Rightarrow 16h = 2880 \Rightarrow h = \frac{2880}{16} = 180$$

Hence, height of tower = 180 metres.



Ex.3 Let α be the solution of $16^{\sin^2 \theta} + 16^{\cos^2 \theta} = 10$ in $(0, \pi/4)$. If the shadow of a vertical pole is $\frac{1}{\sqrt{3}}$ of its height, then the altitude of the sun is-

(1) α (2) $\frac{\alpha}{2}$ (3) 2α (4) $\frac{\alpha}{3}$

Sol. We have $16^{\sin^2 \theta} + 16^{\cos^2 \theta} = 10$

$$\Rightarrow 16^{\sin^2 \theta} + 16^{1-\sin^2 \theta} = 10 \Rightarrow x + \frac{16}{x} = 10, \text{ where } x = 16^{\sin^2 \theta}$$

$$\Rightarrow x = 2, 8 \Rightarrow 16^{\sin^2 \theta} = 2, 8$$

$$\Rightarrow 2^{4\sin^2 \theta} = 2, 2^3$$

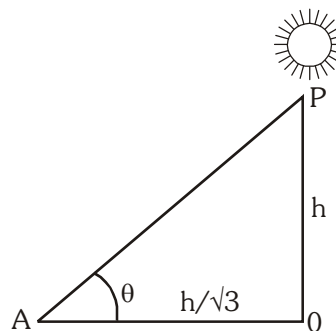
$$\Rightarrow 4\sin^2 \theta = 2, 3$$

$$\Rightarrow \sin^2 \theta = \frac{1}{2}, \left(\frac{\sqrt{3}}{2}\right)^2 \Rightarrow \theta = \frac{\pi}{6}, \frac{\pi}{3}$$

$$\therefore \alpha = \frac{\pi}{6}$$

Let the altitude of the sun be θ . Then,

$$\tan \theta = \frac{h}{h/\sqrt{3}} \Rightarrow \tan \theta = \sqrt{3} \Rightarrow \theta = \frac{\pi}{3} \Rightarrow \theta = 2\alpha$$



Ex.4 A vertical lamp-post of height 9 metres stands at the corner of a rectangular field. The angle of elevation of its top from the farthest corner is 30° , while from another corner it is 45° . The area of the field is-

(1) $81\sqrt{2} \text{ m}^2$ (2) $9\sqrt{2} \text{ m}^2$ (3) $81\sqrt{3} \text{ m}^2$ (4) $9\sqrt{3} \text{ m}^2$

Sol. Let AP be the lamp-post of 9 m standing at corner A of the rectangular field ABCD.

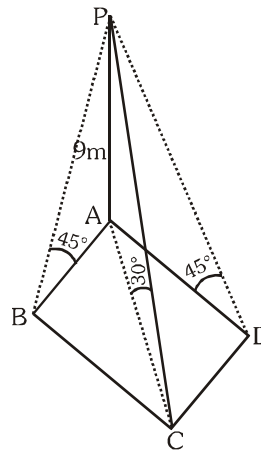
In Δ 's BAP and CAP, we have

$$\tan 45^\circ = \frac{PA}{BA} \text{ and } \tan 30^\circ = \frac{PA}{AC}$$

$$\Rightarrow BA = 9 \text{ m and } AC = 9\sqrt{3} \text{ m}$$

$$\therefore BC = \sqrt{AC^2 - AB^2} = \sqrt{243 - 81} = \sqrt{162} = 9\sqrt{2} \text{ m}$$

$$\text{Hence, area of the field} = AB \times BC = 9 \times 9\sqrt{2} \text{ m}^2 = 81\sqrt{2} \text{ m}^2$$



Ex.5 A vertical tower stands on a horizontal plane and is surmounted by a vertical flag staff of height h . At a point on the plane, the angle of elevation of the bottom and the top of the flag staff are α and β respectively. Prove

that the height of tower is $\frac{h \tan \alpha}{\tan \beta - \tan \alpha}$.

Sol. Let AB be the tower and BC be the flag staff. Let O be a point on the plane containing the foot of the tower such that the angles of elevation of the bottom B and top C of the flagstaff at O are α and β respectively. Let $OA = x$ metres, $AB = y$ metres and $BC = h$ metres.

In $\triangle OAB$, we have

$$\cot \alpha = \frac{OA}{AB} \Rightarrow \cot \alpha = \frac{x}{y} \quad \dots(i)$$

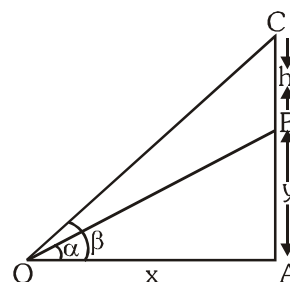
In $\triangle OAC$, we have

$$\cot \beta = \frac{x}{y+h} \Rightarrow x = (y+h) \cot \beta \quad \dots(ii)$$

Equating the values of x from (i) and (ii), we get

$$\begin{aligned} y \cot \alpha &= (y+h) \cot \beta \\ \Rightarrow y \cot \alpha - y \cot \beta &= h \cot \beta \\ \Rightarrow y (\cot \alpha - \cot \beta) &= h \cot \beta \end{aligned}$$

$$\Rightarrow y = \frac{h \cot \beta}{\cot \alpha - \cot \beta} \Rightarrow y = \frac{h / \tan \beta}{\frac{1}{\tan \alpha} - \frac{1}{\tan \beta}} = \frac{h \tan \alpha}{\tan \beta - \tan \alpha}$$



Ex.6 A spherical ball of diameter δ subtends an angle α at the eye of an observer when the elevation of its centre

is β . Prove that the height of the centre of the ball is $\frac{1}{2} \delta \sin \beta \operatorname{cosec} \left(\frac{\alpha}{2} \right)$.

Sol. O is the position of eye.

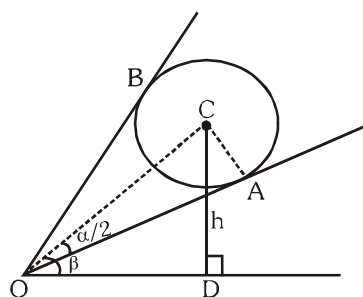
As is clear from figure, from $\triangle ODC$,

$$OC = \frac{h}{\sin \beta}$$

From $\triangle OAC$,

$$\sin \frac{\alpha}{2} = \frac{CA}{OC} = \frac{\frac{\delta}{2}}{h / \sin \beta}$$

$$\Rightarrow h = \frac{1}{2} \delta \sin \beta \cdot \operatorname{cosec} \frac{\alpha}{2}.$$



CHECK YOUR GRASP

HEIGHTS AND DISTANCE

EXERCISE-I

- The angle of elevation of the top of a tower from a point 20 metre away from its base is 45° . Then height of the tower is-
(1) 10 m (2) 20 m (3) 40 m (4) $20\sqrt{3}$ m
- At a point 15 metre away from the base of a 15 metre high house, the angle of elevation of the top is-
(1) 45° (2) 30° (3) 60° (4) 90°
- An aeroplane flying at a height 300 metre above the ground passes vertically above another plane at an instant when the angles of elevation of the two planes from the same point on the ground are 60° and 45° respectively. Then the height of the lower plane from the ground in metres is-
(1) $100\sqrt{3}$ (2) $100/\sqrt{3}$
(3) 50 (4) $150(\sqrt{3} + 1)$.
- If the elevation of the sun is 30° then the length of the shadow cast by a tower of 150 metres height is-
(1) $75\sqrt{3}$ m (2) $200\sqrt{3}$ m
(3) $150\sqrt{3}$ m (4) None of these
- From the top of the cliff 300 metres high, the top of a tower was observed at an angle of depression 30° and from the foot of the tower the top of the cliff was observed at an angle of elevation 45° . The height of the tower is -
(1) $50(3 - \sqrt{3})$ m (2) $200(3 - \sqrt{3})$ m
(3) $100(3 - \sqrt{3})$ m (4) None of these
- The angles of elevation of the top of a tower at the top and the foot of a pole of height 10 m are 30° and 60° respectively. The height of the tower is-
(1) 10 m (2) 15 m
(3) 20 m (4) None of these
- A person, standing on the bank of a river, observes that the angle subtended by a tree on the opposite bank is 60° ; when he retreats 20 m from the bank, he finds the angle to be 30° . The height of the tree and the breadth of the river are -
(1) $10\sqrt{3}$ m, 10 m (2) 10 m ; $10\sqrt{3}$ m
(3) 20 m, 30m (4) None of these
- From the top of a light house 60 m high with its base at sea level the angle of depression of a boat is 15° . The distance of the boat from the light house is-
(1) $60\left(\frac{\sqrt{3}-1}{\sqrt{3}+1}\right)$ m (2) $60\left(\frac{\sqrt{3}+1}{\sqrt{3}-1}\right)$ m
(3) $30\left(\frac{\sqrt{3}-1}{\sqrt{3}+1}\right)$ m (4) $30\left(\frac{\sqrt{3}+1}{\sqrt{3}-1}\right)$ m
- The angle of elevation of the sun when the length of the shadow of a pole is $\sqrt{3}$ times the height of the pole will be-
(1) 30° (2) 60° (3) 90° (4) 45°
- If the length of the shadow of a vertical pole on the horizontal ground is equal to its height, find the angle of elevation of the sun -
(1) 60° (2) 30° (3) 45° (4) 90°
- The angle of elevation of the top of a tower from a point 20m away from its base is 60° . The height of the tower is -
(1) 10 m (2) $20/\sqrt{3}$ m (3) 40 m (4) $20\sqrt{3}$ m
- A flag staff on the top of the tower 80 meter high, subtends an angle $\tan^{-1}\left(\frac{1}{9}\right)$ at a point on the ground 100 meters away from the foot of the tower. Find the height of the flag-staff -
(1) 20 m (2) 30 m (3) 25 m (4) 35 m
- A person walking along a straight road observes that at two points 1 km apart, the angles of elevation of a pole in front of him are 30° and 75° . The height of the pole is -
(1) $250(\sqrt{3} + 1)$ m (2) $250(\sqrt{3} - 1)$ m
(3) $225(\sqrt{2} - 1)$ m (4) $225(\sqrt{2} + 1)$ m
- An observer in a boat finds that the angle of elevation of a tower standing on the top of a cliff is 60° and that of the top of cliff is 30° . If the height of the tower be 60 meters, then the height of the cliff is -
(1) 30 m (2) $60\sqrt{3}$ m
(3) $20\sqrt{3}$ m (4) None of these

15. ABCD is a square plot. The angle of elevation of the top of a pole standing at D from A and C is 30° and that from B is θ , then $\tan \theta$ is equal to -
 (1) $\sqrt{6}$ (2) $1/\sqrt{6}$
 (3) $\sqrt{3}/\sqrt{2}$ (4) $\sqrt{2}/\sqrt{3}$
16. The angle of elevation of a ladder against a wall is 58° and the length of foot of the ladder is 9.6 m from the wall. Then the length of the ladder is -
 $[\cos 58^\circ = 0.5299]$
 (1) 18.11 m (2) 16.11 m
 (3) 17.11 m (4) 19.11 m
17. From the top of a tower, the angle of depression of a point P on the ground is 30° . If the distance of the point P from the tower be 24 meters then height of the tower is.
 (1) 12 m (2) $8\sqrt{3}$ m
 (3) $24\sqrt{3}$ m (4) $12\sqrt{3}$ m
18. A tower subtends an angle of 30° at a point on the same level as the foot of the tower. At a second point, h metre above first, point the depression of the foot of the tower is 60° , the horizontal distance of the tower from the points is
 (1) $h \cos 60^\circ$ (2) $(h/3) \cot 30^\circ$
 (3) $(h/3) \cot 60^\circ$ (4) $h \cot 30^\circ$
19. From a point on the ground 100 m away from the base of a building, the angle of elevation of the top of the building is 60° . Which of the following is the best approximation for the height of the building-
 (1) 172 m (2) 173 m
 (3) 174 m (4) 175 m
20. A kite is flying with the string inclined at 75° to the horizon. If the length of the string is 25 m, the height of the kite is-
 (1) $(25/2)(\sqrt{3}-1)$ (2) $(25/4)(\sqrt{3}+1)$
 (3) $(25/4)(\sqrt{3}+1)^2$ (4) $(25/4)(\sqrt{6}+\sqrt{2})$
21. If a flagstaff 6 metres high placed on the top of a tower throws a shadow of $2\sqrt{3}$ metres along the ground then the angle (in degrees) that the sun makes with the ground is-
 (1) 15° (2) 30°
 (3) 60° (4) $\tan^{-1} 2\sqrt{3}$
22. A 6-ft tall man finds that the angle of elevation of the top of a 24-ft-high pillar and the angle of depression of its base are complementary angles. The distance of the man from the pillar is-
 (1) $2\sqrt{3}$ ft (2) $8\sqrt{3}$ ft
 (3) $6\sqrt{3}$ ft (4) None of these
23. A flagstaff stands vertically on a pillar, the height of the flagstaff being double the height of the pillar. A man on the ground at a distance finds that both the pillar and the flagstaff subtend equal angles at his eyes. The ratio of the height of the pillar and the distance of the man from the pillar is-
 (1) $\sqrt{3}:1$ (2) $1:3$
 (3) $1:\sqrt{3}$ (4) $\sqrt{3}:2$
24. The shadow of a tower of height $(1 + \sqrt{3})$ metre standing on the ground is found to be 2 metre longer when the sun's elevation is 30° , than when the sun's elevation was -
 (1) 30° (2) 45° (3) 60° (4) 75°
25. The angle of elevation of the top of an incomplete vertical pillar at a horizontal distance of 50 m from its base is 45° . If the angle of elevation of the top of the complete pillar the same point is to be 60° , then the height of the incomplete pillar is to be increased by-
 (1) $50(\sqrt{3}-1)$ m (2) $50(\sqrt{3}+1)$ m
 (3) 50 m (4) $25\sqrt{2}$ m.
26. A vertical tower stands on a declivity which is inclined at 15° to the horizon. From the foot of the tower a man ascends the declivity for 80 feet and then finds that tower subtends an angle of 30° . The height of the tower is-
 (1) $20(\sqrt{6}-\sqrt{2})$ (2) $40(\sqrt{6}-\sqrt{2})$
 (3) $40(\sqrt{6}+\sqrt{2})$ (4) None of these

- 27.** AB is a vertical pole. The point A of pole AB is on the level ground. C is the middle point of AB. P is a point on the level ground. The portion BC subtends an angle β at P. If $AP = n AB$, then $\tan \beta$ is equal to-
- (1) $\frac{n}{2n^2 + 1}$ (2) $\frac{n}{n^2 - 1}$
 (3) $\frac{n}{n^2 + 1}$ (4) None of these
- 28.** The top of a hill observed from the top and bottom of a building of height h is at angles of elevation p and q respectively. The height of the hill is -
- (1) $\frac{h \cot q}{\cot q - \cot p}$ (2) $\frac{h \cot p}{\cot p - \cot q}$
 (3) $\frac{h \tan p}{\tan p - \tan q}$ (4) None of these
- 29.** A and B are two points 30 m apart in a line on the horizontal plane through the foot of a tower lying on opposite sides of the tower. If the distance of the top of the tower from A and B are 20 m and 15 m respectively, the angle of elevation of the top of the tower at A is-
- (1) $\cos^{-1}(43/48)$ (2) $\sin^{-1}(43/48)$
 (3) $\cos^{-1}(29/36)$ (4) $\sin^{-1}(29/36)$
- 30.** A vertical pole subtends an angle $\tan^{-1}(1/2)$ at a point P on the ground. The angle subtended by the upper half of the pole at the point P is-
- (1) $\tan^{-1}(1/4)$ (2) $\tan^{-1}(2/9)$
 (3) $\tan^{-1}(1/8)$ (4) $\tan^{-1}(2/3)$
- 31.** The angle of elevation of the top of a tower standing on a horizontal plane from a point A is α . After walking a distance d towards the foot of the tower, the angle of elevation is found to be β . The height of the tower is-
- (1) $\frac{d \sin \alpha \sin \beta}{\sin(\beta - \alpha)}$ (2) $\frac{d \sin \alpha \sin \beta}{\sin(\alpha - \beta)}$
 (3) $\frac{d \sin(\beta - \alpha)}{\sin \alpha \sin \beta}$ (4) $\frac{d \sin(\alpha - \beta)}{\sin \alpha \sin \beta}$
- 32.** The angle of elevation of the top of two vertical towers as seen from the middle point of the line joining the foot of the towers are 60° and 30° respectively. The ratio of the height of the towers is-
- (1) 2 : 1 (2) $\sqrt{3} : 1$
 (3) 3 : 2 (4) 3 : 1
- 33.** A person walking along a st. road towards a hill observes at two points, distance $\sqrt{3}$ km, the angles of elevation of the hill to be 30° and 60° . The height of the hill is-
- (1) $\frac{3}{2}$ km (2) $\sqrt{\frac{2}{3}}$ km
 (3) $\frac{\sqrt{3} + 1}{2}$ km (4) $\sqrt{3}$ km
- 34.** The length of the shadow of a vertical pole of height h , thrown by the sun's rays at three different moments are h , $2h$ and $3h$. The sum of the angles of elevation of the rays at these three moments is equal to-
- (1) $\frac{\pi}{2}$ (2) $\frac{\pi}{3}$ (3) $\frac{\pi}{4}$ (4) $\frac{\pi}{6}$
- 35.** The upper $\frac{3}{4}$ th portion of a vertical pole subtends an angle $\tan^{-1} \frac{3}{5}$ at a point in the horizontal plane through its foot and at a distance 40 m from the foot. A possible height of the vertical pole is-
- [AIEEE-2002]
 (1) 80 m (2) 20 m
 (3) 40 m (4) 60 m
- 36.** A person standing on the bank of a river observes that the angle of elevation of the top of a tree on the opposite bank of the river is 60° and when he retires 40 meters away from the tree the angle of elevation becomes 30° . The breadth of the river is-
- [AIEEE-2004]
 (1) 20 m (2) 30 m
 (3) 40 m (4) 60 m

37. A tower stands at the centre of a circular park. A and B are two points on the boundary of the park such that $AB (=a)$ subtends an angle of 60° at the foot of the tower, and the angle of elevation of the top of the tower from A or B is 30° . The height of the tower is- [AIEEE-2007]

- (1) $2a/\sqrt{3}$ (2) $2a\sqrt{3}$
(3) $a/\sqrt{3}$ (4) $a\sqrt{3}$

38. AB is a vertical pole with B at the ground level and A at the top. A man finds that the angle of elevation of the point A from a certain point C on the ground is 60° . He moves away from the pole along the line BC to a point D such that $CD = 7$ m. From D the angle of elevation of the point A is 45° . Then the height of the pole is- [AIEEE-2008]

- (1) $\frac{7\sqrt{3}}{2} \frac{1}{\sqrt{3}-1}$ m (2) $\frac{7\sqrt{3}}{2} (\sqrt{3}+1)$ m
(3) $\frac{7\sqrt{3}}{2} (\sqrt{3}-1)$ m (4) $\frac{7\sqrt{3}}{2} \frac{1}{\sqrt{3}+1}$ m

39. A man standing on a horizontal plane, observes the angle of elevation of the top of a tower to be α . After walking a distance equal to double the height of the tower, the angle of elevation becomes 2α , then α is-

- (1) $\frac{\pi}{18}$ (2) $\frac{\pi}{12}$ (3) $\frac{\pi}{6}$ (4) $\frac{\pi}{2}$

40. ABCD is a trapezium such that AB and CD are parallel and $BC \perp CD$. If $\angle ADB = \theta$, $BC = p$ and $CD = q$, then AB is equal to [JEE-MAINS-2013]

- (1) $\frac{(p^2 + q^2) \sin \theta}{p \cos \theta + q \sin \theta}$ (2) $\frac{p^2 + q^2 \cos \theta}{p \cos \theta + q \sin \theta}$
(3) $\frac{p^2 + q^2}{p^2 \cos \theta + q^2 \sin \theta}$ (4) $\frac{(p^2 + q^2) \sin \theta}{(p \cos \theta + q \sin \theta)^2}$

41. If the angles of elevation of the top of a tower from three collinear points A, B and C, on a line leading to the foot of the tower, are 30° , 45° and 60° respectively, then the ratio, AB : BC, is : [JEE(Main)-2015]

- (1) $1 : \sqrt{3}$ (2) $2 : 3$
(3) $\sqrt{3} : 1$ (4) $\sqrt{3} : \sqrt{2}$

42. A man is walking towards a vertical pillar in a straight path, at a uniform speed. At a certain point A on the path, he observes that the angle of elevation of the top of the pillar is 30° . After walking for 10 minutes from A in the same direction, at a point B, he observes that the angle of elevation of the top of the pillar is 60° . Then the time taken (in minutes) by him, from B to reach the pillar, is : [JEE(Main)-2016]

- (1) 5 (2) 6 (3) 10 (4) 20

43. Let a vertical tower AB have its end A on the level ground. Let C be the mid-point of AB and P be a point on the ground such that $AP = 2AB$. If $\angle BPC = \beta$, then $\tan \beta$ is equal to :- [JEE(Main)-2017]

- (1) $\frac{4}{9}$ (2) $\frac{6}{7}$ (3) $\frac{1}{4}$ (4) $\frac{2}{9}$

44. PQR is a triangular park with $PQ = PR = 200$ m. A T.V. tower stands at the mid-point of QR. If the angles of elevation of the top of the tower at P, Q and R are respectively 45° , 30° and 30° , then the height of the tower (in m) is- [JEE(Main)-2018]

- (1) 50 (2) $100\sqrt{3}$ (3) $50\sqrt{2}$ (4) 100

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	2	1	1	3	3	2	1	2	1	3	4	1	1	1	2
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	1	2	2	2	4	3	3	3	2	1	2	1	2	1	2
Que.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	
Ans.	1	4	1	1	3	1	3	2	2	1	3	1	4	4	

PRINCIPLE OF MATHEMATICAL INDUCTION

1. THEOREM-I

If $P(n)$ is a statement depending upon n , then to prove $P(n)$ by induction, we proceed as follows :

- (i) Verify the validity of $P(n)$ for $n = 1$
- (ii) Assume that $P(n)$ is true for any positive integer m and then using it establish the validity of $P(n)$ for $n = m + 1$.

Then $P(n)$ is true for each $n \in \mathbb{N}$

2. THEOREM-II

If $P(n)$ is a statement depending upon n but beginning with any positive integer k , then to prove $P(n)$ by Induction, we proceed as follows :

- (i) Verify the validity of $P(n)$ for $n = k$.
- (ii) Assume that the $P(n)$ is true for $n = m \geq k$.

Then using it establish the validity of $P(n)$ for $n = m + 1$.

Then $P(n)$ is true for each $n \geq k$

3. SOME USEFUL RESULT BASED ON PRINCIPLE OF MATHEMATICAL INDUCTION :

For any natural number n

$$(i) \quad 1 + 2 + 3 + \dots + n = \Sigma n = \frac{n(n+1)}{2}$$

$$(ii) \quad 1^2 + 2^2 + 3^2 + \dots + n^2 = \Sigma n^2 = \frac{n(n+1)(2n+1)}{6}$$

$$(iii) \quad 1^3 + 2^3 + 3^3 + \dots + n^3 = \Sigma n^3 = (\Sigma n)^2 = \left\{ \frac{n(n+1)}{2} \right\}^2$$

$$(iv) \quad 2 + 4 + 6 + \dots + 2n = \Sigma 2n = n(n+1)$$

$$(v) \quad 1 + 3 + 5 + \dots + (2n-1) = \Sigma (2n-1) = n^2$$

$$(vi) \quad x^n - y^n = (x-y)(x^{n-1} + x^{n-2}y + x^{n-3}y^2 + \dots + xy^{n-2} + y^{n-1})$$

$$(vii) \quad x^n + y^n = (x+y)(x^{n-1} - x^{n-2}y + x^{n-3}y^2 + \dots - xy^{n-2} + y^{n-1})$$

when n is odd positive integer

4. IMPORTANT TIPS :

- (i) Product of r successive integers is divisible by $r!$
- (ii) For $x \neq y$, $x^n - y^n$ is divisible by
 - (a) $x + y$, if n is even
 - (b) $x - y$, if n is even or odd
- (iii) $x^n + y^n$ is divisible by $x + y$, If n is odd
- (iv) For solving objective question related to natural numbers we find out the correct alternative by negative examination of this principle. If the given statement is $P(n)$, then by putting $n = 1, 2, 3, \dots$ in $P(n)$ we decide the correct answer. We also use the above formulae established by this principle to find the sum of n terms of a given series. For this we first express T_n as a polynomial in n and then for finding S_n , we put Σ before each term of this polynomial and then use above results of $\Sigma n, \Sigma n^2, \Sigma n^3$ etc.

SOLVED EXAMPLES

Ex.1 Use the principle of mathematical induction to show that $5^{2n+1} + 3^{n+2} \cdot 2^{n-1}$ divisible by 19 for all natural numbers n .

Sol. Let $P(n) = 5^{2n+1} + 3^{n+2} \cdot 2^{n-1}$

Step I : For $n = 1$

$$P(1) = 5^{2+1} + 3^{1+2} \cdot 2^{1-1}$$

$$= 125 + 27$$

$$= 152, \text{ which is divisible by 19.}$$

Therefore, the result is true for $n = 1$.

Step II : Assume that the result is true for $n = k$, i.e. $P(k) = 5^{2k+1} + 3^{k+2} \cdot 2^{k-1}$ is divisible by 19.

$\Rightarrow P(k) = 19r$, where r is an integer.

Step III : For $n = k + 1$

$$P(k+1) = 5^{2(k+1)+1} + 3^{k+1+2} \cdot 2^{k+1-1}$$

$$= 5^{2k+3} + 3^{k+3} \cdot 2^k$$

$$= 25 \cdot 5^{2k+1} + 3 \cdot 3^{k+2} \cdot 2 \cdot 2^{k-1}$$

$$= 25 \cdot 5^{2k+1} + 6 \cdot 3^{k+2} \cdot 2^{k-1}$$

$$\text{Now } 25 \cdot 5^{2k+1} + 6 \cdot 3^{k+2} \cdot 2^{k-1} = 25 \cdot (5^{2k+1} + 3^{k-2} \cdot 2^{k-1}) - 19 \cdot 3^{k+2} \cdot 2^{k-1}$$

$$\text{i.e. } P(k+1) = 25 P(k) - 19 \cdot 3^{k+2} \cdot 2^{k-1}$$

But we know that $P(k)$ is divisible by 19. Also $19 \cdot 3^{k+2} \cdot 2^{k-1}$ is clearly divisible by 19.

Hence $P(k+1)$ is divisible by 19. This shows that the result is true for $n = k + 1$. Hence by the principle of mathematical induction, the result is true for all $n \in \mathbb{N}$.

Ex.2. Use the principle of mathematical induction to show that $1 \cdot 3 + 2 \cdot 4 + \dots + n \cdot (n+2) = \frac{1}{6} n(n+1)(2n+7)$.

Sol. Let $P(n) : 1 \cdot 3 + 2 \cdot 4 + \dots + n \cdot (n+2) = \frac{1}{6} n(n+1)(2n+7)$

Step I : For $n = 1$

$$\text{LHS of } P(1) = 1 \cdot 3 = 3 = \frac{1}{6} \cdot 1 \cdot 2 \cdot 9 = \frac{1}{6} \cdot 1(1+1)(2 \cdot 1 + 7) = \text{RHS of } P(1)$$

So $P(1)$ is true

Step II : Now assume $P(k)$ is true, for some natural number k , i.e.

$$1 \cdot 3 + 2 \cdot 4 + \dots + k \cdot (k+2) = \frac{1}{6} k(k+1)(2k+7)$$

Now deduce $P(k+1)$.

$$\text{LHS of } P(k+1) = 1 \cdot 3 + 2 \cdot 4 + \dots + k \cdot (k+2) + (k+1) \cdot (k+1+2)$$

$$= (\text{LHS of } P(k)) + (k+1)(k+3)$$

$$= (\text{RHS of } P(k)) + (k+1)(k+3), \text{ (by inductive assumption)}$$

$$= \frac{1}{6} k(k+1)(2k+7) + (k+1)(k+3)$$

$$= \frac{1}{6} (k+1)(k(2k+7) + 6(k+3))$$

$$= \frac{1}{6} (k+1)(2k^2 + 13k + 18)$$

$$= \frac{1}{6}(k+1)(k+2)(2k+9)$$

$$= \frac{1}{6}(k+1)(k+1+1)(2(k+1)+7)$$

= RHS of $P(k+1)$.

So $P(k+1)$ is true, if $P(k)$ is true.

Hence by induction $P(n)$ is true for all natural numbers n .

Ex.3 Use the principle of mathematical induction to show that for any positive integer number n , $n^3 + 2n$, is divisible by 3.

Sol. Statement $P(n)$ is defined by $n^3 + 2n$ is divisible 3

Step 1 : We first show that $P(1)$ is true. Let $n = 1$ and calculate $n^3 + 2n$

$$1^3 + 2(1) = 3$$

Hence $P(1)$ is true.

Step 2 : We now assume that $P(k)$ is true $k^3 + 2k$ is divisible by 3. is equivalent to $k^3 + 2k = 3M$, where M is a positive integer.

We now consider the algebraic expression $(k+1)^3 + 2(k+1)$; expand it and group like terms.

$$(k+1)^3 + 2(k+1) = k^3 + 3k^2 + 5k + 3$$

$$= [k^3 + 2k] + [3k^2 + 3k + 3]$$

$$= 3M + 3[k^2 + k + 1] = 3[M + k^2 + k + 1]$$

Hence $(k+1)^3 + 2(k+1)$ is also divisible by 3 and therefore statement $P(k+1)$ is true.

Ex.4 Prove that $3^n > n^2$ for $n = 1, n = 2$ and use the mathematical induction to prove that $3^n > n^2$ for n , a positive integer greater than 2.

Sol Statement $P(n)$ is defined by

$$3^n > n^2$$

Step 1 : We first show that $P(1)$ is true. Let $n = 1$ and calculate 3^1 and 1^2 and compare them

$$3^1 = 3$$

$$1^2 = 1$$

3 is greater than 1 and hence $P(1)$ is true.

Let us also show that $P(2)$ is true.

$$3^2 = 9$$

$$2^2 = 4$$

Hence $P(2)$ is also true.

Step 2 : We now assume that $P(k)$ is true

$$3^k > k^2$$

Multiply both sides of the above inequality by 3.

$$3 * 3^k > 3 * k^2$$

The left side is equal to 3^{k+1} . For $k > 2$, we can write

$$k^2 > 2k \text{ and } k^2 > 1$$

We now combine the above inequalities by adding the left hand sides and the right hand sides of the two inequalities.

$$2k^2 > 2k + 1$$

We now add k^2 to both sides of the above inequality to obtain the inequality

$$3k^2 > k^2 + 2k + 1$$

Factor the right side we can write

$$3k^2 > (k+1)^2$$

If $3^k > 3k^2$ and $3^{k+1} > (k+1)^2$ then

$$3^k > (k+1)^2$$

Rewrite the left side as 3^{k+1}

$$3^{k+1} > (k+1)^2$$

* Which proves that $P(k+1)$ is true.

Ex.5. Use mathematical induction to prove De Moivre's theorem

$[R(\cos t + i \sin t)]^n = R^n(\cos nt + i \sin nt)$ for n a positive integer.

Sol. Step 1 : For $n = 1$

$$[R(\cos t + i \sin t)]^1 = R^1(\cos 1.t + i \sin 1.t)$$

It can be easily be seen that the two sides are equal.

Step 2 : We now assume that the theorem is true for $n = k$, hence $[R(\cos t + i \sin t)]^k = R^k(\cos kt + i \sin kt)$

Multiply both sides of the above equation by $R(\cos t + i \sin t)$

$$[R(\cos t + i \sin t)]^k R(\cos t + i \sin t) = R^k(\cos kt + i \sin kt) R(\cos t + i \sin t)$$

Rewrite the above as follows

$$[R(\cos t + i \sin t)]^{k+1} = R^{k+1}[(\cos kt \cos t - \sin kt \sin t) + i(\sin kt \cos t + \cos kt \sin t)]$$

Trigonometric identities can be used to write the trigonometric expressions $(\cos kt \cos t - \sin kt \sin t)$ and $(\sin kt \cos t + \cos kt \sin t)$ as follows

$$(\cos kt \cos t - \sin kt \sin t) = \cos(kt + t) = \cos(k+1)t$$

$$(\sin kt \cos t + \cos kt \sin t) = \sin(kt + t) = \sin(k+1)t$$

Substitute the above into the last equation to obtain.

$$[R(\cos t + i \sin t)]^{k+1} = R^{k+1}[\cos(k+1)t + i \sin(k+1)t]$$

It has been established that the theorem is true for $n = 1$ and that if it assumed true for $n = k$ it is true for $n = k+1$.

CHECK YOUR GRASP PRINCIPLE OF MATHEMATICAL INDUCTION EXERCISE-I

- Let $P(n) : n^2 + n$ is an odd integer. It is seen that truth of $P(n) \Rightarrow$ the truth of $P(n + 1)$. Therefore, $P(n)$ is true for all-
 (1) $n > 1$ (2) n
 (3) $n > 2$ (4) None of these
- If $n \in \mathbb{N}$, then $x^{2n-1} + y^{2n-1}$ is divisible by-
 (1) $x + y$ (2) $x - y$ (3) $x^2 + y^2$ (4) $x^2 + xy$
- If $n \in \mathbb{N}$, then $11^{n+2} + 12^{2n+1}$ is divisible by-
 (1) 113 (2) 123
 (3) 133 (4) None of these
- If $n \in \mathbb{N}$, then $3^{4n+2} + 5^{2n+1}$ is a multiple of-
 (1) 14 (2) 16 (3) 18 (4) 20
- For every positive integer

$$n, \frac{n^7}{7} + \frac{n^5}{5} + \frac{2n^3}{3} - \frac{n}{105}$$
 is-
 (1) an integer
 (2) a rational number which is not an integer
 (3) a negative real number
 (4) an odd integer
- Sum of the infinite series $\frac{1}{1.4} + \frac{1}{4.7} + \frac{1}{7.10} + \dots$ equals-
 (1) $1/3$ (2) 3 (3) $1/4$ (4) 2
- If $a_k = \frac{1}{k(k+1)}$; then $\left(\sum_{k=1}^n a_k\right)^2$ is equal to-
 (1) $\frac{n^2}{(n+1)^2}$ (2) $\frac{n^4}{(n+1)^4}$ (3) $\frac{n^2}{n^2+1}$ (4) $\frac{n}{n+1}$
- The sum of n terms of the series

$$\frac{1}{1^3} \cdot \frac{2}{2} + \frac{2}{1^3+2^3} \cdot \frac{3}{2} + \frac{3}{1^3+2^3+3^3} \cdot \frac{4}{2} + \dots$$
 is-
 (1) $\frac{1}{n(n+1)}$ (2) $\frac{n}{n+1}$ (3) $\frac{n+1}{n}$ (4) $\frac{n+1}{n+2}$
- For all $n \in \mathbb{N}$, $7^{2n} - 48n - 1$ is divisible by-
 (1) 25 (2) 26 (3) 1234 (4) 2304
- For all positive integral values of n , $3^{2n} - 2n + 1$ is divisible by-
 (1) 2 (2) 4 (3) 8 (4) 12
- The smallest positive integer for which the statement $3^{n+1} < 4^n$ holds is-
 (1) 1 (2) 2 (3) 3 (4) 4
- For positive integer n , $10^{n-2} > 81n$ when-
 (1) $n < 5$ (2) $n > 5$ (3) $n \geq 5$ (4) $n > 6$
- If P is a prime number then $n^p - n$ is divisible by p when n is a
 (1) natural number greater than 1
 (2) odd number
 (3) even number
 (4) None of these
- A student was asked to prove a statement by induction. He proved
 (i) $P(5)$ is true and
 (ii) Truth of $P(n) \Rightarrow$ truth of $p(n + 1)$, $n \in \mathbb{N}$
 On the basis of this, he could conclude that $P(n)$ is true for
 (1) no $n \in \mathbb{N}$ (2) all $n \in \mathbb{N}$
 (3) all $n \geq 5$ (4) None of these

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	
Ans.	4	1	3	1	1	1	1	2	4	1	4	3	1	3	

BRAIN TEASERS

PRINCIPLE OF MATHEMATICAL INDUCTION

EXERCISE-II

- If $x > -1$, then the statement $P(n) : (1+x)^n > 1+nx$ is true for-
(1) all $n \in \mathbb{N}$ (2) all $n > 1$
(3) all $n > 1$ and $x \neq 0$ (4) None of these
- For every positive integral value of n , $3^n > n^3$ when-
(1) $n > 2$ (2) $n \geq 3$ (3) $n \geq 4$ (4) $n < 4$
- $P(n) : 3^{2n+2} - 8n - 9$ is divisible by 64, is true for-
(1) all $n \in \mathbb{N} \cup \{0\}$ (2) $n \geq 2, n \in \mathbb{N}$
(3) $n \in \mathbb{N}, n > 2$ (4) None of these
- If m, n are any two odd positive integer with $n < m$, then the largest positive integers which divides all the numbers of the type $m^2 - n^2$ is-
(1) 4 (2) 6 (3) 8 (4) 9
- For all $n \in \mathbb{N}$, $\cos \theta \cos 2\theta \cos 4\theta \dots \cos 2^{n-1}\theta$ equals to-
(1) $\frac{\sin 2^n \theta}{2^n \sin \theta}$ (2) $\frac{\sin 2^n \theta}{\sin \theta}$
(3) $\frac{\cos 2^n \theta}{2^n \cos 2\theta}$ (4) $\frac{\cos 2^n \theta}{2^n \sin \theta}$
- $x(x^{n-1} - na^{n-1}) + a^n(n-1)$ is divisible by $(x-a)^2$ for-
(1) $n > 1$ (2) $n > 2$
(3) all $n \in \mathbb{N}$ (4) None of these
- For any odd integer $n \geq 1$, $n^3 - (n-1)^3 + \dots + (-1)^{n-1} \cdot 1^3$ is equal to-
(1) $\frac{1}{4}(n+1)^2(2n-1)$ (2) $\frac{1}{4}(n-1)^2(2n-1)$
(3) $\frac{1}{2}(n-1)^2(2n-1)$ (4) $\frac{1}{2}(n+1)^2(2n-1)$
- If p and q are respectively. The sum and the sum of squares of n successive integers beginning with a , then $nq - p^2$ is-
(1) independent of a (2) independent of n
(3) dependent on a (4) None of these
- The sum of first n terms of the given series $1^2 + 2.2^2 + 3^2 + 2.4^2 + 5^2 + 2.6^2 + \dots$ is $\frac{n(n+1)^2}{2}$, when n is even. When n is odd, then sum will be-
[AIEEE-2004]
(1) $\frac{n(n+1)^2}{2}$ (2) $\frac{1}{2}n^2(n+1)$
(3) $n(n+1)^2$ (4) None

- Let $S(k) = 1 + 3 + 5 + \dots + (2k-1) = 3 + k^2$, then which of the following is true? [AIEEE-2004]
(1) $S(1)$ is true (2) $S(k) \Rightarrow S(k+1)$
(3) $S(k) \not\Rightarrow S(k+1)$
(4) Principle of mathematical Induction can be used to prove that formula
- The sum of n terms of the series $1 + (1+a) + (1+a+a^2) + (1+a+a^2+a^3) + \dots$, is-
(1) $\frac{n}{1-a} - \frac{a(1-a^n)}{(1-a)^2}$ (2) $\frac{n}{1-a} + \frac{a(1-a^n)}{(1-a)^2}$
(3) $\frac{n}{1-a} + \frac{a(1+a^n)}{(1-a)^2}$ (4) $-\frac{n}{1-a} + \frac{a(1-a^n)}{(1-a)^2}$
- Statement -1** For every natural number $n \geq 2$
 $\frac{1}{\sqrt{1}} + \frac{1}{\sqrt{2}} + \dots + \frac{1}{\sqrt{n}} > \sqrt{n}$
Statement -2 : For every natural number $n \geq 2$, $\sqrt{n(n+1)} < n+1$. [AIEEE-2008]
(1) Statement -1 is false, Statement -2 is true
(2) Statement -1 is true, Statement -2 is false
(3) Statement -1 is true, Statement -2 is true; Statement -2 is a correct explanation for Statement -1
(4) Statement -1 is true, Statement -2 is true; Statement -2 is not a correct explanation for Statement -1
- Statement - 1**: For each natural number n , $(n+1)^7 - n^7 - 1$ is divisible by 7.
Statement - 2: For each natural number n , $n^7 - n$ is divisible by 7. [AIEEE-2011]
(1) Statement-1 is false, statement-2 is true.
(2) Statement-1 is true, statement-2 is true; Statement-2 is correct explanation for statement-1.
(3) Statement-1 is true, statement-2 is true; Statement-2 is not a correct explanation for statement-1.
(4) Statement-1 is true, statement-2 is false.

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13			
Ans.	3	3	1	3	1	3	1	1	2	2	1	3	2			

SETS

SET : A set is a collection of well defined objects which are distinct from each other

Set are generally denoted by capital letters A, B, C, etc. and the elements of the set by a, b, c etc.

If a is an element of a set A, then we write $a \in A$ and say a belongs to A.

If a does not belong to A then we write $a \notin A$,

Ex. The collection of first five prime natural numbers is a set containing the elements 2, 3, 5, 7, 11.

SOME IMPORTANT NUMBER SETS :

N = Set of all natural numbers

$$= \{1, 2, 3, 4, \dots\}$$

W = Set of all whole numbers

$$= \{0, 1, 2, 3, \dots\}$$

Z or I set of all integers

$$= \{\dots -3, -2, -1, 0, 1, 2, 3, \dots\}$$

Z^+ = Set of all +ve integers

$$= \{1, 2, 3, \dots\} = N.$$

Z^- = Set of all -ve integers

$$= \{-1, -2, -3, \dots\}$$

Z_0 = The set of all non-zero integers.

$$= \{\pm 1, \pm 2, \pm 3, \dots\}$$

Q = The set of all rational numbers.

$$= \left\{ \frac{p}{q} : p, q \in I, q \neq 0 \right\}$$

R = the set of all real numbers.

R-Q = The set of all irrational numbers

e.g. $\sqrt{2}$, $\sqrt{3}$, $\sqrt{5}$, π , e, $\log 2$ etc. are all irrational numbers.

METHODS TO WRITE A SET :

(i) Roster Method : In this method a set is described by listing elements, separated by commas and enclose then by curly brackets

Ex. The set of vowels of English Alphabet may be described as $\{a, e, i, o, u\}$

(ii) Set Builder Form : In this case we write down a property or rule p Which gives us all the element of the set

$$A = \{x : P(x)\}$$

Ex. $A = \{x : x \in N \text{ and } x = 2n \text{ for } n \in N\}$

i.e. $A = \{2, 4, 6, \dots\}$

Ex. $B = \{x^2 : x \in Z\}$

i.e. $B = \{0, 1, 4, 9, \dots\}$

TYPES OF SETS :

Null set or Empty set : A set having no element in it is called an Empty set or a null set or void set it is denoted by ϕ or $\{ \}$

Ex. $A = \{x \in N : 5 < x < 6\} = \phi$

A set consisting of at least one element is called a non-empty set or a non-void set.

Singleton : A set consisting of a single element is called a singleton set.

Ex. Then set $\{0\}$, is a singleton set

Finite Set : A set which has only finite number of elements is called a finite set.

Ex. $A = \{a, b, c\}$

Order of a finite set : The number of elements in a finite set is called the order of the set A and is denoted $O(A)$ or $n(A)$. It is also called cardinal number of the set.

Ex. $A = \{a, b, c, d\} \Rightarrow n(A) = 4$

Infinite set : A set which has an infinite number of elements is called an infinite set.

Ex. $A = \{1, 2, 3, 4, \dots\}$ is an infinite set

Equal sets : Two sets A and B are said to be equal if every element of A is a member of B , and every element of B is a member of A .

If sets A and B are equal. We write $A = B$ and A and B are not equal then $A \neq B$

Ex. $A = \{1, 2, 6, 7\}$ and $B = \{6, 1, 2, 7\} \Rightarrow A = B$

Equivalent sets : Two finite sets A and B are equivalent if their number of elements are same
ie. $n(A) = n(B)$

Ex. $A = \{1, 3, 5, 7\}$, $B = \{a, b, c, d\}$
 $n(A) = 4$ and $n(B) = 4 \Rightarrow n(A) = n(B)$

Note : Equal set always equivalent but equivalent sets may not be equal

Subsets : Let A and B be two sets if every element of A is an element B , then A is called a subset of B if A is a subset of B . we write $A \subseteq B$

Example : $A = \{1, 2, 3, 4\}$ and $B = \{1, 2, 3, 4, 5, 6, 7\} \Rightarrow A \subseteq B$

The symbol " $\Rightarrow$ " stands for "implies"

Proper subset : If A is a subset of B and $A \neq B$ then A is a proper subset of B . and we write $A \subset B$

Note-1 : Every set is a subset of itself i.e. $A \subseteq A$ for all A

Note-2 : Empty set ϕ is a subset of every set

Note-3 : Clearly $N \subset W \subset Z \subset Q \subset R \subset C$

Note-4 : The total number of subsets of a finite set containing n elements is 2^n

Universal set : A set consisting of all possible elements which occur in the discussion is called a Universal set and is denoted by U

Note : All sets are contained in the universal set

Ex. If $A = \{1, 2, 3\}$, $B = \{2, 4, 5, 6\}$, $C = \{1, 3, 5, 7\}$ then $U = \{1, 2, 3, 4, 5, 6, 7\}$ can be taken as the Universal set.

Power set : Let A be any set. The set of all subsets of A is called power set of A and is denoted by $P(A)$

Ex.1 Let $A = \{1, 2\}$ then $P(A) = \{\phi, \{1\}, \{2\}, \{1, 2\}\}$

Ex.2 Let $P(\phi) = \{\phi\}$

$\therefore P(P(\phi)) = \{\phi, \{\phi\}\}$

$\therefore P(P(P(\phi))) = \{\phi, \{\phi\}, \{\{\phi\}\}, \{\phi, \{\phi\}\}\}$

Note-1 : If $A = \phi$ then $P(A)$ has one element

Note-2 : Power set of a given set is always non empty

Some Operation on Sets :

- (i) **Union of two sets :** $A \cup B = \{x : x \in A \text{ or } x \in B\}$
e.g. $A = \{1, 2, 3\}$, $B = \{2, 3, 4\}$ then $A \cup B = \{1, 2, 3, 4\}$
- (ii) **Intersection of two sets :** $A \cap B = \{x : x \in A \text{ and } x \in B\}$
e.g. $A = \{1, 2, 3, \}$, $B = \{2, 3, 4\}$ then $A \cap B = \{2, 3\}$

- (iii) **Difference of two sets** : $A - B = \{x : x \in A \text{ and } x \notin B\}$
 e.g. $A = \{1, 2, 3\}$, $B = \{2, 3, 4\}$; $A - B = \{1\}$
- (iv) **Complement of a set** : $A' = \{x : x \notin A \text{ but } x \in U\} = U - A$
 e.g. $U = \{1, 2, \dots, 10\}$, $A = \{1, 2, 3, 4, 5\}$ then $A' = \{6, 7, 8, 9, 10\}$
- (v) **De-Morgan Laws** : $(A \cup B)' = A' \cap B'$; $(A \cap B)' = A' \cup B'$
- (vi) $A - (B \cup C) = (A - B) \cap (A - C)$; $A - (B \cap C) = (A - B) \cup (A - C)$
- (vii) **Distributive Laws** : $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$; $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
- (viii) **Commutative Laws** : $A \cup B = B \cup A$; $A \cap B = B \cap A$
- (ix) **Associative Laws** : $(A \cup B) \cup C = A \cup (B \cup C)$; $(A \cap B) \cap C = A \cap (B \cap C)$
- (x) $A \cap \phi = \phi$; $A \cap U = A$
 $A \cup \phi = A$; $A \cup U = U$
- (xi) $A \cap B \subseteq A$; $A \cap B \subseteq B$
- (xii) $A \subseteq A \cup B$; $B \subseteq A \cup B$
- (xiii) $A \subseteq B \Rightarrow A \cap B = A$
- (xiv) $A \subseteq B \Rightarrow A \cup B = B$

Disjoint Sets :

IF $A \cap B = \phi$, then A, B are disjoint.

e.g. if $A = \{1, 2, 3\}$, $B = \{7, 8, 9\}$ then $A \cap B = \phi$

Note : $A \cap A' = \phi$ $\therefore$ A, A' are disjoint.

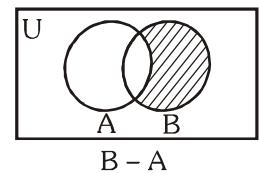
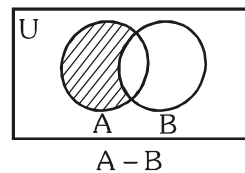
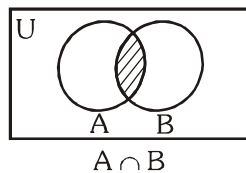
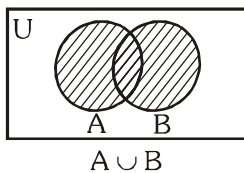
Symmetric Difference of Sets :

$$A \Delta B = (A - B) \cup (B - A)$$

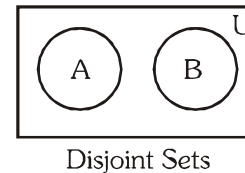
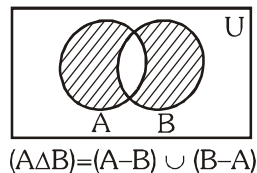
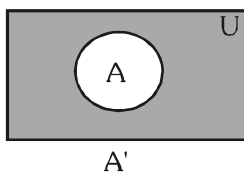
- $(A')' = A$
- $A \subseteq B \Leftrightarrow B' \subseteq A'$

If A and B are any two sets, then

- (i) $A - B = A \cap B'$
 (ii) $B - A = B \cap A'$
 (iii) $A - B = A \Leftrightarrow A \cap B = \phi$
 (iv) $(A - B) \cup B = A \cup B$
 (v) $(A - B) \cap B = \phi$
 (vi) $(A - B) \cup (B - A) = (A \cup B) - (A \cap B)$

Venn Diagramme :

$$\text{Clearly } (A - B) \cup (B - A) \cup (A \cap B) = A \cup B$$



Note : $A \cap A' = \phi$, $A \cup A' = U$

SOME IMPORTANT RESULTS ON NUMBER OF ELEMENTS IN SETS :

If A, B and C are finite sets, and U be the finite universal set, then

$$(i) \quad n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

$$(ii) \quad n(A \cup B) = n(A) + n(B) \Leftrightarrow A, B \text{ are disjoint non-void sets.}$$

$$(iii) \quad n(A - B) = n(A) - n(A \cap B) \text{ i.e. } n(A - B) + n(A \cap B) = n(A)$$

$$(iv) \quad n(A \Delta B) = \text{No. of elements which belong to exactly one of A or B}$$

$$= n((A - B) \cup (B - A))$$

$$= n(A - B) + n(B - A)$$

[$\because (A - B)$ and $(B - A)$ are disjoint]

$$= n(A) - n(A \cap B) + n(B) - n(A \cap B)$$

$$= n(A) + n(B) - 2n(A \cap B)$$

$$= n(A) + n(B) - 2n(A \cap B)$$

$$(v) \quad n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(A \cap C) + n(A \cap B \cap C)$$

$$(vi) \quad \text{Number of elements in exactly two of the sets A, B, C}$$

$$= n(A \cap B) + n(B \cap C) + n(C \cap A) - 3n(A \cap B \cap C)$$

$$(vii) \quad \text{number of elements in exactly one of the sets A, B, C}$$

$$= n(A) + n(B) + n(C) - 2n(A \cap B) - 2n(B \cap C) - 2n(A \cap C) + 3n(A \cap B \cap C)$$

$$(viii) \quad n(A' \cup B') = n((A \cap B)') = n(U) - n(A \cap B)$$

$$(ix) \quad n(A' \cap B') = n((A \cup B)') = n(U) - n(A \cup B)$$

Ex. In a group of 1000 people, there are 750 who can speak Hindi and 400 who can speak Bengali. How many can speak Hindi only? How many can speak Bengali? How many can speak both Hindi and Bengali?

Sol. Let A and B be the sets of persons who can speak Hindi and Bengali respectively.

then $n(A \cup B) = 1000$, $n(A) = 750$, $n(B) = 400$.

Number of persons who can speak both Hindi and Bengali

$$= n(A \cap B) = n(A) + n(B) - n(A \cup B)$$

$$= 750 + 400 - 1000$$

Number of persons who can speak Hindi only = 150

$$= n(A - B) = n(A) - n(A \cap B) = 750 - 150 = 600$$

Number of persons who can speak Bengali only

$$= n(B - A) = n(B) - n(A \cap B) = 400 - 150 = 250$$

SOLVED EXAMPLES

Ex.1 The set $A = \{x : x \in \mathbb{R}, x^2 = 16 \text{ and } 2x = 6\}$ equal-

(1) ϕ

(2) $\{14, 3, 4\}$

(3) $\{3\}$

(4) $\{4\}$

Sol.(1) $x^2 = 16 \Rightarrow x = \pm 4$

$2x = 6 \Rightarrow x = 3$

There is no value of x which satisfies both the above equations.

Thus, $A = \phi$

Hence (1) is the correct answer

Ex.2 Let $A = \{x : x \in \mathbb{R}, |x| < 1\}$; $B = \{x : x \in \mathbb{R}, |x - 1| \geq 1\}$ and $A \cup B = \mathbb{R} - D$, then the set D is-

(1) $\{x : 1 < x \leq 2\}$

(2) $\{x : 1 \leq x < 2\}$

(3) $\{x : 1 \leq x \leq 2\}$

(4) none of these

Sol.(2) $A = \{x : x \in \mathbb{R}, -1 < x < 1\}$

$B = \{x : x \in \mathbb{R} : x - 1 \leq -1 \text{ or } x - 1 \geq 1\}$

$= \{x : x \in \mathbb{R} : x \leq 0 \text{ or } x \geq 2\}$

$\therefore A \cup B = \mathbb{R} - D$

where $D = \{x : x \in \mathbb{R}, 1 \leq x < 2\}$

Thus (2) is the correct answer.

Ex.3 If $a\mathbb{N} = \{ax : x \in \mathbb{N}\}$, then the set $6\mathbb{N} \cap 8\mathbb{N}$ is equal to-

(1) $8\mathbb{N}$

(2) $48\mathbb{N}$

(3) $12\mathbb{N}$

(4) $24\mathbb{N}$

Sol.(4) $6\mathbb{N} = \{6, 12, 18, 24, 30, \dots\}$

$8\mathbb{N} = \{8, 16, 24, 32, \dots\}$

$\therefore 6\mathbb{N} \cap 8\mathbb{N} = \{24, 48, \dots\} = 24\mathbb{N}$

Short cut Method

$6\mathbb{N} \cap 8\mathbb{N} = 24\mathbb{N}$

[24 is the L.C.M. of 6 and 8]

Ex.4 If P, Q and R subsets of a set A , then $R \times (P' \cup Q)'$ =

(1) $(R \times P) \cap (R \times Q)$

(2) $(R \times Q) \cap (R \times P)$

(3) $(R \times P) \cup (R \times Q)$

(4) none of these

Sol.(1,2)

$R \times (P' \cup Q)' = R \times [(P')' \cap (Q)'] = R \times (P \cap Q) = (R \times P) \cap (R \times Q)$

Hence (1) is the correct answer.

Ex.5 If $A = \{x, y\}$, then the power set of A is-

(1) $\{x^y, y^x\}$

(2) $\{\phi, x, y\}$

(3) $\{\phi, \{x\}, \{2y\}\}$

(4) $\{\phi, \{x\}, \{y\}, \{x, y\}\}$

Sol.(4) Clearly $P(A)$ = Power set of A

= set of all subsets of A

$= \{\phi, \{x\}, \{y\}, \{x, y\}\}$

$\therefore$ (4) holds.

CHECK YOUR GRASP

SETS

EXERCISE-I

1. If A and B are two sets, then $A \cap (A \cup B)'$ is equal to-
 (1) A (2) B
 (3) ϕ (4) none of these
2. If A is any set, then-
 (1) $A \cup A' = \phi$ (2) $A \cup A' = U$
 (3) $A \cap A' = U$ (4) none of these
3. If A, B be any two sets, then $(A \cup B)'$ is equal to-
 (1) $A' \cup B'$ (2) $A' \cap B'$
 (3) $A \cap B$ (4) $A \cup B$
4. If A and B be any two sets, then $(A \cap B)'$ is equal to-
 (1) $A' \cap B'$ (2) $A' \cup B'$ (3) $A \cap B$ (4) $A \cup B$
5. Let $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$, $A = \{1, 2, 5\}$, $B = \{6, 7\}$ then $A \cap B'$ is-
 (1) B' (2) A (3) A' (4) B.
6. If A and B are two sets, then $A \cup B = A \cap B$ iff-
 (1) $A \subseteq B$ (2) $B \subseteq A$
 (3) $A = B$ (4) none of these
7. Let A and B be two sets in the universal set. Then $A - B$ equals-
 (1) $A \cap B'$ (2) $A' \cap B$
 (3) $A \cap B$ (4) none of these
8. Two sets A, B are disjoint iff-
 (1) $A \cup B = \phi$ (2) $A \cap B \neq \phi$
 (3) $A \cap B = \phi$ (4) None of these
9. Which of the following is a null set ?
 (1) $\{0\}$
 (2) $\{x : x > 0 \text{ or } x < 0\}$
 (3) $\{x : x^2 = 4 \text{ or } x = 3\}$
 (4) $\{x : x^2 + 1 = 0, x \in \mathbb{R}\}$
10. If $A \subseteq B$, then $A \cap B$ is equal to-
 (1) A (2) B (3) A' (4) B'
11. If A and B are any two sets, then $A \cup (A \cap B)$ is equal to-
 (1) A (2) B (3) A' (4) B'
12. If A and B are not disjoint, then $n(A \cup B)$ is equal to-
 (1) $n(A) + n(B)$
 (2) $n(A) + n(B) - n(A \cap B)$
 (3) $n(A) + n(B) + n(A \cap B)$
 (4) $n(A) \cdot n(B)$
13. If $A = \{2, 4, 5\}$, $B = \{7, 8, 9\}$ then $n(A \times B)$ is equal to-
 (1) 6 (2) 9 (3) 3 (4) 0
14. Let A and B be two sets such that $n(A) = 70$, $n(B) = 60$ and $n(A \cup B) = 110$. Then $n(A \cap B)$ is equal to-
 (1) 240 (2) 20 (3) 100 (4) 120
15. Which set is the subset of all given sets ?
 (1) $\{1, 2, 3, 4, \dots\}$ (2) $\{1\}$
 (3) $\{0\}$ (4) $\{\}$
16. If $Q = \left\{x : x = \frac{1}{y}, \text{ where } y \in \mathbb{N}\right\}$, then-
 (1) $0 \in Q$ (2) $1 \in Q$ (3) $2 \in Q$ (4) $\frac{2}{3} \in Q$
17. $A = \{x : x \neq x\}$ represents-
 (1) $\{0\}$ (2) $\{\}$ (3) $\{1\}$ (4) $\{x\}$
18. Which of the following statements is true ?
 (1) $3 \subseteq \{1, 3, 5\}$ (2) $3 \in \{1, 3, 5\}$
 (3) $\{3\} \in \{1, 3, 5\}$ (4) $\{3, 5\} \in \{1, 3, 5\}$
19. Which of the following is a null set ?
 (1) $A = \{x : x > 1 \text{ and } x < 1\}$
 (2) $B = \{x : x + 3 = 3\}$
 (3) $C = \{\phi\}$
 (4) $D = \{x : x \geq 1 \text{ and } x \leq 1\}$
20. $P(A) = P(B) \Rightarrow$
 (1) $A \subseteq B$ (2) $B \subseteq A$
 (3) $A = B$ (4) none of these

ANSWER KEY

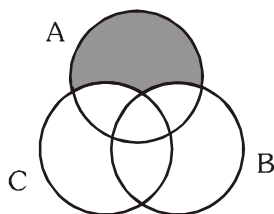
Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	3	2	2	2	2	3	1	3	4	1	1	2	2	2	4
Que.	16	17	18	19	20										
Ans.	2	2	2	1	3										

PREVIOUS YEAR QUESTIONS

SETS

EXERCISE-II

- If A, B, C be three sets such that $A \cup B = A \cup C$ and $A \cap B = A \cap C$, then - **[Roorkee 1991]**
 (1) $A = B$ (2) $B = C$
 (3) $A = C$ (4) $A = B = C$
- Sets A and B have 3 and 6 elements respectively. What can be the minimum number of elements in $A \cup B$? **[Roorkee 1991]**
 (1) 3 (2) 6 (3) 9 (4) 18
- In a college of 300 students, every student reads 5 new spapers and every newspaper is read by 60 students. The number of newspapers is - **[IIT -1998]**
 (1) at least 30 (2) at most 20
 (3) exactly 25 (4) none of these
- The set of intelligent students in a class is - **[A.M.U.-1998]**
 (1) a null set
 (2) a singleton set
 (3) a finite set
 (4) not a well defined collection
- The shaded region in the given figure is -



- (1) $A \cap (B \cup C)$ (2) $A \cup (B \cap C)$
 (3) $A \cap (B - C)$ (4) $A - (B \cup C)$
- Let $n(U) = 700$, $n(A) = 200$, $n(B) = 300$ and $n(A \cap B) = 100$, then $n(A' \cap B') =$
 (1) 400 (2) 600 (3) 300 (4) 200
- If $A = \{1, 2, 3, 4, 5\}$, then the number of proper subsets of A is -
 (1) 120 (2) 30 (3) 31 (4) 32
- Let A and B be two sets such that $n(A) = 0.16$, $n(B) = 0.14$, $n(A \cup B) = 0.25$. Then $n(A \cap B)$ is equal to - **[Jamia Milia Entrance Exam. 2001]**
 (1) 0.3 (2) 0.5
 (3) 0.05 (4) none of these

- If $A = \{x : x^2 - 5x + 6 = 0\}$, $B = \{2, 4\}$, $C = \{4, 5\}$, then $A \times (B \cap C)$ is - **[Kerala P.E.T. 2002]**
 (1) $\{(2, 4), (3, 4)\}$
 (2) $\{(4, 2), (4, 3)\}$
 (3) $\{(2, 4), (3, 4), (4, 4)\}$
 (4) $\{(2, 2), (3, 3), (4, 4), (5, 5)\}$
- If $A = \{(x, y) : x^2 + y^2 = 25\}$ and $B = \{(x, y) : x^2 + 9y^2 = 144\}$ then $A \cap B$ contains -
 (1) one point (2) three points
 (3) two points (4) four points
- A class has 175 students. The following data shows the number of students obtaining one or more subjects. Mathematics 100; Physics 70; Chemistry 40; Mathematics and Physics 30; Mathematics and Chemistry 28; Physics and Chemistry 23; Mathematics, Physics and Chemistry 18. How many students have offered Mathematics alone?
 (1) 35 (2) 48 (3) 60 (4) 22
- The set $S : \{1, 2, 3, \dots, 12\}$ is to be partitioned into three sets A, B, C of equal size. Thus $A \cup B \cup C = S$, $A \cap B = B \cap C = A \cap C = \phi$. The number of ways to partition S is - **[AIEEE - 2007]**
 (1) $12!/3!(4!)^3$ (2) $12!/3!(3!)^4$
 (3) $12!/(4!)^3$ (4) $12!/(3!)^4$
- If A, B and C are three sets such that $A \cap B = A \cap C$ and $A \cup B = A \cup C$, then :- **[AIEEE- 2009]**
 (1) $B = C$ (2) $A \cap B = \phi$
 (3) $A = B$ (4) $A = C$
- Two sets A and B are as under
 $A = \{(a, b) \in \mathbb{R} \times \mathbb{R} : |a - 5| < 1 \text{ and } |b - 5| < 1\};$
 $B = \{(a, b) \in \mathbb{R} \times \mathbb{R} : 4(a - 6)^2 + 9(b - 5)^2 \leq 36\}$. Then :- **[AIEEE-2018]**
 (1) $A \subset B$
 (2) $A \cap B = \phi$ (an empty set)
 (3) neither $A \subset B$ nor $B \subset A$
 (4) $B \subset A$

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	
Ans.	2	2	3	4	4	3	3	3	1	4	3	3	1	1	

RELATIONS

INTRODUCTION :

Let A and B be two sets. Then a relation R from A to B is a subset of $A \times B$.

thus, R is a relation from A to B $\Leftrightarrow R \subseteq A \times B$.

Ex. If $A = \{1, 2, 3\}$ and $B = \{a, b, c\}$, then $R = \{(1, b), (2, c), (1, a), (3, a)\}$ being a subset of $A \times B$, is a relation from A to B. Here $(1, b), (2, c), (1, a)$ and $(3, a) \in R$, so we write $1Rb, 2Rc, 1Ra$ and $3Ra$. But $(2, b) \notin R$, so we write $2 \not R b$

Total Number of Relations : Let A and B be two non-empty finite sets consisting of m and n elements respectively. Then $A \times B$ consists of mn ordered pairs. So, total number of subsets of $A \times B$ is 2^{mn} .

Domain and Range of a relation : Let R be a relation from a set A to a set B. Then the set of all first components or coordinates of the ordered pairs belonging to R is called to domain of R, while the set of all second components or coordinates of the ordered pairs in R is called the range of R.

Thus, $\text{Dom}(R) = \{a : (a, b) \in R\}$
and, $\text{Range}(R) = \{b : (a, b) \in R\}$

It is evident from the definition that the domain of a relation from A to B is a subset of A and its range is a subset of B.

Ex. Let $A = \{1, 3, 5, 7\}$ and $B = \{2, 4, 6, 8\}$ be two sets and let R be a relation from A to B defined by the phrase " $(x, y) \in R \Leftrightarrow x > y$ ". Under this relation R, we have

$3R2, 5R2, 5R4, 7R2, 7R4$ and $7R6$

i.e. $R = \{(3, 2), (5, 2), (5, 4), (7, 2), (7, 4), (7, 6)\}$

$\therefore \text{Dom}(R) = \{3, 5, 7\}$ and $\text{Range}(R) = \{2, 4, 6\}$

Inverse Relation : Let A, B be two sets and let R be a relation from a set A to a set B. Then the inverse of R, denoted by R^{-1} , is a relation from B to A and is defined by

$$R^{-1} = \{(b, a) : (a, b) \in R\}$$

Clearly, $(a, b) \in R \Leftrightarrow (b, a) \in R^{-1}$

Also, $\text{Dom}(R) = \text{Range}(R^{-1})$ and $\text{Range}(R) = \text{Dom}(R^{-1})$

Ex.1 Let A be the set of first ten natural numbers and let R be a relation on A defined by $(x, y) \in R \Leftrightarrow x + 2y = 10$, i.e. $R = \{(x, y) : x \in A, y \in A \text{ and } x + 2y = 10\}$. Express R and R^{-1} as sets of ordered pairs. Determine also (i) domain of R and R^{-1} (ii) range of R and R^{-1}

Sol. We have $(x, y) \in R \Leftrightarrow x + 2y = 10 \Leftrightarrow y = \frac{10-x}{2}, x, y \in A$

where $A = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$

Now, $x = 1 \Rightarrow y = \frac{10-1}{2} = \frac{9}{2} \notin A$.

This shows that 1 is not related to any element in A. Similarly we can observe. that 3, 5, 7, 9 and 10 are not related to any element of A under the defined relation

Further we find that :

$$\text{For } x = 2, y = \frac{10-2}{2} = 4 \in A \quad \therefore (2, 4) \in R$$

$$\text{For } x = 4, y = \frac{10-4}{2} = 3 \in A \quad \therefore (4, 3) \in R$$

$$\text{For } x = 6, y = \frac{10-6}{2} = 2 \in A \quad \therefore (6, 2) \in R$$

$$\text{For } x = 8, y = \frac{10-8}{2} = 1 \in A \quad \therefore (8, 1) \in R$$

Thus, $R = \{(2, 4), (4, 3), (6, 2), (8, 1)\}$

$\Rightarrow R^{-1} = \{(4, 2), (3, 4), (2, 6), (1, 8)\}$

Clearly, $\text{Dom}(R) = \{2, 4, 6, 8\} = \text{Range}(R^{-1})$

and, $\text{Range}(R) = \{4, 3, 2, 1\} = \text{Dom}(R^{-1})$

TYPES OF RELATIONS :

In this section we intend to define various types of relations on a given set A.

Void Relation : Let A be a set. Then $\phi \subseteq A \times A$ and so it is a relation on A. This relation is called the void or empty relation on A.

Universal Relation : Let A be a set. Then $A \times A \subseteq A \times A$ and so it is a relation on A. This relation is called the universal relation on A.

Identity Relation : Let A be a set. Then the relation $I_A = \{(a, a) : a \in A\}$ on A is called the identity relation on A.

In other words, a relation I_A on A is called the identity relation if every element of A is related to itself only.

Ex. The relation $I_A = \{(1, 1), (2, 2), (3, 3)\}$ is the identity relation on set $A = \{1, 2, 3\}$. But relations $R_1 = \{(1, 1), (2, 2)\}$ and $R_2 = \{(1, 1), (2, 2), (3, 3), (1, 3)\}$ are not identity relations on A, because $(3, 3) \notin R_1$ and in R_2 element 1 is related to elements 1 and 3.

Reflexive Relation : A relation R on a set A is said to be reflexive if every element of A is related to itself.

Thus, R on a set A is not reflexive if there exists an element $A \in A$ such that $(a, a) \notin R$.

Ex. Let $A = \{1, 2, 3\}$ be a set. Then $R = \{(1, 1), (2, 2), (3, 3), (1, 3), (2, 1)\}$ is a reflexive relation on A. But $R_1 = \{(1, 1), (3, 3), (2, 1), (3, 2)\}$ is not a reflexive relation on A, because $2 \in A$ but $(2, 2) \notin R_1$.

Note : Every Identity relation is reflexive but every reflexive relation is not identity.

Symmetric Relation : A relation R on a set A is said to be a symmetric relation iff

$(a, b) \in R \Rightarrow (b, a) \in R$ for all $a, b \in A$

i.e. $a R b \Rightarrow b R a$ for all $a, b \in A$.

Ex. Let L be the set of all lines in a plane and let R be a relation defined on L by the rule $(x, y) \in R \Leftrightarrow x$ is perpendicular to y . Then R is a symmetric relation on L, because $L_1 \perp L_2 \Rightarrow L_2 \perp L_1$

i.e. $(L_1, L_2) \in R \Rightarrow (L_2, L_1) \in R$.

Ex. Let $A = \{1, 2, 3, 4\}$ and Let R_1 and R_2 be relation on A given by $R_1 = \{(1, 3), (1, 4), (3, 1), (2, 2), (4, 1)\}$ and $R_2 = \{(1, 1), (2, 2), (3, 3), (1, 3)\}$. Clearly, R_1 is a symmetric relation on A. However, R_2 is not so, because $(1, 3) \in R_2$ but $(3, 1) \notin R_2$

Transitive Relation : Let A be any set. A relation R on A is said to be a transitive relation iff

$(a, b) \in R$ and $(b, c) \in R \Rightarrow (a, c) \in R$ for all $a, b, c \in A$

i.e. $a R b$ and $b R c \Rightarrow a R c$ for all $a, b, c \in A$

Ex. On the set N of natural numbers, the relation R defined by $x R y \Rightarrow x$ is less than y is transitive, because for any $x, y, z \in N$

$x < y$ and $y < z \Rightarrow x < z \Rightarrow x R y$ and $y R z \Rightarrow x R z$

Ex. Let L be the set of all straight lines in a plane. Then the relation 'is parallel to' on L is a transitive relation, because from any $\ell_1, \ell_2, \ell_3 \in L$.

$\ell_1 \parallel \ell_2$ and $\ell_2 \parallel \ell_3 \Rightarrow \ell_1 \parallel \ell_3$

Antisymmetric Relation : Let A be any set. A relation R on set A is said to be an antisymmetric relation iff

$(a, b) \in R$ and $(b, a) \in R \Rightarrow a = b$ for all $a, b \in A$

Ex. Let R be a relation on the set N of natural numbers defined by

$$x R y \Leftrightarrow 'x \text{ divides } y' \text{ for all } x, y \in N$$

This relation is an antisymmetric relation on N . Since for any two numbers $a, b \in N$

$$a \mid b \text{ and } b \mid a \Rightarrow a = b \quad \text{i.e. } a R b \text{ and } b R a \Rightarrow a = b$$

Equivalence Relation : A relation R on a set A is said to be an equivalence relation on A iff

- (i) it is reflexive i.e. $(a, a) \in R$ for all $a \in A$
- (ii) it is symmetric i.e. $(a, b) \in R \Rightarrow (b, a) \in R$ for all $a, b \in A$
- (iii) it is transitive i.e. $(a, b) \in R$ and $(b, c) \in R \Rightarrow (a, c) \in R$ for all $a, b, c \in A$.

Ex. Let R be a relation on the set of all lines in a plane defined by $(\ell_1, \ell_2) \in R \Leftrightarrow$ line ℓ_1 is parallel to line ℓ_2 .

R is an equivalence relation.

Note : It is not necessary that every relation which is symmetric and transitive is also reflexive.

SOLVED EXAMPLES

Ex.1 Three relation R_1, R_2 and R_3 are defined on set $A = \{a, b, c\}$ as follows :

$$(i) R_1 \{(a, a), (a, b), (a, c), (b, b), (b, c), (c, a), (c, b), (c, c)\} \quad (ii) R_2 \{(a, b), (b, a), (a, c), (c, a)\}$$

$$(iii) R_3 \{(a, b), (b, c), (c, a)\}$$

Find whether each of R_1, R_2 and R_3 is reflexive, symmetric and transitive.

Sol. (i) Reflexive : Clearly, $(a, a), (b, b), (c, c) \in R_1$. So, R_1 is reflexive on A .

Symmetric : We observe that $(a, b) \in R_1$ but $(b, a) \notin R_1$. So, R_1 is not symmetric on A .

Transitive : We find that $(b, c) \in R_1$ and $(c, a) \in R_1$ but $(b, a) \notin R_1$. So, R_1 is not transitive on A .

(ii) Reflexive : Since $(a, a), (b, b)$ and (c, c) are not in R_2 . So, it is not a reflexive relation on A .

Symmetric : We find that the ordered pairs obtained by interchanging the components of ordered pairs in R_2 are also in R_2 . So, R_2 is a symmetric relation on A .

Transitive : Clearly $(c, a) \in R_2$ and $(a, b) \in R_2$ but $(c, b) \notin R_2$. So, it is not a transitive relation on R_2 .

(iii) Reflexive : Since none of $(a, a), (b, b)$ and (c, c) is an element of R_3 . So, R_3 is not reflexive on A .

Symmetric : Clearly, $(b, c) \in R_3$ but $(c, b) \notin R_3$. So, it is not symmetric on A .

Transitive : Clearly, $(b, c) \in R_3$ and $(c, a) \in R_3$ but $(b, a) \notin R_3$. So, R_3 is not transitive on A .

Ex.2 Prove that the relation R on the set Z of all integers defined by

$$(x, y) \in R \Leftrightarrow x - y \text{ is divisible by } n$$

is an equivalence relation on Z .

Sol. We observe the following properties

Reflexivity : For any $a \in Z$, we have

$$a - a = 0 = 0 \times n \Rightarrow a - a \text{ is divisible by } n \Rightarrow (a, a) \in R$$

Thus, $(a, a) \in R$ for all $a \in Z$

So, R is reflexive on Z

symmetry : Let $(a, b) \in R$. Then,

$$(a, b) \in R \Rightarrow (a - b) \text{ is divisible by } n$$

$$\Rightarrow a - b = np \text{ for some } p \in Z$$

$$\Rightarrow b - a = n(-p)$$

$$\Rightarrow b - a \text{ is divisible by } n \quad [\because p \in Z \Rightarrow -p \in Z]$$

$$\Rightarrow (b, a) \in R$$

Thus, $(a, b) \in R \Rightarrow (b, a) \in R$ for all $a, b \in Z$

So, R is symmetric on Z .

Transitivity : Let $a, b, c \in Z$ such that $(a, b) \in R$ and $(b, c) \in R$. Then,

$$(a, b) \in R \Rightarrow (a - b) \text{ is divisible by } n$$

$$\Rightarrow a - b = np \text{ for some } p \in Z$$

$$(b, c) \in R \Rightarrow (b - c) \text{ is divisible by } n$$

$$\Rightarrow b - c = nq \text{ for some } q \in Z$$

$$\therefore (a, b) \in R \text{ and } (b, c) \in R$$

$$\Rightarrow a - b = np \text{ and } b - c = nq$$

$$\Rightarrow (a - b) + (b - c) = np + nq$$

$$\Rightarrow a - c = n(p + q)$$

$$\Rightarrow a - c \text{ is divisible by } n$$

$$[\because p, q \in \mathbb{Z} \Rightarrow p + q \in \mathbb{Z}]$$

$$\Rightarrow (a, c) \in R$$

thus, $(a, b) \in R$ and $(b, c) \in R \Rightarrow (a, c) \in R$ for all $a, b, c \in \mathbb{Z}$. so, R is transitive relation in $\mathbb{Z}$.

Ex.3 Shw that the relation is congruent to' on the set of all triangles in a plane is an equivalence relation.

Sol. Let S be the set of all triangles in a plane and let R be the relation on S defined by $(\Delta_1, \Delta_2) \in R \Leftrightarrow$ triangle Δ_1 is congruent to triangle Δ_2 . We observe the following properties.

Reflexivity : For each triangle $\Delta \in S$, we have

$$\Delta \cong \Delta \Rightarrow (\Delta, \Delta) \in R \text{ for all } \Delta \in S \Rightarrow R \text{ is reflexive on } S$$

Symmetry : Let $\Delta_1, \Delta_2 \in S$ such that $(\Delta_1, \Delta_2) \in R$. Then, $(\Delta_1, \Delta_2) \in R \Rightarrow \Delta_1 \cong \Delta_2 \Rightarrow \Delta_2 \cong \Delta_1 \Rightarrow (\Delta_2, \Delta_1) \in R$
So, R is symmetric on S

Transitivity : Let $\Delta_1, \Delta_2, \Delta_3 \in S$ such that $(\Delta_1, \Delta_2) \in R$ and $(\Delta_2, \Delta_3) \in R$. Then,

$$(\Delta_1, \Delta_2) \in R \text{ and } (\Delta_2, \Delta_3) \in R \Rightarrow \Delta_1 \cong \Delta_2 \text{ and } \Delta_2 \cong \Delta_3 \Rightarrow \Delta_1 \cong \Delta_3 \Rightarrow (\Delta_1, \Delta_3) \in R$$

So, R is transitive on S .

Hence, R being reflexive, symmetric and transitive, is an equivalence relation on S .

CHECK YOUR GRASP

RELATIONS

EXERCISE-I

- If R is a relation from a finite set A having m elements to a finite set B having n elements, then the number of relations from A to B is-
(1) 2^{mn} (2) $2^{mn}-1$ (3) $2mn$ (4) m^n
- In the set $A = \{1, 2, 3, 4, 5\}$, a relation R is defined by $R = \{(x, y) \mid x, y \in A \text{ and } x < y\}$. Then R is-
(1) Reflexive (2) Symmetric
(3) Transitive (4) None of these
- For real numbers x and y , we write $x R y \Leftrightarrow x - y + \sqrt{2}$ is an irrational number. Then the relation R is-
(1) Reflexive (2) Symmetric
(3) Transitive (4) none of these
- Let $X = \{1, 2, 3, 4\}$ and $Y = \{1, 3, 5, 7, 9\}$. Which of the following is relations from X to Y -
(1) $R_1 = \{(x, y) \mid y = 2 + x, x \in X, y \in Y\}$
(2) $R_2 = \{(1, 1), (2, 1), (3, 3), (4, 3), (5, 5)\}$
(3) $R_3 = \{(1, 1), (1, 3), (3, 5), (3, 7), (5, 7)\}$
(4) $R_4 = \{(1, 3), (2, 5), (2, 4), (7, 9)\}$
- Let L denote the set of all straight lines in a plane. Let a relation R be defined by $\alpha R \beta \Leftrightarrow \alpha \perp \beta$, $\alpha, \beta \in L$. Then R is-
(1) Reflexive (2) Symmetric
(3) Transitive (4) none of these
- Let R be a relation defined in the set of real numbers by $a R b \Leftrightarrow 1 + ab > 0$. Then R is-
(1) Equivalence relation (2) Transitive
(3) Symmetric (4) Anti-symmetric
- Which one of the following relations on R is equivalence relation-
(1) $x R_1 y \Leftrightarrow |x| = |y|$ (2) $x R_2 y \Leftrightarrow x \geq y$
(3) $x R_3 y \Leftrightarrow x \mid y$ (4) $x R_4 y \Leftrightarrow x < y$
- Two points P and Q in a plane are related if $OP = OQ$, where O is a fixed point. This relation is-
(1) Reflexive but not symmetric
(2) Symmetric but not transitive
(3) An equivalence relation
(4) none of these
- The relation R defined in $A = \{1, 2, 3\}$ by $a R b$ if $|a^2 - b^2| \leq 5$. Which of the following is false-
(1) $R = \{(1, 1), (2, 2), (3, 3), (2, 1), (1, 2), (2, 3), (3, 2)\}$
(2) $R^{-1} = R$
(3) Domain of $R = \{1, 2, 3\}$
(4) Range of $R = \{5\}$
- Let a relation R is the set N of natural numbers be defined as $(x, y) \in R$ if and only if $x^2 - 4xy + 3y^2 = 0$ for all $x, y \in N$. The relation R is-
(1) Reflexive
(2) Symmetric
(3) Transitive
(4) An equivalence relation
- Let $A = \{2, 3, 4, 5\}$ and let $R = \{(2, 2), (3, 3), (4, 4), (5, 5), (2, 3), (3, 2), (3, 5), (5, 3)\}$ be a relation in A . Then R is-
(1) Reflexive and transitive
(2) Reflexive and symmetric
(3) Reflexive and antisymmetric
(4) none of these
- If $A = \{2, 3\}$ and $B = \{1, 2\}$, then $A \times B$ is equal to-
(1) $\{(2, 1), (2, 2), (3, 1), (3, 2)\}$
(2) $\{(1, 2), (1, 3), (2, 2), (2, 3)\}$
(3) $\{(2, 1), (3, 2)\}$
(4) $\{(1, 2), (2, 3)\}$
- Let R be a relation over the set $N \times N$ and it is defined by $(a, b) R (c, d) \Rightarrow a + d = b + c$. Then R is-
(1) Reflexive only
(2) Symmetric only
(3) Transitive only
(4) An equivalence relation
- Let N denote the set of all natural numbers and R be the relation on $N \times N$ defined by $(a, b) R (c, d)$ if $ad(b + c) = bc(a + d)$, then R is-
(1) Symmetric only
(2) Reflexive only
(3) Transitive only
(4) An equivalence relation
- If $A = \{1, 2, 3\}$, $B = \{1, 4, 6, 9\}$ and R is a relation from A to B defined by ' x is greater than y '. Then range of R is-
(1) $\{1, 4, 6, 9\}$ (2) $\{4, 6, 9\}$
(3) $\{1\}$ (4) none of these
- Let L be the set of all straight lines in the Euclidean plane. Two lines ℓ_1 and ℓ_2 are said to be related by the relation R if ℓ_1 is parallel to ℓ_2 . Then the relation R is-
(1) Reflexive (2) Symmetric
(3) Transitive (4) Equivalence

17. A and B are two sets having 3 and 4 elements respectively and having 2 elements in common. The number of relations which can be defined from A to B is-
- (1) 2^5 (2) $2^{10} - 1$
(3) $2^{12} - 1$ (4) 2^{12}
18. For $n, m \in \mathbb{N}$, $n \mid m$ means that n is a factor of m, the relation $\mid$ is-
- (1) reflexive and symmetric
(2) transitive and symmetric
(3) reflexive, transitive and symmetric
(4) reflexive, transitive and not symmetric
19. Let $R = \{(x, y) : x, y \in A, x + y = 5\}$ where $A = \{1, 2, 3, 4, 5\}$ then
- (1) R is not reflexive, symmetric and not transitive
(2) R is an equivalence relation
(3) R is reflexive, symmetric but not transitive
(4) R is not reflexive, not symmetric but transitive
20. Let R be a relation on a set A such that $R = R^{-1}$ then R is-
- (1) reflexive
(2) symmetric
(3) transitive
(4) none of these
21. Let $x, y \in I$ and suppose that a relation R on I is defined by $x R y$ if and only if $x \leq y$ then
- (1) R is partial order relation
(2) R is an equivalence relation
(3) R is reflexive and symmetric
(4) R is symmetric and transitive
22. Let R be a relation from a set A to a set B, then-
- (1) $R = A \cup B$ (2) $R = A \cap B$
(3) $R \subseteq A \times B$ (4) $R \subseteq B \times A$
23. Given the relation $R = \{(1, 2), (2, 3)\}$ on the set $A = \{1, 2, 3\}$, the minimum number of ordered pairs which when added to R make it an equivalence relation is-
- (1) 5 (2) 6 (3) 7 (4) 8
24. Let $P = \{(x, y) \mid x^2 + y^2 = 1, x, y \in \mathbb{R}\}$ Then P is-
- (1) reflexive (2) symmetric
(3) transitive (4) anti-symmetric
25. Let X be a family of sets and R be a relation on X defined by 'A is disjoint from B'. Then R is-
- (1) reflexive (2) symmetric
(3) anti-symmetric (4) transitive
26. In order that a relation R defined in a non-empty set A is an equivalence relation, it is sufficient that R
- (1) is reflexive
(2) is symmetric
(3) is transitive
(4) possesses all the above three properties
27. If R is an equivalence relation in a set A, then R^{-1} is-
- (1) reflexive but not symmetric
(2) symmetric but not transitive
(3) an equivalence relation
(4) none of these
28. Let $A = \{p, q, r\}$. Which of the following is an equivalence relation in A ?
- (1) $R_1 = \{(p, q), (q, r), (p, r), (p, p)\}$
(2) $R_2 = \{(r, q), (r, p), (r, r), (q, q)\}$
(3) $R_3 = \{(p, p), (q, q), (r, r), (p, q)\}$
(4) none of these

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	1	3	1	1	2	3	1	3	4	1	2	1	4	4	3
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28		
Ans.	4	4	4	1	2	1	3	3	2	2	4	3	4		

PREVIOUS YEAR QUESTIONS

RELATIONS

EXERCISE-II

1. Let $R = \{(1, 3), (4, 2), (2, 4), (2, 3), (3, 1)\}$ be a relation on the set $A = \{1, 2, 3, 4\}$. The relation R is- [AIEEE - 2004]
 (1) transitive (2) not symmetric
 (3) reflexive (4) a function
2. Let $R = \{(3, 3), (6, 6), (9, 9), (12, 12), (6, 12), (3, 9), (3, 12), (3, 6)\}$ be relation on the set $A = \{3, 6, 9, 12\}$. The relation is- [AIEEE - 2005]
 (1) reflexive and transitive only
 (2) reflexive only
 (3) an equivalence relation
 (4) reflexive and symmetric only
3. Let W denote the words in the English dictionary. Define the relation R by : $R = \{(x, y) \in W \times W \mid \text{the words } x \text{ and } y \text{ have at least one letter in common}\}$. Then R is- [AIEEE - 2006]
 (1) reflexive, symmetric and not transitive
 (2) reflexive, symmetric and transitive
 (3) reflexive, not symmetric and transitive
 (4) not reflexive, symmetric and transitive
4. Consider the following relations :-
 $R = \{(x, y) \mid x, y \text{ are real numbers and } x = wy \text{ for some rational number } w\}$;
 $S = \{(\frac{m}{n}, \frac{p}{q}) \mid m, n, p \text{ and } q \text{ are integers such that } n, q \neq 0 \text{ and } qm = pn\}$.
 Then : [AIEEE - 2010]
 (1) R is an equivalence relation but S is not an equivalence relation
 (2) Neither R nor S is an equivalence relation
 (3) S is an equivalence relation but R is not an equivalence relation
 (4) R and S both are equivalence relations
5. Let R be the set of real numbers. [AIEEE - 2011]
Statement-1:
 $A = \{(x, y) \in R \times R : y - x \text{ is an integer}\}$ is an equivalence relation on R .
Statement-2:
 $B = \{(x, y) \in R \times R : x = \alpha y \text{ for some rational number } \alpha\}$ is an equivalence relation on R .
 (1) Statement-1 is true, Statement-2 is false.
 (2) Statement-1 is false, Statement-2 is true
 (3) Statement-1 is true, Statement-2 is true; Statement-2 is a correct explanation for Statement-1
 (4) Statement-1 is true, Statement-2 is true; Statement-2 is **not** a correct explanation for Statement-1.

ANSWER KEY

Que.	1	2	3	4	5										
Ans.	2	1	1	3	1										

Mathematics

TARGET : JEE (Main + Advanced)

NURTURE COURSE

Logarithm



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JEE (Advanced) Syllabus :

Logarithms and their properties.

LOGARITHM

1. DEFINITION :

Every positive real number N can be expressed in exponential form as $a^x = N$ where ' a ' is also a positive real number different than unity and is called the base and ' x ' is called an exponent.

We can write the relation $a^x = N$ in logarithmic form as $\log_a N = x$. Hence $a^x = N \Leftrightarrow \log_a N = x$.

Hence logarithm of a number to some base is the exponent by which the base must be raised in order to get that number.

Limitations of logarithm: $\log_a N$ is defined only when

- (i) $N > 0$ (ii) $a > 0$ (iii) $a \neq 1$

Note :

- (i) For a given value of N , $\log_a N$ will give us a unique value.
 (ii) Logarithm of zero does not exist.
 (iii) Logarithm of negative reals are not defined in the system of real numbers.

Illustration 1 : If $\log_4 m = 1.5$, then find the value of m .

Solution : $\log_4 m = 1.5 \Rightarrow m = 4^{3/2} \Rightarrow m = 8$

Illustration 2 : If $\log_5 p = a$ and $\log_2 q = a$, then prove that $\frac{p^4 q^4}{100} = 100^{2a-1}$

Solution : $\log_5 p = a \Rightarrow p = 5^a$

$\log_2 q = a \Rightarrow q = 2^a$

$$\Rightarrow \frac{p^4 q^4}{100} = \frac{5^{4a} \cdot 2^{4a}}{100} = \frac{(10)^{4a}}{100} = \frac{(100)^{2a}}{100} = 100^{2a-1}$$

Illustration 3 : The value of N , satisfying $\log_a [1 + \log_b \{1 + \log_c (1 + \log_p N)\}] = 0$ is -

- (A) 4 (B) 3 (C) 2 (D) 1

Solution : $1 + \log_b \{1 + \log_c (1 + \log_p N)\} = a^0 = 1$

$$\Rightarrow \log_b \{1 + \log_c (1 + \log_p N)\} = 0 \Rightarrow 1 + \log_c (1 + \log_p N) = 1$$

$$\Rightarrow \log_c (1 + \log_p N) = 0 \Rightarrow 1 + \log_p N = 1$$

$$\Rightarrow \log_p N = 0 \Rightarrow N = 1$$

Ans. (D)

Do yourself - 1 :

(i) Express the following in logarithmic form :

(a) $81 = 3^4$ (b) $0.001 = 10^{-3}$ (c) $2 = 128^{1/7}$

(ii) Express the following in exponential form :

(a) $\log_2 32 = 5$ (b) $\log_{\sqrt{2}} 4 = 4$ (c) $\log_{10} 0.01 = -2$

(iii) If $\log_{2\sqrt{3}} 1728 = x$, then find x .

2. FUNDAMENTAL IDENTITIES :

Using the basic definition of logarithm we have 3 important deductions :

- (a) $\log_a 1 = 0$ i.e. logarithm of unity to any base is zero.
 (b) $\log_N N = 1$ i.e. logarithm of a number to the same base is 1.
 (c) $\log_{\frac{1}{N}} N = -1 = \log_N \frac{1}{N}$ i.e. logarithm of a number to the base as its reciprocal is -1 .

Note : $N = (a)^{\log_a N}$ e.g. $2^{\log_2 7} = 7$

Do yourself - 2 :

(i) Find the value of the following :

(a) $\log_{1.43} \frac{43}{30}$ (b) $\left(\frac{1}{2}\right)^{\log_2 5}$

(ii) If $4^{\log_2 2x} = 36$, then find x.

3. THE PRINCIPAL PROPERTIES OF LOGARITHMS :

If m, n are arbitrary positive numbers where $a > 0$, $a \neq 1$ and x is any real number, then-

- (a) $\log_a mn = \log_a m + \log_a n$
 (b) $\log_a \frac{m}{n} = \log_a m - \log_a n$
 (c) $\log_a m^x = x \log_a m$

Illustration 4 : Find the value of $2 \log \frac{2}{5} + 3 \log \frac{25}{8} - \log \frac{625}{128}$

Solution :

$$\begin{aligned} & 2 \log \frac{2}{5} + 3 \log \frac{25}{8} + \log \frac{128}{625} \\ &= \log \frac{2^2}{5^2} + \log \left(\frac{5^2}{2^3}\right)^3 + \log \frac{2^7}{5^4} \\ &= \log \frac{2^2}{5^2} \cdot \frac{5^6}{2^9} \cdot \frac{2^7}{5^4} = \log 1 = 0 \end{aligned}$$

Illustration 5 : If $\log_e x - \log_e y = a$, $\log_e y - \log_e z = b$ & $\log_e z - \log_e x = c$, then find the value of

$$\left(\frac{x}{y}\right)^{b-c} \times \left(\frac{y}{z}\right)^{c-a} \times \left(\frac{z}{x}\right)^{a-b}$$

Solution : $\log_e x - \log_e y = a \Rightarrow \log_e \frac{x}{y} = a \Rightarrow \frac{x}{y} = e^a$

$$\log_e y - \log_e z = b \Rightarrow \log_e \frac{y}{z} = b \Rightarrow \frac{y}{z} = e^b$$

$$\log_e z - \log_e x = c \Rightarrow \log_e \frac{z}{x} = c \Rightarrow \frac{z}{x} = e^c$$

$$\begin{aligned} \therefore (e^a)^{b-c} \times (e^b)^{c-a} \times (e^c)^{a-b} \\ = e^{a(b-c)+b(c-a)+c(a-b)} = e^0 = 1 \end{aligned}$$

Illustration 6 : If $a^2 + b^2 = 23ab$, then prove that $\log \frac{(a+b)}{5} = \frac{1}{2}(\log a + \log b)$.

Solution : $a^2 + b^2 = (a+b)^2 - 2ab = 23ab$
 $\Rightarrow (a+b)^2 = 25ab \Rightarrow a+b = 5\sqrt{ab}$ (i)
 Using (i)

$$\text{L.H.S.} = \log \frac{(a+b)}{5} = \log \frac{5\sqrt{ab}}{5} = \log \sqrt{ab} = \frac{1}{2} \log ab = \frac{1}{2}(\log a + \log b) = \text{R.H.S.}$$

Illustration 7 : If $\log_a x = p$ and $\log_b x^2 = q$, then $\log_x \sqrt{ab}$ is equal to (where $a, b, x \in \mathbb{R}^+ - \{1\}$)-

(A) $\frac{1}{p} + \frac{1}{q}$ (B) $\frac{1}{2p} + \frac{1}{q}$ (C) $\frac{1}{p} + \frac{1}{2q}$ (D) $\frac{1}{2p} + \frac{1}{2q}$

Solution : $\log_a x = p \Rightarrow a^p = x \Rightarrow a = x^{1/p}$.
 similarly $b^q = x^2 \Rightarrow b = x^{2/q}$

$$\text{Now, } \log_x \sqrt{ab} = \log_x \sqrt{x^{1/p} x^{2/q}} = \log_x x^{\left(\frac{1}{p} + \frac{2}{q}\right) \frac{1}{2}} = \frac{1}{2p} + \frac{1}{q}$$

Do yourself - 3 :

(i) Show that $\frac{1}{2} \log 9 + 2 \log 6 + \frac{1}{4} \log 81 - \log 12 = 3 \log 3$

4. BASE CHANGING THEOREM :

Can be stated as "quotient of the logarithm of two numbers is independent of their common base."

Symbolically, $\log_b m = \frac{\log_a m}{\log_a b}$, where $a > 0, a \neq 1, b > 0, b \neq 1$

Note :

(i) $\log_b a \cdot \log_a b = \frac{\log a}{\log b} \cdot \frac{\log b}{\log a} = 1$; hence $\log_b a = \frac{1}{\log_a b}$.

(ii) $a^{\log_b c} = c^{\log_b a}$

(iii) **Base power formula :** $\log_{a^k} m = \frac{1}{k} \log_a m$

(iv) The base of the logarithm can be any positive number other than 1, but in normal practice, only two bases are popular, these are 10 and $e (= 2.718 \text{ approx})$. Logarithms of numbers to the base 10 are named as 'common logarithm' and the logarithms of numbers to the base e are called Natural or Napierian logarithm. **We will consider $\log x$ as $\log_e x$ or $\ln x$.**

(v) Conversion of base e to base 10 & viceversa :

$$\log_e a = \frac{\log_{10} a}{\log_{10} e} = 2.303 \times \log_{10} a; \quad \log_{10} a = \frac{\log_e a}{\log_e 10} = \log_{10} e \times \log_e a = 0.434 \log_e a$$

Illustration 8 : If a, b, c are distinct positive real numbers different from 1 such that

$(\log_b a \cdot \log_c a - \log_a a) + (\log_a b \cdot \log_c b - \log_b b) + (\log_a c \cdot \log_b c - \log_c c) = 0$, then abc is equal to -

- (A) 0 (B) e (C) 1 (D) none of these

Solution : $(\log_b a \log_c a - 1) + (\log_a b \cdot \log_c b - 1) + (\log_a c \log_b c - 1) = 0$

$$\Rightarrow \frac{\log a}{\log b} \cdot \frac{\log a}{\log c} + \frac{\log b}{\log a} \cdot \frac{\log b}{\log c} + \frac{\log c}{\log a} \cdot \frac{\log c}{\log b} = 3$$

$$\Rightarrow (\log a)^3 + (\log b)^3 + (\log c)^3 = 3 \log a \log b \log c$$

$$\Rightarrow (\log a + \log b + \log c) = 0 \quad [\because \text{If } a^3 + b^3 + c^3 - 3abc = 0, \text{ then } a + b + c = 0 \text{ if } a \neq b \neq c]$$

$$\Rightarrow \log abc = \log 1 \Rightarrow abc = 1$$

Illustration 9 : Evaluate : $81^{1/\log_5 3} + 27^{\log_9 36} + 3^{4/\log_7 9}$

Solution : $81^{\log_3 5} + 3^{3 \log_9 36} + 3^{4 \log_9 7}$

$$= 3^{4 \log_3 5} + 3^{\log_3 (36)^{3/2}} + 3^{\log_3 7^2}$$

$$= 625 + 216 + 49 = 890.$$

Do yourself - 4 :

(i) Evaluate : $\frac{\log_3 135}{\log_{15} 3} - \frac{\log_3 5}{\log_{405} 3}$

(ii) Evaluate : $\log_9 27 - \log_{27} 9$

(iii) Evaluate : $2^{\log_3 5} - 5^{\log_3 2}$

(iv) Evaluate : $\log_3 4 \cdot \log_4 5 \cdot \log_5 6 \cdot \log_6 7 \cdot \log_7 8 \cdot \log_8 9$

(v) If $\frac{1}{\log_3 \pi} + \frac{1}{\log_4 \pi} > x$, then x can be -

- (A) 2 (B) 3 (C) 3.5 (D) π

(vi) If $\log_a 3 = 2$ and $\log_b 8 = 3$, then $\log_a b$ is -

- (A) $\log_3 2$ (B) $\log_2 3$ (C) $\log_3 4$ (D) $\log_4 3$

5. POINTS TO REMEMBER :

- (i) If base of logarithm is greater than 1 then logarithm of greater number is greater. i.e. $\log_2 8 = 3$, $\log_2 4 = 2$ etc. and if base of logarithm is between 0 and 1 then logarithm of greater number is smaller. i.e. $\log_{1/2} 8 = -3$, $\log_{1/2} 4 = -2$ etc.

$$\log_a x < \log_a y \Leftrightarrow \begin{cases} x < y & \text{if } a > 1 \\ x > y & \text{if } 0 < a < 1 \end{cases}$$

- (ii) It must be noted that whenever the number and the base are on the same side of unity then logarithm of that number to that base is positive, however if the number and the base are located on different side of unity then logarithm of that number to that base is negative.

e.g. $\log_{10} \sqrt[3]{10} = \frac{1}{3}$; $\log_{\sqrt{7}} 49 = 4$; $\log_{\frac{1}{2}} \left(\frac{1}{8}\right) = 3$; $\log_2 \left(\frac{1}{32}\right) = -5$; $\log_{10}(0.001) = -3$

- (iii) $x + \frac{1}{x} \geq 2$ if x is positive real number and $x + \frac{1}{x} \leq -2$ if x is negative real number

- (iv) $n \geq 2, n \in \mathbb{N}$

$$\sqrt[n]{a} = a^{1/n} \Rightarrow n^{\text{th}} \text{ root of 'a' } \quad ('a' \text{ is a non negative number})$$

Some important values : $\log_{10} 2 \approx 0.3010$; $\log_{10} 3 \approx 0.4771$; $\ln 2 \approx 0.693$, $\ln 10 \approx 2.303$

6. CHARACTERISTIC AND MANTISSA :

For any given number N , logarithm can be expressed as $\log_a N = \text{Integer} + \text{Fraction}$

The integer part is called characteristic and the fractional part is called mantissa. When the value of $\log n$ is given, then to find digits of 'n' we use only the mantissa part. The characteristic is used only in determining the number of digits in the integral part (if $n \geq 1$) or the number of zeros after decimal & before first non-zero digit in the number (if $0 < n < 1$).

Note :

- The mantissa part of logarithm of a number is always non-negative ($0 \leq m < 1$)
- If the characteristic of $\log_{10} N$ be n , then the number of digits in N is $(n + 1)$
- If the characteristic of $\log_{10} N$ be $(-n)$, then there exist $(n - 1)$ zeros after decimal in N .

7. ANTILOGARITHM :

The positive real number 'n' is called the antilogarithm of a number 'm' if $\log n = m$

$$\text{Thus, } \log n = m \Leftrightarrow n = \text{antilog } m$$

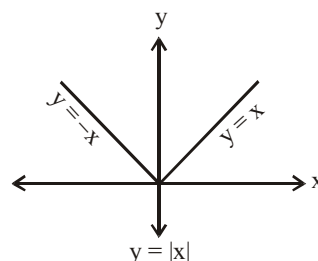
Do yourself - 5 :

- Evaluate : $\log_{10}(0.06)^6$
- Find number of digits in 18^{20}
- Determine number of cyphers (zeros) between decimal & first significant digit in $\left(\frac{1}{6}\right)^{200}$
- Find antilog of $\frac{5}{6}$ to the base 64.

8. ABSOLUTE VALUE FUNCTION/MODULUS FUNCTION :

The symbol of modulus function is $|x|$

$$\text{and is defined as : } y = |x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$



Properties of Modulus :For any $a, b \in \mathbb{R}$

(a) $|a| \geq 0$

(b) $|a| = |-a|$

(c) $|ab| = |a||b|$

(d) $\left| \frac{a}{b} \right| = \frac{|a|}{|b|}$

(e) $\sqrt{a^2} = |a|$

Illustration 10 : If $||x-1| - 2| = 5$, then find x .**Solution :** $|x-1| - 2 = \pm 5$

$|x-1| = 7, -3$

Case-I : When $|x-1| = 7 \Rightarrow x-1 = \pm 7 \Rightarrow x = 8, -6$ **Case-II :** When $|x-1| = -3$ (reject)**Illustration 11 :** If $|x-1| + |x+1| = 2$, then find x .**Solution :** **Case-I :** If $x \leq -1$

$-(x-1) - (x+1) = 2$

$\Rightarrow -x+1-x-1=2$

$\Rightarrow -2x=2 \Rightarrow x=-1$ (i)

Case-II : If $-1 < x < 1$

$-(x-1) + (x+1) = 2$

$\Rightarrow -x+1+x+1=2$

$\Rightarrow 2=2 \Rightarrow -1 < x < 1$ (ii)

Case-III : If $x \geq 1$

$x-1+x+1=2$

$\Rightarrow x=1$ (iii)

Thus from (i), (ii) and (iii) $-1 \leq x \leq 1$ **Do yourself - 6 :**

(i) Solve : $|x+3| = 2(5-x)$

(ii) Solve : $x|x| + 7x - 8 = 0$

Miscellaneous Illustrations :

Illustration 12 : Show that $\log_4 18$ is an irrational number.

Solution : $\log_4 18 = \log_4 (3^2 \times 2) = 2\log_4 3 + \log_4 2 = 2 \frac{\log_2 3}{\log_2 4} + \frac{1}{\log_2 4} = \log_2 3 + \frac{1}{2}$

assume the contrary, that this number $\log_2 3$ is rational number.

$$\Rightarrow \log_2 3 = \frac{p}{q}. \text{ Since } \log_2 3 > 0 \text{ both numbers } p \text{ and } q \text{ may be regarded as natural number}$$

$$\Rightarrow 3 = 2^{p/q} \Rightarrow 2^p = 3^q$$

But this is not possible for any natural number p and q . The resulting contradiction completes the proof.

Illustration 13 : If in a right angled triangle, a and b are the lengths of sides and c is the length of hypotenuse and $c - b \neq 1$, $c + b \neq 1$, then show that

$$\log_{c+b} a + \log_{c-b} a = 2\log_{c+b} a \cdot \log_{c-b} a.$$

Solution : We know that in a right angled triangle

$$c^2 = a^2 + b^2$$

$$c^2 - b^2 = a^2 \quad \dots\dots\dots (i)$$

$$\begin{aligned} \text{LHS} &= \frac{1}{\log_a (c+b)} + \frac{1}{\log_a (c-b)} = \frac{\log_a (c-b) + \log_a (c+b)}{\log_a (c+b) \cdot \log_a (c-b)} \\ &= \frac{\log_a (c^2 - b^2)}{\log_a (c+b) \cdot \log_a (c-b)} = \frac{\log_a a^2}{\log_a (c+b) \cdot \log_a (c-b)} \quad (\text{using (i)}) \\ &= \frac{2}{\log_a (c+b) \cdot \log_a (c-b)} = 2\log_{(c+b)} a \cdot \log_{(c-b)} a = \text{RHS} \end{aligned}$$

ANSWERS FOR DO YOURSELF

- | | | |
|------------------------------------|-------------------------------------|---|
| 1 : (i) (a) $\log_3 81 = 4$ | (b) $\log_{10} (0.001) = -3$ | (c) $\log_{128} 2 = 1/7$ |
| (ii) (a) $32 = 2^5$ | (b) $4 = (\sqrt{2})^4$ | (c) $0.01 = 10^{-2}$ |
| (iii) 6 | | |
| 2 : (i) (a) 1 | (b) $\frac{1}{5}$ | (ii) 3 |
| 4 : (i) 3 | (ii) $5/6$ | (iii) 0 (iv) 2 (v) (A) (vi) (C) |
| 5 : (i) $\bar{8}.6686$ | (ii) 26 | (iii) 155 (iv) 32 |
| 6 : (i) $\frac{7}{3}$ | (ii) $x = 1$ | |

ELEMENTARY EXERCISE

Solve for $x \in \mathbf{R}$ (Q.1 to Q.7)

1. $4^x - 10 \cdot 2^{x-1} = 24$

2. $4 \cdot 2^{2x} - 6^x = 18 \cdot 3^{2x}$

3. $3^{2x-3} - 9^{x-1} + 27^{2x/3} = 675.$

4. $7^{x+2} - \frac{1}{7} \cdot 7^{x+1} - 14 \cdot 7^{x-1} + 2 \cdot 7^x = 48$

5. $\left(\frac{5}{3}\right)^{x+1} \cdot \left(\frac{9}{25}\right)^{x^2+2x-11} = \left(\frac{5}{3}\right)^9$

6. $(3^{x^2-7.2x+3.9} - 9\sqrt{3}) \log(7-x) = 0$

7. $5^{2x} = 3^{2x} + 2 \cdot 5^x + 2 \cdot 3^x$

8. Find the logarithms of the following numbers to the base 2.

- | | | | | |
|------------------------------|-------------------------------|-------------------|--------------------|-------------------|
| (i) 1 | (ii) 2 | (iii) 4 | (iv) 8 | (v) $\frac{1}{2}$ |
| (vi) $\frac{1}{32}$ | (vii) $\frac{1}{16}$ | (viii) $\sqrt{2}$ | (ix) $\sqrt[3]{8}$ | (x) $2\sqrt{2}$ |
| (xi) $\frac{1}{\sqrt[5]{2}}$ | (xii) $\frac{1}{\sqrt[3]{8}}$ | | | |

9. Find the logarithms of the following numbers to the base $\frac{1}{3}$.

- | | | | | |
|---------|---------------------|--------------------------------|------------------|------------------------------|
| (i) 1 | (ii) $\frac{1}{3}$ | (iii) $\frac{1}{9}$ | (iv) 3 | (v) 9 |
| (vi) 81 | (vii) $\sqrt[3]{3}$ | (viii) $\frac{1}{\sqrt[3]{3}}$ | (ix) $9\sqrt{3}$ | (x) $\frac{1}{9\sqrt[4]{3}}$ |

10. Find all values of 'a' for which each of the following equalities hold true.

- | | | |
|------------------------------|--------------------------------|----------------------------|
| (i) $\log_2 a = 2$ | (ii) $\log_a 2 = 1$ | (iii) $\log_a 1 = 0$ |
| (iv) $\log_{10}(a(a+3)) = 1$ | (v) $\log_{1/3}(a^2 - 1) = -1$ | (vi) $\log_2(a^2 - 5) = 2$ |

11. Find the logarithms of the following numbers to the base 3.

- | | | | | |
|-----------------|-----------------------------|---------------------|--------------------|-------------------|
| (i) 1 | (ii) 3 | (iii) 9 | (iv) 81 | (v) $\frac{1}{3}$ |
| (vi) $\sqrt{3}$ | (vii) $\frac{1}{3\sqrt{3}}$ | (viii) $27\sqrt{3}$ | (ix) $\sqrt[3]{9}$ | |

12. Find the logarithms of the following numbers to the base $\frac{1}{2}$.

- | | | | |
|----------------|---------------------------|---------------------|---------------------------------|
| (i) 1 | (ii) $\frac{1}{2}$ | (iii) $\frac{1}{8}$ | (iv) 16 |
| (v) $\sqrt{2}$ | (vi) $\frac{1}{\sqrt{2}}$ | (vii) $2\sqrt{2}$ | (viii) $\frac{1}{4\sqrt[4]{2}}$ |

13. Find all values of 'a' for which the following equalities hold true.

- | | | |
|---------------------|--------------------------|----------------------------|
| (i) $\log_3 a = 2$ | (ii) $\log_{1/3}(a) = 4$ | (iii) $\log_{1/3}(a) = 0$ |
| (iv) $\log_a 1 = 0$ | (v) $\log_a(a+2) = 2$ | (vi) $\log_3(a^2 + 1) = 1$ |

14. Find all values of x for which the following equalities hold true.

- (i) $\log_2 x^2 = 1$ (ii) $\log_3 x = \log_3 (2 - x)$ (iii) $\log_4 x^2 = \log_4 x$
 (iv) $\log_{1/2} (2x + 1) = \log_{1/2} (x + 1)$ (v) $\log_{1/3} (x^2 + 8) = -2$

15. Find all the values of x for which the following equalities hold true.

- (i) $\log_2 x^2 = 2$ (ii) $\log_{1/4} x^2 = 1$
 (iii) $\log_{1/2} x - \log_{1/2} (3 - x) = 0$ (iv) $\log_2 (x + 1) - \log_2 (2x - 3) = 0$

16. Find the logarithms of the following numbers to the base 5.

- (i) 1 (ii) 5 (iii) 25 (iv) 625 (v) $\frac{1}{5}$ (vi) $\frac{1}{25}$
 (vii) $\frac{1}{\sqrt{5}}$ (viii) $\sqrt{\sqrt{5}}$ (ix) $5^{\frac{1}{2}}$ (x) $5^{\frac{1}{3}}$ (xi) $\sqrt[4]{5^3 \sqrt{5}}$

17. Find value of following

- (i) $\log_{\sqrt{3}-\sqrt{2}} (\sqrt{5-2\sqrt{6}})$ (ii) $\log_{\sqrt{5}+1} (6+2\sqrt{5})$
 (iii) $\log_7 3 \cdot \log_5 2 \cdot \log_3 7 \cdot \log_2 (125)$ (iv) $25^{\log_5 3}$
 (v) $6^{\log_6 5} + 3^{\log_9 16}$ (vi) $\log_6 4 + \frac{1}{\log_9 6}$

18. Find the value of following

- (i) $\log_5 \left(\frac{1}{\sqrt{5}} \right)$ (ii) $\log_3 (4 \sin^2(x) + 4 \cos^2(x) - 1)$
 (iii) $\log_{12} 8 + \log_{12} 3 + \log_{12} 6$ (iv) $\log_5 \frac{500}{3} - \log_5 \frac{4}{3}$
 (v) $\log_{39} \frac{15}{7} + \log_{39} \frac{13}{3} - \log_{39} \frac{5}{21}$ (vi) $2 \log_6 2 + 3 \log_6 3 + \log_6 12$

19. If $2 \left(\sqrt{3 + \sqrt{5 - \sqrt{13 + \sqrt{48}}}} \right) = \sqrt{a} + \sqrt{b}$ where a and b are natural number find $(a + b)$.

EXERCISE (O-1)

- If $2^a = 3$ and $9^b = 4$ then value of (ab) is-
 (A) 1 (B) 2 (C) 3 (D) 4
- If $\log_2 (4 + \log_3 (x)) = 3$, then sum of digits of x is -
 (A) 3 (B) 6 (C) 9 (D) 18
- Sum of all the solution(s) of the equation $\log_{10} (x) + \log_{10} (x + 2) - \log_{10} (5x + 4) = 0$ is-
 (A) -1 (B) 3 (C) 4 (D) 5
- The product of all the solutions of the equation $x^{1+\log_{10} x} = 100000x$ is-
 (A) 10 (B) 10^5 (C) 10^{-5} (D) 1
- If x_1 and x_2 are the roots of equation $e^{3/2} \cdot x^{2 \log x} = x^4$, then the product of the roots of the equation is -
 (A) e^2 (B) e (C) $e^{3/2}$ (D) e^{-2}

[MULTIPLE CORRECT CHOICE TYPE]

9. If $(\log_{\beta} \alpha)^2 + (\log_{\alpha} \beta)^2 = 79$, $(\alpha > 0, \beta > 0, \alpha \neq 1, \beta \neq 1)$ then value of $(\log_{\beta} \alpha) + (\log_{\alpha} \beta)$ can be-
 (A) 7 (B) -9 (C) 9 (D) -7
10. Which of the following statements is(are) correct ?
 (A) $7^{1/7} > (42)^{1/14} > 1$ (B) $\log_3(5) \log_7(9) \log_{11}(13) > -2$
 (C) $\sqrt{99+70\sqrt{2}} + \sqrt{99-70\sqrt{2}}$ is rational (D) $\frac{1}{\log_4 3} + \frac{1}{\log_7 3} > 3$

EXERCISE (S-1)

1. Let A denotes the value of $\log_{10} \left(\frac{ab + \sqrt{(ab)^2 - 4(a+b)}}{2} \right) + \log_{10} \left(\frac{ab - \sqrt{(ab)^2 - 4(a+b)}}{2} \right)$
 when $a = 43$ and $b = 57$ and B denotes the value of the expression $(2^{\log_6 18}) \cdot (3^{\log_6 3})$.
 Find the value of (A.B).
2. Compute the following : (a) $\log_{1/3} \sqrt[4]{729 \cdot \sqrt[3]{9^{-1} \cdot 27^{-4/3}}}$ (b) $a^{\frac{\log_b(\log_b N)}{\log_b a}}$
3. Find the square of the sum of the roots of the equation
 $\log_3 x \cdot \log_4 x \cdot \log_5 x = \log_3 x \cdot \log_4 x + \log_4 x \cdot \log_5 x + \log_5 x \cdot \log_3 x$.
4. Calculate : $4^{5 \log_{4\sqrt{2}}(3-\sqrt{6}) - 6 \log_8(\sqrt{3}-\sqrt{2})}$
5. Simplify : $\frac{81^{\frac{1}{\log_5 9}} + 3^{\frac{3}{\log_{\sqrt{6}} 3}}}{409} \cdot \left((\sqrt{7})^{\frac{2}{\log_{25} 7}} - (125)^{\log_{25} 6} \right)$
6. Simplify : $5^{\log_{1/5}(\frac{1}{2})} + \log_{\sqrt{2}} \frac{4}{\sqrt{7} + \sqrt{3}} + \log_{1/2} \frac{1}{10 + 2\sqrt{21}}$
7. Given that $\log_2 a = s$, $\log_4 b = s^2$ and $\log_{c^2}(8) = \frac{2}{s^3 + 1}$. Write $\log_2 \frac{a^2 b^5}{c^4}$ as a function of 's'
 $(a, b, c > 0, c \neq 1)$.
8. Find the value of $49^{(1-\log_7 2)} + 5^{-\log_5 4}$.
9. Prove that $\frac{\log_2 24}{\log_{96} 2} - \frac{\log_2 192}{\log_{12} 2} = 3$.
10. Prove that $a^x - b^y = 0$ where $x = \sqrt{\log_a b}$ & $y = \sqrt{\log_b a}$, $a > 0$, $b > 0$ & $a, b \neq 1$.

8. Solve following equations

(i) $|x - 1| = 5$

(ii) $|x + 1| = |2x - 1|$

(iii) $|x - 1| + 2|x + 1| = 10$

(iv) $|2x - 1| - |x - 3| = 7$

(v) $x + |3x - 2| = 2$

(vi) $|\log_2 x| = 3$

9. Solve the system of equations :

$$\log_a x \log_a (xyz) = 48$$

$$\log_a y \log_a (xyz) = 12, a > 0, a \neq 1$$

$$\log_a z \log_a (xyz) = 84$$

10. Let $y = \sqrt{\log_2 3 \cdot \log_2 12 \cdot \log_2 48 \cdot \log_2 192 + 16} - \log_2 12 \cdot \log_2 48 + 10$. Find $y \in \mathbb{N}$.

11. Let 'L' denotes the antilog of 0.4 to the base 1024.

and 'M' denotes the number of digits in 6^{10} (Given $\log_{10} 2 = 0.3010$, $\log_{10} 3 = 0.4771$)

and 'N' denotes the number of positive integers which have the characteristic 2, when base of the logarithm is 6.

Find the value of LMN.

12. Find the product of the positive roots of the equation $\sqrt{(2017)}(x)^{\log_{2017} x} = x^2$.

13. If (x_1, y_1) and (x_2, y_2) are the solution of the system of equation.

$$\log_{225}(x) + \log_{64}(y) = 4$$

$$\log_x(225) - \log_y(64) = 1,$$

then show that the value of $\log_{30}(x_1 y_1 x_2 y_2) = 12$.

14. (a) Given : $\log_{10} 34.56 = 1.5386$, find $\log_{10} 3.456$; $\log_{10} 0.3456$ & $\log_{10} 0.003456$.

(b) Find the number of positive integers which have the characteristic 3, when the base of the logarithm is 7.

(c) If $\log_{10} 2 = 0.3010$ & $\log_{10} 3 = 0.4771$, find the value of $\log_{10}(2.25)$

(d) Find the antilogarithm of 0.75, if the base of the logarithm is 2401.

15. (a) If $x, y > 0$, $\log_y x + \log_x y = \frac{10}{3}$ and $xy = 144$, then $\frac{x+y}{2} = \sqrt{N}$ where N is a natural number, find the value of N.

(b) If $x = 1 + \log_a bc$, $y = 1 + \log_b ca$ and $z = 1 + \log_c ab$, then prove that $xyz = xy + yz + zx$.

(c) If $\frac{\log_a N}{\log_c N} = \frac{\log_a N - \log_b N}{\log_b N - \log_c N}$ where $N > 0$ & $N \neq 1$, $a, b, c > 0$ & not equal to 1, then prove that $b^2 = ac$.

EXERCISE (JA)

1. $\log_{3/4} \log_8(x^2 + 7) + \log_{1/2} \log_{1/4}(x^2 + 7)^{-1} = -2.$ [REE 2000, 5 out of 100]

2. Number of solutions of $\log_4(x-1) = \log_2(x-3)$ is [JEE 2001 (Screening)]

- (A) 3 (B) 1 (C) 2 (D) 0

3. Let (x_0, y_0) be the solution of the following equations [JEE 2011, 3 (-1)]

$$(2x)^{\ln 2} = (3y)^{\ln 3}$$

$$3^{\ln x} = 2^{\ln y}$$

Then x_0 is

- (A) $\frac{1}{6}$ (B) $\frac{1}{3}$ (C) $\frac{1}{2}$ (D) 6

4. The value of $6 + \log_{\frac{3}{2}} \left(\frac{1}{3\sqrt{2}} \sqrt{4 - \frac{1}{3\sqrt{2}}} \sqrt{4 - \frac{1}{3\sqrt{2}}} \sqrt{4 - \frac{1}{3\sqrt{2}}} \dots \right)$ is [JEE 2012, 4M]

5. If $3^x = 4^{x-1}$, then $x =$ [JEE-Advanced 2013, 4, (-1)]

- (A) $\frac{2\log_3 2}{2\log_3 2 - 1}$ (B) $\frac{2}{2 - \log_2 3}$ (C) $\frac{1}{1 - \log_4 3}$ (D) $\frac{2\log_2 3}{2\log_2 3 - 1}$

ANSWERS

ELEMENTARY EXERCISE

1. $x = 3$ 2. $x = -2$ 3. $x = 3$ 4. $x = 0$ 5. $x = \frac{-7}{2}, 2$ 6. $x = \frac{1}{5}, 6$ 7. $x = 1$
8. (i) 0 (ii) 1 (iii) 2 (iv) 3 (v) -1 (vi) -5 (vii) -4 (viii) $\frac{1}{2}$
(ix) 1 (x) $\frac{3}{2}$ (xi) $-\frac{1}{5}$ (xii) $-\frac{3}{7}$
9. (i) 0 (ii) 1 (iii) 2 (iv) -1 (v) -2 (vi) -4 (vii) $-\frac{1}{3}$ (viii) $\frac{1}{7}$ (ix) $-\frac{5}{2}$ (x) $\frac{9}{4}$
10. (i) 4 (ii) 2 (iii) $a > 0, a \neq 1$ (iv) -5, 2 (v) -2, 2 (vi) -3, 3
11. (i) 0 (ii) 1 (iii) 2 (iv) 4 (v) -1 (vi) $\frac{1}{2}$ (vii) $-\frac{3}{2}$ (viii) $\frac{7}{2}$ (ix) $\frac{2}{7}$
12. (i) 0 (ii) 1 (iii) 3 (iv) -4 (v) $-\frac{1}{2}$ (vi) $\frac{1}{2}$ (vii) $-\frac{3}{2}$ (viii) $\frac{9}{4}$
13. (i) 9 (ii) $\frac{1}{81}$ (iii) 1 (iv) $a > 0, a \neq 1$ (v) 2 (vi) $\sqrt{2}, -\sqrt{2}$
14. (i) $\sqrt{2}, -\sqrt{2}$ (ii) 1 (iii) 1 (iv) 0 (v) 1, -1
15. (i) $x = \pm 2$ (ii) $x = \pm \frac{1}{2}$ (iii) $x = \frac{3}{2}$ (iv) $x = 4$
16. (i) 0 (ii) 1 (iii) 2 (iv) 4 (v) -1 (vi) -2 (vii) $-\frac{1}{2}$
(viii) $\frac{1}{4}$ (ix) $\frac{1}{2}$ (x) $\frac{1}{3}$ (xi) $\frac{1}{3}$
17. (i) 1 (ii) 2 (iii) 3 (iv) 9 (v) 9 (vi) 2
18. (i) $-\frac{1}{2}$ (ii) 1 (iii) 2 (iv) 3 (v) 1 (vi) 4 19. 8

EXERCISE (O-1)

1. A 2. C 3. C 4. D 5. A 6. B 7. C 8. A 9. B 10. B 11. D 12. B

EXERCISE (O-2)

1. C 2. D 3. A 4. A 5. B 6. D 7. C 8. A 9. B, C 10. A, B, D

EXERCISE (S-1)

1. 12 2. (a) -1 (b) $\log_b N$ 3. 3721 4. 9 5. 1 6. 6
7. $2s + 10s^2 - 3(s^3 + 1)$ 8. $\frac{25}{2}$
11. i. $\{1 + \sqrt{3}\}$ ii. $\{3\}$ iii. $\{4\}$ iv. $\{2\}$ v. $\{0\}$ vi. $\{-5\}$
vii. $\{3/2, 10\}$ viii. $\{10^{-1}, 10\}$ ix. $\{\sqrt{5}, 5\}$ x. $\{-(3 - \sqrt{3})/3, 8\}$

- 12.** i. $\{2^{-7}, 2\}$ ii. $\{1/\sqrt{2}, 1, 4\}$ iii. $\{-1, 0\}$ iv. $\{3\}$ v. $\{-9/5, 23\}$
vi. $\{2\}$ vii. $\{1\}$ viii. $\{2\}$ ix. $\{10^{-\sqrt{3}}, 10^{\sqrt{3}}\}$ x. $\{\log_3 4\}$

EXERCISE (S-2)

- 1.** 2 **2.** 1/6 **3.** $\frac{1+2ac}{2c+abc+1}$ **4.** 469 **5.** 4/9 **6.** $xy=2^9$ **7.** 1

8. (i) $x = 6, -4$ (ii) $x = 2, 0$ (iii) $x = 3, \frac{-11}{3}$ (iv) $x = 5, -9$ (v) $x = 1, 0$ (vi) $x = 8, \frac{1}{8}$

9. (a^4, a, a^7) or $\left(\frac{1}{a^4}, \frac{1}{a}, \frac{1}{a^7}\right)$ 10. $y = 6$ 11. 23040 12. $(2017)^2$

- 14.** (a) 0.5386; $\bar{1}.5386$; $\bar{3}.5386$ (b) 2058 (c) 0.3522 (d) 343 **15.** (a) 507

EXERCISE (JA)

- 1.** $x = 3$ or -3 **2.** B **3.** C **4.** 4 **5.** A,B,C

Important Notes